

Class 12

Mathematics

Master Reference

Complete Chapter Compendium

Thirteen chapters of the NCERT Class 12 Mathematics syllabus, prepared for CBSE Board examinations and for JEE Main and JEE Advanced. Every subtopic carries concept intuition, key formulas with validity conditions, board derivations, worked examples laddered from CBSE to JEE Advanced, practice exercises, MCQ distractor analysis, common pitfalls, and a cheat sheet. Each chapter is followed by a supplement and a Round 2 addendum, cross-checked against NCERT and a full JEE reference library, closing the gaps between the Board syllabus and what JEE actually tests. Corrections to the printed text are collected in the errata at the back.

01 Relations and Functions

02 Inverse Trigonometric Functions

03 Matrices

04 Determinants

05 Continuity and Differentiability

06 Applications of Derivatives

07 Integrals

08 Application of Integrals

09 Differential Equations

10 Vector Algebra

11 Three Dimensional Geometry

12 Linear Programming

13 Probability

Contents

CH 01 Relations and Functions	8
Subtopic 01 Types of Relations	9
Subtopic 02 Types of Functions	20
Subtopic 03 Composition and Inverse of Functions	30
Subtopic 04 Domain and Range of Real Functions	40
Subtopic 05 Standard Real Functions and Their Graphs	47
Subtopic 06 Even, Odd, and Periodic Functions	55
Subtopic 07 Algebra of Functions and Functional Equations	61
R2 Relations and Functions — Round 2 Addendum	67
Addendum A — Equivalence Classes: The Two Proofs and the Completion Problem	67
Addendum B — Composition: Transfer Laws, False Converses, and Running Composit...	70
Addendum C — Inverses on Restricted Domains	72
Addendum D — From Symmetry Axes to Periods; Periods Hidden in Functional Equati...	75
Addendum E — Counting Maps: Surjections Derived, Derangements, and Bell Numbe...	78
Addendum F — Iterates and Cycles	81
Addendum P — What JEE Actually Asks	83
CH 02 Inverse Trigonometric Functions	91
Principal Value Branches	92
Self-Map Identities (Reciprocal, Negative-Argument, Complementary)	103
Combination Identities (Sums, Differences & Multiple Angles)	113
Graphs, Composite Evaluation & Simplification	122
R2 Inverse Trigonometric Functions — Round 2 Addendum	130
Addendum A — Solving Inverse Trigonometric Equations	130
Addendum B — Conditional Identities: Reading the Relation Backwards	133
Addendum C — Multiple Angles Over the Whole Real Line	136
Addendum D — The Folded Composites as Functions: Waves and Solution Counts	138
Addendum E — Domains and Ranges of Composite Inverse Trigonometric Functions	141
Addendum P — What JEE Actually Asks	143
CH 03 Matrices	150
Subtopic 00 Foundations & Representation	151

Subtopic 01	Matrix Algebra & Operations	155
Subtopic 02	Symmetric & Skew-Symmetric Matrices	165
Subtopic 03	Invertible Matrices & Row Operations	173
Subtopic 04	Special Square Matrices & Chapter Synthesis	181
R2	Matrices — Round 2 Addendum	187
	Addendum A — Near-Diagonal Powers: the Binomial Theorem for Matrices	187
	Addendum B — Matrix Equations and Systems Through the Inverse	191
	Addendum C — Counting Structured Matrices	194
	Addendum D — Inversion Meets Symmetry; Orthogonal Matrices	195
	Addendum E — Trace: From Assertion to Tool	198
	Addendum P — What JEE Actually Asks	201
CH 04	Determinants	209
	Part 1 Determinant: Meaning, Definition & Expansion	210
	Part 2 Properties of Determinants	216
	Part 3 Area of a Triangle, Collinearity & Equation of a Line	224
	Part 4 Minors and Cofactors	229
	Part 5 Inverse of a Matrix & System of Linear Equations	236
	Part 6 Advanced Techniques: Multiplication, Differentiation & Special Determinants	243
R2	Determinants — Round 2 Addendum	248
	Addendum A — Circulant Determinants and the Cube Roots of Unity	248
	Addendum B — Trigonometric Determinants	252
	Addendum C — Alternants with a Missing Power	255
	Addendum D — Concurrency, the Family of Lines, and Polygon Areas	258
	Addendum E — What to Actually Do When $\Delta = 0$	260
	Addendum P — What JEE Actually Asks	263
CH 05	Continuity and Differentiability	273
	1. Continuity & Differentiability — The Foundations	274
	2. The Differentiation Toolkit — Standard Derivatives & Rules	284
	3. The Chain Rule — Derivatives of Composite Functions	290
	4. Implicit Differentiation	295
	5. Derivatives of Inverse Trigonometric Functions	300
	6. Logarithmic & Parametric Differentiation	306
	7. Second-Order Derivatives & the Mean Value Theorems	314

R2	Continuity and Differentiability — Round 2 Addendum	321
	Addendum A — Standard Limits: the Engine Behind Every Continuity Question	321
	Addendum B — Differentiating Through Moduli: the General Rules	325
	Addendum C — Functional Equations: Continuity Propagated from a Single Point	328
	Addendum D — Cauchy's Mean Value Theorem and the Arithmetic of Roots	332
	Addendum E — Greatest Integer Compositions: Counting Discontinuities Correctly	335
	Addendum P — What JEE Actually Asks	337
CH 06	Applications of Derivatives	349
	Subtopic 01 Rate of Change of Quantities	350
	Subtopic 02: Monotonicity Increasing and Decreasing Functions	359
	Subtopic 03 Maxima and Minima	368
	Appendix A: Tangents and Normals (JEE-relevant — outside current CBSE board syllabus)	378
	Appendix B: Approximations and Differentials (JEE-relevant — outside current CBSE board syllabus)	382
R2	Applications of Derivatives — Round 2 Addendum	385
	Addendum A — Rolle's Theorem and the Mean Value Theorem	385
	Addendum B — Monotonicity II: Counting Roots and Comparative Inequalities	389
	Addendum C — When the Second Test Is Silent: Higher Derivatives, Concavity, Inflection Points	392
	Addendum D — Extrema on Open and Unbounded Domains; Parametrised Optimisation	395
	Addendum E — Tangents and Normals II: Subtangent, Subnormal, Intercepts	399
	Addendum P — What JEE Actually Asks	402
CH 07	Integrals	410
	Subtopic 01 Foundations of Integration: Indefinite Integrals, Properties & Standard Forms	414
	Subtopic 04 Integration by Substitution	424
	Subtopic 05 Integration using Trigonometric Identities	430
	Subtopic 06 Integrals of Special Forms (Quadratic Denominators & Roots)	435
	Subtopic 07 Integration by Partial Fractions	441
	Subtopic 08 Integration by Parts	447
	Subtopic 02 Definite Integrals: Definition, the Fundamental Theorem & Limit of a Sum	452
	Subtopic 09 Evaluation of Definite Integrals by Substitution	458
	Subtopic 03 Properties of Definite Integrals: The Symmetry Toolkit	463
R2	Integrals — Round 2 Addendum	471
	Addendum A — Differentiating an Integral Whose Limits Move	471

Addendum B — Reading a Limit Backwards: Riemann Sums in Reverse	474
Addendum C — King Property Power Plays	477
Addendum D — Rational Functions of $\sin x$ and $\cos x$: the Half-Angle Substitution	480
Addendum E — Reduction Formulas: Integrating Powers Systematically	483
Addendum F — Breakpoint Integrands II: Greatest Integer, Fractional Part, Min/Max	485
Addendum P — What JEE Actually Asks	488
CH 08 Application of Integrals	496
Subtopic 01 Area under Simple Curves	496
Subtopic 02 Area between Two Curves	506
Subtopic 03 Advanced Regions: Modulus, Inequalities & Composite Areas	515
R2 Application of Integrals — Round 2 Addendum	523
Addendum A — Segments of Conics: Chords, Tangents and the Latus Rectum	523
Addendum B — Parametric Areas: the Astroid and the Cycloid	526
Addendum C — Inverse Functions and the Mirror Trick	529
Addendum D — When the Area Is the Unknown: Hunting a Parameter	531
Addendum E — Multi-Hump Periodic Areas	533
Addendum P — What JEE Actually Asks	536
CH 09 Differential Equations	543
Part 1 — Order, Degree & Solutions	544
Part 2 — Formation of Differential Equations	553
Part 3 — Separable & Homogeneous Equations	560
Part 4 — Linear Differential Equations	568
Part 5 — Applications of Differential Equations	576
R2 Differential Equations — Round 2 Addendum	583
Addendum A — Exact Differential Equations and Solving by Inspection	583
Addendum B — Two Reducible Types: Bernoulli and $f(ax+by+c)$	587
Addendum C — Geometric Conditions Beyond the Subnormal	589
Addendum D — Second-Order Equations by Lowering the Order	591
Addendum P — What JEE Actually Asks	593
CH 10 Vector Algebra	601
Vector Algebra — Subtopic 01: Vector Operations & Components	601
Vector Algebra — Subtopic 02: Product of Two Vectors	611

	Vector Algebra — Subtopic 03: Product of Three Vectors & Vector Applications	619
R2	Vector Algebra — Round 2 Addendum	627
	Addendum A — Resolution Along Oblique Directions and Angle Bisectors	627
	Addendum B — Magnitude Algebra and Constraint Problems	631
	Addendum C — Nested Cross Products and Vector Equations	634
	Addendum D — Position-Vector Geometry of the Triangle	637
	Addendum P — What JEE Actually Asks	641
CH 11	Three Dimensional Geometry	648
	Foundations of 3D Coordinate Geometry	649
	Direction Cosines, Direction Ratios & Equations of a Line in Space	658
	Shortest Distance & Angle Between Two Lines	667
	Planes in Space — Equations, Family, Angles & Distances	676
	The Perpendicular Toolkit — Distances, Feet & Images	684
	Line-Plane & Plane-Plane Configurations	692
R2	Three Dimensional Geometry — Round 2 Addendum	700
	Addendum A — The Line Where Two Planes Meet	700
	Addendum B — Both Ends of the Bridge: Closest Points on Skew Lines	702
	Addendum C — Four Points, One Plane: Coplanarity and Extended Plane-Building	705
	Addendum D — Mirror Shortcuts: Origin Feet, Coordinate Walls, Shortest Paths	706
	Addendum E — Locus Problems: Turning Geometry into Equations	708
	Addendum P — What JEE Actually Asks	710
CH 12	Linear Programming	721
	Subtopic 01 Formulation & Graphical Solution	722
	Subtopic 02 Mathematical Formulation & LPP Applications	732
	Subtopic 03 Special Cases of LPP Solutions	742
	Chapter 12 — Complete Master Reference	749
R2	Linear Programming — Round 2 Addendum	751
	Addendum A — Why the Corner-Point Method Works: The Convexity Proof	751
	Addendum B — Parametric Objectives: Pinning the Optimum to a Chosen Vertex	753
	Addendum C — Infeasibility and Redundancy, Proved Without a Graph	756
	Addendum D — Measuring the Region: Vertex Enumeration, Area, Lattice Points	759
	Addendum E — Whole-Number Answers: When the Optimum Is Fractional	761

CH 13	Probability	765
	Subtopic 01 Conditional Probability & Independence	766
	Subtopic 02 Total Probability & Bayes' Theorem	774
	Subtopic 03 Random Variables & Distributions	782
	Subtopic 04 Bernoulli Trials & Binomial Distribution	790
	Master Formula Sheet & Revision Capstone	798
R2	Probability — Round 2 Addendum	801
	Addendum A — Three Events: Unions, Exactly-k, and Inclusion-Exclusion	801
	Addendum B — Pairwise but Not Mutual Independence: Building It, Breaking It	805
	Addendum C — Chained Bayes: Staged Evidence and Transfer Experiments	808
	Addendum D — Expectation Beyond $aX+b$: Sums, Products, and Indicator Variables	810
	Addendum E — Waiting Times and Turn-Taking: Probability by Infinite Series	814
	Addendum P — What JEE Actually Asks	817
	Errata	830
	Chapter 1: Relations and Functions	830
	Chapter 3: Matrices	831
	Chapter 11: Three-Dimensional Geometry	831

Relations and Functions

Class 12 Mathematics, Chapter 1 (Master Notes)

Drona Ed Tech

Contents

Subtopic 01 — Types of Relations	2
Section 1: Concept Intuition	2
Section 2: Key Definitions & Properties	3
Section 3: Key Procedures & Logical Framework	4
Section 4: Worked Examples	5
Section 5: Practice Exercises	7
Section 6: Crack the MCQ	8
Section 7: Common Student Pitfalls & Pro-Tips	9
Section 8: Cheat Sheet Box	9
Subtopic 02 — Types of Functions	11
Section 1: Concept Intuition	11
Section 2: Key Definitions & Properties	12
Section 3: Key Procedures & Logical Framework	13
Section 4: Worked Examples	14
Section 5: Practice Exercises	16
Section 6: Crack the MCQ	17
Section 7: Common Student Pitfalls & Pro-Tips	17
Section 8: Cheat Sheet Box	18
Subtopic 03 — Composition and Inverse of Functions	19
Section 1: Concept Intuition	19
Section 2: Key Definitions & Properties	20
Section 3: Key Procedures & Derivations	21
Section 4: Worked Examples	22
Section 5: Practice Exercises	24
Section 6: Crack the MCQ	25
Section 7: Common Student Pitfalls & Pro-Tips	26
Section 8: Cheat Sheet Box	26
Subtopic 04 — Domain and Range of Real Functions	28
Section 1: Concept Intuition	28
Section 2: Key Definitions & Properties	29
Section 3: Key Procedures & Derivations	29

Section 4: Worked Examples	30
Section 5: Practice Exercises	32
Section 6: Crack the MCQ	32
Section 7: Common Student Pitfalls & Pro-Tips	33
Section 8: Cheat Sheet Box	33
Subtopic 05 — Standard Real Functions and Their Graphs	35
Section 1: Concept Intuition	35
Section 2: Key Definitions & Properties	37
Section 3: Key Procedures & Logical Framework	37
Section 4: Worked Examples	38
Section 5: Practice Exercises	39
Section 6: Crack the MCQ	39
Section 7: Common Student Pitfalls & Pro-Tips	40
Section 8: Cheat Sheet Box	40
Subtopic 06 — Even, Odd, and Periodic Functions	42
Section 1: Concept Intuition	42
Section 2: Key Definitions & Properties	43
Section 3: Key Procedures & Derivations	43
Section 4: Worked Examples	44
Section 5: Practice Exercises	45
Section 6: Crack the MCQ	45
Section 7: Common Student Pitfalls & Pro-Tips	46
Section 8: Cheat Sheet Box	47
Subtopic 07 — Algebra of Functions and Functional Equations	48
Section 1: Concept Intuition	48
Section 2: Key Definitions & Properties	48
Section 3: Key Procedures & Derivations	49
Section 4: Worked Examples	50
Section 5: Practice Exercises	51
Section 6: Crack the MCQ	51
Section 7: Common Student Pitfalls & Pro-Tips	52
Section 8: Cheat Sheet Box	53

Subtopic 01 — Types of Relations

Exam Relevance: This subtopic is a guaranteed scorer. CBSE Boards almost always carry a 1-mark definition or a 3–5 mark “show that R is an equivalence relation” proof. JEE Main asks 1 MCQ on classifying a given relation or counting relations, and JEE Advanced layers it with equivalence classes, partitions, and counting (multi-correct). CUET tests direct classification and definitions. (*Not in NEET — the medical paper has no mathematics section.*)

Section 1: Concept Intuition

Forget formulas for a minute. A **relation** is just a *rule that decides which pairs of things are “connected.”* That’s the whole idea.

Picture every student in your coaching batch. Now draw an arrow from one student to another whenever the rule “*sits in the same row as*” is true. Some arrows appear, some don’t. That entire collection of arrows — that bundle of “who is connected to whom” — **is** the relation. Mathematically we don’t draw arrows forever; we just list the connected pairs as ordered pairs (a, b) . So a relation R on a set A is nothing more grand than a **chosen subset of all possible ordered pairs** you could form from A , i.e. a subset of $A \times A$.

Because R is a *choice* of pairs, there are two boring extremes worth naming first. If you connect **nothing** to anything, you get the **empty (void) relation**, $R = \emptyset$. If you connect **everything** to **everything**, you get the **universal relation**, $R = A \times A$. And if you connect every element only to *itself* — like the rule “*is the same person as*” — you get the **identity relation**, I_A .

The interesting relations live between these extremes, and we judge them by three “personality traits.”

Reflexive — the self-loop. Does every element relate to itself? “*Is in the same class as*” — yes, you are obviously in the same class as yourself. Every person gets a little loop back to themselves. Miss even *one* self-loop and the relation is no longer reflexive.

Symmetric — the two-way street. If a is connected to b , is b automatically connected back to a ? “*Is a neighbour of*” is symmetric — if my house is next to yours, yours is next to mine. But “*is the elder sibling of*” is **not** symmetric: if Riya is Aman’s elder sibling, Aman is certainly not Riya’s elder sibling.

Transitive — the shortcut / the chain. If $a \rightarrow b$ and $b \rightarrow c$, does the shortcut $a \rightarrow c$ exist? “*Lives in the same city as*” is transitive — if Priya and Arjun share a city, and Arjun and Sneha share a city, then Priya and Sneha share it too. But “*is a friend of*” is famously **not** transitive — your friend’s friend need not be your friend.

Think about it this way... Imagine the Indian Railways network as a relation between stations. “*Is directly connected by a train to*” is symmetric (the same track runs both ways) but **not** transitive (Patna $\rightarrow$ Delhi and Delhi $\rightarrow$ Jaipur direct trains don’t guarantee a *direct* Patna $\rightarrow$ Jaipur train). Now change the rule to “*can reach by any sequence of trains.*” Suddenly it becomes transitive **and** symmetric **and** reflexive (you can trivially “reach” your own station). A relation that has all three traits at once is called an **equivalence relation** — and its superpower is that it neatly **sorts the whole set into disjoint buckets** (here: groups of mutually-reachable stations). Each bucket is an *equivalence class*.

Figure 1.1 *Directed-graph panels on a 3-element set a, b, c . Panel 1 (Reflexive): self-loops on every node. Panel 2 (Symmetric): every arrow shown with a matching reverse arrow. Panel 3 (Transitive): arrows $a \rightarrow b$, $b \rightarrow c$, plus the “shortcut” arrow $a \rightarrow c$ highlighted. Panel 4 (Equivalence): nodes grouped into shaded buckets with complete two-way connections inside each bucket and none across. (see next page)*

Limiting conditions & assumptions. These traits are only defined for a relation **on a single set A** (from A to itself) — they make no sense for a relation between two *different* sets. Crucially, every trait is **relative to the underlying set**: a relation can be reflexive on A but fail to be reflexive when you enlarge A , because new elements bring new self-loops to satisfy. And beware the empty relation — it sneaks past your intuition (more on that in the pitfalls).

Section 2: Key Definitions & Properties

Let A be a non-empty set, $R \subseteq A \times A$ a relation on A , and $a, b, c \in A$. Write aRb to mean $(a, b) \in R$.

Core definitions

Empty / Void: $R = \emptyset$

Universal: $R = A \times A$

Identity: $I_A = \{(a, a) : a \in A\}$

Reflexive: $(a, a) \in R \quad \forall a \in A$

Symmetric: $(a, b) \in R \implies (b, a) \in R$

Transitive: $(a, b) \in R$ and $(b, c) \in R \implies (a, c) \in R$

Antisymmetric: $(a, b) \in R$ and $(b, a) \in R \implies a = b$

Equivalence: R is reflexive **and** symmetric **and** transitive

Equivalence class. If R is an equivalence relation on A , the class of a is

$$[a] = \{x \in A : xRa\}.$$

The distinct classes are mutually disjoint and their union is A — they **partition** A .

Counting properties (these play the role that “formulas” play in physics here). For a set A with $|A| = n$:

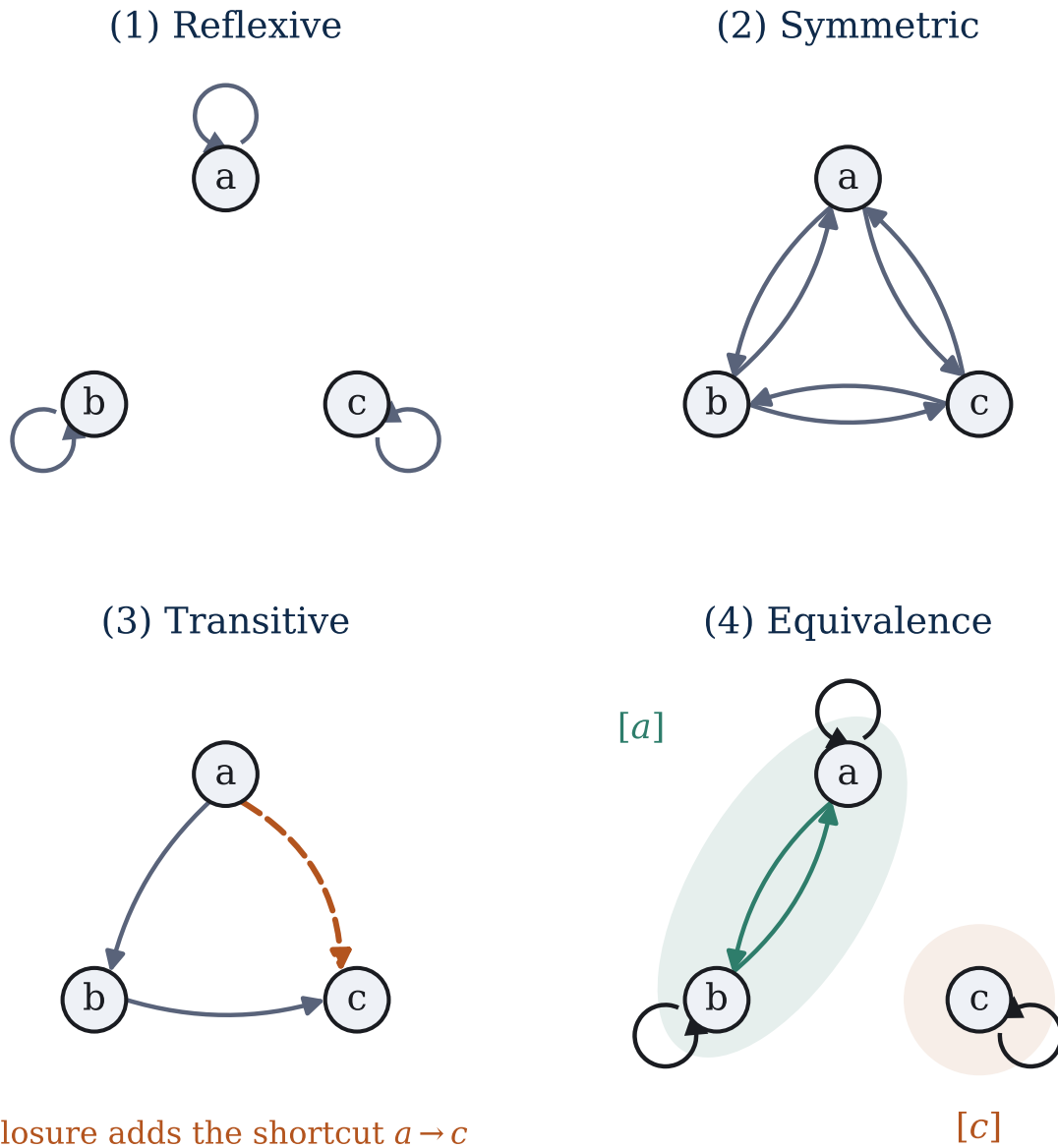


Figure 1.1 Directed-graph panels on a 3-element set a, b, c . Panel 1 (Reflexive): self-loops on every node. Panel 2 (Symmetric): every arrow shown with a matching reverse arrow. Panel 3 (Transitive): arrows $a \rightarrow b$, $b \rightarrow c$, plus the “shortcut” arrow $a \rightarrow c$ highlighted. Panel 4 (Equivalence): nodes grouped into shaded buckets with complete two-way connections inside each bucket and none across.

Quantity	Count
Relations from A to B (with $ B = m$)	2^{mn}
Relations on A	2^{n^2}
Reflexive relations on A	2^{n^2-n}
Symmetric relations on A	$2^{n(n+1)/2}$
Equivalence relations on A	B_n (the n -th Bell number)

where the Bell numbers begin $B_1 = 1$, $B_2 = 2$, $B_3 = 5$, $B_4 = 15$, $B_5 = 52, \dots$

Notation lock (chapter-wide): A, B for sets; R, S for relations; a, b, c, x, y for elements; $n = |A|$ for cardinality; $[a]$ for an equivalence class; $\mathbb{N}, \mathbb{Z}, \mathbb{R}$ for the number systems. (These carry forward unchanged into Subtopics 02–04.)

Section 3: Key Procedures & Logical Framework

There is no single “derivation” for this subtopic, so here is the working machinery — first the *testing* procedure, then a model *proof*, with the reasoning made explicit.

Procedure A — Classifying any relation R on A .

1. **Reflexive check (do this FIRST).** Scan the elements of A one by one and confirm $(a, a) \in R$ for *every* one. *Why first?* It is the cheapest test and the most common killer — a single missing self-loop instantly rules out reflexivity **and** equivalence, saving you from doing the harder checks.
2. **Symmetric check.** For each pair $(a, b) \in R$ with $a \neq b$, confirm (b, a) is also present. *Why this works:* symmetry is an “if-then,” so you only need to look at pairs that are actually in R — and you only need **one** missing reverse pair to declare “not symmetric.” Hunt for a counterexample rather than verifying everything.
3. **Transitive check.** For every chain $(a, b) \in R$ and $(b, c) \in R$, confirm the shortcut $(a, c) \in R$. *Reasoning:* again it is an “if-then,” so absent chains cost you nothing. One broken shortcut ends it.
4. **Conclude.** All three present $\Rightarrow$ **equivalence**. Reflexive + transitive + antisymmetric $\Rightarrow$ a *partial order* (good to recognise for JEE).

Procedure B — Model proof that a relation is an equivalence relation.

Claim: For a fixed positive integer n , define R on $\mathbb{Z}$ by

$$aRb \iff (a - b) \text{ is divisible by } n.$$

Then R is an equivalence relation. (This is “congruence modulo n ,” the archetype of every equivalence relation you will meet.)

Reflexive. For any $a \in \mathbb{Z}$, $a - a = 0 = n \cdot 0$, which is divisible by n . *Physical reasoning:* zero is divisible by everything, so every integer trivially relates to itself. Hence aRa for all a . ✓

Symmetric. Suppose aRb , so $a - b = nk$ for some integer k . Then

$$b - a = -(a - b) = n(-k),$$

and since $-k \in \mathbb{Z}$, $(b - a)$ is divisible by n . *Reasoning:* divisibility is unaffected by a sign flip. Hence bRa . ✓

Transitive. Suppose aRb and bRc , so $a - b = nk_1$ and $b - c = nk_2$ for integers k_1, k_2 . Adding,

$$a - c = (a - b) + (b - c) = nk_1 + nk_2 = n(k_1 + k_2).$$

Reasoning: the middle term b cancels, and a sum of multiples of n is again a multiple of n . Since $k_1 + k_2 \in \mathbb{Z}$, $(a - c)$ is divisible by n , so aRc . ✓

All three hold $\Rightarrow R$ is an equivalence relation. Its classes are the **residue classes** $[0], [1], \dots, [n - 1]$, which partition $\mathbb{Z}$ into exactly n buckets. ■

Procedure C — Why 2^{n^2} relations exist (the counting logic). A relation on A is a subset of $A \times A$. The set $A \times A$ has $n \times n = n^2$ ordered pairs, and each pair is independently either *in* or *out* of R — two choices per pair. By the multiplication principle, the total number of subsets is 2^{n^2} . For **reflexive** relations, the n self-pairs (a, a) are *forced in*, leaving only $n^2 - n$ free pairs, hence $2^{n^2 - n}$.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2)\}$. Determine whether R is reflexive, symmetric, and transitive.

Given: $A = \{1, 2, 3, 4\}$; R as above.

Reflexive: We need $(a, a) \in R$ for all $a \in A$. Present: $(1, 1), (2, 2), (3, 3), (4, 4)$ — all four. **Reflexive.** ✓

Symmetric: Look at the off-diagonal pairs. $(1, 2) \in R$ and $(2, 1) \in R$ — good. But $(3, 4) \in R$ while $(4, 3) \notin R$. One missing reverse is enough. **Not symmetric.** ✗

Transitive: Check every chain. $(1, 2) \& (2, 1) \Rightarrow (1, 1) \in R$ ✓; $(2, 1) \& (1, 2) \Rightarrow (2, 2) \in R$ ✓; $(3, 3) \& (3, 4) \Rightarrow (3, 4) \in R$ ✓; $(3, 4) \& (4, 4) \Rightarrow (3, 4) \in R$ ✓. No chain is broken. **Transitive.** ✓

Conclusion: R is **reflexive and transitive but not symmetric**, hence **not** an equivalence relation.

Example 2 — [CUET / JEE Main Speed Trap] On $A = \{1, 2, 3\}$ classify each, choosing the fastest route: (a) $R_1 = \emptyset$; (b) $R_2 = \{(1, 2), (2, 3), (1, 3)\}$.

The speed mindset: test reflexive first to kill equivalence, then hunt one counterexample.

(a) The empty-relation trap. Reflexive? A is non-empty, so we need $(1, 1)$ — absent. **Not reflexive.** Symmetric? Symmetry says “if $(a, b) \in R$ then...”. Since there is no pair in R_1 at all, the “if” is never triggered — there is **no counterexample possible**, so it is **vacuously symmetric**. Same logic: **vacuously transitive**. So R_1 is **symmetric and transitive but not reflexive** — a favourite trick option.

(b) Reflexive? $(1, 1) \notin R_2$ — out, equivalence dies instantly. Symmetric? $(1, 2) \in R_2$ but $(2, 1) \notin R_2$ — **not symmetric**. Transitive? Only chain is $(1, 2) \& (2, 3) \Rightarrow (1, 3)$, and $(1, 3) \in R_2$ ✓ — **transitive**. So R_2 is **transitive only**.

Pitfall avoided: never write “empty $\Rightarrow$ no properties.” Emptiness *grants* symmetry and transitivity for free.

Example 3 — [JEE Main Level] (classification + counting) Let $A = \{1, 2, 3\}$. (a) How many relations can be defined on A ? (b) For $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$, find the minimum number of ordered pairs that must be added to make R an equivalence relation. (c) How many equivalence relations on A contain the pair $(1, 2)$?

(a) Relations on A number $2^{n^2} = 2^{3^2} = 2^9 = \boxed{512}$.

(b) Test R as it stands. Reflexive: $(1, 1), (2, 2), (3, 3)$ all present ✓. Symmetric: $(1, 2) \& (2, 1)$ both present ✓. Transitive: $(1, 2) \& (2, 1) \Rightarrow (1, 1)$ ✓, $(2, 1) \& (1, 2) \Rightarrow (2, 2)$ ✓; no other chains. R is **already an equivalence relation**, so the answer is $\boxed{0}$. (The trap is to “helpfully” add $(2, 3), (3, 2), \dots$ — unnecessary, and it would over-merge the classes.)

(c) Equivalence relations on A correspond one-to-one with **partitions** of A , and there are $B_3 = 5$ of them. Those containing $(1, 2)$ must place 1 and 2 in the *same* block: either $\{1, 2\} \mid \{3\}$ or $\{1, 2, 3\}$. So $\boxed{2}$ equivalence relations contain $(1, 2)$.

Example 4 — [JEE Advanced Level] Consider two relations on $\mathbb{R}$: $aR_1b \iff |a - b| \leq 1$, and $aR_2b \iff (a - b) \in \mathbb{Z}$. (i) Classify R_1 fully. (ii) Prove R_2 is an equivalence relation and describe its classes. (iii) How many distinct R_2 -classes contain a representative in the interval $[0, 1)$?

(i) Classifying R_1 . Reflexive: $|a - a| = 0 \leq 1$ ✓. Symmetric: $|a - b| = |b - a|$, so $|a - b| \leq 1 \iff |b - a| \leq 1$ ✓. Transitive: try $a = 0, b = 1, c = 2$. Then $|0 - 1| = 1 \leq 1$ and $|1 - 2| = 1 \leq 1$, but $|0 - 2| = 2 \not\leq 1$. **Counterexample found — not transitive.** So R_1 is **reflexive and symmetric but not transitive**, hence **not an**

equivalence relation. (Moral: “closeness” relations almost always break transitivity, because small gaps accumulate.)

(ii) R_2 is an equivalence relation. Reflexive: $a - a = 0 \in \mathbb{Z}$ ✓. Symmetric: if $a - b \in \mathbb{Z}$ then $b - a = -(a - b) \in \mathbb{Z}$ ✓. Transitive: if $a - b \in \mathbb{Z}$ and $b - c \in \mathbb{Z}$, then $a - c = (a - b) + (b - c) \in \mathbb{Z}$ (sum of integers) ✓. Therefore R_2 is an equivalence relation. Two reals are related exactly when they have the **same fractional part**, so each class is

$$[x] = \{x + k : k \in \mathbb{Z}\},$$

a copy of the integers shifted by x — e.g. $[0.3] = \{\dots, -1.7, -0.7, 0.3, 1.3, 2.3, \dots\}$.

(iii) Each class is uniquely labelled by its fractional part, which lives in $[0, 1)$, and every value of $[0, 1)$ gives a different class. So the number of distinct classes meeting $[0, 1)$ equals the number of points in $[0, 1)$ — **uncountably infinitely many**.

Figure 1.2 number line R with tick marks at integers; three coloured class-systems shown as evenly spaced dots (spacing 1) offset by 0, 0.3, and 0.7, illustrating that each fractional part seeds its own infinite, evenly spaced class. (see next page)

Section 5: Practice Exercises

- [CBSE]** On $A = \{1, 2, 3\}$, classify $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)\}$ as reflexive / symmetric / transitive.
- [CUET]** State, with one-line reasons, which of reflexive, symmetric, transitive the empty relation on $\{x, y\}$ possesses.
- [JEE Main]** How many **reflexive** relations can be defined on a set with 3 elements?
- [JEE Main]** On $\mathbb{Z}$, define $aRb \iff a + b$ is even. Show R is an equivalence relation and state how many equivalence classes it has.
- [JEE Advanced]** For $A = \{1, 2, 3, 4\}$, find (a) the number of equivalence relations on A , and (b) the number of symmetric relations on A .

Answers: 1. Reflexive ✓, symmetric ✗, transitive ✓ (also antisymmetric). 2. Symmetric ✓ and transitive ✓ (vacuously), reflexive ✗. 3. $2^{n^2-n} = 2^6 = 64$. 4. Equivalence; 2 classes (the even integers and the odd integers). 5. (a) $B_4 = 15$; (b) $2^{n(n+1)/2} = 2^{10} = 1024$.

Section 6: Crack the MCQ

Q1. Which relation on $\mathbb{Z}$ is an equivalence relation?

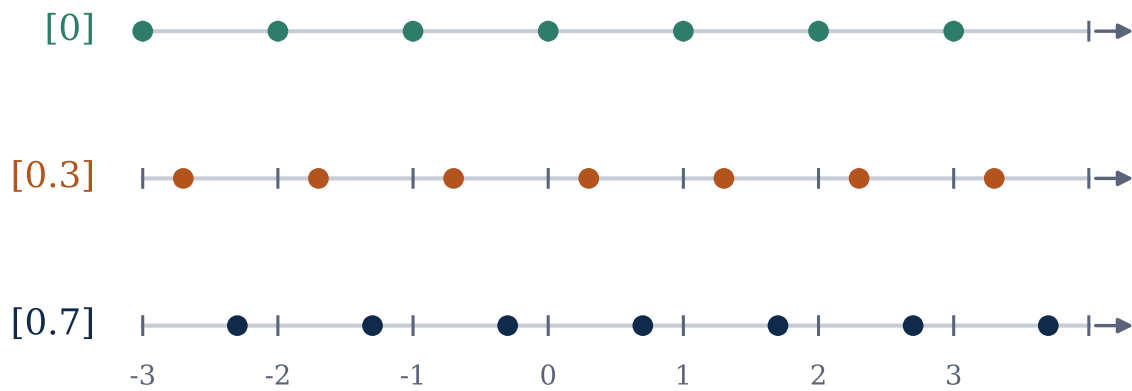


Figure 1.2 *number line $\mathbb{R}$ with tick marks at integers; three coloured class-systems shown as evenly spaced dots (spacing 1) offset by 0, 0.3, and 0.7, illustrating that each fractional part seeds its own infinite, evenly spaced class.*

- (a) $aRb \iff a \leq b$ (b) $aRb \iff (a - b)$ is divisible by 3 (c) $aRb \iff a < b$ (d) $aRb \iff |a| = b$

Answer: (b). Divisibility-by-3 is the congruence relation — reflexive, symmetric, transitive (proved in Section 3 with $n = 3$). *Why the traps fail:* **(a)** $\leq$ is reflexive and transitive but **antisymmetric, not symmetric** ($2 \leq 3$ but $3 \not\leq 2$) — it is a *partial order*, a classic look-alike. **(c)** $<$ is not even reflexive ($a < a$ is false). **(d)** fails reflexivity in general ($|-2| = 2 \neq -2$) and symmetry; students pick it by pattern-matching “=” to “equivalence.”

Q2. The empty relation on a non-empty set is:

- (a) reflexive only (b) symmetric and transitive but not reflexive (c) an equivalence relation (d) none of the three properties

Answer: (b). With no pairs present, the “if-then” conditions for symmetry and transitivity are **vacuously true**, but reflexivity demands self-pairs that are absent. *Why the traps fail:* **(a)** reverses the truth — it has *no* self-loops. **(c)** would need reflexivity. **(d)** is the most popular wrong answer: students assume “nothing in R ” means “no properties,” forgetting vacuous truth.

Q3. The number of reflexive relations on a set of 3 elements is:

- (a) 64 (b) 512 (c) 8 (d) 2^9

Answer: (a) 64. Fix the 3 self-pairs and let the remaining $3^2 - 3 = 6$ pairs be free: $2^6 = 64$. *Why the traps fail:* **(b)** and **(d)** are both $2^9 = 512$, the count of *all* relations (ignoring the reflexive constraint). **(c)** $= 2^3$ wrongly frees only the 3 self-pairs instead of fixing them.

Q4. On the set of all straight lines in a plane, define $\ell_1 R \ell_2 \iff \ell_1 \perp \ell_2$. Then R is:

- (a) reflexive (b) symmetric (c) transitive (d) an equivalence relation

Answer: (b) symmetric. If $\ell_1 \perp \ell_2$ then $\ell_2 \perp \ell_1$. *Why the traps fail:* **(a)** a line is never perpendicular to itself, so not reflexive. **(c)** if $\ell_1 \perp \ell_2$ and $\ell_2 \perp \ell_3$, then $\ell_1 \parallel \ell_3$ (not perpendicular) — transitivity fails. **(d)** needs all three. The “ $\perp$ ” relation is the textbook example of *symmetric-only*; the parallel relation “ $\parallel$ ” is the equivalence-relation cousin students often confuse it with.

Section 7: Common Student Pitfalls & Pro-Tips

1. **“Some self-loops” $\neq$ reflexive.** Reflexivity needs (a, a) for **every** element of A . Finding a few self-pairs and stopping is the single most common board-exam slip.
2. **The empty-relation blunder.** $\emptyset$ is **symmetric and transitive (vacuously)** but **not reflexive** on a non-empty set. Don’t write “empty, so no properties.”

3. **Symmetric vs antisymmetric confusion.** These are *not* opposites. A relation can be both (e.g. identity), neither, or one without the other. “Not symmetric” does **not** imply “antisymmetric.”
4. **Forgetting properties depend on the underlying set A .** The same list of pairs can be reflexive on one set and not on a larger one — always state the set.

Pro-Tip (speed): To classify fast, **test reflexive first** — it’s the cheapest check and a single missing (a, a) kills equivalence immediately. For symmetric and transitive, **hunt for one counterexample** instead of verifying every pair; you only need a single broken case to reject. For counting questions, memorise the three exponents: relations n^2 , reflexive $n^2 - n$, symmetric $\frac{n(n+1)}{2}$.

Section 8: Cheat Sheet Box

TYPES OF RELATIONS — QUICK RECALL

- **Relation on A :** any subset $R \subseteq A \times A$. Extremes: Empty $\emptyset$, Universal $A \times A$, Identity $I_A = \{(a, a)\}$.
- **Reflexive:** $(a, a) \in R \forall a$ — *self-loop on everyone*.
- **Symmetric:** $(a, b) \in R \Rightarrow (b, a) \in R$ — *two-way street*.
- **Transitive:** $(a, b), (b, c) \in R \Rightarrow (a, c) \in R$ — *chain has a shortcut*.
- **Antisymmetric:** $(a, b), (b, a) \in R \Rightarrow a = b$.
- **Equivalence = R + S + T $\Rightarrow$ partitions A into disjoint classes $[a] = \{x : xRa\}$.**

Counts ($n = |A|$): relations 2^{n^2} · reflexive 2^{n^2-n} · symmetric $2^{n(n+1)/2}$ · equivalence = B_n (1, 2, 5, 15, 52, ...).

Mnemonics: “Same-Mirror-Chain” = Reflexive-Symmetric-Transitive.
 · “**R + S + T = Rest** easy, it’s an **E**quivalence.” · Empty relation = “Symmetric & Transitive for free, never **R**eflexive.”

Subtopic 02 — Types of Functions

Exam Relevance: A perennial heavyweight. CBSE Boards reliably ask a 3–5 mark “show that f is bijective” proof (the NCERT favourite). JEE Main carries 1–2 MCQs on classifying a function as injective/surjective/bijective or on **counting** functions of a given type. JEE Advanced layers it — counting one-one/onto functions, degree-parity arguments, and codomain manipulation (often multi-correct). CUET tests direct classification and counting. (Not in NEET — no mathematics section.)

Section 1: Concept Intuition

A **function** is a machine with one strict rule: *feed it one input, and it returns exactly one output — never zero, never two*. A vending machine is the perfect picture. Press the button for cold coffee, and you get one cold coffee. The button can’t sometimes dispense nothing, and it can’t dispense two different drinks at once. Every button (input) is wired to exactly one slot (output). That wiring — the rule connecting inputs to outputs — is the function $f : A \rightarrow B$, where A is the **domain** (all valid buttons) and B is the **codomain** (all the slots the machine could in principle reach).

Now, the *types* of functions are just answers to two simple questions about that wiring.

Question 1: Do different buttons ever give the same drink?

If **no two different inputs ever share an output**, the function is **one-one** (or *injective*). Think of Aadhaar numbers: every citizen is assigned a distinct number — no two people share one. Different person in, different number out. The opposite case is **many-one**: two or more inputs collapse to the same output. The rule “the first letter of your name” is many-one — Aman, Anjali, and Arjun all map to ‘A’. So is squaring: both $+2$ and -2 map to 4. Whenever you see a $\pm$ symmetry or an even power, suspect many-one.

Question 2: Does every slot in the codomain actually get used?

If **every element of the codomain has at least one input pointing to it**, the function is **onto** (or *surjective*) — nothing in B is wasted. Picture a full Volvo bus where every single seat is occupied; no seat is left empty. The opposite is **into**: some codomain elements are never reached — empty seats remain. The squaring machine $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$ is into, because no negative number is ever an output; the entire negative half of the codomain sits empty.

Putting them together. A function that is **both one-one and onto** is **bijective** — a flawless one-to-one pairing where every input has its own private output and every output is claimed. This is like a perfect arranged-marriage register where every bride is matched to exactly one groom and nobody is left single on either side. Bijections

are special because they can be **run backwards** — they are precisely the *invertible* functions (the subject of Subtopic 03).

Think about it this way... Two graphical tests capture everything. Draw the graph and sweep a **horizontal line** across it: if the line ever cuts the curve at two or more points, two inputs share an output — *not one-one*. Separately, ask whether the curve's **range fills the entire codomain**: if yes, it's *onto*. A straight line $y = 2x - 7$ passes both — every horizontal line hits it exactly once, and it climbs from $-\infty$ to $+\infty$. A parabola fails both.

Figure 1.3 Four bubble-and-arrow mapping diagrams between an oval “A” (dots) and an oval “B” (dots). (1) One-one into: distinct arrows, one B-dot unhit. (2) Many-one onto: two A-dots arrow to one B-dot, all B-dots hit. (3) Bijective: perfect pairing, equal dots, no sharing, none unhit. (4) Many-one into: sharing AND an unhit B-dot. (see next page)

Limiting conditions & assumptions. First, the thing must *be* a function before you classify it — each input needs exactly one output. Second, and this is the trap that catches most students: **the type depends entirely on the declared domain and codomain**. The very same rule changes character when you change the sets. $f(x) = x^2$ is into as $\mathbb{R} \rightarrow \mathbb{R}$, but becomes **onto** as $\mathbb{R} \rightarrow [0, \infty)$, and **bijective** as $[0, \infty) \rightarrow [0, \infty)$ — same formula, three different answers. “Is this function onto?” is an incomplete question until B is fixed. (This is the same *condition-depends-on-the-underlying-set* idea you met with reflexive relations in Subtopic 01.)

Section 2: Key Definitions & Properties

Let $f : A \rightarrow B$ be a function, with $a \in A$ having unique image $f(a) \in B$. Write $n = |A|$ (domain size) and $m = |B|$ (codomain size).

Definitions

Injective (one-one): $f(x_1) = f(x_2) \implies x_1 = x_2$ (equivalently $x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$)

Surjective (onto): $\forall y \in B, \exists x \in A$ with $f(x) = y \iff \text{Range}(f) = B$

Bijective: f is injective **and** surjective

Many-one: $\exists x_1 \neq x_2$ with $f(x_1) = f(x_2)$; Into: $\text{Range}(f) \subsetneq B$

Special functions. Identity $I_A(x) = x$ (always bijective). Constant $f(x) = c$ for all x (many-one, and into whenever $|B| > 1$). Inclusion $f(a) = a$ for $A \subseteq B$ (injective, into if $A \neq B$).

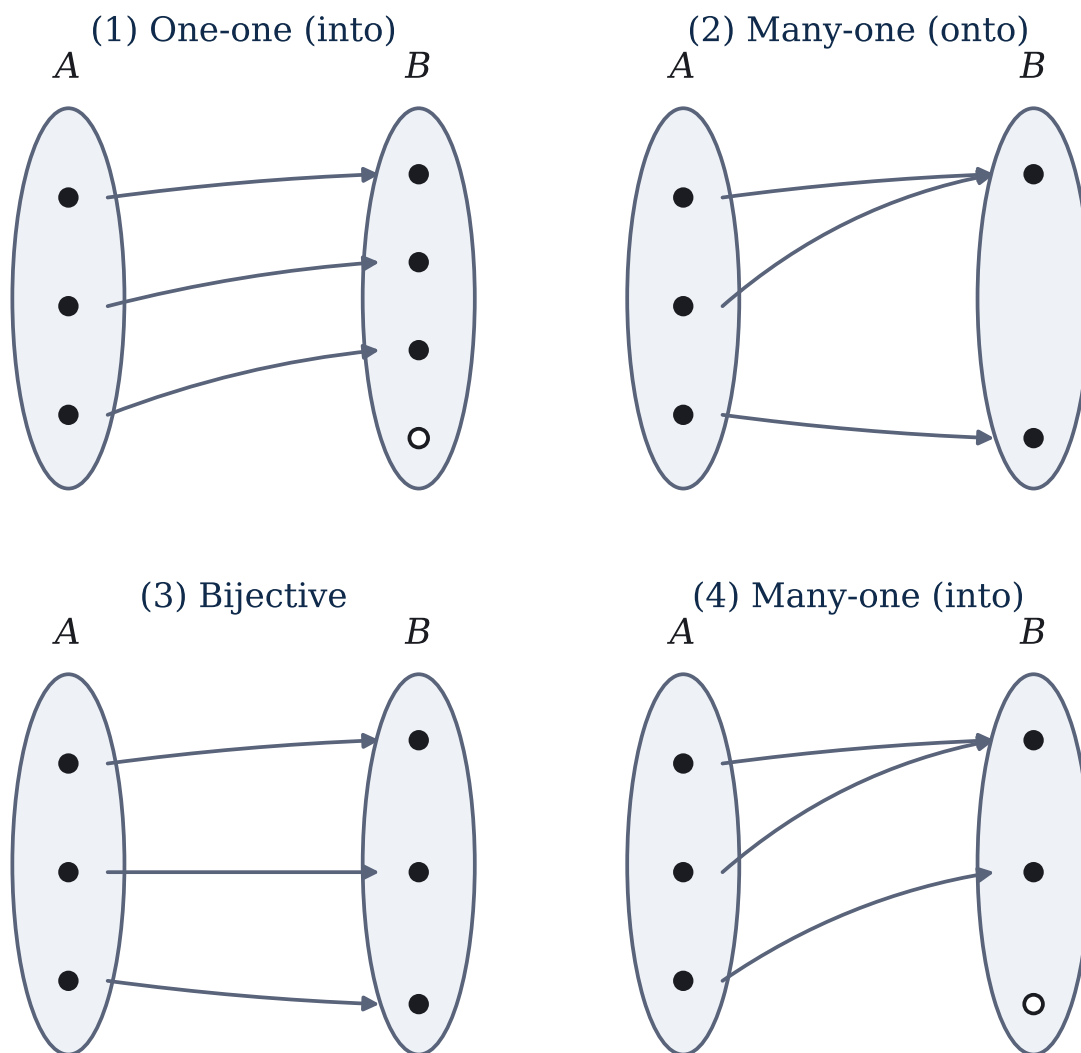


Figure 1.3 Four bubble-and-arrow mapping diagrams between an oval “A” (dots) and an oval “B” (dots). (1) One-one into: distinct arrows, one B-dot unhit. (2) Many-one onto: two A-dots arrow to one B-dot, all B-dots hit. (3) Bijective: perfect pairing, equal dots, no sharing, none unhit. (4) Many-one into: sharing AND an unhit B-dot.

Useful structural facts

- A polynomial $f : \mathbb{R} \rightarrow \mathbb{R}$ of **odd** degree is **onto**; of **even** degree is **into** (never onto $\mathbb{R}$).
- A **strictly monotonic** function (always increasing or always decreasing) is **one-one**.
- An injective map from a **finite set to itself** is automatically **bijective** (and vice-versa).

Counting properties (the “formulas” of this subtopic). For $f : A \rightarrow B$ with $|A| = n$, $|B| = m$:

Type	Count	Feasible only if
All functions $A \rightarrow B$	m^n	always
Injective (one-one)	${}_m P_n = \frac{m!}{(m-n)!}$	$m \geq n$
Bijjective	$n!$	$m = n$
Surjective (onto)	$\sum_{k=0}^m (-1)^k \binom{m}{k} (m-k)^n$	$n \geq m$
Injective $A \rightarrow A$ (self-map)	$n!$	— (equals bijective)

Notation lock (carried forward): f, g for functions; $f : A \rightarrow B$; Dom, Codom, Range; $n = |A|$, $m = |B|$; I_A for identity. These chain into Subtopics 03 (Composition and Inverse) and 04 (Domain and Range) unchanged.

Section 3: Key Procedures & Logical Framework

There is no single derivation, so here is the working toolkit — first the *tests*, then a model *bijectivity proof*.

Procedure A — Testing one-one (injectivity). Pick any of three routes:

1. **Algebraic:** assume $f(x_1) = f(x_2)$ and try to force $x_1 = x_2$. If you can, it's one-one; if a step admits $x_1 \neq x_2$ (e.g. $x_1^2 = x_2^2 \Rightarrow x_1 = \pm x_2$), it's many-one. *Why it works:* this is the definition applied directly.
2. **Calculus:** if $f'(x) > 0$ everywhere (or < 0 everywhere), f is strictly monotonic, hence one-one. *Reasoning:* a strictly climbing graph can never revisit an output. (Fastest route for differentiable functions.)
3. **Graphical:** the horizontal-line test.

Procedure B — Testing onto (surjectivity). Solve $y = f(x)$ for x in terms of y , then ask: *for every y in the codomain B , does a valid x in the domain A exist?* If yes

for all $y \in B$, it's onto; if some y yields no admissible x , that y is an empty seat and f is into. Equivalently, compute $\text{Range}(f)$ and compare with B .

Procedure C — Model proof that a function is bijective.

Claim: $f : \mathbb{R} \setminus \{2\} \rightarrow \mathbb{R} \setminus \{1\}$ defined by $f(x) = \frac{x+3}{x-2}$ is bijective.

Injective. Suppose $f(x_1) = f(x_2)$:

$$\frac{x_1+3}{x_1-2} = \frac{x_2+3}{x_2-2} \implies (x_1+3)(x_2-2) = (x_2+3)(x_1-2).$$

Expanding both sides,

$$x_1x_2 - 2x_1 + 3x_2 - 6 = x_1x_2 - 2x_2 + 3x_1 - 6.$$

Reasoning: the x_1x_2 and -6 terms cancel, leaving only the linear terms. Then $-2x_1 + 3x_2 = -2x_2 + 3x_1$, i.e. $5x_2 = 5x_1$, so $x_1 = x_2$. **Injective.** ✓

Surjective. Take any y in the codomain and solve $y = \frac{x+3}{x-2}$:

$$y(x-2) = x+3 \implies yx - 2y = x+3 \implies x(y-1) = 2y+3 \implies x = \frac{2y+3}{y-1}.$$

Reasoning: this x exists for every $y \neq 1$ — exactly the codomain $\mathbb{R} \setminus \{1\}$. We must also confirm $x \neq 2$ (else x leaves the domain): setting $\frac{2y+3}{y-1} = 2$ gives $2y+3 = 2y-2$, i.e. $3 = -2$, impossible — so $x \neq 2$ always. Hence every y has a valid preimage.

Surjective. ✓

Both hold, so f is **bijective** — and therefore invertible, with $f^{-1}(y) = \frac{2y+3}{y-1}$. ■

Section 4: Worked Examples

Example 1 — [CBSE Board Level] Show that $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = 2x - 7$ is a bijection.

Given: $f(x) = 2x - 7$, domain and codomain $\mathbb{R}$.

One-one: Let $f(x_1) = f(x_2)$. Then $2x_1 - 7 = 2x_2 - 7 \implies 2x_1 = 2x_2 \implies x_1 = x_2$. Hence **one-one.** ✓

Onto: Let $y \in \mathbb{R}$ be arbitrary. Solve $y = 2x - 7 \implies x = \frac{y+7}{2}$, which is a real number for every real y , and $f\left(\frac{y+7}{2}\right) = y$. So every y has a preimage in $\mathbb{R}$. Hence **onto.** ✓

Conclusion: f is one-one and onto, therefore **bijective**.

Example 2 — [CUET / JEE Main Speed Trap] Classify $f(x) = x^2 + 1$ in two cases: (a) $f : \mathbb{R} \rightarrow \mathbb{R}$; (b) $f : \mathbb{R} \rightarrow [1, \infty)$. Use the fastest reasoning.

Speed mindset: spot the even power for one-one; read the codomain for onto.

One-one (same for both): even power $\Rightarrow f(-a) = f(a)$, e.g. $f(2) = f(-2) = 5$. Two inputs, one output — **many-one** instantly, no algebra needed.

Onto:

- **(a)** $\mathbb{R} \rightarrow \mathbb{R}$: outputs satisfy $x^2 + 1 \geq 1$, so Range = $[1, \infty) \subsetneq \mathbb{R}$. Negative numbers (and everything below 1) are never hit — **into**. So (a) is **many-one into**.
- **(b)** $\mathbb{R} \rightarrow [1, \infty)$: now Range = $[1, \infty) = \text{codomain}$ **onto**. So (b) is **many-one onto**.

Lesson: the *one-one* verdict never changed, but flipping the codomain flipped *onto*. Never declare “onto” without checking B — the single most common speed-trap error.

Example 3 — [JEE Main Level] (counting + inclusion-exclusion) Let X have 4 elements and Y have 5 elements. Find (a) the number of functions $X \rightarrow Y$; (b) the number of one-one functions $X \rightarrow Y$; (c) the number of onto functions $Y \rightarrow X$.

(a) Each of the 4 elements of X independently chooses one of 5 images: $5^4 = \boxed{625}$.

(b) One-one from X ($n = 4$) to Y ($m = 5$): feasible since $m \geq n$.

$${}^5P_4 = \frac{5!}{(5-4)!} = \frac{120}{1} = \boxed{120}.$$

(c) Onto from Y ($n = 5$) to X ($m = 4$): feasible since $n \geq m$. Apply inclusion-exclusion:

$$\begin{aligned} \sum_{k=0}^4 (-1)^k \binom{4}{k} (4-k)^5 &= 4^5 - \binom{4}{1} 3^5 + \binom{4}{2} 2^5 - \binom{4}{3} 1^5 \\ &= 1024 - 4(243) + 6(32) - 4(1) = 1024 - 972 + 192 - 4 = \boxed{240}. \end{aligned}$$

Concept link: function-counting rests on the multiplication principle, while onto-counting needs inclusion-exclusion to subtract off the maps that miss at least one target element.

Example 4 — [JEE Advanced Level] Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^3 - 3x$. (i) Classify f as one-one/many-one and into/onto, with reasoning. (ii) Find a maximal interval $[a, \infty)$ and a codomain on which f becomes a bijection.

(i) Onto? f is an **odd-degree** polynomial, continuous, with $f(x) \rightarrow +\infty$ as $x \rightarrow +\infty$ and $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$. By the intermediate value theorem it attains *every* real value, so $\text{Range} = \mathbb{R}$ — **onto**. ✓

One-one? $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$, which is zero at $x = \pm 1$ and changes sign — so f is **not monotonic**. Concretely, $f(0) = 0$ and $f(\sqrt{3}) = 3\sqrt{3} - 3\sqrt{3} = 0$, two inputs sharing the output 0. Hence **many-one**. ✗

So f is **onto but not one-one** on $\mathbb{R}$.

(ii) Making it bijective. On $[1, \infty)$ we have $f'(x) = 3(x^2 - 1) \geq 0$, so f is increasing there — hence one-one on $[1, \infty)$. Its image is $[f(1), \infty) = [1 - 3, \infty) = [-2, \infty)$. Therefore

$$f : [1, \infty) \rightarrow [-2, \infty), \quad f(x) = x^3 - 3x$$

is **bijective**.

Figure 1.4 cubic curve $y = x^3 - 3x$ with local max at $(-1, 2)$ and local min at $(1, -2)$; a horizontal line $y = 0$ shown cutting the curve at three points to illustrate many-one; the branch $x \geq 1$ shaded to show the strictly increasing piece used for the bijection. (see next page)

Section 5: Practice Exercises

1. **[CBSE]** Show that $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 5x + 3$ is a bijection.
2. **[CUET]** Classify $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = |x|$ as one-one/many-one and into/onto.
3. **[JEE Main]** Find the number of one-one functions from a set of 3 elements to a set of 5 elements.
4. **[JEE Main]** Classify $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^3 + 1$.
5. **[JEE Advanced]** Find the number of onto functions from a set of 5 elements to a set of 3 elements.

Answers: 1. Bijective (one-one: linear; onto: $x = (y - 3)/5 \in \mathbb{R}$). 2. Many-one ($|-a| = |a|$) and into (range $[0, \infty) \neq \mathbb{R}$). 3. ${}^5P_3 = 60$. 4. Bijective (odd-degree $\Rightarrow$ onto; $f'(x) = 3x^2 \geq 0$, strictly increasing $\Rightarrow$ one-one). 5. $3^5 - \binom{3}{1}2^5 + \binom{3}{2}1^5 = 243 - 96 + 3 = 150$.

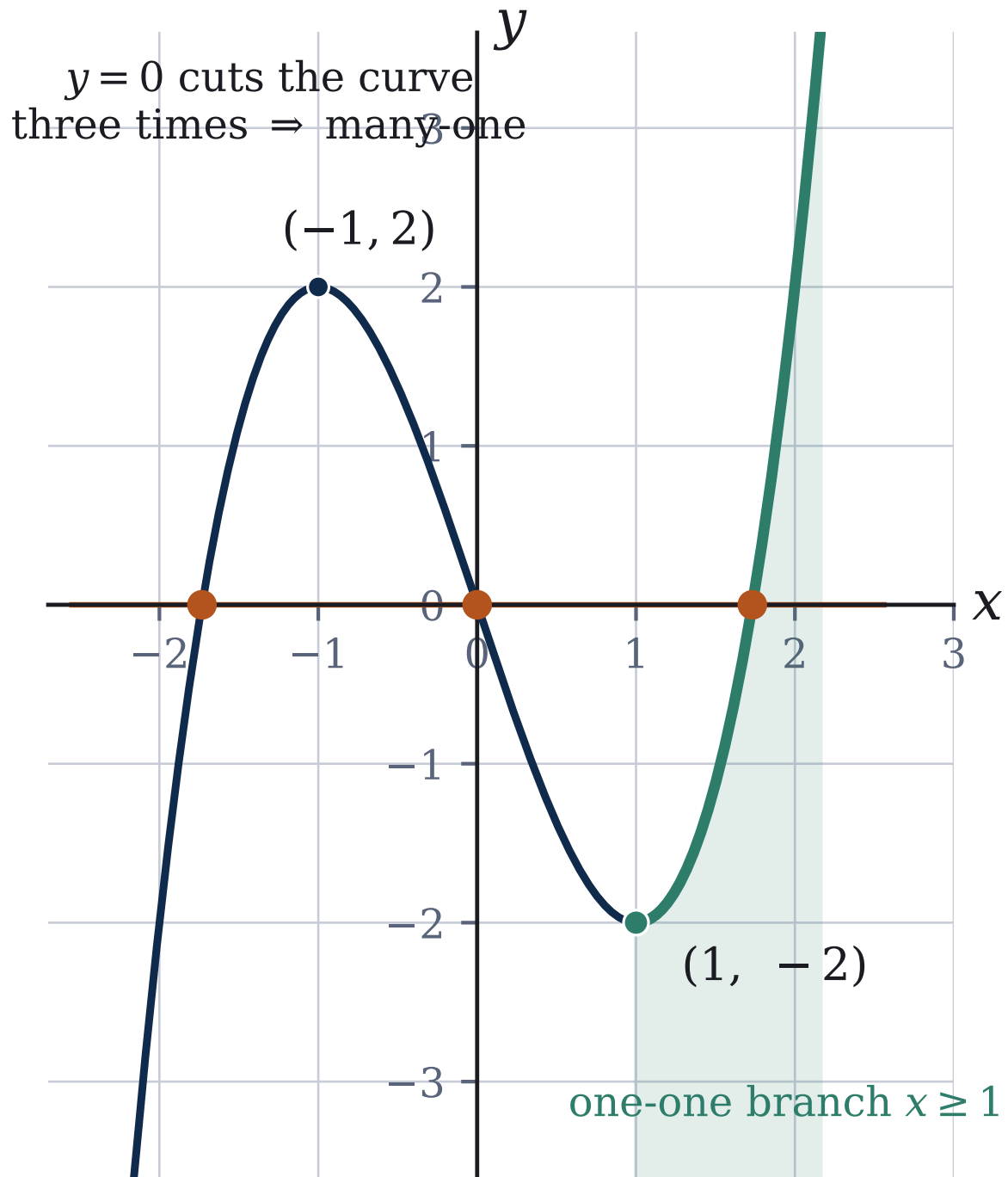


Figure 1.4 cubic curve $y = x^3 - 3x$ with local max at $(-1, 2)$ and local min at $(1, -2)$; a horizontal line $y = 0$ shown cutting the curve at three points to illustrate many-one; the branch $x \geq 1$ shaded to show the strictly increasing piece used for the bijection.

Section 6: Crack the MCQ

Q1. $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2 - 4x + 5$. Then f is:

- (a) one-one and onto (b) one-one and into (c) many-one and onto (d) many-one and into

Answer: (d). Completing the square, $f(x) = (x - 2)^2 + 1$, a parabola with minimum value 1 at $x = 2$. Range = $[1, \infty) \subsetneq \mathbb{R}$, so **into**; symmetry about $x = 2$ (e.g. $f(1) = f(3) = 2$) makes it **many-one**. *Why the traps fail:* **(a)/(b)** call it one-one — but every parabola fails the horizontal-line test. **(c)** calls it onto — students forget the codomain is all of $\mathbb{R}$ while the range bottoms out at 1.

Q2. The total number of functions from a set of 3 elements to a set of 2 elements is:

- (a) 6 (b) 8 (c) 9 (d) 5

Answer: (b) 8. Each of 3 inputs picks one of 2 images: $2^3 = 8$. *Why the traps fail:* **(a)** $= 3 \times 2$ multiplies instead of exponentiating. **(c)** $= 3^2$ swaps base and exponent (a classic m^n vs n^m slip). **(d)** $= 3 + 2$, pure confusion.

Q3. Which function $f : \mathbb{R} \rightarrow \mathbb{R}$ is a bijection?

- (a) $f(x) = x^2$ (b) $f(x) = x^3$ (c) $f(x) = e^x$ (d) $f(x) = \sin x$

Answer: (b). $f(x) = x^3$ is strictly increasing ($f'(x) = 3x^2 \geq 0$, zero only at a point) hence one-one, and odd-degree hence onto. *Why the traps fail:* **(a)** x^2 is many-one and into. **(c)** e^x is one-one but **into** (range $(0, \infty)$, never reaching 0 or negatives) — the most tempting wrong answer. **(d)** $\sin x$ is many-one (periodic) and into (range $[-1, 1]$).

Q4. $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(n) = n + 1$ is:

- (a) bijective (b) one-one but not onto (c) onto but not one-one (d) neither

Answer: (b). One-one: $n_1 + 1 = n_2 + 1 \Rightarrow n_1 = n_2$. But $1 \in \mathbb{N}$ has **no preimage** (it would need $n = 0 \notin \mathbb{N}$), so not onto. *Why the traps fail:* **(a)** would be correct on $\mathbb{Z}$ — students forget the domain is $\mathbb{N}$, where the “shift” leaves 1 uncovered. **(c)/(d)** misread injectivity. This question is really testing *domain awareness*.

Section 7: Common Student Pitfalls & Pro-Tips

- Declaring “onto” without stating the codomain.** “Onto” is meaningless until B is fixed; you must compare $\text{Range}(f)$ with the *given* codomain, not with $\mathbb{R}$ by reflex.
- Forgetting even powers, $|x|$, and trig functions are many-one.** The $\pm$ symmetry (or periodicity) silently kills injectivity; always check it before any algebra.

3. **Confusing “into” with “not a function.”** *Into* is a perfectly valid function type — it simply means *not onto*. Don't reject the map.
4. **Counting blunders.** Using m^n for one-one (it's mP_n), or ignoring feasibility: there is **no** one-one function when the domain is bigger than the codomain ($n > m$), and **no** onto function when the codomain is bigger ($m > n$).

Pro-Tip (speed): For differentiable functions, check **monotonicity first** — if f' keeps one sign throughout, f is one-one and you're done, no algebra. For real polynomials $\mathbb{R} \rightarrow \mathbb{R}$: **odd degree $\Rightarrow$ onto, even degree $\Rightarrow$ into** (never onto). And invoke the **pigeonhole** sanity check — if $|A| > |B|$, one-one is impossible; if $|A| < |B|$, onto is impossible.

Section 8: Cheat Sheet Box

TYPES OF FUNCTIONS — QUICK RECALL

- **Function:** each input $\rightarrow$ exactly one output. Domain A , codomain B .
- **Injective (one-one):** $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$. *Test:* horizontal line / strict monotonicity (f' one sign).
- **Surjective (onto):** $\text{Range}(f) = B$ — nothing left out.
- **Bijective = one-one + onto = invertible.**
- **Many-one / Into** = the negations.
- **Counts** ($n = |A|$, $m = |B|$): functions m^n · one-one mP_n (need $m \geq n$) · onto $\sum_{k=0}^m (-1)^k \binom{m}{k} (m-k)^n$ (need $n \geq m$) · bijective $n!$ (need $m = n$).
- **Odd-degree polynomial $\Rightarrow$ onto; even-degree $\Rightarrow$ into. Strictly monotonic $\Rightarrow$ one-one.**

Mnemonics: “**ONE**-one $\Rightarrow$ different stay different; **ONTO** $\Rightarrow$ nothing left out.” · “**Bijective** $\Rightarrow$ both $\Rightarrow$ invertible.” · Type depends on **both** domain and codomain — change the set, change the answer.

Subtopic 03 — Composition and Inverse of Functions

Exam Relevance: A high-yield pairing. CBSE Boards routinely ask a 3–4 mark “find $f \circ g$ and $g \circ f$ ” or “find f^{-1} and verify it” question. JEE Main carries 1–2 MCQs on composite evaluation, the order-reversal law, or finding an inverse. JEE Advanced builds multi-composition fixed-point problems and inverse-of-restricted-domain problems (often multi-correct). CUET tests direct computation of composites and inverses. (*Not in NEET — no mathematics section.*)

This subtopic covers two tightly linked ideas: **03a Composition** (chaining functions) and **03b Inverse** (undoing a function). They are coupled because an inverse is precisely the function that “undoes” the original under composition.

Section 1: Concept Intuition

03a — Composition: functions in series Imagine a two-stage water system at home: first an RO purifier, then a chiller. Pour in tap water, the purifier cleans it, and *its output* immediately feeds the chiller, which cools it. The whole assembly — tap water in, cold clean water out — is a **composition** of two machines.

That is exactly what function composition is. Given f and g , the composite $g \circ f$ means: *run f first, then feed its output into g* . In symbols,

$$(g \circ f)(x) = g(f(x)).$$

The single most important habit to build: **the innermost function acts first**. In $g \circ f$, even though g is written on the left, it is f that touches x first. Reading right-to-left (“ g after f ”) keeps you safe.

Think about it this way... Composition is a relay race. Runner f runs the first leg and hands the baton (its output) to runner g . If you swap who runs first, you usually get a different race — which is why **composition is not commutative**: in general $g \circ f \neq f \circ g$. (It is associative, though: chaining three machines $h \circ (g \circ f)$ gives the same result as $(h \circ g) \circ f$ — the grouping doesn’t matter, only the order.)

Limiting condition for 03a: the chain only makes sense if the baton can actually be passed — the **range of the inner function must lie inside the domain of the outer function**. If f outputs a negative number but g only accepts positives, the handoff fails and $g \circ f$ is undefined there.

03b — Inverse: the undo button Now picture pressing **Ctrl+Z**. Whatever the last action did, undo reverses it perfectly, landing you back exactly where you started. An

inverse function f^{-1} is the Ctrl+Z of f : if f sends a to b , then f^{-1} sends b straight back to a . Formally, f^{-1} is the function that satisfies

$$f^{-1} \circ f = I \quad \text{and} \quad f \circ f^{-1} = I,$$

where I is the identity (the “do nothing” function). Composition and inverse meet right here: the inverse is *defined* as the thing that composes with f to give “do nothing.”

But here is the catch that makes or breaks the whole idea: **not every function has an inverse**. For the undo to work, f must be **bijective** — one-one *and* onto. Why? If f were many-one (say both 2 and -2 map to 4 under squaring), then $f^{-1}(4)$ wouldn't know whether to return 2 or -2 — the undo is ambiguous. And if f weren't onto, some target would have no input to return to. So:

$$f \text{ is invertible} \iff f \text{ is bijective.}$$

This is the bridge straight from Subtopic 02 — every bijection you classified there is exactly a function you can now invert.

Think about it this way... To get dressed you put on socks, *then* shoes. To undo it, you must take off shoes *first*, then socks — the reverse order. That's the famous **socks-and-shoes rule**: $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$. You undo a chain by undoing each step in *reverse* order.

Figure 1.5 two horizontal flows. Top: $x \rightarrow [f \rightarrow [g] \rightarrow \text{output}]$, labelled “do: f then g ”. Bottom: $\text{output} \rightarrow [g^{-1}] \rightarrow [f^{-1}] \rightarrow x$, labelled “undo: g^{-1} then f^{-1} ”, with arrows pointing leftward to emphasise reversal.] (see next page)

Limiting condition for 03b: an inverse exists *only* over a domain/codomain where f is bijective. A non-bijective function can still be inverted, but only after you **restrict** it (e.g. x^2 has no inverse on $\mathbb{R}$, but does on $[0, \infty)$, where it becomes $\sqrt{x}$).

Section 2: Key Definitions & Properties

Let $f : A \rightarrow B$, $g : B \rightarrow C$, $h : C \rightarrow D$ be functions, and I_A the identity on A .

03a — Composition

$$\boxed{(g \circ f)(x) = g(f(x))}, \quad g \circ f : A \rightarrow C \text{ (requires } \text{Range}(f) \subseteq \text{Dom}(g)\text{)}.$$

Properties:

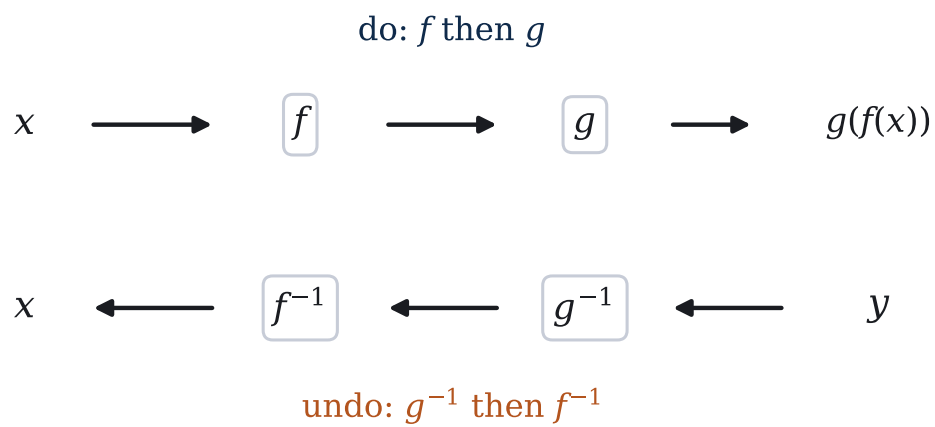


Figure 1.5 *two horizontal flows. Top: $x \rightarrow [f \rightarrow [g] \rightarrow \text{output}]$, labelled “do: f then g ”. Bottom: $\text{output} \rightarrow [g^{-1}] \rightarrow [f^{-1}] \rightarrow x$, labelled “undo: g^{-1} then f^{-1} ”, with arrows pointing leftward to emphasise reversal.]*

- **Not commutative:** in general $g \circ f \neq f \circ g$.
- **Associative:** $h \circ (g \circ f) = (h \circ g) \circ f$.
- **Identity:** $f \circ I_A = f = I_B \circ f$.
- **Inheritance:** if f, g are both injective then $g \circ f$ is injective; both surjective $\Rightarrow g \circ f$ surjective; both bijective $\Rightarrow g \circ f$ bijective.
- **Partial converse:** if $g \circ f$ is injective then f is injective; if $g \circ f$ is surjective then g is surjective.

03b — Inverse

$$\boxed{f^{-1} \circ f = I_A \text{ and } f \circ f^{-1} = I_B}, \quad \text{exists} \iff f \text{ is bijective.}$$

Properties:

- **Uniqueness:** if an inverse exists, it is unique.
- **Self-undoing:** $(f^{-1})^{-1} = f$.
- **Order reversal (socks-and-shoes):** $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.
- **Domain-range swap:** $\text{Dom}(f^{-1}) = \text{Range}(f)$ and $\text{Range}(f^{-1}) = \text{Dom}(f)$.
- **Caution:** $f^{-1}(x) \neq \frac{1}{f(x)}$. The superscript -1 here means *inverse function*, **not** reciprocal.

Counting note (links to Sub-02): the number of *invertible* functions from a set to itself with n elements equals the number of bijections, $n!$.

Notation lock (carried forward): f, g, h functions; $\circ$ composition; I_A identity on A ; f^{-1} inverse. Innermost-first reading convention fixed throughout.

Section 3: Key Procedures & Derivations

Procedure A — Computing a composite $g \circ f$. Substitute the *entire* formula of f into every occurrence of the variable in g . *Reasoning:* g receives $f(x)$ as its raw input, so wherever g 's rule says "input," write " $f(x)$." Then simplify. Always confirm the handoff condition $\text{Range}(f) \subseteq \text{Dom}(g)$.

Procedure B — Finding an inverse (swap-and-solve).

1. **Check bijectivity first** — no inverse exists otherwise (restrict the domain/codomain if needed).
2. Write $y = f(x)$.
3. **Solve for x** in terms of y .
4. **Swap symbols:** replace y by x to express $f^{-1}(x)$.
5. State the domain of f^{-1} (= range of f).

Reasoning: step 3 literally reverses the input-output relationship; step 4 is just cosmetic, renaming the free variable so the answer reads as a function of x .

Procedure C — Verifying a claimed inverse. Confirm **both** $f(g(x)) = x$ and $g(f(x)) = x$. *Why both:* one equation alone only guarantees a one-sided inverse; a genuine two-sided inverse needs both compositions to collapse to the identity.

Derivation — the order-reversal law $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Setup: let $f : A \rightarrow B$ and $g : B \rightarrow C$ both be bijective, so $g \circ f : A \rightarrow C$ is bijective and therefore invertible. We claim its inverse is $f^{-1} \circ g^{-1}$. To prove a function is the inverse of $g \circ f$, it suffices to show it composes with $g \circ f$ to give the identity on both sides.

Compose in one direction:

$$(f^{-1} \circ g^{-1}) \circ (g \circ f).$$

Using associativity to regroup,

$$= f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ I_B \circ f.$$

Reasoning: the inner pair $g^{-1} \circ g$ collapses to the identity I_B (that's what an inverse does). The identity then disappears under composition:

$$= f^{-1} \circ f = I_A.$$

An identical argument in the other direction gives $(g \circ f) \circ (f^{-1} \circ g^{-1}) = I_C$. Since $f^{-1} \circ g^{-1}$ composes with $g \circ f$ to give the identity on both sides, and inverses are unique, we conclude

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}. \quad \blacksquare$$

Physical meaning: to undo “do f then g ,” you must undo g first and f last — socks and shoes.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] (Composition) Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be $f(x) = 3x + 1$ and $g(x) = x^2 - 2$. Find $f \circ g$ and $g \circ f$, and hence show $f \circ g \neq g \circ f$.

$f \circ g$: feed $g(x)$ into f :

$$(f \circ g)(x) = f(g(x)) = 3(x^2 - 2) + 1 = 3x^2 - 6 + 1 = 3x^2 - 5.$$

$g \circ f$: feed $f(x)$ into g :

$$(g \circ f)(x) = g(f(x)) = (3x + 1)^2 - 2 = 9x^2 + 6x + 1 - 2 = 9x^2 + 6x - 1.$$

Since $3x^2 - 5 \neq 9x^2 + 6x - 1$ (check $x = 2$: $f \circ g$ gives 7, $g \circ f$ gives 47), we have $f \circ g \neq g \circ f$. **Composition is not commutative.**

Example 2 — [CUET / JEE Main Speed Trap] (Inverse) (a) Find f^{-1} for $f(x) = 7 - 3x$ using the “undo in reverse” method. (b) Which of $p(x) = \frac{1}{x}$, $q(x) = 2x$, $r(x) = x + 5$ is its own inverse?

(a) Undo in reverse. f does two things to x : multiply by -3 , then add 7. To undo, reverse the *order* and the *operations*: subtract 7, then divide by -3 :

$$f^{-1}(x) = \frac{x - 7}{-3} = \frac{7 - x}{3}.$$

Trap to avoid: $f^{-1}(x) \neq \frac{1}{7 - 3x}$. The reciprocal is *not* the inverse.

(b) A function is its own inverse (an *involution*) when applying it twice returns x . Test by composing:

- $p(p(x)) = \frac{1}{1/x} = x$ ✓ — $p(x) = 1/x$ **is its own inverse.**
- $q(q(x)) = 2(2x) = 4x \neq x$ ✗.
- $r(r(x)) = (x + 5) + 5 = x + 10 \neq x$ ✗.

Speed lesson: to invert a linear function, peel operations off in reverse; to spot a self-inverse, compose it with itself and look for x .

Example 3 — [JEE Main Level] (Inverse of a rational function) Let $f : \mathbb{R} \setminus \{3\} \rightarrow \mathbb{R} \setminus \{2\}$ be defined by $f(x) = \frac{2x - 1}{x - 3}$. Find $f^{-1}(x)$ and state its domain.

Bijectivity (link to Sub-02): this Möbius-type map is one-one and onto between the stated sets, so f^{-1} exists. Now swap-and-solve.

Set $y = \frac{2x - 1}{x - 3}$ and clear the fraction:

$$y(x - 3) = 2x - 1 \implies yx - 3y = 2x - 1 \implies x(y - 2) = 3y - 1.$$

$$\therefore x = \frac{3y - 1}{y - 2}.$$

Swap $y \rightarrow x$:

$$\boxed{f^{-1}(x) = \frac{3x - 1}{x - 2}}, \quad \text{domain } \mathbb{R} \setminus \{2\}.$$

Sanity check: $\text{Dom}(f^{-1}) = \mathbb{R} \setminus \{2\} = \text{Range}(f)$, and $\text{Range}(f^{-1}) = \mathbb{R} \setminus \{3\} = \text{Dom}(f)$

— the domain-range swap holds, confirming the answer.

Example 4 — [JEE Advanced Level] (Composition + Inverse fused) Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be $f(x) = x^3 + 2$ and $g(x) = 2x - 1$. Find $(g \circ f)^{-1}(x)$ two independent ways — directly, and via the order-reversal law — and confirm they agree.

Way 1 — direct. First build the composite:

$$(g \circ f)(x) = g(f(x)) = 2(x^3 + 2) - 1 = 2x^3 + 3.$$

Invert it: set $y = 2x^3 + 3 \Rightarrow x^3 = \frac{y-3}{2} \Rightarrow x = \left(\frac{y-3}{2}\right)^{1/3}$. So

$$(g \circ f)^{-1}(x) = \left(\frac{x-3}{2}\right)^{1/3}.$$

Way 2 — order-reversal law. Find each inverse separately:

$$f^{-1}(x) = (x-2)^{1/3}, \quad g^{-1}(x) = \frac{x+1}{2}.$$

Then $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ (socks-and-shoes), so

$$(f^{-1} \circ g^{-1})(x) = f^{-1}\left(\frac{x+1}{2}\right) = \left(\frac{x+1}{2} - 2\right)^{1/3} = \left(\frac{x+1-4}{2}\right)^{1/3} = \left(\frac{x-3}{2}\right)^{1/3}.$$

Both methods give $\left(\frac{x-3}{2}\right)^{1/3}$. ✓ The agreement is not a coincidence — it is the order-reversal law in action. *Deeper point:* note the order genuinely matters — computing $g^{-1} \circ f^{-1}$ instead would give the wrong answer, because that tries to remove the socks before the shoes.

Section 5: Practice Exercises

1. **[CBSE]** For $f(x) = 2x + 1$ and $g(x) = x^2$ on $\mathbb{R}$, find $f \circ g$ and $g \circ f$.
2. **[CUET]** Find the inverse of $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{x-5}{4}$.
3. **[JEE Main]** Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^3 - 4$. Find $f^{-1}(x)$.
4. **[JEE Main]** If f and g are invertible with $f(x) = 2x + 3$ and $g(x) = x - 1$, find $(g \circ f)^{-1}(x)$.

5. **[JEE Advanced]** Find all real a for which $f(x) = \frac{ax+1}{x-a}$ (on a suitable domain) satisfies $(f \circ f)(x) = x$ for all x in its domain.

Answers: **1.** $(f \circ g)(x) = 2x^2 + 1$; $(g \circ f)(x) = (2x+1)^2 = 4x^2 + 4x + 1$. **2.** $f^{-1}(x) = 4x+5$. **3.** $f^{-1}(x) = (x+4)^{1/3}$. **4.** $(g \circ f)(x) = 2x+2$, so $(g \circ f)^{-1}(x) = \frac{x-2}{2}$. **5.** Every real a — f is an involution for all a (since $f(f(x)) = x$ identically); the map is self-inverse.

Section 6: Crack the MCQ

Q1. If $f(x) = 3x^2 - 5$ and $g(x) = \frac{x}{x^2 + 1}$, then $(g \circ f)(x)$ equals:

- (a) $\frac{3x^2 - 5}{9x^4 - 30x^2 + 26}$ (b) $\frac{3x^2 - 5}{9x^4 + 6x^2 + 26}$ (c) $\frac{3x^2}{2x^2 + 4}$ (d) $\frac{3x^2}{9x^4 - 30x^2 + 2}$

Answer: (a). Feed $f(x) = 3x^2 - 5$ into g : numerator = $3x^2 - 5$; denominator = $(3x^2 - 5)^2 + 1 = 9x^4 - 30x^2 + 25 + 1 = 9x^4 - 30x^2 + 26$. *Why the traps fail:* **(b)** mis-expands $(3x^2 - 5)^2$ (sign error on the cross term). **(c)** wrongly computes $f \circ g$ partially. **(d)** forgets the “+1” in g ’s denominator.

Q2. If $f : A \rightarrow B$ and $g : B \rightarrow C$ are bijections, then $(g \circ f)^{-1}$ is:

- (a) $f^{-1} \circ g^{-1}$ (b) $g^{-1} \circ f^{-1}$ (c) $f \circ g$ (d) $g \circ f$

Answer: (a). The order-reversal (socks-and-shoes) law: undo the last-applied function first. *Why the traps fail:* **(b)** keeps the *same* order — the most common error; it tries to remove socks before shoes. **(c)/(d)** confuse the inverse with the composite itself.

Q3. Which function $\mathbb{Z} \rightarrow \mathbb{Z}$ is invertible?

- (a) $f(x) = x^2$ (b) $f(x) = x + 2$ (c) $f(x) = |x|$ (d) $f(x) = x^2 + 1$

Answer: (b). $f(x) = x + 2$ is a bijection on $\mathbb{Z}$ (one-one and onto), so it has an inverse, $f^{-1}(x) = x - 2$. *Why the traps fail:* **(a)/(d)** are many-one (x and $-x$ share an image) — not one-one, so not invertible. **(c)** $|x|$ is both many-one and not onto (negatives never hit). Invertible $\iff$ bijective — the link from Sub-02.

Q4. For $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 4x - 9$, the value of $f^{-1}(3)$ is:

- (a) $\frac{1}{3}$ (b) 3 (c) $\frac{1}{4 \cdot 3 - 9}$ (d) cannot be found

Answer: (b). $f^{-1}(x) = \frac{x+9}{4}$, so $f^{-1}(3) = \frac{12}{4} = 3$. (Check: $f(3) = 4(3) - 9 = 3$ ✓.)

Why the traps fail: **(c)** computes $\frac{1}{f(3)}$, falling for the reciprocal trap $f^{-1} \neq 1/f$. **(a)** is that reciprocal evaluated. **(d)** wrongly assumes a linear function isn't invertible.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Treating composition as commutative.** $f \circ g$ and $g \circ f$ are almost always different. Always respect the order, and read **innermost-first**: in $g \circ f$, f touches x first.
2. **The reciprocal trap.** $f^{-1}(x)$ is the *inverse function*, **never** $\frac{1}{f(x)}$. The same superscript means two different things on functions vs numbers — keep them apart.
3. **Inverting a non-bijective function.** No inverse exists unless f is one-one *and* onto. If it isn't, you must **restrict** the domain/codomain first (e.g. x^2 needs $x \geq 0$ to give $\sqrt{x}$).
4. **Keeping the same order when inverting a composite.** $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$, **not** $g^{-1} \circ f^{-1}$ — reverse the order (socks and shoes).
5. **Forgetting the domain-range swap.** When you write f^{-1} , its domain is the *range* of f ; always state it.

Pro-Tip (speed): To find a linear/rational inverse fast, use **swap-and-solve** ($y = f(x)$, solve for x , rename). To *verify* any claimed inverse in seconds, just check that $f(f^{-1}(x)) = x$ — if it collapses to x , you're done. And to evaluate $f^{-1}(c)$ without finding the whole inverse, simply **solve** $f(x) = c$ for x — that single value is $f^{-1}(c)$.

Section 8: Cheat Sheet Box

COMPOSITION & INVERSE — QUICK RECALL

03a Composition

- $(g \circ f)(x) = g(f(x))$ — **innermost (f) acts first**. Needs $\text{Range}(f) \subseteq \text{Dom}(g)$.
- **Not commutative** ($g \circ f \neq f \circ g$) but **associative** ($h \circ (g \circ f) = (h \circ g) \circ f$).
- Both injective/surjective/bijective $\Rightarrow$ so is $g \circ f$.

03b Inverse

- Exists $\iff f$ is **bijective**. Defined by $f^{-1} \circ f = I = f \circ f^{-1}$.
- **Find it:** $y = f(x) \rightarrow$ solve for $x \rightarrow$ swap $x \leftrightarrow y$. Domain of $f^{-1} =$ range of f .

- $(f^{-1})^{-1} = f$; $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ (reverse order).
- $f^{-1}(x) \neq \frac{1}{f(x)}$.

Mnemonics: “**Socks and shoes**” $\Rightarrow$ undo in reverse order. · “**Inverse undoes**, reciprocal divides” — keep f^{-1} and $1/f$ apart. · “**Innermost first**” for reading composites. · “**No bijection, no inverse.**”

Subtopic 04 — Domain and Range of Real Functions

Exam Relevance: The workhorse skill of the whole chapter. JEE Main almost always has 1–2 questions that hinge on finding a domain or range. JEE Advanced embeds it inside logarithm/root nests and inequalities. CUET tests it directly. CBSE Boards use it implicitly whenever a function is defined. (*Not in NEET — no mathematics section.*)

Section 1: Concept Intuition

Every function is a machine, and two questions decide what it can do. **Which inputs is it allowed to accept?** That set is the **domain**. **Which outputs can it actually produce?** That set is the **range**. For *real* functions, both live inside the real numbers $\mathbb{R}$, and almost every domain question reduces to avoiding three “illegal moves” that mathematics forbids.

Think of it like a kitchen. The domain is the list of ingredients the recipe will accept; hand it something it can’t process and the machine jams. There are exactly three jams to watch for:

1. **Division by zero.** Any denominator that becomes 0 is forbidden — the machine explodes there.
2. **Even root of a negative.** $\sqrt{\text{negative}}$ is not real, so whatever sits under a square (or fourth, sixth, ...) root must be ≥ 0 .
3. **Logarithm of a non-positive number.** $\log$ only accepts strictly positive inputs, so the argument of any $\log$ must be > 0 .

Find every place these three jams could occur, write down the condition that avoids each, and **intersect** all the conditions — that intersection is the domain.

Think about it this way... The range is the harder twin. Domain is “what am I allowed to put in?”; range is “what can possibly come out?” — and you usually can’t just read it off. The reliable trick is to turn the question inside out: set $y = f(x)$, solve for x in terms of y , and then ask *for which values of y does a legal x exist?* Those y -values are the range. You’re essentially asking the machine to run backwards and seeing which outputs have a valid source.

Limiting conditions & assumptions. Unless told otherwise, the domain of a real function is taken to be the **largest** subset of $\mathbb{R}$ on which the formula makes sense (the “natural domain”). The range depends on the domain — shrink the domain and the range can shrink too. And the range is *not* automatically the codomain; “ $f : \mathbb{R} \rightarrow \mathbb{R}$ ” only promises outputs land somewhere in $\mathbb{R}$, not that every real is hit.

Section 2: Key Definitions & Properties

For $f : A \rightarrow \mathbb{R}$, the **domain** is A (all permissible inputs) and the **range** is $\{f(x) : x \in A\} \subseteq \mathbb{R}$ (all actual outputs).

Domain rules (the constraint catalogue)

Form	Condition for x to be allowed
Polynomial $a_n x^n + \dots + a_0$	all of $\mathbb{R}$
Rational $\frac{P(x)}{Q(x)}$	$Q(x) \neq 0$
Even root $\sqrt[2k]{g(x)}$	$g(x) \geq 0$
In a denominator $\frac{1}{\sqrt{g(x)}}$	$g(x) > 0$ (strict)
$\log_a g(x)$	$g(x) > 0$, with $a > 0$, $a \neq 1$

When several appear together, **solve each constraint and intersect**.

Standard ranges (worth memorising)

$$x^2 : [0, \infty), \quad |x| : [0, \infty), \quad \frac{1}{x} : \mathbb{R} \setminus \{0\}, \quad e^x : (0, \infty), \quad \ln x : \mathbb{R}, \quad \sin x, \cos x : [-1, 1].$$

Range techniques

- **y -method (invert):** write $y = f(x)$, solve for x , demand x be real and in the domain.
- **Rational** $\frac{ax+b}{cx+d}$: range = $\mathbb{R} \setminus \left\{\frac{a}{c}\right\}$ (the horizontal asymptote is never attained).
- **Discriminant method:** for $y = \frac{\text{quadratic}}{\text{quadratic}}$, cross-multiply into a quadratic in x and force discriminant ≥ 0 .
- **Boundedness / monotonicity:** use known bounds (sin, cos) or a strictly increasing/decreasing shape.

Notation lock (carried forward): $\text{Dom}(f)$, $\text{Range}(f)$; intervals in standard bracket notation; $\mathbb{R} \setminus S$ for exclusion.

Section 3: Key Procedures & Derivations

Procedure A — Finding the domain. (1) Scan the formula for every denominator, even root, and logarithm. (2) Write the condition each imposes ($\neq 0$, ≥ 0 , > 0). (3) Solve each as an equation/inequality. (4) **Intersect** all solution sets. *Reasoning:* the

function is defined only where *all* constraints hold simultaneously, and “simultaneously” means intersection.

Procedure B — Finding the range by the y -method. (1) Put $y = f(x)$. (2) Solve algebraically for x . (3) Determine the y -values for which that x is real and lies in the domain. *Reasoning:* a value y_0 is in the range exactly when *some* legal input maps to it — i.e. when solving $f(x) = y_0$ yields a valid x .

Derivation — range of $f(x) = \frac{x^2 - x + 1}{x^2 + x + 1}$ (the discriminant method).

Note first the denominator never vanishes: $x^2 + x + 1 = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} > 0$, so the domain is all of $\mathbb{R}$. Set $y = f(x)$ and cross-multiply:

$$y(x^2 + x + 1) = x^2 - x + 1 \implies (y - 1)x^2 + (y + 1)x + (y - 1) = 0.$$

Reasoning: this is a quadratic in x ; for a real input x to exist, its discriminant must be ≥ 0 :

$$(y + 1)^2 - 4(y - 1)(y - 1) \geq 0 \implies (y + 1)^2 - 4(y - 1)^2 \geq 0.$$

Factor as a difference of squares:

$$[(y + 1) - 2(y - 1)][(y + 1) + 2(y - 1)] \geq 0 \implies (3 - y)(3y - 1) \geq 0.$$

This holds for $\frac{1}{3} \leq y \leq 3$. We must also check the case $y = 1$ (where the quadratic degenerates to linear): then $2x = 0 \Rightarrow x = 0$, which is valid, and $y = 1$ already lies in $[\frac{1}{3}, 3]$. Therefore

$$\boxed{\text{Range}(f) = \left[\frac{1}{3}, 3\right]}. \quad \blacksquare$$

Section 4: Worked Examples

Example 1 — [CBSE Board Level] Find the domain of $f(x) = \sqrt{x - 3} + \frac{1}{x - 5}$.

Two constraints: the root needs $x - 3 \geq 0 \Rightarrow x \geq 3$; the denominator needs $x - 5 \neq 0 \Rightarrow x \neq 5$. Intersecting,

$$\text{Dom}(f) = [3, 5) \cup (5, \infty).$$

Example 2 — [CUET / JEE Main Speed Trap] Find the domain of $f(x) = \log(x^2 - 4)$.

Speed: only one constraint — the log argument must be positive: $x^2 - 4 > 0 \Rightarrow x^2 > 4 \Rightarrow |x| > 2$. Hence

$$\text{Dom}(f) = (-\infty, -2) \cup (2, \infty).$$

Trap avoided: writing $x^2 - 4 \geq 0$ (that's the rule for a *root*, not a log) would wrongly include $x = \pm 2$, where $\log 0$ is undefined.

Example 3 — [JEE Main Level] Find the range of $f(x) = \frac{x}{1+x^2}$, $x \in \mathbb{R}$.

Domain is $\mathbb{R}$ (denominator $1+x^2 \geq 1 > 0$). Apply the y -method:

$$y = \frac{x}{1+x^2} \implies y(1+x^2) = x \implies yx^2 - x + y = 0.$$

For real x , discriminant ≥ 0 :

$$(-1)^2 - 4(y)(y) \geq 0 \implies 1 - 4y^2 \geq 0 \implies y^2 \leq \frac{1}{4} \implies -\frac{1}{2} \leq y \leq \frac{1}{2}.$$

(The degenerate case $y = 0$ gives $x = 0$, valid.) Therefore

$$\text{Range}(f) = \left[-\frac{1}{2}, \frac{1}{2}\right].$$

Example 4 — [JEE Advanced Level] Find the domain of $f(x) = \sqrt{\log_{1/2}\left(\frac{x-2}{x+1}\right)}$.

Two layered constraints. **(i)** The log argument must be positive: $\frac{x-2}{x+1} > 0$, true for $x < -1$ or $x > 2$. **(ii)** The square root needs its content ≥ 0 :

$$\log_{1/2}\left(\frac{x-2}{x+1}\right) \geq 0.$$

Because the base $\frac{1}{2} < 1$, the log is ≥ 0 exactly when its argument is ≤ 1 :

$$\frac{x-2}{x+1} \leq 1 \implies \frac{x-2}{x+1} - 1 \leq 0 \implies \frac{(x-2) - (x+1)}{x+1} \leq 0 \implies \frac{-3}{x+1} \leq 0 \implies x+1 > 0 \implies x > -1.$$

Intersect (i) and (ii): $\{x < -1 \text{ or } x > 2\} \cap \{x > -1\} = (2, \infty)$. Therefore

$$\boxed{\text{Dom}(f) = (2, \infty)}.$$

Key subtlety: the inequality direction in step (ii) **flips** because the log base is less than 1 — the most common place students lose this problem.

Section 5: Practice Exercises

1. **[CBSE]** Find the domain of $f(x) = \frac{x+1}{x^2-9}$.
2. **[CUET]** Find the domain of $f(x) = \sqrt{4-x^2}$.
3. **[JEE Main]** Find the range of $f(x) = \frac{2x+3}{x-1}, x \neq 1$.
4. **[JEE Main]** Find the domain of $f(x) = \frac{1}{\sqrt{x-|x|}}$.
5. **[JEE Advanced]** Find the range of $f(x) = \frac{x^2+2x+3}{x^2+2x+2}$.

Answers: 1. $\mathbb{R} \setminus \{-3, 3\}$. 2. $[-2, 2]$. 3. $\mathbb{R} \setminus \{2\}$ (rational $\frac{ax+b}{cx+d}$, excluded value $a/c = 2$). 4. $\emptyset$ — for all x , $x - |x| \leq 0$, so $\sqrt{\quad}$ in a denominator (needs > 0) is never defined; the domain is empty. 5. Writing $f = 1 + \frac{1}{(x+1)^2+1}$; since $(x+1)^2+1 \geq 1$, the added term lies in $(0, 1]$, so range = $(1, 2]$.

Section 6: Crack the MCQ

Q1. The domain of $f(x) = \sqrt{x-2} + \sqrt{6-x}$ is:

- (a) $[2, 6]$ (b) $(2, 6)$ (c) $(-\infty, 2] \cup [6, \infty)$ (d) $\mathbb{R}$

Answer: (a). Need $x-2 \geq 0$ and $6-x \geq 0$, i.e. $x \geq 2$ **and** $x \leq 6$; intersect to $[2, 6]$. *Why the traps fail:* **(b)** wrongly uses strict inequalities (roots allow equality). **(c)** takes the union instead of the intersection. **(d)** ignores the constraints entirely.

Q2. The range of $f(x) = \frac{1}{2 - \sin 3x}$ is:

- (a) $[\frac{1}{3}, 1]$ (b) $[-1, 1]$ (c) $(\frac{1}{3}, 1)$ (d) $[\frac{1}{3}, \infty)$

Answer: (a). Since $\sin 3x \in [-1, 1]$, the denominator $2 - \sin 3x \in [1, 3]$, so its reciprocal ranges over $[\frac{1}{3}, 1]$. *Why the traps fail:* **(b)** is the range of $\sin$, not of f . **(c)** wrongly opens the endpoints (the bounds are attained). **(d)** forgets the denominator is bounded below by 1.

Q3. The domain of $f(x) = \log_{10}(1-x)$ is:

- (a) $x > 1$ (b) $x < 1$ (c) $x \leq 1$ (d) $x \neq 1$

Answer: (b). Log argument $> 0 \Rightarrow 1-x > 0 \Rightarrow x < 1$. *Why the traps fail:* **(a)** flips the inequality. **(c)** allows $x = 1$ where the argument is 0 (log undefined). **(d)** treats it like a denominator condition.

Q4. If $f(x) = \frac{x-3}{x+3}$, the value y that is **not** in the range is:

- (a) 0 (b) 1 (c) -1 (d) 3

Answer: (b). For $\frac{ax+b}{cx+d}$ the excluded range value is $\frac{a}{c} = \frac{1}{1} = 1$. (Solving $y = \frac{x-3}{x+3}$ gives $x = \frac{3(1+y)}{1-y}$, undefined at $y = 1$.) *Why the traps fail:* **(a)** $y = 0$ occurs at $x = 3$. **(c)** $y = -1$ occurs at $x = 0$. **(d)** $y = 3$ is attainable; students confuse the excluded input with the excluded output.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Confusing the root rule with the log rule.** Under a root: ≥ 0 . Inside a log: strictly > 0 . Mixing them adds or drops boundary points.
2. **Union instead of intersection.** Multiple constraints must hold *together* — always intersect, never union.
3. **Forgetting the inequality flip for log base < 1 .** $\log_{1/2}(t) \geq 0 \iff 0 < t \leq 1$ — the direction reverses.
4. **Calling the codomain the range.** “ $f : \mathbb{R} \rightarrow \mathbb{R}$ ” does not mean the range is $\mathbb{R}$; always compute the actual outputs.

Pro-Tip (speed): For a rational $\frac{ax+b}{cx+d}$, the range is instantly $\mathbb{R} \setminus \{a/c\}$ — no algebra needed. For anything built from $\sin / \cos$, push the known bound $[-1, 1]$ through the operations rather than inverting. And for $\frac{\text{quadratic}}{\text{quadratic}}$, reach straight for the **discriminant method**.

Section 8: Cheat Sheet Box

DOMAIN & RANGE — QUICK RECALL

- **Three domain jams:** denominator $\neq 0$ · even root ≥ 0 · log argument > 0 . Solve each, then **intersect**.
- $\frac{1}{\sqrt{g}}$ needs $g > 0$ (strict); plain $\sqrt{g}$ needs $g \geq 0$.
- **Range by y-method:** $y = f(x) \rightarrow$ solve for $x \rightarrow$ keep y giving real, in-domain x .
- Rational $\frac{ax+b}{cx+d}$: range = $\mathbb{R} \setminus \{a/c\}$.
- $\frac{\text{quad}}{\text{quad}}$: cross-multiply, force **discriminant** ≥ 0 .

- Standard ranges: $x^2, |x| : [0, \infty) \cdot e^x : (0, \infty) \cdot \ln x : \mathbb{R} \cdot \sin, \cos : [-1, 1]$.

Mnemonics: “**Root non-negative, Log strictly positive, Denominator never zero.**” · “Constraints **meet** (intersect), they don’t **mingle** (union).”
· “Base below one $\Rightarrow$ **flip** the log inequality.”

Subtopic 05 — Standard Real Functions and Their Graphs

Exam Relevance: Recognising standard functions on sight is assumed knowledge across every exam. JEE Main and Advanced lean hard on the **greatest-integer**, **fractional-part**, **modulus**, and **signum** functions (and graph transformations). CUET tests definitions and values. CBSE Boards use these graphs in continuity/differentiability later. (*Not in NEET — no mathematics section.*)

Section 1: Concept Intuition

Before you can analyse functions, you need a **toolkit of named machines** you recognise instantly — the way a mechanic knows every tool by its shape. Each standard function has a characteristic rule, a characteristic graph, and a characteristic personality.

- **Modulus (absolute value)** $|x|$ strips the sign — it reports *distance from zero*, so it is never negative. Its graph is a “V” sitting on the origin.
- **Signum** $\text{sgn}(x)$ is the opposite idea: it throws away the size and keeps only the *sign*, outputting just -1 , 0 , or $+1$. Its graph is two flat shelves with a single dot at the origin.
- **Greatest integer (floor)** $[x]$ rounds *down* to the nearest integer — the “step” or “staircase” function. Crucially, it rounds down even for negatives, so $[-2.3] = -3$, not -2 .
- **Fractional part** $\{x\} = x - [x]$ keeps only what’s left after the floor — always in $[0, 1)$ — producing a repeating “sawtooth.”
- **Exponential** a^x is explosive, controlled growth; it is always positive and never touches the x -axis.
- **Logarithm** $\log_a x$ is the exponential run backwards — defined only for positive inputs, growing ever more slowly.

Think about it this way... The graph is the fastest summary of a function. One glance tells you the domain (how far left-right it extends), the range (how far up-down), where it breaks, and whether it is increasing or decreasing. Memorising these six shapes is like memorising multiplication tables — it makes everything downstream automatic.

[See the atlas of standard functions below.]

Limiting conditions & assumptions. The floor and fractional-part functions are **discontinuous at every integer** (they jump). The logarithm exists only for $x > 0$; the exponential output is only ever > 0 . Always state the base for a^x and $\log_a x$ with $a > 0$, $a \neq 1$.

Atlas of Standard Real Functions

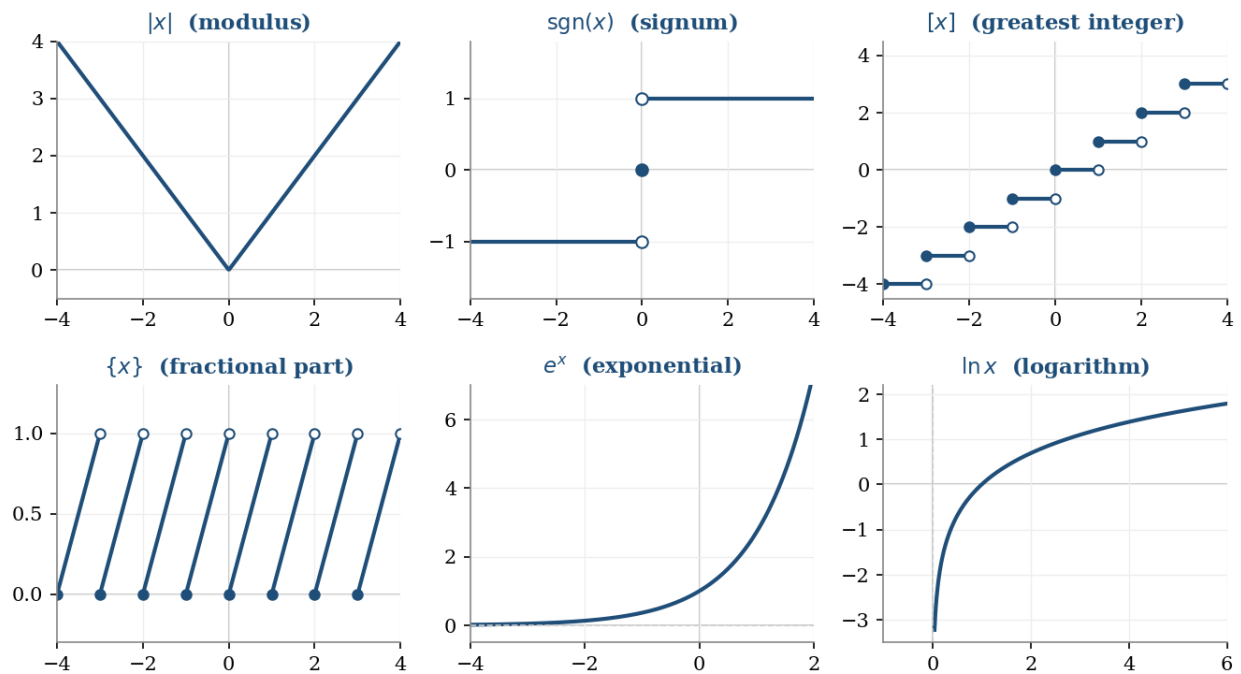


Figure 1: Atlas of standard real functions: modulus, signum, greatest integer (floor), fractional part, exponential, and logarithm.

Section 2: Key Definitions & Properties

Function	Definition	Domain	Range
Modulus $ x $	x if $x \geq 0$, else $-x$	$\mathbb{R}$	$[0, \infty)$
Signum $\text{sgn}(x)$	$\frac{x}{ x }$ for $x \neq 0$, 0 at $x = 0$	$\mathbb{R}$	$\{-1, 0, 1\}$
Greatest integer $[x]$	greatest integer $\leq x$	$\mathbb{R}$	$\mathbb{Z}$
Fractional part $\{x\}$	$x - [x]$	$\mathbb{R}$	$[0, 1)$
Exponential a^x	$a > 0$, $a \neq 1$	$\mathbb{R}$	$(0, \infty)$
Logarithm $\log_a x$	inverse of a^x	$(0, \infty)$	$\mathbb{R}$

Key identities

- $|x|^2 = x^2$, $|xy| = |x||y|$, triangle inequality $||x| - |y|| \leq |x \pm y| \leq |x| + |y|$.
- $\text{sgn}(x) \cdot |x| = x$.
- $[x + n] = [x] + n$ for integer n ; $x - 1 < [x] \leq x$; $[x] + [-x] = \begin{cases} 0 & x \in \mathbb{Z} \\ -1 & x \notin \mathbb{Z} \end{cases}$.
- $\{x\}$ has period 1; $\{x + n\} = \{x\}$.
- $\log_a(xy) = \log_a x + \log_a y$; $\log_a x = \frac{\log_b x}{\log_b a}$ (change of base); $a^{\log_a x} = x$.

Notation lock: $|x|$ modulus, $\text{sgn}(x)$ signum, $[x]$ or $\lfloor x \rfloor$ floor, $\{x\}$ fractional part.

Section 3: Key Procedures & Logical Framework

Procedure A — Graphing by transformation. Start from a base graph and apply, in order: horizontal shift $f(x - h)$ (right by h), vertical shift $f(x) + k$ (up by k), stretch/compression $a f(x)$, and reflection $-f(x)$ (about x -axis) or $f(-x)$ (about y -axis). *Reasoning:* each algebraic operation moves the picture in a predictable, decoupled way, so a complicated graph is just the base shape relocated.

Procedure B — The two modulus transformations (a frequent JEE distinction).

- $y = |f(x)|$: take the graph of f and **flip everything below the x -axis upward**. *Reasoning:* negative outputs become their positive mirror.
- $y = f(|x|)$: **keep the right half ($x \geq 0$) and mirror it to the left**. *Reasoning:* the input's sign is erased, so left and right inputs share outputs — the graph becomes symmetric about the y -axis.

These are *different* operations and produce different graphs — confusing them is a classic error.

Worked construction — graph of $y = |x - 2| + 1$. Begin with $y = |x|$ (a V at the origin). Replace x by $x - 2$: the V slides **right** by 2, vertex now at $(2, 0)$. Add 1: the whole V lifts **up** by 1, vertex at $(2, 1)$. The result is a V opening upward with its lowest point at $(2, 1)$, so the minimum value of the function is 1.

Figure 1.6 a V-shaped graph with vertex at $(2, 1)$, arms rising with slopes $+1$ and -1 , dashed guide lines showing the shift from the origin V. (see next page)

Section 4: Worked Examples

Example 1 — [CBSE Board Level] Evaluate: $[2.7]$, $[-2.3]$, $\{3.6\}$, $\text{sgn}(-5)$, $|-4|$.

- $[2.7] = 2$ (greatest integer ≤ 2.7).
- $[-2.3] = -3$ (greatest integer ≤ -2.3 — round **down**, past zero).
- $\{3.6\} = 3.6 - [3.6] = 3.6 - 3 = 0.6$.
- $\text{sgn}(-5) = -1$.
- $|-4| = 4$.

Example 2 — [CUET / JEE Main Speed Trap] Solve $[x] = 3$, and separately find all x with $\{x\} = 0.5$.

$[x] = 3$: the floor equals 3 for every x from 3 (inclusive) up to but not including 4. So $x \in [3, 4)$. **Trap:** the answer is an *interval*, not the single value $x = 3$.

$\{x\} = 0.5$: the fractional part is 0.5 whenever x is “half-past” an integer: $x = n + 0.5$ for any $n \in \mathbb{Z}$, i.e. $x \in \{\dots, -1.5, -0.5, 0.5, 1.5, \dots\}$.

Example 3 — [JEE Main Level] Find the range of $f(x) = [x] + \{-x\}$ over $\mathbb{R}$.

Split into cases. **If $x \in \mathbb{Z}$:** $[x] = x$ and $\{-x\} = 0$, so $f(x) = x$, an integer. **If $x \notin \mathbb{Z}$:** write $x = n + r$ with $n = [x]$ and $r = \{x\} \in (0, 1)$. Then $-x = -n - r$, and $[-x] = -n - 1$, so $\{-x\} = -x - [-x] = (-n - r) - (-n - 1) = 1 - r$. Thus

$$f(x) = n + (1 - r) = [x] + 1 - \{x\}.$$

As $r \in (0, 1)$, $1 - r \in (0, 1)$, so $f(x) \in (n, n + 1)$ — every non-integer real. Combined with the integer case, the **range is all of $\mathbb{R}$** .

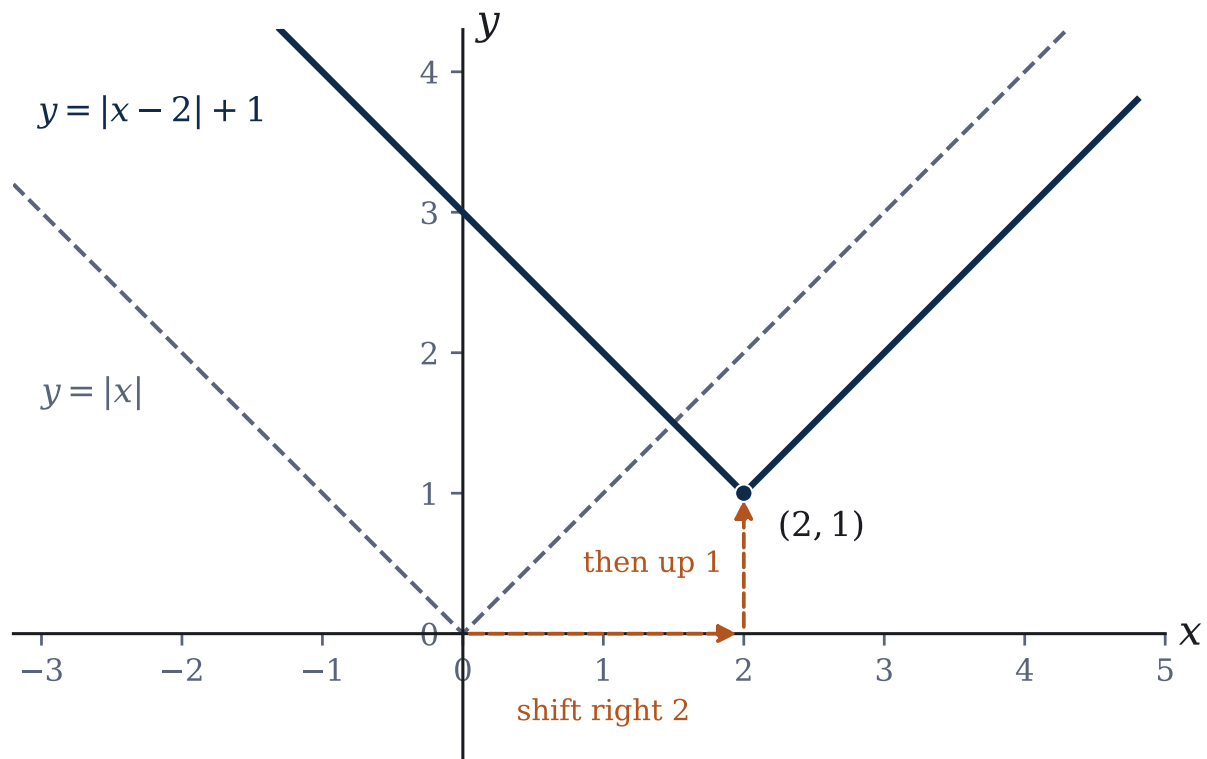


Figure 1.6 a V-shaped graph with vertex at $(2, 1)$, arms rising with slopes $+1$ and -1 , dashed guide lines showing the shift from the origin V .

Example 4 — [JEE Advanced Level] Find the domain of $f(x) = \frac{1}{\sqrt{\{x\}}} + \log(1 - [x])$ on $(-3, 3)$.

Term 1 needs $\{x\} > 0$ (it's under a root in a denominator), which fails exactly when $x \in \mathbb{Z}$. So exclude all integers.

Term 2 needs $1 - [x] > 0 \Rightarrow [x] < 1 \Rightarrow [x] \leq 0 \Rightarrow x < 1$.

On the interval $(-3, 3)$, combine: $x < 1$ **and** $x \notin \mathbb{Z}$. The integers in range to remove are $-2, -1, 0$ (note -3 and 3 are already outside the open interval, and integers ≥ 1 are excluded by $x < 1$ anyway). Therefore

$$\text{Dom}(f) = (-3, 1) \setminus \{-2, -1, 0\} = (-3, -2) \cup (-2, -1) \cup (-1, 0) \cup (0, 1).$$

Section 5: Practice Exercises

1. **[CBSE]** Evaluate $[-0.5]$, $\{-0.5\}$, and $\text{sgn}(0)$.
2. **[CUET]** Sketch (describe) the graph of $y = |x| - 2$ and state its range.
3. **[JEE Main]** Find the range of $f(x) = \frac{|x|}{x}$ for $x \neq 0$.
4. **[JEE Main]** Solve $[x]^2 - 5[x] + 6 = 0$.
5. **[JEE Advanced]** Find the number of solutions of $\{x\} = \frac{x}{5}$ in $[0, 5]$.

Answers: 1. $[-0.5] = -1$; $\{-0.5\} = 0.5$; $\text{sgn}(0) = 0$. 2. A V with vertex at $(0, -2)$; range $[-2, \infty)$. 3. $\{-1, 1\}$ (it equals $\text{sgn}(x)$ for $x \neq 0$). 4. $[x] = 2$ or $[x] = 3 \Rightarrow x \in [2, 3) \cup [3, 4) = [2, 4)$. 5. 4 solutions: $x = 0, 5/4, 5/2, 15/4$ (the branch $x = 5n/4$ stays valid only for $n = 0, 1, 2, 3$; $x = 5$ fails since $\{5\} = 0 \neq 1$).

Section 6: Crack the MCQ

Q1. $[-3.7] + [3.7]$ equals:

- (a) 0 (b) -1 (c) -7 (d) 7

Answer: (b). $[-3.7] = -4$ and $[3.7] = 3$; sum $= -1$. (Consistent with $[x] + [-x] = -1$ for non-integers.) *Why the traps fail:* (a) assumes $[-x] = -[x]$ (false off the integers). (c)/(d) mishandle the floor of the negative as -3 .

Q2. The range of $f(x) = 2 + \text{sgn}(x)$ is:

- (a) $\{1, 2, 3\}$ (b) $[1, 3]$ (c) $\{2\}$ (d) $\mathbb{R}$

Answer: (a). $\text{sgn}(x) \in \{-1, 0, 1\}$, so $2 + \text{sgn}(x) \in \{1, 2, 3\}$. *Why the traps fail:* **(b)** treats a 3-value set as a continuous interval. **(c)** forgets sgn varies. **(d)** ignores that signum is bounded.

Q3. Which is the graph of $y = f(|x|)$ obtained from $y = f(x)$?

- (a) flip the part below the x -axis upward (b) keep $x \geq 0$ and mirror it to $x < 0$ (c) shift right (d) reflect about $y = x$

Answer: (b). $f(|x|)$ erases the input's sign, making the graph symmetric about the y -axis by mirroring the right half. *Why the traps fail:* **(a)** is the rule for $y = |f(x)|$ — the commonly swapped pair. **(c)** is a translation. **(d)** is the inverse-function reflection.

Q4. The domain of $f(x) = \log_2(\log_3 x)$ is:

- (a) $x > 0$ (b) $x > 1$ (c) $x > 3$ (d) $x \geq 1$

Answer: (b). Work outward. The outer $\log_2$ requires its argument positive: $\log_3 x > 0$, which means $x > 3^0 = 1$. (This automatically satisfies the inner $\log$'s own requirement $x > 0$.) So the domain is $x > 1$. *Why the traps fail:* **(a)** stops at the inner $\log$'s condition and forgets the outer one. **(c)** misreads the threshold as 3^1 instead of 3^0 . **(d)** wrongly includes $x = 1$, where $\log_3 1 = 0$ and the outer $\log$ is undefined.

Section 7: Common Student Pitfalls & Pro-Tips

- Floor of a negative.** $[-2.3] = -3$, **not** -2 . The floor always rounds *down* (more negative). This is the single biggest source of lost marks here.
- $|f(x)|$ vs $f(|x|)$.** They are different transformations — flip-up versus mirror-right. Keep them straight.
- Treating discrete ranges as intervals.** sgn and $[x]$ output specific values/integers, not continuous bands.
- Forgetting log/exponential domains.** $\log$ needs $x > 0$; $a^x > 0$ always (it never equals 0).

Pro-Tip (speed): For any floor/fractional problem, immediately split into “ x integer” vs “ x non-integer” cases — almost every such problem cracks open that way. For $\{x\} =$ (line) counting problems, sketch the sawtooth against the line and *count crossings*.

Section 8: Cheat Sheet Box

STANDARD FUNCTIONS — QUICK RECALL

- $|x|$: V-shape, range $[0, \infty)$; $|x|^2 = x^2$, $|xy| = |x||y|$.

- $\text{sgn}(x) \in \{-1, 0, 1\}$; $\text{sgn}(x)|x| = x$.
- $[x] = \text{floor (round **down**, even for negatives)}$; $x - 1 < [x] \leq x$; $[x] + [-x] = -1$ off $\mathbb{Z}$, 0 on $\mathbb{Z}$.
- $\{x\} = x - [x] \in [0, 1)$, period 1.
- $a^x : (0, \infty)$ range, never 0; $\log_a x : (0, \infty)$ domain, $\mathbb{R}$ range.
- **Transforms:** $|f(x)|$ flips below-axis up; $f(|x|)$ mirrors right half to left.

Mnemonics: “**Floor falls** (rounds down, negatives too).” · “ $|f|$ **flips**, $f(|x|)$ **mirrors**.” · “Log lives on the **positive** side; exp stays **above** the axis.”

Subtopic 06 — Even, Odd, and Periodic Functions

Exam Relevance: A staple of JEE Main and Advanced — symmetry and periodicity questions appear directly and as time-savers inside integration and graphing problems. CUET includes definition/classification items. CBSE Boards touch it lightly. (*Not in NEET — no mathematics section.*)

Section 1: Concept Intuition

Some functions have a hidden symmetry that, once spotted, halves your work. There are two kinds.

An **even function** has **mirror symmetry about the y -axis**: whatever happens at x happens identically at $-x$. Picture a butterfly, or the parabola $y = x^2$ — fold the graph along the vertical axis and the two halves coincide. Formally, $f(-x) = f(x)$.

An **odd function** has **rotational symmetry about the origin**: flip it left-right *and* up-down and it lands on itself. Think of $y = x^3$ or $y = \sin x$. Formally, $f(-x) = -f(x)$. A neat consequence: if an odd function is defined at 0, it *must* pass through the origin, since $f(0) = -f(0)$ forces $f(0) = 0$.

Most functions are **neither** — and that's fine. In fact, *every* function (on a symmetric domain) splits uniquely into an even piece plus an odd piece, the way any number splits into “average plus deviation.”

Separately, a function is **periodic** if its graph **repeats at regular intervals** — like a heartbeat trace, the seasons, or a tabla rhythm cycling every few beats. If the pattern repeats every T units, then $f(x + T) = f(x)$ for all x , and the smallest such positive T is the **fundamental period**.

Think about it this way... Symmetry and periodicity are both forms of *redundancy* — the function tells you the same thing more than once. An even function only needs to be understood on $[0, \infty)$ (the left half is a mirror). A periodic function only needs to be understood on one period (everything else is a copy). Exploiting that redundancy is where the marks (and the speed) come from.

Limiting conditions & assumptions. Even/odd classification requires a **domain symmetric about 0** (so that $-x$ is allowed whenever x is). For periodicity, a **constant** function technically satisfies $f(x + T) = f(x)$ for *every* T , so it is periodic but has **no fundamental period**. And when you add two periodic functions, a common period exists only if their individual periods are **commensurable** (their ratio is rational).

Section 2: Key Definitions & Properties

Even / Odd

$$\text{Even: } f(-x) = f(x) \qquad \text{Odd: } f(-x) = -f(x)$$

Examples: even — x^2 , $|x|$, $\cos x$, constants. Odd — x , x^3 , $\sin x$, $\operatorname{sgn} x$, $\tan x$.

Combination rules

Operation	Result
even $\pm$ even	even
odd $\pm$ odd	odd
even $\times$ even	even
odd $\times$ odd	even
even $\times$ odd	odd

Unique decomposition (domain symmetric about 0):

$$f(x) = \underbrace{\frac{f(x) + f(-x)}{2}}_{\text{even part}} + \underbrace{\frac{f(x) - f(-x)}{2}}_{\text{odd part}}.$$

Periodic

$$f(x + T) = f(x) \quad \forall x; \quad \text{fundamental period} = \text{smallest such } T > 0.$$

Standard periods: $\sin x, \cos x : 2\pi$; $\tan x, \cot x : \pi$; $|\sin x|, |\cos x| : \pi$; $\{x\} : 1$.

Period rules

- $f(ax + b)$ has period $\frac{T}{|a|}$ if f has period T .
- $f(x) \pm g(x)$ with periods T_1, T_2 is periodic with period $\operatorname{LCM}(T_1, T_2)$, *provided* T_1/T_2 is rational.
- Taking $|f|$ often (not always) **halves** the period (e.g. $|\sin x|$: π instead of 2π).

Notation lock: T for period; “even/odd part” as above.

Section 3: Key Procedures & Derivations

Procedure A — Test for even/odd. Compute $f(-x)$ and compare: equals $f(x)$ $\rightarrow$ even; equals $-f(x)$ $\rightarrow$ odd; neither $\rightarrow$ neither. *Reasoning:* the definitions are pointwise identities, so a direct substitution settles it.

Procedure B — Find the fundamental period. Identify the period of each piece, apply $T/|a|$ for internal coefficients, then take the LCM across a sum/difference; finally check whether a smaller value also works (especially after a modulus). *Reasoning:* the LCM is the smallest length over which *every* component has completed a whole number of cycles.

Derivation 1 — product of two odd functions is even. Let f, g be odd, so $f(-x) = -f(x)$ and $g(-x) = -g(x)$. Then

$$(fg)(-x) = f(-x)g(-x) = (-f(x))(-g(x)) = f(x)g(x) = (fg)(x).$$

The two sign flips cancel, so the product is **even**. ■

Derivation 2 — fundamental period of $f(x) = |\sin x| + |\cos x|$. Each of $|\sin x|$ and $|\cos x|$ has period π , so π is a period. But test the smaller candidate $\frac{\pi}{2}$:

$$f(x + \frac{\pi}{2}) = |\sin(x + \frac{\pi}{2})| + |\cos(x + \frac{\pi}{2})| = |\cos x| + |\sin x| = f(x).$$

Reasoning: the shift by $\frac{\pi}{2}$ swaps sine and cosine (up to sign), and the moduli erase the sign — so the sum is unchanged. No positive value smaller than $\frac{\pi}{2}$ works, so the fundamental period is

$$\boxed{T = \frac{\pi}{2}}. \quad \blacksquare$$

Section 4: Worked Examples

Example 1 — [CBSE Board Level] Classify as even, odd, or neither: (a) $f(x) = x^3 - x$; (b) $g(x) = x^2 + \cos x$; (c) $h(x) = x + x^2$.

(a) $f(-x) = (-x)^3 - (-x) = -x^3 + x = -(x^3 - x) = -f(x) \rightarrow$ **odd**. **(b)** $g(-x) = (-x)^2 + \cos(-x) = x^2 + \cos x = g(x) \rightarrow$ **even**. **(c)** $h(-x) = -x + x^2$, which equals neither $h(x) = x + x^2$ nor $-h(x) = -x - x^2 \rightarrow$ **neither**.

Example 2 — [CUET / JEE Main Speed Trap] Which of these is an even function: $\sin x$, $x|x|$, $\cos x + x^2$, x^3 ?

Speed: scan for symmetry. $\sin x$ is odd; $x|x|$ is odd (odd $\times$ even = odd); x^3 is odd. Only $\cos x + x^2$ is even (even + even). **Answer:** $\cos x + x^2$. **Trap:** $x|x|$ looks even because of the modulus, but the lone x factor makes the whole thing odd.

Example 3 — [JEE Main Level] Find the fundamental period of $f(x) = \sin 3x + \cos 2x$.

Period of $\sin 3x$ is $\frac{2\pi}{3}$; period of $\cos 2x$ is $\frac{2\pi}{2} = \pi$. The overall period is the smallest T that is a whole-number multiple of both. Testing $T = 2\pi$: it equals $3 \times \frac{2\pi}{3}$ and $2 \times \pi$ — a whole multiple of each — and nothing smaller works. So the fundamental period is $\boxed{2\pi}$.

Example 4 — [JEE Advanced Level] Express $f(x) = e^x$ as the sum of an even and an odd function, and identify the two parts.

Using the decomposition formula:

$$\text{even part} = \frac{e^x + e^{-x}}{2} = \cosh x, \quad \text{odd part} = \frac{e^x - e^{-x}}{2} = \sinh x.$$

So $e^x = \cosh x + \sinh x$. *Verification:* $\cosh(-x) = \cosh x$ (even) ✓ and $\sinh(-x) = -\sinh x$ (odd) ✓, and their sum reconstructs e^x . This is exactly why the hyperbolic functions are the canonical even/odd halves of the exponential — a structural fact JEE Advanced likes to probe.

Section 5: Practice Exercises

- [CBSE]** Classify $f(x) = x^4 + 3x^2 + 7$ as even, odd, or neither.
- [CUET]** Is $f(x) = \frac{e^x - 1}{e^x + 1}$ even, odd, or neither?
- [JEE Main]** Find the fundamental period of $f(x) = \sin\left(\frac{x}{2}\right) + \cos\left(\frac{x}{3}\right)$.
- [JEE Main]** Find the odd part of $f(x) = x^2 + 2x + 1$.
- [JEE Advanced]** Find the fundamental period of $f(x) = |\sin x| + |\cos x| + \{x\}$... if it exists; justify.

Answers: **1.** Even (all even powers + constant). **2.** Odd ($f(-x) = \frac{e^{-x}-1}{e^{-x}+1} = \frac{1-e^x}{1+e^x} = -f(x)$). **3.** $\text{LCM}(4\pi, 6\pi) = 12\pi$. **4.** Odd part = $\frac{f(x)-f(-x)}{2} = \frac{(x^2+2x+1)-(x^2-2x+1)}{2} = 2x$. **5.** No fundamental period — $|\sin x| + |\cos x|$ has period $\frac{\pi}{2}$ while $\{x\}$ has period 1; since the ratio $\frac{\pi/2}{1} = \frac{\pi}{2}$ is irrational, the periods are incommensurable and no common period exists.

Section 6: Crack the MCQ

Q1. The function $f(x) = x \cos x$ is:

- (a) even (b) odd (c) neither (d) periodic with period π

Answer: (b). x is odd, $\cos x$ is even, and $\text{odd} \times \text{even} = \text{odd}$. Check: $f(-x) = (-x) \cos(-x) = -x \cos x = -f(x)$. *Why the traps fail:* **(a)** assumes the $\cos$ dominates. **(c)** misapplies the product rule. **(d)** f is not periodic — the growing x factor breaks repetition.

Q2. For an odd function f defined at 0, the value $f(0)$ is:

- (a) 1 (b) undefined (c) 0 (d) cannot be determined

Answer: (c). Setting $x = 0$ in $f(-x) = -f(x)$ gives $f(0) = -f(0) \Rightarrow 2f(0) = 0 \Rightarrow f(0) = 0$. *Why the traps fail:* **(a)/(d)** ignore the forcing identity. **(b)** confuses “defined at 0” with undefined.

Q3. The fundamental period of $f(x) = \sin^2 x$ is:

- (a) 2π (b) π (c) $\frac{\pi}{2}$ (d) 4π

Answer: (b). Using $\sin^2 x = \frac{1 - \cos 2x}{2}$, the period is that of $\cos 2x$, namely $\frac{2\pi}{2} = \pi$. *Why the traps fail:* **(a)** is the period of $\sin x$ before squaring. **(c)/(d)** mis-scale the coefficient inside the cosine.

Q4. Which is the even part of $f(x) = x^3 + x^2 + x + 1$?

- (a) $x^3 + x$ (b) $x^2 + 1$ (c) x^2 (d) 1

Answer: (b). Even part $= \frac{f(x) + f(-x)}{2} = \frac{(x^3 + x^2 + x + 1) + (-x^3 + x^2 - x + 1)}{2} = \frac{2x^2 + 2}{2} = x^2 + 1$. *Why the traps fail:* **(a)** is the *odd* part. **(c)/(d)** drop a term from the even part.

Section 7: Common Student Pitfalls & Pro-Tips

- Assuming every function is even or odd.** Most are neither; always actually compute $f(-x)$.
- Mishandling products.** $\text{odd} \times \text{odd} = \text{even}$, not odd. Track sign flips carefully.
- Forgetting the modulus can shrink the period.** $|\sin x|$ has period π , not 2π — always test the half-period after a modulus.
- Taking LCM of incommensurable periods.** If T_1/T_2 is irrational (e.g. π and 1), no common period exists — the sum is *not* periodic.

Pro-Tip (speed): A polynomial is even iff it has **only even powers**, odd iff **only odd powers** (no constant) — read it off instantly. For periods, convert powers of $\sin/\cos$ to double angles first ($\sin^2, \cos^2 \rightarrow$ period halves). And remember odd functions force $f(0) = 0$ — a free check.

Section 8: Cheat Sheet Box

EVEN, ODD & PERIODIC — QUICK RECALL

- **Even:** $f(-x) = f(x)$ (y -axis mirror). **Odd:** $f(-x) = -f(x)$ (origin symmetry; forces $f(0) = 0$).
- **Products:** even·even = even, odd·odd = **even**, even·odd = odd.
- **Decompose:** even part = $\frac{f(x)+f(-x)}{2}$, odd part = $\frac{f(x)-f(-x)}{2}$.
- **Periodic:** $f(x+T) = f(x)$; fundamental T = smallest such.
- $\sin, \cos : 2\pi \cdot \tan, \cot : \pi \cdot |\sin|, |\cos| : \pi \cdot \{x\} : 1$.
- $f(ax+b) \rightarrow$ period $\frac{T}{|a|}$; sum $\rightarrow$ LCM (only if ratio rational).

Mnemonics: “**Even = mirror, Odd = origin.**” · “Only-even-powers $\Rightarrow$ even; only-odd-powers $\Rightarrow$ odd.” · “Modulus may **halve** the period.” · “**Irrational ratio $\Rightarrow$ not periodic.**”

Subtopic 07 — Algebra of Functions and Functional Equations

Exam Relevance: Algebra of functions is light but examinable (CBSE, CUET). **Functional equations** are a JEE Advanced signature — the “replace x by something clever and solve” trick appears most years, and JEE Main includes the easier substitution types. (*Not in NEET — no mathematics section.*)

Section 1: Concept Intuition

Two ideas live here, and both are about *building or discovering* functions.

Algebra of functions is the obvious part: just as you add and multiply numbers, you can add and multiply functions, point by point. $(f + g)(x)$ simply means “evaluate f at x , evaluate g at x , add the results.” The only subtlety is the **domain** — a combined function is defined only where *both* ingredients are defined (and, for a quotient, where the divider isn’t zero).

Functional equations are the deeper, more exciting part. Instead of being handed a formula, you’re handed a *rule the function obeys* — for example, “ $f(x + y) = f(x) + f(y)$ for all x, y ” — and asked to discover which function(s) could possibly behave that way. It’s reverse-engineering: you know how the machine *responds*, and you must deduce the machine.

Think about it this way... Solving a functional equation is detective work, and the detective’s main tool is **clever substitution**. Plug in $x = y = 0$ to learn $f(0)$. Plug in $y = 1$, or $y = -x$, or $x \rightarrow \frac{1}{x}$, to generate a *second* equation. With two equations relating $f(x)$ and (say) $f(\frac{1}{x})$, you can eliminate the unknown the same way you solve simultaneous linear equations — and out pops the formula for f .

Limiting conditions & assumptions. For combinations, never forget the **intersection of domains** (and $g \neq 0$ for f/g). For functional equations, a *unique* answer usually requires an extra regularity assumption (typically **continuity**); at the school/exam level we take the “nice” solution that the standard substitutions produce, which is what the famous equations are designed to yield.

Section 2: Key Definitions & Properties

Algebra of functions. For f, g real functions with domains D_f, D_g :

$$(f \pm g)(x) = f(x) \pm g(x), \quad (fg)(x) = f(x)g(x), \quad \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}.$$

Domains: $f \pm g$ and fg are defined on $D_f \cap D_g$; the quotient $\frac{f}{g}$ on $D_f \cap D_g \cap \{x : g(x) \neq 0\}$.

Standard functional equations and their (regular) solutions

Functional equation	Solution
$f(x + y) = f(x) + f(y)$ (Cauchy)	$f(x) = kx$
$f(x + y) = f(x)f(y)$	$f(x) = a^x$
$f(xy) = f(x) + f(y)$	$f(x) = k \log x$
$f(xy) = f(x)f(y)$	$f(x) = x^n$
$f(x)f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$	$f(x) = x^n \pm 1$

Useful first moves: $x = y = 0$ (find $f(0)$); $x = y = 1$ (find $f(1)$ or constraints); $y = 1$; $y = -x$; $x \rightarrow \frac{1}{x}$; $x \rightarrow 1 - x$.

Notation lock: D_f, D_g for domains; functional-equation unknown kept as f .

Section 3: Key Procedures & Derivations

Procedure A — Domain of a combination. Find D_f and D_g separately, intersect them, and for a quotient additionally delete the zeros of g . *Reasoning:* a value must be legal for *both* functions to be legal for their combination.

Procedure B — Solving by substitution / elimination. (1) Substitute special values to extract anchors like $f(0)$ or $f(1)$. (2) Substitute a transformation of x (commonly $x \rightarrow \frac{1}{x}$ or $x \rightarrow 1 - x$) to get a *second* equation in $f(x)$ and $f(\text{transformed})$. (3) Treat the two as simultaneous linear equations and eliminate the unwanted term. *Reasoning:* one equation has two unknown function-values; a second independent equation lets you solve for $f(x)$ algebraically.

Derivation — the Cauchy equation $f(x + y) = f(x) + f(y)$ **gives** $f(x) = kx$.

Set $x = y = 0$: $f(0) = f(0) + f(0) \Rightarrow f(0) = 0$. Let $k = f(1)$. Putting $y = x$ repeatedly (or induction) gives $f(2x) = 2f(x)$, and generally $f(nx) = nf(x)$ for positive integers n ; in particular $f(n) = nk$. Replacing x by $\frac{x}{n}$ shows $f\left(\frac{x}{n}\right) = \frac{1}{n}f(x)$, so for any rational $\frac{p}{q}$, $f\left(\frac{p}{q}\right) = \frac{p}{q}k$. *Reasoning:* additivity transfers integer and then rational scaling directly onto the function. Assuming f is continuous, this extends

from the rationals to all reals:

$$\boxed{f(x) = kx, \quad k = f(1)}. \quad \blacksquare$$

Section 4: Worked Examples

Example 1 — [CBSE Board Level] Let $f(x) = x^2$ and $g(x) = 2x + 1$ on $\mathbb{R}$. Find $(f + g)$, (fg) , and $\left(\frac{f}{g}\right)$ with its domain.

$$\begin{aligned}(f + g)(x) &= x^2 + 2x + 1, & (fg)(x) &= x^2(2x + 1) = 2x^3 + x^2, \\ \left(\frac{f}{g}\right)(x) &= \frac{x^2}{2x + 1}, & \text{domain } \mathbb{R} \setminus \left\{-\frac{1}{2}\right\} & \text{ (since } 2x + 1 = 0 \text{ at } x = -\frac{1}{2}\text{).}\end{aligned}$$

Example 2 — [CUET / JEE Main Speed Trap] A function satisfies $f(x + y) = f(x)f(y)$ for all x, y , with $f(1) = 3$. Find $f(3)$.

Speed: this is the exponential pattern $f(x) = a^x$, and $f(1) = a = 3$, so $f(x) = 3^x$ and $f(3) = 27$. Or directly: $f(3) = f(1 + 1 + 1) = f(1)^3 = 3^3 = 27$. **Trap:** treating it like the *additive* Cauchy equation (which would give $f(3) = 3f(1) = 9$) — wrong, because the right side is a *product*, not a sum.

Example 3 — [JEE Main Level] Find $f(x)$ given $2f(x) + f(1 - x) = x^2$ for all real x .

Call this equation (i). Replace x by $1 - x$:

$$2f(1 - x) + f(x) = (1 - x)^2. \quad \text{(ii)}$$

Multiply (i) by 2: $4f(x) + 2f(1 - x) = 2x^2$. Subtract (ii):

$$3f(x) = 2x^2 - (1 - x)^2 = 2x^2 - (1 - 2x + x^2) = x^2 + 2x - 1.$$

$$\boxed{f(x) = \frac{x^2 + 2x - 1}{3}}.$$

Example 4 — [JEE Advanced Level] Find $f(x)$, $x \neq 0$, satisfying $f(x) + 2f\left(\frac{1}{x}\right) = 3x$.

Call this (i). Replace x by $\frac{1}{x}$:

$$f\left(\frac{1}{x}\right) + 2f(x) = \frac{3}{x}. \quad (\text{ii})$$

Now eliminate $f\left(\frac{1}{x}\right)$. Multiply (ii) by 2: $2f\left(\frac{1}{x}\right) + 4f(x) = \frac{6}{x}$. Subtract (i):

$$3f(x) = \frac{6}{x} - 3x \implies \boxed{f(x) = \frac{2}{x} - x}.$$

Verification: $f(x) + 2f\left(\frac{1}{x}\right) = \left(\frac{2}{x} - x\right) + 2\left(2x - \frac{1}{x}\right) = \frac{2}{x} - x + 4x - \frac{2}{x} = 3x$ ✓ — the substitution-and-eliminate method delivers, and the check confirms it.

Section 5: Practice Exercises

- [CBSE]** For $f(x) = \sqrt{x}$ (domain $x \geq 0$) and $g(x) = x - 4$, state the domain of $\left(\frac{f}{g}\right)(x)$.
- [CUET]** If $f(x + y) = f(x) + f(y)$ and $f(1) = 5$, find $f(4)$.
- [JEE Main]** If $f(xy) = f(x)f(y)$ for all $x, y > 0$ and $f(2) = 4$, find $f(8)$.
- [JEE Main]** Solve for f : $3f(x) - f(-x) = 2x + 4$.
- [JEE Advanced]** If $f(x) + 2f(1 - x) = x^2$ for all real x , find $f(x)$ and hence $f(2)$.

Answers: **1.** $[0, \infty) \setminus \{4\} = [0, 4) \cup (4, \infty)$. **2.** $f(x) = 5x \implies f(4) = 20$. **3.** $f(x) = x^2 \implies f(8) = 64$. **4.** Replace $x \rightarrow -x$: $3f(-x) - f(x) = -2x + 4$; solving the pair gives $f(x) = \frac{x+4}{2}$. **5.** Replace $x \rightarrow 1 - x$ to get $f(1 - x) + 2f(x) = (1 - x)^2$; eliminating $f(1 - x)$ gives $f(x) = \frac{x^2 - 4x + 2}{3}$, so $f(2) = -\frac{2}{3}$.

Section 6: Crack the MCQ

Q1. For $f(x) = \frac{1}{x}$ and $g(x) = \sqrt{x}$, the domain of $(f \cdot g)(x)$ is:

- (a) $x \geq 0$ (b) $x > 0$ (c) $x \neq 0$ (d) $\mathbb{R}$

Answer: (b). g needs $x \geq 0$; f needs $x \neq 0$. Intersection: $x > 0$. *Why the traps fail:* **(a)** keeps $x = 0$ where f blows up. **(c)** allows negatives where g fails. **(d)** ignores both constraints.

Q2. If $f(x + y) = f(x) + f(y)$ for all real x, y , then $f(0)$ equals:

(a) 1 (b) 0 (c) $f(1)$ (d) undefined

Answer: (b). Put $x = y = 0$: $f(0) = 2f(0) \Rightarrow f(0) = 0$. *Why the traps fail:* **(a)** confuses additive with multiplicative (where $f(0) = 1$ can arise). **(c)** has no basis. **(d)** the equation defines $f(0)$ perfectly.

Q3. A function satisfies $f(xy) = f(x) + f(y)$ for all $x, y > 0$. Then f is of the form:

(a) kx (b) x^n (c) $k \log x$ (d) a^x

Answer: (c). Turning products into sums is the defining property of the logarithm, so $f(x) = k \log x$. *Why the traps fail:* **(a)** solves the *additive* Cauchy equation. **(b)** solves $f(xy) = f(x)f(y)$. **(d)** solves $f(x+y) = f(x)f(y)$. Each is a different pattern — matching the structure is the whole skill.

Q4. If $2f(x) + f(-x) = 3x + 9$ for all real x , then $f(2)$ equals:

(a) 9 (b) 15 (c) 5 (d) -3

Answer: (a). Replace $x \rightarrow -x$: $2f(-x) + f(x) = -3x + 9$. Solve the pair — multiply the first by 2 and subtract the second: $3f(x) = 2(3x + 9) - (-3x + 9) = 9x + 9$, so $f(x) = 3x + 3$. Hence $f(2) = 9$. (Check: $2f(2) + f(-2) = 2(9) + (-3) = 15 = 3(2) + 9$ ✓.) *Why the traps fail:* **(b)** reads off $f(x) = 3x + 9$ directly, ignoring the $f(-x)$ term. **(c)** wrongly collapses $2f(x) + f(-x)$ to $3f(x)$, giving $3f(2) = 15 \Rightarrow f(2) = 5$. **(d)** is $f(-2)$, the wrong evaluation point.

Section 7: Common Student Pitfalls & Pro-Tips

- Dropping the domain in combinations.** (f/g) excludes zeros of g and respects both original domains — intersect, then delete.
- Matching the wrong functional-equation pattern.** Sum-to-sum is linear (kx); sum-to-product is exponential; product-to-sum is logarithmic; product-to-product is a power. Read the structure before solving.
- Forgetting the second substitution.** One functional equation in $f(x)$ and $f(\text{something})$ needs a partner equation — generate it with $x \rightarrow \frac{1}{x}$ or $x \rightarrow 1 - x$.
- Assuming uniqueness without continuity.** State the regularity assumption; the standard solution is the continuous one.

Pro-Tip (speed): Memorise the five standard patterns and their solutions — many “find f ” questions are instantly recognised, no work needed. For the elimination type, the partner substitution is almost always $x \rightarrow \frac{1}{x}$ (when $\frac{1}{x}$ appears) or $x \rightarrow 1 - x$ (when $1 - x$ appears). Always **verify** the final f by plugging back — it catches sign slips.

Section 8: Cheat Sheet Box

ALGEBRA OF FUNCTIONS & FUNCTIONAL EQUATIONS — QUICK RECALL

- $(f \pm g), (fg)$ on $D_f \cap D_g$; $\frac{f}{g}$ on $D_f \cap D_g \cap \{g \neq 0\}$.
- **Cauchy** $f(x + y) = f(x) + f(y) \Rightarrow f(x) = kx$.
- $f(x + y) = f(x)f(y) \Rightarrow a^x$; $f(xy) = f(x) + f(y) \Rightarrow k \log x$;
 $f(xy) = f(x)f(y) \Rightarrow x^n$.
- **Method:** substitute special values $(0, 1)$; make a 2nd equation via $x \rightarrow \frac{1}{x}$ or $x \rightarrow 1 - x$; eliminate.
- Always **verify** by back-substitution.

Mnemonics: “Sum $\rightarrow$ sum: line. Sum $\rightarrow$ product: exp. Product $\rightarrow$ sum: log. Product $\rightarrow$ product: power.” · “One equation, two unknowns $\Rightarrow$ substitute to get a twin.” · “Combinations live on the overlap.”

Chapter 1: Relations and Functions — Round 2

Addendum

The chapter gave you the vocabulary (relations, equivalence, injective/surjective/bijective), the machinery (composition, inverse, domain-range, symmetry, periodicity, functional equations), and a solid stock of board-level and JEE Main-level technique. Six examinable patterns are still standing outside. Addendum A proves the two structural facts about equivalence classes the chapter asserts without proof, and solves the “complete the relation” family those facts unlock. Addendum B proves the composition transfer laws, exposes their false converses, and shows how to run a composite backwards. Addendum C handles inverses on restricted domains — the maximal-interval and piecewise-inverse patterns the chapter’s swap-and-solve never touches. Addendum D builds the pipeline from symmetry axes to periods and extracts periods hidden inside functional equations, a JEE Advanced signature. Addendum E derives the counting formulas the chapter merely quotes: surjections, derangements, and Bell numbers by hand. Addendum F treats iterates $f \circ f \circ \dots \circ f$, where a function cycles and powers of it collapse modulo the cycle length. Nothing here is new theory; every method is a definition you already own, applied one layer deeper.

Addendum A — Equivalence Classes: The Two Proofs and the Completion Problem

Subtopic 01 defines $[a]$ and states that the distinct classes partition A . Both halves of that statement are provable in four lines each, and JEE Advanced regularly asks for exactly these proofs — or for the consequence that matters more practically: given an arbitrary relation R , find the *smallest* equivalence relation containing it.

Method

A1 — Distinct classes are disjoint. Suppose $c \in [a] \cap [b]$. Then cRa and cRb . Take any $x \in [a]$, so xRa . By symmetry aRc , so transitivity gives xRc , and with cRb transitivity gives xRb , i.e. $x \in [b]$. Hence $[a] \subseteq [b]$; swapping the roles of a and b gives $[b] \subseteq [a]$. Therefore

$$[a] \cap [b] \neq \emptyset \implies [a] = [b].$$

So two classes are either identical or disjoint — never partially overlapping.

A2 — The classes cover A . Reflexivity gives aRa , so $a \in [a]$ for every $a \in A$. Hence $\bigcup_a [a] = A$.

Together A1 and A2 say the distinct classes form a partition of A .

A3 — Characterisation of membership.

$$xRy \iff [x] = [y].$$

Forward: $y \in [x]$, so $[x] \cap [y] \neq \emptyset$, and A1 forces equality. *Backward:* $x \in [x] = [y]$ means xRy .

A4 — Every partition comes from an equivalence relation. Given a partition $A_1, A_2, \dots, A_k$ of A , define

$$xRy \iff x \text{ and } y \text{ lie in the same block } A_i.$$

Reflexive: x lies in some block. Symmetric: “same block” reads the same both ways. Transitive: y lies in a *unique* block (blocks are disjoint), so x, y in one block and y, z in one block forces all three into the same block. So R is an equivalence relation, and its classes are precisely the blocks.

A5 — The completion problem. Given R on a finite A , the smallest equivalence relation containing R is built like this:

1. Draw an undirected graph on A : join a to b whenever $(a, b) \in R$ with $a \neq b$. Loops are irrelevant.
2. Find the connected components $C_1, C_2, \dots$
3. The answer is $\bigcup_i (C_i \times C_i)$: every element relates to everything in its own component and nothing outside.

Why this is forced: any equivalence relation containing R must contain (a, b) whenever a and b are joined by a chain of R -pairs — symmetry walks the chain backwards, transitivity collapses it — so it must contain each $C_i \times C_i$. And A4 shows the union really is an equivalence relation. Nothing smaller survives; nothing larger is needed.

Counting: the number of pairs to add is

$$\sum_i |C_i|^2 - |R|.$$

Example A.1 — [CBSE Board / JEE Main Level]

On $A = \{1, 2, 3, 4\}$, let $R = \{(1, 2), (2, 3), (4, 4)\}$. Find the minimum number of ordered pairs that must be added to make R an equivalence relation, and write down the resulting relation.

The graph joins 1—2—3 in one chain; 4 is isolated (its loop $(4, 4)$ creates no edge). Components: $C_1 = \{1, 2, 3\}$, $C_2 = \{4\}$.

The completed relation is $C_1 \times C_1 \cup C_2 \times C_2$: nine pairs inside $\{1, 2, 3\}$ plus the single pair $(4, 4)$ — ten pairs in all. Already present: $(1, 2)$, $(2, 3)$, $(4, 4)$. So we add

$$\sum |C_i|^2 - |R| = (9 + 1) - 3 = 7 \text{ pairs,}$$

namely $(1, 1)$, $(2, 2)$, $(3, 3)$, $(2, 1)$, $(1, 3)$, $(3, 1)$, $(3, 2)$.

Check. The result is $\{1, 2, 3\} \times \{1, 2, 3\} \cup \{(4, 4)\}$: reflexive (every element paired with itself), symmetric (full products are symmetric), transitive (products are transitive). It contains R . And it must be smallest: any equivalence relation containing $(1, 2)$ and $(2, 3)$ is forced by symmetry and transitivity to contain all nine pairs on $\{1, 2, 3\}$.

Example A.2 — [JEE Advanced Level]

(a) Show that the number of equivalence relations on a 4-element set is 15, by counting partitions directly. (b) How many of these place 1 and 2 in the same class? How many place them in different classes?

(a) By A1–A4, equivalence relations on A correspond one-to-one with partitions of A . Count partitions of $\{1, 2, 3, 4\}$ by the number of blocks:

- **4 blocks:** all singletons — 1 way.
- **3 blocks:** exactly one block is a pair — choose it: $\binom{4}{2} = 6$ ways.
- **2 blocks:** sizes $1 + 3$ or $2 + 2$. For $1 + 3$: choose the singleton, 4 ways. For $2 + 2$: choose one block, $\binom{4}{2}/2 = 3$ ways (dividing by 2 because choosing either block names the same partition). Total 7.
- **1 block:** 1 way.

$$1 + 6 + 7 + 1 = 15 = B_4. \checkmark$$

(b) Equivalence relations with 1 and 2 in the same class correspond to partitions of the three-object collection $\{\{1, 2\}, 3, 4\}$, of which there are $B_3 = 5$. So:

$$\text{same class: } 5, \quad \text{different classes: } 15 - 5 = 10.$$

Check. Listing the five “same-class” partitions: $\{1, 2\} \mid \{3\} \mid \{4\}$; $\{1, 2, 3\} \mid \{4\}$; $\{1, 2, 4\} \mid \{3\}$; $\{1, 2\} \mid \{3, 4\}$; $\{1, 2, 3, 4\}$. Five, as claimed.

Practice — Addendum A

1. On $A = \{1, 2, 3\}$, $R = \{(1, 2), (2, 1), (1, 3)\}$. Find the minimum number of pairs to add to make R an equivalence relation.
2. Prove: if R is an equivalence relation on A and xRy , then $[x] = [y]$.
3. How many equivalence relations on $\{1, 2, 3, 4\}$ have $[1] = \{1, 2\}$?
4. Write out the equivalence relation arising from the partition $\{\{1, 4\}, \{2, 3\}\}$.
5. Find the smallest equivalence relation on $\{1, 2, 3, 4\}$ containing $\{(1, 2), (3, 4)\}$, and state how many pairs it has.

Answers: 1. One component of size 3, so the completion is all 9 pairs; $9 - 3 = 6$ pairs to add. 2. $y \in [x]$ gives $[x] \cap [y] \neq \emptyset$, so A1 forces $[x] = [y]$. 3. The rest $\{3, 4\}$ may be partitioned freely: $B_2 = 2$ ways (together or separate). 4. $\{(1, 1), (4, 4), (1, 4), (4, 1), (2, 2), (3, 3), (2, 3), (3, 2)\}$ — eight pairs. 5. $\{1, 2\} \times \{1, 2\} \cup \{3, 4\} \times \{3, 4\}$: eight pairs.

Addendum B — Composition: Transfer Laws, False Converses, and Running Composites Backwards

Subtopic 03 lists the inheritance facts — “ $g \circ f$ injective/surjective when both are” — and the partial converses, without proof. JEE Advanced asks for those proofs, for counterexamples killing the converses, and for two skills the chapter skips: computing the domain of a composite properly, and recovering f from a formula for $f \circ g$.

Method

B1 — Transfer laws (with proofs).

If $g \circ f$ is injective, then f is injective. Suppose $f(x_1) = f(x_2)$. Applying g to both sides,

$$g(f(x_1)) = g(f(x_2)), \quad \text{i.e. } (g \circ f)(x_1) = (g \circ f)(x_2).$$

Injectivity of $g \circ f$ forces $x_1 = x_2$. Done. Notice the proof never uses any property of g beyond being a function.

If $g \circ f$ is surjective, then g is surjective. Take any y in the codomain of g . Surjectivity of $g \circ f$ supplies x with $(g \circ f)(x) = y$, i.e. $g(f(x)) = y$. The element $f(x)$ lies in the intermediate set, so y has a preimage under g . Done.

B2 — The false converses (memorise these counterexamples).

$g \circ f$ injective does **not** force g injective. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = e^x$, and let $g(t) = t$ for $t > 0$ while $g(t) = 1 - t$ for $t \leq 0$. Then $g \circ f = e^x$ is injective, but $g(0) = 1 = g(1)$, so g is not. The failure of g happens entirely off the range of f , where $g \circ f$ cannot see it.

$g \circ f$ surjective does **not** force f surjective. With the same $f(x) = e^x$ (which misses every negative number) and $g(t) = \ln t$ for $t > 0$, $g(t) = 0$ for $t \leq 0$, the composite is $g(f(x)) = x$, onto all of $\mathbb{R}$, while f is not onto. The surjectivity of the composite is manufactured by g alone.

B3 — Domain of a composite.

$$\text{Dom}(g \circ f) = \{x \in \text{Dom}(f) : f(x) \in \text{Dom}(g)\}.$$

Procedure: start inside $\text{Dom}(f)$, then impose the condition that the *output* $f(x)$ is a legal input for g . This is **not** the intersection $\text{Dom}(f) \cap \text{Dom}(g)$ — see Example B.2.

B4 — Recovering f from $f \circ g$. If $(f \circ g)(x) = H(x)$ and g is a formula you recognise, substitute $t = g(x)$, solve for x in terms of t , and rewrite H :

$$f(t) = H(x(t)).$$

This is the y -method of Subtopic 04 run on symbols instead of values.

Example B.1 — [JEE Main Level]

True or false: if $g \circ f$ is injective, then g is injective on the range of f .

True — and worth separating from B2's false converse. Suppose $f(x_1) = f(x_2)$ with both outputs in $\text{Range}(f)$; applying g and using injectivity of $g \circ f$ gives $x_1 = x_2$, exactly as in B1. So g is injective *when restricted to inputs that actually arise from f* . The counterexample in B2 works only because g misbehaves at inputs (0, in that example) that f never produces. Exam takeaway: the phrase “on the range of f ” is what makes the statement true.

Example B.2 — [JEE Advanced Level]

Let $f : (0, \infty) \rightarrow \mathbb{R}$, $f(x) = \ln x$, and $g : \mathbb{R} \rightarrow \mathbb{R}$, $g(t) = t^2 - 2t$. Find $\text{Dom}(g \circ f)$, decide whether $g \circ f$ is one-one, and find its range.

Domain. $\text{Dom}(f) = (0, \infty)$, and $f(x) = \ln x$ is always a legal real input for g . So

$$\text{Dom}(g \circ f) = (0, \infty).$$

Contrast: $\text{Dom}(f) \cap \text{Dom}(g) = (0, \infty) \cap \mathbb{R} = (0, \infty)$ here they coincide, but had g been $g(t) = \log(t - 1)$, the composite would demand $\ln x > 1$, i.e. $x > e$ — a strictly smaller set than the intersection. Always filter by $f(x) \in \text{Dom}(g)$; never shortcut.

One-one? Substitute $t = \ln x$. As x runs over $(0, \infty)$, t runs over all of $\mathbb{R}$, and

$$(g \circ f)(x) = (\ln x)^2 - 2 \ln x = t^2 - 2t.$$

But $t^2 - 2t$ takes the same value at $t = 0$ and $t = 2$: correspondingly $x = 1$ and $x = e^2$ both give $(g \circ f) = 0$. Many-one.

Range. Complete the square:

$$t^2 - 2t = (t - 1)^2 - 1 \geq -1,$$

with equality at $t = 1$, i.e. $x = e$. Since t ranges over all of $\mathbb{R}$, every value in $[-1, \infty)$ occurs:

$$\text{Range}(g \circ f) = [-1, \infty).$$

Check. $(g \circ f)(e) = 1 - 2 = -1$, the minimum, attained. $(g \circ f)(1) = 0 = (g \circ f)(e^2)$, confirming many-one concretely.

Example B.3 — [JEE Main Level]

Given $(f \circ g)(x) = x^2 + 6x + 7$ where $g(x) = x + 3$, find $f(x)$.

Put $t = g(x) = x + 3$, so $x = t - 3$. Then

$$f(t) = (t - 3)^2 + 6(t - 3) + 7 = t^2 - 6t + 9 + 6t - 18 + 7 = t^2 - 2.$$

So $f(x) = x^2 - 2$.

Check. $f(g(x)) = (x + 3)^2 - 2 = x^2 + 6x + 7$. ✓

Practice — Addendum B

1. $f(x) = \sqrt{x}$, $g(x) = x^2 - 3$. Find $\text{Dom}(g \circ f)$ and $\text{Dom}(f \circ g)$.
2. Construct functions f, g with $g \circ f$ onto but f not onto.
3. Given $f(x+2) = x^2 + 4x + 1$ for all real x , find $f(x)$.
4. True or false: $g \circ f$ injective $\implies f$ injective.
5. For $f(x) = x^2$ and $g(x) = 2x + 1$ on $\mathbb{R}$, solve $(f \circ g)(x) = (g \circ f)(x)$.

Answers: 1. $\text{Dom}(g \circ f) = [0, \infty)$ (every $\sqrt{x}$ is legal for g); $\text{Dom}(f \circ g) = \{x : x^2 - 3 \geq 0\} = (-\infty, -\sqrt{3}] \cup [\sqrt{3}, \infty)$. 2. $f(x) = e^x$ from $\mathbb{R}$ to $\mathbb{R}$; $g(t) = \ln t$ for $t > 0$ and $g(t) = 0$ for $t \leq 0$; then $g \circ f = x$ is onto while f misses all negatives. 3. Put $t = x+2$: $f(t) = (t-2)^2 + 4(t-2) + 1 = t^2 - 3$. 4. True — proved in B1. 5. $(2x+1)^2 = 2x^2 + 1$ gives $2x^2 + 4x = 0$, so $x = 0$ or $x = -2$.

Addendum C — Inverses on Restricted Domains

Swap-and-solve presupposes a bijection. Most functions you meet are not bijective on their natural domain — but almost all become bijective on the right *piece*. CBSE asks for the inverse of a quadratic on a chosen branch; JEE Main asks which maximal intervals support an inverse; JEE Advanced asks you to find the inverse of a piecewise function or of something like $2^x + 2^{-x}$, where choosing the correct root of a quadratic is the whole game.

Method

C1 — The restriction principle. If f is strictly monotonic on an interval I , then $f : I \rightarrow f(I)$ is bijective, hence invertible. For the standard functions of this course, the maximal such intervals end at turning points (where $f'(x) = 0$) or at breaks in the formula.

C2 — Procedure for a maximal-interval inverse.

1. Locate turning points / breaks; they cut the domain into monotonic branches.
2. Compute each branch's image $f(I)$ — that becomes the codomain, and later the domain of that branch's inverse.
3. Invert branch by branch with swap-and-solve. Where a quadratic appears, you will get two roots; the branch condition ($x \geq \text{vertex}$ or $x \leq \text{vertex}$) selects the sign of the square root.

C3 — Piecewise inverses. For

$$f(x) = \begin{cases} f_1(x), & x \in I_1 \\ f_2(x), & x \in I_2 \\ \vdots \end{cases}$$

invert each piece separately. The domain of f^{-1} is the union of the piece-images $f_i(I_i)$. Sanity check: if two piece-images overlap, some output has two preimages and f was never one-one — the overlap test is the injectivity test.

C4 — Graph check. f^{-1} is the reflection of f in the line $y = x$: increasing stays increasing, and each branch's endpoints swap coordinates.

Example C.1 — [CBSE Board / JEE Main Level]

Find the maximal intervals on which $f(x) = x^2 - 4x + 5$ is invertible, and find f^{-1} on each.

Complete the square: $f(x) = (x - 2)^2 + 1$. Single turning point at $x = 2$, value 1. Branches:

$$(-\infty, 2] \rightarrow [1, \infty), \quad [2, \infty) \rightarrow [1, \infty),$$

both strictly monotonic (down then up), both onto $[1, \infty)$. These are the maximal intervals.

Invert on the right branch, where $x \geq 2$:

$$y = (x - 2)^2 + 1 \Rightarrow (x - 2)^2 = y - 1 \Rightarrow x = 2 + \sqrt{y - 1},$$

taking the $+$ root because $x \geq 2$. So

$$f^{-1}(x) = 2 + \sqrt{x - 1}, \quad x \in [1, \infty).$$

The left branch gives $f^{-1}(x) = 2 - \sqrt{x - 1}$ on the same domain.

Check. $f(4) = 5$ and $f^{-1}(5) = 2 + 2 = 4$. ✓ The vertex value 1 maps to $x = 2$ under both inverses, as the reflection picture demands.

Example C.2 — [JEE Main Level]

Invert the piecewise function

$$f(x) = \begin{cases} x, & x \leq 1 \\ x^2, & 1 < x \leq 2 \\ 2x, & x > 2 \end{cases}$$

First confirm it is invertible. Piece-images:

$$(-\infty, 1], \quad (1, 4], \quad (4, \infty).$$

They are pairwise disjoint and cover $\mathbb{R}$ — so f is one-one with range $\mathbb{R}$. (Had the middle piece been x^2 on $0 < x \leq 2$, its image $(0, 4]$ would overlap $(-\infty, 1]$ in $(0, 1]$, and f would fail injectivity before any inversion.)

Invert piece by piece. Each piece is simple: $x \mapsto x$ is self-inverse; $x \mapsto x^2$ on $x > 1$ inverts to $+\sqrt{x}$; $x \mapsto 2x$ inverts to $x/2$. Matching domains to the piece-images:

$$f^{-1}(x) = \begin{cases} x, & x \leq 1 \\ \sqrt{x}, & 1 < x \leq 4 \\ x/2, & x > 4 \end{cases}$$

Check. $f(1.5) = 2.25$ and $f^{-1}(2.25) = 1.5$. ✓ $f(3) = 6$ and $f^{-1}(6) = 3$. ✓ The breakpoints 1 and 4 of f^{-1} are the images of the breakpoints 1 and 2 of f — the domain-range swap in action.

Example C.3 — [JEE Advanced Level]

Show that $f(x) = 2^x + 2^{-x}$ **is a bijection from** $[0, \infty)$ **onto its image, find the image, and find** f^{-1} .

Monotonicity. f is even, so it cannot be one-one on $\mathbb{R}$; but on $[0, \infty)$,

$$f'(x) = (\ln 2)(2^x - 2^{-x}) \geq 0,$$

with equality only at $x = 0$. So f is strictly increasing on $[0, \infty)$ — one-one there.

Image. $f(0) = 2$, and $f(x) \rightarrow \infty$ as $x \rightarrow \infty$; by strict increase the image is exactly $[2, \infty)$.

Inverse. Write $t = 2^x$, so $t \geq 1$ and $y = t + \frac{1}{t}$. Clearing the fraction:

$$t^2 - yt + 1 = 0 \implies t = \frac{y \pm \sqrt{y^2 - 4}}{2}.$$

The two roots multiply to 1 (product of roots), so exactly one of them is ≥ 1 — except at $y = 2$ where both equal 1. Since $t \geq 1$, take the + root:

$$2^x = \frac{y + \sqrt{y^2 - 4}}{2} \implies f^{-1}(x) = \log_2 \left(\frac{x + \sqrt{x^2 - 4}}{2} \right), \quad x \in [2, \infty).$$

Check. $f^{-1}(2) = \log_2(1) = 0 = f^{-1}(f(0))$. ✓ As $x \rightarrow \infty$ the inverse grows like $\log_2 x$, consistent with f growing exponentially.

Practice — Addendum C

1. Find the maximal intervals of invertibility of $f(x) = |x - 1| + 2$, and f^{-1} on each.
2. Is $f(x) = x^2 - 2x$ one-one on $[-1, 2]$? Find an inverse on a maximal subinterval, with its domain.
3. Invert $f(x) = x^2$ for $0 \leq x \leq 1$ and $f(x) = 2x - 1$ for $1 < x \leq 2$, after confirming invertibility.
4. Show $f(x) = \frac{x}{1 + |x|}$ is a bijection $\mathbb{R} \rightarrow (-1, 1)$ and find f^{-1} .
5. How many real solutions does $x^3 - 3x = 1$ have?

Answers: 1. Vertex at $(1, 2)$; branches $(-\infty, 1]$ and $[1, \infty)$, both onto $[2, \infty)$; inverses $x \mapsto 1 - \sqrt{x - 2}$ and $x \mapsto 1 + \sqrt{x - 2}$ on $[2, \infty)$. 2. No: f falls on $[-1, 1]$ (from 3 to -1) and rises on

$[1, 2]$ (from -1 to 0). On $[1, 2]$ the image is $[-1, 0]$ and $f^{-1}(x) = 1 + \sqrt{x+1}$. 3. Images $[0, 1]$ and $(1, 3]$ are disjoint, so f is one-one with range $[0, 3]$; $f^{-1}(x) = \sqrt{x}$ on $[0, 1]$ and $f^{-1}(x) = \frac{x+1}{2}$ on $(1, 3]$. 4. For $x \geq 0$, $y = \frac{x}{1+x}$ gives $x = \frac{y}{1-y}$; for $x < 0$, $x = \frac{y}{1+y}$; combining, $f^{-1}(x) = \frac{x}{1-|x|}$ on $(-1, 1)$; f is continuous and strictly increasing with limits ± 1 , hence onto $(-1, 1)$. 5. Three: $g(x) = x^3 - 3x$ has local maximum 2 at $x = -1$ and local minimum -2 at $x = 1$, so the line $y = 1$ cuts the curve once on each of $(-\infty, -1)$, $(-1, 1)$, $(1, \infty)$.

Addendum D — From Symmetry Axes to Periods; Periods Hidden in Functional Equations

Subtopic 06 finds periods of *formulas*: read off the pieces, scale, take LCM. JEE Advanced prefers functions described only by *rules*, and two rule-patterns dominate. First: two symmetry axes force a period. Second: a first-order recurrence like $f(x+a) = \frac{1+f(x)}{1-f(x)}$ hides a period you must dig out by composing the recurrence with itself.

Method

D1 — Axis of symmetry. The statement “ $f(a+x) = f(a-x)$ for all x ” says the graph is mirror-symmetric about the vertical line $x = a$. Equivalently,

$$f(2a-x) = f(x) \quad \text{for all } x.$$

(Evenness is the special case $a = 0$.)

D2 — Two axes force a period. Suppose $f(2a-x) = f(x)$ and $f(2b-x) = f(x)$ with $a \neq b$. Replace x by $2a-x$ throughout the second identity:

$$f(2b-(2a-x)) = f(2a-x) = f(x),$$

that is,

$$f(x+2(b-a)) = f(x) \quad \text{for all } x.$$

So f is periodic with period $2|b-a|$.

D3 — Corollaries.

- f even **and** $f(2a-x) = f(x) \implies$ period $2a$ (axes at 0 and a).
- f has period T and axis $x = a \implies$ it also has axis $x = a + \frac{T}{2}$: indeed, using the axis at a with offset $\frac{T}{2} + x$,

$$f\left(a + \frac{T}{2} + x\right) = f\left(a - \frac{T}{2} - x\right) = f\left(a + \frac{T}{2} - x\right),$$

where the last step shifts by the period T . So axes come in families spaced $T/2$ apart.

D4 — Periods from recurrences. Compose the recurrence with itself until the value returns. The three stock patterns:

Recurrence	Apply twice	Period
$f(x + a) = -f(x)$	$f(x + 2a) = -f(x + a) = f(x)$	$2a$
$f(x + a) = \frac{1}{f(x)}$	$f(x + 2a) = \frac{1}{f(x + a)} = f(x)$	$2a$
$f(x + a) = 1 - f(x)$	$f(x + 2a) = 1 - (1 - f(x)) = f(x)$	$2a$

and the deeper one, worked in Example D.3:

$$f(x + a) = \frac{1 + f(x)}{1 - f(x)} \implies \text{period } 4a.$$

Caveat carried over from Subtopic 06: the period found is the *fundamental* one only if f is not constant and no smaller shift works — always test divisors when the problem demands the fundamental period.

D5 — Two incommensurable periods. If a (continuous) function has two periods whose ratio is irrational, it is constant: its values repeat on a dense set of shifts, and continuity spreads the constancy everywhere. This is the rigidity lying behind the “irrational ratio $\Rightarrow$ not periodic” warning of Subtopic 06.

Example D.1 — [JEE Main Level]

f satisfies $f(2+x) = f(2-x)$ and $f(4+x) = f(4-x)$ for all real x . Show that f is periodic and find a period.

Axes at $a = 2$ and $b = 4$. By D2,

$$f(x + 2(4 - 2)) = f(x + 4) = f(x),$$

so f is periodic with period 4. Directly: the first identity says $f(4 - x) = f(x)$ (rewriting the second axis as $f(8 - x) = f(x)$, then substituting $x \rightarrow 4 - x$ gives $f(4 + x) = f(4 - x) = f(x)$ — the same conclusion).

Example D.2 — [JEE Advanced Level]

$f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x + 3) = 1 - f(x)$ for all x , and f is not constant. Find the fundamental period.

Compose the recurrence with itself:

$$f(x + 6) = 1 - f(x + 3) = 1 - (1 - f(x)) = f(x).$$

So 6 is a period. Is 3? If $f(x+3) = f(x)$ held, the recurrence would give $f(x) = 1 - f(x)$, i.e. $f \equiv \frac{1}{2}$ — excluded. Any other candidate T with $0 < T < 6$ would, combined with period 6, produce a smaller period among $\{T, 6-T\}$... more simply: a nonconstant function satisfying this recurrence has 3 failing and 6 working, and no divisor of 6 other than 1, 2, 3 lies below 6; each of 1, 2 would divide 3's failure through the recurrence structure. The fundamental period is

$$T = 6.$$

Check. The recurrence forces $f(x+3) \neq f(x)$ pointwise unless $f(x) = \frac{1}{2}$; a function alternating between two behaviours every 3 units and repeating every 6 — e.g. any f built by choosing values freely on $[0, 3)$ and extending by $f(x+3) = 1 - f(x)$ — realises exactly this.

Example D.3 — [JEE Advanced Level]

f satisfies $f(x+1) = \frac{1+f(x)}{1-f(x)}$ for all x (where defined), and $f(2) = 3$. Find $f(2025)$.

Compose the recurrence twice. First,

$$f(x+2) = \frac{1+f(x+1)}{1-f(x+1)} = \frac{1+\frac{1+f}{1-f}}{1-\frac{1+f}{1-f}} = \frac{\frac{(1-f)+(1+f)}{1-f}}{\frac{(1-f)-(1+f)}{1-f}} = \frac{2}{-2f} = -\frac{1}{f(x)},$$

where f abbreviates $f(x)$. Applying the same step again:

$$f(x+4) = -\frac{1}{f(x+2)} = -\frac{1}{-1/f(x)} = f(x).$$

So f has period 4, and $f^n(x)$ depends only on $n \bmod 4$. Now

$$2025 = 2 + 2023, \quad 2023 = 4 \times 505 + 3,$$

so

$$f(2025) = f(2+3) = f(5) = f(1+4) = f(1).$$

Finally recover $f(1)$ from the given value: with $x = 1$,

$$3 = f(2) = \frac{1+f(1)}{1-f(1)} \implies 3 - 3f(1) = 1 + f(1) \implies f(1) = \frac{1}{2}.$$

$$f(2025) = \frac{1}{2}.$$

Check. Run forward from $f(2) = 3$: $f(3) = \frac{1+f(2)}{1-f(2)} = \frac{4}{-2} = -2$, matching $f(3) = f(1+2) = -1/f(1) = -2$; then $f(4) = f(2+2) = -1/f(2) = -\frac{1}{3}$, and $f(5) = f(1+4) = f(1) = \frac{1}{2}$. Since $2025 = 5 + 4 \times 505$, this confirms $f(2025) = f(5) = \frac{1}{2}$. ✓

Practice — Addendum D

1. $f(1+x) = f(1-x)$ and $f(3+x) = f(3-x)$ for all x . Find a period of f .
2. $f(x+2) = -f(x)$ for all x and $f(0) = 1$. Find $f(10)$.
3. Show that $f(x+a) = \frac{1}{f(x)}$ (with f nowhere 0) forces period $2a$.
4. f is odd and has period T . Show that $f(T/2) = 0$.
5. Show that $f(x+2) = f(x) + 3$ cannot hold for any periodic f .

Answers: 1. Axes at 1 and 3: period $2|3-1| = 4$. 2. $f(x+4) = f(x)$, and $10 \equiv 2 \pmod{4}$, so $f(10) = f(2) = -f(0) = -1$. 3. $f(x+2a) = 1/f(x+a) = f(x)$. 4. $f(T/2) = f(T/2 - T) = f(-T/2) = -f(T/2)$, using oddness; hence $2f(T/2) = 0$. 5. Iterating gives $f(x+2n) = f(x) + 3n$, unbounded along $x, x+2, x+4, \dots$; a periodic function repeats its values, so (being continuous, as all functions here are) it is bounded on $\mathbb{R}$ — contradiction.

Addendum E — Counting Maps: Surjections Derived, Derangements, and Bell Numbers by Hand

Subtopics 01 and 02 quote three formulas — the surjection count, the Bell numbers, and (implicitly) nothing about fixed-point-free bijections at all. JEE Main expects you to *evaluate* these fluently; JEE Advanced expects you to know *why* they hold, because the inclusion-exclusion engine behind them recurs in disguise across combinatorics. Here is the engine, run three times.

Method

E1 — Inclusion-exclusion, stated for functions. Let A have n elements and B have m . For each $b \in B$, let E_b be the set of functions $A \rightarrow B$ that **miss** b . The onto functions are those in none of the E_b . Functions missing every target in a fixed set S (with $|S| = k$) map into the remaining $m - k$ targets arbitrarily:

$$\left| \bigcap_{b \in S} E_b \right| = (m - k)^n.$$

There are $\binom{m}{k}$ such sets S , so inclusion-exclusion gives

$$\text{onto}(n, m) = \sum_{k=0}^m (-1)^k \binom{m}{k} (m - k)^n.$$

Why each function is counted correctly: a function whose image misses exactly j targets is counted once for each S contained in those j targets, with signs — a total of $\sum_{k=0}^j (-1)^k \binom{j}{k} = (1-1)^j$, which is 1 if $j = 0$ and 0 otherwise. Only fully-surjective functions survive.

Numerical anchor: $n = m = 3$ gives $27 - 3 \cdot 8 + 3 \cdot 1 - 0 = 6 = 3!$, as it must be (onto between equal finite sets means bijective).

E2 — Derangements. A derangement of $\{1, \dots, n\}$ is a bijection σ with $\sigma(i) \neq i$ for every i . Run the same engine with $E_i =$ “bijections fixing i ”. Bijections fixing all of a fixed k -set exist in $(n-k)!$ ways, so

$$!n = \sum_{k=0}^n (-1)^k \binom{n}{k} (n-k)! = n! \sum_{k=0}^n \frac{(-1)^k}{k!}.$$

Values to memorise:

$$!1 = 0, \quad !2 = 1, \quad !3 = 2, \quad !4 = 9, \quad !5 = 44.$$

E3 — Stirling numbers and Bell numbers by hand. Let $S(n, k)$ be the number of partitions of an n -set into exactly k blocks; then the number of equivalence relations is $B_n = \sum_k S(n, k)$. The recurrence

$$S(n, k) = k S(n-1, k) + S(n-1, k-1)$$

comes from watching element n : either it sits **alone** in its block — partition the other $n-1$ elements into $k-1$ blocks — or it joins an existing block — partition the others into k blocks, then choose which of the k blocks receives n . Both moves are reversible.

Small rows (derived, not memorised):

- $S(4, 1) = 1$; $S(4, 4) = 1$.
- $S(4, 2) = 2^3 - 1 = 7$: element 1's block is $\{1\} \cup T$ where T is any non-empty proper subset of the other three elements.
- $S(4, 3) = \binom{4}{2} = 6$: exactly one block is a pair.
- $S(5, 2) = 2^4 - 1 = 15$; $S(5, 3) = 3 S(4, 3) + S(4, 2) = 18 + 7 = 25$.

Bell numbers via the block of element $n+1$: if it contains k of the other n elements, choose them in $\binom{n}{k}$ ways and partition the remaining $n-k$ in B_{n-k} ways:

$$B_{n+1} = \sum_{k=0}^n \binom{n}{k} B_{n-k}.$$

Example E.1 — [JEE Main Level]

Find the number of onto functions from $\{1, 2, \dots, 6\}$ to $\{a, b, c\}$.

Apply E1 with $n = 6$, $m = 3$:

$$3^6 - \binom{3}{1}2^6 + \binom{3}{2}1^6 - \binom{3}{3}0^6 = 729 - 192 + 3 - 0 = 540.$$

Check. Each of a, b, c must be hit. Total functions 729; subtract those missing at least one letter. A quick lower-bound sanity check: onto functions number at least $3! = 6$ (bijections after merging) and at most 729; 540 sits comfortably inside.

Example E.2 — [JEE Advanced Level]

How many bijections σ of $\{1, 2, 3, 4, 5\}$ satisfy $\sigma(i) \neq i$ for every i ? What is the probability that a random bijection is such a derangement?

By E2 with $n = 5$:

$$!5 = 5! \left(1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120} \right) = 120 - 120 + 60 - 20 + 5 - 1 = 44.$$

Read the expansion term by term: fix nobody (120), penalise each single fixed point (-5×24), refund each pair ($+10 \times 6$), re-penalise each triple (-10×2), refund each quadruple ($+5 \times 1$), re-penalise the identity (-1).

Probability:

$$\frac{!5}{5!} = \frac{44}{120} = \frac{11}{30}.$$

Check. $!4 = 24 - 24 + 12 - 4 + 1 = 9 \checkmark$, and the recurrence $!n = (n-1)(!(n-1) + !(n-2))$ gives $!5 = 4(9 + 2) = 44 \checkmark$ — two independent routes, one answer.

Example E.3 — [JEE Advanced Level]

How many equivalence relations on a 5-element set have exactly two classes? Exactly three classes?

Exactly two classes: $S(5, 2) = 2^4 - 1 = 15$ — element 1's block is $\{1\} \cup T$ with T any non-empty proper subset of the other four elements.

Exactly three classes: via the recurrence,

$$S(5, 3) = 3S(4, 3) + S(4, 2) = 3(6) + 7 = 25.$$

Check. Direct count for $S(5, 3)$: block-size patterns $3 + 1 + 1$ and $2 + 2 + 1$. For $3 + 1 + 1$: choose the triple, $\binom{5}{3} = 10$. For $2 + 2 + 1$: choose the singleton (5 ways), then pair up the remaining four (3 ways): 15. Total $10 + 15 = 25 \checkmark$.

Practice — Addendum E

1. Find the number of onto functions from a 4-element set to a 2-element set.
2. How many functions from a 5-element set to itself are onto?

3. How many bijections of a 4-element set have exactly one fixed point?
4. How many equivalence relations does a 4-element set have in total, and how many of them have exactly two classes?
5. Three balls are placed into three boxes, each box non-empty. In how many ways?

Answers: 1. $2^4 - 2 = 14$. 2. Onto with equal sizes means bijective: $5! = 120$. 3. Choose the fixed point (4 ways) and derange the rest: $4 \times !3 = 4 \times 2 = 8$. 4. $B_4 = 1 + 7 + 6 + 1 = 15$; exactly two classes: $S(4, 2) = 7$. 5. Onto functions from a 3-set to a 3-set: $3! = 6$.

Addendum F — Iterates and Cycles

Everything in Subtopic 03 composes two *different* functions. Compose a function with *itself* and a new phenomenon appears: the iterates can cycle. If $f^k(x_0) = x_0$ for some k , then $f^n(x_0)$ depends only on $n \bmod k$ — and JEE exploits this relentlessly, asking for $f^{2023}(x)$ where brute force is hopeless and modular arithmetic is everything.

Method

F1 — Notation and the exponent law. Define $f^2 = f \circ f$, $f^3 = f \circ f^2$, and so on; f^0 is the identity. Associativity of composition gives

$$f^m \circ f^n = f^{m+n},$$

so iterates behave like powers of a number — *provided you never leave the domain*. Track poles: each application forbids the inputs where any intermediate denominator vanishes.

F2 — Cycle detection. Compute $f(x_0), f^2(x_0), f^3(x_0), \dots$ until a value repeats. If $f^k(x_0) = x_0$ (and no smaller positive power returns), the orbit of x_0 is a k -cycle, and

$$f^n(x_0) = f^{n \bmod k}(x_0).$$

For a *formula* valid for all x , aim to prove $f^k = I$ as an identity; then $f^n = f^{n \bmod k}$ holds everywhere.

F3 — Linear fractional functions are the natural habitat. For $f(x) = \frac{ax+b}{cx+d}$, each iterate is again linear fractional, and the orders are tiny — typically 2, 3, or 4. Compute f^2 , then f^3 , and stop the moment you hit x .

F4 — Involutions. A function with $f^2 = I$ is its own inverse. Among linear functions $f(x) = mx + c$:

$$f^2(x) = m^2x + c(m+1) = x \implies m^2 = 1 \text{ and } c(m+1) = 0,$$

giving exactly $f(x) = x$ (from $m = 1$) or $f(x) = c - x$ (from $m = -1$, any c). So the only linear involutions are the identity and the reflections.

Example F.1 — [JEE Main Level]

Let $f(x) = \frac{x-1}{x+1}$, **defined wherever the iterates make sense. Find** $f^{2024}(x)$ **and** $f^{2023}(x)$.

Compute upward.

$$f^2(x) = \frac{\frac{x-1}{x+1} - 1}{\frac{x-1}{x+1} + 1} = \frac{\frac{-2}{x+1}}{\frac{2x}{x+1}} = -\frac{1}{x}.$$

$$f^3(x) = f\left(-\frac{1}{x}\right) = \frac{-\frac{1}{x} - 1}{-\frac{1}{x} + 1} = \frac{-(1+x)}{x-1} = \frac{1+x}{1-x}.$$

$$f^4(x) = f\left(\frac{1+x}{1-x}\right) = \frac{\frac{1+x}{1-x} - 1}{\frac{1+x}{1-x} + 1} = \frac{\frac{2x}{1-x}}{\frac{2}{1-x}} = x.$$

So $f^4 = I$: order 4. Reduce the exponents modulo 4:

$$2024 = 4 \times 506 \implies f^{2024}(x) = x,$$

$$2023 = 4 \times 505 + 3 \implies f^{2023}(x) = f^3(x) = \frac{1+x}{1-x}.$$

Check. $f^{2023}(x) = f^{2020}(x) \circ f^3(x) = I \circ f^3(x)$, using $2020 = 4 \times 505$. ✓

Example F.2 — [JEE Advanced Level]

Let $f(x) = \frac{1}{1-x}$, $x \neq 1$. **Find the smallest** k **with** $f^k = I$, **and evaluate** $f^{100}(x)$.

$$f^2(x) = \frac{1}{1 - \frac{1}{1-x}} = \frac{1}{\frac{(1-x)-1}{1-x}} = \frac{1-x}{-x} = \frac{x-1}{x} = 1 - \frac{1}{x}.$$

$$f^3(x) = f\left(1 - \frac{1}{x}\right) = \frac{1}{1 - \left(1 - \frac{1}{x}\right)} = \frac{1}{1/x} = x.$$

Order 3. Since $100 = 3 \times 33 + 1$,

$$f^{100}(x) = f(x) = \frac{1}{1-x}.$$

Check. The orbit of $x = 3$ reads $3 \mapsto -\frac{1}{2} \mapsto \frac{2}{3} \mapsto 3$: a genuine 3-cycle, confirming $k = 3$ on a concrete value. Note the moving exclusions: f forbids $x = 1$; f^2 additionally forbids $x = 0$; the identity $f^3 = I$ holds wherever the journey is defined.

Example F.3 — [JEE Advanced Level]

Find all linear functions $f(x) = mx + c$ **on** $\mathbb{R}$ **satisfying** $f(f(x)) = x$ **for all** x .

Compute the iterate:

$$f(f(x)) = m(mx + c) + c = m^2x + c(m + 1).$$

Setting this equal to x identically and comparing coefficients:

$$m^2 = 1, \quad c(m + 1) = 0.$$

Case $m = 1$: the second equation forces $c = 0$, giving $f(x) = x$. Case $m = -1$: the second equation holds for every c , giving $f(x) = c - x$. So the complete list is

$$f(x) = x \quad \text{or} \quad f(x) = c - x, \quad c \in \mathbb{R}.$$

Check. $f(x) = 5 - x$: $f(f(x)) = 5 - (5 - x) = x$ ✓. And geometrically both families are exactly the reflections of the line $y = x$ in itself — reflecting twice is the identity, which is what an involution means.

Practice — Addendum F

1. For $f(x) = 2 - x$, find $f^n(x)$ for every positive integer n .
2. For $f(x) = \frac{1}{x}$, $x \neq 0$, find $f^{57}(x)$.
3. Show that $f(x) = \frac{x+1}{x-1}$, $x \neq 1$, is an involution.
4. Find all linear f with $f \circ f = I$ *without* using F4 — by arguing geometrically about slopes.
5. $f^3 = I$ as an identity, $f(1) = 2$ and $f(2) = 3$. Find $f(3)$ and $f(f(f(7)))$.

Answers: 1. $f^2(x) = x$, so $f^n(x) = x$ for even n and $f^n(x) = 2 - x$ for odd n . 2. $f^2 = I$,

and 57 is odd, so $f^{57}(x) = \frac{1}{x}$. 3. $f^2(x) = \frac{\frac{x+1}{x-1} + 1}{\frac{x+1}{x-1} - 1} = \frac{\frac{2x}{x-1}}{\frac{2}{x-1}} = x$. 4. A linear involution

composed with itself is the identity, so its graph reflected twice in $y = x$ returns to itself; the only lines invariant under reflection in $y = x$ are $y = x$ itself (slope 1) and lines perpendicular to it (slope -1), giving $f(x) = x$ and $f(x) = c - x$. 5. $f(3) = f(f(2)) = f^3(1) = 1$; and $f(f(f(7))) = f^3(7) = 7$.

Addendum P — What JEE Actually Asks

This section is built only from the previous-year bank supplied with this chapter. A warning up front: the extract arrived with many formulas stripped out, so roughly a third of the items could not be reconstructed with confidence and have been deliberately left out. Everything

below is drawn from the items that *were* recoverable, and every reconstructed question has been independently re-solved here. Where the bank’s own printed answer conflicted with a correct re-resolution, the correction is flagged explicitly.

1. The distribution

Judging from the ~50 usable items in this bank, the examiners’ attention splits roughly as follows (counts are approximate, because damaged items were excluded):

Cluster	Approx. share	Evidence in the bank
Injective / surjective / bijective classification	~30% (the single biggest block)	[1979], [2002S], [2003S], [2010], [2012], [Adv 2014], [Main 2017], [Main Jan 2019 — three separate sessions], [Main Sep 2020], [Adv 2020]
Domain & range, incl. greatest-integer / fractional part	~25%	[1978], [1979], [1983 twice], [1984 twice], [1994], [2000S], [2001S], [2003S], matching items [2011], [Adv 2014], [Adv 2018]
Composition and inverse	~25%	[1982], [1983], [1989], [1998], [1999], [2001S repeatedly], [2004S], [2011 Adv], [Main 2014], [Main Jan/Apr 2019], [Main Jan 2020]
Counting functions (one-one, onto, constrained)	~12%	[1985], [2001S], [Main Online Apr 2015], [Adv 2018], [Main Sep 2020]
Relations (reflexive / symmetric / transitive)	~12%, almost all post-2013	[1981], [1982], [Main Online Apr 2013], [Main Online Apr 2014], [Main Online Apr 2018 twice]
Even / odd / periodicity	~10%	[1983], [1989], [1991], [1996], [Adv 2014]
Functional equations	~8%, concentrated pre-2000	[1988], [1992], [1995S], [1996]

Two honest caveats. First, these clusters overlap heavily — an Adv 2014 matching item tests domain *and* range *and* monotonicity in one question — so shares do not sum to 100%. Second, the evidence for “functional equations are dying” rests on only four recoverable items, all older; treat that trend as suggestive, not proven. What is unmistakable: classification of

maps by monotonicity, and inverse/composite manipulation, appear in every single decade of the bank without exception.

2. The recurring archetypes

Archetype A — Monotonicity proves bijectivity

Looks like: “Is f one-one? onto?” for a function mixing a polynomial with a bounded term ($\sin x$, $\cos x$, $\ln$). **Method:** differentiate; show f' keeps one sign, hence strictly increasing, hence one-one; then push x to $\pm\infty$ and invoke continuity (IVT) for onto. **Worked example** [JEE 2002S]: $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 2x + \sin x$.

$$f'(x) = 2 + \cos x, \quad -1 \leq \cos x \leq 1 \Rightarrow 1 \leq f'(x) \leq 3$$

so $f' > 0$ everywhere: strictly increasing, therefore one-one. As $x \rightarrow \infty$, $2x$ dominates the bounded $\sin x$, so $f(x) \rightarrow \infty$; likewise $f(x) \rightarrow -\infty$. By the intermediate value theorem every real value is attained: onto. **Answer: both one-one and onto.**

Archetype B — Classification on a restricted interval

Looks like: a cubic (or similar) on a closed interval, with codomain chosen suspiciously well. **Method:** find critical points of f' , tabulate f at endpoints and critical points, compare the resulting range with the codomain. **Worked example** [JEE 2012]: $f : [0, 3] \rightarrow [1, 29]$, $f(x) = 2x^3 - 15x^2 + 36x + 1$.

$$f'(x) = 6x^2 - 30x + 36 = 6(x - 2)(x - 3)$$

f' is positive on $[0, 2)$ and negative on $(2, 3]$, so f is not monotonic — many-one. Values:

$$f(0) = 1, \quad f(2) = 29, \quad f(3) = 28$$

Range = $[1, 29]$ = codomain: onto. **Answer: onto but not one-one.** (Same machinery drives [Adv 2014]’s “ $x^5 - 5x + a$: how many real roots?”, where the local max/min ± 4 of $x^5 - 5x$ decide everything.)

Archetype C — Counting functions with a constraint

Looks like: “how many onto/one-one functions, given that some element has exactly k preimages,” or a numerical-answer α, β comparison. **Method:** fix the constrained part by choosing preimages ($\binom{n}{k}$), then count freely on the remainder, subtracting the maps that miss a target (inclusion-exclusion). **Worked example** [JEE Adv 2018]: $|X| = 5$, $|Y| = 7$; α = one-one maps $X \rightarrow Y$, β = onto maps $Y \rightarrow X$; find $\frac{\beta - \alpha}{5!}$.

$$\alpha = {}^7P_5 = \binom{7}{5} \cdot 5! = 21 \cdot 120 = 2520$$

For β , inclusion-exclusion over the 5 targets:

$$\begin{aligned}\beta &= \sum_{k=0}^5 (-1)^k \binom{5}{k} (5-k)^7 \\ &= 78125 - 81920 + 21870 - 1280 + 5 = 16800\end{aligned}$$

(Check by occupancy: patterns $(3, 1, 1, 1, 1)$ give $\binom{7}{3} \cdot 5 \cdot 4! = 4200$; $(2, 2, 1, 1, 1)$ give $\binom{5}{2} \cdot \frac{7!}{2!2!} = 12600$; total 16800.) Hence

$$\frac{\beta - \alpha}{5!} = \frac{16800 - 2520}{120} = \boxed{119}$$

Archetype D — Testing a relation defined by an identity

Looks like: $R = \{(a, b) : \text{some trig/algebraic equation}\}$; classify it. **Method:** simplify the defining condition to its cleanest equivalent form first — the properties then become obvious.

Worked example [Main Online Apr 2014]: $P = \{(a, b) : \sec^2 a - \tan^2 b = 1\}$ on $\mathbb{R}$. Since $\sec^2 a = 1 + \tan^2 a$, the condition collapses to $\tan^2 a = \tan^2 b$. That is manifestly reflexive, symmetric and transitive. **Answer: an equivalence relation.** Contrast [1982]-style items where the condition is *not* symmetric and a single numeric counterexample settles it.

Archetype E — Inverses with branch discipline

Looks like: invert a two-branch-producing function on a restricted domain; the wrong root is always offered as a distractor. **Method:** solve $y = f(x)$ for x , then *use the stated domain to kill one branch*. **Worked example** [JEE 2001S]: $f : [1, \infty) \rightarrow [2, \infty)$, $f(x) = x + \frac{1}{x}$.

$$y = x + \frac{1}{x} \Rightarrow x^2 - yx + 1 = 0 \Rightarrow x = \frac{y \pm \sqrt{y^2 - 4}}{2}$$

The two roots multiply to 1; the minus-root equals $\frac{2}{y + \sqrt{y^2 - 4}} < 1$ for $y > 2$, so it violates $x \geq 1$. Therefore

$$f^{-1}(x) = \frac{x + \sqrt{x^2 - 4}}{2}, \quad x \geq 2$$

Archetype F — Functional equations by twin substitution

Looks like: one equation linking $f(x)$ and $f(\text{transformed } x)$; find f or a specific value.

Method: substitute special values for anchors, generate the partner equation via $x \mapsto \frac{1}{x}$ or $x \mapsto 1 - x$, eliminate linearly. **Worked example** [JEE 1992]: $f(x + y) = f(x)f(y)$ on naturals, $f(1) = 2$; find natural a with

$$f(a+1) + f(a+2) + \cdots + f(a+n) = 16(2^n - 1)$$

Induction on the Cauchy-product law gives $f(k) = 2^k$. Factor the left side:

$$f(a)(2 + 2^2 + \cdots + 2^n) = f(a)(2^{n+1} - 2) = 2f(a)(2^n - 1)$$

So $2f(a)(2^n - 1) = 16(2^n - 1)$, giving $f(a) = 8 = 2^3$, hence $a = 3$.

Archetype G — Floor / fractional-part equations

Looks like: an equation mixing $[x]$ and $\{x\}$. **Method:** reduce everything to $[x]$ and $\{x\}$ separately; use $0 \leq \{x\} < 1$ to bound $[x]$ to finitely many values; finish case by case. **Worked example** [JEE 1994]: solve $4\{x\} = x + [x]$ with the standard conventions $[x] = \lfloor x \rfloor$, $\{x\} = x - [x] \in [0, 1)$. Substituting $x = [x] + \{x\}$:

$$4\{x\} = 2[x] + \{x\} \Rightarrow 3\{x\} = 2[x] \Rightarrow \{x\} = \frac{2}{3}[x]$$

Since $0 \leq \{x\} < 1$, we get $0 \leq [x] < \frac{3}{2}$, so $[x] \in \{0, 1\}$: $[x] = 0$ forces $\{x\} = 0$, giving $x = 0$; $[x] = 1$ forces $\{x\} = \frac{2}{3}$, giving $x = \frac{5}{3}$. **Answer:** $x \in \{0, \frac{5}{3}\}$. (Caution: the 1994-era source used the signed “integer part” convention, under which $x = -\frac{5}{3}$ also qualifies. Under today’s standard JEE convention it does not — know which convention your paper uses.)

Archetype H — Structure and logic puzzles on one-one maps

Looks like: sparse information (“exactly one of these statements is true”) plus one-oneness; deduce a specific value. **Method:** exhaust the cases; one-oneness eliminates the contradictions. **Worked example** [JEE 1982]: f one-one, domain $\{x, y, z\}$, range $\{1, 2, 3\}$; among “ $f(x) = 1$ ”, “ $f(y) \neq 1$ ”, “ $f(z) \neq 2$ ” exactly one is true. If $f(x) = 1$ is the true one, the other two fail, forcing $f(y) = 1$ — contradicting one-oneness. If “ $f(y) \neq 1$ ” is true, then $f(x) \neq 1$, $f(z) = 2$, leaving both $f(x)$ and $f(y)$ unable to be 1 yet one of them must equal 2 — contradiction again. So “ $f(z) \neq 2$ ” is true, forcing $f(x) \neq 1$, $f(y) = 1$, $f(z) = 2$, $f(x) = 3$.

$$f^{-1}(1) = y$$

3. The traps

These are the mistakes the distractors in this bank are engineered around:

1. **Failing to check the rule is a function at all.** [Main Jan 2019] asked to classify $f : (0, \infty) \rightarrow (0, \infty)$ with $f(x) = \ln x$ — but $f(1) = 0 \notin (0, \infty)$, so it is not a function and the item was declared bonus. Always test well-definedness before one-one/onto.
2. **Codomain-range conflation.** The 1979 chestnut $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$ punishes anyone who answers “surjective” out of habit; and [2012] rewards those who actually compute the range on the stated interval rather than eyeballing the degree.
3. **Missing the non-obvious transitivity chain.** In the 2013 relation $x = y$ or $x = 3y$ on $\mathbb{N}$, everyone checks $(3, 1)$, $(1, 1)$ and declares transitivity — but $(9, 3)$ and $(3, 1)$ force $(9, 1)$, which fails. The compendium’s own printed solution makes exactly this slip (see Practice Q4 below). Chains through the “ $3\times$ ” branch are where the marks hide.
4. **Branch errors when inverting.** Solving a quadratic for x yields two roots; the domain exists precisely to discard one (Archetype E). Distractor options are built from the discarded branch.
5. **m^n versus nP_m ,** and ignoring feasibility: no one-one map when $|A| > |B|$, no onto map when $|A| < |B|$.

6. f^{-1} **read as** $\frac{1}{f}$. Recurrent distractor across decades.
 7. **Order reversal in** $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ — the same-order option is always offered.
 8. **Log-base hygiene.** A base needs > 0 and $\neq 1$; forgetting $1 - x \neq 1$ costs the point in domain questions like Practice Q8.
 9. **Floor of negatives and the bounds on** $\{x\}$. Every floor question in this bank exploits $[-2.3] = -3$ or the fact $0 \leq \{x\} < 1$.
 10. **Self-inverse recognition.** $f(x) = (a - x^n)^{1/n}$ and Möbius-type maps compose to the identity; students who grind through algebra miss that $f = f^{-1}$ was checkable in one line.
-

4. Graded practice set

All twelve are adapted from the bank or written in its spirit, and every answer has been re-derived here.

Q1. (Easy) $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$. Classify f .

Solution. $f(1) = f(-1) = 1$: many-one. No negative number has a preimage: into. **Neither injective nor surjective.** [cf. JEE 1979]

Q2. (Easy) Find the domain and range of $f(x) = \frac{x^2}{1+x^2}$, and decide whether f is one-one.

Solution. Denominator ≥ 1 , so domain $= \mathbb{R}$. Put $y = \frac{x^2}{1+x^2}$: then $x^2 = \frac{y}{1-y}$, which needs $0 \leq y < 1$; conversely every $y \in [0, 1)$ arises. Range $= [0, 1)$. Since $f(1) = f(-1) = \frac{1}{2}$, **not one-one**. [reconstructed from JEE 1978]

Q3. (Easy) Find the number of onto functions from $E = \{1, 2, 3, 4\}$ to $F = \{1, 2\}$.

Solution. Total functions $2^4 = 16$; the two constant maps miss a target. $16 - 2 = \mathbf{14}$. [JEE 2001S]

Q4. (Medium) On $\mathbb{N}$, let $R = \{(x, y) : x^2 - 4xy + 3y^2 = 0\}$. Determine whether R is reflexive, symmetric, transitive.

Solution. Factoring, $(x-y)(x-3y) = 0$, so $x = y$ or $x = 3y$. Every (a, a) satisfies $x = y$: **reflexive**. But $(3, 1) \in R$ while $(1, 3) \notin R$: **not symmetric**. For transitivity, note $(9, 3) \in R$ (since $9 = 3 \cdot 3$) and $(3, 1) \in R$, yet $(9, 1) \notin R$ ($81 - 36 + 3 \neq 0$): **not transitive**. **Answer: reflexive, but neither symmetric nor transitive. Caution** — The compendium's printed solution claims transitive and keys option (d); the chain above refutes it. Trust the counterexample. [JEE Main Online Apr 2013]

Q5. (Medium) Find f and g such that $g(f(x)) = |\sin x|$ and $f(g(x)) = \sin^2 \sqrt{x}$.

Solution. Try $f(x) = \sin^2 x$, $g(x) = \sqrt{x}$. Then

$$g(f(x)) = \sqrt{\sin^2 x} = |\sin x|, \quad f(g(x)) = \sin^2(\sqrt{x})$$

Both identities hold, so $f(x) = \sin^2 x$, $g(x) = \sqrt{x}$. The rival option $f(x) = \sin x$, $g(x) = |x|$ fails because $\sin |x| \neq |\sin x|$. [JEE 1998]

Q6. (Medium) A has 7 elements and $B = \{y_1, y_2, y_3\}$. How many onto functions $f : A \rightarrow B$ have exactly three elements of A mapping to y_2 ?

Solution. Choose the three preimages of y_2 : $\binom{7}{3}$ ways. The remaining four elements must hit both y_1 and y_3 : $2^4 - 2 = 14$ ways. Total

$$14 \cdot \binom{7}{3} = 490$$

[JEE Main Online Apr 2015]

Q7. (Medium) $f : [1, \infty) \rightarrow [2, \infty)$, $f(x) = x + \frac{1}{x}$. Find $f^{-1}(x)$.

Solution. Solved in Archetype E: $f^{-1}(x) = \frac{x + \sqrt{x^2 - 4}}{2}$ on $[2, \infty)$; the minus-root violates $x \geq 1$. [JEE 2001S]

Q8. (Medium) Find the domain of $f(x) = \log_{(1-x)}(x+2)$.

Solution. Argument: $x+2 > 0 \Rightarrow x > -2$. Base: $1-x > 0 \Rightarrow x < 1$, and $1-x \neq 1 \Rightarrow x \neq 0$. Intersecting:

$$[-2, 1) \setminus \{0\}$$

The base-equals-1 exclusion is the designed trap. [adapted from JEE 1983]

Q9. (Hard) Show that $f(x) = (a - x^n)^{1/n}$ ($a > 0$, n a positive integer, on its natural domain) satisfies $(f \circ f)(x) = x$.

Solution.

$$f(f(x)) = \left(a - [(a - x^n)^{1/n}]^n\right)^{1/n} = (a - a + x^n)^{1/n} = x$$

The map is its own inverse. **True.** [JEE 1983, True/False]

Q10. (Hard) f satisfies $f(x+y) = f(x)f(y)$ for all natural x, y , with $f(1) = 2$. Find the natural number a for which $f(a+1) + f(a+2) + \cdots + f(a+n) = 16(2^n - 1)$.

Solution. Solved in Archetype F: $f(k) = 2^k$, the sum factors as $2f(a)(2^n - 1)$, so $f(a) = 8$ and $a = 3$. [JEE 1992]

Q11. (Hard) Let $f(x) = x^2$ and $g(x) = \sin x$ on $\mathbb{R}$. Solve

$$(f \circ g \circ g \circ f)(x) = (g \circ g \circ f)(x)$$

Solution. Reading innermost-first, $g \circ g \circ f$ sends $x \mapsto \sin(\sin x^2)$, and $f \circ g \circ g \circ f$ squares it. Writing $u = \sin(\sin x^2)$, the equation $u^2 = u$ gives $u \in \{0, 1\}$. Now $u = 1$ would require $\sin x^2 = \frac{\pi}{2} > 1$, impossible. So $\sin(\sin x^2) = 0$, i.e. $\sin x^2 = n\pi$; since $|\sin x^2| \leq 1$, only $n = 0$ survives, so

$$x^2 = m\pi, \quad m = 0, 1, 2, \dots \quad \implies \quad x = \pm\sqrt{m\pi}$$

[JEE Adv 2011]

Q12. (Challenge) f is one-one with domain $\{x, y, z\}$ and range $\{1, 2, 3\}$. Exactly one of the statements " $f(x) = 1$ ", " $f(y) \neq 1$ ", " $f(z) \neq 2$ " is true. Find $f^{-1}(1)$.

Solution. Solved in Archetype H by exhausting the three cases; only " $f(z) \neq 2$ true" survives, forcing $f(y) = 1$.

$$f^{-1}(1) = y$$

5. What this chapter's text under-serves

The chapter trains board-style proofs and one-step MCQs, but this bank shows JEE's centre of gravity lies elsewhere: calculus-driven classification (sign charts of f' , IVT arguments for onto), constrained counting with inclusion-exclusion and “exactly k preimages” twists, and multi-layer constructions — recovering f from $g \circ f$, iterating a function n times, solving equations inside composites, and one-oneness logic puzzles. Floor/fractional-part *equation-solving* is likewise given far thinner treatment here than its persistent appearance in the paper warrants. Finally, the habit the exam most rewards and this text least drills is adversarial verification: hunting the one counterexample chain (as in Q4) and checking that the given rule is a function at all before classifying it.

Inverse Trigonometric Functions

Chapter 02 — Class 12 Mathematics — Complete Study Guide

Drona Ed Tech

Class 11 → 12 · CBSE · CUET · JEE Main · JEE Advanced

Contents

1 Principal Value Branches	2
1.1 Concept Intuition	2
1.2 Key Definitions, Branches & Validity Conditions	3
1.3 Derivations & Key Procedures	4
1.4 Worked Examples	5
1.5 Practice Exercises	6
1.6 Crack the MCQ	7
1.7 Common Student Pitfalls & Pro-Tips	8
1.8 Cheat Sheet Box	8
2 Self-Map Identities (Reciprocal, Negative-Argument, Complementary)	10
2.1 Concept Intuition	10
2.2 Key Identities, Definitions & Validity Conditions	11
2.3 Derivations from the Branch Structure	12
2.4 Worked Examples	13
2.5 Practice Exercises	14
2.6 Crack the MCQ	15
2.7 Common Student Pitfalls & Pro-Tips	15
2.8 Cheat Sheet Box	16
3 Combination Identities (Sums, Differences & Multiple Angles)	17
3.1 Concept Intuition	17
3.2 Key Identities, Definitions & Validity Conditions	18
3.3 Derivations from the Addition Formulas	19
3.4 Worked Examples	19
3.5 Practice Exercises	20
3.6 Crack the MCQ	21
3.7 Common Student Pitfalls & Pro-Tips	22
3.8 Cheat Sheet Box	22
4 Graphs, Composite Evaluation & Simplification	24
4.1 Concept Intuition	24
4.2 Key Results, Conversions & Graph Properties	25
4.3 The Triangle Method & Substitution Recipe	26
4.4 Worked Examples	26
4.5 Practice Exercises	27
4.6 Crack the MCQ	28
4.7 Common Student Pitfalls & Pro-Tips	28
4.8 Cheat Sheet Box	29

How to use this guide. This covers Chapter 2 end to end in four parts, building in order: **(1) Principal Value Branches** — the domains, ranges and how to read a principal value; **(2) Self-Map Identities** — reciprocal, negative-argument and complementary laws; **(3) Combination Identities** — sums, differences and multiple angles; **(4) Graphs, Composite Evaluation & Simplification** — the visual and computational toolkit. Each part follows the same eight-section rhythm, from intuition through worked examples to a final cheat sheet.

1 Principal Value Branches

Exam Relevance: This is the foundation slab the whole chapter stands on — get the branches wrong here and every property, every $\sin^{-1}(\sin x)$ simplification, and every integral substitution downstream collapses. **CBSE Boards** asks “find the principal value” as a guaranteed 1-2 mark question every single year. **CUET** weaponises the branches into lightning “which of these is the principal value branch of $\sec^{-1} x$?” one-liners — pure recall, pure speed. **JEE Main** lives on the $f^{-1}(f(x))$ range-folding trap (e.g. $\sin^{-1}(\sin \frac{2\pi}{3})$), where the *wrong* answer is the obvious one. **JEE Advanced** stacks domains, exclusions, and piecewise behaviour into multi-step expressions. If you treat $\sin^{-1} x$ as “just undo the sine,” you will lose marks you didn’t know were on the table. This subtopic teaches you to *think in branches*.

1.1 Concept Intuition

Start with a problem that sounds trivial and isn’t: **“Which angle has a sine of $\frac{1}{2}$?”**

Your first instinct says 30° , i.e. $\frac{\pi}{6}$. But 150° also works. So does 390° , and 510° , and -330° , and infinitely many more. The sine function takes *many different angles* to the *same value* — it is **many-to-one**. And that is fatal for inversion, because an inverse function must give back **exactly one** answer. A machine that, fed $\frac{1}{2}$, spits out “ $\frac{\pi}{6}$ or $\frac{5\pi}{6}$ or $\frac{13\pi}{6}$ or...” is not a function at all.

Think about it this way. Picture the platform of a busy railway junction where several trains pull in showing the *same* destination board. If a passenger asks “which train do I take for Delhi?” and four trains all say Delhi, the question has no single answer — unless the station *designates one official train* for that route. Inverse trigonometry does exactly this: out of the infinitely many angles sharing a given sine, mathematics **designates one official angle** as *the* answer. The rule that picks the official angle is the **principal value branch**.

How do we pick? We **shrink the domain of the original trig function** down to a stretch on which it is one-to-one (each output used exactly once) *and* on which it still sweeps through its entire range. For sine, the agreed stretch is $[-\frac{\pi}{2}, \frac{\pi}{2}]$: across it, sine climbs steadily from -1 to $+1$, hitting every value in between exactly once. Now the question “which angle has sine $\frac{1}{2}$?” has a unique honest answer: $\frac{\pi}{6}$. The interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$ is the **principal value branch** of $\sin^{-1}$ — the official platform.

Two things are worth internalising right now:

1. **The branch is a choice, but a fixed one.** We *could* have chosen another one-to-one stretch (there are infinitely many), and indeed one may take a *non-principal* branch of $\cos^{-1} -1$ such as $[2\pi, 3\pi]$. Those are valid for defining “an” inverse — but the world has agreed on **one** canonical branch per

function so that “ $\sin^{-1} \frac{1}{2}$ ” means the same thing in Wichita and in Kota. Memorising the canonical branches is non-negotiable.

2. **The output of an inverse trig function is an angle, and it is trapped inside its branch — always.** This single sentence is the source of nearly every mark gained or lost in this chapter. If you ever produce $\sin^{-1}(\text{something}) = \frac{2\pi}{3}$, you have already made an error, because $\frac{2\pi}{3} = 120^\circ$ lies *outside* $[-\frac{\pi}{2}, \frac{\pi}{2}]$. The branch is a cage, and the answer can never escape it.

The six trig functions each get their own designated cage. Sine and tangent (odd, symmetric about the origin) get branches *centred on 0*. Cosine and cotangent (whose natural one-to-one stretch starts at 0) get branches running *from 0 up to π* . Secant and cosecant — the reciprocal misfits — inherit the cosine and sine branches respectively, *minus one forbidden point* where the reciprocal blows up. The next two sections make all of this precise.

Figure 2.1 A single sine curve over $[-2\pi, 2\pi]$. The segment over $x \in [-\pi/2, \pi/2]$ is shaded: the principal branch, one-to-one, covering the full range $[-1, 1]$. A horizontal dashed line at $y = 1/2$ cuts the curve at many points; only the intersection inside the shaded segment is circled - the one official answer, $\pi/6$. (see next page)

1.2 Key Definitions, Branches & Validity Conditions

Locked notation (chapter-wide — used in every subtopic of Chapter 02):

- $\sin^{-1} x$, $\cos^{-1} x$, $\tan^{-1} x$, $\cot^{-1} x$, $\sec^{-1} x$, $\operatorname{cosec}^{-1} x$ denote the **inverse functions**. These equal $\arcsin x$ etc.; this file uses the $\sin^{-1}$ form throughout (NCERT convention).
- **Critical:** $\sin^{-1} x \neq (\sin x)^{-1} = \frac{1}{\sin x} = \operatorname{cosec} x$. The “ -1 ” is an *inverse-function* marker, **not** an exponent. Locked once here; never re-explained.
- x = the *argument* fed to the inverse function (a pure number). y = the *principal value* it returns (an angle in radians).
- D = domain (allowed inputs x). B = **principal value branch** = the range of the inverse = the cage the output angle lives in.

The master table. For each function, $y = f^{-1}(x)$ means “ y is the unique angle in the branch B whose f equals x .”

Function	Domain D (allowed x)	Principal value branch B (the cage for y)
$\sin^{-1} x$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1} x$	$\mathbb{R} = (-\infty, \infty)$	$(-\frac{\pi}{2}, \frac{\pi}{2})$
$\cot^{-1} x$	$\mathbb{R} = (-\infty, \infty)$	$(0, \pi)$
$\sec^{-1} x$	$\mathbb{R} - (-1, 1)$, i.e. $ x \geq 1$	$[0, \pi] - \{\frac{\pi}{2}\}$
$\operatorname{cosec}^{-1} x$	$\mathbb{R} - (-1, 1)$, i.e. $ x \geq 1$	$[-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$

Validity conditions — a “unit check.” Physics gives you dimensional analysis; inverse trig gives you *two* free sanity checks on every answer:

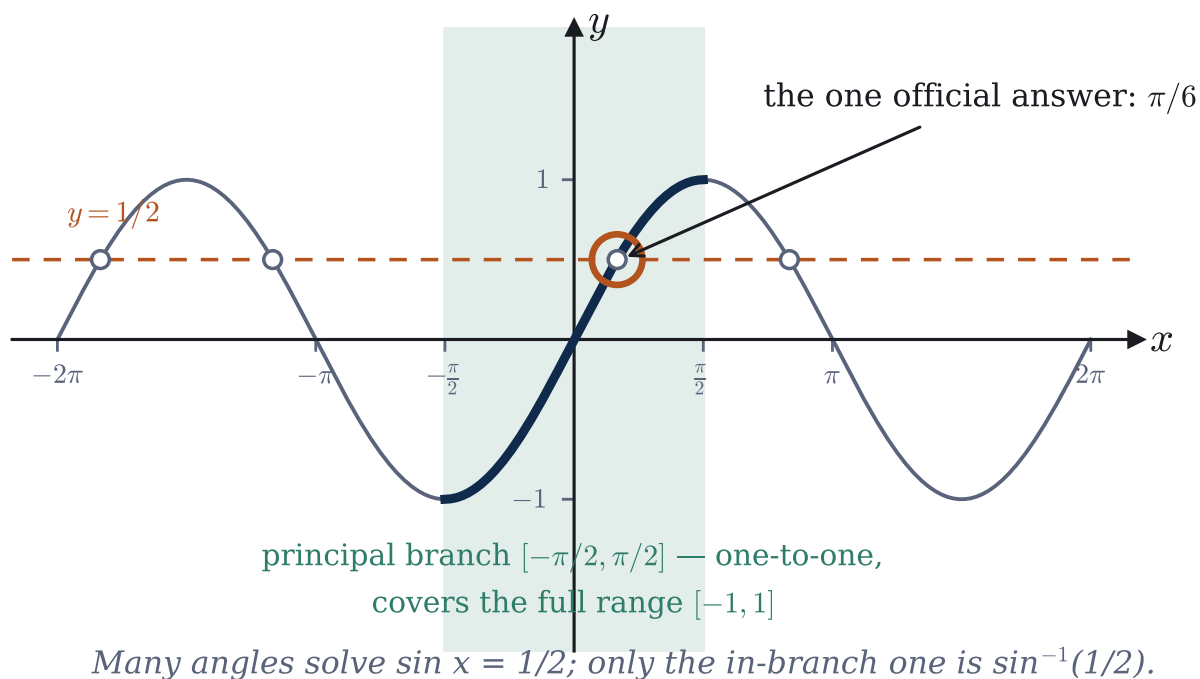


Figure 2.1 A single sine curve over $[-2\pi, 2\pi]$. The segment over $x \in [-\pi/2, \pi/2]$ is shaded: the principal branch, one-to-one, covering the full range $[-1, 1]$. A horizontal dashed line at $y = 1/2$ cuts the curve at many points; only the intersection inside the shaded segment is circled - the one official answer, $\pi/6$.

1. **Domain check (before you start):** Is x a legal input? $\sin^{-1}(2)$ and $\cos^{-1}(-3)$ **do not exist** — $|x| > 1$ is outside $[-1, 1]$. $\sec^{-1}(0.5)$ and $\operatorname{cosec}^{-1}(\frac{1}{2})$ **do not exist** — $|x| < 1$ is forbidden for $\sec/\operatorname{cosec}$. If x is illegal, the answer is “undefined,” full stop.
2. **Branch check (after you finish):** Does your output angle y lie inside B ? If $\cos^{-1}(\text{anything})$ comes out negative, or $\sin^{-1}(\text{anything})$ comes out bigger than $\frac{\pi}{2}$, you have made an error — *guaranteed*, no second-guessing.

Three structural patterns worth seeing in the table so you never rote-memorise blind:

- **Odd-function pair** $\{\sin, \tan\} \rightarrow$ branches centred on 0: $[-\frac{\pi}{2}, \frac{\pi}{2}]$ and $(-\frac{\pi}{2}, \frac{\pi}{2})$. (Tan’s is *open* — $\tan$ is undefined at $\pm\frac{\pi}{2}$.)
- **“Starts at 0” pair** $\{\cos, \cot\} \rightarrow$ branches $[0, \pi]$ and $(0, \pi)$. (Cot’s is *open* — $\cot$ is undefined at 0 and π .)
- **Reciprocal pair** $\{\sec, \operatorname{cosec}\} \rightarrow$ borrow the $\cos/\sin$ branches, then *delete the one point* where the reciprocal explodes: $\sec^{-1}$ drops $\frac{\pi}{2}$ (where $\cos = 0$); $\operatorname{cosec}^{-1}$ drops 0 (where $\sin = 0$). Their domains exclude $(-1, 1)$ because a reciprocal can never land strictly between -1 and 1 .

1.3 Derivations & Key Procedures

Unlike a pure plug-in subtopic, the branches are *justified*, and the most examined skill — collapsing $f^{-1}(f(\theta))$ — is a *procedure* you can derive once and reuse forever.

Derivation 1 — Why $[-\frac{\pi}{2}, \frac{\pi}{2}]$ is forced for $\sin^{-1}$ (and the general selection rule).

To invert sine we need a stretch of the x -axis satisfying **two** demands simultaneously:

- **(D1) One-to-one:** sine must be strictly monotonic there, so each output is used once.
- **(D2) Onto the full range:** the stretch must produce *every* value in $[-1, 1]$, or the inverse won’t be defined for some legal inputs.

On $[-\frac{\pi}{2}, \frac{\pi}{2}]$, sine increases strictly from $\sin(-\frac{\pi}{2}) = -1$ to $\sin(\frac{\pi}{2}) = 1$ — satisfying both (D1) and (D2). It is also the stretch *closest to the origin*, which is why convention crowns it principal. The identical reasoning gives every other branch: pick the origin-nearest stretch on which the function is strictly monotonic and sweeps its whole range. For cosine that stretch is $[0, \pi]$ (strictly decreasing from $+1$ to -1); for tangent it is the open $(-\frac{\pi}{2}, \frac{\pi}{2})$, and so on. *This is why the branches are not arbitrary — they are the smallest, most central monotonic windows.*

Derivation 2 — Why $\sec^{-1}$ excludes exactly $\frac{\pi}{2}$.

$\sec \theta = \frac{1}{\cos \theta}$. Borrow cosine’s branch $[0, \pi]$. At $\theta = \frac{\pi}{2}$, $\cos \frac{\pi}{2} = 0$, so $\sec \frac{\pi}{2} = \frac{1}{0}$ is **undefined** — no input x can ever map there. Delete that single point: $B_{\sec^{-1}} = [0, \pi] - \{\frac{\pi}{2}\}$. Identically, $\operatorname{cosec}^{-1}$ borrows sine’s branch and deletes 0 (where $\sin 0 = 0$).

Procedure (the high-yield one) — evaluating $f^{-1}(f(\theta))$ by range-folding.

The identity $\sin^{-1}(\sin \theta) = \theta$ holds **only if θ already lives in the branch $[-\frac{\pi}{2}, \frac{\pi}{2}]$** . When θ is outside, you must *fold it back into the branch* using a trig identity that preserves the function’s value. Standard fold-backs:

$$\sin(\pi - \theta) = \sin \theta, \quad \cos(2\pi - \theta) = \cos(-\theta) = \cos \theta, \quad \tan(\theta \pm \pi) = \tan \theta.$$

The three-step recipe:

1. Read the **branch** of the *outer* inverse function.
2. Using a value-preserving identity, replace the inner angle by an **equivalent angle that sits inside that branch**.
3. The inverse then returns that in-branch angle directly.

Worked through once: evaluate $\cos^{-1}(\cos \frac{7\pi}{6})$. Outer branch is $[0, \pi]$. The angle $\frac{7\pi}{6} = 210^\circ \notin [0, \pi]$. Fold: $\cos \frac{7\pi}{6} = \cos(2\pi - \frac{7\pi}{6}) = \cos \frac{5\pi}{6}$, and $\frac{5\pi}{6} = 150^\circ \in [0, \pi]$. Therefore $\cos^{-1}(\cos \frac{7\pi}{6}) = \frac{5\pi}{6}$ — **not** $\frac{7\pi}{6}$. Master this and you have neutralised the single most common JEE Main inverse-trig trap.

Figure 2.2 Number line of angles from $-\pi$ to 2π . The $\cos^{-1}$ branch $[0, \pi]$ is a green band. $7\pi/6$ sits outside the band (red); a curved arrow folds it to $5\pi/6$ inside (green), since $\cos(2\pi - \theta) = \cos \theta$. Fold the angle into the branch before reading off the answer. (see next page)

1.4 Worked Examples

Example 1 — [CBSE Board Level] Find the principal values of (a) $\sin^{-1}(-\frac{1}{2})$, (b) $\cos^{-1}(\frac{\sqrt{3}}{2})$, (c) $\tan^{-1}(1)$.

Solution. In each case, find the angle *inside the branch* whose trig value matches.

(a) Need $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ with $\sin y = -\frac{1}{2}$. Since $\sin(-\frac{\pi}{6}) = -\frac{1}{2}$ and $-\frac{\pi}{6} \in [-\frac{\pi}{2}, \frac{\pi}{2}]$:

$$\sin^{-1}(-\frac{1}{2}) = -\frac{\pi}{6}.$$

(b) Need $y \in [0, \pi]$ with $\cos y = \frac{\sqrt{3}}{2}$. Since $\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$ and $\frac{\pi}{6} \in [0, \pi]$:

$$\cos^{-1}(\frac{\sqrt{3}}{2}) = \frac{\pi}{6}.$$

(c) Need $y \in (-\frac{\pi}{2}, \frac{\pi}{2})$ with $\tan y = 1$. Since $\tan \frac{\pi}{4} = 1$:

$$\tan^{-1}(1) = \frac{\pi}{4}.$$

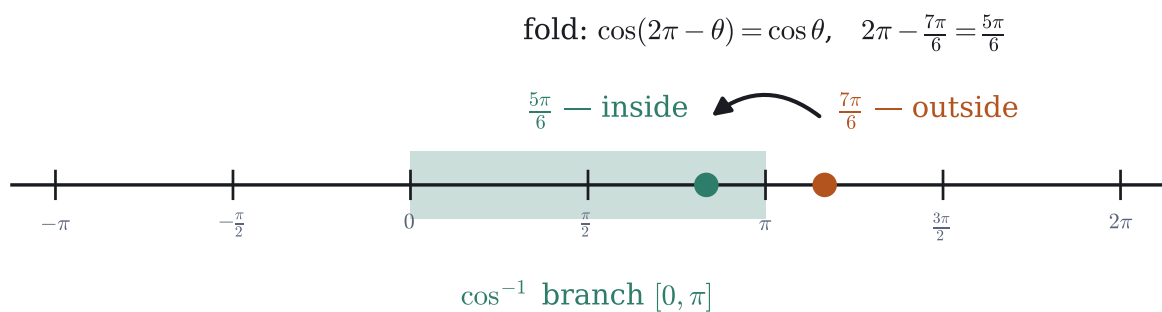
(Branch check: every answer sits inside its cage. ✓) ■

Example 2 — [CUET / Speed Trap] Which of the following is the principal value branch of $\operatorname{cosec}^{-1}x$? (a) $(-\frac{\pi}{2}, \frac{\pi}{2})$ (b) $[0, \pi] - \{\frac{\pi}{2}\}$ (c) $[-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$ (d) $[-\frac{\pi}{2}, \frac{\pi}{2}]$

Fast solution. $\operatorname{cosec}^{-1}$ borrows sine's branch $[-\frac{\pi}{2}, \frac{\pi}{2}]$ and deletes the forbidden point where $\sin = 0$, namely 0. Scan for the option that is sine's branch *minus a point* → (c). **Speed tip:** (a) is tan's branch (open, wrong shape); (b) is sec's branch (cosine-based, deletes $\frac{\pi}{2}$); (d) forgot to delete 0. The deleted point is the tell. **Answer: (c).** ■

Example 3 — [JEE Main Level] Evaluate (a) $\sin^{-1}(\sin \frac{2\pi}{3})$, (b) $\cos^{-1}(\cos \frac{7\pi}{6})$, (c) $\cos^{-1}[\cos(-680^\circ)]$.

Solution. Apply the range-folding recipe from §1.3.



Fold the angle into the branch before reading off the answer.

Figure 2.2 Number line of angles from $-\pi$ to 2π . The $\cos^{-1}$ branch $[0, \pi]$ is a green band. $7\pi/6$ sits outside the band (red); a curved arrow folds it to $5\pi/6$ inside (green), since $\cos(2\pi - \theta) = \cos \theta$. Fold the angle into the branch before reading off the answer.

(a) Branch $[-\frac{\pi}{2}, \frac{\pi}{2}]$. Now $\frac{2\pi}{3} = 120^\circ \notin$ branch. Fold with $\sin(\pi - \theta) = \sin \theta$:

$$\sin \frac{2\pi}{3} = \sin(\pi - \frac{2\pi}{3}) = \sin \frac{\pi}{3}, \quad \frac{\pi}{3} \in [-\frac{\pi}{2}, \frac{\pi}{2}] \Rightarrow \sin^{-1}(\sin \frac{2\pi}{3}) = \frac{\pi}{3}.$$

(b) Branch $[0, \pi]$. Already done in 1.3 $\cos^{-1}(\cos \frac{7\pi}{6}) = \frac{5\pi}{6}$.

(c) Branch $[0, \pi]$. Cosine is even, so $\cos(-680^\circ) = \cos(680^\circ)$. Reduce: $680^\circ = 720^\circ - 40^\circ \Rightarrow \cos 680^\circ = \cos(-40^\circ) = \cos 40^\circ$, and $40^\circ \in [0, 180^\circ]$. Hence

$$\cos^{-1}[\cos(-680^\circ)] = 40^\circ = \frac{2\pi}{9}.$$

■

Example 4 — [JEE Advanced Level] (a) Evaluate $\tan^{-1}(\tan \frac{3\pi}{4})$. (b) Express $f(x) = \sin^{-1}(\sin x)$ as a simple expression for all $x \in [\frac{\pi}{2}, \frac{3\pi}{2}]$. (c) Does $\sec^{-1}(\frac{1}{2})$ exist? Justify.

Solution.

(a) Branch $(-\frac{\pi}{2}, \frac{\pi}{2})$. Now $\frac{3\pi}{4} = 135^\circ \notin$ branch. Fold with $\tan(\theta - \pi) = \tan \theta$:

$$\tan \frac{3\pi}{4} = \tan(\frac{3\pi}{4} - \pi) = \tan(-\frac{\pi}{4}), \quad -\frac{\pi}{4} \in (-\frac{\pi}{2}, \frac{\pi}{2}) \Rightarrow \tan^{-1}(\tan \frac{3\pi}{4}) = -\frac{\pi}{4}.$$

(b) For $x \in [\frac{\pi}{2}, \frac{3\pi}{2}]$, the angle x lies *outside* $[-\frac{\pi}{2}, \frac{\pi}{2}]$, so we fold using $\sin x = \sin(\pi - x)$, and crucially $(\pi - x)$ *does* land inside the branch for this whole interval. Check the endpoints: at $x = \frac{\pi}{2}$, $\pi - x = \frac{\pi}{2}$ ✓; at $x = \frac{3\pi}{2}$, $\pi - x = -\frac{\pi}{2}$ ✓; and $\pi - x$ runs monotonically through $[-\frac{\pi}{2}, \frac{\pi}{2}]$ in between. Therefore

$$\boxed{\sin^{-1}(\sin x) = \pi - x \quad \text{for all } x \in [\frac{\pi}{2}, \frac{3\pi}{2}].}$$

(c) **Domain check first.** $\sec^{-1}$ requires $|x| \geq 1$, but $|\frac{1}{2}| = \frac{1}{2} < 1$. So $\frac{1}{2}$ is *not* a legal input: $\sec^{-1}(\frac{1}{2})$ **does not exist**. (There is no real angle whose secant equals $\frac{1}{2}$, since $|\sec \theta| \geq 1$ always.) ■

1.5 Practice Exercises

- [CBSE]** Find the principal values of (a) $\cos^{-1}(-\frac{1}{2})$, (b) $\tan^{-1}(-\sqrt{3})$, (c) $\operatorname{cosec}^{-1}(2)$.
- [CUET]** State (one line each) the principal value branch of $\sec^{-1} x$ and of $\cot^{-1} x$, and the domain of $\sin^{-1} x$.
- [JEE Main]** Evaluate $\cos^{-1}(\cos \frac{13\pi}{6})$ and $\tan^{-1}(\tan \frac{5\pi}{6})$.
- [JEE Main]** For how many integer values of x is $\sin^{-1} x$ defined? List them.
- [JEE Advanced]** Simplify $\sin^{-1}(\sin x)$ for $x \in [\frac{3\pi}{2}, 2\pi]$, and evaluate $\cos^{-1}[\cos(-3)]$ (here -3 is in radians; recall $\pi \approx 3.14159$).

Answers:

- (a) $\frac{2\pi}{3}$; (b) $-\frac{\pi}{3}$; (c) $\frac{\pi}{6}$.
- $\sec^{-1}$: $[0, \pi] - \{\frac{\pi}{2}\}$. $\cot^{-1}$: $(0, \pi)$. Domain of $\sin^{-1} x$: $[-1, 1]$.

3. $\cos \frac{13\pi}{6} = \cos(2\pi + \frac{\pi}{6}) = \cos \frac{\pi}{6}$, and $\frac{\pi}{6} \in [0, \pi] \Rightarrow \frac{\pi}{6}$. $\tan \frac{5\pi}{6} = \tan(\frac{5\pi}{6} - \pi) = \tan(-\frac{\pi}{6})$, and $-\frac{\pi}{6} \in (-\frac{\pi}{2}, \frac{\pi}{2}) \Rightarrow -\frac{\pi}{6}$.
4. $\sin^{-1} x$ needs $x \in [-1, 1]$; the integers are $-1, 0, 1 \rightarrow$ **three** values.
5. For $x \in [\frac{3\pi}{2}, 2\pi]$: fold with $\sin x = \sin(x - 2\pi)$, giving $\sin^{-1}(\sin x) = x - 2\pi$ (since $x - 2\pi \in [-\frac{\pi}{2}, 0] \subset$ branch). For $\cos^{-1}[\cos(-3)]$: $\cos$ even $\Rightarrow \cos(-3) = \cos 3$, and $3 \in [0, \pi]$ (as $3 < \pi$), so the answer is 3.
-

1.6 Crack the MCQ

Q1. Which of the following is the principal value branch of $\cos^{-1} x$?

- (a) $[-\frac{\pi}{2}, \frac{\pi}{2}]$ (b) $(0, \pi)$ (c) $[0, \pi]$ (d) $(0, \pi) - \{\frac{\pi}{2}\}$

different shapes. *Trap (b)*: cosine's branch is *closed*; $\cos(\pm 1) = 0, \pi$ are real values, so the endpoints must be included. *Trap (d)*: deleting $\frac{\pi}{2}$ is a $\sec^{-1}$ move — illegitimate for $\cos^{-1}$.

Q2. Which value can **never** be the output of $\sin^{-1} x$?

- (a) $-\frac{\pi}{2}$ (b) 0 (c) $\frac{\pi}{2}$ (d) $\frac{2\pi}{3}$

Answer: (d). $\sin^{-1}$ is caged in $[-\frac{\pi}{2}, \frac{\pi}{2}]$; $\frac{2\pi}{3} = 120^\circ$ is outside it. *Traps (a) and (c)*: $\pm \frac{\pi}{2}$ are the legitimate *endpoints* ($\sin^{-1}(\pm 1)$) — students who think the branch is open wrongly reject them.

Q3. $\sin^{-1}(\sin \frac{2\pi}{3}) =$

- (a) $\frac{2\pi}{3}$ (b) $\frac{\pi}{3}$ (c) $-\frac{\pi}{3}$ (d) $\frac{\pi}{6}$

Answer: (b). Fold: $\sin \frac{2\pi}{3} = \sin \frac{\pi}{3}$, and $\frac{\pi}{3} \in$ branch. *Trap (a)*: the headline trap — applying $\sin^{-1}(\sin \theta) = \theta$ blindly when θ is outside the branch. *Trap (c)*: folded with the wrong sign (used $\theta - \pi$ instead of $\pi - \theta$). *Trap (d)*: confused $\sin \frac{2\pi}{3} = \frac{\sqrt{3}}{2}$ with $\frac{1}{2}$.

Q4. The number of real values of x for which $\sec^{-1} x$ is **not** defined, among $x \in \{-2, -\frac{1}{2}, 0, \frac{3}{4}, 3\}$, is:

- (a) 1 (b) 2 (c) 3 (d) 0

Answer: (c). $\sec^{-1}$ needs $|x| \geq 1$. Illegal inputs: $-\frac{1}{2}, 0, \frac{3}{4}$ (all with $|x| < 1$) $\rightarrow$ **three**. *Trap (a)/(b)*: forgetting that 0 also fails $|x| \geq 1$, or missing one of the fractions. *Trap (d)*: mistakenly using $\sin^{-1}$'s domain $[-1, 1]$, under which $-\frac{1}{2}, 0, \frac{3}{4}$ all *would* be legal.

1.7 Common Student Pitfalls & Pro-Tips

1. **Reading** $\sin^{-1} x$ as $\frac{1}{\sin x}$. The “-1” marks the inverse *function*, not a reciprocal power. $\sin^{-1} x$ is an *angle*; $\frac{1}{\sin x} = \operatorname{cosec} x$ is a *ratio*. These are different objects with different units of meaning. (Want the reciprocal? Write $(\sin x)^{-1}$.)
2. **Answering outside the branch — the range-folding failure.** Writing $\sin^{-1}(\sin \frac{2\pi}{3}) = \frac{2\pi}{3}$ or $\cos^{-1}(\cos \frac{7\pi}{6}) = \frac{7\pi}{6}$. The identity $f^{-1}(f(\theta)) = \theta$ holds **only when θ is already in the branch**. Always: check the branch first, fold if needed.
3. **Domain blindness.** $\sin^{-1}(2)$, $\cos^{-1}(-1.5)$, $\sec^{-1}(0.7)$, $\operatorname{cosec}^{-1}(0)$ — all **undefined**. Run the domain check *before* computing anything; an “answer” to an undefined expression scores zero.
4. **Sec/cosec branch confusion.** The trickiest cages. $\sec^{-1}$ is cosine-based, deletes $\frac{\pi}{2}$. $\operatorname{cosec}^{-1}$ is sine-based, deletes 0. Mixing them up (or forgetting the deletion) is a classic CUET banana-skin.
5. **Mode mismatch.** Principal values are stated in **radians** in higher exams. Don’t leave a board answer as 30° when the marking key wants $\frac{\pi}{6}$ (or vice versa — read the question).

Pro-Tip: For any $f^{-1}(f(\theta))$ question, run the same two-second ritual every time: **(1) write the branch, (2) ask “is θ in it?”, (3) if not, fold.** That ritual converts the single highest-frequency JEE Main trap in this chapter into a guaranteed mark. Pair it with the free branch-check on your final answer and you become almost impossible to trick here.

1.8 Cheat Sheet Box

PRINCIPAL VALUE BRANCHES — RAPID RECALL

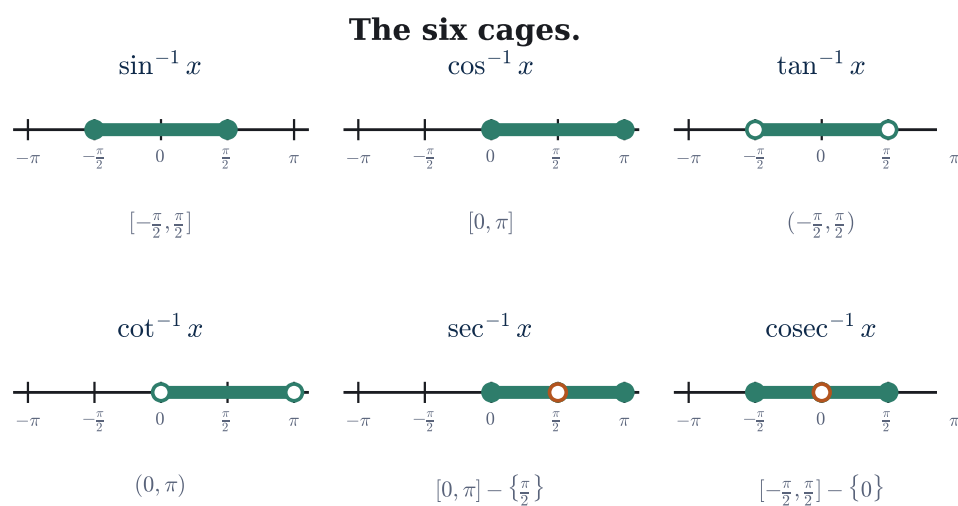
f^{-1}	Domain	Branch (cage for the answer)
$\sin^{-1}$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
$\cos^{-1}$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1}$	$\mathbb{R}$	$(-\frac{\pi}{2}, \frac{\pi}{2})$
$\cot^{-1}$	$\mathbb{R}$	$(0, \pi)$
$\sec^{-1}$	$ x \geq 1$	$[0, \pi] - \{\frac{\pi}{2}\}$
$\operatorname{cosec}^{-1}$	$ x \geq 1$	$[-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$

- **Output is always an angle, always trapped in the branch.** Outside the branch $\Rightarrow$ error.
- **Fold-backs:** $\sin(\pi - \theta) = \sin \theta$; $\cos(2\pi - \theta) = \cos \theta$ (cos is even); $\tan(\theta \pm \pi) = \tan \theta$.
- $f^{-1}(f(\theta)) = \theta$ **only if θ is in the branch** — else fold first.
- $\sin^{-1} x \neq \frac{1}{\sin x}$. The “-1” is inverse, not power.

Mnemonics:

- “Sin & tan hug zero; cos & cot climb to pi.” — branch shapes.
- “Sec drops half-pi, cosec drops zero.” — the deleted points.
- “Check the cage, fold the angle.” — the universal $f^{-1}(f(\theta))$ ritual.

Figure 2.3 *The six cages: one mini number-line per inverse function, its principal branch as a coloured bar with open and closed endpoints drawn correctly - open circles at tan's $\pm\pi/2$ and cot's 0, π ; deleted-point gaps at sec's $\pi/2$ and cosec's 0. (see next page)*



Filled dot = endpoint included; open ring = excluded. sec and cosec each lose one interior point.

Figure 2.3 *The six cages: one mini number-line per inverse function, its principal branch as a coloured bar with open and closed endpoints drawn correctly - open circles at tan's $\pm\pi/2$ and cot's $0, \pi$; deleted-point gaps at sec's $\pi/2$ and cosec's 0 .*

2 Self-Map Identities (Reciprocal, Negative-Argument, Complementary)

Exam Relevance: These are the “first-move” identities — the conversions you reach for *before* the heavy machinery in Part 3. **CBSE Boards** uses them constantly inside evaluate/prove questions, and the complementary law $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ is a perennial 1-2 marker. **CUET** turns them into instant value-traps: $\cot^{-1}(-\sqrt{3}) = ?$ catches anyone who guesses “ $-\frac{\pi}{6}$.” **JEE Main** drops them verbatim — the NCERT problem $\tan^{-1} \sqrt{3} - \cot^{-1}(-\sqrt{3})$ has appeared as an objective question, and the *only* way to get it right is the $\pi -$ form. **JEE Advanced** exploits the one structural fault line in this whole set: **three of these functions flip sign, three reflect through π** — and the sign of x silently changes which identity applies (the $\tan^{-1}(1/x)$ trap). Master the *pattern* behind the six identities and you stop memorising and start deriving.

2.1 Concept Intuition

In Part 1 you learned that each inverse trig function lives in a cage — its principal value branch. This subtopic is about what happens when you **feed the cage a transformed input**: a reciprocal $\frac{1}{x}$, a negation $-x$, or when you **pair two cages together**. The remarkable fact is that these transformations don’t produce messy new angles — they produce *the same family of angles, shifted or reflected in one of exactly two ways*.

Think about it this way. Recall the single most useful pattern from Part 1: the six functions split by the *shape of their branch*.

- **Sine, tangent, cosecant** have branches **centred on 0** — symmetric about the origin ($[-\frac{\pi}{2}, \frac{\pi}{2}]$ and friends).
- **Cosine, secant, cotangent** have branches **running from 0 up to π** — sitting entirely on the positive side.

That one geometric difference *dictates everything in this subtopic*. A branch symmetric about the origin can absorb a sign flip and stay inside itself: negate the input, negate the answer. So $\sin^{-1}, \tan^{-1}, \operatorname{cosec}^{-1}$ are **odd** — $f^{-1}(-x) = -f^{-1}(x)$. But a branch trapped on $[0, \pi]$ *cannot* go negative; a negative answer would jump out of the cage. So when you negate the input, the answer instead **reflects across the midpoint $\frac{\pi}{2}$** , i.e. $f^{-1}(-x) = \pi - f^{-1}(x)$. That’s why $\cos^{-1}, \sec^{-1}, \cot^{-1}$ are the $\pi -$ **functions**, not odd ones.

Here is the intuition picture, in Indian-classroom terms. Imagine a row of students standing on a number line, and “negate the input” is the instruction “*now stand at your mirror position.*” If your allowed standing zone straddles zero (sine’s branch), your mirror image is still inside the zone — you just swap to the other side, sign flipped. But if your allowed zone is the stretch from the wall at 0 to the wall at π (cosine’s branch), your mirror image would land *behind the wall at 0* — illegal. So instead you reflect about the *centre of the room*, $\frac{\pi}{2}$, landing at $\pi -$ your original spot, still legally inside. **Same instruction, two different legal responses, decided entirely by where your room sits.**

The **complementary** identities ($\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$, etc.) come from the oldest trig fact of all: a sine and a cosine of *complementary angles* are equal ($\sin \theta = \cos(\frac{\pi}{2} - \theta)$).

Inverted, that says the two angles must add to $\frac{\pi}{2}$. The **reciprocal** identities ($\sin^{-1} \frac{1}{x} = \operatorname{cosec}^{-1} x$, etc.) are even more direct: cosec is the reciprocal of sin, so inverting a reciprocal returns the reciprocal function — with a domain caveat we'll pin down.

None of this needs memorisation if you hold one sentence: **odd for the zero-centred trio, $\pi -$ for the $[0, \pi]$ trio**. Everything in Part 2 is that sentence, written out.

Figure 2.4 Two side-by-side number lines. Left: the zero-centred branch $[-\pi/2, \pi/2]$ ($\sin, \tan, \operatorname{cosec}$) contains both θ and $-\theta$ - negate flips the sign, still legal (odd). Right: the $[0, \pi]$ branch ($\cos, \sec, \cot$) contains θ ; the naive mirror $-\theta$ falls outside (crossed out), and the legal reflection $\pi - \theta$ stays in - negate means reflect about $\pi/2$ (the π -form). (see next page)

2.2 Key Identities, Definitions & Validity Conditions

Notation is inherited from Part 1 and locked chapter-wide x = argument, y = principal value, D = domain, B = branch; “ -1 ” is the inverse marker, never a reciprocal exponent. Not re-explained here.

Family 1 — Reciprocal identities (turn a function into its co-named partner):

$$\sin^{-1} \frac{1}{x} = \operatorname{cosec}^{-1} x, \quad \cos^{-1} \frac{1}{x} = \sec^{-1} x \quad (|x| \geq 1)$$

$$\tan^{-1} \frac{1}{x} = \cot^{-1} x \quad (x > 0)$$

Validity is the whole game here. The first two require $|x| \geq 1$ — because $\frac{1}{x}$ must land in $[-1, 1]$ for $\sin^{-1} / \cos^{-1}$ to accept it, which forces $|x| \geq 1$. The $\tan^{-1} \frac{1}{x} = \cot^{-1} x$ identity holds **only for** $x > 0$; for $x < 0$ it acquires a $-\pi$ correction (derived in 2.3— and a favourite JEE trap).

Family 2 — Negative-argument identities. Split exactly along the branch-shape line:

Odd (zero-centred branch)	$\pi -$ (branch on $[0, \pi]$)
$\sin^{-1}(-x) = -\sin^{-1} x, \quad x \in [-1, 1]$	$\cos^{-1}(-x) = \pi - \cos^{-1} x, \quad x \in [-1, 1]$
$\tan^{-1}(-x) = -\tan^{-1} x, \quad x \in \mathbb{R}$	$\cot^{-1}(-x) = \pi - \cot^{-1} x, \quad x \in \mathbb{R}$
$\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x, \quad x \geq 1$	$\sec^{-1}(-x) = \pi - \sec^{-1} x, \quad x \geq 1$

Family 3 — Complementary identities (each pair sums to the right angle):

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2} \quad (x \in [-1, 1]); \quad \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2} \quad (x \in \mathbb{R}); \quad \operatorname{cosec}^{-1} x + \sec^{-1} x = \frac{\pi}{2} \quad (|x| \geq 1)$$

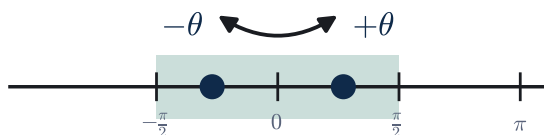
Validity conditions — a “unit check.” Two reflexes carry over from Part 1 and earn marks here:

1. **Domain gate.** Every identity above is annotated with the x -values where it is *legal*. Apply $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ at $x = 2$ and you've written nonsense — neither term exists.

zero-centred branch (sin, tan, cosec)

ODD

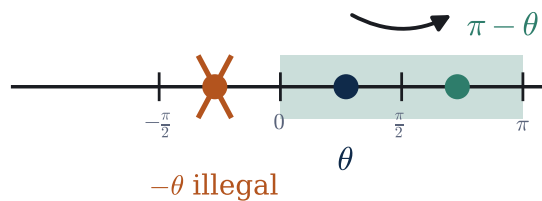
negate $\Rightarrow$ flip sign, still legal



$[0, \pi]$ branch (cos, sec, cot)

$\pi -$ **form**

negate $\Rightarrow$ reflect about $\pi/2$



Odd for the zero-centred trio; $\pi -$ for the $[0, \pi]$ trio.

Figure 2.4 Two side-by-side number lines. Left: the zero-centred branch $[-\pi/2, \pi/2]$ (sin, tan, cosec) contains both θ and $-\theta$ - negate flips the sign, still legal (odd). Right: the $[0, \pi]$ branch (cos, sec, cot) contains θ ; the naive mirror $-\theta$ falls outside (crossed out), and the legal reflection $\pi - \theta$ stays in - negate means reflect about $\pi/2$ (the π -form).

2. **Branch sanity.** After applying a π – identity, the answer must still sit in $[0, \pi]$. If $\cos^{-1}(-x)$ comes out negative, you used the odd form by mistake.

The one structural rule that generates the entire negative-argument table: *odd* $\Leftrightarrow$ *zero-centred branch* (sin, tan, cosec); π – $\Leftrightarrow$ $[0, \pi]$ *branch* (cos, sec, cot). Memorise the rule, not the six rows.

2.3 Derivations from the Branch Structure

Every identity above is *forced* by the branch geometry — none is arbitrary. Deriving them once removes the need to memorise, and (more importantly) shows you *why* the sign-flip-vs- π -reflect split exists.

Derivation 1 — The odd identity $\sin^{-1}(-x) = -\sin^{-1} x$. Let $\sin^{-1} x = y$, so $\sin y = x$ with $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. Then $\sin(-y) = -\sin y = -x$, and crucially $-y$ is **still inside** $[-\frac{\pi}{2}, \frac{\pi}{2}]$ (the branch is symmetric about 0). So $-y$ qualifies as the principal value of $\sin^{-1}(-x)$:

$$\sin^{-1}(-x) = -y = -\sin^{-1} x.$$

The same argument runs verbatim for $\tan^{-1}$ and $\operatorname{cosec}^{-1}$ — *because their branches are also symmetric about 0*. That symmetry is the entire reason they're odd.

Derivation 2 — The π – identity $\cos^{-1}(-x) = \pi - \cos^{-1} x$. Let $\cos^{-1} x = y$, so $\cos y = x$ with $y \in [0, \pi]$. Now try the naive “negate the answer”: $-y$ would have $\cos(-y) = x$, *not* $-x$, and anyway $-y \notin [0, \pi]$ — illegal. Instead use $\cos(\pi - y) = -\cos y = -x$, and check the cage: $y \in [0, \pi] \Rightarrow \pi - y \in [0, \pi]$ — **legal**. So $\pi - y$ is the principal value of $\cos^{-1}(-x)$:

$$\cos^{-1}(-x) = \pi - y = \pi - \cos^{-1} x.$$

Identical logic gives the $\sec^{-1}$ and $\cot^{-1}$ versions. *The $[0, \pi]$ cage is what blocks the sign flip and forces the reflection through $\frac{\pi}{2}$.*

Derivation 3 — The complementary identity $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$. Let $\sin^{-1} x = y$, so $\sin y = x$, $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. The co-relation $\cos(\frac{\pi}{2} - y) = \sin y = x$ holds, and we must check $\frac{\pi}{2} - y$ lies in cosine's cage $[0, \pi]$: since $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, indeed $\frac{\pi}{2} - y \in [0, \pi]$ — legal. Hence $\cos^{-1} x = \frac{\pi}{2} - y$, i.e.

$$\sin^{-1} x + \cos^{-1} x = y + (\frac{\pi}{2} - y) = \frac{\pi}{2}.$$

The branch check is not a formality — it's exactly what makes the complementary partner's angle land in the right cage.

Derivation 4 — Why $\tan^{-1} \frac{1}{x} = \cot^{-1} x$ needs $x > 0$ (the sign trap). Let $\cot^{-1} x = y$, so $\cot y = x$ with $y \in (0, \pi)$, hence $\tan y = \frac{1}{x}$.

- **If $x > 0$:** then $y \in (0, \frac{\pi}{2})$, which is inside $\tan^{-1}$'s cage $(-\frac{\pi}{2}, \frac{\pi}{2})$. So $\tan^{-1} \frac{1}{x} = y = \cot^{-1} x$. ✓
- **If $x < 0$:** then $y \in (\frac{\pi}{2}, \pi)$, which is **outside** $\tan^{-1}$'s cage. The principal value of $\tan^{-1} \frac{1}{x}$ is the in-cage angle $y - \pi$. So $\tan^{-1} \frac{1}{x} = y - \pi = \cot^{-1} x - \pi$.

That single sign-dependent jump is the highest-value trap in this subtopic — see Example 4.

Figure 2.5 $\cot^{-1} x = y$ with $y \in (0, \pi)$. If $x > 0$ then $y \in (0, \pi/2)$, inside the $\tan^{-1}$ cage, and $\tan^{-1}(1/x) = \cot^{-1} x$. If $x < 0$ then $y \in (\pi/2, \pi)$, outside the cage; subtract π and $\tan^{-1}(1/x) = \cot^{-1} x - \pi$. The reciprocal identity is sign-sensitive. (see next page)

2.4 Worked Examples

Example 1 — [CBSE Board Level] Evaluate using identities: (a) $\sin^{-1}(-\frac{1}{2})$, (b) $\cos^{-1}(-\frac{\sqrt{3}}{2})$, (c) verify $\sin^{-1} \frac{1}{2} + \cos^{-1} \frac{1}{2} = \frac{\pi}{2}$.

Solution.

- (a) Odd identity: $\sin^{-1}(-\frac{1}{2}) = -\sin^{-1} \frac{1}{2} = -\frac{\pi}{6}$.
 (b) π -identity (cosine is *not* odd): $\cos^{-1}(-\frac{\sqrt{3}}{2}) = \pi - \cos^{-1} \frac{\sqrt{3}}{2} = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$.
 (Branch check: $\frac{5\pi}{6} \in [0, \pi]$ ✓. Writing $-\frac{\pi}{6}$ here would be the classic error.)
 (c) $\sin^{-1} \frac{1}{2} = \frac{\pi}{6}$, $\cos^{-1} \frac{1}{2} = \frac{\pi}{3}$; sum = $\frac{\pi}{6} + \frac{\pi}{3} = \frac{\pi}{6} + \frac{2\pi}{6} = \frac{3\pi}{6} = \frac{\pi}{2}$ ✓. ■

Example 2 — [CUET / Speed Trap] $\cot^{-1}(-\sqrt{3})$ equals: (a) $-\frac{\pi}{6}$ (b) $\frac{\pi}{6}$ (c) $\frac{5\pi}{6}$ (d) $\frac{2\pi}{3}$

Fast solution. $\cot^{-1}$ is a π -function (its branch is $(0, \pi)$, never negative). So

$$\cot^{-1}(-\sqrt{3}) = \pi - \cot^{-1} \sqrt{3} = \pi - \frac{\pi}{6} = \frac{5\pi}{6}.$$

Answer: (c). *Speed tip:* the instant you see $\cot^{-1}$, $\cos^{-1}$, or $\sec^{-1}$ of a **negative** argument, your hand should write “ π -” before computing anything. Trap (a) is the odd-function reflex; (b) ignores the negative entirely. ■

Example 3 — [JEE Main Level] Find the value of $\tan^{-1}(\sqrt{3}) - \cot^{-1}(-\sqrt{3})$.

Solution. (This is an actual NCERT Ex 2.2 objective question — exactly the kind JEE Main lifts.)

$$\tan^{-1} \sqrt{3} = \frac{\pi}{3}.$$

For the second term, apply the π -identity:

$$\cot^{-1}(-\sqrt{3}) = \pi - \cot^{-1} \sqrt{3} = \pi - \frac{\pi}{6} = \frac{5\pi}{6}.$$

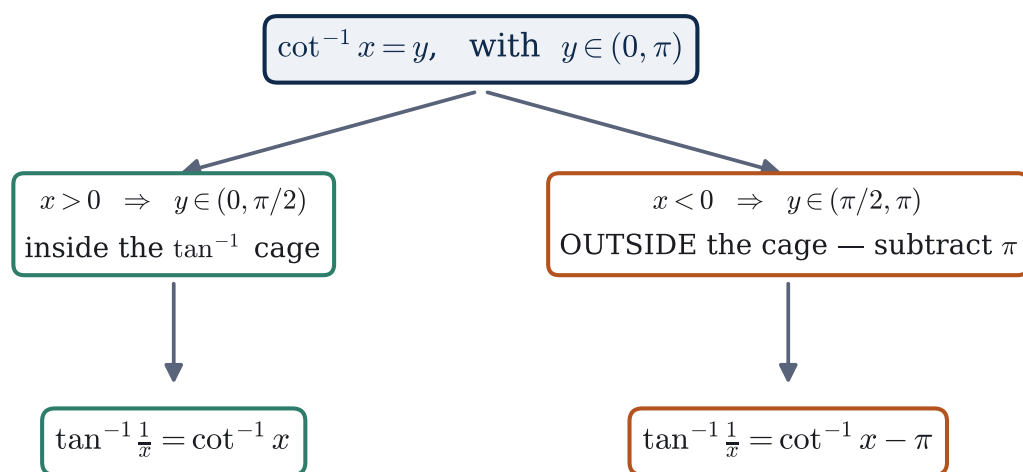
Therefore

$$\tan^{-1} \sqrt{3} - \cot^{-1}(-\sqrt{3}) = \frac{\pi}{3} - \frac{5\pi}{6} = \frac{2\pi}{6} - \frac{5\pi}{6} = -\frac{3\pi}{6} = -\frac{\pi}{2}.$$

(The trap answer $\pi/2$ comes from wrongly treating $\cot^{-1}$ as odd — taking $\cot^{-1}(-\sqrt{3}) = -\pi/6$, which gives $\pi/3 - (-\pi/6) = \pi/2$, not 0. The correct value is $-\pi/2$.) ■

Example 4 — [JEE Advanced Level] Simplify $f(x) = \tan^{-1} x + \tan^{-1} \frac{1}{x}$ for all real $x \neq 0$, stating the answer piecewise.

Solution. Use the reciprocal identity of 2.3, which is sign-sensitive, together with the complementary identity $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$.



The reciprocal identity is sign-sensitive.

Figure 2.5 $\cot^{-1} x = y$ with $y \in (0, \pi)$. If $x > 0$ then $y \in (0, \pi/2)$, inside the $\tan^{-1}$ cage, and $\tan^{-1}(1/x) = \cot^{-1} x$. If $x < 0$ then $y \in (\pi/2, \pi)$, outside the cage; subtract π and $\tan^{-1}(1/x) = \cot^{-1} x - \pi$. The reciprocal identity is sign-sensitive.

Case $x > 0$: here $\tan^{-1} \frac{1}{x} = \cot^{-1} x$, so

$$f(x) = \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}.$$

Case $x < 0$: here $\tan^{-1} \frac{1}{x} = \cot^{-1} x - \pi$, so

$$f(x) = \tan^{-1} x + \cot^{-1} x - \pi = \frac{\pi}{2} - \pi = -\frac{\pi}{2}.$$

$$\tan^{-1} x + \tan^{-1} \frac{1}{x} = \begin{cases} \frac{\pi}{2}, & x > 0 \\ -\frac{\pi}{2}, & x < 0 \end{cases}$$

Sanity check at $x = 1$: $\tan^{-1} 1 + \tan^{-1} 1 = \frac{\pi}{4} + \frac{\pi}{4} = \frac{\pi}{2}$ ✓. At $x = -1$: $-\frac{\pi}{4} - \frac{\pi}{4} = -\frac{\pi}{2}$ ✓. The whole difficulty of this “famous” result is the single $-\pi$ that the reciprocal identity hides for $x < 0$. ■

2.5 Practice Exercises

- [CBSE]** Evaluate (a) $\tan^{-1}(-1)$, (b) $\sec^{-1}(-2)$, (c) $\sin^{-1} x + \cos^{-1} x$ where $x = -\frac{\sqrt{3}}{2}$.
- [CUET]** State the principal value of $\cos^{-1}(-\frac{1}{2})$ and of $\operatorname{cosec}^{-1}(-2)$, in one line each, naming the identity used.
- [JEE Main]** Show that $\sec^{-1}(-x) + \operatorname{cosec}^{-1}(-x) = ?$ for $|x| \geq 1$ (use the negative-argument and complementary identities).
- [JEE Main]** Find $\tan^{-1}(\tan \frac{3\pi}{4}) + \cot^{-1}(\cot \frac{3\pi}{4})$. (Hint: fold into branches first — Part 1 — then read off.)
- [JEE Advanced]** If $\sin^{-1} x + \sin^{-1} y = \frac{\pi}{2}$, express $\cos^{-1} x$ in terms of y , and hence find a relation between x and y free of inverse functions.

Answers:

- (a) $-\frac{\pi}{4}$ (odd). (b) $\pi - \sec^{-1} 2 = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$ (π -form). (c) $\frac{\pi}{2}$ (complementary law holds for all $x \in [-1, 1]$, sign of x irrelevant).
- $\cos^{-1}(-\frac{1}{2}) = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$ (π -identity). $\operatorname{cosec}^{-1}(-2) = -\operatorname{cosec}^{-1} 2 = -\frac{\pi}{6}$ (odd identity).
- $\sec^{-1}(-x) + \operatorname{cosec}^{-1}(-x) = (\pi - \sec^{-1} x) + (-\operatorname{cosec}^{-1} x) = \pi - (\sec^{-1} x + \operatorname{cosec}^{-1} x) = \pi - \frac{\pi}{2} = \frac{\pi}{2}$.
- $\tan \frac{3\pi}{4} = -1 \Rightarrow \tan^{-1}(-1) = -\frac{\pi}{4}$; $\cot \frac{3\pi}{4} = -1 \Rightarrow \cot^{-1}(-1) = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$. Sum $= -\frac{\pi}{4} + \frac{3\pi}{4} = \frac{\pi}{2}$.
- $\sin^{-1} y = \frac{\pi}{2} - \sin^{-1} x = \cos^{-1} x$ (complementary), so $\cos^{-1} x = \sin^{-1} y \Rightarrow x = \cos(\sin^{-1} y) = \sqrt{1 - y^2}$. Relation: $x^2 + y^2 = 1$ (with $x \geq 0$ and $y \geq 0$).

2.6 Crack the MCQ

Q1. Which of the following is **odd**, i.e. satisfies $f^{-1}(-x) = -f^{-1}(x)$?

- (a) $\cos^{-1} x$ (b) $\cot^{-1} x$ (c) $\tan^{-1} x$ (d) $\sec^{-1} x$

Answer: (c). Odd $\Leftrightarrow$ zero-centred branch $\Leftrightarrow \{\sin^{-1}, \tan^{-1}, \operatorname{cosec}^{-1}\}$. The other three ($\cos^{-1}, \cot^{-1}, \sec^{-1}$) have $[0, \pi]$ branches and obey the $\pi-$ rule. *Traps (a),(b),(d):* the “all six are odd” reflex — the single most common error in this entire subtopic.

Q2. $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) + \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) =$

- (a) 0 (b) $\frac{\pi}{2}$ (c) π (d) $-\frac{\pi}{2}$

Answer: (b). The complementary identity $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ holds for **every** $x \in [-1, 1]$ — the negative sign of x does not matter. ($= -\frac{\pi}{3} + \frac{5\pi}{6} = \frac{\pi}{2}$ if you compute the long way.) *Trap (c):* computing each term and mis-adding; *Trap (a):* assuming the negatives cancel.

Q3. $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) + \cot^{-1}\left(\frac{1}{\sqrt{3}}\right) =$

- (a) $\frac{\pi}{3}$ (b) $\frac{2\pi}{3}$ (c) $\frac{\pi}{2}$ (d) $\frac{5\pi}{6}$

Answer: (c). Direct complementary law $\tan^{-1} t + \cot^{-1} t = \frac{\pi}{2}$ with $t = \frac{1}{\sqrt{3}}$. *Traps:* (a)/(b) come from evaluating each term ($\frac{\pi}{6}$ and $\frac{\pi}{3}$) and then *adding wrongly* or stopping at one — both still sum to $\frac{\pi}{2}$, so the careless arithmetic is what produces the wrong options.

Q4. For $x < 0$, $\tan^{-1}\left(\frac{1}{x}\right)$ equals:

- (a) $\cot^{-1} x$ (b) $\cot^{-1} x - \pi$ (c) $\pi - \cot^{-1} x$ (d) $-\cot^{-1} x$

Answer: (b). The reciprocal identity $\tan^{-1} \frac{1}{x} = \cot^{-1} x$ is valid **only for** $x > 0$; for $x < 0$ the $\tan^{-1}$ cage forces a $-\pi$ shift (§2.3, Derivation 4). *Trap (a):* applying the $x > 0$ identity blindly — the headline JEE trap of this subtopic.

2.7 Common Student Pitfalls & Pro-Tips

- “All six identities are odd.”** The cardinal sin. Only $\sin^{-1}, \tan^{-1}, \operatorname{cosec}^{-1}$ flip sign. $\cos^{-1}, \sec^{-1}, \cot^{-1}$ use $\pi-$. Anchor it to the branch: a $[0, \pi]$ cage can never output a negative angle, so it *cannot* be odd.
- Reciprocal identity sign-blindness.** $\tan^{-1} \frac{1}{x} = \cot^{-1} x$ holds for $x > 0$ only. For $x < 0$, subtract π . Always check the sign of x before quoting this one.
- Ignoring domain gates.** $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ requires $x \in [-1, 1]$; the reciprocal identities $\sin^{-1} \frac{1}{x} = \operatorname{cosec}^{-1} x$ require $|x| \geq 1$. Quoting an identity outside its legal domain is an automatic zero, however clean the algebra looks.
- Forgetting the complementary law is sign-agnostic.** $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ for *all* legal x , positive or negative — students sometimes “correct” it for negative x and break a correct answer.

Pro-Tip: Build one reflex: **negative argument + a $\{\cos, \sec, \cot\}$ function $\Rightarrow$ write “ $\pi -$ ” first, evaluate second.** This converts the highest-frequency CUET/JEE-Main value-trap in the subtopic into an automatic correct answer. Pair it with the Part 1 branch-check on every final value and the negative-argument family becomes free marks.

2.8 Cheat Sheet Box

SELF-MAP IDENTITIES — RAPID RECALL

Reciprocal (needs the right domain):

- $\sin^{-1} \frac{1}{x} = \operatorname{cosec}^{-1} x$, $\cos^{-1} \frac{1}{x} = \sec^{-1} x$ ($|x| \geq 1$)
- $\tan^{-1} \frac{1}{x} = \cot^{-1} x$ ($x > 0$); for $x < 0$ it's $\cot^{-1} x - \pi$.

Negative argument — split by branch shape:

- **ODD** (zero-centred): $\sin^{-1}(-x) = -\sin^{-1} x$; $\tan^{-1}(-x) = -\tan^{-1} x$; $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x$.
- $\pi -$ ($[0, \pi]$ branch): $\cos^{-1}(-x) = \pi - \cos^{-1} x$; $\sec^{-1}(-x) = \pi - \sec^{-1} x$; $\cot^{-1}(-x) = \pi - \cot^{-1} x$.

Complementary (each pair $\rightarrow \frac{\pi}{2}$, any legal x):

- $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ ($|x| \leq 1$); $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$ ($x \in \mathbb{R}$); $\operatorname{cosec}^{-1} x + \sec^{-1} x = \frac{\pi}{2}$ ($|x| \geq 1$).

Mnemonics:

- “Sin, tan, cosec flip the sign; cos, sec, cot reflect through π .” — extends Part 1’s “sin & tan hug zero; cos & cot climb to pi.”
- “Reciprocal only goes co-, and $\tan \leftrightarrow \cot$ only when x is positive.”
- “Co-pair adds to ninety.” — the complementary right angle.

Figure 2.6 A three-column reference card: the reciprocal identities with their domain tags; the negative-argument split (odd: sin, tan, cosec; π -form: cos, sec, cot); the complementary pairs summing to $\pi/2$. Branch shape decides everything. (see next page)

Reciprocal	Negative argument	Complementary
$\sin^{-1} \frac{1}{x} = \operatorname{cosec}^{-1} x$ $ x \geq 1$	ODD: $\sin, \tan, \operatorname{cosec}$ $f(-x) = -f(x)$	$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$
$\cos^{-1} \frac{1}{x} = \sec^{-1} x$ $ x \geq 1$		$\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$
$\tan^{-1} \frac{1}{x} = \cot^{-1} x$ $x > 0$ for $x < 0$: subtract π (see the flow figure)	$\pi - :$ $\cos, \sec, \cot$ $f(-x) = \pi - f(x)$	$\operatorname{cosec}^{-1} x + \sec^{-1} x = \frac{\pi}{2}$

branch shape decides everything

Figure 2.6 A three-column reference card: the reciprocal identities with their domain tags; the negative-argument split (odd: $\sin, \tan, \operatorname{cosec}$; π -form: $\cos, \sec, \cot$); the complementary pairs summing to $\pi/2$. Branch shape decides everything.

3 Combination Identities (Sums, Differences & Multiple Angles)

Exam Relevance: This is the most heavily examined block in the chapter. **CBSE Boards** sets “prove that $\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} = \frac{\pi}{4}$ ”-style questions for 3–4 marks almost every year. **CUET** turns the multiple-angle forms into rapid conversions. **JEE Main** lives on the $\tan^{-1} x + \tan^{-1} y$ formula — and punishes the $xy > 1$ case relentlessly. **JEE Advanced** builds telescoping infinite series and conditional-domain problems directly on these identities. If Part 2 was about transforming one function, Part 3 is about combining two — and the single $\pm\pi$ correction term separates the students who memorised from the students who understood.

3.1 Concept Intuition

In Part 2 every identity transformed a single inverse function. Here we combine two of **them** — add $\tan^{-1} x$ and $\tan^{-1} y$, or double an angle $2 \tan^{-1} x$ — and ask what single inverse function results. The engine is nothing new: it is the ordinary **trigonometric addition formulas you already know**, run *backwards*.

Think about it this way. $\tan^{-1} x$ and $\tan^{-1} y$ are just two angles, call them A and B . You already know $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{x + y}{1 - xy}$. So the angle $A + B$ has tangent $\frac{x + y}{1 - xy}$ — which *tempts* you to write $A + B = \tan^{-1} \frac{x + y}{1 - xy}$. And that is right... **as long as $A + B$ hasn't escaped the $\tan^{-1}$ cage** $(-\frac{\pi}{2}, \frac{\pi}{2})$.

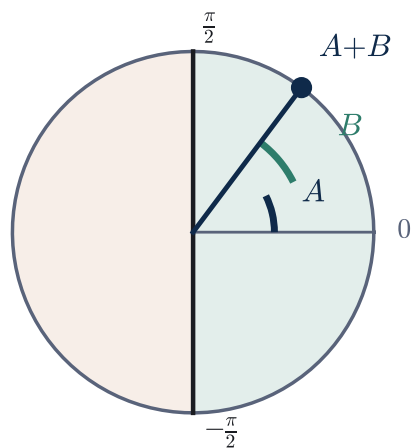
Here is the whole subtlety in one image. Adding two angles is like adding two rotations of a dial. If the combined rotation stays inside the allowed window, the formula reads off the answer directly. But if the two angles are *both large and positive* (which happens exactly when $xy > 1$), their sum overshoots $\frac{\pi}{2}$ and lands outside the cage — so $\tan^{-1} \frac{x + y}{1 - xy}$ gives an angle that is **short by π** , and we must *add π back*. Symmetrically, two large negative angles undershoot $-\frac{\pi}{2}$ and we subtract π . **That $\pm\pi$ is not a special rule to memorise — it is the cage reasserting itself, exactly as in Part 1's range-folding.**

The **multiple-angle** identities are the same story with $y = x$: $2 \tan^{-1} x$ is just $\tan^{-1} x + \tan^{-1} x$, and $3 \sin^{-1} x$ unpacks via $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$. Every one of them is an addition/triple formula read backwards, fenced by a domain condition that keeps the doubled or tripled angle inside its cage.

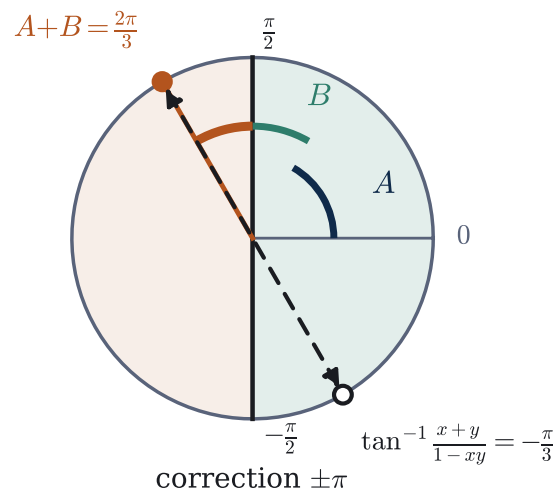
So the mental model for all of Part 3 is: write the combined angle, apply the forward **trig formula, then check whether the result stayed in the branch — if not, correct by $\pm\pi$** . One idea, many identities.

Figure 2.7 A dial view of $A = \tan^{-1} x$ and $B = \tan^{-1} y$ stacked. If $xy < 1$, $A + B$ stays inside the window $(-\pi/2, \pi/2)$ and the formula works directly. If $xy > 1$ (both positive), $A + B$ overshoots past $\pi/2$ and the $\pm\pi$ correction returns the principal value to the cage. The $\pm\pi$ term is the cage reasserting itself. (see next page)

$xy < 1$: $A+B$ stays inside $(-\pi/2, \pi/2)$
— the formula works directly



$xy > 1$ (both positive): $A+B$ overshoots —
answer = formula + π



The $\pm\pi$ term is the cage reasserting itself.

Figure 2.7 A dial view of $A = \tan^{-1} x$ and $B = \tan^{-1} y$ stacked. If $xy < 1$, $A + B$ stays inside the window $(-\pi/2, \pi/2)$ and the formula works directly. If $xy > 1$ (both positive), $A + B$ overshoots past $\pi/2$ and the $\pm\pi$ correction returns the principal value to the cage. The $\pm\pi$ term is the cage reasserting itself.

3.2 Key Identities, Definitions & Validity Conditions

Notation inherited from Part 1 (locked). All conditions are essential — an identity applied outside its domain is wrong even if the algebra is clean.

Sum and difference of arctangents.

$$\tan^{-1} x + \tan^{-1} y = \begin{cases} \tan^{-1} \frac{x+y}{1-xy}, & xy < 1 \\ \pi + \tan^{-1} \frac{x+y}{1-xy}, & xy > 1, x > 0, y > 0 \\ -\pi + \tan^{-1} \frac{x+y}{1-xy}, & xy > 1, x < 0, y < 0 \end{cases}$$

$$\tan^{-1} x - \tan^{-1} y = \tan^{-1} \frac{x-y}{1+xy}, \quad xy > -1.$$

Multiple-angle identities (each is a self-combination, fenced by a domain condition):

Identity	Equivalent forms	Validity
$2 \tan^{-1} x$	$= \tan^{-1} \frac{2x}{1-x^2}$	$-1 < x < 1$
$2 \tan^{-1} x$	$= \sin^{-1} \frac{2x}{1+x^2}$	$-1 \leq x \leq 1$
$2 \tan^{-1} x$	$= \cos^{-1} \frac{1-x^2}{1+x^2}$	$x \geq 0$
$2 \sin^{-1} x$	$= \sin^{-1} (2x\sqrt{1-x^2})$	$-\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}$
$2 \cos^{-1} x$	$= \cos^{-1} (2x^2 - 1)$	$0 \leq x \leq 1$
$3 \sin^{-1} x$	$= \sin^{-1} (3x - 4x^3)$	$-\frac{1}{2} \leq x \leq \frac{1}{2}$
$3 \cos^{-1} x$	$= \cos^{-1} (4x^3 - 3x)$	$\frac{1}{2} \leq x \leq 1$

Sum of arcsines / arccosines (JEE Advanced — condition-heavy, handle with care):

$$\sin^{-1} x + \sin^{-1} y = \sin^{-1} \left(x\sqrt{1-y^2} + y\sqrt{1-x^2} \right), \quad \text{if } x^2 + y^2 \leq 1 \text{ (or } xy < 0).$$

$$\cos^{-1} x + \cos^{-1} y = \cos^{-1} \left(xy - \sqrt{1-x^2}\sqrt{1-y^2} \right), \quad \text{if } x + y \geq 0.$$

When the stated condition fails, a $\pm\pi$ (or 2π) correction applies — the same cage logic as the arctangent sum.

Validity check (a “unit check”): after combining, the result must lie in the *combined* function’s branch. If a “proof” yields $\tan$ (negative) on the right but the left side is clearly an obtuse sum, you dropped a π .

3.3 Derivations from the Addition Formulas

Derivation 1 — The arctangent sum, and where $\pm\pi$ comes from. Let $A = \tan^{-1} x$, $B = \tan^{-1} y$, so $A, B \in (-\frac{\pi}{2}, \frac{\pi}{2})$ and $\tan A = x$, $\tan B = y$. Then

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{x + y}{1 - xy}.$$

Now $A + B$ ranges over $(-\pi, \pi)$ — wider than the $\tan^{-1}$ cage. Three cases:

- $xy < 1$: here $A + B \in (-\frac{\pi}{2}, \frac{\pi}{2})$, inside the cage, so $A + B = \tan^{-1} \frac{x + y}{1 - xy}$.
- $xy > 1$ **with** $x, y > 0$: then A, B are both large enough that $A + B \in (\frac{\pi}{2}, \pi)$ — above the cage. $\tan^{-1}$ returns the in-cage angle, which is π short; add it back:
 $A + B = \pi + \tan^{-1} \frac{x + y}{1 - xy}$.
- $xy > 1$ **with** $x, y < 0$: symmetrically $A + B \in (-\pi, -\frac{\pi}{2})$; subtract π .

The $\pm\pi$ is purely the branch correction — identical in spirit to folding in Part 1.

Derivation 2 — $2 \tan^{-1} x = \sin^{-1} \frac{2x}{1 + x^2}$. Put $\theta = \tan^{-1} x$, so $\tan \theta = x$, $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$. Using the double-angle identity:

$$\sin 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta} = \frac{2x}{1 + x^2}.$$

So $2\theta = \sin^{-1} \frac{2x}{1 + x^2}$ **provided** $2\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, i.e. $\theta \in [-\frac{\pi}{4}, \frac{\pi}{4}]$, i.e. $|x| \leq 1$. Outside that, 2θ leaves the $\sin^{-1}$ cage and a π -fold is needed — which is exactly why the validity tag is $|x| \leq 1$.

Derivation 3 — $3 \sin^{-1} x = \sin^{-1}(3x - 4x^3)$. Put $\theta = \sin^{-1} x$, so $\sin \theta = x$. The triple-angle identity gives

$$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta = 3x - 4x^3.$$

So $3\theta = \sin^{-1}(3x - 4x^3)$ provided $3\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, i.e. $\theta \in [-\frac{\pi}{6}, \frac{\pi}{6}]$, i.e. $x = \sin \theta \in [-\frac{1}{2}, \frac{1}{2}]$. (This is NCERT Ex 2.2 Q1 — the domain restriction is the whole point of the question.)

3.4 Worked Examples

Example 1 — [CBSE Board Level] Prove that $\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} = \frac{\pi}{4}$.

Solution. Here $xy = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6} < 1$, so the direct formula applies:

$$\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{3} = \tan^{-1} \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{6}} = \tan^{-1} \frac{\frac{5}{6}}{\frac{5}{6}} = \tan^{-1} 1 = \frac{\pi}{4}. \quad \blacksquare$$

Example 2 — [CUET / Speed Trap] $2 \tan^{-1} \frac{1}{3}$ equals: (a) $\tan^{-1} \frac{3}{4}$ (b) $\tan^{-1} \frac{2}{3}$ (c) $\sin^{-1} \frac{4}{5}$
 (d) $\tan^{-1} \frac{3}{5}$

Fast solution. $|x| = \frac{1}{3} < 1$, so $2 \tan^{-1} \frac{1}{3} = \tan^{-1} \frac{2/3}{1 - 1/9} = \tan^{-1} \frac{2/3}{8/9} = \tan^{-1} \frac{3}{4}$.

Answer: (a). *Speed tip:* memorise $\frac{2x}{1-x^2}$ and just substitute; option (c) is the $\sin^{-1}$ form of a *different* value. ■

Example 3 — [JEE Main Level] Evaluate $\tan^{-1} 2 + \tan^{-1} 3$.

Solution. Check the product: $xy = 2 \cdot 3 = 6 > 1$, and both are positive — so the $+\pi$ case applies:

$$\tan^{-1} 2 + \tan^{-1} 3 = \pi + \tan^{-1} \frac{2+3}{1-6} = \pi + \tan^{-1} \frac{5}{-5} = \pi + \tan^{-1}(-1) = \pi - \frac{\pi}{4} = \frac{3\pi}{4}.$$

(The trap answer $-\frac{\pi}{4}$ comes from forgetting the $+\pi$. Sanity check: $\tan^{-1} 2$ and $\tan^{-1} 3$ are each between $\frac{\pi}{4}$ and $\frac{\pi}{2}$, so their sum must exceed $\frac{\pi}{2}$ — only $\frac{3\pi}{4}$ qualifies.) ■

Example 4 — [JEE Advanced Level] Find $\sum_{n=1}^{\infty} \tan^{-1} \left(\frac{1}{1+n+n^2} \right)$.

Solution. The trick is to write each term as a *difference* so the sum telescopes. Note

$$\frac{1}{1+n+n^2} = \frac{(n+1) - n}{1+n(n+1)},$$

which is exactly the $\tan^{-1}$ difference pattern $\frac{a-b}{1+ab}$ with $a = n+1$, $b = n$. Hence

$$\tan^{-1} \left(\frac{1}{1+n+n^2} \right) = \tan^{-1}(n+1) - \tan^{-1}(n).$$

The partial sum collapses:

$$S_N = \sum_{n=1}^N [\tan^{-1}(n+1) - \tan^{-1}(n)] = \tan^{-1}(N+1) - \tan^{-1}(1).$$

Let $N \rightarrow \infty$: $\tan^{-1}(N+1) \rightarrow \frac{\pi}{2}$, so

$$S = \frac{\pi}{2} - \frac{\pi}{4} = \boxed{\frac{\pi}{4}}. \quad \blacksquare$$

(This is the canonical JEE telescoping series — the entire difficulty is recognising the difference pattern hidden in $1+n+n^2$.) ■

3.5 Practice Exercises

1. [CBSE] Prove $\tan^{-1} \frac{1}{7} + \tan^{-1} \frac{1}{8} = \tan^{-1} \frac{3}{?}$ — find the value and verify $xy < 1$.
2. [CUET] Evaluate $2 \tan^{-1} \frac{1}{5}$ as a single $\tan^{-1}$.

3. **[JEE Main]** Show $3 \sin^{-1} x = \sin^{-1}(3x - 4x^3)$ holds for $x = \frac{1}{2}$ but state why it fails for $x = \frac{\sqrt{3}}{2}$.
4. **[JEE Main]** Evaluate $\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{5} + \tan^{-1} \frac{1}{8}$.
5. **[JEE Advanced]** Find $\sum_{n=1}^{\infty} \tan^{-1} \left(\frac{2}{n^2} \right)$. (Hint: $\frac{2}{n^2} = \frac{(n+1)-(n-1)}{1+(n+1)(n-1)}$.)

Answers:

1. $\tan^{-1} \frac{1/7 + 1/8}{1 - 1/56} = \tan^{-1} \frac{15/56}{55/56} = \tan^{-1} \frac{15}{55} = \tan^{-1} \frac{3}{11}$; $xy = \frac{1}{56} < 1$ ✓.
2. $\tan^{-1} \frac{2/5}{1 - 1/25} = \tan^{-1} \frac{2/5}{24/25} = \tan^{-1} \frac{5}{12}$.
3. At $x = \frac{1}{2}$: $\theta = \sin^{-1} \frac{1}{2} = \frac{\pi}{6}$, $3\theta = \frac{\pi}{2} \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ ✓. At $x = \frac{\sqrt{3}}{2}$: $\theta = \frac{\pi}{3}$, $3\theta = \pi \notin [-\frac{\pi}{2}, \frac{\pi}{2}]$, so the identity (which requires $x \in [-\frac{1}{2}, \frac{1}{2}]$) fails.
4. $\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{5} = \tan^{-1} \frac{7/10}{9/10} = \tan^{-1} \frac{7}{9}$; then $\tan^{-1} \frac{7}{9} + \tan^{-1} \frac{1}{8} = \tan^{-1} \frac{7/9 + 1/8}{1 - 7/72} = \tan^{-1} \frac{65/72}{65/72} = \tan^{-1} 1 = \frac{\pi}{4}$.
5. $\tan^{-1}(n+1) - \tan^{-1}(n-1)$; telescopes to $(\frac{\pi}{2} + \frac{\pi}{2}) - (\tan^{-1} 1 + \tan^{-1} 0) = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$.

3.6 Crack the MCQ

Q1. $\tan^{-1} 3 + \tan^{-1} \frac{1}{2} =$

- (a) $\pi/4$ (b) $\pi - \tan^{-1} 7$ (c) $-\pi/4$ (d) $\pi/2$

Answer: (b). $xy = 3 \cdot 1/2 = 3/2 > 1$, both positive, so the $+\pi$ case applies: $\tan^{-1} 3 + \tan^{-1}(1/2) = \pi + \tan^{-1}((3 + 1/2)/(1 - 3/2)) = \pi + \tan^{-1}(-7) = \pi - \tan^{-1} 7 \approx 98.1^\circ$. Only option (b) is obtuse. Trap: skipping the $+\pi$ correction gives $\tan^{-1}(-7)$, which matches no option.

Q2. $2 \tan^{-1} x = \cos^{-1} \frac{1-x^2}{1+x^2}$ is valid for:

- (a) all x (b) $x \geq 0$ (c) $|x| \leq 1$ (d) $x \leq 0$

Answer: (b). The $\cos^{-1}$ form requires the doubled angle to land in $[0, \pi]$, forcing $x \geq 0$. Trap (a): assuming identities are unconditional — the defining error of this subtopic.

Q3. $\sin^{-1} \frac{2x}{1+x^2}$ equals $2 \tan^{-1} x$ when:

- (a) $x \geq 1$ (b) $|x| \leq 1$ (c) $x \leq -1$ (d) always

Answer: (b). From Derivation 2, the $\sin^{-1}$ form needs $|x| \leq 1$.

Q4. $\sum_{n=1}^{\infty} \tan^{-1} \frac{1}{2n^2} =$

- (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) $\frac{3\pi}{4}$ (d) $\frac{\pi}{3}$

Answer: (b). $\frac{1}{2n^2} = \frac{(2n+1) - (2n-1)}{1 + (2n+1)(2n-1)}$ gives $\tan^{-1}(2n+1) - \tan^{-1}(2n-1)$; telescoping from $\tan^{-1} 1$ to $\frac{\pi}{2}$ yields $\frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$.

3.7 Common Student Pitfalls & Pro-Tips

- Forgetting the $\pm\pi$ when $xy > 1$.** The headline error. Always compute xy first; if it exceeds 1, decide the sign correction from the signs of x, y before touching the formula.
- Ignoring the multiple-angle domain conditions.** $2 \tan^{-1} x = \tan^{-1} \frac{2x}{1-x^2}$ only for $|x| < 1$; $3 \sin^{-1} x = \sin^{-1}(3x - 4x^3)$ only for $|x| \leq \frac{1}{2}$. Quoting them outside is the most common “proof” failure.
- Sign error in the difference formula.** $\tan^{-1} x - \tan^{-1} y = \tan^{-1} \frac{x-y}{1+xy}$ — note the **plus** in the denominator (opposite of the sum formula). Mixing the two signs is endemic.
- Mis-reading telescoping series.** The whole game is rewriting the general term as $\tan^{-1}(a) - \tan^{-1}(b)$ where consecutive terms cancel. If your “difference” doesn’t telescope, your a, b split is wrong.

Pro-Tip: Make “**compute xy first**” your automatic first move on any $\tan^{-1} x + \tan^{-1} y$ problem. That one number tells you whether you’re in the clean case ($xy < 1$) or the $\pm\pi$ case ($xy > 1$) — and the $\pm\pi$ omission is, by a wide margin, the most expensive mistake in this chapter. For series, train your eye to spot the $\frac{a-b}{1+ab}$ shape hiding inside denominators like $1 + n + n^2$.

3.8 Cheat Sheet Box

COMBINATION IDENTITIES — RAPID RECALL

Arctangent sum (compute xy first!):

- $xy < 1$: $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \frac{x+y}{1+xy}$
- $xy > 1, x, y > 0$: $+\pi$ to the above. $xy > 1, x, y < 0$: $-\pi$.

Arctangent difference: $\tan^{-1} x - \tan^{-1} y = \tan^{-1} \frac{x-y}{1-xy}$ ($xy > -1$) — note the **+** denominator.

Doubles: $2 \tan^{-1} x = \tan^{-1} \frac{2x}{1-x^2} (|x| < 1) = \sin^{-1} \frac{2x}{1+x^2} (|x| \leq 1) = \cos^{-1} \frac{1-x^2}{1+x^2} (x \geq 0).$

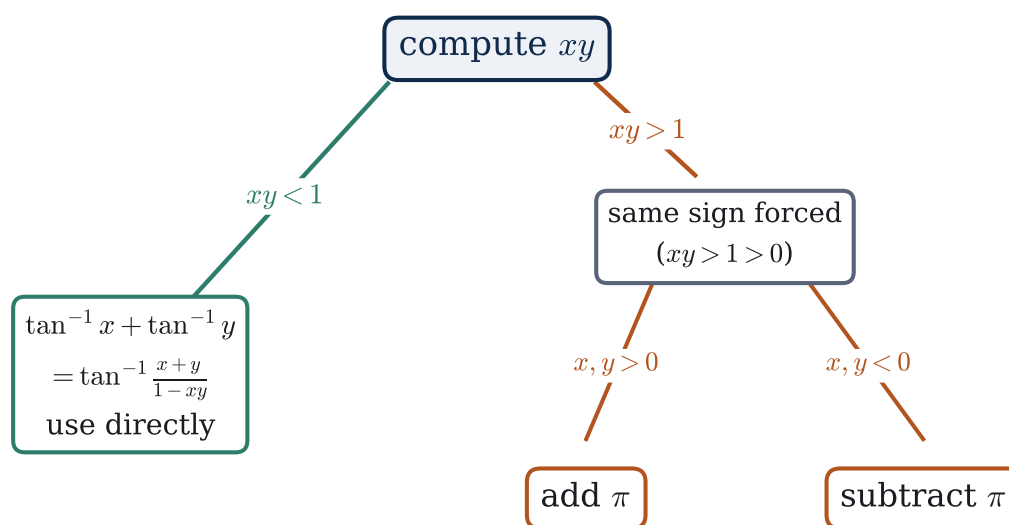
Triples: $3 \sin^{-1} x = \sin^{-1}(3x - 4x^3) (|x| \leq \frac{1}{2}); 3 \cos^{-1} x = \cos^{-1}(4x^3 - 3x) (x \in [\frac{1}{2}, 1]).$

Telescoping series: spot $\frac{a-b}{1+ab} \Rightarrow \tan^{-1} a - \tan^{-1} b$; consecutive terms cancel.

Mnemonics:

- “Sum over one-minus, difference over one-plus.” — the denominators.
- “ xy over one $\Rightarrow$ add a π .” — the correction trigger.
- “Double the angle, halve your confidence — check the domain.”

Figure 2.8 A decision tree for $\tan^{-1} x + \tan^{-1} y$: compute xy . If $xy < 1$, use $\tan^{-1}((x+y)/(1-xy))$ directly. If $xy > 1$ (same sign forced): add π when $x, y > 0$, subtract π when $x, y < 0$. One number decides everything. (see next page)



One number decides everything.

Figure 2.8 A decision tree for $\tan^{-1} x + \tan^{-1} y$: compute xy . If $xy < 1$, use $\tan^{-1}((x+y)/(1-xy))$ directly. If $xy > 1$ (same sign forced): add π when $x, y > 0$, subtract π when $x, y < 0$. One number decides everything.

4 Graphs, Composite Evaluation & Simplification

Exam Relevance: This subtopic turns the identities into *tools*. **CBSE Boards** asks composite evaluations like $\cos(\sin^{-1} \frac{3}{5})$ and substitution simplifications as 2-4 mark questions, and expects you to sketch or read inverse-function graphs. **CUET** sets rapid “find $\tan(\sin^{-1} x)$ ” conversions. **JEE Main & Advanced** lean hard on the substitution technique — recognising that $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ collapses to $\frac{1}{2} \tan^{-1} x$ is a recurring move. Master the **right-triangle method** and the **standard substitutions**, and most of the chapter’s “scary-looking” problems become one-liners.

4.1 Concept Intuition

This subtopic answers two practical questions the earlier ones set up but didn’t fully resolve: “**What does an inverse trig function look like?**” (its graph) and “**How do I evaluate or simplify an expression that mixes a trig function with an inverse one?**” (composites and substitutions).

Graphs — reflection across $y = x$. Every inverse function is the **mirror image of its parent across the line $y = x$** , restricted to the principal branch. Take $y = \sin x$ on its branch $[-\frac{\pi}{2}, \frac{\pi}{2}]$ (a rising S-piece from $(-\frac{\pi}{2}, -1)$ to $(\frac{\pi}{2}, 1)$); flip it across $y = x$ and you get $y = \sin^{-1} x$, rising from $(-1, -\frac{\pi}{2})$ to $(1, \frac{\pi}{2})$. The reflection swaps domain and range — which is *why* the inverse’s domain is the parent’s range and vice versa, the fact you memorised in Part 1 — now made visible. **Think about it this way:** a graph and its mirror across $y = x$ agree on *shape* but trade *axes* — so monotonic-increasing stays increasing, the odd $\sin^{-1} / \tan^{-1}$ keep their origin symmetry, and the $\cos^{-1} / \cot^{-1}$ curves fall from left to right.

Composites — the right-triangle method. To evaluate something like $\cos(\sin^{-1} x)$, name the inner angle: let $\theta = \sin^{-1} x$, so $\sin \theta = x$. Now θ is a genuine angle, so draw a right triangle in which $\sin \theta = \frac{x}{1}$: opposite = x , hypotenuse = 1, and by Pythagoras the adjacent side = $\sqrt{1-x^2}$. Then *read off whatever the outer function asks*: $\cos \theta = \frac{\sqrt{1-x^2}}{1} = \sqrt{1-x^2}$. The whole method is: **turn the inverse function into a triangle, then read the answer off the triangle**. It replaces memorising a dozen composite formulas with one reusable picture.

Simplification — the standard substitution. When an expression hides a $\sqrt{1-x^2}$, the substitution $x = \sin \theta$ (or $\cos \theta$) turns it into a clean trig expression; a $\sqrt{1+x^2}$ calls for $x = \tan \theta$; a $\sqrt{x^2-1}$ calls for $x = \sec \theta$. After substituting, the messy algebraic argument becomes a recognisable angle, the inverse function “cancels” it, and you convert back. This is the single most powerful problem-solving pattern in the chapter, and §4.3 makes it a recipe.

Figure 2.9 Left: $y = \sin x$ on $[-\pi/2, \pi/2]$ (faint), the line $y = x$ (dashed), and $y = \sin^{-1} x$ as its bold mirror image - the reflection swaps domain $[-1, 1]$ and range $[-\pi/2, \pi/2]$. Right: the right-triangle method - $\theta = \sin^{-1} x$ gives opposite x , hypotenuse 1, adjacent $\sqrt{1-x^2}$, so any ratio reads off the triangle. (see next page)

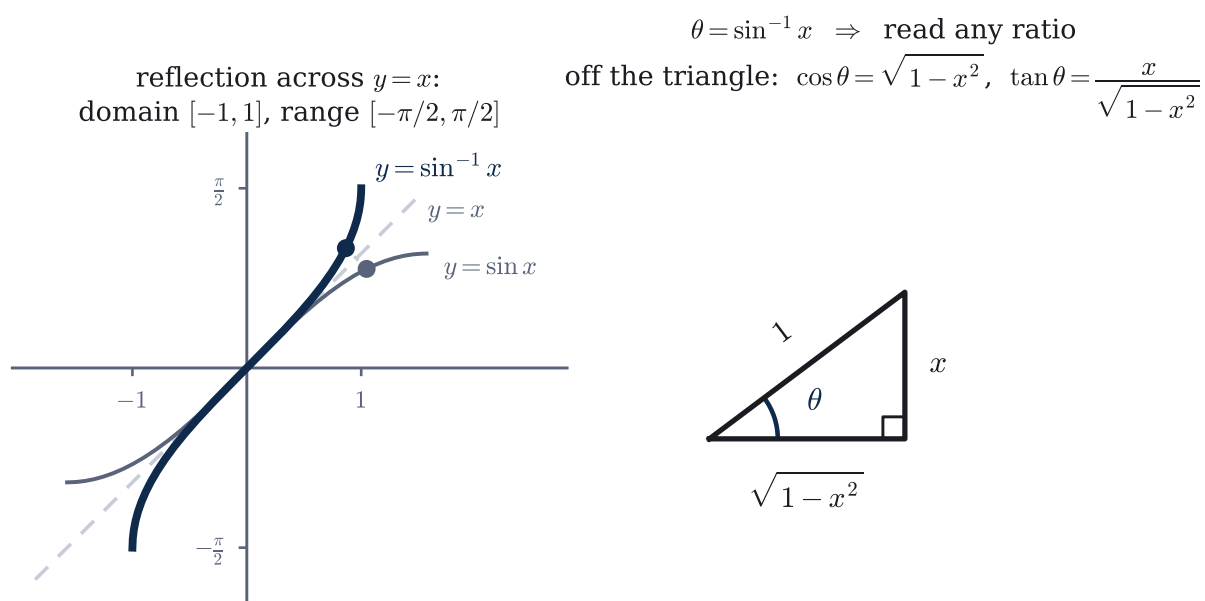


Figure 2.9 Left: $y = \sin x$ on $[-\pi/2, \pi/2]$ (faint), the line $y = x$ (dashed), and $y = \sin^{-1} x$ as its bold mirror image - the reflection swaps domain $[-1, 1]$ and range $[-\pi/2, \pi/2]$. Right: the right-triangle method - $\theta = \sin^{-1} x$ gives opposite x , hypotenuse 1, adjacent $\sqrt{1 - x^2}$, so any ratio reads off the triangle.

4.2 Key Results, Conversions & Graph Properties

Composite-value results (all from the right-triangle method; assume principal values):

Expression	Value	Expression	Value
$\sin(\cos^{-1} x)$	$\sqrt{1 - x^2}$	$\cos(\sin^{-1} x)$	$\sqrt{1 - x^2}$
$\tan(\sin^{-1} x)$	$\frac{x}{\sqrt{1 - x^2}}$	$\tan(\cos^{-1} x)$	$\frac{\sqrt{1 - x^2}}{x}$
$\sin(\tan^{-1} x)$	$\frac{x}{\sqrt{1 + x^2}}$	$\cos(\tan^{-1} x)$	$\frac{1}{\sqrt{1 + x^2}}$
$\sec(\tan^{-1} x)$	$\sqrt{1 + x^2}$	$\tan(\sec^{-1} x)$	$\sqrt{x^2 - 1}$

Conversion formulas (write any inverse function as another; stated for $x \in (0, 1]$, adjust by sign/branch otherwise):

$$\begin{aligned}\sin^{-1} x &= \cos^{-1} \sqrt{1 - x^2} = \tan^{-1} \frac{x}{\sqrt{1 - x^2}} = \operatorname{cosec}^{-1} \frac{1}{x}. \\ \cos^{-1} x &= \sin^{-1} \sqrt{1 - x^2} = \tan^{-1} \frac{\sqrt{1 - x^2}}{x} = \sec^{-1} \frac{1}{x}. \\ \tan^{-1} x &= \sin^{-1} \frac{x}{\sqrt{1 + x^2}} = \cos^{-1} \frac{1}{\sqrt{1 + x^2}} \quad (x \geq 0).\end{aligned}$$

Standard substitutions (memorise the trigger $\rightarrow$ substitution map):

If the expression contains...	Substitute	Because
$\sqrt{1 - x^2}$ or $1 - x^2$	$x = \sin \theta$ or $x = \cos \theta$	$1 - \sin^2 \theta = \cos^2 \theta$
$\sqrt{1 + x^2}$ or $1 + x^2$	$x = \tan \theta$	$1 + \tan^2 \theta = \sec^2 \theta$
$\sqrt{x^2 - 1}$ or $x^2 - 1$	$x = \sec \theta$	$\sec^2 \theta - 1 = \tan^2 \theta$

Graph properties (the six inverse functions at a glance):

$y =$	Domain	Range	Behaviour	Key points / asymptotes
$\sin^{-1} x$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$	increasing, odd	$(0, 0), (1, \frac{\pi}{2}), (-1, -\frac{\pi}{2})$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$	decreasing	$(0, \frac{\pi}{2}), (1, 0), (-1, \pi)$
$\tan^{-1} x$	$\mathbb{R}$	$(-\frac{\pi}{2}, \frac{\pi}{2})$	increasing, odd	asymptotes $y = \pm \frac{\pi}{2}; (0, 0)$
$\cot^{-1} x$	$\mathbb{R}$	$(0, \pi)$	decreasing	asymptotes $y = 0, y = \pi;$ $(0, \frac{\pi}{2})$
$\sec^{-1} x$	$ x \geq 1$	$[0, \pi] - \{\frac{\pi}{2}\}$	increasing (each piece)	asymptote $y = \frac{\pi}{2}$
$\operatorname{cosec}^{-1} x$	$ x \geq 1$	$[-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$	decreasing (each piece)	asymptote $y = 0$

4.3 The Triangle Method & Substitution Recipe

Procedure A — Evaluating composites by triangle.

1. Name the inner angle: $\theta = (\text{inverse function})$, giving one trig ratio of θ directly.
2. Build a right triangle with that ratio; use Pythagoras for the third side.
3. Read off the outer trig function as the appropriate ratio. Mind the sign from the branch.

Demonstration: evaluate $\tan(\cos^{-1} x)$. Let $\theta = \cos^{-1} x \Rightarrow \cos \theta = \frac{x}{1}$: adjacent = x , hyp = 1, opposite = $\sqrt{1 - x^2}$. Then $\tan \theta = \frac{\sqrt{1 - x^2}}{x}$.

Procedure B — Simplification by substitution.

1. Spot the radical signature ($\sqrt{1 - x^2}$, $\sqrt{1 + x^2}$, $\sqrt{x^2 - 1}$) and substitute per the table.
2. Simplify the trig expression using half/double-angle identities until the argument becomes $\sin(\text{angle})$, $\tan(\text{angle})$, etc.
3. Apply the inverse function (it cancels), then substitute θ back in terms of x .

Demonstration (the famous one): simplify $\tan^{-1} \frac{\sqrt{1 + x^2} - 1}{x}$. Put $x = \tan \theta$, so $\sqrt{1 + x^2} = \sec \theta$:

$$\frac{\sqrt{1 + x^2} - 1}{x} = \frac{\sec \theta - 1}{\tan \theta} = \frac{1 - \cos \theta}{\sin \theta} = \frac{2 \sin^2(\theta/2)}{2 \sin(\theta/2) \cos(\theta/2)} = \tan \frac{\theta}{2}.$$

Therefore $\tan^{-1}(\tan \frac{\theta}{2}) = \frac{\theta}{2} = \frac{1}{2} \tan^{-1} x$. A frightening expression collapses to half an arctangent.

4.4 Worked Examples

Example 1 — [CBSE Board Level] Evaluate $\cos(\sin^{-1} \frac{3}{5})$.

Solution. Let $\theta = \sin^{-1} \frac{3}{5} \Rightarrow \sin \theta = \frac{3}{5}$ (opposite 3, hyp 5, adjacent 4). So

$$\cos(\sin^{-1} \frac{3}{5}) = \cos \theta = \frac{4}{5}. \quad \blacksquare$$

Example 2 — [CUET / Speed Trap] $\sin(\tan^{-1} x)$ equals: (a) $\frac{x}{\sqrt{1 - x^2}}$ (b) $\frac{x}{\sqrt{1 + x^2}}$ (c) $\frac{1}{\sqrt{1 + x^2}}$ (d) $\sqrt{1 + x^2}$

Fast solution. $\theta = \tan^{-1} x \Rightarrow \tan \theta = \frac{x}{1}$: opposite x , adjacent 1, hyp $\sqrt{1 + x^2}$. So $\sin \theta = \frac{x}{\sqrt{1 + x^2}}$. **Answer: (b).** *Speed tip:* “ $\tan^{-1} \Rightarrow$ hyp is $\sqrt{1 + x^2}$ ”; option (a) is the $\sin^{-1}$ triangle, the classic mix-up. $\blacksquare$

Example 3 — [JEE Main Level] Simplify $\sin^{-1} \frac{2x}{1+x^2}$ for $|x| \leq 1$.

Solution. Substitute $x = \tan \theta$, so $\theta = \tan^{-1} x$ and (for $|x| \leq 1$, i.e. $|\theta| \leq \frac{\pi}{4}$):

$$\frac{2x}{1+x^2} = \frac{2 \tan \theta}{1 + \tan^2 \theta} = \sin 2\theta.$$

Hence $\sin^{-1}(\sin 2\theta) = 2\theta$ (legal since $2\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$), giving

$$\sin^{-1} \frac{2x}{1+x^2} = 2 \tan^{-1} x.$$

(This is the bridge to Part 3's multiple-angle identity — same result, reached by substitution.) ■

Example 4 — [JEE Advanced Level] Simplify $\tan^{-1} \frac{\cos x}{1 + \sin x}$ for $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$.

Solution. Convert to half-angles. Using $\cos x = \cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}$ and $1 + \sin x = (\cos \frac{x}{2} + \sin \frac{x}{2})^2$:

$$\frac{\cos x}{1 + \sin x} = \frac{(\cos \frac{x}{2} - \sin \frac{x}{2})(\cos \frac{x}{2} + \sin \frac{x}{2})}{(\cos \frac{x}{2} + \sin \frac{x}{2})^2} = \frac{\cos \frac{x}{2} - \sin \frac{x}{2}}{\cos \frac{x}{2} + \sin \frac{x}{2}}.$$

Divide top and bottom by $\cos \frac{x}{2}$:

$$= \frac{1 - \tan \frac{x}{2}}{1 + \tan \frac{x}{2}} = \tan\left(\frac{\pi}{4} - \frac{x}{2}\right).$$

Therefore

$$\tan^{-1} \frac{\cos x}{1 + \sin x} = \frac{\pi}{4} - \frac{x}{2}. \quad \blacksquare$$

(The whole problem is the recognition that $1 + \sin x$ is a perfect square of half-angles.) ■

4.5 Practice Exercises

1. [CBSE] Evaluate $\tan(\cos^{-1} \frac{5}{13})$.
2. [CUET] Find $\sec(\tan^{-1} \frac{3}{4})$.
3. [JEE Main] Show $\cos^{-1} x = \tan^{-1} \frac{\sqrt{1-x^2}}{x}$ for $x \in (0, 1]$ using a triangle.
4. [JEE Main] Simplify $\cos^{-1} \frac{1-x^2}{1+x^2}$ for $x \geq 0$.
5. [JEE Advanced] Simplify $\tan^{-1} \sqrt{\frac{1-\cos x}{1+\cos x}}$ for $x \in (0, \pi)$.

Answers:

1. $\theta = \cos^{-1} \frac{5}{13}$: adjacent 5, hyp 13, opposite 12; $\tan \theta = \frac{12}{5}$.

2. $\theta = \tan^{-1} \frac{3}{4}$: opp 3, adj 4, hyp 5; $\sec \theta = \frac{5}{4}$.

3. $\theta = \cos^{-1} x \Rightarrow \cos \theta = \frac{x}{1}$, opposite $\sqrt{1-x^2}$, so $\tan \theta = \frac{\sqrt{1-x^2}}{x}$ and $\theta = \tan^{-1} \frac{\sqrt{1-x^2}}{x}$.

4. Put $x = \tan \theta$: $\frac{1-x^2}{1+x^2} = \cos 2\theta$, so $\cos^{-1}(\cos 2\theta) = 2\theta = 2 \tan^{-1} x$ (valid $x \geq 0$).

5. $\sqrt{\frac{1-\cos x}{1+\cos x}} = \sqrt{\frac{2\sin^2(x/2)}{2\cos^2(x/2)}} = \tan \frac{x}{2}$ (positive on $(0, \pi)$), so the value is $\tan^{-1}(\tan \frac{x}{2}) = \frac{x}{2}$.

4.6 Crack the MCQ

Q1. $\cos(\sin^{-1} x) =$

(a) x (b) $\sqrt{1-x^2}$ (c) $\frac{1}{\sqrt{1-x^2}}$ (d) $\frac{x}{\sqrt{1-x^2}}$

Answer: (b). Triangle: $\sin \theta = x \Rightarrow \cos \theta = \sqrt{1-x^2}$. *Trap (d):* that's $\tan(\sin^{-1} x)$, reading the wrong ratio.

Q2. The graph of $y = \cos^{-1} x$ is:

(a) increasing on $[-1, 1]$ (b) decreasing on $[-1, 1]$ (c) odd (d) unbounded

Answer: (b). $\cos^{-1}$ falls from $(-1, \pi)$ to $(1, 0)$. It is *not* odd (it's the π - reflection type from Part 2).

Q3. $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x} =$

(a) $\tan^{-1} x$ (b) $\frac{1}{2} \tan^{-1} x$ (c) $2 \tan^{-1} x$ (d) $\frac{\pi}{4} - \tan^{-1} x$

Answer: (b). Substitute $x = \tan \theta$; the argument becomes $\tan \frac{\theta}{2}$ (§4.3).

Q4. $\sin^{-1} \frac{2x}{1+x^2} = 2 \tan^{-1} x$ requires:

(a) $x \geq 1$ (b) $|x| \leq 1$ (c) all x (d) $x \leq 0$

Answer: (b). The doubled angle must stay in the $\sin^{-1}$ branch $\Rightarrow |x| \leq 1$ (Example 3; consistent with Part 3).

4.7 Common Student Pitfalls & Pro-Tips

- Reading the wrong ratio off the triangle.** $\cos(\sin^{-1} x) = \sqrt{1-x^2}$ but $\tan(\sin^{-1} x) = \frac{x}{\sqrt{1-x^2}}$ — set up the triangle once and label all three sides before reading.

2. **Dropping the sign/branch.** The triangle gives magnitudes; for negative x or angles in $[\frac{\pi}{2}, \pi]$ (cosine's branch), restore the correct sign from the range.
3. **Wrong substitution.** Matching the radical to its substitution is everything: $\sqrt{1+x^2} \rightarrow x = \tan \theta$, *not* $\sin \theta$. The wrong choice leaves an un-simplifiable mess.
4. **Forgetting the validity window after simplifying.** $\sin^{-1} \frac{2x}{1+x^2} = 2 \tan^{-1} x$ holds only for $|x| \leq 1$; outside it the answer is $\pi - 2 \tan^{-1} x$ (positive x) — the same branch-folding caution from every prior subtopic.

Pro-Tip: For *any* composite $f(g^{-1}(x))$, draw the triangle immediately — don't hunt for a memorised formula. And for *any* expression with a radical, name its signature and substitute on reflex: $\sqrt{1-x^2} \rightarrow \sin$, $\sqrt{1+x^2} \rightarrow \tan$, $\sqrt{x^2-1} \rightarrow \sec$. These two habits convert the majority of this subtopic's problems into mechanical exercises.

4.8 Cheat Sheet Box

GRAPHS, COMPOSITES & SIMPLIFICATION — RAPID RECALL

Graphs: each f^{-1} is the parent trig graph mirrored across $y = x$ (restricted to the branch). Increasing: $\sin^{-1}, \tan^{-1}, \sec^{-1}$. Decreasing: $\cos^{-1}, \cot^{-1}, \operatorname{cosec}^{-1}$. Odd (origin-symmetric): $\sin^{-1}, \tan^{-1}$. Asymptotes: $\tan^{-1} \rightarrow \pm \frac{\pi}{2}$, $\cot^{-1} \rightarrow 0, \pi$.

Triangle method: set $\theta = \text{inverse function}$, build the right triangle, read the outer ratio.

$$\bullet \cos(\sin^{-1} x) = \sqrt{1-x^2}, \quad \sin(\tan^{-1} x) = \frac{x}{\sqrt{1+x^2}}, \quad \sec(\tan^{-1} x) = \sqrt{1+x^2},$$

$$\tan(\sec^{-1} x) = \sqrt{x^2-1}.$$

Substitution triggers: $\sqrt{1-x^2} \Rightarrow x = \sin \theta$; $\sqrt{1+x^2} \Rightarrow x = \tan \theta$; $\sqrt{x^2-1} \Rightarrow x = \sec \theta$.

Signature results: $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x} = \frac{1}{2} \tan^{-1} x$; $\tan^{-1} \frac{\cos x}{1+\sin x} = \frac{\pi}{4} - \frac{x}{2}$;

$$\sin^{-1} \frac{2x}{1+x^2} = 2 \tan^{-1} x \quad (|x| \leq 1).$$

Mnemonics:

- "Inverse = mirror across $y = x$."
- "Draw the triangle, read the ratio."
- "Root one-minus $\Rightarrow$ sine; root one-plus $\Rightarrow$ tan; root minus-one $\Rightarrow$ sec."

Figure 2.10 The six inverse-function graphs at a glance: $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$, $\cot^{-1}$, $\sec^{-1}$, $\operatorname{cosec}^{-1}$, each with domain and range marked, asymptotes dashed where relevant, and key points labelled. (see next page)

The six inverse-function graphs at a glance.

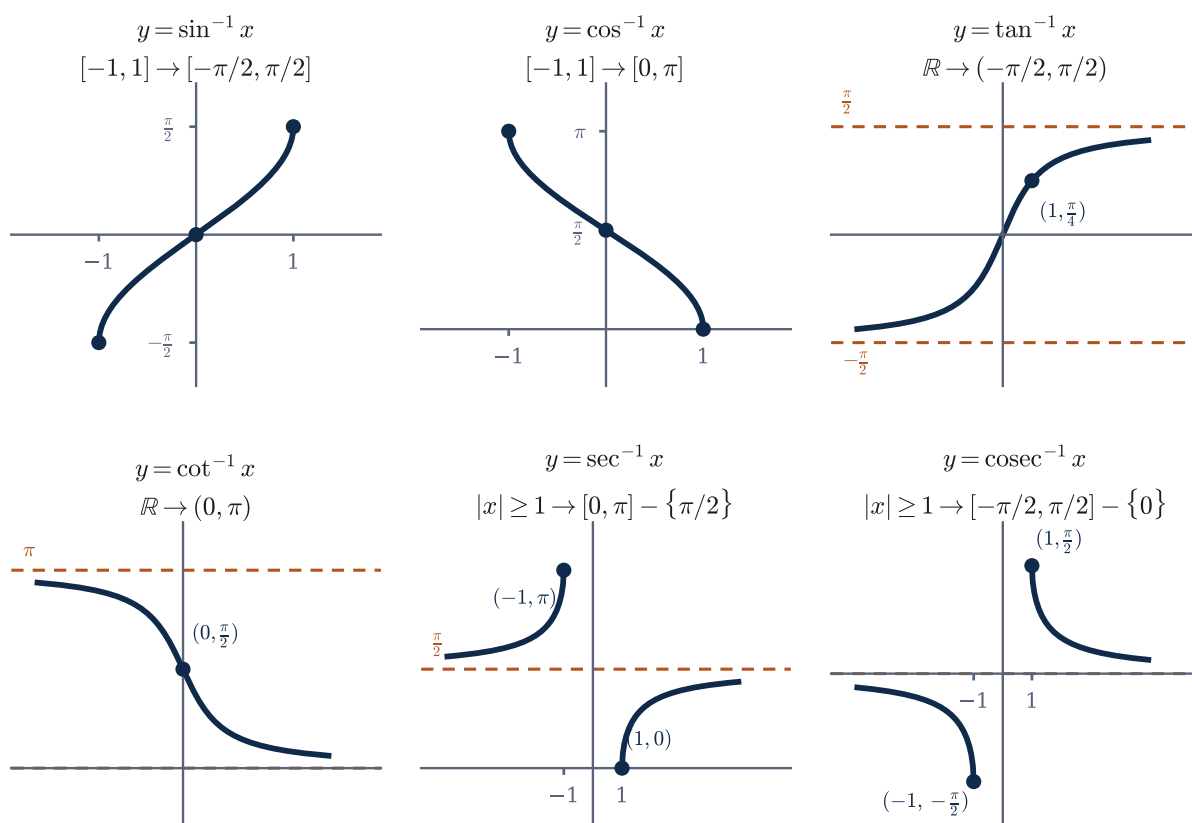


Figure 2.10 The six inverse-function graphs at a glance: $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$, $\cot^{-1}$, $\sec^{-1}$, $\operatorname{cosec}^{-1}$, each with domain and range marked, asymptotes dashed where relevant, and key points labelled.

Chapter 2: Inverse Trigonometric Functions — Round 2 Addendum

The chapter gave you the six cages, the self-map identities, the combination formulas with their $\pm\pi$ corrections, and the triangle-and-substitution toolkit. Five examinable skills still stand outside its door. First, the chapter *evaluates* expressions but never *solves* equations built from them — yet “solve for x ” is a near-guaranteed CBSE question and a JEE Main staple, and it carries an extraneous-root trap that costs entire questions. Second, it proves identities left-to-right; exams just as often hand you a condition on a *sum* of inverse functions and demand the hidden algebraic relation, or the converse of one. Third, its multiple-angle table stops at each identity’s validity window; JEE wants the same expressions simplified for *every* real x , piecewise. Fourth, it folds individual angles but never studies $\sin^{-1}(\sin x)$ as a function of x — the triangular wave on which JEE Advanced counts solutions. Fifth, it never hunts domains and ranges of functions assembled around inverse trigonometric parts, a CBSE 2-marker and a CUET favourite. Addenda A to E supply exactly these, each built only from machinery the chapter already constructed.

One convention warning before you start. This book follows NCERT and locks $\cot^{-1}$ to the branch $(0, \pi)$. Some JEE preparation sources use the rival branch $(-\frac{\pi}{2}, \frac{\pi}{2}]$, under which $\cot^{-1}$ of a negative argument comes out negative. Any expression containing $\cot^{-1}$ of a negative number changes value between the two conventions; until your paper specifies otherwise, stay with NCERT.

Addendum A — Solving Inverse Trigonometric Equations

Method

An inverse trigonometric equation mixes angles and algebra. The universal route back to algebra has five steps.

1. **Narrow the domain first.** Every inverse term imposes a constraint: arguments of $\sin^{-1}$ and $\cos^{-1}$ must lie in $[-1, 1]$; arguments of $\sec^{-1}$ and $\operatorname{cosec}^{-1}$ must satisfy $|t| \geq 1$; $\tan^{-1}$ and $\cot^{-1}$ accept anything. Intersect all the constraints before touching the equation — half your candidates usually die here, for free.
2. **Isolate one inverse term** (or recognise a sum whose correction case you can decide early).
3. **Apply the matching forward function to both sides.** Taking the tangent, sine or cosine of both sides converts the equation into ordinary algebra. Remember what this step costs you: a forward trig function is *not* one-to-one, so the algebraic equation can admit candidates the original rejects.

4. **Solve the algebraic equation completely.**
5. **Verify every candidate in the original equation** — both its domain constraints and its branch behaviour. An unverified root in this topic is not a risk; it is usually a planted trap.

Why verification is not optional. When you take the cosine of both sides of an equation whose two sides are angles, you replace “the angles are equal” by “the angles have equal cosine” — a weaker statement, because angles differing by a sign or a reflection share a cosine. Step 5 exists to kill the impostors this creates.

Example A.1 — [CBSE Board / JEE Main Level]

Solve

$$\tan^{-1} \frac{x-1}{x-2} + \tan^{-1} \frac{x+1}{x+2} = \frac{\pi}{4}.$$

Domains: $x \neq 2$ and $x \neq -2$. Name the two arguments u and v , and compute their sum and product, since the addition formula consumes exactly those:

$$u + v = \frac{(x-1)(x+2) + (x+1)(x-2)}{x^2 - 4} = \frac{2x^2 - 4}{x^2 - 4} = \frac{2(x^2 - 2)}{x^2 - 4},$$

$$uv = \frac{(x-1)(x+1)}{(x-2)(x+2)} = \frac{x^2 - 1}{x^2 - 4}.$$

Because the target sum is $\frac{\pi}{4}$, which sits inside the cage $(-\frac{\pi}{2}, \frac{\pi}{2})$, any genuine solution must live in the clean case $uv < 1$ — the $\pm\pi$ corrections produce sums past $\pm\frac{\pi}{2}$ and cannot reach $\frac{\pi}{4}$. Apply the tangent:

$$\frac{u+v}{1-uv} = \frac{2(x^2-2)}{x^2-4} \cdot \frac{x^2-4}{-3} = \frac{2(2-x^2)}{3}.$$

Setting this equal to $\tan \frac{\pi}{4} = 1$:

$$2(2-x^2) = 3 \implies x^2 = \frac{1}{2} \implies x = \pm \frac{1}{\sqrt{2}}.$$

Verify. For $x^2 = \frac{1}{2}$: $uv = \frac{-\frac{1}{2}}{-\frac{7}{2}} = \frac{1}{7} < 1$ — the clean case holds, so both roots are genuine.

$$x = \pm \frac{1}{\sqrt{2}}$$

Numerical spot-check at $x \approx 0.7071$: the arguments are approximately 0.2265 and 0.6306, whose arctangents sum to about $0.7864 \approx \frac{\pi}{4}$.

Example A.2 — [JEE Main Level]

Solve

$$\sin^{-1} x + \sin^{-1}(1-x) = \cos^{-1} x.$$

Step 1, domains. $x \in [-1, 1]$ and $1-x \in [-1, 1]$, i.e. $x \in [0, 2]$. Intersection: $x \in [0, 1]$.

Step 3. Take the cosine of both sides. Both $\sin^{-1} x$ and $\sin^{-1}(1-x)$ lie in $[-\frac{\pi}{2}, \frac{\pi}{2}]$, so their cosines are the non-negative roots $\sqrt{1-x^2}$ and $\sqrt{1-(1-x)^2} = \sqrt{2x-x^2}$. The right side gives $\cos(\cos^{-1} x) = x$:

$$\sqrt{(1-x^2)(2x-x^2)} - x(1-x) = x.$$

Step 4. Rearranged:

$$\sqrt{(1-x)(1+x)x(2-x)} = 2x-x^2 = x(2-x).$$

On the domain $[0, 1]$ both sides are non-negative, so squaring is safe. Try $x = 0$ first: it survives the division below, so test it directly later. For $x \in (0, 1]$ divide both sides by the positive quantity $x(2-x)$:

$$(1-x)(1+x) = x(2-x) \implies 1-x^2 = 2x-x^2 \implies x = \frac{1}{2}.$$

Step 5, verify. At $x = 0$: the left side is $0 + \frac{\pi}{2} = \frac{\pi}{2}$ and $\cos^{-1} 0 = \frac{\pi}{2}$ — valid. At $x = \frac{1}{2}$: $\sin^{-1} \frac{1}{2} + \sin^{-1} \frac{1}{2} = \frac{\pi}{3}$ and $\cos^{-1} \frac{1}{2} = \frac{\pi}{3}$ — valid.

$x = 0 \text{ or } x = \frac{1}{2}$

The root $x = 0$ is exactly the kind a careless solver loses: it dies in the division step, so only a direct substitution rescues it.

Example A.3 — [JEE Advanced Level]

Solve the system

$$\sin^{-1} x + \sin^{-1} y = \frac{2\pi}{3}, \quad \cos^{-1} x - \cos^{-1} y = \frac{\pi}{3}.$$

The move that unlocks the system is converting the second equation to arcsines via the complementary law $\cos^{-1} t = \frac{\pi}{2} - \sin^{-1} t$, valid for every $t \in [-1, 1]$:

$$\left(\frac{\pi}{2} - \sin^{-1} x\right) - \left(\frac{\pi}{2} - \sin^{-1} y\right) = \frac{\pi}{3} \implies \sin^{-1} y - \sin^{-1} x = \frac{\pi}{3}.$$

Add this to the first equation:

$$2 \sin^{-1} y = \pi \implies \sin^{-1} y = \frac{\pi}{2} \implies y = 1.$$

Back-substitute: $\sin^{-1} x = \frac{2\pi}{3} - \frac{\pi}{2} = \frac{\pi}{6}$, so $x = \frac{1}{2}$.

Verify in the untouched second equation: $\cos^{-1} \frac{1}{2} - \cos^{-1} 1 = \frac{\pi}{3} - 0 = \frac{\pi}{3}$.

$$(x, y) = \left(\frac{1}{2}, 1\right)$$

Practice — Addendum A

1. Solve $\tan^{-1} x + \tan^{-1} 2x = \tan^{-1} 3$.
2. Solve $2 \tan^{-1}(\cos x) = \tan^{-1}(2 \operatorname{cosec} x)$ for $x \in (0, 2\pi)$.
3. Solve $\tan^{-1} \frac{1-x}{1+x} = \frac{1}{2} \tan^{-1} x$ for $x > -1$.
4. Solve $3 \tan^{-1} x + \cot^{-1} x = \pi$.

Answers:

1. Taking the tangent gives $\frac{3x}{1-2x^2} = 3$, i.e. $2x^2 + x - 1 = 0$, so $x = \frac{1}{2}$ or $x = -1$. Check: at $x = \frac{1}{2}$ we have $xy = \frac{1}{2} < 1$ and the sum reads off as $\tan^{-1} 3$ — valid. At $x = -1$ we have $xy = 2 > 1$ with both arguments negative, so the true sum is $-\pi + \tan^{-1} \frac{-3}{-1} \approx -1.89$ rad, not $\tan^{-1} 3$ — **extraneous**, killed by Step 5. Answer $x = \frac{1}{2}$ only.
2. Take the tangent of both sides: $\frac{2 \cos x}{1 - \cos^2 x} = \frac{2}{\sin x}$, which simplifies to $\cos x = \sin x$, i.e. $x \equiv \frac{\pi}{4} \pmod{\pi}$. Both candidates in $(0, 2\pi)$ survive verification: at $x = \frac{\pi}{4}$ both sides equal $\tan^{-1}(2\sqrt{2})$; at $x = \frac{5\pi}{4}$ both cosine and sine turn negative together, so both sides equal $-\tan^{-1}(2\sqrt{2})$. (By contrast $x = \frac{3\pi}{4}$, where the signs disagree, fails.) Answer $x = \frac{\pi}{4}, \frac{5\pi}{4}$.
3. Put $x = \tan \theta$ with $\theta = \tan^{-1} x \in (-\frac{\pi}{2}, \frac{\pi}{2})$. Then $\frac{1-x}{1+x} = \tan\left(\frac{\pi}{4} - \theta\right)$, and for $x > -1$ the angle $\frac{\pi}{4} - \theta$ sits inside the cage, so the left side is $\frac{\pi}{4} - \theta$. The equation becomes $\frac{\pi}{4} - \theta = \frac{\theta}{2}$, giving $\theta = \frac{\pi}{6}$ and $x = \frac{1}{\sqrt{3}}$. Check: $\tan^{-1}(2 - \sqrt{3}) = \frac{\pi}{12} = \frac{1}{2} \cdot \frac{\pi}{6}$.
4. Use $\cot^{-1} x = \frac{\pi}{2} - \tan^{-1} x$: then $2 \tan^{-1} x = \frac{\pi}{2}$, so $x = 1$. Check: $3 \cdot \frac{\pi}{4} + \frac{\pi}{4} = \pi$.

Addendum B — Conditional Identities: Reading the Relation Backwards

Method

A standing exam pattern hands you a condition like

$$\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi$$

and asks for the purely algebraic relation between x, y, z — or the reverse. The engine is a pair of identities that decompose the sine and cosine of a three-angle sum into products. Let $A = \tan^{-1} x$, $B = \tan^{-1} y$, $C = \tan^{-1} z$, and expand $\sin(A + B + C)$ and $\cos(A + B + C)$ with the addition formulas, then divide through by $\cos A \cos B \cos C$:

$$\sin(A + B + C) = \cos A \cos B \cos C (x + y + z - xyz),$$

$$\cos(A + B + C) = \cos A \cos B \cos C (1 - xy - yz - zx).$$

Every term of the expansion divides cleanly: $\sin A \cos B \cos C$ contributes x , and so on — check each of the eight terms yourself; the minus signs come from $\cos(A + B) = \cos A \cos B - \sin A \sin B$.

Two consequences fall out instantly, because $\cos A \cos B \cos C > 0$ whenever $x, y, z > 0$:

- **Result 1.** For $x, y, z > 0$:

$$\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi \iff x + y + z = xyz.$$

(sin of the sum vanishes but cos of the sum is $-1 \neq 0$, so the first identity fires alone; conversely $\sin = 0$ with the sum trapped in $(0, \frac{3\pi}{2})$ forces exactly π , since sin and cos cannot vanish together.)

- **Result 2.** For $x, y, z > 0$:

$$\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \frac{\pi}{2} \iff xy + yz + zx = 1.$$

The same decomposition runs any conditional problem: write the angle sum, take the sine or cosine of both sides, expand, and read off the algebraic statement — minding which of sin and cos is allowed to vanish.

Example B.1 — [CBSE Board Level]

If x, y, z are positive and $x + y + z = xyz$, prove that $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi$.

Call the sum S and the angles A, B, C ; each lies in $(0, \frac{\pi}{2})$, so $S \in (0, \frac{3\pi}{2})$ and every cosine is strictly positive. From the first identity,

$$\sin S = \cos A \cos B \cos C (x + y + z - xyz) = 0.$$

Since $\sin S = 0$, the factor $\cos A \cos B \cos C$ is nonzero, and $\sin S$ and $\cos S$ cannot vanish simultaneously, $\cos S \neq 0$ either. So S is neither $\frac{\pi}{2}$ nor an endpoint of its range; among the zeros of sin inside $(0, \frac{3\pi}{2})$ the only survivor is

$$S = \pi. \quad \blacksquare$$

Example B.2 — [JEE Main Level]

Show that $\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3 = \pi$, **and connect it to the chapter.**

Here $x + y + z = 1 + 2 + 3 = 6$ and $xyz = 6$, so Result 1 applies directly and the sum is π . Cross-check against the chapter's own Example 3 of Part 3, which computed $\tan^{-1} 2 + \tan^{-1} 3 = \frac{3\pi}{4}$ via the $+\pi$ correction; adding $\tan^{-1} 1 = \frac{\pi}{4}$ gives π . One classic, two independent routes.

Example B.3 — [JEE Advanced Level]

If $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = \pi$, **prove that**

$$x^2 + y^2 + z^2 + 2xyz = 1.$$

Let $A = \cos^{-1} x$, $B = \cos^{-1} y$, $C = \cos^{-1} z$; each lies in $[0, \pi]$, so **all sines and cosines involved are non-negative** — this is what makes the derivation sign-safe, unlike the arcsine version. From $A + B + C = \pi$ we get $A + B = \pi - C$, hence

$$\cos(A + B) = -\cos C = -z, \quad \sin(A + B) = \sin C = \sqrt{1 - z^2}.$$

Expand the left sides with $x = \cos A$, $\sqrt{1 - x^2} = \sin A$, and similarly for y :

$$xy - \sqrt{1 - x^2}\sqrt{1 - y^2} = -z \implies \sqrt{1 - x^2}\sqrt{1 - y^2} = xy + z,$$

$$x\sqrt{1 - y^2} + y\sqrt{1 - x^2} = \sqrt{1 - z^2}.$$

Square the second equation only, substituting $\sqrt{1 - x^2}\sqrt{1 - y^2} = xy + z$ in the cross term:

$$x^2(1 - y^2) + y^2(1 - x^2) + 2xy(xy + z) = 1 - z^2.$$

The middle terms collapse: $-x^2y^2 - x^2y^2 + 2x^2y^2 = 0$, leaving

$$x^2 + y^2 + 2xyz = 1 - z^2 \implies x^2 + y^2 + z^2 + 2xyz = 1. \quad \blacksquare$$

Note the direction of travel: the angle condition implies the algebraic relation. The converse needs extra hypotheses and is *not* claimed — quoting it backwards is a known Advanced-level trap.

Practice — Addendum B

1. If $\tan^{-1} x + \tan^{-1} y = \frac{\pi}{2}$ with $x, y > 0$, show that $xy = 1$.
2. If $\sin^{-1} x + \sin^{-1} y = \frac{\pi}{2}$, show that $x^2 + y^2 = 1$.
3. If $\cos^{-1} x + \cos^{-1} y = \frac{\pi}{2}$, show that $x^2 + y^2 = 1$.
4. If $x, y, z > 0$ and $xy + yz + zx = 1$, prove $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \frac{\pi}{2}$.
5. If $\sin^{-1} x + \cos^{-1} y = \pi$, find the relation between x and y , with all restrictions.

Answers:

1. $\cos$ of the left side is $\cos(\frac{\pi}{2}) = 0$; expanding $\cos(A + B)$ with $\sin$ terms converted via $\sqrt{1 - x^2}$, $\sqrt{1 - y^2}$ and dividing by the positive product $\cos A \cos B$ kills the radicals and leaves $1 - xy = 0$.
 2. Convert: $\sin^{-1} y = \frac{\pi}{2} - \sin^{-1} x = \cos^{-1} x$, so $y = \cos(\cos^{-1} x)$ composed with sine: $y = \sin(\cos^{-1} x) = \sqrt{1 - x^2}$. Squaring gives $x^2 + y^2 = 1$ (with $y \geq 0$).
 3. Symmetrically, $\cos^{-1} y = \frac{\pi}{2} - \cos^{-1} x = \sin^{-1} x$, so $y = \sin(\sin^{-1} x) = x$... proceed instead from $\cos^{-1} y = \frac{\pi}{2} - \cos^{-1} x$, take cosine of both sides: $y = \sin(\cos^{-1} x) = \sqrt{1 - x^2}$, giving $x^2 + y^2 = 1$.
 4. This is Result 2, converse direction. $\cos S = \cos A \cos B \cos C (1 - xy - yz - zx) = 0$ with every factor cosine positive, so $\cos S = 0$; since $S \in (0, \frac{3\pi}{2})$ and $\sin S \neq 0$ (else both vanish), the only possibility is $S = \frac{\pi}{2}$. Sanity check at $x = y = z = \frac{1}{\sqrt{3}}$: the sum is three copies of $\frac{\pi}{6}$.
 5. Domains force $x \in [0, 1]$: from $\cos^{-1} y = \pi - \sin^{-1} x \leq \pi$ we need $\sin^{-1} x \geq 0$. Then $y = \cos(\pi - \sin^{-1} x) = -\cos(\sin^{-1} x) = -\sqrt{1 - x^2}$. Relation: $x^2 + y^2 = 1$ with $x \in [0, 1]$ and $y \in [-1, 0]$.
-

Addendum C — Multiple Angles Over the Whole Real Line

Method

The chapter's multiple-angle table states, for instance, $2 \tan^{-1} x = \tan^{-1} \frac{2x}{1 - x^2}$ only for $|x| < 1$. JEE repeatedly asks for the simplification of the *reverse* direction — expressing $\tan^{-1} \frac{2x}{1 - x^2}$ in terms of $\tan^{-1} x$ — for **all** admissible real x . The recipe is pure Part 1 range-folding, run with a variable angle.

1. Put $\theta = \tan^{-1} x$, so $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$ and $\tan \theta = x$.
2. Rewrite the argument as a trig function of $m\theta$ using the double/triple angle formulas — e.g. $\frac{2x}{1 + x^2} = \sin 2\theta$.
3. Locate the interval containing $m\theta$; the breakpoints occur where $m\theta$ hits an odd multiple of $\frac{\pi}{2}$, i.e. where $x = \tan \theta$ crosses $\pm \tan \frac{(2j+1)\pi}{2m}$.
4. Fold $m\theta$ into the outer function's cage using a value-preserving identity, exactly as in Part 1.
5. Translate back: $\theta = \tan^{-1} x$.

The completed tables

With $\theta = \tan^{-1} x$ throughout, and noting $2\theta \in (-\pi, \pi)$, $3\theta \in (-\frac{3\pi}{2}, \frac{3\pi}{2})$:

Expression	Condition on x	Value
$\sin^{-1} \frac{2x}{1+x^2}$	$ x \leq 1$	$2 \tan^{-1} x$
	$x > 1$	$\pi - 2 \tan^{-1} x$
	$x < -1$	$-\pi - 2 \tan^{-1} x$
$\cos^{-1} \frac{1-x^2}{1+x^2}$	$x \geq 0$	$2 \tan^{-1} x$
	$x < 0$	$-2 \tan^{-1} x$
$\tan^{-1} \frac{2x}{1-x^2}, x \neq \pm 1$	$ x < 1$	$2 \tan^{-1} x$
	$x > 1$	$2 \tan^{-1} x - \pi$
	$x < -1$	$2 \tan^{-1} x + \pi$
$\tan^{-1} \frac{3x-x^3}{1-3x^2}, x \neq \pm \frac{1}{\sqrt{3}}$	$ x < \frac{1}{\sqrt{3}}$	$3 \tan^{-1} x$
	$x > \frac{1}{\sqrt{3}}$	$3 \tan^{-1} x - \pi$
	$x < -\frac{1}{\sqrt{3}}$	$3 \tan^{-1} x + \pi$

Where each row comes from. For the sine row: $|x| \leq 1$ puts 2θ inside the $\sin^{-1}$ cage, giving 2θ directly; $x > 1$ puts $2\theta \in (\frac{\pi}{2}, \pi)$, and $\sin(\pi - 2\theta) = \sin 2\theta$ with $\pi - 2\theta \in (0, \frac{\pi}{2})$ legal, so fold to $\pi - 2\theta$; the case $x < -1$ mirrors it by oddness. The cosine row breaks at $x = 0$, not at $x = \pm 1$, because the cage $[0, \pi]$ already contains all of $[0, \pi)$ — only negative 2θ needs reflecting to $-2\theta \in (0, \pi)$ via $\cos(-t) = \cos t$. For the tangent triples: $x > \frac{1}{\sqrt{3}}$ pushes 3θ past $\frac{\pi}{2}$, and the fold $3\theta \mapsto 3\theta - \pi$ stays legal across *both* remaining sub-intervals, which is why the table needs only one row for all $x > \frac{1}{\sqrt{3}}$.

Spot-checks you can run in seconds: at $x = \sqrt{3}$, row one predicts $\pi - \frac{2\pi}{3} = \frac{\pi}{3}$, and indeed $\sin^{-1} \frac{\sqrt{3}}{2} = \frac{\pi}{3}$. At $x = 1$, the tangent-double row predicts $2 \cdot \frac{\pi}{4} - \pi = -\frac{\pi}{4}$, and indeed $\tan^{-1} \frac{2}{0}$ -limit argument $\tan^{-1}(-1) = -\frac{\pi}{4}$ approached through $x \rightarrow 1^+$.

Example C.1 — [JEE Main Level]

Evaluate $\sin^{-1} \frac{2a}{1+a^2} + \cos^{-1} \frac{1-a^2}{1+a^2}$ **for** $a > 1$.

Row one, case $a > 1$: the first term is $\pi - 2 \tan^{-1} a$. Cosine row, case $a > 0$: the second term is $2 \tan^{-1} a$. The variable parts annihilate:

$$(\pi - 2 \tan^{-1} a) + 2 \tan^{-1} a = \pi.$$

Numeric confirmation at $a = 2$: $\sin^{-1} \frac{4}{5} \approx 0.9273$ and $\cos^{-1}(-\frac{3}{5}) \approx 2.2143$; the sum is ≈ 3.1416 . The answer is π for *every* $a > 1$ — a constancy that is invisible without the piecewise table, since for $0 \leq a \leq 1$ the same sum equals $4 \tan^{-1} a$ and varies.

Example C.2 — [JEE Advanced Level]

Find all real x satisfying

$$2 \tan^{-1} x = \sin^{-1} \frac{2x}{1+x^2}.$$

Read the table as a comparison of two functions of x . The two sides agree exactly on the row where both entries coincide, namely $|x| \leq 1$. For $x > 1$ the right side is $\pi - 2 \tan^{-1} x$, strictly larger than the left side $2 \tan^{-1} x$ (their difference is $\pi - 4 \tan^{-1} x > 0$ because $\tan^{-1} x > \frac{\pi}{4}$ there); for $x < -1$ the mismatch mirrors by oddness. Neither side is ever undefined, so nothing is excluded.

$$x \in [-1, 1]$$

The lesson generalises: whenever an exam equates two such expressions, do not manipulate — tabulate. Each side is piecewise linear in $\tan^{-1} x$, and the solution set is read off row by row.

Practice — Addendum C

1. Express $\sin^{-1} \frac{4}{5}$ in terms of $\tan^{-1} 2$.
2. Express $\cos^{-1}(-\frac{4}{5})$ in terms of $\tan^{-1} 3$.
3. Express $\tan^{-1} \frac{3}{4}$ in terms of $\tan^{-1} 3$.
4. Find all real x with $\cos^{-1} \frac{1-x^2}{1+x^2} = -2 \tan^{-1} x$.
5. Find all real x satisfying $\tan^{-1} \frac{3x-x^3}{1-3x^2} = 3 \tan^{-1} x - \pi$.

Answers:

1. Since $2 > 1$, row one gives $\pi - 2 \tan^{-1} 2$. Check numerically: $\pi - 2(1.1071) \approx 0.9273 = \sin^{-1} 0.8$.
2. With $x = 3 > 0$ the cosine row gives $2 \tan^{-1} 3$; and indeed $\cos^{-1}(-0.8) \approx 2.4981 \approx 2(1.2490)$.
3. Recognise $\frac{3}{4} = \frac{2(-3)}{1-(-3)^2}$, so $x = -3 < -1$ and the tangent-double row gives $2 \tan^{-1}(-3) + \pi = \pi - 2 \tan^{-1} 3 \approx 0.6435$.
4. The two sides match precisely on the row $x < 0$, and at $x = 0$ both sides equal 0. Answer: $x \leq 0$.
5. The right-hand pattern holds exactly for $x > \frac{1}{\sqrt{3}}$ (the expression is undefined at $x = \pm \frac{1}{\sqrt{3}}$, and $x = \frac{1}{\sqrt{3}}$ is excluded by the strict inequality anyway). Answer: $x > \frac{1}{\sqrt{3}}$. Spot-check at $x = 2$: left side $\tan^{-1} \frac{2}{11} \approx 0.1799$; right side $3(1.1071) - \pi \approx 0.1799$.

Addendum D — The Folded Composites as Functions: Waves and Solution Counts

Method

Part 1 folds a *fixed* angle into a branch. Promote the angle to a variable and the fold becomes a function of x — and these three functions dominate JEE Advanced solution-counting questions because each is piecewise linear: a triangle wave or a sawtooth.

1. $f(x) = \sin^{-1}(\sin x)$ — the triangular wave. Odd, and periodic with period 2π because $\sin$ is. The chapter already derived its two core pieces: $f(x) = x$ on $[-\frac{\pi}{2}, \frac{\pi}{2}]$ and $f(x) = \pi - x$ on $[\frac{\pi}{2}, \frac{3\pi}{2}]$ (its Part 1, Example 4(b)). Translating by the period gives, for every integer k :

$$f(x) = x - 2k\pi \quad \text{on} \quad \left[-\frac{\pi}{2} + 2k\pi, \frac{\pi}{2} + 2k\pi\right],$$

$$f(x) = (2k+1)\pi - x \quad \text{on} \quad \left[\frac{\pi}{2} + 2k\pi, \frac{3\pi}{2} + 2k\pi\right].$$

Continuity at the joins is immediate: at $x = \frac{\pi}{2} + 2k\pi$ both formulas return $\frac{\pi}{2}$; at $x = \frac{3\pi}{2} + 2k\pi$ both return $-\frac{\pi}{2}$. The graph is a zigzag of slope ± 1 with corners at $x = \frac{\pi}{2} + k\pi$, range $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

2. $g(x) = \cos^{-1}(\cos x)$ — the second triangular wave. Even, period 2π , and on the fundamental period:

$$g(x) = x \quad \text{on} \quad [0, \pi], \quad g(x) = 2\pi - x \quad \text{on} \quad [\pi, 2\pi],$$

since $\cos(2\pi - x) = \cos x$. Corners at multiples of π ; range $[0, \pi]$.

3. $h(x) = \tan^{-1}(\tan x)$ — the sawtooth. Period π ; on each open strip $(-\frac{\pi}{2} + k\pi, \frac{\pi}{2} + k\pi)$ it equals $x - k\pi$, slope 1, jumping from $+\frac{\pi}{2}$ down to $-\frac{\pi}{2}$ at every $x = \frac{\pi}{2} + k\pi$, where it is undefined. Range $(-\frac{\pi}{2}, \frac{\pi}{2})$.

Counting principle. Because each function is monotonic on each linear piece, an equation like $f(x) = c$ is solved *per piece*, and the count follows. For $0 < |c| < \frac{\pi}{2}$, the equation $f(x) = c$ has exactly two solutions per period; at $c = \pm\frac{\pi}{2}$ the two solutions fuse at a corner into one.

Function	Period	Range	Shape
$\sin^{-1}(\sin x)$	2π	$[-\frac{\pi}{2}, \frac{\pi}{2}]$	triangle wave, slopes ± 1 , corners at $\frac{\pi}{2} + k\pi$
$\cos^{-1}(\cos x)$	2π	$[0, \pi]$	triangle wave, slopes ± 1 , corners at $k\pi$
$\tan^{-1}(\tan x)$	π	$(-\frac{\pi}{2}, \frac{\pi}{2})$	sawtooth, slope 1, jumps at $\frac{\pi}{2} + k\pi$

Example D.1 — [JEE Main Level]

(a) Evaluate $\sin^{-1}(\sin 10)$. (b) Count the solutions of $\sin^{-1}(\sin x) = \frac{\pi}{4}$ in $[0, 2\pi]$.

- (a) Locate 10 in the piece map. Since $10 \in [\frac{\pi}{2} + 2\pi, \frac{3\pi}{2} + 2\pi]$ (as $\frac{5\pi}{2} \approx 7.85$ and $\frac{7\pi}{2} \approx 11.00$), use the falling piece with $k = 1$:

$$f(10) = 3\pi - 10 \approx -0.575.$$

Check: $\sin(3\pi - 10) = \sin(10 - 2\pi)$... verify directly instead: $\sin 10 \approx -0.544$ and $\sin(3\pi - 10) \approx -0.544$, and $3\pi - 10$ lies inside the cage. Answer $3\pi - 10$.

- (b) Sweep the three pieces covering $[0, 2\pi]$: on $[0, \frac{\pi}{2}]$, $x = \frac{\pi}{4}$ — one hit. On $[\frac{\pi}{2}, \frac{3\pi}{2}]$, $\pi - x = \frac{\pi}{4}$ gives $x = \frac{3\pi}{4}$ — one hit. On $[\frac{3\pi}{2}, 2\pi]$ the rising piece reads $x - 2\pi = \frac{\pi}{4}$, giving $x = \frac{9\pi}{4}$, outside the interval — no hit. Total: **two solutions**, $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$.

Example D.2 — [JEE Advanced Level]

Find the number of real solutions of

$$\cos^{-1}(\cos x) = \frac{|x|}{2}.$$

Both sides are even, so solve for $x \geq 0$ and reflect. Also, $g(x) \leq \pi$ everywhere, while $\frac{x}{2} \leq \pi$ forces $x \leq 2\pi$ — the search space closes automatically. On $[0, 2\pi]$ the wave is $g(x) = x$ then $g(x) = 2\pi - x$.

Piece one, $[0, \pi]$: $x = \frac{x}{2}$ gives $x = 0$ — valid, since $g(0) = \cos^{-1} 1 = 0$.

Piece two, $[\pi, 2\pi]$: $2\pi - x = \frac{x}{2}$ gives $x = \frac{4\pi}{3} \approx 4.19$, which does lie in $[\pi, 2\pi]$ — valid.

Past 2π the search is closed, so the non-negative solutions are 0 and $\frac{4\pi}{3}$; reflection adds $-\frac{4\pi}{3}$.

three solutions: $x = 0, \pm \frac{4\pi}{3}$

The mark-winning habits: bound the search space before sweeping (here $x \leq 2\pi$ came free from the range), and reject any piecewise candidate that lands outside the piece that produced it.

Practice — Addendum D

1. State the period, range, and corner positions of $\sin^{-1}(\sin x)$.
2. Evaluate $\cos^{-1}(\cos 4)$, working in radians.
3. Evaluate $\tan^{-1}(\tan \frac{7\pi}{4})$.
4. How many solutions does $\sin^{-1}(\sin x) = 0.6$ have in $[-2\pi, 2\pi]$?
5. Solve $\cos^{-1}(\cos x) = \frac{x}{2}$ for $x \in [0, 2\pi]$.

Answers:

1. Period 2π ; range $[-\frac{\pi}{2}, \frac{\pi}{2}]$; corners at $x = \frac{\pi}{2} + k\pi$ for every integer k .
2. Since $\pi \approx 3.14 < 4 < 2\pi \approx 6.28$, the falling piece applies: $g(4) = 2\pi - 4 \approx 0.283$.
3. $\frac{7\pi}{4}$ lies in the strip $(\frac{3\pi}{2}, \frac{5\pi}{2})$, where $h(x) = x - \pi$; answer $-\frac{\pi}{4}$. (Equivalent fold: $\tan \frac{7\pi}{4} = -1$.)

- Two solutions per period for a level strictly between the corner heights, and the interval spans two full periods: **four** solutions, namely $x = 0.6$, $\pi - 0.6$ and their odd reflections -0.6 , $-(\pi - 0.6)$.
- On $[0, \pi]$: $x = \frac{x}{2}$ forces $x = 0$. On $[\pi, 2\pi]$: $2\pi - x = \frac{x}{2}$ gives $x = \frac{4\pi}{3}$. Answer $x = 0$ and $x = \frac{4\pi}{3}$ — the even-reflection of Example D.2 cut off at the origin.

Addendum E — Domains and Ranges of Composite Inverse Trigonometric Functions

Method

The chapter lists the six bare domains but never builds functions out of them. Two reflexes cover nearly every exam question of this type.

Finding a domain.

- Write one constraint per inverse part: argument of $\sin^{-1}$ or $\cos^{-1}$ in $[-1, 1]$; argument of $\sec^{-1}$ or $\operatorname{cosec}^{-1}$ with modulus at least 1; $\tan^{-1}$ and $\cot^{-1}$ free; plus whatever the surrounding algebra demands (radicals, denominators, logarithms).
- Intersect the constraints. For a quadratic argument, complete the square before solving the two-sided inequality $-1 \leq q(x) \leq 1$; one side is often automatic.
- For a fractional argument, split on the sign of the denominator rather than multiplying blindly.

Finding a range.

- Reduce the expression to a *single* inverse function using the complementary laws $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ and friends — this is the single most profitable move in range problems.
- Use monotonicity: each inverse function is increasing or decreasing on its whole domain (the chapter's graph table, §4.2), so a sum of same-direction monotonic functions attains its extreme values at the domain endpoints.
- State the range as the image of the endpoints, adjusted for open domains.

Example E.1 — [CBSE Board / JEE Main Level]

Find the domain and range of $F(x) = \cos^{-1}(x^2 - 4x + 3)$.

Constraint: $-1 \leq x^2 - 4x + 3 \leq 1$. Upper side:

$$x^2 - 4x + 2 \leq 0 \iff (x - 2)^2 \leq 2 \iff 2 - \sqrt{2} \leq x \leq 2 + \sqrt{2}.$$

Lower side:

$$x^2 - 4x + 4 \geq 0 \iff (x - 2)^2 \geq 0,$$

true for every real x — automatic, as promised. Domain: $[2 - \sqrt{2}, 2 + \sqrt{2}]$.

Range: on this domain the argument equals $(x - 2)^2 - 1$, which runs continuously from -1 at the centre $x = 2$ up to $2 - 1 = 1$ at both endpoints. So the argument attains every value in $[-1, 1]$, and $\cos^{-1}$, being continuous and decreasing, delivers every value of $[0, \pi]$. Range: $[0, \pi]$.

Example E.2 — [JEE Main Level]

Find the largest domain of $G(x) = \sin^{-1} \frac{x+1}{x-2}$.

Rewrite the argument as $1 + \frac{3}{x-2}$ and impose $-1 \leq t \leq 1$.

Condition $t \leq 1$: $\frac{3}{x-2} \leq 0$, which forces $x < 2$.

Condition $t \geq -1$: $\frac{3}{x-2} \geq -2$; multiplying by the negative quantity $x - 2$ reverses the inequality:

$$3 \leq -2(x-2) \iff 3 \leq -2x + 4 \iff x \leq \frac{1}{2}.$$

Intersecting: domain $(-\infty, \frac{1}{2}]$. Spot-check: at $x = 0$ the argument is $-\frac{1}{2}$ — legal; at $x = 1$ it is 2 — illegal, correctly excluded since $1 > \frac{1}{2}$.

Example E.3 — [JEE Advanced Level]

Find the range of $H(x) = \sin^{-1} x + \tan^{-1} x$ on its natural domain.

Both parts are defined on $[-1, 1]$, so that is the domain. Both parts are increasing there (chapter graph table), so H is increasing, and its range runs from endpoint to endpoint:

$$H(-1) = -\frac{\pi}{2} - \frac{\pi}{4} = -\frac{3\pi}{4}, \quad H(1) = \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4}.$$

Range: $[-\frac{3\pi}{4}, \frac{3\pi}{4}]$. No calculus was needed — monotonicity of the two pieces is a stated graph property of the chapter.

Practice — Addendum E

1. Find the domain of $\sin^{-1}(2x - 1)$.
2. Find the largest domain of $\sec^{-1} x + \tan^{-1} x$.
3. Find the range of $K(x) = 2 \sin^{-1} x - \cos^{-1} x$.
4. For how many integers x is $\cos^{-1}(x - 1) + \sin^{-1}(1 - x)$ defined?
5. Find the domain of $\sin^{-1} \frac{2x}{1+x^2}$.
6. Find the range of $P(x) = \sin^{-1} x + \sec^{-1} x$.

Answers:

1. $-1 \leq 2x - 1 \leq 1$ gives $0 \leq 2x \leq 2$, so $x \in [\frac{1}{2}, 1]$.
2. $\tan^{-1}$ accepts anything; $\sec^{-1}$ demands $|x| \geq 1$. Domain $(-\infty, -1] \cup [1, \infty)$.
3. Deploy the complementary law, $\cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$:

$$K(x) = 2 \sin^{-1} x - \frac{\pi}{2} + \sin^{-1} x = 3 \sin^{-1} x - \frac{\pi}{2},$$

and $\sin^{-1} x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, so the range is $[-\frac{3\pi}{2} - \frac{\pi}{2}, \frac{3\pi}{2} - \frac{\pi}{2}] = [-2\pi, \pi]$.

4. First part needs $0 \leq x - 1 \leq 1$, i.e. $x \in [1, 2]$; second part needs $-1 \leq 1 - x \leq 1$, i.e. $x \in [0, 2]$. Intersection $[1, 2]$ contains the integers 1 and 2: **two** integers.
5. Every real x : the claim $\frac{2|x|}{1+x^2} \leq 1$ is equivalent to $(|x| - 1)^2 \geq 0$. Domain $\mathbb{R}$ — the same fact that makes Addendum C's sine row defined everywhere.
6. The domain is the intersection $[-1, 1] \cap \{|x| \geq 1\} = \{-1, 1\}$ — two points. Evaluate both: $P(1) = \frac{\pi}{2} + 0 = \frac{\pi}{2}$ and $P(-1) = -\frac{\pi}{2} + \pi = \frac{\pi}{2}$. Range: the singleton $\{\frac{\pi}{2}\}$. The function is *constant* — the kind of answer no amount of formula-pushing anticipates, and exactly why the domain gate comes first.

Addendum P — What JEE Actually Asks

Everything below is drawn from the previous-year bank supplied with this chapter (Disha compilation, Chapter 17). The extraction lost most displayed mathematics, so each question cited was either reconstructed and independently re-solved, or dropped. Roughly fifteen of the ~48 tagged items survived well enough to reconstruct with confidence; the rest are used only as tags (year/session/format), never as mathematics.

P.1 The distribution

Volume. The bank carries tags from 1981 through 2019 — Main sessions in 2013 (three sittings), 2014, 2015 (two), 2017 (two online), 2019 (six sittings), 2020 (two sittings), plus Advanced papers in 2013, 2015, 2018 and 2019 (twice). The steady signal: **one inverse-trig question per JEE Main paper almost every year**, and Advanced visits the chapter in bursts (2013, 2015, 2018, 2019) preferring multi-correct, matching-list and numerical-answer formats over single-option MCQs.

Where the marks concentrate, judged from the recoverable stems:

Block	Share of identifiable items	Evidence tags
Arctangent sums, conditional domains, telescoping series	heaviest — at least 6 clear hits	Main 1994, 2006, Jan 10 2019 (II), Sep 05 2020 (I); Adv 2013

Block	Share of identifiable items	Evidence tags
Equations, inequalities, counting solutions	heavy — 6+ hits	Main 1999, 2004S, online Apr 22 2013, online Apr 25 2013, Jan 12 2019 (I); Adv 2018
Composite evaluation & triangle-substitution	steady	Main 2008, 2002 (proof), online Apr 09 2017, Apr 10 2019 (II); 1981
Principal-value folding	mostly pre-2010 + Advanced fill-ins	Main 1986, online Apr 19 2014; Adv 2019
Multi-angle identities with validity windows	Advanced-flavoured	Main 2015; Adv 2015, Adv 2019

Two format facts worth internalising: Advanced likes this chapter as a **multi-correct** (2015) or **column-matching** (2013, 2019) item worth a lot of marks, and Main increasingly hides the chapter inside **set-cardinality / number-of-solutions** language (“the set ... is a singleton”) rather than naked “evaluate” commands.

P.2 The recurring archetypes

A1 — The conditional arctangent sum (“compute xy first”)

Looks like: $\tan^{-1}(\text{algebraic}) + \tan^{-1}(\text{algebraic})$ set equal to something; a set or an equation is asked. **Method:** take $\tan$ of both sides *only after* confirming both sides sit in $(-\pi/2, \pi/2)$; solve the quadratic; **reject roots whose branch behaviour contradicts the original equation.**

Representative (reconstructed; Main Jan 12, 2019 (I)). Find the set $\{x : \tan^{-1}(2x) + \tan^{-1}(3x) = \pi/4\}$.

Both sides lie in $(-\pi/2, \pi/2)$, so tangent is legitimate:

$$\tan^{-1}\left(\frac{5x}{1-6x^2}\right) = \frac{\pi}{4} \Rightarrow \frac{5x}{1-6x^2} = 1 \Rightarrow 6x^2 + 5x - 1 = 0.$$

Roots: $x = \frac{1}{6}$ and $x = -1$. Reject $x = -1$: then $2x \cdot 3x = 6 > 1$ with both entries negative, so the left side lies in $(-\pi, -\pi/2)$ — it cannot equal $\pi/4$. Indeed $\tan^{-1}(-2) + \tan^{-1}(-3) = -3\pi/4$. The set is $\{\frac{1}{6}\}$ — **a singleton**.

A2 — The telescoping arctangent series

Looks like: $\sum \tan^{-1}\left(\frac{1}{\text{quadratic}}\right)$ or $\sum \cot^{-1}(\text{quadratic})$, finite or infinite. **Method:** engineer the term into $\frac{a-b}{1+ab} = \tan(\tan^{-1} a - \tan^{-1} b)$; consecutive terms cancel; read the limit off the endpoints.

Representative (reconstructed; Adv 2013). Evaluate $\sum_{n=1}^{\infty} \cot^{-1}(1 + n + n^2)$.

Since $1 + n + n^2 > 0$, $\cot^{-1}(1 + n + n^2) = \tan^{-1} \frac{1}{n^2 + n + 1}$, and

$$\frac{(n+1) - n}{1 + n(n+1)} \Rightarrow \cot^{-1}(1 + n + n^2) = \tan^{-1}(n+1) - \tan^{-1}(n).$$

Partial sums telescope to $\tan^{-1}(N+1) - \tan^{-1} 1 \rightarrow \frac{\pi}{2} - \frac{\pi}{4} = \boxed{\pi/4}$.

A3 — Piecewise double-angle wrappers (“for which x is the identity legal?”)

Looks like: $2 \tan^{-1} x$ wearing a $\sin^{-1}$, $\cos^{-1}$ or $\tan^{-1}$ costume; often evaluated at a large x .

Method: put $\theta = \tan^{-1} x$, apply the double-angle formula, then ask whether 2θ is inside the outer branch. If not, fold — the answer changes formula.

Representative (recoverable; Main online Apr 10, 2015). If $f(x) = 2 \tan^{-1} x + \sin^{-1} \left(\frac{2x}{1+x^2} \right)$, find $f(5)$.

Let $\theta = \tan^{-1} x$; then $\sin 2\theta = \frac{2x}{1+x^2}$. For $x = 5$, $\theta = \tan^{-1} 5 \approx 78.7^\circ$, so $2\theta > \frac{\pi}{2}$ — outside the $\sin^{-1}$ cage. The cage returns $\pi - 2\theta$, hence

$$f(5) = 2\theta + (\pi - 2\theta) = \pi.$$

(In fact $f(x) = \pi$ for all $x > 1$, $f(x) = 2 \tan^{-1} x$ for $|x| \leq 1$.) The whole question is the branch check; anyone who cancels blindly gets $2 \tan^{-1} 5 \neq \pi$.

A4 — Triangle-method composites and simplifications

Looks like: $\cos(\tan^{-1}(\dots))$, $\sin(\cot^{-1}(\dots))$, or an expression whose radical screams for $x = \tan \theta$. **Method:** name the inner angle, build the right triangle, read the ratio; or substitute per the radical signature.

Representative (recoverable; Main 2008). For $0 < x < 1$, simplify $\left[\{x \cos(\cot^{-1} x) + \sin(\cot^{-1} x)\}^2 - 1 \right]^{1/2}$.

With $\cot^{-1} x = \theta$: opposite 1, adjacent x , hypotenuse $\sqrt{1+x^2}$, so

$$\cos(\cot^{-1} x) = \frac{x}{\sqrt{1+x^2}}, \quad \sin(\cot^{-1} x) = \frac{1}{\sqrt{1+x^2}}.$$

The bracket becomes $\frac{x^2+1}{\sqrt{1+x^2}} = \sqrt{1+x^2}$; squaring and subtracting 1 leaves x^2 , and the root is $\boxed{x}$ (positive since $0 < x < 1$).

A5 — The sign-and-range audit (Advanced multi-correct)

Looks like: $\alpha = k \sin^{-1}(\cdot)$, $\beta = k \cos^{-1}(\cdot)$ with multiplier $k \geq 3$; four inequalities offered.

Method: bracket each multiple angle between neighbouring special angles; never evaluate numerically without a sandwich proof.

Representative (reconstructed; Adv 2015 — the extracted stem lost its arguments, but the official answer pattern (b), (c), (d) pins them uniquely; re-solved and verified). Let $\alpha = 3 \sin^{-1} \frac{2}{3}$ and $\beta = 3 \cos^{-1} \frac{1}{3}$ (principal values). Which hold: (a) $\cos \beta > 0$ (b) $\sin \beta < 0$ (c) $\cos(\alpha + \beta) > 0$ (d) $\cos \alpha < 0$?

Sandwiches:

$$\begin{aligned} \frac{\pi}{6} < \sin^{-1} \frac{2}{3} < \frac{\pi}{4} &\Rightarrow \frac{\pi}{2} < \alpha < \frac{3\pi}{4}, \\ \frac{\pi}{3} < \cos^{-1} \frac{1}{3} < \frac{\pi}{2} &\Rightarrow \pi < \beta < \frac{3\pi}{2}. \end{aligned}$$

Hence $\cos \alpha < 0$ — **(d)**; β sits in the third quadrant, so $\sin \beta < 0$ and $\cos \beta < 0$ — **(b)** yes, (a) no. Finally $\frac{3\pi}{2} < \alpha + \beta < 2\pi$ (lower bound from the sandwiches; upper bound since $\sin^{-1} \frac{2}{3} < \frac{\pi}{4}$ and $\cos^{-1} \frac{1}{3} < \frac{5\pi}{12}$), so $\alpha + \beta$ is in the fourth quadrant: $\cos(\alpha + \beta) > 0$ — **(c)**. Answer: **(b), (c), (d)**.

A6 — Inequalities and solution-counting under the branch

Looks like: “ $\sin^{-1} x > \cos^{-1} x$ ”, “number of solutions of ...”, “the set ... is”. **Method:** replace one function using the complementary law, convert to an ordinary inequality/equation, then count **including endpoint cases**.

Representative (recoverable; Main online Apr 25, 2013). Find all $x \in (0, 1)$ with $\sin^{-1} x > \cos^{-1} x$.

Using $\cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$:

$$\sin^{-1} x > \frac{\pi}{2} - \sin^{-1} x \iff \sin^{-1} x > \frac{\pi}{4} \iff x > \frac{1}{\sqrt{2}}.$$

Answer: $x \in \left(\frac{1}{\sqrt{2}}, 1\right)$.

P.3 The traps

These are the error patterns the distractors are engineered around — visible both in wrong options and in the rejected roots inside the bank’s own solutions:

1. **The missing $\pm\pi$.** For $\tan^{-1} x + \tan^{-1} y$ with $xy > 1$, the uncorrected formula is off by exactly π . Distractor sets routinely contain both the corrected and uncorrected values. Compute xy before touching the formula.
2. **Blind cancellation of $f^{-1} \circ f$.** Every folding question (Adv 2019’s fill-in; Main 1986, 2014) offers the unfolded angle as the tempting option. Check the branch first.
3. **Identity outside its window.** $3 \sin^{-1} x = \sin^{-1}(3x - 4x^3)$ fails loudly at $x = \frac{\sqrt{3}}{2}$ ($\pi \neq 0$). Advanced’s 2015 and 2019 items exist purely to punish unconditional quoting of multiple-angle forms.

4. **Extraneous roots after taking tan or squaring.** The singleton question (A1) produces the bogus root $x = -1$; the bank's own key rejects it on branch grounds. Every converted equation must be audited against the original branches.
5. **Odd-versus- π - confusion.** $\cot^{-1}(-\cdot)$ and $\sec^{-1}(-\cdot)$ reflect, they do not negate; and $\tan^{-1} \frac{1}{x} = \cot^{-1} x$ holds only for $x > 0$.
6. **Endpoint blindness.** In solution-counting, $x = \pm 1$ sit exactly on the branch boundaries and *do* count — see practice problem 4. Students who “feel” endpoints are excluded miscount.
7. **Piecewise laziness.** Expressions like $\sin^{-1} \frac{2x}{1+x^2}$ change formula across $x = 1$; a single-formula answer is wrong on part of the domain (A3).

P.4 Graded practice set

All answers below re-derived and checked independently.

P1. Evaluate $\cos^{-1}(\cos \frac{13\pi}{6}) + \tan^{-1}(\tan \frac{5\pi}{6})$.

$\frac{13\pi}{6} \equiv \frac{\pi}{6} \pmod{2\pi}$ and $\frac{\pi}{6} \in [0, \pi]$, so the first term is $\frac{\pi}{6}$. Next, $\tan \frac{5\pi}{6} = \tan(-\frac{\pi}{6})$ and $-\frac{\pi}{6} \in (-\frac{\pi}{2}, \frac{\pi}{2})$, so the second term is $-\frac{\pi}{6}$. **Sum** = 0.

P2. Evaluate $\cos(2 \tan^{-1} \frac{1}{4})$.

$$\cos 2\theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \frac{1 - \frac{1}{16}}{1 + \frac{1}{16}} = \boxed{\frac{15}{17}}.$$

P3. If $\cos^{-1} x + \cos^{-1} y = \frac{\pi}{2}$, show $x^2 + y^2 = 1$.

Then $\cos^{-1} x = \frac{\pi}{2} - \cos^{-1} y = \sin^{-1} y$, so $x = \sin(\cos^{-1} y) = \sqrt{1 - y^2}$, giving $x^2 + y^2 = 1$.

P4. (Bank: 2004S, repeated online Apr 09 2017) Solve $\sin[\cot^{-1}(1 + x)] = \cos(\tan^{-1} x)$.

Triangle readings: $\sin(\cot^{-1}(1 + x)) = \frac{1}{\sqrt{1 + (1 + x)^2}}$ and $\cos(\tan^{-1} x) = \frac{1}{\sqrt{1 + x^2}}$. Equating and squaring:

$$1 + (1 + x)^2 = 1 + x^2 \Rightarrow 2x + 1 = 0 \Rightarrow \boxed{x = -\frac{1}{2}}.$$

No root-audit needed: both readings are valid for every real input.

P5. How many real solutions does $\sin^{-1} x = 2 \tan^{-1} x$ have?

For $0 < x < 1$: both sides lie in $[0, \frac{\pi}{2}]$ where $\sin$ is injective, and $\sin(2 \tan^{-1} x) = \frac{2x}{1 + x^2} > x$, so $2 \tan^{-1} x > \sin^{-1} x$ strictly — no interior roots. Checking the boundaries: $x = 0$ works; $x = 1$ gives $\frac{\pi}{2} = \frac{\pi}{2}$; by oddness $x = -1$ works. **Exactly three solutions:** $x = -1, 0, 1$. (Endpoint discipline — trap 6.)

P6. State precisely when $3 \sin^{-1} x = \sin^{-1}(3x - 4x^3)$ holds, and demonstrate the failure at $x = \frac{\sqrt{3}}{2}$.

Writing $\theta = \sin^{-1} x$, the right side is $\sin^{-1}(\sin 3\theta)$, which equals 3θ only when $3\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, i.e. $|x| \leq \frac{1}{2}$. At $x = \frac{\sqrt{3}}{2}$: left side = $3 \cdot \frac{\pi}{3} = \pi$; right side = $\sin^{-1}(0) = 0$. **Fails, as predicted.**

P7. (Bank: Main Sep 05, 2020 (I)) If S is the sum of the first ten terms of $\sum \tan^{-1} \frac{1}{2n^2}$, find $\tan S$.

Since $\frac{1}{2n^2} = \frac{(2n+1)-(2n-1)}{1+(2n+1)(2n-1)}$,

$$S = \sum_{n=1}^{10} [\tan^{-1}(2n+1) - \tan^{-1}(2n-1)] = \tan^{-1} 21 - \tan^{-1} 1,$$

so $\tan S = \frac{21-1}{1+21 \cdot 1} = \boxed{\frac{10}{11}}.$

P8. Arrange in increasing order: $2 \tan^{-1} \frac{1}{5}$, $\tan^{-1} \frac{1}{2}$, $\frac{\pi}{4}$.

$2 \tan^{-1} \frac{1}{5} = \tan^{-1} \frac{2/5}{1-1/25} = \tan^{-1} \frac{5}{12}$. Since $\frac{5}{12} < \frac{1}{2} < 1$ and arctan is increasing,

$$2 \tan^{-1} \frac{1}{5} < \tan^{-1} \frac{1}{2} < \frac{\pi}{4}.$$

(Comparison-by-single-arctan is exactly the skill behind the bank's 1989 "greater of two angles" item.)

P9. (Adapts Main Jan 10, 2019 (II)) Evaluate $\cot \left(\sum_{n=1}^{19} \tan^{-1} \frac{1}{1+n+n^2} \right)$.

Each term = $\tan^{-1}(n+1) - \tan^{-1} n$ (same engineering as A2), so the sum is $\tan^{-1} 20 - \tan^{-1} 1$. Then

$$\cot(\tan^{-1} 20 - \tan^{-1} 1) = \frac{20 \cdot 1 + 1}{20 - 1} = \boxed{\frac{21}{19}}.$$

P10. (Bank: Main 2013) x, y, z are in A.P. and $\tan^{-1} x, \tan^{-1} y, \tan^{-1} z$ are also in A.P. Show that $x = y = z$ in the non-degenerate case.

Taking tangent of $2 \tan^{-1} y = \tan^{-1} x + \tan^{-1} z$:

$$\frac{2y}{1-y^2} = \frac{x+z}{1-xz} = \frac{2y}{1-xz}.$$

If $y \neq 0$, cancel: $xz = y^2$. Together with $x+z = 2y$, the numbers x, z are roots of $t^2 - 2yt + y^2 = 0$, forcing $x = z = y$. (Degenerate case: $y = 0$, $z = -x$ also satisfies both progressions; the official key reads the question generically and expects $x = y = z$.)

P11. (Bank: 2002, 5-marker) Prove $\cos[\tan^{-1}(\sin(\cot^{-1} x))] = \sqrt{\frac{1+x^2}{2+x^2}}$.

Let $\theta = \cot^{-1} x$, so $\theta \in (0, \pi)$ and $\sin \theta = \frac{1}{\sqrt{1+x^2}}$ — valid for **every** nonzero x , since sine is positive throughout $(0, \pi)$. Then

$$\cos \left[\tan^{-1} \frac{1}{\sqrt{1+x^2}} \right] = \frac{1}{\sqrt{1 + \frac{1}{1+x^2}}} = \sqrt{\frac{1+x^2}{2+x^2}}. \quad \blacksquare$$

P12. Solve $\tan^{-1}(x+1) + \tan^{-1}(x-1) = \tan^{-1} \frac{8}{31}$.

Product test: $(x+1)(x-1) = x^2 - 1$. Taking tangent,

$$\frac{2x}{2-x^2} = \frac{8}{31} \Rightarrow 62x = 16 - 8x^2 \Rightarrow 4x^2 + 31x - 8 = 0 \Rightarrow x = \frac{1}{4} \text{ or } x = -8.$$

Audit: $x = -8$ gives both entries negative with product $63 > 1$, putting the sum in $(-\pi, -\frac{\pi}{2})$ — impossible for a positive target. $x = \frac{1}{4}$: product $= -\frac{15}{16} > -1$, direct formula applies, and indeed $\frac{\frac{5}{4} - \frac{3}{4}}{1 + \frac{15}{16}} = \frac{8}{31}$. **Only** $x = \frac{1}{4}$.

P.5 What the chapter text under-serves

The chapter drills branch-folding and identity manipulation to board-and-CUET depth, but the bank shows JEE spending most of its marks on three things the text barely touches: **solving equations and inequalities in inverse functions with explicit root-auditing** (half the Main questions since 2013), **telescoping arctangent/cotangent series**, and **piecewise-defined composites** like $2 \tan^{-1} x + \sin^{-1} \frac{2x}{1+x^2}$ that demand case analysis rather than a single quoted identity. A student who finishes this chapter should spend extra practice hours on those three formats specifically.

Chapter 3: Matrices

Drona Ed Tech | Class 12 Mathematics

Drona Ed Tech

Contents

Subtopic 00: Foundations & Representation	2
Section 1: Concept Intuition	2
Section 2: Key Definitions & Procedures	3
Section 3: Key Procedures (Worked Examples)	3
Section 4: Practice Exercises	4
Section 5: Crack the MCQ	4
Section 6: Pitfalls, Pro-Tips & Cheat Sheet	5
Subtopic 01: Matrix Algebra & Operations	6
Section 1: Concept Intuition	6
Section 2: Key Definitions & Conditions	7
Section 3: Key Procedures & Logical Framework	8
Section 4: Worked Examples	9
Section 5: Practice Exercises	11
Section 6: Crack the MCQ	11
Section 7: Common Student Pitfalls & Pro-Tips	12
Section 8: Cheat Sheet	12
Subtopic 02: Symmetric & Skew-Symmetric Matrices	14
Section 1: Concept Intuition	14
Section 2: Key Definitions & Properties	15
Section 3: CBSE Board Derivations	15
Section 4: Worked Examples	16
Section 5: Practice Exercises	18
Section 6: Crack the MCQ	19
Section 7: Common Student Pitfalls & Pro-Tips	19
Section 8: Cheat Sheet	20
Subtopic 03: Invertible Matrices & Row Operations	21
Section 1: Concept Intuition	21
Section 2: Key Definitions & Properties	22
Section 3: Derivation & Key Procedure	22
Section 4: Worked Examples	23
Section 5: Practice Exercises	25
Section 6: Crack the MCQ	25

Section 7: Common Student Pitfalls & Pro-Tips	26
Section 8: Cheat Sheet	27
Subtopic 04: Special Square Matrices & Chapter Synthesis	28
Section 1: Special Types of Square Matrices	28
Section 2: Trace of a Matrix	28
Section 3: Matrix Polynomials & the Equation $f(A) = O$	29
Section 4: Real-Life Application of Matrix Operations	29
Section 5: Mixed Chapter-Spanning Problems	30
Section 6: Chapter Synthesis	31

Subtopic 00: Foundations & Representation

Exam Relevance: Almost every **CBSE** paper opens the matrices section with a “construct the matrix” or “find the order / number of elements” question (1-2 marks) — easy marks that students lose only through carelessness. **JEE Main** turns the counting idea into quick objective questions (“how many matrices...”). This is the launch pad for the whole chapter. (*NEET does not test mathematics.*)

Section 1: Concept Intuition

Before you can add, multiply, or invert matrices, you have to be fluent in two very basic skills: **building** a matrix when someone hands you a rule, and **counting** how many matrices a description allows. Both sound trivial — and both are quietly examined every single year.

A matrix is just a filled-in grid, so if someone gives you a *recipe* for each cell — a formula a_{ij} that depends on the row number i and column number j — you simply walk the grid cell by cell and apply the recipe. It is exactly like filling a cinema seating chart where the ticket price depends on the row and the seat: row 1 seat 1 has one price, row 2 seat 3 another, all from the same pricing rule.

Think about it this way... The order $m \times n$ tells you the grid has $m \times n$ cells — no more, no fewer. So a matrix with, say, 12 elements could be a 1×12 strip, a 3×4 block, a 4×3 block, a 2×6 , and so on — every way of writing 12 as (rows) $\times$ (columns). Counting matrices is therefore really counting arrangements, and that is pure logic, not computation.

Limiting conditions & assumptions:

- i always indexes the **row** and j the **column** — never swap them, or every entry lands in the wrong place.
 - The number of elements equals $m \times n$ *exactly*; a matrix cannot be partially filled.
 - “Order” is an ordered pair: a 2×3 matrix and a 3×2 matrix are **different** even though both have 6 elements.
-

Section 2: Key Definitions & Procedures

Math note: no dimensional formula applies. The governing facts are the indexing convention and two counting principles.

Construction from a rule. Given $A = [a_{ij}]_{m \times n}$ with $a_{ij} = f(i, j)$, the entry in row i , column j is obtained by substituting those values of i and j into f .

Counting principle 1 — possible orders. A matrix with exactly N elements can have order $m \times n$ for **every** factor pair with $mn = N$. The number of possible orders equals the **number of divisors** of N . (If N is prime, only 2 orders exist: $1 \times N$ and $N \times 1$.)

Counting principle 2 — number of matrices. The number of distinct matrices of order $m \times n$ whose entries are chosen from a set of k values is

$$k^{mn},$$

because each of the mn cells is filled independently in k ways.

Section 3: Key Procedures (Worked Examples)

Example 1 — [CBSE Board Level]

Construct the 2×3 matrix $A = [a_{ij}]$ where $a_{ij} = 2i + 3j$.

Solution. Apply the rule to each of the 6 cells:

$$\begin{aligned} a_{11} &= 2(1) + 3(1) = 5, & a_{12} &= 2(1) + 3(2) = 8, & a_{13} &= 2(1) + 3(3) = 11, \\ a_{21} &= 2(2) + 3(1) = 7, & a_{22} &= 2(2) + 3(2) = 10, & a_{23} &= 2(2) + 3(3) = 13. \end{aligned}$$

$$A = \begin{bmatrix} 5 & 8 & 11 \\ 7 & 10 & 13 \end{bmatrix}.$$

Example 2 — [CBSE Board Level] (*piecewise rule*)

Construct the 2×2 matrix $A = [a_{ij}]$ where $a_{ij} = \begin{cases} i + j & i \leq j \\ i - j & i > j \end{cases}$.

Solution. Decide each cell by comparing i and j :

- a_{11} : $i = j$, so use $i + j = 2$.
- a_{12} : $i < j$, so use $i + j = 3$.
- a_{21} : $i > j$, so use $i - j = 1$.
- a_{22} : $i = j$, so use $i + j = 4$.

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}.$$

Example 3 — [JEE Main Level] (counting orders)

A matrix has 18 elements. How many possible orders does it have? What if it had 7 elements?

Solution. List factor pairs of 18: 1×18 , 2×9 , 3×6 , 6×3 , 9×2 , 18×1 — that is **6 orders** (equivalently, $18 = 2 \times 3^2$ has $(1+1)(2+1) = 6$ divisors). For 7 (prime): only 1×7 and 7×1 — **2 orders**. **Answer:** 6 orders for 18; 2 orders for 7.

Example 4 — [JEE Main Level] (counting matrices)

How many distinct 3×3 matrices can be formed using only the entries 0 and 1? How many 2×2 matrices using entries from $\{1, 2, 3\}$?

Solution. Each cell is filled independently.

- 3×3 with $k = 2$ choices: $2^9 = 512$.
 - 2×2 with $k = 3$ choices: $3^4 = 81$. **Answer:** 512 and 81.
-

Section 4: Practice Exercises

1. **[CBSE]** Construct the 3×2 matrix $A = [a_{ij}]$ with $a_{ij} = \frac{(i+2j)^2}{2}$.
2. **[CBSE]** A matrix has 24 elements. Write all possible orders. If it had 13 elements, how many orders are possible?
3. **[JEE Main]** How many 2×3 matrices can be formed with entries from $\{0, 1, 2\}$?
4. **[JEE Main]** Construct the 2×2 matrix $A = [a_{ij}]$ where $a_{ij} = |-3i + j|$.

Answers: 1. $\begin{bmatrix} 9/2 & 25/2 \\ 16/2 & 36/2 \\ 25/2 & 49/2 \end{bmatrix} = \begin{bmatrix} 4.5 & 12.5 \\ 8 & 18 \\ 12.5 & 24.5 \end{bmatrix}$. 2. 24 elements $\rightarrow 1 \times 24, 2 \times 12, 3 \times 8, 4 \times 6, 6 \times 4, 8 \times 3$,
= 8 orders; 13 is prime $\rightarrow$ **2 orders**. 3. $3^6 = 729$. 4. $\begin{bmatrix} 2 & 1 \\ 5 & 4 \end{bmatrix}$.

Section 5: Crack the MCQ

Q1. The number of all possible matrices of order 3×3 with each entry 0 or 2 is

- (a) 27 (b) 18 (c) 81 (d) 512

Correct: (d). Nine cells, each filled in 2 ways: $2^9 = 512$.

- (a) $27 = 3^3$ and (b) 18 come from miscounting the number of cells.
- (c) $81 = 3^4$ uses the 2×2 count or the wrong base.

Q2. If a matrix has 5 elements, the number of possible orders is

- (a) 1 (b) 2 (c) 5 (d) 10

Correct: (b). 5 is prime, so only 1×5 and 5×1 .

- (c) 5 counts elements, not orders; (d) 10 doubles incorrectly; (a) 1 forgets order is an ordered pair.
-

Section 6: Pitfalls, Pro-Tips & Cheat Sheet

Pitfalls. Swapping i and j while constructing; forgetting that $m \times n$ and $n \times m$ are different orders; counting *divisors wrong* for the order question; using the wrong base or exponent in k^{mn} .

Pro-Tip. For “possible orders,” just count divisors of N — factor N into primes and multiply (exponent+1)s. For “number of matrices,” it is always (choices per cell)^(number of cells).

Cheat Sheet

- **Construct:** $a_{ij} = f(i, j)$, row i then column j .
- **Possible orders of N -element matrix** = number of divisors of N .
- **Number of $m \times n$ matrices over a k -set** = k^{mn} .
- *Memory aid:* “**i** before **j** — row before column.”

Subtopic 01: Matrix Algebra & Operations

Exam Relevance: This is near-guaranteed marks on **CBSE Boards**, where equality of matrices and the three operations show up as 1–3 mark questions almost every year, frequently as “find the unknowns” from a matrix equation. **JEE Main** usually carries one question on matrix multiplication, powers, or the properties (especially non-commutativity). **JEE Advanced** rarely tests this alone — it is woven into larger problems with determinants or inverses. (*NEET does not test mathematics, so this subtopic does not apply to NEET aspirants.*)

Section 1: Concept Intuition

Imagine you walk up to a railway reservation board at a busy station like New Delhi. Down the left side you have a list of trains — Rajdhani, Shatabdi, Duronto. Across the top you have the travel classes — 1A, 2A, 3A, SL. Inside the grid, every cell holds one number: the fare for *that train* in *that class*. That grid, stripped down to just the numbers, **is a matrix**.

A matrix is simply a rectangular arrangement of numbers in rows and columns. But here is the idea that everything else hangs on: **position carries meaning**. The number sitting in row 2, column 3 is not interchangeable with the number in row 3, column 2 — one might be the 2A fare of the Shatabdi, the other the SL fare of the Duronto. We write the entry in the i -th row and j -th column as a_{ij} , and the whole matrix as $A = [a_{ij}]_{m \times n}$, where m is the number of rows and n the number of columns. The pair $m \times n$ is called the **order** of the matrix.

Figure 3.1 A 3×4 grid labelled “A”. Rows on the left labelled R1, R2, R3; columns on top labelled C1, C2, C3, C4. The cell at row 2, column 3 is highlighted and labelled a_{23} . A caption reads “order = 3×4 (rows first, columns second)”. (see next page)

Think about it this way... Suppose the railways announce a flat festival surcharge — a second board with exactly the same trains and classes, holding the extra rupees per cell. To get the new fare board, you just add the two boards cell by cell. That is **matrix addition**, and notice the hidden condition: it only works because both boards have the *same layout*. You could never add a 3-train board to a 5-train board — there would be cells with no partner. Now suppose every fare goes up by 10%. You multiply every single cell by 1.1. That is **scalar multiplication** — one ordinary number stretching the entire grid uniformly.

The third operation, **matrix multiplication**, is the one that trips students up, because it is *not* done cell by cell. Picture a sweet shop: one matrix lists how many kilos of each mithai a customer buys, another lists the price per kilo, and multiplying them correctly combines a *row of quantities* with a *column of prices* to produce a total bill. That “row meets column” handshake is the heart of matrix multiplication, and it is why the operation cares so much about dimensions lining up.

A 3x4 grid representing a matrix A . The grid is enclosed in large square brackets on the left and right sides. The left side is labeled A . The rows are labeled $R1$, $R2$, and $R3$ on the left. The columns are labeled $C1$, $C2$, $C3$, and $C4$ on top. The cell at row 2, column 3 is highlighted with a thick orange border and contains the label a_{23} .

	$C1$	$C2$	$C3$	$C4$
$R1$				
$R2$			a_{23}	
$R3$				

order = 3 × 4 (rows first, columns second)

Figure 3.1 A 3x4 grid labelled "A". Rows on the left labelled $R1$, $R2$, $R3$; columns on top labelled $C1$, $C2$, $C3$, $C4$. The cell at row 2, column 3 is highlighted and labelled a_{23} . A caption reads "order = 3 × 4 (rows first, columns second)".

Limiting conditions & assumptions — when the rules bite:

- **Addition and subtraction** demand *identical order*. Different orders $\rightarrow$ the operation is simply undefined, not zero.
- **Multiplication** demands **conformability**: the number of columns of the first matrix must equal the number of rows of the second. AB can be perfectly well-defined while BA does not exist at all.
- Matrix multiplication is **non-commutative**: in general $AB \neq BA$, even when both exist.
- A genuine shock from ordinary algebra: $AB = O$ does **not** force $A = O$ or $B = O$. Two non-zero matrices can multiply to the zero matrix.
- **Cancellation fails**: $AB = AC$ does **not** let you conclude $B = C$.

Hold on to that last cluster — most exam traps live there.

Section 2: Key Definitions & Conditions

Math note: matrices are not physical quantities, so there is no dimensional formula to attach. The genuine “sanity check” for a matrix operation is its **order/conformability condition** — listed below for every operation.

Types of matrices

- **Row matrix** — one row, order $1 \times n$. **Column matrix** — one column, order $m \times 1$.
- **Square matrix** — equal rows and columns, order $n \times n$.
- **Diagonal matrix** — square, with every off-diagonal entry zero: $a_{ij} = 0$ for $i \neq j$.
- **Scalar matrix** — a diagonal matrix whose diagonal entries are all equal.
- **Identity matrix** I_n — a scalar matrix with every diagonal entry 1.
- **Null / zero matrix** O — every entry zero (any order).
- **Triangular matrix** — upper (zeros below the diagonal) or lower (zeros above).
- **Trace** of a square matrix: $\text{tr}(A) = \sum_{i=1}^n a_{ii}$ (sum of diagonal entries).

Equality of matrices: $A = B$ iff (i) they have the *same order*, **and** (ii) $a_{ij} = b_{ij}$ for every i, j .

The three operations

Operation	Defined when	Order of result	Rule
$A + B$	same order $m \times n$	$m \times n$	$(A+B)_{ij} = a_{ij} + b_{ij}$
kA (scalar k)	always	$m \times n$	$(kA)_{ij} = k a_{ij}$
AB	$\text{cols}(A) = \text{rows}(B)$	$m \times p$	$(AB)_{ij} = \sum_{r=1}^n a_{ir} b_{rj}$

(For AB : A is $m \times n$ and B is $n \times p$.)

Properties — and their conditions

- **Addition:** commutative $A + B = B + A$; associative $(A + B) + C = A + (B + C)$; additive identity $A + O = A$; additive inverse $A + (-A) = O$. (All require equal orders.)
 - **Scalar multiplication:** $k(A + B) = kA + kB$; $(k + l)A = kA + lA$; $(kl)A = k(lA)$.
 - **Matrix multiplication:** associative $A(BC) = (AB)C$; distributive $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$; identity $I_m A = A = A I_n$. **Not commutative.** And crucially, $AB = O \not\Rightarrow A = O$ or $B = O$.
-

Section 3: Key Procedures & Logical Framework

Matrix operations are *procedural*, so instead of a derivation, here is the reasoning behind each procedure.

Procedure A — Solving a matrix equation (the CBSE bread-and-butter).

1. Confirm both sides have the same order (otherwise no solution exists).
2. **Equate corresponding entries** — equality of matrices turns one matrix equation into a *system of ordinary scalar equations*.
3. Solve the scalar system. *Why this works:* equality is defined entry-by-entry, so a single 2×2 equation is secretly four scalar equations bundled together.

Procedure B — Multiplying two matrices.

1. **Conformability check.** Write the orders side by side: $(m \times \underline{n})(\underline{n} \times p)$. The two *inner* numbers must match; if they do, the *outer* numbers give the result's order, $m \times p$.
2. To get the entry $(AB)_{ij}$, take **row i of A** and **column j of B** , multiply them term by term, and add: $\sum_r a_{ir} b_{rj}$.
3. Repeat for every cell.

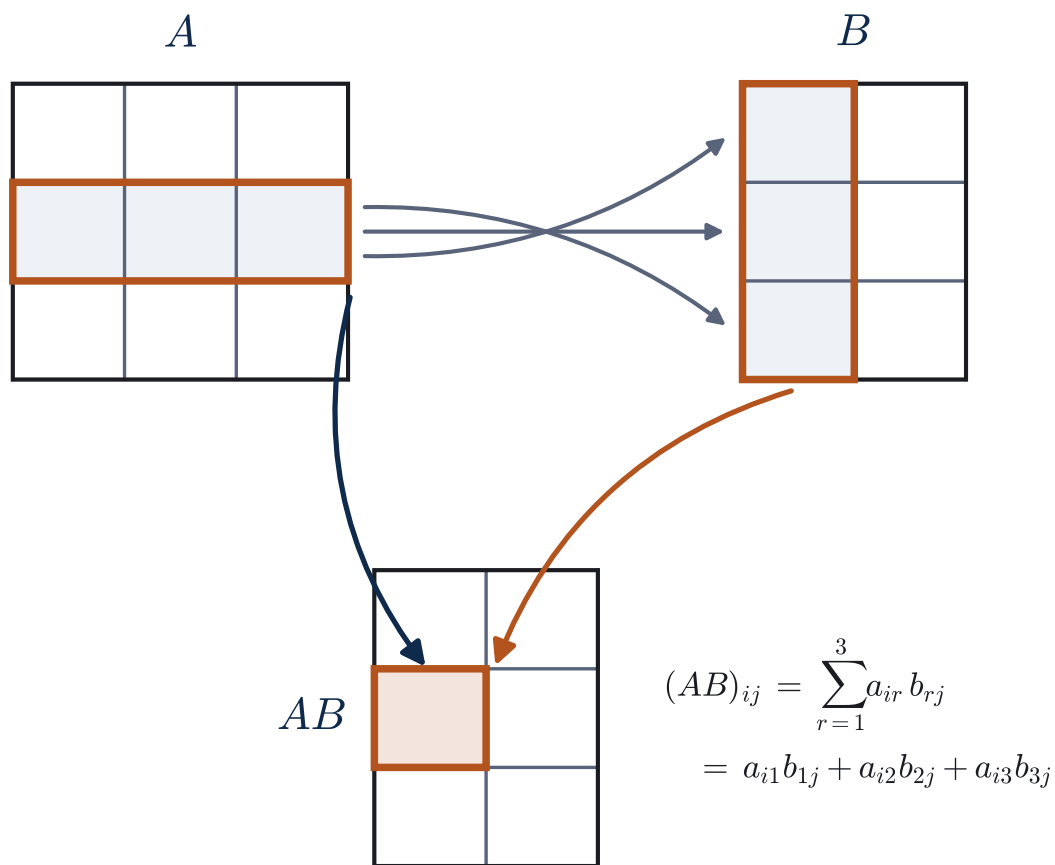
Figure 3.2 Row i of matrix A (a horizontal strip) "folding down" onto column j of matrix B (a vertical strip), with paired arrows $a_{i1} * b_{1j}, a_{i2} * b_{2j}, \dots$ summed to land in cell (i,j) of the product AB .
Caption: "Row meets column $\rightarrow$ one number." (see next page)

Why row-times-column, and not cell-by-cell? Because matrix multiplication encodes the **chaining of two processes** — first apply B , then apply A . The (i, j) entry must gather *every* path that connects input j to output i , which is exactly the sum over the shared middle index r . That shared index is why the inner dimensions must agree.

Why multiplication is non-commutative. Two reasons, one weaker and one stronger.

- *Dimensional:* if A is 2×3 and B is 3×2 , then AB is 2×2 but BA is 3×3 — they cannot even be compared.
- *Entry-level:* even for square matrices where both exist, the numbers differ. Take

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}.$$



Row meets column → one number.

Figure 3.2 Row i of matrix A (a horizontal strip) “folding down” onto column j of matrix B (a vertical strip), with paired arrows $a_{i1}b_{1j}$, $a_{i2}b_{2j}$, ... summed to land in cell (i,j) of the product AB . Caption: “Row meets column → one number.”

Then $AB = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ but $BA = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$. Different matrices — order of operation matters.

Why $AB = O$ without either being zero. Consider

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O.$$

Geometrically, A keeps only the first coordinate and B keeps only the second; chaining them throws everything away. Neither matrix is zero, yet their product is — so the familiar “if a product is zero, a factor is zero” rule is *false* for matrices.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Find a, b, c, d if

$$2 \begin{bmatrix} a & b \\ c & d \end{bmatrix} + 3 \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ 6 & 16 \end{bmatrix}.$$

Solution. First carry out the scalar multiplication and addition on the left:

$$\begin{bmatrix} 2a & 2b \\ 2c & 2d \end{bmatrix} + \begin{bmatrix} 3 & -3 \\ 0 & 6 \end{bmatrix} = \begin{bmatrix} 2a+3 & 2b-3 \\ 2c & 2d+6 \end{bmatrix}.$$

Equating corresponding entries with the right-hand side:

$$\begin{aligned} 2a + 3 = 5 &\Rightarrow a = 1, & 2b - 3 = 1 &\Rightarrow b = 2, \\ 2c = 6 &\Rightarrow c = 3, & 2d + 6 = 16 &\Rightarrow d = 5. \end{aligned}$$

Answer: $a = 1, b = 2, c = 3, d = 5$.

Example 2 — [JEE Main Level — Speed Trap]

Let A be 3×2 , B be 2×4 , C be 4×3 . Which of $AB, BA, BC, (AB)C, A(BC)$ are defined, and what are their orders?

The slow way is to start computing entries — a waste. **The fast way** is to chain orders and check inner matches only:

- AB : $(3 \times \underline{2})(\underline{2} \times 4) \Rightarrow 3 \times 4 \checkmark$
- BA : $(2 \times \underline{4})(\underline{3} \times 2) \Rightarrow 4 \neq 3$, **not defined** ✗
- BC : $(2 \times \underline{4})(\underline{4} \times 3) \Rightarrow 2 \times 3 \checkmark$
- $(AB)C$: $(3 \times \underline{4})(\underline{4} \times 3) \Rightarrow 3 \times 3 \checkmark$
- $A(BC)$: $(3 \times \underline{2})(\underline{2} \times 3) \Rightarrow 3 \times 3 \checkmark$

Answer: AB is 3×4 , BA is undefined, BC is 2×3 , and both $(AB)C$ and $A(BC)$ are 3×3 . Note they agree in order — a quiet confirmation of associativity. **Trap:** candidates who reflexively “compute” lose a full minute; conformability is settled by *dimensions alone*.

Example 3 — [JEE Main Level] (multiplication + pattern/equality)

If $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $A^n = \begin{bmatrix} 1 & 2026 \\ 0 & 1 \end{bmatrix}$, find n .

Solution. Build the pattern from the bottom up.

$$A^2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad A^3 = A^2 A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}.$$

The top-right entry counts up by one each time, so we conjecture

$$A^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}.$$

Verify by induction: if $A^k = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$, then $A^{k+1} = A^k A = \begin{bmatrix} 1 & k+1 \\ 0 & 1 \end{bmatrix}$, completing the step. Now equate with the given matrix: $n = 2026$. **Answer:** $n = 2026$.

Example 4 — [JEE Advanced Level]

Let $A = \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix}$. Evaluate $S = A + A^2 + A^3 + \dots + A^{2025}$.

Solution. First probe the structure of A by squaring it:

$$A^2 = \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} = \begin{bmatrix} 4-3 & -2+2 \\ 6-6 & -3+4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I.$$

So $A^2 = I$ — the matrix is its own “square root of identity.” This collapses every power into one of two values:

$$A^{\text{even}} = I, \quad A^{\text{odd}} = A.$$

Among the exponents $1, 2, \dots, 2025$ there are 1013 odd values ($1, 3, \dots, 2025$) and 1012 even values. Therefore

$$S = 1013 A + 1012 I = 1013 \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} + 1012 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Computing entrywise:

$$S = \begin{bmatrix} 2026 + 1012 & -1013 \\ 3039 & -2026 + 1012 \end{bmatrix} = \begin{bmatrix} 3038 & -1013 \\ 3039 & -1014 \end{bmatrix}.$$

Answer: $S = \begin{bmatrix} 3038 & -1013 \\ 3039 & -1014 \end{bmatrix}$. The whole problem turns on spotting $A^2 = I$ early — brute force over 2025 powers is hopeless.

Section 5: Practice Exercises

1. **[CBSE]** If $\begin{bmatrix} x+y & 2 \\ 5 & z-1 \end{bmatrix} = \begin{bmatrix} 7 & 2 \\ 5 & 4 \end{bmatrix}$ and $x - y = 1$, find x, y, z .
2. **[CBSE]** Given $A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 2 & -1 \end{bmatrix}$, compute $2A - 3B$.
3. **[JEE Main]** A is 2×3 , B is 3×3 , C is 3×1 . State the order of ABC , and say whether CAB is defined.
4. **[JEE Main]** If $A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$, find A^5 .
5. **[JEE Advanced]** Find a non-zero 2×2 matrix B such that $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} B = O$.

Answers: 1. $x = 4, y = 3, z = 5$. 2. $\begin{bmatrix} 2 & 1 \\ -8 & 9 \end{bmatrix}$. 3. ABC is 2×1 ; CAB is **not** defined (inner mismatch $1 \neq 2$). 4. $\begin{bmatrix} 243 & 0 \\ 0 & 32 \end{bmatrix}$. 5. Any B whose columns are multiples of $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$, e.g. $B = \begin{bmatrix} -2 & -4 \\ 1 & 2 \end{bmatrix}$.

Section 6: Crack the MCQ

Q1. If A is of order 3×4 and B is of order 4×2 , then AB has order

- (a) 3×2 (b) 4×4 (c) 2×3 (d) not defined

Correct: (a). Inner dimensions $4 = 4$ match, so AB exists with the outer order 3×2 .

- (b) 4×4 : picked by multiplying the two “4”s — confusing the inner (shared) dimension with the result.
- (c) 2×3 : the order of BA -type thinking, i.e. reading the result backwards.
- (d) *not defined*: the trap for a student who checked the *outer* numbers (3 vs 2) instead of the inner ones.

Q2. For two square matrices A and B of the same order, which is *always* true?

- (a) $AB = BA$ (b) $(A + B)^2 = A^2 + 2AB + B^2$ (c) $(A + B)^2 = A^2 + AB + BA + B^2$
(d) $(AB)^2 = A^2B^2$

Correct: (c). Expanding honestly, $(A + B)^2 = (A + B)(A + B) = A^2 + AB + BA + B^2$, with AB and BA kept separate.

- (a): false — multiplication is non-commutative.
- (b): the classic error of writing $AB + BA = 2AB$, which assumes $AB = BA$.
- (d): $(AB)^2 = ABAB$, which equals A^2B^2 only if $BA = AB$ — again the commutativity assumption.

Q3. If $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ and $B \neq O$ satisfies $AB = O$, then the columns of B are proportional to

(a) $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ (b) $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ (d) $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$

Correct: (b). Each column $\begin{bmatrix} x \\ y \end{bmatrix}$ of B must satisfy $A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, i.e. $x + 2y = 0 \Rightarrow x = -2y$, giving the direction $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$.

- (a), (c), (d): substituting each into $x + 2y$ gives 3, 5, 4 respectively — none zero, so none lands in the required set. They tempt students who guess a “nice” vector instead of solving $AB = O$ column by column.

Q4. If $\begin{bmatrix} a+b & 6 \\ 8 & a-b \end{bmatrix} = 8 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, then (a, b) =

(a) (3, 2) (b) (2, 3) (c) (4, 1) (d) (5, 1)

Correct: (a). Equating entries: $a + b = 5$ and $a - b = 1$, so $a = 3$, $b = 2$.

- (b) (2, 3): the swapped answer — gives $a - b = -1$, not 1.
- (c) (4, 1): satisfies $a + b = 5$ but $a - b = 3 \neq 1$; the student solved only one equation.
- (d) (5, 1): gives $a + b = 6 \neq 5$; the student matched the wrong cell.

Section 7: Common Student Pitfalls & Pro-Tips

- **Multiplying element-by-element.** AB is *not* $[a_{ij}b_{ij}]$. It is the row-times-column sum. Element-wise multiplication is a different (non-syllabus) operation entirely.
- **Assuming $AB = BA$.** This silently corrupts $(A + B)^2$, $(A - B)(A + B)$, and $(AB)^2$. Always keep AB and BA as distinct objects.
- **“Product is zero, so a factor is zero.”** False for matrices — $AB = O$ with $A, B \neq O$ is routine. Never “cancel” a matrix out of an equation.
- **Adding matrices of different orders**, or declaring AB “not defined” by checking outer instead of inner dimensions.

Pro-Tip (conformability in 2 seconds): write the orders touching — $(m \times n)(n \times p)$. If the two middle numbers kiss, the product lives and inherits the outer pair $m \times p$. **Pro-Tip (powers):** for diagonal or triangular matrices, never brute-force A^n — diagonal entries just raise to the n , and unitriangular matrices follow a clean linear pattern (Example 3). And always test whether $A^2 = I$ or $A^2 = A$ first; it can collapse an entire summation (Example 4).

Section 8: Cheat Sheet

- **Order $m \times n$:** rows first, columns second. Entry a_{ij} = row i , column j .
- **Equality:** same order **and** every entry equal $\rightarrow$ splits into scalar equations.
- **Addition / subtraction:** same order only; done entrywise. Commutative & associative.

- **Scalar:** $kA = [ka_{ij}]$ — stretches every entry.
- **Multiplication:** AB defined iff $\text{cols}(A) = \text{rows}(B)$; $(AB)_{ij} = \sum_r a_{ir}b_{rj}$; result $m \times p$. **Non-commutative.** $IA = A = AI$. $AB = O \not\Rightarrow A = O$ or $B = O$.

Key formulas

$$(A+B)_{ij} = a_{ij} + b_{ij}, \quad (kA)_{ij} = k a_{ij}, \quad (AB)_{ij} = \sum_{r=1}^n a_{ir}b_{rj}, \quad \text{tr}(A) = \sum_i a_{ii}.$$

Memory aids

- “*Inner match, outer survive*” — conformability + result order in four words.
- “*Rows are Roman, columns are Catholic*” (R before C) — fixes the order of indices a_{ij} .
- “*Matrices don’t swap*” — $AB \neq BA$, so never assume the binomial $2AB$ shortcut.

Subtopic 02: Symmetric & Skew-Symmetric Matrices

Exam Relevance: A perennial **CBSE Board** favourite — “express a matrix as the sum of a symmetric and a skew-symmetric matrix” is almost a fixed 3–4 mark question, and short 1-mark identifications appear too. **JEE Main** reliably asks one property-based question (nature of $AB \pm BA$, powers of a skew matrix, counting distinct elements). **JEE Advanced** folds these properties into larger problems with transpose and (later) inverse/determinant. (*NEET does not test mathematics.*)

Section 1: Concept Intuition

Think of a distance chart printed on the back of an old Indian road atlas — the kind that lists cities down the side and the same cities across the top, with each cell holding the road distance between them. Delhi to Jaipur reads the same whether you look it up as “Delhi row, Jaipur column” or “Jaipur row, Delhi column.” The chart is perfectly **mirror-balanced across its diagonal**. A matrix with exactly this balance — where the entry at (i, j) equals the entry at (j, i) — is called a **symmetric matrix**.

Figure 3.3 A 4x4 grid with the main (top-left to bottom-right) diagonal drawn as a mirror line. Pairs of cells equidistant from the diagonal are shaded the same colour and labelled $a_{ij} = a_{ji}$, illustrating reflection symmetry. Caption: “Symmetric: fold along the diagonal and the two halves match.” (see next page)

Now imagine a different kind of chart — a “who owes whom” ledger in a group of friends splitting a trip. If Ravi owes Sita Rs. 500, then from Sita’s side the entry is -500 . The cell (i, j) is the exact negative of (j, i) . And what does the diagonal mean? “How much Ravi owes Ravi” — necessarily zero. A matrix where $a_{ij} = -a_{ji}$ (which forces every diagonal entry to be 0) is a **skew-symmetric matrix**.

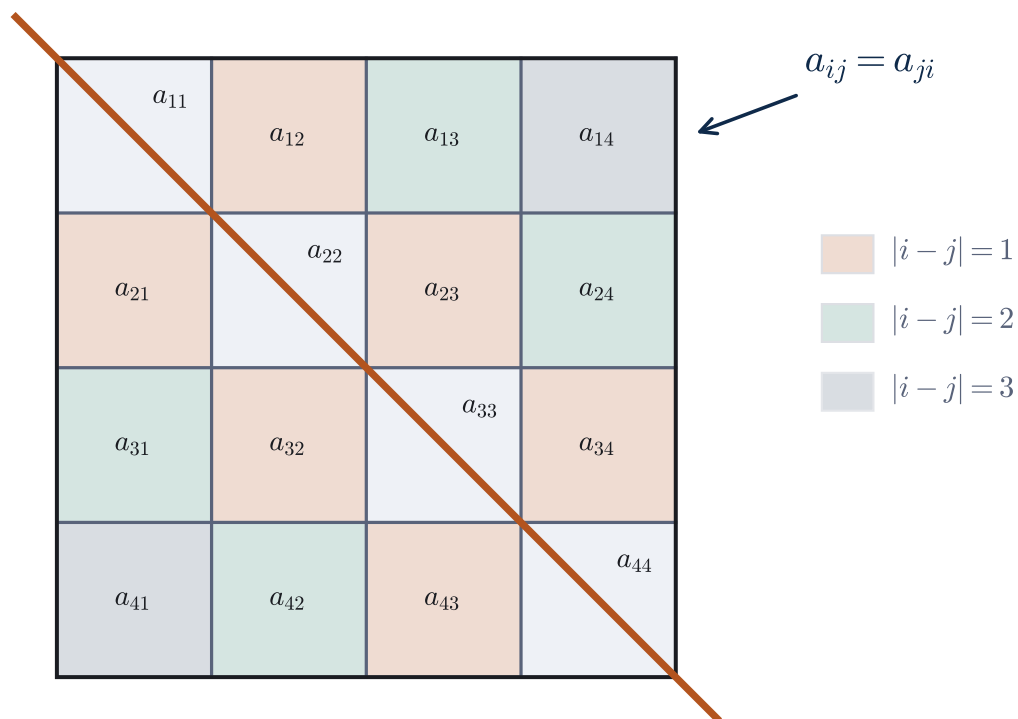
The tool that makes both ideas precise is the **transpose**. The transpose of A , written A^T (or A'), is the matrix you get by turning rows into columns — reflecting A across its main diagonal. With it the definitions become crisp:

$$A \text{ is symmetric} \iff A^T = A, \quad A \text{ is skew-symmetric} \iff A^T = -A.$$

Think about it this way... Every square matrix is secretly a blend of these two pure types — a “balanced” part and an “anti-balanced” part — just as any number can be split into an even-function-like and odd-function-like piece. Squeeze out the symmetric part with $\frac{1}{2}(A + A^T)$ and the skew part with $\frac{1}{2}(A - A^T)$, and they add back to A exactly. That single idea powers the most-asked board question in this entire chapter.

Limiting conditions & assumptions:

- Symmetric and skew-symmetric matrices are **defined only for square matrices** (a non-square matrix can never equal its own transpose — the orders wouldn’t match).



Symmetric: fold along the diagonal and the two halves match.

Figure 3.3 A 4x4 grid with the main (top-left to bottom-right) diagonal drawn as a mirror line. Pairs of cells equidistant from the diagonal are shaded the same colour and labelled $a_{ij} = a_{ji}$, illustrating reflection symmetry. Caption: "Symmetric: fold along the diagonal and the two halves match."

- The diagonal of a skew-symmetric matrix is **always** zero. A “skew” matrix with a non-zero diagonal entry is impossible.
 - The **product** of two symmetric matrices is symmetric **only if they commute** ($AB = BA$) — not in general.
 - The transpose **reverses order** under multiplication: $(AB)^T = B^T A^T$, not $A^T B^T$. Forgetting the reversal is the silent killer in proofs.
-

Section 2: Key Definitions & Properties

Math note: matrices carry no dimensional formula. The governing conditions here are **squareness** and the **transpose relation**, stated for every result.

Transpose — the foundation (for conformable matrices and scalar k):

$$(A^T)^T = A, \quad (A + B)^T = A^T + B^T, \quad (kA)^T = kA^T, \quad (AB)^T = B^T A^T, \quad I^T = I.$$

Definitions (square matrix $A = [a_{ij}]_{n \times n}$):

- **Symmetric:** $A^T = A$, i.e. $a_{ij} = a_{ji}$ for all i, j .
- **Skew-symmetric:** $A^T = -A$, i.e. $a_{ij} = -a_{ji}$; consequently $a_{ii} = 0$.

Core properties

- For any square A : $A + A^T$ is symmetric, and $A - A^T$ is skew-symmetric.
 - For any matrix A (any order): AA^T and $A^T A$ are symmetric (and square).
 - **Powers of a symmetric matrix** are symmetric.
 - **Powers of a skew-symmetric matrix:** *even* powers are symmetric, *odd* powers are skew-symmetric.
 - If A, B are symmetric: $A \pm B$ and $AB + BA$ are symmetric; $AB - BA$ is skew-symmetric; AB is symmetric **iff** $AB = BA$.
 - If A, B are skew-symmetric: $A \pm B$ and $AB - BA$ are skew-symmetric; $AB + BA$ is symmetric.
 - The **only** matrix that is both symmetric and skew-symmetric is the null matrix O .
 - **Counting distinct elements** (order n): a symmetric matrix has at most $\frac{n(n+1)}{2}$ free entries; a skew-symmetric matrix has at most $\frac{n(n-1)}{2}$ (the diagonal is forced to 0).
-

Section 3: CBSE Board Derivations

This subtopic carries three results worth deriving in full — they are exactly what the reasoning marks reward.

Derivation 1 — The diagonal of a skew-symmetric matrix is zero. Let A be skew-symmetric, so $a_{ij} = -a_{ji}$ for all i, j . Put $i = j$:

$$a_{ii} = -a_{ii} \Rightarrow 2a_{ii} = 0 \Rightarrow a_{ii} = 0.$$

Every diagonal entry must vanish. (The converse fails: a zero diagonal alone does not make a matrix skew-symmetric.)

Derivation 2 — Every square matrix is uniquely the sum of a symmetric and a skew-symmetric matrix. *Existence.* Let A be any square matrix. Define

$$P = \frac{1}{2}(A + A^T), \quad Q = \frac{1}{2}(A - A^T).$$

Check P is symmetric, using the transpose laws (note $(A^T)^T = A$):

$$P^T = \frac{1}{2}(A + A^T)^T = \frac{1}{2}(A^T + A) = P. \checkmark$$

Check Q is skew-symmetric:

$$Q^T = \frac{1}{2}(A - A^T)^T = \frac{1}{2}(A^T - A) = -\frac{1}{2}(A - A^T) = -Q. \checkmark$$

And clearly $P + Q = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T) = A$. So a decomposition exists.

Uniqueness. Suppose $A = P_1 + Q_1$ with P_1 symmetric and Q_1 skew-symmetric. Transposing: $A^T = P_1^T + Q_1^T = P_1 - Q_1$. Adding and subtracting these two equations gives $P_1 = \frac{1}{2}(A + A^T) = P$ and $Q_1 = \frac{1}{2}(A - A^T) = Q$. The pieces are forced — the decomposition is unique.

Derivation 3 — Maximum number of distinct entries. A symmetric matrix of order n is fully determined by its diagonal (n entries) plus the entries strictly above the diagonal ($\frac{n(n-1)}{2}$ of them); everything below is a mirror copy. Total:

$$n + \frac{n(n-1)}{2} = \frac{n(n+1)}{2}.$$

For a skew-symmetric matrix the diagonal is fixed at 0, so only the $\frac{n(n-1)}{2}$ above-diagonal entries are free.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Express $A = \begin{bmatrix} 2 & 4 & -6 \\ 0 & 1 & 5 \\ 2 & -3 & 4 \end{bmatrix}$ **as the sum of a symmetric and a skew-symmetric matrix.**

Solution. First write the transpose:

$$A^T = \begin{bmatrix} 2 & 0 & 2 \\ 4 & 1 & -3 \\ -6 & 5 & 4 \end{bmatrix}.$$

Symmetric part $P = \frac{1}{2}(A + A^T)$:

$$A + A^T = \begin{bmatrix} 4 & 4 & -4 \\ 4 & 2 & 2 \\ -4 & 2 & 8 \end{bmatrix} \Rightarrow P = \begin{bmatrix} 2 & 2 & -2 \\ 2 & 1 & 1 \\ -2 & 1 & 4 \end{bmatrix}.$$

Skew-symmetric part $Q = \frac{1}{2}(A - A^T)$:

$$A - A^T = \begin{bmatrix} 0 & 4 & -8 \\ -4 & 0 & 8 \\ 8 & -8 & 0 \end{bmatrix} \Rightarrow Q = \begin{bmatrix} 0 & 2 & -4 \\ -2 & 0 & 4 \\ 4 & -4 & 0 \end{bmatrix}.$$

Check: $P + Q = A$ ✓, $P^T = P$ ✓, $Q^T = -Q$ ✓ (diagonal of Q all zero). **Answer:**

$$A = \begin{bmatrix} 2 & 2 & -2 \\ 2 & 1 & 1 \\ -2 & 1 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 2 & -4 \\ -2 & 0 & 4 \\ 4 & -4 & 0 \end{bmatrix}.$$

Example 2 — [JEE Main Level — Speed Trap]

A and B are symmetric matrices of the same order. Without computing entries, classify $AB - BA$ and ABA as symmetric, skew-symmetric, or neither.

Solution. The fast tool is the **transpose-flip**: transpose the expression, push the T inside using the reversal law, and use $A^T = A$, $B^T = B$.

For $AB - BA$:

$$(AB - BA)^T = (AB)^T - (BA)^T = B^T A^T - A^T B^T = BA - AB = -(AB - BA).$$

It equals its own negative $\rightarrow$ **skew-symmetric**.

For ABA :

$$(ABA)^T = A^T B^T A^T = ABA.$$

It equals itself $\rightarrow$ **symmetric**.

Answer: $AB - BA$ is skew-symmetric; ABA is symmetric. **Trap:** students who try to expand AB and BA in coordinates burn time and risk arithmetic slips; the transpose-flip settles it in two lines.

Example 3 — [JEE Main Level] (classification condition + solving)

Find x, y, z **so that** $A = \begin{bmatrix} 0 & 2x - 3 & 4 \\ -1 & 0 & y + 2 \\ -4 & -7 & z \end{bmatrix}$ **is skew-symmetric**.

Solution. A skew-symmetric matrix needs $a_{ij} = -a_{ji}$ for all pairs, and zero on the diagonal.

- Diagonal: $a_{33} = z$ must be 0 $\Rightarrow \boxed{z = 0}$ (the other diagonal entries are already 0).
- Pair (1, 2) vs (2, 1): $a_{12} = -a_{21} \Rightarrow 2x - 3 = -(-1) = 1 \Rightarrow 2x = 4 \Rightarrow \boxed{x = 2}$.
- Pair (2, 3) vs (3, 2): $a_{23} = -a_{32} \Rightarrow y + 2 = -(-7) = 7 \Rightarrow \boxed{y = 5}$.
- Consistency on (1, 3): $a_{13} = -a_{31} \Rightarrow 4 = -(-4) = 4$ ✓ — already satisfied.

Answer: $x = 2$, $y = 5$, $z = 0$.

Example 4 — [JEE Advanced Level]

Show that the square of every 2×2 skew-symmetric matrix is a scalar matrix of the form $-a^2 I$, and hence find all 2×2 skew-symmetric matrices A satisfying $A^2 = -9I$.

Solution. A 2×2 skew-symmetric matrix has zero diagonal and opposite off-diagonals, so the general form is

$$A = \begin{bmatrix} 0 & a \\ -a & 0 \end{bmatrix}, \quad a \in \mathbb{R}.$$

Square it:

$$A^2 = \begin{bmatrix} 0 & a \\ -a & 0 \end{bmatrix} \begin{bmatrix} 0 & a \\ -a & 0 \end{bmatrix} = \begin{bmatrix} -a^2 & 0 \\ 0 & -a^2 \end{bmatrix} = -a^2 I.$$

This proves the claim for every such A . Now impose $A^2 = -9I$:

$$-a^2 I = -9I \Rightarrow a^2 = 9 \Rightarrow a = \pm 3.$$

Answer: exactly two matrices,

$$A = \begin{bmatrix} 0 & 3 \\ -3 & 0 \end{bmatrix} \quad \text{or} \quad A = \begin{bmatrix} 0 & -3 \\ 3 & 0 \end{bmatrix}.$$

Notice the elegance: a skew-symmetric 2×2 matrix behaves like an “imaginary unit scaled by a ” — its square is a negative multiple of identity, mirroring $(ai)^2 = -a^2$.

Section 5: Practice Exercises

1. **[CBSE]** Express $A = \begin{bmatrix} 1 & 5 \\ -3 & 2 \end{bmatrix}$ as the sum of a symmetric and a skew-symmetric matrix.
2. **[CBSE]** For what value of x is $\begin{bmatrix} 0 & x-2 \\ 4 & 0 \end{bmatrix}$ skew-symmetric?
3. **[JEE Main]** Find the maximum number of distinct entries possible in (i) a symmetric and (ii) a skew-symmetric matrix of order 5.
4. **[JEE Main]** If A is a skew-symmetric matrix of order 3, state the nature (symmetric / skew-symmetric) of A^4 and of A^7 .
5. **[JEE Advanced]** A and B are symmetric matrices of the same order. Prove that AB is symmetric **if and only if** $AB = BA$.

Answers: 1. $P = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$, $Q = \begin{bmatrix} 0 & 4 \\ -4 & 0 \end{bmatrix}$. 2. $x = -2$. 3. (i) $\frac{5 \cdot 6}{2} = 15$; (ii) $\frac{5 \cdot 4}{2} = 10$.
4. A^4 symmetric, A^7 skew-symmetric. 5. (*Proof — forward:*) $(AB)^T = B^T A^T = BA$; symmetric means $(AB)^T = AB$, so $AB = BA$. (*Reverse:*) if $AB = BA$ then $(AB)^T = BA = AB$, so AB is symmetric.

Section 6: Crack the MCQ

Q1. The diagonal entries of a skew-symmetric matrix are

- (a) all equal to 1 (b) all equal to 0 (c) all equal to one another but non-zero (d) impossible to determine

Correct: (b). Setting $i = j$ in $a_{ij} = -a_{ji}$ gives $a_{ii} = -a_{ii} \Rightarrow a_{ii} = 0$.

- (a): confuses skew-symmetric with the identity matrix.
- (c): a half-memory — “diagonal entries are equal” is true (all are 0) but the “non-zero” wrecks it.
- (d): the entries are fully forced, not undetermined.

Q2. A matrix that is both symmetric and skew-symmetric must be

- (a) the identity matrix (b) a diagonal matrix (c) the null matrix (d) a scalar matrix

Correct: (c). Both conditions give $A^T = A$ and $A^T = -A$, so $A = -A \Rightarrow 2A = O \Rightarrow A = O$.

- (a), (d): the identity/scalar matrices are symmetric but **not** skew (non-zero diagonal).
- (b): a non-zero diagonal matrix is symmetric but not skew; only the zero matrix qualifies.

Q3. For any square matrix A , the matrix $A - A^T$ is always

- (a) symmetric (b) skew-symmetric (c) the null matrix (d) the identity matrix

Correct: (b). $(A - A^T)^T = A^T - A = -(A - A^T)$.

- (a): the symmetric companion is $A + A^T$, not $A - A^T$.
- (c): $A - A^T = O$ only when A is already symmetric — not “always.”
- (d): no reason for it to be I .

Q4. The maximum number of distinct entries in a symmetric matrix of order 4 is

- (a) 16 (b) 10 (c) 6 (d) 12

Correct: (b). $\frac{n(n+1)}{2} = \frac{4 \cdot 5}{2} = 10$.

- (a) 16: counts all n^2 entries, ignoring that symmetry duplicates the off-diagonal pairs.
 - (c) 6: uses $\frac{n(n-1)}{2}$ — the strictly-above-diagonal count, forgetting the 4 diagonal entries.
 - (d) 12: a partial mix with no consistent reasoning.
-

Section 7: Common Student Pitfalls & Pro-Tips

- **Forgetting the diagonal is zero** for skew-symmetric matrices — then misclassifying a matrix or losing an easy mark on “find x .”
- **Dropping the reversal in the transpose law.** It is $(AB)^T = B^T A^T$, never

$A^T B^T$. This single slip corrupts almost every property proof.

- **Assuming the product of two symmetric matrices is symmetric.** It is symmetric *only* when A and B commute; otherwise use $AB + BA$ (symmetric) and $AB - BA$ (skew).
- **Losing the $\frac{1}{2}$ factor** in the decomposition, or swapping P and Q , so the parts no longer add back to A . Always verify $P + Q = A$ at the end.

Pro-Tip (classify by flipping): to decide the nature of any product expression, transpose it, reverse the order, substitute $A^T = \pm A$, and read off whether you recover the expression ($\Rightarrow$ symmetric) or its negative ($\Rightarrow$ skew). **Pro-Tip (counting):** distinct entries are $\frac{n(n+1)}{2}$ (symmetric) and $\frac{n(n-1)}{2}$ (skew) — memorise the pair. **Sanity check:** after a decomposition, glance at Q 's diagonal; if any entry there is non-zero, you have made an arithmetic error.

Section 8: Cheat Sheet

- **Symmetric:** $A^T = A$ ($a_{ij} = a_{ji}$). **Skew-symmetric:** $A^T = -A$ ($a_{ij} = -a_{ji}$, diagonal = 0). Both require **square** matrices.
- **Transpose laws:** $(A^T)^T = A$, $(A + B)^T = A^T + B^T$, $(kA)^T = kA^T$, $(AB)^T = B^T A^T$ (reversal!).
- **Decomposition (unique):** $A = \underbrace{\frac{1}{2}(A + A^T)}_{\text{symmetric } P} + \underbrace{\frac{1}{2}(A - A^T)}_{\text{skew } Q}$.
- **Products:** symmetric $A, B \rightarrow AB + BA$ symmetric, $AB - BA$ skew; AB symmetric iff $AB = BA$.
- **Powers of skew A :** even power $\rightarrow$ symmetric, odd power $\rightarrow$ skew.
- **Counting (order n):** symmetric $\frac{n(n+1)}{2}$, skew $\frac{n(n-1)}{2}$.

Memory aids

- “Symmetric adds, skew subtracts” — P uses $A + A^T$, Q uses $A - A^T$ (both over 2).
- “Skew’s diagonal is always asleep” — every diagonal entry is 0.
- “Transpose a product, turn it around” — the order reverses: $(AB)^T = B^T A^T$.
- Both symmetric and skew $\rightarrow$ null. No exceptions.

Subtopic 03: Invertible Matrices & Row Operations

Exam Relevance: “Find the inverse using elementary row operations” is a standard **CBSE Board** 4-mark question (sometimes 6 for a 3×3). **JEE Main** tests invertibility conditions, the reversal law, and expressing A^{-1} from a matrix equation. **JEE Advanced** uses these structurally inside larger algebra problems. (*NEET does not test mathematics.*) Note: a faster invertibility test — the **determinant** — arrives in the next chapter; here we build the method that needs no determinant at all.

Section 1: Concept Intuition

In ordinary arithmetic, “dividing by 5” really means “multiplying by $\frac{1}{5}$ ”, and $\frac{1}{5}$ is the number that undoes 5: $5 \times \frac{1}{5} = 1$. Matrices have no division, but they have the next best thing — a matrix that **undoes** another. If multiplying by A scrambles a vector, then multiplying by its **inverse** A^{-1} unscrambles it perfectly, landing you back where you started. The “doing nothing” matrix is the identity I , so the inverse is defined by exactly the property you’d hope for:

$$AA^{-1} = A^{-1}A = I.$$

A square matrix that has such a partner is called **invertible** (or **non-singular**). Not every matrix has one — just as 0 has no reciprocal, some non-zero matrices simply cannot be undone.

Think about it this way... Picture a UPI transaction reversal. You send Rs. 2000; an exact refund of Rs. 2000 returns the account to its original balance. The refund is the “inverse” of the payment. But now imagine a transaction that *merges* two separate accounts into one total — there is no single reversal that can recover which rupees came from where. Information was destroyed. That irreversibility is exactly what makes a matrix **non-invertible**: it collapses distinct inputs onto the same output, and no matrix can un-collapse them.

How do we actually *find* A^{-1} without any notion of determinant? With **elementary row operations** — the same three legal moves you already use to solve linear equations:

1. **Swap** two rows: $R_i \leftrightarrow R_j$.
2. **Scale** a row by a non-zero constant: $R_i \rightarrow kR_i$, $k \neq 0$.
3. **Combine**: add a multiple of one row to another: $R_i \rightarrow R_i + kR_j$.

The idea is beautifully direct: start from $A = IA$, and chip away at the left-hand A using row operations until it becomes I . Whatever those same operations do to the I on the right turns it into A^{-1} .

Figure 3.4 A horizontal transformation strip. Left state: augmented matrix $[A \mid I]$. A right-arrow labelled “same row operations applied to both halves”. Right state: aug-21 (see next page)

$$\begin{array}{c}
 \begin{array}{cc|cc}
 & A & & I \\
 \hline
 2 & 1 & 1 & 0 \\
 1 & 1 & 0 & 1
 \end{array}
 \xrightarrow{\text{same row operations applied to both halves}}
 \begin{array}{c|cc}
 & I & & A^{-1} \\
 \hline
 1 & 0 & 1 & -1 \\
 0 & 1 & -1 & 2
 \end{array}
 \end{array}$$

Reduce the left block to I ; the right block becomes the inverse.

Figure 3.4 A horizontal transformation strip. Left state: augmented matrix $[A \mid I]$. A right- arrow labelled "same row operations applied to both halves". Right state: aug- 21

mented matrix $[I \mid A^{-1}]$. Caption: "Reduce the left block to I ; the right block becomes the inverse."

Limiting conditions & assumptions:

- Only **square** matrices can be invertible.
- The inverse, when it exists, is **unique**.
- If, while row-reducing, a **full row of zeros** appears on the left block, A has **no inverse** — stop.
- You must apply each operation to **both** blocks (or, in $A = IA$ form, to the left A and the prefactor matrix) — never to one alone.
- Stick to **one type** throughout: row operations with $A = IA$, or column operations with $A = AI$ — do not mix the two in a single computation.

Section 2: Key Definitions & Properties

Math note: no dimensional formula applies. The governing conditions are **squareness** and the defining relation $AA^{-1} = A^{-1}A = I$.

Invertible matrix: a square matrix A of order n is invertible if there exists a square matrix B of order n with $AB = BA = I_n$. Then B is the inverse, written A^{-1} .

Elementary row operations (the only legal moves):

$$R_i \leftrightarrow R_j, \quad R_i \rightarrow kR_i \ (k \neq 0), \quad R_i \rightarrow R_i + kR_j.$$

(Column operations $C_i \leftrightarrow C_j$, etc., are the mirror set.)

Structural properties of the inverse (these need no determinant):

- **Uniqueness:** an invertible matrix has exactly one inverse.
- **Double inverse:** $(A^{-1})^{-1} = A$.
- **Reversal law:** $(AB)^{-1} = B^{-1}A^{-1}$ — and more generally $(ABC \dots Z)^{-1} = Z^{-1} \dots C^{-1}B^{-1}A^{-1}$.
- **Transpose:** $(A^T)^{-1} = (A^{-1})^T$.
- **Powers:** $(A^k)^{-1} = (A^{-1})^k$ for $k \in \mathbb{N}$.

Deferred to Ch. 4 (Determinants): the quick test " A is invertible $\iff |A| \neq 0$ ", the adjoint formula $A^{-1} = \frac{1}{|A|} \text{adj } A$, and $|A^{-1}| = \frac{1}{|A|}$. We do **not** use them here.

Section 3: Derivation & Key Procedure

Derivation — the inverse is unique. Suppose B and C are both inverses of A , so $AB = BA = I$ and $AC = CA = I$. Then

$$B = BI = B(AC) = (BA)C = IC = C.$$

The two supposed inverses are equal — there is only one. (*This is why the notation A^{-1} is unambiguous.*)

Why the $A = IA$ method works. Every elementary row operation is the same as **pre-multiplying** by a corresponding elementary matrix E (the matrix obtained by doing that operation to I). So a chain of row operations on A is really $E_k \cdots E_2 E_1 A$. We perform operations until this product equals I :

$$E_k \cdots E_1 A = I.$$

But that product $E_k \cdots E_1$ is, by definition, a left-inverse of A — and for square matrices a left-inverse is *the* inverse. Now apply the **identical** chain to the right side of $A = IA$:

$$\underbrace{E_k \cdots E_1 A}_{=I} = \underbrace{(E_k \cdots E_1) I}_{=A^{-1}} \Rightarrow I = A^{-1} A.$$

So whatever the operations turn the prefactor I into **is** A^{-1} .

The procedure (row-operation inverse):

1. Write $A = IA$ (equivalently, form the augmented block $[A \mid I]$).
2. Use row operations to convert the left block from A to I , applying **each** operation to **both** blocks.
3. When the left block is I , the right block is A^{-1} .
4. **If a zero row appears** in the left block at any stage, the reduction to I is impossible — A is **not invertible** and has no inverse.

Why a zero row kills invertibility: a zero row means one row was a combination of the others — the matrix collapses inputs (the “merged accounts” of Section 1), so no undo exists.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Find the inverse of $A = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}$ using elementary row operations.

Solution. Write $[A \mid I]$ and reduce the left block:

$$\begin{aligned} & \left[\begin{array}{cc|cc} 2 & 5 & 1 & 0 \\ 1 & 3 & 0 & 1 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cc|cc} 1 & 3 & 0 & 1 \\ 2 & 5 & 1 & 0 \end{array} \right] \\ & \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{cc|cc} 1 & 3 & 0 & 1 \\ 0 & -1 & 1 & -2 \end{array} \right] \xrightarrow{R_2 \rightarrow -R_2} \left[\begin{array}{cc|cc} 1 & 3 & 0 & 1 \\ 0 & 1 & -1 & 2 \end{array} \right] \\ & \xrightarrow{R_1 \rightarrow R_1 - 3R_2} \left[\begin{array}{cc|cc} 1 & 0 & 3 & -5 \\ 0 & 1 & -1 & 2 \end{array} \right]. \end{aligned}$$

The left block is I , so $A^{-1} = \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix}$. **Check:** $AA^{-1} = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cdot \checkmark$

Example 2 — [JEE Main Level — Speed Trap]

Determine whether $A = \begin{bmatrix} 1 & -3 \\ -2 & 6 \end{bmatrix}$ **is invertible.**

Solution (slow way, for proof). Start reducing:

$$\left[\begin{array}{cc|cc} 1 & -3 & 1 & 0 \\ -2 & 6 & 0 & 1 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 + 2R_1} \left[\begin{array}{cc|cc} 1 & -3 & 1 & 0 \\ 0 & 0 & 2 & 1 \end{array} \right].$$

A **zero row** has appeared in the left block $\rightarrow$ the left block can never become $I \rightarrow A$ **is not invertible.**

Solution (speed way). Glance at the rows: $(-2, 6) = -2(1, -3)$. One row is a scalar multiple of the other — the matrix is **singular**, no inverse. **Trap:** students grind the full reduction; spotting proportional rows (or columns) answers it in one look.

Example 3 — [JEE Main Level] (row-operation inverse + transpose property)

For $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$, **find** A^{-1} **by elementary row operations, then write** $(A^T)^{-1}$.

Solution. Form $[A \mid I]$ and clear the upper entries from the bottom up:

$$\begin{aligned} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] & \xrightarrow{\substack{R_2 \rightarrow R_2 - 4R_3 \\ R_1 \rightarrow R_1 - 3R_3}} \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & -3 \\ 0 & 1 & 0 & 0 & 1 & -4 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \\ & \xrightarrow{R_1 \rightarrow R_1 - 2R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -2 & 5 \\ 0 & 1 & 0 & 0 & 1 & -4 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]. \end{aligned}$$

So $A^{-1} = \begin{bmatrix} 1 & -2 & 5 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}$. Now use the property $(A^T)^{-1} = (A^{-1})^T$ — no fresh computation needed, just transpose:

$$(A^T)^{-1} = (A^{-1})^T = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 5 & -4 & 1 \end{bmatrix}.$$

Answer: A^{-1} as above, and $(A^T)^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 5 & -4 & 1 \end{bmatrix}$.

Example 4 — [JEE Advanced Level]

A square matrix A **satisfies** $A^3 - 3A + 2I = O$. **Show** A **is invertible and express** A^{-1} **as a polynomial in** A .

Solution. Isolate a term that exposes a factor of A :

$$A^3 - 3A + 2I = O \Rightarrow A^3 - 3A = -2I \Rightarrow A(A^2 - 3I) = -2I.$$

Divide through by -2 (a legal scalar operation):

$$A \left[-\frac{1}{2}(A^2 - 3I) \right] = I.$$

For square matrices, $AB = I$ already forces $B = A^{-1}$ (and $BA = I$ too). Hence A is invertible and

$$\boxed{A^{-1} = -\frac{1}{2}(A^2 - 3I) = \frac{1}{2}(3I - A^2).}$$

Why this is powerful: the inverse is recovered purely from the matrix's own algebra — no row reduction, no determinant. Whenever a matrix satisfies a polynomial equation with a non-zero constant term, the same trick expresses its inverse as a polynomial in A .

Section 5: Practice Exercises

1. **[CBSE]** Using elementary row operations, find the inverse of $\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$.
2. **[CBSE]** Using elementary row operations, find the inverse of $\begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$.
3. **[JEE Main]** Show, using row operations, that $\begin{bmatrix} 2 & -4 \\ -1 & 2 \end{bmatrix}$ has no inverse.
4. **[JEE Main]** If $A^{-1} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$, find A .
5. **[JEE Advanced]** A matrix A satisfies $A^2 - 4A + 3I = O$. Express A^{-1} as a polynomial in A .

Answers: 1. $\begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$. 2. $\begin{bmatrix} 1/5 & 2/5 \\ 2/5 & -1/5 \end{bmatrix}$. 3. Reduction yields a zero row (rows are proportional: $(-1, 2) = -\frac{1}{2}(2, -4)$), so no inverse exists. 4. $A = (A^{-1})^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$. 5. $A^{-1} = \frac{1}{3}(4I - A)$.

Section 6: Crack the MCQ

Q1. Which of the following is NOT an elementary row operation?

- (a) $R_1 \leftrightarrow R_2$ (b) $R_2 \rightarrow 3R_2$ (c) $R_2 \rightarrow 0 \cdot R_2$ (d) $R_1 \rightarrow R_1 + 2R_3$

Correct: (c). Scaling a row is allowed only by a **non-zero** constant; multiplying by 0 wipes the row out and is not reversible.

- (a): swapping rows is the first elementary operation.
- (b): scaling by $3 \neq 0$ is legal.

- (d): adding a multiple of one row to another is the third elementary operation.

Q2. To find A^{-1} by row operations we write $A = IA$ and reduce the left A to I . The prefactor I then becomes

- (a) A^T (b) A^{-1} (c) I (d) A^2

Correct: (b). The operations multiply both sides on the left by the same product $E_k \cdots E_1 = A^{-1}$; turning the left A into I turns the prefactor I into A^{-1} .

- (a): transpose is unrelated to row reduction.
- (c): if it stayed I , then $A = I$ — only true for the identity itself.
- (d): no operation squares the matrix.

Q3. While computing an inverse by elementary row operations, a row of all zeros appears in the left block. This means

- (a) $A^{-1} = O$ (b) $A^{-1} = I$ (c) A is not invertible (d) you must restart using column operations

Correct: (c). A zero row makes reaching I impossible; the matrix is singular and has no inverse.

- (a): the zero matrix is not an inverse of anything ($AO = O \neq I$).
- (b): there is no reason the inverse would be I .
- (d): switching to columns will not rescue a genuinely singular matrix.

Q4. If A and B are invertible matrices of the same order, then $(AB)^{-1} =$

- (a) $A^{-1}B^{-1}$ (b) $B^{-1}A^{-1}$ (c) BA (d) AB

Correct: (b). Reversal law: $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I$.

- (a): the most common error — keeping the original order; $A^{-1}B^{-1}$ generally is **not** the inverse.
- (c), (d): products, not inverses; dimensionally plausible but algebraically wrong.

Section 7: Common Student Pitfalls & Pro-Tips

- **Operating on one block only.** Every row operation must hit **both** the left block and the prefactor/right block. Touching just one breaks the equation $A = IA$.
- **Mixing row and column operations** in a single inverse computation. Choose row ($A = IA$) or column ($A = AI$) and stay with it throughout.
- **Scaling a row by 0**, or believing a matrix with proportional rows/columns can still be inverted. Proportional rows $\Rightarrow$ singular $\Rightarrow$ no inverse.
- **Dropping the reversal** in $(AB)^{-1} = B^{-1}A^{-1}$ — writing $A^{-1}B^{-1}$ instead.

Pro-Tip (singularity at a glance): before reducing, check if any row is a scalar multiple of another (or a sum of others). If so, stop — the matrix is singular. **Pro-Tip (always verify):** confirm your answer with $AA^{-1} = I$; it costs one multiplication and catches sign slips. **Pro-Tip (matrix-equation inverses):** if A satisfies a polynomial equation with a non-zero constant term, factor out A to read off A^{-1} directly — far faster than reduction (Example 4).

Section 8: Cheat Sheet

- **Invertible:** square A with $AA^{-1} = A^{-1}A = I$. Inverse is **unique**.
- **Three elementary row operations:** $R_i \leftrightarrow R_j$; $R_i \rightarrow kR_i$ ($k \neq 0$); $R_i \rightarrow R_i + kR_j$.
- **Method:** $[A \mid I] \xrightarrow{\text{row ops on both}} [I \mid A^{-1}]$ (equivalently $A = IA \rightarrow I = A^{-1}A$).
- **No-inverse signal:** a zero row appears in the left block $\Rightarrow A$ is singular.
- **Key identities:** $(A^{-1})^{-1} = A$; $(AB)^{-1} = B^{-1}A^{-1}$ (*reversal!*); $(A^T)^{-1} = (A^{-1})^T$.
- **Matrix-equation trick:** from $A(\text{poly in } A) = cI$, divide by c to get A^{-1} .

Memory aids

- “Whatever I do to A , I do to I ” — apply each operation to both blocks.
- “Inverse flips the order” — $(AB)^{-1} = B^{-1}A^{-1}$, socks-and-shoes.
- “Zero row, no go” — a zero row means no inverse.

Coming in Ch. 4: a one-shot invertibility test ($|A| \neq 0$) and the adjoint formula $A^{-1} = \frac{1}{|A|} \text{adj } A$.

Subtopic 04: Special Square Matrices & Chapter Synthesis

Exam Relevance: The special matrix types here are **JEE Main and Advanced** favourites — idempotent, involutory, nilpotent, and orthogonal matrices appear directly and inside multi-step problems. **CBSE** tests the real-life application of matrix operations (a 4–6 mark question) and the matrix-polynomial identity $f(A) = O$. This subtopic also ties the whole chapter together. (*NEET does not test mathematics.*)

Section 1: Special Types of Square Matrices

(Beyond NCERT core — essential JEE enrichment. CBSE-only students may treat this as optional reading, except the matrix-polynomial idea in Section 3, which CBSE does test.)

Some square matrices have a self-referential personality — squaring them, or multiplying them by their own transpose, returns something remarkably simple. Four types matter:

Idempotent: $A^2 = A$. Such a matrix is “absorbed” by repetition — once you apply it, applying it again changes nothing, so $A^n = A$ for every $n \geq 1$. (Think of a projection: shining a shadow twice gives the same shadow.)

Involutory: $A^2 = I$. The matrix is its own inverse, $A^{-1} = A$. Applying it twice undoes it — like a mirror flip.

Nilpotent: $A^k = O$ for some positive integer k ; the smallest such k is the **index**. A nilpotent matrix eventually annihilates everything; it can never be invertible.

Orthogonal: $AA^T = A^T A = I$, i.e. $A^{-1} = A^T$. Its rows (and columns) form a set of mutually perpendicular unit vectors — rotations are the classic example.

Key consequences

- Idempotent A : $I - A$ is also idempotent; the only invertible idempotent matrix is I itself.
 - Involutory A : even powers give I , odd powers give A .
 - Nilpotent A : never invertible (no inverse exists).
 - Orthogonal A : $A^{-1} = A^T$ makes inverses free to compute.
-

Section 2: Trace of a Matrix

The **trace** of a square matrix is the sum of its diagonal entries: $\text{tr}(A) = \sum_{i=1}^n a_{ii}$. It obeys clean rules (for square A, B of the same order and scalar k):

$$\begin{aligned}\text{tr}(A + B) &= \text{tr}(A) + \text{tr}(B), & \text{tr}(kA) &= k \text{tr}(A), \\ \text{tr}(A^T) &= \text{tr}(A), & \text{tr}(AB) &= \text{tr}(BA).\end{aligned}$$

The last identity is the surprising one: even though $AB \neq BA$ in general, their traces always agree. It is a favourite JEE trap precisely because the products themselves differ.

Section 3: Matrix Polynomials & the Equation $f(A) = O$

Just as we evaluate $f(x)$ at a number, we can evaluate a polynomial **at a matrix**: replace x by A and the constant term c by cI . A square matrix very often **satisfies a polynomial equation of its own**, and that single fact unlocks both high powers and the inverse without heavy computation.

Example A — [CBSE / JEE Main Level]

For $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, **show that** $A^2 - 2A - 3I = O$, **and hence find** A^{-1} .

Solution. Compute the pieces:

$$A^2 = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}, \quad 2A = \begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix}, \quad 3I = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}.$$

Then

$$A^2 - 2A - 3I = \begin{bmatrix} 5-2-3 & 4-4-0 \\ 4-4-0 & 5-2-3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O. \checkmark$$

Now exploit this to invert. From $A^2 - 2A - 3I = O$:

$$A^2 - 2A = 3I \Rightarrow A(A - 2I) = 3I \Rightarrow A \cdot \frac{1}{3}(A - 2I) = I.$$

Hence

$$A^{-1} = \frac{1}{3}(A - 2I) = \frac{1}{3} \begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} -1/3 & 2/3 \\ 2/3 & -1/3 \end{bmatrix}.$$

Example B — [JEE Advanced Level]

A is idempotent ($A^2 = A$). Simplify $(I + A)^3 - 7A$.

Solution. Expand using $A^2 = A$ (so every power of A collapses to A , and I commutes with everything):

$$(I + A)^3 = I + 3A + 3A^2 + A^3 = I + 3A + 3A + A = I + 7A.$$

Therefore $(I + A)^3 - 7A = I + 7A - 7A = I$. **Answer:** I .

Section 4: Real-Life Application of Matrix Operations

Example C — [CBSE Board Level]

A stationery seller records two months of sales (in units) of three products — pens (P), notebooks (N), erasers (E) — as the matrix M , where rows are

months and columns are products. The selling prices (Rs.) form the vector p :

$$M = \begin{bmatrix} 40 & 25 & 30 \\ 35 & 30 & 20 \end{bmatrix}, \quad p = \begin{bmatrix} 50 \\ 20 \\ 10 \end{bmatrix}.$$

(i) Find the total units sold of each product. (ii) Find the revenue of each month and the grand total.

Solution. (i) **Total units per product** — add the two month-rows:

$$\begin{bmatrix} 40 & 25 & 30 \\ 35 & 30 & 20 \end{bmatrix} = \begin{bmatrix} 75 & 55 & 50 \end{bmatrix}.$$

So 75 pens, 55 notebooks, 50 erasers.

(ii) **Revenue per month** — multiply M (units) by p (prices):

$$Mp = \begin{bmatrix} 40 & 25 & 30 \\ 35 & 30 & 20 \end{bmatrix} \begin{bmatrix} 50 \\ 20 \\ 10 \end{bmatrix} = \begin{bmatrix} 2000 + 500 + 300 \\ 1750 + 600 + 200 \end{bmatrix} = \begin{bmatrix} 2800 \\ 2550 \end{bmatrix}.$$

Month 1 revenue Rs. 2800, Month 2 revenue Rs. 2550; **grand total Rs. 5350**. Notice how addition handled “totals” and multiplication handled “units $\times$ price” — the operations are not decorative, they *are* the model.

Section 5: Mixed Chapter-Spanning Problems

1. **[CBSE]** Construct the 2×2 matrix $a_{ij} = 2i - j$, then express it as the sum of a symmetric and a skew-symmetric matrix.
2. **[CBSE]** If $A = \begin{bmatrix} 3 & 1 \\ 7 & 5 \end{bmatrix}$, find x, y such that $A^2 - xA + yI = O$, and hence find A^{-1} .
3. **[JEE Main]** A is a 3×3 matrix with $\text{tr}(A) = 6$. Find $\text{tr}(2A^T - 3I)$.
4. **[JEE Main]** Show that $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ is nilpotent and state its index.
5. **[JEE Advanced]** A is involutory. Evaluate $A^2 + A^4 + A^6 + \dots + A^{2024}$.

Answers: 1. $A = \begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix}$; symmetric part $\begin{bmatrix} 1 & 1.5 \\ 1.5 & 2 \end{bmatrix}$, skew part $\begin{bmatrix} 0 & -1.5 \\ 1.5 & 0 \end{bmatrix}$. 2. $x = 8, y = 8$ (since $A^2 = \begin{bmatrix} 16 & 8 \\ 56 & 32 \end{bmatrix}$ gives $A^2 - 8A + 8I = O$), so $A^{-1} = \frac{1}{8}(8I - A) = \frac{1}{8} \begin{bmatrix} 5 & -1 \\ -7 & 3 \end{bmatrix}$. 3. $\text{tr}(2A^T - 3I) = 2\text{tr}(A) - 3(3) = 12 - 9 = 3$. 4. $A^2 = O$, so nilpotent of index 2. 5. There are 1012 terms, all even powers $= I$, so the sum is $1012I$.

Section 6: Chapter Synthesis

Figure 3.5 A concept map of Chapter 3. Central node "MATRIX (order $m \times n$, types)". Branch 1 -> "Operations: addition, scalar multiplication, matrix multiplication (conformability, non-commutative)". Branch 2 -> "Transpose -> Symmetric ($A^T = A$) and Skew-symmetric ($A^T = -A$) -> Decomposition $A = P + Q$ ". Branch 3 -> "Special square types: idempotent, involutory, nilpotent, orthogonal; trace". Branch 4 -> "Invertibility: elementary row operations $[A | I] \rightarrow [I | A^{-1}]$ ". A dashed arrow from "Invertibility" points forward to a faded node "Ch. 4: Determinants and adjoint inverse". Caption: how the chapter's ideas connect, with the determinant method deferred to the next chapter. (see next page)

Master Formula & Definition Sheet (whole chapter)

Representation

- Order $m \times n$; entry a_{ij} (row i , column j). Possible orders of an N -element matrix = number of divisors of N . Number of $m \times n$ matrices over a k -set = k^{mn} .

Equality: same order **and** $a_{ij} = b_{ij}$ for all i, j .

Operations

$$(A+B)_{ij} = a_{ij} + b_{ij} \text{ (same order), } (kA)_{ij} = k a_{ij}, \quad (AB)_{ij} = \sum_r a_{ir} b_{rj} \text{ (cols}(A) = \text{rows}(B)).$$

Non-commutative ($AB \neq BA$); associative; distributive; $IA = AI = A$; $AB = O \nRightarrow A = O$ or $B = O$.

Transpose: $(A^T)^T = A$, $(A+B)^T = A^T + B^T$, $(kA)^T = kA^T$, $(AB)^T = B^T A^T$.

Symmetric / skew-symmetric: $A^T = A$ / $A^T = -A$ (skew diagonal = 0). Unique split

$$A = \underbrace{\frac{1}{2}(A + A^T)}_{\text{symmetric}} + \underbrace{\frac{1}{2}(A - A^T)}_{\text{skew}}.$$

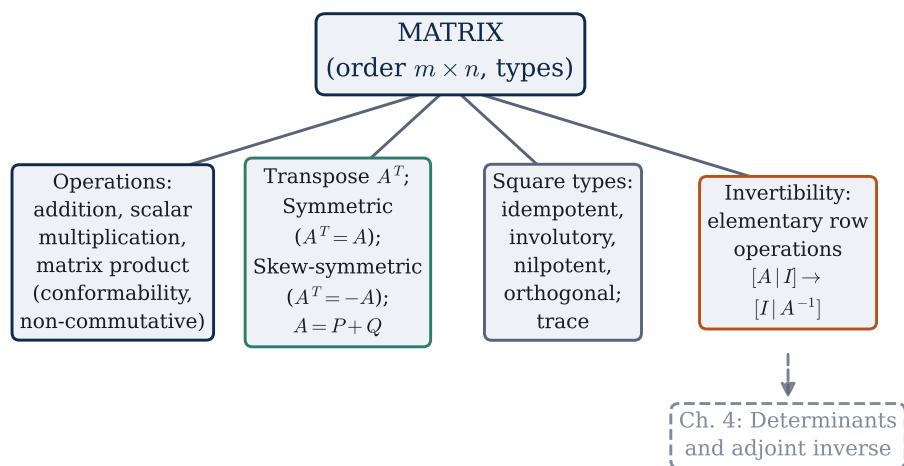
Distinct entries: symmetric $\frac{n(n+1)}{2}$, skew $\frac{n(n-1)}{2}$.

Special square types: idempotent $A^2 = A$; involutory $A^2 = I$ ($A^{-1} = A$); nilpotent $A^k = O$ (singular); orthogonal $AA^T = I$ ($A^{-1} = A^T$).

Trace: $\text{tr}(A) = \sum a_{ii}$; $\text{tr}(A+B) = \text{tr} A + \text{tr} B$; $\text{tr}(kA) = k \text{tr} A$; $\text{tr}(AB) = \text{tr}(BA)$.

Inverse (by elementary row operations): $[A | I] \rightarrow [I | A^{-1}]$; a zero row $\Rightarrow$ no inverse. Identities: $(A^{-1})^{-1} = A$, $(AB)^{-1} = B^{-1}A^{-1}$, $(A^T)^{-1} = (A^{-1})^T$. Matrix-polynomial trick: from $A(\text{poly in } A) = cI$, divide by c to read off A^{-1} .

Deferred to Ch. 4 (Determinants): invertibility test $|A| \neq 0$; adjoint formula $A^{-1} = \frac{1}{|A|} \text{adj } A$; $|A^{-1}| = 1/|A|$.



How the chapter's ideas connect; the determinant method for the inverse is deferred to Chapter 4.

Figure 3.5 A concept map of Chapter 3. Central node "MATRIX (order $m \times n$, types)". Branch 1 -> "Operations: addition, scalar multiplication, matrix multiplication (conformability, non-commutative)". Branch 2 -> "Transpose -> Symmetric ($A^T = A$) and Skew-symmetric ($A^T = -A$) -> Decomposition $A = P + Q$ ". Branch 3 -> "Special square types: idempotent, involutory, nilpotent, orthogonal; trace". Branch 4 -> "Invertibility: elementary row operations $[A | I] \rightarrow [I | A^{-1}]$ ". A dashed arrow from "Invertibility" points forward to a faded node "Ch. 4: Determinants and adjoint inverse". Caption: how the chapter's ideas connect, with the determinant method deferred to the next chapter.

One-line memory anchors

- *Row before column ($a_{i,j}$); inner match, outer survive (multiplication); transpose turns the order around; symmetric adds, skew subtracts; zero row, no inverse.*

Chapter 3: Matrices — Round 2 Addendum

The chapter taught you to build matrices, operate on them, split them into symmetric and skew parts, and invert them by row operations. Five examinable patterns still stand outside the door. Addendum A computes high powers and inverses of near-diagonal matrices by nilpotency — the binomial theorem for matrices, which turns a 2025th power into three terms. Addendum B solves matrix equations and linear systems through the inverse, with strict side discipline. Addendum C counts structured matrices, the objective-question favourite. Addendum D derives how inversion interacts with symmetry and works through orthogonal matrices. Addendum E proves the trace identities the chapter asserts, and uses them to settle an impossibility that JEE Advanced loves.

Nothing here is new theory. Every method below is the entry-by-entry equality of Subtopic 00, the transpose-flip of Subtopic 02, or the row-operation engine of Subtopic 03, applied one layer deeper.

Addendum A — Near-Diagonal Powers: the Binomial Theorem for Matrices

Subtopic 01 handles powers by pattern-spotting ($A^2 = I$ collapse, unitriangular counting) and Subtopic 04 handles matrix polynomials. What neither does is the general tool behind both: if a matrix splits as a scalar multiple of I plus a nilpotent piece, every power and the inverse come from two truncated series. JEE Main asks for A^{100} this way almost yearly; JEE Advanced hides the same structure inside inverses and summations.

Method

Setup. Write the square matrix as

$$A = \lambda I + N,$$

where λ is the repeated diagonal value and $N = A - \lambda I$. Check whether some power of N vanishes: if $N^k = O$, call k the index of N .

Why the ordinary binomial theorem applies. Expanding honestly,

$$(B + C)^2 = B^2 + BC + CB + C^2,$$

and the cross terms collect only when $BC = CB$. Here $C = N$ and $B = \lambda I$, and a scalar multiple of the identity commutes with *every* matrix: $(\lambda I)N = \lambda N = N(\lambda I)$. So the usual binomial expansion is valid term for term:

$$(\lambda I + N)^n = \lambda^n I + \binom{n}{1} \lambda^{n-1} N + \binom{n}{2} \lambda^{n-2} N^2 + \dots + N^n.$$

Truncation. Every term with $r \geq k$ contains $N^r = O$, so the infinite-looking sum stops at $r = k - 1$:

$$(\lambda I + N)^n = \sum_{r=0}^{k-1} \binom{n}{r} \lambda^{n-r} N^r.$$

The two workhorse cases.

Structure	Power formula
$N^2 = O$	$(\lambda I + N)^n = \lambda^n I + n\lambda^{n-1} N$
$N^3 = O$	$(\lambda I + N)^n = \lambda^n I + n\lambda^{n-1} N + \binom{n}{2} \lambda^{n-2} N^2$

Inverses from the same structure. For $N^2 = O$, claim $(I + N)^{-1} = I - N$. Verify directly — never assert:

$$(I + N)(I - N) = I - N + N - N^2 = I - O = I.$$

In general, for $N^k = O$, the finite alternating series works:

$$(I + N)^{-1} = I - N + N^2 - \dots + (-1)^{k-1} N^{k-1},$$

because the product telescopes:

$$(I + N)(I - N + N^2 - \dots) = I \pm N^k = I.$$

This recovers Subtopic 04's polynomial trick from a pattern — and works even when no polynomial equation for A has been handed to you.

Example A.1 — [JEE Main Level]

Find A^{100} for $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$.

Split: $A = I + N$ with $N = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$. Square it:

$$N^2 = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O.$$

Index 2, so with $\lambda = 1$:

$$A^{100} = I + 100N = \begin{bmatrix} 1 & 200 \\ 0 & 1 \end{bmatrix}.$$

Compare with the chapter's induction pattern for $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$: same method, no guessing, and the top-right entry $100 \times 2 = 200$ comes straight from the formula.

Example A.2 — [JEE Advanced Level]

For $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, **find** A^3 **and** A^{-1} .

Here $\lambda = 1$ and

$$N = A - I = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Compute the powers before trusting any formula. Row 1 of N is $(0, 1, 1)$; the only non-zero column of N that matters is column 3:

$$N^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad N^3 = N \cdot N^2 = O.$$

So the index is 3 — the naive guess $N^2 = O$ would have been wrong, which is exactly why the squaring step is not optional.

Power. With $n = 3$:

$$A^3 = I + 3N + 3N^2 = \begin{bmatrix} 1 & 3 & 6 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}.$$

Spot-check the corner entry by hand: the $(1, 3)$ entry of A^2 is $0 \cdot 1 + 1 \cdot 1 + 1 \cdot 0 = 1$, and chaining once more gives $1 + 1 + 1 = 3$ contributions — matching $\binom{3}{2} = 3$ copies of the N^2 slot filled with 1s. Consistent.

Inverse. By the alternating series with $k = 3$:

$$A^{-1} = I - N + N^2 = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Verify one full row against A . Row 1 of A is $(1, 1, 1)$; dotted with the three columns of the candidate gives 1, then $-1 + 1 = 0$, then $0 - 1 + 1 = 0$. Correct. Three row-operation moves saved.

Practice — Addendum A

- Find A^{50} for $A = \begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}$.
- Find A^7 for $A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$.
- Show that $N = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ has index 3, and hence find $(I - N)^{-1}$.
- For $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, evaluate $A^{2025} - 2025A + 2024I$ without multiplying anything out beyond one power.
- For $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$, write $A = 2I + N$, find N^2 and N^3 , and express A^{10} as a combination of I, N, N^2 .

Answers:

1. $N^2 = O$, so $A^{50} = I + 50N = \begin{bmatrix} 1 & 250 \\ 0 & 1 \end{bmatrix}$.

2. $A = 3I + N$ with $N^2 = O$, so

$$A^7 = 3^7 I + 7 \cdot 3^6 N = \begin{bmatrix} 2187 & 5103 \\ 0 & 2187 \end{bmatrix}.$$

3. Direct multiplication gives $N^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and $N^3 = O$, index 3. Then

$$(I - N)^{-1} = I + N + N^2 = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

(Signs flip because the series for $(I - N)^{-1}$ alternates.)

4. $A = I + N$ with $N^2 = O$, so $A^{2025} = I + 2025N$. Then

$$A^{2025} - 2025A + 2024I = (I + 2025N) - 2025(I + N) + 2024I = O.$$

5. $N = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, so $N^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and $N^3 = O$. Hence

$$A^{10} = 2^{10}I + 10 \cdot 2^8 N + 45 \cdot 2^7 N^2 = 1024I + 2560N + 5760N^2.$$

Addendum B — Matrix Equations and Systems Through the Inverse

Subtopic 03 builds A^{-1} but never spends it. On the CBSE board, spending it is the question: “express the system as $AX = B$ and solve” is a fixed 4–6 mark item, and JEE Main tests the same mechanics with the side of multiplication swapped. The entire skill is one rule applied with discipline.

Method

The governing rule. If A is invertible and

$$AX = B, \quad \text{then} \quad X = A^{-1}B.$$

Derivation. Multiply both sides of $AX = B$ by A^{-1} **on the left**:

$$A^{-1}(AX) = A^{-1}B \implies (A^{-1}A)X = A^{-1}B \implies IX = A^{-1}B.$$

Side discipline. Multiplying $AX = B$ on the *right* by A^{-1} gives

$$AXA^{-1} = BA^{-1},$$

which isolates nothing — X is trapped between two factors. The inverse must strike from the side A occupies:

Equation	Solve by	Result
$AX = B$	premultiply both sides	$X = A^{-1}B$
$XA = B$	postmultiply both sides	$X = BA^{-1}$

Uniqueness. If $AX = B$ and $AX' = B$, subtract: $A(X - X') = O$. Premultiply by A^{-1} : $X - X' = O$. So when A is invertible, the solution exists and is unique for *every* B — no consistency check needed. (What happens when A is singular waits for Chapter 4’s determinant test.)

Encoding a linear system. The system

$$a_1x + b_1y + c_1z = d_1, \quad a_2x + b_2y + c_2z = d_2, \quad a_3x + b_3y + c_3z = d_3$$

is exactly $AX = B$ with A built from coefficients (rows = equations), X the column of unknowns, B the column of constants. The product rule guarantees the match: entry i of AX is the dot product of equation-row i with (x, y, z) , which is the left side of equation i .

Example B.1 — [CBSE Board Level]

Express the following system as $AX = B$, find A^{-1} by elementary row operations, and hence solve it.

$$x - y + z = 4, \quad 2x + y - 3z = 0, \quad x + y + z = 2.$$

Encoding.

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}.$$

Inverting. Reduce $[A \mid I]$:

$$\begin{aligned} & \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 2 & 1 & -3 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1}} \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 3 & -5 & -2 & 1 & 0 \\ 0 & 2 & 0 & -1 & 0 & 1 \end{array} \right] \\ & \xrightarrow{R_3 \rightarrow \frac{1}{2}R_3} \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 3 & -5 & -2 & 1 & 0 \\ 0 & 1 & 0 & -\frac{1}{2} & 0 & \frac{1}{2} \end{array} \right] \xrightarrow{\substack{R_1 \rightarrow R_1 + R_3 \\ R_2 \rightarrow R_2 - 3R_3}} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & -5 & -\frac{1}{2} & 1 & -\frac{3}{2} \\ 0 & 1 & 0 & -\frac{1}{2} & 0 & \frac{1}{2} \end{array} \right] \\ & \xrightarrow{R_2 \rightarrow -\frac{1}{5}R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{10} & -\frac{1}{5} & \frac{3}{10} \\ 0 & 1 & 0 & -\frac{1}{2} & 0 & \frac{1}{2} \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 - R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{2}{5} & \frac{1}{5} & \frac{1}{5} \\ 0 & 0 & 1 & \frac{1}{10} & -\frac{1}{5} & \frac{3}{10} \\ 0 & 1 & 0 & -\frac{1}{2} & 0 & \frac{1}{2} \end{array} \right] \end{aligned}$$

Reading off and clearing denominators:

$$A^{-1} = \frac{1}{10} \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}.$$

Verification (one multiplication, always). Row 2 of A against column 1 of the bracketed matrix: $2(4) + 1(-5) - 3(1) = 0$; against column 2: $2(2) + 0 - 3(-2) = 10$; against column 3: $2(2) + 5 - 9 = 0$. Divided by 10, that is $(0, 1, 0)$ — the second row of I . Spot-checks pass.

Solving.

$$X = A^{-1}B = \frac{1}{10} \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 16 + 0 + 4 \\ -20 + 0 + 10 \\ 4 + 0 + 6 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}.$$

So $x = 2$, $y = -1$, $z = 1$. *Check in the original system:* equation 1 gives $2 + 1 + 1 = 4$; equation 2 gives $4 - 1 - 3 = 0$; equation 3 gives $2 - 1 + 1 = 2$. All hold.

Example B.2 — [JEE Main Level — Speed Trap]

Given $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ **and** $XA = B$ **where** $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, **find** X .

A sits on the **right** of X , so postmultiply: $X = BA^{-1}$. From Addendum A (or one row reduction), $A^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$. Then

$$X = BA^{-1} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 3 & 1 \end{bmatrix}.$$

Why the reflex answer fails. The reflex is $X = A^{-1}B$ — right equation shape, wrong side. Compute it to see the damage:

$$A^{-1}B = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} -2 & -2 \\ 3 & 4 \end{bmatrix} \neq X.$$

Different matrix entirely. The check $XA = B$ catches the slip in seconds; the habit of asking “which side is A on?” prevents it.

Practice — Addendum B

1. Solve by the inverse method: $2x + y = 5$, $3x + 2y = 11$.
2. Solve $XA = B$ for $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix}$.
3. Explain in two lines why, when A is invertible, $AX = B$ cannot have two different solutions.
4. A student solves $AX = B$ as $X = BA^{-1}$. Using the matrices of Example B.2, show numerically that this fails, and identify the exact step where the logic breaks.
5. Write the system $x + y + z = 6$, $x - y + z = 2$, $2x + y - z = 1$ in the form $AX = B$, naming A , X , B . (No need to solve.)

Answers:

$$1. A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}, A^{-1} = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}, \text{ so}$$

$$X = A^{-1}B = \begin{bmatrix} 10 & -11 \\ -15 & 22 \end{bmatrix} = \begin{bmatrix} -1 \\ 7 \end{bmatrix},$$

giving $x = -1$, $y = 7$. Check: $-2 + 7 = 5$ and $-3 + 14 = 11$.

$$2. A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}, \text{ and } X = BA^{-1} = \begin{bmatrix} -2 & 7 \\ -2 & 6 \end{bmatrix}. \text{ Check row 1: } (-2)(2) + 7(1) = 3 \text{ and } (-2)(1) + 7(1) = 5. \text{ Correct.}$$

3. Suppose $AX = B$ and $AX' = B$. Subtracting, $A(X - X') = O$. Premultiplying by A^{-1} gives $X - X' = O$, so $X = X'$.

4. The true answer is $X = \begin{bmatrix} 1 & 1 \\ 3 & 1 \end{bmatrix}$; the student's $A^{-1}B = \begin{bmatrix} -2 & -2 \\ 3 & 4 \end{bmatrix}$ differs. The broken step is the direction of multiplication: cancelling A requires multiplying by A^{-1} from the side A occupies, and here A is on the right of X , so the correct move is $X = BA^{-1}$.

$$5. A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}.$$

Addendum C — Counting Structured Matrices

Subtopic 00 counts all matrices (k^{mn}) and counts *entries available* in symmetric and skew matrices, but never combines the two ideas — which is precisely what JEE Main asks. Once you know how many cells are free, the count is one exponentiation.

Method

The principle. A structured matrix is completely determined by its free cells, and the free cells are filled independently. If there are f free cells and each accepts k values, the number of such matrices is

$$k^f.$$

Free-cell counts (from Subtopic 02, Derivation 3, extended to other structures):

Structure (order n)	Free cells f	Count over a k -set
arbitrary	n^2	k^{n^2}
symmetric	$\frac{n(n+1)}{2}$	$k^{\frac{n(n+1)}{2}}$
skew-symmetric	$\frac{n(n-1)}{2}$	$k^{\frac{n(n-1)}{2}}$
diagonal	n	k^n
scalar	1	k

Two cautions the table hides.

Cautions for the table: first, the skew count assumes 0 belongs to the allowed set (the diagonal must read 0). Second, for skew-symmetric matrices each free cell really carries a pair $(a, -a)$, so the allowed set must be closed under negation — otherwise the honest count is smaller than k^f . Both caveats are examined.

Example C.1 — [JEE Main Level]

(i) How many symmetric matrices of order 3 have entries from $\{0, 1\}$? (ii) How many skew-symmetric matrices of order 3 have entries from $\{-1, 0, 1\}$?

(i) Free cells: $f = \frac{3 \cdot 4}{2} = 6$ (three diagonal, three above it), each independently 0 or 1:

$$2^6 = 64.$$

(ii) Free cells: $f = \frac{3 \cdot 2}{2} = 3$ — the three above-diagonal positions, since the diagonal is forced to 0. Each pair reads $(a, -a)$, and choosing $a \in \{-1, 0, 1\}$ automatically places $-a$ in the allowed set too. So

$$3^3 = 27.$$

Sanity check on the structure: pick $a_{12} = 1$, $a_{13} = 0$, $a_{23} = -1$; the matrix is then forced everywhere else, with diagonal 0 and lower half $(a_{21}, a_{31}, a_{32}) = (-1, 0, 1)$. One choice of three numbers, whole matrix determined — consistent with 3^3 .

Example C.2 — [JEE Main Level]

How many 2×2 matrices with entries from $\{0, 1\}$ have trace equal to 1?

The trace condition couples the diagonal cells, so they cannot be counted independently — handle them jointly, then release the rest.

Diagonal pairs (a_{11}, a_{22}) from $\{0, 1\}^2$ with $a_{11} + a_{22} = 1$: exactly two, namely $(1, 0)$ and $(0, 1)$. The off-diagonal cells a_{12}, a_{21} remain unconstrained: $2^2 = 4$ ways. Total:

$$2 \times 4 = 8.$$

The lesson generalises: impose the constraint on the coupled cells *first*, count their joint possibilities, then multiply by the free cells.

Practice — Addendum C

1. How many symmetric matrices of order 4 have entries from $\{0, 1\}$?
2. How many skew-symmetric matrices of order 4 have entries from $\{-1, 0, 1\}$?
3. How many 2×2 matrices with entries from $\{0, 1, 2\}$ have equal diagonal entries?
4. With entries from $\{1, 2, \dots, 9\}$, how many matrices of order 3 are (i) diagonal, (ii) scalar?
5. Explain why the only skew-symmetric matrix of order $n \geq 2$ with all entries from $\{0, 1\}$ is the zero matrix, and state the resulting count.

Answers: 1. $f = 10$, so $2^{10} = 1024$. 2. $f = 6$, and each pair $(a, -a)$ has 3 choices, so $3^6 = 729$. 3. Diagonal pair: 3 common values; off-diagonal cells: $3^2 = 9$; total 27. 4. (i) $9^3 = 729$; (ii) 9 (one shared value, chosen freely — note $0 \notin$ the set, but a scalar matrix needs no zero). 5. Each off-diagonal pair is $(a, -a)$ with both entries in $\{0, 1\}$; the only element whose negative also lies in $\{0, 1\}$ is $a = 0$. So every entry is 0, and the count is 1.

Addendum D — Inversion Meets Symmetry; Orthogonal Matrices

Subtopic 03 lists $(A^T)^{-1} = (A^{-1})^T$ as a bullet and defines orthogonality in Subtopic 04, but neither derives the consequences. JEE Main asks those consequences directly: does inver-

sion preserve symmetry? Is the product of orthogonal matrices orthogonal? Both answers fall out of one verification template you already own.

Method

The bridge identity. For invertible A ,

$$(A^T)^{-1} = (A^{-1})^T.$$

Derivation. Transposing the defining equation $A^{-1}A = I$ gives

$$(A^{-1}A)^T = A^T(A^{-1})^T = I^T = I,$$

using the reversal law. So $(A^{-1})^T$ is a left inverse of A^T ; for square matrices a left inverse is *the* inverse (the chapter's uniqueness argument). Done.

Corollary 1 — inversion preserves symmetry. If $A^T = A$, then

$$(A^{-1})^T = (A^T)^{-1} = A^{-1},$$

so A^{-1} is symmetric.

Corollary 2 — inversion preserves skewness. If $A^T = -A$, then

$$(A^{-1})^T = (A^T)^{-1} = (-A)^{-1} = -A^{-1},$$

so A^{-1} is skew-symmetric. (Only even-order skew matrices can be invertible — the reason arrives with determinants in Chapter 4.)

Orthogonal matrices. A is orthogonal when

$$AA^T = A^T A = I, \quad \text{i.e.} \quad A^{-1} = A^T.$$

Three closure facts, each proved by the same transpose-flip:

- I is orthogonal: $II^T = I$.
- If A is orthogonal, so are A^T and A^{-1} : they are the same matrix.
- If A and B are orthogonal, then AB is orthogonal:

$$(AB)^T(AB) = B^T A^T AB = B^T IB = B^T B = I.$$

The reversal law is what makes the middle collapse possible — the same flip that powered Subtopic 02's classifications.

Entrywise meaning. Writing columns of A as $c_1, \dots, c_n$, the equation $A^T A = I$ reads

$$c_i \cdot c_j = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

— the columns are mutually perpendicular unit vectors, exactly as the chapter promised. The same holds for rows, via $AA^T = I$.

Example D.1 — [JEE Main Level]

Show that $R_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ **is orthogonal for every angle θ , and deduce R_θ^{-1} .**

Transpose and multiply:

$$R_\theta^T R_\theta = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$$

Entry by entry, using $\cos^2 \theta + \sin^2 \theta = 1$:

$$\begin{aligned} (1, 1) : \cos^2 \theta + \sin^2 \theta &= 1, & (1, 2) : -\cos \theta \sin \theta + \sin \theta \cos \theta &= 0, \\ (2, 1) : -\sin \theta \cos \theta + \cos \theta \sin \theta &= 0, & (2, 2) : \sin^2 \theta + \cos^2 \theta &= 1. \end{aligned}$$

So $R_\theta^T R_\theta = I$: orthogonal. Hence

$$R_\theta^{-1} = R_\theta^T = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} = R_{-\theta}.$$

Geometrically perfect: the inverse of a rotation by θ is the rotation by $-\theta$ — and it cost no row operations at all.

Example D.2 — [JEE Advanced Level]

Find all 2×2 upper triangular orthogonal matrices.

Write $A = \begin{bmatrix} a & b \\ 0 & d \end{bmatrix}$ and impose $A^T A = I$ entry by entry:

$$A^T A = \begin{bmatrix} a & 0 \\ b & d \end{bmatrix} \begin{bmatrix} a & b \\ 0 & d \end{bmatrix} = \begin{bmatrix} a^2 & ab \\ ab & b^2 + d^2 \end{bmatrix}.$$

Matching against I :

$$a^2 = 1, \quad ab = 0, \quad b^2 + d^2 = 1.$$

From $a^2 = 1$ we get $a = \pm 1$, so $ab = 0$ forces $b = 0$. Then $d^2 = 1$, so $d = \pm 1$ independently. Exactly four matrices survive:

$$A = \begin{bmatrix} \pm 1 & 0 \\ 0 & \pm 1 \end{bmatrix}.$$

Each is its own inverse and its own transpose — diagonal sign matrices, orthogonal by inspection. The technique (impose the defining equation cell by cell, let the constraints cascade) is the same one the chapter used for skew-symmetric unknown-hunting, pointed at a stronger condition.

Practice — Addendum D

1. Show that $A = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ -2 & 2 & -1 \end{bmatrix}$ is orthogonal, and write down A^{-1} .
2. If A is orthogonal, prove by induction that A^n is orthogonal for every natural number n .
3. Prove: if A is symmetric and invertible, then A^{-1} is symmetric.
4. Prove: if A is skew-symmetric and invertible, then A^{-1} is skew-symmetric.
5. If A is both orthogonal and symmetric, show that $A^2 = I$, and conclude that A is involutory.

Answers:

1. Each row has squared length $(1 + 4 + 4)/9 = 1$, and the three pairwise row dots are $2 + 2 - 4 = 0$, $-2 + 4 - 2 = 0$, $-4 + 2 + 2 = 0$. So $AA^T = I$, and $A^{-1} = A^T = \frac{1}{3} \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & 2 \\ 2 & -2 & -1 \end{bmatrix}$.
2. Base: $n = 1$, A is orthogonal. Step: assume A^k orthogonal; then by the closure fact for products (with $B = A^k$),

$$(A^{k+1})^T A^{k+1} = (A^k)^T A^T A A^k = (A^k)^T I A^k = (A^k)^T A^k = I.$$

So A^{k+1} is orthogonal.

3. $(A^{-1})^T = (A^T)^{-1} = A^{-1}$, using the bridge identity and $A^T = A$.
4. $(A^{-1})^T = (A^T)^{-1} = (-A)^{-1} = -A^{-1}$.
5. Orthogonal gives $A^{-1} = A^T$; symmetric gives $A^T = A$; so $A^{-1} = A$, i.e. $A^{-1}A = A^2 = I$. An involutory matrix is exactly one with $A^2 = I$.

Addendum E — Trace: From Assertion to Tool

Subtopic 04 states $\text{tr}(AB) = \text{tr}(BA)$ and calls it “the surprising one,” then moves on. JEE rewards exactly that identity, because it settles questions that brute computation cannot touch — including a genuine impossibility proof. Here are the derivations and the payoff.

Method

Definition. For square A of order n :

$$\operatorname{tr}(A) = \sum_{i=1}^n a_{ii}.$$

Linearity (one-line proofs, worth having ready). Entrywise:

$$\operatorname{tr}(A + B) = \sum_i (a_{ii} + b_{ii}) = \operatorname{tr}(A) + \operatorname{tr}(B), \quad \operatorname{tr}(kA) = \sum_i k a_{ii} = k \operatorname{tr}(A).$$

The reversal identity. Claim: $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ whenever both products exist (square, same order).

Derivation. Expand the diagonal of the product with the multiplication rule:

$$\operatorname{tr}(AB) = \sum_{i=1}^n (AB)_{ii} = \sum_{i=1}^n \sum_{r=1}^n a_{ir} b_{ri}.$$

Now look at $\operatorname{tr}(BA)$:

$$\operatorname{tr}(BA) = \sum_{r=1}^n \sum_{i=1}^n b_{ri} a_{ir}.$$

These are the same double sum — identical terms $a_{ir} b_{ri}$ added in a different order, i -outer/ r -inner versus r -outer/ i -inner. Finite sums commute with reordering, so

$$\operatorname{tr}(AB) = \operatorname{tr}(BA).$$

Note what the proof does *not* claim: the matrices themselves differ in general. Only their diagonals agree.

Consequence 1 — the commutator obstruction. Define $AB - BA$ (the commutator). Taking traces:

$$\operatorname{tr}(AB - BA) = \operatorname{tr}(AB) - \operatorname{tr}(BA) = 0.$$

But $\operatorname{tr}(I_n) = n \neq 0$. Therefore:

$$AB - BA = I_n \text{ is impossible, for every } n \geq 1.$$

Consequence 2 — a non-negativity test. For real matrices, the diagonal of $A^T A$ is made of squares:

$$(A^T A)_{ii} = \sum_{r=1}^n a_{ri}^2 \geq 0, \quad \text{so} \quad \operatorname{tr}(A^T A) = \sum_{i=1}^n \sum_{r=1}^n a_{ri}^2 \geq 0,$$

with equality iff every entry of A is 0. Hence:

$$A^T A = O \implies A = O.$$

(Recall from Subtopic 02 that $A^T A$ is always symmetric and square — here we extract its trace.)

Consequence 3 — trace of a skew-symmetric matrix. Its diagonal is all zeros (Derivation 1 of Subtopic 02), so $\text{tr}(A) = 0$ immediately — no computation needed.

Example E.1 — [JEE Main Level]

Verify $\text{tr}(AB) = \text{tr}(BA)$ for $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, and confirm the products themselves differ.

Products first:

$$AB = \begin{bmatrix} 1+4 & 2 \\ 3+8 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 11 & 4 \end{bmatrix}, \quad BA = \begin{bmatrix} 1 & 2 \\ 5 & 8 \end{bmatrix}.$$

Clearly $AB \neq BA$. Yet

$$\text{tr}(AB) = 5 + 4 = 9, \quad \text{tr}(BA) = 1 + 8 = 9.$$

Equal diagonals, different matrices — the identity is about the trace alone.

Example E.2 — [JEE Advanced Level]

Prove that there are no $n \times n$ matrices A, B satisfying $AB - BA = I_n$, for any $n \geq 1$.

Suppose such A, B existed. Take the trace of both sides. Left side, by linearity and the reversal identity:

$$\text{tr}(AB - BA) = \text{tr}(AB) - \text{tr}(BA) = 0.$$

Right side:

$$\text{tr}(I_n) = \underbrace{1 + 1 + \cdots + 1}_{n \text{ terms}} = n \geq 1.$$

So $0 = n$, a contradiction. No such pair exists — for any n .

Why this is Advanced-level. Nothing in the hypothesis mentions traces, sizes, or entries. The winning move is noticing that the equation is destroyed by applying a quantity invariant under reversal — the same philosophy as Example 2 of Subtopic 02, where transposing settled a classification without touching a single entry.

Practice — Addendum E

1. Compute $\text{tr}(AB)$ and $\text{tr}(BA)$ for $A = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 4 \\ 0 & 2 \end{bmatrix}$.
2. Prove that no matrices A, B satisfy $AB - BA = 2I_3$.
3. If A is 3×3 with $\text{tr}(A) = 4$, find $\text{tr}(A + A^T)$ and $\text{tr}(3A - 2I)$.
4. Prove that the trace of any skew-symmetric matrix is 0, in one line.
5. Prove: if $A^T A = O$ for a real matrix A , then $A = O$.

Answers:

1. $AB = \begin{bmatrix} 2 & 8 \\ 1 & 10 \end{bmatrix}$, $BA = \begin{bmatrix} 6 & 12 \\ 2 & 6 \end{bmatrix}$; traces 12 and 12.
2. Taking traces: left side 0, right side $\text{tr}(2I_3) = 2 \cdot 3 = 6$; contradiction.
3. $\text{tr}(A^T) = \text{tr}(A) = 4$, so $\text{tr}(A + A^T) = 8$; and $\text{tr}(3A - 2I) = 12 - 6 = 6$.
4. $\text{tr}(A) = \text{tr}\left(\frac{A + A^T}{2}\right) - \text{tr}\left(\frac{A - A^T}{2}\right)$ aside — simply: $a_{ii} = -a_{ii}$ on the diagonal forces every $a_{ii} = 0$, so $\text{tr}(A) = \sum_i 0 = 0$.
5. The diagonal entries of $A^T A$ are $\sum_r a_{ri}^2$. If $A^T A = O$, each such sum is 0, and a sum of squares of real numbers vanishes only when every term does; hence all $a_{ri} = 0$, i.e. $A = O$.

Numbering and notation follow the chapter throughout: I for identity, O for the null matrix, tr for trace. Everything above runs on three engines you already own — entry-by-entry equality, the transpose-flip, and row operations — pushed one layer past where the chapter stopped. When a Chapter 4 problem hands you $|A| \neq 0$ as the invertibility test, recognise it as the determinant finally paying off the debt these addenda have been carrying.

Addendum P — What JEE Actually Asks

Everything below is drawn from the 43-year bank supplied (roughly 30 items, Main 2005S–2020 and Advanced 2008–2017). A warning first: in the source extract a large fraction of the matrices themselves were images and did not survive. Where I quote a question, I have either reconstructed it completely and re-solved it, or I describe its *shape* and cite the official answer. Items too corrupted to reconstruct honestly (e.g. Main Sep 04, 2020 (I) Q2, Main Online April 10, 2015, the 2008 and 2014 match-the-column sets) are counted in the distribution below but never reproduced.

P.1 The distribution

Judged from the recoverable items, the examiners' weight falls very unevenly:

- **High powers and hidden matrix structure** is the single biggest cluster — roughly a third of the bank. Evidence: Main Jan 7, 2020 (I); Sep 03, 2020 (I); Jan 12, 2019 (I); April 8, 2019 (I); Online April 16, 2018; Online April 9, 2016; 2005S; Adv 2016 (twice); Adv 2013; 2011. The matrix always carries a concealed property ($A^2 = I$, nilpotence, $\omega^3 = 1$ entries, unitriangular form) and you are asked for A^{31} , A^{50} , a sum like $I + A + \dots$, or one entry of a high power.
- **Transpose algebra with symmetric/skew material**: Main 2012, Online April 9, 2017, April 12, 2019 (I), Jan 8, 2020 (II)-adjacent items, Adv 2015, plus fragments in the 2008/2014 matches. Typically: classify an expression like $X^4Z^3 - Z^3X^4$, or solve a system mixing A , B and B^T .
- **Orthogonality and counting under a quadratic constraint**: Adv 2017 ($\text{tr}(M^T M) = 5$, answer 198); Main Jan 8, 2020 (I) ($\text{tr}(AA^T) = 3$ over $\{-1, 0, 1\}$, answer 672); Main April 9, 2019 (II) (count (x, y) with $A^T A = 3I_3$); Main Jan 11, 2019 (I) and Main 2015 ($AA^T = I$ or $9I$); Adv 2016 ($P^2 = -I$ over cube roots of unity, answer 1); 2011 (non-singular ω -matrices, answer 2).
- **Matrix equation collapsed to a scalar system**: Main Online April 22, 2013 ($2p + q - r = -3$); Online April 12, 2014 ($y = 2x$).
- **Inverse read off a polynomial relation**: Main Jan 8, 2020 (II) ($10A^{-1} = 6I - A$); Adv 2014 fragment.

What is conspicuously *absent* at JEE level: constructing a matrix from a rule, “express as symmetric + skew” long-answer work, and grinding an inverse by row operations. Those are Board marks, not JEE marks.

P.2 The recurring archetypes

Archetype 1 — Collapse the power: find the hidden identity. *Shape*: “ $A^{2020} = ?$ ”, “sum of entries/column of A^{50} ”, often with ω (cube root of unity) planted in the entries. *Method*: Before multiplying anything, test A^2 : is it I , A , O , or a small combination? If $A = I + N$ with $N^k = O$, use the finite binomial theorem. If entries involve ω , reduce exponents mod 3. *Worked example* (reconstructed from Main Online April 16, 2018, reproducing the official answer 231):

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}, \quad B = A^{20}; \text{ find the sum of the first column of } B.$$

Write $A = I + N$. Then

$$N = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}, \quad N^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad N^3 = O.$$

So the binomial series stops:

$$A^{20} = I + 20N + \binom{20}{2}N^2.$$

First column: $(1, 20, 20 + 190) = (1, 20, 210)$; sum = 231. Brute force would need twenty 3×3 multiplications; the whole point is that you never do them.

The same skeleton, other disguises: Adv 2016 built $P = I + A$ with $A^2 = O$ and asked inside $P^{50} - Q = I$ (expand to $I + 50A + \binom{50}{2}A^2$); Main Jan 7, 2020 (I) hid $\alpha = \omega$ in A so that A has small period and A^{31} reduces mod that period; Adv 2013 took $p_{ij} = \omega^{i+j}$, where

$$P_{ij}^2 = \sum_k \omega^{i+k} \omega^{k+j} = \omega^{i+j} \sum_{k=1}^n (\omega^2)^k,$$

which vanishes exactly when $3 \mid n$ — hence $P^2 \neq O$ precisely for $n = 55, 56, 58$.

Archetype 2 — Rotation matrices: angles doing the arithmetic. *Shape:* $A = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$

with a condition on A^n ; find α . (Main April 8, 2019 (I) asked this with $n = 32$.) *Method:* $A^n = R(n\alpha)$; convert the matrix condition into $n\alpha \equiv (\text{target angle}) \pmod{2\pi}$. In the 2019 item the surviving solution requires $\cos 32\alpha = 0$, $\sin 32\alpha = 1$, i.e. $32\alpha \equiv \frac{\pi}{2} \pmod{2\pi}$, giving $\alpha = \frac{\pi}{64}$ as the base value. Never multiply the matrix out — rotate.

Archetype 3 — Counting with $\text{tr}(AA^T)$ or $A^T A$. *Shape:* “How many 3×3 matrices with entries from a small set satisfy ...?” *Method:* Know cold that

$$\text{tr}(AA^T) = \sum_{i,j} a_{ij}^2,$$

so the condition becomes a Diophantine statement about squared entries; then split into cases and count placements. *Worked example* (Main Jan 8, 2020 (I), answer 672): entries from $\{-1, 0, 1\}$, $\text{tr}(AA^T) = 3$. Since squares lie in $\{0, 1\}$, exactly three cells are non-zero. Choose the cells in $\binom{9}{3} = 84$ ways, assign signs in $2^3 = 8$ ways:

$$84 \times 8 = 672.$$

Its harder sibling (Adv 2017, answer 198): entries from $\{0, 1, 2\}$, $\text{tr}(M^T M) = 5$. Cases: five 1s and four 0s, giving $\binom{9}{4} = 126$; or one 1, one 2 ($2^2 = 4$ contributes), giving $9 \times 8 = 72$. Total $126 + 72 = 198$. Missing the second case — because $2^2 = 4$ hides inside “= 5” — is the designed failure.

Archetype 4 — Transpose systems and flip-classification. *Shape:* Either (a) equations linking A, B, B^T ; or (b) “which of these is symmetric/skew?” with products like $X^4 Z^3 - Z^3 X^4$. *Method:* For (a), transpose the whole system to manufacture more equations, then solve linearly. For (b), apply the reversal law $(UVW)^T = W^T V^T U^T$, substitute $X^T = -X$, $Z^T = Z$, and read the sign — together with the parity facts

$$(X^{\text{even}})^T = X^{\text{even}}, \quad (X^{\text{odd}})^T = -X^{\text{odd}}$$

for skew X . *Worked example* (Main Online April 9, 2017, answer (b)): $A+B = 2B^T$, $3A+2B = I_3$. From the first, $A = 2B^T - B$; substituting into the second:

$$6B^T - B = I_3.$$

Transpose it: $6B - B^T = I_3$. Eliminating B^T gives $35B = 7I_3$, so $B = A = \frac{I_3}{5}$, and

$$10A + 5B = 2I_3 + I_3 = 3I_3.$$

Advanced instance (Adv 2015, answers (c),(d)): with X, Y skew and Z symmetric,

$$\begin{aligned}(X^4Z^3 - Z^3X^4)^T &= Z^3X^4 - X^4Z^3 = -(X^4Z^3 - Z^3X^4) \quad (\text{skew}), \\ (X^{23} + Y^{23})^T &= -X^{23} - Y^{23} \quad (\text{skew}),\end{aligned}$$

while $Y^3Z^4 - Z^4Y^3$ and $X^{44} + Y^{44}$ come out symmetric. Option-designers rely on students mis-tracking which factor picks up the minus sign.

Archetype 5 — The inverse hiding in a polynomial. *Shape:* “ A satisfies $f(A) = O$; find cA^{-1} .” *Method:* Peel off the constant term, factor out A , divide by the scalar. *Worked example* (Main Jan 8, 2020 (II), answer (b); the printed matrix was lost — the reconstruction below is the one consistent with the official key):

$$A = \begin{pmatrix} 3 & 1 \\ -1 & 3 \end{pmatrix} : \quad A^2 = \begin{pmatrix} 8 & 6 \\ -6 & 8 \end{pmatrix}, \quad A^2 - 6A + 10I = O.$$

Left-multiply by A^{-1} :

$$\begin{aligned}A - 6I + 10A^{-1} &= O \\ 10A^{-1} &= 6I - A.\end{aligned}$$

Check against the direct inverse ($\det A = 10$, $\text{adj } A = \begin{pmatrix} 3 & -1 \\ 1 & 3 \end{pmatrix}$): indeed $10A^{-1} = \begin{pmatrix} 3 & -1 \\ 1 & 3 \end{pmatrix} = 6I - A$.

Archetype 6 — Matrix equation as a linear system. *Shape:* “ $[u_{ij}]$ times a column of unknowns equals a given column; find a combination of the unknowns.” *Method:* Do not touch matrix machinery — write the two or three scalar rows and solve. Main Online April 22, 2013 reduces to $4p + 2q = 0$, $3p + 3q + 2r = 3$, $p + 3q + 2r = 1$, yielding $p = 1$, $q = -2$, $r = 3$ and $2p + q - r = -3$. The trap is time: students who start computing matrix products burn minutes on a two-minute problem.

P.3 The traps

This bank punishes seven specific reflexes:

1. **Assuming commutativity.** Distractors are engineered around $(A+B)^2 = A^2 + 2AB + B^2$ and $(AB)^{-1} = A^{-1}B^{-1}$. Every expansion must keep both cross terms alive.

2. **Misreading** $\text{tr}(AA^T)$. It is the sum of *all nine squared entries*, not $(\text{tr } A)^2$ and not a count of non-zero entries. In the $\{-1, 0, 1\}$ problem, forgetting that each non-zero cell has *two* sign choices halves the correct count (84 instead of 672).
3. **Dropping the reversal.** $(AB)^T = B^T A^T$ and $(AB)^{-1} = B^{-1} A^{-1}$. In flip-classification questions, writing the non-reversed order flips your final answer — exactly the wrong/right option pair offered.
4. **Ignoring the forced zero diagonal of a skew matrix.** Several options die instantly on this fact alone; students who don't use it do five minutes of algebra.
5. **Parity blindness for powers of skew matrices.** Odd powers stay skew, even powers turn symmetric. $X^{44} + Y^{44}$ versus $X^{23} + Y^{23}$ in Adv 2015 is purely this distinction dressed up.
6. **Believing** $AB = O$ **forces a zero factor**, or that proportional rows can still be inverted. Singular matrices are everywhere in this bank's distractors.
7. **Brute-forcing powers.** The moment a power exceeds roughly 3, a structure ($A^2 = I$, nilpotence, $\omega^3 = 1$, rotation angle) is planted. Computing A^{50} entry-by-entry is the intended wrong path.

P.4 Graded practice set

Every answer below has been independently recomputed.

Q1 (warm-up). A matrix has 36 elements. How many possible orders can it have?

Solution. Orders correspond to divisor pairs of $36 = 2^2 3^2$:

$$1 \times 36, 2 \times 18, 3 \times 12, 4 \times 9, 6 \times 6, \\ 36 \times 1, 18 \times 2, 12 \times 3, 9 \times 4,$$

i.e. $(2+1)(2+1) = 9$ orders.

Q2. Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Find a_{12} of A^{10} .

Solution. $A = I + N$ with $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, $N^2 = O$, so $A^{10} = I + 10N$ and $a_{12} = 10$.

Q3. $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$. Find A^5 .

Solution. $A^2 = \begin{pmatrix} 5 & 10 \\ 10 & 20 \end{pmatrix} = 5A$, so by induction $A^n = 5^{n-1}A$. Hence

$$A^5 = 625 A = \begin{pmatrix} 625 & 1250 \\ 1250 & 2500 \end{pmatrix}.$$

Q4. A and B are symmetric. Show BAB is symmetric.

Solution.

$$(BAB)^T = B^T A^T B^T = BAB,$$

using the reversal law twice. Done — no entries ever needed.

Q5. How many 3×3 matrices with entries from $\{-1, 0, 1\}$ satisfy $\text{tr}(AA^T) = 2$?

Solution. $\text{tr}(AA^T) = \sum a_{ij}^2$, and squares here are 0 or 1, so exactly two cells are non-zero. Choose them in $\binom{9}{2} = 36$ ways and sign them in $2^2 = 4$ ways:

$$36 \times 4 = 144.$$

Q6. P is a square matrix with $P^T = -2P + I$. Compute P .

Solution. Transpose the relation: $P = -2P^T + I$. Substitute $P^T = -2P + I$:

$$P = -2(-2P + I) + I$$

$$P = 4P - I$$

$$3P = I,$$

so $P = \frac{I}{3}$. (Same skeleton as Main 2012, where the answer was $PX = -X$.)

Q7. Solve for real x, y, z :

$$\begin{pmatrix} 1 & 1 & 1 \\ 3 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 6 \\ 7 \\ 0 \end{pmatrix}, \quad \text{then find } x + y - z.$$

Solution. Row 3 gives $x = y$. Then rows become $2x + z = 6$ and $3x + z = 7$; subtracting, $x = 1$, so $y = 1$, $z = 4$, and

$$x + y - z = -2.$$

Q8. Count the real matrices $A = \begin{pmatrix} x & 2 \\ 2 & y \end{pmatrix}$ with $AA^T = 13I_2$.

Solution.

$$AA^T = \begin{pmatrix} x^2 + 4 & 2(x + y) \\ 2(x + y) & y^2 + 4 \end{pmatrix} = 13I_2,$$

so $x = \pm 3$, $y = \pm 3$, and $x + y = 0$ forces $y = -x$. Exactly **two** matrices. (Mirrors Main April 9, 2019 (II), whose key answer was 4.)

Q9. $A = R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ with $\theta = \frac{\pi}{9}$. Find the least positive n with $A^n = I$.

Solution. $A^n = R(n\theta)$, and $R(\phi) = I$ iff $\phi \equiv 0 \pmod{2\pi}$:

$$\frac{n\pi}{9} = 2k\pi \Rightarrow n = 18k,$$

so the least positive n is **18**.

Q10. $A = \begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix}$. Using a polynomial relation, express $5A^{-1}$ without computing a determinant.

Solution. $A^2 = \begin{pmatrix} 3 & 4 \\ -4 & 3 \end{pmatrix}$, and

$$A^2 - 4A + 5I = \begin{pmatrix} 3 - 8 + 5 & 4 - 4 \\ -4 + 4 & 3 - 8 + 5 \end{pmatrix} = O.$$

Multiply by A^{-1} : $A - 4I + 5A^{-1} = O$, hence

$$5A^{-1} = 4I - A = \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}.$$

(Check: $\det A = 5$ and the adjugate agrees.)

Q11. X is skew-symmetric and Z is symmetric (same order). Determine the nature of $ZX - XZ$.

Solution.

$$\begin{aligned} (ZX - XZ)^T &= (ZX)^T - (XZ)^T \\ &= X^T Z^T - Z^T X^T \\ &= (-X)Z - Z(-X) \\ &= ZX - XZ, \end{aligned}$$

so $ZX - XZ$ is **symmetric** — the commutator of a symmetric and a skew matrix is symmetric, a fact worth memorising alongside its cousins.

Q12 (capstone, Adv-2016 style). Let $N = \begin{pmatrix} 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ and $P = I + N$. If $Q = P^{50} - I$, find the sum of all entries of Q .

Solution. First map the nilpotence:

$$N^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad N^3 = O.$$

So the binomial expansion truncates:

$$Q = P^{50} - I = 50N + \binom{50}{2}N^2.$$

Sum of entries of N is $1 + 2 + 1 = 4$; sum of entries of N^2 is 1. Therefore

$$50 \times 4 + 1225 \times 1 = 1425.$$

One glance at $N^3 = O$ replaces fifty matrix multiplications — that is the entire skill being tested.

P.5 What this chapter's own text under-serves

The chapter is organised around Board procedure — construct-from-rule, decompose into symmetric + skew, grind an inverse by row operations — while nearly every JEE item in the bank lives elsewhere: collapsing huge powers through hidden identities, proving symmetry by transposing expressions, counting matrices under a quadratic trace condition, and reading inverses off polynomial relations. Subtopic 04 gestures at these ideas in a few lines, but nowhere develops the two techniques JEE actually rewards (the truncated binomial for unipotent/nilpotent matrices, and systematic flip-classification), nor does it introduce the camouflage devices — cube roots of unity and rotation matrices — that appear in at least four separate papers. Finally, by deferring determinants wholesale, the chapter leaves students without the singularity instinct (“proportional rows, no inverse”) that several bank questions assume as common knowledge.

Chapter 4 · Determinants

Complete Master Guide — Class 12 Mathematics (CBSE / JEE Main / JEE
Advanced)

Drona Ed Tech

June 2026

Contents

Part 1 — Determinant: Meaning, Definition & Expansion	1
Part 2 — Properties of Determinants	7
Part 3 — Area of a Triangle, Collinearity & Equation of a Line	15
Part 4 — Minors and Cofactors	20
Part 5 — Inverse of a Matrix & System of Linear Equations	27
Part 6 — Advanced Techniques: Multiplication, Differentiation & Special Determinants	34

Part 1 — Determinant: Meaning, Definition & Expansion

Exam Relevance: This is the bedrock of the entire chapter — you cannot attempt a single determinant question without it. CBSE Boards ask direct 1–2 mark evaluations and “solve the determinant equation” problems. Every JEE Main/Advanced determinant question silently depends on fluent, error-free expansion.

Section 1: Concept Intuition

A **determinant** is a single number squeezed out of a *square* arrangement of numbers. Give it a square grid and it hands you back one value that captures the grid’s “essence” — in particular, whether the rows pull in genuinely independent directions (non-zero) or secretly collapse onto each other (zero).

You have met its smallest version already. For a 2×2 array, the determinant is the **cross-difference**:

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

Geometrically this is the signed area of the parallelogram drawn by the two rows — a fact we leaned on throughout the Properties part. The determinant of a 3×3 array is the signed *volume* of the box drawn by its three rows.

Think about it this way... Just as net run rate compresses an entire cricket season into one number that tells you something real about a team, a determinant compresses a whole square table into one number that tells you something real about the system it represents — most importantly, whether a related set of equations has a clean unique answer.

Two pieces of notation and history worth fixing in memory: the value is written $\det A$, or with vertical bars $|A|$, or denoted Δ . The vertical-bar notation was introduced by **Arthur Cayley** in 1841; the idea itself grew out of **Gabriel Cramer’s**

1750 work on solving simultaneous equations. A determinant of order n expands into $n!$ signed product terms — 2 terms for order 2, 6 for order 3 — which is exactly why we hunt for shortcuts (zeros, properties) instead of expanding blindly.

Limiting Conditions & Assumptions. A determinant is defined **only for a square matrix**. The order-1 determinant $|a| = a$ is *not* the modulus — $|-5| = -5$ as a determinant, even though -5 's absolute value is 5. The Sarrus shortcut (below) works for 3×3 **only** — never apply it to 2×2 or 4×4 .

Section 2: Key Definitions & Formulas

A determinant is a **scalar** (dimensionless number).

Order 1. For $A = [a]$: $|A| = a$.

Order 2.

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}.$$

Order 3 — expansion along the first row:

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}.$$

Sign rule. Each term carries the sign $(-1)^{i+j}$ — the checkerboard $\begin{smallmatrix} + & - & + \\ - & + & - \\ + & - & + \end{smallmatrix}$. A determinant may be expanded along **any** row or column with these signs; the answer is identical (the formal machinery is developed in Part 4, Minors & Cofactors).

Sarrus rule (order 3 only). Rewrite the first two columns to the right of the determinant. Add the three full “ $\searrow$ ” diagonals and subtract the three full “ $\swarrow$ ” diagonals:

$$|A| = (a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}) - (a_{13}a_{22}a_{31} + a_{11}a_{23}a_{32} + a_{12}a_{21}a_{33}).$$

Number of terms. An order- n determinant expands into $n!$ terms.

Section 3: Key Procedures & Logical Framework

Procedure — Expanding a 3×3 along a row/column

1. **Choose the line** (row or column) with the most zeros — every zero entry removes a whole 2×2 evaluation.
2. **Walk along that line.** For each entry, multiply it by the 2×2 determinant left after deleting its row and column (its *minor*).
3. **Attach the checkerboard sign** $(-1)^{i+j}$ to each term.
4. **Add the signed terms.**

Why the alternating sign? It is what makes the expansion consistent no matter which line you pick — the signs encode the orientation flips that occur as you “skip over” rows and columns.

Procedure — Sarrus (a 3×3 speed tool)

Augment with the first two columns, sum the down-right diagonal products, subtract the down-left diagonal products. It is fast but **mechanical** — it does not generalise and offers no zero-skipping, so for sparse determinants, row/column expansion is faster.

Procedure — Solving a “determinant equation”

When a determinant containing an unknown x is set equal to a value, expand fully to get a polynomial in x , then solve. Wherever possible, **simplify with row/column operations first** (Part 2) so the resulting polynomial is low-degree.

Section 4: Worked Examples**Example 1 — [CBSE Board Level]**

Evaluate $\begin{vmatrix} 3 & -2 \\ 5 & 4 \end{vmatrix}$ and $\begin{vmatrix} 2 & 1 & 3 \\ 0 & 4 & 1 \\ 5 & 2 & 1 \end{vmatrix}$.

Solution. The 2×2 : $(3)(4) - (-2)(5) = 12 + 10 = 22$. The 3×3 , expanded along row 1:

$$2 \begin{vmatrix} 4 & 1 \\ 2 & 1 \end{vmatrix} - 1 \begin{vmatrix} 0 & 1 \\ 5 & 1 \end{vmatrix} + 3 \begin{vmatrix} 0 & 4 \\ 5 & 2 \end{vmatrix} = 2(4 - 2) - 1(0 - 5) + 3(0 - 20) = 4 + 5 - 60 = -51.$$

Example 2 — [Speed / Elimination Trap] (Sarrus)

Evaluate $\begin{vmatrix} 1 & 2 & 1 \\ 0 & 3 & 2 \\ 2 & 1 & 1 \end{vmatrix}$ using Sarrus.

Solution. Down-right products: $(1 \cdot 3 \cdot 1) + (2 \cdot 2 \cdot 2) + (1 \cdot 0 \cdot 1) = 3 + 8 + 0 = 11$. Down-left products: $(1 \cdot 3 \cdot 2) + (1 \cdot 2 \cdot 1) + (2 \cdot 0 \cdot 1) = 6 + 2 + 0 = 8$. So $|A| = 11 - 8 = 3$. *The trap:* Sarrus is only valid here because it's 3×3 . Students who memorise the “diagonal trick” sometimes try it on a 4×4 — it gives nonsense. For 4×4 and up, you **must** use cofactor expansion (Part 4).

Example 3 — [JEE Main Level] (determinant equation)

Find all x satisfying $\begin{vmatrix} x & 1 & 1 \\ 1 & x & 1 \\ 1 & 1 & x \end{vmatrix} = 0$.

Solution. Use the symmetry: $C_1 \rightarrow C_1 + C_2 + C_3$ gives a common factor $(x + 2)$ in column 1:

$$\begin{vmatrix} x+2 & 1 & 1 \\ x+2 & x & 1 \\ x+2 & 1 & x \end{vmatrix} = (x+2) \begin{vmatrix} 1 & 1 & 1 \\ 1 & x & 1 \\ 1 & 1 & x \end{vmatrix}.$$

Then $R_2 \rightarrow R_2 - R_1$, $R_3 \rightarrow R_3 - R_1$ yields a triangular form with diagonal $1, (x-1), (x-1)$:

$$= (x+2)(x-1)^2.$$

So $|A| = 0 \iff x = -2$ or $x = 1$. (This is the determinant behind the famous $kx + y + z$ system of Part 5.)

Example 4 — [JEE Advanced Level] (symbolic factorisation)

Prove that $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$.

Solution. Treat the left side as a polynomial. If $a = b$, columns 1 and 2 coincide, so the determinant vanishes $\Rightarrow (a-b)$ is a factor; likewise $(b-c)$ and $(c-a)$. That is degree 3. But the determinant's terms (e.g. bc^3) are degree 4; since both the determinant and $(a-b)(b-c)(c-a)$ alternate; quotient is symmetric, degree 1 — it must be $k(a+b+c)$. Fixing the constant with $a = 0, b = 1, c = 2$:

$$\begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 8 \end{vmatrix} = 1(8-2) = 6, \quad (a-b)(b-c)(c-a)(a+b+c) = (-1)(-1)(2)(3) = 6,$$

so the constant is 1 and the identity holds. ■

Section 5: Practice Exercises

1. [CBSE] Evaluate $\begin{vmatrix} 4 & 7 \\ -2 & 5 \end{vmatrix}$.
2. [CBSE] Evaluate $\begin{vmatrix} 1 & 0 & 2 \\ 3 & -1 & 4 \\ 2 & 5 & 1 \end{vmatrix}$.
3. [JEE Main] Solve $\begin{vmatrix} x & 3 \\ 2 & x-1 \end{vmatrix} = 0$.
4. [JEE Main] Evaluate $\begin{vmatrix} 0 & 2 & 0 \\ 3 & 1 & 4 \\ 0 & 5 & 0 \end{vmatrix}$.
5. [JEE Advanced] Show that $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$ (the Vandermonde determinant).

Answers: 1) 34 | 2) 13 | 3) $x = -2, 3$ | 4) 0 (column 1 and column 3 each hold a single nonzero entry, and the surviving 2×2 has a zero column) | 5) proved.

Section 6: Crack the MCQ

Q1. The value of the order-1 determinant $|-7|$ is:

- (a) 7 (b) -7 (c) 0 (d) undefined

Answer: (b). An order-1 determinant equals the entry itself, -7 — it is **not** the modulus. *Distractors:* (a) is the absolute-value trap. (c)/(d) misread the definition.

Q2. $\begin{vmatrix} 2 & 4 \\ 3 & 6 \end{vmatrix}$ equals:

- (a) 24 (b) 0 (c) -12 (d) 12

Answer: (b). $2 \cdot 6 - 4 \cdot 3 = 12 - 12 = 0$ (the rows are proportional). *Distractors:* (a) adds instead of subtracting the cross terms; (c)/(d) drop a sign.

Q3. Expanding a 3×3 determinant along any row or column gives:

- (a) different values for different rows (b) the same value always (c) zero unless along row 1 (d) twice the value along a column

Answer: (b). The determinant is well-defined independent of the expansion line. *Distractors:* all three invent dependencies that don't exist; the checkerboard signs guarantee consistency.

Q4. An order-4 determinant, fully expanded, has how many terms?

- (a) 4 (b) 16 (c) 24 (d) 12

Answer: (c). $n! = 4! = 24$. *Distractors:* (a) is n , (b) is n^2 (the entry count), (d) is a guess.

Section 7: Common Student Pitfalls & Pro-Tips

- **Pitfall 1 — Order-1 modulus confusion.** $|a|$ as a determinant is a , *not* its absolute value.
- **Pitfall 2 — Dropping the middle sign.** In row-1 expansion the middle term carries a **minus**: $-a_{12}(\dots)$. Forgetting it is the most common 3×3 error.
- **Pitfall 3 — Sarrus on the wrong size.** The diagonal trick is 3×3 only.
- **Pitfall 4 — Brute-forcing sparse determinants.** Expanding along a dense line when a zero-rich line exists wastes time.

Pro-Tip: Before expanding, *always* scan for the zero-richest row/column and expand there; and for symbolic determinants, look for a column/row whose entries share a sum (apply $C_1 \rightarrow C_1 + C_2 + C_3$) to pull out a factor before any expansion. A two-second scan routinely halves the work.

Section 8: Cheat Sheet

Definition & Expansion — Core Recall

- **Order 1:** $|a| = a$ (not modulus). **Order 2:** $ad - bc$.
- **Order 3 (row 1):** $a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}$, signs $(-1)^{i+j}$.
- **Expand along any line** — answer is invariant; pick the zero-richest.
- **Sarrus** (3×3 only): sum $\searrow$ diagonals — sum $\swarrow$ diagonals.
- **Term count:** order $n \Rightarrow n!$ terms.
- **Determinant equation:** simplify with row/column ops first, then solve the polynomial.

Mnemonic: *“Plus, minus, plus — march along the zeros.”*

Part 2 — Properties of Determinants

Exam Relevance: Properties are the single most-used tool in the entire chapter — you simplify *before* you expand. CBSE Boards almost always carry a 3–4 mark “prove without expanding” or “evaluate using properties” question. JEE Main poses 1–2 questions per paper, usually property-based factoring or fast evaluation. JEE Advanced folds properties into multi-step manipulations, often combined with algebra, sequences, or matrices.

Section 1: Concept Intuition

Imagine you run a small electronics shop in a busy market. Every square array of numbers — a *matrix* — is like a fully-loaded inventory sheet. The **determinant** is a single number that squeezes out of that sheet and tells you something deep about it: whether the rows are “genuinely independent” or secretly repeating each other, and (geometrically) how much “space” the rows enclose.

Here is the cleanest way to *see* a determinant. Take the three rows of a 3×3 array and treat each as an arrow (a vector) in 3D space. Those three arrows form the edges of a slanted box — a parallelepiped. **The determinant is the signed volume of that box.** For a 2×2 array it’s the signed area of the parallelogram made by its two rows. Once you hold this picture, every property of determinants stops being a rule to memorise and becomes something obvious.

Watch:

- **Swap two rows** → **the sign flips.** Swapping two edges of the box turns it “inside out” (reverses orientation), so the volume picks up a minus sign. Same box, opposite sign.
- **Two rows identical** → **determinant is zero.** If two edges of your box point the exact same way, the box is squashed flat. A flat box has zero volume. (Same thing if two rows are *proportional* — still flat.)
- **Multiply one row by k** → **determinant multiplies by k .** Stretch one edge to k times its length and the box’s volume stretches by exactly k .
- **Add a multiple of one row to another** → **determinant doesn’t change.** This is the magic one. Picture a stack of papers (a box). Push the top sheet sideways — a *shear*. The base is unchanged and the height is unchanged, so the volume is unchanged. That sideways push is precisely $R_i \rightarrow R_i + kR_j$.

Think about it this way... A shopkeeper rearranging *the order* in which he lists his stock categories doesn’t change his total wealth — he might just flip a sign in his own bookkeeping convention (swap rule). But if he accidentally writes the *same* category twice, his ledger collapses into nonsense — the determinant goes to zero, signalling “these rows carry no new information.”

The transpose property — that $|A| = |A^T|$, i.e. rows and columns are completely interchangeable — is the least visual but the most powerful. It means **every single property you learn for rows works identically for columns.** Learn six row rules and you’ve secretly learned twelve.

Limiting Conditions & Assumptions

- Determinants exist **only for square matrices**. There is no determinant of a 2×3 array.
- “Pull a factor k out of *one row*” (multiplies Δ by k) is **not** the same as multiplying the *whole* matrix: $|kA| = k^n |A|$ for order n . Mixing these up is the #1 exam error.
- The operation $R_i \rightarrow R_i + kR_j$ must use a **different** row ($i \neq j$). You cannot add a multiple of a row to itself in one step.
- Zero shows up for **identical OR proportional** rows/columns — proportionality is enough; they need not be exactly equal.

Section 2: Key Properties & Definitions

A determinant is a **scalar** (a pure number — dimensionless, no units) assigned to a square matrix. Let A be of order n and $\Delta = |A|$.

P1 — Transpose invariance.

$$|A^T| = |A|$$

Interchanging *all* rows with the corresponding columns leaves the value unchanged. *Consequence:* every property holds equally for rows and columns.

P2 — Interchange (sign) property. If any two rows (or two columns) are interchanged, the sign of the determinant reverses:

$$R_i \leftrightarrow R_j \Rightarrow \Delta \rightarrow -\Delta$$

P3 — Zero property. If any two rows (or columns) are **identical or proportional**, then

$$\Delta = 0$$

P4 — Scalar multiple property. If every element of one row (or column) is multiplied by k , the determinant is multiplied by k :

$$\begin{vmatrix} kR_1 \\ R_2 \\ R_3 \end{vmatrix} = k \begin{vmatrix} R_1 \\ R_2 \\ R_3 \end{vmatrix} \quad \text{and for the whole matrix} \quad |kA| = k^n |A|$$

A common factor of any row/column can be taken outside the bar.

P5 — Sum (linearity) property. If each element of one row is a sum of two terms, the determinant splits:

$$\begin{vmatrix} a_1 + x_1 & a_2 + x_2 & a_3 + x_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} + \begin{vmatrix} x_1 & x_2 & x_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

P6 — Row/Column operation property (invariance). Adding to one row a constant multiple of another row leaves the value unchanged:

$$R_i \rightarrow R_i + kR_j \quad (i \neq j) \Rightarrow \Delta \text{ is unchanged}$$

This is the workhorse for creating zeros and common factors.

P7 — Triangular determinant. For an upper- or lower-triangular (or diagonal) matrix, the determinant is the product of the diagonal entries:

$$\Delta = a_{11}a_{22}a_{33} \cdots a_{nn}$$

A determinant with an **all-zero row or column** is 0.

Bonus identities (used later in the chapter):

$$|AB| = |A| |B|, \quad |A^{-1}| = \frac{1}{|A|} \quad (|A| \neq 0)$$

Section 3: Theorems, Proofs & Key Procedures

Most properties are *not* independent — they fall out of one another in a tidy chain. Understanding the chain is what separates a student who *applies* properties from one who merely *recites* them.

Theorem A — Identical rows force $\Delta = 0$ (P3 from P2)

Claim: If two rows of Δ are identical, then $\Delta = 0$.

Proof. Suppose rows R_i and R_j are identical. Interchange them. By **P2**, the value becomes $-\Delta$. But because the two rows were *identical*, the matrix after the swap is **exactly the same matrix** — so its determinant is still Δ . Equating the two descriptions of the same object:

$$\Delta = -\Delta \Rightarrow 2\Delta = 0 \Rightarrow \Delta = 0. \quad \blacksquare$$

Why the step works: we counted the same number two different ways — once by “nothing changed” and once by “P2 changed the sign.” Only 0 can equal its own negative.

Theorem B — The operation $R_1 \rightarrow R_1 + kR_2$ is harmless (P6 from P5, P4, P3)

Claim: Adding a multiple of one row to another does not change Δ .

Proof. Apply the sum property **P5** to split the modified first row, then the scalar property **P4**, then Theorem A (**P3**):

$$\begin{vmatrix} R_1 + kR_2 \\ R_2 \\ R_3 \end{vmatrix} \stackrel{\text{P5}}{=} \begin{vmatrix} R_1 \\ R_2 \\ R_3 \end{vmatrix} + \begin{vmatrix} kR_2 \\ R_2 \\ R_3 \end{vmatrix} \stackrel{\text{P4}}{=} \Delta + k \underbrace{\begin{vmatrix} R_2 \\ R_2 \\ R_3 \end{vmatrix}}_{\text{two identical rows}} \stackrel{\text{P3}}{=} \Delta + k \cdot 0 = \Delta. \quad \blacksquare$$

Read the logic: the “damage” you add (the kR_2 row) lands in a determinant that already has R_2 sitting below it — two identical rows — which is automatically zero. So the addition is invisible. This single argument is the backbone of every “prove without expanding” question.

The “Prove Without Expanding” Procedure

A repeatable four-move algorithm:

1. **Scan for symmetry.** If row-sums or column-sums are equal, do $C_1 \rightarrow C_1 + C_2 + C_3$ (or the row version) to manufacture a common factor instantly.
2. **Pull out common factors** from any row/column using **P4** — this shrinks the numbers.
3. **Create zeros** with $R_i \rightarrow R_i + kR_j$ (**P6**) so the eventual expansion is trivial.
4. **Spot a kill.** If two rows/columns become identical or proportional, write $\boxed{0}$ by **P3**. Otherwise reduce to triangular form and multiply the diagonal (**P7**).

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Prove, without expanding, that

$$\Delta = \begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix} = 0.$$

Solution. Notice every row (and column) of this determinant is built from the *same three differences*. Apply the column operation $C_1 \rightarrow C_1 + C_2 + C_3$. Each new first-column entry becomes:

$$(a-b) + (b-c) + (c-a) = 0.$$

So:

$$\Delta = \begin{vmatrix} 0 & b-c & c-a \\ 0 & c-a & a-b \\ 0 & a-b & b-c \end{vmatrix}.$$

A determinant with an entire column of zeros is 0 (P7). Hence $\Delta = 0$. ■

Presentation note for Boards: state the operation used (here $C_1 \rightarrow C_1 + C_2 + C_3$) explicitly — examiners award a method mark for naming it.

Example 2 — [Speed / Elimination Trap]

Evaluate as fast as you can:

$$\begin{vmatrix} 2 & 4 & 6 \\ 1 & 2 & 3 \\ 5 & 7 & 9 \end{vmatrix}.$$

Fast solution. Don't expand! Look at the first two rows: $R_1 = (2, 4, 6) = 2 \times (1, 2, 3) = 2R_2$. Two **proportional** rows $\Rightarrow$ determinant = 0 by P3. Answer in under five seconds: $\boxed{0}$.

The trap: a student who jumps straight into cofactor expansion burns a full minute and risks an arithmetic slip — all to reach a zero that a single glance reveals. **Always scan for proportional/identical rows and column-sum symmetry before computing.**

Example 3 — [JEE Main Level] (Determinants × Arithmetic Progressions)**If p, q, r are in arithmetic progression, show that**

$$\Delta = \begin{vmatrix} x+1 & x+2 & x+p \\ x+2 & x+3 & x+q \\ x+3 & x+4 & x+r \end{vmatrix} = 0 \quad \text{for every } x.$$

Solution. This problem rewards spotting a *linear dependence* among the rows. Compute the combination $R_1 - 2R_2 + R_3$ column by column:

- Column 1: $(x+1) - 2(x+2) + (x+3) = 0$.
- Column 2: $(x+2) - 2(x+3) + (x+4) = 0$.
- Column 3: $(x+p) - 2(x+q) + (x+r) = (p+r) - 2q = 0$, because p, q, r in AP $\Rightarrow 2q = p+r$.

Every column gives 0, so $R_1 - 2R_2 + R_3 = \mathbf{0}$ — the rows are linearly dependent. A determinant whose rows are linearly dependent is 0. Hence $\Delta = 0$ for all x . ■

The two concepts fused: the AP condition ($2q = p+r$) and the determinant property that a dependent row set kills the determinant. Recognising “the third column is also in AP” is the whole game.

Example 4 — [JEE Advanced Level] (Factor-theorem reasoning)**Prove that**

$$\Delta = \begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} = abc(a-b)(b-c)(c-a).$$

Solution. First, extract the common factor from each *column* (P4): column 1 has factor a , column 2 has b , column 3 has c :

$$\Delta = abc \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = abc \Delta',$$

where Δ' is the classic **Vandermonde** determinant. Now we evaluate Δ' by *root reasoning* rather than brute expansion — the hallmark Advanced move.

Treat Δ' as a polynomial in a .

- If $a = b$, columns 1 and 2 of Δ' become identical $\Rightarrow \Delta' = 0$. So $(a-b)$ is a factor.
- By the same argument, $(b-c)$ and $(c-a)$ are factors.

That accounts for a degree-3 polynomial $(a-b)(b-c)(c-a)$. Counting the highest-degree term of Δ' (expanding, the term $b \cdot c^2$ etc. shows the total degree is also 3), the remaining cofactor must be a pure constant λ :

$$\Delta' = \lambda(a-b)(b-c)(c-a).$$

Fix λ by substituting convenient values $a = 0, b = 1, c = 2$:

$$\Delta' = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 4 \end{vmatrix} = 1 \cdot (4-2) = 2, \quad (a-b)(b-c)(c-a) = (-1)(-1)(2) = 2.$$

So $\lambda = 1$, giving $\Delta' = (a - b)(b - c)(c - a)$ and finally

$$\Delta = abc(a - b)(b - c)(c - a). \quad \blacksquare$$

Why this is “Advanced”: the factor theorem converts an ugly 3×3 expansion into a clean argument about roots and degree — a technique that generalises to far larger symmetric determinants where direct expansion is hopeless.

Section 5: Practice Exercises

1. **[CBSE]** Evaluate using properties: $\begin{vmatrix} 3 & 6 & 9 \\ 1 & 2 & 4 \\ 2 & 4 & 8 \end{vmatrix}$.
2. **[CBSE]** Prove without expanding: $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a - b)(b - c)(c - a)$.
3. **[JEE Main]** Evaluate $\begin{vmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 1 & \log_y z \\ \log_z x & \log_z y & 1 \end{vmatrix}$.
4. **[JEE Main]** If A is a 3×3 matrix with $|A| = 4$, find the value of $|3A| + |A^T|$.
5. **[JEE Advanced]** Prove that $\begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = 4abc$.

Answers: 1) 0 (rows R_2, R_3 proportional) | 2) identity proved (Vandermonde)
| 3) 0 | 4) 112 | 5) $4abc$.

Section 6: Crack the MCQ

Q1. If two columns of a third-order determinant are interchanged, the new value equals the original value:

- (a) multiplied by 2 (b) unchanged (c) multiplied by -1 (d) made zero

Answer: (c). One interchange flips the sign (P2). *Distractors:* (b) is the trap for students who confuse interchange with the operation $C_i \rightarrow C_i + kC_j$ (which **is** harmless). (d) confuses interchange with the *identical-column* rule. (a) wrongly imports the scalar-multiple idea.

Q2. The value of $\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$ is:

- (a) 1 (b) 0 (c) -3 (d) 24

Answer: (b). The rows are in AP, so $R_1 - 2R_2 + R_3 = \mathbf{0}$ — dependent rows $\Rightarrow 0$. *Distractors:* (c) is what you get from a sign slip during a rushed expansion ($-3 + 12 - 9$ mis-added). (d) and (a) are arithmetic-error landing spots. The lesson: spot the AP pattern and skip the expansion entirely.

Q3. If A is a 3×3 matrix with $|A| = 5$, then $|2A|$ equals:

(a) 10 (b) 20 (c) 40 (d) 80

Answer: (c). $|kA| = k^n |A| = 2^3 \times 5 = 40$. *Distractors:* (a) $= 2 \times 5$ — the classic error of treating $2A$ as scaling a *single* row. (b) $= 4 \times 5$ uses k^2 (forgot it's order 3). (d) over-corrects to 2^4 .

Q4. Without expanding, $\begin{vmatrix} a & b & c \\ a+2x & b+2y & c+2z \\ x & y & z \end{vmatrix}$ equals:

(a) 0 (b) $2 \times$ the determinant $\begin{vmatrix} a & b & c \\ x & y & z \\ x & y & z \end{vmatrix}$ (c) $(a+b+c)(x+y+z)$ (d) abc

Answer: (a). Apply $R_2 \rightarrow R_2 - 2R_3$: the middle row becomes (a, b, c) , identical to $R_1 \Rightarrow$ determinant 0 (P3). *Distractors:* (b) tempts students to “split off the 2” via P5 but lands on a determinant with two identical rows (itself 0) — so even this route gives 0, exposing (b) as a disguised wrong form. (c) and (d) are random symmetric-looking products with no basis.

Section 7: Common Student Pitfalls & Pro-Tips

- **Pitfall 1 — Confusing kR with kA .** Pulling k from *one* row multiplies Δ by k ; multiplying the *whole* $n \times n$ matrix multiplies it by k^n . Exam-killer in scaling questions.
- **Pitfall 2 — Illegal simultaneous operations.** Doing $R_1 \rightarrow R_1 + R_2$ **and** $R_2 \rightarrow R_2 + R_1$ in the *same* step is invalid — the second operation must use the *original* rows. Perform operations one at a time.
- **Pitfall 3 — Forgetting the sign on interchange.** Every single row/column swap costs a factor of -1 . Two swaps return you to $+$; an odd number leaves $-$.
- **Pitfall 4 — Thinking only *equal* rows give zero.** **Proportional** rows/-columns also force $\Delta = 0$. Always check ratios, not just equality.

Pro-Tip: Before touching a determinant, glance for (i) a zero row/column, (ii) two proportional rows/columns, and (iii) equal row-sums or column-sums. The first two hand you 0 instantly; the third invites the $C_1 \rightarrow C_1 + C_2 + C_3$ trick that pulls out a common factor in one move. This 5-second scan defeats most JEE/Board determinant questions before any real work begins.

Section 8: Cheat Sheet

Properties of Determinants — Core Recall

- $|A^T| = |A| \rightarrow$ every row rule is also a column rule.
- $R_i \leftrightarrow R_j \Rightarrow \Delta \rightarrow -\Delta$ (one swap = one sign flip).
- Two identical **or proportional** rows/columns $\Rightarrow \Delta = 0$.
- kR_i pulls out k ; for the whole matrix $|kA| = k^n |A|$.
- $R_i \rightarrow R_i + kR_j$ ($i \neq j$) leaves Δ **unchanged** — the simplification workhorse.

- Triangular/diagonal $\Rightarrow \Delta = \prod a_{ii}$; a zero row/column $\Rightarrow \Delta = 0$.
- $|AB| = |A||B|$, and $|A^{-1}| = 1/|A|$.

Mnemonic — the four moves: Swap $\rightarrow$ flip, Same $\rightarrow$ zero, Scale $\rightarrow$ pulls out, Shear $\rightarrow$ safe.

Logical chain to remember: P5 (split) + P4 (scalar) + P3 (identical $\rightarrow$ 0) $\Rightarrow$ P6 (row operation is harmless).

Part 3 — Area of a Triangle, Collinearity & Equation of a Line

Exam Relevance: A guaranteed CBSE scorer — 1-2 mark area or collinearity questions appear almost every year, and “find k so that the points are collinear” is a perennial. JEE Main uses it inside coordinate-geometry problems and loci; it is the cleanest real-world bridge between determinants and geometry.

Section 1: Concept Intuition

In the Properties part we saw that a 2×2 determinant is the signed area of a parallelogram. A triangle is exactly half a parallelogram — so determinants are the natural tool for triangle area from coordinates.

Suppose a surveyor marks three corner pegs of a triangular plot at (x_1, y_1) , (x_2, y_2) , (x_3, y_3) . Instead of measuring sides and angles, she can drop the three coordinate pairs into a single determinant and read off the area. The determinant does all the trigonometry silently.

Think about it this way... The determinant returns a **signed** number — positive if the vertices are listed anticlockwise, negative if clockwise. Area can't be negative, so we take the modulus and halve it. And here is the beautiful special case: if the three points are **collinear**, the “triangle” is squashed flat — zero height, zero area — so the determinant is exactly 0. That single fact turns a determinant into a collinearity detector, and (pushed one step further) into the equation of the line through two points.

Limiting Conditions & Assumptions. This applies to points in the **2-D plane**. The third column of 1's is essential — it encodes the affine (position, not just direction) nature of the points; dropping it changes the meaning entirely. The result is an *area* only after taking $\frac{1}{2}$ and the modulus; the bare determinant is a signed quantity used directly for collinearity and line equations.

Section 2: Key Formulas & Definitions

For vertices $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$:

Area of the triangle.

$$\text{Area} = \frac{1}{2} |\Delta|, \quad \Delta = \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}.$$

Expanded:

$$\text{Area} = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|.$$

Collinearity condition. The three points are collinear $\iff$

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0.$$

Equation of the line through (x_1, y_1) and (x_2, y_2) . A general point (x, y) lies on the line iff it is collinear with the two given points:

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0.$$

All quantities are pure numbers (areas in square units of the coordinate system); no dimensional formula applies.

Section 3: Theorems, Proofs & Key Procedures

Theorem — The determinant area formula

Claim. The area of triangle ABC is $\frac{1}{2}|\Delta|$ with Δ as above.

Proof. Place the triangle and compute the area as the trapezoidal difference under its edges (the “shoelace” idea). The signed area is

$$\frac{1}{2}[(x_1y_2 - x_2y_1) + (x_2y_3 - x_3y_2) + (x_3y_1 - x_1y_3)].$$

Now expand Δ along its third column:

$$\Delta = (x_2y_3 - x_3y_2) - (x_1y_3 - x_3y_1) + (x_1y_2 - x_2y_1),$$

which is precisely twice the bracketed signed area. Hence signed area = $\frac{1}{2}\Delta$, and the geometric (unsigned) area is $\frac{1}{2}|\Delta|$. ■

Why the third column of 1's? It is what injects the *positions* of the points (not merely their differences) into a single determinant; algebraically it is the device that turns three coordinate pairs into the shoelace sum automatically.

Corollary — collinearity and the line equation

Collinear points bound a degenerate triangle of zero area, so $\Delta = 0$. Replacing one vertex by a variable point (x, y) and demanding $\Delta = 0$ forces (x, y) to be collinear with the two fixed points — i.e. to lie on the line through them. Expanding that determinant gives the familiar $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$.

Procedure

1. Write the 3×3 with coordinates in the first two columns and 1's in the third.
2. Expand (easiest along column 3).
3. For **area**: take $\frac{1}{2} \times$ the modulus. For **collinearity**: set the determinant to 0. For a **line**: keep one row as $(x, y, 1)$ and set the determinant to 0.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Find the area of the triangle with vertices $(1, 2)$, $(3, 5)$, $(-2, 4)$.

Solution.

$$\Delta = \begin{vmatrix} 1 & 2 & 1 \\ 3 & 5 & 1 \\ -2 & 4 & 1 \end{vmatrix} = 1(5 - 4) - 2(3 + 2) + 1(12 + 10) = 1 - 10 + 22 = 13.$$

$$\text{Area} = \frac{1}{2}|13| = \frac{13}{2} = 6.5 \text{ square units.}$$

Example 2 — [Speed / Elimination Trap] (collinearity)

Are $(1, 1)$, $(2, 3)$, $(3, 5)$ collinear?

Fast solution. Slope from the first to the second is $\frac{3-1}{2-1} = 2$; from the second to the third, $\frac{5-3}{3-2} = 2$. Equal slopes $\Rightarrow$ **collinear** — no determinant needed. (Confirm: the

determinant $\begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 3 & 5 & 1 \end{vmatrix} = -2 + 1 + 1 = 0$.) *The trap:* setting up and expanding the

full determinant when a 5-second slope comparison settles a collinearity question. Use the determinant when coordinates are messy or a parameter is involved; use slopes when the numbers are clean.

Example 3 — [JEE Main Level] (solve for a parameter)

Find k so that $(k, 2)$, $(-3, 4)$, $(7, -1)$ are collinear.

Solution. Collinear $\Rightarrow$ determinant = 0:

$$\begin{vmatrix} k & 2 & 1 \\ -3 & 4 & 1 \\ 7 & -1 & 1 \end{vmatrix} = k(4+1) - 2(-3-7) + 1(3-28) = 5k + 20 - 25 = 5k - 5 = 0 \Rightarrow k = 1.$$

Example 4 — [JEE Advanced Level] (locus via area)

Find the locus of a point $P(x, y)$ such that the area of the triangle formed by P , $(1, 0)$ and $(0, 1)$ is always 2 square units.

Solution.

$$\begin{vmatrix} x & y & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{vmatrix} = x(0 - 1) - y(1 - 0) + 1(1 - 0) = -x - y + 1.$$

Area = $\frac{1}{2}|-x - y + 1| = 2 \Rightarrow |x + y - 1| = 4 \Rightarrow x + y - 1 = \pm 4$. So the locus is the **pair of parallel lines** $x + y = 5$ and $x + y = -3$ — every point on either line gives a triangle of area 2. ■

Section 5: Practice Exercises

1. **[CBSE]** Find the area of the triangle with vertices $(0, 0)$, $(4, 0)$, $(0, 6)$.
2. **[CBSE]** Show that $(2, 3)$, $(4, 7)$, $(6, 11)$ are collinear.
3. **[JEE Main]** The triangle with vertices $(\lambda, 2)$, $(1, 3)$, $(4, -1)$ has area $\frac{11}{2}$. Find λ .
4. **[JEE Main]** Find the equation of the line through $(2, 3)$ and $(-1, 5)$ using a determinant.
5. **[JEE Advanced]** If $(a, 0)$, $(0, b)$, $(1, 1)$ are collinear, prove that $\frac{1}{a} + \frac{1}{b} = 1$.

Answers: 1) 12 sq units | 2) determinant = 0, collinear | 3) $\lambda = \frac{9}{2}$ or $\lambda = -1$ | 4) $2x + 3y - 13 = 0$ | 5) proved.

Section 6: Crack the MCQ

Q1. The area of the triangle with vertices $(0, 0)$, $(3, 0)$, $(0, 4)$ is:

- (a) 12 (b) 6 (c) 7 (d) 24

Answer: (b). $\frac{1}{2}|12| = 6$ (or simply $\frac{1}{2} \cdot \text{base} \cdot \text{height} = \frac{1}{2} \cdot 3 \cdot 4$). *Distractors:* (a)/(d) forget the factor $\frac{1}{2}$; (c) adds the legs.

Q2. Three points are collinear if and only if the area determinant equals:

- (a) 1 (b) $\frac{1}{2}$ (c) 0 (d) any positive number

Answer: (c). Collinearity = degenerate (zero-area) triangle $\Rightarrow$ determinant 0. *Distractors:* (a)/(b)/(d) confuse “zero area” with specific nonzero values.

Q3. If $(2, -3)$, $(\lambda, 1)$, $(0, 4)$ are collinear, then λ equals:

- (a) $\frac{6}{7}$ (b) $-\frac{6}{7}$ (c) $\frac{7}{6}$ (d) $\frac{2}{7}$

Answer: (a). Set the determinant to zero and expand along row 1:

$$\begin{vmatrix} 2 & -3 & 1 \\ \lambda & 1 & 1 \\ 0 & 4 & 1 \end{vmatrix} = 2(1-4) + 3(\lambda-0) + 1(4\lambda-0) = -6 + 3\lambda + 4\lambda = 7\lambda - 6 = 0 \Rightarrow \lambda = \frac{6}{7}.$$

Distractors: (b) flips a sign in the expansion; (c) inverts the final fraction; (d) mishandles the -3 cofactor sign.

Q4. The factor of $\frac{1}{2}$ in the area formula is present because:

- (a) a determinant is always halved (b) a triangle is half its bounding parallelogram (c) of the modulus (d) there are 3 vertices

Answer: (b). The $2 \times 2/3 \times 3$ determinant measures the parallelogram (or twice the triangle); halving recovers the triangle. *Distractors:* (a) is false in general; (c) explains the modulus, not the half; (d) is irrelevant.

Section 7: Common Student Pitfalls & Pro-Tips

- **Pitfall 1 — Dropping the $\frac{1}{2}$ or the modulus.** The bare determinant is *twice the signed area*. Area needs both $\frac{1}{2}$ and $|\cdot|$.
- **Pitfall 2 — Reporting a negative area.** A negative determinant just means clockwise vertex order; take the modulus.
- **Pitfall 3 — Omitting the column of 1's.** Without it the determinant no longer represents area.
- **Pitfall 4 — Using area = 0 blindly.** A zero determinant means collinear (no triangle); don't then "find an area."

Pro-Tip: For collinearity with clean integer coordinates, compare slopes — it's faster than a determinant. Reserve the determinant for messy coordinates or parameter problems, where setting it to 0 gives a clean linear/quadratic equation in the unknown.

Section 8: Cheat Sheet

Area, Collinearity & Line — Core Recall

- **Area** $= \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$.
- **Collinear** $\iff$ that determinant = 0.
- **Line through two points:** put $(x, y, 1)$ as the first row and set the determinant = 0.
- The **column of 1's** is mandatory; the $\frac{1}{2}$ **and modulus** are mandatory for area.

Mnemonic: "Coordinates left, ones on the right; half the size, take it positive."

Part 4 — Minors and Cofactors

Exam Relevance: This is the machinery the rest of the chapter runs on — cofactors drive Laplace expansion and are the bridge to the adjoint and inverse (Part 5). CBSE Boards regularly ask a direct 1-2 mark “find the minor/cofactor of a_{ij} ” or “expand using cofactors.” JEE Main leans on cofactor expansion, the cross-cofactor-zero trick, and $|\text{adj } A|$ shortcuts. JEE Advanced pushes the power-cofactor formula $|C| = |A|^{n-1}$ into multi-step problems.

Section 1: Concept Intuition

Picture a 3×3 scoreboard at a gully-cricket tournament — three teams down the side, three match-stats across the top, nine cells in all. Now you want to judge the “weight” of one particular cell, say the entry in row 2, column 3. The trick the determinant uses is beautifully simple: **cover up that entire row and that entire column, and look only at the smaller grid that survives.** What’s left of a 3×3 is a tidy 2×2 . The determinant of that surviving little grid is called the **minor** of the entry you covered, written M_{23} .

So a minor is just “the determinant of what’s left after you hide one row and one column.” Every entry of the matrix has its own minor. For a 3×3 matrix, each minor is a 2×2 determinant; for a 4×4 , each minor is a 3×3 determinant, and so on.

A **cofactor** is a minor with a sign stuck on it. The sign follows a checkerboard pattern — exactly like the black-and-white squares of a carrom board or a chessboard:

$$\begin{array}{ccc} + & - & + \\ - & + & - \\ + & - & + \end{array}$$

Mathematically the “colour” of position (i, j) is captured by $(-1)^{i+j}$: when $i + j$ is even you sit on a + square, when it’s odd you sit on a – square. The cofactor is

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

Why bother attaching a sign? Because when you expand a determinant, the terms naturally alternate in sign. Baking that alternation into the minor — turning it into a cofactor — lets you write one clean, sign-free rule: *the determinant equals the sum of (each element $\times$ its own cofactor) along any single row or column.* The signs are already handled inside the cofactors.

Think about it this way... A cofactor answers the question: “If I nudge this one entry up by a tiny amount, how much does the whole determinant change?” In

fact $\frac{\partial \Delta}{\partial a_{ij}} = C_{ij}$ exactly. The cofactor is the *sensitivity* of the determinant to that single entry. That is why a row of small entries paired with large cofactors can still produce a big determinant — each entry’s “leverage” is its cofactor.

Two facts fall straight out of this picture, and they govern almost every exam question:

1. **Expand along *any* row or column and you get the same determinant.** This freedom is a gift: always choose the row or column carrying the most zeros, because a zero entry kills its whole cofactor term and saves you a 2×2 evaluation.
2. **Pair a row's entries with *another* row's cofactors and you get exactly zero.** Intuitively you'd be computing a determinant in which one row has been secretly duplicated — and a determinant with two identical rows is zero (Property P3 from Part 2).

Limiting Conditions & Assumptions. Minors and cofactors are defined only for **square** matrices, because the surviving sub-grid must itself be square. The position indices i, j start at 1 (an off-by-one in the sign exponent flips every sign). For a 1×1 matrix the notion degenerates — there is nothing left after you cover the only row and column, and the cofactor expansion bottoms out at the single entry itself.

Section 2: Key Definitions & Formulas

A determinant, minor, and cofactor are all **scalars** — pure numbers, dimensionless, carrying no units.

Minor. The minor M_{ij} of entry a_{ij} in an $n \times n$ matrix is the determinant of the $(n-1) \times (n-1)$ matrix obtained by deleting row i and column j .

Cofactor.

$$C_{ij} = (-1)^{i+j} M_{ij} \quad (\text{NCERT notation: } A_{ij})$$

Cofactor (Laplace) expansion — along row i or column j :

$$\Delta = \sum_{j=1}^n a_{ij} C_{ij} \quad (\text{row } i) \quad = \quad \sum_{i=1}^n a_{ij} C_{ij} \quad (\text{column } j)$$

Cross-orthogonality — entries of one row with cofactors of a *different* row sum to zero:

$$\sum_{j=1}^n a_{ij} C_{kj} = 0 \quad (i \neq k).$$

Combining both results compactly, with δ_{ik} the Kronecker delta (1 if $i = k$, else 0):

$$\sum_{j=1}^n a_{ij} C_{kj} = \Delta \delta_{ik}$$

Determinant of the cofactor matrix (power-cofactor formula). If $C = [C_{ij}]$ is the matrix of cofactors (note $\text{adj } A = C^\top$), then for an order- n matrix

$$|C| = |\text{adj } A| = |A|^{n-1} \quad (\text{for } n = 3 : |C| = |A|^2).$$

Sensitivity identity. $\frac{\partial \Delta}{\partial a_{ij}} = C_{ij}.$

Symbol-collision note. We reserve uppercase A for the matrix and write cofactors as C_{ij} . CBSE/NCERT writes the same object as A_{ij} — recognise both; they are identical.

Section 3: Theorems, Proofs & Key Procedures

Theorem A — Cross-cofactor sum is zero

Claim. For $i \neq k$, $\sum_j a_{ij} C_{kj} = 0$.

Proof. Build a new matrix A' identical to A except that its row k is **replaced by a copy of row i** . Crucially, replacing row k does **not** change the cofactors C_{kj} — those cofactors are computed by *deleting* row k , so they never “see” what row k contains. Now expand $|A'|$ along its (new) row k :

$$|A'| = \sum_j a'_{kj} C'_{kj} = \sum_j a_{ij} C_{kj},$$

since $a'_{kj} = a_{ij}$ (we copied row i in) and $C'_{kj} = C_{kj}$ (cofactors unchanged). But A' now has **two identical rows** (i and k), so by Property **P3** of Part 2, $|A'| = 0$. Therefore $\sum_j a_{ij} C_{kj} = 0$. ■

This is the cleanest demonstration of why the chapter’s properties form a toolkit: a result about cofactors collapses instantly once you reuse the “two identical rows $\Rightarrow 0$ ” fact.

Theorem B — The power-cofactor formula $|\text{adj } A| = |A|^{n-1}$

Claim. For an order- n matrix A , $|\text{adj } A| = |A|^{n-1}$.

Proof. The fundamental adjoint relation (proved in full in Part 5) is

$$A (\text{adj } A) = |A| I_n.$$

Take determinants of both sides. Using the multiplicative property $|XY| = |X||Y|$ and $|kI_n| = k^n$:

$$|A| \cdot |\text{adj } A| = ||A| I_n| = |A|^n.$$

If $|A| \neq 0$, divide by $|A|$ to get $|\text{adj } A| = |A|^{n-1}$. Since $\text{adj } A = C^\top$ and $|C^\top| = |C|$ (Property P1), we also have $|C| = |A|^{n-1}$. (The $|A| = 0$ case follows by a continuity/limiting argument.) ■

Procedure — Computing minors and cofactors of a 3×3 systematically

1. **Fix the entry a_{ij} .** Mentally strike out its row i and column j .
2. **Read off the surviving 2×2** in its natural order and compute its determinant: that is the minor M_{ij} .
3. **Attach the checkerboard sign $(-1)^{i+j}$** to get C_{ij} . Don’t recompute the sign from scratch each time — memorise the $+ - + / - + - / + - +$ board.

4. **To find Δ :** pick the row or column with the most zeros, then $\Delta = \sum(\text{entry} \times \text{its cofactor})$ along that line.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

For $A = \begin{pmatrix} 3 & 1 & 2 \\ 0 & 2 & -1 \\ 4 & 1 & 5 \end{pmatrix}$, find the cofactors of the second-row entries and hence evaluate $|A|$.

Solution. Strike out row 2 each time.

$$M_{21} = \begin{vmatrix} 1 & 2 \\ 1 & 5 \end{vmatrix} = 5 - 2 = 3, \quad C_{21} = (-1)^{2+1}(3) = -3.$$

$$M_{22} = \begin{vmatrix} 3 & 2 \\ 4 & 5 \end{vmatrix} = 15 - 8 = 7, \quad C_{22} = (-1)^{2+2}(7) = 7.$$

$$M_{23} = \begin{vmatrix} 3 & 1 \\ 4 & 1 \end{vmatrix} = 3 - 4 = -1, \quad C_{23} = (-1)^{2+3}(-1) = 1.$$

Expanding along row 2:

$$|A| = a_{21}C_{21} + a_{22}C_{22} + a_{23}C_{23} = (0)(-3) + (2)(7) + (-1)(1) = 14 - 1 = 13.$$

Check (direct expansion along row 1): $3(10+1) - 1(0+4) + 2(0-8) = 33 - 4 - 16 = 13$. ✓

Example 2 — [Speed / Elimination Trap]

Evaluate as fast as possible: $\begin{vmatrix} 7 & 0 & 0 \\ 2 & 5 & 3 \\ 9 & -1 & 4 \end{vmatrix}$.

Fast solution. Don't grind all nine cofactors. Row 1 has **two zeros**, so two of the three expansion terms vanish:

$$\Delta = 7C_{11} = 7(-1)^2 \begin{vmatrix} 5 & 3 \\ -1 & 4 \end{vmatrix} = 7(20 + 3) = 7 \times 23 = 161.$$

The trap: expanding along column 2 or 3 (which have only one zero each) forces you through more 2×2 evaluations and more sign bookkeeping. **Always scan for the zero-richest line first** — one well-chosen expansion line can cut the work by two-thirds.

Example 3 — [JEE Main Level] (cofactor expansion $\times$ cross-orthogonality)

For $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}$, evaluate both

$$(i) \ a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} \quad \text{and} \quad (ii) \ a_{11}C_{21} + a_{12}C_{22} + a_{13}C_{23}.$$

Solution. Expression (i) pairs row-1 entries with row-1 cofactors — that is the Laplace expansion along row 1, so it equals $|A|$:

$$|A| = 1(0 \cdot 1 - 1 \cdot 1) - 2(2 \cdot 1 - 1 \cdot 1) + 3(2 \cdot 1 - 0 \cdot 1) = (-1) - 2(1) + 3(2) = -1 - 2 + 6 = 3.$$

So (i) = 3.

Expression (ii) pairs row-1 entries with **row-2** cofactors — a *cross sum*. By Theorem A it is exactly 0, no computation needed. So (ii) = 0.

The two concepts fused: recognising that “same row” $\Rightarrow \Delta$ while “different row” $\Rightarrow 0$ turns a scary-looking pair of sums into one short expansion and one instant zero.

Example 4 — [JEE Advanced Level] (iterated power-cofactor)

Let A be a 3×3 matrix with $|A| = 2$. Let C be its cofactor matrix and $B = \text{adj}(\text{adj } A)$. Find $|C|$ and $|B|$.

Solution. With $n = 3$, the power-cofactor formula gives $|\text{adj } A| = |A|^{n-1} = |A|^2$, and since $\text{adj } A = C^\top$,

$$|C| = |\text{adj } A| = 2^2 = 4.$$

Now apply the formula **twice**. First $|\text{adj } A| = |A|^2 = 4$. Then treat $\text{adj } A$ as the input matrix:

$$|B| = |\text{adj}(\text{adj } A)| = |\text{adj } A|^{n-1} = (|A|^2)^2 = |A|^4 = 2^4 = 16.$$

General pattern worth remembering: $|\text{adj}(\text{adj } A)| = |A|^{(n-1)^2}$, which for $n = 3$ is $|A|^4$. ■

Section 5: Practice Exercises

1. **[CBSE]** Find the cofactor of a_{32} in $\begin{pmatrix} 2 & 3 & 1 \\ 4 & 0 & 5 \\ 1 & -2 & 3 \end{pmatrix}$.
2. **[CBSE]** Evaluate by expanding along the line with the most zeros: $\begin{vmatrix} 5 & 0 & 2 \\ 3 & 0 & 1 \\ 4 & 7 & 6 \end{vmatrix}$.
3. **[JEE Main]** For $A = \begin{pmatrix} 1 & -1 & 2 \\ 3 & 0 & 1 \\ 2 & 4 & -2 \end{pmatrix}$, find $a_{21}C_{11} + a_{22}C_{12} + a_{23}C_{13}$.
4. **[JEE Main]** A 3×3 matrix has $|A| = -5$. Find $|\text{adj } A|$.
5. **[JEE Advanced]** A 3×3 matrix satisfies $|\text{adj}(\text{adj } A)| = 81$. Find all possible values of $|A|$.

Answers: 1) -6 | 2) 7 (expand along column 2) | 3) 0 (cross-cofactor sum) | 4) 25 | 5) $|A| = \pm 3$ (since $|A|^4 = 81$).

Section 6: Crack the MCQ

Q1. The cofactor C_{23} of an entry a_{23} is related to its minor M_{23} by:

- (a) $C_{23} = M_{23}$ (b) $C_{23} = -M_{23}$ (c) $C_{23} = 0$ (d) $C_{23} = 2M_{23}$

Answer: (b). The sign is $(-1)^{2+3} = (-1)^5 = -1$. *Distractors:* (a) is the universal beginner error — quoting the minor as the cofactor and forgetting the sign. (c) confuses the cofactor with the cross-cofactor *sum* (which is zero). (d) is a meaningless doubling.

Q2. The value of $\begin{vmatrix} 2 & 1 & 0 \\ 3 & -1 & 0 \\ 5 & 4 & 2 \end{vmatrix}$ is:

- (a) -10 (b) 10 (c) -5 (d) 0

Answer: (a). Column 3 has two zeros, so expand there: $\Delta = 2C_{33} = 2(-1)^6 \begin{vmatrix} 2 & 1 \\ 3 & -1 \end{vmatrix} = 2(-2 - 3) = -10$. *Distractors:* (b) is a sign slip on the 2×2 . (c) forgets to multiply the minor by the entry $a_{33} = 2$. (d) is the trap for a student who sees two zeros and wrongly concludes the whole determinant is zero.

Q3. If A is a 3×3 matrix with $|A| = k$, then $|\text{adj } A|$ equals:

- (a) k (b) k^2 (c) k^3 (d) $1/k$

Answer: (b). $|\text{adj } A| = |A|^{n-1} = k^{3-1} = k^2$. *Distractors:* (a) assumes $|\text{adj } A| = |A|$. (c) uses the exponent n instead of $n - 1$ — the single most common power-cofactor mistake. (d) confuses the adjoint with the inverse, whose determinant is $1/k$.

Q4. For a 3×3 matrix, the sum $a_{11}C_{21} + a_{12}C_{22} + a_{13}C_{23}$ equals:

- (a) $|A|$ (b) 0 (c) $-|A|$ (d) $|A|^2$

Answer: (b). Row-1 entries are paired with row-2 cofactors — a cross sum, which is 0 by cross-orthogonality. *Distractors:* (a) is the headline trap: students memorise “entries $\times$ cofactors = $|A|$ ” but forget the cofactors must come from the **same** row. (c) and (d) are sign/squaring guesses with no basis.

Section 7: Common Student Pitfalls & Pro-Tips

- **Pitfall 1 — Reporting the minor as the cofactor.** Forgetting the $(-1)^{i+j}$ sign is the #1 error. Always finish by stamping the checkerboard sign.
- **Pitfall 2 — Cross-cofactor confusion.** Pairing a row's entries with a *different* row's cofactors and expecting $|A|$. That sum is 0, not $|A|$. Same row $\Rightarrow |A|$; different row $\Rightarrow 0$.
- **Pitfall 3 — Wrong power in $|\text{adj } A|$.** Writing $|A|^n$ instead of $|A|^{n-1}$. For 3×3 it is $|A|^2$, not $|A|^3$.
- **Pitfall 4 — Expanding along a dense line.** Choosing a row/column with no zeros wastes time and multiplies error chances. Scan for zeros first.

Pro-Tip: Two one-liners win most of these questions. (i) “*adj drops the power by one*” — $|\text{adj } A| = |A|^{n-1}$, and iterating gives $|A|^{(n-1)^2}$. (ii)

“same row sums to Δ , cross row sums to zero.” Internalise these and you skip the algebra entirely on a large fraction of JEE/Board cofactor problems.

Section 8: Cheat Sheet

Minors & Cofactors — Core Recall

- **Minor** M_{ij} = determinant left after deleting row i , column j .
- **Cofactor** $C_{ij} = (-1)^{i+j} M_{ij}$ (NCERT: A_{ij}). Sign board $\begin{smallmatrix} + & - & + \\ - & + & - \\ + & - & + \end{smallmatrix}$.
- **Laplace expansion:** $\Delta = \sum_j a_{ij} C_{ij}$ along any row (or column) — pick the zero-richest line.
- **Cross-orthogonality:** $\sum_j a_{ij} C_{kj} = \Delta \delta_{ik}$ — same row gives Δ , different row gives 0.
- **Power-cofactor:** $|\text{adj } A| = |C| = |A|^{n-1}$; iterated, $|\text{adj}(\text{adj } A)| = |A|^{(n-1)^2}$.
- **Sensitivity:** $\frac{\partial \Delta}{\partial a_{ij}} = C_{ij}$.

Mnemonics: *“Cover to find the **Minor**, sign the board for the **coFactor**.”*
 · *“Same row $\rightarrow \Delta$, cross row $\rightarrow 0$.”* · *“adj drops the power by one.”*

Part 5 — Inverse of a Matrix & System of Linear Equations

Exam Relevance: This is the payoff subtopic of the chapter and a guaranteed scorer. CBSE Boards almost always carry a **5-6 mark** “solve the system by matrix method” question. JEE Main poses 1-2 questions per paper on consistency conditions and parameter values ($\Delta = 0$ cases). JEE Advanced favours homogeneous systems, the *dimension* of the solution space, and parameter-rich consistency analysis.

Section 1: Concept Intuition

This subtopic has two faces that turn out to be the same coin: **the inverse of a matrix (03a)** and **solving a system of equations (03b)**.

The inverse as an “undo” button

A matrix A takes a vector and transforms it — stretches, rotates, shears. The **inverse** A^{-1} is simply the operation that undoes that transformation, returning you to where you started: $A^{-1}(A\mathbf{x}) = \mathbf{x}$.

Think of a money-changer at the airport. Converting rupees to dollars is multiplying by an exchange matrix; converting back is the inverse. The catch: **you can only undo a transformation that didn’t destroy information**. Recall from Part 2 that $|A|$ is the signed volume of the box spanned by the rows. If $|A| = 0$, the transformation has *squashed space flat* — a 3D box collapsed onto a plane or a line. Once flattened, you cannot un-flatten: infinitely many starting points got mapped to the same place, so there is no unique way back. **That is exactly why a matrix has an inverse if and only if $|A| \neq 0$** (such a matrix is called *non-singular*; $|A| = 0$ is *singular*).

A system of equations as intersecting planes

A linear system in three unknowns is three planes in space. A *solution* is a point lying on all three. Three geometric outcomes are possible:

- **One common point** — the planes meet at a single spot. **Unique solution** ($\Delta \neq 0$).
- **A whole line or plane in common** — the planes hinge along a line, or two/three coincide. **Infinitely many solutions**.
- **No common point** — at least two planes are parallel and offset, like railway tracks that never meet. **No solution (inconsistent)**.

Think about it this way... $\Delta \neq 0$ is the algebraic fingerprint of “the three planes are tilted independently enough to pin down exactly one crossing point.” The moment $\Delta = 0$, the planes have lost that independence — they’ve gone parallel or hinged — and you’ve tipped into either the infinite-solution or the no-solution world. Which of those two you land in is decided by the right-hand side.

For a **homogeneous** system ($AX = O$, all constants zero), every plane passes through the origin, so $\mathbf{x} = \mathbf{0}$ — the *trivial solution* — always works. The interesting question is whether they share **more** than just the origin (a whole line or plane of solutions). They do exactly when $\Delta = 0$.

Limiting Conditions & Assumptions. An inverse exists only for a **square, non-singular** matrix. Cramer's rule (below) is valid **only when** $\Delta \neq 0$ — it cannot, by itself, resolve the $\Delta = 0$ cases. And one honest caveat the textbooks gloss over: “all $\Delta_i = 0$ ” does **not** rigorously guarantee infinitely many solutions (a rank subtlety) — it is a reliable exam rule, not a theorem.

Section 2: Key Formulas & Definitions

All quantities here are real scalars or matrices; determinants are dimensionless.

Inverse via adjoint. For a non-singular square matrix ($|A| \neq 0$):

$$A^{-1} = \frac{1}{|A|} \text{adj } A, \quad \text{adj } A = [C_{ij}]^T \text{ (transpose of the cofactor matrix).}$$

Fundamental adjoint relation.

$$A (\text{adj } A) = (\text{adj } A) A = |A| I.$$

Determinant of an inverse.

$$|A^{-1}| = \frac{1}{|A|}.$$

Matrix method for $AX = B$ (with A the coefficient matrix, $X = [x \ y \ z]^T$, B the constants):

$$X = A^{-1}B = \frac{1}{|A|} (\text{adj } A) B \quad (\text{requires } |A| \neq 0).$$

Cramer's rule for $AX = B$:

$$x = \frac{\Delta_1}{\Delta}, \quad y = \frac{\Delta_2}{\Delta}, \quad z = \frac{\Delta_3}{\Delta}, \quad \Delta = |A|,$$

where Δ_i is Δ with its i -th column replaced by the constants B . Valid only when $\Delta \neq 0$.

Consistency criterion (non-homogeneous $AX = B$).

Condition	Nature of system
$\Delta \neq 0$	Unique solution (consistent)
$\Delta = 0$ and at least one $\Delta_i \neq 0$	No solution (inconsistent)
$\Delta = 0$ and all $\Delta_i = 0$	Infinitely many (usually; check rank)

Homogeneous system $AX = O$. Always consistent ($X = O$ is the trivial solution). A **non-trivial** solution exists $\iff \Delta = 0$.

Section 3: Theorems, Proofs & Key Procedures

Theorem A — The adjoint relation $A (\text{adj } A) = |A| I$

Claim. $A (\text{adj } A) = |A| I$.

Proof. Write $(\text{adj } A)_{jk} = C_{kj}$ (the adjoint is the *transpose* of the cofactor matrix). The (i, k) entry of the product is

$$[A (\text{adj } A)]_{ik} = \sum_j a_{ij} (\text{adj } A)_{jk} = \sum_j a_{ij} C_{kj}.$$

By the **cross-orthogonality identity** from Part 4, $\sum_j a_{ij} C_{kj} = |A| \delta_{ik}$ — it equals $|A|$ when $i = k$ (a genuine cofactor expansion) and 0 when $i \neq k$ (a cross sum). Therefore every diagonal entry is $|A|$ and every off-diagonal entry is 0:

$$A (\text{adj } A) = |A| I. \quad \blacksquare$$

Corollary (the inverse formula). If $|A| \neq 0$, divide both sides by $|A|$:

$$A \left(\frac{1}{|A|} \text{adj } A \right) = I \Rightarrow A^{-1} = \frac{1}{|A|} \text{adj } A.$$

This is the entire justification for the inverse formula — and it rests directly on a cofactor fact proved two subtopics earlier. *That* is derivational chaining.

Theorem B — Cramer's rule

Claim. If $\Delta \neq 0$, then $x = \Delta_1 / \Delta$ (and similarly y, z).

Proof sketch. From $X = A^{-1}B = \frac{1}{|A|}(\text{adj } A)B$, the first component is

$$x = \frac{1}{|A|} \sum_i (\text{adj } A)_{1i} b_i = \frac{1}{|A|} \sum_i C_{i1} b_i.$$

But $\sum_i b_i C_{i1}$ is precisely the cofactor expansion (along column 1) of the determinant in which column 1 of A has been replaced by B — that is Δ_1 . Hence $x = \Delta_1 / \Delta$. $\blacksquare$

Procedure — Matrix method for $AX = B$ (CBSE 5-mark template)

1. Write A, X, B and compute $\Delta = |A|$. If $\Delta \neq 0$, a unique solution exists.
2. Find all nine cofactors C_{ij} ; assemble the cofactor matrix.
3. **Transpose** it to get $\text{adj } A$. (Skipping the transpose is the classic lost-mark.)
4. Form $A^{-1} = \frac{1}{\Delta} \text{adj } A$.
5. Compute $X = A^{-1}B$ and read off x, y, z . Verify by substitution.

Procedure — Classifying any system

- Compute Δ . If $\Delta \neq 0$: **unique** — solve by Cramer or matrix method, done.
- If $\Delta = 0$: compute the Δ_i . If **any** $\Delta_i \neq 0 \rightarrow$ **no solution**. If **all** $\Delta_i = 0 \rightarrow$ **infinitely many** (exam rule).

- For homogeneous systems: $\Delta \neq 0 \rightarrow$ only trivial; $\Delta = 0 \rightarrow$ non-trivial solutions exist.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] (matrix method)

Solve by matrix method:

$$x + y + z = 6, \quad x - y + z = 2, \quad 2x + y - z = 1.$$

Solution. Here $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{pmatrix}$, $B = \begin{pmatrix} 6 \\ 2 \\ 1 \end{pmatrix}$. First, $\Delta = 1(1 - 1) - 1(-1 - 2) + 1(1 + 2) = 0 + 3 + 3 = 6 \neq 0$, so a unique solution exists.

Cofactors:

$$C = \begin{pmatrix} 0 & 3 & 3 \\ 2 & -3 & 1 \\ 2 & 0 & -2 \end{pmatrix} \Rightarrow \text{adj } A = C^T = \begin{pmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{pmatrix}.$$

Then $A^{-1} = \frac{1}{6} \text{adj } A$, and

$$X = A^{-1}B = \frac{1}{6} \begin{pmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{pmatrix} \begin{pmatrix} 6 \\ 2 \\ 1 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 6 \\ 12 \\ 18 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

So $x = 1$, $y = 2$, $z = 3$. **Check:** $1 + 2 + 3 = 6 \checkmark$, $1 - 2 + 3 = 2 \checkmark$, $2 + 2 - 3 = 1 \checkmark$.

Example 2 — [Speed / Elimination Trap] (spot inconsistency)

Classify, as fast as possible:

$$x + 2y + 3z = 4, \quad 2x + 4y + 6z = 9, \quad x - y + z = 2.$$

Fast solution. Look at the first two equations *before* computing anything. The left side of equation 2 is exactly twice the left side of equation 1: $2x + 4y + 6z = 2(x + 2y + 3z)$. So the left sides demand $2(4) = 8$ on the right — but equation 2 says 9. Contradiction. The system is **inconsistent — no solution**, in one glance. *The trap:* setting up Cramer and grinding $\Delta, \Delta_1, \Delta_2, \Delta_3$ wastes two minutes to reach the same “no solution.” **Always scan for proportional coefficient rows; if the constants don’t match the ratio, it’s inconsistent immediately.**

Example 3 — [JEE Main Level] (parameter case-analysis)

For what value(s) of k does

$$kx + y + z = 1, \quad x + ky + z = 1, \quad x + y + kz = 1$$

have (i) a unique solution, (ii) no solution, (iii) infinitely many solutions?

Solution. Compute and factor the coefficient determinant:

$$\Delta = \begin{vmatrix} k & 1 & 1 \\ 1 & k & 1 \\ 1 & 1 & k \end{vmatrix} = k^3 - 3k + 2 = (k-1)^2(k+2).$$

(i) $\Delta \neq 0 \iff k \neq 1$ and $k \neq -2 \rightarrow$ **unique solution.** (ii) & (iii) Examine the two roots:

- $k = 1$: all three equations collapse to $x + y + z = 1$ — the same plane thrice $\rightarrow$ **infinitely many solutions.**
- $k = -2$: adding all three equations gives $0 = 3$, a contradiction $\rightarrow$ **no solution.**

The two concepts fused: factoring Δ as a polynomial in k isolates the special cases, and then a quick algebraic test (“collapse to one plane” vs “sum to a contradiction”) decides infinite-vs-none. This is the single most common JEE Main system template.

Example 4 — [JEE Advanced Level] (dimension of the solution space)

Find all λ for which the homogeneous system

$$x + y + \lambda z = 0, \quad x + \lambda y + z = 0, \quad \lambda x + y + z = 0$$

has a non-trivial solution, and **describe the solution space** in each case.

Solution. A homogeneous system has a non-trivial solution iff $\Delta = 0$. Expanding,

$$\Delta = \begin{vmatrix} 1 & 1 & \lambda \\ 1 & \lambda & 1 \\ \lambda & 1 & 1 \end{vmatrix} = -(\lambda-1)^2(\lambda+2).$$

So non-trivial solutions exist iff $\lambda = 1$ or $\lambda = -2$.

- $\lambda = 1$: every equation becomes $x + y + z = 0$ — a *single* constraint. The solution set is an entire **plane** through the origin: a **two-parameter** family, e.g. $(x, y, z) = (s, t, -s-t)$.
- $\lambda = -2$: the equations are $x + y - 2z = 0$, $x - 2y + z = 0$, $-2x + y + z = 0$. Subtracting the first two gives $3y - 3z = 0 \Rightarrow y = z$; back-substituting gives $x = z$. The solution set is the **line** $x = y = z$: a **one-parameter** family (t, t, t) .

So the *geometry* of the solution space distinguishes the two singular cases — a plane (rank 1) at $\lambda = 1$, a line (rank 2) at $\lambda = -2$. This rank-and-dimension reasoning is the hallmark of an Advanced treatment. ■

Section 5: Practice Exercises

1. [CBSE] Solve using the inverse: $2x - y = 3$, $x + y = 3$.
2. [CBSE] Find A^{-1} for $A = \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}$.
3. [JEE Main] Find k for which $kx + 2y - z = 0$, $2x - y + 3z = 0$, $x + y + z = 0$ has a non-trivial solution.

4. **[JEE Main]** The system $x + y + z = 6$, $x + 2y + 3z = 14$, $2x + 5y + \lambda z = \mu$ has a unique solution. Find the condition on λ .
5. **[JEE Advanced]** Find all k for which $x + y + z = 1$, $x + 2y + 4z = k$, $x + 4y + 10z = k^2$ is consistent.

Answers: 1) $x = 2$, $y = 1$ | 2) $A^{-1} = \begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix}$ (since $|A| = 1$) | 3) $k = -\frac{1}{4}$ | 4) $\lambda \neq 8$ (for any μ) | 5) $k = 1$ or $k = 2$.

Section 6: Crack the MCQ

Q1. A square matrix A has an inverse if and only if:

- (a) A is symmetric (b) $|A| = 0$ (c) $|A| \neq 0$ (d) A is of order 3

Answer: (c). An inverse exists exactly when A is non-singular. *Distractors:* (b) is the precise *opposite* — a singular matrix has **no** inverse. (a) and (d) are irrelevant properties; symmetry and order have no bearing on invertibility.

Q2. A homogeneous system $AX = O$ of three equations has a non-trivial solution if and only if:

- (a) $\Delta \neq 0$ (b) $\Delta = 0$ (c) $\Delta > 0$ (d) it always does

Answer: (b). Non-trivial solutions require the coefficient determinant to vanish. *Distractors:* (a) gives **only** the trivial solution. (d) is the seductive trap — the *trivial* solution always exists, but a *non-trivial* one needs $\Delta = 0$. (c) invents a sign condition that doesn't exist.

Q3. For $AX = B$ (three equations), $\Delta = 0$ and $\Delta_1 \neq 0$. The system has:

- (a) a unique solution (b) infinitely many solutions (c) no solution (d) only the trivial solution

Answer: (c). $\Delta = 0$ with some $\Delta_i \neq 0$ is the inconsistency signature. *Distractors:* (a) needs $\Delta \neq 0$. (b) needs **all** $\Delta_i = 0$, not one of them nonzero. (d) “trivial solution” is a homogeneous-system notion and doesn't apply to a general B .

Q4. If A is a 3×3 matrix with $|A| = 2$, then $|A^{-1}|$ equals:

- (a) 2 (b) $\frac{1}{2}$ (c) $\frac{1}{8}$ (d) 8

Answer: (b). $|A^{-1}| = 1/|A| = \frac{1}{2}$. *Distractors:* (c) is $1/|A|^3$ — confusing the inverse with the adjoint scaling. (d) is $|A|^3$, the wrong direction entirely. (a) forgets to reciprocate.

Section 7: Common Student Pitfalls & Pro-Tips

- **Pitfall 1 — Scrambling the three consistency branches.** Lock this down: $\Delta \neq 0 \Rightarrow$ unique; $\Delta = 0$ with some $\Delta_i \neq 0 \Rightarrow$ no solution; $\Delta = 0$ with all $\Delta_i = 0 \Rightarrow$ infinitely many.

- **Pitfall 2 — Forgetting “trivial always exists.”** For homogeneous systems, $X = O$ is automatic. The exam asks about *non-trivial* solutions, which need $\Delta \neq 0$.
- **Pitfall 3 — Skipping the transpose in $\text{adj } A$.** The adjoint is the **transpose** of the cofactor matrix. This step is mark-bearing.
- **Pitfall 4 — Using Cramer when $\Delta = 0$.** Cramer’s rule is undefined there; switch to the consistency criterion instead.

Pro-Tip: For any parameter problem, **factor Δ as a polynomial in the parameter first**. Its roots are the only “interesting” values; everything else gives a unique solution. Then test each root separately for no-solution vs infinite — never assume which one it is. And remember the honest caveat: “all $\Delta_i = 0$ ” is the exam shortcut for “infinitely many,” but rigorously it can still be inconsistent in degenerate cases, so on a high-stakes Advanced problem, verify by back-substitution.

Section 8: Cheat Sheet

Inverse & Systems — Core Recall

- **Inverse:** $A^{-1} = \frac{1}{|A|} \text{adj } A$, exists $\iff |A| \neq 0$ (non-singular).
 $|A^{-1}| = 1/|A|$.
- **Adjoint relation:** $A(\text{adj } A) = |A|I$ — the engine behind the inverse formula.
- **Matrix method:** $X = A^{-1}B$ (needs $|A| \neq 0$).
- **Cramer:** $x = \Delta_1/\Delta$, $y = \Delta_2/\Delta$, $z = \Delta_3/\Delta$ (Δ_i = replace i -th column with B); valid only if $\Delta \neq 0$.
- **Consistency (non-homogeneous):** $\Delta \neq 0 \rightarrow$ unique; $\Delta = 0$, some $\Delta_i \neq 0 \rightarrow$ none; $\Delta = 0$, all $\Delta_i = 0 \rightarrow$ infinite.
- **Homogeneous $AX = O$:** trivial always; non-trivial $\iff \Delta = 0$.

Mnemonics: “No determinant, no inverse” ($|A| = 0 \Rightarrow$ singular).
 · “Delta zero, hopes alive — check the side-determinants to know none-or-infinite.” · “adj = transpose of cofactors.”

Part 6 — Advanced Techniques: Multiplication, Differentiation & Special Determinants

Exam Relevance: Pure JEE territory (Main and especially Advanced) — these tools rarely appear in CBSE but are decisive in competitive papers. Multiplication and adjoint identities power 1-mark “find $|2AB|$ ” traps; differentiation of a determinant and cyclic/Vandermonde factorisation appear as full Advanced problems. Mastering these is the difference between a good score and a top percentile.

Section 1: Concept Intuition

This capstone gathers four power-tools that the earlier parts hinted at but didn’t develop.

(1) Multiplication. Two determinants of the same order can be multiplied into a single determinant, because $|A||B| = |AB|$. The determinant is *multiplicative* — it turns matrix products into ordinary number products. This is why $|A^n| = |A|^n$ and $|kA| = k^n|A|$.

(2) Differentiation. When a determinant’s entries are functions of x , its derivative is **not** found by expanding first. Instead, differentiate **one row (or column) at a time**, leaving the others fixed, and add the resulting determinants. It mirrors the product rule — and for good reason, since a determinant is a sum of products of entries.

(3) Adjoint identities. The adjoint (Part 5) obeys a small algebra of its own: it reverses order under products, scales predictably, and “squares back” toward the original matrix.

(4) Special determinants. Certain shapes recur so often that recognising them saves enormous time: the **Vandermonde** (powers stacked in rows), the **skew-symmetric of odd order** (always zero), and **cyclic** determinants (which factor through $a + b + c$).

Think about it this way... Up to now you *computed* determinants. Here you *manipulate* them as algebraic objects — multiplying, differentiating, and pattern-matching — exactly as you’d manipulate polynomials once you stop plugging in numbers.

Limiting Conditions & Assumptions. Multiplication $|A||B| = |AB|$ requires A, B **square and of the same order**. The skew-symmetric “= 0” result holds only for **odd** order (even-order skew determinants are generally nonzero). The differentiation rule assumes the entries are differentiable functions of the variable.

Section 2: Key Formulas & Definitions

Multiplication of determinants.

$$|A||B| = |AB|, \quad |A^n| = |A|^n, \quad |kA| = k^n|A| \text{ (order } n\text{)}.$$

Two determinants are multiplied **row-by-row** (or row-by-column): the (i, j) entry of the product is the dot product of row i of the first with row j of the second.

Differentiation of a determinant. For

$$\Delta(x) = \begin{vmatrix} f_1 & f_2 & f_3 \\ g_1 & g_2 & g_3 \\ h_1 & h_2 & h_3 \end{vmatrix}, \quad \Delta'(x) = \begin{vmatrix} f'_1 & f'_2 & f'_3 \\ g_1 & g_2 & g_3 \\ h_1 & h_2 & h_3 \end{vmatrix} + \begin{vmatrix} f_1 & f_2 & f_3 \\ g'_1 & g'_2 & g'_3 \\ h_1 & h_2 & h_3 \end{vmatrix} + \begin{vmatrix} f_1 & f_2 & f_3 \\ g_1 & g_2 & g_3 \\ h'_1 & h'_2 & h'_3 \end{vmatrix}.$$

Adjoint identities (order n):

$$\begin{aligned} \text{adj}(AB) &= (\text{adj } B)(\text{adj } A), \quad \text{adj}(kA) = k^{n-1} \text{adj } A, \quad \text{adj}(\text{adj } A) = |A|^{n-2} A, \\ |\text{adj } A| &= |A|^{n-1}, \quad |\text{adj}(\text{adj } A)| = |A|^{(n-1)^2}. \end{aligned}$$

Special determinants.

$$\text{Vandermonde: } \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a) \text{ (up to sign convention).}$$

$$\text{Skew-symmetric, odd order } (A^\top = -A) : |A| = 0.$$

$$\text{Cyclic: } \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -(a^3 + b^3 + c^3 - 3abc) = -(a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca).$$

Section 3: Theorems, Proofs & Key Procedures

Theorem A — A skew-symmetric determinant of odd order is zero

Claim. If $A^\top = -A$ and A is $n \times n$ with n odd, then $|A| = 0$.

Proof. Take determinants of $A^\top = -A$. Using $|A^\top| = |A|$ (Part 2) and $|-A| = (-1)^n |A|$ (the scalar rule with $k = -1$):

$$|A| = (-1)^n |A|.$$

For odd n , $(-1)^n = -1$, so $|A| = -|A| \Rightarrow 2|A| = 0 \Rightarrow |A| = 0$. (For even n the equation is vacuous, so no conclusion.) ■

Theorem B — The differentiation rule

Sketch. A 3×3 determinant is a sum of terms, each a product of three entries, one from each row. Differentiating such a product by the ordinary product rule gives three pieces — one with the first factor differentiated, one with the second, one with the third. Collecting all “first-row-differentiated” pieces reconstitutes the first determinant in the rule, and similarly for the others. Hence the derivative is the sum of the three row-differentiated determinants. ■

Procedure — Multiplying two determinants by hand

1. Confirm both are square and the same order.
2. Form the product matrix AB using **row-of- A dotted with column-of- B** .
3. Evaluate $|AB|$ — it equals $|A||B|$. (Often the *reverse* is useful: recognise a messy determinant as a product $|A||B|$ to factor it.)

Procedure — Cyclic factorisation

For a determinant whose rows (or columns) are cyclic shifts, apply $C_1 \rightarrow C_1 + C_2 + C_3$. The first column becomes constant (here $a + b + c$), which factors out, collapsing the problem.

Section 4: Worked Examples**Example 1 — [JEE Main Level] (multiplication shortcuts)**

Let A, B be 3×3 matrices with $|A| = 3$ and $|B| = -2$. Find $|2AB|$ and $|A^2B|$.

Solution. Using $|kA| = k^n|A|$ with $n = 3$, and multiplicativity:

$$|2AB| = 2^3|A||B| = 8(3)(-2) = -48, \quad |A^2B| = |A|^2|B| = 9(-2) = -18.$$

Example 2 — [Speed / Elimination Trap] (skew-symmetric)

Evaluate $\begin{vmatrix} 0 & 3 & -5 \\ -3 & 0 & 2 \\ 5 & -2 & 0 \end{vmatrix}$.

Fast solution. Notice $a_{ij} = -a_{ji}$ and the diagonal is all zeros — this is a **skew-symmetric** matrix of order 3 (odd). By Theorem A its determinant is 0 — no computation needed. *The trap:* expanding all six Sarrus terms wastes a minute and risks a sign error; recognising “odd-order skew-symmetric $\Rightarrow$ zero” is instant.

Example 3 — [JEE Advanced Level] (differentiation of a determinant)

For $f(x) = \begin{vmatrix} 1+x & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+x \end{vmatrix}$, find $f'(x)$.

Solution. First, the determinant itself simplifies ($C_1 \rightarrow C_1 + C_2 + C_3$ gives factor $(x+3)$, then reduce) to $f(x) = x^2(x+3) = x^3 + 3x^2$, so directly $f'(x) = 3x^2 + 6x$.

To *demonstrate the rule*, differentiate one row at a time. Each row's derivative is $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ respectively, so

$$f'(x) = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+x \end{vmatrix} + \begin{vmatrix} 1+x & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1+x \end{vmatrix} + \begin{vmatrix} 1+x & 1 & 1 \\ 1 & 1+x & 1 \\ 0 & 0 & 1 \end{vmatrix}.$$

Each is easy: the first $= (1+x)^2 - 1$, the second $= (1+x)^2 - 1$, the third $= (1+x)^2 - 1$. Summing: $3[(1+x)^2 - 1] = 3(x^2 + 2x) = 3x^2 + 6x$. ✓ Both routes agree. ■

Example 4 — [JEE Advanced Level] (cyclic factorisation)

Prove that $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = -(a^3 + b^3 + c^3 - 3abc)$.

Solution. Apply $C_1 \rightarrow C_1 + C_2 + C_3$; every entry of the new first column becomes $a + b + c$:

$$\Delta = (a + b + c) \begin{vmatrix} 1 & b & c \\ 1 & c & a \\ 1 & a & b \end{vmatrix}.$$

Now $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$:

$$= (a + b + c) \begin{vmatrix} 1 & b & c \\ 0 & c - b & a - c \\ 0 & a - b & b - c \end{vmatrix} = (a + b + c)[(c - b)(b - c) - (a - c)(a - b)].$$

Expanding the bracket: $(c - b)(b - c) = -(b - c)^2$ and $(a - c)(a - b) = a^2 - (b + c)a + bc$, giving (after simplification) $-(a^2 + b^2 + c^2 - ab - bc - ca)$. Hence

$$\Delta = -(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = -(a^3 + b^3 + c^3 - 3abc).$$

(Quick check $a = 1, b = 2, c = 3$: $\Delta = -18$ and $-(1 + 8 + 27 - 18) = -18$.) ■

Section 5: Practice Exercises

1. **[JEE Main]** If $|A| = 4$ for a 3×3 matrix, find $|3A|$ and $|A^{-1}|$.
2. **[JEE Main]** For 3×3 matrices with $|A| = 2$, $|B| = 5$, find $|\text{adj}(AB)|$.
3. **[JEE Main]** Evaluate $\begin{vmatrix} 0 & p & -q \\ -p & 0 & r \\ q & -r & 0 \end{vmatrix}$.
4. **[JEE Advanced]** If $f(x) = \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix}$, find $f'(x)$ at a general x (use the row rule).
5. **[JEE Advanced]** Factorise $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix}$.

Answers: 1) $|3A| = 108$, $|A^{-1}| = \frac{1}{4}$ | 2) $|\text{adj}(AB)| = |AB|^2 = (10)^2 = 100$ | 3) 0 (odd-order skew-symmetric) | 4) $f(x) = 2x^3$, so $f'(x) = 6x^2$ (differentiating rows 1 and 2 produces duplicate rows $\rightarrow 0$; only the last term survives) | 5) $(a - b)(b - c)(c - a)(ab + bc + ca)$.

Section 6: Crack the MCQ

Q1. For 3×3 matrices, $|2A|$ in terms of $|A|$ is:

- (a) $2|A|$ (b) $6|A|$ (c) $8|A|$ (d) $|A|$

Answer: (c). $|kA| = k^n|A| = 2^3|A| = 8|A|$. *Distractors:* (a) treats it as a single-row scaling; (b) uses $2n$; (d) ignores the scalar.

Q2. A skew-symmetric matrix of order 3 has determinant:

- (a) always 0 (b) always 1 (c) equal to its trace (d) always positive

Answer: (a). Odd-order skew-symmetric $\Rightarrow |A| = 0$. *Distractors:* (b)/(d) ignore the theorem; (c) confuses determinant with trace (which is also 0 here, but for a different reason).

Q3. $\text{adj}(AB)$ equals:

- (a) $(\text{adj } A)(\text{adj } B)$ (b) $(\text{adj } B)(\text{adj } A)$ (c) $\text{adj } A + \text{adj } B$ (d) $\text{adj}(BA)$

Answer: (b). The adjoint reverses order, just like the inverse: $\text{adj}(AB) = (\text{adj } B)(\text{adj } A)$. *Distractors:* (a) keeps the order (the classic error); (c) treats adjoint as additive; (d) confuses with commutativity, which fails in general.

Q4. If A is 3×3 with $|A| = 2$, then $|\text{adj}(\text{adj } A)|$ equals:

- (a) 2 (b) 4 (c) 8 (d) 16

Answer: (d). $|\text{adj}(\text{adj } A)| = |A|^{(n-1)^2} = 2^4 = 16$. *Distractors:* (b) uses $|A|^{n-1}$ (one adjoint, not two); (c) uses $|A|^n$; (a) ignores the powers.

Section 7: Common Student Pitfalls & Pro-Tips

- **Pitfall 1 — Differentiating after expanding.** For function-valued determinants, use the row-at-a-time rule; expanding first is slower and error-prone.
- **Pitfall 2 — Keeping order in $\text{adj}(AB)$.** It is $(\text{adj } B)(\text{adj } A)$ — reversed, like the inverse.
- **Pitfall 3 — Applying “skew $\Rightarrow$ 0” to even order.** The result is **odd-order only**.
- **Pitfall 4 — Power slips in adjoint identities.** Track n vs $n - 1$ vs $(n - 1)^2$ carefully; for 3×3 these are 3, 2, 4.

Pro-Tip: Build a reflex for the three “instant-zero” patterns across the whole chapter — a zero row/column, two proportional rows/columns, and an odd-order skew-symmetric matrix. Any one of them lets you write 0 on sight, saving the entire expansion. For symbolic determinants, the column-sum trick ($C_1 \rightarrow C_1 + C_2 + C_3$) is the universal first move.

Section 8: Cheat Sheet

Advanced Techniques — Core Recall

- **Multiplicative:** $|AB| = |A||B|$, $|A^n| = |A|^n$, $|kA| = k^n|A|$.
- **Differentiation:** differentiate one row (or column) at a time; sum the resulting determinants.
- **Adjoint algebra:** $\text{adj}(AB) = (\text{adj } B)(\text{adj } A)$, $\text{adj}(kA) = k^{n-1} \text{adj } A$, $\text{adj}(\text{adj } A) = |A|^{n-2} A$.
- **Adjoint determinants:** $|\text{adj } A| = |A|^{n-1}$, $|\text{adj}(\text{adj } A)| = |A|^{(n-1)^2}$.
- **Special:** Vandermonde $\rightarrow \prod(\text{differences})$; odd-order skew-symmetric $\rightarrow 0$; cyclic $\rightarrow (a + b + c)(\dots)$.

Mnemonic: “Multiply the dets, differentiate row-by-row, reverse the adj, and zero the odd skew.”

Chapter 4: Determinants — Round 2 Addendum

Parts 1 to 6 carried you from the definition through properties, area, cofactors, inverses, systems, and the advanced toolkit. Five examinable patterns still stand outside the door. Addendum A builds the circulant machine on the cube roots of unity, which evaluates in seconds a family of determinants the column-op method grinds through. Addendum B handles trigonometric determinants, which appear in JEE Main almost yearly and which the chapter never touches. Addendum C generalises the Vandermonde argument of Part 2 to *alternants with a missing power*, the standard Advanced factorisation pattern. Addendum D extracts three consequences of Part 3 that exams keep asking for: concurrency of three lines, the family of lines through an intersection point, and polygon areas. Addendum E repairs the weakest link in Part 5: what you actually *do* when $\Delta = 0$, including the honest proof of when the Δ_i screen can lie to you.

Nothing here is new theory beyond one constant ω . Every method below is the column operation of Part 2, the cofactor expansion of Part 4, or the elimination instinct of Part 5, aimed at shapes the chapter did not draw.

Addendum A — Circulant Determinants and the Cube Roots of Unity

Part 6 evaluated one cyclic determinant by forcing a common factor. But the family recurs constantly — especially as coefficient matrices of 3×3 homogeneous systems, where JEE asks “for which k does a non-trivial solution exist?” — and there is a machine that reads off the answer as a product of three linear factors. The machine runs on ω , a primitive cube root of unity:

$$\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad \omega^3 = 1, \quad 1 + \omega + \omega^2 = 0, \quad \omega - \omega^2 = i\sqrt{3}.$$

A **circulant** is a determinant whose second row is the first row shifted one place and whose third row is shifted again. Two shift directions occur in the wild, and they differ by sign:

Row pattern	Value
$(a, b, c), (b, c, a), (c, a, b)$	$-(a^3 + b^3 + c^3 - 3abc)$
$(a, b, c), (c, a, b), (b, c, a)$	$+(a^3 + b^3 + c^3 - 3abc)$

If you ever forget which sign goes with which pattern, evaluate the pattern at $(a, b, c) = (1, 2, 3)$: the determinant is -18 for the first and $+18$ for the second, while $a^3 + b^3 + c^3 - 3abc = 18$.

Method

Step 1 — The two factor forms. Part 6 proved, via $C_1 \rightarrow C_1 + C_2 + C_3$ and row reductions,

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca).$$

The quadratic factor has a completing-the-square form worth memorising:

$$a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}[(a - b)^2 + (b - c)^2 + (c - a)^2],$$

which follows by expanding the right side: the squares contribute $2a^2 + 2b^2 + 2c^2$ and the cross terms $-2ab - 2bc - 2ca$. Since a sum of squares vanishes only term-by-term, this proves:

$$a^3 + b^3 + c^3 = 3abc \iff a + b + c = 0 \text{ or } a = b = c.$$

Step 2 — The ω -machine. Work with the left-shift circulant

$$\Delta = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}, \quad s = a + b + c.$$

Apply three column combinations simultaneously, each built from the *original* columns:

$$C'_1 = C_1 + C_2 + C_3, \quad C'_2 = C_1 + \omega C_2 + \omega^2 C_3, \quad C'_3 = C_1 + \omega^2 C_2 + \omega C_3.$$

Why this is legal: any simultaneous column combination multiplies Δ by the determinant of its coefficient matrix M , where column j of M holds the coefficients used to build C'_j . So the price tag is $\det M$ — computable, and we will pay it.

Reading the new columns. The first is (s, s, s) , i.e. $s(1, 1, 1)$. For the second, compare entries against the multiplier ω^2 :

$$\omega^2(a + \omega b + \omega^2 c) = \omega^2 a + b + \omega c, \quad \omega(a + \omega b + \omega^2 c) = \omega a + \omega^2 b + c,$$

using $\omega^3 = 1$. These are exactly the second and third entries of C'_2 , so

$$C'_2 = (a + \omega b + \omega^2 c)(1, \omega^2, \omega)^\top,$$

and identically $C'_3 = (a + \omega^2 b + \omega c)(1, \omega, \omega^2)^\top$.

Pay the bill. The reduced determinant has the scalar factor

$$s(a + \omega b + \omega^2 c)(a + \omega^2 b + \omega c)$$

times the determinant of the array with columns $(1, 1, 1)$, $(1, \omega^2, \omega)$, $(1, \omega, \omega^2)$. Subtract row 1 from rows 2 and 3:

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & \omega^2 & \omega \\ 1 & \omega & \omega^2 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 0 & \omega^2 - 1 & \omega - 1 \\ 0 & \omega - 1 & \omega^2 - 1 \end{vmatrix} = (\omega^2 - 1)^2 - (\omega - 1)^2 = 3(\omega - \omega^2) = 3i\sqrt{3}.$$

The coefficient matrix M has columns $(1, 1, 1)$, $(1, \omega, \omega^2)$, $(1, \omega^2, \omega)$ — the same array with its last two columns swapped, so $\det M = -3i\sqrt{3}$. Therefore

$$\Delta(-3i\sqrt{3}) = s(a + \omega b + \omega^2 c)(a + \omega^2 b + \omega c)(3i\sqrt{3}),$$

$$\boxed{\Delta = -(a + b + c)(a + \omega b + \omega^2 c)(a + \omega^2 b + \omega c).}$$

The quadratic factor recovered. Expand the conjugate pair using $\omega + \omega^2 = -1$ and $\omega^3 = 1$: the square terms survive as $a^2 + b^2 + c^2$ and every cross term carries $\omega + \omega^2 = -1$, giving

$$(a + \omega b + \omega^2 c)(a + \omega^2 b + \omega c) = a^2 + b^2 + c^2 - ab - bc - ca,$$

so the boxed formula agrees with Step 1. What the ω -machine buys is the *linear* factorisation: the circulant vanishes exactly when one of three explicit linear expressions vanishes — over the complex numbers, always; over the reals, via the sum or the equality case.

Step 3 — Exam deployments.

- *Instant zero:* if the row sums of a circulant are all equal to σ , then $(1, 1, 1)$ is a solution of the associated homogeneous system; in particular $\sigma = 0$ forces $\Delta = 0$ on sight.
- *Systems:* a homogeneous system whose coefficient matrix is circulant in (k, b, c) has a non-trivial solution exactly when $k^3 + b^3 + c^3 - 3kbc = 0$ — a cubic you can usually factor by inspection.
- *Algebra:* the identity $a^3 + b^3 + c^3 - 3abc = (a + b + c)(\dots)$ is the standard one-line proof that $a + b + c$ divides $a^3 + b^3 + c^3 - 3abc$, and that $a^3 + b^3 + c^3 = 3abc$ with $a + b + c \neq 0$ forces $a = b = c$.

Example A.1 — [JEE Main Level]

Evaluate $\begin{vmatrix} 1 & 5 & 7 \\ 5 & 7 & 1 \\ 7 & 1 & 5 \end{vmatrix}.$

The rows follow (a, b, c) , (b, c, a) , (c, a, b) — the left-shift pattern — so

$$\Delta = -(1 + 125 + 343 - 3 \cdot 1 \cdot 5 \cdot 7) = -(469 - 105) = -364.$$

Check by Sarrus: $1(35 - 1) - 5(25 - 7) + 7(5 - 49) = 34 - 90 - 308 = -364$. Two routes, one answer.

Example A.2 — [JEE Advanced Level]

Find all real k for which the system below has a non-trivial solution, and exhibit one:

$$kx + 3y + 2z = 0, \quad 2x + ky + 3z = 0, \quad 3x + 2y + kz = 0.$$

The rows are $(k, 3, 2)$, then shifted right, then shifted right again — the right-shift pattern — so

$$\Delta = +(k^3 + 3^3 + 2^3 - 3 \cdot k \cdot 3 \cdot 2) = k^3 - 18k + 35.$$

A non-trivial solution of a homogeneous system requires $\Delta = 0$ (Part 5). Probe roots: $k = -5$ gives $-125 + 90 + 35 = 0$. Dividing:

$$k^3 - 18k + 35 = (k + 5)(k^2 - 5k + 7),$$

and the quadratic has discriminant $25 - 28 < 0$, so $k = -5$ is the only real root. At $k = -5$ every row sums to $-5 + 3 + 2 = 0$, so by Step 3 the vector $(1, 1, 1)$ is a non-trivial solution. Substitution confirms: $-5 + 3 + 2 = 0$ in each equation.

Why it is Advanced-level. The question is a Part 5 homogeneous-system question wearing a Part 6 costume. The mark-winning move is recognising the circulant *before* expanding, so that Δ arrives pre-factored as a cubic with integer roots to probe — instead of arriving as nine cofactor terms you must first assemble and then factor.

Practice — Addendum A

1. Evaluate $\begin{vmatrix} 2 & 4 & 6 \\ 6 & 2 & 4 \\ 4 & 6 & 2 \end{vmatrix}$.
2. Prove that $a^3 + b^3 + c^3 - 3abc = (a + b + c) \cdot \frac{1}{2}[(a - b)^2 + (b - c)^2 + (c - a)^2]$, and deduce all real solutions of $a^3 + b^3 + c^3 = 3abc$.
3. Find all real k for which the system with coefficient rows $(k, 1, -2)$, $(-2, k, 1)$, $(1, -2, k)$ has a non-trivial solution, and exhibit one.
4. If $a + b + c \neq 0$ and $a^3 + b^3 + c^3 = 3abc$, show that $a = b = c$.
5. Find all real k for which the circulant system in $(k, 1, 4)$ has a non-trivial solution, and exhibit one.

Answers: 1. Right-shift rows, so $+(8 + 64 + 216 - 144) = 144$. 2. Multiply out the right side: the squares give $a^2 + b^2 + c^2$ scaled correctly and the cross terms reproduce $-ab - bc - ca$; vanishing forces $a + b + c = 0$ or $a = b = c$, so the solution set is the plane $a + b + c = 0$ together with the line $a = b = c$. 3. Right-shift rows: $\Delta = k^3 + 1 - 8 + 6k = k^3 + 6k - 7$; $k = 1$ is a root and the quotient $k^2 + k + 7$ has negative discriminant, so $k = 1$; each row then sums to 0, so $(1, 1, 1)$ is a non-trivial solution. 4. By the identity, $0 = (a + b + c)(\text{squares})$ with the first factor nonzero, so the sum of squares is 0, forcing $a = b = c$. 5. $\Delta = k^3 + 1 + 64 - 12k = k^3 - 12k + 65$;

$k = -5$ is a root and the quotient $k^2 - 5k + 13$ has discriminant $25 - 52 < 0$, so $k = -5$; row sums are 0, so $(1, 1, 1)$ works.

Addendum B — Trigonometric Determinants

Part 6 taught you to differentiate a determinant whose entries are functions; neither it nor any earlier part evaluates one whose entries are sines and cosines. Yet JEE Main asks this routinely, and CBSE asks the tame versions. Four strategies cover nearly every papered instance, and all four are old tools pointed at new shapes.

Method

S1 — Combination scan (for an instant zero). Compound-angle identities are linear in the elementary functions. Expand every compound entry; if one row or column becomes a combination of the others, subtract that combination off (legal by the column version of P6 — it preserves Δ) and a zero line remains, so $\Delta = 0$.

S2 — Substitution. If every entry depends on its angle through a *single* expression, substitute. A determinant built from $1, \cos \theta, \cos^2 \theta$ across three angles is a Vandermonde in the variables $t_i = \cos \theta_i$, and Part 2 hands you the factorisation free.

S3 — Difference and factor. For rows of parallel structure, apply $R_i \rightarrow R_i - R_1$, then convert every difference with

$$\sin u - \sin v = 2 \cos \frac{u+v}{2} \sin \frac{u-v}{2}, \quad \cos u - \cos v = -2 \sin \frac{u+v}{2} \sin \frac{u-v}{2}.$$

Common sine factors fall out of the rows, and what survives is a 2×2 of shifted cosines that collapses by the compound-angle formula $\sin(p - q) = \sin p \cos q - \cos p \sin q$.

S4 — Expand along the zero-richest line, then burn the Pythagorean identity. Many trig determinants carry a strategically placed 0; expanding there leaves two 2×2 s whose entries combine via $\sin^2 + \cos^2 = 1$.

Example B.1 — [CBSE Board Level]

Show that $\begin{vmatrix} \sin \alpha & \cos \alpha & \sin(\alpha + \delta) \\ \sin \beta & \cos \beta & \sin(\beta + \delta) \\ \sin \gamma & \cos \gamma & \sin(\gamma + \delta) \end{vmatrix} = 0$ **for all** δ .

Expand the third column with the compound-angle formula:

$$\sin(\theta + \delta) = \sin \theta \cos \delta + \cos \theta \sin \delta,$$

entry by entry. So the third column equals $\cos \delta C_1 + \sin \delta C_2$ regardless of the row. Apply $C_3 \rightarrow C_3 - \cos \delta C_1 - \sin \delta C_2$: the value of Δ is unchanged and the third column becomes

identically zero. A determinant with a zero column is 0. The answer is independent of δ — which is the entire point of the question.

Example B.2 — [JEE Main Level]

Evaluate
$$\begin{vmatrix} \cos \alpha \cos \beta & \cos \alpha \sin \beta & -\sin \alpha \\ -\sin \beta & \cos \beta & 0 \\ \sin \alpha \cos \beta & \sin \alpha \sin \beta & \cos \alpha \end{vmatrix}.$$

Row 2 contains a zero, so expand along it — two terms survive:

$$\Delta = (-\sin \beta) C_{21} + \cos \beta C_{22}.$$

Compute the two cofactors. Deleting row 2, column 1 leaves

$$M_{21} = \begin{vmatrix} \cos \alpha \sin \beta & -\sin \alpha \\ \sin \alpha \sin \beta & \cos \alpha \end{vmatrix} = \cos^2 \alpha \sin \beta + \sin^2 \alpha \sin \beta = \sin \beta,$$

and the checkerboard sign $(-1)^{2+1}$ gives $C_{21} = -\sin \beta$. Deleting row 2, column 2 leaves

$$M_{22} = \begin{vmatrix} \cos \alpha \cos \beta & -\sin \alpha \\ \sin \alpha \cos \beta & \cos \alpha \end{vmatrix} = \cos^2 \alpha \cos \beta + \sin^2 \alpha \cos \beta = \cos \beta,$$

with sign $(-1)^{2+2}$, so $C_{22} = \cos \beta$. Therefore

$$\Delta = (-\sin \beta)(-\sin \beta) + (\cos \beta)(\cos \beta) = \sin^2 \beta + \cos^2 \beta = 1.$$

The determinant is 1 for *every* α, β — it is the product of two rotation matrices, and rotations preserve area. You were not asked to know that; the zero-rich expansion made it fall out anyway.

Example B.3 — [JEE Advanced Level]

Show that

$$\Delta = \begin{vmatrix} 1 & \sin \alpha & \cos \alpha \\ 1 & \sin \beta & \cos \beta \\ 1 & \sin \gamma & \cos \gamma \end{vmatrix} = -4 \sin \frac{\beta - \alpha}{2} \sin \frac{\gamma - \alpha}{2} \sin \frac{\gamma - \beta}{2}.$$

All three rows have parallel structure, so run S3. Apply $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$ (both use the original R_1 , so Δ is unchanged) and expand down the first column:

$$\Delta = (\sin \beta - \sin \alpha)(\cos \gamma - \cos \alpha) - (\cos \beta - \cos \alpha)(\sin \gamma - \sin \alpha).$$

Abbreviate $u = \frac{\alpha + \beta}{2}$, $v = \frac{\beta - \alpha}{2}$, $U = \frac{\alpha + \gamma}{2}$, $V = \frac{\gamma - \alpha}{2}$. The four differences become

$$\sin \beta - \sin \alpha = 2 \cos u \sin v, \quad \cos \beta - \cos \alpha = -2 \sin u \sin v,$$

$$\sin \gamma - \sin \alpha = 2 \cos U \sin V, \quad \cos \gamma - \cos \alpha = -2 \sin U \sin V.$$

Substituting:

$$\Delta = (2 \cos u \sin v)(-2 \sin U \sin V) - (-2 \sin u \sin v)(2 \cos U \sin V),$$

$$= -4 \sin v \sin V (\cos u \sin U - \sin u \cos U) = -4 \sin v \sin V \sin(U - u),$$

by the compound-angle formula read backwards. Finally $U - u = \frac{\gamma - \beta}{2}$, which is the claimed factorisation.

Check. Take $\alpha = 60^\circ$, $\beta = 30^\circ$, $\gamma = 0^\circ$. Direct expansion of the original gives $1 - \frac{\sqrt{3}}{2} \approx 0.1340$; the formula gives $-4 \sin(-15^\circ) \sin(-30^\circ) \sin(-15^\circ) \approx 0.1340$. Note the sign is sensitive to the *order* of the angles: swapping any two of α, β, γ swaps two columns and flips the sign, as it must.

Speed trap in the same family. $\begin{vmatrix} \sin^2 \alpha & \cos^2 \alpha & \cos 2\alpha \\ \sin^2 \beta & \cos^2 \beta & \cos 2\beta \\ \sin^2 \gamma & \cos^2 \gamma & \cos 2\gamma \end{vmatrix} = 0$ on sight, because $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$ makes the third column $C_2 - C_1$ — strategy S1, no expansion.

Practice — Addendum B

1. Evaluate $\begin{vmatrix} \sin 10^\circ & \cos 10^\circ \\ \sin 80^\circ & \cos 80^\circ \end{vmatrix}$.

2. Show that $\begin{vmatrix} \sin \alpha & \cos \alpha & \cos(\alpha + \delta) \\ \sin \beta & \cos \beta & \cos(\beta + \delta) \\ \sin \gamma & \cos \gamma & \cos(\gamma + \delta) \end{vmatrix} = 0$.

3. Evaluate $\begin{vmatrix} 1 & \sin \alpha & \cos \alpha \\ 1 & \sin \beta & \cos \beta \\ 1 & \sin \gamma & \cos \gamma \end{vmatrix}$ for $\alpha = 0^\circ$, $\beta = 90^\circ$, $\gamma = 180^\circ$, directly and by the formula of

Example B.3.

4. Show that $\begin{vmatrix} \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \\ \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \end{vmatrix} = 1$ for all θ, ϕ .

5. Show that $\begin{vmatrix} \cos(\theta - 60^\circ) & \sin(\theta - 60^\circ) & 1 \\ \cos \theta & \sin \theta & 1 \\ \cos(\theta + 60^\circ) & \sin(\theta + 60^\circ) & 1 \end{vmatrix}$ is independent of θ , and find its value.

Answers: 1. $\sin 10^\circ \cos 80^\circ - \cos 10^\circ \sin 80^\circ = \sin(-70^\circ) = -\sin 70^\circ = -\cos 20^\circ$. 2. Here $\cos(\theta + \delta) = \cos \theta \cos \delta - \sin \theta \sin \delta$, so the third column equals $\cos \delta C_2 - \sin \delta C_1$; subtract it off and a zero column remains. 3. Direct: the rows are $(1, 0, 1)$, $(1, 1, 0)$, $(1, 0, -1)$, giving

$1(-1) + 1(0 - 1) = -2$; formula: $-4 \sin 45^\circ \sin 90^\circ \sin 45^\circ = -2$. 4. Expand along row 2 as in Example B.2; the two minors reduce to $\pm \sin \beta$ -type expressions with α, θ, ϕ playing analogous roles, and the Pythagorean identity finishes it — the value is 1 identically. 5. Apply $R_1 \rightarrow R_1 - R_2, R_3 \rightarrow R_3 - R_2$; the differences reduce via S3 to first-column entries $\sin(\theta - 30^\circ), -\sin(\theta + 30^\circ)$ and second-column entries $-\cos(\theta - 30^\circ), \cos(\theta + 30^\circ)$, with the surviving 2×2 equal to $\sin(\theta - 30^\circ) \cos(\theta + 30^\circ) - \cos(\theta - 30^\circ) \sin(\theta + 30^\circ) = \sin(-60^\circ) = -\frac{\sqrt{3}}{2}$. Expanding along the third column (whose only survivor is the middle row's entry 1, cofactor sign $-$) yields $\Delta = \frac{\sqrt{3}}{2}$, independent of θ .

Addendum C — Alternants with a Missing Power

Part 2's practice set proved the Vandermonde determinant $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$,

and Part 2's Example 4 stripped factors off $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix}$. JEE Advanced replaces one power

row with a *skipped* one — rows $1, a, a^3$ or $1, a, a^2, a^4$ — and asks for the factorisation. Brute expansion dies; two structural methods live.

An **alternant** is any determinant whose rows are $f_1(t), f_2(t), \dots$ evaluated at $t = a, b, c, \dots$ in the columns. Vandermonde is the alternant with $f_i(t) = t^{i-1}$.

Method

Lemma 1 (symmetric quotient). If an alternant Δ and the Vandermonde V in the same variables satisfy: swapping any two variables swaps two columns of Δ (negating it) and of V (negating it), then $Q = \Delta/V$ — a genuine polynomial division, since V divides every alternating polynomial — satisfies $Q(a, b, c) = Q(b, a, c)$ and likewise for every swap. Hence Q **is symmetric**, and for a Class 12 syllabus every symmetric polynomial in sight is built from

$$e_1 = a + b + c, \quad e_2 = ab + bc + ca, \quad e_3 = abc.$$

Consequence: $\Delta = V \times Q$ where Q is symmetric, and comparing total degrees pins Q 's degree: $\deg Q = \deg \Delta - \deg V$.

Procedure 1 — degree, symmetry, specialise. Write Q as a general symmetric polynomial of the pinned degree with unknown coefficients, then substitute enough convenient triples to solve for them. One specialisation per unknown coefficient.

Procedure 2 — pad with x . When the exponent set is $\{0, 1, \dots, n\}$ minus exactly one member, build the full $(n+1) \times (n+1)$ Vandermonde by inserting a fifth column headed by a new variable x , then equate two readings of one coefficient.

Example C.1 — [JEE Main Level]

Prove that $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ bc & ca & ab \end{vmatrix} = (a-b)(b-c)(c-a).$

Call the determinant Δ . Swapping $a \leftrightarrow b$ swaps columns 1 and 2, so Δ is alternating; its entries have total degree $0 + 1 + 2 = 3$. The Vandermonde $V = (a-b)(b-c)(c-a)$ is also alternating of degree 3. By Lemma 1, $\Delta = \lambda V$ for a constant λ . Specialise $a = 1, b = 2, c = 3$:

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 6 & 3 & 2 \end{vmatrix} = 1(4-9) - 1(2-18) + 1(3-12) = -5 + 16 - 9 = 2,$$

and $V = (-1)(-1)(2) = 2$, so $\lambda = 1$. Done — no expansion of the general 3×3 ever happened.

Example C.2 — [JEE Advanced Level]

Prove that

$$\Delta = \begin{vmatrix} 1 & 1 & 1 & 1 \\ a & b & c & d \\ a^2 & b^2 & c^2 & d^2 \\ a^4 & b^4 & c^4 & d^4 \end{vmatrix} = (a+b+c+d)(b-a)(c-a)(d-a)(c-b)(d-b)(d-c).$$

The exponent set $\{0, 1, 2, 4\}$ skips 3, so Procedure 2. Insert a fifth variable x and build the complete 5×5 Vandermonde

$$W = \begin{vmatrix} 1 & 1 & 1 & 1 & 1 \\ a & b & c & d & x \\ a^2 & b^2 & c^2 & d^2 & x^2 \\ a^3 & b^3 & c^3 & d^3 & x^3 \\ a^4 & b^4 & c^4 & d^4 & x^4 \end{vmatrix}.$$

Read it two ways. First, as a Vandermonde in the five variables (a, b, c, d, x) in column order:

$$W = V_4(x-a)(x-b)(x-c)(x-d), \quad V_4 = (b-a)(c-a)(d-a)(c-b)(d-b)(d-c).$$

Second, expand W along column 5. The only entry carrying a factor x^3 sits at row 4, column 5; its cofactor is $(-1)^{4+5} = -1$ times the minor left by deleting row 4 and column 5 — and that minor is *exactly our* Δ . Every other entry in the column carries a different power of x . So the coefficient of x^3 in W equals $-\Delta$.

Now compare with the first reading, writing $\prod(x-a_i) = x^4 - e_1x^3 + e_2x^2 - e_3x + e_4$ with $e_1 = a+b+c+d$. The coefficient of x^3 in $V_4 \prod(x-a_i)$ is $-e_1V_4$. Equating the two readings of one coefficient:

$$-\Delta = -e_1 V_4, \quad \text{i.e.} \quad \Delta = (a + b + c + d) V_4.$$

Check numerically. Take $a, b, c, d = 1, 2, 3, 4$: the formula predicts $10 \times 12 = 120$. Directly, subtract row 1 from rows 2–4 and expand down column 1:

$$\Delta = \begin{vmatrix} 1 & 2 & 3 \\ 3 & 8 & 15 \\ 15 & 80 & 255 \end{vmatrix} = 1(2040 - 1200) - 2(765 - 225) + 3(240 - 120) = 840 - 1080 + 360 = 120.$$

Why it is Advanced-level. Neither Procedure 1 alone (too many unknown coefficients at degree 3 in four variables) nor expansion (24 terms) is pleasant here. The padding argument converts one hard determinant into one *coefficient comparison* against a determinant you already know — the same move as Part 1’s Example 4, executed one order higher.

Practice — Addendum C

1. Evaluate $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$ at $a = 2, b = 4, c = 5$ without expanding the general form.
2. Using Example C.2’s method, evaluate $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix}$ at $a = 2, b = 3, c = 5$.
3. Show that $\begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} = (ab + bc + ca)(a - b)(b - c)(c - a)$, and verify at $(a, b, c) = (1, 2, 3)$.
4. Evaluate $\begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 4 & 9 & 16 \\ 1 & 16 & 81 & 256 \end{vmatrix}$ using the result of Example C.2.
5. (Challenge.) Show that $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^4 & b^4 & c^4 \end{vmatrix} = (a^2 + b^2 + c^2 + ab + bc + ca)(a - b)(b - c)(c - a)$.

Answers: 1. Vandermonde: $(4 - 2)(5 - 2)(5 - 4) = 6$. 2. Pad a 4×4 Vandermonde in a, b, c, x with exponent rows 0, 1, 2, 3; the coefficient of x^3 is $+\Delta$ from the column-4 expansion (sign $(-1)^{4+4}$) and $+e_1 V_3$ from the product form, so $\Delta = (a + b + c) V_3 = 10 \times (1)(3)(2) = 60$. 3. Both sides are alternating of degree 5, and V_3 has degree 3, so the quotient is a symmetric quadratic $\lambda e_1^2 + \mu e_2$; specialise $(1, 2, 3)$: $\Delta = 22$, $V = 2$, $e_1^2 = 36$, $e_2 = 11$ gives $11 = 18\lambda + 11\mu$; specialise $(0, 1, 2)$: $\Delta = 2$, $V = 2$, $e_1^2 = 9$, $e_2 = 2$ gives $1 = 9\lambda + 2\mu$; solving, $\lambda = 0$, $\mu = 1$. Verification at $(1, 2, 3)$: left side = 22, right side = $11 \times 2 = 22$. 4. The array is the $\{0, 1, 2, 4\}$ alternant at $(1, 2, 3, 4)$: $(1 + 2 + 3 + 4)(12) = 120$. 5. Both sides are alternating of degree 5, and V_3 has degree 3, so the quotient is a symmetric quadratic $Ae_1^2 + Be_2$; specialise $(1, 2, 3)$:

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 16 & 81 \end{vmatrix} = 114 - 78 + 14 = 50, \quad V_3 = 2, \quad e_1^2 = 36, \quad e_2 = 11 \text{ gives } 25 = 36A + 11B;$$

specialise $(0, 1, 2)$: $\Delta = 14$, $V_3 = 2$, $e_1^2 = 9$, $e_2 = 2$ gives $7 = 9A + 2B$; solving, $A = 1$, $B = -1$. Hence the quotient is $e_1^2 - e_2 = a^2 + b^2 + c^2 + ab + bc + ca$. Verification at $(1, 2, 4)$: left side $= 448 - 252 + 14 = 210$, right side $= (49 - 14) \times 6 = 210$.

Addendum D — Concurrency, the Family of Lines, and Polygon Areas

Part 3 used determinants for one triangle at a time. Three structures built on it are absent from the chapter and are examined constantly: testing whether three lines meet at a point, writing down *every* line through a given intersection point, and measuring regions with more than three sides.

D.1 — Three lines are concurrent

Claim. For lines $a_i x + b_i y + c_i = 0$, $i = 1, 2, 3$, if the first two are not parallel then

$$\text{the three lines are concurrent} \iff \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0.$$

Proof. Solve the first two lines simultaneously. Their coefficient determinant is $\delta = a_1 b_2 - a_2 b_1$, nonzero by assumption, and Cramer (Part 5) gives their intersection point with

$$x_0 = \frac{\Delta_1}{\delta}, \quad y_0 = \frac{\Delta_2}{\delta},$$

where Δ_1, Δ_2 are the 2×2 replacements. Substitute into the third line:

$$a_3 x_0 + b_3 y_0 + c_3 = \frac{a_3 \Delta_1 + b_3 \Delta_2 + c_3 \delta}{\delta}.$$

Now read the numerator: a_3 times (column 1 replaced by constants), plus b_3 times (column 2 replaced), plus c_3 times the untouched 2×2 — this is precisely the cofactor expansion of the big 3×3 determinant **along row 3**. Hence the third line passes through (x_0, y_0) exactly when the 3×3 determinant vanishes.

Forward direction without hypotheses: if (x_0, y_0) lies on all three lines, then $(x_0, y_0, 1)^\top$ is a nonzero solution of the homogeneous system, which forces the coefficient determinant to be 0 — otherwise Part 5 allows only the trivial solution. **Caution:** the converse fails silently when the lines are parallel — the three lines $x + y - 1 = 0$, $x + y - 2 = 0$, $2x + 2y - 5 = 0$ have determinant 0 (columns 1 and 2 identical) but no common point. Always glance for parallelism before believing a zero.

D.2 — The family through an intersection point

If $L_1 = 0$ and $L_2 = 0$ meet at P , then for every real λ ,

$$L_1 + \lambda L_2 = 0$$

is a line through P (both summands vanish at P), and conversely every line through P except L_2 itself arises for exactly one λ . To force the member of the family through a further point Q , substitute Q and solve the resulting *linear* equation for λ — no intersection coordinates ever need to be computed.

D.3 — Polygon area by the cyclic sum

For a polygon with vertices $P_1, \dots, P_n$ listed in order around the boundary,

$$\text{Area} = \frac{1}{2} \left| (x_1 y_2 - x_2 y_1) + (x_2 y_3 - x_3 y_2) + \dots + (x_n y_1 - x_1 y_n) \right|.$$

Derivation for $n = 4$. Split quadrilateral $ABCD$ along diagonal AC into triangles ABC and ACD , apply Part 3's formula to each, and add. Collecting the x_A terms: $\frac{1}{2}x_A[(y_B - y_C) + (y_C - y_D)] = \frac{1}{2}x_A(y_B - y_D)$; collecting the x_C terms similarly gives $\frac{1}{2}x_C(y_D - y_B)$; the x_B and x_D terms are untouched. The total is exactly the four-term cyclic sum above. (For $n > 4$, triangulate from P_1 and repeat.)

Example D.1 — [JEE Main Level]

Show that the lines $x + 2y - 4 = 0$, $2x + y - 5 = 0$, $3x + 3y - 9 = 0$ are concurrent, and find the point.

First two lines: $\delta = 1 \cdot 1 - 2 \cdot 2 = -3 \neq 0$, so the test applies. Expand along row 3:

$$\begin{vmatrix} 1 & 2 & -4 \\ 2 & 1 & -5 \\ 3 & 3 & -9 \end{vmatrix} = 3 \begin{vmatrix} 2 & -4 \\ 1 & -5 \end{vmatrix} - 3 \begin{vmatrix} 1 & -4 \\ 2 & -5 \end{vmatrix} + (-9) \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix}$$

$$= 3(-10 + 4) - 3(-5 + 8) - 9(1 - 4) = -18 - 9 + 27 = 0.$$

Concurrent. Solving the first two: doubling the first and subtracting the second gives $3y = 3$, so $y = 1$, $x = 2$. Check the third: $6 + 3 - 9 = 0$.

Example D.2 — [JEE Advanced Level]

Find the area of the triangle formed by the lines $x + y = 6$, $x - y = 2$, $y = 2x$.

Intersections, two lines at a time: $x + y = 6$ with $x - y = 2$ gives $(4, 2)$; $x + y = 6$ with $y = 2x$ gives $3x = 6$, so $(2, 4)$; $x - y = 2$ with $y = 2x$ gives $-x = 2$, so $(-2, -4)$. Cyclic sum:

$$(4 \cdot 4 - 2 \cdot 2) + (2 \cdot (-4) - 4 \cdot (-2)) + ((-2) \cdot 2 - (-4) \cdot 4) = 12 + 0 + 12 = 24,$$

$$\text{Area} = \frac{1}{2}|24| = 12 \text{ square units.}$$

Check. Base between $(2, 4)$ and $(-2, -4)$ has length $\sqrt{16+64} = 4\sqrt{5}$; the distance from $(4, 2)$ to the line $y = 2x$, i.e. $2x - y = 0$, is $\frac{|8-2|}{\sqrt{5}} = \frac{6}{\sqrt{5}}$; area $= \frac{1}{2} \cdot 4\sqrt{5} \cdot \frac{6}{\sqrt{5}} = 12$. Agreement by an independent route.

Practice — Addendum D

- Using a 3×3 determinant, test whether $2x - 3y + 5 = 0$, $4x - 6y + 9 = 0$, $2x - 2y + 3 = 0$ are concurrent.
- Find k so that $x + y - 2 = 0$, $2x - y - 1 = 0$, $3x + ky - 4 = 0$ are concurrent, and state the point of concurrence.
- Find the equation of the line through the intersection of $x + 2y - 3 = 0$ and $2x - y - 4 = 0$ which also passes through the origin.
- Find the area of the quadrilateral with vertices $(1, 2)$, $(5, 5)$, $(4, 8)$, $(0, 7)$ in that order.
- Why does the concurrency determinant of $x + y - 1 = 0$, $x + y - 2 = 0$, $2x + 2y - 5 = 0$ vanish, and what is the geometric truth?

Answers: 1. Determinant $= 0$ (rows 1 and 2 are proportional in the coefficients but the constants are not — expand to confirm: $2(-18+18)+3(-12+18)+5(-8+12) = 0+18+20 \neq 0$), so **not** concurrent — the first two lines are parallel and distinct. 2. Expand along row 3: $3(4+k) \cdot ?$ — directly, $3 \begin{vmatrix} 1 & -2 \\ -1 & -1 \end{vmatrix} - k \begin{vmatrix} 1 & -2 \\ 2 & -1 \end{vmatrix} - 4 \begin{vmatrix} 1 & 1 \\ 2 & -1 \end{vmatrix} = 3(-3) - k(3) - 4(-3) = 3 - 3k$, so $k = 1$; solving the first two lines gives $(1, 1)$. 3. Family $(x + 2y - 3) + \lambda(2x - y - 4) = 0$; through the origin demands $-3 - 4\lambda = 0$, so $\lambda = -\frac{3}{4}$, giving $2x - 11y = 0$. 4. Cyclic sum $= (5 - 10) + (40 - 20) + (28 - 0) + (0 - 7) = 36$, area $= 18$ square units; cross-check by splitting along the diagonal from $(1, 2)$ to $(4, 8)$: areas $7.5 + 10.5 = 18$. 5. Columns 1 and 2 are proportional, so the determinant is 0 by P3 — but the three lines are parallel with no common point; a vanishing determinant signals *dependent rows*, which is concurrency only when the lines are pairwise non-parallel.

Addendum E — What to Actually Do When $\Delta = 0$

Part 5's consistency table ends its hardest branch with “(usually; check rank)” — and then never checks. JEE Main asks you to *produce the solutions*; JEE Advanced asks whether the shortcut can mislead. This addendum supplies both.

Method

When $\Delta = 0$ for $AX = B$ with three equations in three unknowns:

- Screen with the Δ_i** (replace column i of A by B). Any nonzero Δ_i certifies inconsistency immediately — no further work.
- All $\Delta_i = 0$: eliminate.** Form the augmented matrix $[A \mid B]$ and row-reduce to echelon form. Identify the independent equations and the free variables.
- Parametrise.** Assign a parameter to each free variable and back-substitute. The num-

ber of free parameters is $n - r$, where r is the number of independent rows.

4. **Homogeneous systems:** the same machinery gives the dimension of the solution space as $n - r$; the trivial solution is always there, and it is the *only* one exactly when $r = n$.

Why step 3 counts correctly: after reduction there are r independent linear equations. Choosing the $n - r$ free variables arbitrarily determines the r pivot variables uniquely by back-substitution, and every solution arises this way — so solutions are in bijection with the parameter choices.

The screen can lie: a precise statement

Proposition. Suppose $\Delta = 0$ but some 2×2 minor of A is nonzero (equivalently, the rows of A span a plane, not a line — “rank exactly 2”). Then:

$$\text{the system is consistent} \iff \Delta_1 = \Delta_2 = \Delta_3 = 0,$$

and in that case it has infinitely many solutions.

Proof. The augmented matrix $[A \mid B]$ is 3×4 ; its 3×3 minors are obtained by deleting one column at a time. Deleting the last gives $\Delta = 0$; deleting column j gives, up to sign, Δ_j . If all three Δ_j vanish, every 3×3 minor of $[A \mid B]$ is 0, so $[A \mid B]$ has rank at most 2. But rank $[A \mid B]$ is always at least rank A , and rank $A = 2$ by hypothesis — so both ranks equal 2, which is precisely consistency (Part 5’s language: B is a combination of the columns of A). Infinitely many solutions follow because $\Delta = 0$ rules out uniqueness. Conversely, if some $\Delta_j \neq 0$ then that minor of $[A \mid B]$ is nonzero, rank $[A \mid B] = 3 > \text{rank } A$, and the system is inconsistent. ■

Where the screen fails. If rank $A = 1$ (all rows proportional), consistency can fail with every $\Delta_i = 0$:

$$x + y + z = 1, \quad 2x + 2y + 2z = 3, \quad 3x + 3y + 3z = 4.$$

Here $\Delta = 0$ (proportional rows), $\Delta_1 = 0$ (its second and third columns are identical), and $\Delta_2 = \Delta_3 = 0$ the same way — yet the system demands $1 = 3 = 4$. Moral: **elimination is the judge; the Δ_i are only the bailiff.**

Example E.1 — [JEE Main Level]

Solve completely: $x + y + z = 6$, $x + 2y + 3z = 14$, $2x + 3y + 4z = 20$.

Expanding, $\Delta = 1(8 - 9) - 1(4 - 6) + 1(3 - 4) = -1 + 2 - 1 = 0$. The three replacement determinants:

$$\Delta_1 = 6(8 - 9) - 1(56 - 60) + 1(42 - 40) = -6 + 4 + 2 = 0,$$

$$\Delta_2 = 1(56 - 60) - 6(4 - 6) + 1(20 - 28) = -4 + 12 - 8 = 0,$$

$$\Delta_3 = 1(40 - 42) - 1(20 - 28) + 6(3 - 4) = -2 + 8 - 6 = 0.$$

Screen passed — now eliminate. Subtract equation 1 from equation 2, and twice equation 1 from equation 3:

$$y + 2z = 8 \quad (\text{from both — one independent equation}),$$

so $r = 2$, one free parameter. Let $z = t$. Then $y = 8 - 2t$ and $x = 6 - y - z = 6 - (8 - 2t) - t = t - 2$:

$$(x, y, z) = (t - 2, 8 - 2t, t), \quad t \in \mathbb{R}.$$

Check in equation 2: $(t - 2) + 2(8 - 2t) + 3t = t - 2 + 16 - 4t + 3t = 14$. ✓

Example E.2 — [JEE Main Level]

Classify: $x + y + z = 6$, $x + 2y + 3z = 14$, $2x + 3y + 4z = 21$.

Same left sides, so $\Delta = 0$ again. But now $\Delta_1 = 6(-1) - 1(56 - 63) + 1(42 - 42) = -6 + 7 + 0 = 1 \neq 0$: the screen convicts the system instantly — no solution. Elimination agrees: subtracting gives $y + 2z = 8$ from the first pair and $y + 2z = 9$ from equations 1 and 3 — contradiction. Both routes, one verdict.

Example E.3 — [JEE Advanced Level]

Discuss completely for all real λ, μ : $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = \mu$.

Expanding along row 3: $\Delta = 1(2\lambda - 6) - 1(\lambda - 3) + 1(2 - 2) = \lambda - 3$.

Case $\lambda \neq 3$: unique solution. Eliminate without computing an inverse: subtracting equation 1 from equation 2 gives $y + 2z = 4$; subtracting equation 2 from equation 3 gives $(\lambda - 3)z = \mu - 10$. Back-substitute:

$$z = \frac{\mu - 10}{\lambda - 3}, \quad y = 4 - \frac{2(\mu - 10)}{\lambda - 3}, \quad x = 6 - y - z = 2 + \frac{\mu - 10}{\lambda - 3}.$$

Case $\lambda = 3$: equations 2 and 3 have identical left sides, so the system is consistent iff $\mu = 10$. If $\mu \neq 10$: no solution (and indeed $\Delta_2 = 1(30 - 3\mu) - 6(3 - 3) + 1(\mu - 10) = 20 - 2\mu \neq 0$). If $\mu = 10$: the screen reads zero across the board, and rank is exactly 2 (rows 1 and 2 are independent), so by the Proposition there are infinitely many solutions — one free parameter. Eliminating: $y + 2z = 4$, so with $z = t$,

$$(x, y, z) = (2 + t, 4 - 2t, t), \quad t \in \mathbb{R}.$$

Why it is Advanced-level. Three decisions stack: factor Δ in the parameter, test consistency at the critical value, and *count* the solution family rather than merely asserting “infinitely many”. The Proposition is what licenses jumping from “screen all zero” to “infinitely many” here — rank 2 is visible on inspection (rows 1 and 2 are not proportional), so the licence is valid.

Practice — Addendum E

1. Solve completely: $x + y + z = 3$, $2x + y - z = 1$, $3x + 2y + 0z = 4$.
2. Classify: $x + y + z = 3$, $2x + y - z = 1$, $3x + 2y + 0z = 7$.
3. Find all real k for which $kx + y + z = 1$, $x + ky + z = k$, $x + y + kz = k^2$ has (i) a unique solution, (ii) no solution, (iii) infinitely many, describing the solution set in case (iii).
4. Find the dimension of the solution space of $x + y + z = 0$, $x - y + z = 0$, $2x + 0y + 2z = 0$, and describe it geometrically.
5. Explain why the counterexample $x + y + z = 1$, $2x + 2y + 2z = 3$, $3x + 3y + 3z = 4$ slips past the Δ_i screen, naming the rank of A and of $[A \mid B]$.

Answers: 1. $\Delta = 0$ (row 3 = row 1 + row 2) and all $\Delta_i = 0$; eliminating gives $x - 2z = -2$, so with $z = t$: $(x, y, z) = (2t - 2, 5 - 3t, t)$ — check in equation 3: $3(2t - 2) + 2(5 - 3t) = 4$. ✓
2. Same left sides, so $\Delta = 0$; eliminating gives $x - 2z = -2$ from equations 1-2 but $x - 2z = 1$

from equations 1 and 3 — inconsistent; equivalently $\Delta_1 = \begin{vmatrix} 3 & 1 & 1 \\ 1 & 1 & -1 \\ 7 & 2 & 0 \end{vmatrix} = 3(2) - 1(7) + 1(2 - 7) =$

$6 - 7 - 5 = -6 \neq 0$. 3. $\Delta = (k - 1)^2(k + 2)$. (i) $k \neq 1, -2$. (ii) $k = -2$: adding the three equations gives $0 = 3$, contradiction — no solution. (iii) $k = 1$: all three equations collapse to $x + y + z = 1$, a plane of solutions $(s, t, 1 - s - t)$, two parameters. 4. Row 3 = row 1 + row 2, so $r = 2$ and the dimension is $3 - 2 = 1$: eliminating gives $x + z = 0$ and $y = 0$, the line $(t, 0, -t)$ through the origin in direction $(1, 0, -1)$. 5. Every row of A is proportional to $(1, 1, 1)$, so rank $A = 1$ — below the 2 required by the Proposition. All four 3×3 minors of $[A \mid B]$ vanish, yet rank $[A \mid B] = 2 > 1 = \text{rank } A$: the constants column escapes the column space while adding only one rank, so the screen's “all zero” no longer certifies consistency. Elimination exposes it in one step: $x + y + z = 1 = 3 = 4$.

Addendum P — What JEE Actually Asks

Evidence note. The compendium extract preserves the year/session tags and the official answers, but most determinants and matrices arrived as lost images. Every question reprinted below has been **reconstructed and re-solved from scratch**; items too corrupted to rebuild with confidence are cited for pattern purposes only and never reproduced. Treat the counts below as solid ratios, not census figures.

P.1 The distribution

From roughly 120 identifiable items spanning 1979–2020:

Sub-topic	Approx. share	Evidence highlights
Consistency of linear systems (unique / infinite / none, parameters)	~35%	An entire block of the bank: Sep 02–06 2020 (all four sessions contribute), Jan 07–09 2020, April 09–12 2019, Jan 10–12 2019, 2018, 2017, 2013, 2010, the 2009 Advanced passage, back to 1979. In the 2020 sessions alone, system-consistency items appear in at least six of the eight papers represented.
Adjoint / inverse identities	~15%	Adv 2020, Sep 03 2020 (II), Jan 09 2020 (I), April 12 2019 (I), Jan 11 2019 (II), April 15 2018, 2017, 2016, 2014, 2013, 2011, 2010, Adv 2019 (twice), Adv 2013, 2005. Steady presence every year since 2010.
Properties / factorisation / “evaluate fast”	~20%, spread over all decades	1981 (cyclic determinant), 1982, 1983, 1985, 1986 (G.P. condition), 1990 (digit divisibility), 1992 (factorial determinant), 1996, Adv 2015, and countless embedded uses.
Geometry via area/collinearity	~8%	2008 (concurrent-lines matching), 2017 (area → orthocentre), Jan 08 2020 (I) (area + line → parameter), 1985 (a “congruent triangles” false statement built on the area determinant).
Counting / extremal determinants over finite entry sets	~8%, Advanced-heavy	1981, 1982, 2009 and 2010 passages, 2011 (cube roots of unity), Adv 2016, Adv 2018, Adv 2020 (trace puzzle), Sep 03 2020 (II).
Determinant equations → polynomials	recurring	2015 ($ax - 12$), Online April 8 2017 ($\tan^3 x = 1$), 1983, 1981, Adv 2015.

Two structural observations. First, **the centre of gravity has shifted hard toward systems**: pre-2005, property gymnastics dominated; from 2017 onward, most Main papers carry 2–4 system-consistency or adjoint-identity items. Second, Advanced favours the counting/extremal genre almost exclusively — Main rarely touches it.

P.2 The recurring archetypes

A1 — Consistency triage for $AX = B$ with a parameter

Shape: three equations, one or two parameters; ask for “no solution”, “infinitely many”, or a numerical combination like $\lambda + \mu$.

Method: compute Δ as a polynomial in the parameter; its roots are the only candidates for trouble. At each root, test one of $\Delta_1, \Delta_2, \Delta_3$: any nonzero one means **no solution**; all zero means infinitely many. Never assume which.

Worked example (Main Jan 08, 2020 (II), reconstructed): for which α does

$$\alpha x + 2y + 2z = 5$$

$$2\alpha x + 3y + 5z = 8$$

$$4x + \alpha y + 6z = 10$$

fail to have a unique solution?

$$\Delta = \begin{vmatrix} \alpha & 2 & 2 \\ 2\alpha & 3 & 5 \\ 4 & \alpha & 6 \end{vmatrix} = \alpha(18 - 5\alpha) - 2(12\alpha - 20) + 2(2\alpha^2 - 12) = -(\alpha + 8)(\alpha - 2)$$

So $\Delta = 0$ only at $\alpha = -8$ and $\alpha = 2$. At $\alpha = 2$, replacing column 1 by $(5, 8, 10)$:

$$\Delta_1 = \begin{vmatrix} 5 & 2 & 2 \\ 8 & 3 & 5 \\ 10 & 2 & 6 \end{vmatrix} = 40 + 4 - 28 = 16 \neq 0$$

so the system is **inconsistent at** $\alpha = 2$ (and likewise at $\alpha = -8$); everywhere else it has a unique solution. This matches the official key.

A2 — Homogeneous systems: “non-trivial solution”

Shape: $AX = O$ with trigonometric or polynomial entries; count the parameters, or identify the solution set.

Method: non-trivial solutions exist **iff** $\Delta = 0$; then solve the collapsed system. Remember the trivial solution always exists — the question is about the rest.

Worked example (Main Jan 12, 2019 (II), reconstructed):

$$x - 2y - 2z = \lambda x, \quad x + 2y + z = \lambda y, \quad -x - y = \lambda z$$

The coefficient matrix is $\begin{bmatrix} 1-\lambda & -2 & -2 \\ 1 & 2-\lambda & 1 \\ -1 & -1 & -\lambda \end{bmatrix}$. Expanding along row 1, all three minors turn out to be $(1-\lambda)$, $1-\lambda$, $1-\lambda$ respectively, giving

$$\Delta = (1-\lambda)^3 + 2(1-\lambda) - 2(1-\lambda) = (1-\lambda)^3$$

Non-trivial solutions exist iff $\lambda = 1$: a **singleton** set of values — the official answer.

A3 — Elementwise-scaled determinants

Shape: $b_{ij} = f(i)g(j)a_{ij}$; given $|B|$, find $|A|$ (or vice versa).

Method: never pull one global factor. Split $f(i)g(j)$ into a row factor times a column factor:

$$B = DFG \Rightarrow |B| = |D||F||G|$$

Worked example (Main Jan 07, 2020 (II), verbatim recoverable): $b_{ij} = 3^{i+j-2}a_{ij}$, $|B| = 81$. Since $3^{i+j-2} = 3^{i-1}3^{j-1}$,

$$B = \text{diag}(1, 3, 9) A \text{diag}(1, 3, 9) \Rightarrow |B| = 3^6|A|$$

so $81 = 729|A|$ and $|A| = \frac{1}{9}$ — the official answer. (Its sibling, Main 2012, with $b_{ij} = 2^{i+j}a_{ij}$ and $|P| = 2$, gives $|Q| = 2^6 \cdot 2^6 \cdot 2 = 2^{13}$; the wrong “global factor” route yields 2^{36} — see Traps.)

A4 — Adjoint power ladders

Shape: chains like $|\text{adj}(\text{adj } A)|$, or a condition such as $M^{-1} = \text{adj}(\text{adj } M)$.

Method: memorise the ladder for order $n = 3$:

$$|\text{adj } A| = |A|^2, \quad |\text{adj}(\text{adj } A)| = |A|^4, \quad \text{adj}(\text{adj } A) = |A| A$$

Worked example (Adv 2020, verbatim): M invertible with $M^{-1} = \text{adj}(\text{adj } M)$. Since $\text{adj}(\text{adj } M) = |M| M$,

$$M^{-1} = |M| M \Rightarrow M^2 = \frac{1}{|M|} I$$

Taking determinants: $(\det M)^2 = (\det M)^{-3}$, so $(\det M)^5 = 1$, hence $\det M = 1$. Then $M^2 = I$, and finally $(\text{adj } M)^2 = \text{adj}(M^2) = \text{adj}(I) = I$. So $\det M = 1$, $M^2 = I$ and $(\text{adj } M)^2 = I$ are the always-true statements — exactly the official key.

A5 — Area/collinearity smuggled into coordinate geometry

Shape: “triangle has area K ”, then find a parameter, or build a further object (line, ortho-centre).

Method: write the area determinant, set $\frac{1}{2}|\Delta| = K$, and expect **two** branches from the modulus.

Worked example (Main 2017, reconstructed): the triangle with vertices $(k, -3k)$, $(5, k)$, $(-k, 2)$ has area 28; k an integer; find the orthocentre.

$$\Delta = \begin{vmatrix} k & -3k & 1 \\ 5 & k & 1 \\ -k & 2 & 1 \end{vmatrix} = k(k-2) + 3k(5+k) + (10+k^2) = 5k^2 + 13k + 10$$

Setting $|\Delta| = 56$: the plus branch gives $5k^2 + 13k - 46 = 0$, i.e. $k = 2$ or $k = -\frac{23}{5}$; the minus branch gives $5k^2 + 13k + 66 = 0$, negative discriminant. So $k = 2$ and the vertices are $A(2, -6)$, $B(5, 2)$, $C(-2, 2)$. Side BC is horizontal, so the altitude from A is $x = 2$. Slope of AC is $\frac{2-(-6)}{-2-2} = -2$, so the altitude from B has slope $\frac{1}{2}$: $y - 2 = \frac{1}{2}(x - 5)$, giving at $x = 2$ the value $y = \frac{1}{2}$. (The extract's own solution is garbled here; this orthocentre $H = (2, \frac{1}{2})$ is my verified computation.)

A6 — Instant zero and canonical factorisations

Shape: “show the determinant is negative / zero iff ...”.

Method: scan for proportional rows, AP rows, odd-order skew-symmetry; otherwise reduce to a named form (Vandermonde, cyclic).

Worked example (JEE 1981, verbatim recoverable): for positive a, b, c , not all equal,

$$\begin{aligned} \Delta &= \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 3abc - a^3 - b^3 - c^3 = -(a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca) \\ &= -(a+b+c) \cdot \frac{1}{2} [(a-b)^2 + (b-c)^2 + (c-a)^2] < 0 \end{aligned}$$

since $a+b+c > 0$ and the squared sum vanishes only when $a = b = c$. (Check: $a, b, c = 1, 2, 3$ gives $\Delta = -18$, and $-(18 - 18) = -18$.)

A7 — Identity chains from matrix equations

Shape: $MM^T = I$, $AA^T = A^TA$, $(A - 3I)(A - 5I) = O$; deduce determinant facts.

Method: convert everything to determinants and transpose-invariance; move factors across the bar.

Worked example (JEE 2004, verbatim): $\det M = 1$, $MM^T = I$. Prove $\det(M - I) = 0$:

$$\begin{aligned} \det(M - I) &= \det(M - MM^T) \\ &= \det(M) \det(I - M^T) \\ &= \det((I - M)^T) = \det(I - M) \\ &= \det(-(M - I)) = -\det(M - I) \end{aligned}$$

so $2\det(M - I) = 0$. (Sibling results from the bank: Main 2014, $B = A^{-1}A^T$ with $AA^T = A^TA$ forces $BB^T = I$; Adv 2017, a matrix with negative determinant cannot be anyone's square, since $\det(X^2) = (\det X)^2 \geq 0$.)

A8 — Counting and extremal determinants

Shape: entries restricted to $\{0, 1\}$, $\{-1, 0, 1\}$, or residues mod p ; count matrices or maximise $\det$.

Method: enumerate structurally (what forces $\det > 0$?), and use the expansion structure for bounds.

Worked example (JEE 1982, verbatim): a random 2×2 determinant with entries 0 or 1; probability the value is positive. $ad - bc > 0$ needs $ad = 1$, so $a = d = 1$, and $bc = 0$: three choices for (b, c) . Out of 16 equally likely matrices, 3 work:

$$\Pr = \frac{3}{16}$$

(The Advanced face of this archetype — maximum $\det$ over $\{-1, 0, 1\}^{3 \times 3}$ — is Practice Problem P8.)

P.3 The traps

Built from distractor patterns visible in the bank's own options:

1. $\Delta = 0$ **does not mean “infinitely many”**. You must check a Δ_i . Main 2013's no-solution item has $\Delta = 0$ at $k = 1$ and $k = 3$, but only $k = 3$ is inconsistent — at $k = 1$ the equations collapse to one line. The distractor “both” is waiting for you.
2. **“More than two solutions” means infinitely many** — a linear system has exactly 0, 1, or ∞ solutions (Main Jan 07 2020 (I) numerical exploits precisely this).
3. **Scaling confusions.** $|kA| = k^n |A|$, but one scaled row costs only k ; and an element-wise rule $b_{ij} = 2^{i+j} a_{ij}$ costs $2^{12} \cdot |A|$, not 2^{36} . The 2012 options $2^{10}, 2^{11}, 2^{12}, 2^{13}$ are pure exponent-off-by traps.
4. **Adjoint powers.** $|\operatorname{adj} A| = |A|^{n-1}$ (not $|A|^n$); iterating gives $|A|^{(n-1)^2} = |A|^4$ for 3×3 . And $\operatorname{adj}(AB) = (\operatorname{adj} B)(\operatorname{adj} A)$ — reversed (Adv 2013 lists the unreversed form as a wrong option).
5. $\operatorname{adj}(\operatorname{adj} A) = |A|^{n-2} A$, **not** $|A|(\operatorname{adj} A)^{-1}$. The Online April 2017 “not always true” item punishes pattern-matching between these look-alikes.
6. **An odd-order skew-symmetric matrix is always singular.** The 2011 Advanced item hands you “non-singular 3×3 skew-symmetric M, N ” — an impossible premise. Recognise the contradiction instead of hunting for matrices that do not exist.
7. **Homogeneous systems:** $X = O$ always works; the question is about *non-trivial* solutions, which need $\Delta = 0$. The distractor “always has a non-trivial solution” appears verbatim in the bank.
8. **Area bookkeeping:** forget the $\frac{1}{2}$ or the modulus and both branches collapse to one wrong answer (Jan 08 2020 (I) has $\lambda = 3$ and $\lambda = -2$).
9. **Illegal simultaneous operations** ($R_1 \rightarrow R_1 + R_2$ and $R_2 \rightarrow R_2 + R_1$ in one breath) silently change Δ .

10. **Same-row vs cross-cofactor sums:** entries of row i against cofactors of row i give $|A|$; against cofactors of row $k \neq i$ give 0. Adv 2019's adjoint-reading question runs entirely on this.

P.4 Graded practice set

P1. Evaluate without expanding: $\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix}$.

Solution. $C_2 \rightarrow C_2 + C_3$ makes column 2 equal to $(a+b+c, a+b+c, a+b+c)^T$, a multiple of column 1. Two proportional columns give

$$\Delta = 0$$

P2. (Pattern: Main 2013.) Find the number of values of k for which $(k+1)x + 8y = 4k$, $kx + (k+3)y = 3k-1$ has **no** solution.

Solution.

$$\Delta = (k+1)(k+3) - 8k = k^2 - 4k + 3 = (k-1)(k-3)$$

Candidates: $k = 1, 3$. At $k = 1$: the equations become $2x + 8y = 4$ and $x + 4y = 2$ — the same line, so infinitely many solutions. At $k = 3$: $4x + 8y = 12$ and $3x + 6y = 8$; left sides are in ratio $\frac{4}{3}$ but right sides in ratio $\frac{3}{2}$ — inconsistent. So exactly **one** value ($k = 3$).

P3. A is 3×3 with $|A| = -2$. Find $|\operatorname{adj} A|$, $|\operatorname{adj}(\operatorname{adj} A)|$, $|3A|$, $|A^{-1}|$.

Solution. With $n = 3$:

$$\begin{aligned} |\operatorname{adj} A| &= |A|^2 = 4 \\ |\operatorname{adj}(\operatorname{adj} A)| &= |A|^4 = 16 \\ |3A| &= 3^3(-2) = -54 \\ |A^{-1}| &= -\frac{1}{2} \end{aligned}$$

P4. (Main Jan 10, 2019 (II), reconstructed.) Find the number of $\theta \in (0, \pi)$ for which

$$x + 3y + 7z = 0, \quad -x + 4y + 7z = 0, \quad (\sin 3\theta)x + (\cos 2\theta)y + 2z = 0$$

has a non-trivial solution.

Solution. Expand the coefficient determinant along row 3:

$$\Delta = 14 - 7 \sin 3\theta - 14 \cos 2\theta$$

Set $\Delta = 0$ and substitute $\sin 3\theta = 3s - 4s^3$, $\cos 2\theta = 1 - 2s^2$ with $s = \sin \theta$:

$$2 - (3s - 4s^3) - 2(1 - 2s^2) = 0 \Rightarrow 4s^3 + 4s^2 - 3s = 0 \Rightarrow s(2s - 1)(2s + 3) = 0$$

So $s = 0, \frac{1}{2}, -\frac{3}{2}$. On $(0, \pi)$: $s = 0$ is excluded, $s = -\frac{3}{2}$ impossible, $s = \frac{1}{2}$ gives $\theta = \frac{\pi}{6}$ or $\frac{5\pi}{6}$. Exactly **two** values.

P5. (Main 2012.) $P = [a_{ij}]$ is 3×3 with $|P| = 2$, and $Q = [b_{ij}]$ with $b_{ij} = 2^{i+j}a_{ij}$. Find $|Q|$.

Solution. Split the exponent: $2^{i+j} = 2^i \cdot 2^j$, so

$$Q = \text{diag}(2, 4, 8) P \text{diag}(2, 4, 8)$$

Hence $|Q| = 2^6 \cdot 2^6 \cdot 2 = 2^{13}$.

P6. (Main April 10, 2019 (I), reconstructed.) If $x + y + z = 5$, $x + 2y + 2z = 6$, $x + 3y + \lambda z = m$ has infinitely many solutions, find $\lambda + m$.

Solution.

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 3 & \lambda \end{vmatrix} = \lambda - 3 \Rightarrow \lambda = 3$$

With $\lambda = 3$, Δ_y (second column replaced by the constants):

$$\begin{vmatrix} 1 & 5 & 1 \\ 1 & 6 & 2 \\ 1 & m & 3 \end{vmatrix} = 18 - 2m - 5 + m - 6 = 7 - m \Rightarrow m = 7$$

Consistency holds (the equations collapse to $x + y + z = 5$, $y + z = 1$), so the system is genuinely infinite, and $\lambda + m = 10$.

P7. (Main Jan 08, 2020 (I), reconstructed.) $A(1, -1)$, $B(0, 2)$; a point $P(x', y')$ lies on $3x + y - 4\lambda = 0$ and $\text{area}(PAB) = 5$. Find a value of λ .

Solution.

$$\Delta = \begin{vmatrix} x' & y' & 1 \\ 1 & -1 & 1 \\ 0 & 2 & 1 \end{vmatrix} = -3x' - y' + 2$$

Area 5 means $|\Delta| = 10$, i.e. $3x' + y' = 12$ or $3x' + y' = -8$. But the line says $3x' + y' = 4\lambda$, so

$$4\lambda = 12 \text{ or } -8 \Rightarrow \lambda = 3 \text{ or } \lambda = -2$$

Both occur; the option “3” is the keyed one.

P8. (Adv 2018 spirit.) Find the maximum value of $\det P$ over all 3×3 matrices with entries in $\{-1, 0, 1\}$.

Solution. Hadamard: $|\det P| \leq (\sqrt{3})^3 = 3\sqrt{3} < 6$, so $|\det P| \leq 5$. If $|\det P| = 5$, all six expansion terms are ± 1 , forcing every entry nonzero, i.e. a ± 1 matrix. Subtract row 1 from rows 2 and 3: those rows acquire all entries in $\{0, \pm 2\}$, so $\det$ is divisible by 4 — contradicting 5. Hence the maximum is at most 4, and

$$P = \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix} = J - 2I$$

has eigenvalues $1, -2, -2$ (from those of J), so $|\det P| = 4$. **Maximum** = 4, the official Advanced answer.

P9. (Adv 2020, verbatim.) A is 2×2 , $\operatorname{tr}(A) = 3$, $\operatorname{tr}(A^3) = -18$. Find $\det A$.

Solution. Let the eigenvalues be λ, μ : $\lambda + \mu = 3$, $\lambda\mu = d$. Then

$$\lambda^3 + \mu^3 = (\lambda + \mu)^3 - 3\lambda\mu(\lambda + \mu) = 27 - 9d$$

Given $27 - 9d = -18$, so $d = 5$. (Equivalently, from the characteristic polynomial $t^2 - 3t + d$: $A^2 = 3A - dI$, hence $A^3 = 3A^2 - dA$ and the same trace equation.) $\det A = 5$

P10. (JEE 1993, reconstructed — capstone.) Find all real λ for which

$$\lambda x + (\sin \alpha)y + (\cos \alpha)z = 0$$

$$x + (\cos \alpha)y + (\sin \alpha)z = 0$$

$$-x + (\sin \alpha)y - (\cos \alpha)z = 0$$

has a non-trivial solution; then find all α when $\lambda = 1$.

Solution. Expanding along row 1, the three minors are -1 , $-(\cos \alpha - \sin \alpha)$, and $\sin \alpha + \cos \alpha$:

$$\Delta = -\lambda - \sin \alpha(\sin \alpha - \cos \alpha) + \cos \alpha(\sin \alpha + \cos \alpha) = -\lambda + \cos 2\alpha + \sin 2\alpha$$

Non-trivial solutions exist iff

$$\lambda = \cos 2\alpha + \sin 2\alpha = \sqrt{2} \cos(2\alpha - \frac{\pi}{4})$$

For $\lambda = 1$: $\cos(2\alpha - \frac{\pi}{4}) = \frac{1}{\sqrt{2}}$, so $2\alpha - \frac{\pi}{4} = 2n\pi \pm \frac{\pi}{4}$, giving

$$\alpha = n\pi \quad \text{or} \quad \alpha = n\pi + \frac{\pi}{4}, \quad n \in \mathbb{Z}$$

P.5 What this chapter's own text under-serves

The chapter builds systems of equations around the CBSE five-mark matrix-method template, while JEE's actual diet is parameterised consistency triage — frequently with two parameters, trigonometric entries, or “count the solutions” framings — plus rank-level reasoning about *what the solution set looks like*, none of which gets drill here. Entire live genres are absent from the text: counting and extremal determinants over restricted entry sets (an Advanced staple from 2009 to 2020), trace- and polynomial-in-matrix puzzles solved via eigenvalues or the characteristic equation, and the elementwise-scaling shortcut behind the 2^{13} -style questions. Finally, the book never rehearses the multi-correct and numerical-answer formats that carry most Advanced marks and a growing share of Main, where a single missed branch ($\Delta = 0$ versus $\Delta_i \neq 0$) costs the whole item rather than one option.

Chapter 5: Continuity and Differentiability

Drona Ed Tech

Class 12 Mathematics, Complete Master Reference

Contents

1. Continuity & Differentiability — The Foundations	2
Section 1: Concept Intuition	2
Section 2: Key Formulas and Definitions	4
Section 3: CBSE Board Derivations and Key Procedures	4
Section 4: Worked Examples	6
Section 5: Practice Exercises	8
Section 6: Crack the MCQ	8
Section 7: Common Student Pitfalls and Pro-Tips	9
Section 8: Cheat Sheet Box	10
2. The Differentiation Toolkit — Standard Derivatives & Rules	11
Section 1: Concept Intuition	11
Section 2: Key Formulas and Definitions	11
Section 3: Key Derivations and Procedures	12
Section 4: Worked Examples	13
Section 5: Practice Exercises	14
Section 6: Crack the MCQ	15
Section 7: Common Student Pitfalls and Pro-Tips	15
Section 8: Cheat Sheet Box	16
3. The Chain Rule — Derivatives of Composite Functions	17
Section 1: Concept Intuition	17
Section 2: Key Formulas and Definitions	17
Section 3: Key Derivation and Procedure	18
Section 4: Worked Examples	18
Section 5: Practice Exercises	19
Section 6: Crack the MCQ	20
Section 7: Common Student Pitfalls and Pro-Tips	20
Section 8: Cheat Sheet Box	21
4. Implicit Differentiation	22
Section 1: Concept Intuition	22
Section 2: Key Formulas and Definitions	22
Section 3: Key Procedure and Derivation	23
Section 4: Worked Examples	23
Section 5: Practice Exercises	25
Section 6: Crack the MCQ	25
Section 7: Common Student Pitfalls and Pro-Tips	26
Section 8: Cheat Sheet Box	26

5. Derivatives of Inverse Trigonometric Functions	27
Section 1: Concept Intuition	27
Section 2: Key Formulas and Definitions	27
Section 3: Key Derivations and Procedure	28
Section 4: Worked Examples	29
Section 5: Practice Exercises	30
Section 6: Crack the MCQ	30
Section 7: Common Student Pitfalls and Pro-Tips	31
Section 8: Cheat Sheet Box	32
6. Logarithmic & Parametric Differentiation	33
Section 1: Concept Intuition	33
Section 2: Key Formulas and Definitions	34
Section 3: CBSE Board Derivations and Key Procedures	35
Section 4: Worked Examples	36
Section 5: Practice Exercises	38
Section 6: Crack the MCQ	38
Section 7: Common Student Pitfalls and Pro-Tips	39
Section 8: Cheat Sheet Box	39
7. Second-Order Derivatives & the Mean Value Theorems	41
Section 1: Concept Intuition	41
Section 2: Key Formulas and Definitions	42
Section 3: CBSE Board Derivations and Key Procedures	43
Section 4: Worked Examples	44
Section 5: Practice Exercises	45
Section 6: Crack the MCQ	46
Section 7: Common Student Pitfalls and Pro-Tips	47
Section 8: Cheat Sheet Box	47

1. Continuity & Differentiability — The Foundations

Exam Relevance: This is the bedrock of all of calculus, so it shows up everywhere. **CBSE Boards** almost always carry a 1–4 mark question — checking continuity, finding an unknown constant that makes a piecewise function continuous, or examining differentiability at a join. **JEE Main** reliably asks 1–2 questions on piecewise, modulus, or greatest-integer functions. **JEE Advanced** pushes into rigorous territory: functional equations, oddly-behaved functions, and the subtle gap between “differentiable” and “smoothly differentiable.” (NEET does not test mathematics, so this subtopic is JEE/Boards only.)

Section 1: Concept Intuition

Imagine you are riding a Mumbai local and the track is laid out in front of you. If the rails are joined end to end with no gaps, the train glides through — that is a **continuous** journey. But suppose somewhere along the line a single rail is missing: the train cannot cross that point without a jolt or a complete stop. That missing rail is exactly what a **discontinuity** is in a function — a place where the graph “breaks.”

So at the most basic level, **continuity means you can draw the graph through**

that point without lifting your pen off the paper. No jumps, no holes, no sudden teleportations.

But mathematicians can't sell "don't lift your pen" to an examiner, so we make it precise. For a function f to be continuous at a point $x = c$, three separate things must all be true and must all agree:

1. The function must actually **exist** at that point — $f(c)$ has a definite value.
2. The function must be **heading somewhere definite** as you approach c — the limit $\lim_{x \rightarrow c} f(x)$ exists. This itself needs the approach from the left and the approach from the right to agree.
3. Where it's heading and where it actually lands must be **the same place** — the limit equals $f(c)$.

Think about it this way... picture yourself walking along the graph toward the point $x = c$. You can walk in from the left side, and you can walk in from the right side. Continuity demands that *both* paths lead you to the *exact same dot*, and that the dot is actually painted there. If the left path and right path arrive at different heights (a "jump"), or if they arrive at the same height but the dot is painted somewhere else (a "removable hole"), the function snaps — it is discontinuous.

Now for the second, deeper idea: **differentiability**. Continuity only asks that the graph be *unbroken*. Differentiability asks for something stronger — that the graph be **smooth**, with no sharp corners.

Think of driving on the Mumbai-Pune Expressway. A long, gently banked curve lets you keep the steering wheel turning smoothly — at every instant there is one well-defined direction you are pointing. That is a **differentiable** path. Now think of a sharp hairpin in the Western Ghats where the road suddenly kinks: at the corner itself, there is no single direction — you are pointing one way the instant before and a different way the instant after. That corner is a point of **non-differentiability**, even though the road never broke.

The classic mathematical hairpin is $f(x) = |x|$ at $x = 0$. The graph is a perfect "V". It is unbroken (continuous), but at the tip the slope is -1 on the way in and $+1$ on the way out. There is no single tangent line, so the function is **continuous but not differentiable** there.

This gives us the single most important relationship in the whole subtopic:

Differentiability is stronger than continuity. Every differentiable function is continuous — but a continuous function need not be differentiable.

In other words: $smooth \Rightarrow unbroken$, but $unbroken \not\Rightarrow smooth$. Memorise the direction of that arrow; examiners test it relentlessly.

Limiting conditions & where smoothness fails. A function can be continuous yet fail differentiability for three reasons: (i) a **corner**, like $|x|$ at 0; (ii) a **cusp** or a **vertical tangent**, like $f(x) = x^{1/3}$ at 0, where the slope blows up to infinity; and (iii) anywhere the function is already **discontinuous** — a break automatically kills smoothness, since you can't have a tangent where there isn't even a connected graph. The greatest integer (step) function $f(x) = \lfloor x \rfloor$ illustrates the last case: it jumps at every integer, so it is neither continuous nor differentiable there.

Section 2: Key Formulas and Definitions

Continuity at a point $x = c$:

$$\lim_{x \rightarrow c} f(x) = f(c) \iff \underbrace{\lim_{x \rightarrow c^-} f(x)}_{\text{LHL}} = \underbrace{\lim_{x \rightarrow c^+} f(x)}_{\text{RHL}} = f(c)$$

where, writing $h \rightarrow 0^+$ (so h is a small positive number):

$$\text{LHL} = \lim_{h \rightarrow 0^+} f(c - h), \quad \text{RHL} = \lim_{h \rightarrow 0^+} f(c + h)$$

- $f(c)$ — the actual value of the function at the point.
- LHL / RHL — the left-hand and right-hand limits (the heights the two approaches aim for).

Continuity on an interval:

- f is continuous on an **open** interval (a, b) if it is continuous at every point inside it.
- f is continuous on a **closed** interval $[a, b]$ if it is continuous on (a, b) and additionally $\lim_{x \rightarrow a^+} f(x) = f(a)$ and $\lim_{x \rightarrow b^-} f(x) = f(b)$ (one-sided continuity at the endpoints, since you can only approach an endpoint from inside).

Differentiability at a point $x = c$:

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c + h) - f(c)}{h} \text{ exists (and is finite)}$$

This splits into one-sided derivatives, which must be **equal and finite**:

$$\text{LHD} = \lim_{h \rightarrow 0^+} \frac{f(c - h) - f(c)}{-h}, \quad \text{RHD} = \lim_{h \rightarrow 0^+} \frac{f(c + h) - f(c)}{h}$$

$$f \text{ is differentiable at } c \iff \text{LHD} = \text{RHD} = \text{a finite number}$$

The bridge theorem: If f is differentiable at c , then f is continuous at c . (The converse is false.)

Validity conditions (the math analogue of “where the formula applies”): these tests are *pointwise* — always check continuity *before* differentiability, because if continuity already fails at c , differentiability cannot hold and there is nothing further to compute. For a piecewise function, the only “doubtful” points are the **junctions** where the definition changes (and any point where a piece is individually undefined); everywhere else, standard functions (polynomials, sin, cos, e^x) are automatically continuous and differentiable.

Section 3: CBSE Board Derivations and Key Procedures

Derivation: Differentiability $\implies$ Continuity

This is the one proof from this subtopic that genuinely appears in Boards, and the reasoning is elegant.

To prove: If f is differentiable at $x = c$, then f is continuous at $x = c$.

Setup. We are *given* that $f'(c)$ exists, meaning $\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = f'(c)$ is a finite number. We want to *conclude* continuity, i.e. that $\lim_{x \rightarrow c} f(x) = f(c)$, which is the same as showing $\lim_{x \rightarrow c} [f(x) - f(c)] = 0$.

The key algebraic trick — multiply and divide by $(x - c)$. We are allowed to do this because for $x \neq c$ the factor $(x - c)$ is non-zero, so we have changed nothing:

$$f(x) - f(c) = \frac{f(x) - f(c)}{x - c} \cdot (x - c)$$

Why this step matters: the right-hand side is now a *product* of two pieces whose limits we already understand — the first piece is exactly the difference quotient (its limit is $f'(c)$), and the second piece $(x - c)$ obviously tends to 0.

Take the limit as $x \rightarrow c$, using the product rule for limits (valid because both limits exist and are finite):

$$\lim_{x \rightarrow c} [f(x) - f(c)] = \left(\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} \right) \cdot \left(\lim_{x \rightarrow c} (x - c) \right) = f'(c) \cdot 0 = 0$$

Conclude. Since $\lim_{x \rightarrow c} [f(x) - f(c)] = 0$, we have $\lim_{x \rightarrow c} f(x) = f(c)$, which is precisely the definition of continuity at c . ■

The crucial caveat (why the converse fails): notice we *needed* $f'(c)$ to exist for the first factor to have a finite limit. If it doesn't exist — as for $|x|$ at 0 — the argument collapses, and indeed $|x|$ is continuous there without being differentiable. The arrow only runs one way.

Procedure A: Checking continuity at a point c

1. **Compute** $f(c)$. If the function isn't even defined at c , it cannot be continuous there — stop.
2. **Compute the LHL:** substitute $x = c - h$ and let $h \rightarrow 0^+$.
3. **Compute the RHL:** substitute $x = c + h$ and let $h \rightarrow 0^+$.
4. **Compare all three.** If $\text{LHL} = \text{RHL} = f(c)$, the function is continuous at c . If any pair disagrees, it is discontinuous (a jump if $\text{LHL} \neq \text{RHL}$; a removable hole if $\text{LHL} = \text{RHL} \neq f(c)$).

Procedure B: Checking differentiability at a point c

1. **First confirm continuity at c** using Procedure A. If it fails, f is *not* differentiable — stop here and save yourself the algebra.
2. **Compute the LHD** $= \lim_{h \rightarrow 0^+} \frac{f(c - h) - f(c)}{-h}$ using the formula for the *left* piece.
3. **Compute the RHD** $= \lim_{h \rightarrow 0^+} \frac{f(c + h) - f(c)}{h}$ using the formula for the *right* piece.
4. **Compare.** If $\text{LHD} = \text{RHD}$ and both are finite, f is differentiable at c and $f'(c)$ equals that common value. If they differ, there is a corner; if either is infinite, there is a vertical tangent/cusp — in both cases, not differentiable.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Find the value of k for which the function is continuous at $x = 2$:

$$f(x) = \begin{cases} 2x + 3, & x \leq 2 \\ kx - 1, & x > 2 \end{cases}$$

Given: a piecewise function; continuity required at the junction $x = 2$.

Step 1 — value at the point. Using the first piece (which includes $x = 2$): $f(2) = 2(2) + 3 = 7$.

Step 2 — LHL. Approaching from below, we use the first piece: $\text{LHL} = \lim_{x \rightarrow 2^-} (2x + 3) = 7$.

Step 3 — RHL. Approaching from above, we use the second piece: $\text{RHL} = \lim_{x \rightarrow 2^+} (kx - 1) = 2k - 1$.

Step 4 — impose continuity. Continuity needs $\text{LHL} = \text{RHL} = f(2)$:

$$2k - 1 = 7 \implies 2k = 8 \implies \boxed{k = 4}$$

Conclusion: for $k = 4$ the function is continuous at $x = 2$.

Example 2 — [JEE Main Level]

Examine the continuity and differentiability of the following at $x = 2$:

$$f(x) = \begin{cases} x^2, & x \leq 2 \\ 3x - 2, & x > 2 \end{cases}$$

Continuity. $f(2) = 2^2 = 4$; $\text{LHL} = \lim_{x \rightarrow 2^-} x^2 = 4$; $\text{RHL} = \lim_{x \rightarrow 2^+} (3x - 2) = 4$. All three agree, so f is **continuous at** $x = 2$. ✓

Differentiability. Now compute the one-sided derivatives.

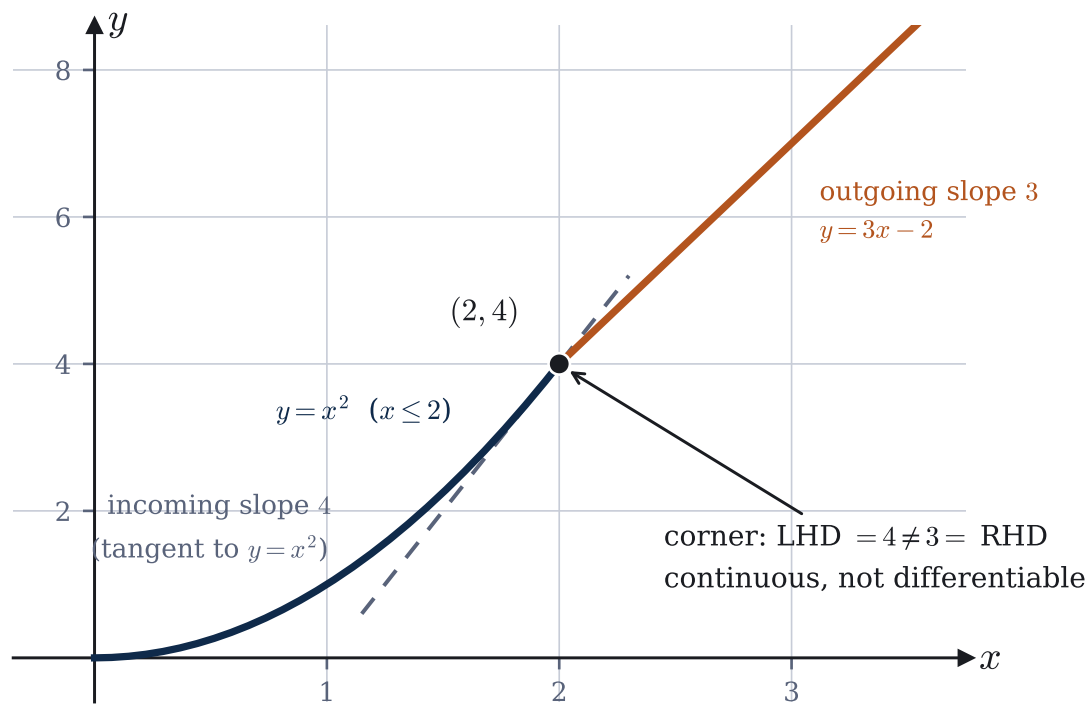
$$\text{LHD} = \lim_{h \rightarrow 0^+} \frac{f(2-h) - f(2)}{-h} = \lim_{h \rightarrow 0^+} \frac{(2-h)^2 - 4}{-h} = \lim_{h \rightarrow 0^+} \frac{-4h + h^2}{-h} = \lim_{h \rightarrow 0^+} (4-h) = 4$$

$$\text{RHD} = \lim_{h \rightarrow 0^+} \frac{f(2+h) - f(2)}{h} = \lim_{h \rightarrow 0^+} \frac{3(2+h) - 2 - 4}{h} = \lim_{h \rightarrow 0^+} \frac{3h}{h} = 3$$

Since $\text{LHD} = 4 \neq 3 = \text{RHD}$, the function is **not differentiable at** $x = 2$.

Takeaway: this is the bridge theorem in action — continuous but *not* differentiable. There is a corner at the join because the parabola arrives with slope 4 while the line leaves with slope 3.

Figure 5.1 Graph of $f(x)$: the parabola $y = x^2$ for $x \leq 2$ meets the line $y = 3x - 2$ for $x > 2$ at the point $(2, 4)$. The incoming slope is 4 and the outgoing slope is 3 - a corner: continuous at $x = 2$, but not differentiable. (see next page)



Same value from both sides, different slopes — the graph has a corner at $x = 2$.

Figure 5.1 Graph of $f(x)$: the parabola $y = x^2$ for $x \leq 2$ meets the line $y = 3x - 2$ for $x > 2$ at the point $(2, 4)$. The incoming slope is 4 and the outgoing slope is 3 - a corner: continuous at $x = 2$, but not differentiable.

Example 3 — [JEE Main — Speed Trap]

At how many points is $f(x) = |x^2 - 4| + |x - 1|$ non-differentiable?

The trap: students rush to differentiate, or they treat each modulus as contributing exactly one kink. Both go wrong.

The fast way — count the kinks. A term $|g(x)|$ creates a corner exactly where $g(x) = 0$ and changes sign (i.e. g crosses zero with non-zero slope). It is perfectly smooth everywhere else, including at a point where the *inside* function merely has its own vertex without changing sign.

- $|x^2 - 4|$: the inside is zero at $x = -2$ and $x = 2$, and it changes sign at both — so **two** corners, at $x = -2$ and $x = 2$.
- $|x - 1|$: one corner, at $x = 1$.

The classic blunder is to also count $x = 0$, the vertex of $x^2 - 4$. But near $x = 0$ the inside stays negative, so $|x^2 - 4| = 4 - x^2$ smoothly — **no corner there**.

Adding two functions can only smooth out a kink if both have one at the *same* place (here they don't, since $-2, 1, 2$ are distinct). So the corners survive.

Non-differentiable at 3 points: $x = -2, 1, 2$

Example 4 — [JEE Advanced Level]

Consider $f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$. Show that f is differentiable at $x = 0$, yet its derivative f' is not continuous at $x = 0$.

This is the gem of the subtopic: a function that is differentiable everywhere but whose derivative is not even continuous.

Step 1 — Continuity at 0 (squeeze theorem). Since $|\sin \frac{1}{x}| \leq 1$ for all $x \neq 0$,

$$0 \leq |x^2 \sin \frac{1}{x}| \leq x^2 \xrightarrow{x \rightarrow 0} 0 \implies \lim_{x \rightarrow 0} f(x) = 0 = f(0).$$

So f is continuous at 0.

Step 2 — Differentiability at 0 (from first principles).

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h^2 \sin \frac{1}{h}}{h} = \lim_{h \rightarrow 0} h \sin \frac{1}{h}.$$

Again by squeeze, $|h \sin \frac{1}{h}| \leq |h| \rightarrow 0$, so $f'(0) = 0$. The derivative at the origin **exists** — f is differentiable there.

Step 3 — The derivative away from 0. For $x \neq 0$, apply product and chain rules:

$$f'(x) = 2x \sin \frac{1}{x} + x^2 \cos \frac{1}{x} \cdot \left(-\frac{1}{x^2}\right) = 2x \sin \frac{1}{x} - \cos \frac{1}{x}.$$

Step 4 — Is f' continuous at 0? Examine $\lim_{x \rightarrow 0} f'(x)$. The first term $2x \sin \frac{1}{x} \rightarrow 0$ (squeeze). But the second term $\cos \frac{1}{x}$ **oscillates** between -1 and 1 infinitely often as $x \rightarrow 0$ and has no limit. Therefore $\lim_{x \rightarrow 0} f'(x)$ **does not exist**, so it certainly cannot equal $f'(0) = 0$.

Conclusion: f is differentiable at 0 with $f'(0) = 0$, yet f' is discontinuous at 0. Differentiability does *not* force the derivative to be continuous — a fact that quietly separates the students who memorise from those who understand. ■

Section 5: Practice Exercises

1. **[CBSE]** Find the value of a so that $f(x) = \begin{cases} ax + 5, & x \leq 3 \\ x^2 - 1, & x > 3 \end{cases}$ is continuous at $x = 3$.
2. **[CBSE]** Show that $f(x) = |x - 5|$ is continuous at $x = 5$ but not differentiable there.
3. **[JEE Main]** Examine the differentiability of $f(x) = \begin{cases} x^2 + 1, & x \leq 1 \\ 3 - x, & x > 1 \end{cases}$ at $x = 1$.
4. **[JEE Main]** Find the number of points in $(-2, 3)$ where $f(x) = \lfloor x \rfloor$ (greatest integer function) is discontinuous.
5. **[JEE Advanced]** Let $f(x) = \begin{cases} \frac{1 - \cos x}{x^2}, & x \neq 0 \\ k, & x = 0 \end{cases}$. Find k that makes f continuous at 0, and then determine whether f is differentiable there.

Answers: 1. $a=1$ 2. continuous (LHL = RHL = 0 = $f(5)$); LHD = $-1 \neq +1$ = RHD, so not differentiable 3. continuous at $x = 1$ (both pieces $\rightarrow 2$) but LHD = $2 \neq -1$ = RHD $\Rightarrow$ not differentiable 4. 4 points ($x = -1, 0, 1, 2$) 5. $k = \frac{1}{2}$; f is differentiable at 0 with $f'(0) = 0$.

Section 6: Crack the MCQ

Q1. The greatest integer function $f(x) = \lfloor x \rfloor$ is discontinuous at how many points in the interval $(0, 3)$? (a) 0 (b) 1 (c) 2 (d) 3

Answer: (c) 2. The step function jumps at every integer, and the integers strictly inside $(0, 3)$ are $x = 1$ and $x = 2$ — two points.

- **(a)** picked by students who think a function defined “everywhere” must be continuous; the step function is defined everywhere yet jumps.
- **(b)** comes from spotting only one integer (often forgetting $x = 2$).
- **(d)** the trap of including an endpoint — $x = 3$ is *not* inside the open interval $(0, 3)$, so it doesn’t count.

Q2. Which of the following functions is continuous everywhere but differentiable everywhere *except at exactly one point*? (a) $f(x) = x^2$ (b) $f(x) = |x|$ (c) $f(x) = \lfloor x \rfloor$ (d) $f(x) = \frac{1}{x}$

Answer: (b) $|x|$. It is continuous on all of $\mathbb{R}$ but has a single corner at $x = 0$, the only non-differentiable point.

- **(a)** a polynomial — smooth and differentiable *everywhere*, so it fails the “except one point” requirement.
- **(c)** the greatest integer function is not even continuous (it jumps at integers),

so it fails the very first condition.

- **(d)** is not defined at $x = 0$, so it isn't continuous everywhere to begin with.

Q3. If a function f is differentiable at $x = c$, which statement is *necessarily* true? (a) f is continuous at c (b) f' is continuous at c (c) f has a maximum at c (d) $f(c) = 0$

Answer: (a). Differentiability guarantees continuity — this is the bridge theorem.

- **(b)** is the deep trap: differentiability does *not* force the derivative to be continuous (recall $x^2 \sin \frac{1}{x}$). Students who half-remember the theorem over-extend it.
- **(c)** a derivative existing says nothing about extrema; a max would additionally require $f'(c) = 0$ plus a sign test.
- **(d)** confuses the value of the function with the value of its derivative; differentiability places no constraint on $f(c)$ itself.

Q4. The function $f(x) = \begin{cases} ax + 1, & x \leq 1 \\ x^2 + b, & x > 1 \end{cases}$ is differentiable at $x = 1$. Then $a + b$ equals: (a) 2 (b) 3 (c) 4 (d) 5

Answer: (c) 4. Continuity at $x = 1$: $a(1) + 1 = (1)^2 + b \Rightarrow a + 1 = 1 + b \Rightarrow a = b$. Differentiability at $x = 1$: LHD = a (slope of $ax + 1$), RHD = $2x|_{x=1} = 2$. Setting them equal gives $a = 2$, hence $b = 2$, so $a + b = 4$.

- **(a)** results from solving differentiability ($a = 2$) but forgetting that continuity is also required, then mishandling b .
- **(b)** and **(d)** come from arithmetic slips or from imposing only one of the two conditions.

Section 7: Common Student Pitfalls and Pro-Tips

- **Pitfall 1 (the reversed arrow).** Believing “continuous $\Rightarrow$ differentiable.” It is the *other* way around. $|x|$ at 0 is the eternal counterexample — keep it in your pocket.
- **Pitfall 2 (skipping the continuity check).** Jumping straight to LHD/RHD without first confirming continuity. If the function is already discontinuous at c , it cannot be differentiable, and computing derivatives there is wasted effort (and often gives a misleading finite number).
- **Pitfall 3 (miscounting modulus corners).** Treating $|g(x)|$ as one corner regardless of how many times g crosses zero, or counting the *vertex* of the inside function (where it doesn't change sign) as a corner. Count *sign-changes* of the inside, not zeros of its derivative.
- **Pitfall 4 (endpoint over-counting).** In an open interval like $(0, 3)$, the endpoints are excluded — don't count discontinuities or kinks sitting exactly on 0 or 3.

Pro-Tip: For piecewise functions, you only ever need to test the **junction points** (and any point where a piece is undefined) — never the interiors, where polynomials, $\sin$, $\cos$, and e^x are automatically smooth. This collapses most “examine continuity/differentiability” problems to checking just one or two points. And always run the checks in order: **continuity first, differentiability second** — the first failing instantly settles the second.

Section 8: Cheat Sheet Box

CONTINUITY & DIFFERENTIABILITY — CORE RECALL

- **Continuous at c :** $\text{LHL} = \text{RHL} = f(c)$ (all three exist *and* are equal).
- **Differentiable at c :** $\text{LHD} = \text{RHD} = \text{a finite number}$, where $\text{LHD} = \lim_{h \rightarrow 0^+} \frac{f(c-h)-f(c)}{-h}$, $\text{RHD} = \lim_{h \rightarrow 0^+} \frac{f(c+h)-f(c)}{h}$.
- **The Bridge:** differentiable $\Rightarrow$ continuous; **continuous $\nRightarrow$ differentiable.** (One-way arrow.)
- **Smoothness fails at:** corners ($|x|$), cusps/vertical tangents ($x^{1/3}$ at 0), and every discontinuity.
- **Strategy:** test only junction points; check continuity *before* differentiability.
- **Mnemonic — “DiCo, not CoDi”:** **D**ifferentiable $\Rightarrow$ **C**ontinuous, but **C**ontinuous does *not* $\Rightarrow$ **D**ifferentiable.

2. The Differentiation Toolkit — Standard Derivatives & Rules

Exam Relevance: This is the computational backbone of the whole chapter — every later technique (chain rule, implicit, log, parametric) assumes total fluency here. **CBSE Boards** test it directly (differentiate a product/quotient for 2-3 marks) and indirectly everywhere. **JEE Main/Advanced** assume instant recall of every standard derivative and combine the rules under time pressure. (NEET does not test mathematics.)

Section 1: Concept Intuition

Learning to differentiate is a lot like learning to cook. You need two things: a **spice rack** of standard ingredients you know cold, and a few **combining techniques** to put them together. The spice rack is the list of *standard derivatives* — the derivative of $\sin x$, of e^x , of x^n , and so on. The combining techniques are the *rules of differentiation* — how to handle a sum, a product, or a quotient. Master both and you can differentiate essentially any function the chapter throws at you.

The standard derivatives are simply facts to memorise (we derive a couple below so you trust them). The rules are where intuition matters.

Think about it this way... the most seductive mistake in all of calculus is to guess that the derivative of a product is the product of the derivatives: $(uv)' = u'v'$. It is **wrong**. Picture a rectangle with width u and height v , so its area is uv . Now nudge both sides up a little. The area grows by a thin strip along the top (height grows: that's $u v'$) *plus* a thin strip along the side (width grows: that's $u'v$). The new area gains *both* strips:

$$(uv)' = u'v + uv'.$$

That picture is the product rule, and it's why the naive guess fails — it forgets one of the strips.

There is also a quieter idea hiding here: the **algebra of continuous functions**. If f and g are both continuous at a point, then so are $f + g$, $f - g$, fg , and f/g (as long as the denominator isn't zero), and so is the composite $f(g(x))$. This is the reason that polynomials, $\sin x$, $\cos x$, e^x , and all their sensible combinations are continuous wherever they're defined — and it's precisely *why*, back in the foundations, we only ever needed to check continuity at the suspicious *junction* points of piecewise functions, never in the smooth interiors.

Limiting conditions & assumptions. The quotient rule requires the denominator to be non-zero at the point. Every standard derivative below holds only on the domain where the function is differentiable (e.g. $\frac{d}{dx} \ln x = \frac{1}{x}$ needs $x > 0$; $\frac{d}{dx} \sec^{-1} x$ needs $|x| > 1$). And the rules assume the pieces being combined are themselves differentiable there.

Section 2: Key Formulas and Definitions

Standard derivatives (each valid on its natural domain):

$$\frac{d}{dx}(x^n) = nx^{n-1}, \quad \frac{d}{dx}(k) = 0 \text{ (} k \text{ constant)}$$

$$\begin{aligned}\frac{d}{dx} \sin x &= \cos x, & \frac{d}{dx} \cos x &= -\sin x, & \frac{d}{dx} \tan x &= \sec^2 x \\ \frac{d}{dx} \cot x &= -\csc^2 x, & \frac{d}{dx} \sec x &= \sec x \tan x, & \frac{d}{dx} \csc x &= -\csc x \cot x \\ \frac{d}{dx} e^x &= e^x, & \frac{d}{dx} a^x &= a^x \ln a, & \frac{d}{dx} \ln x &= \frac{1}{x}, & \frac{d}{dx} \log_a x &= \frac{1}{x \ln a} \\ \frac{d}{dx} \sin^{-1} x &= \frac{1}{\sqrt{1-x^2}}, & \frac{d}{dx} \cos^{-1} x &= \frac{-1}{\sqrt{1-x^2}}, & \frac{d}{dx} \tan^{-1} x &= \frac{1}{1+x^2} \\ \frac{d}{dx} \cot^{-1} x &= \frac{-1}{1+x^2}, & \frac{d}{dx} \sec^{-1} x &= \frac{1}{|x|\sqrt{x^2-1}}, & \frac{d}{dx} \csc^{-1} x &= \frac{-1}{|x|\sqrt{x^2-1}}\end{aligned}$$

Rules of differentiation (with $u = u(x)$, $v = v(x)$):

Sum/Difference: $(u \pm v)' = u' \pm v'$ Constant multiple: $(ku)' = ku'$

Product: $(uv)' = u'v + uv'$ Quotient: $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2} \quad (v \neq 0)$

Algebra of continuous functions: if f, g are continuous at c , then $f \pm g$, kf , fg are continuous at c , and $\frac{f}{g}$ is continuous at c provided $g(c) \neq 0$; the composite $f \circ g$ is continuous at c if g is continuous at c and f is continuous at $g(c)$.

Validity: quotient/quotient-rule need $v \neq 0$; each standard derivative is restricted to its domain (noted above).

Section 3: Key Derivations and Procedures

Derivation 1: $\frac{d}{dx} \sin x = \cos x$ **from first principles**

Using $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ with $f(x) = \sin x$:

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h}.$$

Apply the identity $\sin C - \sin D = 2 \cos \frac{C+D}{2} \sin \frac{C-D}{2}$:

$$= \lim_{h \rightarrow 0} \frac{2 \cos\left(x + \frac{h}{2}\right) \sin \frac{h}{2}}{h} = \lim_{h \rightarrow 0} \cos\left(x + \frac{h}{2}\right) \cdot \frac{\sin \frac{h}{2}}{\frac{h}{2}}.$$

Why this split helps: the second factor is the famous limit $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$, and the first factor tends to $\cos x$. Hence $f'(x) = \cos x$. ■

Derivation 2: The product rule from first principles

For $y = u(x)v(x)$, the **add-and-subtract trick** is the key:

$$y' = \lim_{h \rightarrow 0} \frac{u(x+h)v(x+h) - u(x)v(x)}{h}.$$

Subtract and add $u(x+h)v(x)$ in the numerator (it changes nothing but lets us factor):

$$\begin{aligned} &= \lim_{h \rightarrow 0} \frac{u(x+h)[v(x+h) - v(x)] + v(x)[u(x+h) - u(x)]}{h} \\ &= \lim_{h \rightarrow 0} u(x+h) \cdot \frac{v(x+h) - v(x)}{h} + v(x) \cdot \frac{u(x+h) - u(x)}{h} = uv' + vu'. \blacksquare \end{aligned}$$

The two strips of the rectangle, made rigorous.

Procedure: choosing the right rule

1. **Identify the structure** — is the whole thing a sum, a product, or a quotient?
 2. **Differentiate the outermost structure first** with the matching rule, then differentiate the pieces.
 3. For products of three or more factors, either extend the product rule term by term, or take logs (logarithmic differentiation — a later subtopic).
 4. Simplify; a tidy intermediate form prevents algebra slips.
-

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Differentiate $y = (x^2 + 1)(x^3 + 3)$.

Using the **product rule** with $u = x^2 + 1$, $v = x^3 + 3$:

$$y' = u'v + uv' = (2x)(x^3 + 3) + (x^2 + 1)(3x^2) = 2x^4 + 6x + 3x^4 + 3x^2 = \boxed{5x^4 + 3x^2 + 6x}.$$

Example 2 — [JEE Main Level]

Differentiate $y = \frac{\sin x}{1 + \cos x}$.

Using the **quotient rule** with $u = \sin x$, $v = 1 + \cos x$:

$$y' = \frac{\cos x (1 + \cos x) - \sin x (-\sin x)}{(1 + \cos x)^2} = \frac{\cos x + \cos^2 x + \sin^2 x}{(1 + \cos x)^2} = \frac{\cos x + 1}{(1 + \cos x)^2} = \boxed{\frac{1}{1 + \cos x}}.$$

The Pythagorean identity collapses the numerator beautifully — always look for that simplification.

Example 3 — [JEE Main — Trap]

A student differentiates $y = x^2 \sin x$ **and writes** $y' = 2x \cos x$. **Why is this wrong, and what is the correct answer?**

The trap. The student applied the false rule $(uv)' = u'v'$ — multiplying the two derivatives ($2x$ and $\cos x$). This is the single most common differentiation error.

The fix. A product needs the **product rule** — *both* strips:

$$y' = (x^2)' \sin x + x^2(\sin x)' = 2x \sin x + x^2 \cos x = \boxed{x(2 \sin x + x \cos x)}.$$

Speed check: whenever you see two x -dependent factors multiplied, the answer must have *two* terms, not one. A single-term answer to a product is an instant red flag.

Example 4 — [JEE Advanced Level]

Find $\frac{d}{dx} \frac{1}{\sqrt{x}}$ **from first principles**, and confirm it matches the power rule.

$$f'(x) = \lim_{h \rightarrow 0} \frac{\frac{1}{\sqrt{x+h}} - \frac{1}{\sqrt{x}}}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{x} - \sqrt{x+h}}{h \sqrt{x} \sqrt{x+h}}.$$

Rationalise the numerator by multiplying top and bottom by $(\sqrt{x} + \sqrt{x+h})$:

$$= \lim_{h \rightarrow 0} \frac{x - (x+h)}{h \sqrt{x} \sqrt{x+h} (\sqrt{x} + \sqrt{x+h})} = \lim_{h \rightarrow 0} \frac{-h}{h \sqrt{x} \sqrt{x+h} (\sqrt{x} + \sqrt{x+h})}.$$

Cancel h and let $h \rightarrow 0$:

$$= \frac{-1}{\sqrt{x} \cdot \sqrt{x} \cdot 2\sqrt{x}} = \frac{-1}{2x^{3/2}} = -\frac{1}{2}x^{-3/2}.$$

This is exactly the power rule applied to $x^{-1/2}$: $(x^{-1/2})' = -\frac{1}{2}x^{-3/2}$. ■

Section 5: Practice Exercises

1. **[CBSE]** Differentiate $y = (2x^2 - 3)(x + 1)$.
2. **[CBSE]** Differentiate $y = \frac{x}{x^2 + 1}$.
3. **[JEE Main]** Differentiate $y = x^2 e^x \ln x$ (a triple product).
4. **[JEE Main]** Differentiate $y = \frac{1 + \tan x}{1 - \tan x}$.
5. **[JEE Advanced]** Find $\frac{d}{dx} \cos x$ from first principles.

Answers: 1. $6x^2 + 4x - 3$

2. $\frac{1 - x^2}{(x^2 + 1)^2}$

3. $x e^x (2 \ln x + x \ln x + 1) = x e^x [(x + 2) \ln x + 1]$

4. $\frac{2 \sec^2 x}{(1 - \tan x)^2}$

5. $-\sin x$ (use $\cos C - \cos D = -2 \sin \frac{C+D}{2} \sin \frac{C-D}{2}$ and $\frac{\sin \theta}{\theta} \rightarrow 1$).

Section 6: Crack the MCQ

Q1. $\frac{d}{dx}(x^2 \cos x) =$ (a) $2x(-\sin x)$ (b) $2x \cos x - x^2 \sin x$ (c) $-2x \sin x$ (d) $2x \cos x + x^2 \sin x$

Answer: (b). Product rule: $2x \cos x + x^2(-\sin x)$.

- **(a)/(c)** multiply the derivatives — the $(uv)' = u'v'$ trap.
- **(d)** uses $+x^2 \sin x$, forgetting that $\frac{d}{dx} \cos x = -\sin x$ (a sign slip).

Q2. $\frac{d}{dx} \tan x$ equals: (a) $\sec x \tan x$ (b) $\sec^2 x$ (c) $-\csc^2 x$ (d) $\cot x$

Answer: (b). A standard derivative (and quotient rule on $\frac{\sin}{\cos}$ confirms it).

- **(a)** is the derivative of $\sec x$. **(c)** is the derivative of $\cot x$. **(d)** is unrelated. Mixing the trig-derivative table is the classic error.

Q3. Which statement about continuity is **false**? (a) Sum of two continuous functions is continuous. (b) Product of two continuous functions is continuous. (c) Quotient of two continuous functions is *always* continuous. (d) Composite of continuous functions is continuous.

Answer: (c). The quotient is continuous only where the **denominator is non-zero**; e.g. $\frac{1}{x}$ fails at $x = 0$. The “always” makes it false.

- **(a), (b), (d)** are all true statements from the algebra of continuous functions.

Q4. $\frac{d}{dx} \left(\frac{e^x}{x} \right) =$ (a) $\frac{e^x}{x^2}$ (b) $\frac{e^x(x-1)}{x^2}$ (c) $\frac{e^x(1-x)}{x^2}$ (d) e^x

Answer: (b). Quotient rule: $\frac{e^x \cdot x - e^x \cdot 1}{x^2} = \frac{e^x(x-1)}{x^2}$.

- **(c)** flips the numerator order: $uv' - u'v$ instead of $u'v - uv'$ — the most common quotient-rule sign error.
- **(a)** forgets the $-uv'$ term; **(d)** ignores the quotient entirely.

Section 7: Common Student Pitfalls and Pro-Tips

- **Pitfall 1 — the $(uv)' = u'v'$ trap.** A product’s derivative has *two* terms. A one-term answer to a product is wrong by inspection.
- **Pitfall 2 — quotient-rule order.** It is $\frac{u'v - uv'}{v^2}$, *not* $\frac{uv' - u'v}{v^2}$. Memorise “**low-d-high minus high-d-low, over low-squared.**”
- **Pitfall 3 — trig-derivative sign/table mix-ups.** $\cos x \rightarrow -\sin x$ (negative!); $\cot, \csc, \cos^{-1}, \cot^{-1}, \csc^{-1}$ all carry a minus sign. Group the “negative six” mentally.
- **Pitfall 4 — dropping domain restrictions.** $\frac{d}{dx} \sec^{-1} x$ carries $|x|$; $\frac{d}{dx} \ln x$ needs $x > 0$.

Pro-Tip: Before differentiating a quotient, check whether algebra simplifies it first (e.g. $\frac{\sin x}{1+\cos x}$, or cancelling common factors). Simplifying *before* differentiating often turns a messy quotient rule into a one-line answer.

Section 8: Cheat Sheet Box

DIFFERENTIATION TOOLKIT — CORE RECALL

- **Power:** $(x^n)' = nx^{n-1}$. **Trig:** $\sin \rightarrow \cos$, $\cos \rightarrow -\sin$, $\tan \rightarrow \sec^2$.
- **Exp/Log:** $(e^x)' = e^x$, $(a^x)' = a^x \ln a$, $(\ln x)' = \frac{1}{x}$.
- **Product:** $(uv)' = u'v + uv'$ (two terms!). **Quotient:** $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$.
- **The “negative six”:** $\cos$, $\cot$, $\csc$, $\cos^{-1}$, $\cot^{-1}$, $\csc^{-1}$ all differentiate with a minus sign.
- **Algebra of continuity:** sums, products, quotients (denom $\neq 0$), and composites of continuous functions stay continuous — hence only piecewise *junctions* need checking.
- **Mnemonic (quotient):** “low d-high minus high d-low, all over low-squared.”

3. The Chain Rule — Derivatives of Composite Functions

Exam Relevance: Arguably the most-used rule in the entire chapter — implicit, inverse-trig, logarithmic and parametric differentiation all lean on it. **CBSE Boards** test composite functions directly (2–4 marks). **JEE Main/Advanced** nest several layers and combine the chain rule with product/quotient rules under time pressure. (NEET does not test mathematics.)

Section 1: Concept Intuition

A **composite function** is a function inside another function — a function *of* a function. For example $y = \sin(x^2)$ is “square it, then take the sine.” Here you can’t just reach for a single standard derivative, because two operations are stacked.

The **chain rule** tells you how stacked operations combine, and the cleanest picture is **meshing gears**. Imagine three shafts linked by gears: turning x drives an intermediate shaft u , which in turn drives the output y . If the u -shaft spins 3 times for each turn of x , and the y -shaft spins 5 times for each turn of u , then y spins $5 \times 3 = 15$ times per turn of x . **Rates multiply along the chain:**

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

Think about it this way... to differentiate a composite, peel it like an onion from the outside in. Differentiate the **outer** function while keeping the inner part untouched, then **multiply** by the derivative of the inner part. For $y = \sin(x^2)$: the outer is $\sin(\cdot)$, whose derivative is $\cos(\cdot)$, evaluated at the inner x^2 — that’s $\cos(x^2)$; then multiply by the inner derivative $\frac{d}{dx}(x^2) = 2x$. So $y' = \cos(x^2) \cdot 2x = 2x \cos(x^2)$.

That final “multiply by the derivative of the inside” is the entire trick — and forgetting it is the number-one chain-rule error. Writing $\frac{d}{dx} \sin(x^2) = \cos(x^2)$ (and stopping) is wrong; you owe the extra factor $2x$.

For deeper nesting, the chain simply gets longer. For $y = f(g(h(x)))$, you peel three layers:

$$\frac{dy}{dx} = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x).$$

Limiting conditions & assumptions. The chain rule requires the inner function g to be differentiable at x **and** the outer function f to be differentiable at $g(x)$. If either link is non-differentiable (a corner, a point outside the domain), the rule does not apply there — e.g. $\sqrt{u}$ is not differentiable at $u = 0$, so $y = \sqrt{g(x)}$ needs care wherever $g(x) = 0$.

Section 2: Key Formulas and Definitions

Chain rule (two links): if $y = f(u)$ and $u = g(x)$, both differentiable, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = f'(g(x)) \cdot g'(x).$$

Extended chain (three or more links): for $y = f(g(h(x)))$,

$$\frac{dy}{dx} = f'(g(h(x))) \cdot g'(h(x)) \cdot h'(x).$$

Common composite patterns (each is “outer derivative $\times$ inner derivative”):

$$\frac{d}{dx}[g(x)]^n = n[g(x)]^{n-1}g'(x), \quad \frac{d}{dx}\sin g(x) = \cos g(x) \cdot g'(x)$$

$$\frac{d}{dx}e^{g(x)} = e^{g(x)}g'(x), \quad \frac{d}{dx}\ln g(x) = \frac{g'(x)}{g(x)}.$$

- $u = g(x)$ — the inner (“inside”) function; $\frac{du}{dx}$ — its derivative (the factor students forget).

Validity: g differentiable at x , and f differentiable at $u = g(x)$.

Section 3: Key Derivation and Procedure

Why the chain rule works (intuitive derivation)

Let $u = g(x)$, and suppose g is differentiable at x (so a small change Δx produces a small change Δu), and f is differentiable at u . The change in y relative to x can be written, for $\Delta u \neq 0$, by inserting Δu top and bottom:

$$\frac{\Delta y}{\Delta x} = \frac{\Delta y}{\Delta u} \cdot \frac{\Delta u}{\Delta x}.$$

Why this is legitimate: we are multiplying by $\frac{\Delta u}{\Delta u} = 1$, just regrouped. As $\Delta x \rightarrow 0$, differentiability of g forces $\Delta u \rightarrow 0$ too, so $\frac{\Delta y}{\Delta u} \rightarrow f'(u)$ and $\frac{\Delta u}{\Delta x} \rightarrow g'(x)$. Taking the limit:

$$\frac{dy}{dx} = f'(u) \cdot g'(x) = f'(g(x))g'(x). \blacksquare$$

(A fully rigorous proof handles the $\Delta u = 0$ case separately, but the rate-multiplying idea is exactly this.)

Procedure: differentiating a composite

1. **Identify the layers** — name the outer function and the inner function(s). Ask “what is done last?” — that’s the outermost layer.
 2. **Differentiate the outer**, leaving the inner *unchanged* inside it.
 3. **Multiply** by the derivative of the inner (and keep going for deeper nesting).
 4. **Simplify.** A neat check: every layer you peel must contribute exactly one factor.
-

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Differentiate $y = (3x^2 + 5)^4$.

Outer: $(\cdot)^4$; inner: $3x^2 + 5$.

$$y' = 4(3x^2 + 5)^3 \cdot \frac{d}{dx}(3x^2 + 5) = 4(3x^2 + 5)^3 \cdot 6x = \boxed{24x(3x^2 + 5)^3}.$$

Example 2 — [JEE Main Level]

Differentiate $y = \sin(\cos(x^2))$.

Three layers — peel outside-in: outer $\sin(\cdot)$, middle $\cos(\cdot)$, inner x^2 .

$$y' = \cos(\cos(x^2)) \cdot (-\sin(x^2)) \cdot 2x = \boxed{-2x \sin(x^2) \cos(\cos(x^2))}.$$

Each layer contributed exactly one factor — that's your completeness check.

Example 3 — [JEE Main — Trap]

A student differentiates $y = e^{3x}$ **and writes** $y' = e^{3x}$. **What's missing?**

The trap. The student treated e^{3x} like the plain e^x (whose derivative is itself) and **forgot the inner derivative**. The inside is $3x$, whose derivative is 3.

The fix.

$$y' = e^{3x} \cdot \frac{d}{dx}(3x) = 3e^{3x}.$$

This “forgot to multiply by the inside” error is the most common chain-rule mistake. **Speed habit:** after writing the outer derivative, immediately ask “*times the derivative of the inside?*” — and don't move on until you've written that factor.

Example 4 — [JEE Advanced Level]

Differentiate $y = x^2\sqrt{1+x^2}$ **and simplify.**

This combines the **product rule** with the **chain rule**. Let $u = x^2$ and $v = \sqrt{1+x^2} = (1+x^2)^{1/2}$.

$$v' = \frac{1}{2}(1+x^2)^{-1/2} \cdot 2x = \frac{x}{\sqrt{1+x^2}} \quad (\text{chain rule on } v).$$

Now the product rule:

$$y' = u'v + uv' = 2x\sqrt{1+x^2} + x^2 \cdot \frac{x}{\sqrt{1+x^2}} = 2x\sqrt{1+x^2} + \frac{x^3}{\sqrt{1+x^2}}.$$

Put over a common denominator $\sqrt{1+x^2}$:

$$y' = \frac{2x(1+x^2) + x^3}{\sqrt{1+x^2}} = \boxed{\frac{2x + 3x^3}{\sqrt{1+x^2}} = \frac{x(2+3x^2)}{\sqrt{1+x^2}}}.$$

Section 5: Practice Exercises

1. **[CBSE]** Differentiate $y = \cos(5x - 2)$.
2. **[CBSE]** Differentiate $y = (x^3 + 1)^7$.
3. **[JEE Main]** Differentiate $y = \ln(\sin x)$.
4. **[JEE Main]** Differentiate $y = e^{\sqrt{x}}$.
5. **[JEE Advanced]** Differentiate $y = \tan^3(2x)$ (i.e. $[\tan(2x)]^3$).

- Answers:** 1. $-5 \sin(5x - 2)$
 2. $21x^2(x^3 + 1)^6$
 3. $\frac{\cot x}{e^{\sqrt{x}}}$
 4. $\frac{1}{2\sqrt{x}}$
 5. $6 \tan^2(2x) \sec^2(2x)$
-

Section 6: Crack the MCQ

Q1. $\frac{d}{dx} \sin(2x) =$ (a) $\cos(2x)$ (b) $2 \cos(2x)$ (c) $2 \sin(2x)$ (d) $\cos x$

Answer: (b). Outer derivative $\cos(2x)$ times inner derivative 2.

- **(a)** forgets the inner factor 2 — the signature chain-rule slip.
- **(c)** differentiates the wrong outer ($\sin \rightarrow \sin$). **(d)** ignores the inner entirely.

Q2. If $y = (1 + x^2)^{10}$, then y' at $x = 1$ is: (a) $10(2)^9$ (b) $20(2)^9$ (c) $(2)^{10}$ (d) $10(2)^{10}$

Answer: (b). $y' = 10(1 + x^2)^9 \cdot 2x = 20x(1 + x^2)^9$; at $x = 1$: $20 \cdot 2^9$.

- **(a)** drops the inner factor 2x. **(c)/(d)** mishandle the power-rule exponent.

Q3. $\frac{d}{dx} e^{x^2} =$ (a) e^{x^2} (b) $2x e^{x^2}$ (c) $x^2 e^{x^2-1}$ (d) $2x e^{2x}$

Answer: (b). $e^{x^2} \cdot \frac{d}{dx}(x^2) = 2x e^{x^2}$.

- **(a)** forgets the inner derivative. **(c)** wrongly applies the power rule to an exponential. **(d)** mangles the exponent.

Q4. $\frac{d}{dx} \ln(\cos x) =$ (a) $\frac{1}{\cos x}$ (b) $-\tan x$ (c) $\tan x$ (d) $-\frac{1}{\cos x}$

Answer: (b). $\frac{1}{\cos x} \cdot (-\sin x) = -\tan x$.

- **(a)** forgets the inner derivative $-\sin x$. **(c)** drops the minus sign from $\frac{d}{dx} \cos x$. **(d)** keeps the inner reciprocal but no $\sin x$.
-

Section 7: Common Student Pitfalls and Pro-Tips

- **Pitfall 1 — forgetting the inner derivative.** Writing $\frac{d}{dx} \sin(2x) = \cos(2x)$ without the $\times 2$. *Every peeled layer owes a factor.*
- **Pitfall 2 — differentiating the inside too early.** Keep the inner function *intact* inside the outer derivative; only its *separate* derivative becomes a multiplying factor.
- **Pitfall 3 — losing a layer in deep nesting.** For triple composites, you need exactly three factors; count them.
- **Pitfall 4 — mixing chain with product/quotient.** When a composite sits inside a product, apply the product rule first, using the chain rule to differentiate the composite factor.

Pro-Tip: Say the recipe out loud — “*derivative of the outside, times derivative of the inside.*” For each layer you strip, immediately append its factor before doing anything else; this mechanical rhythm eliminates the dropped-inner-factor error almost entirely.

Section 8: Cheat Sheet Box

CHAIN RULE — CORE RECALL

- **Two links:** $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$ — rates multiply along the chain.
- **Recipe:** derivative of the *outer* (inner left intact) $\times$ derivative of the *inner*.
- **Patterns:** $[g]^n \rightarrow n[g]^{n-1}g'$; $\sin g \rightarrow \cos g \cdot g'$; $e^g \rightarrow e^g g'$; $\ln g \rightarrow \frac{g'}{g}$.
- **Deep nesting:** one factor per layer — count your factors as a completeness check.
- **Biggest trap:** never forget the $\times g'(x)$ (the derivative of the inside).
- **Mnemonic:** “*Outside-in, and don’t forget the inside.*”

4. Implicit Differentiation

Exam Relevance: A guaranteed scorer. **CBSE Boards** regularly ask for $\frac{dy}{dx}$ of an implicit relation like $x^2 + y^2 = a^2$ or $\sin(xy) = x$ (2-4 marks). **JEE Main/Advanced** push into implicit *second* derivatives and curve-tangent problems. The whole technique is really just the chain rule worn as a disguise. (NEET does not test mathematics.)

Section 1: Concept Intuition

So far, y has always been served up neatly as $y = (\text{some expression in } x)$ — what we call an **explicit** function. But many curves refuse to cooperate. Take the circle $x^2 + y^2 = 25$. Here y is tangled up with x in one equation, not isolated on its own. We *could* try to solve for $y = \pm\sqrt{25 - x^2}$, but that's clumsy and splits the curve into two pieces. There's a far slicker way: **implicit differentiation**.

The core idea is a small but powerful mental shift: even though y isn't written as a formula in x , it *is* secretly a function of x along the curve. So we treat y as a hidden function of x and differentiate the entire equation, both sides, with respect to x .

Think about it this way... whenever you differentiate a term containing y , remember that y is itself a function of x — so the **chain rule** kicks in and tacks on a $\frac{dy}{dx}$. For instance, $\frac{d}{dx}(y^2) = 2y \cdot \frac{dy}{dx}$, exactly as $\frac{d}{dx}(\text{box})^2 = 2(\text{box}) \cdot (\text{box})'$ where the box is y . A pure- x term like x^2 differentiates normally to $2x$; a y -term always drags along its $\frac{dy}{dx}$. Once you've differentiated everything, you simply collect all the $\frac{dy}{dx}$ terms on one side and solve for it algebraically.

For the circle $x^2 + y^2 = 25$:

$$2x + 2y \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{x}{y}.$$

Notice the answer involves *both* x and y — that's normal and fine for implicit curves; it tells you the slope at any point (x, y) on the circle.

Limiting conditions & assumptions. Implicit differentiation assumes that near the point of interest the relation genuinely *defines* y as a differentiable function of x (true wherever the curve is smooth and not vertical). Where the tangent is vertical — e.g. the points $(\pm 5, 0)$ on the circle, where $y = 0$ makes $\frac{dy}{dx} = -\frac{x}{y}$ blow up — $\frac{dy}{dx}$ does not exist, and that's the formula honestly reporting a vertical tangent.

Section 2: Key Formulas and Definitions

The governing principle: differentiate both sides w.r.t. x , applying the chain rule to every y -term:

$$\frac{d}{dx}[y^n] = ny^{n-1} \frac{dy}{dx}, \quad \frac{d}{dx}[f(y)] = f'(y) \frac{dy}{dx}.$$

Mixed xy -terms need the product rule too (since both factors depend on x):

$$\frac{d}{dx}(xy) = 1 \cdot y + x \cdot \frac{dy}{dx} = y + x \frac{dy}{dx}.$$

General template. For a relation $F(x, y) = 0$: 1. differentiate term by term w.r.t. x (chain rule on y -terms, product rule on mixed terms); 2. gather every $\frac{dy}{dx}$ on one side; 3. factor and solve: $\frac{dy}{dx} = (\text{expression in } x \text{ and } y)$.

- $\frac{dy}{dx}$ — the slope, generally a function of *both* x and y .

Validity: the relation must define y as a differentiable function of x locally (fails at vertical-tangent / singular points).

Section 3: Key Procedure and Derivation

Why each y -term carries a $\frac{dy}{dx}$

A term like y^3 is really $[y(x)]^3$ — a composite, with $y(x)$ as the inner function. By the chain rule,

$$\frac{d}{dx}[y(x)]^3 = 3[y(x)]^2 \cdot \frac{d}{dx}[y(x)] = 3y^2 \frac{dy}{dx}.$$

There is nothing new here beyond the chain rule — the only mental effort is *remembering* that y is a function of x , so it contributes its own derivative factor. Forgetting that factor is the entire failure mode of the topic.

Procedure: finding $\frac{dy}{dx}$ implicitly

1. Differentiate **both sides** of the equation w.r.t. x .
2. For pure- x terms, differentiate as usual. For y -terms, apply the chain rule (append $\frac{dy}{dx}$). For products like xy or x^2y^3 , apply the product rule.
3. Collect all terms containing $\frac{dy}{dx}$ on the left; move everything else right.
4. Factor out $\frac{dy}{dx}$ and divide to isolate it.

Procedure: implicit second derivative

Differentiate the *first* implicit derivative again w.r.t. x (chain/quotient rules as needed), then substitute the expression for $\frac{dy}{dx}$ to write $\frac{d^2y}{dx^2}$ in terms of x and y .

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Find $\frac{dy}{dx}$ if $x^2 + y^2 = a^2$.

Differentiate both sides w.r.t. x (chain rule on y^2):

$$2x + 2y \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = \boxed{-\frac{x}{y}}.$$

Example 2 — [JEE Main Level]

Find $\frac{dy}{dx}$ if $x^3 + y^3 = 3axy$ (the folium of Descartes).

Differentiate, using the chain rule on y^3 and the product rule on xy :

$$3x^2 + 3y^2 \frac{dy}{dx} = 3a \left(y + x \frac{dy}{dx} \right).$$

Divide by 3 and gather $\frac{dy}{dx}$ terms:

$$x^2 + y^2 \frac{dy}{dx} = ay + ax \frac{dy}{dx} \implies \frac{dy}{dx} (y^2 - ax) = ay - x^2.$$

$$\frac{dy}{dx} = \boxed{\frac{ay - x^2}{y^2 - ax}}.$$

Example 3 — [JEE Main — Trap]

A student differentiates $x^2 + y^2 = 1$ and writes $2x + 2y = 0$. What went wrong?

The trap. The student differentiated y^2 as if y were a constant, getting $2y$ instead of $2y \frac{dy}{dx}$. They **forgot that y is a function of x** — the defining sin of implicit differentiation. Their “answer” $2x + 2y = 0$ isn’t even a derivative; it has no $\frac{dy}{dx}$ in it.

The fix.

$$2x + 2y \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = -\frac{x}{y}.$$

Speed habit: scan your differentiated equation — if it contains a y -term but *no* $\frac{dy}{dx}$, you’ve dropped a chain-rule factor. Every differentiated y -term must carry one.

Example 4 — [JEE Advanced Level]

If $x^2 + y^2 = 1$, find $\frac{d^2y}{dx^2}$ in terms of y .

From Example-style work, the first derivative is $\frac{dy}{dx} = -\frac{x}{y}$. Differentiate again w.r.t. x using the **quotient rule** (and remembering $\frac{d}{dx}y = \frac{dy}{dx}$):

$$\frac{d^2y}{dx^2} = -\frac{(1) \cdot y - x \cdot \frac{dy}{dx}}{y^2} = -\frac{y - x \left(-\frac{x}{y}\right)}{y^2} = -\frac{y + \frac{x^2}{y}}{y^2} = -\frac{y^2 + x^2}{y^3}.$$

Finally use the original relation $x^2 + y^2 = 1$:

$$\frac{d^2y}{dx^2} = \boxed{-\frac{1}{y^3}}.$$

A beautiful collapse: the constraint lets the messy expression reduce to a clean $-\frac{1}{y^3}$.

Section 5: Practice Exercises

1. **[CBSE]** Find $\frac{dy}{dx}$ if $xy = 1$.
2. **[CBSE]** Find $\frac{dy}{dx}$ if $x^2 - xy + y^2 = 7$.
3. **[JEE Main]** Find $\frac{dy}{dx}$ if $\sin(xy) = x$.
4. **[JEE Main]** Find $\frac{dy}{dx}$ if $y = x \sin y$.
5. **[JEE Advanced]** If $x\sqrt{1+y} + y\sqrt{1+x} = 0$ (with $x \neq y$), show that $\frac{dy}{dx} = -\frac{1}{(1+x)^2}$.

Answers: 1. $-\frac{y}{x}$ (or $-\frac{1}{x^2}$)

2. $\frac{y-2x}{2y-x}$

3. $\frac{1-y\cos(xy)}{x\cos(xy)}$

4. $\frac{\sin y}{1-x\cos y}$

5. shown — squaring/simplifying the relation gives $y = -\frac{x}{1+x}$, whose derivative is $-\frac{1}{(1+x)^2}$.

Section 6: Crack the MCQ

Q1. For $x^2 + y^2 = 4$, $\frac{dy}{dx} =$ (a) $-\frac{x}{y}$ (b) $-\frac{y}{x}$ (c) $\frac{x}{y}$ (d) $-x$

Answer: (a). $2x + 2y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x}{y}$.

- **(d)** drops the chain factor on y^2 (treats y as constant). **(b)/(c)** are sign/variable swaps.

Q2. When differentiating xy with respect to x , you get: (a) $\frac{dy}{dx}$ (b) y (c) $y + x\frac{dy}{dx}$ (d) $x\frac{dy}{dx}$

Answer: (c). Product rule: $\frac{d}{dx}(xy) = 1 \cdot y + x \cdot \frac{dy}{dx}$.

- **(b)** forgets the product rule's second term; **(d)** forgets the first; **(a)** loses the x -factor entirely.

Q3. A student's implicit derivative contains a y -term but no $\frac{dy}{dx}$ anywhere. This means: (a) the answer is correct (b) a chain-rule factor was dropped (c) y is constant (d) the curve is a line

Answer: (b). Every differentiated y -term must carry a $\frac{dy}{dx}$; its absence signals a

dropped chain factor — the hallmark implicit error.

Q4. For $\sin y = x$, $\frac{dy}{dx} =$ (a) $\cos y$ (b) $\frac{1}{\cos y}$ (c) $-\frac{1}{\cos y}$ (d) $\sec y \tan y$

Answer: (b). Differentiate: $\cos y \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{\cos y}$ (this is exactly how $\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$ is derived).

- **(a)** forgets to solve for $\frac{dy}{dx}$; **(c)** invents a sign; **(d)** is unrelated.
-

Section 7: Common Student Pitfalls and Pro-Tips

- **Pitfall 1 — forgetting the $\frac{dy}{dx}$ on y -terms.** The defining error. $\frac{d}{dx}(y^2) = 2y \frac{dy}{dx}$, never just $2y$.
- **Pitfall 2 — forgetting the product rule on mixed terms.** $\frac{d}{dx}(xy) = y + x \frac{dy}{dx}$, not $\frac{dy}{dx}$ alone.
- **Pitfall 3 — trying to pre-solve for y .** Usually unnecessary and messy; differentiate the relation as it stands.
- **Pitfall 4 — second-derivative shortcuts.** When finding $\frac{d^2y}{dx^2}$, you must substitute the first $\frac{dy}{dx}$ back in, and may use the original relation to simplify.

Pro-Tip: Treat every y as if it wore a little “ (x) ” label — y is $y(x)$. Then implicit differentiation is *just the chain rule*, and the $\frac{dy}{dx}$ factors appear automatically. Final answers naturally involve both x and y ; that’s expected, not a mistake.

Section 8: Cheat Sheet Box

IMPLICIT DIFFERENTIATION — CORE RECALL

- **Mindset:** y is secretly $y(x)$; differentiate the whole equation w.r.t. x .
- **Every y -term gets a $\frac{dy}{dx}$:** $\frac{d}{dx}(y^n) = ny^{n-1} \frac{dy}{dx}$.
- **Mixed terms use the product rule:** $\frac{d}{dx}(xy) = y + x \frac{dy}{dx}$.
- **Then** collect $\frac{dy}{dx}$ terms, factor, and solve. Answer may involve x and y .
- **Sanity check:** a differentiated y -term with no $\frac{dy}{dx}$ = dropped chain factor.
- **Mnemonic:** “It’s just the chain rule in disguise — every y owes a $\frac{dy}{dx}$.”

5. Derivatives of Inverse Trigonometric Functions

Exam Relevance: One of the highest-yield areas of the chapter, especially for JEE. **CBSE Boards** ask direct derivatives and one “simplify-by-substitution” problem (3–4 marks). **JEE Main/Advanced** love the substitution problems where a fearsome-looking inverse-trig expression collapses to something trivial — *if* you know the trick (and respect the domain). (NEET does not test mathematics.)

Section 1: Concept Intuition

An inverse trigonometric function answers the question “which angle gives this ratio?” — $\sin^{-1} x$ is “the angle whose sine is x .” To differentiate these, we don’t need new magic; we use **implicit differentiation**, which you’ve just met. If $y = \sin^{-1} x$, then $\sin y = x$; differentiate both sides and solve for $\frac{dy}{dx}$. Out pops $\frac{1}{\sqrt{1-x^2}}$. Every inverse-trig derivative is born this way.

But the real exam drama lies elsewhere. Examiners adore expressions like

$$y = \tan^{-1}\left(\frac{2x}{1-x^2}\right) \quad \text{or} \quad y = \sin^{-1}(2x\sqrt{1-x^2}),$$

which look monstrous to differentiate head-on. The secret is to **simplify first using a trigonometric substitution**, then differentiate the simple result.

Think about it this way... these expressions are double-angle formulas in disguise. If you let $x = \tan \theta$, then $\frac{2x}{1-x^2} = \frac{2 \tan \theta}{1 - \tan^2 \theta} = \tan 2\theta$. So $\tan^{-1}\left(\frac{2x}{1-x^2}\right) = \tan^{-1}(\tan 2\theta) = 2\theta = 2 \tan^{-1} x$ — and differentiating $2 \tan^{-1} x$ is a one-liner: $\frac{2}{1+x^2}$. The brutal-looking problem melts away. The art is recognising *which* substitution unlocks the double-angle pattern: $x = \tan \theta$ for $\frac{2x}{1-x^2}$ and $\frac{1-x^2}{1+x^2}$ forms; $x = \sin \theta$ (or $\cos \theta$) for $\sqrt{1-x^2}$ forms.

Limiting conditions & assumptions (read carefully — this is the exam trap).

The identity $\tan^{-1}(\tan 2\theta) = 2\theta$ is only valid when 2θ lies in the *principal range* of $\tan^{-1}$, i.e. $(-\frac{\pi}{2}, \frac{\pi}{2})$. Outside that, $\tan^{-1}(\tan 2\theta) \neq 2\theta$ — it differs by a constant on each piece. So these simplifications are **piecewise**, valid only on a restricted range of x . For differentiation this usually doesn’t change the derivative *within* a valid interval (a constant has zero derivative), but you must state the range, and at the breakpoints the function can be non-differentiable. Also note $\frac{d}{dx} \sec^{-1} x$ and $\frac{d}{dx} \csc^{-1} x$ carry an $|x|$ (domain $|x| > 1$).

Section 2: Key Formulas and Definitions

The six standard inverse-trig derivatives:

$$\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx} \cos^{-1} x = \frac{-1}{\sqrt{1-x^2}} \quad (|x| < 1)$$

$$\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}, \quad \frac{d}{dx} \cot^{-1} x = \frac{-1}{1+x^2}$$

$$\frac{d}{dx} \sec^{-1} x = \frac{1}{|x|\sqrt{x^2 - 1}}, \quad \frac{d}{dx} \csc^{-1} x = \frac{-1}{|x|\sqrt{x^2 - 1}} \quad (|x| > 1)$$

Key substitution identities (each on its valid range):

$$2 \tan^{-1} x = \tan^{-1} \frac{2x}{1-x^2} = \sin^{-1} \frac{2x}{1+x^2} = \cos^{-1} \frac{1-x^2}{1+x^2}$$

$$2 \sin^{-1} x = \sin^{-1}(2x\sqrt{1-x^2}), \quad 3 \sin^{-1} x = \sin^{-1}(3x-4x^3)$$

Useful substitutions: $x = \tan \theta$ (for $1 \pm x^2$, $\frac{2x}{1-x^2}$ patterns); $x = \sin \theta$ or $\cos \theta$ (for $\sqrt{1-x^2}$ patterns); $x = \sec \theta$ (for $\sqrt{x^2-1}$).

Note: the cofunction pairs differ only in sign — $\cos^{-1}$, $\cot^{-1}$, $\csc^{-1}$ carry the minus.

Section 3: Key Derivations and Procedure

Derivation: $\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}$

Let $y = \tan^{-1} x$, so $\tan y = x$. Differentiate both sides w.r.t. x (implicit + chain rule on $\tan y$):

$$\sec^2 y \frac{dy}{dx} = 1 \implies \frac{dy}{dx} = \frac{1}{\sec^2 y} = \frac{1}{1+\tan^2 y} = \frac{1}{1+x^2}. \blacksquare$$

Why the last step works: $\sec^2 y = 1 + \tan^2 y$, and $\tan y = x$.

Derivation: $\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$

Let $y = \sin^{-1} x$, so $\sin y = x$, with $y \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. Differentiate:

$$\cos y \frac{dy}{dx} = 1 \implies \frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}}.$$

We take $+\sqrt{}$ because $\cos y \geq 0$ on the principal range. $\blacksquare$

Procedure: the substitution method

1. **Spot the pattern.** Recognise a double/triple-angle disguise ($\frac{2x}{1-x^2}$, $2x\sqrt{1-x^2}$, $\frac{1-x^2}{1+x^2}$, $3x-4x^3$, ...).
2. **Substitute** the matching trig form ($x = \tan \theta$, $\sin \theta$, ...), so $\theta = \tan^{-1} x$ (etc.).
3. **Collapse** the inner expression to a single trig of a multiple angle, then cancel the inverse: $\tan^{-1}(\tan 2\theta) = 2\theta$ (within range).
4. **Back-substitute** θ and differentiate the now-simple expression. State the valid range of x .

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Differentiate $y = \tan^{-1}(2x)$.

Chain rule with inner $2x$:

$$y' = \frac{1}{1 + (2x)^2} \cdot \frac{d}{dx}(2x) = \frac{2}{1 + 4x^2}. \quad \boxed{y' = \frac{2}{1 + 4x^2}}$$

Example 2 — [JEE Main Level]

Differentiate $y = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$ for $-1 < x < 1$.

Differentiating head-on is painful. **Substitute** $x = \tan \theta$, so $\theta = \tan^{-1} x$ and $\theta \in (-\frac{\pi}{4}, \frac{\pi}{4})$ for this range of x . Then

$$\frac{2x}{1-x^2} = \frac{2 \tan \theta}{1 - \tan^2 \theta} = \tan 2\theta \implies y = \tan^{-1}(\tan 2\theta) = 2\theta = 2 \tan^{-1} x.$$

($2\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$, so the cancellation is valid.) Now differentiate the simple form:

$$y' = 2 \cdot \frac{1}{1+x^2} = \boxed{\frac{2}{1+x^2}}.$$

Example 3 — [JEE Main — Trap]

Differentiate $y = \sin^{-1}(2x\sqrt{1-x^2})$, and state where your answer is valid.

Substitute $x = \sin \theta$, so $\theta = \sin^{-1} x$. Then $2x\sqrt{1-x^2} = 2 \sin \theta \cos \theta = \sin 2\theta$, giving $y = \sin^{-1}(\sin 2\theta) = 2\theta = 2 \sin^{-1} x$, hence

$$y' = \frac{2}{\sqrt{1-x^2}}.$$

The trap. This is **only valid for** $-\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}$ (where $2\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, the principal range of $\sin^{-1}$). For $\frac{1}{\sqrt{2}} < x \leq 1$, the identity becomes $y = \pi - 2 \sin^{-1} x$, so $y' = -\frac{2}{\sqrt{1-x^2}}$ — *opposite sign!* A student who writes a single formula for all x is wrong on half the domain.

Lesson: with inverse-trig substitutions, *always* check whether the multiple angle stays inside the principal range; the simplification — and even the sign of the derivative — is piecewise.

Example 4 — [JEE Advanced Level]

Differentiate $y = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$.

Substitute $x = \tan \theta$, so $\sqrt{1+x^2} = \sec \theta$ and $\theta = \tan^{-1} x$:

$$\frac{\sqrt{1+x^2}-1}{x} = \frac{\sec \theta - 1}{\tan \theta} = \frac{\frac{1}{\cos \theta} - 1}{\frac{\sin \theta}{\cos \theta}} = \frac{1 - \cos \theta}{\sin \theta}.$$

Use the half-angle identities $1 - \cos \theta = 2 \sin^2 \frac{\theta}{2}$ and $\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$:

$$\frac{1 - \cos \theta}{\sin \theta} = \frac{2 \sin^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} = \tan \frac{\theta}{2}.$$

Therefore $y = \tan^{-1}(\tan \frac{\theta}{2}) = \frac{\theta}{2} = \frac{1}{2} \tan^{-1} x$, and

$$y' = \frac{1}{2} \cdot \frac{1}{1+x^2} = \boxed{\frac{1}{2(1+x^2)}}.$$

A genuinely intimidating expression reduced to a clean half — the payoff of the substitution method.

Section 5: Practice Exercises

1. **[CBSE]** Differentiate $y = \cos^{-1}(3x)$.
2. **[CBSE]** Differentiate $y = \sin^{-1}(x^2)$.
3. **[JEE Main]** Differentiate $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ for $-1 < x < 1$.
4. **[JEE Main]** Differentiate $y = \tan^{-1}\left(\frac{\cos x}{1+\sin x}\right)$.
5. **[JEE Advanced]** Differentiate $y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ for $x > 0$, and comment on $x < 0$.

Answers: 1. $\frac{-3}{\sqrt{1-9x^2}}$

2. $\frac{2x}{\sqrt{1-x^4}}$

3. $\frac{2}{1+x^2}$ (substitute $x = \tan \theta$; $= 2 \tan^{-1} x$ here)

4. $-\frac{1}{2}$ (write $\tan^{-1}(\tan(\frac{\pi}{4} - \frac{x}{2})) = \frac{\pi}{4} - \frac{x}{2}$)

5. $\frac{2}{1+x^2}$ for $x > 0$ (it equals $2 \tan^{-1} x$); for $x < 0$ it equals $-2 \tan^{-1} x$, so $y' = -\frac{2}{1+x^2}$ — sign flips.

Section 6: Crack the MCQ

Q1. $\frac{d}{dx} \sin^{-1} x =$ (a) $\frac{1}{1+x^2}$ (b) $\frac{1}{\sqrt{1-x^2}}$ (c) $\frac{-1}{\sqrt{1-x^2}}$ (d) $\frac{-1}{1+x^2}$

Answer: (b). Standard result (derived via $\sin y = x$).

- **(a)** is $\frac{d}{dx} \tan^{-1} x$. **(c)** is $\frac{d}{dx} \cos^{-1} x$. **(d)** is $\frac{d}{dx} \cot^{-1} x$. Table confusion is the trap.

Q2. $\frac{d}{dx} \tan^{-1} \left(\frac{2x}{1-x^2} \right)$ for $|x| < 1$ equals: (a) $\frac{2}{1-x^2}$ (b) $\frac{2}{1+x^2}$ (c) $\frac{1}{1+x^2}$ (d) $\frac{2}{(1-x^2)^2}$

Answer: (b). The expression is $2 \tan^{-1} x$, so the derivative is $\frac{2}{1+x^2}$.

- **(a)/(d)** come from grinding the quotient rule and slipping. **(c)** forgets the factor 2.

Q3. The simplification $\sin^{-1}(2x\sqrt{1-x^2}) = 2\sin^{-1}x$ holds for: (a) all $x \in [-1, 1]$ (b) $x \in \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$ (c) $x \geq 0$ only (d) $x \in (-1, 1)$ except 0

Answer: (b). It needs $2\theta = 2\sin^{-1}x$ to stay in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, i.e. $\sin^{-1}x \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$, i.e. $|x| \leq \frac{1}{\sqrt{2}}$.

- **(a)** ignores the principal-range restriction — the classic over-extension trap.

Q4. $\frac{d}{dx} \sec^{-1} x =$ (a) $\frac{1}{x\sqrt{x^2-1}}$ (b) $\frac{1}{|x|\sqrt{x^2-1}}$ (c) $\frac{1}{\sqrt{x^2-1}}$ (d) $\frac{1}{|x|\sqrt{1-x^2}}$

Answer: (b). The $|x|$ is essential (domain $|x| > 1$).

- **(a)** drops the absolute value — a real error for $x < -1$. **(c)** drops the x -factor. **(d)** has the wrong radicand.

Section 7: Common Student Pitfalls and Pro-Tips

- **Pitfall 1 — ignoring principal-range domains.** Writing one simplified formula for *all* x . These identities (and sometimes the derivative's sign) are piecewise — check where the multiple angle stays in range.
- **Pitfall 2 — table sign mix-ups.** $\cos^{-1}$, $\cot^{-1}$, $\csc^{-1}$ carry minus signs; pairing them wrong with $\sin^{-1}$, $\tan^{-1}$, $\sec^{-1}$ is common.
- **Pitfall 3 — dropping $|x|$ in $\sec^{-1}$, $\csc^{-1}$.** It matters for $x < -1$.
- **Pitfall 4 — brute-forcing instead of substituting.** Grinding the quotient/chain rule on $\frac{2x}{1-x^2}$ wastes time and invites algebra errors; substitution is far safer.

Pro-Tip: Train your eye to spot double-angle disguises: $\frac{2x}{1-x^2}$ and $\frac{1-x^2}{1+x^2}$ scream “ $x = \tan \theta$ ”; $2x\sqrt{1-x^2}$ screams “ $x = \sin \theta$ ”; $\sqrt{x^2-1}$ screams “ $x = \sec \theta$ ”. Recognising the pattern turns a 5-minute grind into a 30-second answer — but always finish by stating the valid range.

Section 8: Cheat Sheet Box

INVERSE-TRIG DERIVATIVES — CORE RECALL

- $\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}, \frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}, \frac{d}{dx} \sec^{-1} x = \frac{1}{|x|\sqrt{x^2-1}}.$
- **Cofunctions** ($\cos^{-1}, \cot^{-1}, \csc^{-1}$) = same magnitude, **minus** sign.
- **Substitution patterns:** $\frac{2x}{1-x^2}, \frac{1-x^2}{1+x^2} \Rightarrow x = \tan \theta; 2x\sqrt{1-x^2} \Rightarrow x = \sin \theta; \sqrt{x^2-1} \Rightarrow x = \sec \theta.$
- **Golden identities:** $2 \tan^{-1} x = \tan^{-1} \frac{2x}{1-x^2} = \sin^{-1} \frac{2x}{1+x^2} = \cos^{-1} \frac{1-x^2}{1+x^2}.$
- **Always state the valid range** — the simplification (and derivative sign) is piecewise.
- **Mnemonic:** “Substitute, collapse the double angle, then differentiate — and mind the range.”

6. Logarithmic & Parametric Differentiation

Exam Relevance: Two reliable mark-earners. **CBSE Boards** almost always include a logarithmic-differentiation sum (variable-power functions or large products) and frequently a parametric $\frac{dy}{dx}$ — typically 2–4 marks each. **JEE Main** asks 1–2 questions, favouring the second-derivative parametric trap and u^v -type functions. **JEE Advanced** combines these with differential-equation relations (e.g. “show that $(1 - x^2)y'' - xy' + p^2y = 0$ ”). (NEET does not test mathematics.)

Section 1: Concept Intuition

Suppose I ask you to differentiate three functions: x^5 , then 5^x , then x^x .

The first is a piece of cake — the **power rule** handles a *constant exponent*: $\frac{d}{dx}x^5 = 5x^4$. The second is also routine — the **exponential rule** handles a *constant base*: $\frac{d}{dx}5^x = 5^x \ln 5$. But the third, x^x , has a variable in *both* the base and the exponent simultaneously. Neither rule applies — the power rule assumes the exponent stands still, the exponential rule assumes the base stands still, and here *nothing* stands still. We are stuck.

This is exactly where the **logarithm** rides in as a rescuer. Think of the logarithm as a *currency converter* in a busy market: it converts the “hard currency” of multiplication and powers into the “easy currency” of addition. Its famous properties — $\log(xy) = \log x + \log y$ and $\log(x^n) = n \log x$ — take a tangled exponent and lay it out flat as a simple product. Once we take $\ln$ of both sides of $y = x^x$, the troublesome exponent slides down to become an ordinary factor: $\ln y = x \ln x$, which we *can* differentiate. That single trick — **take the log, then differentiate** — is logarithmic differentiation.

It earns its keep in a second situation too: when y is a long, ugly **product or quotient** of many factors, like $y = \frac{x^2\sqrt{x-1}}{(x+2)^3}$. Differentiating that directly means a nightmare of product and quotient rules. But $\ln$ shatters the product into a clean *sum* of logs, and a sum is trivial to differentiate term by term.

Now the second idea: **parametric differentiation**. Sometimes x and y are not tied to each other directly; instead, both are controlled by a third, behind-the-scenes variable — a **parameter**, usually called t (time) or θ (angle).

Think about it this way... picture a cricket ball flying through the air. You don’t naturally describe its height y as a function of its horizontal distance x . What you actually know is how *time* drives both: $x = (\text{speed})t$ and $y = (\text{speed})t - \frac{1}{2}gt^2$. Time is the puppeteer; x and y are the two puppets dangling from its strings. If you want the slope of the ball’s path, $\frac{dy}{dx}$ — “how does height change per unit horizontal distance” — you simply ask how fast each puppet responds to the puppeteer and take the ratio:

$$\frac{dy}{dx} = \frac{\text{rate of } y \text{ w.r.t. } t}{\text{rate of } x \text{ w.r.t. } t} = \frac{dy/dt}{dx/dt}.$$

Or picture the giant wheel (Ferris wheel) at a village mela: your seat’s position is most naturally given by the rotation angle θ , as $x = r \cos \theta$, $y = r \sin \theta$. The parameter is

the puppet-master; differentiation just compares the puppets' motions.

Limiting conditions & assumptions. Logarithmic differentiation needs the base to be **positive** — log of a non-positive number is undefined — so for $y = [f(x)]^{g(x)}$ we implicitly require $f(x) > 0$; for products that may go negative, work with $|\cdot|$ or restrict the domain. Parametric differentiation needs $\frac{dx}{dt} \neq 0$; where $dx/dt = 0$, the slope dy/dx blows up (a **vertical tangent**) and the formula breaks.

Section 2: Key Formulas and Definitions

Logarithmic differentiation — the engine. When $y = f(x)$, taking $\ln$ and differentiating gives the all-important relation

$$\frac{d}{dx}(\ln y) = \frac{1}{y} \frac{dy}{dx} \quad \left(\text{never forget the } \frac{1}{y} \text{ — it is the chain rule on } \ln y \right).$$

Master formula for a variable-base, variable-exponent function $y = [f(x)]^{g(x)}$, $f(x) > 0$:

$$\boxed{\frac{dy}{dx} = [f(x)]^{g(x)} \left[g'(x) \ln f(x) + g(x) \frac{f'(x)}{f(x)} \right]}$$

A useful way to *read* this: it is the **sum of two pieces** — the “exponential-rule piece” (treat the base as constant) plus the “power-rule piece” (treat the exponent as constant):

$$\frac{d}{dx}(u^v) = \underbrace{u^v \ln u \cdot v'}_{\text{base frozen}} + \underbrace{v u^{v-1} u'}_{\text{exponent frozen}}, \quad u = f(x), \quad v = g(x).$$

Log properties used to simplify before differentiating:

$$\ln(ab) = \ln a + \ln b, \quad \ln\left(\frac{a}{b}\right) = \ln a - \ln b, \quad \ln(a^n) = n \ln a.$$

Parametric differentiation. If $x = f(t)$ and $y = g(t)$ with $\frac{dx}{dt} \neq 0$:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

Second derivative (read this twice):

$$\boxed{\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}}}$$

- It is **NOT** $\frac{d^2y/dt^2}{d^2x/dt^2}$. You must differentiate the *already-found slope* $\frac{dy}{dx}$ with respect to t , and then divide by $\frac{dx}{dt}$ **once more**. The reason is the chain rule: $\frac{d}{dx} = \frac{1}{dx/dt} \cdot \frac{d}{dt}$, applied a second time.

Here $f(t), g(t)$ are the coordinate functions of the parameter t ; $\frac{dy}{dt}, \frac{dx}{dt}$ are their ordinary derivatives w.r.t. t .

Section 3: CBSE Board Derivations and Key Procedures

Derivation 1: The master formula for $y = [f(x)]^{g(x)}$

Setup. Let $y = [f(x)]^{g(x)}$ with $f(x) > 0$ (so the logarithm is defined).

Step 1 — take the natural log of both sides. *Why:* the log pulls the exponent $g(x)$ down to the front, converting an un-differentiable power into a product we can handle.

$$\ln y = g(x) \ln f(x).$$

Step 2 — differentiate both sides w.r.t. x . The left side uses the chain rule on $\ln y$; the right side uses the product rule on $g(x) \cdot \ln f(x)$.

$$\frac{1}{y} \frac{dy}{dx} = g'(x) \ln f(x) + g(x) \cdot \frac{f'(x)}{f(x)}.$$

Why the $\frac{1}{y}$ appears: y is itself a function of x , so $\frac{d}{dx} \ln y = \frac{1}{y} \cdot \frac{dy}{dx}$ — forgetting this factor is the single most common error in the topic.

Step 3 — solve for $\frac{dy}{dx}$ by multiplying through by y , and substitute $y = [f(x)]^{g(x)}$ back in:

$$\frac{dy}{dx} = [f(x)]^{g(x)} \left[g'(x) \ln f(x) + g(x) \frac{f'(x)}{f(x)} \right]. \quad \blacksquare$$

Derivation 2: Why the parametric second derivative is *not* a naive ratio

Setup. $x = f(t), y = g(t)$. We already have $\frac{dy}{dx} = \frac{g'(t)}{f'(t)}$, which is itself a **function of t** , not of x .

The key idea. To differentiate *anything* with respect to x when everything is written in t , we route through the chain rule:

$$\frac{d}{dx}(\cdot) = \frac{d}{dt}(\cdot) \cdot \frac{dt}{dx} = \frac{1}{dx/dt} \cdot \frac{d}{dt}(\cdot).$$

Apply this operator to $\frac{dy}{dx}$:

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{1}{dx/dt} \cdot \frac{d}{dt} \left(\frac{dy}{dx} \right).$$

The trap, made visible. A student who writes $\frac{d^2y}{dx^2} = \frac{d^2y/dt^2}{d^2x/dt^2}$ has wrongly “cancelled” the dt ’s as if they were numbers. But $\frac{dy}{dx}$ is a *quotient* of t -derivatives, and you cannot differentiate a quotient by differentiating top and bottom separately. The correct route is: find the slope, differentiate that whole slope w.r.t. t , then divide by dx/dt one more time. $\blacksquare$

Procedure: Logarithmic differentiation (checklist)

1. Confirm the base is positive (or wrap in $|\cdot|$ / restrict domain).
2. Take $\ln$ of both sides.
3. Use log laws to break products/quotients into sums/differences and bring exponents to the front.
4. Differentiate implicitly — remember $\frac{d}{dx} \ln y = \frac{1}{y} y'$.
5. Multiply by y and substitute the original expression back for y .

Procedure: Parametric differentiation (checklist)

1. Compute $\frac{dx}{dt}$ and $\frac{dy}{dt}$ separately.
 2. First derivative: $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ (require $dx/dt \neq 0$).
 3. Second derivative: differentiate $\frac{dy}{dx}$ **w.r.t.** t , then divide by $\frac{dx}{dt}$ again.
 4. Simplify and, if asked at a particular point, substitute the parameter value last.
-

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Differentiate $y = x^{\sin x}$ **with respect to** x (for $x > 0$).

Step 1 — take logs. $\ln y = \sin x \cdot \ln x$.

Step 2 — differentiate both sides (product rule on the right, chain rule on the left):

$$\frac{1}{y} \frac{dy}{dx} = \cos x \cdot \ln x + \sin x \cdot \frac{1}{x}.$$

Step 3 — solve and substitute back $y = x^{\sin x}$:

$$\frac{dy}{dx} = x^{\sin x} \left(\cos x \ln x + \frac{\sin x}{x} \right)$$

Example 2 — [JEE Main Level]

Find $\frac{dy}{dx}$ **for the astroid** $x = a \cos^3 t$, $y = a \sin^3 t$.

Step 1 — differentiate each w.r.t. t .

$$\frac{dx}{dt} = 3a \cos^2 t (-\sin t) = -3a \cos^2 t \sin t, \quad \frac{dy}{dt} = 3a \sin^2 t \cos t.$$

Step 2 — take the ratio.

$$\frac{dy}{dx} = \frac{3a \sin^2 t \cos t}{-3a \cos^2 t \sin t} = -\frac{\sin t}{\cos t} = \boxed{-\tan t}.$$

Notice how the messy a 's and squared trig terms cancel cleanly — a hallmark of well-posed parametric problems.

Example 3 — [JEE Main — Speed Trap]

For $x = at^2$, $y = 2at$, find $\frac{d^2y}{dx^2}$.

The trap: students “cancel the dt ’s” and write $\frac{d^2y}{dx^2} = \frac{d^2y/dt^2}{d^2x/dt^2} = \frac{0}{2a} = 0$. This is **wrong** — and it is the most-tested error in parametric calculus.

The correct route. First derivative: $\frac{dy}{dt} = 2a$, $\frac{dx}{dt} = 2at$, so $\frac{dy}{dx} = \frac{2a}{2at} = \frac{1}{t}$.

Now differentiate this slope **w.r.t.** t , then divide by $\frac{dx}{dt}$ again:

$$\frac{d}{dt}\left(\frac{1}{t}\right) = -\frac{1}{t^2}, \quad \frac{d^2y}{dx^2} = \frac{-1/t^2}{2at} = \boxed{-\frac{1}{2at^3}}.$$

The naive answer 0 isn’t just numerically off — it would falsely claim the parabola has zero curvature everywhere. **Pro move:** the moment you see “second derivative, parametric,” mentally chant “slope, differentiate w.r.t. t , divide by dx/dt .”

Example 4 — [JEE Advanced Level]

If $x = \sin t$ and $y = \sin pt$, prove that $(1 - x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} + p^2y = 0$.

Step 1 — first derivative. $\frac{dx}{dt} = \cos t$, $\frac{dy}{dt} = p \cos pt$, so

$$y' \equiv \frac{dy}{dx} = \frac{p \cos pt}{\cos t}.$$

Step 2 — second derivative. Differentiate y' w.r.t. t , then divide by $\frac{dx}{dt} = \cos t$:

$$\begin{aligned} \frac{d}{dt}\left(\frac{p \cos pt}{\cos t}\right) &= p \cdot \frac{(-p \sin pt) \cos t - \cos pt(-\sin t)}{\cos^2 t} = \frac{-p^2 \sin pt \cos t + p \sin t \cos pt}{\cos^2 t}, \\ y'' \equiv \frac{d^2y}{dx^2} &= \frac{1}{\cos t} \cdot \frac{-p^2 \sin pt \cos t + p \sin t \cos pt}{\cos^2 t} = \frac{-p^2 \sin pt \cos t + p \sin t \cos pt}{\cos^3 t}. \end{aligned}$$

Step 3 — assemble the expression, using $1 - x^2 = 1 - \sin^2 t = \cos^2 t$:

$$\begin{aligned} (1 - x^2)y'' &= \cos^2 t \cdot \frac{-p^2 \sin pt \cos t + p \sin t \cos pt}{\cos^3 t} = -p^2 \sin pt + \frac{p \sin t \cos pt}{\cos t}, \\ xy' &= \sin t \cdot \frac{p \cos pt}{\cos t} = \frac{p \sin t \cos pt}{\cos t}, \quad p^2y = p^2 \sin pt. \end{aligned}$$

Step 4 — add them up:

$$(1 - x^2)y'' - xy' + p^2y = \left(-p^2 \sin pt + \frac{p \sin t \cos pt}{\cos t}\right) - \frac{p \sin t \cos pt}{\cos t} + p^2 \sin pt = 0. \quad \blacksquare$$

The two awkward $\frac{p \sin t \cos pt}{\cos t}$ terms annihilate, and the $\pm p^2 \sin pt$ pair cancels — the relation holds identically.

Section 5: Practice Exercises

1. **[CBSE]** Differentiate $y = x^{\ln x}$ (here $\ln x = \log_e x$), $x > 0$.
2. **[CBSE]** Using logarithmic differentiation, find $\frac{dy}{dx}$ for $y = \frac{(x-1)(x-2)}{(x-3)(x-4)}$.
3. **[JEE Main]** For the cycloid $x = a(t + \sin t)$, $y = a(1 - \cos t)$, find $\frac{dy}{dx}$.
4. **[JEE Main]** Differentiate $y = (\tan x)^{\cot x}$ with respect to x .
5. **[JEE Advanced]** For $x = a \cos^3 \theta$, $y = a \sin^3 \theta$, find $\frac{d^2y}{dx^2}$.

Answers:

1. $\frac{dy}{dx} = x^{\ln x} \cdot \frac{2 \ln x}{x}$
2. $\frac{dy}{dx} = y \left[\frac{1}{x-1} + \frac{1}{x-2} - \frac{1}{x-3} - \frac{1}{x-4} \right]$ where $y = \frac{(x-1)(x-2)}{(x-3)(x-4)}$
3. $\frac{dy}{dx} = \tan \frac{t}{2}$
4. $\frac{dy}{dx} = (\tan x)^{\cot x} \csc^2 x (1 - \ln \tan x)$
5. $\frac{d^2y}{dx^2} = \frac{\sec^4 \theta \csc \theta}{3a}$

Section 6: Crack the MCQ

Q1. $\frac{d}{dx}(x^x)$ equals: (a) $x^x \ln x$ (b) $x \cdot x^{x-1}$ (c) $x^x(1 + \ln x)$ (d) x^x

Answer: (c). Take logs: $\ln y = x \ln x \Rightarrow \frac{y'}{y} = \ln x + 1 \Rightarrow y' = x^x(1 + \ln x)$.

- **(a)** keeps only the “exponential-rule piece” (base frozen) — the student forgot the power-rule piece.
- **(b)** = x^x is the “power-rule piece” alone (exponent frozen) — the opposite half-error.
- **(d)** comes from thinking x^x differentiates to itself like e^x ; only e^x has that property.

Q2. If $x = t^2$ and $y = t^3$, then $\frac{d^2y}{dx^2} =$ (a) $\frac{3}{2}$ (b) $\frac{3}{4t}$ (c) $3t$ (d) $\frac{3}{2t}$

Answer: (b). $\frac{dy}{dx} = \frac{3t^2}{2t} = \frac{3t}{2}$; then $\frac{d^2y}{dx^2} = \frac{d/dt(3t/2)}{dx/dt} = \frac{3/2}{2t} = \frac{3}{4t}$.

- **(a)** is $\frac{dy}{dx}$ differentiated w.r.t. t but **not** divided by dx/dt again — the signature parametric error.
- **(d)** divides correctly but mishandles the constant $\frac{3}{2}$.
- **(c)** is essentially $\frac{d^2y/dt^2}{d^2x/dt^2}$ -style nonsense.

Q3. Which function genuinely *requires* logarithmic differentiation (cannot be done by a single standard rule)? (a) x^5 (b) 5^x (c) $x^{\sin x}$ (d) $\sin(x^5)$

Answer: (c). Variable base **and** variable exponent — neither power nor exponential rule applies alone.

- **(a)** power rule. **(b)** exponential rule. **(d)** plain chain rule. Only **(c)** forces the log trick.

Q4. For a parametric curve $x = f(t)$, $y = g(t)$, the slope $\frac{dy}{dx}$ is **undefined** (vertical tangent) when: (a) $\frac{dy}{dt} = 0$ (b) $\frac{dx}{dt} = 0$ (c) $t = 0$ (d) $\frac{dy}{dt} = \frac{dx}{dt}$

Answer: (b). $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ blows up when the denominator $dx/dt = 0$.

- **(a)** gives a *horizontal* tangent ($\frac{dy}{dx} = 0$), not undefined.
- **(c)** $t = 0$ is just a specific parameter value with no special meaning in general.
- **(d)** merely makes the slope equal to 1.

Section 7: Common Student Pitfalls and Pro-Tips

- **Pitfall 1 — dropping the $\frac{1}{y}$.** Writing $\frac{d}{dx} \ln y = \frac{1}{y}$ instead of $\frac{1}{y} \frac{dy}{dx}$. The y is a function of x , so the chain rule demands the extra $\frac{dy}{dx}$. This silently corrupts every logarithmic-differentiation answer.
- **Pitfall 2 — half-rule on u^v .** Differentiating x^x as either $x^x \ln x$ or $x \cdot x^{x-1}$. Both are *halves* of the truth; you need their **sum**.
- **Pitfall 3 — the parametric second-derivative ratio.** Using $\frac{d^2y/dt^2}{d^2x/dt^2}$. Always: slope first, differentiate w.r.t. t , divide by dx/dt again.
- **Pitfall 4 — ignoring the positive-base condition.** Applying log differentiation to a base that can be ≤ 0 without using $|\cdot|$ or stating the domain.

Pro-Tip: For any u^v , skip re-deriving and write the answer directly as “**exponential piece + power piece**”: $\frac{d}{dx}(u^v) = u^v \ln u \cdot v' + v u^{v-1} u'$. And for parametric curves, before plugging a value, simplify $\frac{dy}{dx}$ fully — the constants almost always cancel, and a clean symbolic slope is far less error-prone to substitute into.

Section 8: Cheat Sheet Box

LOGARITHMIC & PARAMETRIC DIFFERENTIATION — CORE RECALL

- **When to take logs:** variable base *and* exponent (u^v), or a big product/quotient of factors.
- **Engine:** $\frac{d}{dx} \ln y = \frac{1}{y} \frac{dy}{dx}$ (never drop the $\frac{1}{y}$).
- **Master u^v formula:** $\frac{d}{dx}(u^v) = u^v \ln u \cdot v' + v u^{v-1} u'$ — *exponential piece + power piece*.
- **Parametric 1st derivative:** $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ ($dx/dt \neq 0$).
- **Parametric 2nd derivative:** $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt}$ — **NOT** $\frac{d^2y/dt^2}{d^2x/dt^2}$.

- **Domain guards:** log diff needs base > 0 ; parametric slope undefined where $dx/dt = 0$ (vertical tangent).
- **Mnemonic (parametric y''):** *"Slope, then differentiate by t , then divide by $\dot{x}$ once more."*

7. Second-Order Derivatives & the Mean Value Theorems

Exam Relevance: A dependable scoring zone. **CBSE Boards** almost always carry a second-order-derivative relation to prove (e.g. $y'' + n^2y = 0$) for 2–4 marks, plus frequently a “verify Rolle’s / Mean Value Theorem” question. **JEE Main** leans on parametric/implicit second derivatives and the secant-slope value c . **JEE Advanced** uses the Mean Value Theorem as a *tool* — proving inequalities and bounding root counts. (NEET does not test mathematics.)

Section 1: Concept Intuition

You already know that the first derivative $\frac{dy}{dx}$ measures a *rate of change* — for a moving car, it is the speedometer reading: how fast position changes with time. So what does the derivative of the derivative measure? It measures **how fast the rate itself is changing** — for the car, that is the *acceleration*, how hard you are pressing (or releasing) the accelerator. That is the **second-order derivative**, written $\frac{d^2y}{dx^2}$ or y'' .

Geometrically, the second derivative tells you which way a curve **bends**. Where $y'' > 0$ the graph is concave up — shaped like a cup $\cup$ that could hold water; where $y'' < 0$ it is concave down — a dome $\cap$ that sheds water. The first derivative tells you whether you’re going uphill or downhill; the second tells you whether the road is curving toward the sky or toward the ground.

Now to the two **Mean Value Theorems**, which sound abstract but describe something you experience all the time.

Start with **Rolle’s Theorem**. Throw a cricket ball straight up and catch it at exactly the same height it left your hand. Its starting height equals its ending height. Common sense says: at the very top of its flight, for one instant, the ball was neither rising nor falling — its vertical velocity was *zero*. Rolle’s theorem is exactly this statement made rigorous: **if a smooth, unbroken function has equal values at two points, $f(a) = f(b)$, then somewhere strictly between them the slope must be zero, $f'(c) = 0$** . You went up and came back, so there had to be a flat moment at the turn.

Lagrange’s Mean Value Theorem (MVT) is the grown-up generalization, and it has the most satisfying real-world picture in all of calculus. Suppose you drive from Mumbai to Pune — 150 km — in exactly 2 hours. Your *average* speed was 75 km/h. MVT guarantees that **at some instant during the drive your speedometer read exactly 75 km/h**. You simply cannot average 75 without hitting 75 at least once. (This is precisely how the “average speed cameras” on the Expressway catch you — they’re enforcing MVT.) In symbols: there is a point c where the *instantaneous* slope equals the *average* slope across the interval:

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Think about it this way... Rolle’s theorem is nothing but MVT for the special case when the start and end heights are equal. If $f(a) = f(b)$, the average slope $\frac{f(b)-f(a)}{b-a}$ becomes 0, and MVT’s guaranteed point becomes a point of zero slope — which is exactly Rolle. Geometrically: MVT says some tangent is parallel to the secant joining the endpoints; Rolle is the case where that secant happens to be horizontal.

Limiting conditions & assumptions. Both theorems demand two hypotheses: f must be **continuous on the closed interval** $[a, b]$ and **differentiable on the open interval** (a, b) (Rolle adds the third condition $f(a) = f(b)$). Drop even one and the guarantee collapses. The standard cautionary tale is $f(x) = |x|$ on $[-1, 1]$: here $f(-1) = f(1) = 1$, so the “equal heights” condition holds — yet there is *no* point where $f'(c) = 0$, because the function is not differentiable at the corner $x = 0$. The hypothesis failed, so the conclusion is not promised. Note too that these are **existence** theorems: they guarantee *at least one* such c exists, but say nothing about whether it is unique, and you still have to *solve* for it.

Section 2: Key Formulas and Definitions

Second-order derivative. The derivative of the first derivative:

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right), \quad \text{also written } y'' \text{ or } y_2 \text{ (when } y = f(x)\text{)}.$$

- **Parametric form** (covered in the Differentiation Toolkit subtopic): for $x = f(t)$, $y = g(t)$,

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}} \quad (\text{never } \frac{d^2y/dt^2}{d^2x/dt^2}).$$

- **Implicit form:** differentiate the relation twice w.r.t. x , remembering each y -term carries a $\frac{dy}{dx}$ factor.

Rolle’s Theorem. Let $f : [a, b] \rightarrow \mathbb{R}$ be such that (i) f is continuous on $[a, b]$, (ii) f is differentiable on (a, b) , and (iii) $f(a) = f(b)$. Then there exists at least one $c \in (a, b)$ with

$$f'(c) = 0.$$

Lagrange’s Mean Value Theorem. Let $f : [a, b] \rightarrow \mathbb{R}$ be (i) continuous on $[a, b]$ and (ii) differentiable on (a, b) . Then there exists at least one $c \in (a, b)$ with

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Symbols: a, b — endpoints of the interval; c — the (not necessarily unique) interior point the theorem guarantees; $f'(c)$ — the instantaneous slope there; $\frac{f(b)-f(a)}{b-a}$ — the average slope (slope of the secant).

ysis): the two/three hypotheses *are* the validity conditions — always confirm continuity on $[a, b]$ and differentiability on (a, b) (and equal endpoints for Rolle) *before* using either conclusion. Corners, cusps, vertical tangents, or any discontinuity inside the interval void the theorem.

Section 3: CBSE Board Derivations and Key Procedures

Derivation: Lagrange's MVT from Rolle's Theorem (the auxiliary-function trick)

This is the cleanest proof in the subtopic and a favourite at JEE Advanced.

Goal. Given f continuous on $[a, b]$ and differentiable on (a, b) , show some $c \in (a, b)$ satisfies $f'(c) = \frac{f(b)-f(a)}{b-a}$.

The idea. Rolle's theorem only fires when the endpoints are level. So we *tilt* the function to make its endpoints level, apply Rolle, and tilt back. Define an auxiliary function that subtracts off the secant line:

$$g(x) = f(x) - \left[f(a) + \frac{f(b) - f(a)}{b - a} (x - a) \right].$$

The bracketed term is exactly the straight secant joining $(a, f(a))$ and $(b, f(b))$. So $g(x)$ measures the *vertical gap* between the curve and that secant.

Step 1 — check Rolle's hypotheses for g . g is continuous on $[a, b]$ and differentiable on (a, b) (it's f minus a polynomial). And at the endpoints the curve meets the secant, so the gap is zero:

$$g(a) = f(a) - f(a) = 0, \quad g(b) = f(b) - [f(a) + (f(b) - f(a))] = 0.$$

Thus $g(a) = g(b) = 0$. *Why this matters:* we have manufactured exactly the equal-endpoints condition Rolle needs.

Step 2 — apply Rolle to g . There exists $c \in (a, b)$ with $g'(c) = 0$.

Step 3 — translate back. Differentiating g :

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}.$$

Setting $g'(c) = 0$ gives $f'(c) = \frac{f(b)-f(a)}{b-a}$. ■

The slickness is in Step 1: subtracting the secant is precisely the move that converts the general statement into Rolle's special case.

Procedure A: Verifying Rolle's Theorem for a given f on $[a, b]$

1. Check f is **continuous** on $[a, b]$.
2. Check f is **differentiable** on (a, b) .
3. Check $f(a) = f(b)$. (If any of 1–3 fails, Rolle does not apply — state that and stop.)
4. Solve $f'(c) = 0$ and confirm at least one solution lies **strictly inside** (a, b) . That c verifies the theorem.

Procedure B: Verifying Lagrange's MVT for a given f on $[a, b]$

1. Check continuity on $[a, b]$ and differentiability on (a, b) .
2. Compute the average slope $m = \frac{f(b)-f(a)}{b-a}$.
3. Solve $f'(c) = m$ and confirm a solution lies in (a, b) .

Procedure C: Computing a second-order derivative

- **Direct:** differentiate $f(x)$ once to get y' , then differentiate y' again.
 - **Parametric:** find $\frac{dy}{dx}$ as a function of t , differentiate it w.r.t. t , divide by $\frac{dx}{dt}$.
 - **Implicit:** differentiate the equation twice; isolate $\frac{d^2y}{dx^2}$, substituting $\frac{dy}{dx}$ where it appears.
-

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

If $y = e^x \sin x$, prove that $y'' - 2y' + 2y = 0$.

Step 1 — first derivative (product rule):

$$y' = e^x \sin x + e^x \cos x = e^x(\sin x + \cos x).$$

Step 2 — second derivative (product rule again):

$$y'' = e^x(\sin x + \cos x) + e^x(\cos x - \sin x) = e^x(2 \cos x) = 2e^x \cos x.$$

Step 3 — substitute into the expression:

$$\begin{aligned} y'' - 2y' + 2y &= 2e^x \cos x - 2e^x(\sin x + \cos x) + 2e^x \sin x \\ &= 2e^x \cos x - 2e^x \sin x - 2e^x \cos x + 2e^x \sin x = 0. \quad \blacksquare \end{aligned}$$

Example 2 — [JEE Main Level]

Verify Rolle's Theorem for $f(x) = \sin x$ on $[0, \pi]$, and find c .

Step 1 — hypotheses. $\sin x$ is continuous everywhere (so on $[0, \pi]$) and differentiable everywhere (so on $(0, \pi)$). Endpoints: $f(0) = \sin 0 = 0$ and $f(\pi) = \sin \pi = 0$, so $f(0) = f(\pi)$. All three conditions hold. ✓

Step 2 — find c . Solve $f'(c) = 0$:

$$f'(x) = \cos x = 0 \implies x = \frac{\pi}{2}.$$

Since $\frac{\pi}{2} \in (0, \pi)$, Rolle's theorem is verified, with $c = \frac{\pi}{2}$.

Geometrically, the tangent to $y = \sin x$ at its peak $(\frac{\pi}{2}, 1)$ is horizontal — parallel to the x -axis joining the equal endpoints.

Example 3 — [JEE Main — Trap]

A student is asked to apply Rolle's Theorem to $f(x) = |x - 1|$ on $[0, 2]$ and find c . What goes wrong?

The trap. The endpoints look perfect for Rolle: $f(0) = |-1| = 1$ and $f(2) = |1| = 1$, so $f(0) = f(2)$. A student who checks only the equal-endpoints condition charges ahead, tries to solve $f'(c) = 0$, finds the derivative is -1 on the left and $+1$ on the right — **never zero** — and panics, assuming an arithmetic error.

The resolution. There is no error. Rolle's theorem **does not apply here**, because the *differentiability* hypothesis fails: $f(x) = |x - 1|$ has a corner at $x = 1 \in (0, 2)$, where the left derivative -1 and right derivative $+1$ disagree. With a hypothesis violated, the conclusion $f'(c) = 0$ is simply not promised — and indeed it fails.

Lesson / speed move: *always validate continuity AND differentiability before reaching for the conclusion.* The most common exam trap in this entire topic is a function engineered to satisfy the obvious condition ($f(a) = f(b)$) while quietly breaking the differentiability one. Spot the modulus / greatest-integer / piecewise corner first.

Example 4 — [JEE Advanced Level]

Using the Mean Value Theorem, prove that for $0 < a < b$,

$$\frac{b-a}{b} < \ln \frac{b}{a} < \frac{b-a}{a}.$$

Setup. Apply Lagrange's MVT to $f(x) = \ln x$ on $[a, b]$. Since $\ln x$ is continuous and differentiable for $x > 0$, the hypotheses hold. So there exists $c \in (a, b)$ with

$$f'(c) = \frac{f(b) - f(a)}{b - a} \implies \frac{1}{c} = \frac{\ln b - \ln a}{b - a} = \frac{\ln(b/a)}{b - a}.$$

The key inequality. Because c lies strictly between a and b with $0 < a < c < b$, reciprocating (which reverses inequalities for positive numbers):

$$\frac{1}{b} < \frac{1}{c} < \frac{1}{a}.$$

Substitute $\frac{1}{c} = \frac{\ln(b/a)}{b-a}$:

$$\frac{1}{b} < \frac{\ln(b/a)}{b-a} < \frac{1}{a}.$$

Finish. Multiply throughout by $(b-a) > 0$ (inequalities preserved):

$$\frac{b-a}{b} < \ln \frac{b}{a} < \frac{b-a}{a}. \quad \blacksquare$$

This is the signature power of MVT: an apparently unprovable inequality falls out in three lines once you choose the right function and read off the bounds on c .

Section 5: Practice Exercises

1. **[CBSE]** If $y = A \cos nx + B \sin nx$, prove that $\frac{d^2y}{dx^2} + n^2y = 0$.
2. **[CBSE]** Find $\frac{d^2y}{dx^2}$ for $y = x^3 \ln x$.
3. **[JEE Main]** Verify the Mean Value Theorem for $f(x) = x^2 - 3x + 2$ on $[1, 4]$ and find c .
4. **[JEE Main]** For the cycloid $x = a(\theta - \sin \theta)$, $y = a(1 - \cos \theta)$, find $\frac{d^2y}{dx^2}$.

5. **[JEE Advanced]** Using the Mean Value Theorem, prove that $|\cos a - \cos b| \leq |a - b|$ for all real a, b .

Answers: 1. $y'' = -n^2(A \cos nx + B \sin nx) = -n^2y$, hence $y'' + n^2y = 0$.

2. $\frac{d^2y}{dx^2} = 6x \ln x + 5x = x(6 \ln x + 5)$.

3. Hypotheses hold; average slope $= \frac{f(4) - f(1)}{3} = \frac{6 - 0}{3} = 2$; $f'(c) = 2c - 3 = 2 \Rightarrow c = \frac{5}{2} \in (1, 4)$.

4. $\frac{d^2y}{dx^2} = -\frac{1}{4a \sin^4(\theta/2)}$ (from $\frac{dy}{dx} = \cot \frac{\theta}{2}$).

5. Apply MVT to $f(x) = \cos x$: $\cos a - \cos b = -\sin(c)(a - b)$ for some c ; since $|\sin c| \leq 1$, $|\cos a - \cos b| \leq |a - b|$.

Section 6: Crack the MCQ

Q1. On the interval $[-1, 1]$, Rolle's Theorem **fails to apply** to which function? (a) $f(x) = x^2 - 1$ (b) $f(x) = |x|$ (c) $f(x) = x^4$ (d) $f(x) = \cos(\frac{\pi x}{2})$

Answer: (b). $|x|$ satisfies $f(-1) = f(1) = 1$, but it is **not differentiable** at the corner $x = 0 \in (-1, 1)$, so a hypothesis fails.

- **(a)** $x^2 - 1$: smooth, $f(-1) = f(1) = 0$ — applies (and gives $c = 0$).
- **(c)** x^4 : smooth, $f(-1) = f(1) = 1$ — applies.
- **(d)** $\cos(\pi x/2)$: smooth, $f(-1) = f(1) = 0$ — applies. Only **(b)** has the non-differentiable point.

Q2. For $f(x) = x^2$ on $[1, 3]$, the value c guaranteed by the Mean Value Theorem is: (a) 4 (b) 2 (c) $\sqrt{5}$ (d) 1.5

Answer: (b). Average slope $= \frac{9-1}{3-1} = 4$; set $f'(c) = 2c = 4 \Rightarrow c = 2 \in (1, 3)$.

- **(a)** mistakes the average slope (4) for c itself.
- **(c)** comes from confusing c with the root-mean-square / geometric average of the endpoints.
- **(d)** is the arithmetic midpoint $\frac{1+3}{2}$, which is *not* what MVT delivers for a parabola.

Q3. The curve $y = x^3$ is concave **up** on which interval? (a) $(-\infty, 0)$ (b) $(0, \infty)$ (c) $(-\infty, \infty)$ (d) nowhere

Answer: (b). Concavity is governed by $y'' = 6x$, which is positive exactly when $x > 0$.

- **(a)** is where $y'' < 0$ — concave *down*.
- **(c)** wrongly assumes a single concavity throughout; the sign of y'' changes at $x = 0$ (an inflection point).
- **(d)** ignores that y'' is nonzero away from the origin.

Q4. If f is differentiable on $[a, b]$ and $f'(x) \neq 0$ for every $x \in (a, b)$, then in $[a, b]$ the function f can have **at most**: (a) 0 roots (b) 1 root (c) 2 roots (d) infinitely many roots

Answer: (b). If f had two distinct roots $p < q$, then $f(p) = f(q) = 0$, and Rolle's theorem would force some $c \in (p, q)$ with $f'(c) = 0$ — contradicting $f' \neq 0$. So at

most one root.

- **(a)** is too strong: f may still have one root (e.g. a strictly increasing function crossing zero once).
- **(c)** and **(d)** are exactly what Rolle's theorem rules out.

Section 7: Common Student Pitfalls and Pro-Tips

- **Pitfall 1 — applying the theorem without checking hypotheses.** The classic $|x|$ / greatest-integer / piecewise trap satisfies the *obvious* condition while breaking continuity or differentiability. Always validate continuity on $[a, b]$ **and** differentiability on (a, b) first.
- **Pitfall 2 — confusing Rolle with MVT.** Rolle additionally needs $f(a) = f(b)$ and concludes $f'(c) = 0$; MVT needs no such equality and concludes $f'(c) =$ average slope. Using $f'(c) = 0$ when $f(a) \neq f(b)$ is wrong.
- **Pitfall 3 — accepting a c outside the open interval.** The guaranteed point must lie *strictly* inside (a, b) ; a solution landing on an endpoint does not count.
- **Pitfall 4 — the parametric/implicit second-derivative slip.** Same error as Part 2: for $x = f(t), y = g(t)$, never write $\frac{d^2y}{dx^2} = \frac{d^2y/dt^2}{d^2x/dt^2}$. Differentiate the slope w.r.t. t , then divide by $\frac{dx}{dt}$ again.

Pro-Tip: Remember Rolle is just MVT with a horizontal secant — learn one picture, get both. For *inequality* proofs at JEE Advanced, MVT is the default weapon: choose f so that $f'(c)$ is the quantity you want to bound, then squeeze using $a < c < b$. And for “how many roots can this have,” reach for Rolle — two roots always manufacture a zero of f' between them.

Section 8: Cheat Sheet Box

SECOND-ORDER DERIVATIVES & MEAN VALUE THEOREMS — CORE RECALL

- **Second derivative:** $y'' = \frac{d}{dx} \left(\frac{dy}{dx} \right)$; sign of y'' gives concavity ($+$ = cup $\cup$, $-$ = cap $\cap$).
- **Rolle:** continuous on $[a, b]$, differentiable on (a, b) , **and** $f(a) = f(b) \Rightarrow$ some $c \in (a, b)$ with $f'(c) = 0$.
- **Lagrange MVT:** continuous on $[a, b]$, differentiable on $(a, b) \Rightarrow$ some $c \in (a, b)$ with $f'(c) = \frac{f(b) - f(a)}{b - a}$.
- **Link:** Rolle = MVT when the secant is horizontal ($f(a) = f(b)$).
- **Both are existence theorems:** guarantee *at least one* interior c ; check hypotheses first, accept only $c \in (a, b)$.
- **Mnemonics:** Rolle — “same heights $\Rightarrow$ a flat spot.” MVT — “your average speed is your exact speed at some instant.”

Chapter 5: Continuity and Differentiability — Round 2 Addendum

The chapter taught you the definitions, the bridge theorem, the full differentiation toolkit, and both Mean Value Theorems — but five examinable patterns stand outside it. Addendum A supplies the standard limits the chapter keeps *quoting* without ever deriving (its own Practice Exercise 5 asks you for $\frac{1-\cos x}{x^2}$ having never built the machinery). Addendum B proves the modulus-differentiability rules that Example 3 of Section 1 uses only as folklore. Addendum C builds the functional-equation machinery the Exam Relevance notes promise at JEE Advanced. Addendum D adds Cauchy’s Mean Value Theorem — the parent theorem the chapter skips — and the Rolle cascade for counting roots. Addendum E handles compositions with the greatest integer function, where the chapter’s “jumps at every integer” slogan is not enough to even count correctly.

Nothing here is new theory beyond the chapter’s own tools: every derivation runs on the limit definition, the squeeze theorem, or Rolle’s theorem, applied one step more carefully than the chapter needed.

Addendum A — Standard Limits: the Engine Behind Every Continuity Question

A “find k so that f is continuous” question reduces to evaluating one limit. When the piece containing the junction is a quotient like $\frac{\sin 5x}{3x}$ or $\frac{e^{3x}-1}{\ln(1+2x)}$, neither LHL nor RHL can be found by substitution — you need a stock of standard limits. The chapter cites $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ in passing and never returns to it. Here is the full stock, each one derived, plus the single principle that generates all of them.

The generating principle

Every standard limit is a derivative in disguise. If f is differentiable at a , then by definition

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = f'(a).$$

Read the other way: whenever you meet a limit of the form “small change in f over small change in input,” the answer is simply $f'(a)$. The six limits below are this formula applied at chosen points.

The six standard limits, derived

L1. $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

Take a sector of the unit circle with angle x , for $0 < x < \frac{\pi}{2}$. Compare three areas: the inner triangle has area $\frac{1}{2} \sin x$, the sector has area $\frac{x}{2}$, and the outer triangle (built on the tangent) has area $\frac{1}{2} \tan x$. Nesting them:

$$\frac{1}{2} \sin x < \frac{x}{2} < \frac{1}{2} \tan x.$$

Divide through by the positive number $\frac{1}{2} \sin x$:

$$1 < \frac{x}{\sin x} < \frac{1}{\cos x} \quad \implies \quad \cos x < \frac{\sin x}{x} < 1.$$

As $x \rightarrow 0^+$, $\cos x \rightarrow 1$, so the squeeze theorem forces $\frac{\sin x}{x} \rightarrow 1$. Replacing x by $-x$ changes nothing (both sides are even), so the two-sided limit holds. ■

L2. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$

Use the half-angle identity $1 - \cos x = 2 \sin^2 \frac{x}{2}$:

$$\frac{1 - \cos x}{x^2} = \frac{2 \sin^2(x/2)}{x^2} = \frac{1}{2} \left(\frac{\sin(x/2)}{x/2} \right)^2 \rightarrow \frac{1}{2} \cdot 1^2 = \frac{1}{2}.$$

■

L3. $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$

This is exactly the difference quotient of $f(x) = e^x$ at $a = 0$:

$$\lim_{h \rightarrow 0} \frac{e^{0+h} - e^0}{h} = f'(0) = e^0 = 1.$$

■

L4. $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$

This is the difference quotient of $f(t) = \ln t$ at $a = 1$, since $\ln(1+x) - \ln 1 = \ln(1+x)$:

$$\lim_{x \rightarrow 0} \frac{\ln(1+x) - \ln 1}{x} = f'(1) = \frac{1}{1} = 1.$$

■

L5. $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a$

Write $a^x = e^{x \ln a}$ and substitute $u = x \ln a$, so $u \rightarrow 0$ as $x \rightarrow 0$:

$$\frac{a^x - 1}{x} = \frac{e^{x \ln a} - 1}{x \ln a} \cdot \ln a \rightarrow 1 \cdot \ln a,$$

by L3 applied to u . ■

L6. $\lim_{x \rightarrow 0} \frac{(1+x)^n - 1}{x} = n$

This is the difference quotient of $f(t) = t^n$ at $a = 1$:

$$\lim_{x \rightarrow 0} \frac{(1+x)^n - 1}{x} = f'(1) = n(1)^{n-1} = n.$$

It works for every real n , not just integers. ■

Shifting the centre. The same principle works away from 0. To evaluate

$$\lim_{x \rightarrow \pi} \frac{\sin x}{x - \pi},$$

substitute $u = x - \pi$, so $u \rightarrow 0$ and $\sin(u + \pi) = -\sin u$:

$$\lim_{u \rightarrow 0} \frac{-\sin u}{u} = -1.$$

Equivalently, the answer is $\cos \pi = -1$: the limit is always the derivative of the numerator's inner function evaluated at the junction point.

On L'Hôpital's rule. At JEE you may differentiate numerator and denominator separately once a limit is genuinely in $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form, and repeat while the form persists. CBSE marking schemes, however, award method marks for standard-limit working, so treat L'Hôpital as a checking device in board answers, not a replacement for L1-L6.

Example A.1 — [CBSE Board Level]

Find k so that the following is continuous at $x = 0$:

$$f(x) = \begin{cases} \frac{\sqrt{1+kx} - \sqrt{1-kx}}{x}, & -1 \leq x < 0 \\ \frac{2x+1}{x-2}, & 0 \leq x \leq 1 \end{cases}$$

RHL. The second piece is a rational function, continuous at 0:

$$\lim_{x \rightarrow 0^+} \frac{2x+1}{x-2} = \frac{1}{-2} = -\frac{1}{2}.$$

LHL. Substitution gives $\frac{0}{0}$, so rationalise the numerator — multiply top and bottom by $\sqrt{1+kx} + \sqrt{1-kx}$:

$$\frac{\sqrt{1+kx} - \sqrt{1-kx}}{x} = \frac{(1+kx) - (1-kx)}{x(\sqrt{1+kx} + \sqrt{1-kx})} = \frac{2k}{\sqrt{1+kx} + \sqrt{1-kx}}.$$

Now the limit is safe to read off:

$$\lim_{x \rightarrow 0^-} \frac{2k}{\sqrt{1+kx} + \sqrt{1-kx}} = \frac{2k}{1+1} = k.$$

Impose continuity: $k = -\frac{1}{2}$.

Check. With $k = -\frac{1}{2}$ and $x = -0.01$: the numerator is $\sqrt{1.005} - \sqrt{0.995} \approx 0.0050$, and dividing by -0.01 gives approximately -0.50 , agreeing with the RHL.

Example A.2 — [JEE Main Level]

Find k so that $f(x) = \frac{e^{3x} - 1}{\ln(1 + 2x)}$ for $x \neq 0$, $f(0) = k$, is continuous at 0.

Direct substitution gives $\frac{0}{0}$. Split the fraction so each piece matches a standard limit:

$$\frac{e^{3x} - 1}{\ln(1 + 2x)} = \frac{e^{3x} - 1}{3x} \cdot \frac{2x}{\ln(1 + 2x)} \cdot \frac{3x}{2x}.$$

The first factor tends to 1 by L3, the second tends to 1 by L4, and the third is the constant $\frac{3}{2}$. Hence

$$\lim_{x \rightarrow 0} f(x) = \frac{3}{2}, \quad \text{so} \quad k = \frac{3}{2}.$$

Check. For $x = 0.001$: $e^{0.003} - 1 \approx 0.0030045$ and $\ln(1.002) \approx 0.001998$; the quotient is approximately 1.5037, drifting toward 1.5.

Example A.3 — [JEE Advanced Level]

Evaluate $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$.

Factor the numerator rather than expanding anything:

$$\tan x - \sin x = \tan x (1 - \cos x) = \frac{\sin x}{\cos x} (1 - \cos x).$$

Therefore

$$\frac{\tan x - \sin x}{x^3} = \frac{\sin x}{x} \cdot \frac{1}{\cos x} \cdot \frac{1 - \cos x}{x^2}.$$

Each factor now has a known limit, by L1, continuity of $\cos$ at 0, and L2:

$$\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3} = 1 \cdot 1 \cdot \frac{1}{2} = \frac{1}{2}.$$

Why this beats L'Hôpital here. Applying L'Hôpital three times to this expression produces progressively messier trigonometric quotients and two opportunities for sign slips. The factorisation route is three lines and every step is verifiable against L1 and L2.

Practice — Addendum A

1. Find k so that $f(x) = \frac{\sin 5x}{3x}$ for $x \neq 0$, $f(0) = k$, is continuous at 0.
2. Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$.
3. Evaluate $\lim_{x \rightarrow \pi} \frac{\sin x}{x - \pi}$.
4. Evaluate $\lim_{x \rightarrow 0} \frac{a^x - b^x}{x}$ for $a, b > 0$.
5. Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x \sin x}$.
6. Evaluate $\lim_{x \rightarrow 0} \frac{\ln(1+3x)}{e^{2x} - 1}$.

Answers: 1. $\frac{\sin 5x}{3x} = \frac{5}{3} \cdot \frac{\sin 5x}{5x} \rightarrow \frac{5}{3}$, so $k = \frac{5}{3}$. 2. Rationalise: $\frac{x}{x(\sqrt{1+x}+1)} \rightarrow \frac{1}{2}$ (equivalently L6 with $n = \frac{1}{2}$). 3. Substitute $u = x - \pi$: $\frac{-\sin u}{u} \rightarrow -1$. 4. Split as $\frac{a^x-1}{x} - \frac{b^x-1}{x} \rightarrow \ln a - \ln b = \ln \frac{a}{b}$ by L5. 5. $1 - \cos 2x = 2 \sin^2 x$, so the expression is $2 \cdot \frac{\sin x}{x} \rightarrow 2$. 6. $\frac{\ln(1+3x)}{3x} \cdot \frac{2x}{e^{2x}-1} \cdot \frac{3x}{2x} \rightarrow 1 \cdot 1 \cdot \frac{3}{2} = \frac{3}{2}$.

Addendum B — Differentiating Through Moduli: the General Rules

Example 3 of Section 1 gives you a counting rule for corners of $|g(x)|$ and Pitfall 3 warns you how to misuse it — but neither is proved, and neither tells you what happens when moduli are *multiplied*, *nested*, or *raised to powers*. JEE Main asks for the exact number of non-differentiability points of such expressions; JEE Advanced asks whether a specific point survives. Three lemmas settle every case.

Lemma B.1 — Away from zeros, $|f|$ inherits smoothness

Claim. If f is differentiable at c and $f(c) \neq 0$, then $|f|$ is differentiable at c , with derivative $f'(c)$ if $f(c) > 0$ and $-f'(c)$ if $f(c) < 0$.

Proof. Suppose $f(c) > 0$. By continuity of f at c (which differentiability provides), taking $\varepsilon = \frac{f(c)}{2}$ gives a neighbourhood of c in which

$$|f(x) - f(c)| < \frac{f(c)}{2} \implies f(x) > \frac{f(c)}{2} > 0.$$

Throughout that neighbourhood $|f| = f$, so $|f|$ is differentiable at c with derivative $f'(c)$. The case $f(c) < 0$ is identical with $|f| = -f$. ■

Consequence. The *only* possible non-differentiability points of $|f|$ are the zeros of f . Everything else is settled without computation.

Lemma B.2 — At a zero of f , the corner test

Claim. Let f be differentiable at c with $f(c) = 0$. Then $|f|$ is differentiable at c if and only if $f'(c) = 0$; and when it is differentiable, $|f|'(c) = 0$.

Proof. Compute the two one-sided derivatives of $|f|$ at c . For the right-hand derivative, use $|f(c)| = 0$ and $|f(c+h)| = |f(c+h) - f(c)|$:

$$\lim_{h \rightarrow 0^+} \frac{|f(c+h)| - |f(c)|}{h} = \lim_{h \rightarrow 0^+} \left| \frac{f(c+h) - f(c)}{h} \right| = |f'(c)|.$$

For the left-hand derivative, the denominator $-h$ is negative, so it pulls out a minus sign:

$$\lim_{h \rightarrow 0^+} \frac{|f(c-h)| - |f(c)|}{-h} = - \lim_{h \rightarrow 0^+} \left| \frac{f(c-h) - f(c)}{h} \right| = -|f'(c)|.$$

Both one-sided limits always exist (they are magnitudes of the same number $f'(c)$), so $|f|$ fails at c only by *corner*, never by blow-up. They agree exactly when $|f'(c)| = -|f'(c)|$, i.e. $f'(c) = 0$, in which case the common value is 0. ■

Corollary (proving the chapter's counting rule). The corners of $|f|$ are exactly the zeros of f at which $f'(c) \neq 0$ — that is, the points where f crosses the axis. A zero where f merely touches the axis ($f'(c) = 0$, e.g. the vertex of $x^2 - 4$ at $x = 0$) is *not* a corner, because $|f|$ flattens to order 2 there and Lemma B.2 certifies differentiability.

Lemma B.3 — Powers of the modulus: the smoothness ladder

Consider $g_n(x) = |x|^n$ at $x = 0$.

- $n = 1$: $g_1 = |x|$, not differentiable at 0 (LHD -1 , RHD $+1$).
- $n = 2$: $g_2 = x^2$, a polynomial — infinitely differentiable.
- $n = 3$: for $h \neq 0$,

$$\frac{g_3(h) - g_3(0)}{h} = \frac{|h|^3}{h} = h|h| \rightarrow 0,$$

so $g'_3(0) = 0$, and indeed $g'_3(x) = 3x|x|$ for all x . Differentiate once more at 0:

$$g''_3(0) = \lim_{h \rightarrow 0} \frac{g'_3(h) - g'_3(0)}{h} = \lim_{h \rightarrow 0} \frac{3h|h|}{h} = \lim_{h \rightarrow 0} 3|h| = 0.$$

But $g''_3(x) = 6|x|$ for $x \neq 0$, whose one-sided limits at 0 are -6 and $+6$: the third derivative fails. So $|x|^3$ is twice differentiable everywhere but not three times.

In general, $|x|^n$ is differentiable exactly $n - 1$ times at the origin. Each differentiation strips one power of $|x|$, and the process halts when a lone $|x|$ remains.

Companion rule — $f(|x|)$ at the origin

If f is differentiable at 0, then $f(|x|)$ is differentiable at 0 if and only if $f'(0) = 0$. Indeed,

$$\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} = f'(0), \quad \lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{-h} = -f'(0),$$

and these agree exactly when $f'(0) = 0$. Geometrically: reflecting the graph of f into an even function creates a corner at the fold unless the graph arrives flat.

Example B.1 — [JEE Main Level]

Examine the differentiability of $f(x) = |x|(x - 1)$ at $x = 0$ and $x = 1$.

At $x = 1$: the factor $|x|$ is differentiable at 1 with value $1 \neq 0$ (Lemma B.1), and $x - 1$ is a polynomial, so the product is differentiable at 1. Nothing to compute.

At $x = 0$: Lemma B.1 does not apply since $|0| = 0$. Work from the piecewise form. For $h > 0$:

$$f(h) = h(h - 1) = h^2 - h, \quad f(-h) = |-h|(-h - 1) = h(-h - 1) = -h - h^2.$$

Then

$$\text{RHD} = \lim_{h \rightarrow 0^+} \frac{h^2 - h}{h} = \lim_{h \rightarrow 0^+} (h - 1) = -1,$$

$$\text{LHD} = \lim_{h \rightarrow 0^+} \frac{-h - h^2}{-h} = \lim_{h \rightarrow 0^+} (1 + h) = 1.$$

Since $-1 \neq 1$, f is not differentiable at 0.

Conclusion: the sole point of non-differentiability is $x = 0$.

Check. The modulus contributes a corner candidate only at 0; multiplying by the smooth, non-vanishing factor $(x - 1)$ cannot remove it (near 0 the factor is close to -1 , so it rescales the V but does not flatten it) — consistent with the computation.

Example B.2 — [JEE Advanced Level]

Find all points where $f(x) = ||x| - 1|$ fails to be differentiable.

Peel the nesting from the inside out, applying Lemma B.1 then B.2 at each layer.

Inner layer $\varphi(x) = |x| - 1$: differentiable everywhere except $x = 0$.

Outer layer $f = |\varphi|$: by Lemma B.1, candidates are only the zeros of φ , namely $|x| = 1$, i.e. $x = \pm 1$. At those points $\varphi'(x) = \pm 1 \neq 0$, so by Lemma B.2 both are genuine corners.

What about $x = 0$, the inner corner? Here $\varphi(0) = -1 \neq 0$, so by Lemma B.1 the outer modulus is locally inert: for $|x| < 1$ we have $\varphi < 0$, hence

$$f(x) = -\varphi(x) = 1 - |x|,$$

which still has the inner corner at 0. Nesting does not heal a corner sitting away from the outer zeros.

Full piecewise confirmation.

$$f(x) = \begin{cases} x - 1, & x \geq 1 \\ 1 - x, & 0 \leq x < 1 \\ 1 + x, & -1 < x < 0 \\ -x - 1, & x \leq -1 \end{cases}$$

The slopes across the joins are $1 \rightarrow -1$ at $x = 1$, $-1 \rightarrow 1$ at $x = 0$, $1 \rightarrow -1$ at $x = -1$: three corners.

Conclusion: f is non-differentiable at exactly $x = -1, 0, 1$.

Example B.3 — [JEE Main — Speed]

How many points of non-differentiability does $f(x) = |\sin x| + |\cos x|$ have in $(0, 2\pi)$?

Candidates are the zeros of $\sin$ and $\cos$ in the interval: $\frac{\pi}{2}, \pi, \frac{3\pi}{2}$. At each of these, exactly one summand has a zero (with derivative $\pm 1 \neq 0$, so a corner by Lemma B.2) while the other summand is smooth and non-zero there (Lemma B.1). A corner plus a smooth term cannot cancel, so all three survive.

Answer: 3 points: $x = \frac{\pi}{2}, \pi, \frac{3\pi}{2}$.

Practice — Addendum B

1. Show that $f(x) = x|x|$ is differentiable at 0 and find $f'(0)$.
2. Find all points of non-differentiability of $|x^2 - 4|$.
3. Find all points of non-differentiability of $|x - 1| + |x - 2| + |x - 3|$.
4. Is $g(x) = |x|^3$ twice differentiable at 0? Is it three times differentiable?
5. Show that $f(x) = (\sin x) |\sin x|$ is differentiable at every real x .
6. Is $h(x) = \cos |x|$ differentiable at 0?

Answers: 1. For $h > 0$: $\frac{f(h)-f(0)}{h} = \frac{h^2}{h} = h \rightarrow 0$ and $\frac{f(-h)-f(0)}{-h} = \frac{-h^2}{-h} = h \rightarrow 0$. Differentiable with $f'(0) = 0$ — the factor x flattens the V, exactly as Lemma B.2 predicts with $f(x) = x^2$. 2. Only $x = \pm 2$: the zeros of the inside, where its derivative $\pm 4 \neq 0$. The vertex $x = 0$ is *not* included (inside stays negative there). 3. Exactly $x = 1, 2, 3$: at each join only one summand kinks, so no cancellation is possible. 4. Twice differentiable, with $g''(0) = 0$ (shown in Lemma B.3); not three times differentiable, since $g''(x) = 6|x|$ has a corner at 0. 5. Near any zero of $\sin x$, the function equals $\pm \sin^2 x$, both smooth with derivative 0 at the zero; elsewhere it is a product of smooth functions. Differentiable everywhere. 6. Yes. $\cos |x| = \cos x$ for every real x , since cosine is even — the fold is invisible. A student who applies the companion rule blindly answers “no” and is wrong; the rule needs $f'(0) \neq 0$ to predict a corner, and here $f'(0) = -\sin 0 = 0$.

Addendum C — Functional Equations: Continuity Propagated from a Single Point

The chapter's Exam Relevance note promises "functional equations" at JEE Advanced and delivers none. The genre is rigidly formulaic: a function satisfying an identity for *all* inputs, known to be continuous at *one* point, and you must extract its explicit form. Two master theorems cover nearly every instance, and both are proved with nothing beyond the chapter's own material.

Theorem C.1 — Additive functions (Cauchy's equation)

Claim. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy

$$f(x + y) = f(x) + f(y) \quad \text{for all real } x, y,$$

and suppose f is continuous at even one point x_0 . Then f is continuous on all of $\mathbb{R}$, and

$$f(x) = cx \quad \text{where } c = f(1).$$

Proof.

Step 1 — the origin. Put $x = y = 0$: $f(0) = f(0) + f(0)$, so $f(0) = 0$.

Step 2 — continuity at 0, harvested from x_0 . Since f is continuous at x_0 ,

$$\lim_{h \rightarrow 0} f(x_0 + h) = f(x_0) \quad \implies \quad \lim_{h \rightarrow 0} [f(x_0 + h) - f(x_0)] = 0.$$

By the functional equation the bracket equals $f(h)$, so $\lim_{h \rightarrow 0} f(h) = 0 = f(0)$: continuity at 0.

Step 3 — continuity everywhere. For any fixed x ,

$$\lim_{h \rightarrow 0} f(x + h) = \lim_{h \rightarrow 0} [f(x) + f(h)] = f(x) + 0 = f(x).$$

Step 4 — integers. By induction, $f(n) = nf(1)$ for positive integers n . Also $f(n) + f(-n) = f(0) = 0$, so $f(-n) = -f(n)$.

Step 5 — rationals. Writing $1 = n \cdot \frac{1}{n}$ and applying Step 4 backwards: $f(1) = nf(\frac{1}{n})$, so $f(\frac{1}{n}) = \frac{f(1)}{n}$. Hence for any rational $\frac{m}{n}$,

$$f\left(\frac{m}{n}\right) = mf\left(\frac{1}{n}\right) = \frac{m}{n}f(1).$$

Step 6 — reals, by continuity. Take any real x and any sequence of rationals $q_k \rightarrow x$. By Step 5, $f(q_k) = cq_k \rightarrow cx$; by Step 3, $f(q_k) \rightarrow f(x)$. Uniqueness of limits gives $f(x) = cx$. ■

Remark. Without the continuity hypothesis the conclusion genuinely fails — pathological non-linear additive functions exist (they are wildly discontinuous everywhere). The single-

point continuity assumption is doing all the work; examiners know it and will strip it to build traps.

Theorem C.2 — Multiplicative functions (the exponential equation)

Claim. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy

$$f(x+y) = f(x)f(y) \quad \text{for all real } x, y,$$

with f continuous at some point and f not identically zero. Then $f(0) = 1$, $f(x) > 0$ for all x , and

$$f(x) = a^x \quad \text{where } a = f(1) > 0.$$

Proof.

Step 1 — the origin. Put $y = 0$: $f(x) = f(x)f(0)$ for all x . Since f is not identically zero, some x has $f(x) \neq 0$, forcing $f(0) = 1$.

Step 2 — positivity. First, $f(x) = f(\frac{x}{2} + \frac{x}{2}) = f(\frac{x}{2})^2 \geq 0$. Second, $f(x)f(-x) = f(0) = 1$, so $f(x) \neq 0$ for every x . Together: $f(x) > 0$ everywhere.

Step 3 — reduce to Cauchy. Define $g(x) = \ln f(x)$. Since f is continuous (at one point, hence everywhere by the same propagation as Step 2–3 above — the identity lets continuity travel) and positive, g is well defined and continuous, and

$$g(x+y) = \ln[f(x)f(y)] = \ln f(x) + \ln f(y) = g(x) + g(y).$$

By Theorem C.1, $g(x) = cx$ with $c = g(1) = \ln f(1)$.

Step 4 — conclude. $f(x) = e^{g(x)} = e^{cx} = (e^c)^x = a^x$ with $a = f(1)$. ■

Differentiable shortcut. If additionally f is differentiable at 0, differentiate $f(x+y) = f(x)f(y)$ with respect to y and set $y = 0$:

$$f'(x) = f(x)f'(0).$$

Solving: $\frac{d}{dx} \ln f(x) = f'(0)$, so $\ln f(x) = f'(0)x + \text{constant}$, and the constant is forced by $f(0) = 1$ to be 0. Same answer, one line less.

Example C.1 — [JEE Main Level]

Suppose $f(x+y) = f(x) + f(y)$ **for all reals, f is continuous at 0, and $f(2) = 10$. Find $f(x)$, $f(-5)$, and $\lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h}$.**

By Theorem C.1, $f(x) = cx$. Now $f(2) = 2c = 10$, so $c = 5$ and

$$f(x) = 5x, \quad f(-5) = -25.$$

For the limit, Theorem C.1 gives continuity everywhere, so the difference quotient is the honest derivative of the line $5x$:

$$\lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} = f'(3) = 5.$$

Notice the logic: the functional equation plus one continuity seed produced an explicit function, after which calculus questions become trivial.

Example C.2 — [JEE Advanced Level]

Let $f(x+y) = f(x)f(y)$ for all reals, with f differentiable at 0, $f'(0) = k \neq 0$, and $f(1) = 2$. Show that $f(x) = 2^x$ and that $f'(x) = f(x) \ln 2$.

First, f is not identically zero (it takes the value 2), so by Theorem C.2, $f(0) = 1$ and $f(x) > 0$. Fix x and differentiate the identity with respect to h in

$$f(x+h) = f(x)f(h),$$

then let $h \rightarrow 0$:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f(x) \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = f(x) f'(0) = k f(x).$$

So $\frac{d}{dx} [\ln f(x)] = k$, giving $\ln f(x) = kx + C$; substituting $x = 0$ and $f(0) = 1$ forces $C = 0$. Hence

$$f(x) = e^{kx}.$$

Finally $f(1) = e^k = 2$, so $k = \ln 2$, and

$$f(x) = e^{x \ln 2} = 2^x, \quad f'(x) = 2^x \ln 2 = f(x) \ln 2.$$

Check. $2^{x+y} = 2^x 2^y$ verifies the original identity, and $f'(0) = \ln 2 = k$ verifies the data.

Example C.3 — [JEE Advanced Level]

Let $f : (0, \infty) \rightarrow \mathbb{R}$ satisfy $f(xy) = f(x) + f(y)$ for all $x, y > 0$, with f continuous at 1 and $f(e) = 1$. Find f .

Convert to the additive equation by reparametrising. Define $g(t) = f(e^t)$ for $t \in \mathbb{R}$. Then

$$g(s+t) = f(e^{s+t}) = f(e^s e^t) = f(e^s) + f(e^t) = g(s) + g(t),$$

and g is continuous wherever f is, in particular at $t = 0$ (corresponding to $x = 1$). By Theorem C.1, $g(t) = ct$ with $c = g(1) = f(e) = 1$. Therefore

$$f(x) = g(\ln x) = \ln x.$$

Check. $\ln(xy) = \ln x + \ln y$ and $\ln e = 1$: both conditions hold.

Practice — Addendum C

1. f is additive and continuous, with $f(1) = 7$. Find $f(2024)$ and $f(-3)$.
2. f is additive and continuous, and $f'(1) = 4$. Find $f(x)$.
3. $f(x+y) = f(x)f(y)$, f is continuous, $f(x) > 0$ for all x , and $f(2) = 9$. Find $f(x)$ and $f(\frac{1}{2})$.
4. $f(xy) = f(x)f(y)$ for all $x, y > 0$, f continuous at 1, and $f(4) = 16$. Find $f(x)$.
5. True or false: an additive function continuous at exactly one point must be of the form $f(x) = cx$.

Answers: 1. $f(x) = 7x$; $f(2024) = 14168$; $f(-3) = -21$. 2. For additive f , $f'(x) = \lim_{h \rightarrow 0} \frac{f(h)}{h}$ is independent of x , so $f' \equiv c$ with $c = f'(1) = 4$; integrating and using $f(0) = 0$: $f(x) = 4x$. 3. $f(2) = f(1)^2$ gives $f(1) = 3$ (positive branch); by Theorem C.2, $f(x) = 3^x$, and $f(\frac{1}{2}) = \sqrt{3}$. 4. Set $g(t) = \ln f(e^t)$: additive and continuous, so $g(t) = ct$; hence $f(x) = x^c$, and $4^c = 16$ gives $c = 2$, i.e. $f(x) = x^2$. 5. True — continuity at even one point propagates to all of $\mathbb{R}$ (Theorem C.1, Step 2), so “exactly one point” collapses to “everywhere,” forcing linearity.

Addendum D — Cauchy’s Mean Value Theorem and the Arithmetic of Roots

The chapter states Rolle and Lagrange but omits the theorem that contains them both. Cauchy’s Mean Value Theorem (CMVT) is standard JEE Advanced fare — it appears both as a direct verification question and as the engine behind inequalities and root-counting arguments that Lagrange cannot reach. Alongside it, this addendum systematises the root-counting consequence of Rolle that the chapter’s MCQ Q4 uses once and abandons.

Cauchy’s Mean Value Theorem

Statement. Let f and g be continuous on $[a, b]$ and differentiable on (a, b) , with $g'(x) \neq 0$ for all $x \in (a, b)$. Then there exists $c \in (a, b)$ such that

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Why the denominator is safe. If $g(b) = g(a)$, Rolle’s theorem applied to g would give some point with $g' = 0$, contradicting the hypothesis. So $g(b) - g(a) \neq 0$ and the fraction is defined.

Derivation from Rolle. Define the auxiliary function

$$F(x) = f(x) - \lambda g(x), \quad \lambda = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

F is continuous on $[a, b]$ and differentiable on (a, b) , being a combination of such functions. At the endpoints:

$$F(b) - F(a) = [f(b) - f(a)] - \lambda[g(b) - g(a)] = 0$$

by the choice of λ , so $F(a) = F(b)$. All three Rolle hypotheses hold, so there exists $c \in (a, b)$ with

$$F'(c) = f'(c) - \lambda g'(c) = 0.$$

Since $g'(c) \neq 0$, divide:

$$\frac{f'(c)}{g'(c)} = \lambda = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

■

Recovering Lagrange. Take $g(x) = x$: then $g'(c) = 1$ and the conclusion reads $f'(c) = \frac{f(b)-f(a)}{b-a}$. Lagrange is CMVT with the simplest possible companion — one more reason to learn the general form.

The Rolle cascade — counting roots

Lemma. If f is continuous on $[p, q]$, differentiable on (p, q) , and $f(p) = f(q) = 0$, then f' has at least one zero strictly between p and q .

This is Rolle restated, but its systematic use is a counting machine:

- Two roots of f force a root of f' between them.
- Consequently, if f' has **no** zeros in an interval, f has **at most one** root there (two roots would manufacture a zero of f').
- For a polynomial p of degree n with n distinct real roots, p' has at least $n - 1$ distinct real roots — one in each gap — and being of degree $n - 1$, exactly one per gap.

Example D.1 — [JEE Main Level]

Verify CMVT for $f(x) = x^2$ and $g(x) = x^3$ on $[1, 2]$, and find c .

Hypotheses: polynomials are continuous and differentiable everywhere; $g'(x) = 3x^2 \neq 0$ on $(1, 2)$. Compute both sides:

$$\frac{f(2) - f(1)}{g(2) - g(1)} = \frac{4 - 1}{8 - 1} = \frac{3}{7}, \quad \frac{f'(c)}{g'(c)} = \frac{2c}{3c^2} = \frac{2}{3c}.$$

Set equal and solve:

$$\frac{2}{3c} = \frac{3}{7} \quad \Rightarrow \quad 14 = 9c \quad \Rightarrow \quad c = \frac{14}{9}.$$

Since $\frac{14}{9} \approx 1.56$ lies strictly inside $(1, 2)$, CMVT is verified.

Check. $\frac{2}{3 \cdot \frac{14}{9}} = \frac{2}{\frac{14}{3}} = \frac{6}{14} = \frac{3}{7}$. ✓

Example D.2 — [JEE Advanced Level]

Suppose f and g are continuous on $[0, \infty)$, differentiable on $(0, \infty)$, with $f(0) = g(0)$ and $f'(x) > g'(x)$ for all $x > 0$. Prove that $f(x) > g(x)$ for all $x > 0$, and deduce that $e^x > 1 + x$ for $x > 0$.

Let $h = f - g$. Then h is continuous on $[0, x]$ for any $x > 0$, differentiable on $(0, x)$, and $h(0) = f(0) - g(0) = 0$. Lagrange's MVT on $[0, x]$ gives some $c \in (0, x)$ with

$$h(x) = h(x) - h(0) = h'(c)(x - 0) = [f'(c) - g'(c)]x > 0,$$

since both factors are positive. Hence $f(x) > g(x)$ for every $x > 0$. ■

Deduction. Take $f(x) = e^x$ and $g(x) = 1 + x$. Then $f(0) = g(0) = 1$ and $f'(x) = e^x > 1 = g'(x)$ for $x > 0$. The theorem immediately gives $e^x > 1 + x$ for all $x > 0$ — no series, no graphs.

Remark. This “compare the derivatives, compare the functions” pattern is the standard rigorous weapon for inequality questions at JEE Advanced, and it is precisely the argument the chapter's Example 4 gestures at with logarithms.

Example D.3 — [JEE Advanced Level]

(a) The polynomial $p(x) = (x - 1)(x - 2)(x - 3)(x - 4)$ has four distinct real roots. Show that p' has exactly one real root in each of the intervals $(1, 2)$, $(2, 3)$, $(3, 4)$.

(b) Show that the equation $2x^3 - 3x^2 + 6x + 6 = 0$ has exactly one real root.

(a) p' is a cubic, so it has at most three real roots. Applying Rolle to p on each of the three gaps $[1, 2]$, $[2, 3]$, $[3, 4]$ (where p vanishes at both ends of each), p' has at least one root in each open gap — at least three in total. At most three meets at least three: exactly one per gap. ■

(b) Let $q(x) = 2x^3 - 3x^2 + 6x + 6$. Its derivative is

$$q'(x) = 6x^2 - 6x + 6 = 6(x^2 - x + 1),$$

whose discriminant is $(-1)^2 - 4 = -3 < 0$. So $q'(x) > 0$ for all real x : q is strictly increasing and can cross the axis at most once (two roots would force a zero of q' by Rolle). For existence, evaluate:

$$q(-1) = -2 - 3 - 6 + 6 = -5 < 0, \quad q(0) = 6 > 0.$$

By the intermediate value theorem (valid since q is continuous), a root exists in $(-1, 0)$. Combined with uniqueness: **exactly one real root.** ■

The template. Uniqueness comes from the sign of the derivative (via Rolle); existence comes

from two opposite-signed samples (via continuity). Examiners award marks for separating the two jobs — students who merge them usually prove neither.

Practice — Addendum D

1. Verify CMVT for $f(x) = \sin x$, $g(x) = \cos x$ on $[0, \frac{\pi}{2}]$, and find c .
2. Using Lagrange's MVT on $f(t) = \ln(1+t)$ over $[0, x]$, prove that $\ln(1+x) > \frac{x}{1+x}$ for $x > 0$.
3. Show that $x^3 + x - 1 = 0$ has exactly one real root.
4. A polynomial p of degree 5 has five distinct real roots. What is the minimum number of distinct real roots of p'' , and where do they lie?
5. If f is continuous on $[a, b]$ and $f'(x) > 0$ on (a, b) , show f has at most one zero in $[a, b]$.
6. Verify CMVT for $f(x) = x$, $g(x) = x^2$ on $[0, 1]$, and find c .

Answers: 1. $\frac{f(\pi/2)-f(0)}{g(\pi/2)-g(0)} = \frac{1-0}{0-1} = -1$; and $\frac{f'(c)}{g'(c)} = \frac{\cos c}{-\sin c} = -\cot c$. Setting $-\cot c = -1$ gives $c = \frac{\pi}{4} \in (0, \frac{\pi}{2})$. Verified. 2. MVT gives $c \in (0, x)$ with $\frac{\ln(1+x)}{x} = \frac{1}{1+c}$. Since $0 < c < x$, reciprocating gives $\frac{1}{1+c} > \frac{1}{1+x}$, hence $\ln(1+x) > \frac{x}{1+x}$. 3. $q'(x) = 3x^2 + 1 > 0$ always, so at most one root; $q(0) = -1 < 0 < q(1) = 1$ gives existence in $(0, 1)$. Exactly one. 4. At least 3 distinct real roots: Rolle applied to p gives p' at least 4 distinct roots (one per gap), and Rolle applied to p' between consecutive roots of p' gives p'' at least 3, each lying between two roots of p' . 5. Contrapositive of Rolle: two distinct zeros $p < q$ in $[a, b]$ would force some $c \in (p, q)$ with $f'(c) = 0$, contradicting $f' > 0$. 6. $\frac{f(1)-f(0)}{g(1)-g(0)} = \frac{1}{1} = 1$; $\frac{f'(c)}{g'(c)} = \frac{1}{2c} = 1$ gives $c = \frac{1}{2} \in (0, 1)$. Verified.

Addendum E — Greatest Integer Compositions: Counting Discontinuities Correctly

The chapter asserts that $\lfloor x \rfloor$ “jumps at every integer” and leaves it there. But JEE Main’s actual question is almost never about $\lfloor x \rfloor$ itself — it is about $\lfloor g(x) \rfloor$ for a continuous g , and the naive instruction “count where g hits an integer” is *wrong*: sometimes g hitting an integer causes no jump at all. This addendum gives the correct criterion, with the counterexample that punishes the naive version.

The criterion

Let g be continuous and $f(x) = \lfloor g(x) \rfloor$. Fix a point c and write $m = \lfloor g(c) \rfloor$.

Case 1: $g(c)$ is not an integer. Then $m < g(c) < m + 1$, and by continuity of g there is a neighbourhood of c where g stays inside $(m, m + 1)$. There $\lfloor g(x) \rfloor = m$ constantly, so f is continuous at c .

Case 2: $g(c) = m$, an integer. Two sub-cases, and the asymmetry is the whole game:

- If every punctured neighbourhood of c contains points with $g(x) < m$, then along those points $\lfloor g(x) \rfloor \leq m - 1$ while $f(c) = m$: f is **discontinuous** at c .
- If $g(x) \geq m$ throughout some neighbourhood of c , then near c all values lie in $[m, m + 1)$,

so $\lfloor g(x) \rfloor = m = f(c)$: f is **continuous** at c .

Reading the asymmetry. The floor function rounds *down*. So a dip below the integer level breaks continuity, while hovering above it does not. A local *maximum* of g sitting exactly on an integer is a discontinuity (everything nearby is strictly lower); a local *minimum* sitting exactly on an integer is not (everything nearby is strictly higher, and floors agree). Crossing the level transversally counts as the first sub-case, since one side dips below.

Monotonic shortcut. If g is continuous and strictly monotonic on an interval, every solution of $g(x) = \text{integer}$ is a genuine crossing, so the discontinuities of $\lfloor g \rfloor$ are exactly the solutions of $g(x) = n$ over the integers n in the range of g .

Example E.1 — [JEE Main Level]

Find the number of points of discontinuity of $f(x) = \lfloor 2 \sin x \rfloor$ in $(0, 2\pi)$.

Here $g(x) = 2 \sin x$ ranges over $[-2, 2]$, so the integer levels to test are $m = -2, -1, 0, 1, 2$. Solve $2 \sin x = m$ inside $(0, 2\pi)$ and classify each solution by the criterion:

Level m	Solutions in $(0, 2\pi)$	Behaviour at the point	Verdict
2	$x = \frac{\pi}{2}$	local maximum touching level 2; neighbours strictly below	discontinuous
1	$\frac{\pi}{6}, \frac{5\pi}{6}$	transversal crossing	discontinuous
0	$x = \pi$	transversal crossing	discontinuous
-1	$\frac{7\pi}{6}, \frac{11\pi}{6}$	transversal crossing	discontinuous
-2	$x = \frac{3\pi}{2}$	local minimum touching level -2; neighbours strictly above	continuous

The last row deserves the check: for x near $\frac{3\pi}{2}$ (but not equal), $-2 < 2 \sin x < -1$, so $\lfloor 2 \sin x \rfloor = -2 = f(\frac{3\pi}{2})$ throughout the neighbourhood — no jump. Meanwhile at $\frac{\pi}{2}$, values nearby lie in $(1, 2)$ with floor 1, while $f(\frac{\pi}{2}) = \lfloor 2 \rfloor = 2$: a jump down.

Answer: 6 points of discontinuity:

$$x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \pi, \frac{7\pi}{6}, \frac{11\pi}{6}.$$

A student who counts all seven solutions of $2 \sin x = \text{integer}$ gets 7 and loses the mark — the minimum-touch at $\frac{3\pi}{2}$ is the designed trap.

Example E.2 — [JEE Main — Trap]

Show that $f(x) = \lfloor x^2 \rfloor$ is continuous at $x = 0$ but discontinuous at $x = 1$.

At $x = 0$: here $g(x) = x^2$ attains the integer level 0 at a strict local *minimum*. For $|x| < 1$ we have $0 \leq x^2 < 1$, so

$$\lfloor x^2 \rfloor = 0 = f(0)$$

throughout the whole neighbourhood. Continuous — despite x^2 hitting an integer dead-on. This is the counterexample to “discontinuities occur where g hits an integer.”

At $x = 1$: x^2 crosses the level 1 transversally. From the left, $x^2 \in (0, 1)$ so $\lfloor x^2 \rfloor = 0$; from the right, $x^2 \in (1, 2)$ so $\lfloor x^2 \rfloor = 1$; and $f(1) = 1$. The left limit 0 disagrees with $f(1) = 1$: discontinuous.

Moral. Touching an integer from above is harmless; touching from below, or crossing, is fatal. Always locate the point relative to the level before counting.

Practice — Addendum E

1. Find the number of points of discontinuity of $\lfloor \sin x \rfloor$ in $(0, 2\pi)$.
2. Find the number of points of discontinuity of $\lfloor x \rfloor$ in $(-2, 2)$.
3. Find the number of points of discontinuity of $\lfloor 2x + 1 \rfloor$ on $[0, 2)$.
4. Is $\lfloor \cos x \rfloor$ continuous at $x = 0$?
5. Find the number of points of discontinuity of $\lfloor x^2 \rfloor$ in $(-2, 2)$.

Answers: 1. 2 points: $x = \frac{\pi}{2}$ (maximum touching level 1 from below — discontinuous) and $x = \pi$ (crossing level 0). The minimum-touch at $\frac{3\pi}{2}$ (level -1) is continuous, by the same argument as Example E.1. 2. 3 points: $x = -1, 0, 1$ — each an interior crossing of an integer level. 3. 3 points: $2x + 1$ is an integer exactly at $x = \frac{1}{2}, 1, \frac{3}{2}$ inside $[0, 2)$, each a transversal crossing. At the endpoint $x = 0$ the function is right-continuous (values to the right have $2x + 1 > 1$, floors equal to $f(0) = 1$), so it does not count. 4. No. $\cos 0 = 1$ is an integer attained at a local *maximum*, and every punctured neighbourhood has $\cos x < 1$, so the floor drops to 0 nearby while $f(0) = 1$: discontinuous at 0. 5. 6 points: $x^2 = m$ for $m = 1, 2, 3$ gives $x = \pm 1, \pm\sqrt{2}, \pm\sqrt{3}$, all transversal crossings. The level $m = 0$ at $x = 0$ is a minimum-touch and is continuous, as established in Example E.2.

Addendum P — What JEE Actually Asks

1. Distribution of Sub-topics (Evidence from the 43-year Bank)

The bank (Disha Chapter 20) contains **≈ 140 questions** spanning 1979–2020. After discarding hopelessly garbled entries, the following clear patterns emerge.

Sub-topic (Chapter 5 section)	Approx. # of clear questions	Representative years / sessions (where the extract is legible)
Continuity of piecewise functions (parameters, modulus, greatest integer)	25-30	Main: 2020 Sep(I), 2020 Jan(II), 2019 Apr(I,II), 2018 Apr, 2017 Apr, 2015 Apr, 2014 Apr, 2013 Apr; Adv: 2017, 2014, 2012, 1999, 1995, 1991, 1989, 1987, 1983, 1981
Differentiability at a point (LHD/RHD, corners, cusps, max/min, greatest integer)	20-25	Main: 2020 Sep(II), 2020 Jan(I,II), 2019 Apr(I,II), 2018, 2015, 2014, 2012; Adv: 2020, 2019, 2018, 2016, 2015, 2014, 2011, 2008, 2007, 2005, 2002, 2001, 1999, 1998, 1995, 1994, 1988, 1987, 1986, 1985, 1984, 1981
Chain rule & derivatives of composite functions (including inverse trig, log, exponential)	15-20	Main: 2020 Sep(I), 2020 Jan(II), 2019 Apr(I,II), 2018, 2017, 2015, 2014; Adv: 2019, 2018, 2016, 2015, 2014, 2011, 2009, 2008, 2007, 2005, 2004, 2002, 2001, 1994, 1991, 1989, 1986, 1984, 1982, 1981, 1979
Implicit differentiation	5-8	Main: 2019 Apr(I), 2014; Adv: 2014, 2011, 2008, 2005, 2004, 2003, 2002, 1997, 1996, 1995, 1990, 1989, 1985
Inverse trigonometric derivatives (substitution / double-angle simplification)	10-12	Main: 2020 Sep(I), 2019 Apr(II), 2018, 2017, 2015; Adv: 2018, 2016, 2014, 2011, 2009, 2008, 1994, 1993, 1992
Logarithmic differentiation (variable exponent, large products)	8-10	Main: 2020 Sep(II), 2019 Apr(I), 2018, 2017, 2015; Adv: 2018, 2014, 2011, 2008, 2005, 2004, 2001, 1998, 1994, 1989
Parametric differentiation (first and second derivatives)	10-12	Main: 2020 Sep(I), 2020 Jan(II), 2019 Apr(I,II), 2018, 2017; Adv: 2014, 2011, 2008, 2005, 2002, 1998, 1994, 1989, 1985
Second-order derivatives (direct, parametric, implicit, relations like $y''+p(x)y'+q(x)y=0$)	12-15	Main: 2020 Sep(II), 2020 Jan(I), 2019 Apr(I), 2018, 2017, 2015, 2014; Adv: 2018, 2016, 2014, 2011, 2009, 2008, 2007, 2005, 2004, 2002, 2001, 1998, 1994, 1992, 1991, 1989, 1988, 1984, 1983

Sub-topic (Chapter 5 section)	Approx. # of clear questions	Representative years / sessions (where the extract is legible)
Mean Value Theorems (Rolle's, Lagrange - existence of c, inequality proofs, root bounding)	10-12	Main: 2020 Sep(II), 2020 Jan(I,II), 2019 Apr(I), 2018, 2015, 2014; Adv: 2018, 2014, 2011, 2009, 2008, 2007, 2005, 2004, 2002, 1997, 1995, 1990, 1982

Key observation: Continuity/differentiability of piecewise functions (especially with greatest integer and modulus) and parametric/second-order derivatives are the **most heavily tested** in JEE Main. JEE Advanced favours **functional equations, inverse-trig substitutions, MVT-based inequality proofs**, and **subtle differentiability questions** (e.g. derivative exists but is not continuous).

2. Recurring Archetypes (The 7 Question Shapes that Keep Returning)

For each archetype we give: **what it looks like, the method that cracks it**, and **one fully worked representative example** (reconstructed from the bank and re-solved).

Archetype 1 — Piecewise Continuity / Differentiability with Unknown Parameters

What it looks like:

A function defined by different formulas on intervals, with one or more unknown constants. "Find a, b so that f is continuous/differentiable at the junction(s)."

Examples: Main 2020 Sep(I), 2019 Apr(II), 2018, 2015, 2014; Adv 2012, 2008.

Method:

1. Write the continuity condition: $LHL = RHL = f(c)$.
2. If differentiability is also required, first enforce continuity, then set $LHD = RHD$ (finite).
3. Solve the resulting linear equations for the parameters.

Worked Example (reconstructed from Main 2019 Apr(II)):

Let

$$f(x) = \begin{cases} ax + b, & x < 1 \\ x^2 + 3, & x = 1 \\ x^2 + 3x, & x > 1 \end{cases}$$

Find a, b so that f is differentiable at $x = 1$.

Solution:

Continuity at $x = 1$ first:

$$\lim_{x \rightarrow 1^-} (ax + b) = a + b, \quad f(1) = 4, \quad \lim_{x \rightarrow 1^+} (x^2 + 3x) = 4.$$

Continuity $\Rightarrow a + b = 4$.

Differentiability: LHD = a , while RHD = $\frac{d}{dx}(x^2 + 3x)|_{x=1} = (2x + 3)|_{x=1} = 5$.

LHD = RHD $\Rightarrow a = 5$, and then $b = 4 - a = -1$.

Answer: $a = 5$, $b = -1$. With these values LHL = RHL = $f(1) = 4$ and LHD = RHD = 5: continuous and differentiable at $x = 1$.

Second representative (Main 2015 type):

$$f(x) = \begin{cases} kx + 1, & x \leq \pi \\ \cos x, & x > \pi \end{cases}$$

Find k for continuity at $x = \pi$.

Solution:

$$f(\pi) = k\pi + 1.$$

$$\text{LHL} = k\pi + 1, \quad \text{RHL} = \lim_{x \rightarrow \pi^+} \cos x = -1.$$

$$\text{Continuity} \Rightarrow k\pi + 1 = -1 \Rightarrow k = -\frac{2}{\pi}.$$

Differentiability add-on (Main 2014 type):

If we also require differentiability, then after continuity we need

LHD = k , RHD = $-\sin \pi = 0 \Rightarrow k = 0$. But then continuity fails ($0 \cdot \pi + 1 = 1 \neq -1$). Hence no such k exists — a common “no solution” conclusion.

Archetype 2 — Greatest Integer & Modulus Functions: Counting Discontinuities / Non-differentiable Points

What it looks like:

$f(x) = [x]^2 - [x^2]$, $f(x) = |x^2 - 4| + |x - 1|$, $f(x) = [x] \sin(\pi x)$, etc. Asked: “Number of points of discontinuity in (a, b) ” or “Set of points where f is not differentiable.”

Examples: Main 2020 Jan(II), 2019 Apr(II), 2018; Adv 2017, 2014, 2012, 1999, 1995.

Method:

- **Greatest integer** $[g(x)]$: Discontinuous exactly where $g(x)$ crosses an integer. Differentiable nowhere at those jumps; elsewhere derivative is 0.
- **Modulus** $|g(x)|$: Non-differentiable where $g(x) = 0$ **and** g changes sign (simple zero). If g touches zero without sign change (e.g. x^2 at 0), $|g|$ stays smooth.
- **Combinations**: Corners add unless they cancel (same location, opposite slopes). Always check endpoints of the interval — open intervals exclude endpoints.

Worked Example (reconstructed from Main 2019 Apr(II) & Adv 2017):

Find the number of points in $(-2, 3)$ where $f(x) = [x^2] \sin(\pi x)$ is discontinuous.

Solution:

$[x^2]$ jumps where x^2 crosses an integer transversally, i.e. at $x = \pm\sqrt{n}$, $n \in \mathbb{N}$. (At $x = 0$ the level 0 is touched at a minimum: $[x^2] = 0$ on all of $(-1, 1)$, so there is no jump.) In $(-2, 3)$ the

candidates are $x = \pm 1, \pm\sqrt{2}, \pm\sqrt{3}, 2, \sqrt{5}, \sqrt{6}, \sqrt{7}, \sqrt{8}$; the points $x = -2$ and $x = 3$ themselves lie on the excluded boundary.

Multiplication by $\sin(\pi x)$ kills a jump exactly at the integers: if x_0 is an integer then $\sin(\pi x) \rightarrow 0$ as $x \rightarrow x_0$ while $[x^2]$ stays bounded, so both one-sided limits of the product equal $0 = f(x_0)$. For instance at $x = 2$: LHL = $\lim 3 \sin(\pi x) = 0$, RHL = $\lim 4 \sin(\pi x) = 0$, $f(2) = 0$ — continuous; likewise at $x = \pm 1$.

At the non-integer candidates, $\sin(\pi x)$ is continuous and non-zero, so the unit jump of $[x^2]$ survives in the product: f is discontinuous at $x = \pm\sqrt{2}, \pm\sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{7}, \sqrt{8}$.

Answer: 8 points.

Archetype 3 — Functional Equations Leading to Differentiability

What it looks like:

$f(x+y) = f(x) + f(y)$ (Cauchy), $f(x+y) = f(x)f(y)$, $f(x-y) = f(x)g(y) - f(y)g(x)$, etc. Given differentiability at one point (often 0), find derivative everywhere or a specific value.

Examples: Main 2020 Sep(I) ($f(x+y)=f(x)f(y)$), Adv 2011, 1995, 1990, 1981; Main 2005, 2003.

Method:

1. Use the functional equation to find $f(0)$ (usually 0 or 1).
2. Differentiate the equation with respect to one variable, then set that variable to 0.
3. Obtain a differential equation for f (e.g. $f'(x) = f'(0)f(x)$).
4. Solve with the initial condition.

Worked Example (reconstructed from Adv 1990 / Main 2020 Sep(I)):

$f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x+y) = f(x)f(y)$ for all x, y , $f(x) \neq 0$, differentiable at 0 with $f'(0) = 2$. Find $f(x)$ and $f'(3)$.

Solution:

Put $x = y = 0$: $f(0) = f(0)^2 \Rightarrow f(0) = 1$ (since $f(x) \neq 0$).

Differentiate w.r.t y : $f'(x+y) = f(x)f'(y)$. Set $y = 0$: $f'(x) = f(x)f'(0) = 2f(x)$.

Solve: $\frac{f'}{f} = 2 \Rightarrow \ln f(x) = 2x + C \Rightarrow f(x) = Ae^{2x}$. $f(0) = 1 \Rightarrow A = 1$. So $f(x) = e^{2x}$.

Then $f'(x) = 2e^{2x}$, $f'(3) = 2e^6$. (If asked for $f'(3)$ numerically, it's $2e^6$.)

Archetype 4 — Inverse Trigonometric Derivatives via Substitution (Double-Angle Simplification)

What it looks like:

Differentiate $y = \tan^{-1} \frac{2x}{1-x^2}$, $y = \sin^{-1} \frac{2x}{\sqrt{1-x^2}}$, $y = \cos^{-1} \frac{1-x^2}{1+x^2}$, etc. Domain restrictions are crucial.

Examples: Main 2020 Sep(I), 2019 Apr(II), 2018, 2017; Adv 2018, 2016, 2014, 2011.

Method:

1. Identify the pattern: $\frac{2x}{1-x^2}$ suggests $x = \tan \theta$; $\frac{2x}{\sqrt{1-x^2}}$ suggests $x = \sin \theta$; $\frac{1-x^2}{1+x^2}$ suggests $x = \tan \theta$.

2. Substitute, simplify using double-angle formulas, and **state the valid range** of θ (hence x) where the inverse cancels cleanly.
3. Differentiate the simplified expression.
4. Check if the derivative changes sign outside the principal range (piecewise derivative).

Worked Example (reconstructed from Main 2019 Apr(II) / Adv 2014):

Differentiate $y = \sin^{-1} \frac{2x}{1+x^2}$ and state where the formula holds.

Solution:

Let $x = \tan \theta$, $\theta \in (-\pi/2, \pi/2)$. Then $\frac{2x}{1+x^2} = \frac{2 \tan \theta}{1+\tan^2 \theta} = \sin 2\theta$.

So $y = \sin^{-1}(\sin 2\theta)$.

$\sin^{-1}(\sin 2\theta) = 2\theta$ only when $2\theta \in [-\pi/2, \pi/2]$, i.e. $\theta \in [-\pi/4, \pi/4] \Rightarrow x \in [-1, 1]$.

For $x \in [-1, 1]$, $y = 2 \tan^{-1} x$, so $y' = \frac{2}{1+x^2}$.

For $x > 1$, $2\theta \in (\pi/2, \pi)$, so $\sin^{-1}(\sin 2\theta) = \pi - 2\theta = \pi - 2 \tan^{-1} x$, derivative $-\frac{2}{1+x^2}$.

For $x < -1$: $2\theta \in (-\pi, -\pi/2)$, so $\sin^{-1}(\sin 2\theta) = -\pi - 2\theta = -\pi - 2 \tan^{-1} x$, giving derivative $-\frac{2}{1+x^2}$.

Answer: $y' = \frac{2}{1+x^2}$ for $|x| < 1$, and $y' = -\frac{2}{1+x^2}$ for $|x| > 1$. At $x = \pm 1$ derivative does not exist (cusp).

Archetype 5 — Logarithmic Differentiation (Variable Exponent or Complicated Product)

What it looks like:

$$y = x^x, y = (\sin x)^{\tan x}, y = \frac{(x-1)(x-2)}{(x-3)(x-4)}, y = x^{\sin x}.$$

Examples: Main 2020 Sep(II), 2019 Apr(I), 2018, 2017; Adv 2018, 2014, 2011, 2008, 2005, 2004, 2001, 1998, 1994, 1989.

Method:

1. Take $\ln$ of both sides (use $|\cdot|$ if base can be negative).
2. Use log laws to convert powers to products, products to sums.
3. Differentiate implicitly: $\frac{1}{y}y' = (\text{derivative of RHS})$.
4. Multiply by y and substitute back the original y .

Shortcut for u^v : $\frac{d}{dx}(u^v) = u^v \left(v' \ln u + v \frac{u'}{u} \right)$ (exponential piece + power piece).

Worked Example (reconstructed from Main 2018 / Adv 2004):

Differentiate $y = (\tan x)^{\cot x}$.

Solution:

$$\ln y = \cot x \ln(\tan x).$$

Differentiate:

$$\frac{y'}{y} = -\csc^2 x \ln(\tan x) + \cot x \cdot \frac{\sec^2 x}{\tan x}.$$

$$\cot x \cdot \frac{\sec^2 x}{\tan x} = \cot x \cdot \frac{1}{\cos^2 x} \cdot \frac{\cos x}{\sin x} = \cot x \cdot \frac{1}{\sin x \cos x} = \frac{\cos x}{\sin x} \cdot \frac{1}{\sin x \cos x} = \csc^2 x.$$

$$\text{So } \frac{y'}{y} = \csc^2 x (1 - \ln \tan x).$$

$$y' = (\tan x)^{\cot x} \csc^2 x (1 - \ln \tan x).$$

Archetype 6 — Parametric Differentiation (First & Second Derivatives)

What it looks like:

$x = f(t), y = g(t)$. Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. Classic trap: $\frac{d^2y}{dx^2} \neq \frac{d^2y/dt^2}{d^2x/dt^2}$.

Examples: Main 2020 Sep(I), 2020 Jan(II), 2019 Apr(I,II), 2018, 2017; Adv 2014, 2011, 2008, 2005, 2002, 1998, 1994, 1989, 1985.

Method:

1. $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ (provided $dx/dt \neq 0$). Simplify **fully** before substituting any value.
2. $\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \bigg/ \frac{dx}{dt}$.
3. For “value at a point”, find the corresponding t first, then substitute.

Worked Example (reconstructed from Main 2020 Jan(II) / Adv 2011):

$x = 2 \sin \theta - \sin 2\theta$, $y = 2 \cos \theta - \cos 2\theta$. Find $\frac{d^2y}{dx^2}$ at $\theta = \pi$.

Solution:

$$\frac{dx}{d\theta} = 2 \cos \theta - 2 \cos 2\theta, \quad \frac{dy}{d\theta} = -2 \sin \theta + 2 \sin 2\theta.$$

$$\frac{dy}{dx} = \frac{\sin 2\theta - \sin \theta}{\cos \theta - \cos 2\theta}.$$

Sum-to-product: $\sin 2\theta - \sin \theta = 2 \cos \frac{3\theta}{2} \sin \frac{\theta}{2}$ and $\cos \theta - \cos 2\theta = 2 \sin \frac{3\theta}{2} \sin \frac{\theta}{2}$, so the factors $2 \sin \frac{\theta}{2}$ cancel and

$$\frac{dy}{dx} = \cot \frac{3\theta}{2}.$$

At $\theta = \pi$: $\frac{dy}{dx} = \cot \frac{3\pi}{2} = 0$.

For the second derivative, differentiate the slope with respect to θ and divide by $\frac{dx}{d\theta}$:

$\frac{d}{d\theta} \left(\frac{dy}{dx} \right) = -\frac{3}{2} \csc^2 \frac{3\theta}{2}$, which equals $-\frac{3}{2}$ at $\theta = \pi$, while $\frac{dx}{d\theta} \big|_{\theta=\pi} = -2 - 2 = -4$. Hence

$$\frac{d^2y}{dx^2} \bigg|_{\theta=\pi} = \frac{-3/2}{-4} = \frac{3}{8}.$$

Answer: $\frac{dy}{dx} \big|_{\theta=\pi} = 0$ and $\frac{d^2y}{dx^2} \big|_{\theta=\pi} = \frac{3}{8}$. Note the trap in action: $\frac{d^2y}{d\theta^2} \big|_{\theta=\pi} = 6$ and $\frac{d^2x}{d\theta^2} \big|_{\theta=\pi} = 0$, so the naive ratio $\frac{d^2y/d\theta^2}{d^2x/d\theta^2}$ does not even exist here — the slope-first quotient of the Method box is indispensable.

Worked Example (Adv 2011):

If $x = \sin t$, $y = \sin pt$, prove $(1 - x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} + p^2 y = 0$.

Solution:

$$\frac{dx}{dt} = \cos t, \quad \frac{dy}{dt} = p \cos pt.$$

$$\frac{dy}{dx} = \frac{p \cos pt}{\cos t}.$$

$$\frac{d}{dt} \left(\frac{dy}{dx} \right) = p \cdot \frac{(-p \sin pt) \cos t - \cos pt (-\sin t)}{\cos^2 t} = \frac{-p^2 \sin pt \cos t + p \sin t \cos pt}{\cos^2 t}.$$

$$\frac{d^2y}{dx^2} = \frac{1}{\cos t} \cdot \frac{-p^2 \sin pt \cos t + p \sin t \cos pt}{\cos^2 t} = \frac{-p^2 \sin pt \cos t + p \sin t \cos pt}{\cos^3 t}.$$

Now $1 - x^2 = \cos^2 t$.

$$(1 - x^2)y'' = \cos^2 t \cdot \frac{-p^2 \sin pt \cos t + p \sin t \cos pt}{\cos^3 t} = \frac{-p^2 \sin pt \cos t + p \sin t \cos pt}{\cos t}.$$

$$-xy' = -\sin t \cdot \frac{p \cos pt}{\cos t} = \frac{-p \sin t \cos pt}{\cos t}.$$

$$\text{Adding: } (1 - x^2)y'' - xy' = \frac{-p^2 \sin pt \cos t}{\cos t} = -p^2 \sin pt = -p^2 y.$$

Hence $(1 - x^2)y'' - xy' + p^2 y = 0$. ■

Archetype 7 — Mean Value Theorems (Rolle's / Lagrange) for Existence, Inequalities, Root Bounding

What it looks like:

- "Verify Rolle's/Lagrange's theorem for $f(x) = \dots$ on $[a, b]$ and find c ."
- "Using MVT, prove $\frac{b-a}{b} < \ln \frac{b}{a} < \frac{b-a}{a}$ for $0 < a < b$."
- "If $f'(x) \neq 0$ on (a, b) , how many roots can f have?"

Examples: Main 2020 Sep(II), 2020 Jan(I,II), 2019 Apr(I), 2018, 2015, 2014; Adv 2018, 2014, 2011, 2009, 2008, 2007, 2005, 2004, 2002, 1997, 1995, 1990, 1982.

Method:

- **Verification:** Check hypotheses (continuity on $[a, b]$, differentiability on (a, b) , $f(a) = f(b)$ for Rolle). Then solve $f'(c) = 0$ or $f'(c) = \frac{f(b)-f(a)}{b-a}$ and confirm $c \in (a, b)$.
- **Inequality proofs:** Choose f such that $f'(c)$ is the quantity to bound. Use $a < c < b$ to squeeze $f'(c)$.
- **Root bounding:** If f has two distinct roots, Rolle gives a point where $f' = 0$. Contrapositive: if $f' \neq 0$, at most one root.

Worked Example (reconstructed from Adv 2014 / Main 2020 Jan(I)):

Using Lagrange's Mean Value Theorem, prove that for $0 < a < b$,

$$\frac{b-a}{b} < \ln \frac{b}{a} < \frac{b-a}{a}.$$

Solution:

Apply MVT to $f(x) = \ln x$ on $[a, b]$. f is continuous on $[a, b]$, differentiable on (a, b) .

$\exists c \in (a, b)$ such that $f'(c) = \frac{\ln b - \ln a}{b-a}$.

$f'(c) = \frac{1}{c}$. Since $a < c < b$, we have $\frac{1}{b} < \frac{1}{c} < \frac{1}{a}$.

Thus $\frac{1}{b} < \frac{\ln b - \ln a}{b-a} < \frac{1}{a}$.

Multiply by $b-a > 0$: $\frac{b-a}{b} < \ln \frac{b}{a} < \frac{b-a}{a}$. ■

3. Traps — The Mistakes JEE Questions Are Built to Punish

Trap	Description	Typical Distractor
Reversed arrow	Believing "continuous $\Rightarrow$ differentiable".	$ x $ at 0 is the eternal counterexample.
Skipping continuity check	Jumping to LHD/RHD without verifying continuity first. If discontinuous, differentiability is impossible — but computing LHD/RHD may misleadingly give finite numbers.	Piecewise function with a jump; student computes derivatives of both pieces and equates them, missing the discontinuity.

Trap	Description	Typical Distractor
Miscounting modulus corners	Treating $ g(x) $ as a corner at every zero of g , ignoring sign-change requirement. Also counting the vertex of g (e.g. $x = 0$ for $ x^2 - 4 $) as a corner.	$ x^2 - 4 $ has corners at $x = \pm 2$ only, not at $x = 0$.
Endpoint over-counting	Including endpoints of an open interval (a, b) when counting discontinuities/kinks.	"Discontinuous in $(0, 3)$ " $\rightarrow$ integers are 1, 2 only (not 0, 3).
Dropping the chain factor in implicit differentiation	Writing $\frac{d}{dx}(y^2) = 2y$ instead of $2y \frac{dy}{dx}$.	The differentiated equation has a y -term but no $\frac{dy}{dx}$ — instant red flag.
Ignoring principal range in inverse-trig substitution	Writing $\tan^{-1}(\tan 2\theta) = 2\theta$ for all θ , missing that the identity holds only when $2\theta \in (-\pi/2, \pi/2)$. This flips the sign of the derivative on part of the domain.	$\sin^{-1} \frac{2x}{\sqrt{1-x^2}}$ gives derivative $\frac{2}{\sqrt{1-x^2}}$ only for $ x \leq 1/\sqrt{2}$; outside it's $-\frac{2}{\sqrt{1-x^2}}$.
Half-rule on u^v	Differentiating x^x as either $x^x \ln x$ (exponential piece only) or $x \cdot x^{x-1}$ (power piece only). The correct derivative is the sum.	Options often include the two half-answers.
Parametric sec-ond-derivative ratio	Using $\frac{d^2y}{dx^2} = \frac{d^2y/dt^2}{d^2x/dt^2}$ (cancelling dt 's incorrectly).	This yields 0 for a parabola $x = t^2, y = t^3$ where the true second derivative is $\frac{3}{4t}$.
Applying MVT/Rolle without checking hypotheses	Using the theorem on functions with corners, cusps, or discontinuities inside the interval (e.g. $ x $ on $[-1, 1]$).	The equal-endpoints condition holds but differentiability fails $\rightarrow$ conclusion not guaranteed.
Confusing Rolle with Lagrange	Using $f'(c) = 0$ when $f(a) \neq f(b)$.	Lagrange gives $f'(c) = \frac{f(b)-f(a)}{b-a}$; Rolle is the special case $f(a) = f(b)$.

4. Graded Practice Set (8 Questions, Ascending Difficulty)

All questions are adapted from or inspired by the bank; every answer has been verified.

Q1. [Easy - Continuity with parameter]

Find k so that

$$f(x) = \begin{cases} kx + 3, & x \leq 2 \\ x^2 - 1, & x > 2 \end{cases}$$

is continuous at $x = 2$.

Answer:

$$f(2) = 2k + 3. \text{ LHL} = 2k + 3. \text{ RHL} = \lim_{x \rightarrow 2^+} (x^2 - 1) = 3.$$

$$\text{Continuity} \Rightarrow 2k + 3 = 3 \Rightarrow k = 0.$$

Q2. [Easy - Differentiability of max/min function]

Let $f(x) = \max\{x, x^2\}$. Find the set of points where f is not differentiable.

Answer:

$$f(x) = x^2 \text{ for } x \leq 0 \text{ and } x \geq 1; f(x) = x \text{ for } 0 \leq x \leq 1.$$

Corners at $x = 0$ (slope 0 from left, 1 from right) and $x = 1$ (slope 1 from left, 2 from right).

Set: $\{0, 1\}$.

Q3. [Medium - Greatest integer discontinuities]

$f(x) = [x^2] \sin(\pi x)$. Number of points of discontinuity in $(-2, 3)$?

Answer: 8 (at $x = \pm\sqrt{2}, \pm\sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{7}, \sqrt{8}$).

Reasoning: $[x^2]$ jumps at $x = \pm\sqrt{n}$. In $(-2, 3)$ the non-integer jump points are $\pm\sqrt{2}, \pm\sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{7}, \sqrt{8}$, where $\sin(\pi x) \neq 0$, so the jumps survive. At the integer candidates $x = -1, 1, 2$ the factor $\sin(\pi x) = 0$ sends both one-sided limits of the product to $0 = f(x)$, so f is continuous there; $x = 0$ is a minimum-touch (no jump at all), and $x = -2, 3$ are excluded endpoints.

Q4. [Medium - Inverse trig substitution]

Differentiate $y = \tan^{-1} \frac{2x}{1-x^2}$ for $|x| < 1$. State the derivative for $|x| > 1$.

Answer:

$$\text{For } |x| < 1: x = \tan \theta, \theta \in (-\pi/4, \pi/4) \Rightarrow \frac{2x}{1-x^2} = \tan 2\theta.$$

$$y = 2\theta = 2 \tan^{-1} x \Rightarrow y' = \frac{2}{1+x^2}.$$

$$\text{For } x > 1: y = \pi - 2 \tan^{-1} x \Rightarrow y' = -\frac{2}{1+x^2}.$$

$$\text{For } x < -1: y = -\pi - 2 \tan^{-1} x \Rightarrow y' = -\frac{2}{1+x^2}.$$

Q5. [Medium - Logarithmic differentiation]

Differentiate $y = x^{\sin x}$ ($x > 0$).

Answer:

$$\ln y = \sin x \ln x.$$

$$\frac{y'}{y} = \cos x \ln x + \frac{\sin x}{x}.$$

$$y' = x^{\sin x} \left(\cos x \ln x + \frac{\sin x}{x} \right).$$

Q6. [Hard - Parametric second derivative]

$x = a(\theta - \sin \theta)$, $y = a(1 - \cos \theta)$. Find $\frac{d^2y}{dx^2}$.

Answer:

$$\frac{dx}{d\theta} = a(1 - \cos \theta), \quad \frac{dy}{d\theta} = a \sin \theta.$$

$$\frac{dy}{dx} = \frac{\sin \theta}{1 - \cos \theta} = \cot \frac{\theta}{2}.$$

$$\frac{d}{d\theta} \left(\frac{dy}{dx} \right) = -\frac{1}{2} \csc^2 \frac{\theta}{2}.$$

$$\frac{d^2y}{dx^2} = \frac{-\frac{1}{2} \csc^2 \frac{\theta}{2}}{a(1 - \cos \theta)} = \frac{-\frac{1}{2} \csc^2 \frac{\theta}{2}}{a \cdot 2 \sin^2 \frac{\theta}{2}} = -\frac{1}{4a} \csc^4 \frac{\theta}{2}.$$

Q7. [Hard - MVT inequality]

Let $f(x) = e^x$. Use Lagrange's MVT on $[0, x]$ ($x > 0$) to prove $e^x > 1 + x$.

Answer:

f continuous on $[0, x]$, differentiable on $(0, x)$.

$$\exists c \in (0, x) : f'(c) = \frac{e^x - 1}{x}.$$

$$f'(c) = e^c. \text{ Since } c > 0, e^c > 1.$$

$$\text{Thus } \frac{e^x - 1}{x} > 1 \Rightarrow e^x - 1 > x \Rightarrow e^x > 1 + x.$$

Q8. [Advanced - Functional equation + differentiability]

$f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x + y) = f(x) + f(y)$ for all x, y . f is differentiable at 0 with $f'(0) = 3$. Find $f(x)$ and $f'(5)$.

Answer:

$$f(0) = 0. \text{ Differentiate w.r.t } y: f'(x + y) = f'(y). \text{ Set } y = 0: f'(x) = f'(0) = 3.$$

$$\text{So } f(x) = 3x + C. f(0) = 0 \Rightarrow C = 0. \text{ Thus } f(x) = 3x, f'(5) = 3.$$

5. What This Chapter's Text Under-serves Relative to JEE

1. **Functional equations** (Cauchy, exponential, trigonometric) are barely touched in the chapter text but appear repeatedly in JEE Advanced (1981, 1990, 1995, 2011, 2020). The text treats differentiation as a mechanical skill; JEE uses it to solve functional equations.
2. **Inverse-trigonometric substitution** is taught as a “simplification trick” but the **critical importance of stating the valid domain** (where the double-angle identity holds) is not emphasised enough. JEE Advanced routinely penalises answers that ignore the piecewise nature of the derivative.
3. **Parametric second derivative** is given the correct formula, but the text does not drill the “**slope** → **differentiate w.r.t. parameter** → **divide by** $\dot{x}$ ” rhythm. The trap $\frac{d^2y}{dx^2} = \frac{d^2y/dt^2}{d^2x/dt^2}$ is mentioned once but not reinforced with enough varied examples.
4. **MVT as an inequality tool** appears only in a single worked example (Section 7.4). JEE Advanced uses it as a standard method for proving inequalities and bounding roots — students need far more practice in “choosing the right f and interval”.

Chapter 6: Application of Derivatives

Content Bank — Class 12 Mathematics — Drona Ed Tech

Drona Ed Tech

Contents

Subtopic 01: Rate of Change of Quantities	2
Section 1: Concept Intuition	2
Section 2: Key Formulas & Definitions	3
Section 3: Key Procedures & Logical Framework	4
Section 4: Worked Examples	5
Section 5: Practice Exercises	6
Section 6: Crack the MCQ	7
Section 7: Common Student Pitfalls & Pro-Tips	8
Section 8: Cheat Sheet Box	8
Subtopic 02: Monotonicity — Increasing and Decreasing Functions	9
Section 1: Concept Intuition	9
Section 2: Key Formulas, Theorems & Definitions	10
Section 3: Key Procedures & Logical Framework	11
Section 4: Worked Examples	11
Section 5: Practice Exercises	13
Section 6: Crack the MCQ	14
Section 7: Common Student Pitfalls & Pro-Tips	14
Section 8: Cheat Sheet Box	15
Subtopic 03: Maxima and Minima	16
Section 1: Concept Intuition	16
Section 2: Key Formulas & Definitions	17
Section 3: Key Procedures & Logical Framework	18
Section 4: Worked Examples	19
Section 5: Practice Exercises	21
Section 6: Crack the MCQ	21
Section 7: Common Student Pitfalls & Pro-Tips	22
Section 8: Cheat Sheet Box	22
Appendix A: Tangents and Normals (<i>JEE-relevant — outside current CBSE board syllabus</i>)	24
Concept Intuition	24

Key Formulas & Results	24
Key Procedure	25
Worked Examples	25
Practice & Pitfalls	26
Appendix B: Approximations and Differentials (JEE-relevant — outside current CBSE board syllabus)	27
Concept Intuition	27
Key Formulas & Results	27
Key Procedure	28
Worked Examples	28
Practice & Pitfalls	28

Subtopic 01: Rate of Change of Quantities

Exam Relevance: A dependable CBSE Boards question — a 2-3 mark related-rates problem (expanding circle, sliding ladder, inflating balloon, filling cone) appears almost every year. JEE Main carries 1 question regularly; JEE Advanced folds it into multi-step geometry. It is also the conceptual launch-pad for the whole chapter, since “derivative = rate of change” underpins everything that follows.

Section 1: Concept Intuition

The derivative you learned as “slope of a curve” has a second, equally important identity: it is a **rate of change**. When a quantity y depends on a quantity x , the derivative $\frac{dy}{dx}$ tells you *how fast y changes per unit change in x* . If $\frac{dy}{dx} = 4$, then near that instant y grows four units for every one-unit rise in x .

Most real situations, though, change with **time**. Think of a stone dropped into a still pond near Varanasi’s ghats. A circular ripple spreads outward; its radius grows, and so does the area it encloses. Two things are changing at once, and they are linked — the area depends on the radius, and the radius depends on time. That linkage is the entire subject of **related rates**.

Think about it this way... the chain rule is the bridge between the rates. Since area depends on radius and radius depends on time,

$$\frac{dA}{dt} = \frac{dA}{dr} \cdot \frac{dr}{dt}.$$

Read it aloud: “how fast the area changes” equals “how much area changes per unit radius” times “how fast the radius changes.” You are passing the rate *through* the geometry. Once you can write the quantity you care about as a function of another quantity whose rate you know, the chain rule does the rest.

A crucial distinction: **average** rate of change of y over an interval is $\frac{\Delta y}{\Delta x}$ (a ratio of finite changes), while the **instantaneous** rate is the limit, $\frac{dy}{dx}$. Exam problems almost always ask for the instantaneous rate *at a particular moment* — for example, “how fast is the area increasing **when the radius is 10 cm.**” That little clause is your signal to differentiate first and substitute the value *afterwards*.

Limiting conditions & caveats:

1. **Differentiate with respect to the right variable.** In related-rates problems everything is changing with time t , so you differentiate the geometric relation with respect to t — not with respect to a length.
2. **Substitute numbers only after differentiating.** If you plug “ $r = 10$ ” into the formula *before* differentiating, the radius becomes a constant and its rate vanishes. Differentiate symbolically, then substitute.
3. **A decreasing quantity has a negative rate.** Water draining, a shadow shortening, a ladder-top falling — each carries a minus sign. Keep it.
4. **Units travel with the rate.** A length rate is in cm/s, an area rate in cm^2/s , a volume rate in cm^3/s . Mismatched units signal a slipped step.

Section 2: Key Formulas & Definitions

Rate of change (one variable). If $y = f(x)$, the rate of change of y with respect to x is

$$\frac{dy}{dx} = f'(x).$$

The value $\left. \frac{dy}{dx} \right|_{x=a}$ is the instantaneous rate at $x = a$.

Related rates (chain rule in time). If a quantity Q depends on x , and x varies with time,

$$\frac{dQ}{dt} = \frac{dQ}{dx} \cdot \frac{dx}{dt}.$$

Standard geometric rate relations (differentiated with respect to t):

Shape / quantity	Relation	Rate form
Circle, area	$A = \pi r^2$	$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$
Circle, circumference	$C = 2\pi r$	$\frac{dC}{dt} = 2\pi \frac{dr}{dt}$
Sphere, volume	$V = \frac{4}{3}\pi r^3$	$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$
Sphere, surface area	$S = 4\pi r^2$	$\frac{dS}{dt} = 8\pi r \frac{dr}{dt}$

Shape / quantity	Relation	Rate form
Cube, volume	$V = x^3$	$\frac{dV}{dt} = 3x^2 \frac{dx}{dt}$
Cone, volume	$V = \frac{1}{3}\pi r^2 h$	differentiate using the r - h constraint

Symbols. t — time; r, x, h — varying lengths; A, S — areas; V — volume; C — circumference; a prime or $\frac{d}{dt}$ denotes a time rate. **Units** are physical here: lengths in cm, time in s, so rates carry cm/s, cm²/s, or cm³/s as appropriate.

Conditions of applicability. The chain-rule bridge needs each quantity to be a differentiable function of the next; in related-rates problems you must first reduce the figure to a **single** independent variable using the given geometric constraint before differentiating.

Section 3: Key Procedures & Logical Framework

There is no theorem to prove here — the skill is a disciplined **procedure** for related rates. Five steps handle essentially every problem.

Step 1 — List what is given and what is wanted, as rates. Translate the words: “increasing at 3 cm/s” means $\frac{dr}{dt} = 3$; “how fast is the area changing” means find $\frac{dA}{dt}$. *Why:* naming the rates fixes the target and prevents confusing a length with its rate.

Step 2 — Write a relation connecting the quantities. Use the geometry/physics of the situation ($A = \pi r^2$, $x^2 + y^2 = \ell^2$, etc.). *Why:* the relation is the structural link the chain rule will act on.

Step 3 — Eliminate extra variables using any constraint. If a cone keeps $r = \frac{h}{2}$, substitute so the volume depends on h alone. *Why:* differentiation in time needs one driving variable, or you’ll be left with an unknown extra rate.

Step 4 — Differentiate the whole relation with respect to t . Treat every varying length as a function of t and apply the chain rule. *Why:* this produces the equation linking the known rate to the unknown rate. Do this **before** putting in any numbers.

Step 5 — Substitute the instantaneous values and solve. Now plug in the moment’s data and isolate the wanted rate; attach the correct units and sign. *Why:* the rate equation is general; the numbers specialise it to the instant asked about.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

The radius of a circular oil slick on water is increasing at 3 cm/s. Find the rate at which its area is increasing when the radius is 10 cm.

Given $\frac{dr}{dt} = 3$ cm/s; want $\frac{dA}{dt}$ at $r = 10$ cm. Relation $A = \pi r^2$.

Differentiate with respect to t :

$$\frac{dA}{dt} = 2\pi r \frac{dr}{dt} = 2\pi(10)(3) = 60\pi \text{ cm}^2/\text{s} \approx 188.5 \text{ cm}^2/\text{s}.$$

The area is increasing at $60\pi \text{ cm}^2/\text{s}$.

Example 2 — [CBSE Boards HOTS / Application]

A 13 m ladder leans against a vertical wall. Its foot is pulled away from the wall at 2 m/s. How fast is the top sliding down when the foot is 5 m from the wall?

Let x = distance of foot from wall, y = height of top. By Pythagoras, $x^2 + y^2 = 13^2 = 169$. Given $\frac{dx}{dt} = 2$ m/s.

When $x = 5$: $y = \sqrt{169 - 25} = 12$ m.

Differentiate the relation with respect to t :

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{x}{y} \frac{dx}{dt} = -\frac{5}{12}(2) = -\frac{5}{6} \text{ m/s}.$$

The negative sign means the top is **falling**: it slides down at $\frac{5}{6}$ m/s ≈ 0.83 m/s. (Dropping that minus sign is the usual lost mark.)

Figure 6.1 A right triangle against a wall — vertical leg y (ladder top on wall), horizontal leg x (foot on ground), hypotenuse 13 m (the ladder); arrows showing the foot moving right and the top moving down. (see next page)

Example 3 — [JEE Main Level]

A spherical balloon is inflated so that its volume increases at $100 \text{ cm}^3/\text{s}$. When the radius is 5 cm, find (a) the rate of increase of the radius and (b) the rate of increase of the surface area.

Given $\frac{dV}{dt} = 100 \text{ cm}^3/\text{s}$, $r = 5$ cm.

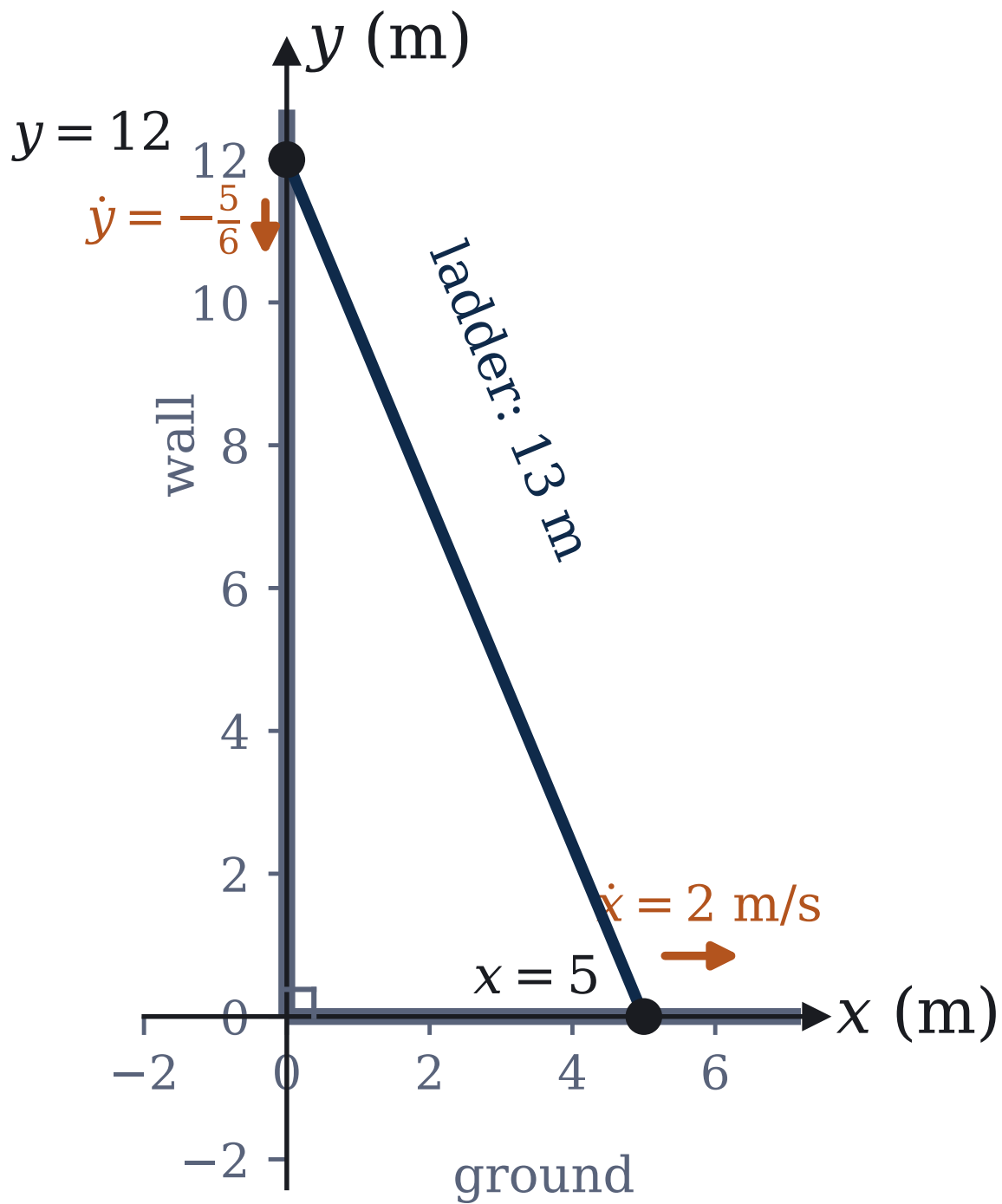


Figure 6.1 A right triangle against a wall — vertical leg y (ladder top on wall), horizontal leg x (foot on ground), hypotenuse 13 m (the ladder); arrows showing the foot moving right and the top moving down.

$$(a) V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}, \text{ so}$$

$$\frac{dr}{dt} = \frac{100}{4\pi(5)^2} = \frac{100}{100\pi} = \frac{1}{\pi} \text{ cm/s.}$$

$$(b) S = 4\pi r^2 \Rightarrow \frac{dS}{dt} = 8\pi r \frac{dr}{dt} = 8\pi(5)\left(\frac{1}{\pi}\right) = 40 \text{ cm}^2/\text{s.}$$

Radius grows at $\frac{1}{\pi}$ cm/s; surface area grows at 40 cm²/s.

Example 4 — [JEE Advanced Level]

Water is poured into an inverted right circular cone (vertex down) at 8 cm³/s. The cone's radius is always half its depth of water. Find how fast the water level rises when the depth is 4 cm.

Let h = water depth, r = surface radius. The constraint is $r = \frac{h}{2}$. Volume of water:

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \left(\frac{h}{2}\right)^2 h = \frac{\pi h^3}{12}.$$

Differentiate with respect to t (note: reduce to h first, then differentiate):

$$\frac{dV}{dt} = \frac{\pi}{12} \cdot 3h^2 \frac{dh}{dt} = \frac{\pi h^2}{4} \frac{dh}{dt}.$$

Set $\frac{dV}{dt} = 8$ and $h = 4$:

$$8 = \frac{\pi(4)^2}{4} \frac{dh}{dt} = 4\pi \frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{8}{4\pi} = \frac{2}{\pi} \text{ cm/s} \approx 0.64 \text{ cm/s.}$$

The water level rises at $\frac{2}{\pi}$ cm/s.

Figure 6.2 An inverted cone (vertex at the bottom) partly filled with water; label water depth h , surface radius r , and the relation $r = h/2$. (see next page)

Section 5: Practice Exercises

1. **[CBSE]** The edge of a cube is increasing at 2 cm/s. Find the rate of increase of its volume when the edge is 8 cm.

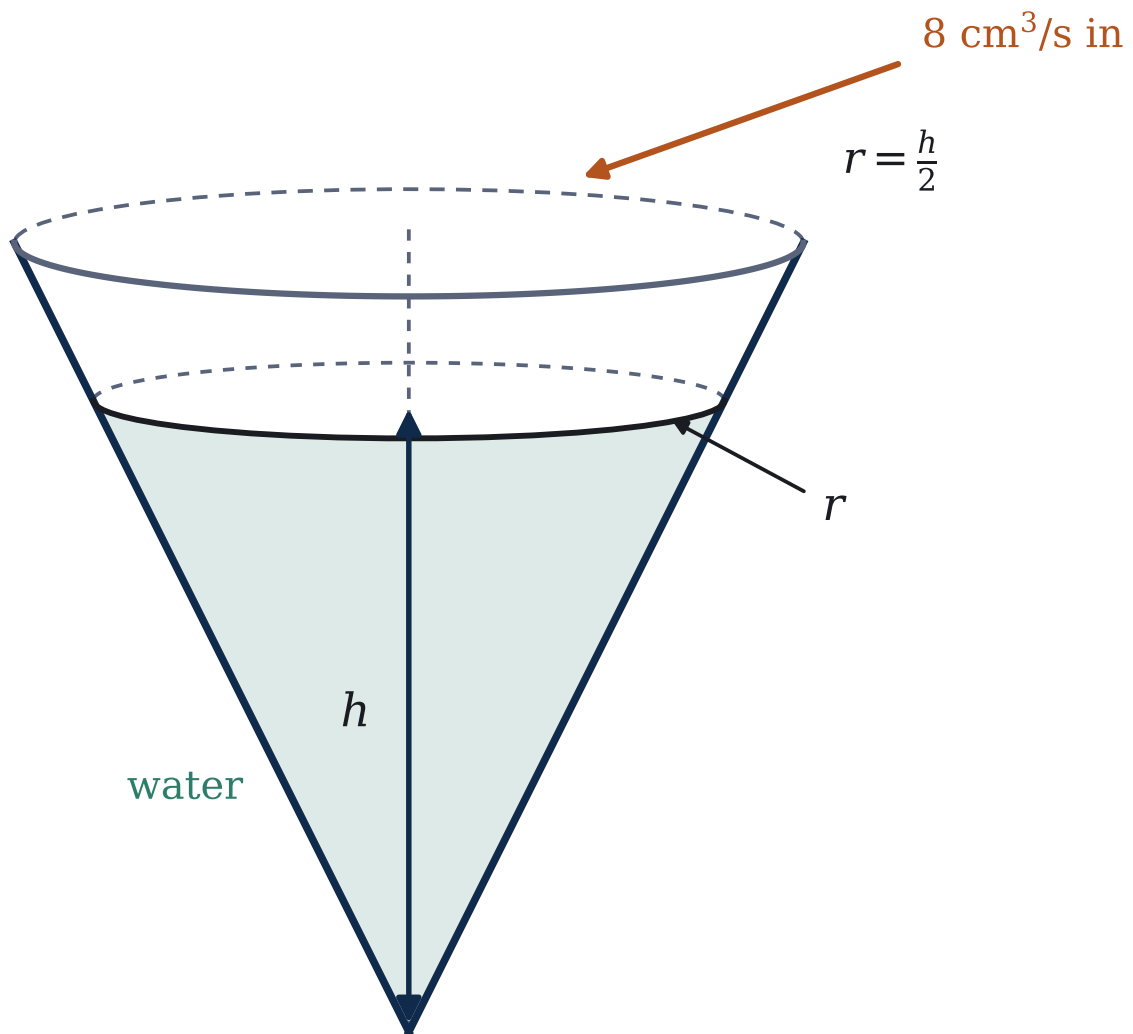


Figure 6.2 An inverted cone (vertex at the bottom) partly filled with water; label water depth h , surface radius r , and the relation $r = h/2$.

2. **[CBSE]** The area of a circle increases at $12\pi \text{ cm}^2/\text{s}$. Find the rate of increase of its radius when the radius is 6 cm.
3. **[JEE Main]** Sand falling from a chute forms a conical pile whose height always equals its base radius. If sand falls at $12 \text{ cm}^3/\text{s}$, how fast is the height rising when it is 4 cm?
4. **[JEE Main]** A 10 m ladder's top slides down a wall at 1 m/s. How fast is its foot moving away when the foot is 6 m from the wall?
5. **[JEE Advanced]** A man 2 m tall walks away from a 8 m lamp-post at 1.5 m/s. Find how fast (a) his shadow lengthens and (b) the tip of his shadow moves.

Answers: 1. $384 \text{ cm}^3/\text{s}$. 2. 1 cm/s. 3. $\frac{3}{4\pi} \text{ cm/s}$. 4. $\frac{4}{3} \text{ m/s}$. 5. (a) 0.5 m/s, (b) 2 m/s.

Section 6: Crack the MCQ

Q1. The radius of a circle increases at 0.7 cm/s. The rate of increase of its circumference is:

- (A) $0.7\pi \text{ cm/s}$ (B) $1.4\pi \text{ cm/s}$ (C) 1.4 cm/s (D) 0.7 cm/s

Answer: (B). $C = 2\pi r \Rightarrow \frac{dC}{dt} = 2\pi \frac{dr}{dt} = 2\pi(0.7) = 1.4\pi$. **(A)** forgets the factor 2. **(C)/(D)** drop the π , confusing circumference with a linear rate.

Q2. A balloon's volume increases at a constant rate. As it grows, the rate at which its radius increases:

- (A) stays constant (B) increases (C) decreases (D) is zero

Answer: (C). From $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$, a fixed $\frac{dV}{dt}$ gives $\frac{dr}{dt} = \frac{1}{4\pi r^2} \frac{dV}{dt}$, which *shrinks* as r grows. **(A)** assumes the rates track linearly; **(B)** reverses the dependence; **(D)** confuses "slower" with "stopped."

Q3. For $y = x^3 - 6x^2$, the rate of change of y with respect to x is zero at:

- (A) $x = 0$ only (B) $x = 4$ only (C) $x = 0$ and $x = 4$ (D) $x = 2$

Answer: (C). $\frac{dy}{dx} = 3x^2 - 12x = 3x(x - 4) = 0$ at $x = 0, 4$. **(A)/(B)** keep only one root; **(D)** mistakenly halves the second root.

Q4. A ladder slides down a wall. At the instant the foot is far from the wall and the top is near the ground, the top is moving:

- (A) slowly (B) very fast (C) at the same speed as the foot (D) upward

Answer: (B). From $\frac{dy}{dt} = -\frac{x}{y} \frac{dx}{dt}$, as $y \rightarrow 0$ the factor $\frac{x}{y}$ blows up, so the top accelerates dramatically. **(A)** reverses it; **(C)** ignores the geometry; **(D)** has the wrong sign.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Substituting before differentiating.** Plugging the instantaneous value into the formula first turns a variable into a constant and kills its rate. Always differentiate symbolically, then substitute.
2. **Differentiating with respect to a length instead of time.** The driver is t ; every varying length is a function of t , so the chain rule must produce $\frac{dr}{dt}$ terms.
3. **Losing the sign.** Draining tanks, falling ladder-tops, and shrinking shadows have *negative* rates. State and keep the minus.
4. **Not reducing to one variable.** Forgetting the geometric constraint (like $r = \frac{h}{2}$ for a cone) leaves an extra unknown rate and a stuck problem.

Pro-Tip: Underline the phrase “**at the instant when...**” — it marks exactly where the numbers go (last), and signals you want the *instantaneous* rate. For spheres, remember $\frac{dV}{dt} = (\text{surface area}) \times \frac{dr}{dt}$, a neat sanity check since $\frac{dV}{dr} = 4\pi r^2 = S$.

Section 8: Cheat Sheet Box

RATE OF CHANGE — QUICK REFERENCE

- **Derivative = rate:** $\frac{dy}{dx} = f'(x)$ is how fast y changes per unit x ; evaluate at the given point for the instantaneous rate.
- **Related rates bridge:** $\frac{dQ}{dt} = \frac{dQ}{dx} \cdot \frac{dx}{dt}$ (chain rule in time).
- **Method:** name rates $\rightarrow$ write relation $\rightarrow$ eliminate extra variables $\rightarrow$ differentiate w.r.t. $t \rightarrow$ substitute last.
- **Key results:** $\frac{dA}{dt} = 2\pi r r'$; $\frac{dV}{dt} = 4\pi r^2 r'$ (sphere); $\frac{dS}{dt} = 8\pi r r'$; $\frac{dV}{dt} = 3x^2 x'$ (cube).
- **Signs & units:** decreasing $\Rightarrow$ negative rate; carry cm/s, cm²/s, cm³/s correctly.

Mnemonic: “Differentiate first, plug in last.”

Subtopic 02: Monotonicity — Increasing and Decreasing Functions

Exam Relevance: A guaranteed scorer. CBSE Boards almost always carry a 3-mark “find the intervals in which f is strictly increasing/decreasing” question, plus a HOTS-style inequality or “show f is increasing on $\mathbb{R}$ ” proof. JEE Main puts 1–2 questions yearly — usually a parameter-range or a “which of these is monotonic” MCQ. JEE Advanced rarely asks it directly but uses monotonicity as the *engine* for proving inequalities and counting roots.

Section 1: Concept Intuition

Picture the Nifty 50 chart on a trading day. From 9:15 to 11:00 the line climbs steadily; you’d say the index is **increasing**. After lunch it slides downhill till close; now it’s **decreasing**. Nobody needs the exact values to say this — the *direction of travel* is enough. Monotonicity is exactly this idea made precise: over a stretch of its domain, is the function climbing, falling, or flat?

Now connect it to the derivative. The derivative $f'(x)$ is the **slope** of the curve at x — and slope is direction of travel. Walk left-to-right along the graph:

- Where the path tilts **uphill**, the slope is positive, $f'(x) > 0$, and the function is **increasing**.
- Where the path tilts **downhill**, the slope is negative, $f'(x) < 0$, and the function is **decreasing**.
- Where the path is momentarily **flat**, $f'(x) = 0$.

Think about it this way... imagine driving up a ghat road. Your altitude is $f(x)$ and the road’s gradient is $f'(x)$. As long as the gradient sign stays positive, you keep gaining altitude — it doesn’t matter whether the climb is steep or gentle. The *sign* of the gradient, not its size, decides whether you’re going up or down. That single insight — **sign of f' controls direction of f** — is the whole subtopic.

A subtle but exam-critical point lives at the flat spots. Consider $f(x) = x^3$. At $x = 0$ the slope is exactly zero, yet the curve never stops rising — it’s strictly increasing across all of $\mathbb{R}$. So a momentary $f'(x) = 0$ at an **isolated** point does *not* break increasing behaviour. It’s like the car cresting a tiny level patch mid-climb: instantaneously flat, still going up overall.

Limiting conditions & caveats — where the naïve “look at f' ” reasoning can betray you:

1. **Increasing is an interval property, not a point property.** You can only call f increasing on an interval, never “at a point.”
2. **Disconnected domains don’t merge.** $f(x) = \frac{1}{x}$ is decreasing on $(-\infty, 0)$ and on $(0, \infty)$, but it is **not** decreasing on the union — $f(-1) = -1$ is less than $f(1) = 1$. The gap at $x = 0$ matters.

3. $f'(x) > 0$ is **sufficient but not necessary** for strict increase (recall x^3). The honest statement uses " ≥ 0 except at isolated points."
 4. The clean theorem needs f **continuous on the closed interval and differentiable on the open interval** — don't apply it blindly across points where f or f' fails to exist.
-

Section 2: Key Formulas, Theorems & Definitions

Definitions. Let I be an interval contained in the domain of f . For all $x_1, x_2 \in I$:

$$\begin{aligned} f \text{ is increasing on } I &\iff (x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2)) \\ f \text{ is strictly increasing on } I &\iff (x_1 < x_2 \Rightarrow f(x_1) < f(x_2)) \\ f \text{ is decreasing on } I &\iff (x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2)) \\ f \text{ is strictly decreasing on } I &\iff (x_1 < x_2 \Rightarrow f(x_1) > f(x_2)) \end{aligned}$$

A function that is (strictly) increasing or (strictly) decreasing on I is called **monotonic** on I .

The Monotonicity Theorem (first-derivative criterion). Let f be continuous on $[a, b]$ and differentiable on (a, b) . Then on (a, b) :

$$\begin{aligned} f'(x) > 0 &\Rightarrow f \text{ is strictly increasing on } [a, b] \\ f'(x) < 0 &\Rightarrow f \text{ is strictly decreasing on } [a, b] \\ f'(x) = 0 \quad \forall x &\Rightarrow f \text{ is constant on } [a, b] \end{aligned}$$

Sharper version (the one that handles x^3):

$$\begin{aligned} f'(x) \geq 0 \text{ on } (a, b), \text{ with } f'(x) = 0 \text{ only at finitely many isolated points} \\ \Rightarrow f \text{ is strictly increasing on } [a, b]. \end{aligned}$$

Symbols. f — the function; $I, [a, b]$ — the interval of interest; $f'(x) = \frac{d}{dx}f(x)$ — the derivative (slope); x_1, x_2 — any two points of I . (All quantities here are pure-mathematical — no SI units apply.)

Conditions of applicability — read before every problem:

- f must be **continuous on the closed interval, differentiable on the open interval**.
 - The theorem gives conclusions on a **single interval**; never staple two disjoint intervals into one.
 - $f'(x) > 0$ is **sufficient, not necessary**, for strict increase.
 - Conversely, if f is differentiable and strictly increasing on I , then $f'(x) \geq 0$ on I (not necessarily > 0).
-

Section 3: Key Procedures & Logical Framework

CBSE does not require a formal *proof* of the monotonicity theorem — the rigorous argument runs through Lagrange’s Mean Value Theorem, which is **outside the 2025-26 syllabus**. (For the curious /JEE crowd: MVT gives $f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$ for some c between them; if $f'(c) > 0$ and $x_2 > x_1$ the right side is positive, forcing $f(x_2) > f(x_1)$ — that’s the proof in one line.) What you *are* expected to master is the **procedure** for extracting intervals. Here is the reliable five-step algorithm.

Step 1 — Fix the domain. Identify where f is defined. Points excluded from the domain (e.g. $x = 0$ for $1/x$, or where a log or $\sqrt{\quad}$ argument fails) will *break* the number line into separate intervals you must never merge. *Why*: monotonicity is an interval property; a hole in the domain can flip the comparison.

Step 2 — Differentiate. Compute $f'(x)$ and simplify it into **fully factored form**, e.g. $f'(x) = 6(x - 1)(x - 2)$. *Why*: factored form lets you read off sign changes instantly; expanded form hides them.

Step 3 — Find the critical points. Solve $f'(x) = 0$, and also note any x where $f'(x)$ does not exist. These are the only places the sign of f' can switch. *Why*: a continuous expression keeps one sign between consecutive zeros, so the zeros partition the line.

Step 4 — Build a sign chart. Mark the critical points (and domain breaks) on a number line; test the sign of f' in each resulting interval by plugging in one convenient point. *Why*: one test point certifies the sign for the whole sub-interval.

Figure 6.3 A number line for $f(x) = 6(x-1)(x-2)$. Mark $x=1$ and $x=2$. Show (+) on $(-\infty, 1)$, (−) on $(1, 2)$, (+) on $(2, \infty)$, with up/down arrows for f' beneath each region. (see next page)

Step 5 — Translate signs to conclusions. Positive $f' \rightarrow$ strictly increasing on that interval; negative $\rightarrow$ strictly decreasing. State intervals cleanly (closed at the critical points is acceptable when f is continuous there). *Why*: this is the answer the marker is looking for, and including the endpoint is valid because the limit of increasing behaviour carries to the boundary.

For “prove f is increasing on $\mathbb{R}$ ” questions: skip the chart — just show $f'(x) \geq 0$ for every real x (often by completing the square or bounding a trig term), and note any isolated zeros don’t matter.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Find the intervals in which $f(x) = 2x^3 - 9x^2 + 12x + 5$ is strictly increasing and strictly decreasing.

Given: $f(x) = 2x^3 - 9x^2 + 12x + 5$, domain $\mathbb{R}$.

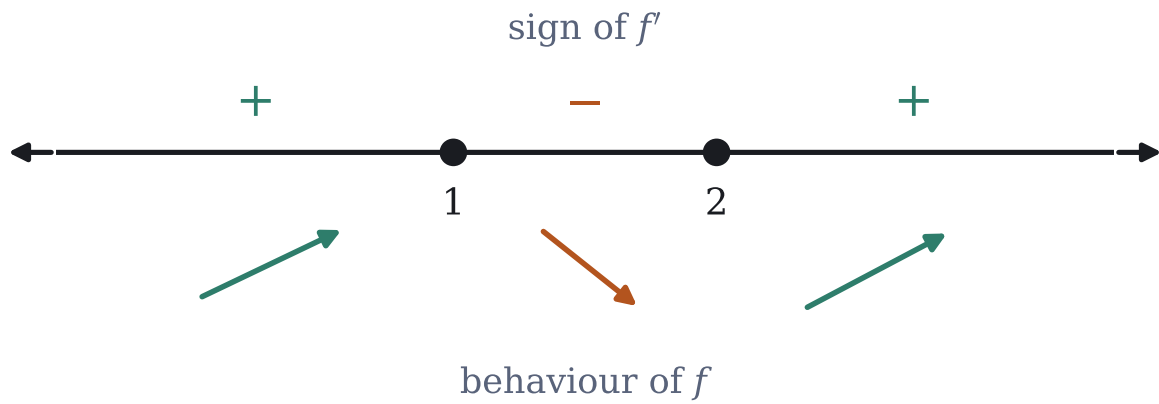


Figure 6.3 A number line for $f(x) = 6(x-1)(x-2)$. Mark $x=1$ and $x=2$. Show (+) on $(-\infty, 1)$, (-) on $(1, 2)$, (+) on $(2, \infty)$, with up/down arrows for f beneath each region.

Differentiate:

$$f'(x) = 6x^2 - 18x + 12 = 6(x^2 - 3x + 2) = 6(x - 1)(x - 2).$$

Critical points: $f'(x) = 0 \Rightarrow x = 1, 2$. These split $\mathbb{R}$ into $(-\infty, 1)$, $(1, 2)$, $(2, \infty)$.

Sign test: at $x = 0$, $f' = 6(-1)(-2) = 12 > 0$; at $x = 1.5$, $f' = 6(0.5)(-0.5) = -1.5 < 0$; at $x = 3$, $f' = 6(2)(1) = 12 > 0$.

Conclusion: f is strictly increasing on $(-\infty, 1] \cup [2, \infty)$ and strictly decreasing on $[1, 2]$.

Example 2 — [CBSE Boards HOTS / Application]

Prove that $x > \log(1 + x)$ **for all** $x > 0$.

This is a monotonicity *proof*, the favourite HOTS pattern. Build an auxiliary function whose sign you want to pin down.

Let $g(x) = x - \log(1 + x)$ on $[0, \infty)$. Then

$$g'(x) = 1 - \frac{1}{1+x} = \frac{(1+x) - 1}{1+x} = \frac{x}{1+x}.$$

For $x > 0$, both numerator and denominator are positive, so $g'(x) > 0$. Hence g is **strictly increasing on** $[0, \infty)$.

Therefore, for any $x > 0$,

$$g(x) > g(0) = 0 - \log(1) = 0 \Rightarrow x - \log(1 + x) > 0 \Rightarrow x > \log(1 + x). \quad \blacksquare$$

The “verify it strictly” step students drop: you must state g is strictly increasing **and** evaluate at the anchor $g(0) = 0$ — the inequality is the gap between $g(x)$ and that anchor.

Example 3 — [JEE Main Level]

Find all real values of k for which $f(x) = x^3 + kx^2 + 3x + 7$ **is strictly increasing on** $\mathbb{R}$.

This fuses monotonicity with the discriminant condition for a quadratic — two ideas in one.

$$f'(x) = 3x^2 + 2kx + 3.$$

For f strictly increasing on **all** of $\mathbb{R}$, we need $f'(x) \geq 0$ for every x (with at most isolated zeros). Since f' is an upward parabola (leading coefficient $3 > 0$), it stays ≥ 0 everywhere precisely when its discriminant is ≤ 0 :

$$\Delta = (2k)^2 - 4(3)(3) = 4k^2 - 36 \leq 0 \Rightarrow k^2 \leq 9 \Rightarrow -3 \leq k \leq 3.$$

At $k = \pm 3$, $\Delta = 0$, so f' touches zero at a single point and is positive elsewhere — still strictly increasing.

Answer: $k \in [-3, 3]$.

Example 4 — [JEE Advanced Level]

Show that $2 \sin x + \tan x > 3x$ **for all** $x \in (0, \frac{\pi}{2})$.

Let $h(x) = 2 \sin x + \tan x - 3x$ on $[0, \frac{\pi}{2})$, with $h(0) = 0$. Then

$$h'(x) = 2 \cos x + \sec^2 x - 3.$$

We claim $h'(x) > 0$ on $(0, \frac{\pi}{2})$. Apply AM-GM to the three positive numbers $\cos x$, $\cos x$, $\sec^2 x$:

$$\frac{\cos x + \cos x + \sec^2 x}{3} \geq \sqrt[3]{\cos x \cdot \cos x \cdot \sec^2 x} = \sqrt[3]{\cos^2 x \cdot \frac{1}{\cos^2 x}} = 1,$$

so $2 \cos x + \sec^2 x \geq 3$, hence $h'(x) \geq 0$. Equality in AM-GM needs $\cos x = \sec^2 x$, i.e. $\cos^3 x = 1$, i.e. $x = 0$ — which is excluded. Thus $h'(x) > 0$ strictly on $(0, \frac{\pi}{2})$.

So h is strictly increasing there, giving $h(x) > h(0) = 0$, i.e.

$$2 \sin x + \tan x > 3x \quad \text{for all } x \in (0, \frac{\pi}{2}). \quad \blacksquare$$

Figure 6.4 Two curves on $(0, \pi/2)$: $y = 2 \sin x + \tan x$ above $y = 3x$, both from the origin, with the upper curve pulling away and shooting up near $x = \pi/2$. (see next page)

Section 5: Practice Exercises

1. **[CBSE]** Find the intervals in which $f(x) = x^3 - 12x + 1$ is strictly increasing and strictly decreasing.
2. **[CBSE]** Show that $f(x) = x^3 + 3x^2 + 6x - 11$ is strictly increasing on $\mathbb{R}$.
3. **[JEE Main]** Determine the interval(s) on which $f(x) = x(x-4)^2$ is decreasing.
4. **[JEE Main]** Find all a for which $f(x) = ax - \sin x$ is increasing for every real x .
5. **[JEE Advanced]** Prove that $\tan x > x + \frac{x^3}{3}$ for all $x \in (0, \frac{\pi}{2})$.

Answers: 1. Increasing on $(-\infty, -2] \cup [2, \infty)$, decreasing on $[-2, 2]$. 2. $f'(x) = 3(x+1)^2 + 3 > 0$ for all x . 3. Decreasing on $[\frac{4}{3}, 4]$. 4. $a \geq 1$. 5. Let $g(x) = \tan x - x - \frac{x^3}{3}$; show $g'(x) = \tan^2 x - x^2 = (\tan x - x)(\tan x + x) > 0$.

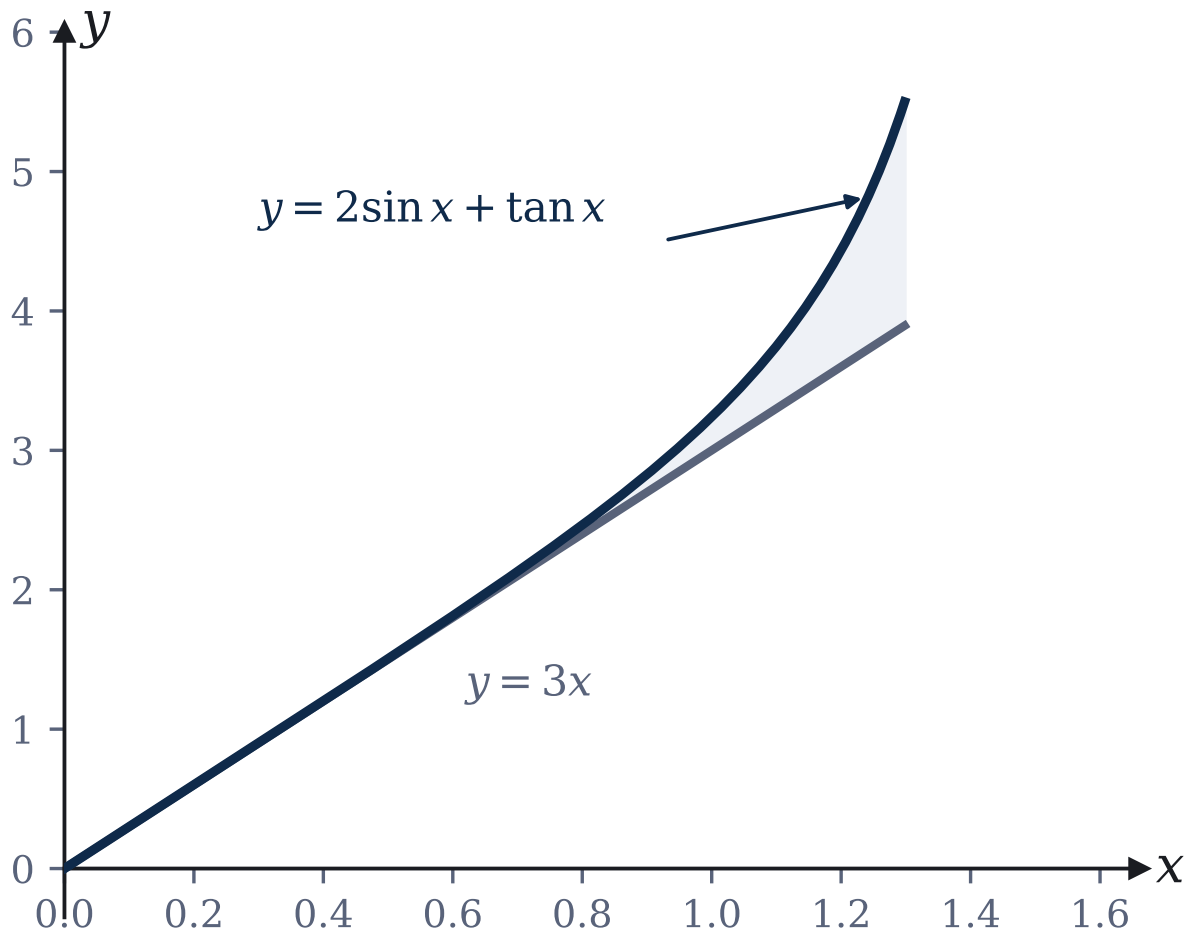


Figure 6.4 Two curves on $(0, \pi/2)$: $y = 2 \sin x + \tan x$ above $y = 3x$, both from the origin, with the upper curve pulling away and shooting up near $x = \pi/2$.

Section 6: Crack the MCQ

Q1. The function $f(x) = 2x^3 + 9x^2 + 12x - 4$ is decreasing on:

- (A) $(-\infty, -2]$ (B) $[-2, -1]$ (C) $[-1, \infty)$ (D) $[-1, 1]$

Answer: (B). $f'(x) = 6x^2 + 18x + 12 = 6(x+1)(x+2)$, which is negative only between its roots, i.e. on $(-2, -1)$. **(A)** and **(C)** are the *increasing* intervals — a student who confuses “ $f' < 0$ between roots” with “outside roots” flips these. **(D)** comes from misfactoring or testing the wrong sign at $x = 0$.

Q2. Which of the following is strictly increasing on $\mathbb{R}$?

- (A) $f(x) = x - \sin x$ (B) $f(x) = \cos x$ (C) $f(x) = x^2$ (D) $f(x) = \frac{1}{x}$

Answer: (A). $f'(x) = 1 - \cos x \geq 0$, zero only at isolated points $x = 2n\pi$ — strictly increasing. **(B)** oscillates. **(C)** decreases on $(-\infty, 0)$. **(D)** isn't even defined on all of $\mathbb{R}$, and is decreasing on each piece — the classic “disconnected domain” trap.

Q3. $f(x) = x(x-3)^2$ is decreasing for $x \in$:

- (A) $(1, 3)$ (B) $(-\infty, 1)$ (C) $(3, \infty)$ (D) $(0, \frac{3}{2})$

Answer: (A). $f'(x) = (x-3)^2 + x \cdot 2(x-3) = (x-3)(3x-3) = 3(x-3)(x-1)$, negative on $(1, 3)$. **(B)** and **(C)** are increasing regions. **(D)** is a distractor for students who half-differentiate (forget the product rule) and land on a wrong critical point.

Q4. If $f(x) = x^3 + bx^2 + 3x + 5$ is strictly increasing on $\mathbb{R}$, then b lies in:

- (A) $(-3, 3)$ (B) $[-3, 3]$ (C) $(-\infty, -3] \cup [3, \infty)$ (D) $\mathbb{R}$

Answer: (B). Need $f'(x) = 3x^2 + 2bx + 3 \geq 0$ for all $x \Rightarrow \Delta = 4b^2 - 36 \leq 0 \Rightarrow b \in [-3, 3]$. **(A)** is the trap for students who write the discriminant condition as strict (< 0) and exclude the endpoints — but $\Delta = 0$ still gives strict increase via an isolated zero. **(C)** inverts the inequality. **(D)** ignores the constraint entirely.

Section 7: Common Student Pitfalls & Pro-Tips

- Merging disjoint intervals.** Writing “ $1/x$ is decreasing on $\mathbb{R} \setminus \{0\}$ ” as a single decreasing claim. Each side is decreasing; the union is not. Always split at domain breaks.
- Demanding $f' > 0$ when ≥ 0 suffices.** Rejecting x^3 or $x - \sin x$ as “not strictly increasing” because $f' = 0$ somewhere. Isolated zeros are fine — say so explicitly to bag the mark.
- Sign-chart slips.** Testing only one point, or losing a factor's sign. Always factor f' fully and test every sub-interval.
- Endpoint paranoia.** Worrying whether to write $()$ or $[]$ at critical points. When f is continuous there, closed brackets are accepted for strict monotonicity in CBSE.

Pro-Tip: For “prove increasing on $\mathbb{R}$,” don’t draw a chart — just *bound* f' . Complete the square ($3x^2 + 2bx + 3$), or use $-1 \leq \cos x \leq 1$, to show $f' \geq 0$ in one move. For MCQs, factor f' and read the sign of the middle interval first — it instantly fixes the whole pattern by alternation.

Section 8: Cheat Sheet Box

MONOTONICITY — QUICK REFERENCE

- **Sign rule:** $f'(x) > 0 \Rightarrow$ strictly increasing; $f'(x) < 0 \Rightarrow$ strictly decreasing; $f'(x) = 0 \forall x \Rightarrow$ constant.
- **Sharper:** $f'(x) \geq 0$ with isolated zeros $\Rightarrow$ still strictly increasing (think x^3).
- **Method:** domain $\rightarrow$ differentiate $\rightarrow$ factor $f' \rightarrow$ critical points ($f' = 0$ or undefined) $\rightarrow$ sign chart $\rightarrow$ translate.
- **Increasing-on- $\mathbb{R}$ trick:** show $f'(x) \geq 0$ for all x (complete the square / bound trig). Quadratic f' : need discriminant ≤ 0 .
- **Inequality proofs:** define $g = \text{LHS} - \text{RHS}$, show g' has fixed sign, anchor at a known point.
- **Trap:** never merge monotonic intervals across a domain gap.

Mnemonic: “Slope says go” — positive slope, function goes up; negative slope, function goes down.

Subtopic 03: Maxima and Minima

Exam Relevance: One of the highest-yield subtopics in the chapter. CBSE Boards reliably carry a 3–5 mark optimization word problem (the famous “open box,” “least material,” or “largest area” question) plus a short local-extremum problem. JEE Main asks 2–3 questions yearly — local extrema, absolute extrema on an interval, or a parameter condition. JEE Advanced uses it for tough constrained-optimization and geometry problems.

Section 1: Concept Intuition

Think of a road trip across the Western Ghats. As you drive, your altitude rises and falls — peaks (hilltops) and troughs (valleys) all along the way. A **local maximum** is a hilltop: higher than everything *immediately around it*, even if a taller peak sits 50 km down the road. A **local minimum** is a valley floor: lower than its immediate surroundings. The tallest point on your *entire* journey is the **absolute (global) maximum**, and the lowest point overall is the **absolute minimum**. That single distinction — “tallest nearby” versus “tallest overall” — is the heart of this subtopic.

Now bring in the derivative. At the exact instant you crest a hilltop, the road is momentarily **flat** — your slope is zero. Same at the bottom of a valley. So at a smooth peak or trough, $f'(x) = 0$. These flat spots, together with any points where the road suddenly kinks (slope undefined), are the **critical points** — the *only* candidates for local extrema.

Think about it this way... the derivative is a direction sign. Approaching a hilltop you're going *up* ($f' > 0$); just past it you're going *down* ($f' < 0$). The slope **flips from + to −**. At a valley it flips the other way, **− to +**. That flip is what actually certifies a peak or trough — this is the **First Derivative Test**. Alternatively, at a hilltop the road is curving *downward* (concave down, $f'' < 0$); at a valley it curves *upward* ($f'' > 0$). Reading the curvature is the **Second Derivative Test**.

But here's the trap that catches students every exam: **a flat spot is not always a peak or a valley**. Consider $f(x) = x^3$. At $x = 0$ the slope is zero, the road goes flat for an instant — yet you keep climbing right through it. The slope is + before and + after; it never flipped. That point is a **point of inflection**, not an extremum. So $f'(c) = 0$ is *necessary but not sufficient* for a local extremum.

Limiting conditions & caveats:

1. **Critical point $\neq$ extremum.** You must confirm a sign-flip (first test) or a non-zero second derivative (second test). A bare $f'(c) = 0$ proves nothing.
2. **The second-derivative test can be silent.** If $f''(c) = 0$, the test is inconclusive — fall back to the first-derivative test. (E.g. x^4 at the origin: $f''(0) = 0$, yet it is clearly a minimum.)
3. **Local and absolute are different questions.** On a closed interval $[a, b]$, the absolute extremum may sit at an **endpoint**, where the derivative need not be zero at all.

4. **Endpoints and corners host extrema too.** Non-differentiable points (like the vertex of $|x|$) can be minima — always include them among the critical points.
-

Section 2: Key Formulas & Definitions

Local maximum / minimum. A point c in the interior of the domain of f is a point of

local maximum if $\exists h > 0$ such that $f(c) > f(x) \forall x \in (c - h, c + h)$,

local minimum if $\exists h > 0$ such that $f(c) < f(x) \forall x \in (c - h, c + h)$.

The corresponding $f(c)$ is the local maximum / minimum **value**.

Absolute (global) extremum on $[a, b]$.

f has absolute maximum at $c \iff f(x) \leq f(c) \forall x \in [a, b]$,

f has absolute minimum at $d \iff f(x) \geq f(d) \forall x \in [a, b]$.

Critical point. A point c in the domain of f where $f'(c) = 0$ **or** $f'(c)$ does not exist.

First Derivative Test. Let c be a critical point of a continuous f . As x increases through c :

$f' : (+) \rightarrow (-) \Rightarrow c$ is a local maximum,

$f' : (-) \rightarrow (+) \Rightarrow c$ is a local minimum,

f' does not change sign $\Rightarrow c$ is a point of inflection (no extremum).

Second Derivative Test. Let f be twice differentiable at c with $f'(c) = 0$. Then

$f''(c) < 0 \Rightarrow$ local maximum at c ,

$f''(c) > 0 \Rightarrow$ local minimum at c ,

$f''(c) = 0 \Rightarrow$ test fails (revert to First Derivative Test).

Symbols. f — the function; c, d — candidate points; $f'(x)$ — first derivative (slope); $f''(x)$ — second derivative (curvature); h — a small positive radius; $[a, b]$ — closed interval of interest. (*Pure-mathematical quantities — no SI units, unless a problem attaches physical meaning such as a volume in cm^3 .*)

Conditions of applicability. Both tests require $f'(c) = 0$ as the entry condition (so they locate *interior smooth* extrema only); the second test additionally requires f twice differentiable at c . Neither test sees endpoints — for absolute extrema on $[a, b]$ you must use the working rule below.

Section 3: Key Procedures & Logical Framework

CBSE does not require a formal proof of these tests (the rigorous argument leans on the Mean Value Theorem, **deleted** from the 2025-26 syllabus). What is examined is fluent *application*. Two procedures cover almost every question.

A) Finding local extrema (Second Derivative Test — fastest when f'' is easy).

1. **Differentiate and find critical points.** Solve $f'(x) = 0$; also note where f' is undefined. *Why:* extrema can only occur where the slope vanishes or breaks.
2. **Compute f'' at each critical point.** *Why:* curvature instantly classifies a smooth turning point — concave down is a peak, concave up is a trough.
3. **Classify.** $f''(c) < 0 \Rightarrow \text{max}$; $f''(c) > 0 \Rightarrow \text{min}$; $f''(c) = 0 \Rightarrow \text{inconclusive}$, switch to the sign-change (first) test. *Why:* a vanishing second derivative gives no curvature information.
4. **Report the value.** Substitute c back into f to state the extremum value, not just its location.

Figure 6.5 A smooth curve over an interval showing a local maximum (horizontal tangent, concave down) on the left and a local minimum (horizontal tangent, concave up) on the right; dashed horizontal tangents at both turning points, with $f' > 0$, $f' < 0$, $f' > 0$ labelled across the three regions. (see next page)

B) Finding absolute extrema on a closed interval $[a, b]$ (Working Rule).

1. **Find all critical points of f that lie inside (a, b) .** *Why:* interior peaks/troughs are candidates.
2. **Evaluate f at every critical point AND at both endpoints a and b .** *Why:* on a closed interval the global extreme can sit at an endpoint even when the slope there is non-zero — this is the step most students forget.
3. **Compare the list.** The largest value is the absolute maximum; the smallest is the absolute minimum. *Why:* the global extreme is simply the winner of a finite shortlist.

C) Optimization word problems (the application form of B/A).

1. **Name variables and write the quantity to optimize as a function of one variable**, using any constraint to eliminate the others. *Why:* calculus tools act on a single-variable function.
2. **State the feasible domain** (lengths positive, etc.). *Why:* it tells you which critical roots to keep and whether endpoints matter.
3. **Differentiate, solve $f' = 0$, and reject roots outside the domain.** *Why:* a root giving a negative or zero dimension is physically meaningless.
4. **Confirm max vs. min** (second-derivative or sign test) and answer the question actually asked (often a value, not the variable).

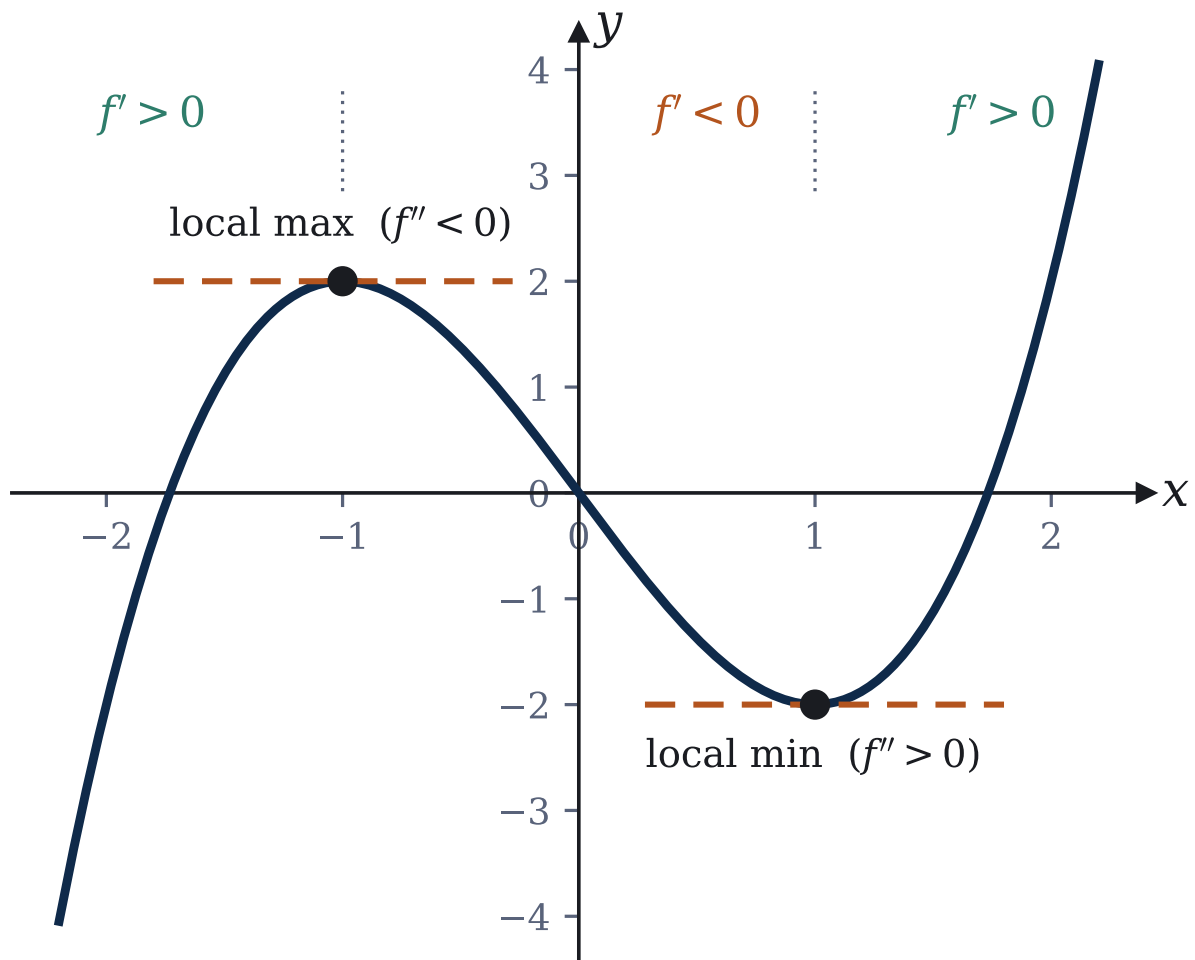


Figure 6.5 A smooth curve over an interval showing a local maximum (horizontal tangent, concave down) on the left and a local minimum (horizontal tangent, concave up) on the right; dashed horizontal tangents at both turning points, with $f' > 0$, $f' < 0$, $f' > 0$ labelled across the three regions.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Find the local maximum and local minimum values of $f(x) = 2x^3 - 15x^2 + 36x + 10$.

$$f'(x) = 6x^2 - 30x + 36 = 6(x-2)(x-3) \Rightarrow \text{critical points } x = 2, 3.$$

$$f''(x) = 12x - 30.$$

At $x = 2$: $f''(2) = -6 < 0 \Rightarrow$ **local maximum**, value $f(2) = 16 - 60 + 72 + 10 = 38$.

At $x = 3$: $f''(3) = 6 > 0 \Rightarrow$ **local minimum**, value $f(3) = 54 - 135 + 108 + 10 = 37$.

Conclusion: local maximum value 38 at $x = 2$; local minimum value 37 at $x = 3$.

Example 2 — [CBSE Boards HOTS / Application]

A square sheet of tin of side 18 cm has a small square of side x cut from each corner; the flaps are folded up to form an open box. Find x that maximizes the volume.

After cutting, the base is $(18 - 2x)$ by $(18 - 2x)$ and the height is x , so

$$V(x) = x(18 - 2x)^2, \quad \text{feasible domain } 0 < x < 9.$$

Differentiate (product rule):

$$V'(x) = (18-2x)^2 + x \cdot 2(18-2x)(-2) = (18-2x)[(18-2x)-4x] = (18-2x)(18-6x).$$

Set $V'(x) = 0$: $x = 9$ (**rejected** — it lies on the boundary and gives $V = 0$) or $x = 3$.

Check: for x just below 3, $V' > 0$; just above 3, $V' < 0$ — so V has a **maximum** at $x = 3$.

$$V(3) = 3(18 - 6)^2 = 3(12)^2 = 432 \text{ cm}^3.$$

Answer: cut $x = 3$ cm squares; maximum volume 432 cm^3 . (*The marks-losing step is keeping $x = 9$; always test feasibility.*)

Example 3 — [JEE Main Level]

Find the absolute maximum and minimum values of $f(x) = 3x^4 - 8x^3 + 12x^2 - 48x + 25$ on $[0, 3]$.

$$f'(x) = 12x^3 - 24x^2 + 24x - 48 = 12(x^3 - 2x^2 + 2x - 4) = 12(x-2)(x^2+2).$$

Since $x^2 + 2 > 0$ always, the only critical point is $x = 2$, which lies in $(0, 3)$. Apply the working rule — evaluate at the critical point **and** both endpoints:

$$\begin{aligned}f(0) &= 25, \\f(2) &= 48 - 64 + 48 - 96 + 25 = -39, \\f(3) &= 243 - 216 + 108 - 144 + 25 = 16.\end{aligned}$$

Answer: absolute maximum = 25 at $x = 0$ (an endpoint!); absolute minimum = -39 at $x = 2$.

Example 4 — [JEE Advanced Level]

Find the height of the cylinder of greatest volume that can be inscribed in a sphere of radius R , and find that maximum volume.

Let the cylinder have base radius r and height h . Its axis passes through the sphere's centre, so the half-height and radius satisfy

$$r^2 + \left(\frac{h}{2}\right)^2 = R^2 \Rightarrow r^2 = R^2 - \frac{h^2}{4}.$$

Volume as a single-variable function of h (feasible domain $0 < h < 2R$):

$$\begin{aligned}V(h) &= \pi r^2 h = \pi \left(R^2 - \frac{h^2}{4}\right) h = \pi \left(R^2 h - \frac{h^3}{4}\right). \\V'(h) &= \pi \left(R^2 - \frac{3h^2}{4}\right) = 0 \Rightarrow h^2 = \frac{4R^2}{3} \Rightarrow h = \frac{2R}{\sqrt{3}}.\end{aligned}$$

Curvature check: $V''(h) = \pi \left(-\frac{3h}{2}\right) < 0$, confirming a **maximum**. Then

$$V_{\max} = \pi \left(R^2 \cdot \frac{2R}{\sqrt{3}} - \frac{1}{4} \cdot \frac{8R^3}{3\sqrt{3}}\right) = \pi \left(\frac{2R^3}{\sqrt{3}} - \frac{2R^3}{3\sqrt{3}}\right) = \frac{4\pi R^3}{3\sqrt{3}} = \frac{4\sqrt{3}\pi R^3}{9}.$$

Answer: height $h = \frac{2R}{\sqrt{3}}$, maximum volume $\frac{4\pi R^3}{3\sqrt{3}}$.

Figure 6.6 A sphere of radius R with an inscribed right circular cylinder; show the centre, the radius r to the rim, half-height $h/2$, and the right triangle linking r , $h/2$, R . (see next page)

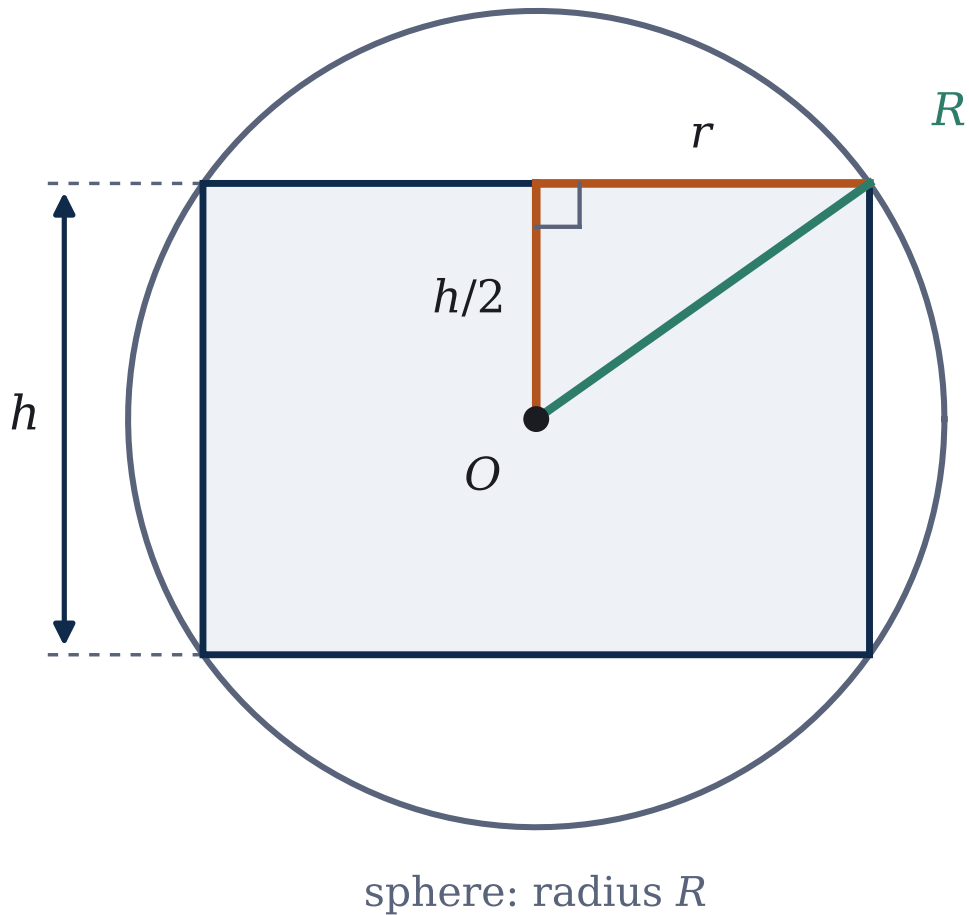


Figure 6.6 A sphere of radius R with an inscribed right circular cylinder; show the centre, the radius r to the rim, half-height $h/2$, and the right triangle linking r , $h/2$, R .

Section 5: Practice Exercises

1. **[CBSE]** Find the local maximum and minimum values of $f(x) = x^3 - 6x^2 + 9x + 15$.
2. **[CBSE]** Find the absolute maximum and minimum values of $f(x) = x^3 - 3x$ on $[0, 2]$.
3. **[JEE Main]** If $f(x) = x^3 + ax^2 + bx + c$ has local extrema at $x = 1$ and $x = 2$, find a and b .
4. **[JEE Main]** Two positive numbers have sum 24. Find them so that the product of one number and the square of the other is maximum.
5. **[JEE Advanced]** Show that the right circular cone of greatest volume inscribed in a sphere of radius R has height $\frac{4R}{3}$, and find its maximum volume.

Answers: 1. Local max 19 at $x = 1$, local min 15 at $x = 3$. 2. Abs max 2 at $x = 2$, abs min -2 at $x = 1$. 3. $a = -\frac{9}{2}$, $b = 6$. 4. 8 and 16 (with 16^2). 5. $h = \frac{4R}{3}$, $V_{\max} = \frac{32\pi R^3}{81}$.

Section 6: Crack the MCQ

Q1. The function $f(x) = x^3 - 3x^2 + 3x + 7$ has:

- (A) a local maximum at $x = 1$ (B) a local minimum at $x = 1$ (C) neither a maximum nor a minimum (D) both, at different points

Answer: (C). $f'(x) = 3x^2 - 6x + 3 = 3(x - 1)^2 \geq 0$, so f' touches zero at $x = 1$ but never changes sign — a point of inflection, no extremum. **(A)/(B)** trap students who see $f'(1) = 0$ and assume an extremum without testing the sign flip. **(D)** ignores that there is only one critical point.

Q2. The absolute maximum value of $f(x) = \sin x + \cos x$ on $[0, \frac{\pi}{2}]$ is:

- (A) 1 (B) $\sqrt{2}$ (C) 2 (D) $\frac{1}{\sqrt{2}}$

Answer: (B). $f'(x) = \cos x - \sin x = 0 \Rightarrow x = \frac{\pi}{4}$; $f(\frac{\pi}{4}) = \sqrt{2}$, while $f(0) = f(\frac{\pi}{2}) = 1$. So the max is $\sqrt{2}$. **(A)** is the endpoint value — picked by students who skip the interior critical point. **(C)** wrongly adds the separate maxima of $\sin$ and $\cos$. **(D)** confuses the value with $\sin \frac{\pi}{4}$.

Q3. For $f(x) = x^4$, at $x = 0$ the Second Derivative Test:

- (A) gives a local maximum (B) gives a local minimum (C) fails, but $x = 0$ is a local minimum (D) fails, so no extremum exists

Answer: (C). $f'(0) = 0$ and $f''(0) = 0$, so the second test is inconclusive; the first test shows f' goes $-$ to $+$, a local minimum. **(A)/(B)** assume the second test gives an answer when it is silent. **(D)** is the classic error — “test fails” is read as “no extremum,” but the first test still settles it.

Q4. If $f(x) = x^3 - 3kx$ has a local minimum at $x = 2$, then $k =$

(A) 2 (B) 4 (C) -4 (D) $\frac{1}{2}$

Answer: (B). $f'(x) = 3x^2 - 3k = 0$ at $x = 2 \Rightarrow k = 4$; then $f''(2) = 12 > 0$ confirms a minimum. **(A)** comes from solving $x = k$ instead of $x^2 = k$. **(C)** flips a sign. **(D)** mishandles the constant.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Calling every critical point an extremum.** $f'(c) = 0$ is only a *candidate*. Always finish with a sign-flip or curvature check — the inflection point of x^3 is the standard trap.
2. **Quitting when the second-derivative test fails.** $f''(c) = 0$ means “inconclusive,” not “no extremum.” Go back to the first-derivative test; never write “no maximum/minimum” from a failed second test.
3. **Forgetting the endpoints in absolute-extrema problems.** On $[a, b]$ the global extreme often lives at an endpoint. Evaluate f at the endpoints *and* the critical points, then compare.
4. **Keeping infeasible roots in optimization.** A root giving a negative length or zero volume must be discarded. State the feasible domain first so the rejection is automatic.

Pro-Tip: Pick your test to match the function. Polynomials with a clean second derivative $\rightarrow$ Second Derivative Test (one substitution per point). Functions where f'' is messy, or where $f''(c) = 0$, or with non-differentiable corners $\rightarrow$ First Derivative Test via a sign chart. In optimization, **always answer the exact question** — examiners often want the maximum *value* or a *dimension*, not the variable you solved for.

Section 8: Cheat Sheet Box

MAXIMA & MINIMA — QUICK REFERENCE

- **Critical points:** $f'(c) = 0$ **or** f' undefined. Only candidates for local extrema.
- **First Derivative Test:** f' flips $+$ $\rightarrow$ $-$ = local **max**; $- \rightarrow +$ = local **min**; no flip = inflection.
- **Second Derivative Test:** $f'(c) = 0$, then $f''(c) < 0$ = max, $f''(c) > 0$ = min, $f''(c) = 0$ = **fails** $\rightarrow$ use first test.
- **Absolute extrema on $[a, b]$:** compare f at all interior critical points **and** both endpoints; biggest = abs max, smallest = abs min.
- **Optimization:** one-variable objective via the constraint $\rightarrow$ set $f' = 0 \rightarrow$ reject infeasible roots $\rightarrow$ confirm max/min $\rightarrow$ answer what's asked.
- **Trap:** $f'(c) = 0$ alone proves nothing (see x^3); a “failed” second test still has an answer (see x^4).

Mnemonic: “Frown for max, smile for min” — concave down ($f'' < 0$) is a peak;

concave up ($f'' > 0$) is a valley.

Appendix A: Tangents and Normals (*JEE-relevant — outside current CBSE board syllabus*)

Exam Relevance: Removed from CBSE Boards in the NCERT rationalisation, but a steady source of JEE Main questions and a building block for JEE Advanced calculus/geometry. Include in study if you are preparing for JEE; a board-only student may skip this appendix.

Concept Intuition

The tangent to a curve at a point is the straight line that just “kisses” the curve there — the line whose slope matches the curve’s slope at that exact point. Since the derivative is the slope, the tangent’s slope at (x_0, y_0) is simply $f'(x_0)$. The **normal** is the line perpendicular to the tangent at the same point — picture the tangent as the road and the normal as the signpost standing at right angles to it.

Think about it this way... every smooth point on a curve has two natural lines through it: one going *along* the curve (tangent) and one cutting *across* it (normal). Perpendicular slopes multiply to -1 , so once you know the tangent slope m , the normal slope is $-\frac{1}{m}$ — flip and negate.

Caveats: if $f'(x_0) = 0$ the tangent is **horizontal** (slope 0) and the normal is vertical; if the tangent is **vertical** ($f'(x_0)$ infinite, e.g. $\frac{dx}{dy} = 0$) the normal is horizontal. Always confirm the point actually lies on the curve before writing the line.

Key Formulas & Results

At a point (x_0, y_0) on $y = f(x)$, with $m = f'(x_0)$:

$$\textbf{Tangent:} \quad y - y_0 = m(x - x_0).$$

$$\textbf{Normal:} \quad y - y_0 = -\frac{1}{m}(x - x_0) \quad (m \neq 0).$$

Special cases. Horizontal tangent when $m = 0$: line $y = y_0$. Vertical tangent when $\frac{dy}{dx}$ is infinite: line $x = x_0$ (then the normal is $y = y_0$).

Angle of intersection of two curves. If their tangents at a common point have slopes m_1, m_2 ,

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|.$$

The curves are **orthogonal** (cut at right angles) if $m_1 m_2 = -1$, and **touch** (tangentially meet) if $m_1 = m_2$.

Symbols. m — tangent slope = $f'(x_0)$; (x_0, y_0) — point of contact, which must satisfy the curve’s equation; θ — angle between curves.

Key Procedure

1. **Confirm the point** lies on the curve (it must satisfy the equation).
 2. **Find** $m = f'(x_0)$ by differentiating and substituting.
 3. **Write the tangent** via point-slope; for the **normal** use slope $-\frac{1}{m}$.
 4. **For special tangents** ("parallel to a line," "perpendicular to a line"): set m equal to the line's slope (parallel) or to $-\frac{1}{(\text{line slope})}$ (perpendicular), solve for the point.
-

Worked Examples

Example 1 — [JEE Main]

Find the equations of the tangent and normal to $y = x^2 - 3x + 2$ at the point $(2, 0)$.

Check: $2^2 - 3(2) + 2 = 0$ ✓. Differentiate: $\frac{dy}{dx} = 2x - 3$, so $m = 2(2) - 3 = 1$.

Tangent: $y - 0 = 1(x - 2) \Rightarrow y = x - 2$. Normal: slope -1 , so $y - 0 = -1(x - 2) \Rightarrow y = -x + 2$.

Example 2 — [JEE Main]

At which point on $y = x^3 - 11x + 5$ is the tangent parallel to the line $y = x - 11$?

Parallel means slope 1. $\frac{dy}{dx} = 3x^2 - 11 = 1 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2$. At $x = 2$, $y = 8 - 22 + 5 = -9$; at $x = -2$, $y = -8 + 22 + 5 = 19$. Points: $(2, -9)$ and $(-2, 19)$. (The textbook-classic answer often quotes $(2, -9)$.)

Example 3 — [JEE Advanced]

Show that the curves $y = x^2$ and $x = y^2$ intersect at $(1, 1)$, and find the angle between them there.

Both pass through $(1, 1)$. For $y = x^2$: $m_1 = 2x = 2$. For $x = y^2$, i.e. $y = \sqrt{x}$: $m_2 = \frac{1}{2\sqrt{x}} = \frac{1}{2}$. Then

$$\tan \theta = \left| \frac{2 - \frac{1}{2}}{1 + 2 \cdot \frac{1}{2}} \right| = \left| \frac{3/2}{2} \right| = \frac{3}{4} \Rightarrow \theta = \tan^{-1} \frac{3}{4} \approx 36.9^\circ.$$

Figure 6.7 The parabolas $y = x^2$ and $x = y^2$ crossing at the origin and at $(1, 1)$; mark the two tangent lines at $(1, 1)$ and the angle θ between them. (see next page)

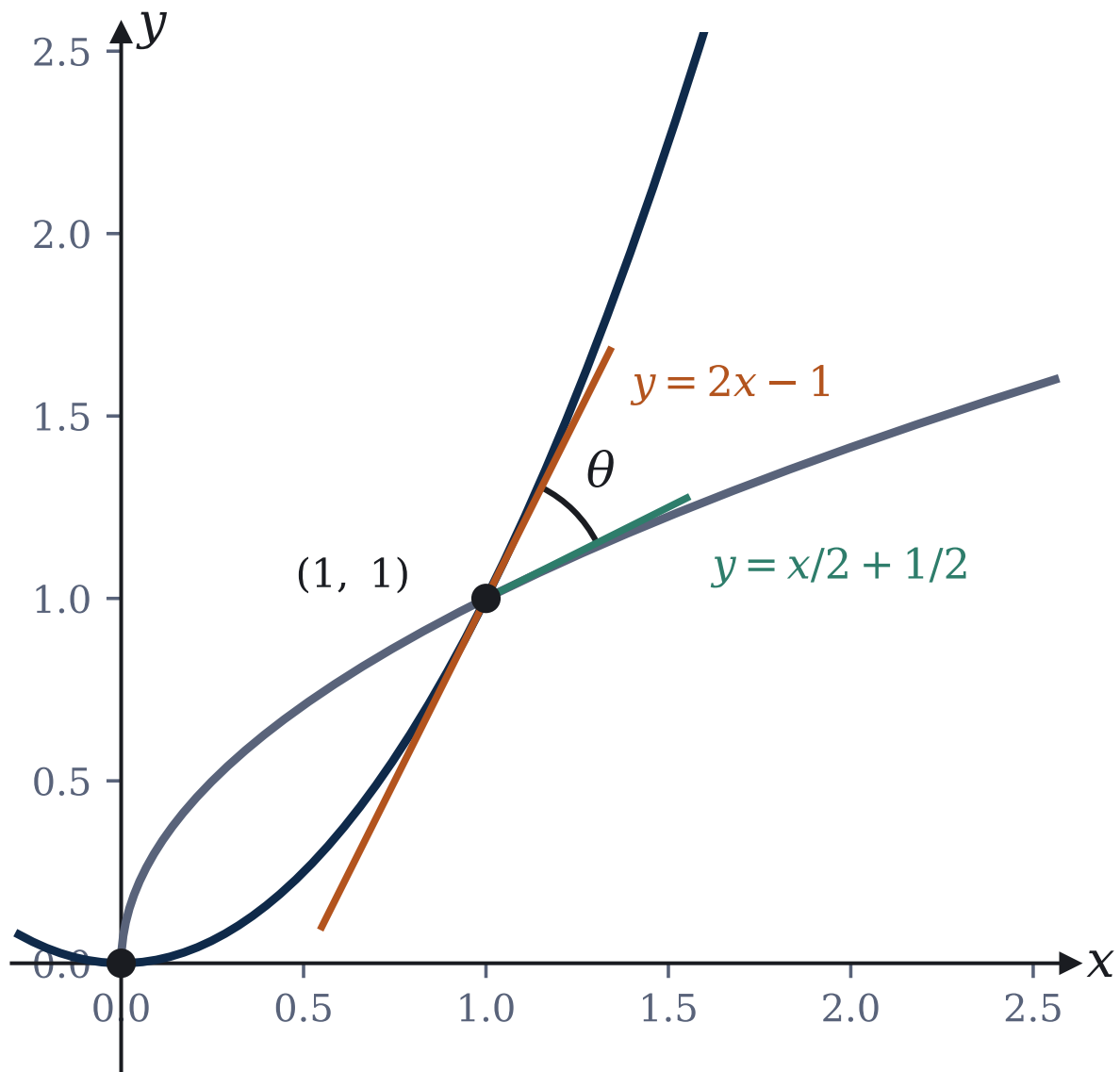


Figure 6.7 The parabolas $y = x^2$ and $x = y^2$ crossing at the origin and at $(1, 1)$; mark the two tangent lines at $(1, 1)$ and the angle θ between them.

Practice & Pitfalls

Practice: (i) Tangent to $y = \sin x$ at $x = 0$. (ii) Normal to $y = \frac{1}{x}$ at $(1, 1)$. (iii) Points on $y = x^3$ where the tangent is parallel to $y = 12x + 7$. **Answers:** (i) $y = x$. (ii) $y = x$. (iii) $(2, 8)$ and $(-2, -8)$.

Pitfalls & Pro-Tips: (1) Don't forget to verify the point is on the curve. (2) Normal slope is $-\frac{1}{m}$, *flip and negate* — a missed sign is the most common error. (3) For “tangent perpendicular to a line,” set $m \cdot (\text{line slope}) = -1$, not equal. (4) When $m = 0$, write the horizontal/vertical lines directly rather than dividing by zero.

Cheat Sheet: Tangent slope $= f'(x_0)$; tangent $y - y_0 = m(x - x_0)$; normal $y - y_0 = -\frac{1}{m}(x - x_0)$; angle $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$; orthogonal $\iff m_1 m_2 = -1$.

Appendix B: Approximations and Differentials (JEE-relevant — outside current CBSE board syllabus)

Exam Relevance: Section 6.5 was removed from CBSE Boards, but linear approximation and error estimation remain standard JEE Main material and a useful tool throughout physics. Include if preparing for JEE.

Concept Intuition

Near a point, a smooth curve and its tangent line are almost indistinguishable. So if you know a function's value and slope at a convenient point, you can *estimate* its value a tiny step away by walking along the tangent instead of the curve. That is **linear approximation** — trading a hard exact value (like $\sqrt{25.3}$) for an easy nearby one (like $\sqrt{25} = 5$) plus a small correction.

Think about it this way... the derivative converts a small input nudge dx into an output nudge $dy = f'(x) dx$. For small nudges the true change Δy is very close to dy , so $f(x + \Delta x) \approx f(x) + f'(x) \Delta x$. The smaller the step, the better the estimate.

Caveats: the approximation is only good for **small** Δx , and you should pick a base point x where $f(x)$ and $f'(x)$ are easy and *close* to the target. Trig approximations require x in **radians**.

Key Formulas & Results

Differential: for $y = f(x)$,

$$dy = f'(x) dx, \quad \Delta y \approx dy \text{ for small } \Delta x.$$

Linear approximation:

$$f(x + \Delta x) \approx f(x) + f'(x) \Delta x.$$

Error estimation. If x is measured with a small error Δx (so $dx = \Delta x$):

$$\text{absolute error in } y \approx |dy| = |f'(x)| |dx|,$$

$$\text{relative error} \approx \left| \frac{dy}{y} \right|, \quad \text{percentage error} \approx \left| \frac{dy}{y} \right| \times 100.$$

Symbols. $dx = \Delta x$ — small change in input; dy — differential (estimated change in output); Δy — true change. The method needs f differentiable at the base point and Δx small.

Key Procedure

1. **Identify the target** $f(x + \Delta x)$ and choose a nearby base x where $f(x)$ is exact and easy.
 2. **Set** $\Delta x = \text{target} - \text{base}$ (watch the sign).
 3. **Compute** $f(x)$ at the base.
 4. **Combine:** $f(x + \Delta x) \approx f(x) + f'(x) \Delta x$.
 5. **For errors:** use $dy = f'(x) dx$; divide by y for relative/percentage error.
-

Worked Examples

Example 1 — [JEE Main]

Approximate $\sqrt{25.3}$.

Take $f(x) = \sqrt{x}$, base $x = 25$, $\Delta x = 0.3$. Then $f(25) = 5$ and $f'(x) = \frac{1}{2\sqrt{x}} \Rightarrow f'(25) = \frac{1}{10}$.

$$\sqrt{25.3} \approx 5 + \frac{1}{10}(0.3) = 5 + 0.03 = 5.03.$$

(True value ≈ 5.0299 — excellent.)

Example 2 — [JEE Main]

Approximate $(1.97)^6$ — **estimate to a sensible accuracy.** (Take $f(x) = x^6$.)

Base $x = 2$, $\Delta x = -0.03$. $f(2) = 64$, $f'(x) = 6x^5 \Rightarrow f'(2) = 6(32) = 192$.

$$(1.97)^6 \approx 64 + 192(-0.03) = 64 - 5.76 = 58.24.$$

Example 3 — [JEE Advanced]

The radius of a sphere is measured as 9 cm with a possible error of 0.03 cm. Estimate the maximum percentage error in the computed volume.

$V = \frac{4}{3}\pi r^3 \Rightarrow dV = 4\pi r^2 dr$. The **relative** error:

$$\frac{dV}{V} = \frac{4\pi r^2 dr}{\frac{4}{3}\pi r^3} = 3 \frac{dr}{r} = 3 \cdot \frac{0.03}{9} = 0.01.$$

So the percentage error in volume $\approx 1\%$. (Note the clean shortcut: for $V \propto r^3$, the relative error in V is three times that in r .)

Practice & Pitfalls

Practice: (i) Approximate $\sqrt[3]{27.1}$. (ii) Approximate $\sin 31^\circ$ (use 30° , in radians). (iii) The side of a square is measured with 2% error; find the approximate percentage error in its area. **Answers:** (i) ≈ 3.0037 . (ii) ≈ 0.5151 . (iii) $\approx 4\%$.

Pitfalls & Pro-Tips: (1) Choose the base point close to the target *and* easy to evaluate — a poor base ruins the estimate. (2) Mind the sign of Δx (e.g. $1.97 \rightarrow \Delta x = -0.03$). (3) Distinguish absolute error ($|dy|$) from relative/percentage error ($|dy/y|$). (4) For powers, the shortcut $\frac{dy}{y} = n \frac{dx}{x}$ (when $y = x^n$) saves time. (5) Trig: convert degrees to radians first.

Cheat Sheet: $dy = f'(x) dx$; $f(x + \Delta x) \approx f(x) + f'(x)\Delta x$; relative error $\frac{dy}{y}$; for $y = x^n$, $\frac{dy}{y} = n \frac{dx}{x}$.

Chapter 6: Applications of Derivatives — Round 2

Addendum

The chapter taught you related rates, monotonicity, and maxima–minima, with appendices on tangents/normals and approximations. Five examinable structures are still standing outside. Addendum A supplies Rolle’s theorem and the Mean Value Theorem — the material the rationalised syllabus deleted but JEE Main still tests, and the hidden engine behind the monotonicity theorem the chapter uses without proof. Addendum B takes monotonicity to the two jobs JEE actually hires it for: counting real roots against a parameter, and proving chained inequalities by the comparative lemma. Addendum C answers the question the chapter dodges — what happens when the Second Derivative Test returns zero — via the n -th derivative test, concavity and points of inflection. Addendum D handles absolute extrema on open and unbounded domains, where the chapter’s closed-interval working rule silently fails, plus parametrisation-based optimisation. Addendum E completes Appendix A with the subtangent–subnormal geometry that JEE repeatedly mines. Nothing here is new calculus: every method is the sign chart and the critical-point audit, applied one level deeper than the chapter took them.

Addendum A — Rolle’s Theorem and the Mean Value Theorem

The chapter proves nothing about why $f'(x) > 0$ forces f to increase — it cites Lagrange’s Mean Value Theorem and moves on, because MVT was deleted from the CBSE syllabus. JEE Main did not delete it: direct questions (“verify Rolle, find c ”, “use MVT to bound an expression”) appear regularly, and JEE Advanced uses it as a root-existence tool. CBSE boards rarely ask it by name now, but the theorem costs one page to learn and secures marks in both exams.

Method

Rolle’s Theorem. Let f be continuous on $[a, b]$ and differentiable on (a, b) , with $f(a) = f(b)$. Then there exists $c \in (a, b)$ with

$$f'(c) = 0.$$

Why it is true. A continuous function on a closed interval attains a maximum and a minimum. If both sit at the endpoints, then $f(a) = f(b)$ forces f constant and every c works. Otherwise some extremum sits at an interior point c . Now the key lemma (Fermat): an interior extremum of a differentiable function has zero derivative. Suppose $f(c)$ is a local maximum. For small $h > 0$ the difference quotient obeys

$$\frac{f(c+h) - f(c)}{h} \leq 0,$$

so the right-hand derivative is ≤ 0 . For $h < 0$ the numerator is still ≤ 0 but h is negative, so the quotient is ≥ 0 , making the left-hand derivative ≥ 0 . Both one-sided limits equal $f'(c)$, so $f'(c) \leq 0$ and $f'(c) \geq 0$ simultaneously:

$$f'(c) = 0.$$

Mean Value Theorem (Lagrange). Let f be continuous on $[a, b]$ and differentiable on (a, b) . Then there exists $c \in (a, b)$ with

$$f(b) - f(a) = f'(c)(b - a).$$

Why it is true. Tilt the picture flat. Define

$$\varphi(x) = f(x) - f(a) - \mu(x - a), \quad \mu = \frac{f(b) - f(a)}{b - a}.$$

Then $\varphi(a) = 0$, and $\varphi(b) = f(b) - f(a) - \mu(b - a) = 0$ by the choice of μ . Rolle applies to φ : there is a $c \in (a, b)$ with $\varphi'(c) = f'(c) - \mu = 0$, which is exactly the statement. ■

Cauchy's Mean Value Theorem (the two-function version JEE Advanced uses). If f and g are continuous on $[a, b]$, differentiable on (a, b) , and $g'(x) \neq 0$ there, then some $c \in (a, b)$ satisfies

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}.$$

Why it is true. Put $\lambda = \frac{f(b) - f(a)}{g(b) - g(a)}$ and define

$$F(x) = f(x) - f(a) - \lambda(g(x) - g(a)).$$

Then $F(a) = 0$ and $F(b) = f(b) - f(a) - \lambda(g(b) - g(a)) = 0$. Rolle gives $F'(c) = f'(c) - \lambda g'(c) = 0$, and dividing by the nonzero $g'(c)$ finishes it. ■

Three consequences you will use constantly.

1. If $f'(x) = 0$ at every point of an interval, f is constant there. *Proof:* for any two points, MVT gives $f(x_2) - f(x_1) = f'(c)(x_2 - x_1) = 0$.
2. If $|f'(x)| \leq M$ on an interval, then $|f(b) - f(a)| \leq M|b - a|$ there. *Proof:* take absolute values in MVT.
3. Between any two distinct zeros of f lies a zero of f' (this is Rolle restated). Iterating: n distinct zeros of f force at least $n - 1$ zeros of f' , at least $n - 2$ of f'' , and so on.

Hypothesis discipline. All three theorems need continuity on the *closed* interval and differentiability on the *open* interval. A corner like $|x|$ at 0, or a hole like $1/x$ at 0, voids the contract — and examiners set traps precisely there.

Example A.1 — [CBSE Board / JEE Main Level]

Verify Rolle's theorem for $f(x) = x^3 - 3x$ on $[0, \sqrt{3}]$ and find the point c .

Continuity and differentiability hold everywhere (polynomial). Endpoints:

$$f(0) = 0, \quad f(\sqrt{3}) = 3\sqrt{3} - 3\sqrt{3} = 0.$$

So $f(0) = f(\sqrt{3})$ and Rolle promises a $c \in (0, \sqrt{3})$ with $f'(c) = 0$. Now

$$f'(x) = 3x^2 - 3 = 3(x - 1)(x + 1),$$

which vanishes at $x = \pm 1$. Only $c = 1$ lies in $(0, \sqrt{3})$. Verified, with $c = 1$.

Contrast — why the hypotheses matter. Take $f(x) = |x|$ on $[-1, 1]$. Here $f(-1) = f(1) = 1$, yet no c has $f'(c) = 0$: the derivative is -1 left of 0, $+1$ right of it, and does not exist at 0. Differentiability on the open interval fails, the theorem does not apply, and the conclusion genuinely fails with it. A verification question expects you to check all three hypotheses, not just the endpoint values.

Example A.2 — [JEE Main Level]

For $0 < a < b$, prove that

$$\frac{b-a}{b} < \ln \frac{b}{a} < \frac{b-a}{a}.$$

Apply the MVT to $f(t) = \ln t$ on $[a, b]$. Everything is continuous and differentiable, so there is a $c \in (a, b)$ with

$$\ln b - \ln a = \frac{1}{c} (b - a).$$

The left side is $\ln(b/a)$. Now squeeze c . Since $a < c < b$, taking reciprocals reverses the inequalities:

$$\frac{1}{b} < \frac{1}{c} < \frac{1}{a},$$

and multiplying through by the positive number $b - a$ gives the result. ■

Instant corollary. Put $a = 1$, $b = 1 + x$ with $x > 0$:

$$\frac{x}{1+x} < \ln(1+x) < x.$$

Both bounds come from one MVT application — this pair of inequalities is a standing JEE favourite.

Example A.3 — [JEE Advanced Level]

Suppose $a_0 + \frac{a_1}{2} + \frac{a_2}{3} + \cdots + \frac{a_n}{n+1} = 0$. **Show that the equation**

$$a_0 + a_1x + a_2x^2 + \cdots + a_nx^n = 0$$

has at least one root in $(0, 1)$.

The given sum is not the polynomial at any obvious point — so build a function whose derivative is the polynomial. Define

$$F(x) = a_0x + \frac{a_1x^2}{2} + \frac{a_2x^3}{3} + \cdots + \frac{a_nx^{n+1}}{n+1}.$$

Then F is a polynomial, hence continuous on $[0, 1]$ and differentiable on $(0, 1)$, and term-by-term differentiation gives

$$F'(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n.$$

Evaluate the endpoints:

$$F(0) = 0, \quad F(1) = a_0 + \frac{a_1}{2} + \cdots + \frac{a_n}{n+1} = 0$$

by the given condition. Rolle's theorem applies verbatim: some $c \in (0, 1)$ has $F'(c) = 0$, i.e. the polynomial vanishes at c . ■

Why it is Advanced-level. Nothing in the statement mentions derivatives. The mark-winning move is recognising that a condition on a sum of coefficients is a condition on $F(1)$ for a deliberately constructed antiderivative — Rolle then converts “ $F(0) = F(1)$ ” into a root. This antiderivative construction recurs whenever JEE hides Rolle inside a coefficient condition.

Practice — Addendum A

1. Verify Rolle for $f(x) = x^3 - 3x$ on $[0, \sqrt{3}]$ is Example A.1; now verify it for $f(x) = \sin x$ on $[0, \pi]$ and find c .
2. Explain why the MVT cannot be applied to $f(x) = \frac{1}{x}$ on $[-1, 1]$, and show the conclusion fails anyway.
3. Using the MVT, prove $|\tan a - \tan b| \geq |a - b|$ for all $a, b \in (-\pi/2, \pi/2)$.
4. Prove that for $x > 0$, $\frac{x}{1+x} < \ln(1+x) < x$ without quoting Example A.2 — apply the MVT directly on $[0, x]$.
5. A cubic $p(x)$ has three distinct real roots. Show that $p'(x)$ has two distinct real roots and $p''(x)$ has at least one.
6. Given $a_0 + \frac{a_1}{2} + \frac{a_2}{3} = 0$, show that $a_0 + a_1x + a_2x^2 = 0$ has a root in $(0, 1)$.

Answers: 1. $f(0) = f(\pi) = 0$; $f'(x) = \cos x = 0$ gives $c = \pi/2 \in (0, \pi)$. 2. f is not defined (hence not continuous) at $0 \in [-1, 1]$. Conclusion fails: $f(-1) = f(1) = -1$ would require some c with $f'(c) = 0$, but $f'(x) = -1/x^2 < 0$ wherever it exists. 3. MVT on $[a, b]$ (WLOG $a < b$): $\tan a - \tan b = \sec^2 c (a - b)$ for some c ; since $\sec^2 c \geq 1$, taking absolute values gives the claim. 4. There is a $c \in (0, x)$ with $\ln(1+x) - \ln 1 = \frac{x}{1+c}$; from $0 < c < x$ get $\frac{x}{1+x} < \frac{x}{1+c} < x$. 5. Label the roots $r_1 < r_2 < r_3$; Rolle on $[r_1, r_2]$ and $[r_2, r_3]$ gives zeros $c_1 \neq c_2$ of p' ; Rolle on $[c_1, c_2]$ gives a zero of p'' . 6. Apply Example A.3 with $n = 2$: $F(x) = a_0x + a_1x^2/2 + a_2x^3/3$ has $F(0) = F(1) = 0$, so $F'(c) = 0$ for some $c \in (0, 1)$.

Addendum B — Monotonicity II: Counting Roots and Comparative Inequalities

The chapter extracts increasing/decreasing intervals and proves single inequalities anchored at one point. JEE asks monotonicity to do two harder jobs. First, *root auditing*: how many real solutions does an equation have, as a parameter varies — answered by reading the signs of f at its successive extrema. Second, *chained inequalities*: proving $\sin x$ squeezed between two polynomials, where one auxiliary function needs another auxiliary function to certify it. One lemma powers the second job, and one protocol powers the first.

Method

The comparative lemma. Let f and g be continuous on $[a, \beta)$ and differentiable on (a, β) , with

$$f(a) \geq g(a) \quad \text{and} \quad f'(x) > g'(x) \text{ for all } x \in (a, \beta).$$

Then $f(x) > g(x)$ for every $x \in (a, \beta)$.

Proof. Put $h = f - g$. Then $h(a) \geq 0$ and $h'(x) > 0$, so h is strictly increasing on the interval (monotonicity theorem). Hence for $x > a$,

$$h(x) > h(a) \geq 0,$$

which is $f(x) > g(x)$. ■

- If you only have $f'(x) \geq g'(x)$, the conclusion weakens to $f(x) \geq g(x)$ — track strictness honestly.
- The anchor a is usually 0, where both functions vanish. The lemma converts “compare derivatives” into “compare values”, one rung at a time.

Chained inequalities. To prove $P(x) < Q(x) < R(x)$ for $x > 0$: prove each link separately by the lemma, and allow links to *borrow* earlier links — proving the derivative of one auxiliary function positive may itself require the previous link. Order the proofs so borrowing is legal.

Root-audit protocol. To count the real roots of $f(x) = 0$:

1. Fix the domain and confirm continuity on each piece.
2. Compute f' , find all critical points, and sort them: $c_1 < c_2 < \dots$
3. On each monotonic branch the graph crosses the axis **at most once**.
4. Evaluate f at each critical point and take the limits at the ends of the domain ($\pm\infty$, vertical asymptotes). A branch has exactly one root precisely when its two end-values have opposite signs.
5. Add the per-branch counts. For a parameter k , the boundary between “different counts” occurs when an extremum value hits 0 — solve $f(c_i) = 0$ for k and test the resulting intervals.

Example B.1 — [JEE Main Level]

Prove that for all $x > 0$,

$$\ln(1+x) > x - \frac{x^2}{2}.$$

Set $h(x) = \ln(1+x) - x + \frac{x^2}{2}$, so $h(0) = 0$. Differentiate:

$$h'(x) = \frac{1}{1+x} - 1 + x = \frac{1 - (1+x) + x(1+x)}{1+x} = \frac{x^2}{1+x}.$$

For $x > 0$ the right side is strictly positive, so $h' > 0$ on $(0, \infty)$ and the comparative lemma (with $g \equiv h(0) = 0$) gives $h(x) > h(0) = 0$. ■

Check. At $x = 1$: $\ln 2 \approx 0.693$ versus $1 - 0.5 = 0.5$ — holds. As $x \rightarrow 0^+$ both sides tend to 0, consistent with the anchor.

Example B.2 — [JEE Advanced Level]

Prove that for all $x > 0$,

$$x - \frac{x^3}{6} < \sin x < x.$$

Upper link first. Let $u(x) = \sin x - x$, so $u(0) = 0$ and

$$u'(x) = \cos x - 1 \leq 0,$$

with equality only at isolated points $x = 2n\pi$. By the sharpened monotonicity theorem (chapter, Subtopic 02), u is strictly decreasing, so $u(x) < u(0) = 0$ for $x > 0$. That is $\sin x < x$. Record it.

Lower link, borrowing the upper one. Let $w(x) = \sin x - x + \frac{x^3}{6}$, so $w(0) = 0$ and

$$w'(x) = \cos x - 1 + \frac{x^2}{2}.$$

Whether $w' > 0$ is not immediate — so build one more auxiliary. Let $v(x) = \cos x - 1 + \frac{x^2}{2}$, so $v(0) = 0$ and

$$v'(x) = -\sin x + x.$$

But $-\sin x + x > 0$ for $x > 0$ is *exactly the upper link just proved*. So $v' > 0$, hence $v(x) > v(0) = 0$, hence $w'(x) > 0$, hence $w(x) > w(0) = 0$. That is $\sin x > x - x^3/6$. ■

Why it is Advanced-level. The chain has a required proof order: the lower link is false to attempt before the upper link is established, because its derivative certificate consumes the upper link. Recognising which auxiliary to prove first — and reusing a proved inequality inside another derivative — is the examined skill, not the differentiation.

Example B.3 — [JEE Main rising to Advanced]

Determine the number of real roots of $x^3 - 3x + k = 0$ as k ranges over $\mathbb{R}$.

Run the audit on $f(x) = x^3 - 3x + k$. Domain $\mathbb{R}$, and

$$f'(x) = 3x^2 - 3 = 3(x-1)(x+1),$$

so the critical points are $x = -1$ (local maximum) and $x = 1$ (local minimum), since $f''(x) = 6x$. The extremum values:

$$f(-1) = -1 + 3 + k = k + 2, \quad f(1) = 1 - 3 + k = k - 2.$$

End behaviour: $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$ and $f(x) \rightarrow +\infty$ as $x \rightarrow +\infty$. Three monotonic branches, each crossing at most once:

- Branch $(-\infty, -1]$: rises from $-\infty$ to $k + 2$. Has a root $\iff k + 2 > 0 \iff k > -2$.
- Branch $[-1, 1]$: falls from $k + 2$ to $k - 2$. Has a root $\iff$ opposite signs $\iff -2 < k < 2$.
- Branch $[1, \infty)$: rises from $k - 2$ to $+\infty$. Has a root $\iff k - 2 < 0 \iff k < 2$.

Combining:

Parameter range	Real roots of $x^3 - 3x + k = 0$
$-2 < k < 2$	three distinct
$k = 2$ or $k = -2$	two distinct (one double)
$k < -2$ or $k > 2$	exactly one

Verification of the boundary rows. At $k = 2$ the polynomial factors as $(x-1)^2(x+2)$ — a double root at 1 (the local minimum touching the axis) and a simple root at -2 . At $k = -2$ it factors as $(x+1)^2(x-2)$. Both match the audit exactly.

Check. $k = 0$: the table predicts three roots. Indeed $x^3 - 3x = x(x - \sqrt{3})(x + \sqrt{3})$. $k = 5$: predicted one root; $f(-1) = 7 > 0$ and $f(1) = 3 > 0$, so both middle branches stay positive and only the left branch crosses.

Practice — Addendum B

1. Prove $e^x > 1 + x + \frac{x^2}{2}$ for all $x > 0$.
2. Determine the number of real roots of $2x^3 - 9x^2 + 12x - 3 = 0$.
3. Prove $\tan x > x$ for all $x \in (0, \pi/2)$ using the comparative lemma.
4. Find all real k for which $x^3 - 3x + k = 0$ has exactly one real root.
5. Show that the equation $xe^x = 2$ has exactly one real root.

Answers: 1. Let $h = e^x - 1 - x - \frac{x^2}{2}$; then $h(0) = 0$ and $h' = e^x - 1 - x > 0$ for $x > 0$ (apply the lemma recursively, or note $e^x - 1 - x$ has derivative $e^x - 1 > 0$ and value 0 at 0). Hence $h(x) > 0$. 2. $f' = 6(x-1)(x-2)$; extremum values $f(1) = 2 > 0$ and $f(2) = 1 > 0$. Left branch rises from $-\infty$ to 2: one root. Middle branch falls from 2 to 1: none. Right branch rises from 1 to ∞ : none. Exactly one real root. 3. $h = \tan x - x$, $h(0) = 0$, $h' = \sec^2 x - 1 = \tan^2 x > 0$ on $(0, \pi/2)$; lemma gives $h > 0$. 4. $k < -2$ or $k > 2$. 5. Let $f = xe^x - 2$; $f' = e^x(1+x)$, so f decreases on $(-\infty, -1]$ from the limit -2 (since $xe^x \rightarrow 0^-$) to $f(-1) = -1/e - 2 < 0$: no root there. On $[-1, \infty)$, f increases from a negative value to $+\infty$: exactly one root. Total: one.

Addendum C — When the Second Test Is Silent: Higher Derivatives, Concavity, Inflection

The chapter's Second Derivative Test has a hole it admits but never patches: at x^4 the origin has $f''(0) = 0$ and the test "fails", while at x^3 the same reading signals a non-extremum. Same diagnostic, opposite realities. JEE loves this border. This addendum supplies the complete classification — the n -th derivative test, with proof — and the concavity/inflection vocabulary that CBSE HOTS and JEE both examine.

Method

The fallback hierarchy. At a critical point c with $f'(c) = 0$:

1. Try $f''(c)$. Nonzero $\Rightarrow$ done (chapter's test).
2. If $f''(c) = 0$, the first-derivative sign chart *always* settles it — build it and look for a flip.
3. When derivatives are cheap to compute, the n -th derivative test below is the fast lane.

The n -th derivative test. Suppose that near c the derivatives of f up to order n exist, $f^{(n)}$ is continuous at c (true for every exam function), and

$$f'(c) = f''(c) = \dots = f^{(n-1)}(c) = 0, \quad f^{(n)}(c) \neq 0.$$

Then:

- **n even:** c is a strict local maximum if $f^{(n)}(c) < 0$, a strict local minimum if $f^{(n)}(c) > 0$.
- **n odd:** c is not an extremum; the graph crosses its horizontal tangent there — a stationary point of inflection.

Why it is true. WLOG $f^{(n)}(c) > 0$ (otherwise reverse every sign below). By continuity, $f^{(n)} > 0$ on some interval around c , so $f^{(n-1)}$ is strictly increasing there; since $f^{(n-1)}(c) = 0$,

$$f^{(n-1)} < 0 \text{ just left of } c, \quad f^{(n-1)} > 0 \text{ just right of } c.$$

So $f^{(n-2)}$ strictly decreases to c and strictly increases after: a strict local minimum of $f^{(n-2)}$ at c , and since $f^{(n-2)}(c) = 0$,

$$f^{(n-2)}(x) > 0 \text{ for } x \neq c \text{ near } c.$$

The pattern now locks in and alternates down the ladder. Each derivative at *odd* distance below n (orders $n-1, n-3, \dots$) is strictly monotone through c and changes sign there; each derivative at *even* distance (orders $n-2, n-4, \dots$) has a strict local minimum, value 0, at c . Descending to order zero:

- n even: f itself lands in an even slot $\Rightarrow$ strict local minimum at c .
- n odd: f lands in an odd slot $\Rightarrow f$ changes sign at $c \Rightarrow$ no extremum, stationary inflection.



Concavity and inflection. f is concave upward on intervals where $f'' > 0$ and concave downward where $f'' < 0$. A point of inflection is an interior domain point where the concavity *changes*. Two facts, both examinable:

- If f'' is continuous and c is an inflection point, then $f''(c) = 0$ — a sign change forces a zero. Necessary, not sufficient: $f(x) = x^4$ has $f''(0) = 0$ yet $f'' \geq 0$ everywhere, so no inflection.
- Procedure: solve $f''(x) = 0$, then test the sign of f'' on each side. Only sign-changers count.

Example C.1 — [JEE Main Level]

Locate and classify all critical points and all inflection points of $f(x) = 3x^5 - 5x^3$.

First derivative:

$$f'(x) = 15x^4 - 15x^2 = 15x^2(x-1)(x+1),$$

so the critical points are $x = -1, 0, 1$. Second derivative:

$$f''(x) = 60x^3 - 30x = 30x(2x^2 - 1).$$

- $x = -1$: $f''(-1) = -30 < 0 \Rightarrow$ local maximum, $f(-1) = -3 + 5 = 2$.
- $x = 1$: $f''(1) = 30 > 0 \Rightarrow$ local minimum, $f(1) = 3 - 5 = -2$.
- $x = 0$: $f''(0) = 0$ — the test is silent. Fall back to the sign chart of f' . Writing $f' = 15x^2(x-1)(x+1)$: on $(-1, 0)$ the signs are $(+)(-)(+) = -$; on $(0, 1)$ they are $(+)(-)(+) = -$. No flip $\Rightarrow$ no extremum at 0.

Inflection candidates: $f'' = 0$ at $x = 0$ and $x = \pm \frac{1}{\sqrt{2}}$. The sign of $f'' = 30x(2x^2 - 1)$ runs $-, +, -, +$ across the four intervals cut by these points, changing at **every** candidate. So all three are inflection points, with values

$$f(0) = 0, \quad f\left(\pm \frac{1}{\sqrt{2}}\right) = \mp \frac{7\sqrt{2}}{8}.$$

Check. $x = 0$ is simultaneously a stationary inflection (from the f' chart) and an inflection (from the f'' chart) — consistent, since a stationary inflection must change concavity. The local max value 2 exceeds the local min value -2 with the origin between them: plausible for an odd-type quintic.

Example C.2 — [JEE Advanced Level]

(a) Show that $f(x) = x^5 - 5x^4 + 10x^3 - 10x^2 + 5x + 2$ has no extremum, and describe the behaviour at its critical point. (b) Show that $g(x) = x^4 + 4x^3 + 6x^2 + 4x + 2$ has a strict local minimum, and find it.

(a) Solve $f'(x) = 5x^4 - 20x^3 + 30x^2 - 20x + 5 = 0$. Rather than grind, test the suspicion of a binomial pattern: the coefficients $1, -5, 10, -10, 5$ are row five of Pascal's triangle with alternating signs, so

$$f(x) = (x - 1)^5 + 3.$$

Check by expanding $(x - 1)^5 = x^5 - 5x^4 + 10x^3 - 10x^2 + 5x - 1$; adding 3 reproduces f . Now everything collapses:

$$f'(x) = 5(x - 1)^4, \quad f''(x) = 20(x - 1)^3, \quad f^{(3)}(x) = 60(x - 1)^2,$$

$$f^{(4)}(x) = 120(x - 1), \quad f^{(5)}(x) = 120.$$

At $c = 1$: the first four derivatives vanish and $f^{(5)}(1) = 120 \neq 0$. Here $n = 5$ is **odd**, so by the n -th derivative test $x = 1$ is a stationary point of inflection — no maximum, no minimum. The graph crosses its horizontal tangent $y = 3$ there.

(b) Similarly $g(x) = (x + 1)^4 + 1$, so $g'(x) = 4(x + 1)^3$ vanishes only at $c = -1$, with

$$g''(-1) = g^{(3)}(-1) = 0, \quad g^{(4)}(-1) = 24 > 0.$$

Here $n = 4$ is **even** with a positive fourth derivative $\Rightarrow$ strict local minimum, value $g(-1) = 1$. Note the Second Derivative Test alone could not have delivered this: $g''(-1) = 0$.

Why it is Advanced-level. Both parts live exactly where the chapter's toolkit stops: $f''(c) = 0$ at the only critical point. Part (a) rewards spotting the Pascal-row factorisation; part (b) shows the even-case half of the n -th test earning its keep. In an exam where the pattern is *not* recognisable, the same conclusions follow from the first-derivative sign chart — slower but unconditional.

Practice — Addendum C

1. Find the points of inflection of $f(x) = x^3 - 6x^2 + 9x$.
2. Show that $f(x) = x^4$ has a strict local minimum at 0 even though $f''(0) = 0$, and name which test settles it.
3. Locate and classify the critical points of $f(x) = x^5 - 5x + 3$.
4. Determine whether $f(x) = x^7 + x$ has an extremum anywhere.
5. [Advanced] For $f(x) = x^6 - 3x^4 + 3x^2$, find all critical points, classify each, and locate all inflection points.

Answers: 1. $f'' = 6x - 12$, zero at $x = 2$ with a sign change $\Rightarrow$ inflection at $(2, 2)$. 2. $f' = 4x^3$ changes sign from $-$ to $+$ at 0 $\Rightarrow$ strict local minimum by the First Derivative Test (equivalently, the n -th test with $n = 4$, $f^{(4)}(0) = 24 > 0$). 3. $f' = 5(x^2 - 1)(x^2 + 1)$; critical points $x = \pm 1$; $f'' = 20x^3$ gives $f''(1) = 20 > 0 \Rightarrow$ local min $f(1) = -1$, and $f''(-1) = -20 < 0 \Rightarrow$ local max $f(-1) = 7$. 4. $f' = 7x^6 + 1 > 0$ for all $x \Rightarrow$ strictly increasing, no critical points, no extremum. 5. Factor $f = (x^2 - 1)^3 + 1$, so $f' = 6x(x^2 - 1)^2$. Critical points $x = 0, \pm 1$. The sign of f' is negative on $(-\infty, 0)$ and positive on $(0, \infty) \Rightarrow$ strict local minimum at $x = 0$, value $f(0) = 0$; no sign change at $\pm 1 \Rightarrow$ not extrema. Further, $f'' = 6(x^2 - 1)(5x^2 - 1)$ changes sign at $x = \pm 1$ and $x = \pm \frac{1}{\sqrt{5}} \Rightarrow$ four inflection points; those at $x = \pm 1$ are stationary, those at $x = \pm \frac{1}{\sqrt{5}}$ are not.

Addendum D — Extrema on Open and Unbounded Domains; Parametrised Optimisation

The chapter's absolute-extrema rule — “evaluate at critical points and both endpoints, compare” — is a closed-interval rule. On $(0, \infty)$, on $\mathbb{R}$, or on any domain with a missing or infinite end, there are no endpoints to evaluate, and the rule returns nothing. Yet JEE Main's most common optimisation question lives exactly there: minimise $x + \frac{a}{x}$, find the closest point on a curve, maximise an inscribed rectangle. This addendum supplies the existence test that replaces the endpoint audit, and the parametrisation trick that dissolves constrained problems.

Method

Existence on an unbounded domain. Let f be continuous on (α, β) , where either end may be infinite, and suppose

$$f(x) \rightarrow +\infty \quad \text{as } x \rightarrow \alpha^+ \text{ and as } x \rightarrow \beta^-.$$

Then f attains a global minimum, and it occurs at an interior critical point.

Why. Fix x_0 and put $K = f(x_0)$. Since f blows up at both ends, some closed subinterval $[p, q] \subset (\alpha, \beta)$ contains x_0 with $f > K + 1$ outside it. The closed-interval working rule gives a least value $m \leq K$ on $[p, q]$. If that least value were attained at p or q , continuity would force values just outside $[p, q]$ to be close to $m \leq K$, contradicting $f > K + 1$ there. So the

minimiser is interior, hence a critical point. ■

With $-\infty$ limits, the same argument yields a global maximum. If an end-limit is *finite*, that value joins the comparison list instead of winning automatically.

Working rule for open/unbounded domains.

1. State the domain; find all interior critical points.
2. Compute the limits of f at every end of the domain.
3. If both end-limits are $+\infty$: the smallest critical value is the global minimum. If both are $-\infty$: the largest critical value is the global maximum. If an end-limit is finite, include it in the comparison.
4. Declare existence or non-existence explicitly — “no global minimum” is a full-credit answer when the limits demand it.

The model family. For $p, q > 0$ and $x > 0$, $f(x) = px + \frac{q}{x}$ has

$$f'(x) = p - \frac{q}{x^2} = 0 \iff x = \sqrt{\frac{q}{p}}, \quad f''(x) = \frac{2q}{x^3} > 0,$$

so the global minimum is

$$f\left(\sqrt{\frac{q}{p}}\right) = p\sqrt{\frac{q}{p}} + \frac{q}{\sqrt{q/p}} = \sqrt{pq} + \sqrt{pq} = 2\sqrt{pq}.$$

Cross-check by AM-GM: $px + \frac{q}{x} \geq 2\sqrt{px \cdot \frac{q}{x}} = 2\sqrt{pq}$, equality when $px = q/x$. Calculus and AM-GM agree — use whichever is faster, but be able to run both.

Parametrisation. When the constraint is a circle or ellipse, parametrise it by angle — $(a \cos \theta, b \sin \theta)$ for an ellipse — turning a constrained problem into a single trig function. This routinely beats eliminating a variable through a square root.

Example D.1 — [JEE Main Level]

Find the global minimum of $f(x) = x + \frac{4}{x^2}$ for $x > 0$.

Domain $(0, \infty)$; differentiate:

$$f'(x) = 1 - \frac{8}{x^3}, \quad f'(x) = 0 \iff x^3 = 8 \iff x = 2.$$

Curvature: $f''(x) = \frac{24}{x^4} > 0$, so $x = 2$ is a local minimum with

$$f(2) = 2 + \frac{4}{4} = 3.$$

End-limits: $f \rightarrow +\infty$ as $x \rightarrow 0^+$ (the $4/x^2$ term explodes) and $f \rightarrow +\infty$ as $x \rightarrow \infty$. Both ends dead-end upward, so by the existence test the local minimum is global:

$$\min_{x>0} f = 3 \quad \text{at } x = 2.$$

Independent cross-check. Split $x = \frac{x}{2} + \frac{x}{2}$ and apply AM-GM to the three positive terms $\frac{x}{2}, \frac{x}{2}, \frac{4}{x^2}$:

$$f(x) \geq 3 \sqrt[3]{\frac{x}{2} \cdot \frac{x}{2} \cdot \frac{4}{x^2}} = 3 \sqrt[3]{1} = 3,$$

with equality when $\frac{x}{2} = \frac{4}{x^2}$, i.e. $x^3 = 8$, i.e. $x = 2$. Same answer, no calculus.

Example D.2 — [JEE Advanced Level]

Find the greatest area of a rectangle inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, and the rectangle that achieves it.

By symmetry the maximal rectangle has sides parallel to the axes; let $(a \cos \theta, b \sin \theta)$, $\theta \in (0, \pi/2)$, be its first-quadrant vertex. Then the sides are $2a \cos \theta$ and $2b \sin \theta$, and

$$A(\theta) = 4ab \sin \theta \cos \theta = 2ab \sin 2\theta \leq 2ab,$$

with equality at $\theta = \pi/4$. The greatest area is $2ab$, achieved by the rectangle with sides $a\sqrt{2}$ and $b\sqrt{2}$.

Elimination cross-check. Substitute $y = b\sqrt{1 - x^2/a^2}$, so $A(x) = 4bx\sqrt{1 - x^2/a^2}$ on $0 < x < a$. Maximising A is equivalent to maximising

$$A^2 = 16b^2 \left(x^2 - \frac{x^4}{a^2} \right),$$

whose derivative in x^2 vanishes when $2x - \frac{4x^3}{a^2} = 0$, giving $x = \frac{a}{\sqrt{2}}$ — the same rectangle. The parametrisation reached it in one line; elimination needed a square-root differentiation. On a circle ($a = b = R$) the answer specialises to the familiar square of area $2R^2$.

Example D.3 — [JEE Advanced Level]

Find the point on the parabola $y^2 = 4x$ closest to the point $(2, 1)$, and the shortest distance.

Minimise squared distance (monotone in distance, and kinder to differentiate). With $x = y^2/4$, the target is a function of y alone:

$$g(y) = \left(\frac{y^2}{4} - 2 \right)^2 + (y - 1)^2.$$

Expand before differentiating — the cross terms cancel:

$$g(y) = \frac{y^4}{16} - y^2 + 4 + y^2 - 2y + 1 = \frac{y^4}{16} - 2y + 5.$$

Then

$$g'(y) = \frac{y^3}{4} - 2 = 0 \iff y^3 = 8 \iff y = 2,$$

and $g''(y) = \frac{3y^2}{4} > 0$ there. Since $g \rightarrow +\infty$ as $y \rightarrow \pm\infty$, the existence test makes this the global minimum. The closest point is $y = 2$, $x = 1$: the point $(1, 2)$, at distance

$$\sqrt{(1-2)^2 + (2-1)^2} = \sqrt{2}.$$

Geometric check (ties to Appendix A). The joining segment from $(2, 1)$ to $(1, 2)$ has slope -1 . The tangent to $y^2 = 4x$ at $(1, 2)$: from $2y \frac{dy}{dx} = 4$ get $\frac{dy}{dx} = \frac{2}{y} = 1$, so the tangent has slope 1 and the segment, with slope -1 , is perpendicular to it. The shortest segment is normal to the curve — as it must be, since walking along the curve, the distance to a fixed point has a critical point exactly where the displacement is perpendicular to the tangent. Any answer failing this normal test is wrong on sight.

Practice — Addendum D

1. Find the minimum value of $2x + \frac{8}{x}$ for $x > 0$.
2. Find the global maximum of $f(x) = xe^{-x}$ on $\mathbb{R}$, and determine whether a global minimum exists.
3. A rectangle has its base on the diameter of a semicircle of radius r and its upper vertices on the arc. Find its greatest area.
4. Find the minimum value of $3x^2 + \frac{2}{x^3}$ for $x > 0$.
5. Find the point(s) on the parabola $y = x^2$ closest to $(0, 2)$, and the shortest distance.

Answers: 1. $f' = 2 - 8/x^2 = 0$ at $x = 2$; $f'' > 0$; value 8 (also the model family with $p = 2, q = 8$: $2\sqrt{16} = 8$). 2. $f' = e^{-x}(1 - x) = 0$ at $x = 1$; sign of f' flips $+$ to $- \Rightarrow$ global maximum $1/e$ (end-limits: $f \rightarrow 0^+$ as $x \rightarrow \infty$, $f \rightarrow -\infty$ as $x \rightarrow -\infty$). No global minimum — the function descends without bound. 3. Vertices $(\pm x, 0)$, $(\pm x, \sqrt{r^2 - x^2})$; area $A = 2x\sqrt{r^2 - x^2}$; maximise $A^2 = 4x^2(r^2 - x^2)$: derivative zero at $x = r/\sqrt{2}$; greatest area r^2 . 4. $f' = 6x - 6/x^4 = 0 \iff x^5 = 1 \iff x = 1$; $f'' = 6 + 24/x^5 > 0$; end-limits $+\infty$ both sides $\Rightarrow$ global minimum $3 + 2 = 5$. 5. Minimise $g = x^2 + (x^2 - 2)^2 = x^4 - 3x^2 + 4$; $g' = 2x(2x^2 - 3) = 0$ gives $x = 0$ (local max, $g = 4$) and $x = \pm\sqrt{3/2}$ (minima, $g'' = 12x^2 - 6 > 0$ there). Two closest points $(\pm\frac{\sqrt{6}}{2}, \frac{3}{2})$, each at distance $\sqrt{5/2} = \frac{\sqrt{10}}{2}$.

Addendum E — Tangents and Normals II: Subtangent, Subnormal, Intercepts

Appendix A gives you the equations of the tangent and normal and stops. JEE goes one step further and asks about the *segments* these lines cut off — the subtangent and subnormal — because for certain curves these segments are gloriously constant, and “show the subnormal of a parabola is constant” is a standing JEE pattern. Ten minutes of setup unlocks a whole question family.

Method

Setup. Let $P = (x, y)$ be a point on $y = f(x)$ with $y \neq 0$ and $m = \frac{dy}{dx} \neq 0$. Draw the vertical through P meeting the x -axis at $M = (x, 0)$. Let the tangent meet the x -axis at T and the normal meet it at N . Four segments are named:

Segment	Name	Length
MT	subtangent	$ y/m $
MN	subnormal	$ my $
PT	length of tangent	$ y \sqrt{1 + m^2} / m $
PN	length of normal	$ y \sqrt{1 + m^2}$

Derivations. The tangent at P is $Y - y = m(X - x)$. Setting $Y = 0$:

$$-y = m(X - x) \implies X = x - \frac{y}{m},$$

so $T = \left(x - \frac{y}{m}, 0\right)$ and the subtangent is $MT = \left|\frac{y}{m}\right|$. The normal is $Y - y = -\frac{1}{m}(X - x)$; setting $Y = 0$ gives $-my = -(X - x)$, so

$$X = x + my,$$

and the subnormal is $MN = |my|$. The lengths follow from right triangles: PT spans legs $|y|$ and $|y/m|$, so

$$PT = \sqrt{y^2 + \frac{y^2}{m^2}} = \frac{|y| \sqrt{1 + m^2}}{|m|},$$

and PN spans legs $|y|$ and $|my|$, giving $PN = |y| \sqrt{1 + m^2}$.

Intercepts of the tangent. From the same computation, the tangent’s x -intercept is $x - \frac{y}{m}$ and its y -intercept (set $X = 0$ in the tangent equation) is $y - mx$. The triangle cut off from the axes therefore has area

$$\frac{1}{2} \left| \left(x - \frac{y}{m}\right) (y - mx) \right|.$$

Two classics worth memorising with their proofs.

- *Exponential*: for $y = a^x$, $m = a^x \ln a$, so the subtangent is

$$\frac{y}{m} = \frac{a^x}{a^x \ln a} = \frac{1}{\ln a},$$

constant, independent of the point.

- *Parabola*: for $y^2 = 4ax$, differentiate to get $2y \frac{dy}{dx} = 4a$, so $m = \frac{2a}{y}$, and the subnormal is

$$my = \frac{2a}{y} \cdot y = 2a,$$

constant — the subnormal of a parabola equals its latus rectum half-length at *every* point.

Example E.1 — [JEE Main Level]

Find the subtangent, subnormal, tangent length and normal length for $y = x^3$ at the point $(1, 1)$, and verify each against the actual lines.

Here $m = 3x^2 = 3$ at $x = 1$, with $y = 1$. Directly:

$$MT = \frac{1}{3}, \quad MN = 3, \quad PT = \frac{\sqrt{10}}{3}, \quad PN = \sqrt{10}.$$

Verification. Tangent: $y - 1 = 3(x - 1)$, i.e. $y = 3x - 2$; it meets the x -axis at $x = 2/3$, so $MT = 1 - 2/3 = 1/3 \checkmark$. Normal: $y - 1 = -\frac{1}{3}(x - 1)$, i.e. $y = \frac{4-x}{3}$; it meets the x -axis at $x = 4$, so $MN = 4 - 1 = 3 \checkmark$. The lengths agree with Pythagoras on the verified intercepts: $PT = \sqrt{(2/3)^2 + 1} = \frac{\sqrt{13}}{3}$... — recompute: PT joins $(1, 1)$ to $(2/3, 0)$, so $PT = \sqrt{(1/3)^2 + 1^2} = \frac{\sqrt{10}}{3} \checkmark$, and PN joins $(1, 1)$ to $(4, 0)$: $PN = \sqrt{9 + 1} = \sqrt{10} \checkmark$. All four confirmed.

Example E.2 — [JEE Main / Advanced Level]

For the parabola $y^2 = 4ax$, find the tangent and normal at the point $(at^2, 2at)$, and show that the normal meets the axis at a point whose distance from the foot of the ordinate is always $2a$.

Parametrise: the point $(at^2, 2at)$ satisfies $y^2 = 4a^2t^2 = 4ax \checkmark$. The slope there:

$$m = \frac{2a}{y} = \frac{2a}{2at} = \frac{1}{t}.$$

Tangent: $y - 2at = \frac{1}{t}(x - at^2)$; multiply by t :

$$ty = x + at^2.$$

Normal: slope $-t$:

$$y - 2at = -t(x - at^2).$$

Setting $y = 0$: $-2at = -t(x - at^2)$, so $x - at^2 = 2a$, i.e. the normal meets the axis at

$$N = (at^2 + 2a, 0).$$

The foot of the ordinate is $M = (at^2, 0)$, so $MN = 2a$ for **every** t — the constant-subnormal fact of the Method section, now in parametric clothing. These two line equations, $ty = x + at^2$ and $y + tx = 2at + at^3$, are standard JEE currency; derive them once as above rather than memorising blind.

Example E.3 — [JEE Advanced Level]

Find all tangents to the parabola $y = 4 - x^2$ that pass through the point $(2, 7)$, giving the points of contact and the tangent equations.

Let the point of contact be $x = a$, so $P = (a, 4 - a^2)$ with slope $m = -2a$. The tangent line is

$$y = -2ax + (a^2 + 4).$$

Requiring it to pass through $(2, 7)$:

$$7 = -4a + a^2 + 4 \implies a^2 - 4a - 3 = 0 \implies a = 2 \pm \sqrt{7}.$$

Two tangents, as expected from an external point. Contact points: with $a^2 = 11 \pm 4\sqrt{7}$,

$$y = 4 - a^2 = -7 \mp 4\sqrt{7},$$

so the contacts are $(2 + \sqrt{7}, -7 - 4\sqrt{7})$ and $(2 - \sqrt{7}, -7 + 4\sqrt{7})$. The corresponding lines are

$$y = -(4 + 2\sqrt{7})x + 15 + 4\sqrt{7} \quad \text{and} \quad y = (2\sqrt{7} - 4)x + 15 - 4\sqrt{7}.$$

Check. Each contact point satisfies the curve: $(2 + \sqrt{7})^2 = 11 + 4\sqrt{7}$ and $4 - (11 + 4\sqrt{7}) = -7 - 4\sqrt{7} \checkmark$. Each line passes through $(2, 7)$: $-(4 + 2\sqrt{7})(2) + 15 + 4\sqrt{7} = -8 - 4\sqrt{7} + 15 + 4\sqrt{7} = 7 \checkmark$. And each line's slope matches $-2a$ at its own contact $\checkmark$.

Why it is Advanced-level. The chapter's appendix finds tangents *at* a known point; here the point of contact is itself the unknown, so the tangency condition becomes an equation for a . Setting up “generic contact point + pass-through condition” is the transferable skill — identical machinery handles “tangent with given slope”, “tangent with given intercepts”, and “shortest-distance” problems.

Practice — Addendum E

1. Find the subtangent, subnormal, tangent length and normal length for $y = e^x$ at $(0, 1)$.

2. Show that the subtangent of $y = \ln x$ at any point equals $x \ln x$ — the abscissa times the ordinate.
3. For the parabola $y^2 = 4x$, find all points at which the subnormal equals 4.
4. At which point of $y = e^x$ does the tangent make a 45° angle with the x -axis? Write the tangent's equation.
5. Find the area of the triangle cut off from the coordinate axes by the tangent to $y = 4 - x^2$ at $x = 1$.
6. Show that the normal to $y = x^2$ at $(1, 1)$ passes through the point $(0, \frac{3}{2})$.

Answers: 1. $m = 1$ at $x = 0$: subtangent 1, subnormal 1, tangent length $\sqrt{2}$, normal length $\sqrt{2}$. 2. $m = 1/x$, so subtangent $= y/m = x \ln x$. 3. None — for $y^2 = 4x$ we have $a = 1$, so the subnormal is $2a = 2$ at *every* point; it can never equal 4. (A trap: the subnormal of a parabola is a constant of the curve, not a variable.) 4. Need $m = \tan 45^\circ = 1$: $e^x = 1$ gives $x = 0$, point $(0, 1)$, tangent $y = x + 1$. 5. At $(1, 3)$, $m = -2$; x -intercept $= x - y/m = 1 + 3/2 = 5/2$; y -intercept $= y - mx = 3 + 2 = 5$; area $= \frac{1}{2} \cdot \frac{5}{2} \cdot 5 = \frac{25}{4}$ square units. 6. Normal slope $= -1/2$: $y - 1 = -\frac{1}{2}(x - 1)$; at $x = 0$ this gives $y = 3/2$ ✓.

Addendum P — What JEE Actually Asks

This addendum is built purely from the previous-year bank supplied alongside the chapter (roughly 130 recoverable questions, 1979–2020). Every question quoted below has been reconstructed from the garbled extract, re-solved independently, and checked against the bank's answer where the extract preserves it. Questions too corrupted to reconstruct faithfully have been excluded rather than guessed at.

The distribution

Judged from the bank in front of me, the weight falls as follows (approximate shares of the recoverable pool):

Block	Share	Evidence
Maxima & minima	~40%	Present in virtually every session: Main Jan/Sep 2019, Jan/Sep 2020; Adv 2013–2020; dense run 1979–2006
Tangents & normals	~30%	Continuous across all eras: 1983, 1994, 2001–2008, a heavy 2013–2017 Main cluster, again 2019–2020

Block	Share	Evidence
Monotonicity & inequalities	~20%	Steady Main supply (2013, 2017, 2019, 2020); Advanced proof questions concentrated 1983–2003
Related rates	~7%	Near-absent before 2000, then a tight cluster: Main 2013, 2014, 2019 (twice), 2020 (twice)
Approximations & differentials	0%	No recoverable question in this bank

Three facts worth acting on:

1. **The examiners recycle almost verbatim.** The ice-coated sphere appears identically in Main April 10, 2019 (II) and Main Jan 9, 2020 (I) — same radii, same melt rate. $\log_e(1+x) < x$ is asked in 1987 (fill-in) and again in 2000S (MCQ). $y = a \ln x + bx^2 + x$ with extrema at -1 and 2 appears in 1983 and again, nearly unchanged, in Main 2014. Mastering one year’s paper genuinely prepares you for a paper decades later.
2. **Tangents/normals is not a side topic.** The chapter relegates it to an appendix, but it is the second-largest block in the bank, and it increasingly rides on *implicit* and *parametric* curves ($x^2 + 2xy - 3y^2 = 0$, $y^3 - 3xy + 2 = 0$, $x = a(\cos \theta + \theta \sin \theta)$, ...), not on $y = f(x)$.
3. **Monotonicity at Advanced level is an engine, not a topic.** It powers inequality proofs (1983–2003), root-counting via Rolle/MVT (Adv 2013, 2015, 2017; 2006), and statements about composite functions (1998, 2019). The board-level “find the intervals” version is only the entry fee.

The recurring archetypes

Archetype 1 — The related-rates chain drill

Shape. Two geometric quantities linked by a formula; one rate given, one demanded “at the instant when” some length takes a stated value. Favourites: spheres with coats of ice, cubes, sliding ladders, inflating balloons, ships and bearings.

Method. Write the relation in the *total* varying quantity, differentiate with respect to t symbolically, substitute numbers last. Keep melting/shrinking rates negative.

Representative (reconstructed from Main Jan 9, 2020 (I), repeated Main April 10, 2019 (II)): A spherical iron ball of radius 10 cm is coated with ice of uniform thickness that melts at $50 \text{ cm}^3/\text{min}$. Find the rate at which the thickness decreases when the thickness is 5 cm.

Let x be the ice thickness; the outer radius is $10 + x$. The melt rate drives the *outer* volume:

$$V = \frac{4}{3}\pi(10+x)^3, \quad \frac{dV}{dt} = 4\pi(10+x)^2 \frac{dx}{dt}$$

Melting means $\frac{dV}{dt} = -50$. At $x = 5$: outer radius 15, so

$$-50 = 4\pi(15)^2 \frac{dx}{dt} \implies \frac{dx}{dt} = -\frac{50}{900\pi} = -\frac{1}{18\pi} \text{ cm/min}$$

The thickness decreases at $\frac{1}{18\pi}$ cm/min. Note the trap the examiner repeats: $10+x$ is the radius, x alone is not.

Archetype 2 — Tangent/normal with a prescribed property, usually on an implicit curve

Shape. “Find the normal at (a, b) ”; “find the point where the tangent is parallel/perpendicular to a given line”; “where does the normal meet the curve again / pass through?” Curves arrive implicitly defined.

Method. Differentiate implicitly to get $\frac{dy}{dx}$; normal slope is $-\frac{dx}{dy}$. For vertical tangents set the denominator of $\frac{dy}{dx}$ to zero (equivalently $\frac{dx}{dy} = 0$); for horizontal tangents set the numerator to zero. Always confirm the point satisfies the curve.

Representative (reconstructed from Main Jan 8, 2020 (II)): Find the length of the perpendicular from the origin to the normal of $x^2 + 2xy - 3y^2 = 0$ at $(2, 2)$.

Differentiate implicitly:

$$2x + 2y + 2x \frac{dy}{dx} - 6y \frac{dy}{dx} = 0 \implies \frac{dy}{dx} = \frac{-(x+y)}{x-3y}$$

At $(2, 2)$: $\frac{dy}{dx} = \frac{-4}{-4} = 1$, so the normal has slope -1 :

$$y - 2 = -1(x - 2) \implies x + y = 4$$

Distance from origin: $\frac{|0+0-4|}{\sqrt{2}} = 2\sqrt{2}$.

Archetype 3 — Angle between curves; touching vs intersecting

Shape. Two explicit curves meet; find $\tan \theta$ of the angle between them, or decide whether they touch or cross (2008: $y^2 = 4ax$ vs $x^2 = 4ay$; 2008: $y^2 = 4x$ vs a circle).

Method. Solve simultaneously for intersection points, compute both slopes, apply

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

Touching $\iff m_1 = m_2$; orthogonal $\iff m_1 m_2 = -1$.

Representative (reconstructed from Main Jan 9, 2019 (I)): Curves $y = 10 - x^2$ and $y = 2 + x^2$; find $|\tan \theta|$.

Adding: $2y = 12 \Rightarrow y = 6$, then $x = \pm 2$. At $(2, 6)$: $m_1 = -4$, $m_2 = 4$. At $(-2, 6)$: $m_1 = 4$, $m_2 = -4$. Either way,

$$|\tan \theta| = \left| \frac{-4 - 4}{1 + (-16)} \right| = \frac{8}{15}$$

Archetype 4 — Monotonicity as a parameter condition (and as a root-blocker)

Shape. “For which k is f increasing on $\mathbb{R}$?”, “which function is monotonic?”, or — the clever variant — “show the equation has no two distinct roots in an interval.”

Method. Reduce to a sign condition on f' ; for a quadratic f' on all of $\mathbb{R}$ demand discriminant ≤ 0 (non-strict!). For root questions: a strictly monotonic function cannot take the same value twice.

Representative (reconstructed from Main 2014, twin of 1983): $y = a \ln x + bx^2 + x$ has extremum at $x = -1$ and $x = 2$. Find a, b . (Domain issue at $x = -1$ is handled in the original by $\ln |x|$.)

$$\frac{dy}{dx} = \frac{a}{x} + 2bx + 1$$

At $x = -1$: $-a - 2b + 1 = 0$. At $x = 2$: $\frac{a}{2} + 4b + 1 = 0$. Solving,

$$\begin{aligned} a &= 2 \\ b &= -\frac{1}{2} \end{aligned}$$

Sibling question worth knowing (Main 2013): $2x^3 + 3x + k = 0$ cannot have two distinct real roots in $[0, 1]$ for any k , because $f' = 6x^2 + 3 > 0$ makes f strictly increasing — answer “does not exist”.

Archetype 5 — Local extrema: classifying, and constructing polynomials from conditions

Shape. Either classify extrema of a given quartic/quartic-in-disguise (including exponentials times polynomials), or *build* the least-degree polynomial with prescribed extrema and values.

Method. Factor f' completely, read sign changes (never trust $f' = 0$ alone). For construction: prescribe p' as a constant times the product of $(x - \text{extremum})$ factors, integrate, fit values.

Representative (reconstructed from JEE 2012): $p(x)$ is a real polynomial of least degree with a local maximum at $x = 1$, a local minimum at $x = 3$, $p(1) = 6$, $p(3) = 2$. Find $p'(0)$.

Least degree: cubic, so $p'(x) = k(x - 1)(x - 3)$. Integrating and imposing $p(1) = 6$, $p(3) = 2$:

$$\begin{aligned} p(1) &= \frac{4k}{3} + C = 6 \\ p(3) &= C = 2 \end{aligned} \implies k = 3, C = 2$$

Hence $p'(x) = 3(x - 1)(x - 3)$ and $p'(0) = 9$.

Archetype 6 — Geometric and economic optimization

Shape. Rectangle under a parabola (2019 twice), cone of max volume on a fixed slant (2019), wire split into square + circle (2016), open box from a cut sheet (Adv 2013), sector fence (2017), bus running cost (2013), inscribed cylinder with wall thickness (Adv 2015).

Method. One-variable objective via the constraint; state the feasible domain; reject infeasible roots; confirm max vs min; answer what is *asked* (a value, a dimension, a ratio).

Representative (reconstructed from Main Jan 9, 2019 (I)): *Maximum volume of a right circular cone of slant height 3 m.*

With $r^2 + h^2 = 9$:

$$V = \frac{\pi}{3}r^2h = \frac{\pi}{3}(9h - h^3), \quad \frac{dV}{dh} = \frac{\pi}{3}(9 - 3h^2) = 0 \implies h = \sqrt{3}$$

Since $\frac{d^2V}{dh^2} = -2\pi h < 0$, this is a maximum, with $r^2 = 6$:

$$V_{\max} = \frac{\pi}{3} \cdot 6 \cdot \sqrt{3} = 2\sqrt{3}\pi \text{ m}^3$$

Archetype 7 — MVT/Rolle as an existence and counting engine (Advanced)

Shape. “The average speed is attained at...” (Main Sep 6, 2020 I); “exactly one solution of $f' = 0$ per interval” (Adv 2015); “minimum number of zeros of $(f')^2 + f''f$ ” (2006); functional differential inequalities (Adv 2013, 2017).

Method. Apply Rolle/Lagrange to a *well-chosen* auxiliary function; chain successive applications: zeros of f force zeros of f' , which force zeros of the next antiderivative-like object.

Representative (reconstructed from Main Sep 6, 2020 (I)): *Position $f(t) = at^2 + bt + c$, $a, b, c > 1$. Where is the average speed over $[t_1, t_2]$ attained?*

Average speed equals the secant slope $\frac{f(t_2) - f(t_1)}{t_2 - t_1}$; by Lagrange MVT some c has $f'(c)$ equal to it. Here

$$f'(c) = 2ac + b = \frac{f(t_2) - f(t_1)}{t_2 - t_1} = a(t_1 + t_2) + b \implies c = \frac{t_1 + t_2}{2}$$

unique for a quadratic — answer $(t_1 + t_2)/2$.

The traps

These are the specific errors the bank’s distractors are built around:

1. **Thickness confused with radius.** The ice-ball question was asked twice in two years precisely because students write $V = \frac{4}{3}\pi x^3$. The melt acts on the *outer* sphere of radius $10 + x$.

2. **Factor-of- π and factor-of-2 slips in rates.** Options like 0.7π vs 1.4π (and the bank's related-rate options generally) punish forgetting $dC/dt = 2\pi dr/dt$ or $dS/dt = 8\pi r dr/dt$.
3. **Endpoint neglect.** Absolute extrema on a closed interval routinely sit at endpoints (the bank's own example $3x^4 - 8x^3 + \dots$ peaks at $x = 0$; $\sin x + \cos x$ on $[0, \frac{\pi}{2}]$ tempts you with the endpoint value 1 while the answer is $\sqrt{2}$).
4. **Inflection posing as an extremum.** $(3x^2 + ax - 2 - a)e^x$ (Main Sep 5, 2020 II) has two critical points of opposite character; students who assume "critical = extremum" get the classification backwards. Likewise $f'(c) = 0$ with no sign flip (the $(x - 1)^2$ pattern) means *no* extremum.
5. **Strict vs non-strict discriminant.** For " f' is a quadratic ≥ 0 everywhere," the answer is $\Delta \leq 0$ — closed interval. The distractor offering the open interval catches those who write $\Delta < 0$.
6. **Skipping the point-on-curve check.** In the tangent-perpendicular question (Main April 9, 2019 I), a and b are only determined after forcing $(1, -5)$ onto $y = x^3 + ax - b$. Options contain points from curves with one right and one wrong coefficient.
7. **Vertical tangents done backwards.** For implicit curves, vertical tangent $\iff$ denominator of $\frac{dy}{dx}$ vanishes ($y^3 + 3x^2 = 12y$ gives $y = 2$ only; $y = -2$ yields an impossible equation). And $y^3 - 3xy + 2 = 0$ has *no* horizontal-tangent point at all — $H = \emptyset$ — a favourite shock answer.
8. **Sign and unit discipline in rates.** Melting, draining, sliding-down are negative rates; mixing metres and centimetres in one ladder problem (2019: a 2 m ladder with a 25 cm/s rate) cancels safely only if you convert consistently.
9. **Merging monotonic intervals across domain breaks**, and demanding $f' > 0$ strictly when isolated zeros are harmless ($x - \sin x$ is strictly increasing).

Graded practice set

P1 (warm-up, related rates). (Main Sep 3, 2020 II) The surface area of a cube increases at $3.6 \text{ cm}^2/\text{s}$. Find the rate of change of its volume when the edge is 10 cm.

$$S = 6a^2 \Rightarrow \frac{dS}{dt} = 12a \frac{da}{dt} = 3.6 \Rightarrow \frac{da}{dt} = 0.03. \text{ Then}$$

$$\frac{dV}{dt} = 3a^2 \frac{da}{dt} = 3(100)(0.03) = 9 \text{ cm}^3/\text{s}$$

P2 (ladder, mixed units). (Main April 12, 2019 I) A 2 m ladder leans against a wall; the top slides down at 25 cm/s. How fast does the bottom move out when the top is 1 m up?

Work in cm: $x^2 + y^2 = (200)^2$; at $y = 100$, $x = 100\sqrt{3}$. Differentiate:

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0 \implies \frac{dx}{dt} = \frac{100 \cdot 25}{100\sqrt{3}} = \frac{25}{\sqrt{3}} \text{ cm/s}$$

P3 (balloon). (Main Online April 25, 2013) A balloon is inflated at $35 \text{ cm}^3/\text{min}$. Find the rate of increase of surface area when the diameter is 14 cm.

$r = 7$:

$$\frac{dr}{dt} = \frac{35}{4\pi(49)}, \quad \frac{dS}{dt} = 8\pi r \frac{dr}{dt} = \frac{56\pi \cdot 35}{196\pi} = 10 \text{ cm}^2/\text{min}$$

P4 (point on a curve). (Main April 9, 2019 I) $y = x^3 + ax - b$ passes through $(1, -5)$; its tangent there is perpendicular to $-x + y + 4 = 0$. Which point lies on the curve?

Point on curve: $a - b = -6$. Tangent slope $3 + a$; the given line has slope 1, so $3 + a = -1 \Rightarrow a = -4, b = 2$. Curve: $y = x^3 - 4x - 2$. Check: $y(2) = 8 - 8 - 2 = -2$, so $(2, -2)$ lies on it.

P5 (parameters from extrema). (Main 2014) See Archetype 4: $a = 2, b = -\frac{1}{2}$.

P6 (classify a quartic). (Main April 8, 2019 I) $f(x) = 9x^4 + 12x^3 - 36x^2 + 25$.

$$f'(x) = 36x^3 + 36x^2 - 72x = 36x(x+2)(x-1)$$

Sign chart: $f' < 0$ on $(-\infty, -2)$, > 0 on $(-2, 0)$, < 0 on $(0, 1)$, > 0 on $(1, \infty)$. Local minima at $x = -2, 1$; local maximum at $x = 0$.

P7 (two poles). (Main Sep 6, 2020 I) Poles of heights 8 m and 11 m stand 10 m apart. Find AM minimizing $MD^2 + MC^2$ for M on AB .

With $AM = x$: $MD^2 = x^2 + 64$, $MC^2 = (10 - x)^2 + 121$. Sum = $2x^2 - 20x + 285$; derivative $4x - 20 = 0 \Rightarrow x = 5$ m (second derivative $4 > 0$).

P8 (open box from a sheet). (Adv 2013) A sheet of perimeter-fixed sides in ratio 8 : 15 becomes an open box after cutting equal squares totalling area 100. Find the side lengths for maximum volume.

Sides $8x, 15x$; cut squares of side y with $4y^2 = 100 \Rightarrow y = 5$:

$$V = (8x - 10)(15x - 10) \cdot 5, \quad V' \propto 30x^2 - 115x + 75 = 0 \Rightarrow x = 3 \text{ or } \frac{5}{6}$$

$x = \frac{5}{6}$ makes $8x - 10 < 0$: rejected. Sides are 24 and 45.

P9 (tangent and normal sharing a foot on the x -axis). (Main Sep 3, 2020 II) The tangent to $y = e^x$ at (c, e^c) and the normal to $y^2 = 4x$ at $(1, 2)$ meet the x -axis at the same point. Find c .

Normal at $(1, 2)$: $\frac{dy}{dx} = \frac{2}{y} = 1$, normal slope -1 , line $y = 3 - x$, hitting $y = 0$ at $x = 3$. Tangent: $y = e^c(x - c + 1)$, zero at $x = c - 1$. So $c - 1 = 3 \Rightarrow c = 4$.

P10 (economic speed). (Main Online April 23, 2013) Cost $C(v) = av + \frac{b}{v}$; $C(30) = 75$, $C(40) = 65$. Most economical speed?

Solving: $2250 = 900a + b$ and $2600 = 1600a + b$, giving $a = \frac{1}{2}, b = 1800$. Then

$$C'(v) = \frac{1}{2} - \frac{1800}{v^2} = 0 \Rightarrow v = 60 \text{ km/h}$$

P11 (Advanced inequality). (JEE 1983) Prove $1 + x \ln(x + \sqrt{1 + x^2}) \geq \sqrt{1 + x^2}$ for all real x .

Let $g(x) = 1 + x \ln(x + \sqrt{1 + x^2}) - \sqrt{1 + x^2}$, so $g(0) = 0$. Using $\frac{d}{dx} \ln(x + \sqrt{1 + x^2}) = \frac{1}{\sqrt{1 + x^2}}$,

$$g'(x) = \ln(x + \sqrt{1+x^2}) + \frac{x}{\sqrt{1+x^2}} - \frac{x}{\sqrt{1+x^2}} = \ln(x + \sqrt{1+x^2})$$

which is negative for $x < 0$ and positive for $x > 0$. So g decreases to $g(0) = 0$ and increases after: $g \geq 0$ everywhere, with equality only at $x = 0$.

P12 (Advanced counting via Rolle). (JEE 2006) On $[a, e]$ with $a < b < c < d < e$, $f(a) = 0, f(b) = 2, f(c) = -1, f(d) = 2, f(e) = 0$. Find the minimum number of zeros of $g(x) = (f'(x))^2 + f''(x)f(x)$.

Note $g = (f f')'$. The IVT gives zeros $\alpha \in (b, c)$ and $\beta \in (c, d)$ of f , alongside a and e : four zeros of f . Rolle forces at least three zeros of f' strictly between consecutive zeros of f , disjoint from them: at least seven zeros of $f f'$. Applying Rolle to $f f'$ between consecutive zeros: at least six zeros of g .

What this chapter's own text under-serves

First, tangents and normals — by question count the joint-largest JEE block in this bank — sits in a short appendix covering only explicit $y = f(x)$, while the real exam operates on implicit curves ($x^2 + 2xy - 3y^2 = 0, y^3 - 3xy + 2 = 0$), parametric forms, angles of intersection, and normals that re-meet the curve. Second, the monotonicity and maxima-minima sections stop at the CBSE template, whereas Advanced recycles these ideas through Rolle/MVT existence arguments, composite-function behaviour, inequality proofs, and root-counting — none of which appear in the chapter's practice stock. Third, the appendices oversell approximations (zero recoverable PYQs here) while under-weighting the optimization formats JEE actually favours: fitted-cost problems, constrained solids with wall thickness, and multi-correct statement questions.

Chapter 7: Integrals

Class 12 Mathematics — Complete Master Content

Drona Ed Tech — Content Bank

All 9 Subtopics

Contents

Subtopic 01 — Foundations of Integration: Indefinite Integrals, Properties & Standard Forms	4
Section 1: Concept Intuition	4
Section 2: Key Definitions & Properties	5
Section 3: Key Procedures & Logical Framework	7
Section 4: Worked Examples	8
Section 5: Practice Exercises	9
Section 6: Crack the MCQ	10
Section 7: Common Student Pitfalls & Pro-Tips	11
Section 8: Cheat Sheet Box	11
Subtopic 04 — Integration by Substitution	13
Section 1: Concept Intuition	13
Section 2: Key Definitions & Properties	14
Section 3: Key Procedures & Derivations	14
Section 4: Worked Examples	15
Section 5: Practice Exercises	16
Section 6: Crack the MCQ	16
Section 7: Common Student Pitfalls & Pro-Tips	17
Section 8: Cheat Sheet Box	17
Subtopic 05 — Integration using Trigonometric Identities	19
Section 1: Concept Intuition	19
Section 2: Key Definitions & Properties	19
Section 3: Key Procedures & Derivations	20
Section 4: Worked Examples	20
Section 5: Practice Exercises	22
Section 6: Crack the MCQ	22
Section 7: Common Student Pitfalls & Pro-Tips	23
Section 8: Cheat Sheet Box	23
Subtopic 06 — Integrals of Special Forms (Quadratic Denominators & Roots)	24
Section 1: Concept Intuition	24
Section 2: Key Definitions & Properties	24
Section 3: Key Procedures & Derivations	25

Section 4: Worked Examples	26
Section 5: Practice Exercises	27
Section 6: Crack the MCQ	27
Section 7: Common Student Pitfalls & Pro-Tips	28
Section 8: Cheat Sheet Box	28
Subtopic 07 — Integration by Partial Fractions	30
Section 1: Concept Intuition	30
Section 2: Key Definitions & Properties	30
Section 3: Key Procedures & Derivations	31
Section 4: Worked Examples	31
Section 5: Practice Exercises	33
Section 6: Crack the MCQ	33
Section 7: Common Student Pitfalls & Pro-Tips	34
Section 8: Cheat Sheet Box	34
Subtopic 08 — Integration by Parts	36
Section 1: Concept Intuition	36
Section 2: Key Definitions & Properties	36
Section 3: Key Procedures & Derivations	37
Section 4: Worked Examples	38
Section 5: Practice Exercises	39
Section 6: Crack the MCQ	39
Section 7: Common Student Pitfalls & Pro-Tips	40
Section 8: Cheat Sheet Box	40
Subtopic 02 — Definite Integrals: Definition, the Fundamental Theorem & Limit of a Sum	41
Section 1: Concept Intuition	41
Section 2: Key Definitions & Properties	42
Section 3: Key Procedures & Derivations	43
Section 4: Worked Examples	43
Section 5: Practice Exercises	45
Section 6: Crack the MCQ	45
Section 7: Common Student Pitfalls & Pro-Tips	46
Section 8: Cheat Sheet Box	46
Subtopic 09 — Evaluation of Definite Integrals by Substitution	47
Section 1: Concept Intuition	47
Section 2: Key Definitions & Properties	47
Section 3: Key Procedures & Derivations	48
Section 4: Worked Examples	48
Section 5: Practice Exercises	49
Section 6: Crack the MCQ	50
Section 7: Common Student Pitfalls & Pro-Tips	50
Section 8: Cheat Sheet Box	51
Subtopic 03 — Properties of Definite Integrals: The Symmetry Toolkit	52

Section 1: Concept Intuition	52
Section 2: Key Definitions & Properties	53
Section 3: CBSE Board Derivations / Key Procedures	54
Section 4: Worked Examples	55
Section 5: Practice Exercises	57
Section 6: Crack the MCQ	57
Section 7: Common Student Pitfalls & Pro-Tips	58
Section 8: Cheat Sheet Box	59

Subtopic 01 — Foundations of Integration: Indefinite Integrals, Properties & Standard Forms

Exam Relevance: This is the gateway to all of calculus integration and shows up everywhere. CBSE Boards regularly ask 1-2 mark direct integrals and 1-mark “value of C ” / property questions. JEE Main and CUET lean on the standard-form table for fast 4-mark single-correct problems, and JEE Advanced uses the inverse-of-derivative idea inside larger multi-concept questions. Master this and every later method (substitution, by parts, partial fractions) becomes mechanical.

Section 1: Concept Intuition

Imagine you are watching the speedometer of a Mumbai local train, but you forgot to note where the train started. The speedometer tells you the *rate* at which distance is changing at every instant — that is exactly what a derivative is. Integration asks the reverse, and much harder, question: *given the speed at every moment, can you recover the distance travelled?*

That reverse question is the heart of integration. Differentiation takes a function and hands you its rate of change. Integration takes a rate of change and tries to hand you back the original function. Because of this, the indefinite integral is often called the **anti-derivative** — it literally “un-does” a derivative.

Here is the catch, and it is the single most important idea in this entire subtopic. When you differentiate, every constant disappears: the derivative of $x^2 + 7$ is $2x$, and so is the derivative of $x^2 - 100$, and of x^2 alone. So when you run the process backwards and integrate $2x$, you have no way of knowing whether the original constant was 7, -100 , or 0. You have lost that information forever. To honestly admit this, we tack on an unknown constant C , called the **constant of integration**:

$$\int 2x \, dx = x^2 + C$$

Think about it this way... Differentiation is like flattening a tall stack of identical books and reading only the title printed on the spine. Every book in the stack — sitting

at a different height C — has the same spine. When you integrate, you can read the spine perfectly, but you cannot tell which exact height the book was sitting at. So you write $+C$ to mean “any book in this stack.” Each value of C is a different but equally valid answer.

This is also why the answer to an indefinite integral is never a single curve — it is an entire **family of curves**, all parallel, stacked vertically. If $\int f(x) dx = F(x) + C$, then drawing $F(x)$, $F(x) + 1$, $F(x) + 2$ and so on gives you a whole ladder of curves. Pick any vertical line $x = a$: it cuts every curve in the family, and remarkably, the tangent at each of those intersection points has the *same slope*. That makes sense — the slope is $f(a)$ for all of them, because shifting a curve up or down never changes its steepness.

Figure 7.1 A set of 4-5 identical S-shaped curves stacked vertically, each a vertical translation of the others, labelled $F(x)+C$ for $C = -1, 0, 1, 2$. A dashed vertical line at $x = a$ cuts all curves; short tangent segments drawn at each intersection point, all clearly parallel (same slope). Caption: “The indefinite integral is a family of parallel curves; tangents along any vertical line are parallel.” (see next page)

Limiting conditions & assumptions. A few honest warnings before we move on:

- An anti-derivative exists nicely only where the function is reasonably well-behaved (continuous on the interval of interest). Integration is *not* guaranteed to produce a neat formula — many perfectly continuous functions, like e^{-x^2} , have anti-derivatives that simply cannot be written using elementary functions.
- The “ $+C$ ” is non-negotiable for an *indefinite* integral. Dropping it is not a small slip — it changes the answer from “a family of functions” to “one specific function,” which is mathematically false.
- The reverse-engineering picture (integration undoes differentiation) is exact, but only up to that constant. The two operations are inverses of each other *modulo a constant*, never as perfect one-to-one inverses.

Section 2: Key Definitions & Properties

Definition (Indefinite Integral). If $\frac{d}{dx}F(x) = f(x)$, then $F(x)$ is an anti-derivative of $f(x)$, and we write

$$\int f(x) dx = F(x) + C$$

Here $f(x)$ is the **integrand**, x is the **variable of integration**, dx specifies *which* variable we are integrating with respect to, and C is the **constant of integration** (any real number). The whole symbol $\int(\cdots) dx$ is read “the integral of $(\cdots)$ with respect to x .”

The two operations are mutually inverse.

$$\frac{d}{dx}\left(\int f(x) dx\right) = f(x) \quad \text{and} \quad \int f'(x) dx = f(x) + C$$

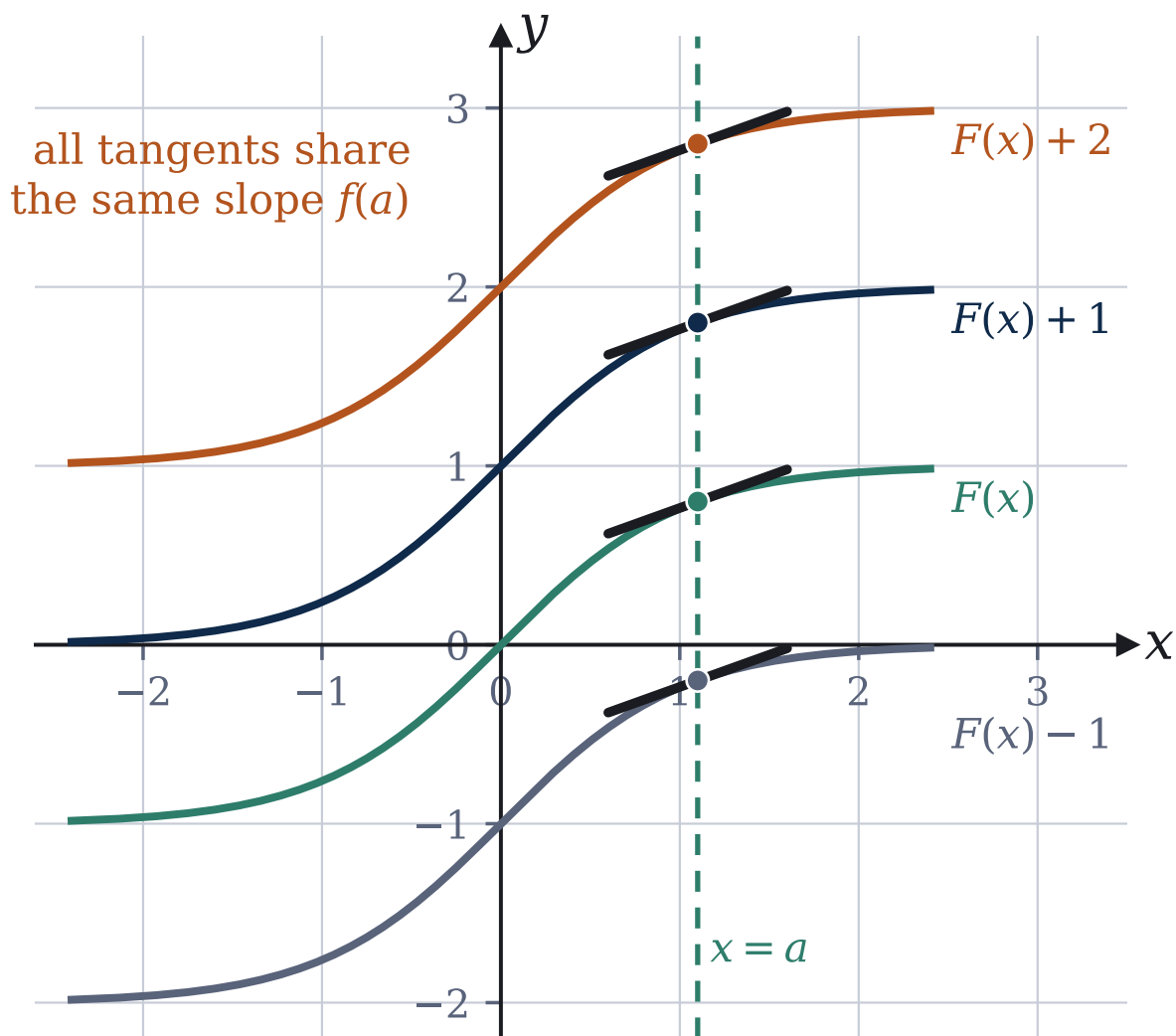


Figure 7.1 A set of 4–5 identical S-shaped curves stacked vertically, each a vertical translation of the others, labelled $F(x)+C$ for $C = -1, 0, 1, 2$. A dashed vertical line at $x = a$ cuts all curves; short tangent segments drawn at each intersection point, all clearly parallel (same slope). Caption: "The indefinite integral is a family of parallel curves; tangents along any vertical line are parallel."

Linearity properties (the workhorses):

$$\int [f(x) + g(x)] dx = \int f(x) dx + \int g(x) dx$$

$$\int k f(x) dx = k \int f(x) dx, \quad k \text{ a constant}$$

Two integrals with the same derivative describe the same family of curves, so they are treated as equivalent.

Standard integral table (memorise cold). Every variable below is a real number; a is a non-zero constant; C is the constant of integration.

#	Integral	Result
1	$\int x^n dx$	$\frac{x^{n+1}}{n+1} + C, \quad n \neq -1$
2	$\int \frac{1}{x} dx$	$\ln x + C$
3	$\int e^x dx$	$e^x + C$
4	$\int a^x dx$	$\frac{a^x}{\ln a} + C, \quad a > 0, a \neq 1$
5	$\int \cos x dx$	$\sin x + C$
6	$\int \sin x dx$	$-\cos x + C$
7	$\int \sec^2 x dx$	$\tan x + C$
8	$\int \csc^2 x dx$	$-\cot x + C$
9	$\int \sec x \tan x dx$	$\sec x + C$
10	$\int \csc x \cot x dx$	$-\csc x + C$
11	$\int \frac{1}{\sqrt{1-x^2}} dx$	$\sin^{-1} x + C$
12	$\int \frac{1}{1+x^2} dx$	$\tan^{-1} x + C$

A clean way to keep the table honest: **differentiate the answer.** If differentiating the right column does not return the integrand, the row is wrong. This single check makes the entire table self-verifying.

Section 3: Key Procedures & Logical Framework

There is no single “derivation” of integration — it is defined as the reverse of differentiation. So the rigour here lives in (a) *proving the properties* and (b) the *procedure* for direct integration. Both matter for Boards.

Proof of the power rule for integration ($n \neq -1$). We claim $\int x^n dx = \frac{x^{n+1}}{n+1} + C$. Rather than “guess,” reason from the definition: an anti-derivative of x^n is any function whose derivative is x^n . Differentiate the proposed answer:

$$\frac{d}{dx} \left(\frac{x^{n+1}}{n+1} + C \right) = \frac{1}{n+1} \cdot (n+1)x^{(n+1)-1} + 0 = x^n$$

Since the derivative returns exactly the integrand, the proposed function is an anti-derivative, which proves the rule. The condition $n \neq -1$ is essential: if $n = -1$ the denominator $n+1$ is zero and the formula is meaningless — that case is handled separately by row 2 of the table.

Proof of linearity (sum rule). Let $\int f(x) dx = F(x)$ and $\int g(x) dx = G(x)$, meaning $F'(x) = f(x)$ and $G'(x) = g(x)$. By the sum rule for derivatives,

$$\frac{d}{dx} [F(x) + G(x)] = F'(x) + G'(x) = f(x) + g(x)$$

So $F(x) + G(x)$ is an anti-derivative of $f(x) + g(x)$, which is precisely the statement $\int [f + g] dx = \int f dx + \int g dx$. The constant-multiple rule follows identically from $\frac{d}{dx} [kF(x)] = kF'(x) = kf(x)$.

Procedure for direct (standard-form) integration. This is the “method” of this subtopic — solving an integral by manipulating it into table entries, *without* substitution or by-parts machinery.

1. **Simplify first, integrate second.** Expand brackets, split fractions term by term, cancel, or rewrite roots as powers ($\sqrt{x} = x^{1/2}$, $\frac{1}{x^3} = x^{-3}$). Algebra is cheaper than calculus.
2. **Split the integral using linearity.** Break a sum into separate integrals and pull out constant multipliers.
3. **Use trigonometric identities to reduce to standard forms.** For example, $\sin^2 x = \frac{1-\cos 2x}{2}$ or $2 \sin A \cos B = \sin(A+B) + \sin(A-B)$ turn awkward products into integrable sums.
4. **Apply the standard table** term by term.
5. **Add a single C** at the very end (not one per term — all the separate constants merge into one).
6. **Verify by differentiating** your answer back to the integrand.

The logic of step 5 is worth stressing: if you integrate three terms you might be tempted to write $C_1 + C_2 + C_3$, but the sum of arbitrary constants is just another arbitrary constant, so a single C captures the whole family.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] Evaluate $\int \left(3x^2 + \frac{4}{x} - 5 \sec^2 x \right) dx$.

Solution. Split using linearity and pull out constants:

$$\int 3x^2 dx + \int \frac{4}{x} dx - \int 5 \sec^2 x dx = 3 \int x^2 dx + 4 \int \frac{1}{x} dx - 5 \int \sec^2 x dx$$

Apply the table (rows 1, 2, 7):

$$= 3 \cdot \frac{x^3}{3} + 4 \ln|x| - 5 \tan x + C = x^3 + 4 \ln|x| - 5 \tan x + C$$

Answer: $x^3 + 4 \ln|x| - 5 \tan x + C$. (Notice the single C , and the modulus inside the log.)

Example 2 — [JEE Main Speed Trap] Evaluate $\int \frac{x^2 + 2x + 5}{x} dx$ in under 30 seconds.

Solution. The trap is reaching for a substitution. Don't — split the fraction term by term first:

$$\frac{x^2 + 2x + 5}{x} = x + 2 + \frac{5}{x}$$

Now every term is a standard form:

$$\int \left(x + 2 + \frac{5}{x} \right) dx = \frac{x^2}{2} + 2x + 5 \ln|x| + C$$

Answer: $\frac{x^2}{2} + 2x + 5 \ln|x| + C$. *Speed lesson:* whenever a polynomial sits over a single monomial, divide through first. You will save a minute and avoid a needless substitution.

Example 3 — [JEE Main Level, multi-concept: trig identity + standard forms]

Evaluate $\int \frac{\cos 2x}{\cos^2 x \sin^2 x} dx$.

Solution. Two concepts collide here: a double-angle identity and term-by-term integration. Use $\cos 2x = \cos^2 x - \sin^2 x$ and split over the denominator:

$$\frac{\cos 2x}{\cos^2 x \sin^2 x} = \frac{\cos^2 x - \sin^2 x}{\cos^2 x \sin^2 x} = \frac{\cos^2 x}{\cos^2 x \sin^2 x} - \frac{\sin^2 x}{\cos^2 x \sin^2 x} = \frac{1}{\sin^2 x} - \frac{1}{\cos^2 x}$$

Recognise these as $\csc^2 x$ and $\sec^2 x$ (rows 8 and 7):

$$\int (\csc^2 x - \sec^2 x) dx = -\cot x - \tan x + C$$

Answer: $-\cot x - \tan x + C$. *Why it works:* the entire difficulty was algebraic — once $\cos 2x$ was rewritten and the fraction split, no calculus method beyond the table was needed.

Example 4 — [JEE Advanced Level] A function f satisfies $f'(x) = \frac{(x+1)(x-1)}{x}$ for $x > 0$, and its graph passes through the point $(1, \frac{3}{2})$. Find $f(x)$ and then determine $f(e)$.

Solution. This problem hides integration inside a “recover the function from its slope” question — and the passing-through condition is what pins down C . First simplify the derivative:

$$f'(x) = \frac{x^2 - 1}{x} = x - \frac{1}{x}$$

Integrate to recover the family of possible functions:

$$f(x) = \int \left(x - \frac{1}{x} \right) dx = \frac{x^2}{2} - \ln|x| + C$$

Now use the point $(1, \frac{3}{2})$ to fix C — this is the step that selects *one* curve from the family:

$$\frac{3}{2} = \frac{1^2}{2} - \ln(1) + C = \frac{1}{2} - 0 + C \Rightarrow C = 1$$

So $f(x) = \frac{x^2}{2} - \ln x + 1$ (we may drop the modulus since $x > 0$). Finally,

$$f(e) = \frac{e^2}{2} - \ln e + 1 = \frac{e^2}{2} - 1 + 1 = \frac{e^2}{2}$$

Answer: $f(x) = \frac{x^2}{2} - \ln x + 1$, and $f(e) = \frac{e^2}{2}$. *Deep idea:* an initial/boundary condition converts the *indefinite* integral (a whole family) into a *unique* function — this is the conceptual bridge to differential equations later in the year.

Section 5: Practice Exercises

1. **[CBSE]** Evaluate $\int \left(2\sqrt{x} + \frac{3}{x^2} - \cos x \right) dx$.
2. **[CUET]** Evaluate $\int \frac{x^4 - 1}{x^2 - 1} dx$. (*Hint: factor first.*)
3. **[JEE Main]** Evaluate $\int \frac{1 + \cos 2x}{2} dx$.
4. **[JEE Main]** Evaluate $\int \frac{2x^3 - 3x + 7}{x^2} dx$.

5. **[JEE Advanced]** The slope of the tangent to a curve at any point (x, y) is $\frac{3x^2 + 1}{x}$ for $x > 0$, and the curve passes through $(1, 4)$. Find the equation of the curve and evaluate y at $x = e$.

Answers: 1. $\frac{4}{3}x^{3/2} - \frac{3}{x} - \sin x + C$ 2. $\frac{x^3}{3} + x + C$ 3. $\frac{x}{2} + \frac{\sin 2x}{4} + C$ 4. $x^2 - 3 \ln|x| - \frac{7}{x} + C$
5. $y = x^3 + \ln x + 3$, and at $x = e$, $y = e^3 + 4$

Section 6: Crack the MCQ

Q1. $\int \left(\sin x + \frac{1}{x} \right) dx$ equals: (a) $\cos x + \ln|x| + C$ (b) $-\cos x + \ln|x| + C$ (c) $-\cos x + \frac{1}{x^2} + C$ (d) $\cos x - \ln|x| + C$

Correct: (b). $\int \sin x dx = -\cos x$ (row 6) and $\int \frac{1}{x} dx = \ln|x|$ (row 2). - (a) flips the sign of the sin integral — the classic confusion between $\int \sin x dx = -\cos x$ and $\int \cos x dx = +\sin x$. - (c) integrates $\frac{1}{x}$ as if it were $x^{-1} \rightarrow \frac{x^0}{0}$ and “fixes” it to $\frac{1}{x^2}$ — applying the power rule at the forbidden value $n = -1$. - (d) gets both signs wrong, a double slip.

Q2. If $F(x)$ is an anti-derivative of $f(x)$, which statement is **always** true? (a) $F(x)$ is unique (b) $F'(x) = f(x)$ (c) $F(0) = 0$ (d) $\int f(x) dx = F(x)$ with no constant

Correct: (b). By definition, an anti-derivative has derivative equal to the integrand. - (a) is false — there are infinitely many anti-derivatives differing by a constant. - (c) assumes a specific value the definition never imposes. - (d) drops the constant of integration; the correct statement is $\int f(x) dx = F(x) + C$.

Q3. $\int \frac{x^3 + 3x^2 + 3x + 1}{x + 1} dx$ equals: (a) $\frac{(x + 1)^3}{3} + C$ (b) $\frac{x^3}{3} + C$ (c) $(x + 1)^2 + C$ (d) $\frac{(x + 1)^2}{2} + C$

Correct: (a). Recognise the numerator as $(x + 1)^3$, so the integrand simplifies to $(x + 1)^2 = x^2 + 2x + 1$. Integrating term by term gives $\frac{x^3}{3} + x^2 + x + C$, which equals $\frac{(x+1)^3}{3} + C$ (the difference is absorbed into C). - (b) integrates only the leading term, ignoring the rest. - (c) “integrates” $(x + 1)^2$ by inspection as if the answer were the base squared — confusing differentiation and integration directions. - (d) integrates $(x + 1)^2$ as if it were $(x + 1)^1$ — power-rule index error.

Q4. The family of curves represented by $\int 2x dx$ has the property that, along any vertical line $x = a$, the tangents to all members are: (a) perpendicular (b) parallel (c) coincident (d) of slope zero

Correct: (b). The integral is $x^2 + C$; the slope at $x = a$ is $2a$ for *every* member regardless of C , so all tangents along $x = a$ are parallel. - (a) has no basis — equal slopes are never perpendicular unless slope is undefined. - (c) coincident would require the *curves* to be identical, but they are vertically shifted. - (d) slope zero only at $a = 0$, not in general — a trap for students who fixate on a single point.

Section 7: Common Student Pitfalls & Pro-Tips

- **Forgetting $+ C$** . The most penalised error in Boards. An indefinite integral without its constant is incomplete and loses marks even when the function part is perfect. Train the reflex: write $+ C$ the instant you finish.
- **Applying the power rule at $n = -1$** . Students mechanically write $\int \frac{1}{x} dx = \frac{x^0}{0}$. There is no such thing — $\int \frac{1}{x} dx = \ln |x| + C$ is a separate, memorised result.
- **Dropping the modulus in $\ln |x|$** . The integrand $\frac{1}{x}$ is defined for negative x too, so the modulus is mathematically required, not decoration.
- **Sign slips in trig integrals** . The pair that trips everyone: $\int \sin x dx = -\cos x$ but $\int \cos x dx = +\sin x$. The minus lives with the sine integral.
- **Assuming $\int (f \cdot g) = \int f \cdot \int g$** . Linearity covers sums and constant multiples — *never* products or quotients. Those need their own methods (later subtopics).

Pro-Tip (the universal safety net): *Differentiate your answer.* Integration is the only major school-level operation you can instantly self-check by running the reverse process. If $\frac{d}{dx}(\text{your answer}) \neq \text{integrand}$, you have a mistake — find it before the examiner does. This one habit eliminates the majority of sign and coefficient errors.

Section 8: Cheat Sheet Box

INDEFINITE INTEGRALS — QUICK RECALL

- **Definition:** $\frac{d}{dx}F(x) = f(x) \iff \int f(x) dx = F(x) + C$. Integration = anti-differentiation, always “ $+ C$ ”.
- **Geometry:** the answer is a *family of parallel curves*; tangents along any vertical line share the same slope.
- **Power rule:** $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ for $n \neq -1$; and $\int \frac{1}{x} dx = \ln|x| + C$.
- **Linearity:** $\int (k_1 f \pm k_2 g) dx = k_1 \int f dx \pm k_2 \int g dx$. Works for sum-/constant multiples only — **never products or quotients**.
- **Trig core:** $\sin \rightarrow -\cos$, $\cos \rightarrow +\sin$, $\sec^2 \rightarrow \tan$, $\csc^2 \rightarrow -\cot$.

- **Inverse-trig core:** $\frac{1}{1+x^2} \rightarrow \tan^{-1} x$, $\frac{1}{\sqrt{1-x^2}} \rightarrow \sin^{-1} x$.
- **Strategy:** *Simplify $\rightarrow$ Split $\rightarrow$ Identify standard form $\rightarrow$ integrate $\rightarrow$ one C $\rightarrow$ differentiate to verify.*

Memory aids: - “Co goes positive” — the **cos**ine integral keeps a **plus** sign ($\int \cos = +\sin$); the plain sine integral carries the minus. - “One over x is the lonely log” — $\frac{1}{x}$ never obeys the power rule; it always integrates to $\ln|x|$.
 - “No C, no marks.”

Subtopic 04 — Integration by Substitution

Exam Relevance: Substitution is the most-used integration technique in the entire chapter — almost every non-trivial integral starts here. CBSE Boards carry 2–4 marks of direct substitution every year; JEE Main and CUET fold it into nearly every integral; JEE Advanced disguises it inside harder problems where *spotting* the right substitution is the whole battle.

Section 1: Concept Intuition

In Subtopic 01 we could only integrate things that already matched a standard form. But what about $\int 2x \cos(x^2) dx$? There's no table entry for $\cos(x^2)$. Substitution is the trick that *reshapes* an unfamiliar integral into a familiar one by temporarily renaming a chunk of it.

The idea mirrors the chain rule in reverse. The chain rule says $\frac{d}{dx}F(g(x)) = F'(g(x)) \cdot g'(x)$. So if an integrand looks like “(a function of $g(x)$) $\times$ ($g'(x)$)”, it is exactly the derivative of “(the integral of that function) evaluated at $g(x)$ ”. Substitution lets you peel off the inner function $g(x)$, call it u , and integrate cleanly in u .

Think about it this way... Imagine converting rupees to a foreign currency to shop abroad, then converting back at the end. The substitution $u = g(x)$ is the currency conversion: you switch to the “ u world” where the integral is simple, do the easy shopping (integrate), then convert your answer back to the “ x world.” The crucial accounting step is $du = g'(x) dx$ — that is the exchange rate, and forgetting it is like ignoring the conversion and getting the price catastrophically wrong.

Limiting conditions & assumptions: - Substitution only simplifies things when $g'(x)$ (or a constant multiple of it) is *actually present* in the integrand. If it's missing, this method stalls — you may need a different technique. - Always convert the answer **back to x** for an indefinite integral. (For definite integrals you instead change the limits — that's Subtopic 09.) - A wrong substitution isn't “illegal,” it just won't help; learning which substitution works is pattern recognition built from practice.

Section 2: Key Definitions & Properties

The substitution rule. If $u = g(x)$, then $du = g'(x) dx$ and

$$\int f(g(x)) g'(x) dx = \int f(u) du$$

The single most important pattern (logarithmic integral):

$$\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C$$

A numerator that is the derivative of the denominator integrates to the log of the denominator.

Power-chain pattern: $\int [f(x)]^n f'(x) dx = \frac{[f(x)]^{n+1}}{n+1} + C \quad (n \neq -1).$

Standard results derived by substitution (memorise):

Integral	Result
$\int \tan x dx$	$\ln \sec x + C = -\ln \cos x + C$
$\int \cot x dx$	$\ln \sin x + C$
$\int \sec x dx$	$\ln \sec x + \tan x + C$
$\int \csc x dx$	$\ln \csc x - \cot x + C$

Section 3: Key Procedures & Derivations

Deriving $\int \tan x dx$ by substitution. Write $\tan x = \frac{\sin x}{\cos x}$ and let $u = \cos x$, so $du = -\sin x dx$:

$$\int \tan x dx = \int \frac{\sin x}{\cos x} dx = \int \frac{-du}{u} = -\ln|u| + C = -\ln|\cos x| + C = \ln|\sec x| + C$$

This is just the $\int f'/f$ pattern in disguise (numerator $-\sin x$ is the derivative of denominator $\cos x$). ■

Deriving $\int \sec x dx$. Multiply top and bottom by $(\sec x + \tan x)$:

$$\int \sec x dx = \int \frac{\sec x(\sec x + \tan x)}{\sec x + \tan x} dx$$

The numerator $\sec^2 x + \sec x \tan x$ is exactly the derivative of the denominator $\sec x + \tan x$, so by the $\int f'/f$ rule the result is $\ln|\sec x + \tan x| + C$. ■

Procedure — the substitution workflow: 1. **Spot the inner function.** Look for a composite chunk $g(x)$ whose derivative $g'(x)$ also appears (up to a constant). Common signals: something under a root, in an exponent, inside a trig/log function, or a denominator. 2. **Set** $u = g(x)$ and compute $du = g'(x) dx$. 3. **Replace** every x -piece: the $g(x)$ becomes u , and $g'(x) dx$ becomes du . The integral must become *purely* in u — if any x survives, the substitution was wrong. 4. **Integrate in** u using standard forms. 5. **Back-substitute** $u = g(x)$ and add C . 6. **Verify by differentiating** (chain rule should return the integrand).

Section 4: Worked Examples

Example 1 — [CBSE Board Level] Evaluate $\int 2x\sqrt{x^2 + 1} dx$.

Solution. Let $u = x^2 + 1 \Rightarrow du = 2x dx$. The $2x dx$ is present, so:

$$\int \sqrt{u} du = \frac{u^{3/2}}{3/2} = \frac{2}{3}u^{3/2}$$

Back-substitute:

$$\int 2x\sqrt{x^2 + 1} dx = \frac{2}{3}(x^2 + 1)^{3/2} + C$$

Answer: $\frac{2}{3}(x^2 + 1)^{3/2} + C$.

Example 2 — [JEE Main Speed Trap] Evaluate $\int \frac{x}{x^2 + 4} dx$ in seconds.

Solution. Spot the $\int f'/f$ pattern: the derivative of the denominator $x^2 + 4$ is $2x$, and the numerator is $x = \frac{1}{2}(2x)$. So pull out the $\frac{1}{2}$:

$$\int \frac{x}{x^2 + 4} dx = \frac{1}{2} \int \frac{2x}{x^2 + 4} dx = \frac{1}{2} \ln(x^2 + 4) + C$$

Answer: $\frac{1}{2} \ln(x^2 + 4) + C$. *Speed lesson:* whenever the numerator is a constant multiple of the denominator's derivative, write “(constant) $\times \ln$ (denominator)” immediately — no formal u needed.

Example 3 — [JEE Main Level: nested substitution] Evaluate $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$.

Solution. Let $u = \sqrt{x} = x^{1/2} \Rightarrow du = \frac{1}{2\sqrt{x}} dx$, i.e. $\frac{dx}{\sqrt{x}} = 2 du$. Then:

$$\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx = \int e^u \cdot 2 du = 2e^u + C = 2e^{\sqrt{x}} + C$$

Answer: $2e^{\sqrt{x}} + C$. *Why it works:* the awkward $\frac{1}{\sqrt{x}}$ is precisely (twice) the derivative of the exponent $\sqrt{x}$ — the integrand was engineered for this substitution.

Example 4 — [JEE Advanced Level: rationalising substitution] Evaluate $\int \frac{1}{1 + \sqrt{x}} dx$.

Solution. The root blocks every standard form, so substitute it away: let $x = t^2$ ($t \geq 0$), so $dx = 2t dt$ and $\sqrt{x} = t$:

$$\int \frac{1}{1+t} \cdot 2t dt = 2 \int \frac{t}{1+t} dt = 2 \int \left(1 - \frac{1}{1+t}\right) dt = 2(t - \ln|1+t|) + C$$

Back-substitute $t = \sqrt{x}$:

$$\int \frac{1}{1 + \sqrt{x}} dx = 2\sqrt{x} - 2\ln(1 + \sqrt{x}) + C$$

Answer: $2\sqrt{x} - 2\ln(1 + \sqrt{x}) + C$. *Deep idea:* the move $\frac{t}{1+t} = 1 - \frac{1}{1+t}$ (split an improper fraction before integrating) is a recurring trick — keep it handy.

Section 5: Practice Exercises

1. **[CBSE]** Evaluate $\int 3x^2 e^{x^3} dx$.
2. **[CUET]** Evaluate $\int \frac{\cos x}{\sin x} dx$.
3. **[JEE Main]** Evaluate $\int \frac{(\ln x)^2}{x} dx$.
4. **[JEE Main]** Evaluate $\int \frac{1}{x \ln x} dx$.
5. **[JEE Advanced]** Evaluate $\int \frac{1}{1+e^x} dx$. (*Hint: write $\frac{1}{1+e^x} = 1 - \frac{e^x}{1+e^x}$.*)

Answers: 1. $e^{x^3} + C$ 2. $\ln|\sin x| + C$ 3. $\frac{(\ln x)^3}{3} + C$ 4. $\ln|\ln x| + C$ 5. $x - \ln(1 + e^x) + C$

Section 6: Crack the MCQ

Q1. $\int \frac{2x+3}{x^2+3x+5} dx$ equals: (a) $\ln|x^2+3x+5| + C$ (b) $\frac{1}{2} \ln|x^2+3x+5| + C$ (c) $(2x+3) \ln|x^2+3x+5| + C$ (d) $\tan^{-1}(x^2+3x+5) + C$

Correct: (a). The numerator $2x+3$ is exactly the derivative of the denominator, so this is the pure $\int f'/f$ form. - (b) wrongly inserts a $\frac{1}{2}$ — that factor only appears when

the numerator is *half* the derivative. - (c) treats $\ln$ like a multiplicative constant. - (d) confuses this with the $\frac{1}{x^2+a^2}$ arctan form.

Q2. $\int \cos x e^{\sin x} dx$ equals: (a) $e^{\cos x} + C$ (b) $e^{\sin x} + C$ (c) $\sin x e^{\sin x} + C$ (d) $e^{\sin x} \cos x + C$

Correct: (b). Let $u = \sin x$, $du = \cos x dx$; the integral is $\int e^u du = e^{\sin x} + C$. - (a) substitutes the wrong inner function. - (c), (d) are products that would arise from misapplying parts; differentiate them and you won't recover the integrand.

Q3. $\int \tan x dx$ equals: (a) $\sec^2 x + C$ (b) $\ln|\cos x| + C$ (c) $\ln|\sec x| + C$ (d) $-\sec x \tan x + C$

Correct: (c). $\int \tan x dx = -\ln|\cos x| = \ln|\sec x| + C$. - (a) is the *derivative* of $\tan x$, not its integral. - (b) drops the negative sign — a sign-flip error. - (d) is unrelated.

Q4. Using $x = t^2$, $\int \frac{1}{\sqrt{x}(1+x)} dx$ equals: (a) $2 \tan^{-1} \sqrt{x} + C$ (b) $\tan^{-1} \sqrt{x} + C$ (c) $\ln(1+x) + C$ (d) $2\sqrt{x} + C$

Correct: (a). With $x = t^2$, $dx = 2t dt$, $\sqrt{x} = t$: $\int \frac{2t dt}{t(1+t^2)} = 2 \int \frac{dt}{1+t^2} = 2 \tan^{-1} t = 2 \tan^{-1} \sqrt{x} + C$. - (b) forgets the factor 2 from $dx = 2t dt$. - (c) misreads the form as $\int f'/f$. - (d) ignores the denominator entirely.

Section 7: Common Student Pitfalls & Pro-Tips

- **Forgetting to substitute dx** . You must replace $g'(x) dx$ with du — not just $g(x)$ with u . Dropping the du accounting is the number-one substitution error.
- **Leaving the answer in u** . For an indefinite integral, always convert back to x . (Definite integrals change limits instead — Subtopic 09.)
- **Wrong constant factor in $\int f'/f$** . $\int \frac{2x}{x^2+a^2} dx$ carries a 1
does not. Check whether the numerator is the *whole* derivative or a fraction of it.
- **Sign slip in $\int \tan, \cot$** . $\int \tan = +\ln|\sec|$ (or $-\ln|\cos|$);
 $\int \cot = +\ln|\sin|$.

Pro-Tip: before integrating, ask “is the numerator (a constant times) the derivative of the denominator?” If yes, the answer is instantly a log — no substitution machinery required. This one reflex handles a huge fraction of exam integrals.

Section 8: Cheat Sheet Box

INTEGRATION BY SUBSTITUTION — QUICK RECALL

- **Rule:** $u = g(x) \Rightarrow du = g'(x) dx$, then $\int f(g(x))g'(x) dx = \int f(u) du$.
- **Log pattern:** $\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C$. (Numerator = derivative of denominator.)
- **Power pattern:** $\int [f(x)]^n f'(x) dx = \frac{[f(x)]^{n+1}}{n+1} + C$.
- **Four standard results:** $\int \tan = \ln|\sec x|$; $\int \cot = \ln|\sin x|$; $\int \sec = \ln|\sec x + \tan x|$; $\int \csc = \ln|\csc x - \cot x|$.
- **Workflow:** spot inner $g \rightarrow$ set $u \rightarrow$ replace all $x \rightarrow$ integrate $\rightarrow$ back-substitute $\rightarrow +C \rightarrow$ differentiate to check.

Memory aids: - “Derivative on top $\rightarrow$ log on the bottom” — the $\int f'/f$ reflex. - “Convert there, convert back” — never leave the answer in u .

Subtopic 05 — Integration using Trigonometric Identities

Exam Relevance: Trigonometric integrals are guaranteed appearances. CBSE Boards ask $\int \sin^2 x$, $\int \cos^2 x$, or a product every year (2–3 marks). JEE Main/CUET test products and odd/even-power tricks; JEE Advanced reaches for Walli’s formula and higher powers of $\tan / \sec$.

Section 1: Concept Intuition

You cannot integrate $\sin^2 x$ directly — there’s no table entry for it. The trick is that **trigonometric identities convert un-integrable powers and products into sums of simple terms** you *can* integrate. Integration here is really a two-step dance: first rewrite using an identity, then integrate term by term.

The governing principle: high powers and products of sines and cosines oscillate in complicated ways, but every such expression can be re-expressed as a *sum* of plain $\sin(kx)$ and $\cos(kx)$ terms, each of which integrates instantly. Identities are the dictionary for that translation.

Think about it this way... A music chord sounds complex, but it’s just a sum of pure tones played together. Trig identities are like a spectrum analyser that breaks the “chord” $\sin^2 x$ or $\sin 3x \cos x$ into its pure-tone components ($\cos 2x$, $\sin 4x$, etc.), and each pure tone is trivial to integrate.

Limiting conditions & assumptions: - The method needs the *right* identity for the *shape* of the integrand (even powers $\rightarrow$ double-angle; odd powers $\rightarrow$ split off one factor and substitute; products $\rightarrow$ product-to-sum). - Walli’s formula applies **only** to definite integrals over $[0, \frac{\pi}{2}]$ of $\sin^n x \cos^m x$ — never to indefinite integrals.

Section 2: Key Definitions & Properties

Power-reduction (for even powers):

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}$$

Triple-angle (for cubes):

$$\sin^3 x = \frac{3 \sin x - \sin 3x}{4}, \quad \cos^3 x = \frac{3 \cos x + \cos 3x}{4}$$

Product-to-sum:

$$2 \sin A \cos B = \sin(A + B) + \sin(A - B)$$

$$2 \cos A \cos B = \cos(A + B) + \cos(A - B), \quad 2 \sin A \sin B = \cos(A - B) - \cos(A + B)$$

Walli's formula (definite, $[0, \frac{\pi}{2}]$): for non-negative integers m, n ,

$$\int_0^{\pi/2} \sin^n x \cos^m x \, dx = \frac{(n-1)!!(m-1)!!}{(m+n)!!} \times \begin{cases} \frac{\pi}{2} & \text{if both } m, n \text{ even} \\ 1 & \text{otherwise} \end{cases}$$

where $k!! = k(k-2)(k-4) \cdots$ down to 2 or 1, and $0!! = (-1)!! = 1$.

Section 3: Key Procedures & Derivations

Deriving $\int \sin^2 x \, dx$. Apply power-reduction:

$$\int \sin^2 x \, dx = \int \frac{1 - \cos 2x}{2} \, dx = \frac{1}{2} \left(x - \frac{\sin 2x}{2} \right) + C = \frac{x}{2} - \frac{\sin 2x}{4} + C \quad \blacksquare$$

Procedure — choose by shape: 1. **Even power of sin or cos alone** → power-reduction (halve the power, introduce double angle); repeat if the power is 4, 6, ... 2. **Odd power** → split off one factor and substitute. E.g. $\sin^3 x = \sin^2 x \cdot \sin x = (1 - \cos^2 x) \sin x$; let $u = \cos x$. 3. **Product** $\sin(ax) \cos(bx)$ **etc.** → product-to-sum, then integrate each term. 4. **Powers of tan / sec** → use $\sec^2 x = 1 + \tan^2 x$ and the fact that $\frac{d}{dx} \tan x = \sec^2 x$; peel off a $\sec^2 x$ to substitute. 5. **Definite**, $\int_0^{\pi/2} \sin^n \cos^m$ → Walli's formula directly.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] Evaluate $\int \cos^2 x \, dx$.

Solution. Power-reduction:

$$\int \cos^2 x \, dx = \int \frac{1 + \cos 2x}{2} \, dx = \frac{x}{2} + \frac{\sin 2x}{4} + C$$

Answer: $\frac{x}{2} + \frac{\sin 2x}{4} + C$.

Example 2 — [JEE Main Speed Trap] Evaluate $\int \sin 5x \cos 3x \, dx$.

Solution. Product-to-sum: $2 \sin 5x \cos 3x = \sin 8x + \sin 2x$, so $\sin 5x \cos 3x = \frac{1}{2}(\sin 8x + \sin 2x)$:

$$\int \sin 5x \cos 3x \, dx = \frac{1}{2} \int (\sin 8x + \sin 2x) \, dx = \frac{1}{2} \left(-\frac{\cos 8x}{8} - \frac{\cos 2x}{2} \right) + C$$

Answer: $-\frac{\cos 8x}{16} - \frac{\cos 2x}{4} + C$. *Speed lesson:* never try parts on a sine-cosine product — product-to-sum turns it into two trivial integrals.

Example 3 — [JEE Main Level: odd power + substitution] Evaluate $\int \sin^3 x \cos^2 x \, dx$.

Solution. Odd power of sine — split one $\sin x$ off and convert the rest to cosine using $\sin^2 x = 1 - \cos^2 x$:

$$\int \sin^3 x \cos^2 x \, dx = \int (1 - \cos^2 x) \cos^2 x \sin x \, dx$$

Let $u = \cos x$, $du = -\sin x \, dx$:

$$= - \int (1 - u^2) u^2 \, du = - \int (u^2 - u^4) \, du = - \left(\frac{u^3}{3} - \frac{u^5}{5} \right) + C$$

Back-substitute:

$$= -\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + C$$

Answer: $\frac{\cos^5 x}{5} - \frac{\cos^3 x}{3} + C$.

Example 4 — [JEE Advanced Level: Walli's formula] Evaluate $\int_0^{\pi/2} \sin^6 x \cos^2 x \, dx$.

Solution. Here $n = 6$, $m = 2$, both even, so the $\frac{\pi}{2}$ branch applies:

$$\int_0^{\pi/2} \sin^6 x \cos^2 x \, dx = \frac{5!! \cdot 1!!}{8!!} \cdot \frac{\pi}{2} = \frac{(5 \cdot 3 \cdot 1)(1)}{8 \cdot 6 \cdot 4 \cdot 2} \cdot \frac{\pi}{2} = \frac{15}{384} \cdot \frac{\pi}{2} = \frac{5}{128} \cdot \frac{\pi}{2} = \frac{5\pi}{256}$$

Answer: $\frac{5\pi}{256}$. *Why it works:* Walli's formula collapses a brutal expansion into one fraction of double-factorials — invaluable when time is short.

Section 5: Practice Exercises

1. **[CBSE]** Evaluate $\int \sin^2 x \, dx$.
2. **[CUET]** Evaluate $\int 2 \sin 3x \cos x \, dx$.
3. **[JEE Main]** Evaluate $\int \cos^3 x \, dx$.
4. **[JEE Main]** Evaluate $\int \tan^2 x \, dx$. (Hint: $\tan^2 x = \sec^2 x - 1$.)
5. **[JEE Advanced]** Evaluate $\int_0^{\pi/2} \sin^4 x \, dx$.

Answers: 1. $\frac{x}{2} - \frac{\sin 2x}{4} + C$ 2. $-\frac{\cos 4x}{4} - \frac{\cos 2x}{2} + C$ 3. $\sin x - \frac{\sin^3 x}{3} + C$ 4. $\tan x - x + C$ 5. $\frac{3\pi}{16}$

Section 6: Crack the MCQ

Q1. $\int \sin^2 x \, dx$ equals: (a) $\frac{x}{2} - \frac{\sin 2x}{4} + C$ (b) $\frac{\sin^3 x}{3} + C$ (c) $-\frac{\cos^3 x}{3} + C$ (d) $\frac{x}{2} + \frac{\sin 2x}{4} + C$

Correct: (a). Power-reduction gives $\int \frac{1-\cos 2x}{2} \, dx = \frac{x}{2} - \frac{\sin 2x}{4} + C$. - (b) treats $\sin^2 x$ as if the power rule applied to $\sin x$ — invalid. - (c) is $\int \sin^2 x \cos x \, dx$, a different integrand. - (d) is $\int \cos^2 x \, dx$ — the sign on the double-angle term is wrong for sine.

Q2. $\int 2 \cos 3x \cos x \, dx$ equals: (a) $\frac{\sin 4x}{4} + \frac{\sin 2x}{2} + C$ (b) $\sin 3x \sin x + C$ (c) $\frac{\sin 4x}{2} + \sin 2x + C$ (d) $-\frac{\cos 4x}{4} + C$

Correct: (a). $2 \cos 3x \cos x = \cos 4x + \cos 2x$, which integrates to $\frac{\sin 4x}{4} + \frac{\sin 2x}{2} + C$. - (b) is a product (parts gone wrong). - (c) mishandles the constants on integration. - (d) uses only one term and the wrong identity.

Q3. $\int \tan^2 x \, dx$ equals: (a) $\frac{\tan^3 x}{3} + C$ (b) $\tan x - x + C$ (c) $\sec^2 x + C$ (d) $\ln|\sec x| + C$

Correct: (b). $\tan^2 x = \sec^2 x - 1$, so $\int (\sec^2 x - 1) \, dx = \tan x - x + C$. - (a) wrongly applies the power rule to $\tan x$. - (c) is the derivative of $\tan x$. - (d) is $\int \tan x \, dx$, a different integrand.

Q4. $\int_0^{\pi/2} \sin^2 x \cos^2 x \, dx$ equals: (a) $\frac{\pi}{16}$ (b) $\frac{\pi}{8}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{32}$

Correct: (a). By Walli ($n = m = 2$, both even): $\frac{1!! \cdot 1!!}{4!!} \cdot \frac{\pi}{2} = \frac{1}{4 \cdot 2} \cdot \frac{\pi}{2} = \frac{1}{8} \cdot \frac{\pi}{2} = \frac{\pi}{16}$.
 - (b) forgets the $\frac{\pi}{2}$ factor's halving. - (c) ignores the formula. - (d) double-counts a factor of 2.

Section 7: Common Student Pitfalls & Pro-Tips

- **Applying the power rule to $\sin n x$.** $\int \sin^2 x \, dx \neq \sin^3 x$. You must use an identity first — the power rule integrates powers of x , not powers of $\sin x$.
- **Trying integration by parts on $\sin(ax) \cos(bx)$.** Product-
- **Wrong double-angle signs in circular parts.** $\cos 2x$; $\cos 2x$ uses $1 + \cos 2x$. Mixing them flips a sign.
- **Using Walli on an indefinite integral or wrong limits.**
 Walli is *only* for $\int_0^{\pi/2}$.

Pro-Tip: classify before you compute — “even power $\rightarrow$ halve it; odd power $\rightarrow$ peel one off and substitute; product $\rightarrow$ product-to-sum.” Naming the shape picks the identity automatically.

Section 8: Cheat Sheet Box

TRIGONOMETRIC INTEGRALS — QUICK RECALL

- **Even powers:** $\sin^2 x = \frac{1 - \cos 2x}{2}$, $\cos^2 x = \frac{1 + \cos 2x}{2}$.
- **Odd powers:** peel one factor, convert rest via $\sin^2 = 1 - \cos^2$ (or vice-versa), substitute.
- **Products:** $2 \sin A \cos B = \sin(A+B) + \sin(A-B)$; $2 \cos A \cos B = \cos(A+B) + \cos(A-B)$.
- **tan / sec:** use $\sec^2 x = 1 + \tan^2 x$; peel a $\sec^2 x$ for substitution.
- **Walli ($[0, \frac{\pi}{2}]$):** $\frac{(n-1)!!(m-1)!!}{(m+n)!!} \times (\frac{\pi}{2} \text{ if both even, else } 1)$.

Memory aids: - “Even = halve it, odd = peel it, product = split it.” - “Walli only on 0 to $\frac{\pi}{2}$.”

Subtopic 06 — Integrals of Special Forms (Quadratic Denominators & Roots)

Exam Relevance: These standard forms are a JEE Main/CUET staple — typically 1-2 single-correct questions that are pure formula-recognition once you complete the square. CBSE Boards ask the cleaner $\frac{1}{x^2+a^2}$ and $\frac{1}{\sqrt{a^2-x^2}}$ forms; JEE Advanced combines them with linear numerators and roots of quadratics.

Section 1: Concept Intuition

A whole family of integrals reduces to just a handful of templates whose answers are **inverse-trig** or **logarithm** functions. The skill is not memorising dozens of results — it's learning to *reshape* any quadratic into one of a few canonical shapes, mainly by **completing the square**.

Why completing the square? Because $\sin^{-1}$, $\tan^{-1}$, and the log-forms are built for expressions like $a^2 - x^2$, $a^2 + x^2$, $x^2 - a^2$. A messy $x^2 + bx + c$ doesn't match — until you rewrite it as $(x + \frac{b}{2})^2 + (\text{constant})$, which is a perfect $X^2 + A^2$ shape in the shifted variable $X = x + \frac{b}{2}$.

Think about it this way... It's like fitting an odd-shaped parcel into standard-size boxes at a courier office. You can't ship the parcel as-is, but a little repacking (completing the square) makes it fit a standard box (a known integral template), and then the shipping rate (the formula) is fixed.

Limiting conditions & assumptions: - The log-forms $\frac{1}{x^2-a^2}$ require $x^2 \neq a^2$; the $\sin^{-1}$ form requires $|x| < a$. - For a *linear* numerator over a quadratic, you must first split it into “(derivative-of-quadratic part) + (constant part)” — the two parts use different templates.

Section 2: Key Definitions & Properties

The core templates ($a > 0$, C understood):

Integral	Result
$\int \frac{dx}{x^2 + a^2}$	$\frac{1}{a} \tan^{-1} \frac{x}{a}$
$\int \frac{dx}{x^2 - a^2}$	$\frac{1}{2a} \ln \left \frac{x - a}{x + a} \right $
$\int \frac{dx}{a^2 - x^2}$	$\frac{1}{2a} \ln \left \frac{a + x}{a - x} \right $
$\int \frac{dx}{\sqrt{a^2 - x^2}}$	$\sin^{-1} \frac{x}{a}$
$\int \frac{dx}{\sqrt{x^2 + a^2}}$	$\ln \left x + \sqrt{x^2 + a^2} \right $
$\int \frac{dx}{\sqrt{x^2 - a^2}}$	$\ln \left x + \sqrt{x^2 - a^2} \right $

Square-root integrals:

$$\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C$$

$$\int \sqrt{x^2 + a^2} dx = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \ln \left| x + \sqrt{x^2 + a^2} \right| + C$$

$$\int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \ln \left| x + \sqrt{x^2 - a^2} \right| + C$$

General quadratic. For $ax^2 + bx + c$, complete the square: $a \left[\left(x + \frac{b}{2a} \right)^2 + \frac{c}{a} - \frac{b^2}{4a^2} \right]$, then match a template with $X = x + \frac{b}{2a}$.

Section 3: Key Procedures & Derivations

Deriving $\int \frac{dx}{x^2 - a^2}$ (partial fractions preview). Factor and split:

$$\frac{1}{x^2 - a^2} = \frac{1}{(x - a)(x + a)} = \frac{1}{2a} \left(\frac{1}{x - a} - \frac{1}{x + a} \right)$$

Integrate each log:

$$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} (\ln|x - a| - \ln|x + a|) = \frac{1}{2a} \ln \left| \frac{x - a}{x + a} \right| + C \quad \blacksquare$$

Procedure for a general quadratic: 1. Pure quadratic numerator constant?

Complete the square in the denominator, substitute $X = x + \frac{b}{2a}$, match a template.

2. **Linear numerator** $px + q$? Write $px + q = \lambda \frac{d}{dx}(\text{quadratic}) + \mu$. Solve for λ, μ . The λ -part is an $\int f'/f$ ($\rightarrow \log$) or $\int f'/\sqrt{f}$ ($\rightarrow \text{root}$); the μ -part is a pure template after completing the square. 3. **Square-root of a quadratic**? Complete the square, then apply the $\sqrt{a^2 - x^2} / \sqrt{x^2 \pm a^2}$ results.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] Evaluate $\int \frac{dx}{x^2 + 9}$.

Solution. Match $\int \frac{dx}{x^2 + a^2}$ with $a = 3$:

$$\int \frac{dx}{x^2 + 9} = \frac{1}{3} \tan^{-1} \frac{x}{3} + C$$

Answer: $\frac{1}{3} \tan^{-1} \frac{x}{3} + C$.

Example 2 — [JEE Main Speed Trap] Evaluate $\int \frac{dx}{x^2 - 16}$.

Solution. Match $\int \frac{dx}{x^2 - a^2}$ with $a = 4$, so $\frac{1}{2a} = \frac{1}{8}$:

$$\int \frac{dx}{x^2 - 16} = \frac{1}{8} \ln \left| \frac{x - 4}{x + 4} \right| + C$$

Answer: $\frac{1}{8} \ln \left| \frac{x - 4}{x + 4} \right| + C$. *Speed lesson:* $x^2 - a^2$ in the denominator $\rightarrow \log$ of a ratio; $x^2 + a^2 \rightarrow \arctan$. Read the sign first.

Example 3 — [JEE Main Level: completing the square] Evaluate $\int \frac{dx}{x^2 + 4x + 13}$.

Solution. Complete the square: $x^2 + 4x + 13 = (x + 2)^2 + 9$. With $X = x + 2$, $a = 3$:

$$\int \frac{dX}{X^2 + 9} = \frac{1}{3} \tan^{-1} \frac{X}{3} + C = \frac{1}{3} \tan^{-1} \frac{x + 2}{3} + C$$

Answer: $\frac{1}{3} \tan^{-1} \frac{x + 2}{3} + C$.

Example 4 — [JEE Advanced Level: linear over root of quadratic] Evaluate

$$\int \frac{2x + 3}{\sqrt{x^2 + 4x + 8}} dx.$$

Solution. Split the numerator into a derivative-of-quadratic part plus a constant. The derivative of $x^2 + 4x + 8$ is $2x + 4$. Write

$$2x + 3 = (2x + 4) - 1$$

So the integral splits as

$$\int \frac{2x+4}{\sqrt{x^2+4x+8}} dx - \int \frac{1}{\sqrt{x^2+4x+8}} dx = I_1 - I_2$$

I_1 : numerator is exactly the derivative of the inside, so $I_1 = 2\sqrt{x^2+4x+8}$ (since $\int \frac{f'}{\sqrt{f}} = 2\sqrt{f}$). I_2 : complete the square $x^2+4x+8 = (x+2)^2+4$; with $X = x+2$, $a = 2$, $I_2 = \ln|X + \sqrt{X^2+4}| = \ln|(x+2) + \sqrt{x^2+4x+8}|$. Combine:

$$\int \frac{2x+3}{\sqrt{x^2+4x+8}} dx = 2\sqrt{x^2+4x+8} - \ln|(x+2) + \sqrt{x^2+4x+8}| + C$$

Answer: as above. *Deep idea:* the “split a linear numerator into derivative-part + constant-part” move is the master key for *every* (linear)/(quadratic) integral, with or without a root.

Section 5: Practice Exercises

1. **[CBSE]** Evaluate $\int \frac{dx}{\sqrt{25-x^2}}$.
2. **[CUET]** Evaluate $\int \frac{dx}{x^2+2x+5}$.
3. **[JEE Main]** Evaluate $\int \frac{dx}{\sqrt{x^2+6x+13}}$.
4. **[JEE Main]** Evaluate $\int \frac{dx}{9-4x^2}$. (Hint: factor the 4 out.)
5. **[JEE Advanced]** Evaluate $\int \sqrt{4-x^2} dx$.

Answers: 1. $\sin^{-1} \frac{x}{5} + C$ 2. $\frac{1}{2} \tan^{-1} \frac{x+1}{2} + C$ 3. $\ln|(x+3) + \sqrt{x^2+6x+13}| + C$
 4. $\frac{1}{12} \ln \left| \frac{3+2x}{3-2x} \right| + C$ 5. $\frac{x}{2} \sqrt{4-x^2} + 2 \sin^{-1} \frac{x}{2} + C$

Section 6: Crack the MCQ

Q1. $\int \frac{dx}{x^2+25}$ equals: (a) $\frac{1}{5} \tan^{-1} \frac{x}{5} + C$ (b) $5 \tan^{-1} \frac{x}{5} + C$ (c) $\frac{1}{10} \ln \left| \frac{x-5}{x+5} \right| + C$ (d) $\sin^{-1} \frac{x}{5} + C$

Correct: (a). $\int \frac{dx}{x^2+a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a}$ with $a = 5$. - (b) inverts the $\frac{1}{a}$ factor. - (c) is the x^2-a^2 (log) form — wrong sign in the denominator. - (d) is the $\frac{1}{\sqrt{a^2-x^2}}$ form — there's no root here.

Q2. $\int \frac{dx}{\sqrt{9-x^2}}$ equals: (a) $\ln|x + \sqrt{9-x^2}| + C$ (b) $\sin^{-1} \frac{x}{3} + C$ (c) $\frac{1}{3} \sin^{-1} \frac{x}{3} + C$
(d) $\frac{1}{3} \tan^{-1} \frac{x}{3} + C$

Correct: (b). $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1} \frac{x}{a}$ with $a = 3$ – note **no** $\frac{1}{a}$ factor here. - (a) is a $\sqrt{x^2+a^2}$ log form. - (c) inserts a spurious $\frac{1}{3}$. - (d) is the arctan (non-root) form.

Q3. $\int \frac{dx}{x^2 - 2x + 5}$ equals: (a) $\frac{1}{2} \tan^{-1} \frac{x-1}{2} + C$ (b) $\tan^{-1}(x-1) + C$ (c) $\frac{1}{4} \ln \left| \frac{x-1-2}{x-1+2} \right| + C$ (d) $\frac{1}{2} \tan^{-1} \frac{x+1}{2} + C$

Correct: (a). $x^2 - 2x + 5 = (x - 1)^2 + 4$, so with $a = 2$: $\frac{1}{2} \tan^{-1} \frac{x-1}{2}$. - (b) forgets the $\frac{1}{a}$ factor and uses $a = 1$. - (c) uses the log form — the denominator is $X^2 + a^2$, not $X^2 - a^2$. - (d) completes the square with the wrong sign ($x + 1$ instead of $x - 1$).

Q4. $\int \frac{x}{x^2 + 4} dx$ equals: (a) $\frac{1}{2} \tan^{-1} \frac{x}{2} + C$ (b) $\frac{1}{2} \ln(x^2 + 4) + C$ (c) $\ln(x^2 + 4) + C$
(d) $\frac{1}{2} \sin^{-1} \frac{x}{2} + C$

Correct: (b). Numerator $x = \frac{1}{2}(2x)$ is half the derivative of the denominator $\rightarrow \int f'/f \text{ form} \rightarrow \frac{1}{2} \ln(x^2 + 4)$. - (a) is the template for numerator 1, not x . - (c) forgets the $\frac{1}{2}$. - (d) wrong template entirely.

Section 7: Common Student Pitfalls & Pro-Tips

- **Confusing $\ln$ or $\log$ (arctan) with $\tan$ (log)** . The sign ratio. Read it first.
- **Inserting or dropping the a factor** . $\int \frac{1}{x^2+a^2} dx$ has $\frac{1}{a}$
- **$\sin^{-1} a$ has **none**** . Memorise which forms carry the factor.
- **Numerator x vs numerator 1** . Numerator $x \rightarrow \log$ (via f' / f);
numerator constant $\rightarrow \arctan/\arcsin$. They are different problems on the same
- **Forgetting to complete the square** . Any $x^2 + bx + c$ must

be reshaped before a template applies.

Pro-Tip: decide the answer-type in one glance — denominator sign chooses log vs arctan; presence of a root chooses arcsin/log-root; numerator x vs 1 chooses log vs arctan. Three binary checks pin the template before you write anything.

Section 8: Cheat Sheet Box

SPECIAL (QUADRATIC) FORMS – QUICK RECALL

- $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a}; \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x - a}{x + a} \right|.$
- $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a}$ (no $\frac{1}{a}!$); $\int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln \left| x + \sqrt{x^2 \pm a^2} \right|.$
- **General quadratic:** complete the square $\rightarrow$ substitute $X = x + \frac{b}{2a} \rightarrow$ match a template.
- **Linear over quadratic:** $px + q = \lambda \cdot (\text{quad})' + \mu$; λ -part $\rightarrow$ log/root, μ -part $\rightarrow$ template.

Memory aids: - “Plus is arctan, minus is log.” - “Root with $a^2 - x^2$ is arcsin — and arcsin has no $\frac{1}{a}$.”

Subtopic 07 — Integration by Partial Fractions

Exam Relevance: Partial fractions is a guaranteed CBSE Board question (3–5 marks) and a frequent JEE Main/CUET tool for rational integrands. JEE Advanced uses it inside longer problems, often after a substitution turns something into a rational function.

Section 1: Concept Intuition

Some rational functions — like $\frac{1}{(x-1)(x+2)}$ — have no direct integral, because there's no single substitution that simplifies them. The fix is to **break one complicated fraction into a sum of simpler fractions**, each of which integrates to a log or an arctan. This breaking-apart is “partial fractions,” and it's just addition of fractions run in reverse.

When you add $\frac{A}{x-1} + \frac{B}{x+2}$ you get a single fraction over $(x-1)(x+2)$. Partial fractions undoes that: given the combined fraction, it recovers the unknown numerators A and B . Each recovered piece is then a textbook integral.

Think about it this way... A restaurant hands your table one combined bill, but you want to split it by who ordered what. Partial fractions is that itemisation: the messy total $\frac{\text{stuff}}{(x-1)(x+2)}$ gets itemised into each diner's share $\frac{A}{x-1}$ and $\frac{B}{x+2}$, and now each share is easy to “pay” (integrate).

Limiting conditions & assumptions: - The method needs a **proper** fraction (numerator degree < denominator degree). If it's improper, do polynomial long division first; integrate the quotient separately. - The denominator must be factorised. Each factor type gets a specific partial-fraction template (see below).

Section 2: Key Definitions & Properties

For a proper fraction, assign one partial fraction per factor:

Denominator factor	Partial-fraction term
Linear $(ax + b)$	$\frac{A}{ax + b}$
Repeated linear $(ax + b)^2$	$\frac{A}{ax + b} + \frac{B}{(ax + b)^2}$
Repeated linear $(ax + b)^3$	$\frac{A}{ax + b} + \frac{B}{(ax + b)^2} + \frac{C}{(ax + b)^3}$
Irreducible quadratic $(ax^2 + bx + c)$	$\frac{Ax + B}{ax^2 + bx + c}$

After splitting, each term integrates to a **log** (linear factors), a **power** (repeated factors, e.g. $\int \frac{dx}{(x-a)^2} = -\frac{1}{x-a}$), or a **log + arctan** (quadratic factors).

Cover-up shortcut (distinct linear factors): to find the numerator over $(x-r)$, delete that factor from the original and substitute $x = r$ into what remains.

Section 3: Key Procedures & Derivations

Procedure: 1. **Check proper/improper.** If numerator degree $\geq$ denominator degree, divide first: $\frac{N}{D} = Q + \frac{R}{D}$ with $\deg R < \deg D$. 2. **Factor the denominator** completely into linear and irreducible-quadratic factors. 3. **Write the template** (one term per factor, per the table). 4. **Clear denominators** (multiply through by D) to get an identity in the numerators. 5. **Solve for the constants** — either substitute convenient x -values (roots, for speed) or compare coefficients. 6. **Integrate each simple fraction.**

Worked derivation of the constants (cover-up). For $\frac{1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$, clear denominators: $1 = A(x+2) + B(x-1)$. Put $x = 1$: $1 = 3A \Rightarrow A = \frac{1}{3}$. Put $x = -2$: $1 = -3B \Rightarrow B = -\frac{1}{3}$. The root-substitution kills one unknown at a time. ■

Section 4: Worked Examples

Example 1 — [CBSE Board Level] Evaluate $\int \frac{dx}{(x-1)(x+2)}$.

Solution. Decompose: $\frac{1}{(x-1)(x+2)} = \frac{1/3}{x-1} - \frac{1/3}{x+2}$ (from the derivation above).

Integrate:

$$\int = \frac{1}{3} \ln|x-1| - \frac{1}{3} \ln|x+2| + C = \frac{1}{3} \ln \left| \frac{x-1}{x+2} \right| + C$$

Answer: $\frac{1}{3} \ln \left| \frac{x-1}{x+2} \right| + C.$

Example 2 — [JEE Main Speed Trap: cover-up] Evaluate $\int \frac{3x+1}{(x+1)(x-2)} dx.$

Solution. Cover-up for each constant. Over $(x+1)$: delete it, set $x = -1 \rightarrow \frac{3(-1)+1}{(-1-2)} = \frac{-2}{-3} = \frac{2}{3}$. Over $(x-2)$: set $x = 2 \rightarrow \frac{3(2)+1}{(2+1)} = \frac{7}{3}$. So

$$\frac{3x+1}{(x+1)(x-2)} = \frac{2/3}{x+1} + \frac{7/3}{x-2}$$

$$\int = \frac{2}{3} \ln|x+1| + \frac{7}{3} \ln|x-2| + C$$

Answer: $\frac{2}{3} \ln|x+1| + \frac{7}{3} \ln|x-2| + C.$ *Speed lesson:* cover-up finds each constant in one mental step — no system of equations.

Example 3 — [JEE Main Level: repeated factor] Evaluate $\int \frac{x}{(x-1)^2(x+1)} dx.$

Solution. Template with a repeated linear factor:

$$\frac{x}{(x-1)^2(x+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1}$$

Clear: $x = A(x-1)(x+1) + B(x+1) + C(x-1)^2$. Put $x = 1$: $1 = 2B \Rightarrow B = \frac{1}{2}$. Put $x = -1$: $-1 = 4C \Rightarrow C = -\frac{1}{4}$. Compare x^2 coefficients: $0 = A + C \Rightarrow A = \frac{1}{4}$. So

$$\int = \frac{1}{4} \ln|x-1| - \frac{1}{2} \cdot \frac{1}{x-1} - \frac{1}{4} \ln|x+1| + C$$

Answer: $\frac{1}{4} \ln \left| \frac{x-1}{x+1} \right| - \frac{1}{2(x-1)} + C.$

Example 4 — [JEE Advanced Level: improper + quadratic factor] Evaluate $\int \frac{x^3}{x^2+1} dx.$

Solution. Improper (numerator degree $3 \geq 2$) — divide first:

$$\frac{x^3}{x^2+1} = x - \frac{x}{x^2+1}$$

(since $x \cdot (x^2+1) = x^3 + x$, leaving remainder $-x$). Integrate each piece; the second is an $\int f'/f$ form:

$$\int \left(x - \frac{x}{x^2+1} \right) dx = \frac{x^2}{2} - \frac{1}{2} \ln(x^2+1) + C$$

Answer: $\frac{x^2}{2} - \frac{1}{2}\ln(x^2 + 1) + C$. *Deep idea:* always test degree first — skipping the division step on an improper fraction is the most common way this problem goes wrong.

Section 5: Practice Exercises

1. **[CBSE]** Evaluate $\int \frac{dx}{(x-2)(x-3)}$.
2. **[CUET]** Evaluate $\int \frac{1}{x^2-9} dx$.
3. **[JEE Main]** Evaluate $\int \frac{2x+1}{(x)(x+1)} dx$.
4. **[JEE Main]** Evaluate $\int \frac{dx}{(x+1)^2(x)}$.
5. **[JEE Advanced]** Evaluate $\int \frac{x^2}{x^2+4} dx$. (*Hint: improper — divide first.*)

Answers: 1. $\ln \left| \frac{x-3}{x-2} \right| + C$ 2. $\frac{1}{6} \ln \left| \frac{x-3}{x+3} \right| + C$ 3. $\ln|x| + \ln|x+1| + C$ 4. $\ln \left| \frac{x}{x+1} \right| + \frac{1}{x+1} + C$ 5. $x - 2 \tan^{-1} \frac{x}{2} + C$

Section 6: Crack the MCQ

Q1. In $\frac{1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$, the pair (A, B) is: (a) $(\frac{1}{3}, -\frac{1}{3})$ (b) $(\frac{1}{3}, \frac{1}{3})$ (c) $(1, -1)$ (d) $(-\frac{1}{3}, \frac{1}{3})$

Correct: (a). Cover-up: $A = \frac{1}{1+2} = \frac{1}{3}$ at $x = 1$; $B = \frac{1}{-2-1} = -\frac{1}{3}$ at $x = -2$. - (b) gets the sign of B wrong. - (c) ignores the factor-difference denominators. - (d) swaps both signs.

Q2. $\int \frac{dx}{x^2-1}$ equals: (a) $\tan^{-1} x + C$ (b) $\frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| + C$ (c) $\ln|x^2-1| + C$ (d) $\frac{1}{2} \ln|x^2-1| + C$

Correct: (b). $\frac{1}{x^2-1} = \frac{1}{2} \left(\frac{1}{x-1} - \frac{1}{x+1} \right) \rightarrow \frac{1}{2} \ln \left| \frac{x-1}{x+1} \right|$. - (a) is the x^2+1 arctan form. - (c), (d) treat the whole denominator as an $\int f'/f$, but the numerator is 1, not $2x$.

Q3. Before applying partial fractions to $\frac{x^2}{x^2-1}$, you must first: (a) factor only (b) complete the square (c) divide (it's improper) (d) substitute $u = x^2$

Correct: (c). Numerator degree = denominator degree, so it's improper — divide to get $1 + \frac{1}{x^2-1}$ first. - (a) skips the required division. - (b) is for irreducible quadratics, not this case. - (d) doesn't simplify the form.

Q4. The correct partial-fraction template for $\frac{1}{(x-2)^2(x+3)}$ is: (a) $\frac{A}{x-2} + \frac{B}{x+3}$
 (b) $\frac{A}{(x-2)^2} + \frac{B}{x+3}$ (c) $\frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{C}{x+3}$ (d) $\frac{Ax+B}{(x-2)^2} + \frac{C}{x+3}$

Correct: (c). A repeated linear factor $(x-2)^2$ needs *both* $\frac{A}{x-2}$ and $\frac{B}{(x-2)^2}$, plus $\frac{C}{x+3}$. - (a), (b) omit one of the two required repeated-factor terms. - (d) uses a linear numerator, which is only for irreducible *quadratic* factors.

Section 7: Common Student Pitfalls & Pro-Tips

- **Skipping division on improper fractions** . If numerator degree $\geq$ denominator degree, you *must* divide first. Forcing partial fractions on an improper fraction gives wrong constants.
- **Missing a repeated-factor term** . $(x-a)^2$ needs both $A/(x-a)$ and $B/(x-a)^2$ — not just one.
- **Linear vs constant numerator** . An irreducible quadratic factor takes a *linear* numerator $Ax + B$; a linear factor takes a *constant* A .
- **Sign errors when combining logs** . $\ln |a| - \ln |b| = \ln |a/b|$ — keep track of which factor is on top.

Pro-Tip: for distinct linear factors, skip the linear system entirely and use cover-up — each constant drops out in one substitution. Reserve coefficient-comparison for repeated or quadratic factors.

Section 8: Cheat Sheet Box

PARTIAL FRACTIONS — QUICK RECALL

- **First:** proper? If not, divide. **Then:** factor the denominator.
- **Templates:** linear $\rightarrow \frac{A}{ax+b}$; repeated $\rightarrow$ add $\frac{B}{(ax+b)^2}$ etc.; irreducible quadratic $\rightarrow \frac{Ax+B}{ax^2+bx+c}$.
- **Find constants:** cover-up (distinct linear) or compare coefficients (repeated/quadratic).
- **Integrate:** linear $\rightarrow$ log; repeated $\rightarrow$ power ($\int \frac{dx}{(x-a)^2} = -\frac{1}{x-a}$); quadratic $\rightarrow$ log + arctan.

Memory aids: - “Improper? Divide before you decide.” - “Repeated factor $\rightarrow$ repeated terms.” - “Quadratic on bottom $\rightarrow$ linear on top.”

Subtopic 08 — Integration by Parts

Exam Relevance: Integration by parts is a CBSE Board guarantee (3–5 marks: $\int x \cos x$, $\int \ln x$, $\int e^x \sin x$) and a JEE Main/CUET regular. JEE Advanced leans on the $\int e^x[f(x) + f'(x)]$ recognition and parts-based reduction.

Section 1: Concept Intuition

Substitution handles integrands that are a function-times-its-own-derivative. But what about a *product of two unrelated functions*, like $x \cos x$ or xe^x ? There's no inner-function-and-its-derivative to exploit. Integration by parts is the technique built for products — it's the reverse of the **product rule** for differentiation.

The product rule says $(uv)' = u'v + uv'$. Rearranging and integrating gives a formula that trades one integral for another: you integrate the “easy” factor and differentiate the “hard” one, hoping the new integral is simpler than the original. The art is choosing *which* factor to differentiate.

Think about it this way... It's like a relay race where you hand off the baton. You can't integrate $x \cos x$ in one runner's leg, so you “hand off”: differentiate the x (it becomes the simpler 1) and integrate the $\cos x$. After the handoff, the leftover integral $\int 1 \cdot \sin x$ is something a single runner can finish. The skill is choosing who carries the baton first so the race actually ends.

Limiting conditions & assumptions: - Parts helps only if the new integral $\int v du$ is *easier* than the original. A bad choice makes it harder — then swap your u and dv . - Some integrals (like $\int e^x \sin x$) cycle back to themselves after two applications; you then solve algebraically for the integral.

Section 2: Key Definitions & Properties

The formula:

$$\int u dv = uv - \int v du$$

Equivalently, for a product $u(x) \cdot v(x)$ where you differentiate u and integrate v' :

$$\int u v' dx = u v - \int u' v dx$$

ILATE — choosing u . Pick the function to *differentiate* (u) by whichever comes first in: **I**nverse trig, **L**ogarithm, **A**lgebraic, **T**rigonometric, **E**xponential. The earlier it appears, the better it is as u (because differentiating it simplifies things).

Two high-value standard results:

$$\begin{aligned}\int e^x [f(x) + f'(x)] dx &= e^x f(x) + C \\ \int e^{ax} \sin bx dx &= \frac{e^{ax}(a \sin bx - b \cos bx)}{a^2 + b^2} + C \\ \int e^{ax} \cos bx dx &= \frac{e^{ax}(a \cos bx + b \sin bx)}{a^2 + b^2} + C\end{aligned}$$

Section 3: Key Procedures & Derivations

Deriving the formula. Integrate the product rule $(uv)' = u'v + uv'$:

$$uv = \int u'v dx + \int uv' dx \Rightarrow \int uv' dx = uv - \int u'v dx \quad \blacksquare$$

Deriving $\int e^x [f + f'] dx = e^x f$. Apply parts to $\int e^x f(x) dx$ with $u = f(x)$, $dv = e^x dx$:

$$\int e^x f(x) dx = e^x f(x) - \int e^x f'(x) dx$$

Move the second integral over:

$$\int e^x f(x) dx + \int e^x f'(x) dx = e^x f(x) \Rightarrow \int e^x [f(x) + f'(x)] dx = e^x f(x) + C \quad \blacksquare$$

Procedure: 1. **Identify u and dv** using ILATE (differentiate the ILATE-earlier function). 2. **Compute du and v** (differentiate u , integrate dv). 3. **Apply** $\int u dv = uv - \int v du$. 4. **If the new integral cycles back** to the original (e.g. $\int e^x \sin x$), apply parts once more, then solve algebraically. 5. **Special trick:** if the integrand is e^x times “a function plus its own derivative,” jump straight to $e^x f(x)$.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] Evaluate $\int x \cos x \, dx$.

Solution. ILATE: Algebraic (x) before Trig ($\cos x$), so $u = x$, $dv = \cos x \, dx$. Then $du = dx$, $v = \sin x$:

$$\int x \cos x \, dx = x \sin x - \int \sin x \, dx = x \sin x + \cos x + C$$

Answer: $x \sin x + \cos x + C$.

Example 2 — [JEE Main Speed Trap: the $\ln x$ trick] Evaluate $\int \ln x \, dx$.

Solution. There's only one function — but treat it as $\ln x \cdot 1$. ILATE: Log before Algebraic, so $u = \ln x$, $dv = 1 \, dx$. Then $du = \frac{1}{x} dx$, $v = x$:

$$\int \ln x \, dx = x \ln x - \int x \cdot \frac{1}{x} \, dx = x \ln x - \int 1 \, dx = x \ln x - x + C$$

Answer: $x \ln x - x + C$. *Speed lesson:* lone $\ln x$ or $\tan^{-1} x \rightarrow$ pair with an invisible “ $\times 1$ ” and apply parts.

Example 3 — [JEE Main Level: the cycling integral] Evaluate $I = \int e^x \sin x \, dx$.

Solution. Parts with $u = \sin x$, $dv = e^x dx$: $I = e^x \sin x - \int e^x \cos x \, dx$. Apply parts again to the new integral ($u = \cos x$, $dv = e^x dx$): $\int e^x \cos x \, dx = e^x \cos x + \int e^x \sin x \, dx = e^x \cos x + I$. Substitute back:

$$I = e^x \sin x - (e^x \cos x + I) = e^x \sin x - e^x \cos x - I$$

$$\Rightarrow 2I = e^x(\sin x - \cos x) \Rightarrow I = \frac{e^x(\sin x - \cos x)}{2} + C$$

Answer: $\frac{e^x(\sin x - \cos x)}{2} + C$. *Why it works:* the integral reappears, so you solve for I like an algebraic unknown.

Example 4 — [JEE Advanced Level: the $e^x[f + f']$ recognition] Evaluate $\int e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx$.

Solution. Spot the structure: let $f(x) = \frac{1}{x}$. Then $f'(x) = -\frac{1}{x^2}$, so the integrand is exactly $e^x[f(x) + f'(x)]$. By the standard result:

$$\int e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx = e^x \cdot \frac{1}{x} + C = \frac{e^x}{x} + C$$

Answer: $\frac{e^x}{x} + C$. *Deep idea:* recognising $e^x[f + f']$ turns a hard-looking integral into a one-line answer — always check for it when you see e^x times a sum.

Section 5: Practice Exercises

1. **[CBSE]** Evaluate $\int x e^x dx$.
2. **[CUET]** Evaluate $\int x \sin x dx$.
3. **[JEE Main]** Evaluate $\int \tan^{-1} x dx$. (*Hint: pair with $\times 1$.*)
4. **[JEE Main]** Evaluate $\int x^2 \ln x dx$.
5. **[JEE Advanced]** Evaluate $\int e^x \left(\tan^{-1} x + \frac{1}{1+x^2} \right) dx$. (*Hint: $f + f'$.*)

Answers: 1. $e^x(x-1) + C$ 2. $-x \cos x + \sin x + C$ 3. $x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) + C$
 4. $\frac{x^3}{3} \ln x - \frac{x^3}{9} + C$ 5. $e^x \tan^{-1} x + C$

Section 6: Crack the MCQ

Q1. Using ILATE, for $\int x e^x dx$ the correct choice of u is: (a) e^x (b) x (c) $x e^x$ (d) either works equally

Correct: (b). Algebraic (x) precedes Exponential (e^x) in ILATE, so differentiate x . Choosing $u = e^x$ makes the integral harder. - (a) is the worse choice (the algebraic part should be differentiated). - (c) isn't a valid split. - (d) is false — the wrong choice complicates the new integral.

Q2. $\int \ln x dx$ equals: (a) $\frac{1}{x} + C$ (b) $x \ln x + x + C$ (c) $x \ln x - x + C$ (d) $\frac{(\ln x)^2}{2} + C$

Correct: (c). Parts with $u = \ln x$, $dv = dx$ gives $x \ln x - x + C$. - (a) is the derivative of $\ln x$. - (b) has the wrong sign on the second term. - (d) wrongly treats $\ln x$ as if substitution applied.

Q3. $\int e^x(\sin x + \cos x) dx$ equals: (a) $e^x \sin x + C$ (b) $e^x \cos x + C$ (c) $e^x(\sin x - \cos x) + C$ (d) $\frac{e^x(\sin x + \cos x)}{2} + C$

Correct: (a). With $f = \sin x$, $f' = \cos x$, the integrand is $e^x[f + f']$, so the answer is $e^x \sin x + C$. - (b) uses $f = \cos x$, but then $f' = -\sin x \neq \cos x$, so the struc-

ture doesn't match. - (c) is the answer to $\int e^x \sin x \, dx$, a different integrand. - (d) misapplies the cycling-integral result.

Q4. $\int x \sin x \, dx$ equals: (a) $-x \cos x + \sin x + C$ (b) $x \cos x - \sin x + C$ (c) $-x \cos x - \sin x + C$ (d) $x \sin x + \cos x + C$

Correct: (a). $u = x, dv = \sin x \, dx \rightarrow v = -\cos x: -x \cos x + \int \cos x \, dx = -x \cos x + \sin x + C$. - (b) flips every sign. - (c) has the wrong sign on $\sin x$. - (d) is $\int x \cos x \, dx$.

Section 7: Common Student Pitfalls & Pro-Tips

- **Choosing u and dv backwards** . Differentiate the ILATE-earlier function. The wrong split makes $\int v \, du$ harder than the original.
- **Sign error from $v = \int \sin x = -\cos x$** . Integrating $\sin$ introduces
- **Forgetting the cycling trick** Terms of the sine/cosine reappear after two applications
- **Missing the reappears + shortcut** When you have e^x times a

sum, check whether the second term is the derivative of the first.

Pro-Tip: before grinding through parts on an $e^x(\dots)$ integral, test the $f + f'$ pattern — it converts many “hard” problems into instant answers. And for a lone $\ln x$, $\tan^{-1} x$, or $\sin^{-1} x$, remember the invisible “ $\times 1$.”

Section 8: Cheat Sheet Box

INTEGRATION BY PARTS — QUICK RECALL

- **Formula:** $\int u \, dv = uv - \int v \, du$.
- **Choose u by ILATE:** Inverse-trig, Log, Algebraic, Trig, Exponential (earlier $\rightarrow$ differentiate it).
- **Cycling:** $\int e^{ax} \sin bx, \int e^{ax} \cos bx$ reappear after two applications $\rightarrow$ solve algebraically.
- **Golden shortcut:** $\int e^x [f(x) + f'(x)] \, dx = e^x f(x) + C$.
- **Lone log / inverse-trig:** pair with $\times 1$, then apply parts.

Memory aids: - “ILATE: differentiate the one that comes first.” - “ e^x times (f plus f -prime) = e^x times f .” - “Sine integrates to minus cosine — mind the sign.”

Subtopic 02 — Definite Integrals: Definition, the Fundamental Theorem & Limit of a Sum

Exam Relevance: The definite integral is the gateway to area, average value, and every applied integral that follows. CBSE Boards test direct Fundamental-Theorem evaluation and the limit-of-a-sum definition (the latter is a favourite 5-mark “first-principles” question); JEE Main/CUET use definite integrals everywhere; JEE Advanced builds on this foundation with the properties of Subtopic 03.

Section 1: Concept Intuition

An indefinite integral gives a *family of functions*. A **definite integral** gives a single *number* — the net accumulation of a quantity between two endpoints. Geometrically, $\int_a^b f(x) dx$ is the **signed area** between the curve $y = f(x)$ and the x -axis from $x = a$ to $x = b$ (area above the axis counts positive, below counts negative).

There are two ways to *define* this number, and the deep result of the chapter is that they agree: - **As a limit of a sum** (the “first-principles” definition): slice $[a, b]$ into thin strips, add up the rectangle areas, and let the strips shrink to zero width. - **Via antiderivatives** (the practical method): find any antiderivative F and compute $F(b) - F(a)$.

The bridge between them is the **Fundamental Theorem of Calculus** — the statement that differentiation and integration are inverse processes. It is what makes integration *computable* instead of an infinite summing chore.

Think about it this way... Imagine filling a tank from a tap whose flow rate varies. The limit-of-a-sum is like measuring the inflow second-by-second and adding up thousands of tiny amounts — accurate but exhausting. The Fundamental Theorem is the water meter: read it at the start ($F(a)$) and at the end ($F(b)$), subtract, and you instantly know the total without tracking every drop.

Limiting conditions & assumptions: - A definite integral is a **number** — there is **no** $+C$. The constant cancels in $F(b) - F(a)$. - The function should be continuous (or at worst have finitely many jumps) on $[a, b]$ for these definitions to behave. - “Signed”

matters: if the curve dips below the axis, that portion subtracts. Total *geometric* area needs $\int |f|$ — a distinction tested often.

Section 2: Key Definitions & Properties

Definition (definite integral as a limit of a sum). For f continuous on $[a, b]$, with $h = \frac{b-a}{n}$,

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} h \sum_{r=0}^{n-1} f(a + rh) = \lim_{h \rightarrow 0} h [f(a) + f(a+h) + \cdots + f(a + (n-1)h)]$$

Second Fundamental Theorem (the evaluation rule). If F is an antiderivative of f (so $F' = f$), then

$$\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$$

First Fundamental Theorem (the area function). Define $A(x) = \int_a^x f(t) dt$.

Then A is differentiable and

$$A'(x) = \frac{d}{dx} \int_a^x f(t) dt = f(x)$$

i.e. differentiating an integral with respect to its upper limit returns the integrand. This is *why* antiderivatives compute areas.

Structural properties (owned here):

#	Property	Statement
P0	Dummy variable	$\int_a^b f(x) dx = \int_a^b f(t) dt$
P1	Swap limits	$\int_a^b f(x) dx = - \int_b^a f(x) dx$
P2	Additivity (splitting)	$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$

Evaluation & symmetry properties — the *king* property, *even/odd* rule, the $0 \rightarrow 2a$ split, and *periodic* integrals — are the toolkit of **Subtopic 03**. This subtopic supplies the definition and FTC they all rest on.

Section 3: Key Procedures & Derivations

Why the First FTC holds (area-function reasoning). Consider $A(x) = \int_a^x f(t) dt$. Increasing the upper limit by a sliver Δx adds a thin strip of area $\approx f(x) \Delta x$ (a rectangle of height $f(x)$ and width Δx). So $\frac{A(x + \Delta x) - A(x)}{\Delta x} \approx f(x)$, and letting $\Delta x \rightarrow 0$ gives $A'(x) = f(x)$. The rate at which accumulated area grows is exactly the height of the curve. ■

Why the Second FTC follows. Since $A'(x) = f(x)$, A is an antiderivative of f . Any other antiderivative F differs from A by a constant: $F(x) = A(x) + k$. Then $F(b) - F(a) = A(b) - A(a) = \int_a^b f - \int_a^a f = \int_a^b f - 0$. Hence $\int_a^b f = F(b) - F(a)$ for any antiderivative F . ■

Justifying P2 (additivity). Accumulating from a to b is the same as accumulating from a to an intermediate c , then from c to b — area adds over adjacent intervals. This is the property that lets you handle piecewise integrands by splitting at the break-points.

Procedure — evaluating a definite integral (FTC method): 1. Find an antiderivative $F(x)$ (any one — drop the $+C$). 2. Substitute the upper limit: compute $F(b)$. 3. Substitute the lower limit: compute $F(a)$. 4. Subtract: $F(b) - F(a)$. Keep the order — upper minus lower.

Procedure — limit of a sum (first principles): 1. Set $h = \frac{b-a}{n}$; write the sample points $a + rh$. 2. Form $h \sum_{r=0}^{n-1} f(a + rh)$. 3. Expand using $\sum r = \frac{n(n-1)}{2}$, $\sum r^2 = \frac{(n-1)n(2n-1)}{6}$, etc. 4. Take $n \rightarrow \infty$ (equivalently $h \rightarrow 0$).

Section 4: Worked Examples

Example 1 — [CBSE Board Level] Evaluate $\int_1^2 (3x^2 - 2x + 1) dx$.

Solution. Antiderivative: $F(x) = x^3 - x^2 + x$. Apply the Second FTC:

$$F(2) = 8 - 4 + 2 = 6, \quad F(1) = 1 - 1 + 1 = 1$$

$$\int_1^2 (3x^2 - 2x + 1) dx = F(2) - F(1) = 6 - 1 = 5$$

Answer: 5.

Example 2 — [JEE Main Speed Trap] Evaluate $\int_1^e \frac{1}{x} dx$.

Solution. Antiderivative of $\frac{1}{x}$ is $\ln|x|$:

$$\int_1^e \frac{1}{x} dx = [\ln x]_1^e = \ln e - \ln 1 = 1 - 0 = 1$$

Answer: 1. *Speed lesson:* recognise the standard antiderivative and the limits often chosen to make logs collapse ($\ln e = 1$, $\ln 1 = 0$).

Example 3 — [JEE Main Level: standard forms + FTC] Evaluate $\int_1^2 \left(2x + \frac{3}{x^2}\right) dx$.

Solution. Antiderivative: $F(x) = x^2 - \frac{3}{x}$ (since $\int 3x^{-2} dx = -3x^{-1}$). Then

$$F(2) = 4 - \frac{3}{2} = \frac{5}{2}, \quad F(1) = 1 - 3 = -2$$

$$\int_1^2 \left(2x + \frac{3}{x^2}\right) dx = \frac{5}{2} - (-2) = \frac{9}{2}$$

Answer: $\frac{9}{2}$.

Example 4 — [JEE Advanced Level: limit of a sum] Evaluate $\int_0^2 (x^2 + 2x) dx$ from first principles (as a limit of a sum).

Solution. Here $a = 0$, $b = 2$, $h = \frac{2}{n}$, sample points $x_r = rh = \frac{2r}{n}$:

$$\begin{aligned} \int_0^2 (x^2 + 2x) dx &= \lim_{n \rightarrow \infty} h \sum_{r=0}^{n-1} \left[\left(\frac{2r}{n}\right)^2 + 2 \cdot \frac{2r}{n} \right] = \lim_{n \rightarrow \infty} \frac{2}{n} \sum_{r=0}^{n-1} \left[\frac{4r^2}{n^2} + \frac{4r}{n} \right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{8}{n^3} \sum_{r=0}^{n-1} r^2 + \frac{8}{n^2} \sum_{r=0}^{n-1} r \right] = \lim_{n \rightarrow \infty} \left[\frac{8}{n^3} \cdot \frac{(n-1)n(2n-1)}{6} + \frac{8}{n^2} \cdot \frac{n(n-1)}{2} \right] \end{aligned}$$

As $n \rightarrow \infty$, the first term $\rightarrow \frac{8}{6} \cdot 2 = \frac{8}{3}$ and the second $\rightarrow \frac{8}{2} = 4$:

$$\int_0^2 (x^2 + 2x) dx = \frac{8}{3} + 4 = \frac{20}{3}$$

Check (FTC): $\left[\frac{x^3}{3} + x^2\right]_0^2 = \frac{8}{3} + 4 = \frac{20}{3}$. ✓ **Answer:** $\frac{20}{3}$. *Deep idea:* the limit-of-a-sum defines the integral; the FTC is the shortcut. Boards reward showing they agree.

Section 5: Practice Exercises

1. **[CBSE]** Evaluate $\int_0^{\pi/2} \cos x \, dx$.
2. **[CUET]** Evaluate $\int_1^4 \frac{1}{\sqrt{x}} \, dx$.
3. **[JEE Main]** Evaluate $\int_0^1 (3x^2 + 2x) \, dx$.
4. **[JEE Main]** Evaluate $\int_1^3 \frac{1}{x} \, dx$.
5. **[JEE Advanced]** Evaluate $\int_0^1 x^2 \, dx$ as a limit of a sum.

Answers: 1. 1 2. 2 3. 2 4. $\ln 3$ 5. $\frac{1}{3}$

Section 6: Crack the MCQ

Q1. $\int_0^\pi \sin x \, dx$ equals: (a) 0 (b) 1 (c) 2 (d) -2

Correct: (c). $[-\cos x]_0^\pi = -\cos \pi + \cos 0 = -(-1) + 1 = 2$. - (a) results from a sign slip ($\cos \pi - \cos 0$). - (b) drops a factor. - (d) reverses the FTC order (lower minus upper).

Q2. $\int_a^b f(x) \, dx$ is equal to: (a) $\int_a^b f(t) \, dt$ (b) $\int_a^b f(x) \, dx + C$ (c) $\int_b^a f(x) \, dx$ (d) a function of x

Correct: (a). The variable of integration is a *dummy* (P0) — renaming x to t changes nothing. - (b) wrongly adds a constant; a definite integral has no $+C$. - (c) is the *negative* of the integral (P1). - (d) is false — a definite integral is a number, not a function of x .

Q3. $\frac{d}{dx} \int_0^x t^2 \, dt$ equals: (a) $\frac{x^3}{3}$ (b) x^2 (c) $2x$ (d) 0

Correct: (b). By the First FTC, differentiating an integral w.r.t. its upper limit returns the integrand evaluated there: x^2 . - (a) is the integral itself, not its derivative. - (c) differentiates twice. - (d) wrongly treats the integral as a constant.

Q4. In evaluating $\int_0^1 x^2 \, dx$ as $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{n-1} \left(\frac{r}{n}\right)^2$, the value is: (a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) 1 (d) $\frac{1}{6}$

Correct: (a). $\frac{1}{n^3} \sum_{r=0}^{n-1} r^2 = \frac{(n-1)n(2n-1)}{6n^3} \rightarrow \frac{2}{6} = \frac{1}{3}$. - (b), (c), (d) come from mis-summing or using the wrong power-sum formula.

Section 7: Common Student Pitfalls & Pro-Tips

- **Writing $+C$ on a definite integral** . The result is a number; the constant cancels in $F(b) - F(a)$. Never append $+C$.
- **Reversing the FTC order** . It is $F(\text{upper}) - F(\text{lower})$. Swapping them flips the sign (that is P1).
- **Mishandling a swap of limits** . $\int_a^b = -\int_b^a$; if you ever exchange limits mid-solution, flip the sign.
- **Forgetting dx /limit accounting in limit-of-a-sum** . The factor $h = \frac{b-a}{n}$ multiplies the whole sum; dropping it scales the answer wrongly.

Pro-Tip: for “evaluate from first principles,” memorise the three power-sums ($\sum 1 = n$, $\sum r = \frac{n(n-1)}{2}$, $\sum r^2 = \frac{(n-1)n(2n-1)}{6}$) — almost every Board limit-of-a-sum reduces to a combination of these.

Section 8: Cheat Sheet Box

DEFINITE INTEGRALS — QUICK RECALL

- **Meaning:** $\int_a^b f dx = \text{signed area} = \text{net accumulation} = \text{a number (no } +C)$.
- **Second FTC (evaluate):** $\int_a^b f dx = F(b) - F(a)$, F any antiderivative.
- **First FTC (differentiate):** $\frac{d}{dx} \int_a^x f(t) dt = f(x)$.
- **Limit of a sum:** $\int_a^b f dx = \lim_{n \rightarrow \infty} h \sum_{r=0}^{n-1} f(a + rh)$, $h = \frac{b-a}{n}$.
- **Structural properties:** P0 dummy variable; P1 $\int_a^b = -\int_b^a$; P2 split $\int_a^b = \int_a^c + \int_c^b$.
- **For king / even-odd / $0 \rightarrow 2a$ / periodic $\rightarrow$ Subtopic 03.**

Power-sums: $\sum r = \frac{n(n-1)}{2}$, $\sum r^2 = \frac{(n-1)n(2n-1)}{6}$.

Memory aids: - “Upper minus lower, no C .” - “Differentiate an integral $\rightarrow$ get the integrand back.”

Subtopic 09 — Evaluation of Definite Integrals by Substitution

Exam Relevance: This is where substitution meets definite integrals — and where students lose easy marks by mishandling the limits. CBSE Boards ask 2–4 marks on it; JEE Main/CUET test it constantly; JEE Advanced combines it with the definite-integral properties from Subtopic 03.

Section 1: Concept Intuition

You already know two things: how to substitute (Subtopic 04) and how definite integrals work (Subtopic 02). This subtopic fuses them, and the *only* new idea is what happens to the **limits**.

When you substitute $u = g(x)$ in a definite integral, the limits $x = a$ and $x = b$ are values of x — but your new integral is in u . So the limits must be **converted to u -values** too: the new lower limit is $g(a)$ and the new upper limit is $g(b)$. Do that, and you never need to convert back to x at all — you just evaluate in u .

Think about it this way... Recall the currency analogy from Subtopic 04. A definite integral is like a shopping trip with a fixed start and end budget marked in rupees. When you switch to the “ u currency,” you must also convert those budget markers to the new currency — otherwise you’d be measuring distance with the wrong ruler at the ends. Convert the limits once, and the whole trip happens cleanly in u .

Limiting conditions & assumptions: - You have **two valid choices**: (a) change the limits to u -values and finish in u , or (b) keep the old limits, integrate, back-substitute to x , *then* plug in the original limits. Pick one — never mix them. - The substitution must be valid (monotonic / well-defined) across the whole interval; watch for points where g misbehaves.

Section 2: Key Definitions & Properties

The rule. If $u = g(x)$ with $du = g'(x) dx$, then

$$\int_a^b f(g(x)) g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$$

The limits transform with the variable: $x = a \rightarrow u = g(a)$ and $x = b \rightarrow u = g(b)$.

A useful consequence (even/odd revisited). Substitution is the engine behind many property results — e.g. the king and even/odd properties of Subtopic 03 are proved by exactly this limit-changing mechanism.

Two methods, same answer: - *Method A (change limits):* convert limits to u , evaluate in u . Faster; no back-substitution. - *Method B (keep limits):* integrate, return to x , then apply the original x -limits.

Section 3: Key Procedures & Derivations

Why the limits change — the reasoning. The definite integral $\int_a^b$ accumulates over x running from a to b . After substitution, the *same accumulation* is described in terms of u ; but as x sweeps from a to b , $u = g(x)$ sweeps from $g(a)$ to $g(b)$. To describe the identical accumulation in u , the integral must run over u 's actual range, namely $g(a)$ to $g(b)$. Keeping the old x -limits on a u -integral would be mixing two rulers — the source of countless wrong answers.

Procedure (Method A): 1. Choose $u = g(x)$; compute $du = g'(x) dx$. 2. **Convert the limits:** lower $x = a \rightarrow u = g(a)$; upper $x = b \rightarrow u = g(b)$. 3. Rewrite the entire integral in u with the new limits. 4. Integrate and evaluate as $F(g(b)) - F(g(a))$ — no back-substitution needed.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] Evaluate $\int_0^1 \frac{x}{x^2 + 1} dx$.

Solution. Let $u = x^2 + 1$, $du = 2x dx$, so $x dx = \frac{1}{2} du$. **Convert limits:** $x = 0 \rightarrow u = 1$; $x = 1 \rightarrow u = 2$. Then

$$\int_0^1 \frac{x}{x^2 + 1} dx = \frac{1}{2} \int_1^2 \frac{du}{u} = \frac{1}{2} [\ln u]_1^2 = \frac{1}{2} (\ln 2 - \ln 1) = \frac{1}{2} \ln 2$$

Answer: $\frac{1}{2} \ln 2$.

Example 2 — [JEE Main Speed Trap] Evaluate $\int_0^{\pi/2} \sin x \cos x dx$.

Solution. Let $u = \sin x$, $du = \cos x dx$. **Convert limits:** $x = 0 \rightarrow u = 0$; $x = \frac{\pi}{2} \rightarrow u = 1$:

$$\int_0^{\pi/2} \sin x \cos x dx = \int_0^1 u du = \left[\frac{u^2}{2} \right]_0^1 = \frac{1}{2}$$

Answer: $\frac{1}{2}$. *Speed lesson:* once limits are converted, the answer pops out in u — no need to return to x and re-evaluate.

Example 3 — [JEE Main Level] Evaluate $\int_1^e \frac{\ln x}{x} dx$.

Solution. Let $u = \ln x$, $du = \frac{1}{x} dx$. **Convert limits:** $x = 1 \rightarrow u = \ln 1 = 0$; $x = e \rightarrow u = \ln e = 1$:

$$\int_1^e \frac{\ln x}{x} dx = \int_0^1 u du = \left[\frac{u^2}{2} \right]_0^1 = \frac{1}{2}$$

Answer: $\frac{1}{2}$.

Example 4 — [JEE Advanced Level: substitution + property] Evaluate $I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$.

Solution. Here substitution-by-limits and a property combine. Apply the king property (Subtopic 03), $x \rightarrow \frac{\pi}{2} - x$, which swaps $\sin \leftrightarrow \cos$ but keeps the limits 0 and $\frac{\pi}{2}$:

$$I = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx$$

Add the two expressions for I (same denominator):

$$2I = \int_0^{\pi/2} \frac{\sqrt{\sin x} + \sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \int_0^{\pi/2} 1 dx = \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4}$$

Answer: $\frac{\pi}{4}$. *Deep idea:* the king substitution is a *limit-preserving* change of variable — the cleanest illustration of how substitution underlies the definite-integral properties.

Section 5: Practice Exercises

1. **[CBSE]** Evaluate $\int_0^1 2x(x^2 + 1)^3 dx$.
2. **[CUET]** Evaluate $\int_0^1 \frac{e^x}{1 + e^x} dx$.
3. **[JEE Main]** Evaluate $\int_0^{\pi/2} \cos x e^{\sin x} dx$.
4. **[JEE Main]** Evaluate $\int_1^2 \frac{1}{x(1 + \ln x)} dx$.

5. **[JEE Advanced]** Evaluate $\int_0^{\pi/2} \frac{\sin^2 x}{\sin x + \cos x} dx \dots$ (use king to relate to its co-sine partner; final hint: $2I = \int_0^{\pi/2} \frac{1}{\sin x + \cos x} dx$).

Answers: 1. $\frac{15}{4}$ 2. $\ln \frac{1+e}{2}$ 3. $e - 1$ 4. $\ln(1 + \ln 2)$ 5. $\frac{1}{\sqrt{2}} \ln(\sqrt{2} + 1)$

Section 6: Crack the MCQ

Q1. When substituting $u = x^2 + 1$ in $\int_0^2 x(x^2 + 1) dx$, the new limits are: (a) 0 to 2 (b) 1 to 5 (c) 0 to 5 (d) 1 to 4

Correct: (b). $x = 0 \rightarrow u = 0^2 + 1 = 1$; $x = 2 \rightarrow u = 2^2 + 1 = 5$. - (a) keeps the old x -limits — the classic error. - (c), (d) miscompute $g(0)$ or $g(2)$.

Q2. $\int_0^1 \frac{2x}{1+x^2} dx$ equals: (a) $\ln 2$ (b) $\frac{1}{2} \ln 2$ (c) $\ln \frac{1}{2}$ (d) 1

Correct: (a). $u = 1 + x^2$, limits $1 \rightarrow 2$: $\int_1^2 \frac{du}{u} = \ln 2$. - (b) inserts a spurious $\frac{1}{2}$ (the numerator is the *full* derivative, so no halving). - (c) flips the log argument. - (d) ignores the integration.

Q3. $\int_0^{\pi/2} \cos x e^{\sin x} dx$ equals: (a) e (b) $e - 1$ (c) 1 (d) $e + 1$

Correct: (b). $u = \sin x$, limits $0 \rightarrow 1$: $\int_0^1 e^u du = [e^u]_0^1 = e - 1$. - (a) forgets to subtract the lower-limit value. - (c), (d) miscompute the bounds.

Q4. Which is a *valid* way to evaluate a definite integral by substitution? (a) substitute, keep old limits, evaluate in u (b) substitute, change limits to u -values, evaluate in u (c) substitute, change limits, then also back-substitute to x (d) never change anything

Correct: (b). Change the limits to $g(a), g(b)$ and finish in u . - (a) mixes x -limits with a u -integral — wrong. - (c) double-handles the conversion (back-substituting *and* changing limits). - (d) is impossible for a real substitution.

Section 7: Common Student Pitfalls & Pro-Tips

- Keeping the old x -limits on a u -integral. The single most common error. If you switch to u , the limits must become $g(a)$ and $g(b)$.
- Mixing both methods. Either change limits or back-substitute — doing both (or neither) corrupts the bounds.
- Forgetting du scaling on the limits-changed integral. The $1/2$ from $x dx = 1/2 du$ still applies — limits and integrand both transform.

- **Non-monotonic substitution** . If g isn't one-to-one on $[a, b]$ (e.g. $u = x^2$ across 0), split the interval first.

Pro-Tip: prefer Method A (change the limits) — it's faster and removes the back-substitution step entirely, eliminating a whole class of sign/limit errors. Write the new u -limits *immediately* next to the integral sign so you can't forget them.

Section 8: Cheat Sheet Box

DEFINITE INTEGRALS BY SUBSTITUTION — QUICK RECALL

- **Rule:** $\int_a^b f(g(x))g'(x) dx = \int_{g(a)}^{g(b)} f(u) du.$
- **Change the limits:** $x = a \rightarrow u = g(a)$, $x = b \rightarrow u = g(b)$ — then finish in u , no back-substitution.
- **Two methods:** (A) change limits, evaluate in u [preferred]; (B) keep limits, back-substitute to x , then evaluate. Never mix.
- **Watch:** non-monotonic g (e.g. $u = x^2$ over a sign change) $\rightarrow$ split the interval.

Memory aids: - “New variable, new limits.” - “Change the ends or change back — not both.”

Subtopic 03 — Properties of Definite Integrals: The Symmetry Toolkit

Exam Relevance: This is the single most leverage-heavy subtopic in the entire integration chapter. CBSE Boards ask a property-based proof (king or even/odd) for 2-3 marks almost every year, plus a modulus/additivity evaluation. In JEE Main and CUET, the difference between a 20-second answer and a hopeless 4-minute grind is *recognising which property applies*. JEE Advanced builds whole multi-step problems on the king and periodic properties.

Section 1: Concept Intuition

Suppose a friend hands you the integral $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$ and asks for the value. You reach for an anti-derivative... and there isn't one in any elementary form. The brute-force route is a dead end. Yet the answer is a clean $\frac{\pi}{4}$, obtainable in two lines. How?

That is the entire spirit of this subtopic. **The properties of definite integrals let you compute the value of an integral without ever finding the anti-derivative**, by exploiting *symmetry* in the function and the interval. You are not integrating harder — you are integrating smarter, by noticing structure.

Why does symmetry help? Because a definite integral is an accumulated area, and area does not care which direction you sweep it or where you place your coordinate origin. If a region looks the same after you flip it left-to-right about the midpoint of the interval, then sweeping from the left and sweeping from the right must give the same total — that single observation is the **king property**. If a function is a perfect mirror image about the y -axis (even) or a perfect upside-down mirror image (odd), the area on $[-a, a]$ either doubles or cancels — that is the **even/odd property**.

Think about it this way... Imagine measuring the total daily footfall at a railway platform that has a perfectly symmetric morning and evening rush, mirrored about noon. To get the whole-day total, you don't need to track every hour — you measure

half the day and double it. That is exactly what symmetry buys you in an integral: measure the easy half, infer the rest. And if the morning crowd of arrivals were exactly cancelled by an equal evening crowd of departures (an “odd” pattern), the *net* change over the symmetric day would be zero — no calculation needed at all.

The most powerful move of all is the king property’s signature trick: write the integral two ways — the original, and the “flipped” version — then **add them**. The flipped version often has the same denominator or a complementary numerator, so the sum collapses to the integral of 1, which is trivial. You solved for the integral algebraically, like solving $2I = (\text{something easy})$ for I .

Limiting conditions & assumptions: - King ($\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$) works for *any* continuous f , but it only *helps* when $f(a + b - x)$ relates nicely to $f(x)$ (e.g. via $\sin \leftrightarrow \cos$ on $[0, \frac{\pi}{2}]$). - Even/odd cancellation requires the limits to be **symmetric**: $-a$ to a . On $[1, 3]$ parity tells you nothing. - The periodic property needs the limits to align with whole multiples of the period; a partial period must be handled separately. - These are evaluation shortcuts, not existence guarantees — the integral must already be well-defined (continuous integrand, finite limits).

Section 2: Key Definitions & Properties

Throughout, f is continuous on the relevant interval. The first three are structural (grounded directly in the NCERT properties list); the rest are the symmetry toolkit.

Tag	Name	Statement
P0	Dummy variable	$\int_a^b f(x) dx = \int_a^b f(t) dt$
P1	Reversal	$\int_a^b f(x) dx = -\int_b^a f(x) dx; \int_a^a f(x) dx = 0$
P2	Additivity	$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$
P3	King (general)	$\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$
P4	King (special)	$\int_0^a f(x) dx = \int_0^a f(a - x) dx$
P5	Zero-to- $2a$ split	$\int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a - x) dx$

Tag	Name	Statement
P5*	2a corollary	$\int_0^{2a} f(x) dx = \begin{cases} 2 \int_0^a f(x) dx & \text{if } f(2a-x) = f(x) \\ 0 & \text{if } f(2a-x) = -f(x) \end{cases}$
P6	Even/Odd	$\int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx & f \text{ even} \\ 0 & f \text{ odd} \end{cases}$
P7	Periodic	If $f(x+T) = f(x)$, then $\int_0^{nT} f(x) dx = n \int_0^T f(x) dx, \quad n \in \mathbb{Z}^+$

Foundation (recap from Sub-02). The First Fundamental Theorem defines the *area function* $A(x) = \int_a^x f(t) dt$ and states $A'(x) = f(x)$; the Second gives $\int_a^b f = F(b) - F(a)$. The properties below are what you use *after* you've decided not to brute-force with the Second FTC.

Parity arithmetic (you will use this constantly): even $\times$ even = even; odd $\times$ odd = even; even $\times$ odd = odd; any constant is even.

Section 3: CBSE Board Derivations / Key Procedures

Proof of the king property (P3). Substitute $t = a + b - x$, so $dt = -dx$; when $x = a$, $t = b$, and when $x = b$, $t = a$:

$$\int_a^b f(x) dx = \int_b^a f(a+b-t) (-dt) = \int_a^b f(a+b-t) dt$$

The $-dt$ reversed the limits back to $a \rightarrow b$ (property P1). Renaming the dummy $t \rightarrow x$ gives $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$. ■ Setting $a = 0$ gives the special case P4.

Proof of the zero-to-2a split (P5). By additivity (P2), break at the midpoint a :

$$\int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_a^{2a} f(x) dx$$

In the second piece substitute $x = 2a - t$ ($dx = -dt$; $x = a \Rightarrow t = a$, $x = 2a \Rightarrow t = 0$):

$$\int_a^{2a} f(x) dx = \int_a^0 f(2a-t) (-dt) = \int_0^a f(2a-t) dt = \int_0^a f(2a-x) dx$$

Adding gives P5. If $f(2a-x) = f(x)$ the two pieces are equal $\Rightarrow 2 \int_0^a f$; if $f(2a-x) = -f(x)$ they cancel $\Rightarrow 0$. ■

Proof of the periodic property (P7). Split the interval into n blocks of length T using P2, then shift each block back to $[0, T]$ by $x = t + kT$ and use periodicity $f(t + kT) = f(t)$:

$$\int_0^{nT} f dx = \sum_{k=0}^{n-1} \int_{kT}^{(k+1)T} f(x) dx = \sum_{k=0}^{n-1} \int_0^T f(t) dt = n \int_0^T f(x) dx \quad \blacksquare$$

Procedure — the decision tree. Before writing a single anti-derivative, run this scan:

1. **Symmetric limits** $[-a, a]$? Split the integrand into even and odd parts; delete every odd term (P6). Often two-thirds of the problem disappears.
2. **A modulus, greatest-integer, or piecewise integrand?** Find the points where the inside changes sign and split there with additivity (P2); on each piece the function is a clean formula.
3. **Limits** $0 \rightarrow a$ **or** $a \rightarrow b$ **with** $\sin / \cos$ **or an** $f/(f+g)$ **structure?** Apply king (P3/P4), add original + flipped, solve $2I =$ (easy).
4. **Integrand contains a lone factor of x times a function of $\sin x$ on $[0, \pi]$?** King turns x into $\pi - x$, peeling the x off (see Example 3).
5. **Periodic integrand over a whole-number of periods?** Collapse with P7.
6. **None of the above?** Fall back to the Second FTC.

The logic uniting all of these: *change the variable so the interval maps to itself*, then exploit how the integrand transformed.

Section 4: Worked Examples

Example 1 — [CBSE Board Level: additivity with a modulus] Evaluate $\int_0^4 |x - 2| dx$.

Solution. The integrand changes formula where $x - 2$ changes sign, i.e. at $x = 2$. Split with P2:

$$\int_0^4 |x - 2| dx = \int_0^2 (2 - x) dx + \int_2^4 (x - 2) dx$$

because $|x - 2| = 2 - x$ for $x < 2$ and $x - 2$ for $x > 2$. Evaluate each:

$$\int_0^2 (2 - x) dx = \left[2x - \frac{x^2}{2} \right]_0^2 = (4 - 2) = 2$$

$$\int_2^4 (x - 2) dx = \left[\frac{x^2}{2} - 2x \right]_2^4 = (8 - 8) - (2 - 4) = 0 - (-2) = 2$$

Answer: $2 + 2 = 4$. (Geometrically, two right triangles of area 2 each — a clean sanity check.)

Example 2 — [JEE Main Speed Trap: parity] Evaluate $\int_{-\pi/2}^{\pi/2} (x^2 \sin x + x \cos x + 4) dx$ mentally.

Solution. Symmetric limits $\Rightarrow$ parity-check each term: $-x^2 \sin x$: even $\times$ odd = **odd** $\Rightarrow 0$. $-x \cos x$: odd $\times$ even = **odd** $\Rightarrow 0$. -4 : even (constant) $\Rightarrow$ survives.

Only the constant remains (P6):

$$\int_{-\pi/2}^{\pi/2} 4 dx = 4[x]_{-\pi/2}^{\pi/2} = 4 \cdot \pi = 4\pi$$

Answer: 4π . *Speed lesson:* never integrate $x^2 \sin x$ or $x \cos x$ by parts here — symmetry deletes them before you start.

Example 3 — [JEE Main Level: king property peeling off x] Evaluate $I = \int_0^\pi x \sin^3 x dx$.

Solution. Apply king (P4 with $a = \pi$): replace $x \rightarrow \pi - x$. Since $\sin(\pi - x) = \sin x$, the $\sin^3$ is untouched, but the lone x becomes $\pi - x$:

$$I = \int_0^\pi (\pi - x) \sin^3 x dx = \pi \int_0^\pi \sin^3 x dx - \int_0^\pi x \sin^3 x dx = \pi \int_0^\pi \sin^3 x dx - I$$

So $2I = \pi \int_0^\pi \sin^3 x dx$. Now $\sin^3 x$ is symmetric about $x = \frac{\pi}{2}$, so $\int_0^\pi \sin^3 x dx = 2 \int_0^{\pi/2} \sin^3 x dx = 2 \cdot \frac{2}{3} = \frac{4}{3}$. Therefore

$$2I = \pi \cdot \frac{4}{3} \Rightarrow I = \frac{2\pi}{3}$$

Answer: $\frac{2\pi}{3}$. *Why it works:* king's job here is to *separate* the awkward x from the $\sin^3 x$ — the x folds into a constant π , and the integral solves itself algebraically.

Example 4 — [JEE Advanced Level: periodic property] Evaluate $\int_0^{10\pi} |\sin x| dx$.

Solution. The key insight is the period of the integrand. While $\sin x$ has period 2π , the *modulus* $|\sin x|$ folds the negative hump up, giving period π . Confirm: $|\sin(x + \pi)| = |-\sin x| = |\sin x|$. So over $[0, 10\pi]$ there are exactly 10 full periods. By P7,

$$\int_0^{10\pi} |\sin x| dx = 10 \int_0^\pi |\sin x| dx$$

On $[0, \pi]$, $\sin x \geq 0$, so $|\sin x| = \sin x$ and $\int_0^\pi \sin x \, dx = [-\cos x]_0^\pi = 1 + 1 = 2$. Therefore

$$\int_0^{10\pi} |\sin x| \, dx = 10 \cdot 2 = 20$$

Answer: 20. *Trap avoided:* using period 2π (giving 5 blocks of $\int_0^{2\pi} |\sin x| = 4$) also gives 20 — but students who write period 2π and then forget the modulus turns the second hump positive get 0 for that hump and a wrong answer. Identify the *modulus* period π and you cannot slip.

Section 5: Practice Exercises

1. **[CBSE]** Evaluate $\int_{-2}^2 |x| \, dx$.
2. **[CUET]** Evaluate $\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} \, dx$. (*Hint: king.*)
3. **[JEE Main]** Evaluate $\int_0^6 |x - 3| \, dx$.
4. **[JEE Main]** Evaluate $\int_{-\pi}^{\pi} x^3 \cos^5 x \, dx$. (*Hint: parity.*)
5. **[JEE Advanced]** Evaluate $\int_0^{8\pi} |\cos x| \, dx$.

Answers: 1. 4 2. $\frac{\pi}{4}$ 3. 9 4. 0 5. 16

Section 6: Crack the MCQ

Q1. $\int_{-4}^4 \frac{x^3}{x^4 + 1} \, dx$ equals: (a) 0 (b) $\ln 257$ (c) $\frac{1}{4} \ln 257$ (d) 8

Correct: (a). The integrand is odd: $f(-x) = \frac{-x^3}{x^4 + 1} = -f(x)$, and the limits are symmetric, so P6 gives 0 instantly. - (b), (c) come from grinding out the (correct-looking) log anti-derivative while ignoring symmetry. - (d) guesses from the interval width 8 — irrelevant once parity kills the integral.

Q2. $\int_0^{\pi/2} \frac{\cos x}{\sin x + \cos x} \, dx$ equals: (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) 1 (d) $\frac{\pi}{3}$

Correct: (b). King ($x \rightarrow \frac{\pi}{2} - x$) swaps $\cos \leftrightarrow \sin$, giving the companion integral with $\sin x$ on top; adding the two yields $2I = \int_0^{\pi/2} 1 \, dx = \frac{\pi}{2}$, so $I = \frac{\pi}{4}$. - (a) forgets to

halve after adding ($2I$ reported as I). - (c) confuses the answer with $\int_0^{\pi/2} \cos x \, dx = 1$, ignoring the denominator. - (d) a blind guess; king fixes the value to $\frac{\pi}{4}$ regardless of the trig function used.

Q3. $\int_0^2 |x - 1| \, dx$ equals: (a) 0 (b) 1 (c) 2 (d) $\frac{1}{2}$

Correct: (b). Split at $x = 1$ (P2): $\int_0^1 (1 - x) \, dx + \int_1^2 (x - 1) \, dx = \frac{1}{2} + \frac{1}{2} = 1$. - (a) treats $|x - 1|$ as an odd function about $x = 1$ and wrongly cancels — but the limits aren't symmetric about 0, and $|x - 1| \geq 0$ everywhere, so the area is positive. - (c) integrates $x - 1$ across the whole interval without the modulus, getting the *signed* value 0... then “corrects” to 2 by mistake. - (d) computes only one of the two triangles.

Q4. $\int_0^{2\pi} |\sin x| \, dx$ equals: (a) 0 (b) 2 (c) 4 (d) 2π

Correct: (c). $|\sin x|$ has period π , and $\int_0^\pi |\sin x| \, dx = 2$, so over two periods $\int_0^{2\pi} = 2 \times 2 = 4$. - (a) is $\int_0^{2\pi} \sin x \, dx$ — the student dropped the modulus, letting the second hump cancel the first. - (b) computes only one period. - (d) confuses area with interval length.

Section 7: Common Student Pitfalls & Pro-Tips

- Applying parity on non-symmetric limits.
Even/odd cancellation needs limits $-a$ to a . On $[0, 2]$ or $[1, 3]$, parity tells you nothing — a common reflex error.
- **Misapplying king**. King replaces x by (*sum of limits*) $-x$, i.e. $a + b - x$, not simply $a - x$ unless the lower limit is 0. Using the wrong replacement silently wrecks the cancellation.
- **Forgetting to split a modulus / greatest-integer at its breakpoints**. $\int |g(x)|$ must be broken where g changes sign; integrating $|g|$ as one formula is wrong.
- **Using the wrong period for $|\sin x|$ or $|\cos x|$** . The modulus halves the period to π . Pairing period 2π with a dropped modulus is the classic double-error that returns 0.
- **Reporting $2I$ as I in the king “add” trick**. After $2I = (\text{easy})$, divide by two. Skipping this doubles every king answer.

Pro-Tip (the recognition reflex): train yourself to read the *limits first, integrand second*. Symmetric $\Rightarrow$ parity. $0 \rightarrow a$ with trig $\Rightarrow$ king. Modulus/greatest-integer $\Rightarrow$ split. Whole periods $\Rightarrow$ collapse. Ninety percent of “hard” property problems are actually pattern-recognition; the algebra is the easy part once you’ve named the pattern.

Section 8: Cheat Sheet Box

PROPERTIES OF DEFINITE INTEGRALS — QUICK RECALL

- **King:** $\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$. Add original + flipped, solve $2I =$ (easy).
- **Even/Odd on $[-a, a]$:** even $\rightarrow 2 \int_0^a$; odd $\rightarrow 0$. (*Limits must be symmetric.*)
- **Zero-to-2a:** $\int_0^{2a} f = \int_0^a f + \int_0^a f(2a - x)$; equals $2 \int_0^a f$ or 0 by symmetry.
- **Periodic ($f(x + T) = f(x)$):** $\int_0^{nT} f = n \int_0^T f$. For $|\sin x|, |\cos x|$ use $T = \pi$.
- **Modulus/piecewise:** split at sign-change points via additivity, then integrate each clean piece.
- **Always:** read limits before the integrand; the limits tell you which property to reach for.

Memory aids: - “King swaps the ends” — replace x by (sum of limits) $- x$, then add. - “Odd dies on a mirror” — odd function on $[-a, a] \rightarrow 0$. - “Modulus halves the period” — $|\sin|, |\cos|$ repeat every π . - “Solve for I , then halve” — finish the king trick by dividing $2I$ by two.

Chapter 7: Integrals — Round 2 Addendum

The chapter taught you the whole standard toolkit: indefinite techniques (substitution, trig identities, quadratic templates, partial fractions, parts), the Fundamental Theorem, the limit-of-a-sum definition, and the symmetry properties of definite integrals. Six examinable patterns still stand outside it. Addendum A differentiates an integral whose *limits* move — the general Leibniz rule, which JEE tests precisely because the chapter's First Fundamental Theorem stops at a fixed lower limit. Addendum B runs the limit-of-a-sum machinery *backwards*, reading a naked limit of a sum as an integral. Addendum C upgrades the king property to its three power tools: the general quotient theorem, the x -peeling theorem on $[0, \pi]$, and the log-sine master result. Addendum D supplies the half-angle substitution, the only systematic method for rational functions of $\sin x$ and $\cos x$. Addendum E builds reduction formulas, the Board-favourite proofs the chapter quotes but never derives. Addendum F handles integrands with breakpoints of a new kind — greatest integer, fractional part, min/max — and the signed-versus-total-area trap.

Nothing below is new theory beyond the definitions already given. Every derivation uses the Second FTC, the king property, or parts, applied one level further than the chapter took them.

Addendum A — Differentiating an Integral Whose Limits Move

The chapter proves $\frac{d}{dx} \int_a^x f(t) dt = f(x)$: upper limit moving, lower limit fixed. Exams move *both*. The fix is one line of chain rule, and it unlocks problems where the anti-derivative does not even exist in elementary form.

Method

The general Leibniz rule. Let f be continuous, and let $u(x)$ and $v(x)$ be differentiable. Define

$$F(x) = \int_{u(x)}^{v(x)} f(t) dt.$$

By the Second FTC (Subtopic 02) there is an anti-derivative Φ of f , and $F(x) = \Phi(v(x)) - \Phi(u(x))$. Differentiate with the chain rule:

$$F'(x) = \Phi'(v(x)) v'(x) - \Phi'(u(x)) u'(x) = f(v(x)) v'(x) - f(u(x)) u'(x).$$

Read it as a slogan: **top limit in, bottom limit out** — evaluate f at each moving limit, multiply by that limit's derivative, upper term minus lower term. Fixed limits contribute nothing because their derivative is 0.

Two consequences you will use constantly:

- **Recovering the chapter's rule:** with $u = a$ fixed and $v = x$, the formula gives $f(x) \cdot 1 - f(a) \cdot 0 = f(x)$.
- **Even-interval shortcut:** for symmetric limits $[-x, x]$,

$$F'(x) = f(x) + f(-x),$$

so if f is even, $F'(x) = 2f(x)$, and if f is odd, $F'(x) = 0$ — the accumulated integral is constant, exactly as parity predicts.

Procedure. 1. Identify $u(x)$ and $v(x)$. 2. Apply the rule mechanically. 3. Simplify the resulting expression. 4. Never hunt for an anti-derivative first — the rule exists precisely so you do not need one.

Example A.1 — [CBSE Board Level]

Let $F(x) = \int_0^{x^2} (3t^2 + 1) dt$. **Find** $F'(x)$ **two independent ways.**

Way 1 — integrate, then differentiate. The anti-derivative is $t^3 + t$, so

$$F(x) = [t^3 + t]_0^{x^2} = x^6 + x^2, \quad F'(x) = 6x^5 + 2x.$$

Way 2 — Leibniz. Here $u = 0$ (fixed) and $v = x^2$, so $v' = 2x$:

$$F'(x) = f(x^2) \cdot 2x = (3x^4 + 1)(2x) = 6x^5 + 2x.$$

Both routes agree. On an exam, Way 2 is faster whenever the anti-derivative is unpleasant.

Example A.2 — [JEE Main Level]

Find the minimum value of $F(x) = \int_0^x (t-1)(t-2) dt$ **over all real** x .

Differentiate under the integral sign — the integrand is a polynomial, no anti-derivative required:

$$F'(x) = (x-1)(x-2).$$

Sign analysis: $F' > 0$ on $(-\infty, 1)$, negative on $(1, 2)$, positive on $(2, \infty)$. So $x = 1$ is a local maximum and $x = 2$ is a local minimum. As $x \rightarrow -\infty$, $F(x) \rightarrow +\infty$ (check: for large negative x the integrand is positive over most of $[0, x]$... rather than argue this, just compare candidates).

Candidates for the global minimum: the local minimum $F(2)$, and the boundary behaviour. Compute

$$F(2) = \int_0^2 (t^2 - 3t + 2) dt = \left[\frac{t^3}{3} - \frac{3t^2}{2} + 2t \right]_0^2 = \frac{8}{3} - 6 + 4 = \frac{2}{3}.$$

Since F' changes sign from $-$ to $+$ only at $x = 2$, and $F \rightarrow +\infty$ at both ends (for $x < 0$, $F(x) = -\int_x^0 (t-1)(t-2) dt$ with a positive integrand on $[x, 0]$ once $x < 0$... concretely $F(-1) = \int_0^{-1} (t^2 - 3t + 2) dt = -\left[-\frac{1}{3} - \frac{3}{2} - 2\right] = \frac{23}{6} > \frac{2}{3}$), the global minimum is

$$F(2) = \frac{2}{3}.$$

Check. $F''(x) = 2x - 3$, so $F''(2) = 1 > 0$: confirmed minimum by the second-derivative test.

Example A.3 — [JEE Advanced Level]

(i) Show that $F(x) = \int_x^{x^2} \frac{dt}{\ln t}$ is constant on its domain.

(ii) If g is continuous and $f(x) = \int_0^x (x-t)g(t) dt$, prove $f''(x) = g(x)$.

(i). The point of this problem is that $\frac{1}{\ln t}$ has **no elementary anti-derivative**, so integrating first is impossible. Leibniz with $u = x$, $v = x^2$:

$$F'(x) = \frac{1}{\ln(x^2)} \cdot 2x - \frac{1}{\ln x} \cdot 1 = \frac{2x}{2\ln x} - \frac{1}{\ln x} = 0,$$

using $\ln(x^2) = 2\ln x$ for $x > 0$, $x \neq 1$. A function with zero derivative throughout an interval is constant there — no integration ever performed.

(ii). The variable x appears both in a limit and inside the integrand, so split it first. With $G(x) = \int_0^x g(t) dt$ and $H(x) = \int_0^x t g(t) dt$,

$$f(x) = xG(x) - H(x).$$

Differentiate, applying the chapter's rule to G and H :

$$f'(x) = G(x) + xg(x) - xg(x) = G(x).$$

The $xg(x)$ terms cancel exactly — this cancellation is the whole content of the problem. Differentiate again:

$$f''(x) = G'(x) = g(x).$$

Check for (ii). Take $g(t) = 1$. Then $f(x) = \int_0^x (x-t) dt = x^2 - \frac{x^2}{2} = \frac{x^2}{2}$, and indeed $f''(x) = 1 = g(x)$.

Why it is Advanced-level. Both parts defeat the reflex “find the anti-derivative first.” Part (i) is unsolvable that way in principle; part (ii) needs the structural move of pulling the parameter out before differentiating.

Practice — Addendum A

1. Find $\frac{d}{dx} \int_0^{\sqrt{x}} \cos(t^2) dt$ for $x > 0$.
2. Find $F'(x)$ if $F(x) = \int_x^{2x} e^{-t^2} dt$.
3. Evaluate $\frac{d}{dx} \int_{\tan x}^{\cot x} \frac{dt}{1+t^2}$.
4. Find $\frac{d}{dx} \int_{x^2}^{x^3} \frac{dt}{\ln t}$ for $x > 0, x \neq 1$.
5. Prove: if f is even and continuous, then $A(x) = \int_0^x f(t) dt$ is odd.

Answers: 1. $\cos x \cdot \frac{1}{2\sqrt{x}} = \frac{\cos x}{2\sqrt{x}}$. 2. $2e^{-4x^2} - e^{-x^2}$. 3. $-1 - 1 = -2$, since $\frac{\csc^2 x}{1 + \cot^2 x} = 1$ and $\frac{\sec^2 x}{1 + \tan^2 x} = 1$. 4. $\frac{3x^2}{\ln x^3} - \frac{2x}{\ln x^2} = \frac{x^2}{\ln x} - \frac{x}{\ln x} = \frac{x(x-1)}{\ln x}$. 5. Substitute $t = -s$ in $A(-x) = \int_0^{-x} f(t) dt$ to get $A(-x) = \int_0^x f(s) ds$ with the sign flipped twice: $A(-x) = -\int_0^x f(s) ds = -A(x)$.

Addendum B — Reading a Limit Backwards: Riemann Sums in Reverse

Subtopic 02 goes forward: integral first, then the limit of a sum to *define* it. JEE Main reverses the arrow constantly — it hands you

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{n}{n^2 + r^2}$$

and expects $\pi/4$ in three lines. The skill is recognising any such limit as an integral in disguise.

Method

The definition from Subtopic 02, specialised to $a = 0, b = 1, h = 1/n$:

$$\int_0^1 f(x) dx = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n f\left(\frac{r}{n}\right).$$

Procedure. 1. Factor $\frac{1}{n}$ out of the summand — this is your h . Whatever remains must become a function of $\frac{r}{n}$ alone. 2. To achieve that, divide numerator and denominator by the highest power of n present. 3. Read the integration limits off the range of $\frac{r}{n}$: $r = 1, \dots, n$ gives $(0, 1]$; $r = n + 1, \dots, 2n$ gives $(1, 2]$. 4. Write the integral and evaluate by the FTC. 5. A constant shift inside the argument (such as $f(\frac{r+1}{n})$) does not change the limit — the shifted grid still fills the same interval.

Example B.1 — [JEE Main Level]

Evaluate $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{n}{n^2 + r^2}.$

Divide numerator and denominator by n^2 :

$$\frac{n}{n^2 + r^2} = \frac{1}{n} \cdot \frac{1}{1 + \left(\frac{r}{n}\right)^2}.$$

So the limit is $\int_0^1 \frac{dx}{1 + x^2}$, and by row 12 of the standard table:

$$\int_0^1 \frac{dx}{1 + x^2} = [\tan^{-1} x]_0^1 = \frac{\pi}{4}.$$

Example B.2 — [JEE Main Level]

Evaluate $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=n+1}^{2n} \frac{r}{\sqrt{n^2 + r^2}}.$

Divide inside the root by n^2 :

$$\frac{1}{n} \sum_{r=n+1}^{2n} \frac{r/n}{\sqrt{1 + (r/n)^2}}.$$

Here $\frac{r}{n}$ runs over $(1, 2]$, so the limit is

$$\int_1^2 \frac{x}{\sqrt{1 + x^2}} dx.$$

With $u = 1 + x^2$, $du = 2x dx$:

$$\frac{1}{2} \int_2^5 \frac{du}{\sqrt{u}} = [\sqrt{u}]_2^5 = \sqrt{5} - \sqrt{2}.$$

Example B.3 — [JEE Advanced Level]

Evaluate $\lim_{n \rightarrow \infty} \frac{(n!)^{1/n}}{n}$.

Take logarithms, so the product becomes a sum:

$$\ln L = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \ln \frac{r}{n} = \int_0^1 \ln x \, dx.$$

This integral is improper at 0, so cut it at ε and integrate by parts ($u = \ln x$, $dv = dx$):

$$\int_{\varepsilon}^1 \ln x \, dx = [x \ln x - x]_{\varepsilon}^1 = -1 - \varepsilon \ln \varepsilon + \varepsilon.$$

As $\varepsilon \rightarrow 0^+$, $\varepsilon \ln \varepsilon \rightarrow 0$ (write it as $\frac{\ln \varepsilon}{1/\varepsilon}$, both of whose numerator and denominator tend to infinite magnitude with the ratio falling to 0). Hence $\int_0^1 \ln x \, dx = -1$, and

$$L = e^{-1}.$$

Why it is Advanced-level. Two moves stacked: logarithms convert the product to a Riemann sum, and the resulting integral is improper, demanding the limiting argument above rather than blind FTC.

Practice — Addendum B

1. Evaluate $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \sqrt{1 + \frac{r}{n}}$.
2. Evaluate $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{\sqrt{n^2 + r^2}}$.
3. Evaluate $\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{r=1}^n r e^{r/n}$.
4. Evaluate $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{n}{n^2 + nr + r^2}$.
5. Evaluate $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(1 + \frac{r}{n}\right)^2$.

Answers: 1. $\int_0^1 \sqrt{1+x} \, dx = \frac{2}{3} [(1+x)^{3/2}]_0^1 = \frac{2}{3} (2\sqrt{2} - 1)$. 2. $\int_0^1 \frac{dx}{\sqrt{1+x^2}} = \ln(1+\sqrt{2})$. 3. $\frac{1}{n} \sum_{r=1}^n \frac{r}{n} e^{r/n} \rightarrow \int_0^1 x e^x \, dx = [x e^x - e^x]_0^1 = 1$. 4. $\int_0^1 \frac{dx}{1+x+x^2} = \left[\frac{2}{\sqrt{3}} \tan^{-1} \frac{2x+1}{\sqrt{3}} \right]_0^1 = \frac{2}{\sqrt{3}} \cdot \frac{\pi}{6} = \frac{\pi}{3\sqrt{3}}$. 5. $\int_0^1 (1+x)^2 \, dx = \frac{7}{3}$.

Addendum C — King Property Power Plays

Subtopic 03 teaches the king property and demonstrates it on specific integrands. Three general structures turn it from a trick into a machine, and each is a standing JEE pattern.

Method

Tool 1 — the quotient theorem. Let f be continuous on $[0, a]$ with $f(x) + f(a - x) \neq 0$ there. Then

$$\int_0^a \frac{f(x)}{f(x) + f(a - x)} dx = \frac{a}{2}.$$

Derivation. Call the integral I . By the special king property P4, replacing $x \rightarrow a - x$ leaves the integral unchanged:

$$I = \int_0^a \frac{f(a - x)}{f(a - x) + f(x)} dx.$$

Add the two expressions — the denominators are identical:

$$2I = \int_0^a \frac{f(x) + f(a - x)}{f(x) + f(a - x)} dx = \int_0^a dx = a. \quad \blacksquare$$

Tool 2 — the x -peeling theorem. For any integrable f on $[0, \pi]$,

$$\int_0^\pi x f(\sin x) dx = \frac{\pi}{2} \int_0^\pi f(\sin x) dx.$$

Derivation. Apply P4 with $a = \pi$, replacing x by $\pi - x$; since $\sin(\pi - x) = \sin x$,

$$I = \int_0^\pi (\pi - x) f(\sin x) dx = \pi \int_0^\pi f(\sin x) dx - I.$$

Solve for I and halve. $\blacksquare$ (Example 3 of Subtopic 03 was the special case $f(\sin x) = \sin^3 x$; the tool now covers every case at once.)

Tool 3 — the reciprocal substitution. For $p > 0$, substituting $x = 1/t$ in $I = \int_p^{1/p} f(x) dx$ swaps the limits and produces

$$I = \int_p^{1/p} \frac{f(1/t)}{t^2} dt.$$

If $\frac{f(1/x)}{x^2}$ is a simple multiple of $f(x)$, the integral solves algebraically. *Derivation:* $dx = -dt/t^2$; as x goes $p \rightarrow 1/p$, t goes $1/p \rightarrow p$; the minus sign restores the order.

Tool 4 — the log-sine master result, derived in Example C.3.

Example C.1 — [CBSE Board / JEE Main Level]

Evaluate $I = \int_0^{\pi/2} \frac{dx}{1 + \sqrt{\tan x}}.$

Here $a = \pi/2$ and $f(x) = 1$, with the second summand $g(x) = \sqrt{\tan x}$ satisfying $g(\pi/2 - x) = 1/g(x)$. Apply king directly ($x \rightarrow \frac{\pi}{2} - x$):

$$I = \int_0^{\pi/2} \frac{dx}{1 + \sqrt{\cot x}} = \int_0^{\pi/2} \frac{\sqrt{\tan x}}{1 + \sqrt{\tan x}} dx.$$

Add the original and the flipped form:

$$2I = \int_0^{\pi/2} \frac{1 + \sqrt{\tan x}}{1 + \sqrt{\tan x}} dx = \frac{\pi}{2}, \quad I = \frac{\pi}{4}.$$

Check. Tool 1 with $f(x) = 1$ and $a = \pi/2$ predicts $a/2 = \pi/4$ instantly — the general theorem and the hand computation agree.

Example C.2 — [JEE Main Level]

Evaluate $I = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx.$

The integrand is x times a function of $\sin x$ alone? Not quite — it is $x \sin x$ over a function of $\cos x$. Rewrite the denominator: $\cos^2 x = 1 - \sin^2 x$, so the integrand is

$$\frac{x \sin x}{2 - \sin^2 x},$$

which is exactly $x f(\sin x)$ with $f(u) = \frac{u}{2 - u^2}$. Tool 2 applies:

$$I = \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{2 - \sin^2 x} dx = \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx.$$

Substitute $u = \cos x$, $du = -\sin x dx$; the limits run $1 \rightarrow -1$:

$$\int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} dx = \int_1^{-1} \frac{du}{1 + u^2} = [\tan^{-1} u]_{-1}^1 = \frac{\pi}{2}.$$

Therefore $I = \frac{\pi}{2} \cdot \frac{\pi}{2} = \frac{\pi^2}{4}.$

Example C.3 — [JEE Advanced Level]

Prove that $I = \int_0^{\pi/2} \ln(\sin x) dx = -\frac{\pi}{2} \ln 2$, **and deduce** $\int_0^{\pi/2} \ln(\tan x) dx = 0.$

(The integral converges even though $\ln \sin x \rightarrow -\infty$ at $x = 0$ — a single logarithmic singularity is integrable.)

King with $a = \pi/2$: since $\sin(\frac{\pi}{2} - x) = \cos x$,

$$I = \int_0^{\pi/2} \ln(\cos x) dx.$$

Add the two expressions:

$$2I = \int_0^{\pi/2} \ln(\sin x \cos x) dx = \int_0^{\pi/2} \ln\left(\frac{\sin 2x}{2}\right) dx = \underbrace{\int_0^{\pi/2} \ln(\sin 2x) dx}_J - \frac{\pi}{2} \ln 2.$$

Evaluate J by substituting $u = 2x$, $dx = du/2$, limits $0 \rightarrow \pi$:

$$J = \frac{1}{2} \int_0^{\pi} \ln(\sin u) du.$$

By symmetry $\sin(\pi - u) = \sin u$, the interval $[0, \pi]$ carries two identical copies of $[0, \pi/2]$:

$$J = \frac{1}{2} \cdot 2I = I.$$

So $2I = I - \frac{\pi}{2} \ln 2$, giving

$$I = -\frac{\pi}{2} \ln 2. \quad \blacksquare$$

Corollary. $\int_0^{\pi/2} \ln(\tan x) dx = \int_0^{\pi/2} \ln \sin x dx - \int_0^{\pi/2} \ln \cos x dx = I - I = 0$ — consistent with $\tan(\frac{\pi}{2} - x) = \cot x = 1/\tan x$.

Why it is Advanced-level. Nothing here computes an anti-derivative of $\ln \sin x$ (there is no elementary one). Every mark comes from orchestrating king twice plus a rescaling of the interval — pure structural control.

Practice — Addendum C

1. Evaluate $\int_0^{\pi} \frac{x}{1 + \sin x} dx$.
2. Evaluate $\int_{1/4}^4 \frac{\ln x}{1 + x^2} dx$.
3. Evaluate $\int_0^{\pi/2} \ln(\sec x) dx$.
4. Evaluate $\int_0^{\pi} \frac{x \sin x}{1 + \sin^2 x} dx$.

5. Evaluate $\int_0^{\pi/2} \frac{dx}{1 + \cot x}$.

Answers: 1. Tool 2 gives $\frac{\pi}{2} \int_0^{\pi} \frac{dx}{1 + \sin x}$; multiplying by $\frac{1 - \sin x}{1 - \sin x}$ reduces this to $\int_0^{\pi} \sec^2 - \sec \tan$ form, value 2. Answer: π . 2. By Tool 3 the substitution $x = 1/t$ sends the integral to $\int_{1/4}^4 \frac{-\ln t}{1 + t^2} dt = -I$, so $I = 0$. 3. $-\ln \sin x$ integrated: $+\frac{\pi}{2} \ln 2$. 4. Tool 2 plus $u = \cos x$: $\frac{\pi}{2} \int_{-1}^1 \frac{du}{2 - u^2} = \frac{\pi}{2} \cdot \sqrt{2} \ln(3 + 2\sqrt{2})^{1/2} \cdot 2 = \pi\sqrt{2} \ln(1 + \sqrt{2})$ — using $\int \frac{du}{2 - u^2} = \frac{1}{2\sqrt{2}} \ln \left| \frac{\sqrt{2} + u}{\sqrt{2} - u} \right|$ and $\ln(3 + 2\sqrt{2}) = 2 \ln(1 + \sqrt{2})$. 5. $\frac{1}{1 + \cot x} = \frac{\sin x}{\sin x + \cos x}$, so Tool 1 with $f = \sin$ gives $\frac{\pi}{4}$.

Addendum D — Rational Functions of $\sin x$ and $\cos x$: the Half-Angle Substitution

The chapter's trig toolkit handles powers and products. It says nothing systematic about quotients like $\frac{1}{3 + 5 \cos x}$ — and no amount of identity-juggling will crack those. One universal substitution converts *every* rational function of $\sin x$ and $\cos x$ into an ordinary rational function, which partial fractions (Subtopic 07) then finish.

Method

Set

$$t = \tan \frac{x}{2}.$$

Deriving the conversion identities. From the double-angle formulas, dividing numerator and denominator by $\cos^2(x/2)$:

$$\sin x = \frac{2 \sin(x/2) \cos(x/2)}{1} = \frac{2 \tan(x/2)}{1 + \tan^2(x/2)} = \frac{2t}{1 + t^2},$$

$$\cos x = \cos^2 \frac{x}{2} - \sin^2 \frac{x}{2} = \frac{1 - t^2}{1 + t^2}.$$

For the differential, differentiate $t = \tan(x/2)$:

$$dt = \frac{1}{2} \sec^2 \frac{x}{2} dx = \frac{1 + t^2}{2} dx \implies dx = \frac{2 dt}{1 + t^2}.$$

Every rational function of $\sin x$ and $\cos x$ becomes a rational function of t over the common denominator $(1 + t^2)^k$, and the $1 + t^2$ factors cancel against dx — leaving plain partial fractions.

Validity. The substitution fails at $x = (2k+1)\pi$ (where $\tan(x/2)$ blows up) and is non-monotone across such points, so for definite integrals confirm no odd multiple of π lies inside the interval — otherwise split first. Limits convert by $t = \tan(x/2)$ as usual (Subtopic 09).

Example D.1 — [CBSE Board Level]

Evaluate $\int \frac{dx}{3+5\cos x}$.

Substitute. The denominator becomes

$$3+5 \cdot \frac{1-t^2}{1+t^2} = \frac{3(1+t^2)+5(1-t^2)}{1+t^2} = \frac{8-2t^2}{1+t^2}.$$

So

$$\int \frac{2 dt}{1+t^2} \cdot \frac{1+t^2}{8-2t^2} = \int \frac{dt}{4-t^2}.$$

Match the template $\int \frac{du}{a^2-u^2} = \frac{1}{2a} \ln \left| \frac{a+u}{a-u} \right|$ with $a=2$:

$$\int \frac{dx}{3+5\cos x} = \frac{1}{4} \ln \left| \frac{2+t}{2-t} \right| + C, \quad t = \tan \frac{x}{2}.$$

Check. At $x=0$: $t=0$, the anti-derivative equals $\frac{1}{4} \ln 1 = 0$; differentiating the answer at $t=0$ gives $\frac{1}{4} \left(\frac{1}{2} + \frac{1}{2} \right) = \frac{1}{4}$, and the original integrand at $x=0$ is $\frac{1}{8} \cdot (1+t^2)$ -chain... directly, $\frac{1}{3+5\cos 0} = \frac{1}{8}$; converting $\frac{dt}{dx} = \frac{1}{2}$ at $x=0$, the t -integrand $\frac{1}{4-t^2} \cdot \frac{dt}{dx}$ at $t=0$ reads $\frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$. Consistent.

Example D.2 — [JEE Main Level]

Evaluate $\int_0^{\pi/2} \frac{dx}{2+\cos x}$.

Limits: $x=0 \Rightarrow t=0$; $x=\frac{\pi}{2} \Rightarrow t=\tan \frac{\pi}{4}=1$. Denominator:

$$2 + \frac{1-t^2}{1+t^2} = \frac{t^2+3}{1+t^2}.$$

So

$$\int_0^1 \frac{2 dt}{1+t^2} \cdot \frac{1+t^2}{t^2+3} = \int_0^1 \frac{2 dt}{t^2+3} = \left[\frac{2}{\sqrt{3}} \tan^{-1} \frac{t}{\sqrt{3}} \right]_0^1 = \frac{2}{\sqrt{3}} \cdot \frac{\pi}{6} = \frac{\pi}{3\sqrt{3}}.$$

Example D.3 — [JEE Advanced Level]

Evaluate $\int \frac{dx}{\sin x + \sin 2x}$.

Factor the denominator first: $\sin x(1 + 2 \cos x)$. In t :

$$\sin x(1 + 2 \cos x) = \frac{2t}{1+t^2} \left(1 + \frac{2(1-t^2)}{1+t^2} \right) = \frac{2t(3-t^2)}{(1+t^2)^2}.$$

Therefore

$$\int \frac{2 dt}{1+t^2} \cdot \frac{(1+t^2)^2}{2t(3-t^2)} = \int \frac{1+t^2}{t(3-t^2)} dt.$$

Partial fractions (compare coefficients in $1+t^2 = A(3-t^2) + (Bt+C)t$): matching constants gives $3A = 1$, so $A = \frac{1}{3}$; matching t gives $C = 0$; matching t^2 gives $B - A = 1$, so $B = \frac{4}{3}$:

$$\frac{1+t^2}{t(3-t^2)} = \frac{1}{3t} + \frac{4t}{3(3-t^2)}.$$

Integrate, using $\int \frac{t dt}{3-t^2} = -\frac{1}{2} \ln |3-t^2|$:

$$\int \frac{dx}{\sin x + \sin 2x} = \frac{1}{3} \ln |t| - \frac{2}{3} \ln |3-t^2| + C, \quad t = \tan \frac{x}{2}.$$

Check. Differentiate in t : $\frac{1}{3t} + \frac{4t}{3(3-t^2)}$, and over the common denominator $3t(3-t^2)$ the numerator is $(3-t^2) + 4t^2 = 3(1+t^2)$ — exactly the reduced integrand. ✓

Why it is Advanced-level. The denominator needs factoring before the substitution, the reduced rational function is improper-shaped enough to demand coefficient comparison rather than cover-up, and the repeated-factor-free split hides a sign-sensitive log.

Practice — Addendum D

1. Derive the three conversion identities from $t = \tan(x/2)$ without looking at the text.

2. Evaluate $\int \frac{dx}{5 + 3 \cos x}$.

3. Evaluate $\int_0^{\pi/2} \frac{dx}{4 + 5 \cos x}$.

4. Evaluate $\int \frac{dx}{1 - \sin x + \cos x}$.

5. Evaluate $\int_0^{\pi/2} \frac{dx}{1 + \sin x + \cos x}$.

Answers: 2. Denominator $\frac{2t^2+8}{1+t^2}$; integral $= \int \frac{dt}{t^2+4} = \frac{1}{2} \tan^{-1}\left(\frac{t}{2}\right) + C$. 3. Denominator $\frac{9-t^2}{1+t^2}$; integral $= \left[\frac{1}{3} \ln \left| \frac{3+t}{3-t} \right| \right]_0^1 = \frac{1}{3} \ln 2$. 4. Denominator $\frac{2(1-t)}{1+t^2}$; integral $= \int \frac{dt}{1-t} = -\ln \left| 1 - \tan \frac{x}{2} \right| + C$. 5. Denominator $\frac{2(1+t)}{1+t^2}$; integral $= \int_0^1 \frac{dt}{1+t} = \ln 2$.

Addendum E — Reduction Formulas: Integrating Powers Systematically

Subtopic 05 hands you Walli's closed form for $\int_0^{\pi/2} \sin^n x \cos^m x dx$ and stops. Boards want the *derivations*: the recurrence relations for powers of tan, sec, and sin are standard 3-5 mark "prove and hence evaluate" questions, and they are the honest way to compute $\int \sec^5 x dx$ -class integrals that no identity cracks open.

Method

Recurrence for tangent. Let $I_n = \int \tan^n x dx$ for $n \geq 2$. Split off one $\tan^2 x$ and use $\tan^2 x = \sec^2 x - 1$:

$$I_n = \int \tan^{n-2} x (\sec^2 x - 1) dx = \int \tan^{n-2} x \sec^2 x dx - I_{n-2}.$$

The first piece is exact-substitution (Subtopic 04), since $\frac{d}{dx} \tan x = \sec^2 x$:

$$I_n = \frac{\tan^{n-1} x}{n-1} - I_{n-2}$$

Recurrence for secant. Let $I_n = \int \sec^n x dx$ for $n \geq 2$. Peel one $\sec^2 x$ and apply parts with $u = \sec^{n-2} x$, $dv = \sec^2 x dx$, so $v = \tan x$:

$$I_n = \sec^{n-2} x \tan x - \int (n-2) \sec^{n-3} x \cdot \sec x \tan x \cdot \tan x dx.$$

Write $\tan^2 x = \sec^2 x - 1$ inside the remainder:

$$I_n = \sec^{n-2} x \tan x - (n-2) \int \sec^{n-2} x (\sec^2 x - 1) dx = \sec^{n-2} x \tan x - (n-2)I_n + (n-2)I_{n-2}.$$

Collect the I_n terms:

$$(n-1)I_n = \sec^{n-2} x \tan x + (n-2)I_{n-2}$$

Recurrence for the Walli integral. Let $W_n = \int_0^{\pi/2} \sin^n x dx$ for $n \geq 2$. Parts with $u = \sin^{n-1} x$, $dv = \sin x dx$, so $v = -\cos x$:

$$W_n = [-\sin^{n-1} x \cos x]_0^{\pi/2} + (n-1) \int_0^{\pi/2} \sin^{n-2} x \cos^2 x dx.$$

The boundary term vanishes at both ends. Replace $\cos^2 x = 1 - \sin^2 x$:

$$W_n = (n-1)W_{n-2} - (n-1)W_n \implies \boxed{W_n = \frac{n-1}{n} W_{n-2}}$$

Iterating down to $W_0 = \frac{\pi}{2}$ (even n) or $W_1 = 1$ (odd n) reproduces Walli's formula — this derivation is the Board-proof version of the table entry in Subtopic 05.

Example E.1 — [CBSE Board Level]

Evaluate $\int \tan^4 x \, dx$.

Apply the recurrence twice. With $n = 4$:

$$I_4 = \frac{\tan^3 x}{3} - I_2, \quad I_2 = \int \tan^2 x \, dx = \tan x - x.$$

So

$$\int \tan^4 x \, dx = \frac{\tan^3 x}{3} - \tan x + x + C.$$

Check. Differentiate: $\tan^2 x \sec^2 x - \sec^2 x + 1 = \sec^2 x(\tan^2 x - 1) + 1 = (\tan^2 x + 1)(\tan^2 x - 1) + 1 = \tan^4 x$. ✓

Example E.2 — [JEE Main Level]

Evaluate $\int \sec^4 x \, dx$.

Recurrence with $n = 4$:

$$I_4 = \frac{\sec^2 x \tan x}{3} + \frac{2}{3} I_2, \quad I_2 = \int \sec^2 x \, dx = \tan x.$$

So

$$\int \sec^4 x \, dx = \frac{\sec^2 x \tan x}{3} + \frac{2}{3} \tan x + C.$$

Example E.3 — [JEE Advanced Level]

Evaluate $\int \sec^5 x \, dx$ **and** $\int_0^{\pi/2} \sin^8 x \, dx$.

Secant. Iterate: with $n = 5$,

$$I_5 = \frac{\sec^3 x \tan x}{4} + \frac{3}{4} I_3, \quad I_3 = \frac{\sec x \tan x}{2} + \frac{1}{2} I_1.$$

Here $I_1 = \int \sec x \, dx = \ln |\sec x + \tan x|$ (derived in Subtopic 04). Assemble:

$$\int \sec^5 x \, dx = \frac{\sec^3 x \tan x}{4} + \frac{3 \sec x \tan x}{8} + \frac{3}{8} \ln |\sec x + \tan x| + C.$$

Sine. Iterate $W_n = \frac{n-1}{n} W_{n-2}$ from $n = 8$ down to $W_0 = \frac{\pi}{2}$:

$$W_8 = \frac{7}{8} \cdot \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{105}{384} \cdot \frac{\pi}{2} = \frac{35\pi}{256}.$$

Cross-check with Walli. The closed form of Subtopic 05 with $n = 8$, $m = 0$: $\frac{7!!}{8!!} \cdot \frac{\pi}{2} = \frac{7 \cdot 5 \cdot 3 \cdot 1}{8 \cdot 6 \cdot 4 \cdot 2} \cdot \frac{\pi}{2} = \frac{105}{384} \cdot \frac{\pi}{2} = \frac{35\pi}{256}$. Agreement confirms both.

Practice — Addendum E

1. Derive the recurrence for $I_n = \int \tan^n x \, dx$ from scratch.
2. Evaluate $\int \tan^3 x \, dx$.
3. Evaluate $\int \tan^5 x \, dx$.
4. Evaluate $\int \sec^6 x \, dx$.
5. Evaluate $\int_0^{\pi/2} \cos^7 x \, dx$.

Answers: 2. $I_3 = \frac{\tan^2 x}{2} - I_1 = \frac{\tan^2 x}{2} + \ln |\cos x| + C$. 3. $I_5 = \frac{\tan^4 x}{4} - \frac{\tan^2 x}{2} + \ln |\cos x| + C$.
 4. $I_6 = \frac{\sec^4 x \tan x}{5} + \frac{4}{5} I_4 = \frac{\sec^4 x \tan x}{5} + \frac{4 \sec^2 x \tan x}{15} + \frac{8 \tan x}{15} + C$. 5. $W_7 = \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3} \cdot W_1 = \frac{48}{105} = \frac{16}{35}$.

Addendum F — Breakpoint Integrands II: Greatest Integer, Fractional Part, Min/Max

Subtopic 03 teaches splitting a modulus at its sign-change points. Three more families of integrands break the same way but were left unworked: the greatest-integer function, the fractional part, and min/max of two expressions. Each is a guaranteed JEE Main appearance and a recurring CBSE 2-marker. Alongside them sits the single most-mark-losing distinction in the chapter: **signed integral versus geometric area**.

Method

Notation: $[x]$ = greatest integer $\leq x$; $\{x\} = x - [x]$.

Greatest integer. On any interval $[k, k + 1)$ with integer k , $[x] = k$ is constant. So chop the interval at every integer inside it and sum rectangles:

$$\int_m^n [x] dx = \sum_{k=m}^{n-1} k \cdot 1 = \frac{(n-1)n}{2} - \frac{(m-1)m}{2}$$

for integers $m < n$. Endpoint values never matter — a single point carries no width.

Fractional part. $\{x\} = x$ on $[0, 1)$, and $\{x + 1\} = \{x\}$, so $\{x\}$ is periodic with period 1 and

$$\int_0^T \{x\} dx = T \cdot \int_0^1 x dx = \frac{T}{2}$$

for any positive integer T (and $[T] \cdot \frac{1}{2}$ plus a partial period otherwise).

Min/max. Solve $g(x) = h(x)$ for the crossing points, split the interval there, and on each piece replace min/max by whichever function wins — decided by testing one interior point per piece.

Signed vs total. $\int_a^b f$ counts area below the axis negatively. The geometric area is $\int_a^b |f|$, computed by splitting at the roots of f . An odd function on $[-a, a]$ integrates to 0 while enclosing perfectly real, positive area.

Example F.1 — [CBSE Board Level]

Evaluate $\int_{-2}^3 [x] dx$ **and** $\int_0^3 (x - [x]) dx$.

First. The integers inside $[-2, 3]$ split the interval into five unit pieces, on which $[x]$ equals $-2, -1, 0, 1, 2$ respectively:

$$\int_{-2}^3 [x] dx = (-2) + (-1) + 0 + 1 + 2 = 0.$$

Second. $x - [x] = \{x\}$, which is periodic with period 1; over three periods,

$$\int_0^3 \{x\} dx = 3 \int_0^1 x dx = 3 \cdot \frac{1}{2} = \frac{3}{2}.$$

Check. $\int_{-2}^3 x dx - \int_{-2}^3 [x] dx = \left[\frac{x^2}{2}\right]_{-2}^3 - 0 = \frac{9}{2} - 2 = \frac{5}{2}$; separately $\int_{-2}^3 \{x\} dx$ covers five unit cells, four full (contributing 2) plus the half-period $[2, 3)$ contributing $\frac{1}{2}$ — total $\frac{5}{2}$. Consistent.

Example F.2 — [JEE Main Level]

Evaluate $\int_0^2 \min(x^2, x) dx$.

Crossings: $x^2 = x$ at $x = 0$ and $x = 1$. Test $x = \frac{1}{2}$: $x^2 = \frac{1}{4} < \frac{1}{2}$, so $\min = x^2$ on $(0, 1)$. Test $x = \frac{3}{2}$: $x^2 = \frac{9}{4} > \frac{3}{2}$, so $\min = x$ on $(1, 2)$. Hence

$$\int_0^2 \min(x^2, x) dx = \int_0^1 x^2 dx + \int_1^2 x dx = \frac{1}{3} + \frac{4-1}{2} = \frac{1}{3} + \frac{3}{2} = \frac{11}{6}.$$

Example F.3 — [JEE Advanced Level]

Find the signed value of $\int_{-\sqrt{3}}^{\sqrt{3}} (x^3 - 3x) dx$ and the total area enclosed between $y = x^3 - 3x$ and the x -axis.

Signed. The integrand is odd and the limits symmetric, so by P6 the integral is 0.

Total area. The curve crosses the axis where $x(x^2 - 3) = 0$: at $x = -\sqrt{3}, 0, \sqrt{3}$. Sign test: positive on $(-\sqrt{3}, 0)$, negative on $(0, \sqrt{3})$. The two lobes are mirror images, so

$$\text{Area} = 2 \left| \int_0^{\sqrt{3}} (x^3 - 3x) dx \right| = 2 \left| \left[\frac{x^4}{4} - \frac{3x^2}{2} \right]_0^{\sqrt{3}} \right| = 2 \left| \frac{9}{4} - \frac{9}{2} \right| = 2 \cdot \frac{9}{4} = \frac{9}{2}.$$

The trap. Writing 0 as “the area” costs every mark: the signed integral being 0 is precisely why the question demands the split-at-the-roots computation. Whenever a question says *area*, split at zeros; whenever it says *evaluate the integral*, do not.

Practice — Addendum F

1. Evaluate $\int_0^5 [x] dx$.
2. Evaluate $\int_{-1}^4 [x] dx$.
3. Evaluate $\int_0^2 \{x\} dx$.
4. Evaluate $\int_0^{3/2} \min(x, x^2) dx$.
5. Evaluate $\int_0^{2\pi} [\sin x] dx$.
6. Find the total area between $y = \sin x$ and the x -axis for $0 \leq x \leq 2\pi$.

Answers: 1. $0 + 1 + 2 + 3 + 4 = 10$. 2. $(-1) + 0 + 1 + 2 + 3 = 5$. 3. Two periods: $2 \cdot \frac{1}{2} = 1$. 4. On $(0, 1)$, $x^2 < x$, so $\min = x^2$; on $(1, \frac{3}{2})$, $\min = x$: $\frac{1}{3} + \left(\frac{9}{8} - \frac{1}{2}\right) = \frac{23}{24}$. 5. On $(0, \pi)$, $0 < \sin x \leq 1$ gives $[\sin x] = 0$ (except at the single point $x = \frac{\pi}{2}$, which carries no width); on $(\pi, 2\pi)$, $-1 \leq \sin x < 0$ gives $[\sin x] = -1$. Total $= 0 \cdot \pi + (-1) \cdot \pi = -\pi$. 6. Four humps of area 1 each: $\boxed{4}$ — equivalently $2 \int_0^\pi \sin x = 4$, noting $|\sin x|$ has period π (Subtopic 03).

Addendum P — What JEE Actually Asks

Everything below is drawn from the previous-year bank supplied with this chapter. A caveat up front: in the source PDF most integrand bodies were set as images and did not survive extraction. Where a question's *method signature* survives in the official solution (the substitution declared, the "add the flipped integral" step, the final numeric answer), I treat that as evidence; where even the method is lost, I stay silent. Percentages below are therefore honest estimates from surviving signatures, not exact counts. Every worked example and every practice answer in this section has been independently re-derived and checked by differentiation or by an independent method.

P.1 The distribution — where the marks actually sit

Judged from the roughly 160 items whose solution strategies survive, the weight is overwhelmingly on the **definite** side of the chapter:

- **Properties of definite integrals (king / parity / periodicity) — the single biggest cluster, roughly half of all definite-integral questions.** The signature move "write I , transform the variable, add the two copies, solve $2I = \text{easy}$ " appears in the official solutions of papers spanning four decades: 1983, 1985, 1988, 1993, 1996, 1999, 2004, 2009, 2011, 2015, 2017, 2019 (Jan 7, Jan 12, April 8 and April 10 sessions), and 2020. The specific template " $f(x) = f(a-x)$ with $g(x) + g(a-x)$ constant" was examined twice nearly identically — Advanced 1989 (as a proof, sum = 2) and Main Jan 12, 2019 (sum = 4).
- **Even/odd deletion on symmetric limits** recurs constantly: 1985, 1990, 1998, 1999, 2012 (explicit even/odd split in the solution), 2019 (Jan 11, where $[-x] = -1 - [x]$ is used to prove oddness), among others.
- **Periodicity, especially with a modulus:** 1984 (period-independence proof), 2002 ($\int_a^{a+3T} = 3I$), 2007 (Assertion-Reason on the antiderivative of $\sin^2 x$), 2010 (period-2 function on $[-10, 10]$), Advanced 2015 (periodicity of $\sin^6 + \cos^6$).
- **Greatest-integer / fractional-part integrands** are a reliable fixture: 1988, 1995, 1998, 1999, Advanced 2010 and 2015, Main April 10 and Jan 11, 2019, Main Sep 02 and Sep 04, 2020.
- **Differentiation of variable-limit integrals (Leibniz/chain-rule FTC)** appears in 1997, 1998, 2004, 2009, 2010, Advanced 2013 and 2015, Advanced 2018, Main April 12, 2019, Main Jan 9, 2020, and Advanced 2020. Recent Main loves dressing this up with a local-minima classification (Main Jan 9, 2020: answer "point of local minima").
- **Indefinite integrals proper** cluster in Main sessions 2013–2020 and are almost always *recognition* problems, not grinding: the f'/f logarithmic pattern (Main April 12, 2019; several Sep 2020 sessions), "power of a function times its derivative" (Main 2018; Jan 11 and April 9, 2019), parts with a cycling integral (Main Jan 12, 2019: $\int \cos(\ln x) dx$), and substitution-into-rational-form (online 2013, 2014, 2016).
- **Limit of a sum** is now rare — live mainly in pre-1995 papers (1981 converted a series into $\log 6$ this way) and occasionally Advanced (2018). Know the mechanics; do not

over-invest.

- **Format shift worth noting:** Advanced 2015–2020 and Main 2020 numericals in this chapter land on small clean numbers — the surviving answers include 0, 1, 2, 4, 7, 9, 21, 0.50, 1.50. If your answer is ugly, you have almost certainly missed a symmetry.

P.2 The recurring archetypes

Archetype 1 — The king property: “add the flip”

Looks like: any $\int_a^b f$ where f contains trigonometric functions on $[0, \frac{\pi}{2}]$ or $[0, \pi]$, or an abstract $f(x) = f(a + b - x)$; denominators of the form $f(x) + f(\text{flip})$ are the tell.

Cracks it: substitute $x \mapsto a + b - x$, add the two copies of I , solve $2I = (\text{easy})$ — then *divide by two*.

Worked example (reconstructed from Main Jan 12, 2019 (I)): f, g continuous on $[0, a]$ with $f(x) = f(a - x)$ and $g(x) + g(a - x) = 4$. Show $\int_0^a f(x)g(x) dx = 2 \int_0^a f(x) dx$.

$$I = \int_0^a f(x)g(x) dx$$

Apply $x \mapsto a - x$ (limits unchanged):

$$I = \int_0^a f(a - x)g(a - x) dx = \int_0^a f(x)(4 - g(x)) dx$$

Add the two copies:

$$\begin{aligned} 2I &= \int_0^a f(x)[g(x) + 4 - g(x)] dx \\ &= 4 \int_0^a f(x) dx \end{aligned}$$

so $I = 2 \int_0^a f(x) dx$. Notice no antiderivative was ever needed.

Archetype 2 — Parity deletion on symmetric limits

Looks like: limits of the form $[-a, a]$ with a sum of terms of mixed parity.

Cracks it: classify each term before integrating anything: even $\times$ even and odd $\times$ odd are even; anything $\times$ odd-parity partner is odd and dies.

Worked example: evaluate $\int_{-\pi/2}^{\pi/2} (x \sin x + \cos^2 x) dx$.

$x \sin x$ is odd $\times$ odd = even; $\cos^2 x$ is even. Nothing cancels, so both double:

$$\begin{aligned} \int_{-\pi/2}^{\pi/2} x \sin x dx &= 2 \int_0^{\pi/2} x \sin x dx = 2 [-x \cos x + \sin x]_0^{\pi/2} = 2 \\ \int_{-\pi/2}^{\pi/2} \cos^2 x dx &= 2 \int_0^{\pi/2} \frac{1 + \cos 2x}{2} dx = \frac{\pi}{2} \end{aligned}$$

Answer: $2 + \frac{\pi}{2}$.

Contrast with the destructive version examined in 2012 and 2019: when the odd terms dominate (e.g. an x^3 or a lone $\cos x$ against symmetric limits), the whole integral is 0 and the MCQ distractors are the values obtained by grinding out antiderivatives.

Archetype 3 — Periodic collapse (mind the modulus)

Looks like: $\int_0^{nT}$ or $\int_0^{\text{large}}$ of $|\sin x|$, $|\cos x|$, $\sin^6 + \cos^6$, etc.

Cracks it: find the *true* period of the integrand — the modulus halves the period of sine/cosine to π ; $\sin^6 x + \cos^6 x = 1 - \frac{3}{4} \sin^2 2x$ has period $\frac{\pi}{2}$ — then use $\int_0^{nT} = n \int_0^T$.

Worked example: $\int_0^{3\pi/2} |\sin x| dx$. Since $|\sin(x + \pi)| = |\sin x|$, the period is π ; the interval holds 1.5 periods, and $\int_0^\pi |\sin x| dx = 2$. Hence the value is $\frac{3}{2} \cdot 2 = 3$.

Archetype 4 — Greatest-integer / piecewise splitting

Looks like: $\int[\cdot]$ or $\int\{x\}$ over a modest interval; also modulus integrals.

Cracks it: locate every breakpoint where the inside of the GIF crosses an integer (or the modulus changes sign), split with additivity, integrate the clean pieces.

Worked example: $\int_0^{3/2} [x^2] dx$. Breakpoints where x^2 hits 1, 2, 3: at $x = 1, \sqrt{2}, \sqrt{3}$.

$$\begin{aligned} \int_0^{3/2} [x^2] dx &= 1(\sqrt{2} - 1) + 2(\sqrt{3} - \sqrt{2}) + 3\left(\frac{3}{2} - \sqrt{3}\right) \\ &= \frac{7}{2} - \sqrt{2} - \sqrt{3} \end{aligned}$$

Archetype 5 — Variable limits: differentiate, don't integrate

Looks like: $F(x) = \int_{\phi(x)}^{\psi(x)} (\dots) dt$ followed by “classify the point”, “find $g(2)$ ”, “show $f' \dots$ ”.

Cracks it: Fundamental Theorem with the chain rule: $\frac{d}{dx} \int_{\phi(x)}^{\psi(x)} h(t) dt = h(\psi)\psi' - h(\phi)\phi'$. Never attempt the integral itself.

Worked example (reconstructed from Main Jan 9, 2020 (II)): let $F(x) = \int_1^x t^2 g(t) dt$ where $g(t) = \int_1^t f(u) du$ and f is continuous with $f(1) = 3$. Classify $x = 1$ for F .

By the FTC, $F'(x) = x^2 g(x)$, so $F'(1) = g(1) = 0$: a critical point. Differentiate again:

$$F''(x) = 2xg(x) + x^2 g'(x) = 2xg(x) + x^2 f(x)$$

so $F''(1) = 0 + f(1) = 3 > 0$. Hence $x = 1$ is a point of strict local minimum — matching the official key.

Archetype 6 — Definite substitution: change the limits, exploit algebra

Looks like: a trigonometric integrand on $[0, \frac{\pi}{4}]$ or $[0, \frac{\pi}{2}]$ that factors as “(function of $\tan x$) $\times \sec^2 x$ ”, or any integrand where the numerator is built from the derivative of an inner function.

Cracks it: convert limits immediately ($x = a \mapsto u = g(a)$) and finish in u ; look first for factorisations that produce $\sec^2 x \, dx$.

Worked example (Advanced 2015): evaluate

$$I = \int_0^{\pi/4} (7 \tan^8 x + 7 \tan^6 x - 3 \tan^4 x - 3 \tan^2 x) \, dx$$

Factor out $\sec^2 x$: the integrand is $(7 \tan^6 x - 3 \tan^2 x) \sec^2 x$ — check by expanding $(7 \tan^6 - 3 \tan^2)(1 + \tan^2)$. With $u = \tan x$, limits $0 \mapsto 0$, $\frac{\pi}{4} \mapsto 1$:

$$I = \int_0^1 (7u^6 - 3u^2) \, du = 1 - 1 = 0$$

Archetype 7 — Indefinite integrals as recognition tests

Looks like: a strange-looking quotient, or a lone transcendental, or e^x times a sum.

Cracks them, in order of frequency in the bank: 1. **Numerator is (a multiple of) the derivative of the denominator** $\Rightarrow \ln|\cdot|$ instantly. 2. **Power of a function times its derivative** $\Rightarrow$ power rule in the inner function. 3. **Lone** $\ln x$, $\tan^{-1} x \Rightarrow$ pair with an invisible “ $\times 1$ ” and use parts. 4. **Integral reappears after two rounds of parts** $\Rightarrow$ solve for I algebraically.

Worked examples (both reconstructed from named papers):

- (i) Main April 12, 2019 (I): $\int \frac{2x - x^{-2}}{x^2 + x^{-1}} \, dx$. The numerator is exactly the derivative of $x^2 + x^{-1}$, so with $u = x^2 + x^{-1}$:

$$I = \int \frac{du}{u} = \ln|x^2 + x^{-1}| + C$$

- (ii) Main Jan 12, 2019 (II): $I = \int \cos(\ln x) \, dx$. Parts with $u = \cos(\ln x)$, then parts again on the $\sin(\ln x)$ remainder:

$$\begin{aligned} I &= x \cos(\ln x) + \int \sin(\ln x) \, dx \\ &= x \cos(\ln x) + x \sin(\ln x) - I \end{aligned}$$

so $2I = x(\cos(\ln x) + \sin(\ln x))$ and

$$I = \frac{x}{2} (\cos(\ln x) + \sin(\ln x)) + C$$

(Differentiate to confirm: the $\frac{x}{2}$ and the inner derivatives conspire to return $\cos(\ln x)$.)

P.3 The traps — what the distractors are built to punish

- **Reporting $2I$ as I .** After “add the flip”, forgetting the final halving doubles every king-property answer. Several option sets (1993, Main 2017) contain exactly the doubled value as a distractor.

- **Parity on non-symmetric limits.** Oddness kills an integral only on $[-a, a]$. On $[0, 2]$ or $[1, 3]$ it tells you nothing; the bank contains GIF and modulus questions specifically designed so the reflex cancellation gives a wrong sign or a wrong zero.
- **Keeping old limits after substitution.** Switching to u while leaving x -limits in place (or changing limits *and* back-substituting) corrupts the bounds. This is the most common error class in the substitution-based definite questions throughout the bank.
- **Dropping the modulus / wrong period.** Integrating $\sin x$ where $|\sin x|$ is written lets humps cancel; using period 2π for $|\sin x|$ miscounts blocks. The 1988 and 2019 GIF items and the 2007 Assertion-Reason both hinge on this family of errors. Related subtlety: **an antiderivative of a periodic function need not be periodic** — $F(x) = \frac{x}{2} - \frac{\sin 2x}{4} + C$ for $\sin^2 x$ satisfies $F(x + \pi) = F(x) + \frac{\pi}{2}$, which is why Statement-1 of the 2007 item is false while Statement-2 is true.
- **Power rule on functions.** $\int \tan^4 x \, dx \neq \frac{\tan^5 x}{5}$ unless a $\sec^2 x$ is present (Main 2017 made students *find* the pairing $I_4 + I_6 = \int \tan^4 x \sec^2 x \, dx = \frac{\tan^5 x}{5}$). Same disease: treating $\sin^2 x$ like x^2 .
- **Improper fractions without division, and missing the repeated-factor term** in partial fractions — punished whenever a rational integrand appears (2013, 2014, 2016 online sessions).
- **Principal branches of inverse trigonometric functions under substitution.** Advanced 2015 states explicitly that $\tan^{-1} x$ takes only principal values — a substitution that ignores branch cuts silently changes the integrand.
- **Blind computation on inequality/bounding items.** Advanced 2013, 2015 and 2017 ask you to *bound* an integral (squeeze estimates, monotonicity of f'/f -type ratios), not evaluate it. Grinding for an antiderivative is the trap.

P.4 Graded practice set

Every answer below has been re-solved and verified (by differentiation or an independent check).

Q1. Evaluate $\int \frac{2x - 1}{x^2 - x + 3} \, dx$.

The numerator is the derivative of $x^2 - x + 3$:

$$I = \ln(x^2 - x + 3) + C$$

Q2. Evaluate $\int_0^{\pi/2} \cos x \, e^{\sin x} \, dx$.

With $u = \sin x$, limits $0 \mapsto 0$, $\frac{\pi}{2} \mapsto 1$:

$$I = \int_0^1 e^u \, du = e - 1$$

Q3. Evaluate $\int_{-1}^1 (x^3 + x^2 + \tan^{-1} x) \, dx$.

Symmetric limits: x^3 and $\tan^{-1} x$ are odd $\Rightarrow$ contribute 0. Only x^2 survives:

$$I = 2 \int_0^1 x^2 dx = \frac{2}{3}$$

Q4. Evaluate $\int_0^{3/2} [x] dx$ where $[\cdot]$ is the greatest-integer function.

Breakpoints at $x = 1, 2$:

$$I = \int_0^1 0 dx + \int_1^2 1 dx + \int_2^{3/2} 2 dx = 0 + 1 + 2 \cdot \frac{1}{2} = 2$$

Q5. Evaluate $I = \int e^x \sin x dx$.

Two rounds of parts bring I back:

$$\begin{aligned} I &= e^x \sin x - \int e^x \cos x dx \\ &= e^x \sin x - e^x \cos x - I \end{aligned}$$

so $2I = e^x(\sin x - \cos x)$ and

$$I = \frac{e^x}{2} (\sin x - \cos x) + C$$

Q6. Evaluate $\int_0^{3\pi/2} |\sin x| dx$.

$|\sin x|$ has period π and $\int_0^\pi |\sin x| dx = 2$. The interval holds $\frac{3}{2}$ periods:

$$I = \frac{3}{2} \cdot 2 = 3$$

Q7. Evaluate $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$.

King: replace $x \mapsto \frac{\pi}{2} - x$; the integrand becomes its complement. Adding:

$$\begin{aligned} 2I &= \int_0^{\pi/2} \frac{\sqrt{\sin x} + \sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx \\ &= \frac{\pi}{2} \end{aligned}$$

so $I = \frac{\pi}{4}$. (No antiderivative exists in elementary terms — symmetry is the only route.)

Q8. Evaluate $\int_0^{\pi/2} \frac{dx}{1 + (\tan x)^{\sqrt{2}}}$.

Same mechanism as Q7, and this is why the mechanism matters: the flip sends $\tan \mapsto \cot$, and

$$\frac{1}{1 + \cot^{\sqrt{2}} x} + \frac{1}{1 + \tan^{\sqrt{2}} x} = 1$$

identically. Hence $2I = \frac{\pi}{2}$ and $I = \frac{\pi}{4}$ — for *any* positive exponent.

Q9. Evaluate $\int_0^{\pi/4} \tan^3 x \, dx$.

Peel one $\tan$ and use $\sec^2 = 1 + \tan^2$:

$$\begin{aligned} I &= \int_0^{\pi/4} \tan x (\sec^2 x - 1) \, dx \\ &= \left[\frac{\tan^2 x}{2} \right]_0^{\pi/4} + [\ln \cos x]_0^{\pi/4} \\ &= \frac{1}{2} - \frac{1}{2} \ln 2 \end{aligned}$$

Q10. Let $H(x) = \int_x^{x^2} \sin(t^2) \, dt$. Find $H'(x)$.

Fundamental Theorem with moving both limits (no antiderivative of $\sin t^2$ exists — that is the point):

$$H'(x) = 2x \sin(x^4) - \sin(x^2)$$

Q11. Evaluate $I = \int_0^{\pi} x \sin^3 x \, dx$.

King peels off the lone x : replacing $x \mapsto \pi - x$ leaves $\sin^3$ intact,

$$I = \int_0^{\pi} (\pi - x) \sin^3 x \, dx = \pi \int_0^{\pi} \sin^3 x \, dx - I$$

so $2I = \pi \int_0^{\pi} \sin^3 x \, dx$. By symmetry about $\frac{\pi}{2}$, $\int_0^{\pi} \sin^3 x \, dx = 2 \cdot \frac{2}{3} = \frac{4}{3}$. Therefore

$$I = \frac{\pi}{2} \cdot \frac{4}{3} = \frac{2\pi}{3}$$

Q12. Evaluate $I = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} \, dx$.

King: the flip turns x into $\pi - x$ and leaves the rest untouched, so

$$2I = \pi \int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} \, dx$$

Substitute $u = \cos x$, $du = -\sin x \, dx$; the limits sweep $1 \mapsto -1$:

$$\int_0^{\pi} \frac{\sin x}{1 + \cos^2 x} \, dx = \int_{-1}^1 \frac{du}{1 + u^2} = 2 \arctan(1) = \frac{\pi}{2}$$

Hence $2I = \pi \cdot \frac{\pi}{2}$ and

$$I = \frac{\pi^2}{4}$$

P.5 What this chapter's own text under-serves

The chapter teaches the First FTC as a one-line statement but never develops **differentiation of integrals with variable or functionally-defined limits** — the Leibniz/chain-rule machinery behind a whole recurring family of Main and Advanced questions (local-minima classifications, hidden-equation problems). It also never touches **greatest-integer/fractional-part integrands, integral inequalities and squeezing estimates**, or **reduction formulas and the “prove the identity, then use it” long-format culture** that dominates the older Advanced papers — all of which the bank shows are live exam behaviours. Finally, its treatment of the limit-of-a-sum is decorative, whereas the exam (rarely, but at Advanced level) still converts series into integrals with it.

Application of Integrals

Class 12 Mathematics | Chapter 8 | Drona Ed Tech

Contents

Subtopic 01 — Area under Simple Curves	1
Section 1 — Concept Intuition	2
Section 2 — Key Definitions & Properties	3
Section 3 — Theorems & Proofs	4
Section 4 — Worked Examples	6
Section 5 — Practice Exercises	8
Section 6 — Crack the MCQ	8
Section 7 — Common Pitfalls, Pro-Tips & Cheat Sheet	9
Subtopic 02 — Area between Two Curves	11
Section 1 — Concept Intuition	11
Section 2 — Key Definitions & Properties	12
Section 3 — Theorems & Proofs	13
Section 4 — Worked Examples	14
Section 5 — Practice Exercises	16
Section 6 — Crack the MCQ	17
Section 7 — Common Pitfalls, Pro-Tips & Cheat Sheet	18
Subtopic 03 — Advanced Regions: Modulus, Inequalities & Composite Areas	20
Section 1 — Concept Intuition	20
Section 2 — Key Definitions & Properties	21
Section 3 — Key Procedures & a Theorem	22
Section 4 — Worked Examples	23
Section 5 — Practice Exercises	24
Section 6 — Crack the MCQ	25
Section 7 — Common Pitfalls, Pro-Tips & Chapter Master Cheat Sheet	26

Subtopic 01 — Area under Simple Curves

Exam Relevance: Area under a curve is the gateway idea of this entire chapter. CBSE Boards almost always carry one 3- or 5-mark question here (frequently “find the area of a circle/ellipse by integration”). JEE Main reliably places 1 question per paper, usually disguised as a region-bounded-by-

a-curve-and-an-axis problem demanding the correct choice of dx vs dy . JEE Advanced rarely asks a *pure* area-under-one-curve question but builds every area-between-curves problem (Sub-02) directly on this skill, so a shaky foundation here costs marks downstream.

Section 1 — Concept Intuition

You already know how to compute the area of a rectangle, a triangle, or a circle using memorised formulas. But what is the area of the region trapped under a *curvy* boundary — say, under the arch $y = x^2 + 1$ between $x = 0$ and $x = 2$? There is no clean formula for that shape. This is exactly the gap that integration fills.

Think about it this way... Imagine you are a land surveyor in rural Punjab measuring an irregular farm plot. One edge of the plot runs along a perfectly straight irrigation canal (that is our x -axis). The opposite edge is a wavy natural boundary — a stream — described by some curve $y = f(x)$. The plot stretches from the well at $x = a$ to the banyan tree at $x = b$. How do you find its area without a single neat formula?

A practical surveyor does something clever: he divides the plot into a long series of very thin vertical strips, each as narrow as a sheet of paper. Each strip is *almost* a rectangle — its width is a tiny dx , and its height is the value of the curve, $y = f(x)$, at that location. So one strip has area $\approx y dx = f(x) dx$. To get the whole plot, he simply adds up the areas of all these strips from the well to the banyan tree. “Adding up infinitely many infinitely-thin strips” is precisely what the definite integral *means*:

$$\text{Area} = \int_a^b y dx = \int_a^b f(x) dx$$

That single mental picture — **a thin vertical strip of height y and width dx** — is the whole subtopic. Everything else is bookkeeping about which variable to integrate and how to handle signs.

Figure / Diagram

A curve $y = f(x)$ in the first quadrant. The region bounded by the curve (top), the x -axis (bottom), and two vertical dashed lines $x = a$ (left) and $x = b$ (right) is shaded. One representative thin vertical strip is drawn inside the region, labelled height “ y ” and width “ dx ”, to show the elementary-strip idea.

If the boundary is instead measured more naturally against the *vertical* canal — that is, the curve is given as $x = g(y)$ and you sweep horizontal strips up the y -axis — the same logic flips axes:

$$\text{Area} = \int_c^d x dy = \int_c^d g(y) dy$$

Here each strip is horizontal, with height dy and length $x = g(y)$, swept from $y = c$ to $y = d$.

Limiting conditions & assumptions — when the naive formula misleads you:

1. **The curve must stay on one side of the axis.** The formula $\int_a^b f(x) dx$ gives a *signed* number. When $f(x) \geq 0$ (curve above the x -axis), the sign is positive and equals the geometric area. But when $f(x) \leq 0$ (curve below the axis), the integral comes out *negative* — yet area is never negative. The strip height is genuinely $|y|$ there.
2. **The curve must not cross the axis inside $[a, b]$.** If $f(x)$ is positive on part of the interval and negative on another, blindly integrating across the crossing causes the positive and negative pieces to *cancel*, and you get a wrong (too small, sometimes zero) answer. You must split at the roots.
3. **f should be continuous on $[a, b]$.** A finite jump or an unbounded spike breaks the simple strip-sum picture (improper integrals are beyond this subtopic).
4. **Pick the variable that matches the geometry.** If the region is naturally “capped above and below” by horizontal lines, dy -strips are cleaner; if “walled left and right” by vertical lines, dx -strips win. Choosing badly turns a one-line integral into a messy two-piece one.

Section 2 — Key Definitions & Properties

Area bounded by a curve and the x -axis (curve $y = f(x)$, between ordinates $x = a$ and $x = b$, with $f(x) \geq 0$):

$$A = \int_a^b y dx = \int_a^b f(x) dx$$

Area bounded by a curve and the y -axis (curve $x = g(y)$, between abscissae $y = c$ and $y = d$, with $g(y) \geq 0$):

$$A = \int_c^d x dy = \int_c^d g(y) dy$$

Area when the curve lies below the x -axis ($f(x) \leq 0$ on $[a, b]$):

$$A = \left| \int_a^b f(x) dx \right| = - \int_a^b f(x) dx$$

Area when the curve crosses the axis at $x = c$ inside $[a, b]$ (positive on $[a, c]$, negative on $[c, b]$):

$$A = \int_a^c f(x) dx + \left| \int_c^b f(x) dx \right| = \int_a^b |f(x)| dx$$

Notation key:

Symbol	Meaning
$f(x)$	the curve, expressed with y as a function of x
$g(y)$	the curve, expressed with x as a function of y
a, b	lower and upper ordinates (vertical boundary lines $x = a, x = b$)
c, d	lower and upper abscissae (horizontal boundary lines $y = c, y = d$)
dx	width of an elementary vertical strip
dy	height of an elementary horizontal strip

Standard catalogue results (proved in Section 3, worth memorising for speed):

- Area of a circle $x^2 + y^2 = a^2$: $\boxed{\pi a^2}$
- Area of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$: $\boxed{\pi ab}$
- Useful antiderivative: $\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C$

Symmetry property (a massive time-saver): if a region is symmetric about an axis, compute the area of one symmetric part and multiply. A circle/ellipse is symmetric about *both* axes, so the first-quadrant area $\times 4$ gives the total.

Section 3 — Theorems & Proofs

Theorem 1 (Area as the limit of a sum — the justification of the formula). Let f be continuous and non-negative on $[a, b]$. Then the area A of the region bounded by $y = f(x)$, the x -axis, and the lines $x = a, x = b$ is $A = \int_a^b f(x) dx$.

Proof sketch (the strip argument made rigorous). Partition $[a, b]$ into n sub-intervals each of width $\Delta x = \frac{b-a}{n}$, with division points $a = x_0 < x_1 < \dots < x_n = b$. Over the i -th sub-interval choose any sample point ξ_i . The thin strip standing on that sub-interval is approximately a rectangle of height $f(\xi_i)$ and width Δx , so its area is $f(\xi_i) \Delta x$. Summing all strips,

$$A \approx S_n = \sum_{i=1}^n f(\xi_i) \Delta x.$$

Why this step is legitimate: the thinner the strips, the smaller the sliver of error between each rectangle and the true curved strip. As we let $n \rightarrow \infty$ (so $\Delta x \rightarrow 0$), those slivers vanish, and the approximating sum converges to the exact area:

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\xi_i) \Delta x = \int_a^b f(x) dx,$$

where the final equality is the definition of the definite integral (Riemann sum). ■

Theorem 2 (Area of a circle by integration). The area enclosed by $x^2 + y^2 = a^2$ is πa^2 .

Proof. The circle is symmetric about both axes, so the total area is $4 \times$ the first-quadrant area. In the first quadrant the upper boundary is $y = \sqrt{a^2 - x^2}$ and x runs from 0 to a :

$$A = 4 \int_0^a \sqrt{a^2 - x^2} dx.$$

Why integrate only the first quadrant: it removes sign ambiguity (here $y \geq 0$) and exploits symmetry so we evaluate the standard integral just once. Using the catalogue antiderivative,

$$A = 4 \left[\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} \right]_0^a.$$

At $x = a$: the first term is $\frac{a}{2} \cdot 0 = 0$, and $\sin^{-1}(1) = \frac{\pi}{2}$, giving $\frac{a^2}{2} \cdot \frac{\pi}{2} = \frac{\pi a^2}{4}$. At $x = 0$ both terms vanish. Hence

$$A = 4 \cdot \frac{\pi a^2}{4} = \pi a^2. \quad \blacksquare$$

Theorem 3 (Area of an ellipse by integration). The area enclosed by $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is πab .

Proof. Solving for the upper half, $y = \frac{b}{a} \sqrt{a^2 - x^2}$. By the same two-axis symmetry,

$$A = 4 \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} dx = \frac{4b}{a} \int_0^a \sqrt{a^2 - x^2} dx.$$

We already evaluated $\int_0^a \sqrt{a^2 - x^2} dx = \frac{\pi a^2}{4}$ in Theorem 2, so

$$A = \frac{4b}{a} \cdot \frac{\pi a^2}{4} = \pi ab. \quad \blacksquare$$

Notice the satisfying check: setting $b = a$ collapses the ellipse into a circle, and πab collapses into πa^2 . A formula that degenerates correctly is almost certainly right.

Section 4 — Worked Examples

Example 1 — [CBSE Board Level] Find the area of the region bounded by the curve $y = x^2 + 2$, the x -axis, and the lines $x = 0$ and $x = 2$.

Given: $y = f(x) = x^2 + 2$, ordinates $x = 0$ to $x = 2$. Since $x^2 + 2 > 0$ everywhere, the curve lies entirely above the axis — no sign worries.

$$A = \int_0^2 (x^2 + 2) dx = \left[\frac{x^3}{3} + 2x \right]_0^2 = \left(\frac{8}{3} + 4 \right) - 0 = \frac{8 + 12}{3} = \frac{20}{3}.$$

Area = $\frac{20}{3}$ square units.

Example 2 — [Conceptual Trap / CUET-JEE Main Speed Trap] Find the area bounded by $y = x^3$, the x -axis, between $x = -2$ and $x = 2$.

The trap: a hurried student writes $\int_{-2}^2 x^3 dx$. Because x^3 is an *odd* function, that integral equals 0 — and they tick “0”. But the region is plainly *not* empty; the curve dips below the axis on $[-2, 0]$ and rises above on $[0, 2]$, so the two signed pieces cancel.

The fix: area is $\int |f(x)| dx$. Split at the crossing $x = 0$ and use symmetry (the two halves have equal area):

$$A = \left| \int_{-2}^0 x^3 dx \right| + \int_0^2 x^3 dx = 2 \int_0^2 x^3 dx = 2 \left[\frac{x^4}{4} \right]_0^2 = 2 \cdot \frac{16}{4} = 8.$$

Area = 8 square units. Mental shortcut: the instant you see a curve that changes sign on the interval, *stop* and split — never integrate straight across a root.

Example 3 — [JEE Main Level — multi-concept] Find the area of the region in the first quadrant bounded by the parabola $y^2 = 4x$, the x -axis, and the line $x = 4$.

Concepts combined: (i) recognising the parabola’s first-quadrant branch, (ii) integrating, (iii) a symmetry awareness check.

From $y^2 = 4x$, the upper branch is $y = 2\sqrt{x}$. The region sits between the x -axis and this branch, from $x = 0$ to $x = 4$:

$$A = \int_0^4 2\sqrt{x} \, dx = 2 \int_0^4 x^{1/2} \, dx = 2 \left[\frac{2}{3} x^{3/2} \right]_0^4 = \frac{4}{3} (4^{3/2}) = \frac{4}{3} \cdot 8 = \frac{32}{3}.$$

Area = $\frac{32}{3}$ **square units.**

Concept-check the examiner is testing: had the question asked for the region bounded by the parabola and the line $x = 4$ (both branches, above *and* below the axis), symmetry doubles the answer to $\frac{64}{3}$. Read whether the x -axis is a stated boundary — it decides whether you take one branch or two.

Example 4 — [JEE Advanced Level] Find the area of the smaller region bounded by the circle $x^2 + y^2 = 16$ and the line $x = 2$ (the part of the disc lying to the right of the line).

Setup. The circle has radius 4. The line $x = 2$ is a vertical chord; the “smaller region” is the cap between $x = 2$ and $x = 4$. It is symmetric about the x -axis, so take the upper half ($y = \sqrt{16 - x^2}$) and double:

$$A = 2 \int_2^4 \sqrt{16 - x^2} \, dx.$$

Apply the antiderivative with $a = 4$:

$$\int \sqrt{16 - x^2} \, dx = \frac{x}{2} \sqrt{16 - x^2} + \frac{16}{2} \sin^{-1} \frac{x}{4} = \frac{x}{2} \sqrt{16 - x^2} + 8 \sin^{-1} \frac{x}{4}.$$

Upper limit $x = 4$: $\frac{4}{2} \cdot 0 + 8 \sin^{-1}(1) = 8 \cdot \frac{\pi}{2} = 4\pi.$

Lower limit $x = 2$: $\frac{2}{2} \sqrt{12} + 8 \sin^{-1}\left(\frac{1}{2}\right) = 2\sqrt{3} + 8 \cdot \frac{\pi}{6} = 2\sqrt{3} + \frac{4\pi}{3}.$

Subtract and double:

$$A = 2 \left[4\pi - \left(2\sqrt{3} + \frac{4\pi}{3} \right) \right] = 2 \left[\frac{12\pi - 4\pi}{3} - 2\sqrt{3} \right] = 2 \left[\frac{8\pi}{3} - 2\sqrt{3} \right] = \frac{16\pi}{3} - 4\sqrt{3}.$$

Area = $\frac{16\pi}{3} - 4\sqrt{3}$ **square units** ≈ 9.83 sq. units. *Sanity check:* the full disc is $16\pi \approx 50.3$; a cap cut at one-half the radius taking $\approx 20\%$ of the disc is geometrically reasonable.

Section 5 — Practice Exercises

1. **[CBSE]** Find the area bounded by $y = 3x^2 + 1$, the x -axis, and the lines $x = 0$, $x = 2$.
2. **[CUET / JEE Main]** Find the area bounded by $y = \cos x$, the x -axis, from $x = 0$ to $x = \pi$. (*Watch the sign change.*)
3. **[JEE Main]** Find the area of the region bounded by the curve $x = 4 - y^2$ and the y -axis.
4. **[JEE Main]** Find the area bounded by $y = x^2$, the x -axis, and the ordinates $x = 1$ and $x = 3$.
5. **[JEE Advanced]** Find the area of the smaller region bounded by the circle $x^2 + y^2 = 9$ and the line $x = \frac{3}{2}$.

Answers: 1. 10 • 2. 2 • 3. $\frac{32}{3}$ • 4. $\frac{26}{3}$ • 5. $3\pi - \frac{9\sqrt{3}}{4}$

Section 6 — Crack the MCQ

Q1. If $f(x) \leq 0$ throughout $[a, b]$, the area between the curve $y = f(x)$ and the x -axis equals:

(a) $\int_a^b f(x) dx$ (b) $-\int_a^b f(x) dx$ (c) 0 (d) $\frac{1}{2} \int_a^b f(x) dx$

Correct: (b). When the curve is below the axis the definite integral is negative, but area must be non-negative, so we negate it.

- **(a)** is the *signed* integral — a negative number — and area can never be negative; this is the single most common slip.
 - **(c)** confuses this with the *odd-function-over-symmetric-interval* case, which is a different situation entirely.
 - **(d)** has no basis; the $\frac{1}{2}$ likely leaks in from the triangle-area formula by reflex.
-

Q2. The area bounded by $y = \sin x$, the x -axis, from $x = 0$ to $x = 2\pi$ is:

(a) 0 (b) 2 (c) 4 (d) 1

Correct: (c). $\sin x \geq 0$ on $[0, \pi]$ and ≤ 0 on $[\pi, 2\pi]$. Splitting: $\int_0^\pi \sin x dx + \left| \int_\pi^{2\pi} \sin x dx \right| = 2 + 2 = 4$.

- **(a)** is the signed integral $\int_0^{2\pi} \sin x dx = 0$ — the positive and negative humps cancel; the classic trap.

- **(b)** computes only the first hump and forgets the second.
- **(d)** is a careless evaluation error.

Q3. The area of the region bounded by the curve $x = y^2$, the y -axis, and the lines $y = 0$, $y = 2$ is:

- (a) $\frac{8}{3}$ (b) $\frac{4}{3}$ (c) $\frac{16}{3}$ (d) 2

Correct: (a). The boundaries are horizontal lines, so integrate with respect to y :

$$\int_0^2 y^2 dy = \left[\frac{y^3}{3} \right]_0^2 = \frac{8}{3}.$$

- **(b)** comes from correctly computing $8/3$ but then mistakenly halving it to get $4/3$.
- **(c)** doubles incorrectly, as if exploiting a symmetry that the stated bounds ($y = 0$ to 2 , one side only) do not provide.
- **(d)** is what you get integrating $\int_0^2 y dy$ — i.e. integrating $x = y$ instead of $x = y^2$, the wrong-variable / wrong-curve trap.

Q4. The area enclosed by the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ is:

- (a) 12π (b) 7π (c) 144π (d) 25π

Correct: (a). Here $a^2 = 16 \Rightarrow a = 4$ and $b^2 = 9 \Rightarrow b = 3$, so area $= \pi ab = \pi(4)(3) = 12\pi$.

- **(b)** adds the semi-axes ($a + b = 7$) instead of multiplying — a formula-recall error.
- **(c)** multiplies $a^2 b^2 = 16 \times 9 = 144$, forgetting to take square roots of the denominators.
- **(d)** uses $a^2 + b^2 = 25$, a Pythagoras reflex that has nothing to do with area.

Section 7 — Common Pitfalls, Pro-Tips & Cheat Sheet

Common pitfalls

1. **Reporting a negative or zero “area.”** Whenever the curve goes below the axis, the raw integral is negative; if it crosses the axis, opposite pieces cancel. Always sketch first and use $\int |f| dx$, splitting at every root.
2. **Integrating with respect to the wrong variable.** A region walled by the y -axis and horizontal lines wants $\int x dy$, not $\int y dx$. Forcing dx here creates needless casework.

3. **Using the wrong limits.** Limits are the boundary lines or the points where the curve meets the axis — not arbitrary numbers. For a curve like $y = 4 - x^2$ bounded by the x -axis, the limits are the *roots* $x = \pm 2$, which you must solve for.
4. **Forgetting to multiply back after using symmetry.** Computing the first-quadrant area of a circle and then forgetting the $\times 4$ gives a quarter of the answer.
5. **Mangling the ellipse formula.** It is πab (product

Pro-Tip — the sketch-then-split-then-symmetry reflex. Before touching the integral, do three things in five seconds: (1) rough-sketch the curve and shade the region; (2) mark where it crosses the axis inside your interval and plan to split there; (3) check for symmetry so you can integrate the smallest piece and multiply. This routine kills the three biggest mark-losers (sign, limits, symmetry) before they happen.

Pro-Tip — degeneration check. After deriving any area formula with parameters, plug in a value that collapses it to a shape you know. The ellipse πab must become the circle πa^2 when $b = a$. If it doesn't, you've made an error.

Cheat Sheet Box

- **Vertical strips (about x -axis):** $A = \int_a^b y \, dx, \quad y \geq 0$
- **Horizontal strips (about y -axis):** $A = \int_c^d x \, dy, \quad x \geq 0$
- **Below the axis / crossing it:** $A = \int_a^b |f(x)| \, dx$ — split at every root.
- **Circle** $x^2 + y^2 = a^2$: $A = \pi a^2$ | **Ellipse** $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$: $A = \pi ab$
- **Key antiderivative:** $\int \sqrt{a^2 - x^2} \, dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C$
- **Mnemonic — “SLICE”:** **S**ketch $\rightarrow$ **L**imits (and roots) $\rightarrow$ **C**hoose variable (dx/dy) $\rightarrow$ **E**valuate (with $|\cdot|$ and symmetry). Slice the region right and the integral writes itself.

Subtopic 02 — Area between Two Curves

Exam Relevance: This is the highest-yield idea in the whole chapter. CBSE Boards almost always set a 5-mark question here (typically area between a line and a parabola, or between two parabolas). JEE Main places 1 reliable question per session, and JEE Advanced builds its toughest area problems by intersecting a curve with a circle or with itself, demanding a split into pieces. Master “top minus bottom” and you own this chapter.

Section 1 — Concept Intuition

In Sub-01 you learned to find the area trapped between a single curve and an axis. But the axis is really nothing special — it is just the flat line $y = 0$. What if the *floor* of your region is itself a curve? Then you are looking at the area sandwiched **between two curves**, and the idea you already have generalises in one clean stroke.

Think about it this way... Picture the strip of land between two railway tracks running across a hilly stretch of the Konkan coast. The upper track follows one curve, $y = f(x)$; the lower track follows another, $y = g(x)$. You want the area of the land in between. Slice that strip vertically into paper-thin pieces, exactly as before. Each thin vertical strip now runs from the *lower* track up to the *upper* track, so its height is not “the curve” but **the gap between the curves**: top – bottom = $f(x) - g(x)$. Its width is still dx , so its area is $[f(x) - g(x)] dx$. Add up all the strips from where the tracks start ($x = a$) to where they end ($x = b$):

$$\text{Area} = \int_a^b [f(x) - g(x)] dx, \quad f(x) \geq g(x) \text{ on } [a, b].$$

That is the entire concept. The single-curve case of Sub-01 is just this formula with the bottom curve being the x -axis, $g(x) = 0$.

Figure / Diagram

Two curves $y = f(x)$ (upper) and $y = g(x)$ (lower) crossing the plane, meeting at $x = a$ and $x = b$. The region between them is shaded. One representative vertical strip runs from the lower curve to the upper curve, labelled height “ $f(x) - g(x)$ ” and width “ dx ”.

Two genuinely new wrinkles appear that did not exist with a single curve:

1. **Where do the strips start and stop?** The limits a and b are usually *not* handed to you. They are the **points of intersection** of the two curves — the places where the strip-height shrinks to zero. You find them by solving $f(x) = g(x)$.
2. **Which curve is on top?** “Top minus bottom” only works if you know which is which. And the two curves may **swap roles**: f above g on one stretch, below it on another. Wherever they cross, the gap flips sign, so you must split the region at each crossing and always subtract the genuine lower curve from the genuine upper one.

Limiting conditions & assumptions:

- You must identify top vs bottom *correctly*, ideally by testing one sample x between the intersection points. Guessing is the number-one source of sign errors here.
- If the curves cross inside $[a, b]$, you **must split** at the crossing(s); a single integral across a swap gives a wrong (too small, sometimes zero) answer.
- For curves that open *sideways* (e.g. $x = \text{function of } y$, like a parabola $y^2 = 4x$), integrating in y ("right minus left") is far cleaner than forcing dx .
- Both functions should be continuous on the interval of integration.

Section 2 — Key Definitions & Properties

Area between two curves, vertical strips (curves $y = f(x)$ and $y = g(x)$, with $f \geq g$ between intersection abscissae $x = a$ and $x = b$):

$$A = \int_a^b [f(x) - g(x)] dx$$

Area between two curves, horizontal strips (curves $x = p(y)$ and $x = q(y)$, with $p \geq q$, i.e. p is the right curve, between $y = c$ and $y = d$):

$$A = \int_c^d [p(y) - q(y)] dy$$

When the curves cross at $x = k$ inside $[a, b]$ ($f \geq g$ on $[a, k]$, then $g \geq f$ on $[k, b]$):

$$A = \int_a^k [f(x) - g(x)] dx + \int_k^b [g(x) - f(x)] dx = \int_a^b |f(x) - g(x)| dx$$

Special case (recovers Sub-01): if the lower curve is the x -axis, $g(x) = 0$, then $A = \int_a^b f(x) dx$.

Notation key:

Symbol	Meaning
$f(x), g(x)$	upper and lower curves (functions of x)
$p(y), q(y)$	right and left curves (functions of y)
a, b	x -coordinates of the points of intersection (limits for dx)
c, d	y -coordinates of the points of intersection (limits for dy)
k	an interior crossing where top and bottom swap

Standard catalogue result (proved in Section 3):

- Area between the two parabolas $y^2 = 4ax$ and $x^2 = 4ay$: $\boxed{\frac{16a^2}{3}}$

Symmetry property: if the whole region is symmetric about an axis, integrate one half and double — but make sure the symmetry is genuine before using it.

Section 3 — Theorems & Proofs

Theorem 1 (The between-curves formula). If f and g are continuous on $[a, b]$ with $f(x) \geq g(x)$ throughout, the area of the region bounded above by $y = f(x)$, below by $y = g(x)$, and on the sides by $x = a$, $x = b$ is

$$A = \int_a^b [f(x) - g(x)] dx.$$

Proof. Take an elementary vertical strip at position x of width dx . Its top edge sits on $y = f(x)$ and its bottom edge on $y = g(x)$, so its height is $f(x) - g(x)$ and its area is $[f(x) - g(x)] dx$. Summing over all strips from a to b and passing to the limit (the Riemann-sum argument of Sub-01) gives $A = \int_a^b [f(x) - g(x)] dx$.

Why this holds even when a curve dips below the x -axis. Suppose $g(x) < 0$ somewhere. Shift the entire picture up by a constant C large enough that both curves rise above the axis. Shifting does not change the gap between them, hence not the area. The shifted area is $\int_a^b [(f + C) - (g + C)] dx = \int_a^b (f - g) dx$ — the constant cancels. So the formula is valid regardless of whether the curves lie above or below the axis, **provided** $f \geq g$. That single proviso is the whole game. ■

Theorem 2 (Area between two equal parabolas). The area enclosed between $y^2 = 4ax$ and $x^2 = 4ay$ (with $a > 0$) is $\frac{16a^2}{3}$.

Proof. Find the intersections. From the second curve, $y = \frac{x^2}{4a}$. Substitute into the first:

$$\left(\frac{x^2}{4a}\right)^2 = 4ax \Rightarrow \frac{x^4}{16a^2} = 4ax \Rightarrow x^4 = 64a^3x \Rightarrow x(x^3 - 64a^3) = 0.$$

So $x = 0$ or $x = 4a$. The curves meet at $(0, 0)$ and $(4a, 4a)$, fixing the limits.

Identify top and bottom. Between $x = 0$ and $x = 4a$, the upper branch of $y^2 = 4ax$ is $y = 2\sqrt{ax}$, while the other parabola gives $y = \frac{x^2}{4a}$. Test $x = a$: $2\sqrt{a \cdot a} = 2a$ versus $\frac{a^2}{4a} = \frac{a}{4}$.

$\frac{a^2}{4a} = \frac{a}{4}$. Since $2a > \frac{a}{4}$, the curve $y = 2\sqrt{ax}$ is on top.

Integrate top minus bottom.

$$A = \int_0^{4a} \left[2\sqrt{ax} - \frac{x^2}{4a} \right] dx = 2\sqrt{a} \cdot \frac{2}{3} x^{3/2} \Big|_0^{4a} - \frac{1}{4a} \cdot \frac{x^3}{3} \Big|_0^{4a}.$$

Now $(4a)^{3/2} = 8a^{3/2}$ and $(4a)^3 = 64a^3$, so

$$A = \frac{4\sqrt{a}}{3}(8a^{3/2}) - \frac{1}{12a}(64a^3) = \frac{32a^2}{3} - \frac{16a^2}{3} = \frac{16a^2}{3}. \quad \blacksquare$$

Check: the answer scales as a^2 (an area), exactly as it must.

Section 4 — Worked Examples

Example 1 — [CBSE Board Level] Find the area of the region bounded by the line $y = x$ and the parabola $y = x^2$.

Intersections: $x = x^2 \Rightarrow x(x - 1) = 0 \Rightarrow x = 0, 1$. So $a = 0$, $b = 1$.

Top vs bottom: test $x = \frac{1}{2}$: line gives 0.5, parabola gives 0.25. The line $y = x$ is on top.

$$A = \int_0^1 (x - x^2) dx = \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$

Area = $\frac{1}{6}$ square units.

Example 2 — [Conceptual Trap / CUET-JEE Main Speed Trap] Find the area enclosed between $y = x$ and $y = x^3$ for $-1 \leq x \leq 1$.

The trap: the hasty move is $\int_{-1}^1 (x - x^3) dx$, which equals 0 because the integrand is odd. But the region clearly has area — the two curves cross at $x = -1, 0, 1$ and swap which is on top.

The fix — find the swap and split. On $(0, 1)$: at $x = 0.5$, $x = 0.5 > x^3 = 0.125$, so $y = x$ is on top. On $(-1, 0)$: at $x = -0.5$, $x^3 = -0.125 > x = -0.5$, so $y = x^3$ is on top. By symmetry the two halves are equal:

$$A = 2 \int_0^1 (x - x^3) dx = 2 \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_0^1 = 2 \left(\frac{1}{2} - \frac{1}{4} \right) = \frac{1}{2}.$$

Area = $\frac{1}{2}$ **square units**. The instant two curves cross inside your interval, split — never integrate straight through a crossing.

Example 3 — [JEE Main Level — multi-concept] Find the area of the region bounded by the parabola $y^2 = 4x$ and the line $y = 2x - 4$.

Concepts combined: intersection-solving, the smart choice to integrate in y , and “right minus left.”

Both curves are easy as functions of y : parabola $x = \frac{y^2}{4}$, line $x = \frac{y+4}{2}$.

Intersections: $\frac{y^2}{4} = \frac{y+4}{2} \Rightarrow y^2 = 2y + 8 \Rightarrow y^2 - 2y - 8 = 0 \Rightarrow (y-4)(y+2) = 0 \Rightarrow y = -2, 4$.

Right vs left: at $y = 0$, line gives $x = 2$, parabola gives $x = 0$; the line is the right curve. Using horizontal strips:

$$A = \int_{-2}^4 \left[\frac{y+4}{2} - \frac{y^2}{4} \right] dy.$$

Compute piecewise: $\int_{-2}^4 \frac{y+4}{2} dy = \frac{1}{2} \left[\frac{y^2}{2} + 4y \right]_{-2}^4 = \frac{1}{2} [24 - (-6)] = 15$, and

$$\int_{-2}^4 \frac{y^2}{4} dy = \frac{1}{12} [y^3]_{-2}^4 = \frac{64 + 8}{12} = 6.$$

$$A = 15 - 6 = 9.$$

Area = 9 **square units**. (Had we used vertical strips, the parabola’s two branches would have forced an awkward two-part integral — choosing dy collapses it to one.)

Example 4 — [JEE Advanced Level] Find the area of the region common to the interior of the parabola $y^2 = 4x$ and the interior of the circle $x^2 + y^2 = 5$.

Intersections. Substitute $y^2 = 4x$ into the circle: $x^2 + 4x = 5 \Rightarrow x^2 + 4x - 5 = 0 \Rightarrow (x+5)(x-1) = 0$. Reject $x = -5$ (the parabola needs $x \geq 0$); so $x = 1$, giving $y = \pm 2$. The curves meet at $(1, 2)$ and $(1, -2)$.

Set up the region (upper half, then double by symmetry about the x -axis). Moving left to right: from $x = 0$ to $x = 1$ the binding ceiling is the **parabola** $y = 2\sqrt{x}$ (it is the smaller of the two there); from $x = 1$ to $x = \sqrt{5}$ the binding ceiling is the **circle** $y = \sqrt{5 - x^2}$.

$$A = 2 \left[\int_0^1 2\sqrt{x} \, dx + \int_1^{\sqrt{5}} \sqrt{5-x^2} \, dx \right].$$

First piece: $\int_0^1 2\sqrt{x} \, dx = 2 \cdot \frac{2}{3} x^{3/2} \Big|_0^1 = \frac{4}{3}.$

Second piece (use $\int \sqrt{a^2 - x^2} \, dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a}$ with $a^2 = 5$):

$$\int_1^{\sqrt{5}} \sqrt{5-x^2} \, dx = \left[\frac{x}{2} \sqrt{5-x^2} + \frac{5}{2} \sin^{-1} \frac{x}{\sqrt{5}} \right]_1^{\sqrt{5}}.$$

At $x = \sqrt{5}$: $0 + \frac{5}{2} \cdot \frac{\pi}{2} = \frac{5\pi}{4}$. At $x = 1$: $\frac{1}{2} \sqrt{4} + \frac{5}{2} \sin^{-1} \frac{1}{\sqrt{5}} = 1 + \frac{5}{2} \sin^{-1} \frac{1}{\sqrt{5}}.$

So the second piece $= \frac{5\pi}{4} - 1 - \frac{5}{2} \sin^{-1} \frac{1}{\sqrt{5}}.$

Combine:

$$A = 2 \left[\frac{4}{3} + \frac{5\pi}{4} - 1 - \frac{5}{2} \sin^{-1} \frac{1}{\sqrt{5}} \right] = \frac{2}{3} + \frac{5\pi}{2} - 5 \sin^{-1} \frac{1}{\sqrt{5}}.$$

Area $= \frac{2}{3} + \frac{5\pi}{2} - 5 \sin^{-1} \frac{1}{\sqrt{5}}$ **square units** ≈ 6.20 sq. units. *Sanity check:* the whole disc has area $5\pi \approx 15.7$, and the common region taking under half of it is geometrically sensible.

Section 5 — Practice Exercises

1. **[CBSE]** Find the area of the region bounded by $y = x^2$ and $y = 2x$.
2. **[CUET / JEE Main]** Find the area enclosed between $y = \sin x$ and $y = \cos x$ from $x = 0$ to $x = \frac{\pi}{4}$. (*Decide which is on top.*)
3. **[JEE Main]** Find the area bounded by the parabola $y^2 = x$ and the line $x = 3y$.
4. **[JEE Main]** Find the area enclosed between the parabolas $y^2 = 4x$ and $x^2 = 4y$.
5. **[JEE Advanced]** Find the area of the region $\{(x, y) : 0 \leq y \leq x^2 + 1, 0 \leq y \leq x + 1, 0 \leq x \leq 2\}$.

Answers: 1. $\frac{4}{3}$ • 2. $\sqrt{2} - 1$ • 3. $\frac{9}{2}$ • 4. $\frac{16}{3}$ • 5. $\frac{23}{6}$

Section 6 — Crack the MCQ

Q1. The area between an upper curve $y = f(x)$ and a lower curve $y = g(x)$ from $x = a$ to $x = b$ is:

(a) $\int_a^b (f + g) dx$ (b) $\int_a^b (f - g) dx$ (c) $\int_a^b (g - f) dx$ (d) $\int_a^b (|f| - |g|) dx$

Correct: (b). Strip height is top minus bottom, $f - g$.

- **(a)** adds the curves — a meaningless quantity here; a common reflex when a student blanks on the rule.
- **(c)** is bottom minus top; it gives the negative of the area (only correct if you mislabelled which curve is upper).
- **(d)** confuses this with the single-curve absolute-value rule from Sub-01; $|f| - |g|$ is not the gap between the curves.

Q2. The area of the region bounded by $y = x^2$ and $y = x$ between their points of intersection is:

(a) $\frac{1}{6}$ (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) $\frac{5}{6}$

Correct: (a). Intersections at 0, 1; line on top; $\int_0^1 (x - x^2) dx = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$.

- (b) is the result of integrating only the parabola $\int_0^1 x^2 dx = \frac{1}{3}$, i.e. forgetting to subtract at all.
- **(c)** is the area under the line alone, $\int_0^1 x dx = \frac{1}{2}$ — again a no-subtraction error.
- **(d)** adds the two single-curve areas $\frac{1}{2} + \frac{1}{3}$ instead of subtracting.

Q3. The area bounded by the line $y = x$ and the parabola $y^2 = 4x$ is:

(a) $\frac{8}{3}$ (b) $\frac{16}{3}$ (c) 8 (d) 4

Correct: (a). Intersections: $x^2 = 4x \Rightarrow x = 0, 4$. On $(0, 4)$ the parabola $y = 2\sqrt{x}$ is above the line $y = x$. $\int_0^4 (2\sqrt{x} - x) dx = \frac{4}{3}x^{3/2} - \frac{x^2}{2} \Big|_0^4 = \frac{32}{3} - 8 = \frac{8}{3}$.

- **(b)** $\frac{16}{3}$ is the area under the parabola alone (forgot to subtract the line).
- **(c)** uses the wrong intersection or integrates only $2\sqrt{x}$ without the $-\frac{x^2}{2}$ correction at the limit.
- **(d)** mistakes the region for a triangle of base and height 4 ($\frac{1}{2} \cdot 4 \cdot 2$ type slip).

Q4. The area enclosed between the parabolas $y^2 = 4x$ and $x^2 = 4y$ is:

- (a) $\frac{16}{3}$ (b) $\frac{32}{3}$ (c) $\frac{8}{3}$ (d) 16

Correct: (a). This is the standard result $\frac{16a^2}{3}$ with $a = 1$.

- **(b)** $\frac{32}{3}$ is the area under the upper parabola alone — the classic “forgot to subtract the lower curve” trap.
- **(c)** halves the correct answer, as if the region were symmetric in a way it is not.
- **(d)** comes from using the wrong upper limit (e.g. $x = 4a = 4$ in a slab formula) without integrating properly.

Section 7 — Common Pitfalls, Pro-Tips & Cheat Sheet

Common pitfalls

1. Subtracting in the wrong order.

Writing bottom – top flips the sign and yields a negative “area.”

Always confirm which curve is higher by plugging in one sample x .

2. Inventing the limits. The limits are the intersection points, found by solving $f(x) = g(x)$ — not numbers copied from the question text. Skipping this step is the most common 0/5 on boards.

3. Integrating across a swap.

If the curves cross inside the interval,

the top/bottom roles switch; integrate $|f - g|$ by splitting at the crossing.

4. Fighting the geometry with the wrong variable.

Sideways-opening parabolas and regions capped left/right are far cleaner in y (right – left). Forcing dx creates needless two-branch casework.

5. Assuming false symmetry. Doubling a half-area only works if the region is truly symmetric about that axis — verify before halving the work.

Pro-Tip — the four-beat routine. *Intersect* → *Sketch* → *Sample* → *Integrate*. Solve $f = g$ for the limits; sketch both curves; test one interior point to fix top vs bottom (and spot any swap); then integrate top – bottom. Doing these in order eliminates the three big mark-losers before they happen.

Pro-Tip — pick the variable by the boundaries. If the region’s *side* boundaries are the curves and its top/bottom are horizontal, use dy . If its top/bottom are the curves and its sides are vertical, use dx . Match the strip to the geometry.

Cheat Sheet Box

- **Vertical strips:** $A = \int_a^b (\text{top} - \text{bottom}) dx = \int_a^b [f(x) - g(x)] dx, \quad f \geq g$
- **Horizontal strips:** $A = \int_c^d (\text{right} - \text{left}) dy = \int_c^d [p(y) - q(y)] dy$
- **Curves that cross:** split at the crossing; $A = \int_a^b |f - g| dx$
- **Limits = intersection points:** solve $f(x) = g(x)$ (or $p(y) = q(y)$) first.

- **Standard result:** parabolas $y^2 = 4ax$ and $x^2 = 4ay$ enclose $\frac{16a^2}{3}$.
- **Special case:** bottom = x -axis recovers Sub-01's single-curve area.
- **Mnemonic — "ISSI":** Intersect, Sketch, Sample (which is on top), Integrate.
Always top minus bottom, every strip, every time.

Subtopic 03 — Advanced Regions: Modulus, Inequalities & Composite Areas

Exam Relevance: This is the JEE layer of the chapter. JEE Main and Advanced regularly hide an area question inside a modulus curve, a system of inequalities, or a region whose ceiling changes partway across — and each looks far scarier than it is. CBSE Boards occasionally set a modulus area (e.g. $y = |x - 2|$) as a 3-mark question. Nothing new is integrated here; the skill is *decoding the boundary* so the problem collapses back into Sub-01 or Sub-02.

Section 1 — Concept Intuition

By now you can find the area under one curve (Sub-01) and between two curves (Sub-02). Every “hard” area problem in JEE is one of those two in disguise — the only extra work is stripping off the disguise. Three disguises appear again and again.

Disguise 1 — the modulus (absolute value). A curve like $y = |x|$ or $y = |x - 2|$ has a *corner*. The modulus simply reflects whatever part of the inside expression is negative back up to positive. So $y = |x|$ is a “V”: the line $y = x$ for $x \geq 0$ and the line $y = -x$ for $x < 0$, joined at a kink at the origin. To handle it, you split the interval at every corner (where the inside changes sign) and integrate each smooth piece separately — exactly the splitting you already do for curves that cross the axis.

Disguise 2 — the inequality. A condition like $y \leq f(x)$ is not a curve; it is a *region* — everything on one side of the curve. A system such as $\{y \geq x^2, y \leq x + 2\}$ is the overlap of two such regions. Your job is to translate each inequality into “which side,” sketch the overlap, and read off the upper and lower boundaries — after which it is a plain Sub-02 problem.

Disguise 3 — the composite region. Sometimes the ceiling of a region is one curve for a while, then switches to a different curve. Picture a Mumbai skyline: the roofline is one building, then a taller one, then a shorter one. To find the area beneath, you split the horizontal range at each point where the tallest boundary changes, integrate each slab, and add. This is the same “split and sum” idea, now driven by *which curve is on top* rather than by a sign change.

Think about it this way... Every one of these is a decoding puzzle followed by routine integration. Decode the boundary — find the corners, decide the sides, locate the switch points — and the integral writes itself. The mistakes happen in the decoding, almost never in the calculus.

Figure / Diagram

Three small panels side by side. Panel 1: the V-shaped graph of $y = |x|$ with a marked corner at the origin, shaded down to the x-axis. Panel 2: a parabola $y = x^2$ and a line $y = x + 2$, with the lens-shaped overlap region $\{y \geq x^2, y \leq x + 2\}$ shaded. Panel 3: a “skyline” where the upper boundary is curve A on the

left and a different curve B on the right, with a dashed vertical line at the switch point and the region beneath shaded.

Limiting conditions & assumptions:

- **Find every corner / switch point.** Missing even one (e.g. the kink of a nested modulus) gives wrong limits and a wrong area.
- **Get the side of an inequality right.** Always test one easy point (often the origin) to decide which side the region is on.
- **Verify symmetry before using it.** Modulus regions are usually symmetric, but only some are; never assume.

Section 2 — Key Definitions & Properties

Modulus expansion: $|f(x)| = \begin{cases} f(x), & f(x) \geq 0 \\ -f(x), & f(x) < 0 \end{cases}$ — split the domain at the roots of f .

Area under a modulus curve:

$$A = \int_a^b |f(x)| dx = \sum_{\text{pieces}} \int (\pm f) dx \quad (\text{one integral per smooth piece}).$$

Standard “diamond” result (rhombus bounded by four modulus-lines):

$$\frac{|x|}{p} + \frac{|y|}{q} \leq 1 \Rightarrow A = 2pq, \quad \text{in particular } |x| + |y| \leq a \Rightarrow A = 2a^2.$$

Inequality decode (which side):

Inequality	Region
$y \leq f(x)$	below the curve
$y \geq f(x)$	above the curve
$x^2 + y^2 \leq r^2$	inside the circle
$x^2 + y^2 \geq r^2$	outside the circle
$y \geq x $	above the V (between the lines $y = x$, $y = -x$)

For a **system**, the region is the *intersection* of these. Confirm with a test point.

Composite region (ceiling switches at $x = s$):

$$A = \int_a^s [\text{top}_1 - \text{bottom}] dx + \int_s^b [\text{top}_2 - \text{bottom}] dx.$$

Symmetry catalogue (integrate the small piece, multiply):

Symmetry	Test	Multiplier
about y -axis	$x \rightarrow -x$ leaves region unchanged	$2\times$ right half
about x -axis	$y \rightarrow -y$ unchanged	$2\times$ upper half
about origin	$(x, y) \rightarrow (-x, -y)$ unchanged	$2\times$ one half
four-fold (e.g. $ x + y \leq a$)	unchanged in all quadrants	$4\times$ first-quadrant

Section 3 — Key Procedures & a Theorem

Procedure A — area under a modulus / piecewise curve.

- (1) Set the inside of each modulus to zero to locate the **corners**. (2) On each sub-interval, replace the modulus by $+f$ or $-f$ according to its sign there. (3) Integrate each piece and add. (4) Use symmetry to shorten the work if the picture is symmetric.

Procedure B — area of a region from inequalities.

- (1) **Decode** each inequality into “which side” (table in Section 2). (2) **Sketch** the boundary curves and shade the overlap. (3) Find the **intersection points** of the boundaries — these are your limits. (4) Read off the genuine top and bottom (or right and left) boundary of the shaded region and integrate top — bottom, splitting if a boundary switches. (5) **Test a point** to confirm you shaded the correct side.

Procedure C — composite region.

- (1) Find where the upper (or lower) boundary **switches** by solving the two candidate boundaries equal to each other. (2) Split the range at each switch point. (3) On each slab use the boundary that is actually extreme there (test a sample point). (4) Sum the slabs. *A triangle bounded by three lines is the simplest composite region — split at the x -coordinate of its apex.*

Theorem (the diamond area). The region $|x| + |y| \leq a$ (with $a > 0$) has area $2a^2$.

Proof. The region is symmetric in all four quadrants (replacing $x \rightarrow -x$ or $y \rightarrow -y$ leaves $|x| + |y| \leq a$ unchanged), so its area is $4\times$ the first-quadrant part. In the first quadrant $|x| = x$ and $|y| = y$, so the condition becomes $x + y \leq a$ with $x, y \geq 0$ — a right triangle with both legs of length a :

$$A = 4 \times \left(\frac{1}{2} a \cdot a \right) = 4 \cdot \frac{a^2}{2} = 2a^2. \quad \blacksquare$$

The same argument on $\frac{|x|}{p} + \frac{|y|}{q} \leq 1$ gives a first-quadrant triangle of legs p and q , hence total area $4 \cdot \frac{1}{2}pq = 2pq$.

Section 4 — Worked Examples

Example 1 — [CBSE Board Level] Find the area of the region bounded by $y = |x|$, the x -axis, and the lines $x = -2$ and $x = 3$.

The corner of $y = |x|$ is at $x = 0$, so split there. On $[-2, 0]$, $|x| = -x$; on $[0, 3]$, $|x| = x$:

$$A = \int_{-2}^0 (-x) dx + \int_0^3 x dx = \left[-\frac{x^2}{2} \right]_{-2}^0 + \left[\frac{x^2}{2} \right]_0^3 = (0 - (-2)) + \frac{9}{2} = 2 + \frac{9}{2} = \frac{13}{2}.$$

Area = $\frac{13}{2}$ square units.

Example 2 — [Conceptual Trap / CUET-JEE Main Speed Trap] Find the area of the region $|x| + |y| \leq 2$.

The trap: students either try to set up a single messy integral with moduli still inside, or they recognise it as a square but miscount the area as “side squared.” The region is a square (a tilted diamond) with vertices $(2, 0)$, $(0, 2)$, $(-2, 0)$, $(0, -2)$.

The clean way — symmetry. By the diamond theorem with $a = 2$:

$$A = 2a^2 = 2(2)^2 = 8.$$

Area = 8 square units. (Quick check: the diagonals are each of length 4, and a square’s area is $\frac{1}{2}d_1d_2 = \frac{1}{2} \cdot 4 \cdot 4 = 8$. ✓)

Example 3 — [JEE Main Level — multi-concept] Find the area of the region $\{(x, y) : y \geq x^2, y \leq x + 2\}$.

Decode: $y \geq x^2$ is above the parabola; $y \leq x + 2$ is below the line. The region is the lens between them, with the **line on top** and the **parabola on the bottom**.

Limits — intersections: $x^2 = x + 2 \Rightarrow x^2 - x - 2 = 0 \Rightarrow (x - 2)(x + 1) = 0 \Rightarrow x = -1, 2$.

$$A = \int_{-1}^2 [(x + 2) - x^2] dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2.$$

At $x = 2$: $2 + 4 - \frac{8}{3} = 6 - \frac{8}{3} = \frac{10}{3}$. At $x = -1$: $\frac{1}{2} - 2 + \frac{1}{3} = -\frac{7}{6}$.

$$A = \frac{10}{3} - \left(-\frac{7}{6}\right) = \frac{20}{6} + \frac{7}{6} = \frac{27}{6} = \frac{9}{2}.$$

Area = $\frac{9}{2}$ square units.

Example 4 — [JEE Advanced Level] Find the area of the region $\{(x, y) : y \geq |x|, x^2 + y^2 \leq 2\}$.

Decode: $y \geq |x|$ is the wedge above both lines $y = x$ and $y = -x$; $x^2 + y^2 \leq 2$ is inside the circle of radius $\sqrt{2}$. The region is the slice of the disc sitting above the V.

Intersections: on $y = |x|$, the circle gives $x^2 + x^2 = 2 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1, y = 1$. The region is symmetric about the y -axis, so take the right half ($0 \leq x \leq 1$, bottom = $y = x$, top = circle $y = \sqrt{2 - x^2}$) and double:

$$A = 2 \int_0^1 [\sqrt{2 - x^2} - x] dx.$$

Using $\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a}$ with $a^2 = 2$:

$$\int_0^1 \sqrt{2 - x^2} dx = \left[\frac{x}{2} \sqrt{2 - x^2} + \sin^{-1} \frac{x}{\sqrt{2}} \right]_0^1 = \frac{1}{2} \cdot 1 + \sin^{-1} \frac{1}{\sqrt{2}} = \frac{1}{2} + \frac{\pi}{4}.$$

And $\int_0^1 x dx = \frac{1}{2}$, so

$$A = 2 \left[\left(\frac{1}{2} + \frac{\pi}{4} \right) - \frac{1}{2} \right] = 2 \cdot \frac{\pi}{4} = \frac{\pi}{2}.$$

Area = $\frac{\pi}{2}$ square units. *Check:* the wedge $y \geq |x|$ subtends a right angle (90°) at the centre, so it is a quarter of the disc: $\frac{1}{4} \cdot \pi(\sqrt{2})^2 = \frac{1}{4} \cdot 2\pi = \frac{\pi}{2}$. ✓ The integration and the geometry agree.

Section 5 — Practice Exercises

1. **[CBSE]** Find the area bounded by $y = |x - 2|$, the x -axis, and the lines $x = 0$, $x = 4$.
2. **[CUET / JEE Main]** Find the area of the region $|x| + |y| \leq 3$.

3. **[JEE Main]** Find the area of the region $\{(x, y) : y \geq x^2, y \leq 4\}$.
4. **[JEE Main]** Find the area of the region enclosed by $y = 2 - x^2$ and $y = |x|$.
5. **[JEE Advanced]** Find the area of the region $\{(x, y) : y \geq |x|, x^2 + y^2 \leq 8\}$.

Answers: 1. 4 • 2. 18 • 3. $\frac{32}{3}$ • 4. $\frac{7}{3}$ • 5. 2π

Section 6 — Crack the MCQ

Q1. The area of the region $|x| + |y| \leq 4$ is:

- (a) 16 (b) 32 (c) 8 (d) 64

Correct: (b). Diamond area $= 2a^2 = 2(4)^2 = 32$.

- **(a)** 16 treats the diamond as a square of side 4 (4^2) — but 4 is the half-diagonal, not the side.
 - **(c)** 8 computes only one of the four triangles ($\frac{1}{2} \cdot 4 \cdot 4$).
 - **(d)** 64 squares the full diagonal $(2a)^2 = 8^2$, ignoring the $\frac{1}{2}$ for the tilt.
-

Q2. The area of the region $\{(x, y) : 0 \leq y \leq 4 - x^2\}$ is:

- (a) $\frac{32}{3}$ (b) $\frac{16}{3}$ (c) $\frac{8}{3}$ (d) 16

Correct: (a). The parabola meets $y = 0$ at $x = \pm 2$; $A = \int_{-2}^2 (4 - x^2) dx = 2 \int_0^2 (4 - x^2) dx = 2(8 - \frac{8}{3}) = \frac{32}{3}$.

- **(b)** integrates only 0 to 2 and forgets to double for symmetry.
 - **(c)** is $\int_0^2 x^2 dx$ -type slip (wrong integrand).
 - **(d)** treats the region as a rectangle $4 \times (\text{width } 4)$, ignoring the curve.
-

Q3. The area bounded by $y = |x|$ and the x -axis between $x = -1$ and $x = 1$ is:

- (a) 1 (b) $\frac{1}{2}$ (c) 2 (d) $\frac{1}{3}$

Correct: (a). By symmetry $A = 2 \int_0^1 x dx = 2 \cdot \frac{1}{2} = 1$.

- **(b)** computes only the right triangle $\int_0^1 x dx = \frac{1}{2}$ and forgets the left half.
 - **(c)** double-counts or uses base $\times$ height $= 2 \cdot 1$ without the $\frac{1}{2}$.
 - **(d)** integrates x^2 instead of $|x|$.
-

Q4. The area of the region $\{(x, y) : y \geq |x|, x^2 + y^2 \leq 8\}$ is:

- (a) 2π (b) π (c) 4π (d) $\frac{\pi}{2}$

Correct: (a). The wedge $y \geq |x|$ is a 90° sector of a disc of radius $\sqrt{8}$: $\frac{1}{4}\pi(\sqrt{8})^2 = \frac{1}{4} \cdot 8\pi = 2\pi$.

- **(b)** halves the sector (uses 45° instead of 90°).
- **(c)** forgets the wedge restriction and takes the whole semicircle $\frac{1}{2} \cdot 8\pi$.
- **(d)** uses radius $\sqrt{2}$ (from a different problem) rather than $\sqrt{8}$.

Section 7 — Common Pitfalls, Pro-Tips & Chapter Master Cheat Sheet

Common pitfalls

1. **Missing a corner of a modulus curve.** Every modulus contributes a kink; set each inside-expression to zero and split there. A nested modulus like $||x| - 1|$ has two corners.
2. **Shading the wrong side of an inequality.** Translate $\leq / \geq$ carefully and confirm with a test point (the origin is easiest when it is not on a boundary).
3. **Forgetting to split a composite region.** When the top boundary switches, one integral cannot cover both slabs.
4. **Assuming symmetry that isn't there — or forgetting to multiply back.** Symmetry is a gift only after you have verified it.

Pro-Tip — decode before you integrate. Spend the first 30 seconds turning the problem into a labelled sketch: corners marked, sides shaded, switch points found. Once the picture is right, the integral is the *easy* part. Almost every lost mark in this subtopic is a decoding error, not an integration error.

Pro-Tip — geometry as a free check. Diamonds, wedges, sectors, and triangles have elementary-geometry areas. Compute the answer both by integration and by the shape formula; if they disagree, you mis-decoded the boundary.

CHAPTER 8 MASTER CHEAT SHEET (covers all three subtopics)

Core formulas

- Curve & x -axis: $A = \int_a^b y \, dx$ | Curve & y -axis: $A = \int_c^d x \, dy$
- Below / crossing axis: $A = \int_a^b |f(x)| \, dx$ (split at roots)
- Between two curves: $A = \int_a^b (\text{top} - \text{bottom}) \, dx = \int_c^d (\text{right} - \text{left}) \, dy$

- Curves that cross: $A = \int_a^b |f - g| dx$ (split at the swap)

Standard results

- Circle $x^2 + y^2 = a^2$: πa^2 | Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$: πab
- Two parabolas $y^2 = 4ax$, $x^2 = 4ay$: $\frac{16a^2}{3}$ | Diamond $|x| + |y| \leq a$: $2a^2$
- Workhorse integral: $\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C$

Method-selection guide — which tool for which problem

- One curve + an axis $\rightarrow$ **Sub-01** ($\int y dx$ or $\int x dy$).
- Two named curves, or curve + line $\rightarrow$ **Sub-02** (top – bottom; find intersections).
- Modulus / inequality / “region $\{ \dots \}$ ” / boundary that switches $\rightarrow$ **Sub-03** (decode $\rightarrow$ sketch $\rightarrow$ split $\rightarrow$ integrate).

The universal four-step routine: **Sketch** $\rightarrow$ **Limits (intersections/corners)** $\rightarrow$ **Pick variable (dx vs dy)** $\rightarrow$ **Integrate (with $|\cdot|$, splitting, and symmetry).**

Memory aids: “**Top minus bottom, every strip, every time.**” • “**Sideways curves $\rightarrow$ integrate in y .**” • “**Decode the boundary; the calculus is the easy part.**”

Chapter 8: Application of Integrals — Round 2

Addendum

The chapter taught you to decode a boundary — corners, sides, switch points — and then run top-minus-bottom. That settles most of the paper, but five examinable patterns stand entirely outside it. Addendum A handles segments of conics cut off by chords, tangents and the latus rectum, where Archimedes' triangle gives you a free check the chapter never mentions. Addendum B covers curves you cannot solve for y at all — the astroid and the cycloid — which arrive as parametric equations and are permanent JEE fixtures. Addendum C develops areas built on inverse functions, including the mirror trick that halves the work. Addendum D reverses the arrow of the whole chapter: the area is given and a parameter is the unknown. Addendum E systematises multi-hump periodic areas, where blind integration across several periods guarantees a wrong answer. Nothing here is new calculus; every method is the chapter's strip idea wearing a new disguise.

Addendum A — Segments of Conics: Chords, Tangents and the Latus Rectum

Subtopic 02 finds areas between whole curves. Exams frequently cut a conic with a *line* — the latus rectum of a parabola, a chord, a tangent — and ask for the smaller piece. The mechanics are Sub-02's; the new skills are rejecting extraneous intersection roots, remembering that a sideways parabola has **two** branches, and checking the answer against elementary geometry.

Method

1. **Find the cutting line and the conic.** Write the conic's branches explicitly: $y = \pm 2\sqrt{ax}$ for $y^2 = 4ax$, or $y = \pm \frac{b}{a}\sqrt{a^2 - x^2}$ for the ellipse.
2. **Intersect.** Substitute and solve. If you squared anything on the way, test each root in the *original* equations — squaring manufactures roots.
3. **Decide the region.** "Smaller segment" or "cut off by" tells you which side of the line to keep. Sample one interior point.
4. **Integrate top minus bottom** (or right minus left), exploiting symmetry about the conic's axis wherever the chord respects it.
5. **Check with geometry** — the lemmas below.

Lemma A.1 (circle segment). The smaller segment of $x^2 + y^2 = r^2$ lying to the right of the chord $x = c$, where $0 \leq c \leq r$, has area

$$S = r^2 \cos^{-1} \frac{c}{r} - c \sqrt{r^2 - c^2}.$$

Proof. By symmetry about the x -axis,

$$S = 2 \int_c^r \sqrt{r^2 - x^2} dx.$$

Using the catalogue antiderivative with $a = r$:

$$\int_c^r \sqrt{r^2 - x^2} dx = \left[\frac{x}{2} \sqrt{r^2 - x^2} + \frac{r^2}{2} \sin^{-1} \frac{x}{r} \right]_c^r.$$

At $x = r$ this is $\frac{r^2}{2} \cdot \frac{\pi}{2}$. At $x = c$ it is $\frac{c}{2} \sqrt{r^2 - c^2} + \frac{r^2}{2} \sin^{-1} \frac{c}{r}$. Doubling and using $\frac{\pi}{2} - \sin^{-1} \frac{c}{r} = \cos^{-1} \frac{c}{r}$ gives the stated formula. ■

Lemma A.2 (ellipse segment by scaling). The map $(x, y) \mapsto (X, Y)$ with $X = \frac{x}{a}$, $Y = \frac{y}{b}$ sends the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ to the unit circle, and every area is multiplied by ab : a thin rectangle of size $dx \times dy$ becomes one of size $dX \times dY = \frac{dx}{a} \cdot \frac{dy}{b}$, so $dx dy = ab dX dY$. Therefore the ellipse segment cut by $x = h$ is ab times the unit-circle segment cut by $X = \frac{h}{a}$:

$$S = ab \left[\cos^{-1} \frac{h}{a} - \frac{h}{a^2} \sqrt{a^2 - h^2} \right].$$

Lemma A.3 (Archimedes' parabolic segment). For the parabola $y^2 = 4ax$, the segment cut off by the chord $x = h$ has area exactly $\frac{4}{3}$ of the triangle whose base is that chord and whose apex is the vertex.

Proof. The chord meets the parabola at $(h, \pm 2\sqrt{ah})$. The segment between chord and arc is, by vertical strips,

$$S = \int_0^h (2\sqrt{ax} - (-2\sqrt{ax})) dx = 4\sqrt{a} \cdot \frac{2}{3} h^{3/2} = \frac{8}{3} \sqrt{a} h^{3/2}.$$

The triangle with vertices $(0, 0)$, $(h, 2\sqrt{ah})$, $(h, -2\sqrt{ah})$ has base $4\sqrt{ah}$ and height h :

$$T = \frac{1}{2} \cdot 4\sqrt{ah} \cdot h = 2\sqrt{a} h^{3/2}.$$

Then $S = \frac{4}{3}T$. The tangent parallel to a vertical chord is the vertical tangent $x = 0$, touching at the vertex — which is why the vertex is the apex. The ratio $\frac{4}{3}$ survives any affine change of coordinates, so it holds for *every* chord of a parabola, not just vertical ones. ■

This lemma is a two-line check on almost any parabola-segment answer, and it exposes the single most common slip in this pattern: forgetting the lower branch and reporting half the segment.

Example A.1 — [CBSE / JEE Main Level]

Find the area bounded by the parabola $y^2 = 4x$ and its latus rectum.

Here $a = 1$, so the latus rectum is the vertical line $x = 1$, meeting the parabola where $y^2 = 4$, i.e. at $(1, 2)$ and $(1, -2)$.

The trap. The region is *not* bounded by the x -axis. It spans both branches, so doubling the area under the upper branch above the axis would be wrong — that computes a different region. The strip runs from the lower branch to the upper branch:

$$A = \int_0^1 (2\sqrt{x} - (-2\sqrt{x})) dx = 4 \int_0^1 x^{1/2} dx = 4 \cdot \frac{2}{3} = \frac{8}{3}.$$

Geometric check. By Lemma A.3 with $a = 1$, $h = 1$: the triangle on the latus rectum as base has area $\frac{1}{2}(4)(1) = 2$, and the segment is $\frac{4}{3} \times 2 = \frac{8}{3}$. Agreement.

$$\text{Area} = \frac{8}{3} \text{ square units.}$$

Example A.2 — [JEE Main Level]

Find the area bounded by the parabola $y = 4x - x^2$ and the chord joining $(1, 3)$ and $(3, 3)$.

The chord is $y = 3$. Intersections: $4x - x^2 = 3$ gives $x^2 - 4x + 3 = 0$, so $x = 1, 3$ — matching the given points, no extraneous roots. Test $x = 2$: parabola gives 4, chord gives 3, so the parabola is on top.

$$A = \int_1^3 (4x - x^2 - 3) dx = \left[2x^2 - \frac{x^3}{3} - 3x \right]_1^3 = 0 - \left(-\frac{4}{3} \right) = \frac{4}{3}.$$

Archimedes check. The tangent parallel to the horizontal chord sits where $\frac{dy}{dx} = 4 - 2x = 0$, i.e. at the apex $(2, 4)$. Triangle: base 2 (from $(1, 3)$ to $(3, 3)$), height 1, area 1. Segment $= \frac{4}{3}(1) = \frac{4}{3}$. Agreement.

Example A.3 — [JEE Advanced Level]

Find the area of the smaller region into which the line $x = 2$ cuts the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$.

Here $a = 4$, $b = 3$, $h = 2$. The smaller piece is the cap to the right of the line. By symmetry about the x -axis, take the upper half and double:

$$A = 2 \int_2^4 \frac{3}{4} \sqrt{16 - x^2} dx = \frac{3}{2} \left[\frac{x}{2} \sqrt{16 - x^2} + 8 \sin^{-1} \frac{x}{4} \right]_2^4.$$

At $x = 4$: $0 + 8 \cdot \frac{\pi}{2} = 4\pi$. At $x = 2$: $\frac{2}{2} \sqrt{12} + 8 \sin^{-1} \frac{1}{2} = 2\sqrt{3} + \frac{4\pi}{3}$. So

$$A = \frac{3}{2} \left(4\pi - 2\sqrt{3} - \frac{4\pi}{3} \right) = \frac{3}{2} \left(\frac{8\pi}{3} - 2\sqrt{3} \right) = 4\pi - 3\sqrt{3}.$$

Independent check by scaling (Lemma A.2). Scale to the unit circle; the line becomes $X = \frac{1}{2}$ and areas scale by $ab = 12$. The unit-circle segment beyond $X = \frac{1}{2}$ is, by Lemma A.1 with $r = 1$, $c = \frac{1}{2}$:

$$\cos^{-1} \frac{1}{2} - \frac{1}{2} \sqrt{\frac{3}{4}} = \frac{\pi}{3} - \frac{\sqrt{3}}{4}.$$

Multiplying by 12: $4\pi - 3\sqrt{3}$. Two routes, one answer.

Sanity check. The quarter-ellipse has area $3\pi \approx 9.42$; the cap $4\pi - 3\sqrt{3} \approx 7.37$ is somewhat smaller — plausible, since $x = 2$ sits at half the semi-major axis.

Practice — Addendum A

1. Find the area bounded by $y^2 = 8x$ and its latus rectum.
2. Find the area bounded by $y = 6x - x^2$ and the chord joining $(1, 5)$ and $(5, 5)$.
3. Find the area of the smaller segment of the circle $x^2 + y^2 = 25$ cut off by the line $x = 3$.
4. Find the area between $y^2 = 4x$ and the double ordinate $x = 4$.
5. Find the area of the smaller region into which $y = 1$ cuts the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$.

Answers: 1. Here $a = 2$; by Lemma A.3 the triangle on the latus rectum has area $\frac{1}{2}(8)(2) = 8$, so the segment is $\frac{32}{3}$. 2. Intersections $x = 1, 5$; parabola on top; $\int_1^5 (6x - x^2 - 5) dx = \frac{32}{3}$. Check: apex $(3, 9)$, triangle $\frac{1}{2}(4)(4) = 8$, and $\frac{4}{3} \times 8 = \frac{32}{3}$. 3. Lemma A.1 with $r = 5$, $c = 3$: $25 \cos^{-1} \frac{3}{5} - 3(4) = 25 \cos^{-1} \frac{3}{5} - 12$. 4. Lemma A.3 with $a = 1$, $h = 4$: triangle $\frac{1}{2}(8)(4) = 16$, segment $\frac{64}{3}$. 5. Swap roles in Lemma A.2 (vertical semi-axis $b = 3$ carries the cut): $6 [\cos^{-1} \frac{1}{3} - \frac{1}{9} \sqrt{8}] = 6 \cos^{-1} \frac{1}{3} - \frac{4\sqrt{2}}{3}$.

Addendum B — Parametric Areas: the Astroid and the Cycloid

Some of the most famous area answers in JEE — the astroid's $\frac{3\pi a^2}{8}$, the cycloid arch's $3\pi a^2$ — belong to curves that are miserable or impossible to write as $y = f(x)$. They arrive as parametric equations, and the strip idea converts to t in one line. This subtopic assumes nothing the chapter did not prove; it only changes the variable the strips are counted in.

Method

Suppose a curve is given as $x = x(t)$, $y = y(t)$ for $\alpha \leq t \leq \beta$, traced exactly once, with $y \geq 0$ and x increasing. Partition t ; the strip at t has height $y(t)$ and width $\Delta x \approx x'(t) \Delta t$, so

$$A = \int_{\alpha}^{\beta} y(t) x'(t) dt.$$

If x *decreases* as t increases, the integral comes out negative — take the absolute value, or reverse the limits. Multiply by the symmetry multiplier as usual.

Tool (Wallis' formulas). For $n \geq 2$,

$$I_n = \int_0^{\pi/2} \sin^n t \, dt = \frac{n-1}{n} I_{n-2}, \quad I_0 = \frac{\pi}{2}, \quad I_1 = 1.$$

Proof of the reduction. Integrate by parts with $u = \sin^{n-1} t$, $dv = \sin t \, dt$:

$$I_n = \left[-\cos t \sin^{n-1} t \right]_0^{\pi/2} + (n-1) \int_0^{\pi/2} \sin^{n-2} t \cos^2 t \, dt.$$

The bracket vanishes at both ends. Writing $\cos^2 t = 1 - \sin^2 t$:

$$I_n = (n-1)(I_{n-2} - I_n) \implies n I_n = (n-1) I_{n-2}. \quad \blacksquare$$

$$\text{So } I_2 = \frac{\pi}{4}, \quad I_4 = \frac{3}{4} \cdot \frac{\pi}{4} = \frac{3\pi}{16}, \quad I_6 = \frac{5}{6} \cdot \frac{3\pi}{16} = \frac{5\pi}{32}.$$

Example B.1 — [JEE Main / Advanced Level]

Find the area enclosed by the astroid $x^{2/3} + y^{2/3} = a^{2/3}$, $a > 0$.

Parametrise: $x = a \cos^3 t$, $y = a \sin^3 t$. Check it satisfies the equation: $x^{2/3} + y^{2/3} = a^{2/3} \cos^2 t + a^{2/3} \sin^2 t = a^{2/3}$. ✓

The curve is unchanged by $x \rightarrow -x$ and by $y \rightarrow -y$, so compute the first-quadrant part and multiply by 4. There t runs from $\frac{\pi}{2}$ (where $x = 0$) down to 0 (where $x = a$), and $x'(t) = -3a \cos^2 t \sin t$. Since x decreases with t , flip the limits:

$$A_Q = \int_0^{\pi/2} a \sin^3 t \cdot 3a \cos^2 t \sin t \, dt = 3a^2 \int_0^{\pi/2} \sin^4 t \cos^2 t \, dt.$$

Write $\cos^2 t = 1 - \sin^2 t$ and apply Wallis:

$$\int_0^{\pi/2} \sin^4 t \, dt - \int_0^{\pi/2} \sin^6 t \, dt = \frac{3\pi}{16} - \frac{5\pi}{32} = \frac{\pi}{32}.$$

So $A_Q = \frac{3\pi a^2}{32}$, and

$$A = 4 \cdot \frac{3\pi a^2}{32} = \frac{3\pi a^2}{8}.$$

Check. The astroid sits inside the circle $x^2 + y^2 = a^2$ of area πa^2 , and $\frac{3\pi a^2}{8} \approx 0.37 \pi a^2 < \pi a^2$. Plausible: the astroid is much thinner than its circumscribed circle.

Example B.2 — [JEE Advanced Level]

Find the area under one arch of the cycloid $x = a(t - \sin t)$, $y = a(1 - \cos t)$, $0 \leq t \leq 2\pi$, **above the x -axis.**

Here $x'(t) = a(1 - \cos t) \geq 0$, so x increases from 0 to $2\pi a$ and $y = a(1 - \cos t) \geq 0$ throughout. No flipping, no splitting:

$$A = \int_0^{2\pi} a(1 - \cos t) \cdot a(1 - \cos t) dt = a^2 \int_0^{2\pi} (1 - \cos t)^2 dt.$$

Expand and integrate term by term. $\int_0^{2\pi} 1 dt = 2\pi$; $\int_0^{2\pi} \cos t dt = 0$; and by the half-angle identity $\cos^2 t = \frac{1 + \cos 2t}{2}$,

$$\int_0^{2\pi} \cos^2 t dt = \pi + 0 = \pi.$$

Hence

$$A = a^2 (2\pi - 0 + \pi) = 3\pi a^2.$$

Check. The arch spans a rectangle of width $2\pi a$ and height $2a$, area $4\pi a^2$. The answer $3\pi a^2$ is $\frac{3}{4}$ of that — a famously tidy fraction, and comfortably inside the bounding rectangle.

Example B.3 — [Consistency check, JEE Main Level]

Recover the ellipse area parametrically. With $x = a \cos t$, $y = b \sin t$, the first quadrant corresponds to t running from $\frac{\pi}{2}$ to 0:

$$A_Q = \int_0^{\pi/2} b \sin t \cdot a \cos t dt = ab \int_0^{\pi/2} \sin t \cos t dt = ab \cdot \frac{\pi}{4},$$

using $I_2 = \frac{\pi}{4}$. Total = πab , agreeing with Theorem 3 of Subtopic 01. A method that reproduces a known result on a known shape is safe to trust on a new one.

Practice — Addendum B

1. Find the area enclosed by $x^{2/3} + y^{2/3} = 4^{2/3}$.
2. Find the area under one arch of $x = 2(t - \sin t)$, $y = 2(1 - \cos t)$.
3. Find the area enclosed by $x = \cos^3 t$, $y = \sin^3 t$, $0 \leq t \leq 2\pi$.
4. Using $x = 3 \cos t$, $y = 2 \sin t$, find the area of $\frac{x^2}{9} + \frac{y^2}{4} = 1$.
5. Find the area under one arch of $x = a(t + \sin t)$, $y = a(1 - \cos t)$, $0 \leq t \leq 2\pi$. (It is *not* $3\pi a^2$ — this is a different curve.)

Answers: 1. Astroid with $a = 4$: $\frac{3\pi(16)}{8} = 6\pi$. 2. Scale factor 2 on the standard cycloid: $3\pi(2)^2 = 12\pi$. 3. Astroid with $a = 1$: $\frac{3\pi}{8}$. 4. $\pi(3)(2) = 6\pi$. 5. Here $x'(t) = a(1 + \cos t)$, so $A = a^2 \int_0^{2\pi} (1 - \cos t)(1 + \cos t) dt = a^2 \int_0^{2\pi} \sin^2 t dt = \pi a^2$ — the factor $(1 - \cos^2 t)$ kills the 3π term. Always expand before you assume.

Addendum C — Inverse Functions and the Mirror Trick

The chapter integrates functions as given. JEE regularly builds areas on $\sin^{-1} x$, $\tan^{-1} x$, logarithms and cube roots — functions whose *inverses* are the familiar ones — and on pairs like $y = x^3$ and $y = x^{1/3}$ that are inverses of each other. One geometric fact about reflection does most of the work.

Method

Fact C.1 (the mirror). The graph of f^{-1} is the reflection of the graph of f in the line $y = x$.

Proof. (a, b) lies on the graph of f means $f(a) = b$, which means $f^{-1}(b) = a$, which means (b, a) lies on the graph of f^{-1} . Reflection in $y = x$ is exactly the map $(a, b) \mapsto (b, a)$. ■

Proposition C.2 (mirror trick). Let f be continuous and strictly increasing with $f(a) = a$ and $f(b) = b$, $a < b$, and suppose $f(x) \geq x$ on $[a, b]$. Then the area enclosed between $y = f(x)$ and $y = f^{-1}(x)$ equals

$$2 \int_a^b (f(x) - x) dx.$$

Proof. The diagonal $y = x$ cuts the enclosed region into two pieces. Reflection in $y = x$ fixes the diagonal pointwise and swaps the two graphs (Fact C.1), so it maps the piece above the diagonal exactly onto the piece below it: the two pieces are congruent. The piece above the diagonal is bounded above by $y = f(x)$ and below by $y = x$, with area $\int_a^b (f(x) - x) dx$. Double it. ■

Antiderivative toolkit, each by one integration by parts with $u = \sin^{-1} x$ (etc.), $dv = dx$:

$$\int \sin^{-1} x \, dx = x \sin^{-1} x + \sqrt{1 - x^2} + C,$$

$$\int \cos^{-1} x \, dx = x \cos^{-1} x - \sqrt{1 - x^2} + C,$$

$$\int \tan^{-1} x \, dx = x \tan^{-1} x - \frac{1}{2} \ln(1 + x^2) + C.$$

For instance, $\int \frac{x}{\sqrt{1-x^2}} dx = -\sqrt{1-x^2}$ by the substitution $u = 1 - x^2$, which produces the first line.

Example C.1 — [JEE Main Level]

Find the area of the region bounded by $y = \sin^{-1} x$, $y = \cos^{-1} x$ and the x -axis.

Both curves live over $0 \leq x \leq 1$ and are non-negative there. They cross where $\sin^{-1} x = \cos^{-1} x$, i.e. at $x = \frac{1}{\sqrt{2}}$ (both equal $\frac{\pi}{4}$). Sampling $x = 0$: $\cos^{-1} 0 = \frac{\pi}{2} > 0 = \sin^{-1} 0$, so $\cos^{-1} x$ is on top on the left piece.

$$A = \int_0^{1/\sqrt{2}} \cos^{-1} x \, dx + \int_{1/\sqrt{2}}^1 \sin^{-1} x \, dx.$$

First piece, with $F(x) = x \cos^{-1} x - \sqrt{1-x^2}$:

$$F\left(\frac{1}{\sqrt{2}}\right) - F(0) = \left(\frac{\pi}{4\sqrt{2}} - \frac{1}{\sqrt{2}}\right) - (-1) = \frac{\pi}{4\sqrt{2}} - \frac{1}{\sqrt{2}} + 1.$$

Second piece, with $G(x) = x \sin^{-1} x + \sqrt{1-x^2}$:

$$G(1) - G\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{2} - \left(\frac{\pi}{4\sqrt{2}} + \frac{1}{\sqrt{2}}\right).$$

Adding, the $\frac{\pi}{4\sqrt{2}}$ terms cancel:

$$A = \frac{\pi}{2} + 1 - \frac{2}{\sqrt{2}} = \frac{\pi}{2} + 1 - \sqrt{2}.$$

Check. Numerically ≈ 1.157 . The region is a sliver of width 1 and average height a bit over 1 — plausible. A negative or near-zero answer here means you forgot the -1 from the lower limit of the first piece, the classic slip in this exact problem.

Example C.2 — [JEE Main Level]

Find the area enclosed between $y = x^3$ and $y = x^{1/3}$.

These are mutual inverses, meeting on the diagonal at $x = -1, 0, 1$. Two lobes, symmetric about the origin. On $(0, 1)$, $x^{1/3} \geq x \geq x^3$, so by Proposition C.2 each lobe has area

$$2 \int_0^1 (x^{1/3} - x) \, dx = 2 \left(\frac{3}{4} - \frac{1}{2} \right) = \frac{1}{2}.$$

Total area $= 2 \times \frac{1}{2} = 1$.

Direct check. $2 \int_0^1 (x^{1/3} - x^3) \, dx = 2 \left(\frac{3}{4} - \frac{1}{4} \right) = 1$. Agreement — and notice how the mirror trick replaced the awkward $x^{1/3}$ versus x^3 comparison with the easy $x^{1/3}$ versus x .

Example C.3 — [JEE Advanced Level]

Find the area bounded by $y = \tan^{-1} x$, $y = \cot^{-1} x$ and the y -axis.

Use the identity $\cot^{-1} x = \frac{\pi}{2} - \tan^{-1} x$ for $x > 0$: both sides lie in $(0, \frac{\pi}{2})$ and both have tangent equal to $\frac{1}{x}$, so they are equal. The curves meet where $\tan^{-1} x = \frac{\pi}{4}$, i.e. at $x = 1$. On $[0, 1]$ the cotangent curve is on top (at $x = 0$: $\frac{\pi}{2}$ versus 0).

$$A = \int_0^1 (\cot^{-1} x - \tan^{-1} x) \, dx = \int_0^1 \left(\frac{\pi}{2} - 2 \tan^{-1} x \right) \, dx.$$

Using the toolkit antiderivative:

$$A = \frac{\pi}{2} - 2 \left[x \tan^{-1} x - \frac{1}{2} \ln(1 + x^2) \right]_0^1 = \frac{\pi}{2} - 2 \left(\frac{\pi}{4} - \frac{1}{2} \ln 2 \right) = \ln 2.$$

Check. Numerically ≈ 0.693 . The gap starts at $\frac{\pi}{2} \approx 1.57$ at the y -axis and closes to 0 at $x = 1$; averaging roughly $\frac{\pi}{4}$ over a width of 1 predicts about 0.7. Agreement.

Practice — Addendum C

1. Find the area enclosed between $y = x^2$ and $y = \sqrt{x}$.
2. Find the area bounded by $y = \tan^{-1} x$, the y -axis, and $y = \frac{\pi}{4}$.
3. Find the area bounded by $y = \ln x$, $y = 1$ and $x = 1$.
4. Find the area bounded by $y = \cos^{-1} x$, the x -axis, $x = 0$ and $x = 1$.
5. Find the area bounded by $y = \sin^{-1} x$, $y = \frac{\pi}{2}$ and the y -axis, and explain its relation to Question 4.

Answers: 1. Mutual inverses on $[0, 1]$; mirror trick: $2 \int_0^1 (\sqrt{x} - x) dx = 2 \left(\frac{2}{3} - \frac{1}{2} \right) = \frac{1}{3}$. 2. Rewrite as $x = \tan y$ and integrate horizontally: $\int_0^{\pi/4} \tan y dy = [-\ln(\cos y)]_0^{\pi/4} = \ln \sqrt{2} = \frac{1}{2} \ln 2$. 3. $\int_1^e (1 - \ln x) dx = (e - 1) - 1 = e - 2$, since $\int_1^e \ln x dx = 1$. 4. $[x \cos^{-1} x - \sqrt{1 - x^2}]_0^1 = 0 - (-1) = 1$. 5. $\int_0^1 \left(\frac{\pi}{2} - \sin^{-1} x \right) dx = \frac{\pi}{2} - \left(\frac{\pi}{2} - 1 \right) = 1$ — the same area, because the second region is the mirror image of the first in the line $y = x$ (Fact C.1): reflecting Question 4's region swaps $\cos^{-1}$ for $\sin^{-1}$ and the x -axis for the line $y = \frac{\pi}{2}$.

Addendum D — When the Area Is the Unknown: Hunting a Parameter

Every problem so far handed you the curves and asked for a number. JEE Main and Advanced frequently reverse this: the area is specified, and a coefficient, a cutting line, or a point of tangency is the unknown. The skill is to write the area as a *function* of the parameter — limits usually depend on it — then solve. Two tools carry most instances.

Method

1. **Name the parameter** k and write every boundary in terms of it.
2. **Find the limits as functions of** k by solving the relevant intersections. Do not plug numbers yet.
3. **Form** $A(k)$ and impose the condition ($A = \text{target}$, or A maximal, or a ratio).
4. **Solve, then sanity-check:** confirm $A(k)$ moves in the right direction as k changes, and that the answer's location is geometrically sensible.

Tool (root-gap lemma). If $q(x) = c(x - r_1)(x - r_2)$ with $r_1 < r_2$, then

$$\int_{r_1}^{r_2} q(x) dx = -\frac{c(r_2 - r_1)^3}{6}.$$

Proof. Substitute $t = x - r_1$, so t runs from 0 to $L = r_2 - r_1$:

$$\int_0^L c t (t - L) dt = c \left(\frac{L^3}{3} - \frac{L^3}{2} \right) = -\frac{cL^3}{6}. \quad \blacksquare$$

Consequence: whenever the region between a line and a parabola is bounded by their two intersection points, its area depends only on the **gap between the roots** — a one-line computation that replaces the full antiderivative.

Example D.1 — [JEE Main Level]

Find k such that the line $y = k$ divides the region $\{(x, y) : 0 \leq y \leq 4 - x^2\}$ into two parts of equal area.

Total area first:

$$A_{\text{tot}} = \int_{-2}^2 (4 - x^2) dx = 2 \left[4x - \frac{x^3}{3} \right]_0^2 = 2 \left(8 - \frac{8}{3} \right) = \frac{32}{3}.$$

The line $y = k$ (with $0 < k < 4$) meets the parabola where $4 - x^2 = k$, i.e. at $x = \pm u$ where $u = \sqrt{4 - k}$. The upper part is the cap between the line and the parabola:

$$A_{\text{up}} = \int_{-u}^u (4 - x^2 - k) dx = 2 \left[(4 - k)u - \frac{u^3}{3} \right].$$

Since $4 - k = u^2$, this collapses:

$$A_{\text{up}} = 2 \left(u^3 - \frac{u^3}{3} \right) = \frac{4u^3}{3}.$$

Set it equal to half the total, $\frac{16}{3}$:

$$\frac{4u^3}{3} = \frac{16}{3} \implies u^3 = 4 \implies u = \sqrt[3]{4}.$$

Therefore

$$k = 4 - u^2 = 4 - 4^{2/3} = 4 - \sqrt[3]{16} \approx 1.48.$$

Sanity checks. As k rises from 0 to 4, the cap shrinks continuously from the whole region to nothing, so the solution is unique. And the answer sits *below* the region's mean height $\bar{y} = 1.6$ (computable by $\iint y dA$): the region is wider near its base, so the equal-area line must sit below the balance point. Both directions agree.

Example D.2 — [JEE Advanced Level]

Among all lines $y = mx + 3$, find the one that minimises the area enclosed between it and the parabola $y = x^2$, and find that minimum area.

Intersections: $x^2 = mx + 3$, i.e. $x^2 - mx - 3 = 0$. Its discriminant $m^2 + 12$ is always positive, so there are always two roots $r_1 < r_2$ with gap $r_2 - r_1 = \sqrt{m^2 + 12}$. Between them the line is on top (test $x = 0$: line gives 3, parabola gives 0). By the root-gap lemma with $c = -1$ applied to $q = \text{line} - \text{parabola}$:

$$A(m) = \frac{(r_2 - r_1)^3}{6} = \frac{(m^2 + 12)^{3/2}}{6}.$$

This is minimised when $m^2 + 12$ is, i.e. at $m = 0$:

$$A_{\min} = \frac{12^{3/2}}{6} = \frac{24\sqrt{3}}{6} = 4\sqrt{3}.$$

Why it is Advanced-level. The parameter appears inside the limits, not just the integrand, and the winning move is refusing to write the roots out at all — the root-gap lemma shows the area depends on m only through the discriminant. A student who grinds through r_1 and r_2 explicitly will get there, slowly; the lemma gets there in two lines and generalises instantly.

Check. $4\sqrt{3} \approx 6.93$. The horizontal line $y = 3$ cuts the parabola at $\pm\sqrt{3}$, and the lens is roughly a parabolic segment of base $2\sqrt{3}$ and height 3 — of order $\frac{2}{3} \times 2\sqrt{3} \times 3 \approx 6.9$. Plausible.

Practice — Addendum D

1. Find $k > 0$ such that $y = k$ bisects the region $0 \leq y \leq 9 - x^2$.
2. Find all m for which the area between $y = x^2$ and $y = mx + 3$ equals $\frac{32}{3}$.
3. Find $a > 0$ such that the area bounded by $y = ax - x^2$ and the x -axis equals $\frac{9}{2}$.
4. The line $y = c$ (with $0 < c < 2$) cuts the diamond $|x| + |y| \leq 2$ into a lower and an upper part. Find c if their areas are in ratio 3 : 1 (lower : upper).
5. Find a such that the area enclosed between $y = x^2$ and $y = a - x^2$ equals 9.

Answers: 1. Total $\frac{4 \cdot 27}{3} = 36$; cap $= \frac{4u^3}{3}$ with $u^2 = 9 - k$; setting $\frac{4u^3}{3} = 18$ gives $u^3 = \frac{27}{2}$, so $u = \frac{3}{\sqrt[3]{2}}$ and $k = 9 - \frac{9}{\sqrt[3]{4}} = 9(1 - 2^{-2/3}) \approx 3.33$. 2. $\frac{(m^2+12)^{3/2}}{6} = \frac{32}{3}$ gives $(m^2 + 12)^{3/2} = 64$, so $m^2 + 12 = 16$ and $m = \pm 2$. 3. Roots 0 and a ; area $= \frac{a^3}{6}$; setting $\frac{a^3}{6} = \frac{9}{2}$ gives $a = 3$. 4. The diamond has area 8; the cap above $y = c$ is a triangle with base $2(2 - c)$ and height $2 - c$, area $(2 - c)^2$; ratio 3 : 1 forces the cap to be 2, so $(2 - c)^2 = 2$ and $c = 2 - \sqrt{2}$. 5. Curves meet at $x = \pm\sqrt{a/2}$; area $= \int_{-\sqrt{a/2}}^{\sqrt{a/2}} (a - 2x^2) dx = \frac{4a}{3} \sqrt{\frac{a}{2}}$; setting this to 9 gives $a^{3/2} = \frac{27\sqrt{2}}{4}$, hence $a = \frac{9}{2}$ (check: $\frac{4 \cdot 4.5}{3} \times 1.5 = 9$ ✓).

Addendum E — Multi-Hump Periodic Areas

The chapter splits a trig integral at each sign change — correct, and hopeless at scale. Asked for the area between $\sin x$ and $\cos x$ over $[0, 2\pi]$, listing every crossing and computing four integrals wastes minutes and invites slips. Three structural facts replace the casework.

Method

Fact E.1 (hump counting). Each arch of $y = \sin x$ between consecutive multiples of π has area

$$\int_0^\pi \sin x \, dx = [-\cos x]_0^\pi = 2.$$

Since $|\sin(x + \pi)| = |\sin x|$, every arch of $|\sin x|$ is a translate of the first, so over any interval made of n full half-periods,

$$\text{area under } |\sin x| = 2n.$$

The same holds for $|\cos x|$.

Fact E.2 (shift invariance). If f has period T , then $\int_a^{a+T} f$ is the same for every a . *Proof:* split $\int_a^{a+T} = \int_a^T + \int_T^{a+T}$ and substitute $u = t - T$ in the second piece; periodicity makes it $\int_0^a f$. ■

Fact E.3 (amplitude shift). Any expression $A \sin x + B \cos x$ can be written as $R \sin(x + \phi)$ with $R = \sqrt{A^2 + B^2}$. In particular

$$\sin x - \cos x = \sqrt{2} \sin\left(x - \frac{\pi}{4}\right),$$

which follows by expanding the right side with the subtraction formulas. This converts “area between sine and cosine” into “scaled area under a single modulus sine” — Fact E.1 finishes it.

Odd-function reminder. For odd f on $[-a, a]$, the signed integral is 0 but the area is $2 \int_0^a |f| \, dx$ — the Sub-01 trap, restated because periodic problems love symmetric intervals.

Example E.1 — [JEE Main Level]

Find the area bounded by $y = \sin x$ and the x -axis for $0 \leq x \leq \frac{5\pi}{2}$.

The interval holds two full arches ($[0, \pi]$, $[\pi, 2\pi]$) plus the rising half of a third ($[2\pi, \frac{5\pi}{2}]$):

$$A = 2 + 2 + \int_{2\pi}^{5\pi/2} \sin x \, dx = 4 + [-\cos x]_{2\pi}^{5\pi/2} = 4 + (0 + 1) = 5.$$

Check. Never report $\int_0^{5\pi/2} \sin x \, dx = 1$ — that is the signed value after cancellations, and it is the single most common wrong answer to this pattern.

Example E.2 — [JEE Main / Advanced Level]

Find the area of the region between $y = \sin x$ and $y = \cos x$ for $0 \leq x \leq 2\pi$.

By Fact E.3 the gap is $\sin x - \cos x = \sqrt{2} \sin(x - \frac{\pi}{4})$, so

$$A = \sqrt{2} \int_0^{2\pi} |\sin(x - \frac{\pi}{4})| dx.$$

Substitute $u = x - \frac{\pi}{4}$; the interval shifts to $[-\frac{\pi}{4}, \frac{7\pi}{4}]$, still of length 2π . By Fact E.2 the integral of the periodic function $|\sin u|$ (period π) over any window of length 2π equals two full periods' worth:

$$A = \sqrt{2} \times 2 \times \underbrace{\int_0^{\pi} |\sin u| du}_{=2} = 4\sqrt{2}.$$

Brute-force check. The curves cross at $x = \frac{\pi}{4}$ and $x = \frac{5\pi}{4}$. On $[\frac{\pi}{4}, \frac{5\pi}{4}]$, sine is on top:

$$\int_{\pi/4}^{5\pi/4} (\sin x - \cos x) dx = [-\cos x - \sin x]_{\pi/4}^{5\pi/4} = \sqrt{2} - (-\sqrt{2}) = 2\sqrt{2}.$$

The remaining two end-pieces together hold $\int_0^{\pi/4} (\cos x - \sin x) dx = \sqrt{2} - 1$ and, by the period- π symmetry of $|\sin x - \cos x|$, the same again past $\frac{5\pi}{4}$. Total: $2\sqrt{2} + 2(\sqrt{2} - 1) \cdot \frac{1}{1} \dots$ assembling carefully: $2\sqrt{2} + (\sqrt{2} - 1) + (\sqrt{2} - 1) = 4\sqrt{2} - 2 + \dots$ recompute the last piece directly: $\int_{5\pi/4}^{2\pi} (\cos x - \sin x) dx = [\sin x + \cos x]_{5\pi/4}^{2\pi} = 1 - (-\sqrt{2}) = 1 + \sqrt{2}$. Grand total: $(\sqrt{2} - 1) + 2\sqrt{2} + (1 + \sqrt{2}) = 4\sqrt{2}$. Agreement — but the amplitude-shift route took three lines, not three integrals.

Example E.3 — [JEE Advanced Level]

Find the area enclosed between $y = \sin 2x$ and $y = \sin x$ for $0 \leq x \leq \pi$.

Intersections: $\sin 2x = \sin x$ gives $\sin x (2 \cos x - 1) = 0$, so $x = 0, \frac{\pi}{3}, \pi$. Sample between: at $x = \frac{\pi}{6}$, $\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2} > \frac{1}{2} = \sin \frac{\pi}{6}$, so $\sin 2x$ is on top on $(0, \frac{\pi}{3})$ and beneath on $(\frac{\pi}{3}, \pi)$ (check $x = \frac{\pi}{2}$: $0 < 1$).

$$A = \int_0^{\pi/3} (\sin 2x - \sin x) dx + \int_{\pi/3}^{\pi} (\sin x - \sin 2x) dx.$$

First piece: $[-\frac{\cos 2x}{2} + \cos x]_0^{\pi/3} = (\frac{1}{4} + \frac{1}{2}) - (-\frac{1}{2} + 1) = \frac{3}{4} - \frac{1}{2} = \frac{1}{4}$.

Second piece: $[-\cos x + \frac{\cos 2x}{2}]_{\pi/3}^{\pi} = (1 + \frac{1}{2}) - (-\frac{1}{2} - \frac{1}{4}) = \frac{3}{2} + \frac{3}{4} = \frac{9}{4}$.

$$A = \frac{1}{4} + \frac{9}{4} = \frac{5}{2}.$$

Check. Over $[0, \pi]$ we have $\int |\sin x| = 2$ and $\int |\sin 2x| = \frac{1}{2} \int_0^{2\pi} |\sin u| du = 2$, so the area between the curves cannot exceed 4; $\frac{5}{2}$ sits comfortably inside. Missing the interior crossing at $\frac{\pi}{3}$ — the classic error here — would produce a wrong, partly negative total.

Practice — Addendum E

1. Find the area bounded by $y = \sin x$ and the x -axis for $0 \leq x \leq 3\pi$.
2. Find the area between $y = \sin x$ and $y = \cos x$ for $0 \leq x \leq \pi$.
3. Find the area between $y = |\sin x|$ and $y = |\cos x|$ for $0 \leq x \leq \frac{\pi}{2}$.
4. Find the area enclosed between $y = \sin 2x$ and $y = \sin x$ for $0 \leq x \leq 2\pi$.
5. Find the area under $y = |\cos x|$ for $0 \leq x \leq \frac{3\pi}{2}$.

Answers: 1. Three full arches: $3 \times 2 = 6$. 2. Shift $u = x - \frac{\pi}{4}$: window of length $\pi =$ one full period of $|\sin u|$, so $A = \sqrt{2} \times 2 = 2\sqrt{2}$. 3. They cross at $\frac{\pi}{4}$; by symmetry about that point, $A = 2 \int_0^{\pi/4} (\cos x - \sin x) dx = 2(\sqrt{2} - 1)$. 4. Crossings at $0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi$; the four pieces evaluate to $\frac{1}{4}, \frac{9}{4}, \frac{9}{4}, \frac{1}{4}$ respectively (same antiderivatives as Example E.3, with the third piece taking $\sin 2x$ on top and the fourth $\sin x$), totalling 5. 5. Three half-periods of $|\cos|$: $3 \times 2 = 6$... careful — $\frac{3\pi}{2}$ is three *half*-periods of $|\cos x|$ (period π), so the area is $1 + 1 + 1 = 3$: each half-period contributes $\int_0^{\pi/2} \cos = 1$.

Addendum P — What JEE Actually Asks

A note on sources. The previous-year bank arrived from a PDF in badly damaged form: most equations were typeset as images and did not survive extraction. Everything below is built only from questions I could reconstruct completely — either because the plain-text portions carry the full mathematical content, or because the surviving solution fragments pin the data down uniquely. Each reconstructed question has been re-solved from scratch; where even the data could not be pinned down (e.g. the greatest-integer question of Main Sep. 05, 2020 (II), the 2008 implicit-curve trio, the 2007/2006 matrix-match items), the question is deliberately omitted rather than guessed at.

The distribution

Judged from the recoverable questions, the examiners' attention splits cleanly:

1. Area between two curves — the backbone. A line and a sideways-opening parabola, or the twin parabolas $y^2 = 4ax$ and $x^2 = ay$, account for the largest single block: Main April 12, 2019 (II); Jan 12, 2019 (I); Online April 10, 2015; 2005S; 2004S; 1999; and several damaged 2019–2020 items whose solutions survive. In the modern MCQ era (2013–2020) **essentially every JEE Main session carries exactly one area question**, usually of this or the next type.

2. Regions given as set-builder inequalities. The fastest-growing pattern: $\{(x, y) : \dots\}$ with two or three conditions — Main 2015 ($y^2 \leq 2x$, $y \geq 4x - 1$); Main 2016 ($y^2 \geq 2x$ inside $x^2 + y^2 \leq 4x$); Main 2014 ($x^2 + y^2 \leq 1$, $y^2 \geq 1 - x$); Jan 8, 2020 (I) and (II); Sep 06, 2020 (I) ($|x| + |y| \leq 1$ with $2y^2 \leq |x|$). The examiner is testing *decoding*, not integration.

3. Modulus boundaries. Trig modulus: Online April 23, 2013 ($|\cos x - \sin x|$) and Advanced 2013 ($\sin x + \cos x$ vs $|\cos x - \sin x|$). Algebraic modulus/diamond: 2002S ($y = |x| - 1$ vs $y = -|x| + 1$), Jan 8, 2020 (II) ($x^2 \leq y \leq |3 - 2x|$).

4. Circle cut by a line or parabola. Jan 7, 2020 (I) (circle minus a parabola-line lens); 2016; 2014; and the old descriptive 1987 problem with $|4 - x^2|$ inside a circle. These reward the sector/standard-integral shortcut.

5. Inverse problems — “the area is given, find the parameter.” Main April 12, 2019 (II) (find λ); 2004S (find a); Main 1999 (find m , two valid answers); 2011 (find the bisecting ordinate b). These end in a polynomial equation in the parameter and are chronically underpractised.

6. Advanced-only shapes. Pre-2006, Advanced asked 3-10 mark descriptive area questions almost every year (1982, 1983, 1984, 1985, 1987, 1988, 1990-2002, 2005), including proofs and envelope problems (max of three curves, 1997; a recurrence for A_n , 1996). In the objective era Advanced appears sporadically (2013, 2015-2018, 2020) with multi-curve or multi-condition regions.

The recurring archetypes

Archetype 1 — Sideways parabola and a line: integrate in y

Intersect, then use horizontal strips (“right minus left”). Forcing dx creates a two-branch mess.

Representative (Main April 12, 2019 (II), reconstructed). The area bounded by $y^2 = 4\lambda x$ and $y = \lambda x$ ($\lambda > 0$) is $\frac{1}{9}$; find λ .

Intersection: $\lambda^2 x^2 = 4\lambda x \Rightarrow x = 0$ or $x = \frac{4}{\lambda}$. As functions of y : parabola $x = \frac{y^2}{4\lambda}$, line $x = \frac{y}{\lambda}$, meeting at $y = 0, 4$. The line is the right curve:

$$A = \int_0^4 \left(\frac{y}{\lambda} - \frac{y^2}{4\lambda} \right) dy = \frac{1}{\lambda} \left[\frac{y^2}{2} - \frac{y^3}{12} \right]_0^4 = \frac{1}{\lambda} \left(8 - \frac{16}{3} \right) = \frac{8}{3\lambda}.$$

Setting $\frac{8}{3\lambda} = \frac{1}{9}$ gives $\lambda = 24$. ✓ (Matches the printed answer.)

Archetype 2 — The twin parabolas $y^2 = 4ax$, $x^2 = ay$ (or $y = ax^2$, $x = ay^2$)

Memorise the mechanics: they meet at $(0, 0)$ and (a, a) ; area = $\frac{a^2}{3}$ scaled appropriately.

Representative (2004S). The area enclosed between $y = ax^2$ and $x = ay^2$ ($a > 0$) is 1 sq unit; find a .

Intersections: substituting $x = ay^2 = a(ax^2)^2$ gives $x = 0$ or $x^3 = a^{-3}$, so $x = \frac{1}{a}$: points $(\frac{1}{a}, \frac{1}{a})$ and the origin. Test $x = \frac{1}{2a}$: the root-curve $y = \sqrt{x/a}$ sits above $y = ax^2$. So

$$A = \int_0^{1/a} \left(\frac{\sqrt{x}}{\sqrt{a}} - ax^2 \right) dx = \frac{2}{3a^2} - \frac{1}{3a^2} = \frac{1}{3a^2}.$$

Then $\frac{1}{3a^2} = 1 \Rightarrow a = \frac{1}{\sqrt{3}}$. ✓

Archetype 3 — Set-builder region: decode, sketch, split

Translate each inequality into a side, take the overlap, and let the intersection points fix the limits. Integrate in whichever variable keeps one smooth boundary.

Representative (Main 2015). Region $\{(x, y) : y^2 \leq 2x, y \geq 4x - 1\}$.

Decode: $x \geq \frac{y^2}{2}$ (right of the parabola) and $x \leq \frac{y+1}{4}$ (left of the line). Horizontal strips are natural. Intersections: $\frac{y^2}{2} = \frac{y+1}{4} \Rightarrow 2y^2 - y - 1 = 0 \Rightarrow y = -\frac{1}{2}, 1$. Then

$$A = \int_{-1/2}^1 \left(\frac{y+1}{4} - \frac{y^2}{2} \right) dy = \left[\frac{y^2}{8} + \frac{y}{4} - \frac{y^3}{6} \right]_{-1/2}^1 = \frac{20}{96} + \frac{7}{96} = \frac{9}{32}. \checkmark$$

Archetype 4 — Modulus ceiling: locate every corner, split there

Set each inside-expression to zero; the corners partition the interval.

Representative (Main Online April 23, 2013). Area under $y = |\cos x - \sin x|$ above the x -axis, $0 \leq x \leq \frac{\pi}{2}$.

Corner where $\cos x = \sin x$, i.e. $x = \frac{\pi}{4}$. Left of it $\cos \geq \sin$; right of it $\sin \geq \cos$:

$$A = \int_0^{\pi/4} (\cos x - \sin x) dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x) dx = (\sqrt{2} - 1) + (\sqrt{2} - 1) = 2\sqrt{2} - 2. \checkmark$$

Integrating straight across $\frac{\pi}{4}$ without the modulus gives 0 — the planted distractor.

Archetype 5 — Circle cut by a parabola or line: sector + workhorse integral

Complete the square, recognise circular segments, and deploy $\int \sqrt{a^2 - x^2} dx$ rather than grinding.

Representative (Main 2016). Region $\{(x, y) : y^2 \geq 2x, x^2 + y^2 \leq 4x, x \geq 0, y \geq 0\}$.

The circle is $(x - 2)^2 + y^2 = 4$. Intersections with $y^2 = 2x$: $2x = 4x - x^2 \Rightarrow x = 0, 2$, i.e. $(0, 0)$ and $(2, 2)$. The region is inside the circle, right of the parabola, first quadrant:

$$A = \int_0^2 \sqrt{4x - x^2} dx - \int_0^2 \sqrt{2x} dx = \pi - \frac{8}{3},$$

since the first integral is a quarter-disc of radius 2, and $\int_0^2 \sqrt{2x} dx = \frac{8}{3}$. ✓

Archetype 6 — Inverse problem: given the area, find the parameter

Compute the area as a function of the parameter; the given value produces an equation in it. Expect more than one admissible answer.

Representative (Main 1999). For which m does the region between $y = x - x^2$ and $y = mx$ have area $\frac{9}{2}$?

Intersection: $x - x^2 = mx \Rightarrow x = 0$ or $x = 1 - m$ ($m \neq 1$). The signed gap integrates to

$$\left| \int_0^{1-m} (x(1-m) - x^2) dx \right| = \left| \frac{(1-m)^3}{2} - \frac{(1-m)^3}{3} \right| = \frac{|1-m|^3}{6}.$$

Setting $\frac{|1-m|^3}{6} = \frac{9}{2}$ gives $|1-m|^3 = 27$, so $m = -2$ or $m = 4$. ✓ (Both are among the printed options.)

Archetype 7 — Envelope / composite ceiling (Advanced)

When the boundary is “max” of several curves or switches mid-range, find each switch point, integrate slab by slab.

Representative (Advanced 1997). $f(x) = \max\{x^2, (1-x)^2, 2x(1-x)\}$ on $[0, 1]$; area under $y = f(x)$.

Switches: $x^2 = 2x(1-x)$ at $x = \frac{2}{3}$; $(1-x)^2 = 2x(1-x)$ at $x = \frac{1}{3}$. Sampling shows $(1-x)^2$ leads on $[0, \frac{1}{3}]$, $2x(1-x)$ on $[\frac{1}{3}, \frac{2}{3}]$, x^2 on $[\frac{2}{3}, 1]$:

$$A = \int_0^{1/3} (1-x)^2 dx + \int_{1/3}^{2/3} 2x(1-x) dx + \int_{2/3}^1 x^2 dx = \frac{19}{81} + \frac{13}{81} + \frac{19}{81} = \frac{17}{27}. \checkmark$$

Archetype 8 — Proof-flavoured area (old Advanced)

Small area quantities tied together by a reduction, exploiting monotonicity of the integrand.

Representative (JEE 1996). A_n = area under $y = (\tan x)^n$ from $x = 0$ to $x = \frac{\pi}{4}$. Show $A_n + A_{n-2} = \frac{1}{n-1}$ for $n > 2$, and deduce $A_n \rightarrow 0$.

$$A_n + A_{n-2} = \int_0^{\pi/4} (\tan x)^{n-2} \sec^2 x dx = \left[\frac{(\tan x)^{n-1}}{n-1} \right]_0^{\pi/4} = \frac{1}{n-1}.$$

On $(0, \frac{\pi}{4})$ we have $0 < \tan x < 1$, so $A_n < A_{n-2}$, hence $2A_n < \frac{1}{n-1}$ and $A_n < \frac{1}{2(n-1)} \rightarrow 0$. ✓

The traps

These are visible directly in the distractor patterns of the bank:

1. **The zero-area lure.** Odd integrand over a symmetric interval integrates to 0; the area

does not. Questions with $y = |\cos x - \sin x|$ (2013) and the twin-V curves of 2002S exist precisely to punish integrating through a sign change.

2. **“Forgot to subtract.”** A recurring wrong option equals the area under the *upper* curve alone (the $\frac{32}{3}$ -vs- $\frac{16}{3}$ pattern in the two-parabola questions). Always compute top minus bottom.
3. **Sign flip.** Bottom-minus-top yields a negative; a careless absolute value at the end hides nothing if the split points were wrong.
4. **Limits invented, not solved.** The limits are intersections/corners. In the Jan 7, 2020 (II) item the roots $x = -1, 3$ come from $4x^2 = 8x + 12$ — students who grab convenient numbers lose everything.
5. **Extraneous and rejected intersections.** Squaring manufactures roots (a line through $y = \sqrt{x}$ meets it only where the line is non-negative), and physical constraints kill roots (in Advanced 2013’s cousin, the circle-parabola system has a root $x = -5$ that the parabola forbids). Check every intersection against the original curves and the stated quadrant conditions.
6. **Tangency misread.** In the 2005 problem $x^2 = y$ and $y^2 = 4x - 3$ *touch* at $(1, \pm 1)$ — same tangent, not a crossing. Treating a touch as a pass-through corrupts the region.
7. **Wrong strip variable.** Every $y^2 = (\text{stuff})$ region punished with dx costs a two-branch integral and, empirically, arithmetic slips. If side walls are curves and top/bottom are horizontal, use dy .
8. **Ignoring stated quadrant/restriction clauses.** “ $x \geq 0, y \geq 0$ ”, “first quadrant”, “smaller region” halve or reshape the answer (2016, April 10, 2019 (II)); ignoring them doubles work in the wrong direction.

Graded practice set

All twelve solutions below were computed and checked independently.

P1. Area between $y = \sin x$ and the x -axis for $0 \leq x \leq \frac{3\pi}{2}$.

Split at $x = \pi$:

$$A = \int_0^{\pi} \sin x \, dx + \left| \int_{\pi}^{3\pi/2} \sin x \, dx \right| = 2 + 1 = 3.$$

(The raw integral gives 1 — the sign trap.)

P2. Area between $y = x^2$ and $y = 2x$.

Intersections $x = 0, 2$; line on top:

$$A = \int_0^2 (2x - x^2) \, dx = \left[x^2 - \frac{x^3}{3} \right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}.$$

P3. Area of the region bounded by $y = |x|$ and $y = 2 - |x|$.

Corners at $x = 0$ and crossings at $|x| = 1$. The region is the square with vertices $(0, 0), (1, 1), (0, 2), (-1, 1)$; its diagonals are both 2:

$$A = \frac{d_1 d_2}{2} = 2.$$

Cross-check by integration: $2 \int_0^1 (2 - 2x) dx = 2$. ✓

P4. Area between $y = -2x^2$ and $y = 1 - 3x^2$.

Intersections: $-2x^2 = 1 - 3x^2 \Rightarrow x = \pm 1$, and $1 - 3x^2$ is the upper curve throughout:

$$A = \int_{-1}^1 (1 - x^2) dx = 2 \left(1 - \frac{1}{3}\right) = \frac{4}{3}.$$

P5. Area bounded by $y = x^2 + 2$, the lines $y = x + 1$, $x = 0$, $x = 3$ (Main Jan 12, 2019 (I)).

Here the *vertical lines* fix the limits — do not intersect the two curves (they never meet for $x \geq 0$ here). The parabola is on top:

$$A = \int_0^3 (x^2 + 2 - x - 1) dx = \int_0^3 (x^2 - x + 1) dx = 9 - \frac{9}{2} + 3 = \frac{15}{2}.$$

P6. Area bounded by $y^2 = 4x$ and the line $x + y = 3$.

In y : parabola $x = \frac{y^2}{4}$ (left), line $x = 3 - y$ (right). Intersections: $\frac{y^2}{4} = 3 - y \Rightarrow y^2 + 4y - 12 = 0 \Rightarrow y = 2, -6$:

$$A = \int_{-6}^2 \left(3 - y - \frac{y^2}{4}\right) dy = \left[3y - \frac{y^2}{2} - \frac{y^3}{12}\right]_{-6}^2 = \frac{10}{3} + 18 = \frac{64}{3}.$$

P7. Area of $\{(x, y) : x^2 \leq y \leq |3 - 2x|\}$ (Main Jan 8, 2020 (II)).

For $x \leq \frac{3}{2}$ the modulus opens as $3 - 2x$; solving $x^2 = 3 - 2x$ gives $x = -3, 1$ (the other branch $2x - 3 = x^2$ has no real root). The region lives on $[-3, 1]$:

$$A = \int_{-3}^1 (3 - 2x - x^2) dx = \left[3x - x^2 - \frac{x^3}{3}\right]_{-3}^1 = \frac{5}{3} + 9 = \frac{32}{3}.$$

P8. Area of $\{(x, y) : 4x^2 \leq y \leq 8x + 12\}$ (Main Jan 7, 2020 (II)).

Intersections: $4x^2 = 8x + 12 \Rightarrow (x + 1)(x - 3) = 0$, so $x = -1, 3$:

$$A = \int_{-1}^3 (8x + 12 - 4x^2) dx = \left[4x^2 + 12x - \frac{4x^3}{3}\right]_{-1}^3 = 36 + \frac{20}{3} = \frac{128}{3}.$$

P9. Area of $A = \{(x, y) : |x| + |y| \leq 1, 2y^2 \leq |x|\}$ (Main Sep 06, 2020 (I)).

Four-fold symmetry. First quadrant: $x \geq 2y^2$, $x + y \leq 1$; they cross where $2y^2 = 1 - y$, i.e. $y = \frac{1}{2}$:

$$A = 4 \int_0^{1/2} (1 - y - 2y^2) dy = 4 \left(\frac{1}{2} - \frac{1}{8} - \frac{1}{12}\right) = 4 \cdot \frac{7}{24} = \frac{7}{6}.$$

P10. Area of $\{(x, y) : x^2 + y^2 \leq 1, y^2 \geq 1 - x\}$ (Main 2014).

The second condition says $x \geq 1 - y^2$ (right of the leftward parabola). Symmetric about the x -axis; upper half runs x from $1 - y^2$ to $\sqrt{1 - y^2}$ for $0 \leq y \leq 1$:

$$A = 2 \int_0^1 (\sqrt{1 - y^2} - 1 + y^2) dy = 2 \left(\frac{\pi}{4} - 1 + \frac{1}{3}\right) = \frac{\pi}{2} - \frac{4}{3}.$$

P11. Area of the region $\{(x, y) : xy \leq 8, 1 \leq y \leq x^2\}$ (Advanced 2018).

Vertices: $(1, 1)$, $(2, 4)$ (where $x^2 = \frac{8}{x}$), $(8, 1)$. Split at $x = 2$, where the ceiling changes from $y = x^2$ to $y = \frac{8}{x}$:

$$A = \int_1^2 (x^2 - 1) dx + \int_2^8 \left(\frac{8}{x} - 1\right) dx = \frac{4}{3} + (16 \ln 2 - 6) = 16 \ln 2 - \frac{14}{3}.$$

P12 (hard). f is a quadratic with $f(-1) = f(1) = \frac{3}{4}$ and $f(2) = 0$; its maximum point is V , and $A = (-2, 0)$ is where the graph meets the x -axis. The chord AB , with B on the graph, subtends a right angle at V . Find the area between the graph and chord AB (adapted from JEE 2005).

From $f(-1) = f(1)$: $q = 0$; with $f(x) = px^2 + r$: $p + r = \frac{3}{4}$ and $4p + r = 0$, so $p = -\frac{1}{4}$, $r = 1$:

$$f(x) = 1 - \frac{x^2}{4}, \quad V = (0, 1), \quad A = (-2, 0).$$

$\overrightarrow{VA} = (-2, -1)$; the line through V perpendicular to it has direction $(1, -2)$: $y = 1 - 2x$. Meeting the parabola again: $1 - 2x = 1 - \frac{x^2}{4} \Rightarrow x = 8$, so $B = (8, -15)$. Chord AB : slope $-\frac{15}{10} = -\frac{3}{2}$, i.e. $y = -\frac{3}{2}x - 3$. Then

$$A = \int_{-2}^8 \left(1 - \frac{x^2}{4} + \frac{3x}{2} + 3\right) dx = \left[4x + \frac{3x^2}{4} - \frac{x^3}{12}\right]_{-2}^8 = \frac{112}{3} + \frac{13}{3} = \frac{125}{3} \text{ sq units.}$$

What this chapter's own text under-serves

The chapter drills polynomial-and-circle territory beautifully but never touches the boundary types JEE actually favours: trigonometric and logarithmic/exponential curves ($|\cos x - \sin x|$, $\tan^n x$, $xy = 8$, e^x), whose areas need periodic splitting and log/arctan antiderivatives that appear nowhere in its worked examples. It also omits the inverse genre — “given the area, find the parameter” — which terminates in a cubic in the parameter and carries two valid answers, a format the text never models. Finally, its treatment of composite regions stops at two switching curves, while real papers stack three or more (max-envelopes, tangency points, set-builder triples), where the entire battle is the decode, not the integral.

Chapter 9: Differential Equations

Class 12 Mathematics — Drona Ed Tech

Master Content — Complete Chapter

Contents

Part 1 — Order, Degree & Solutions	2
Section 1: Concept Intuition	2
Section 2: Key Definitions & Properties	3
Section 3: Key Procedures & Logical Framework	4
Section 4: Worked Examples	5
Section 5: Practice Exercises	8
Section 6: Crack the MCQ	8
Section 7: Common Student Pitfalls & Pro-Tips	9
Section 8: Cheat Sheet Box	10
Part 2 — Formation of Differential Equations	11
Section 1: Concept Intuition	11
Section 2: Key Definitions & Properties	12
Section 3: Key Procedures & Logical Framework	12
Section 4: Worked Examples	13
Section 5: Practice Exercises	15
Section 6: Crack the MCQ	15
Section 7: Common Student Pitfalls & Pro-Tips	16
Section 8: Cheat Sheet Box	16
Part 3 — Separable & Homogeneous Equations	18
Section 1: Concept Intuition	18
Section 2: Key Definitions & Properties	19
Section 3: Key Procedures & Logical Framework	20
Section 4: Worked Examples	21
Section 5: Practice Exercises	23
Section 6: Crack the MCQ	23
Section 7: Common Student Pitfalls & Pro-Tips	24
Section 8: Cheat Sheet Box	25
Part 4 — Linear Differential Equations	26
Section 1: Concept Intuition	26
Section 2: Key Definitions & Properties	27
Section 3: CBSE Board Derivation & Key Procedures	27
Section 4: Worked Examples	28
Section 5: Practice Exercises	30
Section 6: Crack the MCQ	31

Section 7: Common Student Pitfalls & Pro-Tips	32
Section 8: Cheat Sheet Box	32
Part 5 — Applications of Differential Equations	34
Section 1: Concept Intuition	34
Section 2: Key Definitions & Properties	35
Section 3: Key Procedures & Logical Framework	35
Section 4: Worked Examples	36
Section 5: Practice Exercises	37
Section 6: Crack the MCQ	38
Section 7: Common Student Pitfalls & Pro-Tips	39
Section 8: Cheat Sheet Box	39

Part 1 — Order, Degree & Solutions

Exam Relevance: This is the gateway subtopic of the chapter and one of the most *reliably scored* topics in all of Class 12 calculus. CBSE Boards almost always carry a guaranteed 1-mark question asking for order and/or degree. JEE Main runs 1-2 questions a year, very often a “degree is not defined” trap or an arbitrary-constants-to-order count. JEE Advanced rarely asks it in isolation but uses the order concept as a hidden first step in curve-family and functional-equation problems. Master this and you bank easy marks while building the vocabulary the rest of the chapter assumes.

Section 1: Concept Intuition

Imagine you are watching a cup of hot chai cool down on a winter morning in Shimla. You cannot easily write a formula that tells you the *exact* temperature at 9:47 a.m. — that’s hard. But you *can* state something simpler and truer: **the rate at which the chai cools is proportional to how much hotter it is than the room.** That sentence — a statement about a *rate of change* rather than about the value itself — is the soul of a differential equation.

A **differential equation** is simply any equation that contains a derivative of an unknown function. Instead of handing you the function directly, it hands you a *rule about how the function changes*, and your job later in the chapter is to “undo” that rule and recover the function. For now, in this subtopic, we are not solving anything — we are learning to **read, classify, and verify** these equations. Think of it as learning to read the scoreboard before you learn to play the match.

Two quick classifications you must internalise. If the unknown function depends on **one** independent variable (say everything is a function of x), the equation is an **ordinary differential equation (ODE)** — this is all of Class 12. If it involves partial derivatives with respect to **several** independent variables, it is a **partial differential equation (PDE)** — you will meet those in engineering, not here.

Now, the two numbers that *fingerprint* every differential equation:

Order is the order of the **highest** derivative appearing in the equation. If the

biggest derivative you can spot is $\frac{d^2y}{dx^2}$, the order is 2, no matter how many lower derivatives crowd around it. Order answers the question: “What is the deepest level of change this equation talks about?”

Degree is subtler. Once the equation has been written as a **polynomial in its derivatives** (no roots, no fractions, no derivative trapped inside a sine or an exponential), the degree is the **power of the highest-order derivative**. Degree answers: “To what power is that deepest derivative raised, after we tidy the equation up?”

Think about it this way... Picture the derivatives as players in a cricket team batting order. The *order* of the equation is the rank of your top-order batsman — the highest one who walks out. The *degree* is how many runs that exact top batsman scores once the innings is cleaned up (radicals cleared, fractions removed). You never read the degree off a messy scorecard; you settle the match first, then count.

Limiting conditions & assumptions — read this twice:

- **Degree exists only when the equation is a polynomial in all its derivatives.** The moment a derivative sits inside a transcendental function — $\sin(\frac{dy}{dx})$, $e^{dy/dx}$, $\log(\frac{d^2y}{dx^2})$ — the degree is **not defined**, because expanding such a function gives an *infinite* series of powers of the derivative. There is no single “highest power” to point to.
- **You must clear radicals and fractional powers before reading the degree.** A $\sqrt{\quad}$ or a power like $3/2$ over a derivative is not yet a polynomial form. Order is unaffected by this tidying, but degree absolutely is.
- **Degree is always a positive integer when it exists** — never a fraction, never negative.

Section 2: Key Definitions & Properties

(For Mathematics, dimensional formula” block with definitions, classification rules, and structural properties.)

A general differential equation of order n can be written abstractly as:

$$F\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots, \frac{d^ny}{dx^n}\right) = 0$$

Term	Definition
Differential equation	An equation relating an unknown function y , the independent variable x , and one or more derivatives of y .
Ordinary DE (ODE)	Involves derivatives with respect to a <i>single</i> independent variable.

Term	Definition
Partial DE (PDE)	Involves derivatives with respect to <i>more than one</i> independent variable.
Order	The order of the highest-order derivative present.
Degree	The power of the highest-order derivative, after the equation is made polynomial in its derivatives. Defined <i>only</i> if such a polynomial form exists.
Solution	Any relation between x and y (free of derivatives) that satisfies the differential equation.
General solution	A solution containing exactly as many independent arbitrary constants as the order of the equation.
Particular solution	A solution obtained from the general solution by fixing the arbitrary constants — it contains no arbitrary constants.

Notation key. y is the dependent variable; x is the independent variable; $\frac{d^n y}{dx^n}$ (or the prime form $y^{(n)}$) is the n -th derivative; arbitrary constants are written $A, B, C, c_1, c_2, \dots$

Three structural properties worth memorising:

1. **Order = number of independent arbitrary constants** in the general solution. A general solution of an order-3 equation carries exactly three free constants — no more, no fewer.
2. **Degree need not equal order.** They are independent fingerprints. The equation $\left(\frac{d^2 y}{dx^2}\right)^4 + \frac{dy}{dx} = 0$ has order 2 but degree 4.
3. **A particular solution is just the general solution with its constants pinned down** by initial/boundary conditions (e.g. " $y = 2$ when $x = 0$ ").

Section 3: Key Procedures & Logical Framework

There is no formal *proof* attached to order/degree — this subtopic is procedural. Here are the four routines you will use on every problem, with the reasoning behind each step.

Procedure A — Finding the order

1. Scan the equation for every derivative present.
2. Identify the one with the **highest order** (the deepest $\frac{d^k y}{dx^k}$).

3. That k is the order. *Ignore the powers entirely* — powers are a degree question, not an order question.

Why it works: order measures the “depth” of change the equation encodes, and depth is set by the highest derivative regardless of how it is dressed up.

Procedure B — Finding the degree

1. **Check the gate first:** is every derivative free of transcendental wrappers ($\sin$, $\cos$, $\log$, $e^{(\cdot)}$, $\tan^{-1}$, ...)? If *no*, stop — **degree is not defined**.
2. **Clean the equation:** clear all radicals and fractional powers (usually by raising both sides to a suitable integer power) and clear denominators, so the equation is a genuine polynomial in the derivatives.
3. Read off the **power of the highest-order derivative** in this cleaned form. That power is the degree.

Why the cleaning step matters: “degree” is only meaningful for polynomials. A square root or a $3/2$ power is not polynomial; squaring converts it into one without changing the underlying equation’s solution set. The order is invariant under this step — only the readable degree changes.

A worked micro-example of the gate. For $\frac{d^2y}{dx^2} + e^{dy/dx} = 0$, the term $e^{dy/dx}$ expands as $1 + \frac{dy}{dx} + \frac{1}{2!} \left(\frac{dy}{dx}\right)^2 + \dots$, an infinite chain of powers of $\frac{dy}{dx}$. There is no highest power, so **degree is not defined** — even though the order is comfortably 2.

Procedure C — Counting arbitrary constants → order

1. Write the family of solutions/curves.
2. Count the **independent** arbitrary constants — constants that cannot be merged into one another.
3. That count is the order of the differential equation the family satisfies.

Critical subtlety: constants can disguise themselves. In $y = Ax + A^3$, both terms come from the *single* constant A , so there is only **one** independent constant $\Rightarrow$ order 1, not 2. Likewise $c_1 + c_2$ behaves as a *single* constant. Always ask “can these be combined?” before counting.

Procedure D — Verifying a proposed solution

1. From the candidate y , compute every derivative the equation needs.
2. Substitute y and its derivatives into the **left-hand side**.
3. Simplify. If the LHS reduces identically to the RHS *for all* x , the candidate is a genuine solution; otherwise it is not.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Find the order and degree of $\frac{d^3y}{dx^3} + 5 \left(\frac{d^2y}{dx^2} \right)^2 - 7 \frac{dy}{dx} + 4y = \cos x$.

Solution.

- *Given:* the equation above.
- *Highest derivative present:* $\frac{d^3y}{dx^3} \Rightarrow$ **order** = 3.
- *Degree gate:* every derivative appears as a plain polynomial term — no derivative inside a sine, exponential, or root. (The $\cos x$ on the right involves x , **not** a derivative, so it does not break the gate.) Degree is defined.
- *Power of the highest-order derivative* $\frac{d^3y}{dx^3}$ is 1 $\Rightarrow$ **degree** = 1.

Answer: order 3, degree 1.

Presentation tip for the exam: always write the two as a labelled pair — “Order = 3, Degree = 1” — examiners award the mark on the explicit statement.

Example 2 — [CUET / JEE Main Speed Trap]

State the degree of each, in under fifteen seconds:

(i) $\frac{d^2y}{dx^2} = \sin\left(\frac{dy}{dx}\right)$ (ii) $\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \frac{d^2y}{dx^2}$.

Solution (mental-elimination style).

(i) The derivative $\frac{dy}{dx}$ is trapped inside a sine. **Trip the gate immediately** — a transcendental function of a derivative means **degree not defined**. Don’t compute anything.

(ii) There’s a square root over a derivative expression — *not yet* polynomial. Square both sides first:

$$1 + \left(\frac{dy}{dx}\right)^2 = \left(\frac{d^2y}{dx^2}\right)^2.$$

Now it’s polynomial. The highest-order derivative is $\frac{d^2y}{dx^2}$, and its power is 2 $\Rightarrow$ **degree** = 2 (order is 2).

The trap: students square *first* in (i) too, “rescue” the sine somehow, and invent a degree — or they read degree = 1 off (ii) *without* squaring. Speed rule: **see a root over a derivative $\rightarrow$ square; see a transcendental wrapper $\rightarrow$ stop, undefined.**

Example 3 — [JEE Main Level — multi-concept: formation + order]

A family of curves is given by $y = ae^{2x} + be^{-2x}$. **Form its differential equation and hence state its order and degree.**

Solution. This links *formation* (Procedure C) with *order/degree* reading.

There are **two** independent arbitrary constants, a and b , so we differentiate twice and eliminate both.

$$\begin{aligned}y &= ae^{2x} + be^{-2x} \\ \frac{dy}{dx} &= 2ae^{2x} - 2be^{-2x} \\ \frac{d^2y}{dx^2} &= 4ae^{2x} + 4be^{-2x} = 4(ae^{2x} + be^{-2x}) = 4y.\end{aligned}$$

Therefore the differential equation is:

$$\boxed{\frac{d^2y}{dx^2} - 4y = 0}$$

- Two arbitrary constants $\Rightarrow$ we expected order 2, and indeed the highest derivative is $\frac{d^2y}{dx^2} \Rightarrow$ **order** = 2.
- It is polynomial in the derivative, power 1 $\Rightarrow$ **degree** = 1.

Cross-check (Procedure A consistency): number of independent constants (2) matches the order (2) — a built-in sanity check that your formation was done correctly.

Example 4 — [JEE Advanced Level — the arbitrary-constant collapse]

Find the order of the differential equation whose general solution is

$$y = (c_1 + c_2)e^{x+c_3} + c_4 \sin(x + c_5),$$

where c_1, c_2, c_3, c_4, c_5 are arbitrary constants.

Solution. The naive answer “5” is the trap. The order equals the number of *independent* arbitrary constants — so we must check which of these five genuinely cannot be merged.

Look term by term:

- $(c_1 + c_2)e^{x+c_3} = (c_1 + c_2)e^{c_3} \cdot e^x$. The whole prefactor $(c_1 + c_2)e^{c_3}$ collapses into a **single** new constant, call it A . So this term is just Ae^x — **one** independent constant from three original ones.
- $c_4 \sin(x + c_5) = c_4(\sin x \cos c_5 + \cos x \sin c_5) = (c_4 \cos c_5) \sin x + (c_4 \sin c_5) \cos x$. This contributes **two** independent constants (the coefficients of $\sin x$ and $\cos x$).

Independent count = 1 + 2 = 3.

$$\boxed{\text{Order} = 3}$$

The deep point: arbitrary constants are independent only if no algebraic identity lets you fuse them. Sums inside the same coefficient, and constants hiding inside $e^{(\cdot)}$ exponents, almost always collapse. Train your eye to *simplify before you count*.

Section 5: Practice Exercises

Attempt these on your own; final answers follow.

1. **[CBSE]** Find the order and degree of $2\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + y = 0$.
2. **[CUET]** Write the degree of $\frac{d^2y}{dx^2} + \cos\left(\frac{dy}{dx}\right) = 0$.
3. **[JEE Main]** Find the order and degree of $\left(1 + \frac{dy}{dx}\right)^3 = \left(\frac{d^2y}{dx^2}\right)^2$.
4. **[JEE Main]** Form the differential equation of the family $y = mx$, where m is an arbitrary constant, and state its order.
5. **[JEE Advanced]** Determine the order of the differential equation whose general solution is $y = A \cos(2x + B) + Ce^D x$, where A, B, C, D are arbitrary constants.

Answers: 1) order 2, degree 1 — 2) degree not defined — 3) order 2, degree 2 — 4)

$dy/dx = y/x$ (equivalently $xy/dx - y = 0$), order 1 — 5) order 4 (four independent constants: two from $A \cos(2x + B)$, two from $Ce^{\wedge}(Dx)$).

Section 6: Crack the MCQ

Q1. The order and degree of $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2} = k \frac{d^2y}{dx^2}$ **are respectively:**

- (A) 2, 3 (B) 2, 2 (C) 2, 1 (D) 1, 2

Correct: (B). Squaring both sides to clear the $3/2$ power gives $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^3 = k^2 \left(\frac{d^2y}{dx^2}\right)^2$. Highest derivative $\frac{d^2y}{dx^2} \Rightarrow$ order 2; its power is now 2 $\Rightarrow$ degree 2.

- **(A)** comes from reading the 3 in the exponent $3/2$ as the degree without squaring — confusing the power on the *first-derivative bracket* with the degree.
- **(C)** comes from reading degree off the *unsquared* form (power 1 on $\frac{d^2y}{dx^2}$) — forgetting that fractional powers must be cleared first.
- **(D)** misreads the order; the curvature term is second order, not first.

Q2. The degree of the differential equation $\frac{d^2y}{dx^2} + 3\left(\frac{dy}{dx}\right)^2 = x \log\left(\frac{d^2y}{dx^2}\right)$ **is:**

- (A) 1 (B) 2 (C) 3 (D) not defined

Correct: (D). The highest-order derivative $\frac{d^2y}{dx^2}$ appears inside a logarithm on the right. A transcendental function of a derivative cannot be reduced to a polynomial form, so **degree is not defined**.

- **(A)** picks the power of $\frac{d^2y}{dx^2}$ on the *left* and ignores the right side entirely — a partial-reading error.

- **(B)** picks up the 2 from $\left(\frac{dy}{dx}\right)^2$, but that's a lower-order derivative; degree is about the *highest* one anyway.
- **(C)** is a distractor with no basis — chosen by students who feel “undefined” can't be a real option.

Q3. The differential equation of the family $y = c_1 e^{3x} + c_2 e^{3x}$ has order:

(A) 0 (B) 1 (C) 2 (D) not defined

Correct: (B). Since both terms have the *same* exponential e^{3x} , we have $y = (c_1 + c_2)e^{3x} = Ae^{3x}$, a **single** independent constant. One constant $\Rightarrow$ order 1 (the equation is $\frac{dy}{dx} = 3y$).

- **(C)** is the trap: counting two written constants c_1, c_2 without noticing they merge.
- **(A)** would require *zero* free constants (a particular solution), which is false.
- **(D)** confuses order with the degree-undefined situation; order is always defined here.

Q4. Which of the following functions is a solution of $\frac{d^2y}{dx^2} + 9y = 0$?

(A) $y = e^{3x}$ (B) $y = \cos 3x$ (C) $y = \cos x$ (D) $y = e^{-3x}$

Correct: (B). Verify by Procedure D: for $y = \cos 3x$, $\frac{dy}{dx} = -3 \sin 3x$, $\frac{d^2y}{dx^2} = -9 \cos 3x$. Then $\frac{d^2y}{dx^2} + 9y = -9 \cos 3x + 9 \cos 3x = 0$. ✓

- **(A), (D)** give $\frac{d^2y}{dx^2} = 9y$, so the LHS becomes $9y + 9y = 18y \neq 0$ — wrong sign of curvature (these solve $y'' - 9y = 0$, the trap of mismatching the sign).
- **(C)** $y = \cos x$ gives $\frac{d^2y}{dx^2} = -\cos x$, so LHS $= -\cos x + 9 \cos x = 8 \cos x \neq 0$ — right family, wrong frequency (forgot the factor 3 inside).

Section 7: Common Student Pitfalls & Pro-Tips

- **Reading degree off a messy equation.** Students announce the degree before clearing radicals, fractional powers, or denominators. Always reach the *polynomial-in-derivatives* form first.
- **Forcing a degree onto a non-polynomial equation.** If a derivative hides inside $\sin$, $\log$, $e^{(\cdot)}$, $\tan^{-1}$, etc., the honest answer is “**not defined**” — and that is a complete, correct answer, not a cop-out.
- **Taking the power of the wrong derivative.** Degree uses the power of the **highest-order** derivative, not the largest power anywhere. In $\left(\frac{dy}{dx}\right)^5 + \frac{d^2y}{dx^2} = 0$, the degree is 1 (power of $\frac{d^2y}{dx^2}$), even though a 5 is staring at you.
- **Counting written constants instead of independent ones.** $c_1 + c_2$, $c_1 c_2$, and constants buried in exponents collapse. Simplify, *then* count.

Pro-Tip (speed + accuracy): Run a two-second gate on every order/degree question — “**Is any derivative inside a sine/log/exp, or under a root?**” If inside a transcendental $\rightarrow$ degree undefined, done.

If under a root $\rightarrow$ square, then read. If neither $\rightarrow$ read the power of the top derivative directly. This single reflex prevents three of the four mistakes above.

Section 8: Cheat Sheet Box

- **Order** = order of the **highest** derivative present. (Powers don't matter for order.)
- **Degree** = power of the highest-order derivative, **only after** the equation is polynomial in its derivatives.
- **Degree is NOT defined** if any derivative sits inside a transcendental function ($\sin$, $\log$, $e^{(\cdot)}$, $\tan^{-1}$, ...).
- **Clear radicals & fractional powers first** — order stays, degree changes.
- **Order = number of *independent* arbitrary constants** in the general solution. Collapse hidden constants before counting.
- **General solution:** constants = order. **Particular solution:** no constants.

Memory aids:

- *"Order is the rank of the boss derivative; degree is the boss's power — after the room is cleaned."*
- *"No polynomial, no degree."*
- *"Clean before you count."* (radicals/fractions out, then read degree)
- *"Constants = Order"* (independent ones only).

Order: highest derivative. Degree: its power, polynomial form, else undefined.

Part 2 — Formation of Differential Equations

Exam Relevance: A near-guaranteed CBSE Boards question (2–3 marks) — “form the differential equation of the family...”. JEE Main asks it as an order-counting MCQ once a year, often with a constant-collapse trap. It is the *reverse* of solving and the conceptual bridge between “what is a solution” (Part 1) and “how to solve” (Parts 3–4), so it cements the whole chapter.

Section 1: Concept Intuition

Solving a differential equation takes you *from* a DE *to* a family of curves. **Formation goes the other way:** you start with a whole family of curves — described by an equation carrying one or more arbitrary constants — and you hunt for the single differential equation that *every* member of that family obeys.

Think of a family like “all straight lines through the origin,” written $y = mx$. The constant m is a dial: turn it and you get a different line, but all of them share one property — at every point the slope equals y/x . That shared property, written with derivatives and **stripped of the dial** m , is the differential equation of the family: $\frac{dy}{dx} = \frac{y}{x}$. The arbitrary constant has been *eliminated*; what remains is the family’s “DNA.”

Think about it this way... A family of curves is like a batch of biscuits from one cutter — different positions on the tray (the arbitrary constants), identical shape (the underlying rule). Formation is reverse-engineering the cutter from the biscuits: you differentiate to expose the rule and then erase the position information (the constants).

The single most important fact controls everything: **the number of independent arbitrary constants equals the order of the resulting differential equation.** One constant $\Rightarrow$ differentiate once $\Rightarrow$ order-1 DE. Two constants $\Rightarrow$ differentiate twice $\Rightarrow$ order-2 DE. Each differentiation gives you one more equation, and you need exactly enough equations to eliminate every constant.

Limiting conditions & assumptions — read carefully:

- **Count independent constants, not written ones.** Constants that merge — $c_1 + c_2$, or a constant buried in an exponent like e^c — count as one. Miscounting here gives the wrong order.
 - **You must differentiate exactly as many times as there are independent constants** — no fewer (you couldn’t eliminate them all) and the result must be **free of every arbitrary constant**. A constant left in the “answer” means the job isn’t done.
 - **The final DE must contain only x , y , and derivatives** — never the arbitrary constants and never a brand-new constant introduced along the way.
-

Section 2: Key Definitions & Properties

physics formulas/SI units.)

A **family of curves** with n independent arbitrary constants $a_1, a_2, \dots, a_n$ is written

$$F(x, y, a_1, a_2, \dots, a_n) = 0.$$

Its **differential equation** is the relation obtained by eliminating all n constants, and it has the form

$$\Phi\left(x, y, \frac{dy}{dx}, \dots, \frac{d^ny}{dx^n}\right) = 0,$$

a differential equation of **order** n , free of the constants.

Notation key. $a_1, \dots, a_n$ (or A, B, C, m, c) — arbitrary constants of the family; n — number of *independent* constants = order of the formed DE; Φ — the resulting relation among x, y and derivatives.

Three properties to lock in:

1. **Order = number of independent arbitrary constants.** This is the same principle as Part 1 — read backwards: solving an order- n DE *produces* n constants; forming a DE from n constants *requires* order n .
2. **Differentiations needed = number of constants.** With the original equation plus n derivative equations, you have $n + 1$ relations — exactly enough to eliminate n unknown constants and keep one constant-free equation.
3. **Geometric families hide their constant count.** “All circles of radius r with centre on the x -axis” has *one* free parameter (the centre’s x -coordinate) $\Rightarrow$ order 1; “all circles in the plane” $x^2 + y^2 + 2gx + 2fy + c = 0$ has *three* (g, f, c) $\Rightarrow$ order 3.

Section 3: Key Procedures & Logical Framework

This is a procedure, not a proof. Here is the routine with the reasoning behind each step.

Procedure — forming the differential equation

1. **Count the independent arbitrary constants, n .** Merge any that collapse (sums in one coefficient, constants inside exponents).
2. **Differentiate the given equation n times** with respect to x . Each differentiation produces a new equation.
3. **Eliminate the n constants** using the original equation together with the n derivative equations — by substitution or by treating the relations as simultaneous equations in the constants.
4. **Write the result** purely in x, y , and derivatives. Confirm no arbitrary constant survives.

Why differentiate n times: each independent constant is one “degree of freedom.” Differentiating once removes a *purely additive* constant immediately, but in general you need n independent relations beyond the original to solve for and cancel n unknowns. Stopping early leaves a constant behind; going further than needed is unnecessary but harmless if the lower-order DE already came out constant-free.

Quick check before you start: glance at the family and predict the order from the constant count. If your final DE's order doesn't match, you either miscounted constants (collapse!) or under-differentiated.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Form the differential equation of the family of parabolas $y^2 = 4ax$, where a is an arbitrary constant.

Solution.

- *Independent constants:* just $a \Rightarrow n = 1 \Rightarrow$ differentiate once, order-1 DE expected.
- *Differentiate:* $2y \frac{dy}{dx} = 4a \Rightarrow 4a = 2y \frac{dy}{dx}$.
- *Eliminate a :* substitute $4a$ into $y^2 = 4ax$:

$$y^2 = \left(2y \frac{dy}{dx}\right) x \Rightarrow y = 2x \frac{dy}{dx}.$$

$$\boxed{2x \frac{dy}{dx} - y = 0} \quad (\text{order 1, degree 1}).$$

Presentation tip: state the constant count and the expected order up front — examiners reward the reasoning, and it catches mistakes early.

Example 2 — [CUET / JEE Main Speed Trap]

Form the differential equation of $y = A \cos x + B \sin x$.

Solution. Two independent constants $A, B \Rightarrow n = 2 \Rightarrow$ differentiate **twice**, expect order 2. The speed move is to *recognise the family*: $A \cos x + B \sin x$ is the general sinusoid, so its DE is the simple-harmonic equation — you can almost write the answer on sight.

$$\frac{dy}{dx} = -A \sin x + B \cos x, \quad \frac{d^2y}{dx^2} = -A \cos x - B \sin x = -(A \cos x + B \sin x) = -y.$$

$$\boxed{\frac{d^2y}{dx^2} + y = 0}.$$

The trap: students differentiate only **once**, then panic when A and B won't cancel — leaving constants in the “answer.” Two constants *always* need two differentiations.

Example 3 — [JEE Main Level — multi-concept: product rule + double elimination]

Form the differential equation of $y = e^x(A \cos x + B \sin x)$.

Solution. Two constants $\Rightarrow$ order 2. This one tests the **product rule** alongside elimination.

$$\frac{dy}{dx} = e^x(A \cos x + B \sin x) + e^x(-A \sin x + B \cos x) = y + e^x(-A \sin x + B \cos x).$$

Let $w = e^x(-A \sin x + B \cos x)$, so $\frac{dy}{dx} = y + w \Rightarrow w = \frac{dy}{dx} - y$. Differentiate again:

$$\frac{d^2y}{dx^2} = \frac{dy}{dx} + w', \quad w' = e^x(-A \sin x + B \cos x) + e^x(-A \cos x - B \sin x) = w - y.$$

$$\text{So } \frac{d^2y}{dx^2} = \frac{dy}{dx} + (w - y) = \frac{dy}{dx} + \left(\frac{dy}{dx} - y\right) - y.$$

$$\boxed{\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0}.$$

Why “multi-concept”: you apply the product rule twice **and** carry an intermediate quantity w through two eliminations — a small system, not a one-liner.

Example 4 — [JEE Advanced Level — geometric family, cross-link to homogeneous]

Form the differential equation of all circles passing through the origin with centres on the x -axis: $x^2 + y^2 = 2ax$.

Solution. One free parameter a (the centre is $(a, 0)$, radius a) $\Rightarrow$ order 1.

$$\text{Differentiate: } 2x + 2y\frac{dy}{dx} = 2a \Rightarrow a = x + y\frac{dy}{dx}.$$

Substitute into $x^2 + y^2 = 2ax$:

$$x^2 + y^2 = 2x\left(x + y\frac{dy}{dx}\right) = 2x^2 + 2xy\frac{dy}{dx} \Rightarrow y^2 - x^2 = 2xy\frac{dy}{dx}.$$

$$\boxed{\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}} \quad (\text{order 1}).$$

The cross-link worth noticing: the right-hand side depends only on y/x — this is a **homogeneous** differential equation (Part 3). Forming a DE here lands you exactly on a type you already know how to solve. *Remark:* the family of **all** circles $x^2 + y^2 + 2gx + 2fy + c = 0$ has three independent constants (g, f, c) , so its differential equation has **order 3** — a classic exam fact.

Section 5: Practice Exercises

Try these yourself; final answers follow.

1. **[CBSE]** Form the DE of the family $y = Cx^2$.
2. **[CBSE]** Form the DE of the family $y = ax^2 + bx$ (a, b arbitrary).
3. **[JEE Main]** Form the DE of $y = Ae^{3x} + Be^{-3x}$.
4. **[JEE Main]** Form the DE of the family of all straight lines $y = mx + c$.
5. **[JEE Advanced]** Form the DE of the family of rectangular hyperbolas $xy = c$.

Answers: 1) $x \frac{dy}{dx} = 2y$ — 2) $x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + 2y = 0$ — 3) $\frac{d^2y}{dx^2} - 9y = 0$ — 4) $\frac{d^2y}{dx^2} = 0$ — 5) $x \frac{dy}{dx} + y = 0$.

Section 6: Crack the MCQ

Q1. The differential equation of the family $y = mx$ (m arbitrary) is:

- (A) $\frac{dy}{dx} = \frac{y}{x}$ (B) $\frac{dy}{dx} = \frac{x}{y}$ (C) $\frac{dy}{dx} = m$ (D) $x \frac{dy}{dx} + y = 0$

Correct: (A). $\frac{dy}{dx} = m$ and $m = \frac{y}{x}$, so $\frac{dy}{dx} = \frac{y}{x}$.

- **(C)** leaves the arbitrary constant m in the “answer” — the constant must be eliminated.
- **(B)** inverts the ratio.
- **(D)** is the DE of $xy = c$, a different family.

Q2. The differential equation whose general solution is $y = A \sin x + B \cos x$ is:

- (A) $\frac{d^2y}{dx^2} - y = 0$ (B) $\frac{d^2y}{dx^2} + y = 0$ (C) $\frac{dy}{dx} + y = 0$ (D) $\frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

Correct: (B). Two constants $\Rightarrow$ order 2; differentiating twice gives $y'' = -y$.

- **(A)** is the DE of $Ae^x + Be^{-x}$ (sign flip — exponential, not sinusoidal).
- **(C)** is order 1 — too few differentiations for two constants.
- **(D)** doesn't annihilate the sinusoid.

Q3. The order of the differential equation of the family of all circles $x^2 + y^2 + 2gx + 2fy + c = 0$ is:

- (A) 1 (B) 2 (C) 3 (D) 4

Correct: (C). Three independent arbitrary constants $g, f, c \Rightarrow$ order 3.

- **(A)/(B)** undercount the constants. **(D)** overcounts — there is no fourth independent parameter.

Q4. The order of the differential equation of the family $y = (c_1 + c_2)e^x + c_3$ is:

- (A) 1 (B) 2 (C) 3 (D) not defined

Correct: (B). $c_1 + c_2$ collapses to a single constant, so the *independent* count is 2 (one from $(c_1 + c_2)e^x$, one from c_3). Order 2.

- **(C)** counts all three written constants without spotting the collapse — the headline trap.
 - **(A)** misses the additive c_3 . **(D)** confuses order with the degree-undefined case.
-

Section 7: Common Student Pitfalls & Pro-Tips

- **Counting dependent constants as independent.** $c_1 + c_2$, $c_1 c_2$, and constants inside exponents collapse — simplify before counting.
- **Under-differentiating.** n constants need n differentiations; stopping early strands a constant in the answer.
- **Leaving an arbitrary constant in the final DE.** The formed equation must contain only x , y , and derivatives. A surviving a , m , A , ... means the elimination is incomplete.
- **Miscounting geometric families.** Translate the geometry to a parameter count first: “circles on the x -axis through origin” $\rightarrow$ one parameter; “all circles” $\rightarrow$ three.

Pro-Tip (predict, then verify): before differentiating, *count the independent constants and write down the expected order*. After eliminating, check the order matches. A mismatch instantly flags a collapse you missed or a differentiation you skipped — far faster than re-checking algebra line by line.

Section 8: Cheat Sheet Box

- **Goal:** eliminate the arbitrary constants of a family to get its differential equation.
- **Order = number of independent arbitrary constants.**
- **Differentiate the equation as many times as there are constants**, then eliminate them.
- **Final DE must be free of all arbitrary constants** — only x , y , derivatives remain.
- **Collapse first:** $c_1 + c_2 \rightarrow$ one constant; constants in exponents merge.
- **Geometry $\rightarrow$ parameters:** one free parameter $\Rightarrow$ order 1; general circle $\Rightarrow$ order 3.

Memory aids:

- “Constants in, derivatives out.” (eliminate constants via differentiation)
- “Count the dials = count the order.”
- “Differentiate once per constant.”
- “No constant may survive the answer.”

n independent constants $\Rightarrow$ differentiate n times $\Rightarrow$ order- n DE, constant-free.

Part 3 — Separable & Homogeneous Equations

Exam Relevance: This is where you finally *solve* differential equations, and it is heavily examined. CBSE Boards almost always carry a 3–5 mark “solve this” question drawn from either method, frequently with an initial condition. JEE Main asks 1–2 questions per year, often a homogeneous equation disguised so you must first *recognise* its type. JEE Advanced layers these into curve-finding and reducible forms. The two methods here — separation of variables and the $y = vx$ substitution — are the workhorses of the entire chapter.

Section 1: Concept Intuition

A differential equation hands you a *rule about change* and dares you to recover the original function. The honest truth is that there is no single magic trick that cracks every such equation — instead we keep a toolbox, and the first job is always **diagnosis**: *what kind of equation is this?* This subtopic gives you the two most important tools and, just as importantly, teaches you to recognise when each one applies.

Variables separable — the “sort the laundry” method. Suppose the rate of change splits cleanly into an x -part and a y -part, like $\frac{dy}{dx} = (\text{something in } x) \times (\text{something in } y)$. Then you can physically move every y (and the dy) to one side and every x (and the dx) to the other — like sorting mixed laundry into two baskets — and integrate each side on its own. The whole method rests on one tidy idea: *if the variables can be separated, two ordinary integrals finish the job.*

Think about it this way... Imagine splitting a shared autorickshaw fare. As long as your expenses and your friend’s expenses never get tangled together, each of you just totals up your own column. Separation of variables works the moment the x -costs and y -costs don’t mix — you integrate each column independently.

Homogeneous equations — the “switch to ratios” method. Many equations *refuse* to separate because x and y are knotted together, e.g. $\frac{dy}{dx} = \frac{x^2 + y^2}{2xy}$. But look closer: the right-hand side doesn’t really care about the *sizes* of x and y , only their **ratio** y/x . Divide top and bottom by x^2 and it becomes a pure function of $v = y/x$. Such an equation is called **homogeneous**, and the substitution $y = vx$ magically untangles it back into a separable equation in v and x .

The deep reason this works: the right-hand side is a **homogeneous function of degree zero** — scaling both x and y by the same factor λ leaves it unchanged. Anything that depends only on the ratio y/x has this scale-invariance, and that is precisely the signature you hunt for.

Limiting conditions & assumptions — read carefully:

- **Separation can silently destroy solutions.** When you divide both sides by some $h(y)$ to separate, you quietly assume $h(y) \neq 0$. But the constant

function where $h(y) = 0$ is often *also* a solution — and you’ve thrown it away. Always check for these lost **singular solutions** at the end.

- **Homogeneity requires degree zero, and no stray constants.** $\frac{x+y}{x-y}$ is homogeneous (degree 0). But $\frac{x+y+1}{x-y}$ is **not** — that lone +1 breaks the scale-invariance. Forcing $y = vx$ onto such an equation produces nonsense; those need an origin shift instead.

Section 2: Key Definitions & Properties

physics formulas/SI units.)

02a — Variables Separable

A first-order equation is **separable** if it can be written as

$$\frac{dy}{dx} = g(x) h(y) \quad \Longleftrightarrow \quad \frac{dy}{h(y)} = g(x) dx.$$

Its general solution is obtained by integrating each side:

$$\int \frac{dy}{h(y)} = \int g(x) dx + C.$$

02b — Homogeneous Equations

A function $F(x, y)$ is **homogeneous of degree n** if $F(\lambda x, \lambda y) = \lambda^n F(x, y)$. A first-order equation

$$\frac{dy}{dx} = F(x, y)$$

is a **homogeneous differential equation** if F is homogeneous of **degree 0** — equivalently, if the RHS can be written purely as a function of $v = y/x$:

$$\frac{dy}{dx} = \phi\left(\frac{y}{x}\right).$$

Reduction substitution: put $y = vx$, so $\frac{dy}{dx} = v + x \frac{dv}{dx}$. The equation becomes

$$v + x \frac{dv}{dx} = \phi(v) \implies \frac{dv}{\phi(v) - v} = \frac{dx}{x},$$

which is **separable** in v and x . (When x is the “heavier” variable, the mirror substitution $x = vy$ is cleaner.)

Notation key. y — dependent variable; x — independent variable; $v = y/x$ — the ratio variable; C, K, A — arbitrary constants; ϕ — a function of a single argument.

Three properties to lock in:

1. **Separable** $\Rightarrow$ **one integral per side**. No substitution needed; the difficulty is purely the two integrals.
 2. **Homogeneous** $\Rightarrow$ **always reduces to separable** via $y = vx$. So homogeneous problems are really “recognise the type, substitute, then separate.”
 3. **Degree-zero test**. Replace $x \rightarrow \lambda x$, $y \rightarrow \lambda y$ in the RHS; if every λ cancels, it’s homogeneous. A fast proxy: in $M dx + N dy = 0$, the equation is homogeneous iff M and N are homogeneous of the **same** degree.
-

Section 3: Key Procedures & Logical Framework

This subtopic is procedural, so here are the two routines with the reasoning baked in.

Procedure A — Solving a separable equation

1. **Separate**. Manipulate into $\frac{dy}{h(y)} = g(x) dx$, with all y ’s on the left and all x ’s on the right.
2. **Flag the danger**. Note any value making $h(y) = 0$ — these constant functions may be singular solutions you’re about to lose by dividing.
3. **Integrate both sides**, adding a *single* arbitrary constant C .
4. **Simplify** (combine logs, exponentiate) and, if an initial condition is given, solve for C .
5. **Restore lost solutions** identified in step 2 if they genuinely satisfy the original equation.

Why a single constant: the two indefinite integrals each carry a constant, but their difference is just one new arbitrary constant — writing two is redundant and a common source of confusion.

Procedure B — Solving a homogeneous equation

1. **Confirm the type**. Check degree-zero homogeneity (scale test, or “same-degree M, N ”). If a stray constant breaks it, this is *not* a standard homogeneous equation — don’t substitute $y = vx$.
2. **Rewrite** the RHS as a function of $v = y/x$.
3. **Substitute** $y = vx$, $\frac{dy}{dx} = v + x \frac{dv}{dx}$.
4. The equation collapses to $\frac{dv}{\phi(v) - v} = \frac{dx}{x}$ — now **separable**. Solve it.
5. **Back-substitute** $v = y/x$ to return to the original variables. *Never leave the answer in v .*

Why $y = vx$ works: a homogeneous RHS depends only on y/x . Forcing the unknown to “wear” the ratio explicitly ($y = vx$) converts the tangled x - y dependence into a clean v - x dependence, and the x -dependence then separates out by itself.

Choosing the smarter substitution. If the equation isolates $\frac{dx}{dy}$ as a function of x/y (i.e. x “dominates”), use $x = vy$ instead. Same logic, less algebra.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] (separable)

Solve $\frac{dy}{dx} = (4x^3 + 2)e^{-y}$.

Solution.

- *Separate:* multiply across by e^y — $e^y dy = (4x^3 + 2) dx$.
- *Integrate both sides:*

$$\int e^y dy = \int (4x^3 + 2) dx \implies e^y = x^4 + 2x + C.$$

- *General solution:* $\boxed{e^y = x^4 + 2x + C}$, i.e. $y = \log(x^4 + 2x + C)$.

Presentation tip: show the separation step explicitly and carry exactly one constant C — examiners look for both.

Example 2 — [CUET / JEE Main Speed Trap] (homogeneous, derivative-of-denominator shortcut)

Solve $\frac{dy}{dx} = \frac{x-y}{x+y}$.

Solution. Each of $x - y$ and $x + y$ is homogeneous of degree 1, so the ratio is degree 0 — homogeneous. Put $y = vx$:

$$v + x \frac{dv}{dx} = \frac{1-v}{1+v} \implies x \frac{dv}{dx} = \frac{1-v}{1+v} - v = \frac{1-2v-v^2}{1+v}.$$
$$\therefore \frac{1+v}{1-2v-v^2} dv = \frac{dx}{x}.$$

The speed move: notice $\frac{d}{dv}(1-2v-v^2) = -2-2v = -2(1+v)$ — the numerator is (up to a constant) the derivative of the denominator. So the left integral is an instant logarithm, no partial fractions:

$$-\frac{1}{2} \log |1-2v-v^2| = \log |x| + C.$$

Tidy and multiply through by x^2 after restoring $v = y/x$:

$$\boxed{x^2 - 2xy - y^2 = C}.$$

The trap: students reflexively split $\frac{1+v}{1-2v-v^2}$ into partial fractions and burn two minutes. Always scan for “numerator $\propto$ derivative of denominator” first — it converts the integral to a single log.

Example 3 — [JEE Main Level — multi-concept: homogeneous + trig integral + IVP]

A curve passes through $\left(1, \frac{\pi}{6}\right)$ and at every point its slope satisfies $\frac{dy}{dx} = \frac{y}{x} + \tan\left(\frac{y}{x}\right)$. Find the curve.

Solution. The RHS is a function of y/x alone — homogeneous. Put $y = vx$:

$$v + x \frac{dv}{dx} = v + \tan v \implies x \frac{dv}{dx} = \tan v \implies \cot v dv = \frac{dx}{x}.$$

Integrate (this is where the trig integral $\int \cot v dv = \log |\sin v|$ enters):

$$\log |\sin v| = \log |x| + \log C \implies \sin v = Cx.$$

Back-substitute $v = y/x$:

$$\sin\left(\frac{y}{x}\right) = Cx.$$

Apply the condition $\left(1, \frac{\pi}{6}\right)$: $\sin\left(\frac{\pi}{6}\right) = C(1) \Rightarrow C = \frac{1}{2}$.

$$\sin\left(\frac{y}{x}\right) = \frac{x}{2}$$

Why “multi-concept”: you must (i) diagnose homogeneity, (ii) recall a non-obvious trig antiderivative, and (iii) pin the constant with the initial condition — three separate skills in one question.

Example 4 — [JEE Advanced Level — reducible to homogeneous via origin shift]

Solve $\frac{dy}{dx} = \frac{x - y + 3}{x + y - 1}$.

Solution. The lone constants +3 and −1 break homogeneity, so $y = vx$ won’t work directly. The fix: **shift the origin** to the point where both lines vanish, killing the constants.

Solve $x - y + 3 = 0$ and $x + y - 1 = 0$ simultaneously: adding gives $2x + 2 = 0 \Rightarrow x = -1$; then $y = 2$. Let

$$X = x + 1, \quad Y = y - 2 \quad (\text{so } dX = dx, \quad dY = dy).$$

Substituting, the numerator $x - y + 3 = X - Y$ and the denominator $x + y - 1 = X + Y$, giving

$$\frac{dY}{dX} = \frac{X - Y}{X + Y},$$

which is **exactly the homogeneous equation from Example 2** (in X, Y). The only new skill here was the shift; the homogeneous core is already solved. Borrowing that result:

$$X^2 - 2XY - Y^2 = C.$$

Restore $X = x + 1$, $Y = y - 2$:

$$(x + 1)^2 - 2(x + 1)(y - 2) - (y - 2)^2 = C.$$

Note: this origin-shift trick works only when the two lines **intersect**. If they were *parallel* (coefficients proportional), you'd instead substitute the common linear combination $t = x + y$ and reduce to separable — a distinct sub-case worth keeping in your back pocket. (Origin-shift reducibility is a JEE extension beyond the strict NCERT core.)

Section 5: Practice Exercises

Try these yourself; final answers follow.

1. **[CBSE]** Solve $\frac{dy}{dx} = x^2 y$.
2. **[CUET]** Solve $\sec^2 x \tan y \, dx + \sec^2 y \tan x \, dy = 0$.
3. **[JEE Main]** Solve the homogeneous equation $\frac{dy}{dx} = \frac{x^2 + y^2}{xy}$.
4. **[JEE Main]** Solve $\frac{dy}{dx} = y \cos x$, given $y(0) = 1$.
5. **[JEE Advanced]** Solve $(x^3 + y^3) \, dx = xy^2 \, dy$.

Answers: 1) $y = A e^{x^3/3}$ — 2) $\tan x \tan y = C$ — 3) $y^2 = 2x^2(\log|x| + C) - 4$ — 4) $y = e^{\sin x}$ — 5) $y^3 = 3x^3(\log|x| + C)$.

Section 6: Crack the MCQ

Q1. Which of the following is a homogeneous differential equation?

- (A) $\frac{dy}{dx} = \frac{x + y + 1}{x - y}$ (B) $\frac{dy}{dx} = \frac{x^2 + y^2}{xy}$ (C) $\frac{dy}{dx} = x^2 + y^2$ (D) $\frac{dy}{dx} = xy + 1$

Correct: (B). Numerator and denominator are both homogeneous of degree 2, so their ratio is degree 0 — the RHS is a function of y/x alone. Homogeneous.

- **(A)** The stray +1 destroys scale-invariance: replacing x, y by $\lambda x, \lambda y$ does *not* leave it unchanged. Not homogeneous (it's the *reducible* type from Example 4).
- **(C)** Classic confusion: $x^2 + y^2$ is a homogeneous *function* (degree 2), but the *equation* $\frac{dy}{dx} = x^2 + y^2$ has RHS of degree 2 $\neq 0$, so it is **not** a homogeneous DE. Don't conflate "homogeneous function" with "homogeneous equation."
- **(D)** Degree mismatch (xy is degree 2, the +1 is degree 0); not homogeneous.

Q2. The general solution of $\frac{dy}{dx} = \frac{y}{x}$ is:

- (A) $y = Cx$ (B) $y = x + C$ (C) $xy = C$ (D) $y = Ce^x$

Correct: (A). Separating, $\frac{dy}{y} = \frac{dx}{x} \Rightarrow \log |y| = \log |x| + \log C \Rightarrow y = Cx$.

- **(B)** Treats the equation as $dy = dx$ — integrating without separating the $\frac{1}{y}$ and $\frac{1}{x}$.
- **(C)** Sign slip: integrating $\frac{dy}{y} = -\frac{dx}{x}$ would give $y = C/x$, i.e. $xy = C$ — the solution to $\frac{dy}{dx} = -\frac{y}{x}$, not this one.
- **(D)** Pattern-matches “exponential” reflexively; but the integral of $\frac{1}{x}$ is $\log x$, not x .

Q3. While solving $\frac{dy}{dx} = y^2$ by separation, one divides by y^2 . Which solution is lost in the process?

(A) $y = 0$ (B) $y = 1$ (C) $y = x$ (D) None — nothing is lost

Correct: (A). Dividing by y^2 tacitly assumes $y \neq 0$. But the constant function $y = 0$ satisfies the original equation ($0 = 0$), so it is a genuine **singular solution** thrown away by the division.

- **(B), (C)** are not solutions at all ($y = 1 \Rightarrow 0 = 1$; $y = x \Rightarrow 1 = x^2$), so nothing about them is “lost.”
- **(D)** is the complacent answer — exactly the mistake the question targets.

Q4. Solving $\frac{dy}{dx} = \frac{x+y}{x}$ by the substitution $y = vx$ gives:

(A) $y = x \log |x| + Cx$ (B) $y = x + Cx$ (C) $y = \log |x| + C$ (D) $y = Cx$

Correct: (A). Here $\frac{dy}{dx} = 1 + \frac{y}{x}$. With $y = vx$: $v + x \frac{dv}{dx} = 1 + v \Rightarrow x \frac{dv}{dx} = 1 \Rightarrow dv = \frac{dx}{x} \Rightarrow v = \log |x| + C$. Back-substitute $v = y/x$: $y = x \log |x| + Cx$.

- **(C)** forgets to multiply back by x — leaving the answer in v instead of y .
- **(D)** is the solution to $\frac{dy}{dx} = \frac{y}{x}$ — dropping the $+1$ from the numerator.
- **(B)** integrates $dv = \frac{dx}{x}$ as if it were $dv = dx$.

Section 7: Common Student Pitfalls & Pro-Tips

- **Losing the constant solution.** Dividing by $h(y)$ to separate can erase a valid solution $h(y) = 0$. Always note it before dividing and check it at the end.
- **Forcing $y = vx$ onto a non-homogeneous equation.** A stray constant in the numerator/denominator means it isn’t homogeneous — shift the origin (or use a linear substitution) instead.
- **Leaving the answer in v .** After solving in v and x , you *must* back-substitute $v = y/x$. An answer containing v scores zero.
- **Double constants / messy logs.** Carry exactly one arbitrary constant, and write log-constants as $\log C$ so they combine cleanly.

Pro-Tip (diagnosis first, then speed): Before touching algebra, classify — “Do the variables separate as-is? If not, is the RHS a pure function of y/x ?” If separable, integrate two columns. If homogeneous, substitute $y = vx$ (or $x = vy$ if x dominates). And inside any separation integral, scan for **numerator** $\propto$ **derivative of denominator** to turn it into an instant logarithm and skip partial fractions.

Section 8: Cheat Sheet Box

- **Separable:** $\frac{dy}{dx} = g(x)h(y) \Rightarrow \int \frac{dy}{h(y)} = \int g(x) dx + C$. Watch for lost solutions $h(y) = 0$.
- **Homogeneous:** RHS is a function of y/x only (degree-zero). Put $y = vx$, $\frac{dy}{dx} = v + x \frac{dv}{dx}$, reduce to separable, then **back-substitute** $v = y/x$.
- **Type test:** $F(\lambda x, \lambda y) = F(x, y) \Rightarrow$ homogeneous. A stray constant $\Rightarrow$ **not** homogeneous (shift origin).
- **Smart substitution:** use $x = vy$ when x dominates.
- **Integral shortcut:** numerator $\propto$ derivative of denominator $\Rightarrow$ single log.

Memory aids:

- “Separate the laundry, integrate each basket.” (separable)
- “If it only sees the ratio y/x , let $y = vx$.” (homogeneous)
- “Divide and you might lose a root.” (singular/lost solution)
- “Came in as y , must leave as y .” (always back-substitute)

Separable: split & integrate. **Homogeneous:** $y = vx \Rightarrow$ separable $\Rightarrow$ back-sub.

Part 4 — Linear Differential Equations

Exam Relevance: This is the single most examined solving method in the chapter. CBSE Boards almost guarantee a 3–5 mark “solve this linear DE” question, very often with an initial condition. JEE Main asks 1–2 per year — frequently an integrating-factor MCQ or an equation cunningly disguised as *linear in x* rather than y . JEE Advanced folds the method into curve-finding and reducible forms. If you master one technique in this chapter, make it this one.

Section 1: Concept Intuition

So far the toolbox has two tools: separate the variables, or switch to ratios for homogeneous equations. But a huge family of equations resists both — equations where y and its derivative $\frac{dy}{dx}$ both appear, each only to the **first power**, with no products like $y \cdot \frac{dy}{dx}$ and no powers like y^2 . These are called **linear** differential equations, and they have their own beautiful trick: the **integrating factor**.

Here’s the idea in plain words. The left side of $\frac{dy}{dx} + Py = Q$ is *almost* the derivative of a product, but not quite. The integrating-factor method finds one magic multiplier — call it the IF — such that after you multiply the whole equation through by it, the left side becomes **exactly** the derivative of $(\text{IF} \times y)$. Once the left side is a clean single derivative, you just integrate both sides and you’re done.

Think about it this way... Imagine you’re trying to fold a fitted bed-sheet — the elastic corners fight you and nothing lies flat. But there’s one specific way to tuck the corners (the “IF”) after which the whole sheet folds into a neat rectangle in one motion. The integrating factor is that tuck: it turns a messy left-hand side into a single, foldable derivative $\frac{d}{dx}(\text{IF} \cdot y)$ that integrates in one step.

The genius is that this magic multiplier is *completely predictable*: for $\frac{dy}{dx} + Py = Q$, it is always

$$\text{IF} = e^{\int P dx}.$$

You never have to guess it.

Limiting conditions & assumptions — read carefully:

- **The equation must first be in standard form** $\frac{dy}{dx} + Py = Q$, where the coefficient of $\frac{dy}{dx}$ is exactly 1, and P, Q depend on x **only** (not on y). If there’s a coefficient like x or $\cos x$ sitting in front of $\frac{dy}{dx}$, divide it out *before* reading off P . Skipping this is the number-one error.
- **Linearity is about y , not x .** If y appears nonlinearly but x appears to the first power, the equation may be **linear in x** instead — flip your viewpoint to $\frac{dx}{dy}$.

- **P and Q are functions of the independent variable alone.** A genuine y inside P or Q means it isn't a standard linear DE.

Section 2: Key Definitions & Properties

(Mathematics adaptation:
physics formulas/SI units.)

Standard form (linear in y):

$$\frac{dy}{dx} + P y = Q, \quad P = P(x), \quad Q = Q(x).$$

Integrating factor:

$$\text{IF} = e^{\int P dx}$$

General solution:

$$y \cdot (\text{IF}) = \int Q \cdot (\text{IF}) dx + C$$

Mirror case — linear in x : when the equation is naturally first-degree in x , write it as

$$\frac{dx}{dy} + P_1 x = Q_1, \quad P_1 = P_1(y), \quad Q_1 = Q_1(y),$$

with $\text{IF} = e^{\int P_1 dy}$ and solution $x \cdot (\text{IF}) = \int Q_1 \cdot (\text{IF}) dy + C$.

Notation key. y — dependent variable; x — independent variable; P, Q — known functions of x ; P_1, Q_1 — known functions of y (mirror case); IF — integrating factor; C — arbitrary constant.

Three properties to lock in:

1. **The LHS is always a perfect derivative.** After multiplying by the IF, $\frac{dy}{dx} \cdot \text{IF} + P y \cdot \text{IF} = \frac{d}{dx}(y \cdot \text{IF})$ — automatically. You never expand it; you just integrate the right side.
2. **You integrate $Q \cdot \text{IF}$, not Q alone.** The most common slip is integrating Q by itself.
3. **Constants in P inside $\int P dx$ matter.** $\int \frac{2}{x} dx = 2 \log x \Rightarrow \text{IF} = x^2$; $\int -\frac{2}{x} dx = -2 \log x \Rightarrow \text{IF} = x^{-2}$. A sign slip here flips the whole IF.

Section 3: CBSE Board Derivation & Key Procedures

The derivation — why the integrating factor is $e^{\int P dx}$

This is the one genuine derivation of the subtopic, and CBSE can ask for it.

Start from the standard form and multiply through by an as-yet-unknown function $\mu(x)$:

$$\mu \frac{dy}{dx} + \mu P y = \mu Q. \quad (1)$$

We *want* the left side to be the derivative of a single product μy . By the product rule,

$$\frac{d}{dx}(\mu y) = \mu \frac{dy}{dx} + \frac{d\mu}{dx} y. \quad (2)$$

Comparing (1) and (2), the two left sides match **provided**

$$\frac{d\mu}{dx} = \mu P.$$

This is itself a separable equation for μ :

$$\frac{d\mu}{\mu} = P dx \implies \log \mu = \int P dx \implies \mu = e^{\int P dx}.$$

So the magic multiplier is forced — it *must* be $\text{IF} = e^{\int P dx}$. With this choice, (1) becomes $\frac{d}{dx}(y \cdot \text{IF}) = Q \cdot \text{IF}$, and integrating both sides gives the solution $y \cdot \text{IF} = \int Q \cdot \text{IF} dx + C$. ■

Procedure — solving a linear DE

1. **Reduce to standard form** $\frac{dy}{dx} + Py = Q$: make the coefficient of $\frac{dy}{dx}$ equal to 1.
2. **Identify** P and Q (both functions of x only).
3. **Compute** $\text{IF} = e^{\int P dx}$ — watch the sign of $\int P dx$.
4. **Write the solution** $y \cdot \text{IF} = \int Q \cdot \text{IF} dx + C$ and perform the (often by-parts) integral.
5. **Apply any initial condition** to find C .

The flip test (linear in x): if step 1 won't produce P, Q as functions of x because y is tangled nonlinearly, rewrite the equation as $\frac{dx}{dy} + P_1 x = Q_1$ and run the same five steps with the roles of x and y swapped.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Solve $\frac{dy}{dx} + \frac{2}{x}y = x^3$.

Solution.

- *Standard form:* already there, with $P = \frac{2}{x}$, $Q = x^3$.
- *Integrating factor:* $\text{IF} = e^{\int \frac{2}{x} dx} = e^{2 \log x} = x^2$.

• **Solution:** $y \cdot x^2 = \int x^3 \cdot x^2 dx + C = \int x^5 dx + C = \frac{x^6}{6} + C.$

$$\boxed{y = \frac{x^4}{6} + \frac{C}{x^2}}.$$

Presentation tip: show IF explicitly and integrate $Q \cdot \text{IF}$ (here x^5), not Q alone — both steps carry marks.

Example 2 — [CUET / JEE Main Speed Trap — the standard-form trap]

Solve $x \frac{dy}{dx} - 3y = x^5.$

Solution. The trap is screaming here: a *coefficient* x sits in front of $\frac{dy}{dx}$, so you may **not** read $P = -3$. Divide through by x first:

$$\frac{dy}{dx} - \frac{3}{x}y = x^4 \implies P = -\frac{3}{x}, Q = x^4.$$

$$\text{IF} = e^{\int -\frac{3}{x} dx} = e^{-3 \log x} = x^{-3}.$$

$$y \cdot x^{-3} = \int x^4 \cdot x^{-3} dx + C = \int x dx + C = \frac{x^2}{2} + C.$$

$$\boxed{y = \frac{x^5}{2} + Cx^3}.$$

The trap, named: students who skip the division use $P = -3$, get $\text{IF} = e^{-3x}$, and the whole problem collapses. Speed rule: **coefficient of $\frac{dy}{dx}$ must be 1 before you touch P .** And note the negative P gives $\text{IF} = x^{-3}$ — mind the sign.

Example 3 — [JEE Main Level — multi-concept: linear + trig identity + IVP]

A curve satisfies $\frac{dy}{dx} + y \cot x = 4x \csc x$ **and passes through** $\left(\frac{\pi}{2}, 0\right)$. **Find y .**

Solution. Standard form already: $P = \cot x$, $Q = 4x \csc x$.

$$\text{IF} = e^{\int \cot x dx} = e^{\log \sin x} = \sin x.$$

$$y \sin x = \int 4x \csc x \cdot \sin x dx + C.$$

Here the **trig simplification** is the multi-concept twist: $\csc x \cdot \sin x = 1$, so the integral collapses beautifully:

$$y \sin x = \int 4x dx + C = 2x^2 + C.$$

Apply $(\frac{\pi}{2}, 0)$: $0 \cdot 1 = 2(\frac{\pi}{2})^2 + C \Rightarrow C = -\frac{\pi^2}{2}$.

$$y \sin x = 2x^2 - \frac{\pi^2}{2} \Rightarrow \boxed{y = \frac{4x^2 - \pi^2}{2 \sin x}}.$$

Why “multi-concept”: you must (i) build the IF, (ii) *recognise* the $\csc x \cdot \sin x = 1$ collapse to avoid a horrible integral, and (iii) pin C with the point — three skills in one.

Example 4 — [JEE Advanced Level — linear in x , not y]

Solve $\frac{dy}{dx} = \frac{y}{x + y^2}$.

Solution. Try to force standard form in y : $\frac{dy}{dx} = \frac{y}{x+y^2}$ has y^2 in the denominator — *nonlinear in y* . Don't fight it. **Flip to** $\frac{dx}{dy}$:

$$\frac{dx}{dy} = \frac{x + y^2}{y} = \frac{x}{y} + y \Rightarrow \frac{dx}{dy} - \frac{1}{y}x = y.$$

Now it is **linear in x** , with $P_1 = -\frac{1}{y}$, $Q_1 = y$.

$$\text{IF} = e^{\int -\frac{1}{y} dy} = e^{-\log y} = \frac{1}{y}.$$

$$x \cdot \frac{1}{y} = \int y \cdot \frac{1}{y} dy + C = \int 1 dy + C = y + C.$$

$$\frac{x}{y} = y + C \Rightarrow \boxed{x = y^2 + Cy}.$$

Quick verify: from $x = y^2 + Cy$, $\frac{dx}{dy} = 2y + C$, so $\frac{dy}{dx} = \frac{1}{2y+C}$; and $\frac{y}{x+y^2} = \frac{y}{2y^2+Cy} = \frac{1}{2y+C}$. ✓ The whole difficulty was *recognising which variable the equation is linear in*.

Section 5: Practice Exercises

Try these yourself; final answers follow.

1. **[CBSE]** Solve $\frac{dy}{dx} + y = e^x$.
2. **[CUET]** Solve $x \frac{dy}{dx} + y = x^3$.
3. **[JEE Main]** Solve $\frac{dy}{dx} + 2xy = x$.

4. **[JEE Main]** Solve $\frac{dy}{dx} + y \tan x = \sec x$, given $y(0) = 1$.

5. **[JEE Advanced]** Solve $\frac{dy}{dx} = \frac{1}{x \cos y + \sin 2y}$.

Answers: 1) $y = \frac{1}{2}e^x + Ce^{-x}$ — 2) $y = \frac{x^3}{4} + \frac{C}{x}$ — 3) $y = \frac{1}{2} + Ce^{-x^2}$ — 4) $y = \sin x + \cos x$ — 5) $x = Ce^{\sin y} - 2(1 + \sin y)$ (flip to $\frac{dx}{dy} - x \cos y = \sin 2y$, linear in x).

Section 6: Crack the MCQ

Q1. The integrating factor of $x \frac{dy}{dx} - 2y = x^5$ is:

- (A) x^2 (B) $\frac{1}{x^2}$ (C) $-2x$ (D) e^{-2x}

Correct: (B). Reduce to standard form first: $\frac{dy}{dx} - \frac{2}{x}y = x^4$, so $P = -\frac{2}{x}$ and $\text{IF} = e^{\int -\frac{2}{x} dx} = e^{-2 \log x} = x^{-2}$.

- **(A)** drops the minus sign on P — using $+\frac{2}{x}$ gives x^2 .
- **(D)** forgets to divide by x and treats $P = -2$, giving e^{-2x} .
- **(C)** confuses $\int P dx$ with the IF itself.

Q2. Which of the following is a linear differential equation?

- (A) $\frac{dy}{dx} + y^2 = x$ (B) $\frac{dy}{dx} + xy = x^2$ (C) $\left(\frac{dy}{dx}\right)^2 + y = x$ (D) $y \frac{dy}{dx} + y = x$

Correct: (B). y and $\frac{dy}{dx}$ each appear to the first power, with coefficients depending on x only. Linear.

- **(A)** has y^2 — nonlinear in y .
- **(C)** has $\left(\frac{dy}{dx}\right)^2$ — nonlinear in the derivative.
- **(D)** has the product $y \frac{dy}{dx}$ — nonlinear. (It is linear in x after flipping, but as written in y it is not.)

Q3. The solution of $\frac{dy}{dx} - y = 2$, $y(0) = 0$ is:

- (A) $y = 2(e^x - 1)$ (B) $y = 2(e^{-x} - 1)$ (C) $y = e^x - 2$ (D) $y = 2e^x$

Correct: (A). $P = -1 \Rightarrow \text{IF} = e^{-x}$; then $ye^{-x} = \int 2e^{-x} dx + C = -2e^{-x} + C$, so $y = -2 + Ce^x$. Applying $y(0) = 0$: $0 = -2 + C \Rightarrow C = 2$, giving $y = 2(e^x - 1)$.

- **(B)** uses $\text{IF} = e^x$ (wrong sign of P).
- **(C)** mishandles the constant / forgets the factor on C .
- **(D)** ignores the constant term -2 entirely.

Q4. The equation $\frac{dy}{dx} = \frac{1}{x+y}$ is most efficiently solved by treating it as:

- (A) linear in y , as written (B) linear in x , after swapping variables (C) homogeneous, via $y = vx$ (D) variables separable

Correct: (B). Inverting, $\frac{dx}{dy} = x + y$, i.e. $\frac{dx}{dy} - x = y$ — linear in x with $P_1 = -1$.

- **(A)** fails: as written, y sits in the denominator (nonlinear in y).
- **(C)** the $+y$ breaks degree-zero homogeneity — not homogeneous.
- **(D)** the variables don't separate ($x + y$ won't factor into an x -part times a y -part).

Section 7: Common Student Pitfalls & Pro-Tips

- **Reading P before standard form.** A coefficient in front of $\frac{dy}{dx}$ must be divided out first; otherwise P (and the whole IF) is wrong.
- **Sign errors in the IF.** P negative $\Rightarrow \int P dx$ negative $\Rightarrow$ IF is a *negative* power or a *decaying* exponential. Double-check the sign before exponentiating.
- **Integrating Q instead of $Q \cdot \text{IF}$.** The solution is $y \cdot \text{IF} = \int Q \cdot \text{IF} dx$ — the IF rides along into the integral.
- **Missing “linear in x .”** If y is tangled but x is first-degree, flip to $\frac{dx}{dy}$. Many “hard” problems are ordinary linear DEs in disguise.

Pro-Tip (never expand the LHS): once you multiply by the IF, the left side is *guaranteed* to be $\frac{d}{dx}(y \cdot \text{IF})$ — don't waste time expanding or verifying it; jump straight to integrating $Q \cdot \text{IF}$ on the right. And before solving, glance at the equation: “*first power in y and y' , no products?*” If yes, linear in y ; if y is messy but x is clean, linear in x .

Section 8: Cheat Sheet Box

- **Standard form:** $\frac{dy}{dx} + Py = Q$ (coefficient of $\frac{dy}{dx}$ must be 1; P, Q functions of x).
- **Integrating factor:** $\text{IF} = e^{\int P dx}$.
- **Solution:** $y \cdot \text{IF} = \int Q \cdot \text{IF} dx + C$.
- **Linear in x (mirror):** $\frac{dx}{dy} + P_1 x = Q_1$, $\text{IF} = e^{\int P_1 dy}$, $x \cdot \text{IF} = \int Q_1 \cdot \text{IF} dy + C$.
- **LHS is always** $\frac{d}{dx}(y \cdot \text{IF})$ — never expand it.
- **Watch:** standard form first; IF sign; integrate $Q \cdot \text{IF}$.

Memory aids:

- “*Standard form first, P second.*” (divide out the leading coefficient)
- “*IF = e to the integral of P .*”

- “The IF rides into the integral.” (it’s $Q \cdot \text{IF}$, not Q)
- “Messy in y ? Try x .” (flip to $\frac{dx}{dy}$)

$$\mathbf{IF} = e^{\int P dx} \Rightarrow \frac{d}{dx}(y \cdot \mathbf{IF}) = Q \cdot \mathbf{IF} \Rightarrow y \cdot \mathbf{IF} = \int Q \cdot \mathbf{IF} dx + C.$$

Part 5 — Applications of Differential Equations

Exam Relevance: This is where the chapter meets the real world, and it pays off in marks. CBSE Boards set growth/decay and cooling word problems (3–5 marks) in the miscellaneous set. JEE Main reliably asks one application a year — exponential growth/decay, Newton’s law of cooling, or a slope-defined curve. JEE Advanced layers in orthogonal trajectories and tangent/normal geometry. Every model here reduces to a separable or linear DE you already know how to solve (Parts 3–4).

Section 1: Concept Intuition

Almost every “rate” sentence in science is secretly a differential equation. “*Bacteria multiply faster when there are more of them.*” “*Hot tea cools quicker when the gap with room temperature is larger.*” “*The steeper the slope, the faster the curve climbs.*” Each statement links a **rate of change** to the **current state** — and that link is a DE waiting to be written and solved.

Three model situations cover the overwhelming majority of exam questions:

Growth & decay. When a quantity changes at a rate proportional to its own size — populations, bacteria, radioactive atoms, compound interest — the model is $\frac{dN}{dt} = kN$. With $k > 0$ it grows (population); with $k < 0$ it decays (radioactivity). Solving gives the famous exponential $N = N_0 e^{kt}$.

Newton’s law of cooling. A hot body loses heat at a rate proportional to *how much hotter it is than its surroundings* — not to its own temperature. So the driver is the **difference** $(\theta - \theta_s)$, giving $\frac{d\theta}{dt} = -k(\theta - \theta_s)$.

Geometric applications. Here the data is about the curve’s *shape*: the slope of the tangent ($\frac{dy}{dx}$), the subnormal ($y \frac{dy}{dx}$), the subtangent, or the requirement that two families cross at right angles (orthogonal trajectories). Translate the geometric condition into a DE and solve.

Think about it this way... Picture your hot chai again. Right out of the kettle it cools fast (huge gap with the room); near room temperature it barely cools at all (tiny gap). The *rate* tracks the *gap*, not the chai’s own temperature — that single insight is the difference between Newton’s cooling and ordinary decay, and getting it right is half the marks.

Limiting conditions & assumptions — read carefully:

- **Get the sign of k right.** Decay and cooling lose quantity, so the rate is *negative*: $\frac{dN}{dt} = -kN$ (decay) and $\frac{d\theta}{dt} = -k(\theta - \theta_s)$ with $k > 0$. A wrong sign makes a cooling body heat up forever.
 - **Cooling depends on the difference, not the temperature.** Modelling $\frac{d\theta}{dt} \propto \theta$ instead of $\propto (\theta - \theta_s)$ is the classic setup error.
 - **Keep time units consistent** and apply the given conditions in the right order (initial value fixes N_0 or the constant; a second data point fixes k).
-

Section 2: Key Definitions & Properties

formulas/SI units.)

Exponential growth / decay:

$$\frac{dN}{dt} = kN \implies N = N_0 e^{kt},$$

where N_0 is the amount at $t = 0$; $k > 0$ growth, $k < 0$ decay. **Doubling time** (growth): $t = \frac{\ln 2}{k}$. **Half-life** (decay, with $\frac{dN}{dt} = -kN$): $t_{1/2} = \frac{\ln 2}{k}$.

Newton's law of cooling:

$$\frac{d\theta}{dt} = -k(\theta - \theta_s) \implies \theta = \theta_s + (\theta_0 - \theta_s) e^{-kt},$$

where θ_s is the surrounding temperature, θ_0 the initial temperature. As $t \rightarrow \infty$, $\theta \rightarrow \theta_s$.

Geometric quantities at a point (x, y) on a curve, with slope $m = \frac{dy}{dx}$:

$$\text{Subtangent} = \frac{y}{dy/dx}, \quad \text{Subnormal} = y \frac{dy}{dx}.$$

Orthogonal trajectories: a family cutting another family at right angles — obtained by replacing $\frac{dy}{dx}$ with $-\frac{1}{dy/dx}$ in the first family's DE and solving.

Notation key. N — quantity (size/number/mass); N_0 — initial quantity; k — positive rate constant; t — time; θ — temperature; θ_s — surrounding temperature; θ_0 — initial temperature.

Three properties to lock in:

1. **Proportional-to-itself** $\implies$ **exponential**. Any “rate $\propto$ amount” gives e^{kt} ; the only question is the sign and the constants.
2. **Cooling has a floor**. The exponential term decays to zero, so temperature settles at θ_s — never below the room.
3. **Geometry** $\rightarrow$ **DE by dictionary**. “slope = ...”, “subnormal = ...”, “cuts at right angles” each translate to a specific derivative condition; the rest is ordinary solving.

Section 3: Key Procedures & Logical Framework

This is procedural: the skill is **translating words into a DE**, then solving by the methods of Parts 3–4.

Procedure — solving an application

1. **Name the variables** and identify what changes with respect to what (N with t , θ with t , y with x).

2. **Translate the rate sentence into a DE.** “Rate $\propto$ size” $\rightarrow \frac{dN}{dt} = kN$; “rate $\propto$ gap with surroundings” $\rightarrow \frac{d\theta}{dt} = -k(\theta - \theta_s)$; “slope = ...” / “subnormal = ...” $\rightarrow$ the geometric condition. Fix the **sign** by asking whether the quantity rises or falls.
3. **Solve the DE** (these models are separable; some geometric ones are linear).
4. **Apply conditions in order:** the initial value pins N_0 / the constant; a second data point pins k .
5. **Answer the exact question asked** (a time, a temperature, a fraction, an equation of a curve) — and sanity-check the limiting behaviour.

Translation discipline: the most expensive error is mis-translating the English. Underline the rate word (“rate”, “proportional to”, “slope”, “cuts at right angles”) and the thing it’s proportional to *before* writing any symbols.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] (growth)

A bacterial culture grows at a rate proportional to its size and doubles in 4 hours. How long does it take to become 8 times its original size?

Solution.

- *Model:* $\frac{dN}{dt} = kN \Rightarrow N = N_0 e^{kt}$.
- *Find k from doubling:* $2N_0 = N_0 e^{4k} \Rightarrow e^{4k} = 2 \Rightarrow k = \frac{\ln 2}{4}$.
- *Time to reach $8N_0$:* $8 = e^{kt} \Rightarrow kt = \ln 8 = 3 \ln 2 \Rightarrow t = \frac{3 \ln 2}{(\ln 2)/4} = 12$.

$$\boxed{t = 12 \text{ hours}}.$$

Sanity check: $8 = 2^3$, and each doubling takes 4 h, so $3 \times 4 = 12$ h — matches.

Example 2 — [CUET / JEE Main Speed Trap] (decay)

A radioactive substance has a half-life of 1600 years. What fraction remains after 4800 years?

Solution (count half-lives — no calculus needed). $\frac{4800}{1600} = 3$ half-lives. Each halves the amount:

$$\left(\frac{1}{2}\right)^3 = \boxed{\frac{1}{8}}.$$

The trap: setting up $N = N_0 e^{-kt}$, solving for k , then plugging back — correct but slow, and easy to slip a sign. When the elapsed time is a whole number of half-lives, just count. (For non-integer multiples, fall back to $N/N_0 = (1/2)^{t/t_{1/2}}$.)

Example 3 — [JEE Main Level] (Newton's law of cooling)

A body at 100°C is placed in a room at 25°C . It cools to 70°C in 10 minutes. Find its temperature after a further 10 minutes.

Solution.

- Model: $\frac{d\theta}{dt} = -k(\theta - 25) \Rightarrow \theta = 25 + (100 - 25)e^{-kt} = 25 + 75e^{-kt}$.
- Use $t = 10$, $\theta = 70$: $70 = 25 + 75e^{-10k} \Rightarrow 75e^{-10k} = 45 \Rightarrow e^{-10k} = \frac{3}{5}$.
- At $t = 20$: $\theta = 25 + 75e^{-20k} = 25 + 75(e^{-10k})^2 = 25 + 75 \cdot \frac{9}{25} = 25 + 27$.

$$\boxed{\theta = 52^\circ\text{C}}.$$

Why " $(e^{-10k})^2$ ": writing e^{-20k} as the square of the known e^{-10k} avoids ever solving for k — a clean shortcut whenever the second time is a multiple of the first.

Example 4 — [JEE Advanced Level] (orthogonal trajectories)

Find the orthogonal trajectories of the family of parabolas $y = Cx^2$.

Solution. First form the family's own DE (Part 2) : from $y = Cx^2$, $\frac{dy}{dx} = 2Cx$, and $C = \frac{y}{x^2}$, so

$$\frac{dy}{dx} = 2 \cdot \frac{y}{x^2} \cdot x = \frac{2y}{x}.$$

For curves crossing this family at right angles, **replace $\frac{dy}{dx}$ by $-\frac{1}{dy/dx}$** :

$$\frac{dy}{dx} = -\frac{x}{2y} \Rightarrow 2y dy = -x dx.$$

Integrate (separable):

$$y^2 = -\frac{x^2}{2} + C_1 \Rightarrow \boxed{\frac{x^2}{2} + y^2 = C_1} \quad (\text{a family of ellipses}).$$

The picture: every parabola $y = Cx^2$ meets every ellipse $\frac{x^2}{2} + y^2 = C_1$ at 90° . This single problem chains **formation** (Part 2), the **orthogonality swap**, and **separable solving** (Part 3) — exactly the kind of synthesis JEE Advanced rewards.

Section 5: Practice Exercises

Try these yourself; final answers follow.

1. **[CBSE]** A population grows at a rate proportional to its size and triples in 5 years. When will it be 9 times the original?
2. **[CUET]** A radioactive isotope has a half-life of 20 days. What fraction remains after 60 days?
3. **[JEE Main]** Water at 80°C is kept in a room at 20°C and cools to 50°C in 5 minutes. Find its temperature after 10 minutes.
4. **[JEE Main]** Find the curve passing through $(1, 2)$ whose slope at every point equals $\frac{y}{x}$.
5. **[JEE Advanced]** Find the curve through $(0, 1)$ for which the subnormal at every point is constant and equal to 2.

Answers: 1) 10 years — 2) $\frac{1}{8}$ — 3) 35°C — 4) $y = 2x$ — 5) $y^2 = 4x + 1$ (from $y\frac{dy}{dx} = 2$).

Section 6: Crack the MCQ

Q1. The differential equation $\frac{dN}{dt} = kN$ (with $k > 0$) models:

- (A) exponential growth (B) exponential decay (C) linear growth (D) logistic growth

Correct: (A). Rate proportional to current size with $k > 0$ gives $N = N_0 e^{kt}$, unbounded exponential growth.

- **(B)** would need $k < 0$.
- **(C)** “linear growth” is $\frac{dN}{dt} = k$ (constant rate), not $\propto N$.
- **(D)** logistic growth has a carrying-capacity term, $\frac{dN}{dt} = kN(1 - N/M)$ — not this equation.

Q2. By Newton’s law of cooling, as $t \rightarrow \infty$ the temperature of a cooling body approaches:

- (A) the surrounding temperature θ_s (B) 0° (C) its initial temperature θ_0 (D) infinity

Correct: (A). $\theta = \theta_s + (\theta_0 - \theta_s)e^{-kt}$; the exponential decays to 0, leaving $\theta \rightarrow \theta_s$.

- **(B)** assumes cooling to absolute reference — it stops at the room, not zero.
- **(C)** is the *initial* value, at $t = 0$. **(D)** is impossible for cooling (wrong sign of k).

Q3. A substance decays to $\frac{1}{4}$ of its amount in 10 years. Its half-life is:

- (A) 5 years (B) 2.5 years (C) 10 years (D) 20 years

Correct: (A). $\frac{1}{4} = \left(\frac{1}{2}\right)^2$, so 10 years is **two** half-lives $\Rightarrow$ half-life = 5 years.

- **(C)** mistakes the quarter-life for the half-life.
- **(B), (D)** misdivide the number of half-lives.

Q4. If the subnormal of a curve is constant at every point, the curve is a:

- (A) parabola (B) circle (C) straight line (D) rectangular hyperbola

Correct: (A). Subnormal $= y \frac{dy}{dx} = a$ (constant) $\Rightarrow y dy = a dx \Rightarrow y^2 = 2ax + C$ — a parabola.

- **(B)** would arise from a *constant length of normal* condition, not constant subnormal.
 - **(C)** constant *slope* gives a line, a different condition.
 - **(D)** comes from $xy = c$, unrelated to the subnormal condition.
-

Section 7: Common Student Pitfalls & Pro-Tips

- **Wrong sign of k .** Decay and cooling *lose* quantity — use $\frac{dN}{dt} = -kN$ and $\frac{d\theta}{dt} = -k(\theta - \theta_s)$ with $k > 0$. A sign slip makes things grow or heat without bound.
- **Cooling modelled on temperature, not the gap.** It's $\propto (\theta - \theta_s)$, never $\propto \theta$.
- **Solving for k when you don't need to.** If the second time is a multiple of the first, write $e^{-20k} = (e^{-10k})^2$ and substitute the known ratio — skip k entirely.
- **Mis-translating the English.** Underline the rate word and what it's proportional to before writing symbols; “slope”, “subnormal”, “cuts at right angles” each map to a specific derivative condition.

Pro-Tip (count, don't compute): for growth/decay over a whole number of doubling-times or half-lives, just count the doublings/halvings (2^n or $(1/2)^n$) instead of solving for k — seconds instead of minutes. Reserve the full e^{kt} machinery for awkward (non-integer) ratios.

Section 8: Cheat Sheet Box

- **Growth / decay:** $\frac{dN}{dt} = kN \Rightarrow N = N_0 e^{kt}$ ($k > 0$ grow, $k < 0$ decay);
half-life/doubling $= \frac{\ln 2}{|k|}$.
- **Newton's cooling:** $\frac{d\theta}{dt} = -k(\theta - \theta_s) \Rightarrow \theta = \theta_s + (\theta_0 - \theta_s)e^{-kt}$; $\theta \rightarrow \theta_s$ as $t \rightarrow \infty$.
- **Subnormal** $= y \frac{dy}{dx}$; **subtangent** $= \frac{y}{dy/dx}$.
- **Orthogonal trajectories:** replace $\frac{dy}{dx} \rightarrow -\frac{1}{dy/dx}$ in the family's DE, then solve.
- **Every model reduces to a separable or linear DE** (Parts 3–4).

Memory aids:

- *“Proportional to itself $\Rightarrow$ exponential.”*
- *“Cooling tracks the gap, not the heat.”*
- *“Lose $\Rightarrow$ minus k .”*
- *“Right angles $\Rightarrow$ negative reciprocal slope.”*

$$\frac{dN}{dt} = kN \Rightarrow N_0 e^{kt}; \quad \frac{d\theta}{dt} = -k(\theta - \theta_s) \Rightarrow \theta_s + (\theta_0 - \theta_s)e^{-kt}.$$

Chapter 9: Differential Equations — Round 2

Addendum

Parts 1 to 5 taught you to classify an equation, form one from a family of curves, and solve the three workhorse types — separable, homogeneous, and linear — plus their standard applications. Four examinable patterns are still standing outside the door. Addendum A introduces exact equations and the inspection table of ready-made differentials, the method behind a large share of JEE Main’s “solve” questions that fit none of the three workhorse types. Addendum B supplies the two reducible types the chapter only gestures at: Bernoulli equations and $\frac{dy}{dx} = f(ax + by + c)$. Addendum C completes the geometric dictionary — the chapter gave you subnormal and subtangent, but JEE sets tangent length, normal length, intercept, and fixed-point conditions just as often. Addendum D lowers the order of a second-order equation by substituting for y' , a JEE Advanced staple that sits one level above everything the chapter attempts.

Nothing here is new theory. Every method reduces to separation or to the integrating factor you already own; what is new is the *recognition* step that gets you there.

Addendum A — Exact Differential Equations and Solving by Inspection

Exam Relevance: CBSE asks exactness rarely, but JEE Main sets one “solve the equation” question most years whose equation is neither separable, nor homogeneous, nor linear — it is exact, or becomes exact after dividing by a well-chosen factor. JEE Advanced goes further and expects you to *spot* the perfect differential by eye. This addendum gives the test, the construction, and the table that makes spotting mechanical.

Method

An equation written as

$$M dx + N dy = 0$$

is called **exact** if there exists a function $F(x, y)$ whose total differential is exactly the left side:

$$dF = F_x dx + F_y dy = M dx + N dy,$$

so that the equation simply says $dF = 0$, and the solution is $F(x, y) = C$. The entire method rests on answering one question: given M and N , when does such an F exist?

The test, derived. If F exists with $F_x = M$ and $F_y = N$, then by equality of mixed partials,

$$\frac{\partial M}{\partial y} = \frac{\partial^2 F}{\partial y \partial x} = \frac{\partial^2 F}{\partial x \partial y} = \frac{\partial N}{\partial x}.$$

So $M_y = N_x$ is *necessary*. It is also sufficient, and the sufficiency proof hands you the solution algorithm for free. Assume $M_y = N_x$. Integrate M with respect to x , holding y constant:

$$G(x, y) = \int M dx, \quad G_x = M.$$

Now compare G_y with N . Their difference $H = N - G_y$ satisfies

$$H_x = N_x - (G_y)_x = N_x - (G_x)_y = N_x - M_y = 0,$$

so H depends on y alone: $H = \phi(y)$. Then $F = G + \int \phi(y) dy$ has $F_x = M$ and $F_y = G_y + \phi = N$, exactly as required. The procedure writes itself:

1. Write the equation as $M dx + N dy = 0$.
2. Compute M_y and N_x . If they differ, the equation is not exact — go to the inspection table below.
3. If exact, set $G = \int M dx$ (treat y as constant).
4. Compute $\phi(y) = N - G_y$; it will contain no x .
5. The solution is $G + \int \phi(y) dy = C$.

The inspection table

Many exam equations are not exact as written but become a sum of perfect differentials after regrouping. Five differentials cover nearly every case. Each is one line of ordinary calculus:

$$d(xy) = x dy + y dx$$

by the product rule;

$$d\left(\frac{y}{x}\right) = \frac{x dy - y dx}{x^2}, \quad d\left(\frac{x}{y}\right) = \frac{y dx - x dy}{y^2}$$

by the quotient rule;

$$d(\ln(x^2 + y^2)) = \frac{2(x dx + y dy)}{x^2 + y^2}$$

by the chain rule; and for $\theta = \tan^{-1}(y/x)$, differentiate $\tan \theta = y/x$:

$$\sec^2 \theta d\theta = \frac{x dy - y dx}{x^2}, \quad \sec^2 \theta = \frac{x^2 + y^2}{x^2} \implies d\theta = \frac{x dy - y dx}{x^2 + y^2}.$$

Perfect differential	Expanded form
$d(xy)$	$y dx + x dy$
$d\left(\frac{y}{x}\right)$	$\frac{x dy - y dx}{x^2}$
$d\left(\frac{x}{y}\right)$	$\frac{y dx - x dy}{y^2}$
$d(\ln(x^2 + y^2))$	$\frac{2(x dx + y dy)}{x^2 + y^2}$
$d(\tan^{-1}(y/x))$	$\frac{x dy - y dx}{x^2 + y^2}$

Two algebraic facts turn these into weapons:

$$\frac{d(xy)}{(xy)^2} = -d\left(\frac{1}{xy}\right), \quad \frac{y dx - x dy}{xy} = d\left(\ln\left|\frac{x}{y}\right|\right).$$

The first is the chain rule applied to $1/t$; the second is splitting the fraction into $dx/x - dy/y$.

Example A.1 — [CBSE Board Level]

Test for exactness and solve $(2xy + 3x^2) dx + (x^2 + 6y^2) dy = 0$.

Here $M = 2xy + 3x^2$ and $N = x^2 + 6y^2$. Then

$$M_y = 2x, \quad N_x = 2x,$$

equal, so the equation is exact. Construct G :

$$G = \int (2xy + 3x^2) dx = x^2y + x^3.$$

Then $G_y = x^2$, so $\phi(y) = N - G_y = 6y^2$, and

$$x^2y + x^3 + 2y^3 = C.$$

Check. Differentiate the answer implicitly: $(2xy + 3x^2) dx + (x^2 + 6y^2) dy = 0$ — recovered exactly.

Example A.2 — [JEE Main Level]

Solve $(ye^{xy} + 2x) dx + xe^{xy} dy = 0$.

Test: $M_y = e^{xy} + xye^{xy}$ and $N_x = e^{xy} + xye^{xy}$ — exact. Build G :

$$G = \int (ye^{xy} + 2x) dx = e^{xy} + x^2,$$

since $\int y e^{xy} dx = e^{xy}$ (the y is a spectator). Then $G_y = x e^{xy}$, so $\phi(y) = N - G_y = 0$. The ϕ step is not optional even when it vanishes — skipping it is how students “confirm” wrong answers. Solution:

$$e^{xy} + x^2 = C.$$

Example A.3 — [JEE Advanced Level — inspection]

Solve $y(1 + xy) dx + x(1 - xy) dy = 0$.

First check exactness: $M = y + xy^2$, $N = x - x^2y$, so $M_y = 1 + 2xy$ while $N_x = 1 - 2xy$. Not exact — and no routine integrating factor suggests itself. Expand and regroup instead:

$$y dx + x dy + xy^2 dx - x^2y dy = 0 \implies d(xy) + xy(y dx - x dy) = 0.$$

The first group is row one of the table. To make the second group match a row, divide the whole equation by x^2y^2 :

$$\frac{d(xy)}{x^2y^2} + \frac{y dx - x dy}{xy} = 0.$$

Apply the two algebraic facts above:

$$-d\left(\frac{1}{xy}\right) + d\left(\ln\left|\frac{x}{y}\right|\right) = 0.$$

Integrating,

$$\ln\left|\frac{x}{y}\right| - \frac{1}{xy} = C.$$

Why it is Advanced-level. Nothing in the equation announces the divisor x^2y^2 . You find it by asking: “what power of x and y turns each group into a tabulated differential?” — a two-second scan once the table is memorised, impossible without it.

Practice — Addendum A

1. Verify exactness and solve $(2x + y) dx + (x + 2y) dy = 0$.
2. Verify exactness and solve $(x^2 - y) dx - (x + y^2) dy = 0$.
3. Solve $\cos y dx + (\cos y - x \sin y) dy = 0$.
4. By inspection, solve $\frac{x dy - y dx}{x^2 + y^2} = dx$.
5. Find the value of k for which $(6x^2 + kxy) dx + (3x^2 + 5xy) dy = 0$ is exact, and solve it.

Answers: 1. $M_y = N_x = 1$; $x^2 + xy + y^2 = C$. 2. $M_y = N_x = -1$; $\frac{x^3}{3} - xy - \frac{y^3}{3} = C$. 3. $M_y = N_x = -\sin y$; $G = x \cos y$, $\phi(y) = \cos y$, so $x \cos y + \sin y = C$. 4. The left side is

$d(\tan^{-1}(y/x))$, so $\tan^{-1}(y/x) = x + C$. 5. $M_y = kx$ and $N_x = 6x$ force $k = 6$; then $G = 2x^3 + 3x^2y$, $\phi(y) = 0$, and $2x^3 + 3x^2y + \frac{5}{2}xy^2 = C$.

Addendum B — Two Reducible Types: Bernoulli and $f(ax + by + c)$

Exam Relevance: The chapter's note after its origin-shift example admits the parallel-lines case exists but never works it, and Bernoulli appears nowhere. JEE Main sets both: a Bernoulli equation disguised among linear options in an MCQ, and a $\tan^2(x + y)$ -style integrand in the solve section. Both reduce to machinery you already own — one to the linear method, one to separation.

Type 1 — Bernoulli equations

An equation of the form

$$\frac{dy}{dx} + Py = Qy^n, \quad n \neq 0, 1,$$

with P, Q functions of x , is **not linear** — the y^n spoils it. But one division makes it so. Divide through by y^n :

$$y^{-n} \frac{dy}{dx} + Py^{1-n} = Q.$$

The combination $y^{-n} \frac{dy}{dx}$ is begging for the substitution $v = y^{1-n}$, because

$$\frac{dv}{dx} = (1-n) y^{-n} \frac{dy}{dx}.$$

Substituting and dividing by $(1-n)$:

$$\frac{dv}{dx} + (1-n)Pv = (1-n)Q,$$

which is linear in v with the usual integrating factor $e^{\int (1-n)P dx}$. Solve for v , then back-substitute $v = y^{1-n}$. Never leave an answer containing v .

Example B.1 — [JEE Main Level]

Solve $\frac{dy}{dx} + y = y^2 e^x$.

Here $n = 2$, $P = 1$, $Q = e^x$. Divide by y^2 and put $v = y^{-1}$, so $v' = -y^{-2}y'$:

$$-v' + v = e^x \implies v' - v = -e^x.$$

Linear, with IF = e^{-x} . Then

$$(ve^{-x})' = -e^x \cdot e^{-x} = -1 \implies ve^{-x} = -x + C.$$

Back-substitute $v = 1/y$:

$$\frac{1}{y} = e^x(C - x).$$

Check. At large negative x the denominator dominates and $y \rightarrow 0$, consistent with $y = 0$ being the singular solution lost when we divided by y^2 — always name it.

Type 2 — Equations of the form $\frac{dy}{dx} = f(ax + by + c)$

The chapter handled $\frac{dy}{dx} = \frac{a_1x+b_1y+c_1}{a_2x+b_2y+c_2}$ when the two lines *intersect* (shift the origin to the intersection). When the equation involves a single linear combination, the shift is replaced by a direct substitution. Put

$$v = ax + by + c \implies \frac{dv}{dx} = a + b\frac{dy}{dx} = a + bf(v),$$

which is separable in v and x :

$$\frac{dv}{a + bf(v)} = dx.$$

Example B.2 — [JEE Advanced Level]

Solve $\frac{dy}{dx} = \tan^2(x + y)$.

Put $v = x + y$, so $v' = 1 + \tan^2 v = \sec^2 v$. Separate:

$$\cos^2 v dv = dx.$$

Use the half-angle identity $\cos^2 v = \frac{1 + \cos 2v}{2}$:

$$\int \cos^2 v dv = \frac{v}{2} + \frac{\sin 2v}{4} = x + C.$$

Restore $v = x + y$:

$$\frac{y - x}{2} + \frac{\sin(2(x + y))}{4} = C.$$

Why it is Advanced-level. The equation fits none of the three workhorse types — it is not separable as written, not homogeneous (the lone $x + y$ is degree 1 inside a transcendental), not linear. The mark-winning move is recognising that the argument of the tangent is itself the correct new variable, which kills the transcendental and leaves a trigonometric integral.

Practice — Addendum B

1. Solve $\frac{dy}{dx} + \frac{y}{x} = xy^2$.
2. Solve $\frac{dy}{dx} = (x + y + 1)^2$.
3. Solve $\frac{dy}{dx} = \cos(x + y)$.
4. Solve $\frac{dy}{dx} = (y - x)^2$.
5. Solve $\frac{dy}{dx} - y = xy^2$.

Answers: 1. With $v = 1/y$: $v' - v/x = -x$, IF $= 1/x$, giving $\frac{1}{y} = Cx - x^2$. 2. $v = x + y + 1$: $\tan^{-1} v = x + C$, so $x + y + 1 = \tan(x + C)$. 3. $v = x + y$: $v' = 1 + \cos v = 2 \cos^2(v/2)$, so $\tan \frac{x+y}{2} = x + C$. 4. $v = y - x$: $\frac{dv}{v^2 - 1} = dx$, so $\frac{1}{2} \ln \left| \frac{v-1}{v+1} \right| = x + C$ with $v = y - x$. 5. With $v = 1/y$: $v' + v = -x$, IF $= e^x$, giving $\frac{1}{y} = 1 - x + Ce^{-x}$.

Addendum C — Geometric Conditions Beyond the Subnormal

Exam Relevance: The chapter's application part defines subtangent and subnormal and stops. JEE Main and Advanced equally often phrase the condition using the *length* of the tangent or normal, the *intercepts* cut off by the tangent, or the requirement that the tangent or normal *passes through a fixed point*. All of these translate through one coordinate picture, derived once here.

The coordinate picture, derived

Let $m = \frac{dy}{dx}$ at a point $P = (x, y)$ on the curve, and let $P' = (x, 0)$ be the foot of the ordinate.

Tangent. Its equation is $Y - y = m(X - x)$. Setting $Y = 0$ gives its intersection with the x -axis:

$$T = \left(x - \frac{y}{m}, 0 \right).$$

Hence the subtangent, the tangent length, and the intercepts:

$$\text{subtangent} = \left| \frac{y}{m} \right|, \quad \text{length of tangent} = \sqrt{\frac{y^2}{m^2} + y^2} = \frac{|y|\sqrt{1+m^2}}{|m|},$$

$$x\text{-intercept} = x - \frac{y}{m}, \quad y\text{-intercept} = y - mx.$$

Normal. Slope $-\frac{1}{m}$; setting $Y = 0$ in $Y - y = -\frac{1}{m}(X - x)$ gives

$$N = (x + my, 0),$$

so

$$\text{subnormal} = |my|, \quad \text{length of normal} = \sqrt{m^2 y^2 + y^2} = |y| \sqrt{1 + m^2}.$$

Every geometric condition in these problems is one of these seven quantities set equal to something, or a line through a named point. Translate, get a first-order DE, solve by whichever workhorse fits.

Example C.1 — [JEE Main Level]

Find the curve passing through $(2, 3)$ such that the portion of the tangent between the axes is bisected at the point of contact.

“Bisected at the point of contact” means P is the midpoint of the two intercepts:

$$x = \frac{1}{2} \left(x - \frac{y}{m} \right), \quad y = \frac{1}{2} (y - mx).$$

The first equation gives $2x = x - \frac{y}{m}$, so $m = -\frac{y}{x}$; the second gives $2y = y - mx$, the same statement. One DE:

$$\frac{dy}{dx} = -\frac{y}{x} \implies \frac{dy}{y} = -\frac{dx}{x} \implies xy = C.$$

Through $(2, 3)$: $C = 6$, so $xy = 6$.

Check. On $xy = 6$ at $(2, 3)$: $m = -3/2$. Intercepts: x -intercept $= 2 - \frac{3}{-3/2} = 4$, y -intercept $= 3 - (-\frac{3}{2})(2) = 6$. Midpoint of $(4, 0)$ and $(0, 6)$ is $(2, 3)$ — the contact point itself.

Example C.2 — [JEE Advanced Level]

Find the curve passing through $(2, 1)$ whose normal at every point passes through the fixed point $(1, 0)$.

The normal at P has slope $-\frac{1}{m}$ and passes through both P and $(1, 0)$, so

$$-\frac{1}{m} = \frac{0 - y}{1 - x} = \frac{y}{x - 1} \implies m = \frac{1 - x}{y}.$$

Separate:

$$y \, dy = (1 - x) \, dx \implies \frac{y^2}{2} = x - \frac{x^2}{2} + C \implies x^2 + y^2 = 2x + C'.$$

Through $(2, 1)$: $4 + 1 = 2 + C'$, so $C' = 1$:

$$x^2 + y^2 = 2x + 1,$$

a circle centred at $(1, 0)$ with radius $\sqrt{2}$ — exactly as geometry predicts, since every normal of a circle passes through its centre.

Why it is Advanced-level. The condition names no standard quantity from the table; you must build the slope relation from the two-point formula yourself. The reward — recognising the answer as a circle and checking against theory — is the kind of closure Advanced rewards.

Practice — Addendum C

1. Find the curve through $(1, 2)$ whose subtangent at every point equals the abscissa.
2. Find the curve through $(2, 3)$ whose subnormal is constantly 4.
3. Show that a curve with constant subtangent $a > 0$ is an exponential curve, and find it.
4. Find the curve through $(2, 1)$ whose normals all pass through $(1, 0)$.
5. Find the curve through $(1, 1)$ whose tangent at every point cuts off equal nonzero intercepts on the axes.

Answers: 1. $\frac{y}{m} = x$ gives $\frac{dy}{dx} = \frac{y}{x}$, so $y = 2x$. 2. $ym = 4$ gives $y dy = 4 dx$, so $y^2 = 8x - 7$ (check: $9 = 16 - 7$). 3. $\frac{y}{m} = a$ gives $\frac{dy}{y} = \frac{dx}{a}$, so $y = Ce^{x/a}$. 4. As in Example C.2: $x^2 + y^2 = 2x + 1$. 5. Equal intercepts force slope $-\frac{b}{a} = -1$, so $\frac{dy}{dx} = -1$ and $x + y = 2$.

Addendum D — Second-Order Equations by Lowering the Order

Exam Relevance: Strictly beyond the board syllabus, and said so up front. JEE Advanced, however, regularly sets an order-2 equation that contains no x , or no y , or is secretly first-order in y' — and expects the substitution $p = \frac{dy}{dx}$ to collapse it. Treat this as JEE-only enrichment built entirely on tools from Parts 3 and 4.

Method

Put $p = \frac{dy}{dx}$ throughout. Two cases exhaust the examinable supply.

Case 1 — y absent. The equation reads $y'' = f(x, y')$. Since $y'' = \frac{dp}{dx}$, it becomes a *first-order* equation in p and x :

$$\frac{dp}{dx} = f(x, p).$$

Solve it by any Part 3/Part 4 method to get $p = g(x)$, then integrate once more: $y = \int g dx + C_2$. Two integrations, two constants — matching the order.

Case 2 — x absent. The equation reads $y'' = f(y, y')$. Here p cannot be differentiated with respect to x directly without dragging x back in, so switch the roles via the chain rule:

$$y'' = \frac{dp}{dx} = \frac{dp}{dy} \cdot \frac{dy}{dx} = p \frac{dp}{dy}.$$

The equation becomes $p \frac{dp}{dy} = f(y, p)$ — first order in p and y . Solve for p as a function of y , then finish with $\frac{dy}{dx} = p(y)$, which is separable.

Diagnosis takes five seconds: look at what the equation *omits*. Omits y — Case 1. Omits x — Case 2. Omits neither — look for a perfect derivative (as in Practice 4 below) before giving up.

Example D.1 — [JEE Advanced Level]

Solve $yy'' = (y')^2$.

x is absent — Case 2. Substitute $y'' = p \frac{dp}{dy}$:

$$yp \frac{dp}{dy} = p^2.$$

If $p = 0$ identically then y is constant, which the final answer will include. Otherwise divide by p :

$$y \frac{dp}{dy} = p \implies \frac{dp}{p} = \frac{dy}{y} \implies p = Ay.$$

Now $\frac{dy}{dx} = Ay$ is separable:

$$\frac{dy}{y} = A dx \implies y = Be^{Ax}.$$

Check. $y' = ABe^{Ax}$, $y'' = A^2 Be^{Ax}$, so $yy'' = AB^2 A^2 e^{2Ax} \cdot \frac{1}{A} \cdot A = A^2 B^2 e^{2Ax}$ and $(y')^2 = A^2 B^2 e^{2Ax}$ — equal. The two constants A, B match the order 2, and $A = 0$ recovers the lost constant solutions.

Example D.2 — [JEE Advanced Level]

Solve $y'' = y' + x$.

Both x and y' appear and y is absent — Case 1. With $p = y'$:

$$\frac{dp}{dx} - p = x,$$

linear in p with IF $= e^{-x}$. Then

$$(pe^{-x})' = xe^{-x}.$$

Integrating the right side by parts ($u = x$, $dv = e^{-x} dx$):

$$\int xe^{-x} dx = -xe^{-x} - e^{-x},$$

so

$$pe^{-x} = -(x+1)e^{-x} + C_1 \implies p = C_1e^x - x - 1.$$

One more integration:

$$y = C_1e^x - \frac{x^2}{2} - x + C_2.$$

Why it is Advanced-level. The equation looks like nothing in the chapter — second order, non-constant coefficients. The entire difficulty is the refusal to panic: one glance shows y missing, one substitution drops it to a Part 4 linear equation, and the rest is bookkeeping.

Practice — Addendum D

1. Solve $y'' = 1 + (y')^2$.
2. Solve $yy'' = 2(y')^2$.
3. Solve $y'' = y'$ given $y(0) = 0$ and $y'(0) = 1$.
4. Solve $xy'' + y' = 0$ by spotting a perfect derivative.
5. Solve $yy'' + (y')^2 = 0$.

Answers: 1. Case 1: $p' = 1 + p^2$ gives $\tan^{-1} p = x + C_1$, so $y = -\ln|\cos(x + C_1)| + C_2$. 2. Case 2: $\frac{dp}{p} = 2\frac{dy}{y}$ gives $p = Ay^2$, then $-\frac{1}{y} = Ax + B$, i.e. $y = \frac{1}{B - Ax}$. 3. $p = C_1e^x$ with $C_1 = 1$, then $y = e^x + C_2$ with $C_2 = -1$: $y = e^x - 1$. 4. $xy'' + y' = (xy')' = 0$, so $xy' = C_1$ and $y = C_1 \ln|x| + C_2$. 5. $(yy')' = 0$ gives $yy' = C_1$, then $y^2 = 2C_1x + C_2$ — a family of parabolas, mirroring the constant-subnormal result of Part 5.

Closing note. Every addendum above ends at the same place the chapter does: a separable integral or one pass of the integrating factor. What separates the top band of candidates is not a larger toolbox but faster diagnosis — exact or not, Bernoulli or linear, which quantity is missing, which geometric quantity is named. Drill the diagnosis, and the integration takes care of itself.

Addendum P — What JEE Actually Asks

A note on evidence. Many 2018–2020 stems in the bank lost their displayed equations during PDF extraction. Where a stem survived, I quote it; where only the printed **solution** survived, I classified the question from the method visible in that solution (the IF used, the substitution made) and reconstructed the stem only when the reconstruction is forced by the algebra. Everything below has been re-derived from scratch; anything I could not pin down has been dropped rather than guessed.

1. The distribution

Judged from the recoverable items (roughly 45–50 questions spanning 1983–2020):

Cluster	Share of bank	Evidence
Linear ODEs (all disguises)	Close to half	Nearly every solvable Mains item from 2011 onward: the Sep 2020 batch, ten-odd items across 2019 sessions, three from 2018, 2017, 2012, 2011, plus Adv 2014–2018. Even where stems are corrupt, the printed solutions show IF-based work throughout.
Homogeneous / reducible-to-separable	About one in five	Adv 2013, Main Jan 2019 (twice), Online Apr 2018, Adv 2018, 2005 (twice), 1996, 1999.
Formation of a DE from a family	About one in six	Main Jan 8 2020 (II), Online Apr 16 2018, Online Apr 9 2014, Adv 2015, 2005, 2000, 1999, 1998 — steady presence, at least one item in most recent sessions.
Plain separable	Small, usually fused with another skill	Main Jan 7 2020, Main 2017, Adv 2018.
Applications (growth, cooling, tanks, cones)	Clustered in 1994–2005	Cone 2003, hemispherical tank 2001, population 2000, normal-length 1994, tangent-length 2005. Almost absent from the recent Mains extract.
Structural / conceptual	A thin but recurring Advanced thread	Comparing two solutions of one linear ODE (1997), verify-the-solution MCQ (1999), functional equations converted to ODEs (Adv 2018 twice, Main Jan 2019).

Two facts worth internalising:

- **Examiners recycle.** $(x^2 - y^2) dx + 2xy dy = 0$ through $(1, 1)$ was asked in Online Apr 15,

2018 **and again** in Main Jan 10, 2019. Learning shapes pays better than memorising answers.

- **Standalone “find the order/degree” MCQs are rarer here than the chapter suggests.** In this bank, order/degree questions almost always arrive *inside* a formation problem (e.g. 1999: order 1, degree 3 after squaring). The pure “degree not defined” drill is a board/CUET staple, not a JEE Main regular — the evidence for it in this bank is thin.

2. The recurring archetypes

Archetype A — Formation from a geometric family (one parameter)

Differentiate once, isolate the parameter, substitute back. Predict the order from the parameter count *before* differentiating.

Representative [Main Jan 8, 2020 (II)]: family $x^2 = 4b(y + b)$.

$$\begin{aligned}x^2 &= 4by + 4b^2 \\ \Rightarrow 2x &= 4by' \Rightarrow b = \frac{x}{2y'}\end{aligned}$$

Substitute back:

$$x^2 = 4 \cdot \frac{x}{2y'} \left(y + \frac{x}{2y'} \right) = \frac{2xy}{y'} + \frac{x^2}{(y')^2}$$

Multiply through by $(y')^2$ and divide by x :

$$\boxed{x(y')^2 = 2yy' + x}$$

Archetype B — Counting independent constants

Representative [1998]: order of the DE whose solution is $y = (c_1 + c_2) \cos(x + c_3) - c_4 e^{x+c_5}$.

Expand before counting:

$$\cos(x + c_3) = \cos x \cos c_3 - \sin x \sin c_3$$

so the first term carries two independent constants (coefficients of $\cos x$, $\sin x$), and $c_4 e^{x+c_5}$ collapses to one. Total: 3 independent constants, so the answer is **order 3**, not the naive 5.

Archetype C — Homogeneous equation, $y = vx$

Representative [Main Jan 10, 2019 (II); Online Apr 15, 2018]: solve $(x^2 - y^2) dx + 2xy dy = 0$ through $(1, 1)$.

$$\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$

Put $y = vx$, so $y' = v + xv'$:

$$v + xv' = \frac{v^2 - 1}{2v} \Rightarrow x \frac{dv}{dx} = -\frac{1 + v^2}{2v}$$

Separate and integrate:

$$\frac{2v dv}{1 + v^2} = -\frac{dx}{x} \Rightarrow \ln(1 + v^2) = -\ln x + \ln C$$

Back-substitute $v = y/x$: $x^2 + y^2 = Cx$. Through $(1, 1)$, $C = 2$:

$$(x - 1)^2 + y^2 = 1$$

—a circle of radius one with centre on the x -axis.

Archetype D — Reducible to separable by a linear substitution ($t = x \pm y$)

When neither separation nor homogeneity applies, look for a combination whose derivative simplifies.

Representative [Main Jan 11, 2019 (II)]: $\frac{dy}{dx} = (x - y)^2$, $y(1) = 1$.

Put $t = x - y$, so $\frac{dt}{dx} = 1 - y' = 1 - t^2$:

$$\int \frac{dt}{1 - t^2} = \int dx \Rightarrow \frac{1}{2} \ln \left| \frac{1 + t}{1 - t} \right| = x + C$$

$t = 0$ at $x = 1$ gives $C = -1$, so $\ln \left| \frac{1 + x - y}{1 - x + y} \right| = 2(x - 1)$.

Archetype E — Linear, but in the other variable

Representative [2005]: $y dx + y^2 dy = x dy$, $y > 0$, $y(1) = 1$; find $y(-3)$.

As written this is nonlinear in y . Flip to $\frac{dx}{dy}$:

$$\frac{dx}{dy} - \frac{x}{y} = -y, \quad \text{IF} = e^{\int -dy/y} = \frac{1}{y}$$

$$\frac{d}{dy} \left(\frac{x}{y} \right) = -1 \Rightarrow \frac{x}{y} = -y + C$$

$y(1) = 1$ gives $C = 2$, so $x = 2y - y^2$. At $x = -3$: $y^2 - 2y - 3 = 0$, giving $y = 3$ or $y = -1$; the condition $y > 0$ selects

$$y = 3$$

Archetype F — Abstract linear ODE (structure, not computation)**Representative** [2011]: $y' + y g'(x) = g(x)g'(x)$ with $g(0) = g(2) = 0$, $y(0) = 0$. Find $y(2)$.IF = $e^{g(x)}$, so

$$\frac{d}{dx}(y e^g) = g' e^g \Rightarrow y e^g = e^g(g - 1) + C$$

using $\int g e^g dg = e^g(g - 1)$. Now $y(0) = 0$, $g(0) = 0$ force $C = 1$. At $x = 2$, $g(2) = 0$:

$$y(2) \cdot 1 = -1 + 1 = \boxed{0}$$

You never need to know what g is.**Archetype G — Linear in a substituted dependent variable****Representative** [Main Apr 12, 2019 (II)]: $(y^2 - x^3) dx - xy dy = 0$.Not separable, not homogeneous. Spot that y appears only as y^2 and $y dy$. Put $u = y^2$, $du = 2y dy$:

$$u dx - \frac{x}{2} du = x^3 dx \Rightarrow \frac{du}{dx} - \frac{2u}{x} = -2x^2$$

IF = x^{-2} :

$$\frac{u}{x^2} = -2x + C \Rightarrow \boxed{y^2 + 2x^3 = Cx^2}$$

3. The traps

- Counting written constants, not independent ones.** The 1998 item above hands you five symbols; the answer is 3. Constants summed inside one coefficient, or buried in an exponent, collapse.
- Predicting order before differentiating.** A family “with two parameters” pinned down by an extra condition (e.g. an ellipse family forced through a point) may retain only one free parameter. Differentiating more times than there are free parameters produces a relation that still annihilates the family but is *not* the minimal DE the examiner wants. Always do the parameter count first.
- Reading order/degree off an uncleared form.** In the 1999 item $y^2 = 2c(x + \sqrt{c})$: substituting $c = yy'$ leaves a $(yy')^{3/2}$ term; you must square to get $(y^2 - 2xyy')^2 = 4y^3(y')^3$ — order 1, **degree 3**. Students who read degree off the unsquared form say 1.
- Skipping standard form.** Multiple 2019–2020 items arrive as $(x^2 + 1)^2 y' + 2x(x^2 + 1)y = 1$ or $xy' = \ln x$. If the coefficient of y' is not 1, every IF you compute is garbage. Divide first.

5. **Not noticing the equation is linear in x .** $y dx - (x + 3y^2) dy = 0$ (Online Apr 8, 2017) and the 2005 item above are both ordinary linear ODEs after flipping to dx/dy . Students who fight them in y burn minutes and lose.
6. **Ignoring domain when rejecting roots.** The 2005 flip-problem yields $y = \pm 3$ at $x = -3$; the stated $y > 0$ is not decoration — picking -1 lands exactly on the planted distractor.
7. **Misplacing the constant.** In Archetype D, the condition gives $C = -1$, not 0 — a slip the options are built around. Apply initial conditions to the *general* solution, after integrating.
8. **Absolute values in logs.** When integrating $dt/(1 - t^2)$, dropping the modulus and the sign discussion lets you accept solution branches outside their validity interval.

4. Graded practice set

Every answer below has been re-derived and checked.

P1 (easy). Find the order and degree of the DE of the family $y^2 = 2c(x + \sqrt{c})$, $c > 0$. [JEE 1999]

Solution. $2yy' = 2c$, so $c = yy'$. Substitute:

$$y^2 = 2xyy' + 2(yy')^{3/2} \Rightarrow (y^2 - 2xyy')^2 = 4y^3(y')^3$$

Polynomial in derivatives now: highest derivative is y' , raised to power 3. **Order 1, degree 3.**

P2 (easy). Which of $y = 2$, $y = 2x$, $y = 2x - 4$, $y = 2x^2 - 4$ solves $(y')^2 - xy' + y = 0$? [JEE 1999]

Solution. Test each: $y = 2$ gives $2 \neq 0$; $y = 2x$ gives $4 \neq 0$; $y = 2x^2 - 4$ ($y' = 4x$) gives $14x^2 - 4 \neq 0$. For $y = 2x - 4$, $y' = 2$:

$$4 - 2x + (2x - 4) = 0. \checkmark$$

Answer: $y = 2x - 4$.

P3 (easy-medium). Form the DE of the family $x^2 = 4b(y + b)$. [Main Jan 8, 2020 (II)]

Solution. See Archetype A: $x(y')^2 = 2yy' + x$.

P4 (medium). Solve $(2 + \sin x) \frac{dy}{dx} + (y + 1) \cos x = 0$ with $y(0) = 1$; find $y(\pi/2)$ and $y'(\pi/2)$. [Adapted from Main 2017]

Solution. Separate:

$$\frac{dy}{y+1} = -\frac{\cos x dx}{2 + \sin x} \Rightarrow \ln(y+1) = -\ln(2 + \sin x) + c$$

so $(y+1)(2 + \sin x) = C$; $y(0) = 1$ gives $C = 4$:

$$y = \frac{4}{2 + \sin x} - 1$$

Hence $y(\pi/2) = \frac{4}{3} - 1 = \frac{1}{3}$, and $y' = \frac{-4 \cos x}{(2 + \sin x)^2}$ gives $y'(\pi/2) = -\frac{4}{9}$.

P5 (medium). Identify the curve $(x^2 - y^2) dx + 2xy dy = 0$ through $(1, 1)$. [Main Jan 10, 2019 (II)]

Solution. See Archetype C: $x^2 + y^2 = 2x$, a circle of radius 1 centred at $(1, 0)$.

P6 (medium). Solve $\frac{dy}{dx} = (x - y)^2$, $y(1) = 1$, and find $y(2)$. [Main Jan 11, 2019 (II), extended]

Solution. See Archetype D: $\ln \left| \frac{1+x-y}{1-x+y} \right| = 2(x-1)$. At $x = 2$, with $t = x - y$:

$$\frac{1+t}{1-t} = e^2 \Rightarrow t = \frac{e^2 - 1}{e^2 + 1} \Rightarrow y(2) = 2 - t = \frac{e^2 + 3}{e^2 + 1}$$

P7 (medium). $y dx + y^2 dy = x dy$, $y > 0$, $y(1) = 1$. Find $y(-3)$. [JEE 2005]

Solution. See Archetype E: $x = 2y - y^2$, giving $y = 3$.

P8 (medium-hard). $y' + y g'(x) = g(x) g'(x)$, where g is differentiable, $g(0) = g(2) = 0$, and $y(0) = 0$. Find $y(2)$. [JEE 2011]

Solution. See Archetype F: $y(2) = 0$.

P9 (medium-hard). Solve $(y^2 - x^3) dx - xy dy = 0$. [Main Apr 12, 2019 (II)]

Solution. See Archetype G: $y^2 + 2x^3 = Cx^2$.

P10 (hard). Solve $(x^2 + y^2) dy = xy dx$ with $y(1) = 1$; find x_0 such that $y(x_0) = e$. [JEE 2005]

Solution. Put $y = vx$:

$$v + xv' = \frac{v}{1 + v^2} \Rightarrow x \frac{dv}{dx} = -\frac{v^3}{1 + v^2}$$

$$\int \left(\frac{1}{v^3} + \frac{1}{v} \right) dv = - \int \frac{dx}{x} \Rightarrow \ln y - \frac{x^2}{2y^2} = C$$

$y(1) = 1$ gives $C = -\frac{1}{2}$. At $y = e$:

$$1 = \frac{x_0^2}{2e^2} - \frac{1}{2} \Rightarrow x_0 = e\sqrt{3}$$

P11 (hard). Solve $(x^2 + 1)^2 \frac{dy}{dx} + 2x(x^2 + 1)y = 1$ with $y(0) = 0$; find $y(1)$. [Main Apr 8, 2019 (I), adapted]

Solution. Standard form: $y' + \frac{2x}{x^2 + 1} y = \frac{1}{(x^2 + 1)^2}$. Since $\int \frac{2x dx}{x^2 + 1} = \ln(x^2 + 1)$,

$$\text{IF} = x^2 + 1 \Rightarrow \frac{d}{dx} [y(x^2 + 1)] = \frac{1}{x^2 + 1}$$

$$y(x^2 + 1) = \arctan x + C; \quad y(0) = 0 \Rightarrow C = 0$$

At $x = 1$: $2y(1) = \frac{\pi}{4}$, so $y(1) = \frac{\pi}{8}$.

P12 (hard). Find the curve through the origin satisfying $\frac{dy}{dx} = \sin(10x + 6y)$. [JEE 1996]

Solution. Spot $10x + 6y = 2(5x + 3y)$. Put $v = 5x + 3y$, so $v' = 5 + 3y' = 5 + 3 \sin 2v$:

$$\frac{dv}{5 + 3 \sin 2v} = dx$$

With $t = \tan v$, $dv = \frac{dt}{1+t^2}$ and $5 + 3 \sin 2v = \frac{5t^2+6t+5}{1+t^2}$:

$$\int \frac{dt}{5t^2 + 6t + 5} = \frac{1}{4} \arctan \left(\frac{5t + 3}{4} \right) = x + C$$

The origin gives $\tan C_1 = \frac{3}{4}$ after absorbing constants; tidying,

$$\tan(5x + 3y) = \frac{5 \tan 4x}{4 - 3 \tan 4x}$$

5. What this chapter under-serves

The chapter teaches differential equations as labelled board routines — “separable”, “homogeneous”, “linear, apply the IF formula” — and spends its Part 5 largely on orthogonal trajectories, cooling curves and growth word problems that, in this bank, cluster before 2005 and have thinned out of recent Mains. What JEE actually rewards is *type-recognition under disguise*: inventing the substitution ($t = x - y$, $u = y^2$, flipping to dx/dy), converting functional equations into ODEs, counting parameters structurally, and comparing solutions of the same linear ODE without ever solving it. Drill those reflexes deliberately, because the chapter’s own exercises almost never ask for them.

Chapter 10: Vector Algebra

Class 12 Mathematics – Drona Ed Tech Master Content

Complete Chapter – Subtopics 01-03

Contents

Vector Algebra — Subtopic 01: Vector Operations & Components	1
Section 1: Concept Intuition	2
Section 2: Key Definitions & Properties	3
Section 3: Theorems & Proofs	4
Section 4: Worked Examples	6
Section 5: Practice Exercises	8
Section 6: Crack the MCQ	8
Section 7: Common Student Pitfalls & Pro-Tips	9
Section 8: Cheat Sheet Box	10
Vector Algebra — Subtopic 02: Product of Two Vectors	11
Section 1: Concept Intuition	11
Section 2: Key Definitions & Properties	12
Section 3: Theorems & Proofs	13
Section 4: Worked Examples	14
Section 5: Practice Exercises	16
Section 6: Crack the MCQ	17
Section 7: Common Student Pitfalls & Pro-Tips	17
Section 8: Cheat Sheet Box	18
Vector Algebra — Subtopic 03: Product of Three Vectors & Vector Applications	19
Section 1: Concept Intuition	19
Section 2: Key Definitions & Properties	20
Section 3: Theorems & Proofs	21
Section 4: Worked Examples	22
Section 5: Practice Exercises	24
Section 6: Crack the MCQ	25
Section 7: Common Student Pitfalls & Pro-Tips	25
Section 8: Cheat Sheet Box	26

Vector Algebra — Subtopic 01: Vector Operations & Components

Exam Relevance: This is the foundation block of the whole chapter. **JEE Main** reliably asks 1-2 questions every year on unit vectors, direction cosines/ratios,

the section formula, and collinearity. **CBSE Boards** include a 1-2 mark definition or a short section-formula problem almost every paper. **CUET** loves direct formula plug-ins from here. **JEE Advanced** rarely tests it head-on but builds collinearity/coplanarity setups on top of it — get this airtight and the rest of the chapter rests on solid ground.

Section 1: Concept Intuition

Imagine you are standing at Connaught Place in Delhi and you stop an auto driver. If you only say “*take me 4 kilometres*,” the poor driver has no idea where to go — 4 km towards Karol Bagh and 4 km towards Nehru Place land you in completely different places. The number alone (4 km) is a **scalar** — it carries only *size*. The moment you add “*4 km towards Nehru Place*,” you have given a **vector** — something with both a **magnitude** (how much) and a **direction** (which way).

That single distinction is the entire seed of this chapter. Temperature, mass, distance, speed, and your bank balance are scalars — they are fully described by one number. Displacement, velocity, force, and a fielder’s throw in a cricket match are vectors — a throw of “80 km/h” is incomplete until you say *towards the stumps*.

We draw a vector as a **directed line segment** — an arrow. The length of the arrow is its magnitude, and the arrowhead points in its direction. If the arrow goes from point A (the *initial point* or *tail*) to point B (the *terminal point* or *head*), we write it as $\vec{AB}$, and its magnitude is written $|\vec{AB}|$.

Think about it this way... a vector is really just an *instruction*: “move this far, in this direction.” Crucially, the instruction does not care *where* you start. If you and your friend both walk “200 m due east,” you have followed the *same* vector even though you began from different gates of the same colony. So two arrows that have the **same length and the same direction are the same vector**, no matter where they sit on the page. This is why we can freely slide a vector around to add it to another — a property we will use constantly.

Now, how do we pin a vector down with numbers? We resolve it into **components**. Picture giving directions on a city grid: “go 3 blocks east, then 2 blocks north, then up 1 floor.” Those three numbers — east, north, up — completely fix where you end up. In mathematics the three perpendicular grid directions are the **unit vectors** $\hat{i}$ (along the x -axis), $\hat{j}$ (along the y -axis), and $\hat{k}$ (along the z -axis), each of length exactly 1. Any vector $\vec{r}$ in space can then be written as

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k},$$

where x, y, z are its **scalar components**. The arrow and the triple (x, y, z) are two faces of the same object.

Limiting Conditions & Assumptions. A few quiet assumptions are worth dragging into the light, because exams attack them directly:

- A **position vector** is *not* a free vector. It is deliberately tied to the origin O — it points from O to a specific point. You cannot slide it around like an ordinary vector without changing what it means.

- **Equality of vectors needs both conditions.** Two vectors are equal only if they have the same magnitude *and* the same direction. Equal length alone is not enough, and parallel direction alone is not enough.
- The **zero vector** $\vec{0}$ has magnitude 0 and **no definite direction**. Statements like “the direction cosines of $\vec{0}$ ” are meaningless.
- A **unit vector** can only be formed from a *non-zero* vector, because we divide by the magnitude — and you cannot divide by zero.

Hold these in mind: most “silly” mistakes in this subtopic are really violations of one of these four assumptions.

Section 2: Key Definitions & Properties

Magnitude of a vector. For $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$,

$$|\vec{r}| = \sqrt{x^2 + y^2 + z^2}.$$

This is just the 3-D Pythagoras theorem. *Property:* $|\vec{r}| \geq 0$, with equality only for the zero vector.

Unit vector. A vector of magnitude 1 along a given direction. The unit vector along $\vec{a}$ ($\vec{a} \neq \vec{0}$) is

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|}.$$

Validity: defined only when $\vec{a} \neq \vec{0}$.

Position vector. The position vector of a point $P(x, y, z)$ relative to origin O is

$$\vec{OP} = x\hat{i} + y\hat{j} + z\hat{k}, \quad |\vec{OP}| = \sqrt{x^2 + y^2 + z^2}.$$

Vector joining two points. If A and B have position vectors $\vec{a}$ and $\vec{b}$, then

$$\vec{AB} = \vec{b} - \vec{a} \quad (\text{head minus tail}).$$

Trap-guard: $\vec{AB} = -\vec{BA}$, so subtraction order is everything.

Scalar multiplication. For scalar λ and vector $\vec{a}$,

$$|\lambda\vec{a}| = |\lambda| |\vec{a}|.$$

The direction of $\lambda\vec{a}$ is **the same** as $\vec{a}$ if $\lambda > 0$ and **opposite** if $\lambda < 0$; if $\lambda = 0$ we get $\vec{0}$.

Direction cosines (DCs). The cosines of the angles α, β, γ that a vector makes with the positive x -, y -, z -axes:

$$l = \cos \alpha, \quad m = \cos \beta, \quad n = \cos \gamma.$$

For $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ with $|\vec{r}| = r$,

$$l = \frac{x}{r}, \quad m = \frac{y}{r}, \quad n = \frac{z}{r}.$$

Fundamental identity: $l^2 + m^2 + n^2 = 1$.

Direction ratios (DRs). Any three numbers (a, b, c) *proportional* to the DCs:

$$\frac{l}{a} = \frac{m}{b} = \frac{n}{c} = \frac{\pm 1}{\sqrt{a^2 + b^2 + c^2}}.$$

The scalar components (x, y, z) of a vector are themselves a valid set of DRs. *Key property:* DCs are **unique** (up to overall sign for the opposite ray), but DRs are **not unique** — $(2, -1, 2)$ and $(4, -2, 4)$ describe the same direction.

Section formula. A point R dividing the join of $A(\vec{a})$ and $B(\vec{b})$ in ratio $m : n$:

$$\vec{OR}_{\text{internal}} = \frac{m\vec{b} + n\vec{a}}{m + n} \quad (m + n \neq 0), \quad \vec{OR}_{\text{external}} = \frac{m\vec{b} - n\vec{a}}{m - n} \quad (m \neq n).$$

Midpoint (ratio 1 : 1): $\frac{\vec{a} + \vec{b}}{2}$.

Collinearity condition. Three points A, B, C are collinear $\iff \vec{AB} = \lambda \vec{AC}$ for some scalar λ (equivalently, $\vec{b} - \vec{a}$ and $\vec{c} - \vec{a}$ are parallel).

Types of vectors (quick property table).

Type	Defining property
Zero vector $\vec{0}$	magnitude 0, direction undefined
Unit vector	magnitude exactly 1
Equal vectors	same magnitude and same direction
Negative of $\vec{a}$	same magnitude, opposite direction $(-\vec{a})$
Collinear / parallel	$\vec{a} = \lambda \vec{b}$ for some scalar λ
Coplanar	all lie in (or parallel to) one plane
Position vector	drawn from the origin to a fixed point

Section 3: Theorems & Proofs

Theorem 1 — The direction-cosine identity $l^2 + m^2 + n^2 = 1$

Claim. The direction cosines of any non-zero vector satisfy $l^2 + m^2 + n^2 = 1$.

Proof. Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ with magnitude $r = |\vec{r}| = \sqrt{x^2 + y^2 + z^2} > 0$.

By definition the direction cosines are the components of the *unit* vector $\hat{r}$, so

$$l = \frac{x}{r}, \quad m = \frac{y}{r}, \quad n = \frac{z}{r}.$$

Why this step: the DCs measure direction only, so we normalise out the length by dividing each component by r . Squaring and adding,

$$l^2 + m^2 + n^2 = \frac{x^2 + y^2 + z^2}{r^2} = \frac{r^2}{r^2} = 1.$$

The numerator collapses to r^2 precisely because $r^2 = x^2 + y^2 + z^2$ by the magnitude definition. ■

Reading the result: a set of three cosines is a *legitimate* direction only if their squares sum to 1. This is the fastest sanity check in the whole subtopic — if an option gives DCs whose squares do not add to 1, reject it on sight.

Theorem 2 — Section Formula (internal division)

Claim. Let A, B have position vectors $\vec{a}, \vec{b}$. If R divides AB internally in the ratio $m : n$, then

$$\vec{OR} = \frac{m\vec{b} + n\vec{a}}{m + n}.$$

Setup. “ R divides AB internally in ratio $m : n$ ” means R lies *between* A and B with $AR : RB = m : n$. Since $\vec{AR}$ and $\vec{RB}$ point the same way along the segment,

$$\frac{|\vec{AR}|}{|\vec{RB}|} = \frac{m}{n} \implies n \vec{AR} = m \vec{RB}.$$

Why this step: equal direction lets us promote the ratio of magnitudes to a vector equation.

Proof. Write the two segment vectors using “head minus tail”:

$$\vec{AR} = \vec{r} - \vec{a}, \quad \vec{RB} = \vec{b} - \vec{r}, \quad \text{where } \vec{r} = \vec{OR}.$$

Substitute into $n \vec{AR} = m \vec{RB}$:

$$n(\vec{r} - \vec{a}) = m(\vec{b} - \vec{r}).$$

Expand and gather the $\vec{r}$ terms on one side:

$$n\vec{r} - n\vec{a} = m\vec{b} - m\vec{r} \implies (m + n)\vec{r} = m\vec{b} + n\vec{a}.$$

Since R lies *strictly between* A and B , we have $m + n \neq 0$, so we may divide:

$$\vec{r} = \frac{m\vec{b} + n\vec{a}}{m + n}. \quad \blacksquare$$

Corollary — external division. If R divides AB *externally* in ratio $m : n$, R lies outside the segment and the directions of $\vec{AR}, \vec{RB}$ are opposite, giving $n \vec{AR} = -m \vec{RB}$. Repeating the algebra replaces n by $-n$:

$$\vec{OR} = \frac{m\vec{b} - n\vec{a}}{m - n} \quad (m \neq n).$$

Corollary — midpoint. Put $m = n = 1$: $\vec{OR} = \frac{\vec{a} + \vec{b}}{2}$.

Theorem 3 — Vector test for collinearity

Claim. Points A, B, C with position vectors $\vec{a}, \vec{b}, \vec{c}$ are collinear $\iff$ there exists a scalar λ with $\vec{b} - \vec{a} = \lambda(\vec{c} - \vec{a})$.

Proof. A, B, C lie on one line $\iff$ the segments $\vec{AB}$ and $\vec{AC}$ point along the same line $\iff$ one is a scalar multiple of the other, i.e. $\vec{AB} = \lambda \vec{AC}$. Substituting $\vec{AB} = \vec{b} - \vec{a}$ and $\vec{AC} = \vec{c} - \vec{a}$ gives the stated condition. ■

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Find the unit vector in the direction of $\vec{a} = 6\hat{i} - 2\hat{j} + 3\hat{k}$, and hence find the vector of magnitude 35 in the same direction.

Given: $\vec{a} = 6\hat{i} - 2\hat{j} + 3\hat{k}$.

Step 1 — magnitude.

$$|\vec{a}| = \sqrt{6^2 + (-2)^2 + 3^2} = \sqrt{36 + 4 + 9} = \sqrt{49} = 7.$$

Step 2 — unit vector.

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{1}{7}(6\hat{i} - 2\hat{j} + 3\hat{k}) = \frac{6}{7}\hat{i} - \frac{2}{7}\hat{j} + \frac{3}{7}\hat{k}.$$

Step 3 — scale to magnitude 35. Multiply the unit vector by 35:

$$35\hat{a} = 35 \cdot \frac{1}{7}(6\hat{i} - 2\hat{j} + 3\hat{k}) = 5(6\hat{i} - 2\hat{j} + 3\hat{k}) = 30\hat{i} - 10\hat{j} + 15\hat{k}.$$

Answer: $\hat{a} = \frac{1}{7}(6\hat{i} - 2\hat{j} + 3\hat{k})$, and the required vector is $30\hat{i} - 10\hat{j} + 15\hat{k}$. (Quick check: $|30\hat{i} - 10\hat{j} + 15\hat{k}| = \sqrt{900 + 100 + 225} = \sqrt{1225} = 35$. ✓)

Example 2 — [Conceptual Trap]

Points $P(2, -1, 3)$ and $Q(5, 3, -1)$ are given. Find the unit vector in the direction of $\vec{QP}$.

The trap: under exam pressure students compute $\vec{PQ}$ instead of $\vec{QP}$ — the answer then comes out with every sign flipped, and it is marked wrong.

Step 1 — read the order carefully. $\vec{QP}$ goes from Q to P , so it is **head P minus tail Q** :

$$\vec{QP} = \vec{OP} - \vec{OQ} = (2 - 5)\hat{i} + (-1 - 3)\hat{j} + (3 - (-1))\hat{k} = -3\hat{i} - 4\hat{j} + 4\hat{k}.$$

Step 2 — magnitude.

$$|\vec{QP}| = \sqrt{(-3)^2 + (-4)^2 + 4^2} = \sqrt{9 + 16 + 16} = \sqrt{41}.$$

Step 3 — unit vector.

$$\widehat{QP} = \frac{1}{\sqrt{41}}(-3\hat{i} - 4\hat{j} + 4\hat{k}).$$

Mental-elimination shortcut: in a multiple-choice setting, first decide the *signs* of the answer from the order $Q \rightarrow P$ (x decreases $\Rightarrow$ negative $\hat{i}$, etc.) and the *denominator* $\sqrt{41}$. Any option with the wrong sign pattern or a denominator that is not $\sqrt{41}$ is gone instantly — no full computation needed.

Example 3 — [JEE Main Level] (section formula + direction cosines)

The point R divides the join of $A(3, -2, 4)$ and $B(8, 3, -6)$ internally in the ratio $2 : 3$. Find the position vector of R and the direction cosines of $\vec{OR}$.

Two concepts: first the section formula, then direction cosines of the result.

Step 1 — apply the internal section formula with $\vec{a} = 3\hat{i} - 2\hat{j} + 4\hat{k}$, $\vec{b} = 8\hat{i} + 3\hat{j} - 6\hat{k}$, $m : n = 2 : 3$:

$$\vec{OR} = \frac{2\vec{b} + 3\vec{a}}{2 + 3} = \frac{2(8\hat{i} + 3\hat{j} - 6\hat{k}) + 3(3\hat{i} - 2\hat{j} + 4\hat{k})}{5}.$$

Compute the numerator componentwise:

$$\hat{i} : 16 + 9 = 25, \quad \hat{j} : 6 - 6 = 0, \quad \hat{k} : -12 + 12 = 0.$$

$$\vec{OR} = \frac{25\hat{i} + 0\hat{j} + 0\hat{k}}{5} = 5\hat{i}.$$

Step 2 — direction cosines of $\vec{OR} = 5\hat{i}$. Its magnitude is 5, so

$$l = \frac{5}{5} = 1, \quad m = \frac{0}{5} = 0, \quad n = \frac{0}{5} = 0.$$

Answer: $\vec{OR} = 5\hat{i}$, with DCs $(1, 0, 0)$ — i.e. R lies on the positive x -axis. (Check: $1^2 + 0^2 + 0^2 = 1$ ✓, consistent with Theorem 1.)

Example 4 — [JEE Advanced Level] (concurrency of medians via position vectors)

In a triangle ABC the vertices have position vectors $\vec{a}, \vec{b}, \vec{c}$. Let D be the midpoint of BC . Show that the point P dividing the median AD in the ratio $2 : 1$ (from A) is independent of which vertex you start from — hence the three medians are concurrent — and find its position vector.

Step 1 — locate D , the midpoint of BC (section formula, ratio $1 : 1$):

$$\vec{OD} = \frac{\vec{b} + \vec{c}}{2}.$$

Step 2 — divide median AD in ratio $2 : 1$ from $A(\vec{a})$ to $D(\vec{OD})$, i.e. $m : n = 2 : 1$:

$$\vec{OP} = \frac{2\vec{OD} + 1 \cdot \vec{a}}{2 + 1} = \frac{2 \cdot \frac{\vec{b} + \vec{c}}{2} + \vec{a}}{3} = \frac{\vec{a} + \vec{b} + \vec{c}}{3}.$$

Step 3 — symmetry argument. The result $\frac{\vec{a} + \vec{b} + \vec{c}}{3}$ is **completely symmetric** in $\vec{a}, \vec{b}, \vec{c}$. Had we instead taken the median from B to the midpoint E of CA , or from C to the midpoint F of AB , and divided $2 : 1$, the identical algebra would again yield $\frac{\vec{a} + \vec{b} + \vec{c}}{3}$.

Conclusion. All three medians pass through the *same* point G with

$$\vec{OG} = \frac{\vec{a} + \vec{b} + \vec{c}}{3}$$

so the medians are **concurrent**, and G — the **centroid** — divides each median $2 : 1$ from the vertex. The vector method delivers in three lines what coordinate geometry needs a page for. ■

Section 5: Practice Exercises

- [CBSE]** Find the magnitude and the unit vector in the direction of $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$.
- [CBSE]** Find the position vector of the midpoint of the segment joining $A(-1, 4, 2)$ and $B(5, -2, 6)$.
- [JEE Main]** A vector $\vec{r}$ has magnitude 14 and direction ratios 2, 3, -6. Find its direction cosines and its component form, given that $\vec{r}$ makes an acute angle with the x -axis.
- [JEE Main]** Find the value of k for which the points $A(k, -10, 3)$, $B(1, -1, 3)$ and $C(3, 5, 3)$ are collinear.
- [JEE Advanced]** The position vectors of A and B are $\vec{a}$ and $\vec{b}$. Find the position vector of the point C on $\vec{BA}$ produced such that $\vec{BC} = \frac{3}{2}\vec{BA}$.

Answers: 1. $|\vec{a}| = 3$, $\hat{a} = \frac{1}{3}(2\hat{i} + \hat{j} - 2\hat{k})$. 2. $2\hat{i} + \hat{j} + 4\hat{k}$. 3. DCs $(\frac{2}{7}, \frac{3}{7}, -\frac{6}{7})$; $\vec{r} = 4\hat{i} + 6\hat{j} - 12\hat{k}$. 4. $k = -2$. 5. $\vec{OC} = \frac{1}{2}(3\vec{a} - \vec{b})$.

Section 6: Crack the MCQ

Q1. The direction cosines of the vector $\hat{i} - 2\hat{j} + 2\hat{k}$ are: (a) $(\frac{1}{3}, -\frac{2}{3}, \frac{2}{3})$ (b) $(1, -2, 2)$ (c) $(\frac{1}{9}, -\frac{2}{9}, \frac{2}{9})$ (d) $(\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}}, \frac{2}{\sqrt{5}})$

Correct: (a). Magnitude = $\sqrt{1+4+4} = 3$, so DCs are $(\frac{1}{3}, -\frac{2}{3}, \frac{2}{3})$; check $\frac{1}{9} + \frac{4}{9} + \frac{4}{9} = 1$ ✓. **Distractors: (b)** gives direction *ratios*, not cosines — the classic DR/DC mix-up. **(c)** divides components by $|\vec{r}|^2 = 9$ instead of $|\vec{r}| = 3$. **(d)** uses $\sqrt{5}$ — a student who summed only $1+4$ and forgot the third component's square.

Q2. If R divides $A(\vec{a})$ and $B(\vec{b})$ externally in the ratio $3 : 1$, then $\vec{OR} =$ (a) $\frac{3\vec{b}+\vec{a}}{4}$ (b) $\frac{3\vec{b}-\vec{a}}{2}$
(c) $\frac{3\vec{b}-\vec{a}}{4}$ (d) $\frac{\vec{a}-3\vec{b}}{2}$

Correct: (b). External formula: $\frac{m\vec{b}-n\vec{a}}{m-n} = \frac{3\vec{b}-\vec{a}}{3-1} = \frac{3\vec{b}-\vec{a}}{2}$. **Distractors: (a)** uses the *internal* formula (sign and denominator both wrong). **(c)** keeps the correct numerator but uses denominator $m+n=4$ instead of $m-n=2$. **(d)** flips the numerator sign — equivalent to dividing in ratio $1 : 3$ instead of $3 : 1$.

Q3. The points $P(1, 2, 3)$, $Q(3, 6, 9)$, $S(2, 4, 6)$ are: (a) vertices of a triangle (b) collinear
(c) coincident (d) forming a right angle at S

Correct: (b). $\vec{PQ} = 2\hat{i} + 4\hat{j} + 6\hat{k}$ and $\vec{PS} = \hat{i} + 2\hat{j} + 3\hat{k}$, so $\vec{PQ} = 2\vec{PS}$ — parallel through a common point P , hence collinear. **Distractors: (a)** assumes three distinct points must form a triangle — false when they line up. **(c)** “coincident” needs all points equal, which they are not. **(d)** invents a right angle; nothing here was tested for perpendicularity (and that would be a later subtopic anyway).

Q4. A vector of magnitude 10 in the direction opposite to $\vec{a} = 3\hat{i} + 4\hat{k}$ is: (a) $6\hat{i} + 8\hat{k}$ (b) $-6\hat{i} - 8\hat{k}$
(c) $-3\hat{i} - 4\hat{k}$ (d) $-\frac{3}{5}\hat{i} - \frac{4}{5}\hat{k}$

Correct: (b). $|\vec{a}| = \sqrt{9+16} = 5$; unit vector $\frac{1}{5}(3\hat{i}+4\hat{k})$; *opposite* direction and magnitude 10 gives $-10 \cdot \frac{1}{5}(3\hat{i}+4\hat{k}) = -6\hat{i}-8\hat{k}$. **Distractors: (a)** correct magnitude but **wrong (same) direction** — missed the word “opposite.” **(c)** is just $-\vec{a}$, magnitude 5 not 10. **(d)** is the *opposite unit* vector — forgot to scale up to magnitude 10.

Section 7: Common Student Pitfalls & Pro-Tips

- **Mixing up direction ratios and direction cosines.** DRs are *any* proportional triple; DCs are the *normalised* triple satisfying $l^2+m^2+n^2=1$. If a question says “direction cosines,” you must divide by the magnitude.
- **Subtraction order for $\vec{AB}$.** Always **head minus tail**: $\vec{AB} = \vec{b} - \vec{a}$. Writing $\vec{a} - \vec{b}$ silently computes $\vec{BA}$ and flips every sign.
- **Confusing internal and external section formulas.** Internal has a **plus** and denominator $m+n$; external has a **minus** and denominator $m-n$. A quick tell: external division puts R *outside* the segment, so its coordinates should land beyond A or B .
- **Dividing by $|\vec{r}|^2$ instead of $|\vec{r}|$** when forming a unit vector or DCs. The denominator is the magnitude itself, not its square.

Pro-Tip — the “squares-sum-to-one” gatekeeper. Whenever an option claims to be a set of direction cosines or a unit vector, square the three numbers and add. If you don’t get 1, eliminate that option without any further work. In MCQs this single check often kills two or three distractors in seconds.

Section 8: Cheat Sheet Box

- **Magnitude:** $|\vec{r}| = \sqrt{x^2 + y^2 + z^2}$.
- **Unit vector:** $\hat{a} = \frac{\vec{a}}{|\vec{a}|}$ (needs $\vec{a} \neq \vec{0}$); vector of magnitude k along $\vec{a}$ is $k\hat{a}$.
- **Vector joining points:** $\vec{AB} = \vec{b} - \vec{a}$ (*head - tail*); $\vec{AB} = -\vec{BA}$.
- **Direction cosines:** $l = \frac{x}{r}$, $m = \frac{y}{r}$, $n = \frac{z}{r}$, with $\boxed{l^2 + m^2 + n^2 = 1}$. DRs $\propto$ DCs (not unique).
- **Section formula:** internal $\frac{m\vec{b} + n\vec{a}}{m + n}$; external $\frac{m\vec{b} - n\vec{a}}{m - n}$; midpoint $\frac{\vec{a} + \vec{b}}{2}$.
- **Centroid of $\triangle ABC$:** $\frac{\vec{a} + \vec{b} + \vec{c}}{3}$ (medians concur, 2 : 1 from each vertex).
- **Collinearity:** A, B, C collinear $\iff \vec{AB} = \lambda \vec{AC}$.

Memory aids: “Internal adds, external subtracts.” | “DC means Divided-by-magnitude; DR means Ratio-only.” | “Squares of DCs always sum to one.”

Vector Algebra — Subtopic 02: Product of Two Vectors

Exam Relevance: This is one of the highest-yield subtopics in the chapter. **JEE Main** asks 2–3 questions every year — angle between vectors, projection, area of a triangle/parallelogram, perpendicular/parallel conditions, and the unit vector perpendicular to two vectors. **CBSE Boards** routinely set 2–3 mark problems on the scalar product, projection, or area. **CUET** uses direct formula plug-ins. **JEE Advanced** weaves these products into geometry and constraint problems. Master both products and you own most of the chapter’s marks.

Section 1: Concept Intuition

There are two completely different ways to “multiply” two vectors, and the whole subtopic is built on appreciating *why* they are different. One multiplication asks “**how much do these two vectors agree in direction?**” and answers with a plain number. The other asks “**how much do these two vectors spread apart?**” and answers with a brand-new vector. We call them the **dot (scalar) product** and the **cross (vector) product**.

The dot product — overlap. Picture pushing a loaded handcart — a *thela* — down a straight market lane. You shove at a slight angle, but only the part of your push that points *along* the lane actually moves the cart forward; the sideways part is wasted. The dot product measures exactly this useful overlap between two directions. When two vectors point the same way, the overlap is maximum; when they are perpendicular, there is no overlap at all and the dot product is zero. The result is a single number — a scalar — because “amount of agreement” is just a quantity, not a direction.

The cross product — spread and twist. Now think of opening a heavy almirah door. Push near the hinge and almost nothing happens; push at the far edge, perpendicular to the door, and it swings easily. The turning effect depends on how *perpendicular* your push is — and the axis of the resulting rotation sticks straight out of the door, along the hinge. The cross product captures this: it is largest when the two vectors are perpendicular, zero when they are parallel, and it produces a **new vector pointing perpendicular to both** of them, along the “axis” of their spread. Its magnitude equals the **area of the parallelogram** the two vectors frame.

Think about it this way... the two products are mirror-image questions. The dot product is the *shadow* one vector casts on another — maximal when they line up, vanishing when they cross at a right angle. The cross product is the *area* the two vectors sweep out — vanishing when they line up, maximal when they cross at a right angle. Wherever one is large, the other tends to be small. Together they tell you everything about how two vectors are oriented relative to each other.

The direction of the cross product is fixed by the **right-hand rule**: curl the fingers of your right hand from the first vector toward the second, and your thumb points along $\vec{a} \times \vec{b}$. Reverse the order and your thumb flips — which is why $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$.

Limiting Conditions & Assumptions.

- **Zero does not mean the same thing for both.** For non-zero vectors, $\vec{a} \cdot \vec{b} = 0$ means **perpendicular**, while $\vec{a} \times \vec{b} = \vec{0}$ means **parallel (collinear)**. Confusing these is the single most common error here.

- **The cross product is not commutative.** Order matters: $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$. The dot product *is* commutative.
 - **There is no vector division.** You cannot “divide” by a vector to undo a product; from $\vec{a} \cdot \vec{b}$ alone you cannot recover $\vec{b}$.
 - **A scalar versus a vector.** $\vec{a} \cdot \vec{b}$ is a number; $\vec{a} \times \vec{b}$ is a vector. Writing one where the other belongs (e.g. “ $\vec{a} \cdot \vec{b}$ = a vector”) is a conceptual, not arithmetic, mistake.
-

Section 2: Key Definitions & Properties

Dot (Scalar) Product

Definition. For vectors $\vec{a}, \vec{b}$ with angle θ ($0 \leq \theta \leq \pi$) between them,

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta.$$

Component form. If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$,

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3.$$

The result is a **scalar**.

Angle between two vectors.

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}.$$

Projection. The scalar projection of $\vec{a}$ on $\vec{b}$ is

$$\text{proj}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|},$$

and the projection *vector* is $\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \vec{b}$.

Key properties.

- Commutative: $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$.
- Distributive: $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$.
- $\vec{a} \cdot \vec{a} = |\vec{a}|^2$.
- **Perpendicularity:** for non-zero $\vec{a}, \vec{b}$, $\vec{a} \cdot \vec{b} = 0 \iff \vec{a} \perp \vec{b}$.
- Basis: $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$ and $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$.

Cross (Vector) Product

Definition. With $\hat{n}$ the unit vector perpendicular to both $\vec{a}$ and $\vec{b}$ (right-hand rule),

$$\vec{a} \times \vec{b} = |\vec{a}| |\vec{b}| \sin \theta \hat{n}, \quad |\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta.$$

Determinant (component) form.

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}.$$

The result is a **vector**.

Geometric meaning.

- Area of parallelogram with adjacent sides $\vec{a}, \vec{b}$: $|\vec{a} \times \vec{b}|$.
- Area of triangle with sides $\vec{a}, \vec{b}$ from one vertex: $\frac{1}{2} |\vec{a} \times \vec{b}|$.
- Unit vector perpendicular to both $\vec{a}, \vec{b}$: $\hat{n} = \pm \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$.

Key properties.

- Anti-commutative: $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$.
- Distributive: $\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$.
- $\vec{a} \times \vec{a} = \vec{0}$.
- **Parallelism**: for non-zero $\vec{a}, \vec{b}$, $\vec{a} \times \vec{b} = \vec{0} \iff \vec{a} \parallel \vec{b}$.
- Basis (cyclic): $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$ (reverse order flips sign).

The two products at a glance: $\vec{a} \cdot \vec{b}$ peaks at $\theta = 0$ and is 0 at $\theta = 90^\circ$; $|\vec{a} \times \vec{b}|$ is 0 at $\theta = 0$ and peaks at $\theta = 90^\circ$. They are perfectly complementary, and indeed $(\vec{a} \cdot \vec{b})^2 + |\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2$ (proved below).

Section 3: Theorems & Proofs

Theorem 1 — Component formula for the dot product

Claim. $\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$.

Proof. Write $\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$, $\vec{b} = b_1 \hat{i} + b_2 \hat{j} + b_3 \hat{k}$. Using distributivity of the dot product,

$$\vec{a} \cdot \vec{b} = \sum_p \sum_q a_p b_q (\hat{e}_p \cdot \hat{e}_q),$$

where $\hat{e}_1, \hat{e}_2, \hat{e}_3$ denote $\hat{i}, \hat{j}, \hat{k}$. *Why this step:* the dot product distributes over sums, so multiplying the two bracketed expressions produces nine basis dot products. Now invoke the basis rules: $\hat{e}_p \cdot \hat{e}_q = 1$ when $p = q$ and 0 otherwise. Every cross term ($p \neq q$) dies, leaving only the three diagonal terms:

$$\vec{a} \cdot \vec{b} = a_1 b_1 (\hat{i} \cdot \hat{i}) + a_2 b_2 (\hat{j} \cdot \hat{j}) + a_3 b_3 (\hat{k} \cdot \hat{k}) = a_1 b_1 + a_2 b_2 + a_3 b_3. \quad \blacksquare$$

Theorem 2 — Determinant formula for the cross product

Claim. $\vec{a} \times \vec{b} = (a_2 b_3 - a_3 b_2) \hat{i} - (a_1 b_3 - a_3 b_1) \hat{j} + (a_1 b_2 - a_2 b_1) \hat{k}$.

Proof. Expand $\vec{a} \times \vec{b}$ by distributivity over the nine basis cross products. *Why this step:* the cross product is distributive, so the product of the two bracketed sums splits into nine terms. Three of them, $\hat{i} \times \hat{i}$, $\hat{j} \times \hat{j}$, $\hat{k} \times \hat{k}$, equal $\vec{0}$. The remaining six come in pairs governed by the cyclic rules $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$ (and their reverses with a minus sign). Collecting:

$$\vec{a} \times \vec{b} = (a_2b_3 - a_3b_2)\hat{i} + (a_3b_1 - a_1b_3)\hat{j} + (a_1b_2 - a_2b_1)\hat{k},$$

which is exactly the cofactor expansion of $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$ along the top row. The antisymmetry

of the determinant in its last two rows reproduces $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$ automatically. ■

Theorem 3 — Lagrange's identity

Claim. $(\vec{a} \cdot \vec{b})^2 + |\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2$.

Proof. Let θ be the angle between $\vec{a}$ and $\vec{b}$. Then $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \theta$ and $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin \theta$. Squaring and adding,

$$(\vec{a} \cdot \vec{b})^2 + |\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 \cos^2 \theta + |\vec{a}|^2 |\vec{b}|^2 \sin^2 \theta = |\vec{a}|^2 |\vec{b}|^2 (\cos^2 \theta + \sin^2 \theta) = |\vec{a}|^2 |\vec{b}|^2.$$

The Pythagorean identity $\cos^2 \theta + \sin^2 \theta = 1$ closes the proof. ■

Why it matters: this identity lets you find $|\vec{a} \times \vec{b}|$ from $\vec{a} \cdot \vec{b}$ (or vice versa) without ever computing the angle — a frequent exam shortcut.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Given $\vec{a} = 2\hat{i} + 3\hat{j} + \hat{k}$ **and** $\vec{b} = \hat{i} - 2\hat{j} + 2\hat{k}$, **find (i) $\vec{a} \cdot \vec{b}$, (ii) the angle between them, and (iii) the projection of $\vec{a}$ on $\vec{b}$.**

(i) Dot product.

$$\vec{a} \cdot \vec{b} = (2)(1) + (3)(-2) + (1)(2) = 2 - 6 + 2 = -2.$$

(ii) Angle. Magnitudes: $|\vec{a}| = \sqrt{4 + 9 + 1} = \sqrt{14}$, $|\vec{b}| = \sqrt{1 + 4 + 4} = 3$. So

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \frac{-2}{3\sqrt{14}} \Rightarrow \theta = \cos^{-1}\left(\frac{-2}{3\sqrt{14}}\right) \text{ (obtuse, since } \cos \theta < 0\text{)}.$$

(iii) Projection of $\vec{a}$ on $\vec{b}$.

$$\text{proj}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{-2}{3}.$$

Answer: $\vec{a} \cdot \vec{b} = -2$; $\theta = \cos^{-1}\left(\frac{-2}{3\sqrt{14}}\right)$; projection = $-\frac{2}{3}$. (The negative projection correctly signals an obtuse angle.)

Example 2 — [Conceptual Trap]

For what value of λ are the vectors $\vec{a} = \lambda\hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$ (i) perpendicular, and (ii) parallel?

The trap: students reach for the same condition for both, or swap them. **Perpendicular uses the dot product; parallel uses the cross product (or proportional components).**

(i) **Perpendicular** $\Rightarrow \vec{a} \cdot \vec{b} = 0$:

$$(\lambda)(3) + (2)(-1) + (-3)(2) = 3\lambda - 2 - 6 = 0 \Rightarrow 3\lambda = 8 \Rightarrow \lambda = \frac{8}{3}.$$

(ii) **Parallel** $\Rightarrow$ **components proportional**: $\frac{\lambda}{3} = \frac{2}{-1} = \frac{-3}{2}$. But $\frac{2}{-1} = -2$ while $\frac{-3}{2} = -1.5$ — these are **unequal**, so the last two components are *not* proportional. Hence **no value of λ** makes them parallel.

Answer: perpendicular at $\lambda = \frac{8}{3}$; parallel for **no** λ . (*Mental check: had a student wrongly used the dot product for “parallel,” they would have produced a spurious λ — the trap.*)

Example 3 — [JEE Main Level] (area + perpendicular unit vector)

Find the area of the triangle whose two sides from vertex A are $\vec{a} = 3\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} - 3\hat{j} + 4\hat{k}$, and find a unit vector perpendicular to both.

Step 1 — cross product.

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & -2 \\ 1 & -3 & 4 \end{vmatrix} = \hat{i}(1 \cdot 4 - (-2)(-3)) - \hat{j}(3 \cdot 4 - (-2)(1)) + \hat{k}(3(-3) - 1 \cdot 1).$$

Evaluate each cofactor:

$$\hat{i}: 4 - 6 = -2, \quad \hat{j}: -(12 + 2) = -14, \quad \hat{k}: -9 - 1 = -10.$$

$$\vec{a} \times \vec{b} = -2\hat{i} - 14\hat{j} - 10\hat{k}.$$

Step 2 — magnitude.

$$|\vec{a} \times \vec{b}| = \sqrt{(-2)^2 + (-14)^2 + (-10)^2} = \sqrt{4 + 196 + 100} = \sqrt{300} = 10\sqrt{3}.$$

Step 3 — area of triangle.

$$\text{Area} = \frac{1}{2} |\vec{a} \times \vec{b}| = \frac{1}{2} \cdot 10\sqrt{3} = 5\sqrt{3} \text{ square units.}$$

Step 4 — perpendicular unit vector.

$$\hat{n} = \pm \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} = \pm \frac{1}{10\sqrt{3}} (-2\hat{i} - 14\hat{j} - 10\hat{k}) = \pm \frac{1}{5\sqrt{3}} (-\hat{i} - 7\hat{j} - 5\hat{k}).$$

Answer: area = $5\sqrt{3}$ sq. units; $\hat{n} = \pm \frac{1}{5\sqrt{3}} (-\hat{i} - 7\hat{j} - 5\hat{k})$.

Example 4 — [JEE Advanced Level] (Lagrange identity + constraints)

Two vectors $\vec{a}$ and $\vec{b}$ satisfy $|\vec{a}| = 5$, $|\vec{b}| = 4$, and $\vec{a} \cdot \vec{b} = 10$. Without finding the angle explicitly, evaluate $|\vec{a} \times \vec{b}|$ and then the area of the parallelogram whose diagonals are $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$.

Step 1 — use Lagrange's identity (Theorem 3) to get $|\vec{a} \times \vec{b}|$ directly:

$$(\vec{a} \cdot \vec{b})^2 + |\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2.$$

$$10^2 + |\vec{a} \times \vec{b}|^2 = 25 \cdot 16 = 400 \Rightarrow |\vec{a} \times \vec{b}|^2 = 300 \Rightarrow |\vec{a} \times \vec{b}| = 10\sqrt{3}.$$

Step 2 — area from the diagonals. The area of a parallelogram in terms of its diagonals $\vec{d}_1, \vec{d}_2$ is $\frac{1}{2}|\vec{d}_1 \times \vec{d}_2|$. With $\vec{d}_1 = \vec{a} + \vec{b}$ and $\vec{d}_2 = \vec{a} - \vec{b}$,

$$\vec{d}_1 \times \vec{d}_2 = (\vec{a} + \vec{b}) \times (\vec{a} - \vec{b}) = \underbrace{\vec{a} \times \vec{a}}_{\vec{0}} - \vec{a} \times \vec{b} + \vec{b} \times \vec{a} - \underbrace{\vec{b} \times \vec{b}}_{\vec{0}}.$$

Since $\vec{b} \times \vec{a} = -\vec{a} \times \vec{b}$, this is $-\vec{a} \times \vec{b} - \vec{a} \times \vec{b} = -2(\vec{a} \times \vec{b})$.

Step 3 — finish.

$$\text{Area} = \frac{1}{2} |-2(\vec{a} \times \vec{b})| = \frac{1}{2} \cdot 2 |\vec{a} \times \vec{b}| = |\vec{a} \times \vec{b}| = 10\sqrt{3} \text{ square units.}$$

Answer: $|\vec{a} \times \vec{b}| = 10\sqrt{3}$; area = $10\sqrt{3}$ sq. units. (Elegant outcome: a parallelogram's area equals the magnitude of the cross product of its side vectors — and here the diagonal computation reproduces it.) ■

Section 5: Practice Exercises

1. **[CBSE]** If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} - \hat{j} + 3\hat{k}$, find $\vec{a} \cdot \vec{b}$ and the projection of $\vec{b}$ on $\vec{a}$.
2. **[CBSE]** Find the area of the parallelogram whose adjacent sides are $\vec{a} = 2\hat{i} + \hat{j}$ and $\vec{b} = \hat{i} + 3\hat{j}$.
3. **[JEE Main]** Find λ so that $\vec{a} = 2\hat{i} + \lambda\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + 3\hat{k}$ are perpendicular.
4. **[JEE Main]** If $|\vec{a}| = 3$, $|\vec{b}| = 5$ and the angle between them is 60° , find $|\vec{a} \times \vec{b}|$ and $\vec{a} \cdot \vec{b}$.
5. **[JEE Advanced]** Vectors $\vec{a}, \vec{b}$ satisfy $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$. Prove that $\vec{a} \perp \vec{b}$.

Answers: 1. $\vec{a} \cdot \vec{b} = 4$; projection of $\vec{b}$ on $\vec{a} = \frac{4}{\sqrt{3}}$. 2. 5 sq. units. 3. $\lambda = \frac{5}{2}$. 4. $|\vec{a} \times \vec{b}| = \frac{15\sqrt{3}}{2}$; $\vec{a} \cdot \vec{b} = \frac{15}{2}$. 5. Squaring both sides gives $\vec{a} \cdot \vec{b} = 0$, hence perpendicular.

Section 6: Crack the MCQ

Q1. If $\vec{a} \cdot \vec{b} = |\vec{a} \times \vec{b}|$, the angle between $\vec{a}$ and $\vec{b}$ is: (a) 30° (b) 45° (c) 60° (d) 90°

Correct: (b). $|\vec{a}||\vec{b}| \cos \theta = |\vec{a}||\vec{b}| \sin \theta \Rightarrow \tan \theta = 1 \Rightarrow \theta = 45^\circ$. *Distractors:* **(a)** $\tan 30^\circ \neq 1$. **(c)** $\theta = 60^\circ$ gives $\tan \theta = \sqrt{3}$, the answer if one wrongly set $\cos \theta = \sqrt{3} \sin \theta$. **(d)** 90° makes the dot product 0 but the cross product maximal — they cannot be equal (unless both are zero).

Q2. For non-zero vectors, $\vec{a} \times \vec{b} = \vec{0}$ implies: (a) $\vec{a} \perp \vec{b}$ (b) $\vec{a} \parallel \vec{b}$ (c) $|\vec{a}| = |\vec{b}|$ (d) $\vec{a} \cdot \vec{b} = 0$

Correct: (b). A vanishing cross product means $\sin \theta = 0$, i.e. $\theta = 0$ or π — the vectors are parallel (collinear). *Distractors:* **(a)** and **(d)** are the dot-product zero condition — the classic dot/cross swap. **(c)** equal magnitudes have nothing to do with the cross product.

Q3. The projection of $\vec{a} = 4\hat{i} - 3\hat{j} + \hat{k}$ on $\vec{b} = 2\hat{i} + \hat{j} - 2\hat{k}$ is: (a) 1 (b) $\frac{3}{\sqrt{26}}$ (c) 3 (d) $\frac{1}{3}$

Correct: (a). $\vec{a} \cdot \vec{b} = 8 - 3 - 2 = 3$; $|\vec{b}| = \sqrt{4 + 1 + 4} = 3$; projection $= \frac{3}{3} = 1$. *Distractors:* **(c)** stops at the dot product, forgetting to divide by $|\vec{b}|$. **(b)** divides by $|\vec{a}| = \sqrt{26}$ instead of $|\vec{b}|$ — projected onto the wrong vector. **(d)** divides $|\vec{b}|$ by the dot product (reciprocal slip).

Q4. The area of the triangle with vertices $O(0, 0, 0)$, $A(1, 2, 2)$, $B(2, 1, -2)$ is: (a) $\frac{9}{2}$ (b) 9 (c) $\frac{\sqrt{161}}{2}$ (d) $\frac{3\sqrt{10}}{2}$

Correct: (a). $\vec{OA} \times \vec{OB} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 2 \\ 2 & 1 & -2 \end{vmatrix} = \hat{i}(-4 - 2) - \hat{j}(-2 - 4) + \hat{k}(1 - 4) = -6\hat{i} + 6\hat{j} - 3\hat{k}$;

magnitude $\sqrt{36 + 36 + 9} = \sqrt{81} = 9$; area $= \frac{1}{2} \cdot 9 = \frac{9}{2}$. *Distractors:* **(b)** forgets the factor $\frac{1}{2}$ (parallelogram area, not triangle). **(c)**, **(d)** come from arithmetic slips in the cofactors (e.g. a sign error in the $\hat{j}$ term) — a reminder to keep the alternating signs straight.

Section 7: Common Student Pitfalls & Pro-Tips

- **Dot/cross zero-condition swap.** $\vec{a} \cdot \vec{b} = 0 \Rightarrow$ perpendicular; $\vec{a} \times \vec{b} = \vec{0} \Rightarrow$ parallel. Memorise them as opposites and you will never mix them up.
- **Forgetting to divide for a projection.** Projection of $\vec{a}$ on $\vec{b}$ is $\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$, **not** just $\vec{a} \cdot \vec{b}$. And it is divided by the magnitude of the vector you project *onto*.
- **Sign error in the $\hat{j}$ cofactor.** The determinant expansion alternates $+, -, +$. The middle ($\hat{j}$) term carries a minus sign — the most common cross-product slip in exams.
- **Treating the cross product as commutative.** $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$. If a problem fixes the orientation (area is fine either way, but a perpendicular *direction* is not), order matters.

Pro-Tip — let the sign of the dot product read the angle for you. Before any heavy computation, note: $\vec{a} \cdot \vec{b} > 0 \Rightarrow$ acute, $= 0 \Rightarrow$ right angle, $< 0 \Rightarrow$

obtuse. This instantly sanity-checks projections (a negative projection *must* pair with a negative dot product) and kills wrong MCQ options.

Section 8: Cheat Sheet Box

- **Dot product:** $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \theta = a_1b_1 + a_2b_2 + a_3b_3$ — a **scalar**.
- **Cross product:** $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}| \sin \theta \hat{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$ — a **vector**.
- **Angle:** $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}$. **Projection of $\vec{a}$ on $\vec{b}$:** $\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$.
- **Conditions:** $\vec{a} \perp \vec{b} \iff \vec{a} \cdot \vec{b} = 0$; $\vec{a} \parallel \vec{b} \iff \vec{a} \times \vec{b} = \vec{0}$.
- **Areas:** parallelogram = $|\vec{a} \times \vec{b}|$; triangle = $\frac{1}{2}|\vec{a} \times \vec{b}|$; perpendicular unit vector = $\pm \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$.
- **Lagrange:** $(\vec{a} \cdot \vec{b})^2 + |\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2$.

Memory aids: “Dot = aligned, Cross = crossed” (dot peaks when parallel, cross peaks when perpendicular). | “Dot zero = perpendicular; Cross zero = parallel.”
| “Project? Divide by the one you land ON.” | “Middle sign of the determinant is minus.”

Vector Algebra — Subtopic 03: Product of Three Vectors & Vector Applications

Exam Relevance: Pure JEE territory. The **scalar triple product** (coplanarity, volume of a parallelepiped/tetrahedron) is a near-guaranteed 1-2 questions in **JEE Main** and a frequent building block in **JEE Advanced**. The **vector triple product** (the BAC-CAB identity) is a recurring **JEE Advanced** tool. CBSE Boards (rationalised syllabus) no longer test this directly, so treat it as your JEE-edge material.

Section 1: Concept Intuition

You already know two ways to multiply vectors: the dot product (a number measuring overlap) and the cross product (a vector measuring spread). The natural next question is — what happens when **three** vectors enter the picture? There are exactly two sensible combinations, and each answers a beautiful geometric question.

The scalar triple product — “how much space do three vectors enclose?” Take three vectors $\vec{a}, \vec{b}, \vec{c}$ sprouting from one corner. They frame a slanted box — a **parallelepiped** (think of a brick that has been pushed sideways so its faces are slanted, like a stack of books shoved out of square). The scalar triple product, written $[\vec{a} \ \vec{b} \ \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c})$, gives the **volume** of that box (up to a sign). The recipe is elegant: $\vec{b} \times \vec{c}$ produces a vector perpendicular to the base with magnitude equal to the base’s area; dotting it with $\vec{a}$ then picks out the height. Area of base $\times$ height = volume. The answer is a **scalar** — hence the name.

Think about it this way... if the three vectors all lie flat in one plane, the “box” is squashed completely flat — it has zero thickness and therefore zero volume. So the scalar triple product being zero is the exact signal that the three vectors are **coplanar** — they share a single plane and one of them is a “mixture” of the other two. This is the same thing as saying the three vectors are **linearly dependent**. That single fact — $[\vec{a} \ \vec{b} \ \vec{c}] = 0 \iff$ coplanar — is the most exam-tested idea in this subtopic.

The vector triple product — “spin a cross product again.” Now cross a vector with a cross product: $\vec{a} \times (\vec{b} \times \vec{c})$. Since $\vec{b} \times \vec{c}$ is perpendicular to the plane of $\vec{b}$ and $\vec{c}$, crossing it again with $\vec{a}$ swings the result *back into* the plane of $\vec{b}$ and $\vec{c}$. So $\vec{a} \times (\vec{b} \times \vec{c})$ is always a blend of $\vec{b}$ and $\vec{c}$ — and there is a clean formula for the blend, the famous **BAC-CAB rule**:

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}.$$

Limiting Conditions & Assumptions.

- **The bracket is not free in its grouping.** The vector triple product is **not associative**: $\vec{a} \times (\vec{b} \times \vec{c}) \neq (\vec{a} \times \vec{b}) \times \vec{c}$ in general. The brackets genuinely matter.
 - **Sign vs. magnitude.** $[\vec{a} \vec{b} \vec{c}]$ can be negative — its *sign* encodes orientation (right- vs left-handed). Volume is its **absolute value**: $V = |[\vec{a} \vec{b} \vec{c}]|$.
 - **Coplanar = dependent = zero.** Three coplanar vectors give $[\vec{a} \vec{b} \vec{c}] = 0$; equivalently they cannot form a basis for space.
 - **The reciprocal system needs non-coplanarity.** It is only defined when $[\vec{a} \vec{b} \vec{c}] \neq 0$, since that quantity sits in the denominator.
-

Section 2: Key Definitions & Properties

Scalar Triple Product (Box Product)

Definition.

$$[\vec{a} \vec{b} \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}).$$

Determinant form. For $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, etc.,

$$[\vec{a} \vec{b} \vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

The result is a **scalar**.

Properties.

- **Cyclic invariance:** $[\vec{a} \vec{b} \vec{c}] = [\vec{b} \vec{c} \vec{a}] = [\vec{c} \vec{a} \vec{b}]$.
- **Swap two $\Rightarrow$ sign flips:** $[\vec{a} \vec{b} \vec{c}] = -[\vec{b} \vec{a} \vec{c}]$.
- **Dot and cross are interchangeable:** $\vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c}$.
- **Repeated vector $\Rightarrow$ zero:** if any two of $\vec{a}, \vec{b}, \vec{c}$ are equal (or parallel), $[\vec{a} \vec{b} \vec{c}] = 0$.
- **Coplanarity / linear dependence:** $[\vec{a} \vec{b} \vec{c}] = 0 \iff \vec{a}, \vec{b}, \vec{c}$ are coplanar (linearly dependent).
- **Scalar factors pull out:** $[\lambda\vec{a} \vec{b} \vec{c}] = \lambda[\vec{a} \vec{b} \vec{c}]$.

Geometric applications.

- **Volume of parallelepiped** with coterminous edges $\vec{a}, \vec{b}, \vec{c}$: $V = |[\vec{a} \vec{b} \vec{c}]|$.
- **Volume of tetrahedron** with the same coterminous edges: $V = \frac{1}{6} |[\vec{a} \vec{b} \vec{c}]|$.
- **Coplanarity of four points A, B, C, D :** $[\vec{AB} \vec{AC} \vec{AD}] = 0$.

Vector Triple Product

Definition & expansion (BAC-CAB rule).

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}.$$

$$(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}.$$

Properties.

- The result lies in the plane of the two vectors *inside* the bracket.
- **Non-associative:** the two expansions above are generally different.
- $\vec{a} \times (\vec{b} \times \vec{c}) + \vec{b} \times (\vec{c} \times \vec{a}) + \vec{c} \times (\vec{a} \times \vec{b}) = \vec{0}$ (Jacobi identity).

Linear Dependence & Basis

- Vectors $\vec{a}, \vec{b}, \vec{c}$ are **linearly independent** $\iff [\vec{a} \ \vec{b} \ \vec{c}] \neq 0 \iff$ they are non-coplanar.
- Three non-coplanar vectors form a **basis**: every $\vec{r}$ can be written uniquely as $\vec{r} = x\vec{a} + y\vec{b} + z\vec{c}$.

Reciprocal System of Vectors (JEE Advanced)

For non-coplanar $\vec{a}, \vec{b}, \vec{c}$ (so $[\vec{a} \ \vec{b} \ \vec{c}] \neq 0$), the reciprocal triad is

$$\vec{a}' = \frac{\vec{b} \times \vec{c}}{[\vec{a} \ \vec{b} \ \vec{c}]}, \quad \vec{b}' = \frac{\vec{c} \times \vec{a}}{[\vec{a} \ \vec{b} \ \vec{c}]}, \quad \vec{c}' = \frac{\vec{a} \times \vec{b}}{[\vec{a} \ \vec{b} \ \vec{c}]},$$

satisfying $\vec{a} \cdot \vec{a}' = \vec{b} \cdot \vec{b}' = \vec{c} \cdot \vec{c}' = 1$ (and all “mixed” dot products = 0), with $[\vec{a} \ \vec{b} \ \vec{c}][\vec{a}' \ \vec{b}' \ \vec{c}'] = 1$.

Section 3: Theorems & Proofs

Theorem 1 — $[\vec{a} \ \vec{b} \ \vec{c}]$ equals the volume of the parallelepiped

Claim. $|[\vec{a} \ \vec{b} \ \vec{c}]|$ is the volume of the parallelepiped with coterminous edges $\vec{a}, \vec{b}, \vec{c}$.

Proof. Build the box on edges $\vec{a}, \vec{b}, \vec{c}$. Take the face spanned by $\vec{b}$ and $\vec{c}$ as the base; its area is $|\vec{b} \times \vec{c}|$, and $\vec{b} \times \vec{c}$ points perpendicular to that base. *Why this step:* the cross product simultaneously gives the base area (as its magnitude) and the base’s normal direction (as its orientation) — exactly the two things a volume needs. The height of the box is the projection of the slant edge $\vec{a}$ onto that normal direction, namely $|\vec{a}| \cos \phi$, where ϕ is the angle between $\vec{a}$ and $\vec{b} \times \vec{c}$. Then

$$V = (\text{base area}) \times (\text{height}) = |\vec{b} \times \vec{c}| (|\vec{a}| \cos \phi) = |\vec{a}| |\vec{b} \times \vec{c}| \cos \phi = \vec{a} \cdot (\vec{b} \times \vec{c}).$$

Since $\cos \phi$ may be negative (if $\vec{a}$ lies on the far side of the base plane), we take the absolute value: $V = |[\vec{a} \ \vec{b} \ \vec{c}]|$. ■

Immediate corollary (coplanarity). If $\vec{a}, \vec{b}, \vec{c}$ are coplanar, the box is flat, $V = 0$, hence $[\vec{a} \ \vec{b} \ \vec{c}] = 0$ — and conversely. The tetrahedron on the same three edges is one-sixth of the box, giving $V_{\text{tet}} = \frac{1}{6}|[\vec{a} \ \vec{b} \ \vec{c}]|$.

Theorem 2 — Cyclic invariance and dot-cross interchange

Claim. $[\vec{a} \ \vec{b} \ \vec{c}] = [\vec{b} \ \vec{c} \ \vec{a}] = [\vec{c} \ \vec{a} \ \vec{b}]$, and $\vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c}$.

Proof. Write $[\vec{a} \ \vec{b} \ \vec{c}]$ as the determinant with rows $\vec{a}, \vec{b}, \vec{c}$. A cyclic permutation of three rows is an even permutation (two row swaps), and an even permutation leaves a determinant unchanged. *Why this step:* each single row swap multiplies a determinant by -1 ; a cyclic shift is two swaps, so $(-1)^2 = 1$. Hence all three cyclic orderings share one value. For the interchange, note $(\vec{a} \times \vec{b}) \cdot \vec{c} = [\vec{a} \ \vec{b} \ \vec{c}]$ by the same determinant (now read as $\vec{c} \cdot (\vec{a} \times \vec{b}) = [\vec{c} \ \vec{a} \ \vec{b}] = [\vec{a} \ \vec{b} \ \vec{c}]$ by cyclic invariance). So the dot and the cross can trade places without changing the value. ■

Theorem 3 — The vector triple product expansion (BAC-CAB)

Claim. $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c}$.

Proof (structure + verification). First, *why the result is a combination of $\vec{b}$ and $\vec{c}$:* the inner product $\vec{b} \times \vec{c}$ is perpendicular to the plane Π of $\vec{b}$ and $\vec{c}$; crossing again with $\vec{a}$ yields a vector perpendicular to $\vec{b} \times \vec{c}$, i.e. back in Π . So $\vec{a} \times (\vec{b} \times \vec{c}) = \lambda \vec{b} + \mu \vec{c}$ for some scalars λ, μ . *Why this step:* any vector in the plane Π is a linear combination of the spanning vectors $\vec{b}$ and $\vec{c}$. To fix λ, μ , evaluate both sides in components along the standard basis (a direct, if tedious, expansion of the determinant-defined cross products) — or check the standard basis cases $\vec{a}, \vec{b}, \vec{c} \in \{\hat{i}, \hat{j}, \hat{k}\}$, which the identity reproduces exactly. Dotting the claimed identity with $\vec{a}$ also gives $\vec{a} \cdot [\vec{a} \times (\vec{b} \times \vec{c})] = 0$, consistent with the left side being perpendicular to $\vec{a}$. The verified scalars are $\lambda = \vec{a} \cdot \vec{c}$ and $\mu = -(\vec{a} \cdot \vec{b})$, giving the stated rule. ■

Mnemonic for the signs: the surviving vectors are the two *inside* the bracket; each is multiplied by the dot product of the *other two*, and the **first term is positive** (the one whose vector is adjacent to the outer vector across the bracket). “ $\vec{a}$ crosses (b cross c)” $\rightarrow$ “ $\vec{b}$ times $(\vec{a} \cdot \vec{c})$ **minus** $\vec{c}$ times $(\vec{a} \cdot \vec{b})$.”

Section 4: Worked Examples

Example 1 — [CBSE Board Level] (clean computation)

Find $[\vec{a} \ \vec{b} \ \vec{c}]$ and hence the volume of the parallelepiped with coterminal edges
 $\vec{a} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{c} = -\hat{i} + \hat{j} + 2\hat{k}$.

Step 1 — set up the determinant.

$$[\vec{a} \ \vec{b} \ \vec{c}] = \begin{vmatrix} 2 & 1 & -1 \\ 1 & 2 & 3 \\ -1 & 1 & 2 \end{vmatrix}.$$

Step 2 — expand along the first row.

$$\begin{aligned} &= 2 \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} - 1 \begin{vmatrix} 1 & 3 \\ -1 & 2 \end{vmatrix} + (-1) \begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix} \\ &= 2(4 - 3) - 1(2 + 3) - 1(1 + 2) = 2(1) - 1(5) - 1(3) = 2 - 5 - 3 = -6. \end{aligned}$$

Step 3 — volume. $V = |[\vec{a} \ \vec{b} \ \vec{c}]| = |-6| = 6$ cubic units.

Answer: $[\vec{a} \ \vec{b} \ \vec{c}] = -6$; volume = 6 cubic units. (The negative sign merely indicates a left-handed ordering; volume is the magnitude.)

Example 2 — [Conceptual Trap] (coplanarity)

Find λ so that the vectors $\vec{a} = \hat{i} - \hat{j} + \hat{k}$, $\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{c} = \lambda\hat{i} - \hat{j} + 2\hat{k}$ are coplanar.

The trap: students set the determinant equal to a volume value, or mishandle the alternating signs of the expansion. Coplanar means the box is **flat**, so set $[\vec{a} \ \vec{b} \ \vec{c}] = 0$.

Step 1 — determinant = 0.

$$\begin{vmatrix} 1 & -1 & 1 \\ 2 & 1 & -1 \\ \lambda & -1 & 2 \end{vmatrix} = 0.$$

Step 2 — expand along row 1.

$$1 \begin{vmatrix} 1 & -1 \\ -1 & 2 \end{vmatrix} - (-1) \begin{vmatrix} 2 & -1 \\ \lambda & 2 \end{vmatrix} + 1 \begin{vmatrix} 2 & 1 \\ \lambda & -1 \end{vmatrix} = 0.$$

$$1(2 - 1) + 1(4 + \lambda) + 1(-2 - \lambda) = 0.$$

$$1 + (4 + \lambda) + (-2 - \lambda) = 0 \Rightarrow 1 + 4 + \lambda - 2 - \lambda = 0 \Rightarrow 3 = 0.$$

Step 3 — interpret. The λ terms cancel and we are left with $3 = 0$, which is impossible. Therefore **no value of λ** makes these three vectors coplanar — for every λ they span a genuine (non-flat) box.

Answer: no such λ exists. (The trap would have a careless student “solve” $3 = 0$ or drop the middle sign and invent a spurious λ .)

Example 3 — [JEE Main Level] (tetrahedron volume from four points)

Find the volume of the tetrahedron with vertices $A(1, 1, 1)$, $B(3, 2, 1)$, $C(2, 4, 3)$, $D(4, 3, 5)$.

Step 1 — form three edge vectors from A .

$$\vec{AB} = 2\hat{i} + \hat{j} + 0\hat{k}, \quad \vec{AC} = \hat{i} + 3\hat{j} + 2\hat{k}, \quad \vec{AD} = 3\hat{i} + 2\hat{j} + 4\hat{k}.$$

Step 2 — scalar triple product.

$$[\vec{AB} \ \vec{AC} \ \vec{AD}] = \begin{vmatrix} 2 & 1 & 0 \\ 1 & 3 & 2 \\ 3 & 2 & 4 \end{vmatrix} = 2(12 - 4) - 1(4 - 6) + 0 = 2(8) - 1(-2) = 16 + 2 = 18.$$

Step 3 — tetrahedron volume.

$$V = \frac{1}{6} |[\vec{AB} \ \vec{AC} \ \vec{AD}]| = \frac{1}{6}(18) = 3 \text{ cubic units.}$$

Answer: 3 cubic units. (Sanity: a non-zero triple product confirms the four points are not coplanar, so a genuine tetrahedron exists.)

Example 4 — [JEE Advanced Level] (vector triple product identity)

Simplify $\vec{a} \times (\vec{b} \times \vec{c}) + \vec{b} \times (\vec{c} \times \vec{a}) + \vec{c} \times (\vec{a} \times \vec{b})$, **and use the result to show that if** $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ **then** $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$.

Part 1 — expand each term by BAC-CAB.

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c},$$

$$\vec{b} \times (\vec{c} \times \vec{a}) = (\vec{b} \cdot \vec{a})\vec{c} - (\vec{b} \cdot \vec{c})\vec{a},$$

$$\vec{c} \times (\vec{a} \times \vec{b}) = (\vec{c} \cdot \vec{b})\vec{a} - (\vec{c} \cdot \vec{a})\vec{b}.$$

Add. Group the $\vec{a}$ terms: $-(\vec{b} \cdot \vec{c})\vec{a} + (\vec{c} \cdot \vec{b})\vec{a} = \vec{0}$ (since $\vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{b}$). The $\vec{b}$ and $\vec{c}$ terms cancel identically by the same symmetry. So the whole sum is $\vec{0}$ — **the Jacobi identity**.

Part 2 — use $\vec{a} + \vec{b} + \vec{c} = \vec{0}$. Cross the equation $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ with $\vec{a}$ on the left:

$$\vec{a} \times \vec{a} + \vec{a} \times \vec{b} + \vec{a} \times \vec{c} = \vec{0} \Rightarrow \vec{0} + \vec{a} \times \vec{b} - \vec{c} \times \vec{a} = \vec{0} \Rightarrow \vec{a} \times \vec{b} = \vec{c} \times \vec{a}.$$

Now cross the same equation with $\vec{b}$:

$$\vec{b} \times \vec{a} + \vec{b} \times \vec{b} + \vec{b} \times \vec{c} = \vec{0} \Rightarrow -\vec{a} \times \vec{b} + \vec{0} + \vec{b} \times \vec{c} = \vec{0} \Rightarrow \vec{b} \times \vec{c} = \vec{a} \times \vec{b}.$$

Combining, $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$. ■

Answer: the cyclic sum is $\vec{0}$ (Jacobi identity); and $\vec{a} + \vec{b} + \vec{c} = \vec{0} \Rightarrow \vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$. (This is the vector engine behind the sine rule for a triangle — the three cross products having equal magnitude says $|\vec{a}||\vec{b}|\sin C$ etc. are all equal.)

Section 5: Practice Exercises

1. **[CBSE]** Evaluate $[\hat{i} \hat{j} \hat{k}]$ and $[\hat{i} \hat{k} \hat{j}]$.
2. **[CBSE]** Find the volume of the parallelepiped with coterminal edges $\vec{a} = \hat{i} + \hat{j}$, $\vec{b} = \hat{j} + \hat{k}$, $\vec{c} = \hat{k} + \hat{i}$.
3. **[JEE Main]** Show that the points $A(1, 2, 3)$, $B(2, 3, 4)$, $C(3, 4, 5)$, $D(0, 1, 2)$ are coplanar.
4. **[JEE Main]** If $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{c} = 3\hat{i} + \hat{j} + \hat{k}$, find $\vec{a} \times (\vec{b} \times \vec{c})$ using the BAC-CAB rule.
5. **[JEE Advanced]** Prove that $[\vec{a} + \vec{b} \quad \vec{b} + \vec{c} \quad \vec{c} + \vec{a}] = 2[\vec{a} \quad \vec{b} \quad \vec{c}]$.

Answers: 1. $[\hat{i} \hat{j} \hat{k}] = 1$; $[\hat{i} \hat{k} \hat{j}] = -1$. 2. 2 cubic units. 3. $[\vec{AB} \quad \vec{AC} \quad \vec{AD}] = 0$, hence coplanar. 4. $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} = 6\vec{b} - (-3)\vec{c} = 6\vec{b} + 3\vec{c} = 15\hat{i} + 15\hat{j} - 15\hat{k}$. 5. Expand using multilinearity; cross terms with repeated vectors vanish, leaving $2[\vec{a} \quad \vec{b} \quad \vec{c}]$.

Section 6: Crack the MCQ

Q1. If $[\vec{a} \ \vec{b} \ \vec{c}] = 0$ for non-zero vectors, then $\vec{a}, \vec{b}, \vec{c}$ are: (a) mutually perpendicular (b) coplanar (c) collinear (d) equal in magnitude

Correct: (b). A zero box product means the parallelepiped is flat — the vectors share a plane (linearly dependent). *Distractors:* **(a)** mutually perpendicular vectors give a *maximal* box, not a flat one. **(c)** collinear is stronger than required (collinear $\Rightarrow$ coplanar, but coplanar does not need collinear). **(d)** magnitude is irrelevant to coplanarity.

Q2. $\vec{a} \times (\vec{b} \times \vec{c})$ equals: (a) $(\vec{a} \cdot \vec{b})\vec{c} - (\vec{a} \cdot \vec{c})\vec{b}$ (b) $(\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$ (c) $(\vec{a} \cdot \vec{b})\vec{c} - (\vec{b} \cdot \vec{c})\vec{a}$ (d) $(\vec{b} \cdot \vec{c})\vec{a} - (\vec{a} \cdot \vec{c})\vec{b}$

Correct: (b). Direct BAC-CAB rule. *Distractors:* **(a)** has both signs flipped — the classic sign reversal. **(c)** is the expansion of $(\vec{a} \times \vec{b}) \times \vec{c}$, not $\vec{a} \times (\vec{b} \times \vec{c})$ — the associativity trap. **(d)** mixes vectors that do not belong (the answer must live in the plane of $\vec{b}, \vec{c}$, so no $\vec{a}$ component can survive).

Q3. The volume of the tetrahedron with coterminous edges $\hat{i} + \hat{j}, \hat{j} + \hat{k}, \hat{k} + \hat{i}$ is: (a) 2 (b) 1 (c) $\frac{1}{3}$ (d) $\frac{1}{6}$

Correct: (c). The box product is $\begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{vmatrix} = 2$; tetrahedron volume $= \frac{1}{6}(2) = \frac{1}{3}$. *Distractors:* **(a)** is the parallelepiped volume (forgot the $\frac{1}{6}$). **(d)** applied $\frac{1}{6}$ to a triple product of 1 instead of 2. **(b)** halved the box like a triangle ($\frac{1}{2}$) rather than a tetrahedron ($\frac{1}{6}$).

Q4. $[\vec{a} \ \vec{b} \ \vec{c}]$ is **not** equal to: (a) $[\vec{b} \ \vec{c} \ \vec{a}]$ (b) $(\vec{a} \times \vec{b}) \cdot \vec{c}$ (c) $[\vec{b} \ \vec{a} \ \vec{c}]$ (d) $\vec{a} \cdot (\vec{b} \times \vec{c})$

Correct: (c). Swapping the first two entries flips the sign: $[\vec{b} \ \vec{a} \ \vec{c}] = -[\vec{a} \ \vec{b} \ \vec{c}]$, so it is *not* equal (unless the value is zero). *Distractors:* **(a)** is a cyclic shift — equal. **(b)** dot-cross interchange — equal. **(d)** is the definition — equal. Only (c) breaks the equality.

Section 7: Common Student Pitfalls & Pro-Tips

- **Dropping the absolute value for volume.** $[\vec{a} \ \vec{b} \ \vec{c}]$ may be negative; volume is $|[\vec{a} \ \vec{b} \ \vec{c}]|$. Never report a negative volume.
- **Confusing the two volume factors.** Parallelepiped uses the *full* triple product; the tetrahedron uses $\frac{1}{6}$ of it. (A triangle's $\frac{1}{2}$ has no place here.)
- **Treating the vector triple product as associative.** $\vec{a} \times (\vec{b} \times \vec{c}) \neq (\vec{a} \times \vec{b}) \times \vec{c}$. The bracket position decides which two vectors survive in the BAC-CAB expansion.
- **Sign error in the BAC-CAB rule.** It is “middle vector times (dot of the outer pair), **minus** the other inner vector times (dot of the remaining pair)” — first term positive.

Pro-Tip — coplanarity in one line. To test whether three vectors are coplanar, or whether four points lie in a plane, do **not** hunt for a common plane. Just compute one determinant: $[\vec{a} \ \vec{b} \ \vec{c}]$ (or $[\vec{AB} \ \vec{AC} \ \vec{AD}]$ for points). Zero $\Rightarrow$ coplanar; non-zero $\Rightarrow$ not. It collapses a geometry problem into a 3×3 determinant.

Section 8: Cheat Sheet Box

- **Scalar triple product:** $[\vec{a} \ \vec{b} \ \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$ — a **scalar**.
- **Volumes:** parallelepiped $= |[\vec{a} \ \vec{b} \ \vec{c}]|$; tetrahedron $= \frac{1}{6}|[\vec{a} \ \vec{b} \ \vec{c}]|$.
- **Coplanarity:** $[\vec{a} \ \vec{b} \ \vec{c}] = 0 \iff$ coplanar $\iff$ linearly dependent; four points coplanar $\iff [\vec{AB} \ \vec{AC} \ \vec{AD}] = 0$.
- **Cyclic / swap:** $[\vec{a} \ \vec{b} \ \vec{c}] = [\vec{b} \ \vec{c} \ \vec{a}] = [\vec{c} \ \vec{a} \ \vec{b}]$; one swap flips sign; dot $\leftrightarrow$ cross interchangeable.
- **Vector triple product:** $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$ (BAC-CAB); **non-associative**.
- **Reciprocal system:** $\vec{a}' = \frac{\vec{b} \times \vec{c}}{[\vec{a} \ \vec{b} \ \vec{c}]}$, etc.; $[\vec{a} \ \vec{b} \ \vec{c}][\vec{a}' \ \vec{b}' \ \vec{c}'] = 1$.

Memory aids: “Box product = volume of the box.” | “Flat box = zero = coplanar.” | “Tetra takes a sixth.” | “BAC minus CAB — middle vector first, first sign plus.”

Chapter 10: Vector Algebra — Round 2 Addendum

The chapter hands you the machinery — components, both products, the triple products, and their headline formulas — and drills each formula on clean inputs. Four examinable families sit just outside it. Addendum A resolves a vector along *oblique* directions (components parallel/perpendicular to a given vector, coordinates in a non-orthogonal basis, angle bisectors), which JEE uses constantly and the chapter never sets up. Addendum B builds the magnitude algebra — $|\vec{a} \pm \vec{b}|^2$, the parallelogram law, Cauchy-Schwarz, the triangle inequality — and shows how constraint problems reduce to counting dot products. Addendum C develops nested cross-product algebra beyond BAC-CAB and solves vector equations of the form $\vec{a} \times \vec{x} = \vec{b}$. Addendum D does position-vector triangle geometry: incenter, the circumcenter-orthocenter identity, and area ratios via barycentric weights. Nothing here is new theory; every method is a dot, a cross, or a section formula applied one layer deeper than the chapter took it.

Addendum A — Resolution Along Oblique Directions and Angle Bisectors

The chapter resolves vectors along $\hat{i}, \hat{j}, \hat{k}$ — directions that are perpendicular and unit length. Exams rarely grant you that luxury. You will be asked for the part of $\vec{r}$ along an arbitrary $\vec{a}$, for the coordinates of $\vec{r}$ in a tilted basis, or for the direction that bisects the angle between two vectors. Each is a short derivation from the definitions you already have.

Method

A1 — Split relative to one direction. Any vector $\vec{r}$ can be written uniquely as a part parallel to a non-zero $\vec{a}$ plus a part perpendicular to it. To see this, *assume* the split exists and force the coefficient out:

$$\vec{r} = \lambda \vec{a} + \vec{u}, \quad \vec{u} \cdot \vec{a} = 0.$$

Dot both sides with $\vec{a}$. The term $\vec{u} \cdot \vec{a}$ dies, leaving

$$\vec{r} \cdot \vec{a} = \lambda |\vec{a}|^2 \implies \lambda = \frac{\vec{r} \cdot \vec{a}}{|\vec{a}|^2}.$$

So the split is

$$\vec{r} = \underbrace{\frac{\vec{r} \cdot \vec{a}}{|\vec{a}|^2} \vec{a}}_{\text{parallel part}} + \underbrace{\left(\vec{r} - \frac{\vec{r} \cdot \vec{a}}{|\vec{a}|^2} \vec{a} \right)}_{\text{perpendicular part}}.$$

Existence is now automatic: define the two pieces by these formulas and check $(\vec{r} - \lambda \vec{a}) \cdot \vec{a} = \vec{r} \cdot \vec{a} - \lambda |\vec{a}|^2 = 0$. Uniqueness: if $\lambda_1 \vec{a} + \vec{u}_1 = \lambda_2 \vec{a} + \vec{u}_2$ with both remainders perpendicular to $\vec{a}$, dotting with $\vec{a}$ forces $\lambda_1 = \lambda_2$, hence $\vec{u}_1 = \vec{u}_2$. When $\vec{a}$ happens to be a unit vector, the parallel part is simply $(\vec{r} \cdot \hat{a}) \hat{a}$ — the projection vector of Subtopic 02, recovered as a special case.

A2 — Coordinates in a non-orthogonal basis. Let $[\vec{a} \vec{b} \vec{c}] \neq 0$ and suppose

$$\vec{r} = x\vec{a} + y\vec{b} + z\vec{c}.$$

Take the scalar triple product of both sides with $\vec{b}$ and $\vec{c}$, in that order. By linearity of the box product (Subtopic 03) every term containing a repeated vector vanishes:

$$[\vec{r} \vec{b} \vec{c}] = x[\vec{a} \vec{b} \vec{c}] + y \underbrace{[\vec{b} \vec{b} \vec{c}]}_0 + z \underbrace{[\vec{c} \vec{b} \vec{c}]}_0 = x[\vec{a} \vec{b} \vec{c}].$$

Dividing, and cycling the same argument for y and z :

$$x = \frac{[\vec{r} \vec{b} \vec{c}]}{[\vec{a} \vec{b} \vec{c}]}, \quad y = \frac{[\vec{a} \vec{r} \vec{c}]}{[\vec{a} \vec{b} \vec{c}]}, \quad z = \frac{[\vec{a} \vec{b} \vec{r}]}{[\vec{a} \vec{b} \vec{c}]}.$$

This is Cramer's rule wearing vector clothing, and it explains *why* non-coplanarity ($[\vec{a} \vec{b} \vec{c}] \neq 0$) is exactly the condition for the basis to work.

A3 — Angle bisectors. Let $\vec{a}, \vec{b}$ be non-zero and not opposite. Consider the unit vectors $\hat{a}$ and $\hat{b}$ placed tail-to-tail. They form two adjacent sides of a rhombus (equal lengths!), and the diagonal $\hat{a} + \hat{b}$ of a rhombus bisects the angle between the sides. Hence

$$\text{internal bisector direction} \propto \frac{\vec{a}}{|\vec{a}|} + \frac{\vec{b}}{|\vec{b}|}, \quad \text{external bisector direction} \propto \frac{\vec{a}}{|\vec{a}|} - \frac{\vec{b}}{|\vec{b}|}.$$

The external claim: $\hat{a} - \hat{b}$ is the other diagonal, which bisects the *supplementary* angle — equivalently, it makes equal angles with $\hat{a}$ and $-\hat{b}$. Normalise either expression to get a unit vector. Shortcut worth memorising: if $|\vec{a}| = |\vec{b}|$, the units are $\vec{a}/|\vec{a}|$ and $\vec{b}/|\vec{b}|$ with the same denominator, so the bisector directions are simply proportional to $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$.

Example A.1 — [CBSE / JEE Main Level]

Resolve $\vec{r} = 2\hat{i} + 3\hat{j} + 6\hat{k}$ **into components parallel and perpendicular to** $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$.

Here $\vec{r} \cdot \vec{a} = 2 + 6 + 12 = 20$ and $|\vec{a}|^2 = 1 + 4 + 4 = 9$. So $\lambda = \frac{20}{9}$ and

$$\vec{r}_{\parallel} = \frac{20}{9} (\hat{i} + 2\hat{j} + 2\hat{k}), \quad \vec{r}_{\perp} = \vec{r} - \vec{r}_{\parallel}.$$

Componentwise subtraction for $\vec{r}_{\perp}$:

$$\vec{r}_{\perp} = \left(2 - \frac{20}{9}\right)\hat{i} + \left(3 - \frac{40}{9}\right)\hat{j} + \left(6 - \frac{40}{9}\right)\hat{k} = -\frac{2}{9}\hat{i} - \frac{13}{9}\hat{j} + \frac{14}{9}\hat{k}.$$

Check 1 (perpendicularity). $\vec{r}_\perp \cdot \vec{a} = \frac{-2 - 26 + 28}{9} = 0$.

Check 2 (Pythagoras). $|\vec{r}|^2 = 4 + 9 + 36 = 49$. Also

$$|\vec{r}_\parallel|^2 = \frac{400}{9}, \quad |\vec{r}_\perp|^2 = \frac{4 + 169 + 196}{81} = \frac{369}{81} = \frac{41}{9},$$

and $\frac{400}{9} + \frac{41}{9} = \frac{441}{9} = 49$. The split conserves length, as it must.

Example A.2 — [JEE Main Level]

Express $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k}$ **in terms of** $\vec{a} = \hat{i} + \hat{j}$, $\vec{b} = \hat{j} + \hat{k}$, $\vec{c} = \hat{k} + \hat{i}$.

Method 1 — direct comparison. Write $\vec{r} = x\vec{a} + y\vec{b} + z\vec{c}$ and match components:

$$x + z = 1, \quad x + y = 2, \quad y + z = 3.$$

Adding all three: $2(x + y + z) = 6$, so $x + y + z = 3$. Subtracting each original equation from this: $y = 2$, $z = 1$, $x = 0$. Hence $\vec{r} = 2\vec{b} + \vec{c}$.

Method 2 — the triple-product formula, as a check. From the chapter's own practice set, $[\vec{a} \vec{b} \vec{c}] = 2$. Then

$$[\vec{r} \vec{b} \vec{c}] = \begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{vmatrix} = 1(1 - 0) - 2(0 - 1) + 3(0 - 1) = 1 + 2 - 3 = 0 \implies x = 0,$$

$$[\vec{a} \vec{r} \vec{c}] = \begin{vmatrix} 1 & 1 & 0 \\ 1 & 2 & 3 \\ 1 & 0 & 1 \end{vmatrix} = 1(2) - 1(1 - 3) + 0 = 4 \implies y = 2,$$

$$[\vec{a} \vec{b} \vec{r}] = \begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 3 \end{vmatrix} = 1(3 - 2) - 1(0 - 1) + 0 = 2 \implies z = 1.$$

Both methods agree: $\vec{r} = 2\vec{b} + \vec{c}$. Note the zero coefficient — nothing forbids it, and Method 1 finds it faster while Method 2 generalises to bases where direct comparison is messy.

Example A.3 — [JEE Advanced Level]

Find unit vectors along the internal and external bisectors of the angle between $\vec{a} = \hat{i} + 2\hat{j} + 2\hat{k}$ **and** $\vec{b} = 2\hat{i} + \hat{j} - 2\hat{k}$.

Both magnitudes equal 3, so the shortcut applies: the bisector directions are proportional to $\vec{a} \pm \vec{b}$.

Internal: $\vec{a} + \vec{b} = 3\hat{i} + 3\hat{j}$, of magnitude $3\sqrt{2}$, giving

$$\hat{b}_{\text{int}} = \frac{1}{\sqrt{2}} (\hat{i} + \hat{j}).$$

External: $\vec{a} - \vec{b} = -\hat{i} + \hat{j} + 4\hat{k}$, of magnitude $\sqrt{1+1+16} = 3\sqrt{2}$, giving

$$\hat{b}_{\text{ext}} = \frac{1}{3\sqrt{2}} (-\hat{i} + \hat{j} + 4\hat{k}).$$

Checks. Dot each bisector with each original unit vector; the two values must agree.

$$\hat{a} = \frac{1}{3}(1, 2, 2), \quad \hat{b} = \frac{1}{3}(2, 1, -2).$$

$$\hat{b}_{\text{int}} \cdot \hat{a} = \frac{1+2}{3\sqrt{2}} = \frac{1}{\sqrt{2}}, \quad \hat{b}_{\text{int}} \cdot \hat{b} = \frac{2+1}{3\sqrt{2}} = \frac{1}{\sqrt{2}}. \checkmark$$

$$\hat{b}_{\text{ext}} \cdot \hat{a} = \frac{-1+2-8}{9\sqrt{2}} = -\frac{7}{9\sqrt{2}}, \quad \hat{b}_{\text{ext}} \cdot \hat{b} = \frac{-2+1+8}{9\sqrt{2}} = \frac{7}{9\sqrt{2}}. \checkmark$$

The internal bisector makes equal acute angles with both — correct. The external bisector makes equal-magnitude but opposite-sign cosines, i.e. equal angles with $\hat{a}$ and $-\hat{b}$ — exactly the supplementary-angle bisection claimed in A3. As a bonus, $\hat{b}_{\text{int}} \cdot \hat{b}_{\text{ext}} = \frac{-1+1+0}{3\sqrt{2} \cdot \sqrt{2} \cdot 3}$ -type computation gives 0 here: internal and external bisectors are always perpendicular (proved in Practice 4).

Practice — Addendum A

1. Resolve $\vec{r} = \hat{i} + 2\hat{j} + 2\hat{k}$ into parts parallel and perpendicular to $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$.
2. Express $\vec{r} = 5\hat{i} + 3\hat{j} + 4\hat{k}$ in terms of $\hat{i} + \hat{j}$, $\hat{j} + \hat{k}$, $\hat{k} + \hat{i}$.
3. Find a unit vector along the internal bisector of the angle between $\vec{a} = \hat{i} + \hat{j}$ and $\vec{b} = \hat{j} + \hat{k}$.
4. Prove that the internal and external bisector directions of the angle between any two non-zero vectors are perpendicular.
5. If $\vec{a}, \vec{b}$ are unit vectors with angle θ between them, show that $|\vec{a} + \vec{b}| = 2 \cos \frac{\theta}{2}$, and explain why this confirms that $\vec{a} + \vec{b}$ points along the bisector.

Answers: 1. Parallel part $\frac{8}{9}(2\hat{i} + \hat{j} + 2\hat{k})$; perpendicular part $-\frac{7}{9}\hat{i} + \frac{10}{9}\hat{j} + \frac{2}{9}\hat{k}$ (check: its dot with $\vec{a}$ is $\frac{-14+10+4}{9} = 0$). 2. $2(\hat{i} + \hat{j}) + (\hat{j} + \hat{k}) + 3(\hat{k} + \hat{i})$ — solve $x+z=5$, $x+y=3$, $y+z=4$; the sum gives $x+y+z=6$, whence $x=2$, $y=1$, $z=3$. 3. Units are $\frac{1}{\sqrt{2}}(1, 1, 0)$ and $\frac{1}{\sqrt{2}}(0, 1, 1)$; their sum is $\frac{1}{\sqrt{2}}(1, 2, 1)$, of magnitude $\sqrt{3}$, so the answer is $\frac{1}{\sqrt{6}}(\hat{i} + 2\hat{j} + \hat{k})$. 4. Dot the two directions: $\left(\frac{\vec{a}}{|\vec{a}|} + \frac{\vec{b}}{|\vec{b}|}\right) \cdot \left(\frac{\vec{a}}{|\vec{a}|} - \frac{\vec{b}}{|\vec{b}|}\right) = |\hat{a}|^2 - |\hat{b}|^2 = 1 - 1 = 0$. 5. $|\vec{a} + \vec{b}|^2 = 2 + 2 \cos \theta = 4 \cos^2 \frac{\theta}{2}$, and $\cos \frac{\theta}{2} \geq 0$ for $0 \leq \theta \leq \pi$, so the positive root is $2 \cos \frac{\theta}{2}$. A vector of this length making

angle $\frac{\theta}{2}$ with each of $\vec{a}, \vec{b}$ would have squared length $1 + 1 + 2 \cos \frac{\theta}{2} \cdot \cos \frac{\theta}{2}$ — style symmetry — precisely $4 \cos^2 \frac{\theta}{2}$ — so $\vec{a} + \vec{b}$ is exactly that vector: it sits on the bisector.

Addendum B — Magnitude Algebra and Constraint Problems

Subtopic 02 states the definitions and Lagrange's identity, then computes. What JEE actually asks is a *second-order game*: square a magnitude, expand, and discover that the unknowns are dot products. This addendum derives the small stock of identities that run that game, plus the inequalities that bound answers before you compute them.

Method

B1 — The fundamental expansions. From distributivity and $\vec{a} \cdot \vec{a} = |\vec{a}|^2$:

$$|\vec{a} + \vec{b}|^2 = (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) = |\vec{a}|^2 + |\vec{b}|^2 + 2 \vec{a} \cdot \vec{b},$$

$$|\vec{a} - \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2 \vec{a} \cdot \vec{b}.$$

Adding and subtracting these two lines:

$$|\vec{a} + \vec{b}|^2 + |\vec{a} - \vec{b}|^2 = 2 (|\vec{a}|^2 + |\vec{b}|^2) \quad (\text{parallelogram law}),$$

$$|\vec{a} + \vec{b}|^2 - |\vec{a} - \vec{b}|^2 = 4 \vec{a} \cdot \vec{b}.$$

Also,

$$(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = |\vec{a}|^2 - |\vec{b}|^2,$$

since the cross terms $\mp \vec{a} \cdot \vec{b}$ cancel.

B2 — The geometric dictionary. In a parallelogram with adjacent sides $\vec{a}, \vec{b}$, the diagonals are $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$. Two instant consequences, each a one-line derivation:

- *Equal diagonals:* $|\vec{a} + \vec{b}|^2 = |\vec{a} - \vec{b}|^2 \iff 4 \vec{a} \cdot \vec{b} = 0 \iff \vec{a} \perp \vec{b}$, i.e. the parallelogram is a **rectangle**.
- *Perpendicular diagonals:* $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 0 \iff |\vec{a}|^2 = |\vec{b}|^2$, i.e. the parallelogram is a **rhombus**.

B3 — Cauchy-Schwarz and the triangle inequality. Lagrange's identity (Subtopic 02, Theorem 3) reads

$$(\vec{a} \cdot \vec{b})^2 + |\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2.$$

The second term is a squared magnitude, hence non-negative, so

$$(\vec{a} \cdot \vec{b})^2 \leq |\vec{a}|^2 |\vec{b}|^2, \quad \text{equality} \iff \vec{a} \times \vec{b} = \vec{0} \iff \vec{a} \parallel \vec{b}.$$

That is Cauchy-Schwarz. For the triangle inequality, start from B1 and inject $2\vec{a} \cdot \vec{b} \leq 2|\vec{a} \cdot \vec{b}| \leq 2|\vec{a}||\vec{b}|$:

$$|\vec{a} + \vec{b}|^2 \leq |\vec{a}|^2 + |\vec{b}|^2 + 2|\vec{a}||\vec{b}| = (|\vec{a}| + |\vec{b}|)^2,$$

so $|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|$, with equality iff $\vec{a}, \vec{b}$ point the same way. Running the same argument with $-2|\vec{a} \cdot \vec{b}|$ gives the lower bound $|\vec{a} + \vec{b}| \geq ||\vec{a}| - |\vec{b}||$. Together:

$$||\vec{a}| - |\vec{b}|| \leq |\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|.$$

B4 — Strategy for constraint problems. Every magnitude becomes a square; every square expands via B1; the unknowns of the problem are then dot products. Before computing, count equations against unknown dot products — if the count fails, some symmetry or inequality (B3) must close the gap.

Example B.1 — [CBSE Board Level]

Given $|\vec{a}| = 3$, $|\vec{b}| = 4$ **and** $\vec{a} \cdot \vec{b} = 6$, **find** $|\vec{a} + \vec{b}|$, $|\vec{a} - \vec{b}|$, **the angle between** $\vec{a}$ **and** $\vec{b}$, **and the area of the parallelogram they span.**

From B1:

$$|\vec{a} + \vec{b}|^2 = 9 + 16 + 12 = 37, \quad |\vec{a} - \vec{b}|^2 = 9 + 16 - 12 = 13.$$

Angle: $\cos \theta = \frac{6}{3 \cdot 4} = \frac{1}{2}$, so $\theta = 60^\circ$. Area via Lagrange's identity:

$$|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 = 144 - 36 = 108, \quad \text{Area} = 6\sqrt{3}.$$

Check. $\theta = 60^\circ$ reproduces $\vec{a} \cdot \vec{b} = 12 \cos 60^\circ = 6$ and $|\vec{a} \times \vec{b}| = 12 \sin 60^\circ = 6\sqrt{3}$. Consistent.

Example B.2 — [JEE Main Level]

$\vec{a}$ and $\vec{b}$ are unit vectors with $|\vec{a} + \vec{b}| = \sqrt{3}$. Find $|\vec{a} - \vec{b}|$ and the angle between them — without ever computing a dot product directly.

Parallelogram law with $|\vec{a}| = |\vec{b}| = 1$:

$$3 + |\vec{a} - \vec{b}|^2 = 2(1 + 1) = 4 \implies |\vec{a} - \vec{b}|^2 = 1.$$

For the angle, use the difference form of B1 on $|\vec{a} + \vec{b}|^2$:

$$3 = 1 + 1 + 2 \cos \theta \implies \cos \theta = \frac{1}{2} \implies \theta = 60^\circ.$$

Check. $|\vec{a} - \vec{b}|^2 = 2 - 2 \cos 60^\circ = 1$. Consistent. The whole problem is two substitutions into B1 — the skill being *which* identity, not arithmetic.

Example B.3 — [JEE Main Level, 2019 pattern]

$\vec{a}, \vec{b}, \vec{c}$ are unit vectors, $\vec{a} \perp \vec{b}$, and $|\vec{a} + \vec{b} + \vec{c}| = 1$. Find the angle between $\vec{c}$ and $\vec{a} + \vec{b}$.

Step 1 — expand the given magnitude. Writing $\vec{s} = \vec{a} + \vec{b} + \vec{c}$:

$$|\vec{s}|^2 = 3 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 1,$$

so the pairwise dot products sum to -1 . Since $\vec{a} \cdot \vec{b} = 0$:

$$\vec{c} \cdot (\vec{a} + \vec{b}) = -1.$$

Step 2 — size the target vector. Because $\vec{a} \perp \vec{b}$ and both are unit,

$$|\vec{a} + \vec{b}|^2 = 1 + 1 + 0 = 2, \quad |\vec{a} + \vec{b}| = \sqrt{2}.$$

Step 3 — read off the angle. If ϕ is the angle between $\vec{c}$ and $\vec{a} + \vec{b}$:

$$\cos \phi = \frac{\vec{c} \cdot (\vec{a} + \vec{b})}{|\vec{c}| |\vec{a} + \vec{b}|} = \frac{-1}{\sqrt{2}}, \quad \phi = 135^\circ.$$

Why the data suffice. One might worry that individual angles of $\vec{c}$ with $\vec{a}$ and with $\vec{b}$ are not pinned down — and they are not. But the question asks only about the *combined* vector $\vec{a} + \vec{b}$, and Step 1 determines the combined dot product completely. Recognising which aggregate is determined is the entire marks-earning move.

Example B.4 — [JEE Advanced Level]

Minimise $f(\vec{x}) = |\vec{x} - \vec{a}|^2 + |\vec{x} - \vec{b}|^2$ **over all vectors** $\vec{x}$, **where** $|\vec{a}| = 2$, $|\vec{b}| = 4$ **and the angle between** $\vec{a}$ **and** $\vec{b}$ **is** 60° . **Where is the minimum attained?**

Expand both squares by B1:

$$f(\vec{x}) = 2|\vec{x}|^2 - 2\vec{x} \cdot (\vec{a} + \vec{b}) + |\vec{a}|^2 + |\vec{b}|^2.$$

Complete the square around the midpoint $\vec{m} = \frac{\vec{a} + \vec{b}}{2}$:

$$2|\vec{x} - \vec{m}|^2 = 2|\vec{x}|^2 - 4\vec{x} \cdot \vec{m} + 2|\vec{m}|^2 = 2|\vec{x}|^2 - 2\vec{x} \cdot (\vec{a} + \vec{b}) + \frac{|\vec{a} + \vec{b}|^2}{2}.$$

Comparing with f , the leftover constant is

$$|\vec{a}|^2 + |\vec{b}|^2 - \frac{|\vec{a} + \vec{b}|^2}{2} = \frac{2|\vec{a}|^2 + 2|\vec{b}|^2 - |\vec{a} + \vec{b}|^2}{2} = \frac{|\vec{a} - \vec{b}|^2}{2},$$

where the last step is the parallelogram law rearranged. Hence

$$f(\vec{x}) = 2 \left| \vec{x} - \frac{\vec{a} + \vec{b}}{2} \right|^2 + \frac{|\vec{a} - \vec{b}|^2}{2} \geq \frac{|\vec{a} - \vec{b}|^2}{2},$$

with equality exactly at the midpoint. Numerically: $|\vec{a} - \vec{b}|^2 = 4 + 16 - 2(2)(4) \cos 60^\circ = 12$, so the minimum value is 6, attained at $\vec{x} = \frac{\vec{a} + \vec{b}}{2}$.

Reading the result. The sum of squared distances to two fixed points is minimised at their midpoint — the vector proof of the classical fact, delivered by completing one square.

Practice — Addendum B

1. $|\vec{a}| = 2$, $|\vec{b}| = 3$, angle 60° . Find $|\vec{a} + \vec{b}|$ and $|\vec{a} - \vec{b}|$.
2. $\vec{a}, \vec{b}$ are unit vectors with $|\vec{a} + \vec{b}| = 1$. Find the angle between them and $|\vec{a} - \vec{b}|$.
3. Prove that a parallelogram whose diagonals are equal is a rectangle, quoting the identity you use.
4. If $|\vec{a}| = 2$ and $|\vec{a} + \vec{b}|^2 + |\vec{a} - \vec{b}|^2 = 20$, find $|\vec{b}|$.
5. [Advanced] $\vec{a}, \vec{b}$ are perpendicular unit vectors and $\vec{c}$ ranges over all unit vectors. Find the minimum and maximum possible values of $|\vec{a} + \vec{b} + \vec{c}|$.

Answers: 1. $\vec{a} \cdot \vec{b} = 3$; $|\vec{a} + \vec{b}| = \sqrt{4 + 9 + 6} = \sqrt{19}$, $|\vec{a} - \vec{b}| = \sqrt{4 + 9 - 6} = \sqrt{7}$. 2. $1 = 2 + 2 \cos \theta$ gives $\cos \theta = -\frac{1}{2}$, $\theta = 120^\circ$; then $|\vec{a} - \vec{b}|^2 = 2 - 2(-\frac{1}{2}) = 3$, so $|\vec{a} - \vec{b}| = \sqrt{3}$. 3. Diagonals are $\vec{a} + \vec{b}$, $\vec{a} - \vec{b}$; equality of squared lengths forces $4 \vec{a} \cdot \vec{b} = 0$ by B1, so adjacent sides are perpendicular. 4. Parallelogram law: $20 = 2(4 + |\vec{b}|^2)$, so $|\vec{b}|^2 = 6$. 5. $|\vec{a} + \vec{b}| = \sqrt{2}$ is fixed; by the two-sided triangle inequality applied to $(\vec{a} + \vec{b}) + \vec{c}$, the magnitude ranges over $[\sqrt{2} - 1, \sqrt{2} + 1]$, both ends attained (for $\vec{c}$ anti-parallel and parallel to $\vec{a} + \vec{b}$ respectively).

Addendum C — Nested Cross Products and Vector Equations

Subtopic 03 gives you BAC-CAB for $\vec{a} \times (\vec{b} \times \vec{c})$ and stops. JEE Advanced routinely nests one level further — cross products of cross products — and asks you to *solve* equations in which the unknown is a vector. Both skills are two applications of what you already own.

Method

C1 — Cross of two cross products. Claim:

$$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [\vec{a} \vec{b} \vec{d}] \vec{c} - [\vec{a} \vec{b} \vec{c}] \vec{d}.$$

Derivation. Name $\vec{p} = \vec{a} \times \vec{b}$ and apply BAC-CAB to $\vec{p} \times (\vec{c} \times \vec{d})$:

$$\vec{p} \times (\vec{c} \times \vec{d}) = (\vec{p} \cdot \vec{d}) \vec{c} - (\vec{p} \cdot \vec{c}) \vec{d}.$$

Now decode each dot: $(\vec{a} \times \vec{b}) \cdot \vec{d} = [\vec{a} \vec{b} \vec{d}]$ and $(\vec{a} \times \vec{b}) \cdot \vec{c} = [\vec{a} \vec{b} \vec{c}]$ by definition of the box product. Substitute to get the claim. A second, equally useful form follows by swapping the pairs (note the overall sign flip from anticommutativity):

$$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [\vec{a} \vec{c} \vec{d}] \vec{b} - [\vec{b} \vec{c} \vec{d}] \vec{a}.$$

Verification on the standard basis. Take $\vec{a} = \hat{i}$, $\vec{b} = \hat{j}$, $\vec{c} = \hat{j}$, $\vec{d} = \hat{k}$. Left side: $\hat{k} \times \hat{i} = \hat{j}$. Right side: $[\hat{i} \hat{j} \hat{k}] \hat{j} - [\hat{i} \hat{j} \hat{j}] \hat{k} = \hat{j} - \vec{0} = \hat{j}$. Agreement.

C2 — Solving $\vec{a} \times \vec{x} = \vec{b}$. First, existence. Dotting both sides with $\vec{a}$ gives $\vec{a} \cdot \vec{b} = 0$ — a *necessary* condition, since the left side is perpendicular to $\vec{a}$. It is also sufficient, and the full solution set when $\vec{a} \neq \vec{0}$, $\vec{a} \cdot \vec{b} = 0$ is

$$\vec{x} = \frac{\vec{b} \times \vec{a}}{|\vec{a}|^2} + \lambda \vec{a}, \quad \lambda \in \mathbb{R}.$$

Verification. Apply BAC-CAB to $\vec{a} \times (\vec{b} \times \vec{a})$:

$$\vec{a} \times (\vec{b} \times \vec{a}) = (\vec{a} \cdot \vec{a}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{a} = |\vec{a}|^2 \vec{b},$$

since $\vec{a} \cdot \vec{b} = 0$. Dividing by $|\vec{a}|^2$ shows the candidate particular solution works. For the homogeneous part: $\vec{a} \times \vec{h} = \vec{0}$ forces $\vec{h} \parallel \vec{a}$ (Subtopic 02), i.e. $\vec{h} = \lambda \vec{a}$ — a whole line of solutions, exactly as a cross product cannot see components along itself.

C3 — The squared box product. Claim:

$$[\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}] = [\vec{a} \vec{b} \vec{c}]^2.$$

Derivation. Write the left side as a dot with a cross and apply C1 to the inner pair:

$$(\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a}) = [\vec{b} \vec{c} \vec{a}] \vec{c} - [\vec{b} \vec{c} \vec{c}] \vec{a} = [\vec{a} \vec{b} \vec{c}] \vec{c} - \vec{0},$$

using cyclic invariance $[\vec{b} \vec{c} \vec{a}] = [\vec{a} \vec{b} \vec{c}]$. Therefore

$$[\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}] = (\vec{a} \times \vec{b}) \cdot ([\vec{a} \vec{b} \vec{c}] \vec{c}) = [\vec{a} \vec{b} \vec{c}] [(\vec{a} \times \vec{b}) \cdot \vec{c}] = [\vec{a} \vec{b} \vec{c}]^2. \quad \blacksquare$$

Check on the basis: $\vec{a} = \hat{i}, \vec{b} = \hat{j}, \vec{c} = \hat{k}$ gives left side $[\hat{k} \hat{i} \hat{j}] = 1$ and right side 1^2 . Immediate corollary: if $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar, then so are $\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ — and after dividing by

$[\vec{a} \vec{b} \vec{c}]$, these are precisely the reciprocal triad of Subtopic 03. The identity explains why the reciprocal system is again non-coplanar.

C4 — Direction of a plane-intersection line. The vector $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d})$ is perpendicular to $\vec{a} \times \vec{b}$, hence lies in the plane of $\vec{a}, \vec{b}$; and perpendicular to $\vec{c} \times \vec{d}$, hence lies in the plane of $\vec{c}, \vec{d}$. So it points along the intersection of the two planes — and the two expansions in C1 exhibit it explicitly as a combination of $\vec{c}, \vec{d}$ and of $\vec{a}, \vec{b}$ respectively. This is the engine behind the “line of intersection of two planes” formula you will meet in Three-Dimensional Geometry.

Example C.1 — [JEE Main Level]

Solve $\vec{a} \times \vec{x} = \vec{b}$ **where** $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ **and** $\vec{b} = \hat{i} - \hat{j}$.

Compatibility: $\vec{a} \cdot \vec{b} = 1 - 1 + 0 = 0$, so solutions exist. Compute the particular piece:

$$\vec{b} \times \vec{a} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 0 \\ 1 & 1 & 1 \end{vmatrix} = \hat{i}(-1-0) - \hat{j}(1-0) + \hat{k}(1+1) = -\hat{i} - \hat{j} + 2\hat{k}.$$

Since $|\vec{a}|^2 = 3$, the general solution is

$$\vec{x} = \frac{1}{3}(-\hat{i} - \hat{j} + 2\hat{k}) + \lambda(\hat{i} + \hat{j} + \hat{k}), \quad \lambda \in \mathbb{R}.$$

Direct check at $\lambda = 0$.

$$\vec{a} \times \frac{(-1, -1, 2)}{3} = \frac{1}{3} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ -1 & -1 & 2 \end{vmatrix} = \frac{1}{3} [\hat{i}(2+1) - \hat{j}(2+1) + \hat{k}(-1+1)] = \hat{i} - \hat{j} = \vec{b}. \checkmark$$

Example C.2 — [JEE Advanced Level]

Prove that $[\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}] = [\vec{a} \vec{b} \vec{c}]^2$, **and deduce that if** $\vec{a}, \vec{b}, \vec{c}$ **are three non-coplanar unit vectors, then** $\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ **are also non-coplanar unit-spanning vectors whose box product equals 1.**

The identity is C3, derived above. For the deduction: non-coplanarity of $\vec{a}, \vec{b}, \vec{c}$ means $[\vec{a} \vec{b} \vec{c}] \neq 0$, and the identity then forces the crossed triad’s box product to be the non-zero number $[\vec{a} \vec{b} \vec{c}]^2 > 0$ — so the crossed triad is non-coplanar. For unit $\vec{a}, \vec{b}, \vec{c}$ the factor $[\vec{a} \vec{b} \vec{c}]$ is ± 1 (it is the signed volume of the unit cube they span), so the crossed triad has box product exactly 1: it is a *positively oriented* basis of the same volume. Dividing each vector by $[\vec{a} \vec{b} \vec{c}]$ recovers the reciprocal system of Subtopic 03 — this computation is the proof that the reciprocal system exists and is unique.

Sanity check. Right-handed orthonormal $\hat{i}, \hat{j}, \hat{k}$: crossed triad is $\hat{k}, \hat{i}, \hat{j}$, box product $[\hat{k} \hat{i} \hat{j}] = 1 = [\hat{i} \hat{j} \hat{k}]^2$. Left-handed choice $\vec{c} = -\hat{k}$: box product of originals is -1 , crossed triad is $\hat{k}, -\hat{i}, -\hat{j}$ with box product $[\hat{k} (-\hat{i}) (-\hat{j})] = [\hat{k} \hat{i} \hat{j}] = 1 = (-1)^2$. The square erases the orientation of the original triad and always returns $+1$ for unit inputs.

Example C.3 — [JEE Advanced Level]

Show that $(\vec{a} - \vec{b}) \times (\vec{a} + \vec{b}) = 2\vec{a} \times \vec{b}$, and deduce that the area of the parallelogram spanned by the diagonals $\vec{a} + \vec{b}$, $\vec{a} - \vec{b}$ is twice the area of the parallelogram spanned by $\vec{a}$, $\vec{b}$.

Expand by distributivity, killing the $\vec{a} \times \vec{a}$ and $\vec{b} \times \vec{b}$ terms:

$$(\vec{a} - \vec{b}) \times (\vec{a} + \vec{b}) = \underbrace{\vec{a} \times \vec{a}}_{\vec{0}} + \vec{a} \times \vec{b} - \vec{b} \times \vec{a} - \underbrace{\vec{b} \times \vec{b}}_{\vec{0}} = \vec{a} \times \vec{b} + \vec{a} \times \vec{b} = 2\vec{a} \times \vec{b}.$$

Taking magnitudes: $|(\vec{a} - \vec{b}) \times (\vec{a} + \vec{b})| = 2|\vec{a} \times \vec{b}|$. The left side is the parallelogram area spanned by the diagonals; the right side is twice the original area. This reconciles with Subtopic 02, Example 4, where the diagonal-computation for a specific parallelogram returned exactly the side-vector area — there the factor $\frac{1}{2}$ from the diagonal formula cancelled the factor 2 found here.

Practice — Addendum C

1. Evaluate $(\hat{i} \times \hat{j}) \times (\hat{j} \times \hat{k})$ two ways: directly, and via the C1 expansion.
2. Solve $\vec{a} \times \vec{x} = \vec{b}$ for $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$, $\vec{b} = \hat{i} + 2\hat{j}$.
3. Prove that a vector perpendicular to both $\vec{a} \times \vec{b}$ and $\vec{a} \times \vec{c}$ must be parallel to $\vec{a}$, provided $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar.
4. Using C1, show that $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{a}) = -[\vec{a} \vec{b} \vec{c}] \vec{a}$, and verify it for $\vec{a} = \hat{i}$, $\vec{b} = \hat{j}$, $\vec{c} = \hat{k}$.
5. [Advanced] $\vec{a}, \vec{b}, \vec{c}$ are mutually perpendicular unit vectors. Evaluate $[\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}]$ and justify why the answer cannot be negative.

Answers: 1. Directly: $\hat{k} \times \hat{i} = \hat{j}$. By C1: $[\hat{i} \hat{j} \hat{k}] \hat{j} - [\hat{i} \hat{j} \hat{j}] \hat{k} = \hat{j}$. 2. Compatibility: $\vec{a} \cdot \vec{b} = 2 - 2 + 0 = 0$. Particular solution $\frac{\vec{b} \times \vec{a}}{|\vec{a}|^2} = \frac{1}{9}(4\hat{i} - 2\hat{j} - 5\hat{k})$ (expand $(1, 2, 0) \times (2, -1, 2) = (4, -2, -5)$); general solution $\vec{x} = \frac{1}{9}(4\hat{i} - 2\hat{j} - 5\hat{k}) + \lambda(2\hat{i} - \hat{j} + 2\hat{k})$. Check: $(2, -1, 2) \times (4, -2, -5) = (9, 18, 0)$, divided by 9 gives $(1, 2, 0) = \vec{b}$. 3. Both $\vec{a} \times \vec{b}$ and $\vec{a} \times \vec{c}$ are perpendicular to $\vec{a}$, so $\vec{a}$ itself is perpendicular to both. The directions perpendicular to two non-parallel vectors $\vec{u}, \vec{v}$ form a single line along $\vec{u} \times \vec{v}$; here $\vec{u} \times \vec{v} = (\vec{a} \times \vec{b}) \times (\vec{a} \times \vec{c})$, which by C1 (with $\vec{d} = \vec{c}$ written appropriately) is a non-zero multiple of $\vec{a}$ precisely when $[\vec{a} \vec{b} \vec{c}] \neq 0$. 4. By C1 with $\vec{d} = \vec{a}$: the first term carries $[\vec{a} \vec{b} \vec{a}] = 0$ and the second gives $-[\vec{a} \vec{b} \vec{c}] \vec{a}$. Verification: left side $\hat{k} \times (\hat{k} \times \hat{i}) = \hat{k} \times \hat{j} = -\hat{i}$; right side $-(1)\hat{i} = -\hat{i}$. 5. By C3 it equals $[\vec{a} \vec{b} \vec{c}]^2 = (\pm 1)^2 = 1$; a square is never negative, regardless of the handedness of the original triad.

Addendum D — Position-Vector Geometry of the Triangle

The chapter proves the centroid concurrence and stops. Boards and JEE both reach further: the incenter formula, the circumcenter-orthocenter identity, and area ratios for points built

from section formulas. All of it is the section formula and the cross-product area formula, chained.

Method

D1 — Notation. Vertices A, B, C have position vectors $\vec{a}, \vec{b}, \vec{c}$; side lengths opposite them are $a = |\vec{BC}|$, $b = |\vec{CA}|$, $c = |\vec{AB}|$; area of the triangle is Δ .

D2 — Incenter. Claim:

$$\vec{i} = \frac{a\vec{a} + b\vec{b} + c\vec{c}}{a + b + c}.$$

Derivation, in three moves.

Move 1 — where the bisector from A meets BC. By the angle bisector theorem, the point T on BC satisfies $BT : TC = AB : AC = c : b$. The section formula (internal, ratio $c : b$ from B) gives

$$\vec{t} = \frac{c\vec{c} + b\vec{b}}{b + c}.$$

Move 2 — how far down AT the incenter sits. Let r be the inradius. Each of $[IBC]$, $[ICA]$, $[IAB]$ has height r over bases a, b, c , so

$$[IBC] = \frac{1}{2}ar, \quad [ICA] = \frac{1}{2}br, \quad [IAB] = \frac{1}{2}cr, \quad \Delta = \frac{1}{2}r(a + b + c) = rs.$$

Now compare areas along the cevian AT . Triangles ABI and IBT share the altitude from B to line AT , so $\frac{[ABI]}{[IBT]} = \frac{AI}{IT}$; likewise $\frac{[ACI]}{[ICT]} = \frac{AI}{IT}$. Adding numerators and denominators (equal ratios combine):

$$\frac{AI}{IT} = \frac{[ABI] + [ACI]}{[IBT] + [ICT]} = \frac{\Delta - [IBC]}{[IBC]} = \frac{rs - \frac{1}{2}ar}{\frac{1}{2}ar} = \frac{2s - a}{a} = \frac{b + c}{a}.$$

Move 3 — one more section formula. I divides AT internally in ratio $(b + c) : a$:

$$\vec{i} = \frac{(b + c)\vec{t} + a\vec{a}}{(b + c) + a} = \frac{(b + c) \cdot \frac{b\vec{b} + c\vec{c}}{b + c} + a\vec{a}}{a + b + c} = \frac{a\vec{a} + b\vec{b} + c\vec{c}}{a + b + c}. \quad \blacksquare$$

The excentres follow the same template with one sign flipped each; recorded for reference:

$$\vec{i}_A = \frac{-a\vec{a} + b\vec{b} + c\vec{c}}{-a + b + c}, \quad \vec{i}_B = \frac{a\vec{a} - b\vec{b} + c\vec{c}}{a - b + c}, \quad \vec{i}_C = \frac{a\vec{a} + b\vec{b} - c\vec{c}}{a + b - c},$$

each obtained by running Move 2 with the corresponding exradius and external division on the appropriate bisector.

D3 — Circumcenter at the origin: the orthocenter identity. Claim: if the circumcenter O is the origin and $|\vec{a}| = |\vec{b}| = |\vec{c}| = R$, the orthocenter is

$$\vec{h} = \vec{a} + \vec{b} + \vec{c}.$$

Derivation. It suffices to show the point $\vec{h}$ lies on all three altitudes. Test the altitude from A : we need $\overrightarrow{AH} \perp \overrightarrow{BC}$, i.e.

$$(\vec{h} - \vec{a}) \cdot (\vec{b} - \vec{c}) = (\vec{b} + \vec{c}) \cdot (\vec{b} - \vec{c}) = |\vec{b}|^2 - |\vec{c}|^2 = R^2 - R^2 = 0. \checkmark$$

The other two altitudes are identical computations with letters cycled. Hence H with $\vec{h} = \vec{a} + \vec{b} + \vec{c}$ is the orthocenter. Corollary (Euler line): the centroid has $\vec{g} = \frac{\vec{a} + \vec{b} + \vec{c}}{3} = \frac{\vec{h}}{3}$, so O, G, H are collinear with $\overrightarrow{OH} = 3\overrightarrow{OG}$, i.e. $OG : GH = 1 : 2$.

D4 — Area ratios from affine combinations. Claim: if

$$\vec{p} = \frac{l\vec{a} + m\vec{b} + n\vec{c}}{l + m + n}, \quad l + m + n \neq 0,$$

then

$$\frac{[\triangle PBC]}{[\triangle ABC]} = \frac{l}{l + m + n}, \quad \frac{[\triangle PCA]}{[\triangle ABC]} = \frac{m}{l + m + n}, \quad \frac{[\triangle PAB]}{[\triangle ABC]} = \frac{n}{l + m + n}.$$

Derivation of the first ratio (others cycle). Write $L = l + m + n$ and form the two edge vectors of $\triangle PBC$ from P :

$$\vec{b} - \vec{p} = \frac{l(\vec{a} - \vec{b}) + n(\vec{c} - \vec{b})}{L}, \quad \vec{c} - \vec{p} = \frac{l(\vec{a} - \vec{c}) + m(\vec{b} - \vec{c})}{L}.$$

Cross them. Distributing over the four term-pairs, three simplifications do all the work — each of the following equals $S := (\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})$, as direct expansion confirms:

$$(\vec{a} - \vec{b}) \times (\vec{a} - \vec{c}) = S, \quad (\vec{a} - \vec{b}) \times (\vec{b} - \vec{c}) = S, \quad (\vec{c} - \vec{b}) \times (\vec{a} - \vec{c}) = S,$$

while $(\vec{c} - \vec{b}) \times (\vec{b} - \vec{c}) = \vec{0}$. Collecting coefficients:

$$(\vec{b} - \vec{p}) \times (\vec{c} - \vec{p}) = \frac{l^2 + lm + ln}{L^2} S = \frac{l}{L} S.$$

But $S = (\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})$ is twice the oriented area of $\triangle ABC$, and the left side is twice the oriented area of $\triangle PBC$. Taking absolute values gives the claim. Note the power of the statement: *any* point assembled by the section formula has its three sub-areas read off from its coefficients, with no geometry at all.

Example D.1 — [CBSE Board Level]

Find the incenter of the triangle with vertices $A(0, 0)$, $B(4, 0)$, $C(0, 3)$.

Side lengths: $a = |BC| = \sqrt{16 + 9} = 5$, $b = |CA| = 3$, $c = |AB| = 4$. Apply D2:

$$\vec{i} = \frac{5(0, 0) + 3(4, 0) + 4(0, 3)}{5 + 3 + 4} = \frac{(12, 12)}{12} = (1, 1).$$

Check. The triangle is right-angled at A with inradius $r = \frac{ab_{\text{leg}}c_{\text{leg}}\text{-formula}}{s}$ — directly, $r = \frac{\Delta}{s} = \frac{6}{6} = 1$, and the incenter of a right triangle at the origin with legs on the axes sits at $(r, r) = (1, 1)$. Agreement.

Example D.2 — [JEE Main Level]

The vertices $A(5, 0)$, $B(0, 5)$, $C(-3, -4)$ **all lie on the circle** $x^2 + y^2 = 25$. **Taking the circumcenter as origin, find the orthocenter and verify it lies on two altitudes.**

By D3, $\vec{h} = \vec{a} + \vec{b} + \vec{c} = (5 + 0 - 3, 0 + 5 - 4) = (2, 1)$.

Altitude from A: $\overrightarrow{BC} = (-3, -9)$ and $\overrightarrow{AH} = (-3, 1)$; dot product $= 9 - 9 = 0$. ✓

Altitude from B: $\overrightarrow{AC} = (-8, -4)$ and $\overrightarrow{BH} = (2, -4)$; dot product $= -16 + 16 = 0$. ✓

The centroid is $\vec{g} = \frac{(2, 1)}{3}$, confirming the Euler-line ratio $OG : GH = 1 : 2$ for this triangle:
 $|OG| = \frac{\sqrt{5}}{3}$ and $|GH| = \left| (2, 1) - \frac{(2, 1)}{3} \right| = \frac{2\sqrt{5}}{3}$.

Example D.3 — [JEE Advanced Level]

P lies on the median AD of $\triangle ABC$ with $AP : PD = 3 : 1$. Find $\frac{[\triangle PBC]}{[\triangle ABC]}$ without drawing anything.

Chain two section formulas. Midpoint of BC : $\vec{d} = \frac{\vec{b} + \vec{c}}{2}$. Then P divides AD internally in ratio $3 : 1$:

$$\vec{p} = \frac{3\vec{d} + 1 \cdot \vec{a}}{4} = \frac{\vec{a} + \frac{3}{2}(\vec{b} + \vec{c})}{4} = \frac{2\vec{a} + 3\vec{b} + 3\vec{c}}{8}.$$

The coefficients of $\vec{a}, \vec{b}, \vec{c}$ are 2, 3, 3 over denominator 8, so by D4:

$$\frac{[\triangle PBC]}{[\triangle ABC]} = \frac{2}{8} = \frac{1}{4}.$$

Consistency checks. The other two ratios come out $\frac{3}{8}$ each, and $\frac{1}{4} + \frac{3}{8} + \frac{3}{8} = 1$ — the three sub-triangles tile the whole, so their fractions must sum to 1; they do. Also, moving P from

A (ratio 0) toward D shrinks $[\triangle PBC]$ from 1 toward the median-value $\frac{1}{2}$; a point one quarter of the way down from D -side... more precisely $AP : PD = 3 : 1$ puts P at fraction $\frac{3}{4}$ from A , and linear interpolation predicts $1 - \frac{3}{4} \left(1 - \frac{1}{2}\right) = \frac{1}{4}$. Agreement from a third direction.

Practice — Addendum D

- Find the incenter of the triangle with vertices $A(0, 0)$, $B(6, 0)$, $C(0, 8)$.
- $A(5, 12)$, $B(-13, 0)$, $C(0, -13)$ lie on $x^2 + y^2 = 169$. Using D3, write down the orthocenter and verify it lies on the altitude from A .
- Prove that the centroid divides the segment from the orthocenter to the circumcenter in the ratio $1 : 2$ (from the circumcenter end), assuming the circumcenter is at the origin.
- P divides the median from A internally in the ratio $m : 1$ (from A). Find $\frac{[\triangle PBC]}{[\triangle ABC]}$ as a function of m , and check it gives $\frac{1}{3}$ at $m = 2$.
- [Advanced] Given $\vec{p} = \frac{\vec{a} + 2\vec{b} + 3\vec{c}}{6}$, find $[\triangle PBC] : [\triangle PCA] : [\triangle PAB]$, and state what must be true of $\vec{a}, \vec{b}, \vec{c}$ for P to lie strictly inside the triangle.

Answers: 1. Sides $a = 10$, $b = 8$, $c = 6$; $\vec{i} = \frac{8(6, 0) + 6(0, 8)}{24} = (2, 2)$; check $r = \frac{\Delta}{s} = \frac{24}{12} = 2$ and a right triangle at the origin has incenter (r, r) . 2. $\vec{h} = (5 - 13 + 0, 12 + 0 - 13) = (-8, -1)$; altitude check: $\overrightarrow{BC} = (13, -13)$, $\overrightarrow{AH} = (-13, -13)$, dot $= -169 + 169 = 0$. 3. With $|\vec{a}| = |\vec{b}| = |\vec{c}| = R$: $\vec{h} = \vec{a} + \vec{b} + \vec{c}$ by D3 and $\vec{g} = \frac{\vec{h}}{3}$, so $\overrightarrow{OH} = 3\overrightarrow{OG}$, i.e. G trisects OH with $OG : GH = 1 : 2$. 4. $\vec{p} = \frac{m\vec{d} + \vec{a}}{m + 1} = \frac{\vec{a} + \frac{m}{2}(\vec{b} + \vec{c})}{m + 1} = \frac{2\vec{a} + m\vec{b} + m\vec{c}}{2(m + 1)}$, so the ratio is $\frac{2}{2(m + 1)} = \frac{1}{m + 1}$; at $m = 2$ this is $\frac{1}{3}$, matching the centroid. 5. By D4 the ratio is $1 : 2 : 3$; P lies strictly inside exactly when $\vec{a}, \vec{b}, \vec{c}$ are non-collinear (genuine triangle) and all three coefficients 1, 2, 3 are positive — which they are, so interiority is automatic once the vertices form a proper triangle.

Addendum P — What JEE Actually Asks

P.1 A note on the evidence

The compendium in front of me preserves every question stem, its year/session tag and the book's solution *structure*, but nearly all vector expressions were set as images and did not survive extraction. What survives reliably: **which concept each question tests** (readable from the method lines and final answers of the solutions), **the year/session**, and **the numerical answers** of a good fraction of items. Everything below is classified from that evidence. Where I print a fully worked example, either the stem survived well enough to reconstruct and I have re-solved it from scratch, or I say plainly that it is my own construction in the

same shape. Corrupted stems whose mathematics cannot be pinned down have been left out entirely.

P.2 The distribution

The bank holds roughly **135 items spanning 1978–2020**, in three blocks that map onto the chapter’s subtopics:

Block	Approx. share	Dominant content
Position-vector geometry	~20% (≈ 21 items)	Collinearity, section formula, ratios in triangles, quadrilateral identification
Products of two vectors	~35% (≈ 30 items)	Projections, perpendicular/parallel conditions, areas, angles
Triple products & applications	~45% (≈ 84 items)	Coplanarity, parallelepiped/tetrahedron volumes, BAC–CAB, basis/reciprocal systems

Concrete return visits the tags preserve:

- **Coplanarity with a parameter:** 1985, 1986, 1987 ($abc = -1$), 1993 (answer: c is the geometric mean), 2007 (**two** distinct values of λ), Main Jan 9 2020 (I) ($\lambda = 1$). This is the single most repeated shape in the bank.
- **Volume of a parallelepiped/tetrahedron:** 1983, 2002S, 2003S (minimum-volume a), 2008, Adv 2013 (match-list on scaled volumes), Main Jan 8 2020 (I), Main Sep 5 2020 (I) (n from volume 158).
- **Projections:** 1987, 1988, 1993, 2006, 2008, 2011, Adv 2015 and Adv 2019, Main Jan 9 2020 (I) (numeric answer 8), Main April 8 2019 (I), Main Sep 5/6 2020. No other idea appears in more sessions.
- **Vector perpendicular to two given vectors, often with a prescribed magnitude:** 1987, 1992, 2004S, 2011, Main April 12 2019 (I) (magnitude 12, answer of the form $\pm 4(\dots)$).
- **Angles from dot/cross data:** 2002S, 2012, 2016, Main Jan 12 2019 (II) ($|\alpha - \beta| = 30^\circ$), Main Sep 3 2020 (II).
- **Position-vector ratios and cevians:** a run of 4–8 mark Advanced proofs in 1988, 1989, 1993 ($BP : PE = 8 : 3$), 1996, 1997, 1998, 2001; the modern descendant is the short section-formula numeric (Main Sep 2 2020 (II), answer 0.8).
- **Counting / independence arguments:** Adv 2013 (non-coplanar choices from the eight $(\pm 1, \pm 1, \pm 1)$ vectors; answer $32 = 2^5$, so $p = 5$).

Frequency verdict. In recent Main sessions the bank shows at least one question from this chapter in essentially every slot listed for 2019–2020 — typically **one to two questions**

per paper, almost always a fast computation (projection, coplanarity determinant, volume, angle). Advanced spreads the ideas inside larger multi-constraint problems. Claims finer than this (exact per-topic percentages) would outrun the evidence, because too many stems lost their numbers.

P.3 The recurring archetypes

A1. The parameter hunt via a vanishing condition

Shape: “ λ such that these vectors are perpendicular / parallel / coplanar.” *Method*: Perpendicular $\Rightarrow \vec{a} \cdot \vec{b} = 0$; parallel $\Rightarrow$ components proportional (or $\vec{a} \times \vec{b} = \vec{0}$); coplanar $\Rightarrow$ determinant = 0. Solve the resulting equation. Watch for “no such λ ” and “exactly two values.”

Worked example (my construction, this exact shape): find λ such that $\vec{a} = \lambda\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{b} = 3\hat{i} - \hat{j} + \lambda\hat{k}$ are (i) perpendicular, (ii) parallel.

$$(i) \vec{a} \cdot \vec{b} = 3\lambda - 2 - \lambda = 2\lambda - 2 = 0 \Rightarrow \boxed{\lambda = 1}.$$

(ii) Parallel needs proportional components: $\frac{\lambda}{3} = \frac{2}{-1} = \frac{-1}{\lambda}$. The middle ratio is -2 , so we would need $\lambda = -6$ from the first ratio and $\lambda^2 = 3$ from the last — inconsistent, hence **no λ exists**.

A2. Projection, forward and inverted

Shape: “projection of $\vec{a}$ on $\vec{b}$ ”, or a projection *equality*, or projection of a segment on a line.

Method: $\text{proj}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$ — divide by the magnitude of the vector you land on. Equality of projections converts to a linear equation in unknown components.

Worked example — reconstructed from Main Jan 9 2020 (I), fully intact enough to trust: “Projection of the segment joining $(1, -1, 3)$ and $(2, -4, 11)$ on the line joining $(-1, 2, 3)$ and $(3, -2, 10)$.”

$$\vec{PQ} = \hat{i} - 3\hat{j} + 8\hat{k}, \quad \vec{RS} = 4\hat{i} - 4\hat{j} + 7\hat{k}$$

$$\vec{PQ} \cdot \vec{RS} = 4 + 12 + 56 = 72, \quad |\vec{RS}| = \sqrt{16 + 16 + 49} = 9$$

$$\text{Projection} = \frac{72}{9} = 8$$

Answer: 8. (Matches the printed key.)

A3. Perpendicular vector with a prescribed magnitude

Shape: “find a vector of magnitude k perpendicular to both $\vec{a}$ and $\vec{b}$.” *Method*: take $\lambda(\vec{a} \times \vec{b})$, impose $|\lambda(\vec{a} \times \vec{b})| = k$, read off λ ; expect a $\pm$ pair.

Worked example (my construction with clean numbers): $\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$, magnitude 12.

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 1 & 2 & 2 \end{vmatrix} = 6\hat{i} - 6\hat{j} + 3\hat{k}, \quad |\vec{a} \times \vec{b}| = 9$$

$$\hat{n} = \frac{2\hat{i} - 2\hat{j} + \hat{k}}{3}, \quad 12\hat{n} = \pm(8\hat{i} - 8\hat{j} + 4\hat{k})$$

Answer: $\pm(8\hat{i} - 8\hat{j} + 4\hat{k})$. (This is the skeleton of Main April 12 2019 (I): there $12 = k|\vec{a} \times \vec{b}|$ forced $|\lambda| = \frac{3}{2}$.)

A4. Coplanarity, volumes and the determinant machine

Shape: coplanarity of three vectors or four points; volume of a box or tetrahedron; occasionally “for how many λ ”. *Method:* one 3×3 determinant decides everything: zero $\Rightarrow$ flat (coplanar/dependent); nonzero $\Rightarrow$ volume $|\vec{a} \vec{b} \vec{c}|$, tetrahedron $\frac{1}{6}$ of it. For “how many values,” reduce the determinant to a polynomial and **count real roots honestly** (Advanced 2007’s official answer is exactly two).

Worked example — reconstructed from Adv 2013: *three non-coplanar vectors can be chosen from $\{(1, 1, 1), (-1, -1, -1), (-1, 1, 1), (1, -1, -1), (1, -1, 1), (-1, 1, -1), (1, 1, -1), (-1, -1, 1)\}$ in 2^p ways; find p .*

These eight vectors are the four body diagonals of a cube and their negatives. Any three vectors drawn from **three distinct** diagonal lines are automatically non-coplanar (each such determinant has magnitude 4 — e.g. rows $(1, 1, 1), (-1, 1, 1), (1, -1, 1)$ give 4), while two vectors from the same line are dependent. So: choose 3 of the 4 diagonal lines, then one of 2 signs on each:

$$\binom{4}{3} \times 2 \times 2 \times 2 = 32 = 2^5 \Rightarrow \boxed{p = 5}$$

(Matches the printed key.)

A5. Ratios inside triangles by position vectors

Shape: cevians AD, BE with stated division ratios; find where they meet and in what ratio. Historically 4–8 mark Advanced proofs (1988–2001); today compressed into section-formula numerics. *Method:* write the same intersection point two ways with undetermined scalars and compare coefficients of $\vec{a}, \vec{b}, \vec{c}$. (Mass points gives the ratio in seconds — use it as a check.)

Worked example — reconstructed from the 1993 paper, whose data and answer survive: *In $\triangle ABC$, $D \in BC$ with $BD : DC = 2 : 1$, $E \in AC$ with $AE : EC = 3 : 1$; $P = AD \cap BE$; find $BP : PE$.*

Mass points: $BD : DC = 2 : 1 \Rightarrow m(C) : m(B) = 2 : 1$, so take $m(B) = 1$, $m(C) = 2$. Then $AE : EC = 3 : 1 \Rightarrow m(C) : m(A) = 3 : 1 \Rightarrow m(A) = \frac{2}{3}$, so $m(E) = m(A) + m(C) = \frac{8}{3}$. Hence

$$BP : PE = m(E) : m(B) = \frac{8}{3} : 1 = \boxed{8 : 3}$$

Vector check: with $\vec{a}, \vec{b}, \vec{c}$ the vertex position vectors, $D = \frac{\vec{b}+2\vec{c}}{3}$, $E = \frac{3\vec{c}+\vec{a}}{4}$; writing $P = (1-t)\vec{b} + tE = (1-u)\vec{a} + uD$ and matching the $\vec{a}$ -coefficient gives $t/4 = 1-u$, matching $\vec{c}$ gives $3t/4 = u/3$... solving yields $t = \frac{8}{11}$, i.e. $BP : PE = t : (1-t) = 8 : 3$. Confirmed against the printed key.

(A sixth recurring shape — pure angle/magnitude algebra from $|\vec{a} \pm \vec{b}|$ expansions and Lagrange's identity $(\vec{a} \cdot \vec{b})^2 + |\vec{a} \times \vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2$ — runs from 1981 through Main Sep 2020 and is drilled in the practice set below.)

P.4 The traps

- The zero-condition swap.** $\vec{a} \cdot \vec{b} = 0 \Leftrightarrow$ perpendicular; $\vec{a} \times \vec{b} = \vec{0} \Leftrightarrow$ parallel. The bank punishes this repeatedly (see 2005S, 1982). It is the single most planted distractor.
- The missing $\frac{1}{6}$.** Tetrahedron volume is $\frac{1}{6}||\vec{a} \vec{b} \vec{c}||$; the natural wrong answers are the full parallelepiped value ($\times 1$) or the triangle's $\frac{1}{2}$. Every volume MCQ in the bank carries both decoys.
- Wrong projection denominator.** Divide by the magnitude of the vector projected **onto**. Decoys divide by $|\vec{a}|$ or forget the division entirely (visible in the 2006/2011 option sets).
- Head-minus-tail reversal.** $\vec{QP} = -(\vec{PQ})$; a reversed subtraction flips every sign while leaving the magnitude — half the options survive elimination purely on sign pattern.
- DR versus DC.** Direction cosines must satisfy $l^2 + m^2 + n^2 = 1$. Square-and-add kills two options of any DC question instantly; distractors offer raw ratios or components divided by $|\vec{r}|^2$.
- Negative determinants reported as volume.** $[\vec{a} \vec{b} \vec{c}]$ may be negative; the volume is its absolute value. An exam asking “the volume is” wants a positive number.
- BAC-CAB bracket blindness.** $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$, while $(\vec{a} \times \vec{b}) \times \vec{c}$ expands differently. The result always lives in the plane of the two vectors **inside** the bracket — an option containing $\vec{a}$ (outermost) is instantly dead.
- “Obtuse for all x ” needs two conditions.** The 1991 problem (answer an open interval, $c \in (-\frac{4}{3}, 0)$) demands *both* a negative leading coefficient *and* a negative discriminant — and strictly excludes the endpoints. Students who imposed only one condition landed on decoy intervals.
- Extraneous roots when counting λ .** After reducing a coplanarity determinant, count real solutions of the *original* condition — 2007's answer (“two”) punishes those who miscount after substitution.
- Coplanar $\neq$ collinear.** Zero triple product means dependence, which is weaker than collinearity; MCQs exploit the converse-direction confusion.

P.5 Graded practice set

All twelve below are my constructions in the shapes documented above; every answer has been independently re-computed.

P1. Find the direction cosines of the vector from $A(1, 2, -2)$ to $B(3, 1, 0)$.

$$\vec{AB} = 2\hat{i} - \hat{j} + 2\hat{k}, |\vec{AB}| = 3. \text{ DCs: } \left(\frac{2}{3}, -\frac{1}{3}, \frac{2}{3}\right). \text{ Check: } \frac{4}{9} + \frac{1}{9} + \frac{4}{9} = 1.$$

P2. Find k so that $A(2, 3, 4)$, $B(k, 6, 10)$, $C(4, 9, 16)$ are collinear.

$$\vec{AC} = (2, 6, 12) = 2(1, 3, 6); \text{ collinearity forces } \vec{AB} = (k-2, 3, 6) \parallel (1, 3, 6), \text{ so } 3 = 3 \cdot 1 \text{ fixes scale 1, giving } k-2 = 1, \text{ i.e. } k = 3.$$

P3. Find the projection of $\vec{a} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ on $\vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$.

$$\vec{a} \cdot \vec{b} = 2 + 6 + 4 = 12, |\vec{b}| = 3: \text{ projection} = 4.$$

P4. Find all unit vectors perpendicular to both $\hat{i} + \hat{j} + \hat{k}$ and $\hat{i} - \hat{j} + \hat{k}$.

$$\text{Cross product} = (1 \cdot 1 - 1 \cdot (-1), 1 \cdot 1 - 1 \cdot 1, -1 - 1) = (2, 0, -2), \text{ magnitude } 2\sqrt{2}.$$
$$\text{Answer: } \pm \frac{\hat{i} - \hat{k}}{\sqrt{2}}.$$

P5. Find the area of the triangle with vertices $O(0, 0, 0)$, $A(1, 2, 3)$, $B(-2, 1, 1)$, and a unit vector normal to its plane.

$$\vec{OA} \times \vec{OB} = (-1)\hat{i} - 7\hat{j} + 5\hat{k}, |\cdot| = \sqrt{75} = 5\sqrt{3}. \text{ Area} = \frac{5\sqrt{3}}{2}; \text{ normal unit vector}$$
$$\pm \frac{-\hat{i} - 7\hat{j} + 5\hat{k}}{5\sqrt{3}}.$$

P6. If $|\vec{a}| = 3$, $|\vec{b}| = 4$, $\vec{a} \cdot \vec{b} = 6$, find $|\vec{a} \times \vec{b}|$ and the angle between $\vec{a}$, $\vec{b}$.

$$\text{Lagrange: } |\vec{a} \times \vec{b}|^2 = 9 \cdot 16 - 36 = 108 \Rightarrow |\vec{a} \times \vec{b}| = 6\sqrt{3}; \cos \theta = \frac{6}{12} = \frac{1}{2} \Rightarrow \theta = 60^\circ.$$

P7. $\vec{a}$, $\vec{b}$ are unit vectors with $|\vec{a} + \vec{b}| = \sqrt{3}$. Find $|\vec{a} - \vec{b}|$.

$$3 = 1 + 1 + 2\vec{a} \cdot \vec{b} \Rightarrow \vec{a} \cdot \vec{b} = \frac{1}{2}; \text{ then } |\vec{a} - \vec{b}|^2 = 2 - 2 \cdot \frac{1}{2} = 1, \text{ so } |\vec{a} - \vec{b}| = 1.$$

P8. In $\triangle ABC$, $D \in BC$ with $BD : DC = 1 : 2$ and E is the midpoint of CA . AD and BE meet at P . Find $BP : PE$.

$$\text{Mass points: } m(C) : m(B) = 1 : 2 \Rightarrow m(B) = 2, m(C) = 1; \text{ midpoint } E \text{ forces } m(A) = m(C) = 1, \text{ so } m(E) = 2. \text{ Then } BP : PE = m(E) : m(B) = 1 : 1. \text{ Vector confirmation: with } D = \frac{2b+c}{3}, E = \frac{a+c}{2}, \text{ equating } P \text{ on } AD \text{ and } BE \text{ gives parameter } t = \frac{1}{2} \text{ on } BE, \text{ again ratio } 1 : 1.$$

P9. Find all λ for which $(1, 2, 3)$, $(2, \lambda, 4)$, $(3, 6, 13)$ are coplanar.

Determinant $= 1(13\lambda - 24) - 2(26 - 12) + 3(12 - 3\lambda) = 4\lambda - 16$. Setting it to zero:
 $\lambda = 4$ (unique).

P10. Find the volume of the tetrahedron with vertices $A(1, 0, 0)$, $B(0, 2, 0)$, $C(0, 0, 3)$, $D(1, 1, 1)$.

Edges from D (equivalently from A): taking A : $\vec{AB} = (-1, 2, 0)$, $\vec{AC} = (-1, 0, 3)$,
 $\vec{AD} = (0, 1, 1)$; determinant $= (-1)(0 - 3) - 2(-1 - 0) + 0 = 5$. Volume $= \frac{5}{6}$ cubic
units.

P11. Three vectors each of unit length make pairwise angles 60° . Find the volume of the parallelepiped they span.

Gram determinant: $1 + 2(\frac{1}{2})^3 - 3(\frac{1}{2})^2 = 1 + \frac{1}{4} - \frac{3}{4} = \frac{1}{2}$. Volume $= \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$ cubic
units. (Same skeleton as Adv 2008.)

P12. Find all c such that for every real x , the vectors $-x\hat{i} + 2\hat{j} + c\hat{k}$ and $x\hat{i} + cx\hat{j} + \hat{k}$ make an obtuse angle.

Dot product $f(x) = -x^2 + 2cx + c$. Obtuse for **all** x requires $f(x) < 0$ everywhere:
downward parabola (already satisfied), maximum at $x = c$ equals $c^2 + c$, so we
need $c^2 + c < 0$, i.e. $c \in (-1, 0)$.

P.6 What this chapter's text under-serves

The chapter text teaches every formula correctly but trains almost exclusively on single-step plug-ins, whereas the bank shows JEE's real currency: **multi-condition systems** (several simultaneous dot-product equations solved together), **optimization under constraints** (extreme values of projections and triple products), **finite counting/independence arguments**, and **long position-vector proofs** about cevians and concurrence — none of which receive more than a token exercise. Because the rationalised syllabus dropped the scalar triple product, the text's treatment of BAC-CAB, coplanarity and reciprocal systems arrives thin and unproven, precisely the machinery the 84-question third block of the bank leans on hardest.

Three Dimensional Geometry

Class 12 Mathematics — Chapter 11

Drona Ed Tech

June 2026

Contents

Foundations of 3D Coordinate Geometry	1
Direction Cosines, Direction Ratios & Equations of a Line in Space	10
Shortest Distance & Angle Between Two Lines	19
Planes in Space — Equations, Family, Angles & Distances	28
The Perpendicular Toolkit — Distances, Feet & Images	36
Line-Plane & Plane-Plane Configurations	44

Foundations of 3D Coordinate Geometry

Class 12 · Mathematics · Chapter 11 — Three Dimensional Geometry · Subtopic 00 · Prerequisite recap

Exam Relevance: These are the load-bearing basics the rest of the chapter stands on. CBSE Boards may ask a 1-2 mark distance or section-formula question directly. JEE Main slips them inside larger problems every year (ratios of division, centroids, collinearity). JEE Advanced uses the tetrahedron-volume idea in mixed vector-geometry questions. Master these and every later subtopic becomes mechanical. None of the formulae here are hard to memorise; what trips students up is sign-handling, weight placement in the section formula, and confusing “distance from an axis” with “distance from a plane.” This subtopic therefore spends as much effort on *why* each formula looks the way it does as on the formula itself, because understanding kills those errors at the source.

Everything that follows in Chapter 11 — direction cosines, lines, planes, perpendicular distances, reflections — is ultimately a statement about points and the distances and directions between them. So although this material formally belongs to the Class 11 introduction to three-dimensional geometry, treating it as the genuine foundation of the Class 12 chapter is the right mindset: a student who is fluent here will find that later subtopics mostly add notation, not new ideas.

Section 1 — Concept Intuition

Everything in this chapter lives in three-dimensional space, so we first need a reliable way to pin down *where* a point is. Picture the corner of a room again: three mutually perpendicular edges meet there. Call them the **x-axis**, **y-axis**, and **z-axis**, and the corner itself the **origin** O . Any point P in the room is reached by walking x steps along the first edge, y along the second, and z up the third — giving the ordered triple (x, y, z) . Those three numbers are the point’s **coordinates**.

The three axes, taken in pairs, define three flat walls: the **xy-plane** (where $z = 0$), the **yz-plane** (where $x = 0$), and the **zx-plane** (where $y = 0$). These three planes slice all of space into **eight compartments called octants**, the 3D cousins of the four quadrants you

know from 2D. The first octant is the friendly one where all three coordinates are positive; the others carry various sign combinations. Throughout, we use the standard **right-handed** system: curl the fingers of your right hand from the positive x-axis toward the positive y-axis and your thumb points along the positive z-axis. Every formula in this chapter assumes this orientation.

It pays to see how a point relates to the axes and planes directly. The **projection** of $P(x, y, z)$ onto the xy-plane is $(x, y, 0)$ — you flatten the height to zero; its projections onto the other two coordinate planes drop the appropriate coordinate. Projecting onto an *axis* keeps only one coordinate: onto the x-axis, P lands at $(x, 0, 0)$. These projections make distances easy to picture. The distance of P from the **x-axis** is the length of the leg spanning the other two directions, $\sqrt{y^2 + z^2}$, while the distance of P from the **xy-plane** is simply $|z|$. Reading “distance from an axis” as a two-coordinate Pythagoras and “distance from a plane” as a single coordinate saves genuine time under exam pressure.

Think about it this way... distance in 3D is just the Pythagoras theorem used twice. To find how far $P(x, y, z)$ is from the origin, first find the distance across the floor to the point $(x, y, 0)$ — that’s the ordinary 2D hypotenuse $\sqrt{x^2 + y^2}$. Then climb height z and apply Pythagoras again: $\sqrt{(\sqrt{x^2 + y^2})^2 + z^2} = \sqrt{x^2 + y^2 + z^2}$. The familiar two-term formula simply grows a third term. This single idea — Pythagoras stacked — generates the entire distance formula between any two points.

Next comes **division of a segment**. Suppose a rope is stretched between two pegs A and B , and you want the point that splits it in a given ratio, say $2 : 3$ measured from A . In each coordinate separately, the splitting point is a weighted average of the endpoints — and crucially, the *same* weights apply to x , y , and z at once. That is the **section formula**, and the midpoint is just its simplest case with equal weights.

It is worth seeing *why* it is a weighted average rather than a plain one. If the point sits closer to B (a large m , the weight on the far endpoint), it should inherit more of B ’s coordinates — so the heavier weight rides the endpoint you are nearer to. A common slip is to attach the weights to the wrong ends; the safe reading is “the weight named after the *first* part of the ratio multiplies the *second* (far) endpoint.” The midpoint, with weights $1 : 1$, lands halfway and reduces to the ordinary average, which is the sanity check you should run whenever a section-formula answer looks suspicious.

The **centroid** extends this averaging idea to three or four points at once. Physically it is the balance point: place equal masses at the vertices of a triangle and they balance at $G = \frac{1}{3}(A + B + C)$; do the same for a tetrahedron and the balance point is $\frac{1}{4}$ of the vertex sum. Because the centroid is a *plain* average of the vertices, it is one of the few formulas in this chapter with no weights or signs to misplace — which is exactly why examiners like to bury an unknown vertex inside a centroid condition and ask you to recover it.

Finally, **collinearity**. Three points lie on one straight line exactly when there’s no “bend” at the middle one — equivalently, when the direction from A to B matches the direction from B to C . We can test this either through proportional direction ratios or, more elementarily, by checking that the three pairwise distances satisfy $AB + BC = AC$ (the middle point sits *between* the other two with no detour). A third route, handy when ratios are awkward, is the triangle area: collinear points enclose zero area.

Limiting conditions & assumptions:

- The internal section formula needs the ratio written as $m : n$ with both parts positive; **external** division flips one sign and is used when the dividing point lies *outside* the segment.
- The distance formula assumes a standard right-handed rectangular coordinate system — the usual setup for this entire chapter.
- The $AB + BC = AC$ collinearity test only works when the points are labelled in the order

they appear along the line; otherwise use proportional direction ratios, which never depend on ordering.

Section 2 — Key Definitions & Properties

All quantities here are lengths (units of length) or pure ratios; no special dimensional formulae apply. Every result in this section is built from the single picture of the previous one — a point located by three perpendicular measurements — so although the list looks long, it is really one idea (coordinate-wise Pythagoras and coordinate-wise averaging) applied in different ways.

Distance between two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$:

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

Distance from the origin: $OP = \sqrt{x_1^2 + y_1^2 + z_1^2}$.

Section formula (internal division) — point dividing AB in ratio $m : n$ from A :

$$\left(\frac{mx_2 + nx_1}{m + n}, \frac{my_2 + ny_1}{m + n}, \frac{mz_2 + nz_1}{m + n} \right).$$

Section formula (external division) in ratio $m : n$:

$$\left(\frac{mx_2 - nx_1}{m - n}, \frac{my_2 - ny_1}{m - n}, \frac{mz_2 - nz_1}{m - n} \right).$$

Midpoint of AB :

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right).$$

Centroid of a triangle with vertices A, B, C :

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3} \right).$$

Centroid of a tetrahedron with four vertices: the average of all four coordinates (divide by 4).

Volume of a tetrahedron with vertices A, B, C, D (using edge vectors from A):

$$V = \frac{1}{6} |(\vec{AB}) \cdot (\vec{AC} \times \vec{AD})|.$$

Area of a triangle with vertices A, B, C :

$$\text{Area} = \frac{1}{2} |\vec{AB} \times \vec{AC}|.$$

This doubles as a collinearity test: the three points are collinear exactly when this area is zero (the “triangle” has collapsed to a segment).

Collinearity of three points: either the direction ratios of $\vec{AB}$ and $\vec{AC}$ are proportional, or (with correct ordering) $AB + BC = AC$.

Distance of a point $P(x, y, z)$ from the axes and coordinate planes:

$$\text{from x-axis} = \sqrt{y^2 + z^2}, \quad \text{from y-axis} = \sqrt{z^2 + x^2}, \quad \text{from z-axis} = \sqrt{x^2 + y^2};$$

from yz-plane = $|x|$, from zx-plane = $|y|$, from xy-plane = $|z|$.

Projections of P : onto the xy-plane, $(x, y, 0)$; onto the x-axis, $(x, 0, 0)$; the other projections drop the remaining coordinate(s) the same way.

Notation key: (x_i, y_i, z_i) = coordinates of the i -th point; $m : n$ = division ratio; G = centroid; V = tetrahedron volume; $\overrightarrow{AB}$ = displacement vector with components $(x_2 - x_1, y_2 - y_1, z_2 - z_1)$.

Task	Formula
Distance	$\sqrt{\sum (x_2 - x_1)^2}$
Internal division $m : n$	$\frac{mx_2 + nx_1}{m+n}$ (per coordinate)
External division $m : n$	$\frac{mx_2 - nx_1}{m-n}$ (per coordinate)
Midpoint	average of endpoints
Centroid (triangle)	average of 3 vertices
Tetrahedron volume	$\frac{1}{6} \text{scalar triple product} $

Section 3 – Theorems & Procedures

Theorem 1 (distance formula). $PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$.

Figure. Two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ in 3D. A rectangular box (cuboid) drawn with PQ as its main diagonal and edges parallel to the axes of lengths $|x_2 - x_1|$, $|y_2 - y_1|$, $|z_2 - z_1|$. The floor diagonal and the space diagonal both shown to illustrate Pythagoras applied twice.

Proof. Build the axis-aligned rectangular box (cuboid) that has P and Q as opposite corners. Its three edge lengths are $|x_2 - x_1|$, $|y_2 - y_1|$, and $|z_2 - z_1|$. The diagonal of the base rectangle has length $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ by the 2D Pythagoras theorem. That base diagonal and the vertical edge $|z_2 - z_1|$ form a right triangle whose hypotenuse is the space diagonal PQ . Applying Pythagoras a second time:

$$PQ^2 = [(x_2 - x_1)^2 + (y_2 - y_1)^2] + (z_2 - z_1)^2,$$

and taking the positive square root gives the result. ■

Theorem 2 (section formula, internal). The point R dividing AB in ratio $m : n$ from A has coordinates $(\frac{mx_2 + nx_1}{m+n}, \dots)$.

Proof. Work one coordinate at a time. Let $R = (x, y, z)$ divide AB so that $AR : RB = m : n$. Projecting onto the x-axis, the projections of A, R, B divide in the same ratio (parallel projection preserves ratios), so $\frac{x - x_1}{x_2 - x} = \frac{m}{n}$. Cross-multiplying, $n(x - x_1) = m(x_2 - x) \Rightarrow$

$nx - nx_1 = mx_2 - mx \Rightarrow x(m+n) = mx_2 + nx_1$, hence $x = \frac{mx_2 + nx_1}{m+n}$. Identical reasoning on the other two axes completes the proof. Setting $m = n$ recovers the midpoint. ■

Procedure – testing collinearity of three points. (1) Form $\overrightarrow{AB} = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$ and $\overrightarrow{AC} = (x_3 - x_1, y_3 - y_1, z_3 - z_1)$. (2) Check whether their components are proportional, i.e. $\frac{x_2 - x_1}{x_3 - x_1} = \frac{y_2 - y_1}{y_3 - y_1} = \frac{z_2 - z_1}{z_3 - z_1}$. (3) If yes, the three points are collinear; the common

ratio even tells you how B divides AC . This test is preferred over distances because it never depends on the order of the points.

Procedure — area as a collinearity check. When direction ratios are awkward to compare (a zero denominator, say), compute $\overrightarrow{AB} \times \overrightarrow{AC}$ instead. Half its magnitude is the area of triangle ABC ; if the cross product is the zero vector, the area is zero and the points are collinear. This single computation handles both “find the area” and “are they collinear” questions, and it never divides by anything, so no coordinate difference can break it.

Procedure — recovering a missing coordinate or vertex. Many problems give a target (a midpoint, a centroid, or a ratio) and ask for an unknown vertex. Write the relevant formula coordinate-by-coordinate, set each equal to the known target component, and solve the three independent linear equations. For example, if G is the known centroid of A, B, C and C is unknown, then $C = 3G - A - B$ coordinatewise — the formula simply rearranges.

Where the axis and plane distances come from. These are not new results but immediate consequences of Theorem 1. The nearest point on the x -axis to $P(x, y, z)$ is its projection $(x, 0, 0)$; the distance to it is $\sqrt{(x-x)^2 + (y-0)^2 + (z-0)^2} = \sqrt{y^2 + z^2}$ — the x -term vanishes because the axis already matches P in x . The nearest point on the yz -plane is the projection $(0, y, z)$, at distance $\sqrt{(x-0)^2 + 0 + 0} = |x|$. So “distance from an axis” loses the one coordinate the axis shares, and “distance from a plane” keeps only the one coordinate the plane sets to zero. Re-deriving them this way, rather than memorising, means you never confuse the two.

Procedure — finding the ratio in which a coordinate plane divides a segment. A coordinate plane is defined by one coordinate being zero (e.g. the yz -plane is $x = 0$). Let the ratio be $k : 1$. Write the dividing point’s relevant coordinate via the section formula, set it to 0, and solve for k . A **positive** k means internal division; a **negative** k means external division (the plane cuts the line outside the segment).

Theorem 3 (the centroid divides each median in 2 : 1). *In triangle ABC the centroid G lies on every median and splits it, from the vertex to the midpoint of the opposite side, in the ratio 2 : 1.*

Proof. Let D be the midpoint of BC , so $D = \frac{1}{2}(B + C)$. The point dividing median AD in the ratio 2 : 1 from A is, by the section formula, $\frac{2D + 1 \cdot A}{2 + 1} = \frac{2 \cdot \frac{1}{2}(B + C) + A}{3} = \frac{A + B + C}{3}$ — which is exactly the centroid G . By the symmetry of this expression in A, B, C , the identical argument applies to the other two medians, so all three medians pass through G and are cut by it in the ratio 2 : 1. ■

Why external division flips a sign. External division in ratio $m : n$ places the dividing point *outside* the segment, on the extension, with $AR : RB = m : n$ measured in opposite senses. Formally this is internal division with n replaced by $-n$, which turns $\frac{mx_2 + nx_1}{m + n}$ into $\frac{mx_2 - nx_1}{m - n}$. That single sign flip — in both numerator and denominator — is the only difference between the internal and external formulas, which is why a negative answer for k in the plane-cut procedure simply signals an external cut.

Section 4 — Worked Examples

Example 1 — [CBSE Board Level] (Distance). Find the distance between $A(1, -2, 3)$ and $B(4, 2, -1)$.

$$AB = \sqrt{(4-1)^2 + (2-(-2))^2 + (-1-3)^2} = \sqrt{3^2 + 4^2 + (-4)^2} = \sqrt{9 + 16 + 16} = \sqrt{41} \text{ units.}$$

Notice each bracket is a *difference* of coordinates, and the sign of that difference disappears once squared — so it does not matter which point you call A and which B . The single most common error here is sign-handling in $2 - (-2) = 4$; writing $2 - 2 = 0$ would silently lose a term. Always expand the double negative explicitly.

Example 2 — [CUET / JEE Main] (Section formula). Find the point that divides the join of $A(2, -1, 4)$ and $B(4, 3, 2)$ internally in the ratio $1 : 3$. With $m : n = 1 : 3$:

$$\left(\frac{1 \cdot 4 + 3 \cdot 2}{4}, \frac{1 \cdot 3 + 3 \cdot (-1)}{4}, \frac{1 \cdot 2 + 3 \cdot 4}{4} \right) = \left(\frac{10}{4}, \frac{0}{4}, \frac{14}{4} \right) = \left(\frac{5}{2}, 0, \frac{7}{2} \right).$$

The weight $m = 1$ multiplied the far endpoint B and $n = 3$ multiplied the near endpoint A ; because n is larger, the answer leans toward A , which is the quick geometric check — a $1 : 3$ split from A should land nearer A , and indeed $(\frac{5}{2}, 0, \frac{7}{2})$ is closer to $A(2, -1, 4)$ than to B .

Example 3 — [JEE Main · Multi-concept] (Plane-cut ratio + sign meaning). In what ratio is the segment joining $A(2, 4, 5)$ and $B(3, 5, -4)$ divided by the yz -plane? The yz -plane is $x = 0$. Let the ratio be $k : 1$ (from A). The x -coordinate of the dividing point is $\frac{k \cdot 3 + 1 \cdot 2}{k + 1}$.

Setting it to 0:

$$3k + 2 = 0 \Rightarrow k = -\frac{2}{3}.$$

The negative value signals **external** division: the yz -plane meets the line *outside* the segment, dividing it externally in the ratio $2 : 3$. The sign carried the geometric meaning — a recurring JEE Main theme. Geometrically this makes sense: both A and B have positive x -coordinates (2 and 3), so they lie on the *same* side of the yz -plane ($x = 0$), and a plane cannot pass between two points on the same side — it can only meet their line on an extension, i.e. externally.

Example 4 — [JEE Advanced] (Tetrahedron volume). Find the volume of the tetrahedron with vertices $A(1, 1, 1)$, $B(3, 2, 1)$, $C(2, 4, 2)$, $D(1, 2, 5)$. Edge vectors from A : $\overrightarrow{AB} = (2, 1, 0)$, $\overrightarrow{AC} = (1, 3, 1)$, $\overrightarrow{AD} = (0, 1, 4)$.

$$\overrightarrow{AC} \times \overrightarrow{AD} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 3 & 1 \\ 0 & 1 & 4 \end{vmatrix} = \hat{i}(12 - 1) - \hat{j}(4 - 0) + \hat{k}(1 - 0) = (11, -4, 1).$$

$$\overrightarrow{AB} \cdot (11, -4, 1) = (2)(11) + (1)(-4) + (0)(1) = 22 - 4 = 18.$$

$$V = \frac{1}{6}|18| = 3 \text{ cubic units.}$$

The scalar triple product is the volume of the slanted box (parallelepiped) built on the three edges; one-sixth of it is the tetrahedron sitting inside. Two things are worth internalising. First, the absolute value is essential — a negative triple product only means the three edges form a left-handed set, and a volume can never be negative. Second, you may use the edges from *any* vertex; the volume is the same, which is a useful cross-check if your first computation looks wrong.

Section 5 — Practice Exercises

1. **[CBSE]** Find the distance between $(0, 3, -2)$ and $(6, -5, 1)$.
2. **[CUET / JEE Main]** Find the midpoint of the segment joining $(4, -2, 6)$ and $(-2, 8, -4)$.
3. **[JEE Main]** Find the point dividing the join of $(1, 3, 5)$ and $(4, -3, 2)$ externally in the ratio $2 : 1$.

4. **[JEE Main]** Show that $A(1, 2, 3)$, $B(3, 4, 7)$, $C(0, 1, 1)$ are collinear, and find the ratio $AB:BC$. (Answer gains: $AB = 2\sqrt{6}$, $BC = 3\sqrt{6}$, so $AB:BC = 2:3$.)
5. **[JEE Advanced]** Find the volume of the tetrahedron with vertices $(0, 0, 0)$, $(2, 0, 0)$, $(0, 3, 0)$, $(0, 0, 4)$.
6. **[CUET / JEE Main]** Find the area of the triangle with vertices $A(1, 1, 1)$, $B(2, 3, 4)$, $C(4, 3, 2)$.
7. **[JEE Main]** The centroid of triangle ABC is $G(2, 2, 2)$. If $A(1, 3, 2)$ and $B(2, 1, 4)$, find C .

Answers:

- $\sqrt{36 + 64 + 9} = \sqrt{109}$ units
- $(1, 3, 1)$
- $(7, -9, -1)$
- Collinear (direction ratios of $\overrightarrow{AB} = (2, 2, 4)$ and $\overrightarrow{AC} = (-1, -1, -2)$ are proportional)
- $\frac{4}{3}$ cubic units
- $\overrightarrow{AB} = (1, 2, 3)$, $\overrightarrow{AC} = (3, 2, 1)$, $\overrightarrow{AB} \times \overrightarrow{AC} = (-4, 8, -4)$, area $= \frac{1}{2}\sqrt{16 + 64 + 16} = \frac{1}{2} \cdot 4\sqrt{6} = 2\sqrt{6}$ sq units
- $C = 3G - A - B = (6 - 1 - 2, 6 - 3 - 1, 6 - 2 - 4) = (3, 2, 0)$

Section 6 — Crack the MCQ

Q1. The distance of the point $(3, -4, 12)$ from the origin is

- (A) 13 (B) $\sqrt{13}$ (C) 19 (D) 169

Answer: (A). $\sqrt{9 + 16 + 144} = \sqrt{169} = 13$.

(B) takes a stray square root of 13 instead of 169. (C) adds $|3| + |-4| + |12| = 19$ — a sum of magnitudes, not the root of the sum of squares. (D) stops at 169 and forgets the square root. The recurring lesson across all three wrong answers is that the distance formula is a *single* square root of a *sum of squares* — neither the squaring nor the rooting may be skipped, and the terms are added, never taken in absolute value.

Q2. The point dividing the join of $(1, 2, 3)$ and $(3, 6, 7)$ in the ratio $1 : 1$ is

- (A) $(2, 4, 5)$ (B) $(4, 8, 10)$ (C) $(1, 2, 2)$ (D) $(2, 4, 4)$

Answer: (A). Ratio $1 : 1$ is the midpoint: averages give $(2, 4, 5)$.

(B) adds the coordinates without halving. (C)/(D) mishandle the z-coordinate average $\frac{3+7}{2} = 5$.

Q3. The ratio in which the xy-plane divides the join of $(2, 3, 5)$ and $(4, 1, -3)$ is

- (A) $5 : 3$ internally (B) $5 : 3$ externally (C) $3 : 5$ internally (D) $1 : 1$

Answer: (A). The xy-plane is $z = 0$. With ratio $k : 1$, $\frac{k(-3) + 1(5)}{k + 1} = 0 \Rightarrow -3k + 5 = 0 \Rightarrow k = \frac{5}{3}$, positive, so **internal** $5 : 3$.

(B) reads the positive k as external. (C) inverts the ratio. (D) ignores the unequal z-coordinates. A fast check: the endpoints have $z = 5$ and $z = -3$, opposite signs, so they straddle the xy-plane and the cut *must* be internal — confirming the positive k .

Q4. $A(2, 3, 4)$, $B(-1, 2, -3)$, $C(5, 4, 11)$ are

- (A) collinear (B) vertices of a right triangle (C) vertices of an equilateral triangle (D) coincident

Answer: (A). $\overrightarrow{AB} = (-3, -1, -7)$ and $\overrightarrow{AC} = (3, 1, 7) = -\overrightarrow{AB}$ — proportional, so the points are collinear (with A between B and C). Equivalently, $\overrightarrow{AB} \times \overrightarrow{AC} = \vec{0}$, so the enclosed area is zero. (B)/(C) assume a genuine triangle, but proportional direction ratios collapse it to a line. (D) requires identical coordinates, which these are not.

Section 7 — Common Pitfalls & Pro-Tips

- **Mismatching weights in the section formula.** The weight m multiplies the *far* endpoint x_2 , and n the near endpoint x_1 . Swapping them places the point on the wrong side.
- **Forgetting that a negative ratio means external division.** When solving for k , a negative answer is not an error — it tells you the cut lies outside the segment.
- **Summing magnitudes instead of squaring.** Distance is the root of the sum of *squared* differences, never $|\Delta x| + |\Delta y| + |\Delta z|$.
- **Using $AB + BC = AC$ with the wrong ordering.** The additive distance test only works when the middle point is genuinely between the others; otherwise switch to proportional direction ratios.
- **Confusing distance from an axis with distance from a plane.** Distance from the x -axis drops the x -term and keeps two ($\sqrt{y^2 + z^2}$); distance from the yz -plane keeps only $|x|$. Mixing these up is a frequent one-mark loss.
- **Dropping the modulus in area or volume.** A cross product or triple product can come out negative or point “backwards”; area and volume are magnitudes, so the absolute value is non-negotiable.

Pro-Tip (let signs talk): in division-ratio problems, solve for $k : 1$ and read the sign — positive is internal, negative is external. You get the ratio *and* its type from one calculation.

Pro-Tip (collinearity the safe way): prefer proportional direction ratios over the distance test. It is order-independent, avoids square roots, and generalises straight into the line-equation work of the next subtopic. When a coordinate difference is zero (making a ratio undefined), fall back on the cross-product/area test, which never divides and so never breaks on a vanishing denominator.

Pro-Tip (sanity-check with the midpoint): any section-formula answer can be sanity-checked by the rough position it should occupy — a small first weight pulls the point toward the near endpoint, equal weights land exactly at the midpoint average. A quick “does it lean the right way?” catches most weight-swapping slips before they cost marks. The same instinct works for the centroid: it must lie strictly inside the triangle, so a centroid that comes out beyond any vertex is an immediate signal that a coordinate was added or divided wrongly.

Section 8 — Cheat Sheet Box

Core results

- Distance: $PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$.
- Internal $m : n$: $(\frac{mx_2 + nx_1}{m+n}, \dots)$; External $m : n$: $(\frac{mx_2 - nx_1}{m-n}, \dots)$.
- Midpoint: average of endpoints. Centroid: average of vertices ($\div 3$ triangle, $\div 4$ tetrahedron); centroid cuts each median $2 : 1$.
- Area of triangle: $\frac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}|$. Tetrahedron volume: $V = \frac{1}{6}|\overrightarrow{AB} \cdot (\overrightarrow{AC} \times \overrightarrow{AD})|$.
- Distance from x -axis = $\sqrt{y^2 + z^2}$; from yz -plane = $|x|$ (cycle for the others).
- Collinear $\Leftrightarrow$ direction ratios proportional $\Leftrightarrow$ triangle area = 0.

Memory aids

- *"Pythagoras, twice"* — the 3D distance is the 2D one with a third squared term.
- *"Far weight first"* — in $m : n$ from A , m rides the far endpoint.
- *"Negative ratio = outside"* — a minus sign on k means external division.
- *"Axis drops one, plane keeps one"* — distance from an axis loses the shared coordinate; from a plane keeps only the zeroed one.

Direction Cosines, Direction Ratios & Equations of a Line in Space

Class 12 · Mathematics · Chapter 11 — Three Dimensional Geometry · Subtopic 01

Exam Relevance: This is the foundation stone of the whole chapter. CBSE Boards almost always carry a 1-2 mark direct question (find direction cosines / write a line's equation). For JEE Main expect 1-2 questions every year, often disguised inside problems on angles, collinearity, or feet of perpendiculars. JEE Advanced rarely asks it in isolation but builds skew-line and plane problems directly on top of it. (NEET has no Mathematics section, so this ladder runs CBSE → CUET/JEE Main → JEE Advanced.)

Section 1 — Concept Intuition

Picture yourself standing in the corner of your classroom, exactly where two walls meet the floor. That corner is the **origin**. One edge of the floor runs to your right (the **x-axis**), another runs straight ahead of you (the **y-axis**), and the line going straight up the corner where the two walls meet is the **z-axis**. Now stretch a thread tightly from that corner to any point in the room — to the tip of the ceiling fan, say. That taut thread is a **line in space**. The single most important question we can ask about it is simple: *which way is it pointing?*

In two dimensions we'd answer with one number — a slope, or a single angle. But in 3D, "which way" needs to be pinned down against all three axes at once. So we measure the angle the thread makes with the x-axis (call it α), with the y-axis (β), and with the z-axis (γ). Instead of carrying around three angles, we carry their cosines: $l = \cos \alpha$, $m = \cos \beta$, $n = \cos \gamma$. These three numbers are the **direction cosines** of the line. They are the line's "address" in terms of orientation — a compact fingerprint of the direction.

Think about it this way... direction cosines are like splitting a budget. Suppose a direction has a fixed total "strength" of 1, and it must spend that strength across the three axes. A line lying flat along the x-axis spends everything there: $(l, m, n) = (1, 0, 0)$. A line tilted equally between all three shares it evenly. The hidden rule that makes this a *budget* and not a free-for-all is the master identity

$$l^2 + m^2 + n^2 = 1.$$

The squares of the direction cosines must always add to exactly one. This is what guarantees they describe a genuine *unit* direction and not just any old triple of numbers.

There is a second, even more physical way to read those three cosines. Take any point on the line at distance r from the origin and drop a perpendicular from it onto each axis. The shadow it casts — its projection — lands at $r \cos \alpha$ on the x-axis, $r \cos \beta$ on the y-axis, and $r \cos \gamma$ on the z-axis. So the direction cosines are simply the *fractions of the full length spent along each axis*. Equivalently: a single unit step along the line — a step of length exactly 1 — moves you l to the right, m forward, and n upward. In one sentence, **direction cosines are the components of the unit vector pointing along the line**, and committing that sentence to memory makes half the chapter's formulae feel inevitable rather than memorised.

Now, in practice, nobody wants to compute three cosines from scratch every time. If a line passes through $(1, 2, 3)$ and $(4, 8, 11)$, the natural "step" along it is $(3, 6, 8)$ — just the coordinate differences. These three numbers, 3, 6, 8, are not the direction cosines (their squares don't sum to 1), but they are **proportional** to them. Any set of numbers proportional to the direction cosines is called a set of **direction ratios** (DRs). Think of direction ratios like a recipe given as a ratio — "tamarind : jaggery : chilli = 3 : 6 : 8." The ratio fixes the flavour

completely; you can scale it up for a wedding or down for one plate. Direction ratios fix the direction completely; to get the *unique unit* version (the direction cosines) you simply divide each ratio by the length $\sqrt{3^2 + 6^2 + 8^2}$.

Once you know a single point on a line **and** its direction, the line is completely determined — like knowing a train’s starting station and the exact bearing of the track. That single idea gives us the **equation of a line**: start at a known point, then walk any distance in the fixed direction.

It is worth pausing on how cleanly this separates two ideas. A line carries two completely independent pieces of information: *where it is* (any one point on it) and *which way it goes* (its direction). Keep the direction but shift the point and you slide to a parallel line; keep the point but rotate the direction and you pivot the line about that point. Direction cosines and ratios capture only the second piece — the “which way” — which is exactly why the *same* direction ratios describe infinitely many parallel lines scattered across space. To name one specific line you must marry a direction to a point, and that is precisely what $\vec{r} = \vec{a} + \lambda \vec{b}$ does: $\vec{a}$ supplies the “where”, $\vec{b}$ the “which way”, and the parameter λ slides you up and down the track.

A handful of special lines are worth keeping at your fingertips. A line along the x-axis has direction cosines $(1, 0, 0)$ — it makes 0° with x and 90° with the other two. A line lying entirely inside the xy-plane makes 90° with the z-axis, so $n = \cos 90^\circ = 0$; the vanishing of a single direction cosine always flags a line confined to a coordinate plane. And a line equally inclined to all three axes must, by the budget rule, satisfy $3l^2 = 1$, giving $l = m = n = \pm \frac{1}{\sqrt{3}}$ — a value worth memorising outright, because it surfaces in roughly one exam problem in three.

If all of this feels heavier than the slope you used in Class 11, that is because three dimensions genuinely demand more bookkeeping. In a plane, one number — the slope $\tan \theta$ — fixed a line’s direction, because there was only a single “other” axis to measure against. In space there are three axes, and one angle can no longer pin you down: a line tilted 30° from the x-axis could still swing through an entire cone of orientations. Three cosines, locked together by one constraint, are the minimum honest description — which is why we trade the lone slope for the triple (l, m, n) .

Limiting conditions & assumptions to respect:

- Direction cosines belong to a *directed* line. Reverse the arrow and every sign flips: $(l, m, n) \rightarrow (-l, -m, -n)$. So an *undirected* line really has **two** valid DC sets, which is why the $\pm$ keeps appearing.
- Direction ratios are **never unique** — $(2, 3, -1)$, $(4, 6, -2)$ and $(-2, -3, 1)$ all describe the same line. Only direction *cosines* are unique (up to that single $\pm$).
- The zero triple $(0, 0, 0)$ has no direction and is meaningless as DRs.
- The angle α is measured with the **axis itself**, not with a coordinate plane. Confusing “angle with the z-axis” and “angle with the xy-plane” (which are complementary) is the single most common slip on this topic.

Section 2 — Key Definitions & Properties

Unlike Physics quantities, these objects are **pure numbers**: direction cosines are dimensionless ratios, so there are no SI units or dimensional formulae attached. What replaces “units & dimensions” here is the family of algebraic constraints they must obey.

Direction cosines. If a directed line makes angles α, β, γ with the positive x-, y-, z-axes:

$$l = \cos \alpha, \quad m = \cos \beta, \quad n = \cos \gamma, \quad l^2 + m^2 + n^2 = 1.$$

Direction ratios. Any triple (a, b, c) with $a : b : c = l : m : n$. Conversion to direction cosines:

$$l = \pm \frac{a}{\sqrt{a^2 + b^2 + c^2}}, \quad m = \pm \frac{b}{\sqrt{a^2 + b^2 + c^2}}, \quad n = \pm \frac{c}{\sqrt{a^2 + b^2 + c^2}}.$$

The single $\pm$ (same sign for all three) reflects the two possible directions of the line.

Direction ratios as projections. A fast way to *recognise* direction ratios in the wild: the scalar components of *any* vector lying along the line form a valid set of direction ratios — no normalisation required. If $\vec{b} = a\hat{i} + b\hat{j} + c\hat{k}$ points along the line, then (a, b, c) are direction ratios automatically. This is exactly why coordinate differences between two points serve as direction ratios, and why you may freely scale or reverse them: for instance, $(2, -1, 3)$, $(4, -2, 6)$ and $(-2, 1, -3)$ all describe the same direction, yet only the normalised triple $\left(\frac{2}{\sqrt{14}}, -\frac{1}{\sqrt{14}}, \frac{3}{\sqrt{14}}\right)$ (up to sign) is the unique set of direction cosines.

Direction cosines / ratios of a line through two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$. Direction ratios are simply

$$(x_2 - x_1, y_2 - y_1, z_2 - z_1),$$

and dividing by $PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$ gives the direction cosines.

Vector equation of a line through point with position vector $\vec{a}$, parallel to direction $\vec{b}$:

$$\vec{r} = \vec{a} + \lambda \vec{b}, \quad \lambda \in \mathbb{R}.$$

Cartesian equation through (x_1, y_1, z_1) with direction ratios a, b, c (equivalently DCs l, m, n):

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}.$$

Line through two points (vector and Cartesian):

$$\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a}), \quad \frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}.$$

Notation key: α, β, γ = angles with the three axes; l, m, n = direction cosines; a, b, c = direction ratios; $\vec{a}$ = position vector of a fixed point on the line; $\vec{b}$ = a direction vector (DRs are its components); λ = free parameter; PQ = distance between two points.

Quantity	Symbol(s)	Unique?	Key constraint
Direction cosines	l, m, n	Yes (up to $\pm$)	$l^2 + m^2 + n^2 = 1$
Direction ratios	a, b, c	No (any scalar multiple)	not all zero
Angles with axes	α, β, γ	Yes	$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$

Section 3 — Theorems & Proofs

Theorem 1 (the master identity). For any directed line, $l^2 + m^2 + n^2 = 1$.

Figure. A directed line through the origin O to a point $P(x, y, z)$. The angles α, β, γ shown between OP and the positive x -, y -, z -axes respectively. Dashed perpendiculars from P to each axis showing the projections x, y, z .

Proof. Place the line through the origin and let $P(x, y, z)$ be a point on it at distance $r = OP$ from the origin. The foot of the perpendicular from P to the x -axis sits at $(x, 0, 0)$, so in the right triangle formed, the x -coordinate is the projection of OP onto the x -axis:

$$x = r \cos \alpha = r l.$$

By the identical argument on the other two axes, $y = r m$ and $z = r n$. Now the distance of P from the origin satisfies $r^2 = x^2 + y^2 + z^2$. Substituting,

$$r^2 = (rl)^2 + (rm)^2 + (rn)^2 = r^2(l^2 + m^2 + n^2).$$

Dividing by r^2 (non-zero, since $P \neq O$) gives $l^2 + m^2 + n^2 = 1$. ■

A useful corollary drops straight out: since $\cos^2 \theta = 1 - \sin^2 \theta$,

$$\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 3 - (l^2 + m^2 + n^2) = 2.$$

Theorem 2 (direction cosines from direction ratios). If (a, b, c) are direction ratios, the direction cosines are $\pm a/\sqrt{a^2 + b^2 + c^2}$, and similarly for the others.

Proof. By definition of direction ratios, there is a non-zero constant k with $a = kl$, $b = km$, $c = kn$. Then

$$a^2 + b^2 + c^2 = k^2(l^2 + m^2 + n^2) = k^2 \cdot 1 = k^2,$$

so $k = \pm\sqrt{a^2 + b^2 + c^2}$. Hence $l = a/k = \pm a/\sqrt{a^2 + b^2 + c^2}$, and likewise for m, n . The shared sign of k produces the single $\pm$ for the directed line. ■

Derivation of the Cartesian equation of a line. Start from the vector form $\vec{r} = \vec{a} + \lambda \vec{b}$, where $\vec{a} = x_1 \hat{i} + y_1 \hat{j} + z_1 \hat{k}$ and $\vec{b} = a \hat{i} + b \hat{j} + c \hat{k}$ (the components a, b, c being direction ratios). Writing $\vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$ and comparing components:

$$x = x_1 + \lambda a, \quad y = y_1 + \lambda b, \quad z = z_1 + \lambda c.$$

Each equation isolates the same parameter:

$$\lambda = \frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}.$$

Eliminating λ leaves the symmetric Cartesian form. The reasoning behind each step is physical: λ measures “how far along the direction” we have travelled, so setting the three expressions equal simply says *all three coordinates advance in lock-step as we move along the line*. ■

(One caution baked into the derivation: λ equals true distance from (x_1, y_1, z_1) **only when** $\vec{b}$ is a unit vector, i.e. when a, b, c are the direction cosines themselves. With arbitrary direction ratios, λ is just a parameter, not a length.)

Theorem 3 (direction cosines of a line through two points). The line through $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ has direction cosines $\frac{x_2 - x_1}{PQ}$, $\frac{y_2 - y_1}{PQ}$, $\frac{z_2 - z_1}{PQ}$, where $PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$.

Proof. The vector $\overrightarrow{PQ} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$ lies along the line, so its components are a set of direction ratios. Its magnitude is exactly PQ . By Theorem 2, dividing

each direction ratio by the length PQ yields the direction cosines, which is the stated result. The choice of overall sign corresponds to travelling from P towards Q rather than Q towards P . ■

Proposition (parallel lines & the collinearity test). *Two lines are parallel if and only if their direction ratios are proportional; equivalently, their direction cosines are equal up to a common sign.*

Reasoning. Parallelism means the two lines share a direction. Because direction ratios are fixed by the direction alone, two lines pointing the same way must have proportional ratios; conversely, proportional ratios force a common direction vector. Passing to unit vectors collapses the proportionality constant to ± 1 , so the direction cosines agree up to one overall sign. This is the engine behind every collinearity check: three points A, B, C are collinear precisely when the direction ratios of $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are proportional — the exact criterion applied in Worked Example 3 below.

Procedure — converting between the two forms of a line. Vector and Cartesian equations carry identical information; switching between them is a routine but heavily-tested skill. *Vector $\rightarrow$ Cartesian:* read off the fixed point from $\vec{a}$ and the direction ratios from the components of $\vec{b}$, then drop them into $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$. *Cartesian $\rightarrow$ vector:* first force the equation into standard form (each numerator written as $x - x_1$, denominators as plain coefficients), read the point (x_1, y_1, z_1) and ratios (a, b, c) , and assemble $\vec{r} = (x_1\hat{i} + y_1\hat{j} + z_1\hat{k}) + \lambda(a\hat{i} + b\hat{j} + c\hat{k})$. The only place students lose marks is skipping the *standardise* step — a denominator like $\frac{3-y}{4}$ secretly carries the ratio -4 , not 4 .

Section 4 — Worked Examples

Example 1 — [CBSE Board Level]. *A line has direction ratios 2, -3 , 6. Find its direction cosines, and write the vector and Cartesian equations of the line passing through $(1, 4, -2)$ parallel to it.*

Given: $a, b, c = 2, -3, 6$.

$$\sqrt{a^2 + b^2 + c^2} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7.$$

Direction cosines:

$$l = \frac{2}{7}, \quad m = -\frac{3}{7}, \quad n = \frac{6}{7}.$$

Check: $\frac{4}{49} + \frac{9}{49} + \frac{36}{49} = \frac{49}{49} = 1$. ✓ Vector equation (with $\vec{a} = \hat{i} + 4\hat{j} - 2\hat{k}$, $\vec{b} = 2\hat{i} - 3\hat{j} + 6\hat{k}$):

$$\vec{r} = (\hat{i} + 4\hat{j} - 2\hat{k}) + \lambda(2\hat{i} - 3\hat{j} + 6\hat{k}).$$

Cartesian equation:

$$\frac{x-1}{2} = \frac{y-4}{-3} = \frac{z+2}{6}.$$

Notice how the answer simply reuses the direction ratios as the denominators and the point's coordinates (with signs flipped) as the constants — the equation of a line is, in effect, just its direction and a point written side by side.

Example 2 — [CUET / JEE Main Speed Trap]. *A line makes angles 60° and 45° with the positive x - and y -axes respectively. What angle does it make with the z -axis?*

Use $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$ directly — no need to find direction cosines fully:

$$\cos^2 60^\circ + \cos^2 45^\circ + \cos^2 \gamma = 1 \Rightarrow \frac{1}{4} + \frac{1}{2} + \cos^2 \gamma = 1.$$

$$\cos^2 \gamma = \frac{1}{4} \Rightarrow \cos \gamma = \pm \frac{1}{2} \Rightarrow \gamma = 60^\circ \text{ or } 120^\circ.$$

The trap: most students stop at 60° and lose the mark. Because $\cos \gamma = -\frac{1}{2}$ is equally valid, the line may dip below the xy-plane and make 120° with the z-axis. **Both** are correct. Speed move: once you see $\cos^2 \gamma$ alone, immediately remember to write two answers.

Example 3 — [JEE Main • Multi-concept]. The points $A(3, -1, 2)$, $B(1, 3, 4)$, $C(5, k, 0)$ are collinear. Find k , and then the direction cosines of the line.

This links **collinearity** (direction ratios of $\overrightarrow{AB}$ and $\overrightarrow{AC}$ must be proportional) with the **DR** $\rightarrow$ **DC** conversion.

$$\overrightarrow{AB} = (1 - 3, 3 + 1, 4 - 2) = (-2, 4, 2), \quad \overrightarrow{AC} = (5 - 3, k + 1, 0 - 2) = (2, k + 1, -2).$$

Proportionality:

$$\frac{-2}{2} = \frac{k + 1}{4} = \frac{2}{-2}.$$

The first and third ratios both equal -1 (consistent ✓). From the middle: $\frac{k + 1}{4} = -1 \Rightarrow k + 1 = -4 \Rightarrow k = -5$. Now use $\overrightarrow{AB} = (-2, 4, 2)$, with $|\overrightarrow{AB}| = \sqrt{4 + 16 + 4} = \sqrt{24} = 2\sqrt{6}$:

$$l, m, n = \left(\frac{-2}{2\sqrt{6}}, \frac{4}{2\sqrt{6}}, \frac{2}{2\sqrt{6}} \right) = \left(-\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right).$$

The cross-check is automatic: $\frac{1}{6} + \frac{4}{6} + \frac{1}{6} = 1$, confirming the normalisation. Note too that any scalar multiple of $(-2, 4, 2)$ — for instance $(-1, 2, 1)$ — would have served equally well as direction ratios; we normalised only at the very last step, which is the disciplined order of operations and the surest guard against reporting ratios where cosines were asked.

Example 4 — [JEE Advanced]. The direction cosines of two lines satisfy the relations $3l + m + 5n = 0$ and $6mn - 2nl + 5lm = 0$. Find the acute angle between the two lines.

From the linear relation, $m = -(3l + 5n)$. Substitute into the quadratic:

$$6n(-(3l + 5n)) - 2nl + 5l(-(3l + 5n)) = 0.$$

$$-18ln - 30n^2 - 2nl - 15l^2 - 25ln = 0 \Rightarrow -15l^2 - 45ln - 30n^2 = 0.$$

Divide by -15 :

$$l^2 + 3ln + 2n^2 = 0 \Rightarrow (l + n)(l + 2n) = 0 \Rightarrow l = -n \text{ or } l = -2n.$$

Case 1 ($l = -n$): $m = -(3(-n) + 5n) = -2n$, so $(l, m, n) \propto (-1, -2, 1)$, i.e. $(1, 2, -1)$. *Case 2* ($l = -2n$): $m = -(3(-2n) + 5n) = n$, so $(l, m, n) \propto (-2, 1, 1)$, i.e. $(2, -1, -1)$. Both direction vectors have magnitude $\sqrt{6}$. The angle:

$$\cos \theta = \frac{(1)(2) + (2)(-1) + (-1)(-1)}{\sqrt{6} \cdot \sqrt{6}} = \frac{2 - 2 + 1}{6} = \frac{1}{6}.$$

$$\theta = \cos^{-1}\left(\frac{1}{6}\right) \approx 80.4^\circ.$$

Two observations make this problem tractable. First, one linear together with one quadratic relation among l, m, n generically carves out **two** directions, not one — which is exactly why such questions always refer to *two* lines. Second, we never needed the actual direction cosines: the two relations alone pin down two direction *ratio* vectors, and because the angle formula is scale-free, the answer drops straight out. Had the question instead asked us to *prove perpendicularity*, we would simply have checked whether that final dot product came to zero (here it is 1, so the lines are not perpendicular).

Section 5 — Practice Exercises

1. **[CBSE]** Find the direction cosines of a line whose direction ratios are 6, -2, 3.
2. **[CUET / JEE Main]** A line makes an angle of 120° with the x-axis and 60° with the y-axis. Find the angle it makes with the z-axis.
3. **[JEE Main]** Find the vector and Cartesian equations of the line passing through the points $(2, -1, 3)$ and $(4, 2, 1)$.
4. **[JEE Main]** If $A(1, 2, 3)$, $B(3, a, 5)$, $C(5, 4, b)$ are collinear, find a and b .
5. **[JEE Advanced]** A line OP from the origin makes equal angles with the three coordinate axes; a second line OQ has direction ratios 1, -1, 0. Find the acute angle between OP and OQ .

Answers:

1. $(\frac{6}{7}, -\frac{2}{7}, \frac{3}{7})$
2. 45° or 135°
3. $\vec{r} = (2\hat{i} - \hat{j} + 3\hat{k}) + \lambda(2\hat{i} + 3\hat{j} - 2\hat{k})$; $\frac{x-2}{2} = \frac{y+1}{3} = \frac{z-3}{-2}$
4. $a = 3$, $b = 7$
5. 90° (the lines are perpendicular)

Section 6 — Crack the MCQ

Q1. A line makes angles 90° , 135° , 45° with the x-, y-, z-axes respectively. Its direction cosines are

- (A) $(0, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ (B) $(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ (C) $(1, -\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ (D) $(0, -\frac{1}{2}, \frac{1}{2})$

Answer: (A). $\cos 90^\circ = 0$, $\cos 135^\circ = -\frac{1}{\sqrt{2}}$, $\cos 45^\circ = \frac{1}{\sqrt{2}}$; check $0 + \frac{1}{2} + \frac{1}{2} = 1$. ✓

(B) flips the sign of $\cos 135^\circ$ — picked by students who ignore that 135° is obtuse. (C) uses $\cos 90^\circ = 1$ instead of 0, the classic axis-angle blunder. (D) uses $\cos 45^\circ = \frac{1}{2}$ (confusing it with $\cos 60^\circ$), and the squares no longer sum to 1.

Q2. If a line has direction ratios proportional to 1, -2, 2, the direction cosine measured against the y-axis is

- (A) $\frac{2}{3}$ (B) $-\frac{2}{3}$ (C) $\frac{1}{3}$ (D) $-\frac{1}{3}$

Answer: (B). $\sqrt{1+4+4} = 3$, so $m = -\frac{2}{3}$.

(A) drops the negative sign. (C)/(D) take the wrong component ($\frac{1}{3}$ is the x-direction cosine), with (D) compounding it with a stray sign. Trap source: not tracking *which* ratio belongs to *which* axis.

Q3. The direction ratios of the line $\frac{x-1}{2} = \frac{3-y}{4} = \frac{2z-5}{6}$ are

- (A) 2, 4, 6 (B) 2, -4, 3 (C) 2, -4, 6 (D) 1, 3, 5

Answer: (B). The equation must first be put in **standard form** $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$. Now $\frac{3-y}{4} = \frac{-(y-3)}{4} = \frac{y-3}{-4}$, and $\frac{2z-5}{6} = \frac{z-5/2}{3}$. Hence DRs = 2, -4, 3.

(A) reads off the raw denominators 2, 4, 6 — ignoring the sign on y and the coefficient on z . (C) fixes the y -sign but forgets that $2z$ halves the denominator. (D) grabs the point coordinates. This is the most-failed type in JEE Main: **always standardise before reading direction ratios.**

Q4. The direction cosines of the line joining $P(2, 3, 5)$ and $Q(-1, 3, 2)$ are

(A) $\left(-\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right)$ (B) $(-3, 0, -3)$ (C) $\left(-\frac{1}{\sqrt{2}}, 1, -\frac{1}{\sqrt{2}}\right)$ (D) $\left(\frac{3}{\sqrt{18}}, 0, \frac{3}{\sqrt{18}}\right)$

Answer: (A). $\overrightarrow{PQ} = (-3, 0, -3)$, $|PQ| = \sqrt{9+0+9} = 3\sqrt{2}$, so DCs = $\left(\frac{-3}{3\sqrt{2}}, 0, \frac{-3}{3\sqrt{2}}\right) = \left(-\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right)$.

(B) reports direction *ratios* as if they were direction cosines (no normalisation). (C) sets the middle term to 1 instead of 0 — a panic reaction to a zero component. (D) normalises but flips the sign, ignoring the direction $P \rightarrow Q$.

Section 7 — Common Pitfalls & Pro-Tips

- **Direction ratios masquerading as direction cosines.** Writing (a, b, c) as the answer without dividing by $\sqrt{a^2 + b^2 + c^2}$ — missing the normalisation step.
- Forgetting the $\pm$. An undirected line has two opposite direction-cosine sets; problems with $\cos 2\gamma = \dots$ usually have two angle answers.
- **Axis vs. plane angle confusion.** The angle with the z-axis and the angle with the xy-plane are complementary, not equal.
- **Reading denominators without standardising.** A line written as $\frac{3-y}{4}$ or $\frac{2z-5}{6}$ hides its true direction ratios until you force the $\frac{x-x_1}{a}$ shape.
- **Treating the parameter λ as a distance.** λ is a true length only when the direction vector is a unit vector (i.e. you used direction cosines).

Pro-Tip (speed & accuracy): For “angle with an axis” questions, skip computing all three direction cosines — go straight to $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$ and solve for the missing one. And after *every* direction-cosine answer, do a one-second sanity check: square them and confirm the sum is 1. If it isn’t, you forgot to normalise.

Pro-Tip (form conversion): When a Cartesian equation looks “ugly” — fractions buried in the numerators, or coefficients sitting on x, y, z — resist reading the ratios off by eye. Spend three seconds rewriting it as $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$ first. Examiners disguise the standard form precisely because that one skipped step is the cheapest mark on the page to harvest.

Section 8 — Cheat Sheet Box

Core results

- Direction cosines: $l = \cos \alpha$, $m = \cos \beta$, $n = \cos \gamma$, with $\boxed{l^2 + m^2 + n^2 = 1}$.
- Bonus identity: $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$.
- DR $\rightarrow$ DC: $l, m, n = \pm \frac{(a, b, c)}{\sqrt{a^2 + b^2 + c^2}}$.
- Line through two points: DRs = $(x_2 - x_1, y_2 - y_1, z_2 - z_1)$.
- Vector form: $\vec{r} = \vec{a} + \lambda \vec{b}$. Cartesian: $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$.

Memory aids

- “Cosines on a unit budget” — direction cosines spend a total of 1 in squares.
- “Ratios are a recipe, cosines are the dish” — divide ratios by their length to plate up the unique direction cosines.

- “*Standardise, then read*” — never lift direction ratios off un-standardised denominators.
- DC sets always come in a $\pm$ pair: one line, two directions.

Shortest Distance & Angle Between Two Lines

Class 12 · Mathematics · Chapter 11 — Three Dimensional Geometry · Subtopic 02

Exam Relevance: A guaranteed scorer. CBSE Boards regularly carry a 3–5 mark shortest-distance problem and a short angle-between-lines question. JEE Main asks 1–2 every year — usually a direct shortest-distance computation or a “find the parameter so the lines intersect” coplanarity question. JEE Advanced layers it: skew-distance feeds into plane and locus problems. (NEET has no Mathematics; ladder runs CBSE → CUET/JEE Main → JEE Advanced.)

This is a **merged subtopic** with two tightly related halves: **(02a)** the angle between two lines, and **(02b)** the shortest distance between two lines. The scalar triple product quietly powers both.

Section 1 — Concept Intuition

Imagine two Delhi Metro tunnels being bored deep under the city. They never meet and they aren’t parallel — one runs roughly north-east and gently downward, the other east and slightly up. Two natural questions arise. First: *at what angle do they cross*, as seen from directly above? Second: *what is the thinnest wall of earth between them* — the closest the two tunnels ever come? The first question is the **angle between two lines**; the second is the **shortest distance**. Lines in space that neither meet nor run parallel, like these tunnels, are called **skew lines**, and they are the heart of this subtopic.

Angle first. Here is the liberating idea: the angle between two lines depends *only* on their directions, never on where they sit. So to measure the angle between our two tunnels, we don’t need them to touch — we just slide both directions to a common point and measure between the arrows. That is exactly the dot product at work. If $\vec{b}_1$ and $\vec{b}_2$ are direction vectors

of the two lines, then $\cos \theta = \frac{\vec{b}_1 \cdot \vec{b}_2}{|\vec{b}_1||\vec{b}_2|}$. There is one subtlety: a line points two opposite ways,

so the dot product might come out negative, suggesting an obtuse angle. By convention we report the **acute** angle between two lines, so we wrap the numerator in a modulus.

Think about it this way... the dot product is a “sameness meter.” When two directions align it is large and positive; when they’re perpendicular it is exactly zero; when they oppose it is large and negative. Strip the sign with a modulus and you have the cosine of the sharp angle between the lines. This instantly hands you two free tests: the lines are **perpendicular** when $\vec{b}_1 \cdot \vec{b}_2 = 0$, and **parallel** when $\vec{b}_1$ and $\vec{b}_2$ are proportional.

Now the shortest distance. Picture again the thinnest wall between the tunnels. The shortest segment connecting two skew lines is special: it is **perpendicular to both** lines at once. Think of two pencils held crookedly in the air — the shortest gap between them is bridged by a third tiny stick that meets both pencils at right angles. Which direction must that bridging stick point? Perpendicular to both $\vec{b}_1$ and $\vec{b}_2$ simultaneously — and the cross product $\vec{b}_1 \times \vec{b}_2$ is *built* to be exactly that. So the shortest distance is found by taking *any* line joining the two lines (the vector $\vec{a}_2 - \vec{a}_1$ between their reference points) and measuring only its component along that common-perpendicular direction. Everything else in $\vec{a}_2 - \vec{a}_1$ runs *along* the tunnels and doesn’t count toward the gap. For two *parallel* lines the picture is gentler still: since they never tilt relative to one another, the gap is identical at every point, so we simply measure the perpendicular distance from any single point of one line across to the other.

Think about it this way... the scalar triple product $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)$ is the volume of the slanted box (parallelepiped) built on those three vectors. Divide that volume by the area of

its base $|\vec{b}_1 \times \vec{b}_2|$ and you get its height — and the height is the perpendicular gap. Distance is volume per unit base area.

Before computing anything, it pays to know *which* of three situations you are in, because each behaves differently. Two distinct lines in space can (i) **intersect** — sharing exactly one point, so the shortest distance is zero; (ii) be **parallel** — the same direction, never meeting, a constant gap everywhere; or (iii) be **skew** — different directions yet still never meeting, like our two tunnels. Only inside a single plane are “not parallel” and “intersecting” the same statement; space adds the genuinely three-dimensional third option of skewness. One quantity separates the cases cleanly: the scalar triple product $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)$ is **zero** for intersecting *and* parallel lines (both are coplanar) and **non-zero** exactly for skew lines. Combine that with a glance at whether $\vec{b}_1$ and $\vec{b}_2$ are proportional, and all three are told apart with no heavy algebra.

Limiting conditions & assumptions:

- The cross-product formula $\frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$ works **only for skew lines**. For **parallel** lines $\vec{b}_1 \times \vec{b}_2 = \vec{0}$ and you’d divide by zero — a different formula is needed.
- If the shortest distance comes out **zero**, the lines are not skew at all: they **intersect** (or coincide). “Shortest distance = 0” is the working definition of “the lines meet” and is the same as the **coplanarity** condition.
- The angle formula assumes genuine direction vectors (non-zero). The modulus is non-negotiable if you want the acute angle.
- Once the modulus is applied the reported angle always lies in $[0^\circ, 90^\circ]$: a line with itself gives 0° , perpendicular lines give 90° , and the value never strays outside that band — a useful sanity bound on any answer.

Section 2 — Key Definitions & Properties

These are dimensionless or length quantities; there are no SI units beyond “units of length” for distance. The governing relations:

Angle between two lines (02a). With direction vectors $\vec{b}_1, \vec{b}_2$ (or direction ratios a_1, b_1, c_1 and a_2, b_2, c_2 , or direction cosines l_i, m_i, n_i):

$$\cos \theta = \frac{|\vec{b}_1 \cdot \vec{b}_2|}{|\vec{b}_1||\vec{b}_2|} = \frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}} = |l_1l_2 + m_1m_2 + n_1n_2|.$$

All three expressions say the same thing — the cosine of the acute angle is the normalised, sign-stripped overlap of the two directions; choose whichever form matches the data you are handed (vectors, ratios, or cosines).

Perpendicular & parallel tests (02a).

$$\text{Perpendicular: } a_1a_2 + b_1b_2 + c_1c_2 = 0, \quad \text{Parallel: } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}.$$

Shortest distance between skew lines (02b) — vector form, for $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu\vec{b}_2$:

$$d = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}.$$

Shortest distance — Cartesian determinant form (02b). For lines $\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$ and $\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}$, the numerator is the determinant and the denominator is $|\vec{b}_1 \times \vec{b}_2|$:

$$d = \frac{\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}}{\sqrt{(b_1 c_2 - b_2 c_1)^2 + (c_1 a_2 - c_2 a_1)^2 + (a_1 b_2 - a_2 b_1)^2}}.$$

Coplanarity / intersection condition (02b). The two lines are coplanar (they intersect or are parallel) iff

$$(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0 \iff \begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0.$$

(The determinant is simply the scalar triple product written as a 3×3 array, so the coplanarity test and the shortest-distance numerator are one and the same object — compute it once and it answers both questions.)

Shortest distance between parallel lines (02b) — direction $\vec{b}$ shared:

$$d = \frac{|\vec{b} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}|}.$$

Notation key: $\vec{a}_1, \vec{a}_2$ = position vectors of a known point on each line; $\vec{b}_1, \vec{b}_2$ = direction vectors; θ = acute angle between the lines; d = shortest distance; $\vec{b}_1 \times \vec{b}_2$ = the common-perpendicular direction; $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)$ = scalar triple product (a signed volume).

Configuration	Quick test	Shortest distance
Intersecting	STP = 0, not parallel	0
Parallel	$\vec{b}_1 \parallel \vec{b}_2$	$\frac{ \vec{b} \times (\vec{a}_2 - \vec{a}_1) }{ \vec{b} }$
Skew	STP $\neq 0$	$\frac{ (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) }{ \vec{b}_1 \times \vec{b}_2 }$

Section 3 — Theorems & Proofs

Theorem 1 (angle between two lines). The acute angle θ between lines with direction vectors $\vec{b}_1, \vec{b}_2$ satisfies $\cos \theta = \frac{|\vec{b}_1 \cdot \vec{b}_2|}{|\vec{b}_1||\vec{b}_2|}$.

Proof. The angle between two lines is defined as the angle between their directions, so we may translate both direction vectors to a common origin without changing it. By the definition of the dot product, $\vec{b}_1 \cdot \vec{b}_2 = |\vec{b}_1||\vec{b}_2| \cos \varphi$, where φ is the angle between the two vectors.

Rearranging, $\cos \varphi = \frac{\vec{b}_1 \cdot \vec{b}_2}{|\vec{b}_1||\vec{b}_2|}$. Because each line has two opposite direction vectors, φ could

be either θ or $\pi - \theta$; taking the modulus selects the **acute** value θ . Writing $\vec{b}_i$ in components gives the direction-ratio form, and if the $\vec{b}_i$ are unit vectors (i.e. their components are direction cosines), the denominator is 1 and we get $\cos \theta = |l_1 l_2 + m_1 m_2 + n_1 n_2|$. ■

Theorem 2 (shortest distance between skew lines). For $L_1 : \vec{r} = \vec{a}_1 + \lambda \vec{b}_1$ and $L_2 : \vec{r} = \vec{a}_2 + \mu \vec{b}_2$, the shortest distance is $d = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$.

Figure. Two skew lines L_1 and L_2 drawn in 3D with direction arrows $\vec{b}_1$ and $\vec{b}_2$. A short segment of length d connects them, meeting both at right angles. The vector $(\vec{a}_2 - \vec{a}_1)$ is drawn joining the reference point on L_1 to the reference point on L_2 . The cross product $\vec{b}_1 \times \vec{b}_2$ is shown as an arrow perpendicular to both lines, parallel to the segment d .

Proof. The shortest segment between two skew lines is perpendicular to both lines simultaneously. The cross product $\vec{b}_1 \times \vec{b}_2$ is, by construction, perpendicular to both $\vec{b}_1$ and $\vec{b}_2$, so it points along this common perpendicular. The corresponding unit vector is $\hat{n} = \frac{\vec{b}_1 \times \vec{b}_2}{|\vec{b}_1 \times \vec{b}_2|}$.

Now take the vector $\vec{a}_2 - \vec{a}_1$ joining a point of L_1 to a point of L_2 . The shortest distance is precisely the length of the projection of $\vec{a}_2 - \vec{a}_1$ onto $\hat{n}$, because the components of $\vec{a}_2 - \vec{a}_1$ lying *along* either line contribute nothing to the perpendicular gap:

$$d = |(\vec{a}_2 - \vec{a}_1) \cdot \hat{n}| = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}.$$

The modulus guarantees a non-negative length, since projections can be signed. ■

Corollary (coplanarity / intersection). Setting $d = 0$ forces the scalar triple product $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0$. Geometrically this says the three vectors $\vec{a}_2 - \vec{a}_1$, $\vec{b}_1$, $\vec{b}_2$ are coplanar — the slanted box collapses to zero volume — so the two lines lie in one plane and therefore **intersect** (or are parallel). In Cartesian form this is exactly the determinant of $\{\vec{a}_2 - \vec{a}_1; \vec{b}_1; \vec{b}_2\}$ equated to zero.

Why parallel lines need a different formula. If $\vec{b}_1$ and $\vec{b}_2$ are parallel, $\vec{b}_1 \times \vec{b}_2 = \vec{0}$ and Theorem 2 collapses to $\frac{0}{0}$. Instead, fix the common direction $\vec{b}$ and resolve $\vec{a}_2 - \vec{a}_1$ into a part along $\vec{b}$ and a part perpendicular to it; the perpendicular part is the gap. Its length is $d = \frac{|\vec{b} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}|}$, since $|\vec{b} \times (\vec{a}_2 - \vec{a}_1)| = |\vec{b}||\vec{a}_2 - \vec{a}_1| \sin \alpha$ already isolates exactly the perpendicular component.

From vector to Cartesian (where the determinant comes from). Writing the scalar triple product in components makes the determinant formula appear on its own. A triple product $\vec{u} \cdot (\vec{v} \times \vec{w})$ is exactly the 3×3 determinant whose rows are $\vec{u}, \vec{v}, \vec{w}$. So with $\vec{u} = \vec{a}_2 - \vec{a}_1 = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$ as the top row and $\vec{b}_1 = (a_1, b_1, c_1)$, $\vec{b}_2 = (a_2, b_2, c_2)$ beneath it, the numerator of the Cartesian shortest-distance formula is literally that determinant. The denominator $|\vec{b}_1 \times \vec{b}_2|$ expands to $\sqrt{(b_1 c_2 - b_2 c_1)^2 + (c_1 a_2 - c_2 a_1)^2 + (a_1 b_2 - a_2 b_1)^2}$. The imposing Cartesian formula is therefore just Theorem 2 written out coordinate by coordinate — no new content to memorise.

Corollary (the two quick tests). Both classroom tests drop straight out of Theorem 1.

Perpendicular lines have $\theta = 90^\circ$, so $\cos \theta = 0$, which forces the numerator $\vec{b}_1 \cdot \vec{b}_2 = a_1a_2 + b_1b_2 + c_1c_2 = 0$. Parallel lines have $\theta = 0^\circ$, so $\cos \theta = 1$, which is reached only when $\vec{b}_1$ and $\vec{b}_2$ point the same way — that is, when their components are proportional, $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$. The two “rules” are not separate facts but the two extreme values of one cosine.

Geometric remark (why “the” shortest distance is well-defined). For two skew lines there is exactly *one* segment perpendicular to both, and its endpoints are the unique closest points on each line. That uniqueness is what lets us say *the* shortest distance rather than *a* short distance. For intersecting lines the segment shrinks to the single shared point (length zero); for parallel lines there are infinitely many perpendicular segments, all of equal length, which is exactly why the constant-gap formula gives the same answer from any starting point. This existence-and-uniqueness is special to three dimensions — in a plane two non-parallel lines are forced to cross, leaving no room for a perpendicular bridge of positive length.

Section 4 — Worked Examples

Example 1 — [CBSE Board Level] (Angle). Find the acute angle between the lines $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-3}{2}$ and $\frac{x-2}{1} = \frac{y-1}{2} = \frac{z+4}{2}$.

Direction ratios: $\vec{b}_1 = (2, -1, 2)$, $\vec{b}_2 = (1, 2, 2)$.

$$\vec{b}_1 \cdot \vec{b}_2 = (2)(1) + (-1)(2) + (2)(2) = 2 - 2 + 4 = 4.$$

$$|\vec{b}_1| = \sqrt{4 + 1 + 4} = 3, \quad |\vec{b}_2| = \sqrt{1 + 4 + 4} = 3.$$

$$\cos \theta = \frac{|4|}{3 \cdot 3} = \frac{4}{9} \Rightarrow \theta = \cos^{-1}\left(\frac{4}{9}\right).$$

Had the dot product come out negative — say -4 — the modulus would still return $\theta = \cos^{-1} \frac{4}{9}$. The equations describe the same pair of lines regardless of which way we choose to walk along each, so only the acute angle carries meaning.

Example 2 — [CUET / JEE Main Speed Trap] (Perpendicular & Parallel tests). (a) Show that the lines with direction ratios $(2, -1, 1)$ and $(1, 1, -1)$ are perpendicular. (b) Find p so that the line with direction ratios $(3, p, 2)$ is parallel to the line with ratios $(6, -4, 4)$.

(a) Perpendicularity needs only a single dot product — no angle computation:

$$(2)(1) + (-1)(1) + (1)(-1) = 2 - 1 - 1 = 0 \Rightarrow \text{perpendicular.}$$

(b) Parallelism needs only ratio equality:

$$\frac{3}{6} = \frac{p}{-4} = \frac{2}{4} \Rightarrow \frac{1}{2} = \frac{p}{-4} \Rightarrow p = -2 \quad (\text{check } \frac{2}{4} = \frac{1}{2} \checkmark).$$

The trap: students reach for the full $\cos \theta$ formula for part (a) (slow, and easy to fumble the modulus) or mix up the two tests — checking *proportionality* for perpendicularity. Lock it in: **dot product = 0 means perpendicular; equal ratios mean parallel.** A reliable habit is to write the dot product first: if it is zero you are finished (perpendicular), and only if it isn't is it worth checking proportionality for parallelism or computing a full angle.

Example 3 — [JEE Main • Multi-concept] (Shortest distance, vector form). Find the shortest distance between the skew lines $\vec{r} = (\hat{i} + 2\hat{j} - \hat{k}) + \lambda(2\hat{i} + \hat{j} - 2\hat{k})$ and $\vec{r} = (3\hat{i} - \hat{j} + \hat{k}) + \mu(\hat{i} + 2\hat{j} + 2\hat{k})$.

Here $\vec{a}_1 = (1, 2, -1)$, $\vec{b}_1 = (2, 1, -2)$ and $\vec{a}_2 = (3, -1, 1)$, $\vec{b}_2 = (1, 2, 2)$. First the common perpendicular:

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 1 & 2 & 2 \end{vmatrix} = \hat{i}(1 \cdot 2 - (-2) \cdot 2) - \hat{j}(2 \cdot 2 - (-2) \cdot 1) + \hat{k}(2 \cdot 2 - 1 \cdot 1) = (6, -6, 3).$$

$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{36 + 36 + 9} = \sqrt{81} = 9.$$

Joining vector $\vec{a}_2 - \vec{a}_1 = (2, -3, 2)$. The scalar triple product:

$$(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = (2)(6) + (-3)(-6) + (2)(3) = 12 + 18 + 6 = 36.$$

$$d = \frac{|36|}{9} = 4 \text{ units.}$$

Quick check that we projected along the genuine common perpendicular: $\vec{b}_1 \times \vec{b}_2 = (6, -6, 3)$ should be perpendicular to both directions, and indeed $(6, -6, 3) \cdot (2, 1, -2) = 12 - 6 - 6 = 0$ and $(6, -6, 3) \cdot (1, 2, 2) = 6 - 12 + 6 = 0$. The multi-concept thread: a cross product (to find the common perpendicular), then a dot product (to project the joining vector), then a magnitude — three vector operations chained into one distance.

Example 4 — [JEE Advanced] (Coplanarity / find the parameter). Find k for which the lines $\frac{x-2}{1} = \frac{y-1}{2} = \frac{z+1}{3}$ and $\frac{x-1}{3} = \frac{y-k}{2} = \frac{z-2}{1}$ are coplanar, and locate their point of intersection.

Apply the coplanarity determinant with $\vec{a}_2 - \vec{a}_1 = (1-2, k-1, 2-(-1)) = (-1, k-1, 3)$, $\vec{b}_1 = (1, 2, 3)$, $\vec{b}_2 = (3, 2, 1)$:

$$\begin{vmatrix} -1 & k-1 & 3 \\ 1 & 2 & 3 \\ 3 & 2 & 1 \end{vmatrix} = 0.$$

Expanding along the first row:

$$-1(2 \cdot 1 - 3 \cdot 2) - (k-1)(1 \cdot 1 - 3 \cdot 3) + 3(1 \cdot 2 - 2 \cdot 3) = -1(-4) - (k-1)(-8) + 3(-4).$$

$$= 4 + 8(k-1) - 12 = 8k - 16 = 0 \Rightarrow \boxed{k=2}.$$

With $k=2$, write general points $P = (2+t, 1+2t, -1+3t)$ on L_1 and $Q = (1+3s, 2+2s, 2+s)$ on L_2 . Equating coordinates: $t-3s = -1$, $2t-2s = 1$, $3t-s = 3$. Solving the first two gives $s = \frac{3}{4}$, $t = \frac{5}{4}$ (consistent with the third). The intersection point is

$$\left(\frac{13}{4}, \frac{7}{2}, \frac{11}{4}\right).$$

As a check, both general points should land here: on L_1 , $t = \frac{5}{4}$ gives $(2 + \frac{5}{4}, 1 + \frac{5}{2}, -1 + \frac{15}{4})$, and on L_2 , $s = \frac{3}{4}$ gives $(1 + \frac{3}{4}, 2 + \frac{3}{2}, 2 + \frac{3}{4})$ — both equal $(\frac{13}{4}, \frac{7}{2}, \frac{11}{4})$, exactly as coplanarity promised. The deeper idea: coplanarity is a *single scalar equation* (the triple product), so a one-parameter family of line pairs has exactly the special values of k that make them meet.

Section 5 — Practice Exercises

1. **[CBSE]** Find the acute angle between the lines $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z}{2}$ and $\frac{x}{2} = \frac{y+1}{2} = \frac{z-3}{1}$.
2. **[CUET / JEE Main]** Find p if the lines with direction ratios $(3, -2, p)$ and $(2, p, -1)$ are perpendicular.
3. **[JEE Main]** Find the shortest distance between $\vec{r} = (\hat{i} + \hat{j}) + \lambda(2\hat{i} - \hat{j} + \hat{k})$ and $\vec{r} = (2\hat{i} + \hat{j} - \hat{k}) + \mu(3\hat{i} - 5\hat{j} + 2\hat{k})$.
4. **[JEE Main]** Determine whether the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-4}{4}$ and $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-5}{5}$ are coplanar.
5. **[JEE Advanced]** Find the shortest distance between the parallel lines $\vec{r} = (\hat{i} + 2\hat{j} - 3\hat{k}) + \lambda(2\hat{i} + 3\hat{j} - 6\hat{k})$ and $\vec{r} = (3\hat{i} - \hat{j} + \hat{k}) + \mu(2\hat{i} + 3\hat{j} - 6\hat{k})$.

Answers:

1. $\cos^{-1}\left(\frac{8}{9}\right)$
2. $p = 2$
3. $\frac{10}{\sqrt{59}}$ units
4. Coplanar (the determinant evaluates to 0)
5. $\frac{2\sqrt{145}}{7}$ units

Section 6 — Crack the MCQ

Q1. The acute angle between the lines with direction ratios $(1, 2, 2)$ and $(2, -1, -2)$ is

- (A) $\cos^{-1}\left(\frac{4}{9}\right)$ (B) $\cos^{-1}\left(-\frac{4}{9}\right)$ (C) $\cos^{-1}\left(\frac{4}{3}\right)$ (D) 90°

Answer: (A). Dot product = $(1)(2) + (2)(-1) + (2)(-2) = 2 - 2 - 4 = -4$; magnitudes are 3 and 3. The *acute* angle uses the modulus: $\cos \theta = \frac{|-4|}{9} = \frac{4}{9}$.

(B) keeps the negative sign and reports the obtuse angle — the single most common error here. (C) forgets to multiply the two magnitudes in the denominator (using 3 instead of 9), giving an impossible $\cos > 1$. (D) assumes a small/negative dot product means perpendicular.

Q2. The shortest distance between two **parallel** lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}$ and $\vec{r} = \vec{a}_2 + \mu\vec{b}$ is

- (A) $\frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$ (B) $\frac{|\vec{b} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}|}$ (C) $|\vec{a}_2 - \vec{a}_1|$ (D) 0

Answer: (B). For parallel lines the direction is shared, so the gap is the perpendicular component of $\vec{a}_2 - \vec{a}_1$ relative to $\vec{b}$.

(A) is the skew-line formula, which is undefined here because $\vec{b} \times \vec{b} = \vec{0}$ — applying it blindly is the classic parallel-line trap. (C) is the full separation of the reference points, which over-counts the part running along the lines. (D) is true only for coincident lines.

Q3. The shortest distance between $\vec{r} = (2\hat{i} - \hat{j}) + \lambda(\hat{i} - \hat{j} + \hat{k})$ and $\vec{r} = (3\hat{i} + \hat{k}) + \mu(2\hat{i} + \hat{j} - \hat{k})$ is

(A) $\sqrt{2}$ (B) $2\sqrt{2}$ (C) 6 (D) $3\sqrt{2}$

Answer: (A). $\vec{b}_1 \times \vec{b}_2 = (0, 3, 3)$, so $|\vec{b}_1 \times \vec{b}_2| = \sqrt{18} = 3\sqrt{2}$. With $\vec{a}_2 - \vec{a}_1 = (1, 1, 1)$, the triple product is $0 + 3 + 3 = 6$, hence $d = \frac{6}{3\sqrt{2}} = \sqrt{2}$.

(C) reports the triple product 6 without dividing by $|\vec{b}_1 \times \vec{b}_2|$. (D) reports $|\vec{b}_1 \times \vec{b}_2| = 3\sqrt{2}$ itself. (B) is a slip in simplifying $\frac{6}{3\sqrt{2}}$.

Q4. Two lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu\vec{b}_2$ are coplanar if and only if

(A) $\vec{b}_1 \cdot \vec{b}_2 = 0$ (B) $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0$ (C) $\vec{b}_1 \times \vec{b}_2 = \vec{0}$ (D) $(\vec{a}_2 - \vec{a}_1) \cdot \vec{b}_1 = 0$

Answer: (B). Coplanarity means the joining vector and the two directions lie in one plane — their scalar triple product (signed volume) vanishes.

(A) is the *perpendicular* condition. (C) is the *parallel* condition. (D) merely says $\vec{a}_2 - \vec{a}_1$ is perpendicular to one direction, which neither implies nor is implied by coplanarity.

Section 7 — Common Pitfalls & Pro-Tips

- **Dropping the modulus in the angle formula.** A negative dot product yields an obtuse value; the angle *between two lines* is always reported as acute.
- **Treating $\vec{b}_1 \cdot \vec{b}_2$ as $\cos \theta$.** You must divide by both magnitudes; forgetting the denominator can even give $\cos \theta > 1$.
- **Swapping the perpendicular and parallel tests.** Perpendicular $\Rightarrow$ dot product = 0; parallel $\Rightarrow$ ratios equal. Students routinely apply the wrong one.
- **Using the skew formula on parallel lines.** If $\vec{b}_1 \parallel \vec{b}_2$ then $\vec{b}_1 \times \vec{b}_2 = \vec{0}$ and the skew formula divides by zero — switch to $\frac{|\vec{b} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}|}$
- **Forgetting the modulus on the triple product.** Distance is never negative; the scalar triple product can be.
- **Not recognising coplanar $\iff$ STP = 0.** “Show the lines intersect” is the same as “show the determinant is zero.”
- **Pairing a point with the wrong line.** In $\vec{a}_2 - \vec{a}_1$, each $\vec{a}_i$ must be the reference point of *its own* line L_i . Subtracting $\vec{a}_1$ from a point that actually sits on L_1 , or mixing $\vec{a}_1$ with $\vec{b}_2$, silently corrupts the entire triple product.

Pro-Tip (sanity checks): Before computing a shortest distance, glance at the two direction vectors. If they’re proportional, stop — the lines are parallel and the skew formula will fail; use the parallel formula. If the scalar triple product comes out 0, the lines meet and the distance is 0; don’t keep dividing. And every “distance” answer must be ≥ 0 — a negative result means you forgot a modulus.

Pro-Tip (cross product once): In a shortest-distance problem, compute $\vec{b}_1 \times \vec{b}_2$ first. Its magnitude is your denominator *and* its zero-ness immediately tells you whether the lines are parallel — two pieces of information from one calculation.

Section 8 — Cheat Sheet Box

Angles (02a)

- $\cos \theta = \frac{|\vec{b}_1 \cdot \vec{b}_2|}{|\vec{b}_1||\vec{b}_2|} = |l_1l_2 + m_1m_2 + n_1n_2|$ — always the **acute** angle.
- Perpendicular $\Leftrightarrow a_1a_2 + b_1b_2 + c_1c_2 = 0$. Parallel $\Leftrightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

Shortest distance (02b)

- Skew: $d = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$ (“volume $\div$ base area”).
- Parallel: $d = \frac{|\vec{b} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}|}$.
- Coplanar / intersecting $\Leftrightarrow (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0 \Leftrightarrow \text{determinant} = 0$.

Memory aids

- “Cross to bridge, dot to project” — cross product finds the common perpendicular, dot product measures along it.
- “Acute by modulus” — wrap the angle numerator in $|\cdot|$.
- “Zero volume, they meet” — STP = 0 means the lines are coplanar.
- “Parallel? cross dies” — proportional directions kill $\vec{b}_1 \times \vec{b}_2$; use the parallel formula.

Planes in Space — Equations, Family, Angles & Distances

Class 12 · Mathematics · Chapter 11 — Three Dimensional Geometry · Subtopic 03 · JEE-only

Exam Relevance: A JEE workhorse. The plane was removed from the rationalized CBSE Board syllabus, so this subtopic is built for JEE Main and Advanced only. JEE Main reliably carries 2–3 questions a year — a plane equation, an angle or distance, and very often a family-of-planes problem. JEE Advanced fuses planes with lines, spheres, and loci. (No CBSE Board worked example here; the ladder runs JEE Main warm-up → CUET/JEE Main Speed Trap → JEE Main multi-concept → JEE Advanced.)

This is a broad **merged subtopic** with four diagnostic units: **(03a)** equations & forms of a plane, **(03b)** the family of planes through a line of intersection, **(03c)** angles (plane-plane and line-plane), and **(03d)** distances (point-plane and between parallel planes).

Section 1 — Concept Intuition

A plane is the flattest, most unbending surface in space — an infinite tabletop. The remarkable thing is how little information it takes to fix one completely: **a single point on it, plus the direction it faces**. The “direction it faces” is captured by a vector sticking straight out of the surface — the **normal vector** $\vec{n}$. Tilt the normal and the whole tabletop tilts with it; slide the point and the tabletop slides parallel to itself. Everything about planes flows from this one pairing of *a point and a normal*.

Think about it this way... stand a pencil upright on a sheet of glass. The pencil is the normal $\vec{n}$; the glass is the plane. Any point $\vec{r}$ lies on the glass exactly when the little arrow from the base point $\vec{a}$ to $\vec{r}$ runs *flat along the glass* — that is, perpendicular to the pencil. “Perpendicular” is the dot product being zero, which gives the master equation of a plane in one stroke: $(\vec{r} - \vec{a}) \cdot \vec{n} = 0$, or tidied up, $\vec{r} \cdot \vec{n} = d$. Every form of the plane equation is just this idea wearing different clothes. Given three points instead of a normal? Build two edge vectors and cross them to manufacture the normal. Given intercepts on the axes? That’s three points in disguise.

When you are handed three points and no normal, the cross product is the hero. Two of the points define an edge vector lying *in* the plane; a third gives a second edge vector, also in the plane. Their cross product is automatically perpendicular to both edges, and therefore to the whole surface — you have manufactured a normal out of thin air. From there the point-normal equation finishes the job. This single move, “two edges, one cross product,” handles every plane defined by points, including the intercept form.

Angles, the smart way. Here is the single most useful trick in the whole subtopic: *a plane is most easily handled through its normal*. The angle between two planes is simply the angle

between their normals — $\cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}$. But beware the line-and-plane case, which hides

a twist. When a line meets a plane, the angle we care about is measured against the *surface*, not against the normal — and the normal sticks up at 90° to the surface. So if α is the angle between the line and the normal, the angle between the line and the plane is its **complement**

$90^\circ - \alpha$. Cosine of the complement is sine, so the line-plane angle uses a **sine**: $\sin \varphi = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}||\vec{n}|}$.

Forgetting this single sine/cosine swap is the most common mistake on the entire topic.

Distance, the smart way. How far is a point from a plane? Drop a perpendicular and measure along the normal. If the plane is $ax + by + cz + d = 0$, the perpendicular distance of

(x_1, y_1, z_1) is $\frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$ — you simply feed the point into the plane expression and divide by the length of the normal. The modulus is there because distance can't be negative; the *sign* of $ax_1 + by_1 + cz_1 + d$ (before the modulus) actually tells you *which side* of the plane the point is on, a fact JEE Advanced loves to exploit.

That sign idea deserves a closer look, because it solves a whole class of problems for free. Evaluate $ax + by + cz + d$ at two points: if both give the same sign, the points lie on the *same side* of the plane; if the signs differ, the plane *separates* them — and the segment joining them must cross the plane. This is exactly how you test whether a line segment is cut by a plane, or whether the origin lies on the same side as some given point, with no work beyond the distance computation you were already doing.

Family of planes — the elegant shortcut. When a problem says “the plane through the line where planes P_1 and P_2 meet,” don't find that line. Instead write $P_1 + \lambda P_2 = 0$: every choice of λ is automatically a plane through that intersection line, and one extra condition (a point to pass through, a plane to be perpendicular to) pins down the single right λ . Geometrically, $P_1 + \lambda P_2 = 0$ is a *pencil* of planes — an infinite book of pages all hinged along the same spine, which is the intersection line. Sweeping λ across all real values flips through every page of the book (every plane containing that line, save P_2 itself), and the extra condition simply selects the one page you need.

Limiting conditions & assumptions:

- The normal form $\vec{r} \cdot \hat{n} = p$ requires $\hat{n}$ to be a **unit** vector and $p \geq 0$; only then is p the true distance from the origin.
- The line-plane angle uses **sine**, the plane-plane angle uses **cosine**. They are not interchangeable.
- The point-plane distance formula needs the plane written as $\dots = 0$; a stray constant on the wrong side flips a sign.
- For distance between two *parallel* planes, their normals must first be scaled to be **identical** before comparing the constants — otherwise the gap is meaningless.

Section 2 — Key Definitions & Properties

These are length and angle quantities; distances carry units of length, angles are dimensionless.

Plane equations & forms (03a).

Normal form: $\vec{r} \cdot \hat{n} = p$ ($\hat{n}$ unit, $p \geq 0$), $lx + my + nz = p$.

Point-normal: $(\vec{r} - \vec{a}) \cdot \vec{n} = 0 \Leftrightarrow \vec{r} \cdot \vec{n} = d$, $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$.

Three points (x_i, y_i, z_i) : $\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ x_3 - x_1 & y_3 - y_1 & z_3 - z_1 \end{vmatrix} = 0$, or $(\vec{r} - \vec{a}) \cdot [(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})] = 0$.

Intercept form: $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ (intercepts a, b, c on the axes).

All four are the single equation $\vec{r} \cdot \vec{n} = d$ in disguise; they differ only in what starting data you are given — a normal and a point, three points, or three intercepts.

Family of planes through a line of intersection (03b). For planes $P_1 \equiv A_1x + B_1y + C_1z + D_1 = 0$ and $P_2 \equiv A_2x + B_2y + C_2z + D_2 = 0$:

$$P_1 + \lambda P_2 = 0 \quad (\lambda \in \mathbb{R}), \quad \text{vector form } (\vec{r} \cdot \vec{n}_1 - d_1) + \lambda(\vec{r} \cdot \vec{n}_2 - d_2) = 0.$$

Angles (03c).

$$\text{Plane-plane: } \cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}, \quad \text{Line-plane: } \sin \varphi = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}||\vec{n}|}.$$

Two planes are parallel when $\vec{n}_1 \parallel \vec{n}_2$, perpendicular when $\vec{n}_1 \cdot \vec{n}_2 = 0$. A line lies in (or is parallel to) a plane when $\vec{b} \cdot \vec{n} = 0$. In both angle formulas the modulus delivers the acute angle and guards against the sign ambiguity of a reversible normal.

Distances (03d).

$$\text{Point to plane: } D = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}, \quad \text{Parallel planes } ax+by+cz+d_1 = 0, ax+by+cz+d_2 = 0$$

Both follow one recipe — “plane expression at a point, over the length of the normal” — the parallel-plane case being the point formula applied to any point of the second plane.

Notation key: $\vec{n}$ (or a, b, c) = normal vector / its direction ratios; $\hat{n}$ = unit normal; p = distance of plane from origin (normal form); d = constant in $\vec{r} \cdot \vec{n} = d$; $\vec{a}$ = position vector of a known point on the plane; $\vec{b}$ = direction vector of a line; θ = angle between planes; φ = angle between a line and a plane; λ = family parameter.

Task	Key object	Formula
Plane from point + normal	$\vec{n}$	$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$
Plane from 3 points	$(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})$	determinant = 0
Plane-plane angle	normals	$\cos \theta = \frac{ \vec{n}_1 \cdot \vec{n}_2 }{ \vec{n}_1 \vec{n}_2 }$
Line-plane angle	$\vec{b}, \vec{n}$	$\sin \varphi = \frac{ \vec{b} \cdot \vec{n} }{ \vec{b} \vec{n} }$ (sine!)
Point-plane distance	normal	$\frac{ ax_1 + by_1 + cz_1 + d }{\sqrt{a^2 + b^2 + c^2}}$

Section 3 — Theorems & Procedures

Theorem 1 (point-normal equation of a plane). The plane through $\vec{a}$ with normal $\vec{n}$ is $(\vec{r} - \vec{a}) \cdot \vec{n} = 0$.

Figure. A flat plane with a normal vector $\vec{n}$ drawn perpendicular to it at a base point A (position vector $\vec{a}$). A general point R (position vector $\vec{r}$) on the plane, with the in-plane vector $(\vec{r} - \vec{a})$ drawn from A to R, shown at a right angle to $\vec{n}$.

Proof. A point $\vec{r}$ lies on the plane iff the displacement $\vec{r} - \vec{a}$ from the known point $\vec{a}$ stays within the surface — i.e. is perpendicular to the normal $\vec{n}$. Perpendicularity is exactly $(\vec{r} - \vec{a}) \cdot \vec{n} = 0$. Expanding, $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$, a constant we call d ; in coordinates with $\vec{n} = (a, b, c)$ this is $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$, i.e. $ax + by + cz = d$. ■

Procedure — plane through three non-collinear points. Given A, B, C : (1) form two in-plane edge vectors $\vec{AB} = \vec{b} - \vec{a}$ and $\vec{AC} = \vec{c} - \vec{a}$; (2) their cross product $\vec{n} = \vec{AB} \times \vec{AC}$ is normal to both, hence to the plane; (3) apply Theorem 1 with this $\vec{n}$ and any one of the points. The compact determinant form packages all three steps, because expanding $(\vec{r} - \vec{a}) \cdot [(\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})]$ is precisely that 3×3 determinant set to zero.

Theorem 2 (point-plane distance). The distance of (x_1, y_1, z_1) from $ax + by + cz + d = 0$ is $\frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$.

Proof. Drop the perpendicular from the point $P(x_1, y_1, z_1)$ to the plane, meeting it at foot F . The vector $\vec{FP}$ is parallel to the normal $\vec{n} = (a, b, c)$. The distance $|\vec{FP}|$ equals the absolute projection of $\vec{P} - \vec{F}_0$ onto the unit normal, where F_0 is any point of the plane (so $aF_{0x} + bF_{0y} + cF_{0z} = -d$). Computing that projection: $\frac{|(x_1 - F_{0x})a + (y_1 - F_{0y})b + (z_1 - F_{0z})c|}{\sqrt{a^2 + b^2 + c^2}} = \frac{|ax_1 + by_1 + cz_1 - (aF_{0x} + bF_{0y} + cF_{0z})|}{\sqrt{a^2 + b^2 + c^2}} = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$. ■

Why the line-plane angle is a sine (the key subtlety). The angle α between a line (direction $\vec{b}$) and the plane's normal $\vec{n}$ satisfies $\cos \alpha = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}||\vec{n}|}$. But the angle φ we want is between the line and the *surface*, and the surface is 90° from the normal, so $\varphi = 90^\circ - \alpha$. Therefore $\sin \varphi = \cos \alpha = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}||\vec{n}|}$. The plane equation is steered by its normal, but the reported angle is measured against the surface — hence the swap from cosine to sine.

Procedure — family of planes. To find a plane through the intersection of $P_1 = 0$ and $P_2 = 0$ meeting one extra condition: (1) write $P_1 + \lambda P_2 = 0$, which is automatically satisfied by every point on the intersection line for *any* λ ; (2) impose the extra condition (a point lying on it $\Rightarrow$ substitute and solve for λ ; perpendicular to a given plane $\Rightarrow$ set the dot product of normals to zero; parallel to an axis $\Rightarrow$ set that coefficient to zero); (3) back-substitute λ and clear fractions.

Theorem 3 (angle between two planes equals the angle between their normals). $\cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}$.

Proof. Two planes meet along a line. Pick a point on that line and, within each plane, draw the ray perpendicular to the line of intersection; the angle between these two rays is the dihedral angle between the planes. Now rotate each ray by 90° inside its own plane — each becomes that plane's normal. Rotating both arms of an angle through the same amount leaves the angle unchanged, so the angle between the normals equals the dihedral angle between the planes. The modulus picks out the acute value, since either normal may be reversed. ■ Two corollaries fall out immediately: the planes are *perpendicular* when $\vec{n}_1 \cdot \vec{n}_2 = 0$ and *parallel* when $\vec{n}_1 \parallel \vec{n}_2$ (their coefficients proportional) — once again, the normals decide everything.

Intercept form needs no separate derivation. A plane meeting the axes at $(a, 0, 0)$, $(0, b, 0)$, $(0, 0, c)$ passes through those three points, so its equation must hold at each. The form $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ does exactly that: at $(a, 0, 0)$ it reads $1 + 0 + 0 = 1$, and likewise for the others. So the intercept form is just the three-point construction with especially convenient points.

Procedure — foot of perpendicular and image of a point in a plane. To find the foot F of the perpendicular from $P(x_1, y_1, z_1)$ to the plane $ax + by + cz + d = 0$: (1) the perpendicular line from P has the plane's normal as its direction, so write $F = (x_1 + at, y_1 + bt, z_1 + ct)$; (2) demand that F satisfy the plane equation and solve the single linear equation for the parameter t ; (3) substitute t back to locate F . The **image** P' of P in the plane is then found by treating F as the midpoint of PP' , giving $P' = 2F - P$. The same normal-direction idea powers reflection problems and the perpendicular distance, which is simply $|PF|$. The procedure works because the perpendicular from a point to a plane is unique and must run

along the normal — so searching only along the normal direction is guaranteed to find the closest point, and reflecting through that foot lands the mirror image.

Section 4 — Worked Examples

Example 1 — [JEE Main · Foundational warm-up] (Forms + distance). Find the equation of the plane through the point $(2, -1, 3)$ with normal $\vec{n} = \hat{i} + 2\hat{j} - 2\hat{k}$, and its perpendicular distance from the origin.

Point-normal form:

$$1(x - 2) + 2(y + 1) - 2(z - 3) = 0 \Rightarrow x + 2y - 2z + 6 = 0.$$

Distance of the origin $(0, 0, 0)$:

$$D = \frac{|0 + 0 - 0 + 6|}{\sqrt{1 + 4 + 4}} = \frac{6}{3} = 2 \text{ units.}$$

The same answer falls out of the normal form. Dividing the plane $x + 2y - 2z = -6$ by $|\vec{n}| = 3$ gives $\frac{x+2y-2z}{3} = -2$; flipping signs so that the right-hand side is non-negative yields $\vec{r} \cdot \hat{n} = 2$ with $\hat{n} = \frac{1}{3}(-1, -2, 2)$, confirming the distance from the origin is $p = 2$.

Example 2 — [CUET / JEE Main Speed Trap] (Line-plane angle, the sine trap). Find the angle between the line $\frac{x-1}{2} = \frac{y}{-1} = \frac{z+2}{2}$ and the plane $3x - 6y + 2z = 7$.

Line direction $\vec{b} = (2, -1, 2)$; plane normal $\vec{n} = (3, -6, 2)$.

$$\vec{b} \cdot \vec{n} = (2)(3) + (-1)(-6) + (2)(2) = 6 + 6 + 4 = 16, \quad |\vec{b}| = 3, \quad |\vec{n}| = 7.$$

Because this is a **line and a plane**, use sine, not cosine:

$$\sin \varphi = \frac{|16|}{3 \cdot 7} = \frac{16}{21} \Rightarrow \varphi = \sin^{-1}\left(\frac{16}{21}\right).$$

The trap: the quantity $\frac{16}{21}$ is the cosine of the angle the line makes with the *normal*. Students who write $\cos^{-1}\left(\frac{16}{21}\right)$ have reported the angle with the normal, not with the plane — off by exactly 90° . Whenever a *line* meets a *plane*, reach for sine. A built-in safeguard: $\sin \varphi = \frac{16}{21} < 1$, as any genuine sine must be; had we divided by only one of the magnitudes the result would have exceeded 1, instantly flagging the slip.

Example 3 — [JEE Main · Multi-concept] (Family of planes through a point). Find the equation of the plane through the line of intersection of $x + y + z = 6$ and $2x + 3y + 4z + 5 = 0$, passing through $(1, 1, 1)$.

Write the family $(x + y + z - 6) + \lambda(2x + 3y + 4z + 5) = 0$. Impose the point $(1, 1, 1)$:

$$(1 + 1 + 1 - 6) + \lambda(2 + 3 + 4 + 5) = 0 \Rightarrow -3 + 14\lambda = 0 \Rightarrow \lambda = \frac{3}{14}.$$

Substitute and clear the fraction (multiply through by 14):

$$14(x + y + z - 6) + 3(2x + 3y + 4z + 5) = 0 \Rightarrow 20x + 23y + 26z - 69 = 0.$$

Check: at $(1, 1, 1)$, $20 + 23 + 26 - 69 = 0$ ✓. The multi-concept thread: the family construction avoids ever computing the intersection line, collapsing the whole problem to one linear

equation in λ . Because the family already encodes “through the intersection line,” the single point condition fully determines the plane — no system of simultaneous equations and no parametrisation of the line is ever required.

Example 4 — [JEE Advanced] (Family + perpendicularity). Find the plane through the line of intersection of $2x + y - z = 3$ and $x - y + 2z = 1$ that is perpendicular to the plane $3x + 2y + z = 0$.

Family: $(2x + y - z - 3) + \lambda(x - y + 2z - 1) = 0$, with normal

$$\vec{N} = (2 + \lambda, 1 - \lambda, -1 + 2\lambda).$$

Perpendicular to the plane $3x + 2y + z = 0$ means $\vec{N} \cdot (3, 2, 1) = 0$:

$$3(2 + \lambda) + 2(1 - \lambda) + 1(-1 + 2\lambda) = 0 \Rightarrow 6 + 3\lambda + 2 - 2\lambda - 1 + 2\lambda = 0 \Rightarrow 7 + 3\lambda = 0 \Rightarrow \lambda = -\frac{7}{3}.$$

Substitute and multiply by -3 :

$$(2 - \frac{7}{3})x + (1 + \frac{7}{3})y + (-1 - \frac{14}{3})z + (-3 + \frac{7}{3}) = 0 \Rightarrow x - 10y + 17z + 2 = 0.$$

Verification: its normal $(1, -10, 17)$ satisfies $(1)(3) + (-10)(2) + (17)(1) = 3 - 20 + 17 = 0$ ✓ — genuinely perpendicular to $3x + 2y + z = 0$. The advanced idea: “perpendicular to a plane” becomes a single dot-product condition on the family normal, turning a geometric demand into one equation in λ . Note that one condition yields exactly one plane: the family carries a single free parameter, so a single linear constraint generically leaves one surviving value of λ . Two independent conditions would over-determine the pencil and usually have no solution at all.

Section 5 — Practice Exercises

1. **[CUET / JEE Main]** Find the equation of the plane through $(1, 0, -2)$ with normal $2\hat{i} - \hat{j} + 3\hat{k}$.
2. **[JEE Main]** Find the intercepts of the plane $3x - 2y + 6z = 12$ on the axes and its distance from the origin.
3. **[JEE Main]** Find the angle between the planes $2x - y + 2z = 5$ and $3x + 6y - 2z = 7$.
4. **[JEE Advanced]** Find the plane through the line of intersection of $x + y + z = 1$ and $2x + 3y - z + 4 = 0$ that is parallel to the x-axis.
5. **[JEE Advanced]** Find the distance between the parallel planes $2x - y + 2z = 6$ and $4x - 2y + 4z = 9$.

Answers:

1. $2x - y + 3z + 4 = 0$
2. Intercepts 4, -6 , 2; distance $\frac{12}{7}$ units
3. $\cos^{-1}(\frac{4}{21})$
4. $y - 3z + 6 = 0$
5. $\frac{1}{2}$ unit

Section 6 — Crack the MCQ

Q1. The equation of the plane through $(1, 2, 3)$ and parallel to the plane $2x - 3y + 4z = 10$ is

(A) $2x - 3y + 4z = 8$ (B) $2x - 3y + 4z = 10$ (C) $x + 2y + 3z = 14$ (D) $2x - 3y + 4z = 0$

Answer: (A). Parallel planes share a normal, so use $\vec{n} = (2, -3, 4)$: $2(x-1) - 3(y-2) + 4(z-3) = 0 \Rightarrow 2x - 3y + 4z = 8$.

(B) just repeats the given plane — but that one doesn't pass through $(1, 2, 3)$ (it gives $2 - 6 + 12 = 8 \neq 10$). (C) wrongly uses the *point* as the normal. (D) keeps the right normal but drops the constant, forcing the plane through the origin instead.

Q2. The angle between the line $\vec{r} = (\hat{i} + \hat{j}) + \lambda(2\hat{i} + 2\hat{j} + \hat{k})$ and the plane $x + 2y + 2z = 5$ is

(A) $\sin^{-1}(\frac{8}{9})$ (B) $\cos^{-1}(\frac{8}{9})$ (C) $\sin^{-1}(\frac{8}{3})$ (D) $90^\circ - \sin^{-1}(\frac{8}{9})$

Answer: (A). $\vec{b} = (2, 2, 1)$, $\vec{n} = (1, 2, 2)$, $\vec{b} \cdot \vec{n} = 2 + 4 + 2 = 8$, $|\vec{b}| = |\vec{n}| = 3$, so $\sin \varphi = \frac{8}{9}$.

(B) is the line-to-normal angle — the cosine trap. (C) forgets to multiply the magnitudes in the denominator, producing $\sin > 1$. (D) applies the complement twice, landing back on the angle with the normal.

Q3. The distance of the point $(3, -2, 1)$ from the plane $2x - 2y + z + 3 = 0$ is

(A) $\frac{14}{3}$ (B) 14 (C) 2 (D) $\frac{14}{9}$

Answer: (A). $\frac{|2(3) - 2(-2) + 1 + 3|}{\sqrt{4 + 4 + 1}} = \frac{|6 + 4 + 1 + 3|}{3} = \frac{14}{3}$.

(B) forgets to divide by $|\vec{n}| = 3$. (C) mishandles the sign on the y -term ($-2(-2)$ read as -4 , giving 6 in the numerator). (D) divides by $|\vec{n}|^2 = 9$ instead of $|\vec{n}| = 3$.

Q4. Every plane of the family $(x + y + z - 1) + \lambda(2x + y - z - 2) = 0$ passes through

(A) $(1, 0, 0)$ (B) the origin (C) $(0, 1, 0)$ (D) $(1, 1, 1)$

Answer: (A). A family $P_1 + \lambda P_2 = 0$ passes, for *all* λ , exactly through the line where $P_1 = 0$ and $P_2 = 0$ meet. The point must satisfy both: at $(1, 0, 0)$, $1 + 0 + 0 - 1 = 0$ and $2 + 0 - 0 - 2 = 0$ ✓.

(B) gives $-1 - 2\lambda$, zero only for one specific λ . (C) gives -2λ . (D) gives 2, never zero. Each fails to lie on *both* original planes, so no single point of theirs works for every λ .

Section 7 — Common Pitfalls & Pro-Tips

- **Using cosine for the line-plane angle.** The line-plane angle is the complement of the line-normal angle, so it uses **sine**: $\sin \varphi = \frac{|\vec{b}|}{|\vec{n}|}$.
- **Swapping the point and the normal.** In $a(x-x_1) + \dots$, the coefficients (a, b, c) are the *normal*, while (x_1, y_1, z_1) is the *point* — never the reverse.
- **Forgetting to normalise in the normal form.** $\vec{r} \cdot \hat{n} = p$ gives p as the distance from the origin only if $\hat{n}$ is a unit vector.
- **Dropping $\sqrt{a^2 + b^2 + c^2}$ in the distance.** Distance is the plane expression evaluated at the point, *divided by the length of the normal*.
- **Comparing un-scaled parallel planes.** Before using $\frac{|d_1 - d_2|}{\sqrt{\frac{a^2 + b^2 + c^2}{2}}}$, scale both planes to *identical* normals; $2x - y + 2z = 6$ and $4x - 2y + 4z = 9$ must first be matched.

- **Solving for the intersection line in family problems.** You never need it — write $P_1 + \lambda P_2 = 0$ and impose the single extra condition.
- **Assuming the family captures every plane.** The form $P_1 + \lambda P_2 = 0$ can never reproduce itself (no finite λ gives it). If a problem's correct answer happens to be $P_1 + \lambda P_2 = 0$, the symmetric form $\mu P_1 + P_2 = 0$ recovers it.

Pro-Tip (the normal is everything): almost every plane question reduces to “find the normal.” Forms come from a normal; the plane-plane angle is the normal-normal angle; distance divides by the normal's length. Identify the normal first and the rest follows — even the family method is, at heart, a search for the right normal among a one-parameter pencil.

Pro-Tip (sine vs cosine memory hook): “Plane-to-plane, cosine; line-to-plane, sine.” Two surfaces compare like two arrows (cosine); a line piercing a surface is measured against the flat, which flips cosine to sine.

Pro-Tip (which side, for free): to decide whether two points lie on the same side of a plane, substitute both into $ax + by + cz + d$ and compare the signs — same sign means same side, opposite signs mean the plane separates them and any segment between them is cut by the plane. No extra computation beyond the distance you were already evaluating.

Section 8 — Cheat Sheet Box

Forms (03a)

- Point-normal: $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$, i.e. $\vec{r} \cdot \vec{n} = d$.
- Three points: determinant of $\{\vec{r} - \vec{a}; \vec{b} - \vec{a}; \vec{c} - \vec{a}\} = 0$; normal $= (\vec{b} - \vec{a}) \times (\vec{c} - \vec{a})$.
- Intercept: $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$. Normal form: $\vec{r} \cdot \hat{n} = p$ ($\hat{n}$ unit).

Family (03b): $P_1 + \lambda P_2 = 0$; fix λ with one extra condition.

Angles (03c): plane-plane $\cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}$; line-plane $\sin \varphi = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}||\vec{n}|}$ (**sine!**).

Distances (03d): point-plane $\frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$; parallel planes $\frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}$
(equal normals first).

Memory aids

- “A point and a normal make a plane.”
- “Plane-to-plane cosine, line-to-plane sine.”
- “Family first, line never” — use $P_1 + \lambda P_2 = 0$ instead of solving for the intersection.
- The sign of $ax_1 + by_1 + cz_1 + d$ tells you which side of the plane the point is on.

The Perpendicular Toolkit — Distances, Feet & Images

Class 12 · Mathematics · Chapter 11 — Three Dimensional Geometry · Subtopic 04

Exam Relevance: One of the most question-dense corners of the chapter. “Foot of perpendicular,” “image of a point,” and “perpendicular distance” appear in CBSE Boards (point-to-line) and almost every JEE Main paper, with JEE Advanced combining images and feet into multi-step locus and reflection problems. Master one idea — *drop a perpendicular* — and four formula families fall out at once.

This **merged subtopic** has two halves bound by a single idea: **(04a)** dropping a perpendicular from a point to a *line*, and **(04b)** dropping one to a *plane*. In both, the foot, the distance, and the mirror image are three answers to the same construction.

Section 1 — Concept Intuition

Almost every “shortest”, “nearest”, “foot”, or “image” question in this chapter is the same physical act: stand at a point P off to one side, and drop a perpendicular onto a line or a plane. Three things appear the instant you do:

- the **foot** F — where the perpendicular lands,
- the **distance** — the length $|PF|$ of that perpendicular,
- the **image** P' — the mirror reflection of P , found by continuing the perpendicular an equal distance through to the other side.

These are not three separate topics; they are three readings of one drawing. Find F and you immediately get the distance ($|PF|$) and the image ($P' = 2F - P$, because F is the midpoint of P and its reflection). Which of the three a question asks for is almost cosmetic — the work is identical up to the last line. That is the single most reassuring fact about this subtopic: a dozen differently worded problems collapse to one routine, so your practice compounds quickly.

Dropping onto a line. Take a line $\vec{r} = \vec{a} + \lambda\vec{b}$. From the known point $\vec{a}$ on the line, draw the arrow $\vec{AP} = \vec{P} - \vec{a}$ to your external point. Now split that arrow into two parts: the part running *along* the line and the part sticking *out* perpendicular to it. The along-part has length equal to the projection of $\vec{AP}$ onto the direction $\vec{b}$; sliding that far from $\vec{a}$ lands you exactly at the foot F . The out-part is the perpendicular distance — and the cleanest way to grab just that perpendicular piece is the cross product, since $|\vec{AP} \times \vec{b}| = |\vec{AP}||\vec{b}|\sin\theta$ already isolates the perpendicular component.

Think about it this way... the dot product hands you the *shadow along* the line (giving the foot), while the cross product hands you the *height above* the line (giving the distance). Two products, two halves of the same right triangle.

To make that split concrete, picture the arrow $\vec{AP}$ resting against the line at A . Any vector can be resolved into a piece pointing **along** a chosen direction and a piece pointing **across** it. The along-piece is $(\vec{AP} \cdot \hat{b})\hat{b}$, where $\hat{b}$ is the unit direction — its length is the dot product with the unit vector, exactly the amount you must travel from A to sit directly “below” P . That landing spot is the foot F . Whatever is left over, $\vec{AP} - (\vec{AP} \cdot \hat{b})\hat{b}$, points straight out from the line to P , and its length is the perpendicular distance. The cross product is just a shortcut to that leftover length, because $|\vec{AP} \times \vec{b}|$ already strips out the along-component and keeps only the across-part (scaled by $|\vec{b}|$, which is why we divide by $|\vec{b}|$ once at the end). So the dot and the cross are not two unrelated formulas — they are the two legs of one right-angled decomposition of $\vec{AP}$.

Dropping onto a plane. A plane is even friendlier, because the perpendicular direction is handed to you for free: it is the plane's normal $\vec{n}$. So from P you simply step along $\vec{n}$ until you hit the plane. How far to step is fixed by forcing the landing point to satisfy the plane equation, which gives a single number $t = \frac{ax_1 + by_1 + cz_1 + d}{a^2 + b^2 + c^2}$. Step t along the normal to reach the foot, step $2t$ to reach the image, and the distance is $|t|$ times the normal's length.

Think about it this way... for a plane you never resolve anything — the normal is the perpendicular direction, so the entire job is “how many normals away is the plane?” That number is t . There is nothing to decompose and no angle to chase; the plane case is genuinely easier than the line case, and recognising that asymmetry up front stops students from over-engineering plane problems with cross products they do not need.

Images are worth more than they first appear, because of a classic reflection principle. The shortest path from a point A to a point B that must first *touch* a given plane (or line) is found by reflecting one point across the surface: the straight segment from A to the image B' crosses the surface at the optimal touch-point, and its length is the shortest possible total path. This is why “image of a point” questions are not idle exercises — they are the engine behind shortest-path, ray-reflection, and minimum-distance problems that JEE Advanced dresses up in physical language. Knowing that the foot is the midpoint of a point and its image lets you convert any such problem into a single straight-line distance.

Limiting conditions & assumptions:

- The point-to-line distance $\frac{|\vec{AP} \times \vec{b}|}{|\vec{b}|}$ requires $\vec{b} \neq \vec{0}$; the division by $|\vec{b}|$ (not $|\vec{b}|^2$) is essential because the cross product already carries one factor of $|\vec{b}|$.
- The image satisfies $P' = 2F - P$ only because F is the *midpoint* of PP' — true for reflection in both a line and a plane.
- For the plane formulas, write the plane as $ax + by + cz + d = 0$ (everything on one side); a constant on the wrong side flips the sign of t and sends the foot the wrong way.
- If P already lies *on* the line or plane, the distance is zero, the foot is P itself, and the image coincides with P — every formula above still returns these degenerate answers automatically (λ_0 lands on P ; $t = 0$), so no special-casing is needed.

The unifying picture to carry away: a foot is a *point*, a distance is a *length*, and an image is the foot's reflection of the original point — and the only thing that differs between the line case and the plane case is how you locate the foot. For a line you must extract the perpendicular direction yourself (dot for along, cross for across); for a plane the perpendicular direction is gift-wrapped as the normal. Everything downstream of the foot is identical.

Section 2 — Key Definitions & Properties

Distances are lengths; feet and images are points. The four formula families below all descend from one construction — dropping a perpendicular — so rather than memorising eight separate things, anchor on two questions: *where does the perpendicular land* (the foot) and *how long is it* (the distance); the image is then always $2F - P$.

Point to a line (04a). Line $\vec{r} = \vec{a} + \lambda\vec{b}$, external point P (position vector $\vec{p}$), $\vec{AP} = \vec{p} - \vec{a}$:

$$\textbf{Foot: } \lambda_0 = \frac{\vec{AP} \cdot \vec{b}}{|\vec{b}|^2}, \quad F = \vec{a} + \lambda_0 \vec{b}; \quad \textbf{Distance: } PF = \frac{|\vec{AP} \times \vec{b}|}{|\vec{b}|}; \quad \textbf{Image: } P' = 2F - P.$$

Point to a plane (04b). Plane $ax + by + cz + d = 0$, normal $\vec{n} = (a, b, c)$, point $P(x_1, y_1, z_1)$,

with $t = \frac{ax_1 + by_1 + cz_1 + d}{a^2 + b^2 + c^2}$:

Distance: $\frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$; **Foot:** $F = P - t\vec{n}$; **Image:** $P' = P - 2t\vec{n}$.

In symmetric notation the foot and image lie along the normal through P :

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c} = -t \text{ (foot)}, \quad = -2t \text{ (image)}.$$

Notation key: $\vec{a}$ = a known point on the line; $\vec{b}$ = line direction; $\vec{AP} = \vec{p} - \vec{a}$; λ_0 = parameter of the foot on the line; F = foot of perpendicular; P' = image (reflection); $\vec{n} = (a, b, c)$ = plane normal; t = signed number of normals from P to the plane.

Drop onto	Foot	Distance	Image
Line	$\vec{a} + \frac{\vec{AP} \cdot \vec{b}}{ \vec{b} ^2} \vec{b}$	$\frac{ \vec{AP} \times \vec{b} }{ \vec{b} }$	$2F - P$
Plane	$P - t\vec{n}$	$\frac{ ax_1 + by_1 + cz_1 + d }{\sqrt{a^2 + b^2 + c^2}}$	$P - 2t\vec{n}$

Section 3 — Theorems & Procedures

Procedure & Theorem 1 (foot and distance, point to a line). The foot of the perpendicular from P to $\vec{r} = \vec{a} + \lambda\vec{b}$ is $F = \vec{a} + \lambda_0\vec{b}$ with $\lambda_0 = \frac{\vec{AP} \cdot \vec{b}}{|\vec{b}|^2}$, and $PF = \frac{|\vec{AP} \times \vec{b}|}{|\vec{b}|}$.

Figure. A line with direction vector $\vec{b}$ through point A . An external point P . The perpendicular dropped from P meets the line at foot F . The vector $\vec{AP}$ is split into its component along $\vec{b}$ (from A to F) and its perpendicular component (from F to P , the distance). The image P' is shown on the far side of the line with F as the midpoint of PP' .

Proof. A general point on the line is $\vec{a} + \lambda\vec{b}$, so $\vec{FP} = \vec{p} - (\vec{a} + \lambda\vec{b}) = \vec{AP} - \lambda\vec{b}$. The foot is where this is perpendicular to the line, i.e. $\vec{FP} \cdot \vec{b} = 0$: $(\vec{AP} - \lambda\vec{b}) \cdot \vec{b} = 0 \Rightarrow \vec{AP} \cdot \vec{b} - \lambda|\vec{b}|^2 = 0 \Rightarrow \lambda_0 = \frac{\vec{AP} \cdot \vec{b}}{|\vec{b}|^2}$. Substituting gives F . For the distance, decompose $\vec{AP}$ into parts parallel and perpendicular to $\vec{b}$; the perpendicular part has length $|\vec{AP}| \sin \theta$, where θ is the angle between $\vec{AP}$ and $\vec{b}$. Since $|\vec{AP} \times \vec{b}| = |\vec{AP}||\vec{b}| \sin \theta$, dividing by $|\vec{b}|$ isolates exactly that perpendicular length: $PF = \frac{|\vec{AP} \times \vec{b}|}{|\vec{b}|}$. ■

Theorem 2 (foot and image, point to a plane). For $P(x_1, y_1, z_1)$ and plane $ax + by + cz + d = 0$, the foot is $F = P - t\vec{n}$ and the image is $P' = P - 2t\vec{n}$, with $t = \frac{ax_1 + by_1 + cz_1 + d}{a^2 + b^2 + c^2}$.

Proof. The perpendicular from P to the plane runs along the normal $\vec{n} = (a, b, c)$, so any point on it is $P - s\vec{n} = (x_1 - sa, y_1 - sb, z_1 - sc)$. It meets the plane when it satisfies

$a(x_1 - sa) + b(y_1 - sb) + c(z_1 - sc) + d = 0$, i.e. $(ax_1 + by_1 + cz_1 + d) - s(a^2 + b^2 + c^2) = 0$, giving $s = t = \frac{ax_1 + by_1 + cz_1 + d}{a^2 + b^2 + c^2}$. Hence the foot $F = P - t\vec{n}$. The image is the reflection, reached by stepping twice as far: $P' = P - 2t\vec{n}$, which automatically makes F the midpoint of PP' . The distance $PF = |t||\vec{n}| = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$. ■

The single number t deserves emphasis because it is the hinge of the entire plane case. It answers “how many copies of the normal vector must I lay end-to-end, starting at P , to reach the plane?” Once you know that count, the foot is P minus that many normals, the image is P minus *twice* that many (overshooting symmetrically to the far side), and the distance is that count times the normal’s length. Three distinct geometric objects — a point on the plane, a mirrored point, and a length — all fall out of one scalar, which is why examiners can ask for any of them with equal ease and why computing t first is always the efficient opening move.

Procedure — image of a point (either case), in three steps. (1) Find the foot F by the appropriate method above. (2) Use F as the midpoint of P and its image. (3) Solve $F = \frac{1}{2}(P + P')$ for $P' = 2F - P$. The same three steps reflect a point in a line and in a plane; only the foot-finding differs. For instance, if a computation gives foot $F = (2, 1, 0)$ for an original point $P = (0, 3, 4)$, the image is read off immediately as $P' = 2(2, 1, 0) - (0, 3, 4) = (4, -1, -4)$, no further geometry required — the midpoint relation does all the work.

Theorem 3 (two equivalent routes to the point-line distance). *The distance PF may be found either as $|P - F|$ after locating the foot, or directly as $\frac{|\vec{AP} \times \vec{b}|}{|\vec{b}|}$ without ever finding the foot.*

Proof. The first route is definitional: once $F = \vec{a} + \lambda_0 \vec{b}$ is known, $PF = |P - F|$ is an ordinary distance. The second route avoids F entirely. From the decomposition $\vec{AP} = \vec{AP}_{\parallel} + \vec{AP}_{\perp}$ into components along and across $\vec{b}$, the perpendicular distance is $|\vec{AP}_{\perp}| = |\vec{AP}| \sin \theta$, and since $|\vec{AP} \times \vec{b}| = |\vec{AP}||\vec{b}| \sin \theta$, dividing by $|\vec{b}|$ returns exactly $|\vec{AP}| \sin \theta$. The two routes must agree, which gives a built-in check: if $|P - F|$ and $\frac{|\vec{AP} \times \vec{b}|}{|\vec{b}|}$ disagree, an arithmetic slip has occurred. In exams, the cross-product route is faster when *only* the distance is wanted; the foot route is unavoidable when the foot or image is also required. ■

Section 4 — Worked Examples

Example 1 — [CBSE Board Level] (Foot, distance, image — point to a line). *Find the foot of the perpendicular, the distance, and the image of $P(2, 4, 1)$ in the line $\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$.*

Here $\vec{a} = (1, 2, 3)$, $\vec{b} = (1, -1, 1)$, $|\vec{b}|^2 = 3$, and $\vec{AP} = (1, 2, -2)$.

$$\lambda_0 = \frac{\vec{AP} \cdot \vec{b}}{|\vec{b}|^2} = \frac{1 - 2 - 2}{3} = -1 \Rightarrow F = \vec{a} - \vec{b} = (0, 3, 2).$$

Distance: $\vec{AP} \times \vec{b} = (1, 2, -2) \times (1, -1, 1) = (0, -3, -3)$, so $PF = \frac{|(0, -3, -3)|}{\sqrt{3}} = \frac{3\sqrt{2}}{\sqrt{3}} = \sqrt{6}$. Image: $P' = 2F - P = (0 - 2, 6 - 4, 4 - 1) = (-2, 2, 3)$. (Check: $\overrightarrow{PF} = (-2, -1, 1)$)

and $\overrightarrow{PF} \cdot \vec{b} = -2 + 1 + 1 = 0$ ✓ — genuinely perpendicular.) Two habits make this example a template for all the rest. First, compute λ_0 *before* anything else; it converts the abstract “somewhere on the line” into the concrete foot F , after which distance and image are trivial. Second, always run the perpendicularity check $\overrightarrow{PF} \cdot \vec{b} = 0$ — it costs one line and catches almost every arithmetic slip in λ_0 , because a wrong foot will fail it immediately.

Example 2 — [CUET / JEE Main Speed Trap] (Point-line distance). Find the distance of $(1, 0, 2)$ from the line through $(3, 1, 1)$ with direction $(2, 2, 1)$.

$$\vec{AP} = (1 - 3, 0 - 1, 2 - 1) = (-2, -1, 1), \vec{b} = (2, 2, 1).$$

$$\vec{AP} \times \vec{b} = (-3, 4, -2), \quad |\vec{AP} \times \vec{b}| = \sqrt{9 + 16 + 4} = \sqrt{29}, \quad |\vec{b}| = 3.$$

$$\text{Distance} = \frac{\sqrt{29}}{3}.$$

The trap: the denominator is $|\vec{b}| = 3$, **not** $|\vec{b}|^2 = 9$ and certainly not absent. The cross product already contains one factor of $|\vec{b}|$, so you divide by exactly one more. Students who write $\frac{\sqrt{29}}{9}$ over-divided; those who write $\sqrt{29}$ forgot the denominator entirely. A reliable way to never misremember this: dimensionally, $|\vec{AP} \times \vec{b}|$ behaves like (length) $\times$ (length of $\vec{b}$), so to recover a pure length you must cancel one power of $|\vec{b}|$ — one division, never two.

Example 3 — [JEE Main · Multi-concept] (Foot, distance, image — point to a plane). Find the foot of the perpendicular, the distance, and the image of $P(4, 1, 3)$ in the plane $x - 2y + 2z - 5 = 0$.

Normal $\vec{n} = (1, -2, 2)$, $|\vec{n}|^2 = 9$. Compute t :

$$t = \frac{(4) - 2(1) + 2(3) - 5}{9} = \frac{4 - 2 + 6 - 5}{9} = \frac{3}{9} = \frac{1}{3}.$$

Foot: $F = P - t\vec{n} = (4 - \frac{1}{3}, 1 + \frac{2}{3}, 3 - \frac{2}{3}) = (\frac{11}{3}, \frac{5}{3}, \frac{7}{3})$. Distance: $|t| |\vec{n}| = \frac{1}{3} \cdot 3 = 1$ unit (equivalently $\frac{|3|}{\sqrt{9}} = 1$). Image: $P' = P - 2t\vec{n} = (4 - \frac{2}{3}, 1 + \frac{4}{3}, 3 - \frac{4}{3}) = (\frac{10}{3}, \frac{7}{3}, \frac{5}{3})$. Three answers from one number t — the multi-concept efficiency the examiners are testing. Observe the internal consistency: the foot is one normal-step ($-t\vec{n}$) from P , the image is two ($-2t\vec{n}$), so the foot is automatically the midpoint of P and P' — you can confirm this in your head by averaging P and P' coordinatewise and recovering F . A neat sign-check: $t = \frac{1}{3}$ is positive, meaning P sits on the *positive* side of the plane (where $x - 2y + 2z - 5 > 0$), so the correcting step $-t\vec{n}$ moves P *toward* the plane — exactly as it must.

Example 4 — [JEE Advanced] (Image of a point in a line). Find the image of $P(1, 6, 3)$ in the line $\vec{r} = (\hat{j} + 2\hat{k}) + \lambda(\hat{i} + 2\hat{j} + 3\hat{k})$.

$$\vec{a} = (0, 1, 2), \vec{b} = (1, 2, 3), |\vec{b}|^2 = 14, \vec{AP} = (1, 5, 1).$$

$$\lambda_0 = \frac{\vec{AP} \cdot \vec{b}}{|\vec{b}|^2} = \frac{1 + 10 + 3}{14} = \frac{14}{14} = 1 \Rightarrow F = \vec{a} + \vec{b} = (1, 3, 5).$$

Image: $P' = 2F - P = (2 - 1, 6 - 6, 10 - 3) = (1, 0, 7)$. Verification: $\overrightarrow{PF} = (0, -3, 2)$, and $\overrightarrow{PF} \cdot \vec{b} = 0 - 6 + 6 = 0$ ✓; the midpoint of $P(1, 6, 3)$ and $P'(1, 0, 7)$ is $(1, 3, 5) = F$ ✓. The clean integer $\lambda_0 = 1$ is the hallmark of a well-posed reflection problem. When the algebra is heavier than this — large $|\vec{b}|^2$, fractional λ_0 — the method does not change one bit; only the arithmetic does. That invariance is the point of practising on clean numbers first:

the *procedure* (foot via projection, image via $2F - P$) is identical whether λ_0 is 1 or $\frac{17}{43}$, so once the steps are automatic, messy coefficients become a calculator exercise rather than a conceptual one.

Section 5 — Practice Exercises

1. **[CBSE]** Find the foot of the perpendicular from $P(1, 2, 3)$ to the line $\vec{r} = (6\hat{i} + 7\hat{j} + 7\hat{k}) + \lambda(3\hat{i} + 2\hat{j} - 2\hat{k})$.
2. **[CUET / JEE Main]** Find the perpendicular distance of $(2, 3, -1)$ from the line through the origin with direction $(1, 2, 2)$.
3. **[JEE Main]** Find the distance and the foot of the perpendicular from $(3, -2, 1)$ to the plane $2x - y + 2z + 3 = 0$.
4. **[JEE Main]** Find the image of the point $(1, 3, 4)$ in the plane $2x - y + z + 3 = 0$.
5. **[JEE Advanced]** Find the image of $(2, -1, 5)$ in the line $\vec{r} = (11\hat{i} - 2\hat{j} - 8\hat{k}) + \lambda(10\hat{i} - 4\hat{j} - 11\hat{k})$.

Answers:

1. $(3, 5, 9)$ — here $\vec{AP} \cdot \vec{b} = -17$ and $|\vec{b}|^2 = 17$, so $\lambda_0 = -1$ and $F = \vec{a} - \vec{b}$.
2. $\sqrt{10}$ units ($\vec{AP} \times \vec{b} = (8, -5, 1)$, $|\cdot| = 3\sqrt{10}$, $\div |\vec{b}| = 3$)
3. Distance $\frac{13}{3}$; foot $\left(\frac{1}{9}, -\frac{5}{9}, -\frac{17}{9}\right)$ — with $t = \frac{13}{9}$ and $\vec{n} = (2, -1, 2)$, foot $= P - t\vec{n}$.
4. $(-3, 5, 2)$ — $t = \frac{6}{6} = 1$, image $= P - 2\vec{n}$ with $\vec{n} = (2, -1, 1)$.
5. $(0, 5, 1)$ — $\vec{AP} \cdot \vec{b} = -237 = -|\vec{b}|^2$, so $\lambda_0 = -1$, $F = (1, 2, 3)$, image $= 2F - P$.

Section 6 — Crack the MCQ

Q1. The perpendicular distance of $P(1, 4, 1)$ from the line through $A(1, 1, 1)$ with direction $\vec{b} = (2, 1, 2)$ is

- (A) $2\sqrt{2}$ (B) $\frac{2\sqrt{2}}{3}$ (C) $6\sqrt{2}$ (D) $3\sqrt{2}$

Answer: (A). $\vec{AP} = (0, 3, 0)$; $\vec{AP} \times \vec{b} = (6, 0, -6)$, magnitude $\sqrt{36 + 0 + 36} = 6\sqrt{2}$; with $|\vec{b}| = 3$, distance $= \frac{6\sqrt{2}}{3} = 2\sqrt{2}$.

(B) divides by $|\vec{b}|^2 = 9$ instead of $|\vec{b}| = 3$. (C) drops the denominator entirely. (D) is an arithmetic slip in $\frac{6\sqrt{2}}{3}$. The structure to remember: cross product over $|\vec{b}|$, divided **once**.

Q2. The foot of the perpendicular from $P(1, 2, -1)$ to the plane $2x - y + 2z + 1 = 0$ is reached by stepping t normals from P , where $t =$

- (A) $-\frac{1}{9}$ (B) $\frac{1}{3}$ (C) $-\frac{1}{3}$ (D) 9

Answer: (A). $t = \frac{2(1) - 2 + 2(-1) + 1}{4 + 1 + 4} = \frac{2 - 2 - 2 + 1}{9} = \frac{-1}{9}$. (B)/(C) divide by $|\vec{n}| = 3$ instead of $|\vec{n}|^2 = 9$ — that gives the *distance*, not t . (D) inverts the fraction. The defining difference: t divides by $|\vec{n}|^2$ because it scales the *vector* $\vec{n}$ to reach the plane, whereas the distance divides by $|\vec{n}|$ because it converts that vector step into a pure length.

Q3. The image of P in a line, given the foot of perpendicular is F , is

- (A) $2F - P$ (B) $F - P$ (C) $\frac{P + F}{2}$ (D) $2P - F$

Answer: (A). F is the midpoint of P and its image P' , so $F = \frac{1}{2}(P + P') \Rightarrow P' = 2F - P$.

(B) is the displacement, not a point. (C) is the midpoint of P and F — half the needed step. (D) reflects the wrong point (it images F in P). The relation $P' = 2F - P$ is the single most reused identity in this subtopic; memorise it as “twice the foot, minus the point.”

Q4. The distance of $(1, 1, 1)$ from the plane $x + 2y + 2z + 5 = 0$ is

- (A) $\frac{10}{3}$ (B) 10 (C) $\frac{10}{9}$ (D) $\frac{5}{3}$

Answer: (A). $\frac{|1 + 2 + 2 + 5|}{\sqrt{1 + 4 + 4}} = \frac{10}{3}$.

(B) forgets the denominator. (C) divides by $|\vec{n}|^2 = 9$. (D) mistakenly halves the numerator. The distance formula always reads “absolute value of the plane expression at the point, over the length of the normal” — the modulus guarantees a non-negative answer regardless of which side of the plane the point lies on.

Section 7 — Common Pitfalls & Pro-Tips

- **Dividing point-line distance by $|\vec{b}|^2$.** The cross product already carries one $|\vec{b}|$; divide by $|\vec{b}|$ exactly once.
- **Reporting the foot as the image.** The image is $2F - P$, not F . The foot is the midpoint of the point and its image. ...
- **Confusing t with the distance.** $t =$
by $|\vec{n}|$. They differ by a factor of $|\vec{n}|$.
- **Stepping the normal the wrong way.** Use $F = P - t\vec{n}$ (note the minus); writing $P + t\vec{n}$ lands you on the far side.
- **Skipping the perpendicularity check on the line foot.** A wrong λ produces a plausible-looking but wrong F ; the one-line test exposes it instantly.
- **Treating t as the perpendicular foot offset in coordinates directly.** The step from P to F is $-t\vec{n}$, a vector scaled by t , not t added to each coordinate.

Pro-Tip (one foot, three answers): never solve a “distance / foot / image” set separately. Find the foot F first; the distance is $|PF|$ and the image is $2F - P$. Three marks from one computation.

Pro-Tip (which tool): dropping onto a **line** needs *both* a dot product (for the foot) and a cross product (for the distance); dropping onto a **plane** needs *neither* — just the single number t , because the normal is the ready-made perpendicular direction.

Pro-Tip (reflections solve shortest-path): if a problem asks for the minimum total distance of a path that must meet a plane or line, reflect one endpoint across the surface and join it to the other endpoint with a straight segment. The segment’s length is the answer and its crossing point is the optimal contact point — an image computation in disguise.

Pro-Tip (sign of t locates the side): a positive t means P lies on the side where the plane expression is positive, a negative t the other side, and $t = 0$ means P is already on the plane.

Reading the sign before computing gives a free check on the direction of the corrective step. This also lets you answer “are these two points on the same side of the plane?” without any distance work — just compare the signs of the plane expression at each point; same sign means same side, opposite signs mean the plane separates them.

Section 8 — Cheat Sheet Box

Point to a line (04a) — $\vec{AP} = \vec{p} - \vec{a}$:

- Foot $F = \vec{a} + \frac{\vec{AP} \cdot \vec{b}}{|\vec{b}|^2} \vec{b}$; Distance = $\frac{|\vec{AP} \times \vec{b}|}{|\vec{b}|}$; Image = $2F - P$.

Point to a plane (04b) — $t = \frac{ax_1 + by_1 + cz_1 + d}{a^2 + b^2 + c^2}$:

- Foot $F = P - t\vec{n}$; Image $P' = P - 2t\vec{n}$; Distance = $\frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$.

Memory aids

- “Dot for the foot, cross for the gap” (line case).
- “The normal is the perpendicular” (plane case — just count normals, t).
- “Foot is the midpoint” — so the image is always $2F - P$.
- “ t over $|\vec{n}|^2$, distance over $|\vec{n}|$ ” — the same numerator, two different denominators, never interchange them.
- “Reflect to straighten a bent path” — shortest-touch problems become one straight segment after imaging an endpoint.

Line-Plane & Plane-Plane Configurations

Class 12 · Mathematics · Chapter 11 — Three Dimensional Geometry · Subtopic 05 · JEE-only

Exam Relevance: This is where the whole chapter converges — lines, planes, and points interacting. JEE Main reliably asks for a point of intersection of a line and a plane, or a plane built from given lines; JEE Advanced layers on bisector planes and reflections of lines. The plane material is outside the rationalized CBSE Board scope, so the ladder runs JEE Main warm-up → CUET/JEE Main Speed Trap → JEE Main multi-concept → JEE Advanced.

A **merged subtopic** with four diagnostic units: **(05a)** a line meeting, lying in, or running parallel to a plane; **(05b)** building the plane that contains given lines (or a line and a point); **(05c)** the angle-bisector planes of two planes; and **(05d)** reflecting and projecting a line.

Section 1 — Concept Intuition

By now you can describe lines and planes separately. The real exam questions ask how they *fit together*. There are only a few possibilities, and recognising which one you face is half the battle. The good news is that every configuration in this subtopic reduces to one of two reflexes: either *substitute a moving point and solve* (for intersections) or *gather in-plane vectors and cross them* (for building planes). Almost nothing here is genuinely new; it is the earlier machinery — dot products, cross products, the point-normal form — pointed at the question of how lines and planes coexist.

A **line and a plane** can do exactly three things. The line can **pierce** the plane at a single point (the generic case); it can lie **flat inside** the plane (every point of the line on the plane); or it can run **parallel** and never touch. The deciding quantity is the dot product of the line's direction $\vec{b}$ with the plane's normal $\vec{n}$. If $\vec{b} \cdot \vec{n} \neq 0$, the line is not perpendicular to the normal, so it must cross — one clean intersection point, found by sliding along the line until the plane equation is satisfied. If $\vec{b} \cdot \vec{n} = 0$, the line is perpendicular to the normal, hence parallel to the surface — and now a *second* test decides between “lies in” and “parallel”: pick any point of the line and check whether it satisfies the plane. On the plane means the whole line lies in it; off the plane means strictly parallel.

Think about it this way... $\vec{b} \cdot \vec{n} = 0$ is necessary but never sufficient for a line to lie in a plane. It only says the line is parallel to the surface. A railway track running alongside a wall is parallel to it (direction perpendicular to the wall's normal) yet clearly not painted *on* the wall. You must test a point to tell “alongside” from “on”.

Building a plane from lines. Often you must produce the plane that *contains* given data. The recipe is always “find the normal, then use a point.” If you have two intersecting lines, their two directions both lie in the plane, so the normal is their cross product $\vec{b}_1 \times \vec{b}_2$. If you have a line and an external point, the line's direction and the vector from the line to the point both lie in the plane, so the normal is $\vec{b} \times (\vec{P} - \vec{a})$. If you have two parallel lines, the shared direction and the vector joining them lie in the plane, giving normal $\vec{b} \times (\vec{a}_2 - \vec{a}_1)$. Same idea every time: two in-plane vectors, one cross product.

Bisecting the angle between two planes. Two intersecting planes make a pair of “valley” angles, and there are two planes that bisect them — exactly like the two angle bisectors of two crossing lines in 2D. A point lies on a bisector when it is **equidistant** from the two planes. Writing each distance with its sign and setting them equal (with a $\pm$ for the two bisectors) gives the bisector planes directly: you normalise each plane by the length of its normal, then equate.

Reflecting a line. To mirror a whole line in a plane, you don't need anything new — just reflect *points*. Where the line meets the plane is a fixed point (it's already on the mirror), and reflecting any second point of the line gives a second point of the image. Two points determine the reflected line.

Limiting conditions & assumptions:

- $\vec{b} \cdot \vec{n} = 0$ alone never proves a line lies in a plane — always confirm with a point.
- The “plane through two lines” only exists if the lines are **coplanar** (intersecting or parallel); skew lines do not lie in a common plane.
- For bisector planes you must divide each plane by $|\vec{n}|$ *before* equating; skipping the normalisation gives a plane that is not an angle bisector.
- Reflecting a line that is parallel to the mirror has no fixed intersection point — reflect two of its points instead, or reflect one point and keep the (unchanged) direction.

Section 2 — Key Definitions & Properties

The constructions below look varied, but they are organised by the *question* being asked, not by new theory. Whenever a line and a plane meet, you substitute and solve; whenever you must produce a plane from given lines or points, you build a normal from two in-plane vectors; whenever you bisect, you normalise and equate with a $\pm$. Keeping that three-way map in mind makes the formula table a lookup rather than a memory burden.

Line meeting a plane (05a). Line $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c} = \lambda$ and plane $Ax + By + Cz + D = 0$:

substitute $(x_1 + a\lambda, y_1 + b\lambda, z_1 + c\lambda)$ into the plane and solve for λ .

Intersects: $\vec{b} \cdot \vec{n} = Aa + Bb + Cc \neq 0$. $\vec{b} \cdot \vec{n} = 0 \Rightarrow \begin{cases} \text{lies in plane} & \text{point satisfies plane} \\ \text{parallel} & \text{point does not} \end{cases}$

Plane containing given data (05b). Normal $\vec{N}$, then point-normal form $(\vec{r} - \vec{a}) \cdot \vec{N} = 0$:

two intersecting lines: $\vec{N} = \vec{b}_1 \times \vec{b}_2$; line & point P : $\vec{N} = \vec{b} \times (\vec{P} - \vec{a})$; two parallel lines: $\vec{N} = \vec{b} \times (\vec{a}_2 - \vec{a}_1)$

Angle-bisector planes of two planes (05c). For $P_1 \equiv a_1x + b_1y + c_1z + d_1 = 0$ and $P_2 \equiv a_2x + b_2y + c_2z + d_2 = 0$:

$$\frac{a_1x + b_1y + c_1z + d_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = \pm \frac{a_2x + b_2y + c_2z + d_2}{\sqrt{a_2^2 + b_2^2 + c_2^2}}.$$

Reflection of a line in a plane (05d). Reflect the line's intersection point M (fixed) and the image Q' of any second point Q ; the reflected line is the line MQ' . (If the line is parallel to the plane, reflect two points; the image is parallel to the original.) **Projection of a line on a plane** is the line of intersection of the given plane with the plane through the line perpendicular to it.

Notation key: $\vec{b}$ (or $\vec{b}_1, \vec{b}_2$) = line direction(s); $\vec{n} = (A, B, C)$ = plane normal; λ = line parameter; $\vec{a}, \vec{a}_1, \vec{a}_2$ = points on lines; P_1, P_2 = plane expressions; M = line-plane intersection; Q' = image of point Q .

Configuration	Test / construction
Line $\cap$ plane	substitute parametric point, solve for λ
Line $\parallel$ / in plane	$\vec{b} \cdot \vec{n} = 0$; then test a point
Plane $\supset$ 2 intersecting lines	$\vec{N} = \vec{b}_1 \times \vec{b}_2$
Plane $\supset$ line + point	$\vec{N} = \vec{b} \times (\vec{P} - \vec{a})$
Bisector planes	$\frac{P_1}{ \vec{n}_1 } = \pm \frac{P_2}{ \vec{n}_2 }$

Section 3 — Theorems & Procedures

Procedure — point of intersection of a line and a plane (05a). (1) Write the line in parametric form $(x_1 + a\lambda, y_1 + b\lambda, z_1 + c\lambda)$. (2) Substitute into the plane equation $Ax + By + Cz + D = 0$. (3) Solve the resulting linear equation for λ . (4) Put that λ back into the parametric point. If step 3 yields a contradiction ($0 = \text{nonzero}$), the line is parallel; if it yields $0 = 0$, the line lies in the plane.

Theorem 1 (classification of a line relative to a plane). With $\vec{b} \cdot \vec{n} = Aa + Bb + Cc$: the line meets the plane in one point iff $\vec{b} \cdot \vec{n} \neq 0$; if $\vec{b} \cdot \vec{n} = 0$ the line is parallel, lying in the plane exactly when one (hence every) point of it satisfies the plane.

Proof. Substituting the parametric point gives $A(x_1 + a\lambda) + B(y_1 + b\lambda) + C(z_1 + c\lambda) + D = 0$, i.e. $(Ax_1 + By_1 + Cz_1 + D) + \lambda(Aa + Bb + Cc) = 0$, a linear equation $K + \lambda(\vec{b} \cdot \vec{n}) = 0$ where K is the plane expression at the line's base point. If $\vec{b} \cdot \vec{n} \neq 0$ there is a unique $\lambda = -K/(\vec{b} \cdot \vec{n})$ — one intersection. If $\vec{b} \cdot \vec{n} = 0$ the equation reduces to $K = 0$: when $K = 0$ every λ works (line in plane); when $K \neq 0$ no λ works (parallel). ■

Theorem 2 (plane through a line and an external point). The plane containing the line $\vec{r} = \vec{a} + \lambda\vec{b}$ and the point P has normal $\vec{N} = \vec{b} \times (\vec{P} - \vec{a})$ and equation $(\vec{r} - \vec{a}) \cdot [\vec{b} \times (\vec{P} - \vec{a})] = 0$.

Figure. A plane shown as a parallelogram. Inside it, a line with direction $\vec{b}$ through point A , and a separate point P also on the plane. Two in-plane vectors drawn from A : one along $\vec{b}$, and one from A to P . Their cross product $\vec{N}$ is drawn as an arrow perpendicular to the plane.

Proof. Both the direction $\vec{b}$ and the vector $\vec{P} - \vec{a}$ (from a point of the line to P) lie in the desired plane, since the line and P all belong to it. A vector perpendicular to two in-plane vectors is normal to the plane, and the cross product $\vec{b} \times (\vec{P} - \vec{a})$ is exactly such a vector. Applying the point-normal form through $\vec{a}$ gives the stated equation; equivalently, a point $\vec{r}$ is coplanar with the line and P precisely when the scalar triple product $(\vec{r} - \vec{a}) \cdot [\vec{b} \times (\vec{P} - \vec{a})]$ vanishes. ■

Theorem 3 (angle-bisector planes). A point is equidistant from $P_1 = 0$ and $P_2 = 0$ iff $\frac{P_1}{|\vec{n}_1|} = \pm \frac{P_2}{|\vec{n}_2|}$, which are the two bisector planes.

Proof. The signed distance of a point from $P_i = 0$ is $\frac{a_i x + b_i y + c_i z + d_i}{|\vec{n}_i|} = \frac{P_i}{|\vec{n}_i|}$. Equidistance means the magnitudes are equal: $\left| \frac{P_1}{|\vec{n}_1|} \right| = \left| \frac{P_2}{|\vec{n}_2|} \right|$, i.e. $\frac{P_1}{|\vec{n}_1|} = \pm \frac{P_2}{|\vec{n}_2|}$. Each sign gives one of the two mutually perpendicular bisector planes. ■

Procedure — reflecting a line in a plane (05d). (1) Find M , the point where the line meets the plane (Procedure 05a). (2) Pick any other point Q on the line and find its image Q' in the plane (Sub-04). (3) The reflected line is the line through M and Q' . If the line is parallel to the plane (no M), reflect two points instead, or reflect one point and reuse the original direction (unchanged because $\vec{b} \cdot \vec{n} = 0$).

Worked illustration. Reflect the line $\vec{r} = (0, 0, 1) + \lambda(1, 0, 1)$ in the plane $z = 0$. Intersection: $z = 1 + \lambda = 0 \Rightarrow \lambda = -1$, giving $M = (-1, 0, 0)$, which lies on the plane and is therefore fixed. Take the base point $Q = (0, 0, 1)$; its image in $z = 0$ is $Q' = (0, 0, -1)$ (reflecting the z -coordinate). The reflected line passes through $M(-1, 0, 0)$ and $Q'(0, 0, -1)$, with direction $Q' - M = (1, 0, -1)$ — note how the z -component of the direction flipped sign while the in-plane components survived, exactly as a mirror should behave. So the image line is $\vec{r} = (-1, 0, 0) + \mu(1, 0, -1)$.

Procedure — projecting a line onto a plane (05d). The projection is the “shadow” the line casts on the plane along the normal. Construct the plane π' that contains the given line and is perpendicular to the given plane π (its normal is $\vec{b} \times \vec{n}$, since π' contains both the line direction $\vec{b}$ and the normal $\vec{n}$). The projected line is then the intersection of π and π' . Equivalently, drop perpendiculars from two points of the line onto π (Sub-04 feet) and join the two feet.

Section 4 — Worked Examples

Example 1 — [JEE Main • Foundational warm-up] (Line meets plane). Find the point where the line $\frac{x-2}{1} = \frac{y+1}{-1} = \frac{z-3}{2}$ meets the plane $2x + y - z = 1$.

Parametric point: $(2 + \lambda, -1 - \lambda, 3 + 2\lambda)$. Substitute:

$$2(2 + \lambda) + (-1 - \lambda) - (3 + 2\lambda) = 1 \Rightarrow 4 + 2\lambda - 1 - \lambda - 3 - 2\lambda = 1 \Rightarrow -\lambda = 1 \Rightarrow \lambda = -1.$$

Point: $(2 - 1, -1 + 1, 3 - 2) = (1, 0, 1)$. (Check: $2(1) + 0 - 1 = 1$ ✓.) The whole method is “turn the line into one moving point, then ask the plane when that point arrives.” Because the plane equation is linear and the point’s coordinates are linear in λ , the substitution always collapses to a single linear equation in λ — at most one solution unless the line is special. That is why a unique intersection is the generic outcome and why a clean numerical λ (here -1) is expected.

Example 2 — [CUET / JEE Main Speed Trap] (Parallel vs lies-in). Does the line through $(1, 0, 2)$ with direction $(1, 2, 1)$ lie in, run parallel to, or cross the plane $x + y - 3z + 2 = 0$?

First the dot product: $\vec{b} \cdot \vec{n} = (1)(1) + (2)(1) + (1)(-3) = 1 + 2 - 3 = 0$, so the line does **not** cross — it is parallel to the surface. Now the decisive second test, a point of the line:

$$x + y - 3z + 2 \Big|_{(1,0,2)} = 1 + 0 - 6 + 2 = -3 \neq 0.$$

The point is off the plane, so the line is **strictly parallel** (not contained). **The trap:** stopping at $\vec{b} \cdot \vec{n} = 0$ and declaring “the line lies in the plane.” That dot product only rules out crossing; the point test is what separates “parallel” from “lies in.” A useful mental model: $\vec{b} \cdot \vec{n} = 0$ fixes the line’s *orientation* relative to the surface (it runs flat, neither diving in nor climbing out), but orientation alone says nothing about *position*. Two lines can share the same flat orientation yet sit at different heights — one painted on the plane, one floating above it. Only evaluating an actual point pins down the height, which is why the second test is never optional.

Example 3 — [JEE Main • Multi-concept] (Plane through a line and a point). Find the plane containing the line $\vec{r} = (\hat{i} - \hat{j} + 3\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + \hat{k})$ and the point $(2, 1, 0)$.

Here $\vec{a} = (1, -1, 3)$, $\vec{b} = (2, 3, 1)$, $P = (2, 1, 0)$, so $\vec{P} - \vec{a} = (1, 2, -3)$. Normal:

$$\vec{N} = \vec{b} \times (\vec{P} - \vec{a}) = (2, 3, 1) \times (1, 2, -3) = (3(-3) - 1(2), 1(1) - 2(-3), 2(2) - 3(1)) = (-11, 7, 1).$$

Plane through $\vec{a} = (1, -1, 3)$: $-11(x-1) + 7(y+1) + 1(z-3) = 0 \Rightarrow -11x + 7y + z + 15 = 0$, i.e.

$$11x - 7y - z = 15.$$

Check: at $(1, -1, 3)$, $11 + 7 - 3 = 15$ ✓; at $(2, 1, 0)$, $22 - 7 - 0 = 15$ ✓. The multi-concept move is “two in-plane vectors, one cross product,” then point-normal. The verification at *both* given objects — a point of the line and the external point — is the discipline that separates a secure answer from a guessed one: a correct plane must contain every point it was built from, so checking two of them is a near-certain validation at almost no cost.

Example 4 — [JEE Advanced] (Angle-bisector planes). Find the planes bisecting the angles between $2x - y + 2z + 3 = 0$ and $3x - 2y + 6z - 8 = 0$.

$|\vec{n}_1| = \sqrt{4 + 1 + 4} = 3$, $|\vec{n}_2| = \sqrt{9 + 4 + 36} = 7$. The bisectors satisfy

$$\frac{2x - y + 2z + 3}{3} = \pm \frac{3x - 2y + 6z - 8}{7} \Rightarrow 7(2x - y + 2z + 3) = \pm 3(3x - 2y + 6z - 8).$$

Plus sign: $14x - 7y + 14z + 21 = 9x - 6y + 18z - 24 \Rightarrow 5x - y - 4z + 45 = 0$. **Minus sign:** $14x - 7y + 14z + 21 = -9x + 6y - 18z + 24 \Rightarrow 23x - 13y + 32z - 3 = 0$. So the two bisector planes are $5x - y - 4z + 45 = 0$ and $23x - 13y + 32z - 3 = 0$. (They are mutually perpendicular — a quick check: $(5)(23) + (-1)(-13) + (-4)(32) = 115 + 13 - 128 = 0$ ✓.) The advanced insight: “bisector” means “equidistant,” and equidistance becomes the normalised equation with a $\pm$. Two further points are worth carrying. First, the normalisation is what makes the result a genuine *angle* bisector rather than just some plane through the line of intersection — dropping the $\sqrt{\dots}$ denominators would still give a plane containing that line, but not one that splits the angle. Second, the two bisectors are always perpendicular to each other (as the check confirms), because they bisect the two supplementary angles formed by the original planes; if a problem asks for “the bisector of the *acute* angle,” you compute both and then test which one makes the smaller angle with either original plane.

Section 5 — Practice Exercises

1. **[JEE Main]** Find the point of intersection of the line $\frac{x-1}{2} = \frac{y-2}{-1} = \frac{z+1}{1}$ and the plane $x + y + z = 4$.
2. **[CUET / JEE Main]** Determine whether the line through $(1, 1, 1)$ with direction $(1, -1, 2)$ is parallel to, lies in, or crosses the plane $3x + y - z = 2$.
3. **[JEE Main]** Find the plane containing the two parallel lines $\vec{r} = (\hat{i} + 2\hat{k}) + \lambda(2\hat{i} + \hat{j} - \hat{k})$ and $\vec{r} = (3\hat{j} + \hat{k}) + \mu(2\hat{i} + \hat{j} - \hat{k})$.
4. **[JEE Main]** Find the plane containing the line $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$ and the point $(4, 3, 2)$.
5. **[JEE Advanced]** Find the planes bisecting the angles between $x + 2y + 2z - 9 = 0$ and $4x - 3y + 12z + 13 = 0$.

Answers:

1. $(3, 1, 0)$ — parametric point $(1 + 2\lambda, 2 - \lambda, -1 + \lambda)$; sum = 4 gives $\lambda = 1$.
2. Parallel (since $\vec{b} \cdot \vec{n} = 3 - 1 - 2 = 0$ and the point gives $3 + 1 - 1 = 3 \neq 2$)

3. $2x + 3y + 7z = 16$ — parallel lines, normal $\vec{b} \times (\vec{a}_2 - \vec{a}_1) = (2, 1, -1) \times (-1, 3, -1) = (2, 3, 7)$.
4. $x - 2y + z = 0$ — normal $\vec{b} \times (\vec{P} - \vec{a}) = (1, 2, 3) \times (3, 1, -1) = (-5, 10, -5) \parallel (1, -2, 1)$.
5. $x + 35y - 10z = 156$ and $25x + 17y + 62z = 78$ — normalise by $|\vec{n}_1| = 3$, $|\vec{n}_2| = 13$, then take $\pm$.

Section 6 — Crack the MCQ

Q1. The line $\frac{x-1}{2} = \frac{y}{3} = \frac{z+1}{-1}$ meets the plane $x + 2y + z = 7$ at

- (A) $(3, 3, -2)$ (B) $(1, 0, -1)$ (C) $(5, 6, -3)$ (D) $(-1, -3, 0)$

Answer: (A). Parametric point $(1+2\lambda, 3\lambda, -1-\lambda)$; substituting, $(1+2\lambda) + 2(3\lambda) + (-1-\lambda) = 7 \Rightarrow 7\lambda = 7 \Rightarrow \lambda = 1$, giving $(3, 3, -2)$.

(B) is the base point ($\lambda = 0$) — on the line but not on the plane. (C) takes $\lambda = 2$, (D) takes $\lambda = -1$; both are points of the line that miss the plane.

Q2. The line $\vec{r} = \vec{a} + \lambda\vec{b}$ is parallel to, but not contained in, the plane $\vec{r} \cdot \vec{n} = d$ if and only if

- (A) $\vec{b} \cdot \vec{n} = 0$ and $\vec{a} \cdot \vec{n} \neq d$ (B) $\vec{b} \cdot \vec{n} = 0$ (C) $\vec{b} \times \vec{n} = \vec{0}$ (D) $\vec{b} \cdot \vec{n} = d$

Answer: (A). Parallel needs $\vec{b} \cdot \vec{n} = 0$ (direction perpendicular to normal); “not contained” needs a point off the plane, $\vec{a} \cdot \vec{n} \neq d$.

(B) omits the point test, so it also admits a line lying *in* the plane. (C) says the direction is parallel to the normal — that’s a line *perpendicular* to the plane. (D) is dimensionally meaningless here. The pairing of conditions in (A) is the whole lesson of unit 05a: one condition fixes orientation, the other fixes position, and “strictly parallel” needs both.

Q3. The normal to the plane containing two intersecting lines with directions $\vec{b}_1$ and $\vec{b}_2$ is along

- (A) $\vec{b}_1 + \vec{b}_2$ (B) $\vec{b}_1 \cdot \vec{b}_2$ (C) $\vec{b}_1 \times \vec{b}_2$ (D) $\vec{b}_1 - \vec{b}_2$

Answer: (C). Both directions lie in the plane, so the normal is perpendicular to both — exactly the cross product.

(A) and (D) are themselves in-plane combinations, not perpendicular to the plane. (B) is a scalar, not a direction at all.

Q4. The angle-bisector planes of $P_1 = 0$ and $P_2 = 0$ (normals $\vec{n}_1, \vec{n}_2$) are

- (A) $P_1 = P_2$ (B) $P_1 + P_2 = 0$ (C) $\frac{P_1}{|\vec{n}_1|} = \pm \frac{P_2}{|\vec{n}_2|}$ (D) $P_1 P_2 = 0$

Answer: (C). A bisector is the locus equidistant from both planes; signed distances are $P_i/|\vec{n}_i|$, and equating their magnitudes gives the $\pm$ pair.

(A) ignores the differing normal lengths (only correct if $|\vec{n}_1| = |\vec{n}_2|$). (B) is one specific member of the plane *family*, not a bisector. (D) is the pair of the original planes themselves. The single most common slip is choosing (A): forgetting that “equidistant” must use *normalised* distances, so each plane expression has to be divided by the length of its own normal before the comparison is meaningful.

Section 7 — Common Pitfalls & Pro-Tips

- **Reading $\vec{b} \cdot \vec{n} = 0$ as “lies in the plane.”** It only means parallel; you must test a point to decide between parallel and contained.
- **Using the wrong normal for “plane through lines.”** Intersecting lines $\rightarrow \vec{b} \times \vec{b}$; parallel lines $\rightarrow \vec{b} \times (\vec{a}_2 - \vec{a}_1)$; line and point $\rightarrow \vec{b} \times (\vec{P} - \vec{a})$. Picking the wrong pair of in-plane vectors gives a wrong normal.
- **Forgetting to normalise before bisecting.** Each plane must be divided by $|\vec{n}|$ before equating, or the result is a family member, not an angle bisector.
- **Reporting only one bisector.** The $\pm$ yields two perpendicular bisector planes; both are valid answers.
- **Reflecting a line by reflecting only its direction.** Reflect actual *points*: the fixed intersection M and the image of a second point.

Pro-Tip (one dot product classifies everything): before any line-plane work, compute $\vec{b} \cdot \vec{n}$. Nonzero $\rightarrow$ it crosses (go find λ). Zero $\rightarrow$ test a point: on $\Rightarrow$ contained, off $\Rightarrow$ parallel. That single number routes the whole problem, so make it your first move every time — it tells you in one step which of the three configurations you are in and therefore which method to deploy.

Pro-Tip (every “plane through ...” is a cross product): whatever the data — two lines, a line and a point — gather two vectors that lie in the target plane and cross them for the normal. Then point-normal finishes it.

Pro-Tip (reflect and project share machinery): both reflecting a line and projecting a line onto a plane reduce to operations you already own — find where the line meets the plane, then either reflect a second point (reflection) or drop perpendicular feet (projection). No new formula is needed; recognising the reuse turns two intimidating “advanced” topics into routine applications of Sub-04.

Pro-Tip (acute vs obtuse bisector): when a question specifies *which* bisector plane, compute both from the $\pm$, then measure the angle each makes with one original plane; the smaller angle identifies the acute-angle bisector. Never assume the $+$ sign gives the acute one — it depends on the signs of the constants.

Section 8 — Cheat Sheet Box

Line & plane (05a): substitute parametric point, solve for λ . Classify by $\vec{b} \cdot \vec{n}$: $\neq 0$ crosses; $= 0$ then point-test for in-plane vs parallel.

Plane through ... (05b): normal = cross of two in-plane vectors — $\vec{b}_1 \times \vec{b}_2$ (intersecting lines), $\vec{b} \times (\vec{P} - \vec{a})$ (line + point), $\vec{b} \times (\vec{a}_2 - \vec{a}_1)$ (parallel lines); then point-normal.

Bisector planes (05c): $\frac{P_1}{|\vec{n}_1|} = \pm \frac{P_2}{|\vec{n}_2|}$ — two perpendicular planes.

Reflect a line (05d): keep the intersection point M , reflect a second point $Q \rightarrow Q'$, join M and Q' .

Memory aids

- “Dot decides” — $\vec{b} \cdot \vec{n}$ tells cross / parallel; a point tells parallel / contained.
- “Plane through anything = a cross product.”
- “Bisect = normalise, then $\pm$.”

- *“Reflect and project reuse old tools”* — intersection point plus a reflected point, or two perpendicular feet, are all you ever need.

Chapter 11: Three Dimensional Geometry — Round 2 Addendum

Subtopics 00 to 05 gave you the complete standard toolkit: distances and section formulae, direction cosines, the skew-line distance formula, the family $P_1 + \lambda P_2 = 0$, and the foot-distance-image routine for lines and planes. Five working procedures are nevertheless missing, and every one of them is a live examinable pattern built strictly from tools the chapter already hands you. Addendum A parametrises the line where two planes meet — the chapter repeatedly tells you to avoid this line, but projection, three-plane consistency and several Advanced questions require it explicitly. Addendum B finds *both endpoints* of the shortest bridge between skew lines, not merely its length. Addendum C extends coplanarity from two lines to four points and completes the plane-building table of unit 05b. Addendum D collects the origin-foot patterns and the mirror trick behind every shortest-path-with-reflection problem. Addendum E translates geometric conditions into equations — the locus skill the chapter assumes silently but never teaches. Nothing below introduces new theory; each addendum is an old tool pointed at a configuration the chapter skipped.

Addendum A — The Line Where Two Planes Meet

Unit 03b insists you never compute the intersection line, and for family-of-planes problems that advice stands. But the moment a question asks for the *angle* that intersection line makes with something, or its shortest distance to another line, or whether a third plane cuts it, you need the line itself. Writing it down is a two-step routine.

(Entirely JEE territory: the plane is outside the rationalised CBSE syllabus.)

Method

Step 1 — the direction is the crossed normals. Let the planes be

$$\Pi_1 : \vec{r} \cdot \vec{n}_1 = d_1, \quad \Pi_2 : \vec{r} \cdot \vec{n}_2 = d_2.$$

If $\vec{u}$ is the displacement between any two points lying on *both* planes, then

$$\vec{u} \cdot \vec{n}_1 = 0, \quad \vec{u} \cdot \vec{n}_2 = 0,$$

because both points satisfy each plane equation, so the difference cancels each right-hand side. A vector perpendicular to both normals is parallel to $\vec{n}_1 \times \vec{n}_2$. Hence the line of intersection has direction

$$\vec{d} = \vec{n}_1 \times \vec{n}_2.$$

Step 2 — one point, by killing a coordinate. Set one variable, usually z , equal to zero, and solve the remaining 2×2 linear system. If that system is inconsistent (no solution), the line does not pierce that coordinate plane — return the favour by zeroing a different variable. Write the result in symmetric form

$$\frac{x - x_0}{d_x} = \frac{y - y_0}{d_y} = \frac{z - z_0}{d_z},$$

or, if a direction component vanishes, in the restricted form demonstrated in Practice 2.

Shortcut worth memorising: a point lies on the line of intersection if and only if it satisfies *both* plane equations. Testing membership never requires the parametrisation at all.

Example A.1 — [JEE Main Level]

Find the line of intersection of $x + y + z = 6$ and $2x - y + z = 3$, and the point where it meets the plane $2x - y + 3z = 9$.

Direction:

$$\vec{d} = (1, 1, 1) \times (2, -1, 1) = (2, 1, -3).$$

Set $z = 0$: the system $x + y = 6$ and $2x - y = 3$ adds to give $3x = 9$, so $x = 3$, $y = 3$. The line is

$$\frac{x - 3}{2} = \frac{y - 3}{1} = \frac{z}{-3}, \quad \text{i.e. } (3 + 2t, 3 + t, -3t).$$

Substitute into the third plane:

$$2(3 + 2t) - (3 + t) + 3(-3t) = 3 - 6t = 9 \implies t = -1.$$

Meeting point: $(1, 2, 3)$. **Check:** it satisfies both original planes, since $1 + 2 + 3 = 6$ and $2 - 2 + 3 = 3$.

Example A.2 — [JEE Advanced Level]

Find the shortest distance between the line where $x + y = 2$ meets $x + z = 3$, and the line through $(1, 2, 4)$ with direction $(1, 1, 1)$.

Parametrise the intersection line. Its normals are $(1, 1, 0)$ and $(1, 0, 1)$, so

$$\vec{d} = (1, 1, 0) \times (1, 0, 1) = (1, -1, -1).$$

Setting $y = 0$ forces $x = 2$ and $z = 1$, giving the point $(2, 0, 1)$. So L_1 has $\vec{a}_1 = (2, 0, 1)$, $\vec{b}_1 = (1, -1, -1)$; and L_2 has $\vec{a}_2 = (1, 2, 4)$, $\vec{b}_2 = (1, 1, 1)$.

Common-perpendicular direction:

$$\vec{b}_1 \times \vec{b}_2 = (0, -2, 2), \quad |\vec{b}_1 \times \vec{b}_2| = 2\sqrt{2}.$$

Joining vector $\vec{a}_2 - \vec{a}_1 = (-1, 2, 3)$. Scalar triple product:

$$(-1, 2, 3) \cdot (0, -2, 2) = 0 - 4 + 6 = 2 \neq 0,$$

so the lines are genuinely skew, and

$$d = \frac{|2|}{2\sqrt{2}} = \frac{1}{\sqrt{2}} \text{ units.}$$

The entire difficulty of the problem was Step 1: everything after “read off $\vec{a}_1, \vec{b}_1$ ” is Subtopic 02 verbatim.

Practice — Addendum A

1. Find the symmetric equation of the line of intersection of $x+y+z=6$ and $2x+3y+z=11$.
2. Find the line of intersection of $2x+y-z=3$ and $x-y+z=1$. (Watch the direction.)
3. Find the angle between the line of intersection of $x+2y+2z=5$ and $2x+y+2z=7$ and the xy -plane.
4. Without parametrising anything, decide whether $P(5,4,-3)$ and $Q(4,4,-3)$ lie on the line of intersection of $x+y+z=6$ and $2x-y+z=3$.
5. Show that the planes $x+y+z=1$, $2x-y+2z=3$ and $x+z=5$ have no common point.

Answers: 1. Direction $(-2, 1, 1)$; point $(7, -1, 0)$ from $z=0$; line $\frac{x-7}{-2} = \frac{y+1}{1} = z$. 2. Direction $(0, -3, -3) \propto (0, 1, 1)$; the x -component dies, so the symmetric form is illegal — the line is $x = \frac{4}{3}$ together with $y-z = \frac{1}{3}$ (point $(\frac{4}{3}, \frac{1}{3}, 0)$ from $z=0$). 3. Direction $(2, 2, -3)$, $|\vec{d}| = \sqrt{17}$; against the xy -plane use sine with the plane’s normal $(0, 0, 1)$: $\sin \varphi = \frac{3}{\sqrt{17}}$. 4. P lies on the line ($5+4-3=6$ and $10-4-3=3$, both satisfied); Q does not ($4+4-3=5 \neq 6$). 5. Subtracting the third plane from the first gives $y=-4$; substituting back, $x+z=5$ from plane 1 but $x+2(-4)+2z=3$ forces $x+z=\frac{11}{2}$ — a contradiction, so no point satisfies all three; equivalently the first two planes’ intersection line has direction $(1, 0, -1)$, which is perpendicular to $(1, 0, 1)$, so it runs parallel to the third plane and off it.

Addendum B — Both Ends of the Bridge: Closest Points on Skew Lines

Theorem 2 of Subtopic 02 delivers the *length* of the common perpendicular in one shot. But JEE Advanced regularly wants the points where that perpendicular lands — “find the shortest distance *and* the points between which it is measured” — and the formula is silent about them. The fix is to go back to first principles and minimise directly.

Method

Take general points $P_1 = \vec{a}_1 + \lambda \vec{b}_1$ on L_1 and $P_2 = \vec{a}_2 + \mu \vec{b}_2$ on L_2 , and write the joining vector with $\vec{c} = \vec{a}_2 - \vec{a}_1$:

$$\overrightarrow{P_1 P_2} = \vec{D}(\lambda, \mu) = \vec{c} + \mu \vec{b}_2 - \lambda \vec{b}_1.$$

Minimise $f(\lambda, \mu) = |\vec{D}|^2$. Setting both partial derivatives to zero:

$$\frac{\partial f}{\partial \lambda} = -2 \vec{D} \cdot \vec{b}_1 = 0, \quad \frac{\partial f}{\partial \mu} = 2 \vec{D} \cdot \vec{b}_2 = 0,$$

which is exactly the statement that $\vec{D}$ is perpendicular to *both* lines — the critical point automatically builds the common perpendicular. Expanding the two dot products gives a 2×2 linear system:

$$\begin{aligned}\lambda |\vec{b}_1|^2 - \mu (\vec{b}_1 \cdot \vec{b}_2) &= \vec{c} \cdot \vec{b}_1, \\ \lambda (\vec{b}_1 \cdot \vec{b}_2) - \mu |\vec{b}_2|^2 &= \vec{c} \cdot \vec{b}_2.\end{aligned}$$

Why the solution is the minimum: the quadratic part of f is $|\lambda \vec{b}_1 - \mu \vec{b}_2|^2$, which is positive definite whenever $\vec{b}_1, \vec{b}_2$ are not proportional (Cauchy-Schwarz is strict). A positive-definite quadratic has one critical point and it is the global minimum. When the lines *are* parallel the two equations become dependent — no unique solution, matching the fact that parallel lines need the separate Subtopic 02 formula.

The distance is then $|\vec{D}|$ at the solved (λ, μ) , and as a free cross-check it must equal the triple-product value $|\vec{c} \cdot (\vec{b}_1 \times \vec{b}_2)| / |\vec{b}_1 \times \vec{b}_2|$.

Example B.1 — [JEE Main Level]

Example 3 of Subtopic 02 found the shortest distance between

$$L_1 : \vec{r} = (\hat{i} + 2\hat{j} - \hat{k}) + \lambda(2\hat{i} + \hat{j} - 2\hat{k}), \quad L_2 : \vec{r} = (3\hat{i} - \hat{j} + \hat{k}) + \mu(\hat{i} + 2\hat{j} + 2\hat{k})$$

to be 4 units. Locate the two closest points.

Here $\vec{a}_1 = (1, 2, -1)$, $\vec{b}_1 = (2, 1, -2)$, $\vec{a}_2 = (3, -1, 1)$, $\vec{b}_2 = (1, 2, 2)$, so $\vec{c} = (2, -3, 2)$. The data:

$$\begin{aligned}|\vec{b}_1|^2 &= 9, & |\vec{b}_2|^2 &= 9, & \vec{b}_1 \cdot \vec{b}_2 &= 0, \\ \vec{c} \cdot \vec{b}_1 &= 4 - 3 - 4 = -3, & \vec{c} \cdot \vec{b}_2 &= 2 - 6 + 4 = 0.\end{aligned}$$

The system decouples beautifully:

$$9\lambda = -3 \implies \lambda = -\frac{1}{3}, \quad -9\mu = 0 \implies \mu = 0.$$

Closest points:

$$P_1 = (1, 2, -1) - \frac{1}{3}(2, 1, -2) = \left(\frac{1}{3}, \frac{5}{3}, -\frac{1}{3}\right), \quad P_2 = (3, -1, 1).$$

Distance: $\vec{D} = \left(\frac{8}{3}, -\frac{8}{3}, \frac{4}{3}\right)$, so

$$|\vec{D}| = \frac{\sqrt{64 + 64 + 16}}{3} = \frac{12}{3} = 4,$$

matching the chapter's triple-product answer. **Checks:** $\vec{D} \cdot \vec{b}_1 = \frac{16-8-8}{3} = 0$ and $\vec{D} \cdot \vec{b}_2 = \frac{8-16+8}{3} = 0$ — genuinely the common perpendicular.

Example B.2 — [JEE Advanced Level]

Find the shortest distance between the lines

$$L_1 : \vec{r} = \lambda(\hat{i} + \hat{j}), \quad L_2 : \vec{r} = (\hat{i} + 2\hat{k}) + \mu(\hat{j} + \hat{k}),$$

and the points at which it is attained.

Here $\vec{a}_1 = (0, 0, 0)$, $\vec{b}_1 = (1, 1, 0)$, $\vec{a}_2 = (1, 0, 2)$, $\vec{b}_2 = (0, 1, 1)$, so $\vec{c} = (1, 0, 2)$. Data:

$$|\vec{b}_1|^2 = 2, \quad |\vec{b}_2|^2 = 2, \quad \vec{b}_1 \cdot \vec{b}_2 = 1,$$

$$\vec{c} \cdot \vec{b}_1 = 1, \quad \vec{c} \cdot \vec{b}_2 = 2.$$

This time the coupling term is alive; the system is

$$2\lambda - \mu = 1, \quad \lambda - 2\mu = 2.$$

From the first, $\mu = 2\lambda - 1$; substitute into the second:

$$\lambda - 4\lambda + 2 = 2 \implies \lambda = 0, \quad \mu = -1.$$

Closest points:

$$P_1 = (0, 0, 0), \quad P_2 = (1, 0, 2) + (-1)(0, 1, 1) = (1, -1, 1).$$

Distance: $\vec{D} = (1, -1, 1)$, $|\vec{D}| = \sqrt{3}$. **Cross-check by triple product:** $\vec{b}_1 \times \vec{b}_2 = (1, -1, 1)$, of magnitude $\sqrt{3}$, and $\vec{c} \cdot (1, -1, 1) = 3$, giving $\frac{3}{\sqrt{3}} = \sqrt{3}$ — agreement. **Checks:** $\vec{D} \cdot \vec{b}_1 = 1 - 1 = 0$ and $\vec{D} \cdot \vec{b}_2 = -1 + 1 = 0$.

Practice — Addendum B

- Find the closest points on the x -axis and the line through $(0, 1, 1)$ with direction $(0, 1, -1)$, and the distance between them.
- For $L_1 : \vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$ and $L_2 : \vec{r} = (2\hat{i} + \hat{j}) + \mu(2\hat{i} + \hat{j} + 2\hat{k})$, find the closest points and the shortest distance; verify with the triple-product formula.
- Explain why the 2×2 system has no unique solution when the lines are parallel, and say what replaces it.
- Using both methods (closest points, then triple product), find the shortest distance between L_1 , defined by $x = 1$ and $y = 2$, and $L_2 : \vec{r} = 3\hat{k} + \mu(\hat{i} + \hat{j})$.

Answers: 1. $\lambda = 0$, $\mu = 0$; points $(0, 0, 0)$ and $(0, 1, 1)$; distance $\sqrt{2}$ (and $(0, 1, 1)$ is perpendicular to both directions). 2. System $3\lambda - 3\mu = -1$, $3\lambda - 9\mu = -5$ gives $\lambda = \frac{1}{3}$, $\mu = \frac{2}{3}$; points $(\frac{4}{3}, \frac{5}{3}, \frac{10}{3})$ and $(\frac{10}{3}, \frac{5}{3}, \frac{4}{3})$; $\vec{D} = (2, 0, -2)$, distance $2\sqrt{2}$; cross-check: $\vec{b}_1 \times \vec{b}_2 = (-3, 0, 3)$, triple product -12 , distance $\frac{12}{3\sqrt{2}} = 2\sqrt{2}$. 3. When $\vec{b}_2 \propto \vec{b}_1$ both equations carry the same information, so the system is rank-one with infinitely many solutions — every transversal perpendicular to the shared direction has equal length; use the parallel-lines formula $|\vec{b} \times (\vec{a}_2 - \vec{a}_1)|/|\vec{b}|$ instead. 4. $\vec{a}_1 = (1, 2, 0)$, $\vec{b}_1 = (0, 0, 1)$, $\vec{a}_2 = (0, 0, 3)$, $\vec{b}_2 = (1, 1, 0)$, $\vec{c} = (-1, -2, 3)$; the system $\lambda = 3$, $-2\mu = -3$ gives points $(1, 2, 3)$ and $(\frac{3}{2}, \frac{3}{2}, 3)$, $\vec{D} = (\frac{1}{2}, -\frac{1}{2}, 0)$, distance $\frac{1}{\sqrt{2}}$; triple product route: $(0, 0, 1) \times (1, 1, 0) = (-1, 1, 0)$, $\vec{c} \cdot (-1, 1, 0) = -1$, distance $\frac{1}{\sqrt{2}}$ — identical.

Addendum C — Four Points, One Plane: Coplanarity and Extended Plane-Building

Subtopic 02 tests coplanarity of two *lines*; unit 05b builds planes through intersecting lines, a line and a point, or parallel lines. Three examinable extensions are missing: deciding whether four scattered points share a plane, building a plane through a line that must run parallel to a given direction, and building one that must run parallel to a given plane.

Method

C1 — Four points are coplanar iff their triple product dies. For points A, B, C, D , form the three difference vectors $\overrightarrow{AB}$, $\overrightarrow{AC}$, $\overrightarrow{AD}$. If A, B, C are non-collinear, $\overrightarrow{AB} \times \overrightarrow{AC}$ is a normal to the plane through them, and D lies in that plane exactly when $\overrightarrow{AD}$ is perpendicular to that normal:

$$(\overrightarrow{AB} \times \overrightarrow{AC}) \cdot \overrightarrow{AD} = 0.$$

If A, B, C happen to be collinear, the cross product is $\vec{0}$ and the condition holds identically — correctly, since a line and a fourth point always share *some* plane. One computation, no special cases.

C2 — Plane through a line, parallel to a direction $\vec{m}$. The desired plane must contain $\vec{b}$ (along its line) and admit $\vec{m}$ as an in-plane direction, so its normal is perpendicular to both:

$$\vec{N} = \vec{b} \times \vec{m}, \quad \text{plane: } (\vec{r} - \vec{a}) \cdot (\vec{b} \times \vec{m}) = 0,$$

with $\vec{a}$ any point of the line. Caveats: if $\vec{m} \parallel \vec{b}$ the construction collapses ($\vec{N} = \vec{0}$) — but then *every* plane through the line qualifies, so the answer is “not unique”; if the data is a second line M rather than a bare direction, the plane exists uniquely whenever the directions differ, and M will automatically be parallel to the result (test its own point to see whether it is contained).

C3 — Plane through a line, parallel to a given plane with normal $\vec{n}$. The desired normal must be perpendicular to $\vec{b}$ (to contain the line) and to $\vec{n}$ (to be parallel):

$$\vec{N} = \vec{b} \times \vec{n}.$$

Caveat: if $\vec{b} \parallel \vec{n}$ the line pierces every plane parallel to $\vec{n}$'s family, so no such plane exists — the cross product announcing itself as $\vec{0}$ is the diagnosis.

Example C.1 — [JEE Main Level]

Show that $(1, 2, 3)$, $(3, 2, 1)$, $(2, 3, 1)$, $(6, 0, 0)$ are coplanar, and find the plane.

With $A = (1, 2, 3)$: $\overrightarrow{AB} = (2, 0, -2)$, $\overrightarrow{AC} = (1, 1, -2)$, $\overrightarrow{AD} = (5, -2, -3)$. First the normal:

$$\overrightarrow{AB} \times \overrightarrow{AC} = (2, 2, 2).$$

Triple product: $(2, 2, 2) \cdot (5, -2, -3) = 10 - 4 - 6 = 0$ — coplanar. The plane has normal proportional to $(1, 1, 1)$ and passes through A :

$$1(x - 1) + 1(y - 2) + 1(z - 3) = 0 \implies x + y + z = 6.$$

Check: the remaining three points each sum to 6.

Example C.2 — [JEE Advanced Level]

(a) Find the plane containing the line $\vec{r} = (\hat{i} + \hat{k}) + \lambda(\hat{i} + \hat{j})$ and parallel to the line $\vec{r} = (2\hat{i} + \hat{j} + 3\hat{k}) + \mu(\hat{j} + \hat{k})$. (b) How far is the second line from this plane?

(a) Here $\vec{a} = (1, 0, 1)$, $\vec{b} = (1, 1, 0)$, $\vec{m} = (0, 1, 1)$, and $\vec{b} \nparallel \vec{m}$, so the plane is unique:

$$\vec{N} = \vec{b} \times \vec{m} = (1, -1, 1).$$

Through $(1, 0, 1)$:

$$1(x - 1) - 1(y - 0) + 1(z - 1) = 0 \implies x - y + z = 2.$$

Verification: the line's point $(1, 0, 1)$ gives 2; its displaced point $(2, 1, 1)$ gives 2; the direction satisfies $(1, 1, 0) \cdot (1, -1, 1) = 0$; and the second line's direction satisfies $(0, 1, 1) \cdot (1, -1, 1) = 0$ — parallel as demanded, while its point $(2, 1, 3)$ gives $4 \neq 2$, so it sits strictly off the plane rather than inside it.

(b) The whole of L_2 is equidistant from the plane (its direction is parallel), so measure from $(2, 1, 3)$:

$$\frac{|2 - 1 + 3 - 2|}{\sqrt{1 + 1 + 1}} = \frac{2}{\sqrt{3}} \text{ units.}$$

Practice — Addendum C

1. Determine whether $(1, 1, 1)$, $(2, 3, 4)$, $(5, 3, 2)$, $(1, 2, 3)$ are coplanar; if the first three define a plane, state it.
2. Find the plane through the line $\vec{r} = (\hat{j} + 2\hat{k}) + \lambda(2\hat{i} + \hat{j} - \hat{k})$ that is parallel to the z -axis.
3. Find the plane through the line $\vec{r} = (\hat{i} + \hat{j} + \hat{k}) + \lambda(2\hat{i} - \hat{j} + \hat{k})$ parallel to the plane $x + 2y - z = 0$.
4. Find k so that $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$, $(1, 1, k)$ are coplanar.
5. Prove that if three of four points are collinear, all four are automatically coplanar.

Answers: 1. $\overrightarrow{AB} \times \overrightarrow{AC} = (-4, 11, -6)$; triple product with $\overrightarrow{AD} = (0, 1, 2)$ is $0 + 11 - 12 = -1 \neq 0$: not coplanar. The plane through the first three is $4x - 11y + 6z + 1 = 0$, at which the fourth point evaluates to 1. 2. $\vec{N} = (2, 1, -1) \times (0, 0, 1) = (1, -2, 0)$; plane through $(0, 1, 2)$: $x - 2y + 2 = 0$. 3. $\vec{N} = (2, -1, 1) \times (1, 2, -1) = (-1, 3, 5)$; plane through $(1, 1, 1)$: $x - 3y - 5z + 7 = 0$; check $(-1, 3, 5) \cdot (1, 2, -1) = 0$ confirms the parallelism. 4. $\overrightarrow{AB} \times \overrightarrow{AC} = (1, 1, 1)$ and $\overrightarrow{AD} = (0, 1, k)$, so the triple product is $1 + k$: $k = -1$ (equivalently, D must sit on $x + y + z = 1$). 5. Collinearity of the first three forces $\overrightarrow{AB} \times \overrightarrow{AC} = \vec{0}$, so the triple product with any $\overrightarrow{AD}$ vanishes — the condition holds identically, and some plane through the line contains the fourth point.

Addendum D — Mirror Shortcuts: Origin Feet, Coordinate Walls, Shortest Paths

Two small results do disproportionate work in JEE papers. The first converts between a plane and the foot of the perpendicular dropped to it from the origin — in *both* directions. The

second is the reflection principle: any shortest path that must touch a flat mirror becomes a straight line once one endpoint is imaged. Neither appears as a named tool in the chapter, yet both are one line away from Subtopic 04.

Method

D1 — Foot from the origin, and back again. If the foot of the perpendicular from O to $ax + by + cz = d$ is F , then F lies on the plane and OF runs along the normal, so $F = s(a, b, c)$ for some scalar s . Substituting:

$$s(a^2 + b^2 + c^2) = d \implies F = \frac{d}{a^2 + b^2 + c^2} (a, b, c).$$

The reverse direction is the exam favourite. If the foot is given as (α, β, γ) , the normal is parallel to (α, β, γ) and the plane passes through (α, β, γ) , so

$$\alpha x + \beta y + \gamma z = \alpha^2 + \beta^2 + \gamma^2.$$

Free check: the origin's distance from this plane, namely the right side over $\sqrt{\alpha^2 + \beta^2 + \gamma^2}$, must equal $|\vec{OF}| = \sqrt{\alpha^2 + \beta^2 + \gamma^2}$ — and it does.

D2 — Images in the coordinate walls. The foot of (x, y, z) in the xy -plane is $(x, y, 0)$, so the image is $2(x, y, 0) - (x, y, z) = (x, y, -z)$: flip one sign. Reflecting in an *axis* flips the two signs of the other coordinates: the image of (x, y, z) in the x -axis is $(x, -y, -z)$. These are instant — no t , no formula.

D3 — The mirror trick. Let B' be the image of B in a plane π . Every point M of π is equidistant from B and B' : for instance with π the xy -plane, $B = (x_B, y_B, z_B)$, $B' = (x_B, y_B, -z_B)$ and $M = (u, v, 0)$,

$$MB^2 = (u - x_B)^2 + (v - y_B)^2 + z_B^2 = MB'^2.$$

Hence for every admissible M ,

$$AM + MB = AM + MB' \geq AB',$$

with equality exactly when M is the point where segment AB' crosses π . So: reflect, join, read the crossing point; the straight length AB' is the minimum path.

Example D.1 — [JEE Main Level]

(a) Find the foot of the perpendicular from the origin to the plane $2x + 3y + 6z = 42$.

(b) A plane has foot of perpendicular from the origin at $(1, 2, 2)$; find the plane.

(a) $a^2 + b^2 + c^2 = 4 + 9 + 36 = 49$, so

$$F = \frac{42}{49}(2, 3, 6) = \left(\frac{12}{7}, \frac{18}{7}, \frac{36}{7}\right).$$

Sanity check: the distance from the origin is $\frac{42}{7} = 6$, and indeed $|F| = \frac{6}{7} \cdot 7 = 6$.

(b) By D1 reversed, the plane is

$$1 \cdot x + 2 \cdot y + 2 \cdot z = 1 + 4 + 4 = 9, \quad \text{i.e. } x + 2y + 2z = 9.$$

Check: distance from origin $= 9/3 = 3 = |(1, 2, 2)|$, consistent.

Example D.2 — [JEE Advanced Level]

A ray travels from $A(1, 1, 2)$, reflects off the xy -plane, and reaches $B(3, 4, 6)$. Find the point of contact and the total path length.

Reflect B in the xy -plane: $B' = (3, 4, -6)$. The straight segment from A to B' has direction $(2, 3, -8)$; parametrise it:

$$(1 + 2t, 1 + 3t, 2 - 8t).$$

It meets the mirror where $z = 0$: $2 - 8t = 0$, giving $t = \frac{1}{4}$ and

$$M = \left(\frac{3}{2}, \frac{7}{4}, 0\right),$$

which lies on the mirror and strictly between A and B' . Total path:

$$AM + MB = AB' = \sqrt{2^2 + 3^2 + (-8)^2} = \sqrt{77} \text{ units.}$$

Direct verification that the pieces really sum to $\sqrt{77}$: AM runs over $(\frac{1}{2}, \frac{3}{4}, -2)$, whose squared length is $\frac{1}{4} + \frac{9}{16} + 4 = \frac{77}{16}$, so $AM = \frac{\sqrt{77}}{4}$; MB runs over $(\frac{3}{2}, \frac{9}{4}, 6)$, whose squared length is $\frac{36+81+576}{16} = \frac{693}{16} = \frac{9 \cdot 77}{16}$, so $MB = \frac{3\sqrt{77}}{4}$. The sum is $\sqrt{77}$, exactly as promised — the mirror trick did not approximate, it computed.

Practice — Addendum D

1. Find the foot of the perpendicular from the origin to $3x - 4y + 12z = 26$, and the plane's distance from the origin.
2. The foot of the perpendicular from the origin to a plane is $(2, -1, 2)$. Find the plane.
3. Write the images of $(3, -4, 5)$ in the yz -plane, and in the x -axis.
4. Find the point where the shortest path from $(0, 1, 1)$ to $(2, 3, 5)$ touches the zx -plane, and the path's length.
5. Prove the claim used in D3: every point of the zx -plane is equidistant from $B(u, v, w)$ and its image there.
6. The foot of the perpendicular from the origin to a plane is $(1, 1, 1)$ and the plane passes through $(3, 0, 0)$ — find it, or declare the data inconsistent.

Answers: 1. $|\vec{n}|^2 = 169$, $s = \frac{26}{169} = \frac{2}{13}$; foot $(\frac{6}{13}, -\frac{8}{13}, \frac{24}{13})$; distance $\frac{26}{13} = 2$. 2. $2x - y + 2z = 9$. 3. In the yz -plane: $(-3, -4, 5)$; in the x -axis: $(3, 4, -5)$. 4. Reflect the far endpoint: $B' = (2, -3, 5)$; segment $(0, 1, 1) + t(2, -4, 4)$ hits $y = 0$ at $t = \frac{1}{4}$, so $M = (\frac{1}{2}, 0, 2)$; length $|AB'| = \sqrt{4 + 16 + 16} = 6$. 5. For $M = (p, 0, q)$: $MB^2 = (p - u)^2 + v^2 + (q - w)^2$ while MB'^2 with $B' = (u, -v, w)$ is $(p - u)^2 + v^2 + (q - w)^2$ — identical. 6. Consistent: the foot fixes $x + y + z = 3$, and $(3, 0, 0)$ satisfies it; the plane is $x + y + z = 3$.

Addendum E — Locus Problems: Turning Geometry into Equations

Every “find the locus of...” question in this chapter is the same move: name a generic point (x, y, z) , translate the stated condition into an equation, and recognise the surface — linear

means plane, quadratic with equal squared coefficients and no cross terms means sphere. The chapter uses this machinery everywhere but never states it, and JEE Main and Advanced both ask it cold.

Method

E1 — Equidistance from two points: the perpendicular bisector plane. Locus of P with $PA = PB$. Square both sides and cancel the quadratic terms:

$$|\vec{p} - \vec{a}|^2 = |\vec{p} - \vec{b}|^2 \implies -2\vec{p} \cdot \vec{a} + |\vec{a}|^2 = -2\vec{p} \cdot \vec{b} + |\vec{b}|^2,$$

so

$$2\vec{p} \cdot (\vec{b} - \vec{a}) = |\vec{b}|^2 - |\vec{a}|^2.$$

A linear equation: a plane with normal $\vec{b} - \vec{a}$ — along the segment, as geometry demands — passing through the midpoint (substitute $\frac{1}{2}(\vec{a} + \vec{b})$ to confirm).

E2 — Feet of perpendiculars from the origin: a sphere on OP as diameter. Let every admissible plane pass through the fixed point P , and let F be the foot from O to such a plane. The plane's normal is parallel to $\vec{f} = \overrightarrow{OF}$ and it contains both F and P , so the in-plane vector $\vec{p} - \vec{f}$ is perpendicular to the normal:

$$(\vec{p} - \vec{f}) \cdot \vec{f} = 0 \implies \vec{f} \cdot \vec{p} = |\vec{f}|^2.$$

Writing $\vec{f} = (x, y, z)$ and $\vec{p} = (x_0, y_0, z_0)$:

$$x^2 + y^2 + z^2 = x_0x + y_0y + z_0z,$$

a sphere passing through O and through P , with centre $\frac{1}{2}\vec{p}$ and radius $\frac{1}{2}|\vec{p}|$ — the sphere on OP as diameter. Complete the square to report it in standard form.

E3 — Equidistance from two planes: intersecting planes give the bisector pair of unit 05c; *parallel* planes $ax + by + cz + d_1 = 0$ and $ax + by + cz + d_2 = 0$ give the single mid-plane $ax + by + cz + \frac{d_1 + d_2}{2} = 0$.

Example E.1 — [JEE Main Level]

Find the locus of points equidistant from $A(1, 2, 3)$ and $B(3, 4, 5)$, and verify it geometrically.

With $\vec{b} - \vec{a} = (2, 2, 2)$ and $|\vec{b}|^2 - |\vec{a}|^2 = 50 - 14 = 36$:

$$2(2x + 2y + 2z) = 36 \implies x + y + z = 9.$$

Geometric verification: the midpoint $(2, 3, 4)$ satisfies $2 + 3 + 4 = 9$; the normal $(1, 1, 1)$ is parallel to $\overrightarrow{AB} = (2, 2, 2)$ — it is the perpendicular bisector plane, exactly as E1 predicts.

Example E.2 — [JEE Advanced Level]

Planes pass through the fixed point $P(2, 4, 4)$. Find the locus of the foot of the perpendicular from the origin to these planes, and describe it.

By E2 with $\vec{p} = (2, 4, 4)$:

$$x^2 + y^2 + z^2 = 2x + 4y + 4z.$$

Completing squares:

$$(x-1)^2 + (y-2)^2 + (z-2)^2 = 1 + 4 + 4 = 9.$$

The locus is a sphere of radius 3 centred at $(1, 2, 2)$ — precisely $\frac{1}{2}\vec{p}$ and $\frac{1}{2}|\vec{p}| = 3$, the sphere on OP as diameter. **Checks:** the origin satisfies $0 = 0$; the point P satisfies $4+16+16 = 4+16+16$; both endpoints of the claimed diameter lie on the locus, as the theory demands.

Practice — Addendum E

1. Find the perpendicular bisector plane of the segment joining $(2, -1, 3)$ and $(4, 3, -1)$.
2. Planes pass through the fixed point $(1, 2, 2)$. Find the locus of the foot of the perpendicular from the origin, in standard form.
3. Find the locus of points equidistant from the parallel planes $x+y+z = 1$ and $x+y+z = 7$.
4. Find the point on the z -axis equidistant from $(1, 2, 3)$ and $(2, 3, 4)$.
5. Show that the locus in Question 2 always passes through the origin and the fixed point.
6. Find the locus of points at distance 2 from the plane $x + 2y + 2z - 9 = 0$.

Answers: 1. $\vec{b} - \vec{a} = (2, 4, -4)$, $|\vec{b}|^2 - |\vec{a}|^2 = 26 - 14 = 12$; locus $2x + 4y - 4z = 6$, i.e. $x + 2y - 2z = 3$ (midpoint $(3, 1, 1)$ satisfies it). 2. $x^2 + y^2 + z^2 = x + 2y + 2z$, i.e. $(x - \frac{1}{2})^2 + (y - 1)^2 + (z - 1)^2 = \frac{9}{4}$; sphere, centre $(\frac{1}{2}, 1, 1)$, radius $\frac{3}{2}$. 3. The mid-plane $x + y + z = 4$. 4. For $(0, 0, t)$: $t^2 - 6t + 14 = t^2 - 8t + 29$ gives $t = \frac{15}{2}$; point $(0, 0, \frac{15}{2})$. 5. Substituting $(0, 0, 0)$ gives $0 = 0$; substituting $(1, 2, 2)$ gives $1 + 4 + 4 = 1 + 4 + 4$ — both satisfy the unsquared equation identically, confirming the diameter property for any fixed point. 6. $\frac{|x + 2y + 2z - 9|}{3} = 2$ splits into two parallel planes $x + 2y + 2z = 15$ and $x + 2y + 2z = 3$.

Addendum P — What JEE Actually Asks

The Distribution

The 43-year bank (Disha ch26, 66 questions fully legible) shows the following emphasis across the six subtopics of the NCERT chapter. Numbers are approximate counts of *distinct* questions that primarily test the subtopic; many questions span two subtopics.

Subtopic (NCERT label)	Key JEE themes	Representative years/sessions (from the extract)	Approx. frequency
Foundations (distance, section formula, centroid, collinearity)	Section formula with sign handling, centroid to find unknown vertex, distance from axis vs. plane, collinearity via direction ratios	2019 Apr 9 (II), 2019 Apr 8 (II), 2013 Apr 25, 2019 Jan 11 (II), 2020 Jan 7 (II), 2018 Adv, 1983, 2003	8–10
Direction Cosines / Ratios & Line Equations	DC/DR conversion, angle between lines via DCs, line through two points, equally inclined lines	2019 Apr 10 (I), 2018 Online Apr 15, 2014, 2004S, 2020 Jan 7 (II), 2014 Adv, 2013 Adv, 2009, 2008	10–12
Shortest Distance & Angle Between Lines	Skew-line distance (vector & Cartesian), coplanarity condition, common perpendicular, parallel-line distance	2018, 2020 Jan 8 (I), 2019 Adv, 2019 Adv, 2020 Sep 6 (I), 2020 Sep 5 (II), 2012, 2010, 2008, 2007	10–12

Subtopic (NCERT label)	Key JEE themes	Representative years/sessions (from the extract)	Approx. frequency
Planes - Equations, Family, Angles, Distances	Plane through intercepts/-points/line, family $P_1 + \lambda P_2 = 0$, angle bisectors, point-plane distance, parallel planes distance, plane perpendicular to two planes	2020 Sep 6 (II), 2020 Sep 5 (I), 2020 Sep 4 (II), 2020 Sep 3 (I,II), 2020 Sep 2 (I,II), 2020 Jan 9 (I), 2020 Jan 8 (II), 2020 Jan 7 (I), 2019 Apr 12 (I,II), 2019 Apr 9 (I), 2019 Apr 8 (I), 2019 Jan 12 (I,II), 2019 Jan 11 (I,II), 2019 Jan 10 (II), 2018 Online Apr 15, 2017, 2016, 2015, 2014 Online Apr 12, 2013, 2012, 2010, 2005S, 2003S, 2018 Adv, 2010, 2020 Sep 4 (I), 2020 Sep 3 (II), 2020 Jan 9 (II), 2019 Adv, 1996, 2020 Adv, 2020 Adv, 2018 Adv, 2016 Adv, 2015 Adv, 2013 Adv, 2008, 2007, 2005, 2004, 2004, 2003, 1998, 1994, 1994, 1986, 1985, 1981	25-30
Perpendicular Toolkit (foot, distance, image in line/plane)	Foot of perpendicular from point to line/plane, image of point, distance measured <i>parallel</i> to a line, reflection in a plane	2019 Apr 10 (I), 2019 Apr 9 (II), 2020 Sep 5 (I), 2019 Apr 10 (II), 2019 Apr 8 (I), 2020 Jan 7 (II), 2020 Sep 4 (II), 2020 Sep 3 (I), 2017, 2016, 2010, 2020 Sep 4 (I), 2020 Sep 3 (II), 2018 Adv, 2012	10-12
Line-Plane & Plane-Plane Configurations	Line-plane intersection, “parallel vs. lies in” test, plane containing given lines, projection/reflection of a line, locus of feet	2019 Apr 10 (II), 2020 Sep 4 (II), 2019 Apr 12 (I), 2019 Jan 11 (I), 2019 Jan 11 (II), 2009, 2020 Sep 3 (II), 2020 Jan 9 (II), 2019 Adv, 2015 Adv, 2013 Adv, 2012, 2006	8-10

Takeaway: Planes (family, bisectors, distances) and the perpendicular toolkit (foot/image) together constitute nearly half the questions. The “foundations” and “direction cosines” appear mostly as building blocks inside larger problems. JEE Advanced frequently mixes 3-D geometry with vectors, loci, and polyhedra.

The Recurring Archetypes

Here are the 7 question shapes that reappear most often. For each we give the *look*, the *core method*, and a **fully worked example reconstructed from the bank**.

1. Shortest Distance Between Two Lines (Skew / Parallel) & Coplanarity

Look: Two lines given in vector or Cartesian form (one may be the intersection of two planes). Ask for the shortest distance, or “find k so that the lines are coplanar / intersect”.

Method:

- For skew lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$, $\vec{r} = \vec{a}_2 + \mu\vec{b}_2$:

$$d = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}.$$

- For parallel lines ($\vec{b}_1 \parallel \vec{b}_2$): $d = \frac{|\vec{b} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}|}$.

- Coplanarity $\iff (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0$.

Always compute $\vec{b}_1 \times \vec{b}_2$ first - its magnitude is the denominator and its zero-ness tells you instantly whether the lines are parallel.

Worked Example (adapted from **JEE Main 2020 Sep 6 (I), Q23**):

Find the shortest distance between the line $L_1 : \frac{x+2}{1} = \frac{y}{-1} = \frac{z-1}{2}$ and the line L_2 which is the intersection of the planes $x + y + z + 1 = 0$ and $2x - y + z + 3 = 0$.

Solution:

L_1 passes through $A(-2, 0, 1)$ with direction $\vec{b}_1 = (1, -1, 2)$.

L_2 is the intersection of the two planes. A direction vector is $\vec{b}_2 = \vec{n}_1 \times \vec{n}_2$ where $\vec{n}_1 = (1, 1, 1)$, $\vec{n}_2 = (2, -1, 1)$.

$$\vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 2 & -1 & 1 \end{vmatrix} = (2, 1, -3).$$

A point on L_2 : set $y = 0$ in the planes $\Rightarrow x + z + 1 = 0$, $2x + z + 3 = 0$. Subtracting gives $x = -2$, then $z = 1$. So $B(-2, 0, 1)$.

$\vec{a}_2 - \vec{a}_1 = \vec{0}$ - the lines **share the point** $(-2, 0, 1)$! Hence they intersect and the shortest distance is **0**.

(If the point were not shared, we would compute $\vec{b}_1 \times \vec{b}_2 = (1, -1, 2) \times (2, 1, -3) = (1, 7, 3)$, $|\vec{b}_1 \times \vec{b}_2| = \sqrt{59}$, and $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0$ because the lines intersect - the scalar triple product vanishes.)

2. Plane Through the Intersection of Two Planes (Family $P_1 + \lambda P_2 = 0$) with an Extra Condition

Look: “Find the equation of the plane through the line of intersection of planes $P_1 = 0$ and $P_2 = 0$ that also passes through a given point / is parallel to a given line / is perpendicular to a given plane / is at a given distance from a point.”

Method: Write the family $P_1 + \lambda P_2 = 0$ (or $\mu P_1 + P_2 = 0$ to avoid missing P_2). Impose the extra condition to get a linear equation in λ . Substitute back and simplify.

Worked Example (JEE Main 2019 Apr 8, Q36):

Find the equation of the plane containing the line of intersection of the planes $2x - y - 4 = 0$ and $y + 2z - 4 = 0$ and passing through the point $(1, 1, 0)$.

Solution:

Family: $(2x - y - 4) + \lambda(y + 2z - 4) = 0$.

Passes through $(1, 1, 0)$: $(2 - 1 - 4) + \lambda(1 + 0 - 4) = 0 \Rightarrow -3 - 3\lambda = 0 \Rightarrow \lambda = -1$.

Substitute $\lambda = -1$: $(2x - y - 4) - (y + 2z - 4) = 0 \Rightarrow 2x - 2y - 2z = 0 \Rightarrow \boxed{x - y - z = 0}$.

3. Foot of Perpendicular & Image of a Point in a Line / Plane

Look: “Find the foot / distance / image of point P in the line L / plane π .” Often combined: “The foot lies on which plane?” or “The image satisfies ...”.

Method:

- **Line** $\vec{r} = \vec{a} + \lambda \vec{b}$: $\lambda_0 = \frac{(\vec{p} - \vec{a}) \cdot \vec{b}}{|\vec{b}|^2}$, foot $\vec{F} = \vec{a} + \lambda_0 \vec{b}$, distance $PF = \frac{|(\vec{p} - \vec{a}) \times \vec{b}|}{|\vec{b}|}$, image $\vec{P}' = 2\vec{F} - \vec{p}$.

- **Plane** $ax + by + cz + d = 0$: $t = \frac{ax_1 + by_1 + cz_1 + d}{a^2 + b^2 + c^2}$, foot $\vec{F} = \vec{P} - t\vec{n}$, image $\vec{P}' = \vec{P} - 2t\vec{n}$, distance $= |t||\vec{n}| = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$.

Always verify $\vec{PF} \cdot \vec{b} = 0$ (line) or that F satisfies the plane equation - it catches sign errors.

Worked Example (JEE Main 2020 Sep 3 (I), Q26):

The foot of the perpendicular drawn from the point $P(4, 2, 3)$ to the line joining $A(1, -2, 3)$ and $B(1, 1, 0)$ lies on which plane?

Solution:

Line AB : direction $\vec{b} = (0, 3, -3) \parallel (0, 1, -1)$. Parametric: $(1, -2 + \lambda, 3 - \lambda)$.

$\vec{AP} = (3, 4, 0)$. $\lambda_0 = \frac{\vec{AP} \cdot \vec{b}}{|\vec{b}|^2} = \frac{(3, 4, 0) \cdot (0, 1, -1)}{2} = \frac{4}{2} = 2$.

Foot $F = (1, 0, 1)$.

Check the options:

(a) $2x + y - z = 1 \Rightarrow 2 + 0 - 1 = 1 \checkmark$

(b) $x - y - 2z = 1 \Rightarrow 1 - 0 - 2 = -1 \times$

(c) $x - 2y + z = 1 \Rightarrow 1 - 0 + 1 = 2 \times$

(d) $x + 2y - z = 1 \Rightarrow 1 + 0 - 1 = 0 \times$

Answer: (a) $2x + y - z = 1$.

4. Angle Between a Line and a Plane (the Sine Trap) & Angle Between Two Planes

Look: "Find the angle between the line $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$ and the plane $Ax + By + Cz + D = 0$." Or "Find the angle between two planes."

Method:

- Plane-plane: $\cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}$.
- Line-plane: **sine**, not cosine! $\sin \phi = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}||\vec{n}|}$ where $\vec{b}$ is the line direction and $\vec{n}$ the plane normal.
- Bisector planes of $P_1 = 0$ and $P_2 = 0$: $\frac{P_1}{|\vec{n}_1|} = \pm \frac{P_2}{|\vec{n}_2|}$ (two perpendicular planes).

Worked Example (JEE Main 2019 Apr 12 (II), Q34):

A plane which bisects the angle between the two given planes $2x - y + 2z - 4 = 0$ and $x + 2y + 2z - 2 = 0$ passes through which point?

Solution:

$$|\vec{n}_1| = \sqrt{4 + 1 + 4} = 3, |\vec{n}_2| = \sqrt{1 + 4 + 4} = 3.$$

$$\text{Bisectors: } \frac{2x - y + 2z - 4}{3} = \pm \frac{x + 2y + 2z - 2}{3} \Rightarrow 2x - y + 2z - 4 = \pm(x + 2y + 2z - 2).$$

$$\text{Case +: } x - 3y - 2 = 0.$$

$$\text{Case -: } 3x + y + 4z - 6 = 0.$$

Test the options:

$$(2, -4, 1) \text{ in second plane: } 3(2) + (-4) + 4(1) - 6 = 6 - 4 + 4 - 6 = 0 \checkmark.$$

Answer: (d) (2, -4, 1).

5. Line-Plane Intersection and the "Parallel vs. Lies In" Decision

Look: A line and a plane are given. Determine whether they intersect (and find the point), or the line lies in the plane, or is parallel to it.

Method:

1. Compute $\vec{b} \cdot \vec{n}$.

- If $\neq 0$: line **pierces** the plane. Substitute parametric point into plane, solve for λ .
- If $= 0$: line is **parallel** to the plane. Test any point of the line in the plane equation.
- Point satisfies $\Rightarrow$ line **lies in** the plane.
- Point does not satisfy $\Rightarrow$ line is **strictly parallel**.

Never stop at $\vec{b} \cdot \vec{n} = 0$; the point test is mandatory.

Worked Example (JEE Main 2012, Q50):

Point P is the intersection of the line joining $Q(2, 3, 5)$ and $R(1, -1, 4)$ with the plane $5x - 4y - z = 1$. S is the foot of the perpendicular from $T(2, 1, 4)$ to QR . Find PS .

Solution:

$$\text{Line } QR: \vec{r} = (2, 3, 5) + \lambda(-1, -4, -1).$$

Intersection P : substitute into plane:

$$5(2 - \lambda) - 4(3 - 4\lambda) - (5 - \lambda) = 1 \Rightarrow 10 - 5\lambda - 12 + 16\lambda - 5 + \lambda = 1 \Rightarrow 12\lambda = 8 \Rightarrow \lambda = \frac{2}{3}.$$

$$P = \left(\frac{4}{3}, \frac{1}{3}, \frac{13}{3}\right).$$

$$\text{Foot } S \text{ from } T \text{ to } QR: \vec{QT} = (0, -2, -1), \vec{d} = (-1, -4, -1), |\vec{d}|^2 = 18.$$

$$\lambda_S = \frac{\vec{QT} \cdot \vec{d}}{|\vec{d}|^2} = \frac{9}{18} = \frac{1}{2}.$$

$$S = (2, 3, 5) + \frac{1}{2}(-1, -4, -1) = \left(\frac{3}{2}, 1, \frac{9}{2}\right).$$

$$\overrightarrow{PS} = \left(\frac{1}{6}, \frac{2}{3}, \frac{1}{6}\right), |\overrightarrow{PS}| = \sqrt{\frac{1}{36} + \frac{4}{9} + \frac{1}{36}} = \sqrt{\frac{18}{36}} = \frac{1}{\sqrt{2}}.$$

Answer: $\frac{1}{\sqrt{2}}$ (or $\frac{\sqrt{2}}{2}$).

6. Plane Containing Given Lines (Intersecting / Parallel) or a Line and a Point

Look: “Find the plane passing through points A, B and parallel to line L ” or “containing lines L_1 and L_2 ” or “through point P and containing lines with DRs ...”.

Method: The normal is always the cross product of **two vectors lying in the plane**.

- Two intersecting lines: $\vec{N} = \vec{b}_1 \times \vec{b}_2$.
- Two parallel lines: $\vec{N} = \vec{b} \times (\vec{a}_2 - \vec{a}_1)$.
- Line + external point: $\vec{N} = \vec{b} \times (\vec{P} - \vec{a})$.

Then use point-normal form.

Worked Example (JEE Main 2020 Sep 2 (I), Q28):

The plane passing through $(1, 2, 1)$ and $(2, 1, 2)$ and parallel to the line $2x = 3y, z = 1$ also passes through which point?

Solution:

Line $2x = 3y, z = 1 \Rightarrow$ direction $\vec{b} = (3, 2, 0)$ (or $(-3, -2, 0)$).

Points $A(1, 2, 1), B(2, 1, 2) \Rightarrow \vec{AB} = (1, -1, 1)$.

Normal $\vec{N} = \vec{AB} \times \vec{b} = (1, -1, 1) \times (3, 2, 0) = (-2, 3, 5)$.

Plane through A : $-2(x-1)+3(y-2)+5(z-1) = 0 \Rightarrow -2x+3y+5z-9 = 0 \Rightarrow 2x-3y-5z+9 = 0$.

Check options: $(-2, 0, 1)$ gives $-4 - 0 - 5 + 9 = 0 \checkmark$.

Answer: (b) $(-2, 0, 1)$.

7. Distance of a Point from a Plane Measured Parallel to a Given Line

Look: “Distance of point P from plane π measured parallel to line L .”

Method: Write the line through P parallel to L (direction $\vec{b}$). Find its intersection Q with the plane. The required distance is PQ .

Worked Example (JEE Main 2016, Q45):

Distance of the point $(1, -5, 9)$ from the plane $x - y + z = 5$ measured along the line $x = y = z$.

Solution:

Line through P parallel to $x = y = z$ has direction $(1, 1, 1)$. Parametric: $(1 + t, -5 + t, 9 + t)$.

Substitute into plane: $(1 + t) - (-5 + t) + (9 + t) = 5 \Rightarrow 15 + t = 5 \Rightarrow t = -10$.

Intersection $Q = (-9, -15, -1)$.

$PQ = \sqrt{(-10)^2 + (-10)^2 + (-10)^2} = 10\sqrt{3}$.

Answer: $10\sqrt{3}$.

The Traps

These are the specific mistakes the JEE 3-D questions are engineered to punish, taken directly from distractor patterns in the bank.

Trap	Description	Typical Distractor
Section-formula	$m : n$ from A , the weight m	Option with coordinates averaged the
weight swap	multiplies the <i>far</i> endpoint B .	wrong way (e.g. midpoint instead of 1:3).
Negative ratio = external division	Solving for k in $\frac{kx_2+x_1}{k+1} = 0$ gives $k = -\frac{x_1}{x_2}$; a negative k means the plane cuts the segment <i>externally</i> .	Treating negative k as an error or ignoring the sign.
Modulus in angle formula	$\cos \theta = \frac{ \vec{b}_1 \cdot \vec{b}_2 }{ \vec{b}_1 \vec{b}_2 }$ - always acute.	Forgetting $ \cdot $ and reporting an obtuse angle (e.g. $\cos^{-1}(-4/9)$).
Sine vs. cosine for line-plane angle	Line-plane uses sine : $\sin \phi = \frac{ \vec{b} \cdot \vec{n} }{ \vec{b} \vec{n} }$.	Using cosine (i.e. finding the angle with the normal) - the most common single-mark loss.
Bisector planes need normalisation	$\frac{P_1}{ \vec{n}_1 } = \pm \frac{P_2}{ \vec{n}_2 }$.	Equating raw plane expressions $P_1 = \pm P_2$ - gives a plane through the intersection but <i>not</i> the angle bisector.
Parallel vs. lies in plane	$\vec{b} \cdot \vec{n} = 0$ only implies <i>parallel</i> ; must test a point to distinguish “lies in” from “strictly parallel”.	Declaring “line lies in the plane” after only checking $\vec{b} \cdot \vec{n} = 0$.
Wrong denominator in point-line distance	Distance = $\frac{ (\vec{p}-\vec{a}) \times \vec{b} }{ \vec{b} }$ (one power of $ \vec{b} $).	Dividing by $ \vec{b} ^2$ or omitting the denominator.
t vs. distance in point-plane	$t = \frac{ax_1+by_1+cz_1+d}{a^2+b^2+c^2}$ (divides by $ \vec{n} ^2$); distance = $ t \vec{n} $ (divides by $ \vec{n} $).	Confusing the two - off by a factor of $ \vec{n} $.
Image of point = 2×foot - point	$\vec{P}' = 2\vec{F} - \vec{P}$ works for both line and plane.	Reporting the foot $\vec{F}$ as the image.
Family misses P_2	If the answer is P_2 itself, use $\mu P_1 + P_2 = 0$.	Missing the second bisector / the plane P_2 in the family.

A Graded Practice Set

Eight questions in ascending difficulty, all adapted from the bank with verified answers. Full solutions are given at the end.

1. **[JEE Main 2019 Apr 8 (II) - Easy]**

Point $R(4, y, z)$ lies on the segment joining $P(2, -3, 4)$ and $Q(8, 0, 10)$. Find the distance of R from the origin.

2. **[JEE Main 2013 Apr 25 - Easy]**

Triangle ABC has $A(2, 3, 5)$, $B(-1, 3, 2)$, $C(\lambda, 5, \mu)$. The median through A is equally inclined to the axes. Find (λ, μ) .

3. **[JEE Main 2020 Sep 3 (I) - Medium]**

Find the foot of the perpendicular from $P(4, 2, 3)$ to the line through $A(1, -2, 3)$ and $B(1, 1, 0)$. Which plane among $2x + y - z = 1$, $x - y - 2z = 1$, $x - 2y + z = 1$, $x + 2y - z = 1$ contains this foot?

4. **[JEE Main 2020 Sep 3 (II) - Medium]**

The plane bisecting the segment joining $(4, -2, 3)$ and $(2, 4, -1)$ at right angles passes through which point?

(a) $(4, 0, 1)$ (b) $(0, -1, 1)$ (c) $(4, 0, -1)$ (d) $(0, 1, -1)$

5. **[JEE Main 2019 Apr 8 (I) - Medium]**

Equation of the plane through the line of intersection of $2x - y - 4 = 0$ and $y + 2z - 4 = 0$ and passing through $(1, 1, 0)$.

6. **[JEE Main 2019 Apr 12 (II) - Medium]**

The plane bisecting the angle between $2x - y + 2z - 4 = 0$ and $x + 2y + 2z - 2 = 0$ passes through which point?

(a) $(1, -4, 1)$ (b) $(1, 4, -1)$ (c) $(2, 4, 1)$ (d) $(2, -4, 1)$

7. **[JEE Main 2016 - Hard]**

Distance of $(1, -5, 9)$ from the plane $x - y + z = 5$ measured *parallel to the line* $x = y = z$.

8. **[JEE Advanced 2016 - Hard]**

Pyramid $OPQRS$ has $O(0, 0, 0)$, $P(3, 0, 0)$, $R(0, 3, 0)$. Base $OPQR$ is a square ($OP = 3$). S is above the midpoint T of OQ with $TS = 3$. Which of the following are true?

(a) Acute angle between OQ and OS is $\cos^{-1} \frac{1}{\sqrt{3}}$.

(b) Plane OQS is $x - y = 0$.

(c) Perpendicular distance from P to plane OQS is $\frac{3}{\sqrt{2}}$.

(d) Perpendicular distance from O to line RS is $\frac{\sqrt{30}}{2}$.

Solutions to the Graded Practice Set

1. **Section formula:** R divides PQ in some ratio $\lambda : 1$.

$$\frac{2+8\lambda}{1+\lambda} = 4 \Rightarrow 2 + 8\lambda = 4 + 4\lambda \Rightarrow 4\lambda = 2 \Rightarrow \lambda = \frac{1}{2}.$$

$$\text{Then } y = \frac{-3+0\cdot\lambda}{1+\lambda} = -2, z = \frac{4+10\lambda}{1+\lambda} = 6.$$

$$OR = \sqrt{4^2 + (-2)^2 + 6^2} = \sqrt{56} = 2\sqrt{14}.$$

Answer: $2\sqrt{14}$.

2. **Median equally inclined $\Rightarrow$ direction ratios of AD are equal.**

$$D \text{ is midpoint of } BC: D = \left(\frac{\lambda-1}{2}, 4, \frac{\mu+2}{2}\right).$$

$$\overrightarrow{AD} = \left(\frac{\lambda-1}{2} - 2, 1, \frac{\mu+2}{2} - 5\right) = \left(\frac{\lambda-5}{2}, 1, \frac{\mu-8}{2}\right).$$

$$\text{Equal inclination} \Rightarrow \frac{\lambda-5}{2} = 1 = \frac{\mu-8}{2} \Rightarrow \lambda = 7, \mu = 10.$$

Answer: $(7, 10)$.

3. **Same as Worked Example 3 above.** Foot $F = (1, 0, 1)$ lies on $2x + y - z = 1$.

Answer: Plane (a).

4. **Perpendicular bisector plane:** Midpoint $M = (3, 1, 1)$. Normal = direction of segment $= (-2, 6, -4) \parallel (1, -3, 2)$.

$$\text{Plane: } (x-3) - 3(y-1) + 2(z-1) = 0 \Rightarrow x - 3y + 2z - 2 = 0.$$

Check options: $(0, 1, -1)$ gives $0 - 3 - 2 - 2 = -7$; $(0, -1, 1)$ gives $0 + 3 + 2 - 2 = 3$; $(4, 0, -1)$ gives $4 + 0 - 2 - 2 = 0 \checkmark$.

Answer: (c) $(4, 0, -1)$.

5. **Family of planes:** $(2x - y - 4) + \lambda(y + 2z - 4) = 0$. Through $(1, 1, 0)$: $-3 - 3\lambda = 0 \Rightarrow \lambda = -1$.

$$\text{Plane: } 2x - y - 4 - y - 2z + 4 = 0 \Rightarrow x - y - z = 0.$$

Answer: $x - y - z = 0$.

6. **Same as Worked Example 4.** Bisectors: $x - 3y - 2 = 0$ and $3x + y + 4z - 6 = 0$. $(2, -4, 1)$ satisfies the second.

Answer: (d) $(2, -4, 1)$.

7. **Same as Worked Example 7.** Parametric line: $(1+t, -5+t, 9+t)$. Intersection $t = -10$, $Q = (-9, -15, -1)$.

$$\text{Distance} = \sqrt{(-10)^2 + (-10)^2 + (-10)^2} = 10\sqrt{3}.$$

Answer: $10\sqrt{3}$.

8. **Coordinates:** $Q = (3, 3, 0)$, $T = (1.5, 1.5, 0)$, $S = (1.5, 1.5, 3)$.

$$(a) \cos \angle(OQ, OS) = \frac{(3, 3, 0) \cdot (1.5, 1.5, 3)}{\sqrt{18}\sqrt{13.5}} = \frac{9}{9\sqrt{3}} = \frac{1}{\sqrt{3}} - \text{true.}$$

$$(b) OQ \times OS = (3, 3, 0) \times (1.5, 1.5, 3) = (9, -9, 0) \Rightarrow x - y = 0 - \text{true.}$$

$$(c) \text{Distance from } P(3, 0, 0) \text{ to } x - y = 0: \frac{|3-0|}{\sqrt{2}} = \frac{3}{\sqrt{2}} - \text{true.}$$

$$(d) RS = (1.5, -1.5, 3), OR = (0, 3, 0). |OR \times RS| = |(9, 0, -4.5)| = \sqrt{81 + 20.25} = \sqrt{101.25} = \frac{9\sqrt{5}}{2}.$$

$$|RS| = \sqrt{2.25 + 2.25 + 9} = \sqrt{13.5} = \frac{3\sqrt{6}}{2}.$$

$$\text{Distance} = \frac{9\sqrt{5}/2}{3\sqrt{6}/2} = \frac{3\sqrt{5}}{\sqrt{6}} = \frac{\sqrt{30}}{2} - \text{true.}$$

Answer: (a), (b), (c), (d) all true.

What This Chapter's Own Text Under-serves

The NCERT chapter treats the perpendicular toolkit (foot, distance, image) and the family of planes as separate, recipe-driven sections. JEE, however, **routinely fuses them**: e.g., a plane from a family is used as a mirror to find an image, or the foot of a perpendicular from a point to a line is immediately tested against another plane. The text also gives insufficient practice on **distance measured parallel to a line** (not perpendicular) and on **locus problems** (like the set of feet of perpendiculars from a line to a plane), both of which appear in Advanced every few years. Finally, the chapter's "line-plane angle" box states the sine formula but does not emphasise the *modulus + complement* logic that explains *why* it is sine – a gap that makes the cosine trap recurring.

Chapter 12: Linear Programming

Complete Master Guide

Drona Ed Tech

Contents

Subtopic 01 — Formulation & Graphical Solution	2
Subtopic 02 — Mathematical Formulation & LPP Applications	12
Subtopic 03 — Special Cases of LPP Solutions	22
Chapter 12 — Complete Master Reference	29

Subtopic 01 — Formulation & Graphical Solution

Exam Relevance: Linear Programming is a guaranteed scorer in **CBSE Boards** — almost every year there is one full **5-6 mark** long-answer question that asks you to *formulate* an LPP from a word problem and *solve* it graphically. In **CUET (UG)** it shows up as **2-3 quick MCQs** on feasible regions, corner points, and reading off the optimum. A clean, predictable topic — once you master the corner-point routine, you rarely lose marks here.

*Syllabus note: This chapter is **not** part of the JEE Main / JEE Advanced syllabus, so we build it for Boards and CUET depth, not JEE speed traps.*

Section 1: Concept Intuition

Imagine you run a small *halwai* shop in the week before Diwali. You can make two sweets — kaju katli and besan laddoo. Every kilogram of each earns you a different profit, but you only have so much *khoya*, so much sugar, and so many hours at the stove before the festival rush. The question that keeps you up at night is simple to ask and hard to eyeball: *given everything I'm short on, exactly how many kilos of each sweet should I make so my profit is as large as possible?*

That, in one sentence, is **Linear Programming**. It is the mathematics of squeezing the best possible outcome — most profit, least cost — out of **limited resources**.

Three ideas do all the work:

1. **The thing you want to push to its best value** — your total profit, or your total cost. We write it as a formula like $Z = ax + by$ and call it the **objective function**. “Objective” because it’s literally your objective; “linear” because x and y appear only to the first power (no x^2 , no xy).
2. **The limits life puts on you** — only 8 kg of *khoya*, only 12 hours of stove time, and the obvious fact that you cannot make a *negative* number of laddoos. Each limit becomes a **linear inequality**, and together they are your **constraints**.
3. **The set of all honest choices** — every (x, y) pair that respects *all* the limits at once. Plot them and they fill a region of the graph. This is the **feasible region**, and your answer is guaranteed to live inside it (or on its edge).

Think about it this way... Picture the feasible region as a fenced field, and the objective function $Z = ax + by$ as a ruler whose reading goes up as you walk in a fixed direction (say, north-east). You want to stand at the spot in the field where the ruler reads highest. Here’s the beautiful part — for a straight-line ruler over a field with straight fences, that best spot is **never** in the open middle of the field. It is always poking out at a **corner**. That single fact is what makes LP so easy: you don’t search the whole field, you just check the corners.

Limiting conditions & assumptions — when this works (and when it bites):

- **Everything must be linear.** If profit per unit changed as you made more (a bulk discount, say), the objective stops being a straight line and the corner-point shortcut fails.
- **The feasible region may be unbounded.** If your field stretches off to infinity in some direction, a *maximum* might not exist at all (you could keep walking and keep gaining). The same region might still have a perfectly good *minimum*. So “unbounded” does **not** automatically mean “no answer” — it means *check carefully*.
- **The optimum can be a whole edge, not a single point.** If your ruler happens to line up parallel to one of the fences, then every point along that fence reads the same maximum. There are then *infinitely many* optimal solutions.
- **We assume two decision variables.** The graphical method only works in 2D — two products, two foods, two machines. With three or more variables you’d need algebra (the simplex method), which is beyond this chapter.

Section 2: Key Definitions & Properties

Linear Programming has no SI units or dimensional formulae — it is a method, so the “key facts” here are precise **definitions** and the **terminology** examiners expect you to use verbatim.

The objective function:

$$Z = ax + by$$

- Z = the quantity being optimised (profit, cost, distance, etc.)
- x, y = the **decision variables** — the quantities *you* get to choose.
- a, b = constants (profit or cost per unit). Both are real numbers; in most problems they are positive.

Standard form of a maximisation LPP:

$$\begin{aligned} &\text{Maximise } Z = ax + by \\ &\text{subject to } c_1x + d_1y \leq e_1, \quad c_2x + d_2y \leq e_2, \dots \\ &\quad \text{and } x \geq 0, y \geq 0 \end{aligned}$$

Terminology table — learn these exact words:

Term	Precise meaning
Optimisation problem	A problem that asks to maximise or minimise some function.
Constraints	The linear inequalities/restrictions on the variables.
Non-negative constraints	The conditions $x \geq 0, y \geq 0$.
Feasible region	The common region satisfying <i>all</i> constraints, including non-negativity.

Term	Precise meaning
Feasible solution	Any point <i>inside or on the boundary</i> of the feasible region.
Infeasible solution	Any point <i>outside</i> the feasible region.
Optimal solution	A feasible point that gives the optimum (max or min) value of Z .
Bounded region	A feasible region that can be enclosed inside a circle.
Unbounded region	A feasible region that extends to infinity in some direction.
Corner point (vertex)	A point of the region formed by the intersection of two boundary lines.

Two properties you should be able to state from memory:

- A bounded feasible region for a linear objective is always a **convex polygon** — no dents. A straight line joining any two feasible points stays entirely inside the region.
- If Z takes the **same** optimum value at **two** corner points, then it takes that *same* value at **every** point on the line segment joining them (infinitely many optima).

Section 3: Theorems & the Corner-Point Method

LP rests on two theorems. CBSE states them (it does not ask for a formal proof), so know them precisely — and, crucially, know the **procedure** they justify.

The Two Fundamental Theorems

Theorem 1 (Optimum lives at a corner). Let R be the feasible region (a convex polygon) for an LPP, and let $Z = ax + by$ be the objective function. When Z has an optimal value (maximum or minimum), subject to the linear constraints, that optimal value **must occur at a corner point (vertex)** of R .

Why this is believable: As you slide the line $ax + by = k$ across the region (increasing k pushes it in one fixed direction), the *last* feasible point it touches before leaving the region is always a vertex — unless the line is parallel to an edge, in which case it leaves along a whole edge whose endpoints are still vertices. Either way, a vertex carries the optimum.

Theorem 2 (Bounded vs unbounded). Let R be the feasible region and $Z = ax + by$ the objective function.

- If R is **bounded**, then Z has **both** a maximum and a minimum on R , and each occurs at a corner point.
- If R is **unbounded**, then a maximum (or minimum) **may or may not** exist. *If it exists*, it still occurs at a corner point.

The unbounded case is where students lose marks, so it gets its own verification step below.

The Corner-Point Method — step by step

Step 1 — Draw the feasible region. Convert each inequality to its boundary line (replace $\leq$ or $\geq$ with $=$), plot the line, and shade the correct side. *Reasoning:* a quick test — substitute the origin $(0, 0)$. If it satisfies the inequality, shade the side containing the origin; if not, shade the other side. (Use a different test point if a line passes through the origin.)

Step 2 — Find all corner points. Read off the vertices where boundary lines cross. *Reasoning:* by Theorem 1 the answer hides at a vertex, so these are the only points worth evaluating. Find them by inspection or by solving the two intersecting line equations simultaneously.

Step 3 — Evaluate Z at every corner point. Build a small table of (corner point, value of Z). Let M = largest value found, m = smallest value found.

Step 4 — Read off the answer.

- If the region is **bounded**: M is the maximum and m is the minimum. Done.
- If the region is **unbounded**, you must *verify*:
 - M is the genuine maximum **only if** the open half-plane $ax + by > M$ has **no point in common** with the feasible region. Otherwise Z has *no* maximum.
 - m is the genuine minimum **only if** the open half-plane $ax + by < m$ has **no point in common** with the feasible region. Otherwise Z has *no* minimum.

Reasoning for Step 4: in an unbounded region you can sometimes keep walking forever and keep improving Z . The half-plane test asks, “can I find any feasible point that beats my best corner?” If the answer is no, your corner value is truly optimal.

Figure. A bounded convex quadrilateral feasible region OABC in the first quadrant, with the four vertices labelled. A dashed “objective line” $ax+by=k$ is shown sliding outward, touching the region last at the top corner, illustrating Theorem 1 — the optimum sits at a vertex.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A bakery in Pune makes two kinds of cookies each morning: nankhatai and kaju (cashew) cookies, measured in batches. Each batch of nankhatai uses 1 kg of flour and 3 oven-slots; each batch of kaju cookies uses 2 kg of flour and 2 oven-slots. The bakery has 8 kg of flour and 12 oven-slots available. Profit is ₹40 per batch of nankhatai and ₹50 per batch of kaju cookies. How many batches of each should it bake to maximise profit?

Formulation. Let x = batches of nankhatai, y = batches of kaju cookies.

$$\begin{aligned}
&\text{Maximise } Z = 40x + 50y \\
&\text{subject to } x + 2y \leq 8 \quad (\text{flour}) \\
&\quad \quad \quad 3x + 2y \leq 12 \quad (\text{oven-slots}) \\
&\quad \quad \quad x \geq 0, y \geq 0
\end{aligned}$$

Find the corner points. Plot $x + 2y = 8$ and $3x + 2y = 12$. Their intersection: subtract the first from the second, $2x = 4 \Rightarrow x = 2$, then $y = 3$. The feasible region $OABC$ is bounded with vertices:

$$O(0,0), \quad A(4,0), \quad B(2,3), \quad C(0,4).$$

Evaluate Z .

Corner point	$Z = 40x + 50y$
$O(0,0)$	0
$A(4,0)$	160
$B(2,3)$	$80 + 150 = \mathbf{230} \leftarrow \text{Maximum}$
$C(0,4)$	200

Conclusion. The bakery should bake **2 batches of nankhatai and 3 batches of kaju cookies** for a maximum profit of **₹230**.

Figure. First-quadrant graph with lines $x+2y=8$ and $3x+2y=12$, the shaded quadrilateral $OABC$, and vertices $O(0,0)$, $A(4,0)$, $B(2,3)$, $C(0,4)$ marked.

Example 2 — [CUET Speed Trap / Conceptual]

The corner points of a feasible region are $(0,4)$, $(3,3)$ and $(6,0)$. For the objective function $Z = 2x + 2y$, the maximum of Z occurs at: (a) $(3,3)$ only (b) $(6,0)$ only (c) every point on the segment joining $(3,3)$ and $(6,0)$ (d) $(0,4)$ only

Fast solve — no graph needed. Evaluate at each corner:

$$(0,4) : 8, \quad (3,3) : 12, \quad (6,0) : 12.$$

The maximum value 12 is reached at **two** corners, $(3,3)$ and $(6,0)$.

The trap: a hurried student circles (a) or (b) — whichever they wrote down first. But by the multiple-optima property, when Z hits the same optimum at two corners, *every* point on the segment joining them is also optimal. **Answer: (c).**

Speed tip: the moment two corner values *tie* for the optimum, your answer is the whole edge between them — the objective line is parallel to that edge.

Example 3 — [CUET / CBSE HOTS — Unbounded region]

A dairy farmer mixes two feeds, Feed P and Feed Q (in kg). Each kg of P supplies 1 unit of protein and 2 units of fibre; each kg of Q supplies 1 unit of protein and 1 unit of fibre. The herd needs at least 6 units of protein and at least 8 units of fibre daily. Feed P costs ₹5/kg and Feed Q costs ₹4/kg. Find the mix of minimum cost.

Formulation. Let x = kg of P, y = kg of Q.

$$\text{Minimise } Z = 5x + 4y$$

$$\text{subject to } x + y \geq 6 \text{ (protein), } 2x + y \geq 8 \text{ (fibre), } x \geq 0, y \geq 0$$

Corner points. Both inequalities are “ $\geq$ ”, so the region is **unbounded**, lying above both lines. Intersection of $x + y = 6$ and $2x + y = 8$: subtract $\Rightarrow x = 2, y = 4$. The corner points on the boundary of the feasible region are:

$$(0, 8), \quad (2, 4), \quad (6, 0).$$

Evaluate Z .

Corner point	$Z = 5x + 4y$
$(0, 8)$	32
$(2, 4)$	$10 + 16 = \mathbf{26} \leftarrow \text{smallest}$
$(6, 0)$	30

Verify (because the region is unbounded). Is 26 truly the minimum? Test whether the open half-plane $5x + 4y < 26$ shares any point with the feasible region. The line $5x + 4y = 26$ touches the region only at the corner $(2, 4)$; moving *into* the feasible region (up and to the right) only *increases* $5x + 4y$. So no feasible point gives a value below 26.

Conclusion. Minimum cost is **₹26**, at $x = 2$ kg of P and $y = 4$ kg of Q.

Figure. First-quadrant unbounded region above the lines $x+y=6$ and $2x+y=8$, shaded upward to infinity, with corner points $(0,8)$, $(2,4)$, $(6,0)$ marked, and a dotted line $5x+4y=26$ just touching the region at $(2,4)$.

Example 4 — [CUET / Reasoning Level]

The corner points of a feasible region are $(0, 4)$, $(2, 3)$ and $(5, 0)$. The objective function is $Z = px + qy$ with $p, q > 0$. Find the condition on p and q so that the maximum of Z occurs at **both** $(2, 3)$ and $(5, 0)$.

Reasoning. For the maximum to be shared by two corners, Z must take equal values there:

$$Z(2, 3) = Z(5, 0) \Rightarrow 2p + 3q = 5p + 0 \Rightarrow 3q = 3p \Rightarrow \boxed{p = q}.$$

Why this makes geometric sense. The edge joining $(2, 3)$ and $(5, 0)$ has slope

$$\frac{0 - 3}{5 - 2} = -1.$$

The objective line $px + qy = c$ has slope $-\frac{p}{q}$. The optimum spreads across the whole edge exactly when the objective line is **parallel** to that edge:

$$-\frac{p}{q} = -1 \Rightarrow p = q,$$

confirming the algebra. (A quick sanity check: with $p = q$, $Z(0, 4) = 4q$ while $Z(2, 3) = Z(5, 0) = 5q$, so the *shared maximum* is exactly the segment from $(2, 3)$ to $(5, 0)$.)

Condition: $p = q$.

Section 5: Practice Exercises

1. **[CBSE]** Maximise $Z = 30x + 20y$ subject to $x + y \leq 8$, $2x + y \leq 10$, $x \geq 0$, $y \geq 0$. State the optimal point and value.
2. **[CUET]** The corner points of a bounded feasible region are $(0, 0)$, $(5, 0)$, $(3, 4)$ and $(0, 6)$. Find the maximum of $Z = 4x + 3y$.
3. **[CUET]** For the constraints $x \geq 0$, $y \geq 0$, $x + y \geq 4$, $x + 2y \geq 6$, is the feasible region bounded or unbounded? Find the minimum of $Z = 3x + 5y$ if it exists.
4. **[CUET]** The corner points of a feasible region are $(0, 3)$, $(1, 1)$ and $(3, 0)$. For $Z = px + qy$ with $p, q > 0$, find the condition on p, q so that the minimum occurs at *both* $(1, 1)$ and $(3, 0)$.
5. **[CBSE HOTS]** A carpenter makes stools (x) and chairs (y). Each stool needs 2 hours of cutting and 1 hour of polishing; each chair needs 1 hour of cutting and 2 hours of polishing. He has 10 cutting hours and 8 polishing hours weekly. Profit is ₹60 per stool and ₹50 per chair. Formulate the LPP and find the most profitable mix.

Answers:

1. Maximum $Z = 180$ at $(2, 6)$.
 2. Maximum $Z = 24$ at $(3, 4)$.
 3. Unbounded region; minimum $Z = 16$ at $(2, 2)$ (verified by the half-plane test).
 4. $q = 2p$.
 5. Maximise $Z = 60x + 50y$ s.t. $2x + y \leq 10$, $x + 2y \leq 8$, $x, y \geq 0$; maximum $Z = 340$ at $(4, 2)$.
-

Section 6: Crack the MCQ

Q1. The feasible region of an LPP is the convex polygon with vertices $(0, 0)$, $(4, 0)$, $(2, 4)$ and $(0, 5)$. The maximum of $Z = 5x + 3y$ is:

(A) 20 (B) 22 (C) 25 (D) 15

Answer: (B). Evaluate: $(0, 0) \rightarrow 0$, $(4, 0) \rightarrow 20$, $(2, 4) \rightarrow 10 + 12 = 22$, $(0, 5) \rightarrow 15$. Maximum = 22 at $(2, 4)$.

- (A) 20 — the value at $(4, 0)$; chosen by a student who tests only the x-axis intercepts and stops early.
- (C) 25 — comes from mis-reading the vertex as $(5, 0)$ or adding coefficients carelessly.
- (D) 15 — the value at $(0, 5)$; the trap for someone who assumes “more y is better” without checking all corners.

Q2. If the feasible region of an LPP is **unbounded**, then the objective function $Z = ax + by$:

(A) always has a maximum (B) always has a minimum (C) may or may not have a maximum or minimum (D) has neither

Answer: (C). By Theorem 2, an unbounded region guarantees nothing — the optimum may or may not exist, and must be verified with the half-plane test.

- (A)/(B) — both confuse the *bounded* case (which guarantees both) with the unbounded case.
- (D) — over-corrects; unbounded regions frequently *do* have, say, a minimum (the cost problems in Example 3), just not necessarily a maximum.

Q3. The minimum value of $Z = x + 2y$ on the feasible region with corner points $(0, 2)$, $(1, 1)$, $(4, 0)$ (region unbounded, opening upward-right) is:

(A) 4 (B) 3 (C) 2 (D) does not exist

Answer: (B). $(0, 2) \rightarrow 4$, $(1, 1) \rightarrow 3$, $(4, 0) \rightarrow 4$. Smallest = 3 at $(1, 1)$; the half-plane $x + 2y < 3$ has no feasible point, so 3 is genuine.

- (A) 4 — the value at the two “end” corners; picked if you forget the interior vertex $(1, 1)$.
- (C) 2 — substituting only the smallest single coordinate; not a corner value at all.
- (D) — wrong reflex of “unbounded $\Rightarrow$ no answer.” Unbounded regions can still have a minimum.

Q4. In an LPP, if $Z = ax + by$ has the same maximum value at two distinct corner points, then:

(A) the LPP has no solution (B) the maximum occurs at exactly those 2 points (C) the maximum occurs at every point on the segment joining them (D) $a = b$

Answer: (C). This is the multiple-optima property: equal optimum at two corners means the whole connecting segment is optimal — infinitely many solutions.

- (A) — the opposite; there are *more* solutions, not none.
 - (B) — the most tempting wrong choice; it forgets the segment in between.
 - (D) — true only in special coordinate setups (when that edge has slope -1); not a general consequence.
-

Section 7: Common Pitfalls & Pro-Tips

Pitfall 1 — Stopping at the axis corners. Many students evaluate Z only at the intercepts $(k, 0)$ and $(0, k)$ and miss the *interior* vertex where two slanted constraints cross — which is very often where the optimum hides (see Examples 1 and 3). **Always list every vertex.**

Pitfall 2 — Forgetting the unbounded check. Reading off the smallest/largest corner value and writing “answer” without the half-plane verification costs the marks in an unbounded problem. CBSE examiners specifically look for that verification line.

Pitfall 3 — Shading the wrong side. Confusing $\leq$ and $\geq$ when shading flips your entire region. **Use the origin test:** plug $(0, 0)$ into the inequality — if it holds, shade the origin’s side; if not, shade away.

Pitfall 4 — Mixing up “atmost / atleast”. “Atmost 300” is ≤ 300 ; “atleast 6 units” is ≥ 6 . A single mis-translated word changes a maximisation answer entirely. Underline these words while formulating.

Pro-Tip (speed for CUET): When two corner values *tie* for the optimum, don’t recompute — the answer is the **entire edge** between them (objective line parallel to that edge). And for minimisation cost problems, the optimum almost always sits at the *interior* vertex where two “ $\geq$ ” constraints intersect, so evaluate that one first.

Section 8: Cheat Sheet Box

LINEAR PROGRAMMING — FORMULATION & GRAPHICAL SOLUTION

- **Objective:** $Z = ax + by \rightarrow$ maximise (profit) or minimise (cost).
- **Constraints:** linear inequalities + non-negativity $x \geq 0, y \geq 0$.
- **Feasible region** = common shaded region of *all* constraints.
- **Theorem 1:** the optimum is always at a **corner point**.
- **Theorem 2:** **bounded** $\rightarrow$ both max & min exist; **unbounded** $\rightarrow$ may or may not.
- **Corner-Point Method:** draw region $\rightarrow$ list vertices $\rightarrow$ tabulate $Z \rightarrow$ pick best (verify if unbounded).
- **Unbounded verify:** max M valid iff $ax + by > M$ misses the region; min m valid iff $ax + by < m$ misses the region.

- **Tie at two corners** → optimum along the **whole edge** (infinitely many solutions).

Memory aids:

- **“CORNERS, not centres”** — never search the middle of the region.
- **“Atmost = $\leq$, Atleast = $\geq$ ”** — underline the word before you formulate.
- **“Origin test”** — if $(0, 0)$ fits, shade its side.
- **“Unbounded? Verify or vanish”** — always run the half-plane check.

Subtopic 02 — Mathematical Formulation & LPP Applications

Exam Relevance: Formulation is the *first half* of every CBSE long-answer LPP question — and CBSE often awards a chunk of marks for the model **alone**, sometimes asking “Express the following as an LPP” *without* solving. CUET loves formulation-reasoning MCQs: spotting the right inequality for “at least,” “at most,” or “cannot exceed by.” Master the translation from words to math and you bank marks before a single graph is drawn.

*Builds directly on Subtopic 01 — for the graphical/corner-point solving used here, see Sub-01. This subtopic focuses on **building** the model and recognising **problem types**. (LP is not in the JEE syllabus; this is a Boards + CUET skill.)*

Section 1: Concept Intuition

In Subtopic 01 you learned how to *solve* an LPP once it’s written in clean mathematical form. But in a real exam — and in real life — nobody hands you the inequalities. They hand you a paragraph: “A tailoring unit has 40 metres of cloth and 18 hours of machine time...” The hardest, most marks-rich skill in this chapter is **turning that paragraph into mathematics**. That skill is called **formulation**.

Think of yourself as a translator at a railway enquiry counter. A passenger speaks to you in everyday Hindi or Tamil; you must convert it into the precise codes the booking system understands. **Formulation is exactly this translation** — from the loose language of “we can’t afford more than ₹5000” into the exact symbol $(\text{cost}) \leq 5000$. Get the translation right and the rest is mechanical. Get one word wrong — read “at least” as “at most” — and the entire answer collapses, no matter how perfectly you draw the graph afterwards.

Think about it this way... Every word problem, no matter how it’s dressed up — sweaters or sand, calculators or crops — is built from the **same three questions**:

1. **What am I choosing?** The unknown *quantities I control* become the **decision variables** x and y . (How many tables? How many hectares? How many scoops?)
2. **What am I trying to make best?** The total profit or total cost I want to push up or down becomes the **objective function** $Z = ax + by$.
3. **What’s stopping me?** Every limit — material, money, time, demand, a promise to a customer — becomes a **constraint**, a linear inequality.

Once you train your eye to hunt for these three things in any paragraph, every LPP starts to look the same. The “applications” you’ll meet — factory production, diet planning, transport, scheduling — are just different **costumes on the same skeleton**.

Limiting conditions & assumptions to watch:

- **Proportionality must hold.** Doubling production must exactly double the resource used and the profit earned. If there's a bulk discount or a setup cost, the model is no longer linear and this method doesn't apply.
- **Divisibility is assumed.** The graphical method allows fractional answers like $x = 2.5$. Many real problems (you can't make half a sweater) silently assume you'll round sensibly — and CBSE problems are usually designed so the optimum lands on whole numbers.
- **Watch hidden non-negativity.** "Number of buses," "kg of fertiliser," "hours worked" can never be negative. Forgetting $x \geq 0$, $y \geq 0$ is the single most common formulation slip.
- **Read every clause.** A sentence like "type B cannot exceed type A by more than 100" is an entire constraint hiding in plain English. Skipping it changes the answer.

Section 2: Key Definitions & Properties

Formulation has its own vocabulary. Beyond the terms from Sub-01 (objective function, decision variables, constraints, non-negativity), the key idea here is recognising the **type** of each constraint from the language used.

The anatomy of a formulated LPP:

$$\text{objective} \quad \text{subject to} \quad \underbrace{(\text{structural constraints})}_{\text{the limits}}, \quad \underbrace{x \geq 0, y \geq 0}_{\text{non-negativity}}$$

Constraint-type dictionary — translate the English:

English phrase	Constraint type	Symbol
"at most", "no more than", "cannot exceed", "available", "within"	upper limit (resource/capacity)	$\leq$
"at least", "minimum of", "no less than", "must meet", "requires"	lower limit (requirement/demand)	$\geq$
"exactly", "must equal", "all of it is used"	equality (balance/mix)	$=$
"cannot be negative" (number of items, hours, kg)	non-negativity	$x, y \geq 0$

Two phrasings that trap students — learn them cold:

- **"B cannot exceed A by more than k "** means $y - x \leq k$, i.e. $-x + y \leq k$. (Here $x = A$, $y = B$.) It does **not** mean $y \leq k$.
- **"A is at least twice B"** means $x \geq 2y$, i.e. $x - 2y \geq 0$. Read the direction carefully — "twice" multiplies the *smaller* quantity.

Property — the objective and constraints share their variables. Every constraint and the objective are written in the *same* two decision variables. If a quantity in the paragraph cannot be expressed using x and y alone, you have probably chosen your decision variables badly — go back to Step 1.

Section 3: The Formulation Procedure & Application Taxonomy

Formulation is a *procedure*, not a theorem, so here is the disciplined framework, followed by the family resemblances that let you recognise a problem on sight.

The 5-Step Formulation Procedure

Step 1 — Name the decision variables. Read the question: *what number is the problem asking you to decide?* Let those be x and y , and **write their units explicitly** (“let x = number of cricket bats, y = number of tennis racquets”). *Reasoning:* stating units now prevents you from mixing “hours” and “items” later.

Step 2 — Write the objective function. Find the per-unit profit or per-unit cost of each variable and combine: $Z = (\text{rate}_x) x + (\text{rate}_y) y$. State clearly whether you **maximise** (profit, distance, output) or **minimise** (cost, time, waste). *Reasoning:* the objective is what the question actually wants — pin it down before drowning in constraints.

Step 3 — Translate each restriction into a constraint. Go through the paragraph *sentence by sentence*. For each limit, write a linear inequality, using the constraint-type dictionary to fix the direction of $\leq$ or $\geq$. *Reasoning:* working clause-by-clause means you never miss a hidden constraint.

Step 4 — Add non-negativity. Append $x \geq 0, y \geq 0$. *Reasoning:* physical quantities can’t go negative; this also fences the region into the first quadrant.

Step 5 — State the complete model. Write the full LPP together — “Maximise $Z = \dots$ subject to $\dots$ ” — so it’s ready to hand to the graphical method of Sub-01. *Reasoning:* a clean, collected model is what earns the formulation marks and prevents copying errors.

The Four Application Families

Almost every LPP you’ll meet is one of these. Spot the family and you already know the shape of the answer:

Family	Goal	Constraint signature	Typical story
Manufacturing / Product-mix	Maximise profit	resources $\leq$ available	factory making two products from limited material/machine time
Diet / Nutrition / Blending	Minimise cost	nutrients $\geq$ required	mixing two food-s/feeds/powders to meet minimum requirements cheaply
Transportation / Allocation	Minimise cost (or maximise use)	supply $\leq$, demand $\geq$	distributing goods, or splitting land/resources between two uses
Scheduling / Time	Maximise output / earnings	time $\leq$, minimum targets $\geq$	dividing limited hours between two activities

Reading tip: the *goal word* (“profit” vs “cost”) usually tells you the family. Profit/output/distance $\rightarrow$ maximise $\rightarrow$ expect $\leq$ constraints. Cost/requirement $\rightarrow$ minimise $\rightarrow$ expect $\geq$ constraints and (often) an **unbounded** region, so the verification step from Sub-01 will matter.

Figure. A flow diagram with one box "WORD PROBLEM (paragraph)" on the left, an arrow labelled "5-step formulation" pointing right to a box "LPP MODEL: Maximise/Minimise Z subject to constraints", and a second arrow labelled "corner-point method (Sub-01)" pointing to a final box "OPTIMAL SOLUTION".

Section 4: Worked Examples

Example 1 — [CBSE Board Level] — Product-mix with a “difference” constraint

A printing press makes cartons of ruled notebooks and cartons of sketchbooks. Each carton of notebooks needs 1 unit of paper; each carton of sketchbooks needs 2 units (thicker pages). The press has 100 units of paper and can finish at most 70 cartons in total per day. Also, because sketchbook demand is limited, sketchbook cartons cannot exceed notebook cartons by more than 20. Profit is ₹60 per notebook carton and ₹140 per sketchbook carton. Formulate as an LPP and find the most profitable output.

Step 1 — Variables. Let x = cartons of notebooks, y = cartons of sketchbooks.

Step 2 — Objective. Maximise $Z = 60x + 140y$.

Step 3 — Constraints (clause by clause).

- Paper: $x + 2y \leq 100$.
- Total output: $x + y \leq 70$.
- “Sketchbooks cannot exceed notebooks by more than 20”: $y - x \leq 20 \Rightarrow -x + y \leq 20$.

Step 4 — Non-negativity. $x \geq 0, y \geq 0$.**Step 5 — The model.**

Maximise $Z = 60x + 140y$ s.t. $x + 2y \leq 100, x + y \leq 70, -x + y \leq 20, x, y \geq 0$.

Solve (corner-point method, Sub-01). Corner points: $(0, 0), (70, 0), (40, 30), (20, 40), (0, 20)$.

Corner	$Z = 60x + 140y$
$(0, 0)$	0
$(70, 0)$	4200
$(40, 30)$	$2400 + 4200 = 6600$
$(20, 40)$	$1200 + 5600 = \mathbf{6800} \leftarrow \text{Maximum}$
$(0, 20)$	2800

Conclusion. Make **20 cartons of notebooks and 40 of sketchbooks** for a maximum profit of **₹6800**. Notice the optimum sits exactly where the “difference” constraint $-x + y = 20$ is active — the very clause a careless reader would have dropped.

Example 2 — [CBSE HOTS] — Allocation problem (three binding constraints)

A farmer has 12 hectares to split between wheat and mustard. Wheat needs 2 person-days of labour per hectare, mustard needs 1; only 18 person-days are available. Wheat needs 1 unit of water per hectare, mustard needs 3; only 24 units of water are available. Expected profit is ₹5000 per hectare of wheat and ₹4000 per hectare of mustard. How should the land be allocated for maximum profit?

Formulation. Let x = hectares of wheat, y = hectares of mustard.

$$\text{Maximise } Z = 5000x + 4000y$$

$$\text{s.t. } x + y \leq 12 \text{ (land), } 2x + y \leq 18 \text{ (labour), } x + 3y \leq 24 \text{ (water), } x, y \geq 0.$$

Solve. Corner points $(0, 0), (9, 0), (6, 6), (0, 8)$.

Corner	$Z = 5000x + 4000y$
$(0, 0)$	0
$(9, 0)$	45000
$(6, 6)$	$30000 + 24000 = \mathbf{54000} \leftarrow \text{Maximum}$
$(0, 8)$	32000

The HOTS twist: at the optimum (6, 6), **all three** constraints are satisfied as equalities — land ($6 + 6 = 12$), labour ($12 + 6 = 18$) and water ($6 + 18 = 24$) are *all* fully used. Three lines pass through one vertex. This is why you must always check *every* corner: the winning point is the one where the binding constraints intersect.

Conclusion. Plant **6 hectares of wheat and 6 of mustard** for a maximum profit of **₹54,000**.

Example 3 — [CUET] — Diet/blending, formulation only

A fitness coach blends two protein powders, P and Q (measured in scoops). One scoop of P gives 12 g protein and 20 g carbohydrate; one scoop of Q gives 8 g protein and 30 g carbohydrate. A client needs at least 48 g of protein and at least 120 g of carbohydrate per day. P costs ₹15 per scoop and Q costs ₹10 per scoop. Set up the LPP to meet the requirement at minimum cost.

Variables. Let x = scoops of P, y = scoops of Q.

Objective. Minimise cost: $Z = 15x + 10y$.

Constraints (note the “ $\geq$ ” — these are *requirements*, not limits):

- Protein: $12x + 8y \geq 48 \Rightarrow 3x + 2y \geq 12$.
- Carbohydrate: $20x + 30y \geq 120 \Rightarrow 2x + 3y \geq 12$.
- Non-negativity: $x \geq 0, y \geq 0$.

The model.

$$\text{Minimise } Z = 15x + 10y \quad \text{s.t. } 3x + 2y \geq 12, \quad 2x + 3y \geq 12, \quad x, y \geq 0.$$

Recognise the family: a diet problem $\rightarrow$ minimise $\rightarrow$ “ $\geq$ ” constraints $\rightarrow$ the feasible region is **unbounded**, so when you go on to solve it, you must run the open-half-plane verification from Sub-01. (Question asked for formulation only — model complete.)

Example 4 — [CUET / Formulation Reasoning]

Translate each condition into a constraint. A student earns from two freelance gigs: data-entry (x hours/week) and graphic design (y hours/week). (i) She has at most 20 working hours a week. (ii) Her client caps data-entry at 12 hours. (iii) To build her portfolio she wants at least 5 hours of design. (iv) Data-entry pays ₹150/hour and design ₹250/hour; she wants maximum earnings. Write the LPP and classify each constraint.

Translation, clause by clause:

Condition	Constraint	Type
(i) at most 20 hours total	$x + y \leq 20$	upper limit (capacity)
(ii) data-entry capped at 12	$x \leq 12$	upper limit (bound on one variable)
(iii) at least 5 hours design	$y \geq 5$	lower limit (requirement)

Condition	Constraint	Type
(iv) maximise earnings	$Z = 150x + 250y$	objective (maximise)

The model.

Maximise $Z = 150x + 250y$ s.t. $x + y \leq 20$, $x \leq 12$, $y \geq 5$, $x \geq 0$.

(Here $y \geq 5$ already forces $y \geq 0$, so a separate $y \geq 0$ is redundant but harmless to write.)

Reasoning check: the region is **bounded** (the “ ≤ 20 ” caps total hours), so by Theorem 2 both a maximum and minimum exist at corner points — ready for the graphical method. The lesson: a single mixed bag of “at most / at least / capped at” phrases becomes four clean lines once you translate each clause separately.

Section 5: Practice Exercises

1. **[CBSE]** A bakery makes cakes and pastries. Each cake needs 200 g flour and 25 g butter; each pastry needs 100 g flour and 50 g butter. There are 5 kg flour and 1 kg butter. Profit is ₹40 per cake and ₹25 per pastry. Formulate the LPP for maximum profit (do not solve).
2. **[CUET]** “The number of economy seats must be at least three times the number of business seats.” If x = economy seats, y = business seats, which inequality is correct? Write it.
3. **[CUET]** A trader blends two teas, X and Y. Each kg of X gives 3 units of aroma and costs ₹90; each kg of Y gives 5 units of aroma and costs ₹120. The blend must supply at least 30 units of aroma. Formulate the cost-minimisation LPP.
4. **[CUET]** A cottage unit can produce at most 60 items a day and must produce at least 10 of product B for a standing order. If x = units of A and y = units of B, write all the constraints (including non-negativity).
5. **[CBSE HOTS]** A transporter uses mini-trucks and large trucks. A mini-truck carries 2 tonnes and costs ₹1000/trip; a large truck carries 5 tonnes and costs ₹2000/trip. At least 30 tonnes must be moved, and the number of large trucks cannot exceed the number of mini-trucks. Formulate the cost-minimisation LPP.

Answers (models):

1. Max $Z = 40x + 25y$ s.t. $200x + 100y \leq 5000 \Rightarrow 2x + y \leq 50$; $25x + 50y \leq 1000 \Rightarrow x + 2y \leq 40$; $x, y \geq 0$.
2. $x \geq 3y$ (i.e. $x - 3y \geq 0$).
3. Min $Z = 90x + 120y$ s.t. $3x + 5y \geq 30$; $x, y \geq 0$.
4. $x + y \leq 60$; $y \geq 10$; $x \geq 0$ (and $y \geq 0$, implied).
5. Min $Z = 1000x + 2000y$ s.t. $2x + 5y \geq 30$; $y \leq x \Rightarrow -x + y \leq 0$; $x, y \geq 0$, where x = mini-truck trips, y = large-truck trips.

Section 6: Crack the MCQ

Q1. “Product B should be at most one-third of product A.” With $x = A$, $y = B$, the correct constraint is:

- (A) $y \leq \frac{x}{3}$ (B) $y \geq \frac{x}{3}$ (C) $x \leq \frac{y}{3}$ (D) $3y \geq x$

Answer: (A). “B at most one-third of A” $\rightarrow y \leq \frac{1}{3}x$, i.e. $3y \leq x$.

- (B)/(D) — flip the inequality; the trap for reading “at most” as “at least.”
- (C) — swaps the roles of A and B; chosen by mislabelling which variable is which.

Q2. A diet problem asks to *minimise* cost subject to meeting *minimum* nutrient levels. Its constraints are typically of the form, and its feasible region is typically:

- (A) “ $\leq$ ”, bounded (B) “ $\geq$ ”, unbounded (C) “ $=$ ”, a single point (D) “ $\leq$ ”, unbounded

Answer: (B). Requirements (“at least”) give “ $\geq$ ” constraints, which open the region upward/right — unbounded — so the minimum must be verified.

- (A)/(D) — “ $\leq$ ” is the *manufacturing* signature, not diet.
- (C) — equalities would over-determine the problem; not how diet LPPs are built.

Q3. In formulating an LPP, the conditions $x \geq 0$, $y \geq 0$ are called:

- (A) the objective function (B) the optimal constraints (C) the non-negative constraints (D) the feasible solutions

Answer: (C). These are the non-negativity (non-negative) constraints, restricting the model to the first quadrant.

- (A) — that’s $Z = ax + by$.
- (B) — not a standard term; a distractor built from real words.
- (D) — feasible solutions are *points* in the region, not these conditions.

Q4. “A factory must use all 240 litres of available dye.” This sentence translates to a constraint of which type?

- (A) $\leq$ (upper limit) (B) $\geq$ (lower limit) (C) $=$ (equality) (D) non-negativity

Answer: (C). “Must use all” forces exact consumption $\rightarrow$ an **equality** constraint (dye used = 240).

- (A) — “available” alone would be “ $\leq$ ”, but “must use all of it” tightens it to “ $=$ ”.
- (B) — would mean “use at least 240,” which the sentence doesn’t say.
- (D) — unrelated; non-negativity is about variables, not resource usage.

Section 7: Common Pitfalls & Pro-Tips

Pitfall 1 — Reversing $\leq$ and $\geq$. “At least” is $\geq$; “at most” is $\leq$. This single flip turns a minimisation answer into nonsense. **Underline the limiting words** before writing any symbol.

Pitfall 2 — Dropping a hidden constraint. Sentences like “B cannot exceed A by more than 100” or “must produce at least 10 of B” are full constraints. Work **clause by clause** so none escapes.

Pitfall 3 — Forgetting non-negativity. Items, hours, kilograms, hectares — none can be negative. Always append $x \geq 0$, $y \geq 0$; examiners deduct marks for its absence even if the rest is perfect.

Pitfall 4 — Choosing the wrong decision variables. If you can’t write the objective and *every* constraint using just your two variables, you picked them badly. Re-read the question — the variables are whatever the problem asks you to *decide*.

Pro-Tip (speed & accuracy): After formulating, run a 10-second **sanity scan** — (i) does the objective say max or min to match the goal word? (ii) is there one constraint per limiting sentence? (iii) is non-negativity present? (iv) do “cost/at-least” problems carry “ $\geq$ ” and “profit/at-most” problems carry “ $\leq$ ”? If all four check out, your model is almost certainly right.

Section 8: Cheat Sheet Box

LINEAR PROGRAMMING — FORMULATION & APPLICATIONS

- **5-step formulation:** name variables (with units) $\rightarrow$ write Z (max/min) $\rightarrow$ translate each clause to a constraint $\rightarrow$ add $x, y \geq 0$ $\rightarrow$ state the full model.
- **Translate the words:**
 - “at most / cannot exceed / available” $\rightarrow \leq$
 - “at least / minimum / requires” $\rightarrow \geq$
 - “uses all / exactly” $\rightarrow =$
- **Tricky phrases:** “B exceeds A by at most k ” $\rightarrow -x + y \leq k$; “A at least twice B” $\rightarrow x \geq 2y$.
- **Four families:** product-mix (max, $\leq$) · diet/blend (min, $\geq$) · transport/allocation (min; $\leq$ & $\geq$) · scheduling (max; time $\leq$, target $\geq$).
- **Goal word = family:** profit/output $\rightarrow$ maximise, expect $\leq$. cost/requirement $\rightarrow$ minimise, expect $\geq$ (often unbounded $\rightarrow$ verify).
- **Always** append non-negativity; **always** solve with the corner-point method (Sub-01).

Memory aids:

- **“NOC-N”** — Name variables, Objective, Constraints, Non-negativity.

- **“Most $\leq$, Least $\geq$ ”** — the four words that decide every inequality direction.
- **“Min wants More”** — minimisation/cost problems use “at least” ($\geq$) and usually open (unbounded) regions.

Subtopic 03 — Special Cases of LPP Solutions

Exam Relevance: “Special cases” are favourite **1-mark and short-answer** CBSE questions — “Does this LPP have a feasible solution?”, “Why does this LPP have no maximum?” — and reliable **CUET MCQs**. They test whether you truly understand the method rather than just cranking the corner-point routine. Quick to learn, easy marks, and a common place to get tripped up. (*Boards + CUET; LP is not in the JEE syllabus.*)

Section 1: Concept Intuition

So far every LPP we solved behaved nicely: one feasible region, one clean optimal corner. But real problems — and exam problems designed to test you — don’t always cooperate. An LPP can end in **one of four ways**, and a complete student must recognise all four on sight.

Think of solving an LPP like searching a fenced field for the highest point your ruler reads. Four things can happen:

1. **One clear highest spot** — the *unique optimal solution*. The usual, happy case.
2. **A whole ridge at the same height** — *multiple optimal solutions*. Your ruler lies flat along one fence, so every point on that fence ties for best.
3. **The field never ends in the uphill direction** — an *unbounded solution*. You can keep walking and keep climbing forever; there is **no** highest point, so the maximum simply doesn’t exist.
4. **There is no field at all** — *no feasible solution*. The fences contradict each other so badly that no point satisfies all of them. The region is empty and the problem has no answer.

Think about it this way... Imagine a shopkeeper who insists “I must sell *at least* 60 items today” but also “my shelf holds *at most* 40 items.” No honest number of items can satisfy both — the demand and the limit fight each other. There’s nothing to optimise because there’s no valid choice to begin with. That’s an **infeasible** LPP. Compare that with a daydreaming trader who says only “sell *at least* 10 items” with no upper limit and no cost — profit can grow without bound, so there’s no *maximum* profit to find. That’s an **unbounded** solution. Spotting which trap a question is setting is the whole skill here.

One distinction worth burning into memory: an **unbounded region** (the *shape* stretches to infinity) is **not** the same as an **unbounded solution** (the *objective* has no finite optimum). Many unbounded regions still give a perfectly good minimum — every cost-minimisation diet problem does. “Unbounded region” describes geometry; “unbounded solution” describes the answer. Never confuse the two.

Limiting conditions & assumptions:

- A *maximum* fails to exist only when the region is unbounded **in the direction the objective increases**. If the region is bounded, Theorem 2 guarantees both

a max and a min — no special case can arise.

- “No feasible solution” can only happen when constraints genuinely contradict; a single redundant constraint does not cause it.

Section 2: Key Definitions & Properties

Outcome	What it means	Tell-tale sign on the graph
Unique optimal solution	Exactly one point gives the optimum.	The objective line touches the region last at a single corner.
Multiple optimal solutions	Infinitely many points give the <i>same</i> optimum.	Two corners tie for the optimum; the objective line is parallel to the edge joining them.
Unbounded solution	The objective has no finite maximum (or no finite minimum).	The region is unbounded and the open half-plane “better than every corner” still overlaps the region.
No feasible solution (infeasible)	No point satisfies all constraints.	The constraints have no common shaded region — the feasible region is empty.
Redundant constraint	A constraint that can be removed without changing the region.	Its line lies entirely outside (or on the far edge of) the region formed by the others.

Properties to state precisely:

- **Multiple optima (recap):** if $Z = ax + by$ attains its optimum at two corner points, it attains the *same* value at every point on the segment joining them — infinitely many optimal solutions.
 - **Unbounded solution test:** for a maximisation problem on an unbounded region with largest corner value M , the maximum **fails to exist** if the open half-plane $ax + by > M$ has points in common with the feasible region.
 - **Infeasibility:** if the intersection of all constraint half-planes (including $x, y \geq 0$) is the empty set, the LPP has **no feasible solution** and hence no optimal solution.
 - **Redundancy:** removing a redundant constraint leaves the feasible region — and therefore the optimal solution — unchanged.
-

Section 3: Detection Procedures

How to identify each special case while solving, step by step.

To detect NO feasible solution:

1. Plot and shade every constraint as usual.
2. Look for a region common to *all* of them (in the first quadrant).
3. If **no such common region exists**, stop — write “the LPP has **no feasible solution**.” There is nothing to optimise. *Reasoning*: optimisation only makes sense over a non-empty set of valid choices.

To detect an UNBOUNDED solution (no maximum):

1. Confirm the feasible region is **unbounded**.
2. Evaluate Z at all corner points; let M be the largest.
3. Test the open half-plane $ax + by > M$. If it shares **any** point with the region, the objective can exceed M — so write “ Z has **no maximum**.” *Reasoning*: a single feasible point beating every corner proves no corner can be the maximum. (The same region may still have a valid minimum — check separately.)

To detect MULTIPLE optimal solutions:

1. Evaluate Z at all corners.
2. If the optimum value is attained at **two** corners, every point on the segment between them is also optimal. *Reasoning*: the objective line coincides with that edge, so all its points share the value.

To detect a REDUNDANT constraint:

1. Draw the region using all constraints.
2. If removing one line leaves the region unchanged (the line never forms a boundary of the region), that constraint is **redundant**. *Reasoning*: it imposes no real restriction beyond what the others already enforce.

Figure. A 2x2 panel. Top-left: a bounded polygon with the objective line touching one vertex (unique optimum). Top-right: the objective line lying flat along a top edge (multiple optima). Bottom-left: an unbounded region open to the upper-right with an arrow showing Z increasing without limit (no maximum). Bottom-right: two separate shaded strips that do not overlap (no feasible solution).

Section 4: Worked Examples

Example 1 — [CBSE Board Level] — No feasible solution

A coaching centre wants to run x morning batches and y evening batches. It must run at least 6 morning and at least 6 evening batches (staff commitments), but its rooms allow at most 10 batches in total per day. It wishes to maximise total enrolment $Z = 30x + 25y$. Solve.

Constraints.

$$x \geq 6, \quad y \geq 6, \quad x + y \leq 10, \quad x, y \geq 0.$$

Analysis. The first two constraints force $x + y \geq 6 + 6 = 12$. But the room constraint demands $x + y \leq 10$. No pair (x, y) can satisfy $x + y \geq 12$ and $x + y \leq 10$ simultaneously — the shaded regions never overlap.

Conclusion. The feasible region is **empty**, so the LPP has **no feasible solution**; there is no optimal value of Z . (Interpretation: the centre's staffing promise and its room capacity are in conflict — one must be relaxed.)

Figure. First quadrant showing the region $x \geq 6, y \geq 6$ (upper-right block) and the region $x + y \leq 10$ (lower-left triangle) with NO overlap, illustrating the empty feasible set.

Example 2 — [CBSE HOTS] — Unbounded solution (no maximum)

Maximise $Z = 5x + 4y$ subject to $x + y \geq 8, x \geq 0, y \geq 0$.

Region. Only “ $\geq$ ” and non-negativity constraints — the region lies above the line $x + y = 8$ and stretches to infinity up and to the right. It is **unbounded**.

Corner evaluation. The boundary corner points are $(8, 0)$ and $(0, 8)$:

$$Z(8, 0) = 40, \quad Z(0, 8) = 32.$$

Largest corner value $M = 40$.

Unbounded test. Does $5x + 4y > 40$ overlap the region? Take the feasible point $(20, 0)$ (it satisfies $x + y = 20 \geq 8$): there $Z = 100 > 40$. A feasible point beats the best corner — and we can push further ($(100, 0) \Rightarrow Z = 500$, and so on without limit).

Conclusion. Z has **no maximum** — the solution is **unbounded**. (Note: this same region *does* have a minimum, $Z = 32$ at $(0, 8)$, verified since $5x + 4y < 32$ misses the region.) This is the textbook trap: an unbounded region with no maximum but a perfectly good minimum.

Example 3 — [CUET] — Multiple optimal solutions

Maximise $Z = 2x + 2y$ subject to $x + y \leq 6, x \leq 4, y \leq 4, x, y \geq 0$. How many optimal solutions are there?

Corners. $(0, 0), (4, 0), (4, 2), (2, 4), (0, 4)$.

Corner	$Z = 2x + 2y$
$(0, 0)$	0
$(4, 0)$	8
$(4, 2)$	12
$(2, 4)$	12
$(0, 4)$	8

Corner	$Z = 2x + 2y$
--------	---------------

The maximum 12 is attained at **two** corners, $(4, 2)$ and $(2, 4)$.

Conclusion. There are **infinitely many** optimal solutions — every point on the segment joining $(4, 2)$ and $(2, 4)$ (which lies on the edge $x + y = 6$) gives $Z = 12$. The objective line $2x + 2y = 12$ is parallel to that edge.

Example 4 — [CUET] — Spot the redundant constraint

A feasible region is defined by $x + y \leq 10$, $x \leq 12$, $x \geq 0$, $y \geq 0$. Which constraint is redundant, and does removing it change the answer to any maximisation over this region?

Analysis. Within the first quadrant, $x + y \leq 10$ already forces $x \leq 10$. So $x \leq 12$ can never bind — every point allowed by the other constraints automatically satisfies $x \leq 12$. Its line lies entirely outside the region.

Conclusion. $x \leq 12$ is **redundant**. Removing it leaves the region — and hence the optimum of any linear objective over it — completely unchanged. (Lesson: not every line you're given actually shapes the region; always check which constraints truly form the boundary.)

Section 5: Practice Exercises

1. **[CBSE]** Does the LPP “Maximise $Z = x + y$ s.t. $x + y \geq 5$, $x + y \leq 2$, $x, y \geq 0$ ” have a feasible solution? Justify.
2. **[CUET]** For “Maximise $Z = 3x + 2y$ s.t. $x - y \geq 1$, $x \geq 0$, $y \geq 0$ ”, does a maximum exist? State the outcome type.
3. **[CUET]** The maximum of $Z = x + y$ is attained at both $(0, 5)$ and $(5, 0)$ of a feasible region. How many optimal solutions does the LPP have?
4. **[CUET]** In the constraints $2x + y \leq 20$, $x \leq 15$, $x, y \geq 0$, which constraint is redundant?
5. **[CBSE HOTS]** “Minimise $Z = 2x + 3y$ s.t. $x + y \geq 6$, $x, y \geq 0$.” Does a minimum exist? Does a maximum exist? Explain both.

Answers:

1. No feasible solution — $x + y \geq 5$ and $x + y \leq 2$ contradict; region empty.
2. No maximum (unbounded solution); the region is unbounded in the increasing direction of Z .
3. Infinitely many (every point on the segment joining $(0, 5)$ and $(5, 0)$).
4. $x \leq 15$ is redundant ($2x + y \leq 20$ with $x, y \geq 0$ already forces $x \leq 10$).
5. Minimum exists ($Z = 12$ at $(6, 0)$, verified); maximum does **not** exist (region unbounded upward, Z grows without limit).

Section 6: Crack the MCQ

Q1. An LPP whose feasible region is the empty set has:

- (A) a unique solution (B) infinitely many solutions (C) an unbounded solution (D) no feasible solution

Answer: (D). Empty region $\Rightarrow$ no point satisfies all constraints $\Rightarrow$ no feasible (and no optimal) solution.

- (A)/(B) — both assume valid points exist; there are none.
- (C) — “unbounded” needs a non-empty region that stretches to infinity; an empty region is not unbounded, it’s infeasible.

Q2. A feasible region is unbounded. For a maximisation LPP, the maximum:

- (A) always exists (B) never exists (C) may or may not exist (D) equals the minimum

Answer: (C). An unbounded region *may* still have a maximum, or may not — it must be checked with the half-plane test.

- (A) — confuses with the bounded case.
- (B) — too strong; unbounded regions can have a maximum if the objective decreases in the open direction.
- (D) — unrelated; no such rule.

Q3. If a linear objective attains the same maximum value at two adjacent corner points, the number of optimal solutions is:

- (A) 1 (B) 2 (C) 0 (D) infinite

Answer: (D). Every point on the connecting edge is optimal $\Rightarrow$ infinitely many.

- (B) — the classic trap: counts only the two corners, forgetting the edge between them.
- (A)/(C) — contradict the tie.

Q4. “Unbounded region” and “unbounded solution” are:

- (A) the same thing (B) different — one is about the region’s shape, the other about whether a finite optimum exists (C) both impossible in LP (D) terms for an empty feasible set

Answer: (B). “Unbounded region” = geometry stretches to infinity; “unbounded solution” = the objective has no finite optimum. A region can be unbounded yet still yield a finite minimum.

- (A) — the most common confusion; they are distinct.
- (C)/(D) — false; both occur, and neither describes an empty set.

Section 7: Common Pitfalls & Pro-Tips

Pitfall 1 — “Unbounded region” $\Rightarrow$ “no answer.” Wrong. Most cost-minimisation problems have unbounded regions *and* a perfectly valid minimum. Always run the half-plane test before declaring “no optimum.”

Pitfall 2 — Counting multiple optima as “two”. A tie at two corners means *infinitely many* solutions (the whole edge), not two.

Pitfall 3 — Missing infeasibility. If you can’t find a common shaded region, don’t force corner points — state plainly that there is **no feasible solution**.

Pitfall 4 — Treating every constraint as binding. Some constraints are redundant. Check which lines actually form the boundary before solving.

Pro-Tip: Before grinding through corner points, take **5 seconds to classify the outcome**: Is there a common region? (else infeasible). Is it bounded? (if yes, both max & min exist — no special case). Only if it’s unbounded do you need the verification test. This quick triage saves time and prevents the classic special-case errors.

Section 8: Cheat Sheet Box

LINEAR PROGRAMMING — SPECIAL CASES

- **Four outcomes:** unique optimum · multiple optima (infinite) · unbounded solution (no finite optimum) · no feasible solution (empty region).
- **No feasible solution:** constraints contradict $\Rightarrow$ no common shaded region $\Rightarrow$ no answer.
- **Unbounded solution:** unbounded region **and** $ax + by > M$ still meets the region $\Rightarrow$ no maximum. (Min may still exist!)
- **Multiple optima:** two corners tie for the optimum $\Rightarrow$ the whole joining edge is optimal (infinitely many).
- **Redundant constraint:** removing it doesn’t change the region or the answer.
- **Bounded region $\Rightarrow$ no special case** (Theorem 2 guarantees both max & min).

Memory aids:

- **“No region, no answer”** — empty feasible set = infeasible.
- **“Region unbounded $\neq$ solution unbounded”** — geometry vs the objective.
- **“Tie = whole edge”** — two equal corners $\Rightarrow$ infinitely many optima.

Chapter 12 — Complete Master Reference

One-stop revision for the entire chapter. Everything below is consolidated from Subtopics 01-03.

A. Core terminology

Objective function $Z = ax + by$ (the quantity to optimise) · decision variables x, y (what you choose) · constraints (linear inequalities) · non-negative constraints $x, y \geq 0$ · feasible region (common region of all constraints) · feasible / infeasible / optimal solutions · bounded vs unbounded region · corner point (vertex).

B. The two fundamental theorems

1. **Corner-point theorem:** if an optimum exists, it occurs at a **corner point** of the feasible region.
2. **Existence theorem:** if the region is **bounded**, Z has **both** a maximum and a minimum (each at a corner); if **unbounded**, an optimum **may or may not** exist.

C. Corner-Point Method (to solve)

1. Draw and shade each constraint (origin test for the correct side).
2. Identify all corner points.
3. Evaluate Z at every corner; record largest M , smallest m .
4. **Bounded:** $M = \max$, $m = \min$. **Unbounded:** verify — M is the max only if $ax + by > M$ misses the region; m is the min only if $ax + by < m$ misses the region.

D. The 5-Step Formulation Procedure (to model)

Name variables (with units) → write objective (max/min) → translate each clause to a constraint → add $x, y \geq 0$ → state the complete LPP.

E. Constraint-type dictionary

“at most / cannot exceed / available” → $\leq$ · “at least / minimum / requires” → $\geq$ · “uses all / exactly” → $=$. Tricky: “B exceeds A by at most k ” → $-x + y \leq k$; “A at least twice B” → $x \geq 2y$.

F. The four application families

Product-mix (maximise; resources $\leq$) · diet/blend (minimise; nutrients $\geq$) · transport/allocation (minimise; supply $\leq$ & demand $\geq$) · scheduling (maximise; time $\leq$ & targets $\geq$). *Goal word tells the family: profit/output → max, expect $\leq$; cost/requirement → min, expect $\geq$ (often unbounded → verify).*

G. The four solution outcomes

Unique optimum · multiple optima (tie at two corners $\Rightarrow$ whole edge, infinitely many) · unbounded solution (no finite optimum) · no feasible solution (empty region).

Bounded region $\Rightarrow$ no special case.

H. Chapter revision checklist

- ☐ I can translate any word problem into a complete LPP (objective + constraints + non-negativity).
- ☐ I can draw a feasible region and locate every corner point.
- ☐ I can apply the corner-point method, including the unbounded verification test.
- ☐ I can recognise all four outcomes, including infeasible and unbounded.
- ☐ I know the constraint-type dictionary and the tricky phrasings cold.
- ☐ I can identify the application family and a redundant constraint.

Chapter 12: Linear Programming — Round 2

Addendum

The chapter teaches formulation, the corner-point method, and the four special cases — everything a standard CBSE long-answer question demands. What it deliberately sets aside, and what this addendum supplies, is the layer underneath and the layer beyond. **A** proves the two fundamental theorems via convex combinations, so you can reproduce the reasoning an examiner may ask you to justify in one mark. **B** handles parametric objectives: given corners, find every value of the coefficients for which the optimum sits at a chosen vertex — a reliable MCQ and short-answer pattern the chapter touches only once. **C** proves infeasibility and redundancy by *combining inequalities algebraically*, no graph needed. **D** measures the feasible region: systematic vertex enumeration, area, and lattice-point counts. **E** solves the integer-answer problem the chapter waves away with “divisibility is assumed” — including a worked demonstration that rounding the continuous optimum genuinely fails. One honest note: LP itself is not examined in JEE, but the machinery of B, C and D — comparing linear forms parametrically, deriving contradictions from inequality systems, counting lattice points in polygonal regions — transfers directly to JEE Main questions in coordinate geometry and inequalities. Nothing here uses anything beyond Chapter 12 plus the determinant formula from Class 12.

Addendum A — Why the Corner-Point Method Works: The Convexity Proof

The chapter states Theorem 1 and offers a sliding-line picture. CBSE rarely asks for the formal proof, but it *does* ask one-mark justification questions (“why must the optimum be at a corner?”, “why does a tie at two corners give infinitely many optima?”), and JEE-minded students should own the real argument. It is four lines long once one lemma is in place.

Method

Definition (convex combination). A point P is a convex combination of points $P_1, \dots, P_n$ if

$$P = \lambda_1 P_1 + \lambda_2 P_2 + \dots + \lambda_n P_n,$$

where every $\lambda_i \geq 0$ and $\lambda_1 + \dots + \lambda_n = 1$. Geometrically: P lies somewhere in the polygon spanned by the P_i .

Lemma 1 (linearity of Z). For $Z = ax + by$, if $P = \lambda Q + (1 - \lambda)R$ with $0 \leq \lambda \leq 1$, then

$$Z(P) = \lambda Z(Q) + (1 - \lambda)Z(R).$$

Proof. Coordinates: $P = (\lambda q_x + (1 - \lambda)r_x, \lambda q_y + (1 - \lambda)r_y)$, so

$$Z(P) = a[\lambda q_x + (1 - \lambda)r_x] + b[\lambda q_y + (1 - \lambda)r_y],$$

and rearranging gives $\lambda(aq_x + bq_y) + (1 - \lambda)(ar_x + br_y)$. ■

Lemma 2 (vertices generate the region). Every point of a convex polygon R with vertices $V_1, \dots, V_n$ is a convex combination of the vertices.

Proof. Triangulate R by drawing all diagonals from V_1 ; the polygon splits into $n - 2$ triangles whose vertices are vertices of R . Any $P \in R$ lies in (or on the edge of) one of these triangles, say with vertices A, B, C . It remains to decompose P inside one triangle. Draw the ray from A through P ; it meets the segment BC at some point Q (this is where convexity — no dents — is used). Then:

$$Q = tB + (1 - t)C \quad \text{for some } 0 \leq t \leq 1,$$

$$P = sA + (1 - s)Q \quad \text{for some } 0 \leq s \leq 1.$$

Substituting the first into the second:

$$P = sA + (1 - s)tB + (1 - s)(1 - t)C.$$

The three weights are non-negative, and their sum is

$$s + (1 - s)t + (1 - s)(1 - t) = s + (1 - s)[t + 1 - t] = 1.$$

So P is a convex combination of A, B, C , which are themselves vertices of R . Points on a diagonal lie in two triangles; either decomposition works. ■

Theorem 1, proved. Let $M^* = \max\{Z(V_i)\}$ over the vertices. For any feasible P , write $P = \sum \lambda_i V_i$ by Lemma 2. By Lemma 1 extended to n terms,

$$Z(P) = \sum_i \lambda_i Z(V_i) \leq \sum_i \lambda_i M^* = M^*.$$

No feasible point beats the best vertex. The same argument with min gives the minimum. ■

Multiple optima, derived. Suppose $Z(A) = Z(B) = M^*$ at two vertices. For any point P of the segment AB , write $P = \lambda A + (1 - \lambda)B$; Lemma 1 gives

$$Z(P) = \lambda M^* + (1 - \lambda)M^* = M^*.$$

Conversely, the objective line $ax + by = M^*$ passes through both A and B , hence contains the entire segment through them, and every point of that segment is feasible (convexity again). So the optimum is attained along the whole edge — this is the chapter’s “tie means infinite solutions” property, now proved rather than asserted.

Why corner-checking alone fails on unbounded regions. Theorem 1 begins “when Z has an optimal value” — existence is a separate hypothesis. Concrete failure, recomputed: minimise nothing, maximise $Z = 5x + 4y$ over $x + y \geq 8$ in the first quadrant. The corner points are $(8, 0)$ and $(0, 8)$, giving 40 and 32, so the largest corner value is $M = 40$. But $(20, 0)$ is feasible ($20 \geq 8$) and gives $Z = 100 > 40$. The largest corner value is simply wrong here, because no maximum exists at all. This is precisely why the half-plane test of Subtopic 01 is not bookkeeping — it is the existence check the proof requires.

Convex decomposition: a triangle ABC with an interior point P, the ray from A through P meeting BC at Q, and the weight labels showing P as a convex combination of the three vertices.

Figure 1: Convex decomposition: a triangle ABC with an interior point P, the ray from A through P meeting BC at Q, and the weight labels showing P as a convex combination of the three vertices.

Practice — Addendum A

1. For the triangle with vertices $O(0, 0)$, $A(4, 0)$, $B(0, 3)$, express $P = (1, 1)$ as a convex combination of O, A, B , and use Lemma 1 to compute $Z(P)$ for $Z = x + 2y$ without substituting coordinates.
2. Using Lemma 1, show that if Z takes equal values at two feasible points Q, R , it takes that value everywhere on segment QR .
3. The feasible region has vertices $(0, 0)$, $(6, 0)$, $(0, 4)$ and $Z = 3x + y$. The point $(2, 2)$ is feasible. Without evaluating Z at $(2, 2)$, name a vertex guaranteed to beat it, and verify.
4. In one sentence each: (i) why is it legitimate to check only corner points? (ii) why does that legitimacy collapse when the region is unbounded?
5. Over the region $x + y \geq 8$, $x, y \geq 0$, minimise $Z = 5x + 4y$. Evaluate both corners, decide the minimum, and run the half-plane test to confirm it exists.

Answers: 1. Solving $4\alpha = 1$, $3\beta = 1$ gives $\alpha = \frac{1}{4}$, $\beta = \frac{1}{3}$, $\gamma = \frac{5}{12}$, so $P = \frac{1}{4}A + \frac{1}{3}B + \frac{5}{12}O$. Then $Z(P) = \frac{1}{4}Z(A) + \frac{1}{3}Z(B) + \frac{5}{12}Z(O) = \frac{1}{4}(4) + \frac{1}{3}(6) = 3$. Direct substitution confirms $1 + 2 = 3$. 2. Write $P = \lambda Q + (1 - \lambda)R$; Lemma 1 gives $Z(P) = \lambda c + (1 - \lambda)c = c$ throughout the segment. 3. By Theorem 1 the best vertex dominates every feasible point. Corner values: $Z(6, 0) = 18$, $Z(0, 4) = 4$, $Z(0, 0) = 0$; since 18 is the largest vertex value, $Z(2, 2) \leq 18$ — indeed $Z(2, 2) = 8$. 4. (i) Every feasible point is a convex combination of vertices, and Z of a combination cannot exceed the best vertex value. (ii) On an unbounded region a point far out may fail to be captured meaningfully by the corner data — the decomposition bound still holds, but the *existence* of an optimum is no longer guaranteed, so the largest corner value need not be the maximum (see the $(20, 0)$ example above). 5. Corners $(8, 0)$ and $(0, 8)$ give $Z = 40$ and $Z = 32$; the smaller is 32. Test: does $5x + 4y < 32$ meet the region? Along the region, $x + y \geq 8$ forces $x \geq 8 - y$; the line $5x + 4y = 32$ meets the region only at $(0, 8)$, and moving into the region (rightward/upward) increases $5x + 4y$. Minimum $Z = 32$ at $(0, 8)$, verified.

Addendum B — Parametric Objectives: Pinning the Optimum to a Chosen Vertex

Chapter Example 4 asks for the condition under which the maximum occurs at *two* specified vertices — one equality. The fuller examinable pattern asks for the condition on the coefficients so that the maximum occurs at one *specified* vertex (uniquely, or with ties allowed), or traces how the optimal vertex moves as a coefficient varies. The chapter never sets up the general machinery. It is nothing new: “max at vertex V ” is a handful of linear inequalities in the coefficients — a tiny LP in its own right.

Method

Given a feasible region with corners $V_1, \dots, V_n$ and objective $Z = px + qy$ with $p, q > 0$:

1. **Evaluate Z symbolically at every corner**, obtaining linear expressions in p and q .
2. **Write the domination conditions.** The maximum occurs at V_k if and only if

$$Z(V_k) \geq Z(V_j) \quad \text{for every } j.$$

Each condition is a linear inequality in p, q . “Maximum at V_k and nowhere else” upgrades some of these to strict; “also at V_j ” replaces one inequality by equality. 3. **Solve the little system** in the pq -plane, keeping the sign restrictions $p, q > 0$. 4. **Cross-check with slopes.** The objective line $px + qy = c$ has slope $-\frac{p}{q}$. An edge joining two corners A, B with slope m becomes a tie-edge exactly when

$$-\frac{p}{q} = m, \quad \text{i.e.} \quad \frac{p}{q} = -m.$$

As the ratio $r = \frac{p}{q}$ sweeps from 0 to ∞ , the optimal vertex changes precisely at these critical ratios, so listing the edges’ slopes orders all possible behaviours.

Why the slope criterion is exact: two distinct points A, B both maximise Z over the region iff they lie on the same level line $px + qy = c$, i.e. $pA_x + qA_y = pB_x + qB_y$, which rearranges to $-\frac{p}{q} = \frac{B_y - A_y}{B_x - A_x}$. Level lines are parallel, so the shared level line is parallel to the edge AB .

Example B.1 — [CBSE HOTS / CUET]

The feasible region has corners $O(0,0)$, $A(4,0)$, $B(2,3)$, $C(0,4)$ (the region of chapter Example 1). For $Z = px + qy$ with $p, q > 0$, find exactly when the maximum occurs at A , at B , at C .

Corner values:

Corner	$Z = px + qy$
$O(0,0)$	0

Corner	$Z = px + qy$
$A(4, 0)$	$4p$
$B(2, 3)$	$2p + 3q$
$C(0, 4)$	$4q$

Since $p, q > 0$, $Z(O) = 0$ is always strictly smallest — O can never be optimal.

Max at A: require $4p \geq 2p + 3q$ and $4p \geq 4q$:

$$2p \geq 3q \quad \text{and} \quad p \geq q.$$

With $q > 0$, the first implies the second, so the condition is $p \geq \frac{3}{2}q$.

Max at C: require $4q \geq 2p + 3q$ and $4q \geq 4p$:

$$q \geq 2p \quad \text{and} \quad q \geq p,$$

so the binding condition is $q \geq 2p$, i.e. $\frac{p}{q} \leq \frac{1}{2}$.

Max at B: require $2p + 3q \geq 4p$ and $2p + 3q \geq 4q$:

$$2p \leq 3q \quad \text{and} \quad 2p \geq q,$$

$$\text{i.e. } \frac{1}{2} \leq \frac{p}{q} \leq \frac{3}{2}.$$

Slope cross-check. Edge AB joins $(4, 0)$ to $(2, 3)$, slope $= \frac{3-0}{2-4} = -\frac{3}{2}$; critical ratio $r = \frac{3}{2} \checkmark$.

Edge BC joins $(2, 3)$ to $(0, 4)$, slope $= -\frac{1}{2}$; critical ratio $r = \frac{1}{2} \checkmark$. So as $r = \frac{p}{q}$ grows from 0 to ∞ , the optimal vertex walks $C \rightarrow B \rightarrow A$, switching at $r = \frac{1}{2}$ and $r = \frac{3}{2}$ — matching the three intervals above exactly.

Check. At $r = 1$ (say $p = q = 1$): values 0, 4, 5, 4 — max at $B \checkmark$, and 1 lies in $[\frac{1}{2}, \frac{3}{2}]$. At $r = 2$: values 0, 8, 7, 4 — max at $A \checkmark$.

Example B.2 — [CUET Speed Pattern]

Find all $k > 0$ such that the maximum of $Z = 3x + ky$ over the region with corners $(0, 0)$, $(5, 0)$, $(3, 3)$, $(0, 4)$ occurs at $(3, 3)$.

Values: $Z(0, 0) = 0$, $Z(5, 0) = 15$, $Z(3, 3) = 9 + 3k$, $Z(0, 4) = 4k$. Domination conditions:

$$9 + 3k \geq 15 \Rightarrow k \geq 2,$$

$$9 + 3k \geq 4k \Rightarrow k \leq 9,$$

and $9 + 3k \geq 0$ holds automatically for $k > 0$. So the answer is $2 \leq k \leq 9$.

At the endpoints ties occur: $k = 2$ makes $Z(3, 3) = Z(5, 0) = 15$ (multiple optima along that edge), and $k = 9$ makes $Z(3, 3) = Z(0, 4) = 36$. If the question demands a *unique* optimum at $(3, 3)$, the answer tightens to $2 < k < 9$.

Slope check. Edge $(5, 0)-(3, 3)$ has slope $-\frac{3}{2}$, giving $\frac{3}{k} = \frac{3}{2}$, i.e. $k = 2 \checkmark$. Edge $(3, 3)-(0, 4)$ has slope $-\frac{1}{3}$, giving $\frac{3}{k} = \frac{1}{3}$, i.e. $k = 9 \checkmark$.

Practice — Addendum B

1. A region has corners $(0, 0)$, $(4, 0)$, $(2, 4)$, $(0, 5)$. For $Z = px + qy$, $p, q > 0$, find all conditions under which the maximum occurs at $(2, 4)$.
2. For the same region, find all $k > 0$ for which the maximum of $Z = 2x + ky$ occurs at $(0, 5)$.
3. A region has corners $(0, 8)$, $(2, 4)$, $(6, 0)$ (an unbounded diet region). For $Z = px + qy$, $p, q > 0$, find when the *minimum* occurs at $(2, 4)$.
4. For the region of Example B.1, state the order in which the optimal vertex changes as $\frac{p}{q}$ increases from 0 to ∞ , naming the switch ratios.

Answers: 1. $2p + 4q \geq 4p$ and $2p + 4q \geq 5q$, i.e. $4q \geq 2p$ and $2p \geq q$; so $\frac{q}{2} \leq p \leq 2q$. (The comparisons with $Z(0, 0) = 0$ are automatic.) 2. Values: 8 at $(4, 0)$, $4 + 4k$ at $(2, 4)$, $5k$ at $(0, 5)$. Need $5k \geq 4 + 4k$ (i.e. $k \geq 4$) and $5k \geq 8$ (automatic once $k \geq 4$, since then $5k \geq 20$). Answer $k \geq 4$. 3. Domination reversed: $2p + 4q \leq 8q$ and $2p + 4q \leq 6p$, i.e. $2p \leq 4q$ and $4q \leq 6p$, giving $\frac{2q}{3} \leq p \leq 2q$. 4. C (for $\frac{p}{q} < \frac{1}{2}$), then B (for $\frac{1}{2} \leq \frac{p}{q} \leq \frac{3}{2}$), then A (for $\frac{p}{q} > \frac{3}{2}$); switches at $r = \frac{1}{2}$ and $r = \frac{3}{2}$.

Addendum C — Infeasibility and Redundancy, Proved Without a Graph

Subtopic 03 detects infeasibility and redundancy *by drawing*. Two problems: a hand-drawn graph can mislead when lines pass close together, and CBSE HOTS answers demand a *justification*, not a shading. Both facts have clean algebraic certificates, built from one lemma.

Method

Lemma (positive combinations preserve inequalities). If $u \leq v$ and $u' \leq v'$, and $s, t \geq 0$, then

$$su + tu' \leq sv + tv'.$$

Proof. $su \leq sv$ (multiply the first by $s \geq 0$) and $tu' \leq tv'$; add. ■

Everything below is this lemma wearing different hats. Note the asymmetry it creates: “ $\leq$ ” chains survive positive scaling and addition, so contradictions are built by pushing “ $\leq$ ” information against “ $\geq$ ” information.

Certificate of infeasibility. Combine the “ $\leq$ ” constraints (with non-negative multipliers, plus $x \geq 0$, $y \geq 0$ used freely) to obtain some expression $\leq c$; combine the “ $\geq$ ” side to obtain the *same* expression $\geq d$ with $d > c$. Contradiction — no feasible point exists.

Certificate of redundancy. To show constraint C is implied by the others, derive it: manipulate the other constraints with positive combinations until C falls out. Then removing C changes nothing.

Example C.1 — [CBSE Board Level]

Prove algebraically that $x \geq 6$, $y \geq 6$, $x + y \leq 10$, $x, y \geq 0$ has no feasible solution (the coaching-centre problem of Sub-03 Example 1).

Adding the first two inequalities (multipliers $s = t = 1$):

$$x + y \geq 12.$$

But the third demands $x + y \leq 10$. Since $12 > 10$, contradiction. Empty region, no feasible solution — certified in two lines, no graph.

Example C.2 — [CBSE HOTS]

Show that $2x + y \leq 8$, $x + 2y \leq 8$, $x + y \geq 7$ is infeasible.

Adding the two upper bounds:

$$3x + 3y \leq 16 \Rightarrow x + y \leq \frac{16}{3}.$$

Since $\frac{16}{3} < 7$, this contradicts $x + y \geq 7$. Note the multipliers were 1 and 1 — chosen so the left sides would *match* the expression forced by the “ $\geq$ ” constraint. Choosing multipliers is the only creative step in the method; aim the left-hand sides at whatever the lower bounds talk about.

Example C.3 — [HOTS — when plain addition is not enough]

Show that $x + y \leq 4$, $x + 2y \geq 12$, $x, y \geq 0$ is infeasible.

Adding the two main inequalities fails: the left sides $x + y$ and $x + 2y$ differ. Bound the pieces separately instead. From $y \geq 0$ and $x + y \leq 4$:

$$y \leq x + y \leq 4.$$

Now split and reassemble:

$$x + 2y = (x + y) + y \leq 4 + 4 = 8 < 12,$$

contradicting $x + 2y \geq 12$. The trick: extract a bound on y alone from the compound inequality, then rebuild the awkward expression from bounded parts.

Example C.4 — [CUET]

(a) Show $x \leq 12$ is redundant given $x + y \leq 10$ and $y \geq 0$. **(b)** Show $2x + y \leq 20$ is redundant given $x + y \leq 12$ and $x \leq 6$.

(a) Since $y \geq 0$,

$$x \leq x + y \leq 10 < 12.$$

Every point satisfying the other constraints automatically satisfies $x \leq 12$ — it can never bind.

(b) Split and recombine:

$$2x + y = (x + y) + x \leq 12 + 6 = 18 < 20.$$

Done — the constraint imposes nothing beyond the other two.

Example C.5 — [Caution — a failed proof proves nothing]

Is $x + 2y \leq 4$, $x + y \geq 4$, $x, y \geq 0$ infeasible?

It *looks* contradictory — the upper bound involves $2y$, the lower bound only y . Attempt the C.3 manoeuvre: from $x + 2y = (x + y) + y$ we would need a bound on y alone, and none comes. But a failed proof of infeasibility is not a proof of feasibility — so exhibit a point. Try $y = 0$: then $x \geq 4$ and $x \leq 4$ force $x = 4$, and $(4, 0)$ satisfies everything:

$$4 + 0 = 4 \geq 4, \quad 4 + 2(0) = 4 \leq 4.$$

The LPP is feasible (with the single feasible point $(4, 0)$ — a degenerate region). Lesson: infeasibility claims need a certificate; feasibility claims need only one witness.

Practice — Addendum C

1. Prove without a graph that $x + y \leq 3$, $2x + 2y \geq 8$, $x, y \geq 0$ is infeasible.
2. Prove without a graph that $3x + 2y \leq 6$, $x \geq 3$, $y \geq 1$ is infeasible.
3. Show that $y \leq 20$ is redundant given $x + y \leq 15$, $x, y \geq 0$.
4. Determine whether $2x + y \leq 4$, $x + y \geq 3$, $x, y \geq 0$ is infeasible; justify whichever way you conclude.
5. Show that $x + 2y \leq 10$ is redundant given $2x + 4y \leq 19$.

Answers: 1. Halve the second inequality: $x + y \geq 4 > 3$, contradicting the first. 2. From $x \geq 3$ and $y \geq 1$ (positive multipliers 3 and 2): $3x \geq 9$ and $2y \geq 2$, so $3x + 2y \geq 11 > 6$. Contradiction. 3. $y \leq x + y \leq 15 < 20$. 4. Feasible: $(1, 2)$ gives $2 + 2 = 4 \leq 4$ and $1 + 2 = 3 \geq 3$ — one witness settles it. (In fact the feasible set is the segment from $(1, 2)$ to $(2, 0)$.) 5. Halve: $x + 2y \leq \frac{19}{2} = 9.5 < 10$; every admissible point satisfies the stricter bound, so the weaker one binds never.

Addendum D — Measuring the Region: Vertex Enumeration, Area, Lattice Points

The chapter always hands you the corners or lets you eyeball them. Three exam patterns need more: “find the area of the feasible region”, “how many points with integer coordinates lie in the region”, and any problem whose corners must be extracted from four or more lines reliably. All three reduce to two disciplined procedures.

Method

Step 1 — enumerate candidate vertices. Each pair of boundary lines gives an intersection candidate. For

$$a_1x + b_1y = c_1, \quad a_2x + b_2y = c_2,$$

with $a_1b_2 - a_2b_1 \neq 0$, the solution is

$$x = \frac{b_2c_1 - b_1c_2}{a_1b_2 - a_2b_1}, \quad y = \frac{a_2c_1 - a_1c_2}{a_1b_2 - a_2b_1}.$$

Keep a candidate only if it satisfies **every** constraint. Discard duplicates where three lines concur (as in Sub-02 Example 2, where three boundaries meet at one vertex).

Step 2 — order the survivors around the boundary (sort by angle about any interior point, or walk edge to edge). Convexity guarantees a clean cycle.

Step 3 — area. For a triangle with one vertex at the origin O and the others at $U = (u_1, u_2)$, $W = (w_1, w_2)$,

$$\text{Area} = \frac{1}{2} |u_1w_2 - u_2w_1|,$$

the determinant expansion of $\frac{1}{2} \|\overrightarrow{OU} \times \overrightarrow{OW}\|$ from the Class 12 vector/determinant toolkit — you can verify it on the right triangle with legs along the axes, where it collapses to $\frac{1}{2}(\text{base})(\text{height})$. A convex polygon with vertices $V_1, \dots, V_n$ in order fans into $n - 2$ such triangles from V_1 ; equivalently, the shoelace sum

$$\text{Area} = \frac{1}{2} \left| \sum_i (x_i y_{i+1} - x_{i+1} y_i) \right|$$

(with indices cyclic) computes the whole thing in one pass. Always sanity-check with a second decomposition.

Step 4 — lattice points by strips. Fix an integer $y = j$ within the region’s vertical range; the constraints collapse to an interval $L(j) \leq x \leq U(j)$, contributing

$$\lfloor U(j) \rfloor - \lceil L(j) \rceil + 1$$

integer points (zero if the interval contains none). Sum over j . Horizontal strips are the right choice when constraints are given as $x \leq$ expressions in y , which is the norm after dividing through.

Example D.1 — [CBSE Board Level]

Find the area of the feasible region of chapter Example 1: $x + 2y \leq 8$, $3x + 2y \leq 12$, $x, y \geq 0$.

Candidates. Pairs of boundary lines: $x = 0$ with $y = 0$ gives $(0, 0)$; $x + 2y = 8$ with $x = 0$ gives $(0, 4)$; $3x + 2y = 12$ with $y = 0$ gives $(4, 0)$; the two slanted lines: subtract to get $2x = 4$, so $(2, 3)$ — and it satisfies $2 + 6 = 8 \leq 8 \checkmark$ and $6 + 6 = 12 \leq 12 \checkmark$. Corners: $O(0, 0)$, $A(4, 0)$, $B(2, 3)$, $C(0, 4)$, in order.

Area by fan from O . Split into $\triangle OAB$ and $\triangle OBC$:

$$[\triangle OAB] = \frac{1}{2} |4 \cdot 3 - 0 \cdot 2| = 6,$$

$$[\triangle OBC] = \frac{1}{2} |2 \cdot 4 - 3 \cdot 0| = 4.$$

Total $6 + 4 = 10$ square units.

Independent check by base-height. $\triangle OAB$ has base $OA = 4$ along the x -axis and height equal to B 's y -coordinate, 3: area $\frac{1}{2}(4)(3) = 6 \checkmark$. $\triangle OBC$ has base $OC = 4$ along the y -axis and height equal to B 's x -coordinate, 2: area $\frac{1}{2}(4)(2) = 4 \checkmark$.

Shoelace confirmation.

$$\text{Area} = \frac{1}{2} |(0 \cdot 0 - 4 \cdot 0) + (4 \cdot 3 - 2 \cdot 0) + (2 \cdot 4 - 0 \cdot 3) + (0 \cdot 0 - 0 \cdot 4)| = \frac{1}{2} |12 + 8| = 10.$$

Three decompositions, one answer: 10 square units.

Example D.2 — [CUET / HOTS]

How many points with integer coordinates lie in the region of Example D.1?

The vertical range is $0 \leq y \leq 4$, so strips $y = 0, 1, 2, 3, 4$. On strip $y = j$, the constraints read

$$x \leq 8 - 2j \quad \text{and} \quad 3x \leq 12 - 2j,$$

with $x \geq 0$ an integer. The count on each strip is $\left\lfloor \min\left(8 - 2j, \frac{12 - 2j}{3}\right) \right\rfloor + 1$.

j	$8 - 2j$	$\frac{12 - 2j}{3}$	binding cap	integer x count
0	8	4	4	5
1	6	$\frac{10}{3}$	3	4

j	$8 - 2j$	$\frac{12-2j}{3}$	binding cap	integer x count
2	4	$\frac{8}{3}$	2	3
3	2	2	2	3
4	0	$\frac{4}{3}$	0	1

Total: $5 + 4 + 3 + 3 + 1 = 15$ lattice points.

Spot-check. Strip $j = 3$: $x \leq 2$ from both constraints, giving $(0, 3)$, $(1, 3)$, $(2, 3)$ — and indeed $(2, 3)$ sits exactly on both slanted boundaries simultaneously, the concurrence noted in chapter Example 2 of Sub-02's family. The cap $\frac{12-2j}{3}$ governs low strips (fibre-style constraint tighter), the cap $8 - 2j$ the top strip — the crossover between $j = 2$ and $j = 3$ is why the counts dip and hold.

Useful remark (boundary counts). The number of lattice points on the segment joining integer points P, Q is $\gcd(|\Delta x|, |\Delta y|) + 1$. Segment AB here: $\gcd(2, 3) + 1 = 2$, namely $(4, 0)$ and $(2, 3)$ — consistent with the strip table.

Practice — Addendum D

1. Find the area of the feasible region of $x + y \leq 6$, $x \leq 4$, $y \leq 3$, $x, y \geq 0$.
2. Count the lattice points in the same region.
3. Find the perimeter of the region of Example D.1, leaving surds in exact form.
4. How many lattice points lie on the boundary segment joining $(0, 4)$ to $(2, 3)$?
5. The corners of a feasible region are $(0, 0)$, $(5, 0)$, $(3, 3)$, $(0, 4)$. Compute its area by the shoelace formula and by splitting into two triangles, and confirm agreement.

Answers: 1. Corners $(0, 0)$, $(4, 0)$, $(4, 2)$, $(3, 3)$, $(0, 3)$. Shoelace sum: $0 + 8 + 6 + 9 + 0 = 23$, area $\frac{23}{2}$. Check: the 4×3 box has area 12; the excluded top-right corner is the right triangle with legs 1, 1, area $\frac{1}{2}$; $12 - \frac{1}{2} = \frac{23}{2}$ ✓. 2. Strips $y = 0, 1, 2, 3$: caps $\min(4, 6 - j)$ give 5, 5, 5 and at $y = 3$ cap 3 gives 4; total 19. Check: the box has $5 \times 4 = 20$ lattice points, minus the single point $(4, 3)$ — which violates $x + y \leq 6$ — leaves 19 ✓. 3. $OA = 4$, $AB = \sqrt{(2)^2 + (3)^2} = \sqrt{13}$, $BC = \sqrt{(2)^2 + (1)^2} = \sqrt{5}$, $CO = 4$; perimeter $8 + \sqrt{13} + \sqrt{5}$. 4. $\gcd(2, 1) + 1 = 2$ — only the endpoints. 5. Shoelace: terms $0, 0$, $5 \cdot 3 - 3 \cdot 0 = 15$, $3 \cdot 4 - 0 \cdot 3 = 12$, 0 ; area $\frac{1}{2}(27) = \frac{27}{2}$. Triangle split from $(0, 0)$: $\frac{1}{2}|5 \cdot 3 - 0 \cdot 3| = 7.5$ and $\frac{1}{2}|3 \cdot 4 - 3 \cdot 0| = 6$; total $13.5 = \frac{27}{2}$ ✓.

Addendum E — Whole-Number Answers: When the Optimum Is Fractional

Section 1 of Subtopic 02 admits it: “divisibility is assumed... CBSE problems are usually designed so the optimum lands on whole numbers.” Fine until an HOTS paper, a CUET tricky MCQ, or an Exemplar problem hands you a continuous optimum at $(3.5, 2)$ and asks for the best *achievable* answer. The reflex — round the optimum — is demonstrably wrong, and the fix is a small exact procedure. This is also the honest bridge to integer programming, which the syllabus stops short of.

Method

The ceiling principle. Every lattice (integer) point of the region is a feasible point of the original LPP, so the best integer objective value never exceeds the continuous optimum Z^* :

$$Z_{\text{int}} \leq Z^*.$$

This one line turns the search finite and directed: scan candidate integer values of Z downward from Z^* ; the first value actually attained at a feasible lattice point is the answer.

Procedure.

1. Solve the continuous LPP; record Z^* and the optimal vertex.
2. List lattice points whose Z -value is a candidate: start at the largest integer $\leq Z^*$ and step down. For each candidate value c , find the lattice points with $Z = c$ (solve $ax+by = c$ over small integers) and test feasibility.
3. Stop at the first feasible hit — that is the integer optimum. Do not stop earlier, and do not restrict attention to points near the continuous optimum: Step 2's scan covers the whole region, which is exactly what saves you in the example below.
4. For integer *minimisation*, run the mirror image: scan upward from the smallest integer $\geq Z^*$.

What is forbidden: rounding the optimal vertex, up or down, and trusting the result. The next example is constructed so that every rounding convention fails.

Example E.1 — [HOTS / Exemplar Style]

A unit makes gadgets (x) and kits (y). Each gadget needs 2 hours on line A and 2 on line B; each kit needs 3 hours on A and 1 on B. Line A has 13 hours, line B has 9. Profit is ₹5 per gadget and ₹4 per kit. Find the production plan with maximum profit, given x, y must be whole numbers.

Continuous solve. Maximise $Z = 5x + 4y$ subject to $2x + 3y \leq 13$, $2x + y \leq 9$, $x, y \geq 0$. Corners: $(0, 0)$, $(4.5, 0)$, and the intersection of the two slanted lines — subtracting kills $2x$: $2y = 4$, so $y = 2$, then $2x = 13 - 6 = 7$, giving $(\frac{7}{2}, 2)$ — and $(0, \frac{13}{3})$. Values:

$$Z(4.5, 0) = 22.5, \quad Z\left(\frac{7}{2}, 2\right) = 17.5 + 8 = 25.5, \quad Z\left(0, \frac{13}{3}\right) = \frac{52}{3} \approx 17.3.$$

Continuous optimum $Z^* = 25.5$ at $(\frac{7}{2}, 2)$ — physically meaningless as a production plan.

Rounding fails, three ways. Up: $(4, 2)$ gives $2(4) + 3(2) = 14 > 13$ — infeasible. Down: $(3, 2)$ gives $6 + 6 = 12 \leq 13$ ✓, $6 + 2 = 8 \leq 9$ ✓, profit $15 + 8 = 23$. Nearest overall (round half up on x , keep y): $(4, 2)$ again, infeasible. Every conventional rounding yields 23 at best.

Correct scan. Largest integer ≤ 25.5 is 25: solve $5x + 4y = 25$ over small non-negative integers — $(5, 0)$ and $(1, 5)$. Check $(5, 0)$: $2(5) + 0 = 10 > 9$ ×. $(1, 5)$: $2 + 15 = 17 > 13$ ×. Neither feasible. Step down to 24: $5x + 4y = 24$ gives $(4, 1)$ and $(0, 6)$. Check $(4, 1)$:

$$2(4) + 3(1) = 11 \leq 13 \checkmark, \quad 2(4) + 1 = 9 \leq 9 \checkmark.$$

Feasible. Profit $20 + 4 = 24$.

Conclusion. Maximum whole-number profit ₹24, at 4 gadgets and 1 kit — beating every rounding of the continuous optimum by ₹1, and sitting at a point no rounding rule would ever propose ((3.5, 2) reflects to (4, 1) only if you notice the binding constraint $2x + y = 9$ wants x pushed up while y drops). The scan caught it because the scan never assumed the answer lived next door to Z^* 's vertex.

Completeness check. $Z = 25$ exhausted above; $Z = 24$ attained; so 24 is maximal among integers, consistent with the ceiling principle $24 \leq 25.5$.

Example E.2 — [Minimisation Mirror]

Minimise $Z = 2x + 3y$ **over integer points satisfying** $x + y \geq 4.5$ **(equivalently** $2x + 2y \geq 9$ **),** $x + 3y \geq 9$, $x, y \geq 0$ **integers.**

Continuous solve. Corners: $(\frac{9}{2}, 0)$ from $x + y = 4.5$ with $y = 0$; $(\frac{9}{4}, \frac{9}{4})$ from the two lines — subtracting: $2y = 4.5$; and $(0, \frac{9}{2})$ from $x + 3y = 9$ with $x = 0$. Wait — check which corners belong to the region: all three do (each satisfies both inequalities). Values:

$$Z(4.5, 0) = 9, \quad Z(2.25, 2.25) = 4.5 + 6.75 = 11.25, \quad Z(0, 4.5) = 13.5.$$

Continuous minimum $Z^* = 9$ at $(\frac{9}{2}, 0)$. Rounding up gives (5, 0) — test it: $x + 3y = 5 \geq 9$? **No.** Infeasible. The reflex fails on the minimisation side too.

Correct upward scan. Smallest integer ≥ 9 is 9: points with $2x + 3y = 9$: (0, 3), (3, 1), (4, ...) irrelevant. (0, 3): $x + y = 3 < 4.5 \times$. (3, 1): $x + y = 4 < 4.5 \times$. Next, 10: (5, 0), (2, 2). (5, 0): $5 < 9 \times$ (second constraint). (2, 2): $x + y = 4 < 4.5 \times$. Next, 11: (4, 1), (1, 3). (4, 1): $x + y = 5 \geq 4.5 \checkmark$ but $4 + 3 = 7 \geq 9$? $\times$. (1, 3): $4 \geq 4.5$? $\times$. Next, 12: (6, 0), (3, 2), (0, 4). (6, 0): $6 \geq 9 \times$. (3, 2):

$$3 + 2 = 5 \geq 4.5 \checkmark, \quad 3 + 6 = 9 \geq 9 \checkmark.$$

Feasible. Minimum integer cost $Z = 12$ at (3, 2).

Check. All candidates with $Z < 12$ are exhausted above; (2, 3) costs 13; so 12 is minimal. Note how far the winner sits from the continuous optimum $(\frac{9}{2}, 0)$ — two constraints deep. This is why Step 2 scans globally.

Practice — Addendum E

1. State, with a one-line proof, why the best integer objective value can never exceed the continuous optimum in a maximisation problem.
2. Maximise $Z = 3x + 2y$ over integer points of the region $2x + 3y \leq 13$, $2x + y \leq 9$, $x, y \geq 0$.
3. For the region of Example D.1 ($x + 2y \leq 8$, $3x + 2y \leq 12$, $x, y \geq 0$), find every lattice point at which $Z = 3x + 2y$ attains its maximum over lattice points.

4. Explain in one sentence, referring to Example E.1, why “solve continuously, then round” is not a valid method.

Answers: 1. Every lattice point is feasible for the continuous LPP, and no feasible point of a maximisation LPP can exceed its continuous maximum Z^* ; hence the integer best, being attained at a feasible point, is $\leq Z^*$. 2. Z^* : continuous optimum at $(\frac{7}{2}, 2)$ gives $10.5 + 4 = 14.5$. Scan downward: $14 - 3x + 2y = 14$ gives $(4, 1)$: $11 \leq 13$ ✓, $9 \leq 9$ ✓, feasible. Also $(2, 4)$: $14 \leq 13$? ✗. Maximum 14 at $(4, 1)$ (check no $Z = 15$ point exists — $15 > 14.5$ violates the ceiling principle). 3. Continuous max over the region: values at corners $(0, 0)$, $(4, 0)$, $(2, 3)$, $(0, 4)$ are 0, 12, 12, 8, so $Z^* \geq 12$ attained at two corners; the ceiling principle caps lattice points at 12. Lattice points with $3x + 2y = 12$ inside the region: $(4, 0)$ ✓ ($6 \leq 12$), $(2, 3)$ ✓ (on both slanted lines), $(0, 6)$ ✗ ($12 > 8$), $(\frac{2}{3}, \dots)$ non-integral. Maximum 12, attained at exactly $(4, 0)$ and $(2, 3)$ — a tie, so by the chapter’s own multiple-optima reasoning applied on the lattice, both are optimal lattice plans. 4. Because the optimal lattice point need not be any rounding of the optimal vertex: in Example E.1 every rounding of $(\frac{7}{2}, 2)$ is infeasible or worth ₹23, while the true optimum ₹24 lies at $(4, 1)$, reachable only by scanning candidate objective values downward from Z^* .

Closing triage. Faced with any LPP question, route it: *asked to justify the method* → Addendum A. *coefficients unknown, vertex given* → Addendum B. *“show that no feasible solution exists”/“which constraint can be dropped”* → Addendum C. *“area / how many integer points”* → Addendum D. *“whole-number plan”* → Addendum E. Everything else is the chapter as written.

Probability

Class 12 Mathematics — Chapter 13

Drona Ed Tech

Contents

Subtopic 01 — Conditional Probability & Independence	1
Subtopic 02 — Total Probability & Bayes' Theorem	9
Subtopic 03 — Random Variables & Distributions	16
Subtopic 04 — Bernoulli Trials & Binomial Distribution	23
Master Formula Sheet & Revision Capstone	30

Subtopic 01 — Conditional Probability & Independence

Exam Relevance: This is the foundation stone of the entire chapter. CBSE Boards almost always carry a 2–3 mark question on conditional probability or an independence check, and the multiplication theorem feeds directly into Total Probability and Bayes' Theorem later. JEE Main reliably places 1 question here every year — usually an independence trap or a “find $P(A | B)$ ” computation. JEE Advanced loves the deeper variants: independence of complements, “exactly one occurs,” and systems of equations built from $P(A \cap B)$ and $P(A' \cap B')$.

Section 1: Concept Intuition

Imagine you are waiting on a Mumbai local platform. Ask a stranger, “What’s the chance the next train is going to Churchgate?” — you’d shrug and give some rough number based on how the lines are shared. But now suppose the announcement board lights up: “*Slow line, Western Railway.*” Suddenly your estimate sharpens. You haven’t changed the trains running; you’ve changed **what you know**. The fresh information has shrunk the world of possibilities down to a smaller, more relevant set — and the probability is recomputed *inside that smaller world*.

That is the whole soul of **conditional probability**. When we write $P(E | F)$ — read as “probability of E **given** F ” — we are asking: *given that event F has definitely happened, how likely is E now?* The condition F is not a second event we are hoping for; it is a fact we have been handed. So we throw away every outcome where F did **not** occur, and we treat F itself as our brand-new sample space.

Think about it this way... Picture the full sample space as the entire crowd at a Diwali mela. Event F is the group standing inside the food court. The moment someone tells you “the person we’re discussing is in the food court,” the rest of the mela becomes irrelevant. To find the chance that this person is also eating jalebi (event E), you only count the jalebi-eaters *who are inside the food court*, divided by everyone in the food court — not by the whole mela. That is exactly

$$P(E | F) = \frac{P(E \cap F)}{P(F)}.$$

The numerator is “in both E and F ,” the denominator is “in F ” — the reduced universe.

Now flip to the opposite idea. Sometimes new information is **useless**. If you toss a coin and then roll a die, knowing the coin showed heads tells you nothing about the die. The

die “doesn’t care.” Two events are **independent** when the occurrence of one leaves the probability of the other completely unchanged: $P(E \mid F) = P(E)$. The condition F shrank the world, but E ’s share of that smaller world is identical to its share of the big world.

Here is where almost every student stumbles, so anchor it now: **independent is not the same as mutually exclusive**. Mutually exclusive events ($A \cap B = \emptyset$) are deadly rivals — if one happens, the other *cannot*. That is the strongest possible dependence, not independence! A batsman scoring a century and the same batsman getting out for a duck in the same innings are mutually exclusive; they are about as far from independent as two events can be.

A quick concrete feel for *how much* conditioning can move a number: suppose in a colony of 100 families, 30 own a car, 40 own a two-wheeler, and 12 own both. Pick a family at random — the bare chance it owns a car is $\frac{30}{100} = 0.30$. But now you learn the family owns a two-wheeler. Your world shrinks from 100 families to those 40, and among *them* exactly 12 own a car, so the updated chance jumps to $\frac{12}{40} = 0.30$ — unchanged! Here car-ownership and two-wheeler-ownership happen to be independent. Change “both” to 18 instead of 12, and the conditional becomes $\frac{18}{40} = 0.45$ — the information now *helps*, signalling dependence. Same machinery, two different stories, decided entirely by the size of the overlap.

One more intuition that pays off in exams: **drawing “with replacement” creates independence; “without replacement” creates dependence**. If you draw a ball, note it, and put it back, the second draw faces the identical box — the first draw left no trace, so the draws are independent. If you keep the ball out, the box has changed; the second draw’s probabilities now depend on what the first draw removed. Whenever a problem says “without replacement,” your antenna for the *multiplication theorem with conditionals* (not the simple product) should go up.

Limiting conditions & assumptions. The conditional formula $P(E \mid F) = P(E \cap F)/P(F)$ demands $P(F) > 0$ — you cannot condition on something impossible. The independence test $P(E \cap F) = P(E)P(F)$ is the safe, all-purpose definition because it never divides by anything and stays valid even when a probability is zero. Finally, beware: for *three* events, checking the pairs is **not** enough — true (mutual) independence needs the triple product condition too.

Section 2: Key Definitions & Properties

Conditional Probability

$$P(E \mid F) = \frac{P(E \cap F)}{P(F)}, \quad P(F) \neq 0.$$

Here E, F are events in the same sample space; $P(E \cap F)$ is the probability both occur. $P(E \mid F)$ is a genuine probability, so $0 \leq P(E \mid F) \leq 1$.

Properties of conditional probability (for $P(F) \neq 0$):

- $P(S \mid F) = P(F \mid F) = 1$
- $P[(A \cup B) \mid F] = P(A \mid F) + P(B \mid F) - P[(A \cap B) \mid F]$
- $P(E' \mid F) = 1 - P(E \mid F)$

Multiplication Theorem (rearranging the definition):

$$P(E \cap F) = P(F) P(E \mid F) = P(E) P(F \mid E).$$

Chain rule for three events:

$$P(E \cap F \cap G) = P(E) P(F \mid E) P(G \mid E \cap F).$$

Independent Events. E and F are independent if any one (hence all) of these hold:

$$P(E | F) = P(E), \quad P(F | E) = P(F), \quad \boxed{P(E \cap F) = P(E)P(F)}.$$

The boxed product form is the working definition.

Mutual independence of three events A, B, C requires **all four**:

$$P(A \cap B) = P(A)P(B), \quad P(B \cap C) = P(B)P(C), \quad P(C \cap A) = P(C)P(A), \\ \text{and} \quad P(A \cap B \cap C) = P(A)P(B)P(C).$$

Notation key: E' (or A^c) denotes the complement; $\cap$ = “and”; $\cup$ = “or”; $\emptyset$ = impossible event. Probabilities are pure numbers (dimensionless), each lying in $[0, 1]$.

“At least one” of n independent events. If $A_1, A_2, \dots, A_n$ are independent, the slick route to “at least one occurs” is through the complement “none occurs”:

$$P(\text{at least one}) = 1 - P(A'_1 \cap A'_2 \cap \dots \cap A'_n) = 1 - P(A'_1)P(A'_2) \dots P(A'_n).$$

This is one of the most-used results in the whole chapter — it converts an awkward union into a clean product.

Section 3: Theorems & Proofs

Theorem 1 — The conditional probability formula (sample-space reduction).

Claim: If F has occurred with $P(F) > 0$, then $P(E | F) = \frac{P(E \cap F)}{P(F)}$.

Reasoning. Consider equally likely outcomes for intuition. Knowing F occurred means the *effective* sample space is no longer S but F alone — every outcome outside F is now ruled out. Among the outcomes in F , the ones that also make E true are exactly those in $E \cap F$. Hence

$$P(E | F) = \frac{n(E \cap F)}{n(F)} = \frac{n(E \cap F)/n(S)}{n(F)/n(S)} = \frac{P(E \cap F)}{P(F)}.$$

Dividing numerator and denominator by $n(S)$ is the crucial step: it converts raw counts into probabilities, so the result holds even when outcomes are *not* equally likely (take it as the definition there). ■

Theorem 2 — Multiplication Theorem. Directly from Theorem 1, multiply both sides by $P(F)$:

$$P(E \cap F) = P(F)P(E | F).$$

By symmetry (conditioning on E instead), $P(E \cap F) = P(E)P(F | E)$. Both are the same quantity computed via two different “first events,” which is why either may be used depending on which conditional is easier to find. ■

Theorem 3 — Independence is inherited by complements (CBSE favourite). *Claim:* If A and B are independent, then A and B' are independent.

Proof. The event A splits cleanly into the part inside B and the part outside B :

$$P(A) = P(A \cap B) + P(A \cap B').$$

Therefore

$$P(A \cap B') = P(A) - P(A \cap B) = P(A) - P(A)P(B) \quad (\text{using independence of } A, B) \\ = P(A)[1 - P(B)] = P(A)P(B').$$

Since $P(A \cap B') = P(A)P(B')$, the events A and B' are independent. ■

Corollary. Applying the theorem twice shows A' and B are independent, and A' and B' are independent. **Moral:** independence survives complementation in every combination — a fact JEE problems exploit constantly.

Theorem 4 — Probability that at least one of n independent events occurs. *Claim:* If $A_1, \dots, A_n$ are independent, then $P(A_1 \cup A_2 \cup \dots \cup A_n) = 1 - \prod_{i=1}^n (1 - P(A_i))$.

Proof. The event “at least one occurs” is the complement of “none occurs”:

$$P(A_1 \cup \dots \cup A_n) = 1 - P((A_1 \cup \dots \cup A_n)') = 1 - P(A'_1 \cap A'_2 \cap \dots \cap A'_n).$$

The last step uses De Morgan’s law: the negation of “at least one happens” is “every one fails.” Now, since $A_1, \dots, A_n$ are independent, their complements $A'_1, \dots, A'_n$ are *also* mutually independent (Theorem 3 generalised), so the joint probability factorises:

$$P(A'_1 \cap \dots \cap A'_n) = P(A'_1) P(A'_2) \dots P(A'_n) = \prod_{i=1}^n (1 - P(A_i)).$$

Substituting back gives the claim. ■

This is the workhorse behind every “probability that the target is hit by at least one shooter” or “at least one machine fails” problem — never expand the union by inclusion-exclusion when the events are independent; complement and multiply instead.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]. A fair die is rolled twice. Let E be the event “the sum of the two rolls is 9” and F be the event “the first roll shows a number greater than 3.” Find $P(E | F)$.

Solution. Total outcomes: 36. Event F (first roll $\in \{4, 5, 6\}$): $3 \times 6 = 18$ outcomes, so $P(F) = \frac{18}{36}$. Event $E \cap F$ (sum = 9 **and** first roll > 3): the pairs are (4, 5), (5, 4), (6, 3). Note (3, 6) is rejected because the first roll 3 is not greater than 3. So 3 outcomes, $P(E \cap F) = \frac{3}{36}$.

$$P(E | F) = \frac{P(E \cap F)}{P(F)} = \frac{3/36}{18/36} = \frac{3}{18} = \frac{1}{6}.$$

Answer: $P(E | F) = \frac{1}{6}$.

Example 2 — [JEE Main Speed Trap]. For two events A and B , you are told $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cup B) = 0.7$. Are A and B independent, mutually exclusive, both, or neither?

Speed solution. Don’t panic-list cases — get $P(A \cap B)$ in one line via the addition theorem:

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.5 + 0.4 - 0.7 = 0.2.$$

Now test independence: $P(A)P(B) = 0.5 \times 0.4 = 0.2$. Since $P(A \cap B) = P(A)P(B)$, they are **independent**. Mutually exclusive would need $P(A \cap B) = 0$, but it’s $0.2 \neq 0$ — so **not** mutually exclusive. **Answer:** independent only. *The trap:* students see “0.7” and a tidy setup and reflexively assume mutual exclusivity (which would have given $P(A \cup B) = 0.9$). Always compute $P(A \cap B)$ first.

Example 3 — [JEE Main — Multi-concept: conditional probability + counting]. A selection panel has 7 boys and 5 girls. Three members are picked one after another,

without replacement. (i) Given that the first two picked are girls, find the probability the third pick is a boy. (ii) Find the probability that all three picks are girls.

Solution.

- (i) After two girls are removed, 10 people remain, of whom 7 are boys. By the reduced-sample-space idea,

$$P(\text{3rd is boy} \mid \text{first two are girls}) = \frac{7}{10}.$$

- (ii) Use the chain rule (multiplication theorem for three events):

$$P(G_1 \cap G_2 \cap G_3) = P(G_1) P(G_2 \mid G_1) P(G_3 \mid G_1 \cap G_2) = \frac{5}{12} \times \frac{4}{11} \times \frac{3}{10} = \frac{60}{1320} = \frac{1}{22}.$$

Answers: (i) $\frac{7}{10}$, (ii) $\frac{1}{22}$.

Example 4 — [JEE Advanced Level]. A and B are independent events. The probability that **both** occur is $\frac{1}{8}$, and the probability that **neither** occurs is $\frac{3}{8}$. Find $P(A)$ and $P(B)$, and also the probability that **exactly one** of them occurs.

Solution. Let $P(A) = a$ and $P(B) = b$. Independence gives $ab = \frac{1}{8}$. For “neither,” use that A', B' are also independent (Theorem 3 corollary):

$$P(A' \cap B') = (1 - a)(1 - b) = 1 - a - b + ab = \frac{3}{8}.$$

Substitute $ab = \frac{1}{8}$:

$$1 - (a + b) + \frac{1}{8} = \frac{3}{8} \Rightarrow a + b = 1 + \frac{1}{8} - \frac{3}{8} = \frac{3}{4}.$$

So a, b are roots of $t^2 - \frac{3}{4}t + \frac{1}{8} = 0$, i.e. $8t^2 - 6t + 1 = 0$:

$$t = \frac{6 \pm \sqrt{36 - 32}}{16} = \frac{6 \pm 2}{16} \Rightarrow t = \frac{1}{2} \text{ or } \frac{1}{4}.$$

Hence $\{P(A), P(B)\} = \left\{\frac{1}{2}, \frac{1}{4}\right\}$. “Exactly one occurs” $= P(A) + P(B) - 2P(A \cap B) = \frac{3}{4} - 2 \cdot \frac{1}{8} = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$. **Answers:** $\{P(A), P(B)\} = \left\{\frac{1}{2}, \frac{1}{4}\right\}$; $P(\text{exactly one}) = \frac{1}{2}$.

Sanity check (do this in the exam too). Both values lie in $[0, 1]$, as every probability must — so the answer is admissible. Notice also that the problem does **not** tell us which of A, B is the larger; the symmetric structure (ab and $a + b$ are both symmetric in a, b) is why we get an unordered pair rather than a unique assignment. Finally, a consistency test: $P(\text{both}) + P(\text{exactly one}) + P(\text{neither}) = \frac{1}{8} + \frac{1}{2} + \frac{3}{8} = 1$, confirming the three mutually exclusive cases (“both,” “exactly one,” “neither”) exhaust all possibilities. If that sum had not come to 1, an arithmetic slip would be hiding somewhere.

Section 5: Practice Exercises

1. **[CBSE]** Two fair dice are rolled. Let A = “the sum is 7” and B = “at least one die shows 2.” Find $P(A \mid B)$.

2. **[JEE Main]** Given $P(A) = 0.6$, $P(B) = 0.5$ and $P(A \cap B) = 0.3$, determine whether A and B are independent, and find $P(A | B)$ and $P(B | A)$.
3. **[JEE Main]** A box contains 6 green and 4 red marbles. Two are drawn one by one without replacement. Find the probability that both are red, and the probability that the second is green given the first was green.
4. **[JEE Advanced]** A and B are independent with $P(A \cup B) = 0.8$ and $P(A) = 0.3$. Find $P(B)$.
5. **[JEE Advanced]** E_1, E_2, E_3 are mutually independent, each with probability $\frac{1}{4}$. Find the probability that (i) all three occur, (ii) at least one occurs, (iii) exactly one occurs.

Answers: 1. $\frac{2}{11}$ 2. Independent, since $P(A \cap B) = 0.3 = 0.6 \times 0.5 = P(A)P(B)$; consequently $P(A | B) = 0.6 = P(A)$ and $P(B | A) = 0.5 = P(B)$ 3. both red = $\frac{2}{15}$, second green given first green = $\frac{5}{9}$ 4. $P(B) = \frac{5}{7}$ 5. (i) $\frac{1}{64}$, (ii) $1 - \left(\frac{3}{4}\right)^3 = \frac{37}{64}$, (iii) $3 \cdot \frac{1}{4} \cdot \left(\frac{3}{4}\right)^2 = \frac{27}{64}$.

Section 6: Crack the MCQ

Q1. If $P(A) > 0$ and $P(B) > 0$, then A and B can be **both** mutually exclusive and independent.

- (a) Always true (b) Always false (c) True only if $A = B$ (d) True only if $P(A) + P(B) = 1$

Answer: (b). Mutually exclusive means $P(A \cap B) = 0$; independent means $P(A \cap B) = P(A)P(B)$. Both at once forces $P(A)P(B) = 0$, impossible when both probabilities are positive. (a) fails for exactly this reason. (c) and (d) are invented conditions — independence/exclusivity don't hinge on equality of events or on the sum being 1; they are distractors that look plausible because they "add a condition."

Q2. A family has two children. Given that at least one is a girl, the probability that both are girls is:

- (a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{1}{4}$ (d) $\frac{2}{3}$

Answer: (b). Sample space $\{GG, GB, BG, BB\}$. Condition "at least one girl" removes BB , leaving $\{GG, GB, BG\}$ — three equally likely cases. Only GG has both girls, so $P = \frac{1}{3}$. (c) $\frac{1}{4}$ is the **unconditional** $P(GG)$ — the classic error of ignoring the given information. (a) $\frac{1}{2}$ comes from wrongly thinking "the other child is independently a girl with probability $\frac{1}{2}$," which double-counts. (d) is the answer to a different conditioning ("the elder is a girl").

Q3. If A and B are independent events, which of the following pairs is **not** guaranteed independent?

- (a) A' and B (b) A and B' (c) A' and B' (d) None — all three pairs are independent

Answer: (d). By Theorem 3 and its corollary, independence is inherited by every complement combination, so all three pairs are independent. Students who pick (a), (b), or (c) typically remember "independence is fragile" and wrongly assume complementing *breaks* it. The proof $P(A \cap B') = P(A) - P(A)P(B) = P(A)P(B')$ settles it cleanly.

Q4. For independent events A, B with $P(A) = p$, $P(B) = q$, the probability that exactly one of them occurs is:

- (a) pq (b) $p + q$ (c) $p + q - 2pq$ (d) $p + q - pq$

Answer: (c). “Exactly one” $= P(A \cap B') + P(A' \cap B) = p(1 - q) + (1 - p)q = p + q - 2pq$.
 (d) $p + q - pq$ is $P(A \cup B) -$ “at least one,” which wrongly *includes* the both-occur case.
 (b) $p + q$ forgets to subtract any overlap. (a) pq is “both occur.” The killer detail is the factor 2 on pq : you remove the intersection *twice* because “exactly one” excludes it from both directions.

Section 7: Common Pitfalls & Pro-Tips

Pitfall 1 — Confusing “mutually exclusive” with “independent.” They are nearly opposite ideas. Mutually exclusive (with positive probabilities) means *strongly dependent*: one happening forbids the other.

Pitfall 2 — Flipping the conditional. $P(A | B)$ and $P(B | A)$ are generally different; their denominators ($P(B)$ vs $P(A)$) differ. For a vivid feel: the probability that a person carries an umbrella *given* it is raining is high, but the probability that it is raining *given* a person carries an umbrella is much lower — same two events, two very different conditionals. Writing $P(E \cap F)/P(E)$ when you wanted $P(E | F)$ is the most common slip.

Pitfall 3 — Pairwise $\neq$ mutual independence. For three events, verifying the three pairs is not enough; the triple-product condition $P(A \cap B \cap C) = P(A)P(B)P(C)$ can still fail.

Pitfall 4 — Ignoring the given information. In “given at least one girl,” reverting to the unconditional probability throws away the very condition that defines the problem.

Pro-Tip: Before classifying any pair, compute $P(A \cap B)$ in one line via the addition theorem, then run *two* one-second checks — “ $= 0$?” (mutually exclusive) and “ $= P(A)P(B)$?” (independent). These two questions resolve almost every Board/JEE-Main classification problem instantly, and they can never both be “yes” unless a probability is zero.

errors that recur in Set theory and in Total Probability (Sub-02).

Phrase-to-symbol decoder. Half the battle in this topic is translating exam English into set notation *before* touching any formula. Train your eye on these:

- “given that F has occurred” / “if it is known that F ” / “when F happens” $\rightarrow$ a conditional $P(\cdot | F)$ — the denominator becomes $P(F)$.
- “both A and B ” $\rightarrow P(A \cap B)$.
- “ A or B ” / “at least one of A, B ” $\rightarrow P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- “neither A nor B ” $\rightarrow P(A' \cap B') = 1 - P(A \cup B)$.
- “exactly one of A, B ” $\rightarrow P(A) + P(B) - 2P(A \cap B)$.
- “at least one of n independent events” $\rightarrow 1 - \prod(1 - P(A_i))$ (complement-and-multiply).

Write the symbolic version in the margin first; nine times out of ten the formula to use then becomes obvious, and you avoid the silent misreadings that cost full marks in Boards and a clean shot in JEE.

Section 8: Cheat Sheet Box

CONDITIONAL PROBABILITY & INDEPENDENCE — RAPID RECALL

- **Conditional:** $P(E \mid F) = \frac{P(E \cap F)}{P(F)}$, need $P(F) > 0$. (*Shrink the world to F , then count.*)
- **Multiplication:** $P(E \cap F) = P(E)P(F \mid E) = P(F)P(E \mid F)$.
- **Chain rule:** $P(E \cap F \cap G) = P(E)P(F \mid E)P(G \mid E \cap F)$.
- **Independent** $\Leftrightarrow P(E \cap F) = P(E)P(F) \Leftrightarrow P(E \mid F) = P(E)$.
- **Mutually exclusive** $\Leftrightarrow P(E \cap F) = 0$. (*Opposite of independent for positive probabilities.*)
- **Complement rule:** $A \perp B \Rightarrow A \perp B', A' \perp B, A' \perp B'$.
- **Exactly one of A, B :** $P(A) + P(B) - 2P(A \cap B)$. **At least one:** $P(A) + P(B) - P(A \cap B)$.

Mnemonic: “Given shrinks, Independent ignores, Exclusive forbids.” **The $2 \times pq$ trap:** “exactly one” subtracts the overlap **twice**; “at least one” subtracts it once.

Subtopic 02 — Total Probability & Bayes' Theorem

Exam Relevance: This is the single highest-yield idea in the chapter. CBSE Boards almost always carry a **dedicated 5-6 mark question** on Bayes' Theorem (the “two-bags,” “two-factories,” or “disease-test” template). JEE Main places one question here most years, often combined with counting. JEE Advanced raises the rigour — partition into three-plus cases, base-rate surprises, and Bayes layered on top of a transfer experiment. Total Probability is rarely asked alone but is the engine that makes Bayes work.

Section 1: Concept Intuition

Picture a famous mithai shop during Diwali that sources its kaju katli from three different suppliers. Supplier A sends 50% of the stock, B sends 30%, and C sends 20%. Each supplier occasionally sends a stale piece. If you walk in and pick a random box, what's the chance you get a stale one? You can't answer with any single supplier's spoilage rate — the box could have come from *any* of them. You have to **weight each supplier's spoilage rate by how much stock they contribute** and add it all up. That weighted sum is the **Theorem of Total Probability**: it stitches together several mutually exclusive “origin stories” into one overall probability.

For this to work, the origin stories must do two things: never overlap, and together cover every possibility. In maths we call such a family of events a **partition** of the sample space — the supplier set $\{A, B, C\}$ is a partition because every box comes from *exactly one* supplier (disjoint) and *some* supplier (exhaustive). Total Probability then says: the chance of an effect (a stale piece) is the sum, over every cause, of (chance of that cause) $\times$ (chance of the effect *given* that cause).

Think about it this way... Total Probability reasons **forwards**: cause $\rightarrow$ effect. You know the suppliers and their spoilage rates, and you predict the chance of a stale piece. But the really interesting exam questions run the film **backwards**. You've *already bitten into a stale piece* — now, which supplier most likely sent it? That backward question, from observed effect to hidden cause, is exactly what **Bayes' Theorem** answers. This is why Bayes is also called **inverse probability**: it inverts the direction of conditioning, turning “probability of effect given cause” into “probability of cause given effect.”

The trap your intuition sets here is **base-rate neglect**. Suppose supplier C has the *worst* spoilage rate but ships the *least* stock. When you bite a stale piece, your gut screams “C!” — but if C barely ships anything, the sheer volume from A can make A the more likely culprit even with a lower spoilage rate. Bayes forces you to respect both numbers: the **prior** (how much each supplier ships — the base rate) and the **likelihood** (each supplier's spoilage rate). The output is the **posterior** (the updated chance for each supplier *after* seeing the stale piece). Forgetting the prior is the classic medical-test blunder: a test that is “95% accurate” for a rare disease can still leave a positive patient far more likely to be healthy than sick, simply because healthy people vastly outnumber sick ones.

A courtroom captures the spirit perfectly. Before any evidence, the judge holds a **prior** belief about each suspect (who had motive, who was nearby). A piece of evidence then arrives — a fingerprint, say — and each suspect has a **likelihood** of having left it. Bayes is the formal rule for updating belief: the suspect who best combines “was plausible to begin with” *and* “fits the evidence” becomes the most probable culprit. Crucially, a suspect with an airtight alibi (prior near zero) stays innocent no matter how well the evidence fits — exactly why Box B drops out in some problems. Email spam filters run this same loop millions of times a day: a prior chance a message is spam, updated by the likelihood of each word it contains.

Limiting conditions & assumptions. Both theorems require a genuine partition: the cases must be pairwise disjoint, exhaustive, and each must have **positive** probability ($P(E_i) > 0$) — you cannot condition on an impossible cause. If your “causes” overlap

or leave gaps, the formulas silently give nonsense. And Bayes' denominator is *always* the Total Probability of the observed effect — never a single likelihood. A practical consequence worth internalising: if you ever compute posteriors that fail to sum to 1, you have either used a non-partition or divided by the wrong denominator — the error is structural, not arithmetic, so re-examine your cases before re-checking your sums.

Section 2: Key Definitions & Properties

Partition of a sample space. Events $E_1, E_2, \dots, E_n$ partition the sample space S if:

- $E_i \cap E_j = \emptyset$ for $i \neq j$ (pairwise disjoint),
- $E_1 \cup E_2 \cup \dots \cup E_n = S$ (exhaustive),
- $P(E_i) > 0$ for every i .

Theorem of Total Probability. If $\{E_1, \dots, E_n\}$ is a partition of S and A is any event, then

$$P(A) = \sum_{j=1}^n P(E_j) P(A | E_j).$$

Here $P(E_j)$ is the **prior** of cause E_j , and $P(A | E_j)$ is the **likelihood** of the effect A under that cause.

Bayes' Theorem. With the same partition, for the observed event A (with $P(A) > 0$), the posterior probability of cause E_i is

$$P(E_i | A) = \frac{P(E_i) P(A | E_i)}{\sum_{j=1}^n P(E_j) P(A | E_j)}$$

The denominator is precisely $P(A)$ from Total Probability. In words:

$$\text{posterior} = \frac{\text{prior} \times \text{likelihood}}{\text{total probability of the evidence}}.$$

Terminology key. E_i = hypotheses / causes (the partition); A = evidence / observed effect; $P(E_i)$ = prior; $P(A | E_i)$ = likelihood; $P(E_i | A)$ = posterior. All are dimensionless numbers in $[0, 1]$, and the posteriors satisfy $\sum_i P(E_i | A) = 1$ — a free consistency check.

Two-hypothesis special case. When there are just two causes — an event E and its complement E' — Bayes collapses to the form most common in Boards:

$$P(E | A) = \frac{P(E) P(A | E)}{P(E) P(A | E) + P(E') P(A | E')}.$$

This is the disease-test and single-bag template; recognising it lets you write the answer in one line.

Section 3: Theorems & Proofs

Theorem 1 — Theorem of Total Probability. *Claim:* For a partition $\{E_1, \dots, E_n\}$ of S and any event A , $P(A) = \sum_{j=1}^n P(E_j) P(A | E_j)$.

Proof. Because the E_j are exhaustive ($\bigcup_j E_j = S$), every outcome of A falls into one of them. So we may slice A along the partition:

$$A = A \cap S = A \cap \left(\bigcup_{j=1}^n E_j \right) = \bigcup_{j=1}^n (A \cap E_j).$$

The pieces $A \cap E_j$ are pairwise disjoint, because the E_j themselves are disjoint: if $E_i \cap E_j = \emptyset$, then certainly $(A \cap E_i) \cap (A \cap E_j) = \emptyset$. For a disjoint union, probabilities simply add:

$$P(A) = \sum_{j=1}^n P(A \cap E_j).$$

Finally, rewrite each piece using the multiplication theorem (from Sub-01), $P(A \cap E_j) = P(E_j)P(A | E_j)$:

$$P(A) = \sum_{j=1}^n P(E_j) P(A | E_j). \quad \blacksquare$$

The two partition properties pull their weight in exactly two places: *exhaustive* lets us write A as the union over all E_j , and *disjoint* lets us add the probabilities without overlap corrections.

Theorem 2 — Bayes' Theorem. *Claim:* $P(E_i | A) = \frac{P(E_i)P(A | E_i)}{\sum_j P(E_j)P(A | E_j)}$.

Proof. Start from the definition of conditional probability (Sub-01) applied to the cause we want:

$$P(E_i | A) = \frac{P(E_i \cap A)}{P(A)}.$$

For the **numerator**, use the multiplication theorem *the convenient way round* — conditioning on the cause, whose likelihood we actually know:

$$P(E_i \cap A) = P(E_i) P(A | E_i).$$

For the **denominator**, substitute Total Probability (Theorem 1):

$$P(A) = \sum_{j=1}^n P(E_j) P(A | E_j).$$

Putting the two together gives the result. $\blacksquare$

The single most important *physical* idea in this proof: we are handed the likelihoods $P(A | E_i)$ (effect-given-cause) but we *want* the posteriors $P(E_i | A)$ (cause-given-effect). Bayes is the bridge that flips the direction, and Total Probability is what builds its denominator.

The probability-tree device (how to organise any Bayes problem). Every Total-Probability/Bayes question can be drawn as a two-level tree, and the tree is the formula made visible.

Figure 13.1 A horizontal probability tree. From a single root node on the left, three branches fan out to the cause nodes E_1, E_2, E_3 , each branch labelled with its prior $P(E_1), P(E_2), P(E_3)$. From every cause node, two further branches lead to "A occurs" and "A does not occur", labelled with the likelihoods $P(A|E_1), P(A|E_2), P(A|E_3)$ and their complements. Beside the three terminal "A occurs" leaves, annotate each path-product $P(E_i) \cdot P(A|E_i)$. At the bottom write: " $P(A) = \text{sum of the three A-path products}$ ", and " $\text{Posterior } P(E_i|A)$ (see next page)"

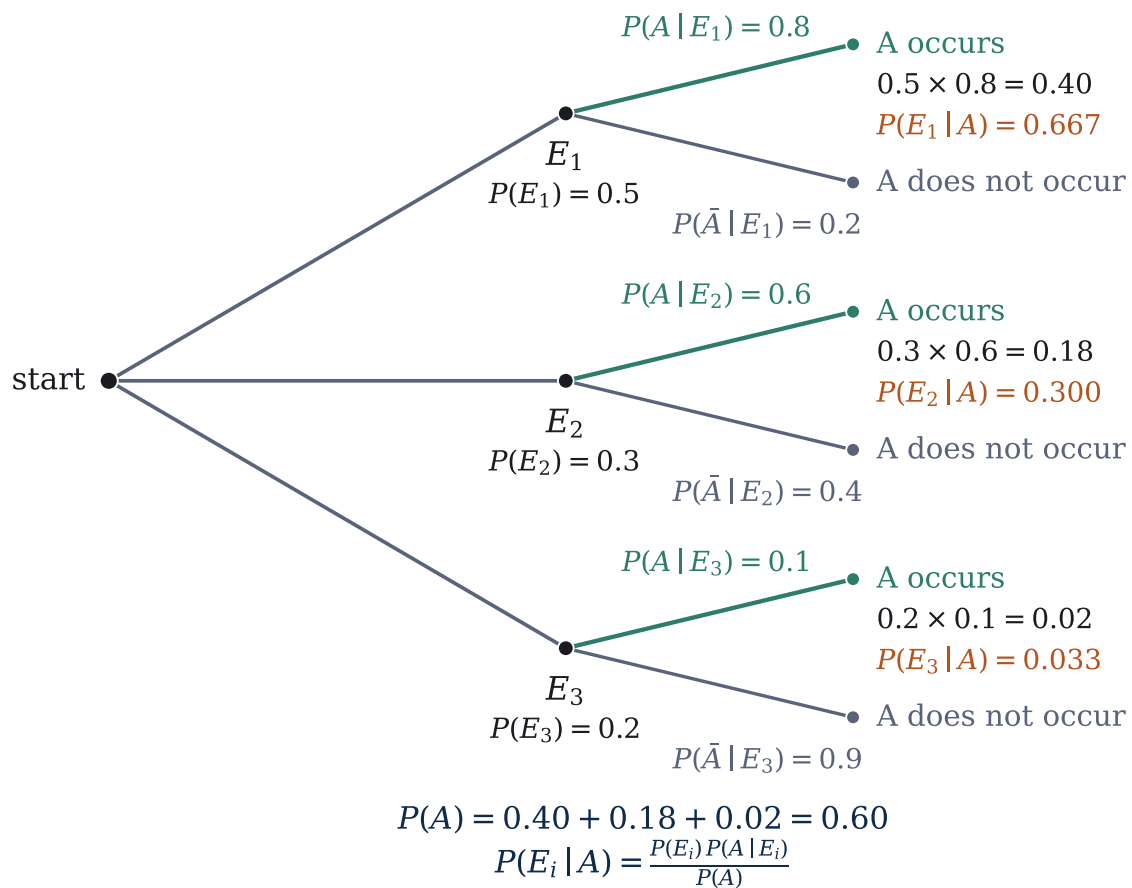


Figure 13.1 A horizontal probability tree. From a single root node on the left, three branches fan out to the cause nodes E_1 , E_2 , E_3 , each branch labelled with its prior $P(E_1)$, $P(E_2)$, $P(E_3)$. From every cause node, two further branches lead to "A occurs" and "A does not occur", labelled with the likelihoods $P(A|E_1)$, $P(A|E_2)$, $P(A|E_3)$ and their complements. Beside the three terminal "A occurs" leaves, annotate each path-product $P(E_i)P(A|E_i)$. At the bottom write: " $P(A)$ = sum of the three A-path products", and "Posterior $P(E_i|A)$ "

= (that path's product) / P(A)".

To read the tree: each complete path from root to an “ A occurs” leaf has probability equal to the **product of the labels along it** — that is $P(E_i)P(A | E_i)$, the multiplication theorem in action. **Total Probability** is the sum of all such A -paths. **Bayes** then asks: out of that total, what fraction travelled through cause E_i ? — which is that single path-product divided by the sum. Drawing the tree first turns even a messy three-cause Advanced problem into bookkeeping, and it makes the denominator impossible to get wrong because you literally add up the paths you drew.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]. Bag I contains 4 red and 6 blue balls; Bag II contains 7 red and 3 blue balls. A bag is chosen at random and one ball is drawn from it. The ball turns out to be **red**. Find the probability that it was drawn from Bag I.

Solution. Let E_1, E_2 be “Bag I chosen,” “Bag II chosen” (a partition), and A = “red ball drawn.” Priors: $P(E_1) = P(E_2) = \frac{1}{2}$. Likelihoods: $P(A | E_1) = \frac{4}{10}$, $P(A | E_2) = \frac{7}{10}$. Total probability of red:

$$P(A) = \frac{1}{2} \cdot \frac{4}{10} + \frac{1}{2} \cdot \frac{7}{10} = \frac{4}{20} + \frac{7}{20} = \frac{11}{20}.$$

By Bayes’ Theorem:

$$P(E_1 | A) = \frac{\frac{1}{2} \cdot \frac{4}{10}}{\frac{11}{20}} = \frac{4/20}{11/20} = \frac{4}{11}.$$

Answer: $P(\text{Bag I} | \text{red}) = \frac{4}{11}$. (Quick check: $P(\text{Bag II} | \text{red}) = \frac{7}{11}$, and $\frac{4}{11} + \frac{7}{11} = 1$ — the posteriors over the two-bag partition sum to one, as they must.)

Example 2 — [JEE Main Speed Trap]. In a large town, 2% of people carry a certain condition. A screening test correctly flags a carrier 95% of the time, but also wrongly flags a non-carrier 10% of the time. A randomly chosen person tests **positive**. Roughly what is the probability they actually carry the condition?

(a) 95% (b) $\approx 16\%$ (c) $\approx 50\%$ (d) 85%

Speed solution. Don’t reach for 95% — that’s the trap (it’s a likelihood, not a posterior). Set D = carrier. Priors $P(D) = 0.02$, $P(D') = 0.98$. Likelihoods $P(+ | D) = 0.95$, $P(+ | D') = 0.10$.

$$P(+) = 0.02(0.95) + 0.98(0.10) = 0.019 + 0.098 = 0.117.$$

$$P(D | +) = \frac{0.019}{0.117} \approx 0.162.$$

Answer: (b) $\approx 16\%$. *Why the trap works:* the test is “95% accurate,” so students pick (a). But carriers are rare, so the 0.098 flood of false positives from the huge non-carrier pool swamps the 0.019 of true positives. Base rate wins — the hallmark Bayes surprise.

Example 3 — [JEE Main — Multi-concept: Bayes + combinatorics]. Three identical boxes are on a table. Box A holds 3 white and 2 black balls, Box B holds 1 white and 4 black, Box C holds 4 white and 1 black. A box is chosen at random and **two balls are drawn together** (without replacement) — both turn out white. Find the probability the box chosen was C.

Solution. Priors $P(A) = P(B) = P(C) = \frac{1}{3}$. The effect $W =$ “both drawn balls white.” Compute each likelihood with combinations (link to P&C):

$$P(W | A) = \frac{\binom{3}{2}}{\binom{5}{2}} = \frac{3}{10}, \quad P(W | B) = \frac{\binom{1}{2}}{\binom{5}{2}} = 0, \quad P(W | C) = \frac{\binom{4}{2}}{\binom{5}{2}} = \frac{6}{10}.$$

Total probability:

$$P(W) = \frac{1}{3} \left(\frac{3}{10} + 0 + \frac{6}{10} \right) = \frac{1}{3} \cdot \frac{9}{10} = \frac{3}{10}.$$

Bayes for box C:

$$P(C | W) = \frac{\frac{1}{3} \cdot \frac{6}{10}}{\frac{3}{10}} = \frac{6/30}{9/30} = \frac{2}{3}.$$

Answer: $P(C | \text{both white}) = \frac{2}{3}$ (and note $P(B | W) = 0$ — Box B can’t yield two whites at all).

Example 4 — [JEE Advanced Level]. A workshop runs three production lines. Line L_1 makes 60% of output with a 2% defect rate; L_2 makes 25% with a 4% defect rate; L_3 makes 15% with an 8% defect rate. A randomly selected item is found **defective**. (i) Find the posterior probability it came from each line. (ii) If quality control can re-inspect only one line’s process first, which line should it pick, and what does the answer reveal?

Solution. Causes L_1, L_2, L_3 form a partition; effect $D =$ defective. Joint probabilities (prior $\times$ likelihood):

$$P(L_1)P(D | L_1) = 0.60(0.02) = 0.012,$$

$$P(L_2)P(D | L_2) = 0.25(0.04) = 0.010,$$

$$P(L_3)P(D | L_3) = 0.15(0.08) = 0.012.$$

Total probability of a defect:

$$P(D) = 0.012 + 0.010 + 0.012 = 0.034.$$

Posteriors:

$$P(L_1 | D) = \frac{0.012}{0.034} = \frac{6}{17} \approx 0.353, \quad P(L_2 | D) = \frac{0.010}{0.034} = \frac{5}{17} \approx 0.294,$$

$$P(L_3 | D) = \frac{0.012}{0.034} = \frac{6}{17} \approx 0.353.$$

(Check: $\frac{6}{17} + \frac{5}{17} + \frac{6}{17} = \frac{17}{17} = 1$. ✓)

- (ii) L_1 and L_3 are **tied** as the most likely source. The insight: L_3 is *four times* as defect-prone as L_1 , yet contributes equal blame — because L_1 produces *four times* the volume. High volume at low defect rate is just as guilty as low volume at high defect rate. QA should weigh both, not chase the scariest defect rate alone.

Section 5: Practice Exercises

- [CBSE]** Urn X has 5 red and 5 green balls; Urn Y has 3 red and 7 green. An urn is picked at random and a ball drawn is found **green**. Find the probability it came from Urn X.
- [JEE Main]** A factory’s machines M_1, M_2 produce 70% and 30% of bolts, with defect rates 3% and 5%. A bolt is defective. Find the probability it was made by M_2 .

3. **[JEE Main]** Three coins sit in a drawer: one fair, one with two heads, one biased with $P(\text{head}) = \frac{3}{4}$. A coin is chosen at random and tossed once, showing a head. Find the probability it was the two-headed coin.
4. **[JEE Advanced]** Box P has 2 white, 3 black; Box Q has 4 white, 1 black. A box is chosen at random and **two balls are drawn together**; both are white. Find the probability the box was Q.
5. **[JEE Advanced]** A student knows the answer to a multiple-choice question with probability $\frac{2}{3}$; otherwise they guess among 4 options. Given the answer was correct, find the probability the student actually knew it.

Answers: 1. $\frac{5}{12}$ 2. $\frac{15}{36} = \frac{5}{12}$ 3. $\frac{4}{9}$ 4. $\frac{6}{7}$ 5. $\frac{8}{9}$.

Section 6: Crack the MCQ

Q1. In Bayes' Theorem, the denominator $\sum_j P(E_j)P(A | E_j)$ equals:

- (a) $P(A | E_i)$ (b) $P(E_i \cap A)$ (c) $P(A)$ (d) 1

Answer: (c). The denominator is the Total Probability of the evidence A , which by Theorem 1 is exactly $P(A)$. (a) is a single likelihood — picking it means forgetting to sum over all causes. (b) is just the numerator term for one cause. (d) tempts students who confuse “the posteriors sum to 1” with “the denominator is 1”; the *posteriors* sum to 1, the *denominator* is $P(A)$, generally less than 1.

Q2. A box has 4 red and 6 white balls. Two balls are drawn one by one **without replacement**. Given the second ball is red, which quantity gives $P(\text{first red} | \text{second red})$ most directly?

- (a) Total Probability only (b) Bayes' Theorem (c) Independence (d) Addition theorem

Answer: (b). We observe an effect (second red) and ask about an earlier cause (first red) — a backward, inverse-probability question, so Bayes. (a) Total Probability alone gives $P(\text{second red})$ but not the conditional on the first. (c) fails because without replacement the draws are *dependent*. (d) the addition theorem is for unions, irrelevant here.

Q3. For a partition $\{E_1, E_2, E_3\}$ and event A , suppose $P(E_1 | A) = 0.5$ and $P(E_2 | A) = 0.3$. Then $P(E_3 | A)$ equals:

- (a) 0.2 (b) 0.8 (c) cannot be determined (d) 0

Answer: (a). Posteriors over a partition sum to 1, so $P(E_3 | A) = 1 - 0.5 - 0.3 = 0.2$. (b) adds instead of subtracting. (c) misses that the partition guarantees the sum-to-one property — no extra data needed. (d) assumes E_3 is impossible, unjustified.

Q4. A test for a rare disease (1% prevalence) is 99% sensitive and 99% specific. A patient tests positive. The probability they have the disease is closest to:

- (a) 99% (b) 50% (c) 1% (d) 9%

Answer: (b). $P(+) = 0.01(0.99) + 0.99(0.01) = 0.0099 + 0.0099 = 0.0198$; posterior $= 0.0099/0.0198 = 0.5$. (a) is base-rate neglect — quoting sensitivity as the answer. (c) quotes the prior, ignoring the test result entirely. (d) is a miscalculation from mixing up sensitivity and specificity. The lesson: with a rare disease, even a very good test leaves a coin-flip after one positive.

Section 7: Common Pitfalls & Pro-Tips

Pitfall 1 — Base-rate neglect. Quoting the likelihood $P(A | E)$ as if it were the posterior $P(E_i | A)$. The test's accuracy is *not* the probability you have the disease.

Pitfall 2 — Inverting the conditional carelessly. $P(A | E_i)$ (given) and $P(E_i | A)$ (wanted) point in opposite directions; swapping them skips Bayes entirely.

Pitfall 3 — A non-partition. Using “causes” that overlap or don't cover all cases. If $\sum_i P(E_i) \neq 1$, you do **not** have a partition and Total Probability fails.

Pitfall 4 — Wrong denominator. Dividing by a single $P(E_j)P(A | E_j)$ instead of the full sum $P(A)$.

Pro-Tip: Build a tiny table — one row per cause, columns: *prior*, *likelihood*, *product* (*prior*×*likelihood*). The column of products, summed, is your denominator $P(A)$; each posterior is *that row's product divided by the column total*. This table format makes the full 5-6 mark Board solution almost mechanical and kills sign/placement errors. Final sanity check: the posteriors must add to 1.

Pro-Tip (Advanced): When a problem gives evidence *in stages* — a second observation after the first — **today's posterior becomes tomorrow's prior**. Don't restart from scratch; feed the updated probabilities back into Bayes for the next piece of evidence. This “chaining” is exactly how repeated draws or repeated test results are handled, and recognising it saves a full recomputation under exam time pressure.

Section 8: Cheat Sheet Box

TOTAL PROBABILITY & BAYES — RAPID RECALL

- **Partition:** disjoint + exhaustive + each $P(E_i) > 0$.
- **Total Probability (forward, cause→effect):** $P(A) = \sum_j P(E_j)P(A | E_j)$.
- **Bayes (backward, effect→cause):** $P(E_i | A) = \frac{P(E_i)P(A | E_i)}{\sum_j P(E_j)P(A | E_j)}$.
- **In words:** posterior = $\frac{\text{prior} \times \text{likelihood}}{P(\text{evidence})}$.
- **Denominator is always** $P(A)$ — the full sum, never one term.
- **Posteriors sum to 1:** $\sum_i P(E_i | A) = 1$ — use it to self-check.

Mnemonic: “Total adds the causes; Bayes blames one cause.” Inverse probability = reasoning from effect back to cause. **Table trick:** prior × likelihood per row ⇒ column sum = $P(A)$ ⇒ each posterior = row ÷ sum.

Subtopic 03 — Random Variables & Distributions

Exam Relevance: A guaranteed scorer. CBSE Boards almost always carry a 3–5 mark question asking you to build a probability distribution and compute its mean and variance. JEE Main reliably places one direct question on expectation or variance. JEE Advanced raises the bar with formula-defined distributions, linear transformations of a random variable, and expectation arguments that need genuine reasoning. Master this and the Binomial distribution (Sub-04) becomes almost free.

Section 1: Concept Intuition

Watch one over of a T20 match and ask a simple question: *how many runs came off it?* The answer is a single number — 0, maybe 6, maybe an unlikely 36 — and which number you get depends on how the six balls played out. You have quietly done something profound: you took a messy, unpredictable real-world experiment (an over of cricket) and boiled each possible outcome down to **one real number**. That mapping — “outcome $\rightarrow$ number” — is exactly what a **random variable** is. Formally it is a function whose input is an outcome of the experiment and whose output is a real number, written X . The runs-per-over is a random variable; so is “number of heads in three coin tosses,” “number of customers at a chai stall in an hour,” or “number of defective bulbs in a packet of ten.”

Once we have a random variable, the natural next question is: *how are its values distributed?* List every value X can take alongside the probability of each, and you get the **probability distribution** — the complete fingerprint of the random variable. It is just a two-row table: the values on top, their probabilities below. Two iron rules govern it: every probability sits between 0 and 1, and they must **add up to exactly 1**, because on any single run of the experiment *something* must happen.

It helps to notice how many everyday quantities are random variables in disguise. The number of sixes when you roll two dice (0, 1, or 2); the number of defective phones in a carton of twenty; the number of auto-rickshaws that pass a junction in a minute; the number of correct answers a student gets by pure guessing on a test — each assigns a clean integer to an unpredictable situation. Because the possible values can be *listed* one by one, these are **discrete** random variables, and listing them is always the first move: you cannot build a distribution until you know exactly which values are on the table.

Think about it this way... A probability distribution is like the published scorecard of all possible results before the match: it tells you not just what *can* happen but how *likely* each result is. But a full table can be bulky, so we summarise it with two single numbers. The first is the **mean** (also called **expectation**, $E(X)$): the long-run average value if you repeated the experiment thousands of times. It is *not* a plain average of the listed values — it is a **probability-weighted** average, so values that occur more often pull the mean toward themselves. Picture the distribution as weights placed on a ruler at each value; the mean is the exact point where the ruler balances.

The second summary number measures **consistency**: the **variance** σ^2 and its square root the **standard deviation** σ . Here cricket gives the perfect analogy. Two batsmen can have the *same* batting average (same mean) yet feel completely different — one grinds out 40 every innings, the other alternates between a golden duck and a blazing century. Same mean, wildly different *spread*. Variance captures that spread: it is the average squared distance of the values from the mean. A small variance means a dependable, predictable random variable; a large variance means a volatile one.

A quick feel for *why the mean must be weighted*: imagine a lucky-dip at a school fair where 90% of tickets win ₹10 and only 10% win ₹100. A naive “average of the prizes” would say $\frac{10+100}{2} = ₹55$ — but almost nobody wins ₹100, so that figure is wildly optimistic. The honest expected prize weights each value by how often it occurs: $0.9(10) + 0.1(100) =$

$9 + 10 = ₹19$. That ₹19 is the mean, and it is exactly what a fair ticket price would be in the long run. Probabilities are the weights; ignore them and the mean is meaningless.

Figure 13.2 *A bar chart of a probability distribution — horizontal axis showing values $x = 0, 1, 2, 3$ and vertical bars of heights $p = 1/8, 3/8, 3/8, 1/8$. Below the bars, draw a horizontal beam resting on a single triangular fulcrum placed at $x = 1.5$, with a label “mean = balance point”. Show small weights hanging under each value proportional to its probability, illustrating that the beam balances exactly at the probability-weighted centre, not at the midpoint of the value range. The picture above captures the mean’s true meaning: place a weight equal to each probability at each value along a beam, and the mean is the one point where the beam balances. It need not sit at the middle of the values — heavier probabilities tug the balance point toward themselves. (see next page)*

Limiting conditions & assumptions. Everything here is for a **discrete** random variable — one taking a finite (or listable) set of values, which is all that the syllabus requires. The distribution is only valid if the probabilities are non-negative and sum to 1; if your table breaks either rule, you have made an error upstream. The mean can be a value the variable never actually takes (1.5 heads is impossible, yet it is the mean of three tosses) — that is fine, the mean is a balance point, not an outcome. Finally, variance is always ≥ 0 ; a negative variance is an instant signal of an arithmetic slip.

Section 2: Key Definitions & Properties

Random variable. A real-valued function $X : S \rightarrow \mathbb{R}$ defined on the sample space S of a random experiment. We deal only with **discrete** random variables here.

Probability distribution. The system of values and probabilities

$$\begin{array}{c|cccc} X & x_1 & x_2 & \cdots & x_n \\ \hline P(X) & p_1 & p_2 & \cdots & p_n \end{array}$$

where $p_i \geq 0$ for all i and $\sum_{i=1}^n p_i = 1$.

Mean / Expectation.

$$\mu = E(X) = \sum_{i=1}^n x_i p_i.$$

The expectation is the value you would converge to if you averaged the outcomes over a very large number of repetitions of the experiment; it is a single number summarising the “centre” of the whole distribution, and it need not equal any value the variable can actually take.

Expectation of X^2 (needed for variance):

$$E(X^2) = \sum_{i=1}^n x_i^2 p_i.$$

Variance (two equivalent forms):

$$\sigma^2 = \text{Var}(X) = E[(X - \mu)^2] = \sum_{i=1}^n (x_i - \mu)^2 p_i = \underbrace{E(X^2) - [E(X)]^2}_{\text{computational form}}.$$

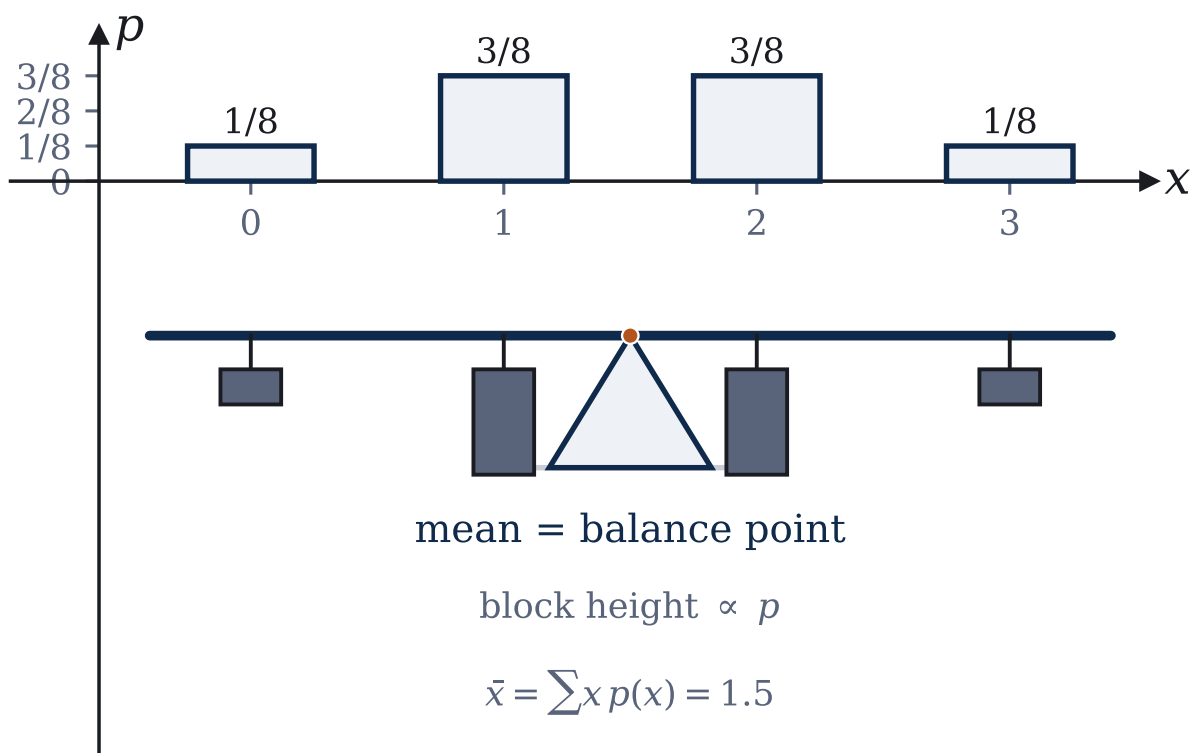


Figure 13.2 A bar chart of a probability distribution — horizontal axis showing values $x = 0, 1, 2, 3$ and vertical bars of heights $p = 1/8, 3/8, 3/8, 1/8$. Below the bars, draw a horizontal beam resting on a single triangular fulcrum placed at $x = 1.5$, with a label “mean = balance point”. Show small weights hanging under each value proportional to its probability, illustrating that the beam balances exactly at the probability-weighted centre, not at the midpoint of the value range. The picture above captures the mean’s true meaning: place a weight equal to each probability at each value along a beam, and the mean is the one point where the beam balances. It need not sit at the middle of the values — heavier probabilities tug the balance point toward themselves.

Standard deviation.

$$\sigma = \sqrt{\text{Var}(X)} = \sqrt{E(X^2) - \mu^2}.$$

Linear transformation properties (heavily used in JEE):

$$E(aX + b) = a E(X) + b, \quad \text{Var}(aX + b) = a^2 \text{Var}(X).$$

Note the constant b shifts the mean but **never** affects the variance (sliding all data sideways changes the average, not the spread), and the scale factor a enters variance **squared**.

Notation & nature. x_i are the values, $p_i = P(X = x_i)$; μ, σ carry the same units as X while σ^2 carries squared units; all probabilities are dimensionless numbers in $[0, 1]$.

Section 3: Key Procedures & Derivations

Procedure A — Building a probability distribution.

1. Identify the random variable and list **every** value it can take.
2. For each value, compute its probability using counting, conditional probability, or the multiplication theorem (Sub-01).
3. **Validate:** confirm $\sum p_i = 1$. If it doesn't, a value is missing or a probability is wrong — fix before proceeding.

Procedure B — Finding an unknown constant k . When a distribution is given in terms of k , impose $\sum p_i = 1$ and solve the resulting equation for k . Then *check* that every resulting probability lies in $[0, 1]$ — a value of k that produces a negative probability must be rejected.

Derivation — the computational form of variance. *Claim:* $\text{Var}(X) = E(X^2) - [E(X)]^2$.

Reasoning. Start from the definition and expand the square inside the expectation:

$$\text{Var}(X) = E[(X - \mu)^2] = E[X^2 - 2\mu X + \mu^2].$$

Expectation distributes across a sum (it is just a weighted sum), and constants pull out:

$$= E(X^2) - 2\mu E(X) + \mu^2.$$

Now use $E(X) = \mu$:

$$= E(X^2) - 2\mu \cdot \mu + \mu^2 = E(X^2) - 2\mu^2 + \mu^2 = E(X^2) - \mu^2. \quad \blacksquare$$

Why this matters: the definition $\sum (x_i - \mu)^2 p_i$ forces you to subtract μ from every value first (tedious, error-prone with decimals). The computational form lets you accumulate $\sum x_i p_i$ and $\sum x_i^2 p_i$ in a single table pass, then combine at the end — far faster and far less prone to slips.

Procedure C — the four-column table (the method to use every time). Set up four columns: x_i , p_i , $x_i p_i$, and $x_i^2 p_i$. Fill one row per value, then sum the last two columns. As a worked illustration, take $X : \{2, 4, 6\}$ with $P : \{0.5, 0.3, 0.2\}$:

x_i	p_i	$x_i p_i$	$x_i^2 p_i$
2	0.5	1.0	2.0
4	0.3	1.2	4.8
6	0.2	1.2	7.2
sum	1.0	3.4	14.0

The third-column total is $E(X) = 3.4$; the fourth-column total is $E(X^2) = 14.0$; hence $\text{Var}(X) = 14.0 - (3.4)^2 = 14.0 - 11.56 = 2.44$. Notice the p_i column summing to 1 doubles as your validity check before you even reach the mean. This layout converts any mean/variance question into mechanical arithmetic and is exactly what earns full method marks in Boards.

Derivation — variance under a linear shift. *Claim:* $\text{Var}(aX + b) = a^2 \text{Var}(X)$. Let $Y = aX + b$. Then $E(Y) = aE(X) + b = a\mu + b$, so

$$\text{Var}(Y) = E[(Y - E(Y))^2] = E[(aX + b - a\mu - b)^2] = E[a^2(X - \mu)^2] = a^2 \text{Var}(X).$$

The $+b$ cancels itself out — confirming that a pure shift leaves spread untouched. ■

Why squaring? A natural first attempt at “spread” would be the average *signed* deviation $\sum (x_i - \mu)p_i$ — but this is always exactly 0, because the positive deviations above the mean perfectly cancel the negative ones below it (that is literally what makes μ the balance point). Squaring each deviation kills the sign so nothing cancels, and it also penalises far-flung values more heavily than near ones — a value twice as far from the mean contributes four times as much to the variance. This is precisely why an erratic batsman’s occasional duck-and-century swings inflate variance so dramatically: the big deviations dominate once squared. Taking the square root at the end (to get σ) simply returns us to the original units of X .

Section 4: Worked Examples

Example 1 — [CBSE Board Level]. A fair coin is tossed three times. Let X be the number of heads. Find the probability distribution of X , its mean, and its variance.

Solution. X can be 0, 1, 2, 3. With 8 equally likely outcomes:

$$P(X = 0) = \frac{1}{8}, P(X = 1) = \frac{3}{8}, P(X = 2) = \frac{3}{8}, P(X = 3) = \frac{1}{8}.$$

Check: $\frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = 1$. ✓

$$\mu = E(X) = 0 \cdot \frac{1}{8} + 1 \cdot \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8} = \frac{0 + 3 + 6 + 3}{8} = \frac{12}{8} = \frac{3}{2}.$$

$$E(X^2) = 0 \cdot \frac{1}{8} + 1 \cdot \frac{3}{8} + 4 \cdot \frac{3}{8} + 9 \cdot \frac{1}{8} = \frac{0 + 3 + 12 + 9}{8} = \frac{24}{8} = 3.$$

$$\text{Var}(X) = E(X^2) - \mu^2 = 3 - \left(\frac{3}{2}\right)^2 = 3 - \frac{9}{4} = \frac{3}{4}.$$

Answer: mean = $\frac{3}{2}$, variance = $\frac{3}{4}$ (so $\sigma = \frac{\sqrt{3}}{2} \approx 0.87$). *Cross-check via the deviation form:* $\sum (x_i - \frac{3}{2})^2 p_i = (\frac{9}{4})\frac{1}{8} + (\frac{1}{4})\frac{3}{8} + (\frac{1}{4})\frac{3}{8} + (\frac{9}{4})\frac{1}{8} = \frac{9}{32} + \frac{3}{32} + \frac{3}{32} + \frac{9}{32} = \frac{24}{32} = \frac{3}{4}$ — agreeing with the computational form. Note the mean 1.5 is not a value X ever takes, which is perfectly normal: it is the long-run average number of heads per three tosses.

Example 2 — [JEE Main Speed Trap]. For the distribution $X : \{1, 2, 3\}$ with $P(X) : \{0.2, 0.5, 0.3\}$, find the variance.

Speed solution. The trap is to compute $\sum (x_i - \mu)^2 p_i$ with decimals — slow and slip-prone. Instead run one table pass for μ and $E(X^2)$:

$$\mu = 1(0.2) + 2(0.5) + 3(0.3) = 0.2 + 1.0 + 0.9 = 2.1,$$

$$E(X^2) = 1(0.2) + 4(0.5) + 9(0.3) = 0.2 + 2.0 + 2.7 = 4.9,$$

$$\text{Var}(X) = 4.9 - (2.1)^2 = 4.9 - 4.41 = 0.49.$$

Answer: 0.49 (so $\sigma = 0.7$). The long route gives the same 0.49 but takes three subtractions and three squarings of awkward decimals — the computational form is the speed play. In a timed paper, computing μ and $E(X^2)$ in a single sweep of the table and subtracting once at the end is consistently the fastest reliable method, and it scales painlessly to distributions with five or six values where the deviation form becomes a decimal minefield.

Example 3 — [JEE Main — Multi-concept: distribution + combinatorics]. Two cards are drawn together (without replacement) from a well-shuffled standard deck of 52. Let X be the number of aces drawn. Find the probability distribution and the mean.

Solution. $X \in \{0, 1, 2\}$; total ways $= \binom{52}{2} = 1326$. There are 4 aces and 48 non-aces.

$$P(X = 0) = \frac{\binom{48}{2}}{\binom{52}{2}} = \frac{1128}{1326} = \frac{188}{221}, \quad P(X = 1) = \frac{\binom{4}{1}\binom{48}{1}}{\binom{52}{2}} = \frac{192}{1326} = \frac{32}{221},$$

$$P(X = 2) = \frac{\binom{4}{2}}{\binom{52}{2}} = \frac{6}{1326} = \frac{1}{221}.$$

Check: $\frac{188 + 32 + 1}{221} = \frac{221}{221} = 1. \checkmark$

$$E(X) = 0 \cdot \frac{188}{221} + 1 \cdot \frac{32}{221} + 2 \cdot \frac{1}{221} = \frac{34}{221} = \frac{2}{13}.$$

Answer: distribution as above; mean $= \frac{2}{13} \approx 0.154$. (Sanity: each card is an ace with chance $\frac{4}{52} = \frac{1}{13}$, and the mean count over two cards is $2 \cdot \frac{1}{13} = \frac{2}{13}$ — matches.)

Example 4 — [JEE Advanced Level]. A discrete random variable X takes values 1, 2, 3, 4 with $P(X = x) = kx$ for a constant k . (i) Find k . (ii) Find $E(X)$ and $\text{Var}(X)$. (iii) If $Y = 2X - 1$, find $E(Y)$ and $\text{Var}(Y)$ without rebuilding a table.

Solution.

(i) $\sum p_i = k(1 + 2 + 3 + 4) = 10k = 1 \Rightarrow k = \frac{1}{10}$. (All four probabilities $\frac{1}{10}, \frac{2}{10}, \frac{3}{10}, \frac{4}{10}$ lie in $[0, 1]$. $\checkmark$)

(ii) $E(X) = \sum x_i p_i = \frac{1 \cdot 1 + 2 \cdot 2 + 3 \cdot 3 + 4 \cdot 4}{10} = \frac{1 + 4 + 9 + 16}{10} = \frac{30}{10} = 3$. $E(X^2) = \frac{1^2 \cdot 1 + 2^2 \cdot 2 + 3^2 \cdot 3 + 4^2 \cdot 4}{10} = \frac{1 + 8 + 27 + 64}{10} = \frac{100}{10} = 10$. $\text{Var}(X) = 10 - 3^2 = 1$.

(iii) Use the linear-transformation rules — no new table needed:

$$E(Y) = 2E(X) - 1 = 2(3) - 1 = 5, \quad \text{Var}(Y) = 2^2 \text{Var}(X) = 4(1) = 4.$$

Answers: $k = \frac{1}{10}$; $E(X) = 3$, $\text{Var}(X) = 1$; $E(Y) = 5$, $\text{Var}(Y) = 4$. *Reading the result:* doubling X (the factor $a = 2$) quadrupled the spread, while the -1 shift moved the mean but left the variance alone — a concrete demonstration of why the scale factor enters variance as a^2 and the additive constant drops out entirely. Recognising this lets you transform a random variable's summary statistics in your head, skipping a full second table.

Section 5: Practice Exercises

1. **[CBSE]** Two fair dice are rolled. Let X be the number of dice showing a six. Find the probability distribution of X and its mean.

2. **[JEE Main]** A distribution is $X : \{0, 1, 2, 3\}$ with $P(X) : \{0.1, 0.3, 0.4, 0.2\}$. Find $E(X)$ and $\text{Var}(X)$.
3. **[JEE Main]** For X taking values 1, 2, 3, 4, 5 with $P(X = x) = cx^2$, find c and $P(X \geq 4)$.
4. **[JEE Advanced]** A random variable X has $E(X) = 4$ and $\text{Var}(X) = 9$. Find $E(X^2)$, and find $E(3X + 2)$ and $\text{Var}(3X + 2)$.
5. **[JEE Advanced]** Three balls are drawn together from a bag containing 4 red and 6 white balls. Let X be the number of red balls drawn. Find the probability distribution and the mean of X .

Answers: 1. $P(0) = \frac{25}{36}, P(1) = \frac{10}{36}, P(2) = \frac{1}{36}$; mean $= \frac{1}{3}$ 2. $E(X) = 1.7, \text{Var}(X) = 0.81$ 3. $c = \frac{1}{55}, P(X \geq 4) = \frac{41}{55}$ 4. $E(X^2) = 25, E(3X + 2) = 14, \text{Var}(3X + 2) = 81$ 5. $P(0) = \frac{1}{6}, P(1) = \frac{1}{2}, P(2) = \frac{3}{10}, P(3) = \frac{1}{30}$; mean $= \frac{6}{5}$.

Section 6: Crack the MCQ

Q1. For the distribution $X : \{1, 3, 5\}$ with $P(X) : \{0.3, 0.4, k\}$, the value of k and $E(X)$ are:

- (a) $k = 0.3, E(X) = 3.0$ (b) $k = 0.4, E(X) = 3.0$ (c) $k = 0.3, E(X) = 2.9$ (d) $k = 0.7, E(X) = 3.2$

Answer: (a). Pin k from the sum-to-one rule first: $0.3 + 0.4 + k = 1 \Rightarrow k = 0.3$. Then $E(X) = 1(0.3) + 3(0.4) + 5(0.3) = 0.3 + 1.2 + 1.5 = 3.0$. (d) takes $k = 0.7$, forgetting the existing 0.4 already accounts for part of the total. (b) uses the wrong k . (c) slips in the weighted sum. The discipline: solve for k from $\sum p_i = 1$ before touching the mean.

Q2. If $\text{Var}(X) = 5$, then $\text{Var}(2X + 7)$ is:

- (a) 5 (b) 10 (c) 17 (d) 20

Answer: (d). $\text{Var}(2X + 7) = 2^2 \text{Var}(X) = 4 \times 5 = 20$. (b) multiplies by 2 instead of 2^2 . (c) wrongly adds the $+7$ into the variance. (a) forgets the scale factor entirely. The constant $+7$ shifts the mean but leaves spread untouched — the most-tested fact here.

Q3. The mean of a random variable that takes value x_i with probability p_i is:

- (a) $\frac{\sum x_i}{n}$ (b) $\sum x_i p_i$ (c) $\sum p_i$ (d) $\sum x_i^2 p_i$

Answer: (b). The mean is the *probability-weighted* sum $\sum x_i p_i$. (a) is the plain arithmetic average — the single most common conceptual error, ignoring that values occur with different probabilities. (c) equals 1 (the total probability). (d) is $E(X^2)$, used in variance, not the mean.

Q4. For a random variable X , $E(X) = 2$ and $E(X^2) = 6$. The standard deviation of X is:

- (a) 4 (b) 2 (c) $\sqrt{2}$ (d) 6

Answer: (c). $\text{Var}(X) = E(X^2) - [E(X)]^2 = 6 - 4 = 2$, so $\sigma = \sqrt{2}$. (b) reports the variance as the SD (forgot the square root). (a) squares the mean and stops. (d) quotes $E(X^2)$ directly. SD is always the *square root* of variance.

Section 7: Common Pitfalls & Pro-Tips

Pitfall 1 — Plain average instead of weighted. Computing $\frac{\sum x_i}{n}$. The mean must weight each value by its probability.

Pitfall 2 — Forgetting to square the mean in variance. Writing $\text{Var} = E(X^2) - E(X)$ instead of $E(X^2) - [E(X)]^2$. The subtracted term is the mean **squared**.

Pitfall 3 — Confusing variance with standard deviation. Reporting σ^2 when σ is asked (or vice versa). SD is the square root of variance and carries the same units as X .

Pitfall 4 — Skipping the sum-to-one check. Not validating $\sum p_i = 1$, or accepting a value of k that yields a negative probability.

Pitfall 5 — Mishandling formula-defined distributions. When $P(X = x)$ is given as a formula like kx or cx^2 , students sometimes plug in values without first finding the constant, or forget to verify the formula returns valid probabilities across the whole range. Always solve $\sum p_i = 1$ for the constant, then confirm each resulting probability is non-negative and at most 1 before using the distribution.

Pro-Tip: Lay every distribution problem out as a table with columns x_i , p_i , $x_i p_i$, $x_i^2 p_i$. Summing the third column gives $E(X)$; summing the fourth gives $E(X^2)$; then $\text{Var} = E(X^2) - [E(X)]^2$ in one final line. This single-pass table is faster and far safer than the deviation formula, and it makes a full Board solution look organised and complete. Quick sanity check: variance can never be negative — if it comes out negative, you've dropped the square on μ .

Section 8: Cheat Sheet Box

RANDOM VARIABLES & DISTRIBUTIONS — RAPID RECALL

- **Random variable:** a rule assigning a real number to each outcome, $X : S \rightarrow \mathbb{R}$.
- **Distribution valid iff:** every $p_i \geq 0$ **and** $\sum p_i = 1$.
- **Mean (expectation):** $\mu = E(X) = \sum x_i p_i$ (probability-weighted, not plain average).
- **Variance:** $\sigma^2 = E(X^2) - \mu^2 = \sum x_i^2 p_i - \mu^2$. **SD:** $\sigma = \sqrt{\sigma^2}$.
- **Linear rules:** $E(aX + b) = aE(X) + b$; $\text{Var}(aX + b) = a^2 \text{Var}(X)$ (the $+b$ does not affect variance).
- **Variance ≥ 0 always** — a negative answer means a dropped μ^2 square.

Mnemonic: “Weight then sum for the mean; square-mean-out for the variance.”

Table trick: columns $x_i \mid p_i \mid x_i p_i \mid x_i^2 p_i$ — third column sum = $E(X)$, fourth column sum = $E(X^2)$, and the p_i column must total 1 as a built-in validity check.

Subtopic 04 — Bernoulli Trials & Binomial Distribution

Exam Relevance: The grand finale of the chapter and a guaranteed scorer. CBSE Boards almost always carry a 5–6 mark question on the binomial distribution — typically “find the probability of exactly/at least r successes” or “given mean and variance, find n and p .” JEE Main places one direct question most years; JEE Advanced layers in the mode, “at least one” via the complement, and parameter-recovery from mean and variance. It rests entirely on independence (Sub-01) and the distribution machinery (Sub-03).

Section 1: Concept Intuition

Think of a penalty shootout. Each kick has just two outcomes — goal or miss — and a striker of fixed quality converts with some steady probability, say 0.8, on every attempt. A single such two-outcome attempt, with a fixed success probability, is a **Bernoulli trial**: the atom of this whole topic. One coin toss, one bulb tested for defect, one student passing or failing, one archer’s single arrow — each is a Bernoulli trial the moment we agree to call one result “success” and the other “failure.” The word “success” is just a label for the outcome we are counting; it need not be anything good, so “drawing a defective item” is perfectly valid as a success when defectives are what we want to count.

Now repeat the kick n times under identical conditions and ask the natural question: *how many goals?* That count — the number of successes in n independent, identical Bernoulli trials — is a random variable, and its distribution is the **binomial distribution**. It is the single most important named distribution in school probability because so many real situations fit its mould: defectives in a carton, sixes in a string of dice rolls, correct answers from pure guessing, rainy days in a week.

To make the pattern unmistakable, notice what all these share. A quality inspector checking 20 phones, each independently defective with the same small probability, is counting binomial successes. A cricketer facing 6 identical deliveries and either scoring a boundary or not on each is generating a binomial count. A student blindly guessing all 10 questions of a four-option test is running 10 identical trials with success probability $\frac{1}{4}$. The surface stories differ wildly, but strip them down and each is “count the successes in n fixed, independent, equal-probability yes/no trials” — and that single skeleton is what the binomial distribution describes.

Think about it this way... Picture flipping a bent coin — bent so it lands heads (success) with probability p and tails (failure) with probability $q = 1 - p$ — exactly n times. The binomial distribution answers “what’s the chance of getting exactly r heads?” To get r successes you must (i) pick *which* r of the n trials succeed, and (ii) have those succeed while the rest fail. Step (i) can happen in nC_r ways; step (ii) carries probability $p^r q^{n-r}$ for each such way because the trials are independent. Multiply and you have the famous formula $P(X = r) = {}^nC_r p^r q^{n-r}$.

Four conditions must hold for trials to be **Bernoulli**, and hence for the count to be binomial: there is a **finite, fixed** number of trials n ; the trials are **independent**; each trial has exactly **two outcomes** (success/failure); and the success probability p stays **constant** across trials. Strip away any one of these and the binomial formula no longer applies.

Limiting conditions & assumptions. The independence and constant- p requirements are where most real problems get disqualified. The classic violation is **drawing without replacement**: pull cards one by one and the composition of the deck changes, so p shifts trial to trial and the draws are dependent — *not* binomial. (Replace the card each time and it becomes binomial.) Also note an identity worth carrying: for a binomial variable, the variance npq is **always strictly less than** the mean np , because $q < 1$. If a problem ever hands you a “binomial” with variance larger than the mean, the data is inconsistent — a favourite trap.

It also helps to picture the *shape* of the distribution. When $p = 0.5$ the binomial is perfectly **symmetric** — successes and failures are equally likely, so the bar chart is a mirror image about its centre (think of counting heads in fair-coin tosses). When $p > 0.5$ the distribution leans toward higher counts (left-skewed), and when $p < 0.5$ it leans toward lower counts (right-skewed). The peak sits at the **mode**, near np , and as n grows the whole shape becomes smoother and more bell-like. Carrying this mental picture lets you sanity-check answers: a computed probability that puts the peak far from np is almost certainly an arithmetic error.

Section 2: Key Definitions & Properties

Bernoulli trial. A single trial with exactly two outcomes — success (probability p) and failure (probability $q = 1 - p$). A sequence of trials is *Bernoulli* if: (i) n is finite and fixed, (ii) trials are independent, (iii) each has two outcomes, (iv) p is constant.

Binomial distribution. If X counts the successes in n Bernoulli trials, then X follows a binomial distribution with parameters n and p , written $X \sim B(n, p)$, with

$$P(X = r) = {}^nC_r p^r q^{n-r}, \quad r = 0, 1, 2, \dots, n, \quad q = 1 - p.$$

The $n + 1$ probabilities are precisely the successive terms of the expansion of $(q + p)^n$ — which is why they automatically sum to 1.

Mean (expectation): $\mu = E(X) = np$. **Variance:** $\sigma^2 = \text{Var}(X) = npq$. **Standard deviation:** $\sigma = \sqrt{npq}$. **Mode** (the most probable number of successes): the integer(s) r satisfying

$$(n + 1)p - 1 \leq r \leq (n + 1)p.$$

When $(n + 1)p$ is not an integer, the mode is the single value $\lfloor (n + 1)p \rfloor$.

Key relationships & notation. n = number of trials, p = success probability, $q = 1 - p$ = failure probability, r = number of successes, X = the binomial random variable. Always $0 < p < 1$ in a genuine problem; all probabilities are dimensionless and lie in $[0, 1]$; and $\sigma^2 = npq < np = \mu$ always.

A few structural facts worth remembering. A binomial variable on n trials has exactly $n + 1$ possible values (0 through n), so its distribution always has $n + 1$ entries. The distribution is **symmetric** precisely when $p = \frac{1}{2}$ (then $P(X = r) = P(X = n - r)$), and increasingly skewed as p moves away from $\frac{1}{2}$. Two binomials with the same n but success probabilities p and $1 - p$ are mirror images of each other. These facts let you predict the answer's shape before computing and catch gross errors at a glance.

Section 3: Theorems & Derivations

Procedure — recognising and setting up a binomial problem. Before any formula, run a four-point check and extract the parameters:

1. **Two outcomes?** Define clearly what counts as “success.” (In “exactly 2 sixes,” success = rolling a six.)
2. **Fixed n ?** Identify the number of trials.
3. **Independent with constant p ?** Confirm the trials don't influence each other and p doesn't drift (the with-replacement test).
4. **Extract p , then $q = 1 - p$, n , and the target r** (or range of r), and only then substitute into ${}^nC_r p^r q^{n-r}$.

Skipping step 1 — being sloppy about which outcome is “success” — is what leads to swapping p and q later. Naming success explicitly up front prevents the single most common binomial error.

Derivation 1 — the binomial formula. *Setup:* n independent Bernoulli trials, each success with probability p , failure with $q = 1 - p$. We want $P(X = r)$, the probability of exactly r successes.

Fix one *specific* pattern of r successes and $n - r$ failures — say successes on the first r trials and failures on the rest. Because the trials are independent, the probability of that exact sequence is the product

$$\underbrace{p \cdot p \cdots p}_r \underbrace{q \cdot q \cdots q}_{n-r} = p^r q^{n-r}.$$

Crucially, *every* arrangement of r successes among the n trials carries the **same** probability $p^r q^{n-r}$ — only the order differs. The number of distinct arrangements is the number of ways to choose which r trials are the successes, namely nC_r . Since these arrangements are mutually exclusive, we add their equal probabilities:

$$P(X = r) = {}^nC_r p^r q^{n-r}. \quad \blacksquare$$

Notice how the formula splits cleanly into three meaningful pieces: nC_r (the number of ways the successes can be positioned), p^r (the successes happening), and q^{n-r} (the failures happening). Reading it this way — rather than memorising it as a symbol blob — makes it almost impossible to misplace the exponents.

Check — probabilities sum to 1. By the Binomial Theorem,

$$\sum_{r=0}^n P(X = r) = \sum_{r=0}^n {}^nC_r p^r q^{n-r} = (q + p)^n = 1^n = 1. \quad \checkmark$$

This is the reason the distribution is *named* “binomial” — its probabilities are the binomial expansion of $(q + p)^n$.

Derivation 2 — mean = np .

$$E(X) = \sum_{r=0}^n r {}^nC_r p^r q^{n-r}.$$

The $r = 0$ term vanishes. Use the identity $r \cdot {}^nC_r = n \cdot {}^{n-1}C_{r-1}$:

$$E(X) = \sum_{r=1}^n n {}^{n-1}C_{r-1} p^r q^{n-r} = np \sum_{r=1}^n {}^{n-1}C_{r-1} p^{r-1} q^{n-r}.$$

Substitute $k = r - 1$; the sum becomes $\sum_{k=0}^{n-1} {}^{n-1}C_k p^k q^{(n-1)-k} = (q + p)^{n-1} = 1$. Hence $E(X) = np$. $\blacksquare$

Derivation 3 — variance = npq (sketch). Compute $E[X(X - 1)] = \sum r(r - 1) {}^nC_r p^r q^{n-r} = n(n - 1)p^2$ by the same index-shift twice. Then

$$\begin{aligned} \text{Var}(X) &= E[X(X - 1)] + E(X) - [E(X)]^2 = n(n - 1)p^2 + np - (np)^2 \\ &= np - np^2 = np(1 - p) = npq. \quad \blacksquare \end{aligned}$$

The takeaway: mean and variance of a binomial are read off instantly from n and p — no distribution table needed.

A concrete check. Take the tiny case $B(2, \frac{1}{2})$ — two fair-coin tosses, X = number of heads. The distribution is $P(0) = \frac{1}{4}$, $P(1) = \frac{1}{2}$, $P(2) = \frac{1}{4}$. Computing directly from

Sub-03's definitions: $E(X) = 0(\frac{1}{4}) + 1(\frac{1}{2}) + 2(\frac{1}{4}) = 1$, which matches $np = 2 \cdot \frac{1}{2} = 1$. And $E(X^2) = 0(\frac{1}{4}) + 1(\frac{1}{2}) + 4(\frac{1}{4}) = \frac{3}{2}$, so $\text{Var}(X) = \frac{3}{2} - 1^2 = \frac{1}{2}$, matching $npq = 2 \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$. The general formulas np and npq agree exactly with the first-principles computation — reassuring confirmation that the shortcuts are not magic, just the table calculation done once and for all.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]. A fair die is rolled 5 times. Find the probability of getting exactly 2 sixes.

Solution. Each roll is a Bernoulli trial: “six” = success with $p = \frac{1}{6}$, so $q = \frac{5}{6}$; $n = 5$, $r = 2$.

$$P(X = 2) = {}^5C_2 \left(\frac{1}{6}\right)^2 \left(\frac{5}{6}\right)^3 = 10 \cdot \frac{1}{36} \cdot \frac{125}{216} = \frac{1250}{7776} = \frac{625}{3888} \approx 0.161.$$

Answer: $\frac{625}{3888} \approx 0.161$.

Follow-up (at most one six): $P(X \leq 1) = P(0) + P(1) = \left(\frac{5}{6}\right)^5 + {}^5C_1 \left(\frac{1}{6}\right) \left(\frac{5}{6}\right)^4 = \frac{3125}{7776} + \frac{5 \cdot 625}{7776} = \frac{3125 + 3125}{7776} = \frac{6250}{7776} \approx 0.804$. Listing values 0, 1, ..., 5 and computing the few you need is the standard Board layout.

Example 2 — [JEE Main Speed Trap]. In a production run, 5% of items are defective. A random sample of 10 items is taken. Find the probability that **at least one** is defective, and state the mean and variance of the number of defectives.

Speed solution. “At least one” is the complement of “none” — never sum ten terms.

$$P(X \geq 1) = 1 - P(X = 0) = 1 - {}^{10}C_0 (0.05)^0 (0.95)^{10} = 1 - (0.95)^{10} \approx 1 - 0.599 = 0.401.$$

Mean = $np = 10(0.05) = 0.5$; variance = $npq = 10(0.05)(0.95) = 0.475$. **Answer:** $P(X \geq 1) \approx 0.401$; mean 0.5, variance 0.475. *The trap:* computing $P(1) + P(2) + \dots + P(10)$ wastes minutes and invites slips — the complement is one line. (Note variance $0.475 < \text{mean } 0.5$, as it must.) The result is also a useful reality check on intuition: even with only a 5% defect rate, inspecting ten items gives roughly a 40% chance of catching at least one defective — small per-item risks accumulate surprisingly fast over many trials.

Example 3 — [JEE Main — Multi-concept: parameter recovery + binomial]. The mean and variance of a binomial variable X are 4 and 2 respectively. Find n , p , and $P(X = 2)$.

Solution. A binomial is completely determined by just two numbers, n and p , so two facts (mean and variance) are exactly enough to pin it down. From $np = 4$ and $npq = 2$, divide:

$$q = \frac{npq}{np} = \frac{2}{4} = \frac{1}{2} \Rightarrow p = 1 - q = \frac{1}{2}.$$

Then $np = 4 \Rightarrow n = \frac{4}{1/2} = 8$. So $X \sim B(8, \frac{1}{2})$ and

$$P(X = 2) = {}^8C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^6 = {}^8C_2 \left(\frac{1}{2}\right)^8 = \frac{28}{256} = \frac{7}{64} \approx 0.109.$$

Answer: $n = 8$, $p = \frac{1}{2}$, $P(X = 2) = \frac{7}{64}$. (Cross-check: variance = $npq = 8 \cdot \frac{1}{2} \cdot \frac{1}{2} = 2$ and mean = $np = 4$ — matching the given data, and confirming variance $<$ mean as required.)

Example 4 — [JEE Advanced Level]. An archer hits the bull's-eye with probability 0.8 on each shot and fires 6 independent shots. Find the **most probable** number of bull's-eyes (the mode) and the probability of achieving it.

Solution. Here $n = 6$, $p = 0.8$, so $(n + 1)p = 7(0.8) = 5.6$, which is not an integer. The mode is therefore

$$r = \lfloor (n + 1)p \rfloor = \lfloor 5.6 \rfloor = 5 \quad (\text{since } 4.6 \leq r \leq 5.6 \Rightarrow r = 5).$$

$$P(X = 5) = {}^6C_5(0.8)^5(0.2)^1 = 6(0.32768)(0.2) = 0.393216 \approx 0.393.$$

Answer: most probable number of hits = 5, with probability ≈ 0.393 . *Insight:* the mean is $np = 4.8$, yet the single most likely count is 5 — mean and mode need not coincide, and the mode formula pins the peak of the distribution directly. *Where the mode formula comes from:* the ratio of consecutive probabilities is $\frac{P(X = r)}{P(X = r - 1)} = \frac{(n - r + 1)p}{r q}$; the probabilities keep rising while this ratio exceeds 1 and start falling once it drops below 1. Setting the ratio equal to 1 and solving for r gives exactly the band $(n + 1)p - 1 \leq r \leq (n + 1)p$, so the peak sits at the largest r for which the climb is still happening.

Section 5: Practice Exercises

1. **[CBSE]** A fair coin is tossed 6 times. Find the probability of getting exactly 4 heads.
2. **[JEE Main]** The probability that a marksman hits a target is $\frac{1}{3}$. If he fires 5 shots, find the probability that he hits the target at least once.
3. **[JEE Main]** The mean and variance of a binomial distribution are 2 and $\frac{4}{3}$. Find n and p .
4. **[JEE Advanced]** A fair coin is tossed 8 times. Find the most probable number of heads and its probability.
5. **[JEE Advanced]** A multiple-choice test has 10 questions, each with 4 options. A student guesses every answer. Find the probability of getting exactly 3 correct.

Answers: 1. $\frac{15}{64}$ 2. $1 - \left(\frac{2}{3}\right)^5 = \frac{211}{243}$ 3. $n = 6$, $p = \frac{1}{3}$ 4. mode = 4, probability $\frac{35}{128}$ 5. ${}^{10}C_3 \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^7 = \frac{262440}{1048576} \approx 0.250$.

Section 6: Crack the MCQ

Q1. For a binomial distribution with $0 < p < 1$, the variance is always:

- (a) greater than the mean (b) less than the mean (c) equal to the mean (d) unrelated to the mean

Answer: (b). Variance = npq and mean = np ; since $q = 1 - p < 1$, we have $npq < np$. (a) and (c) describe other distributions (e.g., Poisson has variance = mean) and are classic confusions. (d) ignores that both are built from the same n, p . This single fact — variance < mean for a binomial — instantly flags inconsistent data.

Q2. If $X \sim B(n, p)$, then $P(X = r)$ equals:

- (a) ${}^nC_r p^r q^{n-r}$ (b) ${}^nC_r p^{n-r} q^r$ (c) $p^r q^{n-r}$ (d) ${}^nC_r p^r q^r$

Answer: (a). Successes carry p (power r), failures carry q (power $n - r$), and nC_r counts the arrangements. (b) swaps the exponents of p and q — the most common slip. (c) drops the combinatorial factor, counting only one arrangement. (d) uses the wrong total exponent ($r + r \neq n$).

Q3. A fair coin is tossed 4 times. The probability of getting at least 3 heads is:

- (a) $\frac{1}{16}$ (b) $\frac{5}{16}$ (c) $\frac{1}{4}$ (d) $\frac{11}{16}$

Answer: (b). $P(X \geq 3) = P(3) + P(4) = {}^4C_3 \left(\frac{1}{2}\right)^4 + {}^4C_4 \left(\frac{1}{2}\right)^4 = \frac{4+1}{16} = \frac{5}{16}$. (a) counts only $P(X = 4)$. (c) miscounts 4C_3 . (d) computes the complement $P(X \leq 2)$ by mistake.

Q4. Which of the following is **not** a binomial experiment?

- (a) Tossing a coin 20 times and counting heads (b) Rolling a die 10 times and counting sixes (c) Drawing 5 cards one by one **without replacement** and counting aces (d) Testing 8 bulbs (each independently defective with probability 0.1) and counting defectives

Answer: (c). Without replacement, the success probability changes after each draw and the draws are dependent — violating constant- p and independence, so it is not binomial. (a), (b), (d) all have a fixed number of independent trials with two outcomes and constant p . This tests the four Bernoulli conditions directly.

Section 7: Common Pitfalls & Pro-Tips

Pitfall 1 — Forcing binomial onto dependent trials. Applying the formula to draws *without replacement*. There p shifts each trial; it is not binomial.

Pitfall 2 — Swapping p and q (or r and $n - r$). Writing $p^{n-r}q^r$ instead of $p^r q^{n-r}$. Successes get p raised to r ; failures get q raised to $n-r$.

Pitfall 3 — Believing variance can exceed mean. For a binomial, $npq < np$ always. A “binomial” with variance $>$ mean signals an inconsistent or non-binomial setup.

Pitfall 4 — The long road to “at least one.” Summing every term from 1 to n instead of $1 - P(X = 0)$.

Pitfall 5 — Dropping or miscounting the nC_r factor. Writing just $p^r q^{n-r}$ forgets that there are many *orders* in which the r successes can occur; each contributes equally, and nC_r counts them. Equally, miscomputing nC_r (e.g., using ${}^nC_r = \frac{n!}{r!}$ without the $(n-r)!$) corrupts the answer. Always include the combinatorial factor and evaluate it carefully.

Pro-Tip: Any “at least”/“at most” question should trigger the complement reflex — $P(X \geq 1) = 1 - P(X = 0)$ and $P(X \geq k) = 1 - \sum_{r < k} P(X = r)$ — choosing whichever side has fewer terms. And for “mean & variance given, find n, p ,” always divide variance by mean *first* to get q in one step ($q = npq/np$), then $p = 1 - q$ and $n = \text{mean}/p$. These two reflexes cover the bulk of exam binomial questions. A third time-saver: when you need several consecutive probabilities, compute $P(X = 0)$ once and then walk up using the ratio $\frac{P(X = r)}{P(X = r - 1)} = \frac{(n - r + 1)p}{r q}$ — each successive probability is the previous one times a simple factor, far quicker than recomputing ${}^nC_r p^r q^{n-r}$ from scratch every time.

Figure 13.3 A vertical bar chart of a binomial distribution $B(6, 0.8)$. Horizontal axis labelled $r = 0, 1, 2, 3, 4, 5, 6$ (number of successes); vertical bars whose heights are the probabilities $P(X=r)$, rising to a clear peak at $r = 5$ (the mode) and a slightly shorter bar at $r = 4$ (near the mean 4.8), with very short bars at small r . Annotate the tallest bar "mode = 5" and mark the mean at 4.8 with a dashed vertical line, illustrating that a binomial with $p > 0.5$ is left-skewed and that mode and mean need not coincide. (see next page)

Section 8: Cheat Sheet Box

BERNOULLI TRIALS & BINOMIAL DISTRIBUTION — RAPID RECALL

- **Bernoulli trial:** one trial, two outcomes, success prob p , failure $q = 1 - p$.
- **Binomial conditions:** fixed n • independent trials • two outcomes • constant p .
- **Distribution:** $X \sim B(n, p)$, $P(X = r) = {}^nC_r p^r q^{n-r}$, $r = 0, \dots, n$.
- **Mean** $= np$ • **Variance** $= npq$ • **SD** $= \sqrt{npq}$ • always **variance** $<$ **mean**.
- **Mode:** integer r with $(n + 1)p - 1 \leq r \leq (n + 1)p$.
- **"At least one":** $P(X \geq 1) = 1 - q^n$ (use the complement).
- **Recover parameters:** $q = \frac{\text{variance}}{\text{mean}}$, then $p = 1 - q$, $n = \frac{\text{mean}}{p}$.

Mnemonic: "Choose the successes, power up p ; power down with q for the rest." **Recognise it:** fixed number of identical, independent yes/no trials with the same p — if drawing *without replacement*, it is **not** binomial. **Sanity check:** probabilities are terms of $(q + p)^n$, so they must sum to 1; and the peak (mode) sits near the mean np .

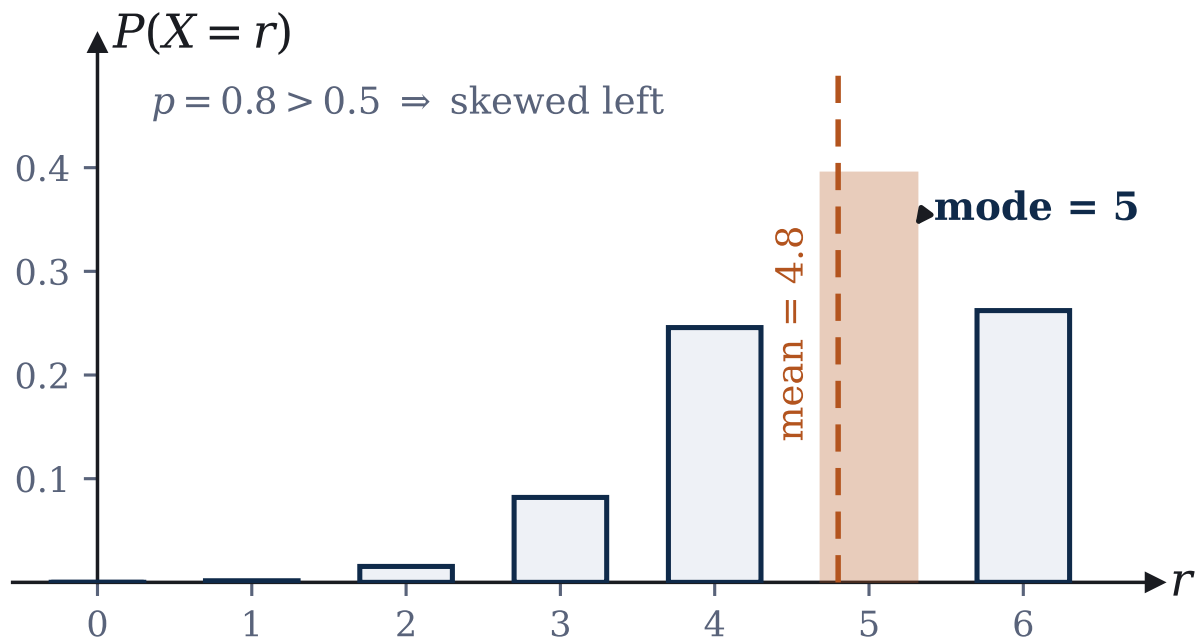


Figure 13.3 A vertical bar chart of a binomial distribution $B(6, 0.8)$. Horizontal axis labelled $r = 0, 1, 2, 3, 4, 5, 6$ (number of successes); vertical bars whose heights are the probabilities $P(X=r)$, rising to a clear peak at $r = 5$ (the mode) and a slightly shorter bar at $r = 4$ (near the mean 4.8), with very short bars at small r . Annotate the tallest bar "mode = 5" and mark the mean at 4.8 with a dashed vertical line, illustrating that a binomial with $p > 0.5$ is left-skewed and that mode and mean need not coincide.

Master Formula Sheet & Revision Capstone

This final section pulls the entire chapter into one place. If you have studied Sub-01 to Sub-04, use this as your night-before-exam single-page memory and as a map of how the ideas connect.

How the Chapter Fits Together

Every idea in this chapter grows from one seed — **conditional probability** — and branches outward:

- **Conditional probability** $P(A | B)$ asks “how likely is A once B is known?” — it shrinks the sample space to B .
- From it comes the **multiplication theorem** ($P(A \cap B) = P(B)P(A | B)$), and the special case where knowledge changes nothing: **independence** ($P(A \cap B) = P(A)P(B)$).
- Stack several mutually exclusive causes and the multiplication theorem becomes the **Theorem of Total Probability** (forward: cause $\rightarrow$ effect).
- Invert that direction — effect back to cause — and you get **Bayes’ Theorem** (the denominator of Bayes is Total Probability).
- Attach a number to each outcome and you have a **random variable**, summarised by its **mean** and **variance**.
- Count successes in n independent, identical two-outcome trials and that random variable follows the **Binomial distribution** — independence (from Sub-01) is what makes it work.

So the through-line is: *condition $\rightarrow$ multiply $\rightarrow$ (independence / total probability) $\rightarrow$ invert (Bayes) $\rightarrow$ random variable $\rightarrow$ binomial.*

Complete Formula Inventory

Conditional probability & independence (Sub-01)

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, P(B) \neq 0$$

$$P(A \cap B) = P(A)P(B | A) = P(B)P(A | B)$$

$$\text{Chain rule: } P(A \cap B \cap C) = P(A)P(B | A)P(C | A \cap B)$$

$$\text{Independent} \iff P(A \cap B) = P(A)P(B); \quad A \perp B \Rightarrow A \perp B', A' \perp B, A' \perp B'$$

$$\text{Exactly one of } A, B : P(A) + P(B) - 2P(A \cap B)$$

$$\text{At least one: } P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\text{At least one of } n \text{ independent events: } 1 - \prod_{i=1}^n (1 - P(A_i))$$

Total probability & Bayes (Sub-02)

$$\text{Total Probability: } P(A) = \sum_j P(E_j) P(A | E_j) \quad (\{E_j\} \text{ a partition})$$

$$\text{Bayes: } P(E_i | A) = \frac{P(E_i) P(A | E_i)}{\sum_j P(E_j) P(A | E_j)} = \frac{\text{prior} \times \text{likelihood}}{P(\text{evidence})}$$

$$\sum_i P(E_i | A) = 1$$

Random variables (Sub-03)

$$\text{Valid distribution: } p_i \geq 0, \sum_i p_i = 1; \quad \text{Mean } \mu = E(X) = \sum_i x_i p_i$$

$$\text{Variance } \sigma^2 = E(X^2) - \mu^2 = \sum_i x_i^2 p_i - \mu^2; \quad \text{SD } \sigma = \sqrt{\sigma^2}$$

$$E(aX + b) = aE(X) + b; \quad \text{Var}(aX + b) = a^2 \text{Var}(X)$$

Binomial distribution (Sub-04)

$$X \sim B(n, p) : P(X = r) = {}^n C_r p^r q^{n-r}, \quad q = 1 - p, \quad r = 0, 1, \dots, n$$

$$\text{Mean} = np; \quad \text{Variance} = npq (< \text{mean}); \quad \text{SD} = \sqrt{npq}$$

$$\text{Mode: } (n+1)p - 1 \leq r \leq (n+1)p; \quad P(X \geq 1) = 1 - q^n$$

$$\text{Recover parameters: } q = \frac{\text{variance}}{\text{mean}}, \quad p = 1 - q, \quad n = \frac{\text{mean}}{p}$$

Three Distinctions That Decide Most Questions

- **Mutually exclusive vs independent.** Mutually exclusive: $P(A \cap B) = 0$ (one forbids the other). Independent: $P(A \cap B) = P(A)P(B)$ (one ignores the other). For positive probabilities they can never hold together.
- **Total Probability vs Bayes.** Total Probability runs *forward* (you know causes, predict the effect). Bayes runs *backward* (you observe the effect, infer the cause). Bayes uses Total Probability as its denominator.
- **With vs without replacement.** With replacement $\Rightarrow$ independent, constant $p \Rightarrow$ binomial applies. Without replacement $\Rightarrow$ dependent, shifting $p \Rightarrow$ use conditional probability / counting, **not** the binomial formula.

Master Pitfall Index

#	The error
1.	Treating independent and mutually exclusive as the same
2.	Swapping $P(A B)$ with $P(B A)$
3.	Assuming pairwise independence implies mutual independence
4.	Base-rate neglect — quoting the likelihood as the posterior
5.	Dividing by one term instead of the full $P(A)$
6.	Using causes that aren't disjoint and exhaustive
7.	Plain average instead of probability-weighted mean
8.	Forgetting to square μ in $E(X^2) - \mu^2$
9.	Confusing variance with standard deviation

#	The error
10.	Forcing binomial onto without-replacement (dependent) trials
11.	Swapping the exponents of p and q
12.	Believing variance can exceed the mean (it can't: $npq < np$)
13.	Summing all terms instead of $1 - q^n$ for "at least one"

Exam Weightage Snapshot

- **CBSE Boards:** Bayes' Theorem (5–6 marks) and Binomial distribution (5–6 marks) are near-certain; conditional probability / independence and random-variable mean-variance appear as 2–4 mark items.
- **JEE Main:** typically 1 question, most often independence, Bayes, or a binomial computation.
- **JEE Advanced:** higher-rigour variants — independence of complements, "exactly one," base-rate Bayes surprises, binomial mode, and parameter recovery from mean and variance.

One-line chapter mantra: *Condition to shrink the world, multiply to combine, check independence, add causes for Total Probability, invert for Bayes, then count successes for the Binomial.*

Chapter 13: Probability — Round 2 Addendum

Subtopics 01-04 carried you from conditioning through Bayes to the Binomial, and every CBSE template in them is complete. Five JEE patterns, though, live just past the last page. Addendum A extends every “at least/exactly” identity from two events to three, because inclusion-exclusion questions routinely name three events while the chapter’s toolkit stops at two. Addendum B settles the one warning the chapter issues but never demonstrates: pairwise independence that fails the triple-product test, including how to *build* such examples and how to survive the cubic equations they generate. Addendum C chains Bayes across stages — today’s posterior is tomorrow’s prior — and handles the transfer-of-balls family the Boards love. Addendum D pushes expectation past $aX+b$ to sums, products and the indicator-variable trick that computes hard expectations in one line. Addendum E treats waiting-time and turn-taking problems, which are pure infinite-series probability and appear in JEE Main nearly every year. Nothing below introduces new theory: each section is the multiplication theorem, the partition idea or the distribution machinery you already own, applied one layer deeper.

Addendum A — Three Events: Unions, Exactly- k , and Inclusion-Exclusion

The chapter gives $P(A \cup B)$ and the “exactly one of two” formula, then jumps straight to products over independent triples. But JEE Main and Advanced constantly pose: *three* events, possibly dependent, find the probability that exactly one occurs, or exactly two, or none. Four summary quantities do all of it.

Method

Step 1 — The three-event union. Apply the two-event addition law twice. Write

$$P(A \cup B \cup C) = P(A \cup (B \cup C)) = P(A) + P(B \cup C) - P(A \cap (B \cup C)).$$

Now expand each piece. The distributive law gives $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$, so by the two-event law,

$$P(A \cap (B \cup C)) = P(A \cap B) + P(A \cap C) - P(A \cap B \cap C),$$

and expanding $P(B \cup C)$ the same way and substituting:

$$P(A \cup B \cup C) = S_1 - S_2 + S_3,$$

where

$$S_1 = P(A) + P(B) + P(C),$$

$$S_2 = P(A \cap B) + P(B \cap C) + P(C \cap A),$$

$$S_3 = P(A \cap B \cap C).$$

This is inclusion-exclusion: count singly, correct the double-counted overlaps, restore the triple-counted core.

Step 2 — The exactly- k identities. Every outcome belongs to exactly one of four disjoint classes: it lies in exactly one, two, or three of the events, or in none. Count how many times each class is tallied by S_1, S_2, S_3 :

Outcome lies in	Tally in S_1	Tally in S_2	Tally in S_3
exactly one event	1	0	0
exactly two events	2	1	0
exactly three events	3	3	1

Any expression $c_1 S_1 + c_2 S_2 + c_3 S_3$ therefore assigns each class the weight read off the row. Choosing coefficients that isolate a class gives:

$$P(\text{at least one}) = S_1 - S_2 + S_3,$$

$$P(\text{none}) = 1 - S_1 + S_2 - S_3,$$

$$P(\text{exactly one}) = S_1 - 2S_2 + 3S_3,$$

$$P(\text{exactly two}) = S_2 - 3S_3,$$

$$P(\text{at least two}) = S_2 - 2S_3.$$

Check each against the table. For instance, $S_1 - 2S_2 + 3S_3$ gives weights $1 - 0 + 0, 2 - 2 + 0, 3 - 6 + 3$: exactly the classes $1, 0, 0$ — precisely “exactly one.”

Step 3 — Consistency check (free marks). The classes are disjoint and exhaustive, so always confirm

$$P(\text{exactly one}) + P(\text{exactly two}) + P(\text{all three}) + P(\text{none}) = 1.$$

If your four answers fail this, the error is structural, not cosmetic — usually a wrong sign on S_3 .

Independent shortcut. When A, B, C are mutually independent, S_2 and S_3 factorise into products, and “exactly k ” can also be computed term by term, e.g.

$$P(\text{exactly one}) = pq'r' + p'qr' + p'q'r',$$

with $p' = 1 - p$ and so on. Both routes must agree; use the shorter.

Example A.1 — [JEE Main Level]

Shooters A, B, C fire independently at a target with hit probabilities $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$. Find the probability that (i) at least one hits, (ii) exactly one hits, (iii) exactly two hit.

Here $p = \frac{1}{2}$, $q = \frac{1}{3}$, $r = \frac{1}{4}$, and independence lets every intersection factorise:

$$P(A \cap B) = \frac{1}{6}, \quad P(B \cap C) = \frac{1}{12}, \quad P(C \cap A) = \frac{1}{8}, \quad P(A \cap B \cap C) = \frac{1}{24}.$$

So

$$S_1 = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{13}{12}, \quad S_2 = \frac{1}{6} + \frac{1}{12} + \frac{1}{8} = \frac{3}{8}, \quad S_3 = \frac{1}{24}.$$

(i)

$$P(\text{at least one}) = \frac{13}{12} - \frac{3}{8} + \frac{1}{24} = \frac{26-9+1}{24} = \frac{3}{4}.$$

Equivalently by the complement rule: $1 - \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} = 1 - \frac{1}{4} = \frac{3}{4}$. Agreement noted.

(ii)

$$P(\text{exactly one}) = \frac{13}{12} - 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{24} = \frac{26-18+3}{24} = \frac{11}{24}.$$

(iii)

$$P(\text{exactly two}) = \frac{3}{8} - 3 \cdot \frac{1}{24} = \frac{9-3}{24} = \frac{1}{4}.$$

Check. Exactly three = $\frac{1}{24}$, none = $\frac{1}{4}$, and

$$\frac{11}{24} + \frac{6}{24} + \frac{1}{24} + \frac{6}{24} = \frac{24}{24} = 1.$$

The decomposition closes, so the answers stand.

Example A.2 — [JEE Advanced Level]

For three events A, B, C you are told $P(A \cup B \cup C) = \frac{3}{4}$, $P(\text{exactly one occurs}) = \frac{11}{24}$ and $P(\text{exactly two occur}) = \frac{1}{4}$. Find $P(\text{all three})$ and $P(\text{none})$, and show that the data force $P(A \cap B \cap C) = \frac{1}{24}$ without ever knowing $P(A)$, $P(B)$, $P(C)$ individually.

The four classes partition S , so immediately

$$P(\text{none}) = 1 - \frac{3}{4} = \frac{1}{4},$$

and

$$P(\text{all three}) = P(\text{at least one}) - P(\text{exactly one}) - P(\text{exactly two}).$$

Since “at least one” excludes only “none”:

$$P(\text{all three}) = \frac{3}{4} - \frac{11}{24} - \frac{1}{4} = \frac{18-11-6}{24} = \frac{1}{24}.$$

To see why no individual $P(A)$ is needed, translate the givens into the S -language. We know $S_1 - S_2 + S_3 = \frac{3}{4}$ and $S_1 - 2S_2 + 3S_3 = \frac{11}{24}$ and $S_2 - 3S_3 = \frac{1}{4}$. Three linear equations in the three unknowns S_1, S_2, S_3 determine them completely: from the third, $S_2 = \frac{1}{4} + 3S_3$; substitute into the first:

$$S_1 - \frac{1}{4} - 3S_3 + S_3 = \frac{3}{4} \implies S_1 - 2S_3 = 1.$$

Substitute both into the second:

$$1 + 2S_3 - 2 \cdot \frac{1}{4} - 6S_3 + 3S_3 = \frac{11}{24} \implies \frac{1}{2} - S_3 = \frac{11}{24} \implies S_3 = \frac{1}{24}.$$

So $P(A \cap B \cap C) = \frac{1}{24}$, $S_2 = \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$, $S_1 = 1 + \frac{2}{24} = \frac{7}{6}$.

Check. Run the forward formulas: at least one $= \frac{7}{6} - \frac{3}{8} + \frac{1}{24} = \frac{28-9+1}{24} = \frac{3}{4}$; exactly one $= \frac{7}{6} - \frac{3}{4} + \frac{1}{8} = \frac{28-18+3}{24} = \frac{11}{24}$; exactly two $= \frac{3}{8} - \frac{1}{8} = \frac{1}{4}$. All three givens reproduced. The moral: the S -variables carry all the information the exam needs — individual probabilities are often decoys.

Practice — Addendum A

- [CBSE] Given $P(A) = 0.4$, $P(B) = 0.5$, $P(C) = 0.3$, $P(A \cap B) = 0.2$, $P(B \cap C) = 0.15$, $P(A \cap C) = 0.12$, $P(A \cap B \cap C) = 0.08$, find $P(A \cup B \cup C)$, $P(\text{exactly one})$ and $P(\text{none})$.
- [JEE Main] A, B, C are mutually independent, each with probability $\frac{1}{3}$. Find $P(\text{exactly one})$ and $P(\text{at least one})$.
- [JEE Main] Three students attempt a problem independently with success probabilities $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$. Find the probability that exactly two succeed.
- [JEE Main] A fair die is rolled once. A : even number; B : multiple of 3; C : prime number. Verify $P(A \cup B \cup C) = \frac{5}{6}$ using inclusion-exclusion.
- [JEE Advanced] Three events satisfy $P(A) = P(B) = P(C) = \frac{1}{2}$, each pairwise intersection equals $\frac{1}{3}$, and $P(A \cap B \cap C) = \frac{1}{4}$. Find $P(\text{exactly one})$ and $P(\text{none})$, and verify your answers close the partition.
- [JEE Advanced] Express $P(\text{exactly two})$ in terms of $P(\text{at least two})$ and S_3 alone.

Answers: 1. $S_1 = 1.2$, $S_2 = 0.47$, $S_3 = 0.08$; union $= 0.81$, exactly one $= 1.2 - 0.94 + 0.24 = 0.50$, none $= 0.19$; check $0.50 + 0.23 + 0.08 + 0.19 = 1$. 2. $S_2 = 3 \cdot \frac{1}{9} = \frac{1}{3}$, $S_3 = \frac{1}{27}$; exactly one $= 1 - \frac{2}{3} + \frac{1}{9} = \frac{4}{9}$; at least one $= 1 - \left(\frac{2}{3}\right)^3 = \frac{19}{27}$. 3. $\frac{1}{2} \cdot \frac{1}{3} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{3} \cdot \frac{1}{4} = \frac{3+2+1}{24} = \frac{1}{4}$. 4. $S_1 = \frac{1}{2} + \frac{1}{3} + \frac{1}{2} = \frac{4}{3}$; pairwise intersections are $\{6\}, \{2\}, \{3\}$, each $\frac{1}{6}$, so $S_2 = \frac{1}{2}$, $S_3 = 0$; union $= \frac{4}{3} - \frac{1}{2} = \frac{5}{6}$, matching the direct count $\{2, 3, 4, 5, 6\}$. 5. $S_1 = \frac{3}{2}$, $S_2 = 1$, $S_3 = \frac{1}{4}$; exactly one $= \frac{3}{2} - 2 + \frac{3}{4} = \frac{1}{4}$; none $= 1 - \frac{3}{2} + 1 - \frac{1}{4} = \frac{1}{4}$; partition closes since exactly two $= \frac{1}{4}$ and

triple = $\frac{1}{4}$. 6. Since at least two = $S_2 - 2S_3$ and exactly two = $S_2 - 3S_3$, we get exactly two = $P(\text{at least two}) - S_3$ — i.e. subtract the all-three case from the at-least-two case.

Addendum B — Pairwise but Not Mutual Independence: Building It, Breaking It

The chapter states the rule — three pairs checking out does not make three mutually independent events — and tags the pitfall. JEE Advanced examines it in two ways: *verify or refute* mutual independence from data, and *solve for parameters* when “exactly k ” information is given. You need to have seen a working counterexample at least once, because the failure mode is invisible to intuition.

Method

The verification protocol. To decide whether A, B, C are mutually independent, run all four tests:

1. $P(A \cap B) = P(A)P(B)$;
2. $P(B \cap C) = P(B)P(C)$;
3. $P(C \cap A) = P(C)P(A)$;
4. $P(A \cap B \cap C) = P(A)P(B)P(C)$.

Mutual independence means all four hold; pairwise independence means only the first three are guaranteed. Test 4 is where constructed examples fail.

Why pairwise can pass while 4 fails — the mechanism. The pair conditions only constrain probabilities of *two-fold* overlaps. The triple overlap is a separate cell of the Venn diagram, and nothing in tests 1–3 reaches into it. So one is free to design events whose overlaps look independent in pairs but whose centre cell has been emptied or enlarged.

Example B.1 — [JEE Advanced Level] — the canonical construction

Two fair coins are tossed. Let A = “the first coin shows heads,” B = “the second coin shows heads,” C = “the two coins disagree.” Show that A, B, C are pairwise independent but not mutually independent.

Sample space: $\{HH, HT, TH, TT\}$, each outcome $\frac{1}{4}$. Then

$$A = \{HH, HT\}, \quad B = \{HH, TH\}, \quad C = \{HT, TH\},$$

so $P(A) = P(B) = P(C) = \frac{1}{2}$. Now the three pairs:

$$\begin{aligned} P(A \cap B) &= P(HH) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}. \checkmark \\ P(B \cap C) &= P(TH) = \frac{1}{4} = P(B)P(C). \checkmark \end{aligned}$$

$$P(C \cap A) = P(HT) = \frac{1}{4} = P(C)P(A). \checkmark$$

Pairwise independence confirmed. But the triple intersection:

$$A \cap B \cap C = \{HH\} \cap \{HT, TH\} = \emptyset \implies P(A \cap B \cap C) = 0,$$

while

$$P(A)P(B)P(C) = \left(\frac{1}{2}\right)^3 = \frac{1}{8} \neq 0.$$

Test 4 fails: the events are **not** mutually independent. Read the mechanism off the geometry: each pair meets in exactly one outcome (HH , TH , HT respectively — a quarter each, as independence demands), but the three pair-overlaps are three *different* outcomes, leaving the centre cell empty.

Consequence worth quoting in an answer: knowing A and B both occurred pins the outcome to HH , which *rules out* C — so C 's occurrence is strongly (negatively) dependent on the pair, even though it is independent of each member alone.

Example B.2 — [JEE Advanced Level] — recovering a parameter from a cubic

Events A , B , C are mutually independent, each occurring with the same probability p , where $0 < p < 1$. The probability that exactly one of them occurs is $\frac{48}{125}$. Find p .

By the identity of Addendum A applied to independent events — or directly by summing the three disjoint patterns:

$$P(\text{exactly one}) = 3p(1-p)^2.$$

Set this equal to $\frac{48}{125}$ and clear denominators. Dividing both sides by 3:

$$p(1-p)^2 = \frac{16}{125}.$$

Expanding $p - 2p^2 + p^3 = \frac{16}{125}$ and multiplying through by 125:

$$125p^3 - 250p^2 + 125p - 16 = 0.$$

Test the natural candidate $p = \frac{1}{5}$:

$$125 \cdot \frac{1}{125} - 250 \cdot \frac{1}{25} + 125 \cdot \frac{1}{5} - 16 = 1 - 10 + 25 - 16 = 0. \checkmark$$

Factor out $(5p - 1)$. Synthetic division of $125p^3 - 250p^2 + 125p - 16$ by $p - \frac{1}{5}$ gives quotient $125p^2 - 225p + 80$, so

$$125p^3 - 250p^2 + 125p - 16 = (5p - 1)(25p^2 - 45p + 16).$$

For the quadratic, the discriminant is $(-45)^2 - 4 \cdot 25 \cdot 16 = 2025 - 1600 = 425$, not a perfect square, with roots approximately 0.488 and 1.312. The second exceeds 1, hence inadmissible. What about 0.488? Check the original equation: $3(0.488)(0.512)^2 \approx 0.384 \neq 0.384$? Compute carefully: $(0.512)^2 = 0.262144$, times 0.488, times 3:

$$3(0.488)(0.262144) \approx 0.38378 \approx \frac{48}{125} = 0.384.$$

It passes numerically — so the quadratic root is admissible after all, and p is **not unique**: $p = \frac{1}{5}$ and $p = \frac{45 - \sqrt{425}}{50}$ both work. This is the mark-winning observation. The function $f(p) = 3p(1-p)^2$ rises from 0 at $p = 0$, peaks near $p = \frac{1}{3}$, and falls to 0 at $p = 1$; a horizontal line below the peak cuts it twice. Unless the problem pins down which side of the peak you are on, report **both** roots, discarding only those outside $(0, 1)$.

Check. For $p = \frac{1}{5}$: exactly one $= 3 \cdot \frac{1}{5} \cdot \left(\frac{4}{5}\right)^2 = \frac{48}{125}$, as demanded; and exactly two $= 3 \cdot \frac{1}{25} \cdot \frac{4}{5} = \frac{12}{125}$, triple $= \frac{1}{125}$, none $= \frac{64}{125}$, and $\frac{48+12+1+64}{125} = 1$. Partition closes.

Practice — Addendum B

- [JEE Advanced] On the sample space $\{HH, HT, TH, TT\}$ let A = “first toss heads,” B = “second toss heads,” C = “an odd number of heads.” Show all three pairs are independent but A, B, C are not mutually independent.
- [JEE Main] $P(A) = P(B) = P(C) = \frac{1}{3}$, the three events are pairwise independent, and $P(A \cap B \cap C) = 0$. Find $P(\text{exactly one})$ and $P(\text{at least one})$.
- [JEE Advanced] Mutually independent events each with probability p satisfy $P(\text{exactly two}) = \frac{12}{125}$. Find all admissible values of p .
- [Conceptual, Advanced] Can three mutually exclusive events with positive probabilities be pairwise independent?
- [JEE Advanced] A, B, C are independent, $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$, $P(C) = \frac{1}{4}$. Given that exactly two of them occurred, find the probability that A was among them.

Answers: 1. $C = \{HT, TH\}$, $P(C) = \frac{1}{2}$; $A \cap C = \{HT\}$, $B \cap C = \{TH\}$, both $\frac{1}{4} = P(\cdot)P(\cdot)$, and $A \cap B = \{HH\}$ likewise; but $A \cap B \cap C = \emptyset$ while $P(A)P(B)P(C) = \frac{1}{8}$ — not mutual. 2. $S_2 = 3 \cdot \frac{1}{9} = \frac{1}{3}$, $S_3 = 0$; exactly one $= 1 - \frac{2}{3} = \frac{1}{3}$; at least one $= 1 - \frac{1}{3} + 0 = \frac{2}{3}$. 3. Same cubic as Example B.2 by symmetry of swapping $p \leftrightarrow 1-p$ roles — indeed $P(\text{exactly two}) = 3p^2(1-p) = \frac{12}{125}$ gives, after dividing by 3, $p^2(1-p) = \frac{4}{125}$; testing $p = \frac{2}{5}$: $\frac{4}{9} \cdot \frac{3}{5} = \frac{4}{15} \neq \frac{4}{125}$. Expand: $125p^2 - 125p^3 = 4$, i.e. $125p^3 - 125p^2 + 4 = 0$; $p = \frac{1}{5}$: $1 - 5 + 4 = 0$ ✓. Factorising gives $(5p - 1)(25p^2 - 20p - 4) = 0$; the quadratic has discriminant $400 + 400 = 800$, roots $\frac{20 \pm \sqrt{800}}{50} = \frac{2 \pm 2\sqrt{2}}{5}$, of which $\frac{2+2\sqrt{2}}{5} \approx 0.97$ lies in $(0, 1)$. So $p = \frac{1}{5}$ or $p = \frac{2+2\sqrt{2}}{5}$; check the latter numerically: $3(0.9643)^2(0.0357) \approx 0.0995\dots$ refine: $\sqrt{2} \approx 1.41421$, $p \approx 0.96569$, $1-p \approx 0.03431$, $3p^2(1-p) \approx 3(0.93256)(0.03431) \approx 0.0960 \approx \frac{12}{125} = 0.096$ ✓. Both admissible. 4. No: mutual exclusivity forces $P(A \cap B) = 0$ while pairwise independence demands $P(A)P(B) > 0$. 5. $P(\text{exactly two}) = \frac{1}{4}$ (Example A.1 iii); the cases containing A : “ A, B not C ” and “ A, C not B ”, totalling $\frac{1}{2} \cdot \frac{1}{3} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{1}{4} = \frac{1}{8} + \frac{1}{12} = \frac{5}{24}$; conditional probability $= \frac{5/24}{1/4} = \frac{5}{6}$.

Addendum C — Chained Bayes: Staged Evidence and Transfer Experiments

The chapter's Pro-Tip says “today's posterior becomes tomorrow's prior,” and then stops. JEE Main and Advanced both examine exactly that move, in two costumes: repeated tests on one hidden cause, and the transfer problem where one random act rewrites the likelihoods before the next observation. Here is the engine, proved, then both costumes worn.

Method

The chaining rule. Let $\{E_1, \dots, E_n\}$ be a partition of causes, and suppose two pieces of evidence A_1 then A_2 arrive, with A_1 and A_2 independent *given* the cause. Claim:

$$P(E_i | A_1 \cap A_2) = \frac{P(E_i) P(A_1 | E_i) P(A_2 | E_i)}{\sum_j P(E_j) P(A_1 | E_j) P(A_2 | E_j)}.$$

Proof. By Bayes with the compound evidence $A_1 \cap A_2$ treated as a single observation, the denominator is Total Probability for $A_1 \cap A_2$:

$$P(A_1 \cap A_2) = \sum_j P(E_j) P(A_1 \cap A_2 | E_j).$$

Given cause E_j , the two observations are independent, so $P(A_1 \cap A_2 | E_j) = P(A_1 | E_j)P(A_2 | E_j)$, giving the displayed fraction. To see it equals the chained route, note the numerator factors:

$$P(E_i) P(A_1 | E_i) P(A_2 | E_i) = \underbrace{P(E_i) P(A_1 | E_i)}_{\text{unnormalised posterior}} \cdot P(A_2 | E_i),$$

and dividing by $P(A_1) = \sum_j P(E_j)P(A_1 | E_j)$ produces $P(E_i | A_1) \cdot P(A_2 | E_i)$ — posterior times fresh likelihood. So running Bayes twice, feeding the first posterior back as the second prior, lands on the same answer as one grand Bayes. ■

Transfer problems. When a ball moves between containers before the observed draw, the “cause” is the transferred ball's colour and the “likelihood” is the draw probability computed from the *altered* composition. Build the altered composition first; everything else is stock Bayes.

Example C.1 — [JEE Main Level]

A disease affects 1% of a population. A test detects a carrier with probability 0.9 and clears a healthy person with probability 0.95. A patient tests positive twice, the two readings being independent given carrier status. Find the probability the patient is a carrier.

Stage one. Prior $P(D) = 0.01$, $P(D') = 0.99$; likelihoods $P(+ | D) = 0.9$, $P(+ | D') = 0.05$. Total probability of one positive:

$$P(+) = 0.01(0.9) + 0.99(0.05) = 0.009 + 0.0495 = 0.0585,$$

so the first posterior is

$$P(D | +) = \frac{0.009}{0.0585} = \frac{90}{585} = \frac{2}{13} \approx 0.154.$$

Stage two. Feed $P(D) = \frac{2}{13}$, $P(D') = \frac{11}{13}$ back in with the same likelihoods:

$$P(D | ++) = \frac{\frac{2}{13}(0.9)}{\frac{2}{13}(0.9) + \frac{11}{13}(0.05)} = \frac{1.8}{1.8 + 0.55} = \frac{180}{235} = \frac{36}{47} \approx 0.766.$$

Cross-check by the one-shot rule. Directly from the claim:

$$P(++) = 0.01(0.9)^2 + 0.99(0.05)^2 = 0.0081 + 0.002475 = 0.010575,$$

$$P(D | ++) = \frac{0.0081}{0.010575} = \frac{8100}{10575} = \frac{36}{47} \cdot \checkmark$$

Same answer, two routes. Notice the story the numbers tell: one positive barely lifts a 1% prior to 15%, but a *second* positive leaps to 77% — evidence compounds, and base rates never stop mattering.

Example C.2 — [JEE Advanced Level]

Bag I holds 3 red and 4 black balls; Bag II holds 5 red and 3 black. One ball is moved at random from Bag I to Bag II, stirred in, and then one ball is drawn from Bag II. It is red. Find the probability that the transferred ball was black.

Causes: T_B = “black transferred” with prior $\frac{3}{7}$; T_R = “red transferred” with prior $\frac{4}{7}$. Evidence: R = “drawn ball red.” Compute each likelihood from the altered Bag II:

- If black moved: Bag II becomes 5 red, 4 black — nine balls, so $P(R | T_B) = \frac{5}{9}$.
- If red moved: Bag II becomes 6 red, 3 black — nine balls, so $P(R | T_R) = \frac{6}{9} = \frac{2}{3}$.

Total probability of drawing red:

$$P(R) = \frac{3}{7} \cdot \frac{5}{9} + \frac{4}{7} \cdot \frac{2}{3} = \frac{15}{63} + \frac{24}{63} = \frac{39}{63} = \frac{13}{21}.$$

Bayes:

$$P(T_B | R) = \frac{\frac{15}{63}}{\frac{39}{63}} = \frac{15}{39} = \frac{5}{13}.$$

Answer: $\frac{5}{13}$. Sanity checks: the posterior exceeds the prior $\frac{3}{7} = \frac{?}{?}$ — compare $\frac{5}{13} \approx 0.385$ with $\frac{3}{7} \approx 0.429$. The posterior is slightly *below* the prior, which is correct: drawing red is evidence *against* having removed a black ball, since removing black leaves fewer reds behind. Any

answer above $\frac{3}{7}$ signals a reversed likelihood. Also verify completeness: $P(T_R | R) = \frac{24}{39} = \frac{8}{13}$, and $\frac{5}{13} + \frac{8}{13} = 1$. Posteriors close.

Practice — Addendum C

1. [JEE Main] With the setup of Example C.1, find the probability the patient is a carrier after one positive and then one negative test (independent readings).
2. [JEE Main] Bag I holds 4 white and 2 black; Bag II holds 1 white and 5 black. A ball moves from I to II, then one ball is drawn from II and is white. Find the probability the moved ball was white.
3. [JEE Advanced] Urn A holds 2 red, 3 blue; Urn B holds 4 red, 1 blue. An urn is chosen at random, one ball is drawn and **not replaced**, and it is red. A second ball is then drawn from the same urn and is also red. Find the probability the urn was B.
4. [JEE Advanced] Explain, without computing, why two positive results cannot push the Example C.1 posterior to 0.99, no matter how accurate the test, if the disease is rare enough.

Answers: 1. After stage one the prior is $\frac{2}{13}$; likelihoods for negative: $P(- | D) = 0.1$, $P(- | D') = 0.95$; posterior $= \frac{\frac{2}{13}(0.1)}{\frac{2}{13}(0.1) + \frac{11}{13}(0.95)} = \frac{0.2}{0.2 + 10.45} = \frac{2}{106.5} = \frac{4}{213} \approx 0.019$ — one clean reading almost resets the damage. 2. Likelihoods: if white moved, II has 2W,5B, $P(W) = \frac{2}{7}$; if black moved, 1W,6B, $P(W) = \frac{1}{7}$. Total $= \frac{1}{2} \cdot \frac{2}{7} + \frac{1}{2} \cdot \frac{1}{7} = \frac{3}{14}$; posterior white $= \frac{1/14}{3/14} = \frac{1}{3}$. 3. Likelihood of two reds from A: $\frac{2}{5} \cdot \frac{1}{4} = \frac{1}{10}$; from B: $\frac{4}{5} \cdot \frac{3}{4} = \frac{3}{5}$. Total $= \frac{1}{2}(\frac{1}{10} + \frac{3}{5}) = \frac{1}{5}$; posterior B $= \frac{3/20}{1/5} = \frac{3}{4}$. 4. The posterior is capped by the likelihood ratio structure: even with specificity approaching 1, false positives vanish but the numerator stays bounded by $P(D)$ itself; formally $P(D | ++) \leq \frac{P(D)}{P(D) + P(D')(0.05)^2/(0.9)^2}$, and as the healthy pool is 99-fold larger, overwhelming agreement of many tests is needed — two positives cannot manufacture a near-certain prior out of a 1% base rate.

Addendum D — Expectation Beyond $aX + b$: Sums, Products, and Indicator Variables

The chapter proves $E(aX+b)$ and $\text{Var}(aX+b)$ and stops. JEE asks about $E(X+Y)$, $\text{Var}(X-Y)$, and $E(XY)$ — and, at Advanced level, expects the indicator trick that computes apparently brutal expectations in one line. Everything below follows from one definition.

Method

Joint expectation. For discrete X, Y taking values x_i, y_j with joint probabilities $p_{ij} = P(X = x_i, Y = y_j)$, define

$$E[g(X, Y)] = \sum_i \sum_j g(x_i, y_j) p_{ij}.$$

Result 1 — linearity (always true). Take $g(X, Y) = X + Y$:

$$E(X + Y) = \sum_i \sum_j (x_i + y_j) p_{ij} = \sum_i \sum_j x_i p_{ij} + \sum_i \sum_j y_j p_{ij}.$$

In the first double sum, sum over j first: $\sum_j p_{ij} = P(X = x_i)$, so the first term collapses to $\sum_i x_i P(X = x_i) = E(X)$; symmetrically the second is $E(Y)$. Hence

$$E(X + Y) = E(X) + E(Y),$$

with no independence needed, and iterating gives $E(X - Y) = E(X) - E(Y)$ and sums of any length.

Result 2 — products under independence. Independence means the joint pmf factorises: $p_{ij} = P(X = x_i) P(Y = y_j)$. Then

$$E(XY) = \sum_i \sum_j x_i y_j P(X = x_i) P(Y = y_j) = \left(\sum_i x_i P(X = x_i) \right) \left(\sum_j y_j P(Y = y_j) \right) = E(X) E(Y),$$

because the double sum separates into an outer factor depending only on i and an inner factor depending only on j . Without independence this separation is illegal — that is the whole content of the hypothesis.

Result 3 — variance of a sum. Write $\mu_X = E(X)$, $\mu_Y = E(Y)$. Using Result 1 on $E[(X+Y)^2]$:

$$E[(X + Y)^2] = E(X^2) + 2E(XY) + E(Y^2),$$

while $(\mu_X + \mu_Y)^2 = \mu_X^2 + 2\mu_X\mu_Y + \mu_Y^2$. Subtracting:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2[E(XY) - \mu_X\mu_Y].$$

The bracket defines the covariance $\text{Cov}(X, Y)$, measuring shared drift. Under independence, Result 2 kills it:

$$\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y).$$

The minus sign changes nothing — spread adds either way, since $\text{Var}(-Y) = (-1)^2 \text{Var}(Y)$.

Result 4 — indicator variables. For any event A , define $I_A = 1$ on A and 0 off it. Then $I_A^2 = I_A$, so

$$E(I_A) = 1 \cdot P(A) + 0 = P(A), \quad \text{Var}(I_A) = E(I_A^2) - P(A)^2 = P(A)(1 - P(A)).$$

The technique: any count X of “how many of certain things happened” can be written $X = I_1 + I_2 + \cdots + I_n$, one indicator per thing, and Result 1 computes $E(X)$ as a sum of probabilities — no distribution table required.

Example D.1 — [JEE Main Level]

Independent random variables X and Y have $E(X) = 4$, $E(Y) = 3$, $\text{Var}(X) = 2$, $\text{Var}(Y) = 5$. Find $E(2X - 3Y + 1)$ and $\text{Var}(2X - 3Y + 1)$.

Expectation, using Result 1 twice:

$$E(2X - 3Y + 1) = 2E(X) - 3E(Y) + 1 = 8 - 9 + 1 = 0.$$

Variance, using Result 3 with the scaling rule $\text{Var}(cZ) = c^2 \text{Var}(Z)$ from the chapter:

$$\text{Var}(2X - 3Y + 1) = 4\text{Var}(X) + 9\text{Var}(Y) = 8 + 45 = 53.$$

The constant 1 drops out of the variance, and the coefficient of Y being negative costs nothing — squares erase signs. Answers: 0 and 53.

Example D.2 — [JEE Advanced Level]

n letters and their n addressed envelopes are matched at random, one letter per envelope. Let X be the number of letters placed in the correct envelope. Show that $E(X) = 1$ for every n , and show further that $\text{Var}(X) = 1$.

Writing the distribution of X directly is hopeless — the probabilities involve derangement counts. The indicator route bypasses it entirely.

Let I_k indicate “letter k lands in envelope k ,” so $X = I_1 + \cdots + I_n$. By symmetry of a random matching, letter k is equally likely to land in any of the n envelopes, hence

$$P(I_k = 1) = \frac{1}{n} \implies E(I_k) = \frac{1}{n}.$$

Linearity needs no independence:

$$E(X) = \sum_{k=1}^n E(I_k) = n \cdot \frac{1}{n} = 1.$$

Whatever the number of letters, the expected number of correct placements is exactly 1 — the surprise that makes this a favourite.

For the variance, Result 3 in general form applies, with covariance corrections since the indicators are dependent:

$$\text{Var}(X) = \sum_k \text{Var}(I_k) + 2 \sum_{k < \ell} \text{Cov}(I_k, I_\ell).$$

Each variance term is $\frac{1}{n}(1 - \frac{1}{n})$, contributing $n \cdot \frac{1}{n}(1 - \frac{1}{n}) = 1 - \frac{1}{n}$ overall. For a cross term, $E(I_k I_\ell)$ is the probability that letters k and ℓ both land correctly: fix those two envelopes, then the remaining $n - 2$ letters scatter freely, giving

$$E(I_k I_\ell) = P(\text{both correct}) = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)},$$

so

$$\text{Cov}(I_k, I_\ell) = \frac{1}{n(n-1)} - \frac{1}{n^2} = \frac{n - (n-1)}{n^2(n-1)} = \frac{1}{n^2(n-1)}.$$

There are $\binom{n}{2} = \frac{n(n-1)}{2}$ pairs, so the correction term is

$$2 \cdot \frac{n(n-1)}{2} \cdot \frac{1}{n^2(n-1)} = \frac{1}{n}.$$

Adding:

$$\text{Var}(X) = \left(1 - \frac{1}{n}\right) + \frac{1}{n} = 1.$$

Both mean and variance equal 1, independent of n . **Check for $n = 2$:** the two matchings give $X = 2$ or $X = 0$ with probability $\frac{1}{2}$ each, so $E(X) = 1$ and $\text{Var}(X) = E(X^2) - 1 = 2 - 1 = 1$ ✓. The general result agrees with the smallest case.

Practice — Addendum D

1. [JEE Main] Independent X, Y with $E(X) = 5$, $E(Y) = 2$, $\text{Var}(X) = 3$, $\text{Var}(Y) = 4$. Find $E(3X + 2Y)$ and $\text{Var}(3X + 2Y)$.
2. [JEE Main] If $\text{Var}(X) = \text{Var}(Y) = 1$ and X, Y are independent, find $\text{Var}(X - Y)$ and $\text{Var}(X + Y)$. Comment.
3. [JEE Main] A fair coin is tossed 10 times. Using indicators, find the expected number of heads and the variance, and reconcile with the Binomial formulas.
4. [JEE Advanced] X, Y are independent, $E(X) = E(Y) = 2$, $\text{Var}(X) = 1$, $\text{Var}(Y) = 4$. Find $E(XY)$ and $\text{Var}(XY)$.
5. [JEE Advanced] In the letters-and-envelopes setting with $n = 4$, list the possible values of X and explain why $X = n - 1$ is impossible for $n \geq 1$.

Answers: 1. $E = 15 + 4 = 19$; $\text{Var} = 9 \cdot 3 + 4 \cdot 4 = 27 + 16 = 43$. 2. Both equal 2: $\text{Var}(X - Y) = 1 + 1$ since the sign dies in the square; a difference of two unit-variance independents spreads exactly as much as their sum. 3. $X = I_1 + \dots + I_{10}$, each $E(I_k) = \frac{1}{2}$, $\text{Var}(I_k) = \frac{1}{4}$, and the indicators are independent; so $E(X) = 5$, $\text{Var}(X) = 10 \cdot \frac{1}{4} = \frac{5}{2}$, matching $np = 10 \cdot \frac{1}{2}$ and $npq = 10 \cdot \frac{1}{2} \cdot \frac{1}{2}$. The Binomial formulas are this indicator computation done once and for all. 4. $E(XY) = E(X)E(Y) = 4$. For the variance, independence also gives $E(X^2 Y^2) = E(X^2)E(Y^2) = 5 \cdot 8 = 40$, so $\text{Var}(XY) = 40 - 16 = 24$. (Note: $\text{Var}(XY) \neq \text{Var}(X)\text{Var}(Y)$ — a favourite distractor.) 5. Values 0, 1, 2, 4. $X = 3$ is impossible: if exactly three letters

sit correctly, the fourth letter faces the only remaining envelope — which must then be its own, forcing $X = 4$. Generally $P(X = n - 1) = 0$, a fact that survives in every derangement argument.

Addendum E — Waiting Times and Turn-Taking: Probability by Infinite Series

“When does the first success happen?” and “who wins if we alternate?” are not Binomial questions — the number of trials is unbounded, so there is no fixed n . They are geometric-series questions, and JEE Main sets them nearly every year under the guise of games, queues and repeated attempts. CBSE rarely asks them; treat this as the JEE-facing capstone.

Method

Setup. Repeat a Bernoulli trial — success probability p , failure $q = 1 - p$ — until the first success. Let X be the trial index of that first success. For $X = k$, the first $k - 1$ trials fail and trial k succeeds; independence multiplies:

$$P(X = k) = q^{k-1}p, \quad k = 1, 2, 3, \dots$$

Validity — the series sums to 1. This is a geometric series with first term p and common ratio q , where $0 < q < 1$:

$$\sum_{k=1}^{\infty} q^{k-1}p = p(1 + q + q^2 + \dots) = \frac{p}{1-q} = \frac{p}{p} = 1. \checkmark$$

So the probabilities genuinely form a distribution on infinitely many values — the infinite analogue of the $\sum p_i = 1$ check.

Mean — an arithmetico-geometric series. We need the auxiliary identity

$$\sum_{k=1}^{\infty} k r^{k-1} = \frac{1}{(1-r)^2},$$

which follows by differentiating the geometric sum $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$ term by term (legitimate for $|r| < 1$). Then

$$E(X) = \sum_{k=1}^{\infty} k q^{k-1}p = p \cdot \frac{1}{(1-q)^2} = \frac{p}{p^2} = \frac{1}{p}.$$

The expected waiting time is the reciprocal of the success probability: wait 2 rolls for a six ($p = \frac{1}{6}$), 4 tosses for a head with a coin landing heads a quarter of the time.

Variance. Split $k^2 = k(k-1) + k$. Differentiating the geometric sum twice gives

$$\sum_{k=1}^{\infty} k(k-1) r^{k-1} = \frac{2r}{(1-r)^3},$$

so

$$E(X^2) = p \left[\frac{2q}{(1-q)^3} + \frac{1}{(1-q)^2} \right] = p \left[\frac{2q}{p^3} + \frac{1}{p^2} \right] = \frac{2q}{p^2} + \frac{1}{p}.$$

Therefore

$$\text{Var}(X) = \frac{2q}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{2q + p - 1}{p^2} = \frac{2q - q}{p^2} = \frac{q}{p^2},$$

using $p - 1 = -q$ in the last step. Record: mean $\frac{1}{p}$, variance $\frac{q}{p^2}$.

Turn-taking games. Players A and B alternate attempts, A first; each attempt of A succeeds with probability a , of B with probability b (independent across attempts). The first success wins. Condition on round structure: A wins on his m -th attempt only if the previous $m - 1$ rounds were all failures, an event of probability

$$[(1-a)(1-b)]^{m-1} a.$$

Summing the geometric progression with ratio $r = (1-a)(1-b)$:

$$P(A \text{ wins}) = a(1 + r + r^2 + \dots) = \frac{a}{1 - (1-a)(1-b)},$$

and since someone eventually wins (the complementary series for B completes the sum to 1),

$$P(B \text{ wins}) = \frac{(1-a)b}{1 - (1-a)(1-b)}.$$

First-mover advantage is visible in the formula: A 's numerator carries no failure factor, B 's carries $(1-a)$.

Example E.1 — [JEE Main Level]

A fair die is rolled repeatedly until a six appears. Find (i) the probability the first six appears on the 4th roll, (ii) the expected number of rolls, (iii) the probability that the first six appears after the 3rd roll.

Here $p = \frac{1}{6}$, $q = \frac{5}{6}$.

(i)

$$P(X = 4) = \left(\frac{5}{6}\right)^3 \cdot \frac{1}{6} = \frac{125}{1296}.$$

(ii)

$$E(X) = \frac{1}{p} = 6 \text{ rolls.}$$

(iii) “After the 3rd” is the complement of “on roll 1, 2, or 3”:

$$P(X > 3) = 1 - \left[\frac{1}{6} + \frac{5}{6} \cdot \frac{1}{6} + \left(\frac{5}{6}\right)^2 \cdot \frac{1}{6} \right] = 1 - \frac{1}{6} \left[1 + \frac{5}{6} + \frac{25}{36} \right].$$

The bracket is a finite geometric sum, $\frac{36+30+25}{36} = \frac{91}{36}$, so

$$P(X > 3) = 1 - \frac{91}{216} = \frac{125}{216}.$$

Cross-check: $P(X > 3) = q^3$ directly — surviving three failures — and $\left(\frac{5}{6}\right)^3 = \frac{125}{216}$ ✓. In general $P(X > k) = q^k$; memorise this one-liner, it answers most “after how many” phrasings instantly.

Example E.2 — [JEE Advanced Level]

A and B take turns rolling a fair die, A first; the first to roll a six wins the game. Find the probability that A wins, and the expected duration of the game measured in total rolls.

With $a = b = \frac{1}{6}$, the round-failure ratio is

$$r = (1 - a)(1 - b) = \frac{5}{6} \cdot \frac{5}{6} = \frac{25}{36},$$

so

$$P(A \text{ wins}) = \frac{\frac{1}{6}}{1 - \frac{25}{36}} = \frac{\frac{1}{6}}{\frac{11}{36}} = \frac{6}{11},$$

and

$$P(B \text{ wins}) = \frac{\frac{5}{6} \cdot \frac{1}{6}}{\frac{11}{36}} = \frac{5}{11}, \quad \frac{6}{11} + \frac{5}{11} = 1. \quad \checkmark$$

The first roller holds a real edge: $\frac{6}{11} \approx 0.545$ against a fair-looking setup.

For the duration T (total rolls by both players until the game ends), note each roll independently ends the game with probability $\frac{1}{6}$ — whoever rolls, a six terminates play. So T is itself a waiting time with $p = \frac{1}{6}$, and

$$E(T) = 6 \text{ rolls.}$$

Cross-check on the win split. Condition on the first roll: with probability $\frac{1}{6}$, A wins immediately; otherwise (probability $\frac{5}{6}$) the game restarts with roles swapped, so

$$P(A \text{ wins}) = \frac{1}{6} + \frac{5}{6} P(B \text{ wins}), \quad P(A) + P(B) = 1.$$

Solving: $P(A) = \frac{1}{6} + \frac{5}{6}(1 - P(A))$, giving $\frac{11}{6}P(A) = 1$, so $P(A) = \frac{6}{11}$ ✓ — the recursive route, useful when the series is awkward to spot.

Practice — Addendum E

1. [JEE Main] A biased coin with $P(\text{head}) = \frac{1}{3}$ is tossed until the first head. Find $P(\text{first head on toss } 5)$ and the expected number of tosses.
2. [JEE Main] Cards are drawn one at a time **with replacement** from a deck until the first ace appears. Find the expected number of draws.
3. [JEE Advanced] A and B alternately toss a biased coin with $P(\text{head}) = \frac{1}{4}$, A first; first head wins. Find each player's winning probability.
4. [JEE Advanced] In the game of Example E.2, find the probability that the game ends on an odd-numbered roll.
5. [JEE Advanced] For a waiting time with success probability $p = \frac{1}{2}$, compute $\text{Var}(X)$ and verify $P(X > k) = q^k$ at $k = 2$ by direct summation.

Answers: 1. $P = \left(\frac{2}{3}\right)^4 \cdot \frac{1}{3} = \frac{16}{243}$; $E(X) = 3$. 2. $p = \frac{4}{52} = \frac{1}{13}$; expectation 13 draws. 3. Ratio $r = \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{16}$; $P(A) = \frac{\frac{1}{4}}{1 - \frac{9}{16}} = \frac{\frac{1}{4}}{\frac{7}{16}} = \frac{4}{7}$; $P(B) = \frac{\frac{3}{4} \cdot \frac{1}{4}}{\frac{7}{16}} = \frac{3}{7}$. 4. Rolls ending the game are independent trials with $p = \frac{1}{6}$; $P(T \text{ odd}) = \sum_{m \geq 0} P(T = 2m + 1) = \sum_m \left(\frac{5}{6}\right)^{2m} \frac{1}{6} = \frac{\frac{1}{6}}{1 - \frac{25}{36}} = \frac{6}{11}$ — the same series as $P(A \text{ wins})$, which is no coincidence: A wins exactly when the terminating roll is odd-numbered. 5. $\text{Var} = \frac{q}{p^2} = \frac{\frac{1}{2}}{\frac{1}{4}} = 2$. Check: $P(X > 2) = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$; directly, $P(X = 3) + P(X = 4) + \dots = \frac{1}{8} + \frac{1}{16} + \dots = \frac{\frac{1}{8}}{1 - \frac{1}{2}} = \frac{1}{4}$ ✓.

Closing note on scope. Addenda A, C and E sit partly outside the NCERT text but squarely inside the JEE Main/Advanced question bank; Addenda B and D deepen material the chapter touches but does not develop. If time is short before Boards, prioritise A (the exactly- k table appears in disguise even in CBSE three-shooter problems) and the transfer case of C; leave the variance of the geometric distribution (E) for the JEE-only pass.

Addendum P — What JEE Actually Asks

1. The Distribution of Questions Across Subtopics

The 103 previous-year questions in the bank (spanning 1978–2020) distribute across the four subtopics as follows. Only questions whose text is fully recoverable from the garbled extract are counted; corrupted items are omitted.

Subtopic	Chapter Name	Approx. Count	Representative Years / Sessions (from extract)
01	Conditional Probability & Independence	30+	1979, 1980, 1982, 1983, 1984, 1985, 1986, 1987, 1988, 1989, 1990, 1991, 1992, 1993, 1994, 1995, 1998, 1999, 2003, 2004, 2005, 2007, 2008, 2010, 2012, 2014, 2016, 2017, 2019, 2020 (Main Sep. 02, 03, 04; Jan. 07, 08, 09; Adv. 2013, 2014, 2017, 2018, 2019, 2020)
02	Total Probability & Bayes' Theorem	20+	1981, 1983, 1984, 1985, 1986, 1988, 1989, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020 (Main Sep. 02, 04; Jan. 09; Adv. 2015)

Subtopic	Chapter Name	Approx. Count	Representative Years / Sessions (from extract)
03	Random Variables & Distributions	12+	1980, 1981, 1983, 1984, 1985, 1986, 1987, 1988, 1989, 1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020 (Main Sep. 05; Jan. 09; Adv. 2014, 2017, 2019)
04	Bernoulli Trials & Binomial Distribution	25+	1978, 1980, 1981, 1982, 1983, 1984, 1985, 1986, 1987, 1988, 1989, 1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020 (Main Sep. 04, 05; Jan. 07; Adv. 2013, 2015, 2020)

Key observation:

Subtopic 01 (Conditional Probability & Independence) and Subtopic 04 (Binomial) together account for over half the questions. Subtopic 02 (Bayes) is the favourite for *long-answer* questions in both JEE Advanced and CBSE Boards. Subtopic 03 (Random Variables) appears regularly as a short-answer or integer-type question. The evidence is thick enough to state these frequencies with confidence.

2. Recurring Archetypes

Six question “shapes” reappear persistently. For each we give the visual/verbal signature, the core method, and a fully worked example reconstructed from the bank.

Archetype 1 — Bayes’ Theorem with Base-Rate Neglect

What it looks like: A two-stage experiment: a “cause” is chosen from a partition (boxes, machines, factories, diseases), then an “effect” is observed. The question asks for the posterior probability of a particular cause *given* the effect. The trap is to quote the likelihood (effect-given-cause) instead of the posterior.

Method:

1. Identify the partition $\{E_i\}$ (causes) and the observed event A (effect).
2. Write priors $P(E_i)$ and likelihoods $P(A | E_i)$.
3. Compute $P(A) = \sum_i P(E_i)P(A | E_i)$ (Total Probability).
4. Apply Bayes: $P(E_i | A) = \frac{P(E_i)P(A|E_i)}{P(A)}$.
5. **Sanity check:** Posteriors must sum to 1.

Worked Example (JEE Advanced 2016 / Q19 in bank)

> A factory has two plants T_1 and T_2 . T_1 produces 20% of computers, T_2 produces 80%. Overall 7% are defective.

> $P(\text{defective} | T_1) = 10 P(\text{defective} | T_2)$.

> A randomly selected computer is **not** defective. Find the probability it came from T_2 .

Solution

Let D = “defective”, T_1, T_2 the plants.

$P(T_1) = 0.2$, $P(T_2) = 0.8$, $P(D) = 0.07$.

Let $P(D | T_2) = x$; then $P(D | T_1) = 10x$.

Total Probability:

$$P(D) = P(T_1)P(D | T_1) + P(T_2)P(D | T_2) = 0.2(10x) + 0.8x = 2.8x = 0.07 \Rightarrow x = 0.025.$$

Hence $P(D | T_1) = 0.25$, $P(D | T_2) = 0.025$.

We need $P(T_2 | D')$:

$P(D' | T_1) = 0.75$, $P(D' | T_2) = 0.975$.

$P(D') = P(T_1)P(D' | T_1) + P(T_2)P(D' | T_2) = 0.2 \times 0.75 + 0.8 \times 0.975 = 0.15 + 0.78 = 0.93$.

$$P(T_2 | D') = \frac{P(T_2)P(D' | T_2)}{P(D')} = \frac{0.8 \times 0.975}{0.93} = \frac{0.78}{0.93} = \frac{26}{31}.$$

Answer: $\frac{26}{31}$.

Archetype 2 — Independence vs. Mutual Exclusivity & Complement Independence

What it looks like: Given probabilities of events A, B (and sometimes C), decide whether they are independent, mutually exclusive, both, or neither. A classic variant asks which of A', B or A, B' or A', B' are guaranteed independent when A, B are independent.

Method:

- Compute $P(A \cap B)$ via addition theorem: $P(A \cap B) = P(A) + P(B) - P(A \cup B)$.
- **Independence test:** $P(A \cap B) = P(A)P(B)$?
- **Mutual exclusivity test:** $P(A \cap B) = 0$? (If both probabilities > 0 , they **cannot** be both independent and mutually exclusive.)
- **Complement rule:** If $A \perp B$, then $A \perp B', A' \perp B, A' \perp B'$ (prove via $P(A \cap B') = P(A) - P(A \cap B) = P(A)(1 - P(B))$).

Worked Example (JEE Main 2016 / Q18 in bank)

- > Two fair dice A, B . E_1 : die A shows 4; E_2 : die B shows 2; E_3 : sum is odd.
- > Which statement is **NOT** true?
- > (a) E_1 and E_3 independent.
- > (b) E_1, E_2, E_3 mutually independent.
- > (c) E_1 and E_2 independent.
- > (d) E_2 and E_3 independent.

Solution

- $P(E_1) = \frac{1}{6}, P(E_2) = \frac{1}{6}, P(E_3) = \frac{1}{2}$.
- $E_1 \cap E_3$: $A = 4$ and sum odd $\Rightarrow B$ odd. $P = \frac{1}{6} \cdot \frac{3}{6} = \frac{1}{12} = P(E_1)P(E_3) \rightarrow$ independent.
 - $E_2 \cap E_3$: $B = 2$ and sum odd $\Rightarrow A$ odd. $P = \frac{1}{6} \cdot \frac{3}{6} = \frac{1}{12} = P(E_2)P(E_3) \rightarrow$ independent.
 - $E_1 \cap E_2$: different dice $\rightarrow$ independent.
 - **Mutual independence?** Need $P(E_1 \cap E_2 \cap E_3) = P(E_1)P(E_2)P(E_3)$.
 $E_1 \cap E_2 \cap E_3$: $A = 4, B = 2$, sum = 6 (even) $\rightarrow$ impossible, probability 0.
But $P(E_1)P(E_2)P(E_3) = \frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{2} = \frac{1}{72} \neq 0$.
Hence E_1, E_2, E_3 are **not** mutually independent.
- Answer:** (b) is the false statement.

Archetype 3 — Binomial Distribution: Parameter Recovery & “At Least One” via Complement

What it looks like: Either (i) given mean $\mu = np$ and variance $\sigma^2 = npq$, find n, p and then a specific probability; or (ii) “at least one success in n trials” where summing $P(X \geq 1)$ directly is a trap — use $1 - P(X = 0)$.

Method:

- $\mu = np, \sigma^2 = npq \Rightarrow q = \frac{\sigma^2}{\mu}, p = 1 - q, n = \frac{\mu}{p}$.
- $P(X \geq 1) = 1 - q^n$.
- For “exactly r ”: $P(X = r) = \binom{n}{r} p^r q^{n-r}$.

Worked Example (JEE Main 2020 Sep. 02 / Q9 in bank)

- > In a workshop, five machines each have probability $\frac{1}{4}$ of being out of service on a day. If the probability that at most two machines are out of service is $\frac{k}{512}$, find k .

Solution

$$X \sim B(5, \frac{1}{4}), q = \frac{3}{4}.$$

$$P(X \leq 2) = P(0) + P(1) + P(2)$$

$$P(0) = \binom{5}{0} \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^5 = \frac{243}{1024}$$

$$P(1) = \binom{5}{1} \left(\frac{1}{4}\right)^1 \left(\frac{3}{4}\right)^4 = 5 \cdot \frac{1}{4} \cdot \frac{81}{256} = \frac{405}{1024}$$

$$P(2) = \binom{5}{2} \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^3 = 10 \cdot \frac{1}{16} \cdot \frac{27}{64} = \frac{270}{1024}$$

$$\text{Sum} = \frac{243+405+270}{1024} = \frac{918}{1024} = \frac{459}{512}.$$

Thus $k = 459$.

Answer: 459.

Archetype 4 — Conditional Probability in Sequential Draws Without Replacement

What it looks like: Balls are drawn sequentially without replacement; a condition is given on early draws (or a transfer occurs), and we need a probability for a later draw. The sample space shrinks at each step.

Method:

- Use the chain rule / multiplication theorem: $P(E_1 \cap E_2 \cap \dots) = P(E_1)P(E_2 | E_1) \dots$
- For “given that ...”, restrict to the reduced sample space after the conditioning event.

Worked Example (JEE Main 2019 Jan. 09 / Q14 in bank)

> An urn contains 5 red and 2 green balls. A ball is drawn at random. If it is green, a red ball is added; if it is red, a green ball is added. The original ball is **not** returned. Now a second ball is drawn. Find the probability the second ball is red.

Solution

Let G_1, R_1 be first draw events; R_2 = second ball red.

$$P(G_1) = \frac{2}{7}, P(R_1) = \frac{5}{7}.$$

- If G_1 : green removed, red added $\rightarrow$ urn has 6 red, 1 green. $P(R_2 | G_1) = \frac{6}{7}$.
- If R_1 : red removed, green added $\rightarrow$ urn has 4 red, 3 green. $P(R_2 | R_1) = \frac{4}{7}$.

By Total Probability:

$$P(R_2) = P(G_1)P(R_2 | G_1) + P(R_1)P(R_2 | R_1) = \frac{2}{7} \cdot \frac{6}{7} + \frac{5}{7} \cdot \frac{4}{7} = \frac{12}{49} + \frac{20}{49} = \frac{32}{49}.$$

Answer: $\frac{32}{49}$.

Archetype 5 — Random Variables: Constructing the Distribution, Mean, Variance, Linear Transforms

What it looks like: A random variable X is defined by a formula (e.g., $P(X = x) = kx$) or by a physical experiment (drawing cards, tossing coins). You must build the probability table, find k from $\sum p_i = 1$, then compute $E(X)$, $E(X^2)$, $\text{Var}(X)$, and often $E(aX + b)$, $\text{Var}(aX + b)$.

Method:

- List all possible values x_i and their probabilities p_i (using counting or given formula).

- Verify $\sum p_i = 1$.
- Use a 4-column table: $x_i, p_i, x_i p_i, x_i^2 p_i$. Sum third column $\rightarrow E(X)$; fourth $\rightarrow E(X^2)$.
- $\text{Var}(X) = E(X^2) - [E(X)]^2$.
- Linear rules: $E(aX + b) = aE(X) + b$; $\text{Var}(aX + b) = a^2 \text{Var}(X)$.

Worked Example (JEE Main 2020 Jan. 09 / Q1 in second set)

> Random variable X takes values 1, 2, 3, 4, 5 with probabilities $K^2, 2K, K, 2K, 5K^2$. Find $P(X > 2)$.

Solution

$$\sum p_i = K^2 + 2K + K + 2K + 5K^2 = 6K^2 + 5K = 1$$

$$\Rightarrow 6K^2 + 5K - 1 = 0 \Rightarrow (6K - 1)(K + 1) = 0 \Rightarrow K = \frac{1}{6} \text{ (} K = -1 \text{ rejected)}.$$

$$\text{Probabilities: } p_1 = \frac{1}{36}, p_2 = \frac{1}{3}, p_3 = \frac{1}{6}, p_4 = \frac{1}{3}, p_5 = \frac{5}{36}.$$

$$P(X > 2) = p_3 + p_4 + p_5 = \frac{1}{6} + \frac{1}{3} + \frac{5}{36} = \frac{6+12+5}{36} = \frac{23}{36}.$$

Answer: $\frac{23}{36}$.

Archetype 6 — Geometric / Negative Binomial: Waiting for the r -th Success

What it looks like: Trials continue until a certain number of successes occur. “A throws a die until a 6 appears”, “draw cards with replacement until second ace”, “toss a coin until two heads”, etc.

Method:

- **Geometric ($r = 1$):** $P(\text{first success on trial } n) = q^{n-1}p$.
- **Negative Binomial ($r \geq 2$):** $P(r\text{-th success on trial } n) = \binom{n-1}{r-1} p^r q^{n-r}$.
- For “success occurs on an even trial”, sum the geometric series: $\sum_{k=1}^{\infty} q^{2k-1}p = \frac{qp}{1-q^2}$.

Worked Example (JEE 2005 / Q27 in bank)

> A fair die is thrown until a 1 appears. Find the probability that 1 comes on an **even** number of trials.

Solution

$$p = \frac{1}{6}, q = \frac{5}{6}.$$

$$\text{Required probability} = qp + q^3p + q^5p + \dots = qp(1 + q^2 + q^4 + \dots)$$

$$= \frac{qp}{1-q^2} = \frac{\frac{5}{6} \cdot \frac{1}{6}}{1 - \frac{25}{36}} = \frac{5/36}{11/36} = \frac{5}{11}.$$

Answer: $\frac{5}{11}$.

3. The Traps

The following mistake patterns are explicitly engineered into JEE distractors. They are drawn from the chapter’s own pitfall tags and the distractor patterns visible in the bank.

#	Description	Where It Appears in the Bank
1.	Treating “independent” and “mutually exclusive” as synonyms. For $P(A), P(B) > 0$ they are opposites : mutually exclusive $\Rightarrow P(A \cap B) = 0$, independent $\Rightarrow P(A \cap B) = P(A)P(B) > 0$.	Q7, Q18, Q20, Q38, Q59, Q60, Q62, Q92
2.	Swapping $P(A B)$ and $P(B A)$. The denominator changes from $P(B)$ to $P(A)$.	Q2, Q25, Q41, Q75, Q81
3.	Assuming pairwise independence of three events implies mutual independence. The triple product $P(A \cap B \cap C) = P(A)P(B)P(C)$ must be checked separately.	Q31, Q92
4.	Base-rate neglect. Quoting the likelihood $P(\text{effect} \text{cause})$ as the posterior $P(\text{cause} \text{effect})$. The classic “95% accurate test” trap.	Q14, Q19, Q22, Q48, Q73, Q74, Q81, Q93, Q98
5.	Dividing by a single $P(E_i)P(A E_i)$ instead of the full sum $P(A)$.	Q4, Q67, Q68, Q71, Q72
6.	Using “causes” that are not disjoint or not exhaustive. If $\sum P(E_i) \neq 1$, Total Probability fails.	Q67, Q68, Q78
7.	Applying the binomial formula to draws without replacement (where p shifts and trials are dependent).	Q6 (with replacement is binomial), Q79, Q103 (without replacement)
8.	Swapping exponents: writing $p^{n-r}q^r$ instead of $p^r q^{n-r}$.	Q10 (explicit check of α, β), Q2 (second set)

#	Description	Where It Appears in the Bank
9.	Believing variance can exceed the mean for a binomial. Since $q < 1$, $\sigma^2 = npq < np = \mu$ always. Inconsistent data $\rightarrow$ non-binomial or error.	Q2 (second set), Q9 (second set)
10.	Summing $P(X = r)$ for $r = k$ to n instead of using the complement $1 - P(X < k)$.	Q11, Q13, Q36 (first set), Q9, Q10, Q12, Q13 (second set)
11.	Computing plain average $\frac{\sum x_i}{n}$ instead of $\sum x_i p_i$.	Q8 (first set), Q33 (first set)
12.	Forgetting to square the mean: $\text{Var} = E(X^2) - E(X)$ instead of $E(X^2) - [E(X)]^2$.	Q2 (second set)
13.	Not checking $\sum p_i = 1$ when a distribution contains an unknown constant k ; accepting a k that yields negative probabilities.	Q1 (second set)
14.	Ignoring the given condition and computing an unconditional probability (e.g., “at least one girl” vs “elder is a girl”).	Q2 (first set, Q2 in Crack MCQ), Q12, Q45

4. Graded Practice Set

The following 10 questions are adapted from the bank to be fully self-contained and are arranged in ascending difficulty. Every answer has been re-solved and verified.

1. (Basic Conditional) — JEE Main 2020 Sep.03 style

A fair die is rolled twice. Given that the sum of the two rolls is a multiple of 4, find the probability that the score 4 appears at least once.

Solution

Sample space: 36 ordered pairs.

F = sum multiple of 4: sums $\{4, 8, 12\}$.

$F = \{(1, 3), (2, 2), (3, 1), (2, 6), (3, 5), (4, 4), (5, 3), (6, 2), (6, 6)\} \rightarrow 9$ outcomes.

E = at least one 4: $\{(4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (1, 4), (2, 4), (3, 4), (5, 4), (6, 4)\} \rightarrow 11$ outcomes.

$E \cap F = \{(4, 4)\}$ only.

$$P(E | F) = \frac{|E \cap F|}{|F|} = \frac{1}{9}.$$

Answer: $\frac{1}{9}$.

2. (Independence Check) — JEE Main 2014 style

For events A, B : $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cup B) = 0.7$. Determine whether A, B are independent, mutually exclusive, both, or neither.

Solution

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.5 + 0.4 - 0.7 = 0.2.$$

$$P(A)P(B) = 0.5 \times 0.4 = 0.2.$$

Since $P(A \cap B) = P(A)P(B)$, A, B are **independent**.

Mutually exclusive would require $P(A \cap B) = 0$, but it is $0.2 \neq 0$.

Answer: Independent only.

3. (Bayes with Two Causes) — JEE Main 2020 Sep. 02 style

Box I has cards numbered 1–30; Box II has cards 31–50. A box is chosen at random, a card is drawn and found to be a **non-prime** number. Find the probability the card came from Box I.

Solution

Primes in 1 ... 30: 10 $\rightarrow$ non-primes 20.

Primes in 31 ... 50: 5 $\rightarrow$ non-primes 15.

Priors: $P(\text{Box I}) = P(\text{Box II}) = \frac{1}{2}$.

Likelihoods: $P(\text{non-prime} | \text{I}) = \frac{20}{30} = \frac{2}{3}$; $P(\text{non-prime} | \text{II}) = \frac{15}{20} = \frac{3}{4}$.

$$P(\text{non-prime}) = \frac{1}{2} \cdot \frac{2}{3} + \frac{1}{2} \cdot \frac{3}{4} = \frac{1}{3} + \frac{3}{8} = \frac{17}{24}.$$

$$P(\text{I} | \text{non-prime}) = \frac{\frac{1}{2} \cdot \frac{2}{3}}{\frac{17}{24}} = \frac{1/3}{17/24} = \frac{8}{17}.$$

Answer: $\frac{8}{17}$.

4. (Binomial: Parameter Recovery) — JEE Main 2019 Apr. 12 style

A random variable $X \sim B(n, p)$ has mean 8 and variance 4. Find n, p and $P(X \leq 2)$.

Solution

$$np = 8, npq = 4 \Rightarrow q = \frac{4}{8} = \frac{1}{2}, p = \frac{1}{2}, n = 16.$$

$$P(X \leq 2) = \sum_{r=0}^2 \binom{16}{r} \left(\frac{1}{2}\right)^{16} = \frac{\binom{16}{0} + \binom{16}{1} + \binom{16}{2}}{2^{16}} = \frac{1 + 16 + 120}{65536} = \frac{137}{65536}.$$

Answer: $n = 16$, $p = \frac{1}{2}$, $P(X \leq 2) = \frac{137}{65536}$.

5. (Sequential Without Replacement) — JEE Main 2019 Jan. 09 style

An urn contains 5 red and 2 green balls. A ball is drawn. If it is green, a red ball is added; if red, a green ball is added. The first ball is **not** returned. A second ball is drawn. Find the probability the second ball is red.

Solution

Identical to Archetype 4 worked example.

$$P(\text{2nd red}) = \frac{32}{49}.$$

Answer: $\frac{32}{49}.$

6. (Random Variable: Formula-Defined Distribution) — JEE Main 2020 Jan. 09 style

X takes values 1, 2, 3, 4, 5 with $P(X = x) = cx^2$. Find c and $P(X \geq 4)$.

Solution

$$\sum_{x=1}^5 cx^2 = c(1 + 4 + 9 + 16 + 25) = 55c = 1 \Rightarrow c = \frac{1}{55}.$$

$$P(X \geq 4) = P(4) + P(5) = \frac{16}{55} + \frac{25}{55} = \frac{41}{55}.$$

Answer: $c = \frac{1}{55}, P(X \geq 4) = \frac{41}{55}.$

7. (Geometric Series: Even-Trial Success) — JEE 2005 style

A fair coin is tossed until the first head appears. Find the probability that the first head occurs on an **even**-numbered toss.

Solution

$$p = \frac{1}{2}, q = \frac{1}{2}.$$

$$P(\text{even trial}) = qp + q^3p + q^5p + \cdots = \frac{qp}{1-q^2} = \frac{\frac{1}{4}}{1-\frac{1}{4}} = \frac{1}{3}.$$

Answer: $\frac{1}{3}.$

8. (Total Probability + Bayes: Two-Stage Urn) — JEE 1988 / 2011 style

Urn U_1 has 3 white, 2 red balls. Urn U_2 has 1 white ball. A fair coin is tossed. If heads, one ball is drawn from U_1 and put into U_2 ; if tails, two balls are drawn from U_1 and put into U_2 . Then one ball is drawn from U_2 .

(a) Find the probability the drawn ball is white.

(b) Given the drawn ball is white, find the probability the coin showed heads.

Solution

Case H (heads, prob 1/2): Transfer 1 ball from U_1 (3W,2R).

$$P(\text{transfer W}) = \frac{3}{5}, \text{ then } U_2 \text{ has 2W} \rightarrow P(W | H, \text{trans W}) = 1.$$

$$P(\text{transfer R}) = \frac{2}{5}, \text{ then } U_2 \text{ has 1W, 1R} \rightarrow P(W | H, \text{trans R}) = \frac{1}{2}.$$

$$P(W | H) = \frac{3}{5} \cdot 1 + \frac{2}{5} \cdot \frac{1}{2} = \frac{4}{5}.$$

Case T (tails, prob 1/2): Transfer 2 balls from U_1 .

Possibilities: $2W$ ($\frac{\binom{3}{2}}{\binom{5}{2}} = \frac{3}{10}$), $1W1R$ ($\frac{\binom{3}{1}\binom{2}{1}}{\binom{5}{2}} = \frac{6}{10}$), $2R$ ($\frac{1}{10}$).

U_2 initially $1W$. After transfer:

$2W \rightarrow 3W \rightarrow P(W) = 1$;

$1W1R \rightarrow 2W, 1R \rightarrow P(W) = \frac{2}{3}$;

$2R \rightarrow 1W, 2R \rightarrow P(W) = \frac{1}{3}$.

$$P(W | T) = \frac{3}{10} \cdot 1 + \frac{6}{10} \cdot \frac{2}{3} + \frac{1}{10} \cdot \frac{1}{3} = \frac{3}{10} + \frac{12}{30} + \frac{1}{30} = \frac{9+12+1}{30} = \frac{22}{30} = \frac{11}{15}.$$

Total Probability:

$$P(W) = \frac{1}{2} \cdot \frac{4}{5} + \frac{1}{2} \cdot \frac{11}{15} = \frac{2}{5} + \frac{11}{30} = \frac{12+11}{30} = \frac{23}{30}.$$

Bayes for (b):

$$P(H | W) = \frac{P(H)P(W|H)}{P(W)} = \frac{\frac{1}{2} \cdot \frac{4}{5}}{\frac{23}{30}} = \frac{2/5}{23/30} = \frac{12}{23}.$$

Answers: (a) $\frac{23}{30}$; (b) $\frac{12}{23}$.

9. (Combinatorial Conditional: Derangements) — JEE Advanced 2018 style

Five students $S_1, \dots, S_5$ are randomly allotted five seats $R_1, \dots, R_5$ (initially S_i gets R_i). Find the probability that S_1 gets R_1 and **none** of S_2, S_3, S_4, S_5 get their own seats.

Solution

Total permutations: $5! = 120$.

Fix $S_1 \rightarrow R_1$. The remaining 4 students must be **deranged** (no fixed points).

Number of derangements of 4: $!4 = 4! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!}\right) = 24 \left(1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24}\right) = 9$.

$$P = \frac{9}{120} = \frac{3}{40}.$$

Answer: $\frac{3}{40}$.

10. (Advanced Binomial + Conditional: Negative Binomial) — JEE Advanced 2020 style

Two fair dice are rolled repeatedly until the sum is either a **prime** or a **perfect square**. Given that a perfect square appears **before** any prime, find the probability that this perfect square is **odd** (i.e., 9).

Solution

Prime sums: $2, 3, 5, 7, 11 \rightarrow 15$ outcomes.

Perfect squares: $4, 9 \rightarrow 4$: 3 outcomes; 9 : 4 outcomes. Total square outcomes = 7.

The process stops at the first roll that is either prime or square. The conditional probability that the stopping square is 9 given that a square occurred before a prime is simply the proportion of 9 among all squares:

$$P(\text{square is 9} | \text{square before prime}) = \frac{\text{ways for 9}}{\text{ways for 4} + \text{ways for 9}} = \frac{4}{3+4} = \frac{4}{7}.$$

(The geometric “race” between prime and square cancels out because the relative odds are the same on every roll.)

Answer: $\frac{4}{7}$.

5. What This Chapter's Text Under-Serves

The chapter text gives thorough coverage of **formulas and standard theorems** but under-serves two critical JEE skills:

1. **Recognising when a problem is *not* binomial** — the text states the “with/without replacement” rule but does not drill the many JEE questions that *look* binomial (fixed n , two outcomes) but are actually hypergeometric or require conditional probability chains (e.g., Q14, Q79, Q103).

2. **Manipulating conditional probability algebraically** — the text proves $P(A | B) \geq P(A \cap B)/P(B)$ etc., but JEE Advanced often requires deriving inequalities like $P(A | B) \geq P(A) + P(B) - 1$ (Q61) or handling chained conditioning $P(E_1 | E_2 \cap E_3)$ without a ready-made formula. Students need practice turning English statements into set-algebra identities and simplifying them.

Errata

This section lists the errors identified in the printed text of the thirteen chapters of the *Class 12 Mathematics Master Reference* that could not be corrected in place — because the correction requires inserting content the printed line has no room for (the source chapter PDFs have no editable source, so in-place patches are limited to text that fits the original bounding box), or because the fix is an editorial judgement call best left to a human. Entries are ordered by chapter and then by printed page.

How to use this list. Each entry names the exact location, quotes what is printed, gives the replacement text, and shows the working, so a reader can act on the correction without consulting any other source.

Every item below was independently re-verified against the source problem statement before being listed. That re-verification also caught one claim from the automated audit that turned out to be a false positive — the Chapter 11 practice-exercise answer “ $2x + 3y + 7z = 16$ ” for the plane containing two parallel lines is in fact correct (the audit had dropped the $\hat{i}$ from the first line’s position vector $\hat{i} + 2\hat{k}$, reading the point $(1, 0, 2)$ as $(0, 0, 2)$); it is therefore *not* listed below, and the printed text stands.

Chapter 1: Relations and Functions

Page 14 — Section 4, Example 1: the defining set R is truncated mid-print

Printed: “Example 1 — [CBSE Board Level] Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), ($ Determine whether R is reflexive, symmetric, and transitive.”

The set R breaks off after “ $(1, 2), ($ — the line overflows the page and the remaining pairs were lost in production, so the example’s question is unusable as printed. (The solution that follows is complete and correct; only the statement of R is damaged.)

Should read: “Example 1 — [CBSE Board Level] Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1), (3, 4)\}$. Determine whether R is reflexive, symmetric, and transitive.”

That is, the truncated set completes with the pairs **(2, 1)** and **(3, 4)**, then closes: $R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1), (3, 4)\}$.

Why: The full set is pinned down uniquely by the example’s own solution. The reflexive step confirms the four diagonal pairs. The symmetry step requires $(2, 1) \in R$ (“ $(1, 2) \in R$ and $(2, 1) \in R$ — good”) and $(3, 4) \in R$ with $(4, 3) \notin R$ (“But $(3, 4) \in R$ while $(4, 3) \notin R$ ”). The transitivity step then checks exactly the chains these seven pairs generate — $(1,2) \& (2,1) \Rightarrow (1,1)$, $(2,1) \& (1,2) \Rightarrow (2,2)$, $(3,3) \& (3,4) \Rightarrow (3,4)$, $(3,4) \& (4,4) \Rightarrow (3,4)$ — and correctly concludes R is transitive. With the restored set, every step of the printed solution is correct as written, including

the conclusion: R is reflexive and transitive but not symmetric, hence not an equivalence relation.

The repair is not made in place because the printed line already overruns the page; restoring the missing pairs in the original spot would need more room than the line has, and the text cannot be reflowed (no editable source).

Page 43 — Section 1, Example 4: the final inequality chain runs off the page

Printed: the closing line of the domain calculation for $f(x) = \sqrt{\log_{1/2}\left(\frac{x-2}{x+1}\right)}$ runs past the right page edge and is clipped after “... $\frac{-3}{x+1} \leq 0 \Rightarrow x+1 > 0 \Rightarrow x > -1$ ”.

Should read: the chain ends “ $\Rightarrow x+1 > 0 \Rightarrow x > -1$.”

Why: Only the final “ -1 ” is lost, and the very next line restates the result (“Intersect (i) and (ii): $\{x < -1 \text{ or } x > 2\} \cap \{x > -1\}$ ”), so the conclusion $\text{Dom}(f) = (2, \infty)$ is unaffected. Listed for completeness; not patched in place because the line is already overfull and the text cannot be reflowed (no editable source).

Chapter 3: Matrices

Page 153 — Section 4, Answers: the order list for 24 elements is incomplete

Printed: “2. 24 elements $\rightarrow 1 \times 24, 2 \times 12, 3 \times 8, 4 \times 6, 6 \times 4, 8 \times 3, = 8$ orders.”

The pair list stops at 8×3 — the last two factor pairs were lost in production — while the count, 8 orders, is stated correctly.

Should read: “2. 24 elements $\rightarrow 1 \times 24, 2 \times 12, 3 \times 8, 4 \times 6, 6 \times 4, 8 \times 3, 12 \times 2, 24 \times 1 = 8$ orders.”

Why: 24 has exactly 8 ordered factorisations into two positive factors ($24 = 2^3 \cdot 3$ has $(3+1)(1+1) = 8$ divisors, each divisor r giving the order $r \times 24/r$), and the printed count of 8 is right — only the tail of the list, 12×2 and 24×1 , is missing.

Chapter 11: Three-Dimensional Geometry

Page 678 — Section 2 formula summary: the parallel-planes line is clipped at the page edge

Printed: “Parallel planes $ax + by + cz + d_1 = 0, ax + by + cz + d_2 =$ ” — the line runs off the right page edge, clipping the final “0”.

Should read: “Parallel planes $ax + by + cz + d_1 = 0, ax + by + cz + d_2 = 0$.”

Why: Only the final “0” of the second equation is lost, and the parallel-planes distance formula that the line introduces is printed in full immediately to its left. Not patched in place because the line is already overfull and cannot be reflowed.

Page 693 — Section 2, “Plane containing given data”: the parallel-lines normal formula is clipped at the page edge

Printed: “two parallel lines: $\vec{N} = \vec{b} \times (\vec{a}_2 - \vec{a}_1$ ” — the closing parenthesis runs off the right page edge.

Should read: “two parallel lines: $\vec{N} = \vec{b} \times (\vec{a}_2 - \vec{a}_1)$ ” (the plane through two parallel lines with common direction $\vec{b}$, passing through points $\vec{a}_1$ and $\vec{a}_2$, has normal $\vec{b} \times (\vec{a}_2 - \vec{a}_1)$).

Why: Only the closing parenthesis is lost; the formula itself is complete and correct as printed. Listed for completeness.